

6D SUGRA Bases — Non-unimodular ($|\det| > 1$)

Total SUGRA bases: 25241 (from 2990 distinct LST bases, $T_{\text{LST}} \in [0, 20]$).
Filters: NHC, $c(k) \leq 8$, $H_n \geq 0$. Deduplicated by graph invariants.
Red = attached external curve. $[\# = n]$ = intersection number n . $\Delta = H_{\text{charged}} - V + 29T - 273 \leq 0$.
Each entry shows the b_0Q extended matrix after blowdown.

T	Count
2	8
3	26
4	67
5	159
6	248
7	371
8	572
9	875
10	1084
11	1218
12	1502
13	1666
14	1842
15	1810
16	2131
17	2318
18	2261
19	2286
20	2277
21	2520

$T = 2$ (8 bases)

Base: 11

$$\begin{aligned}
11\bar{3}_{[\# = 9]} \quad \Delta = -223, n_+ = 1, n_- = 2, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
11\bar{3}_{[\# = 8]} \quad \Delta = -223, n_+ = 1, n_- = 2, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
11\bar{3}_{[\# = 6]} \quad \Delta = -223, n_+ = 1, n_- = 2, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
11\bar{3}_{[\# = 5]} \quad \Delta = -223, n_+ = 1, n_- = 2, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
11\bar{3}_{[\# = 4]} \quad \Delta = -223, n_+ = 1, n_- = 2, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
11\bar{3}_{[\# = 3]} \quad \Delta = -223, n_+ = 1, n_- = 2, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
11\bar{3}_{[\# = 2]} \quad \Delta = -223, n_+ = 1, n_- = 2, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
11\bar{3}_{[\# = 7]} \quad \Delta = -223, n_+ = 1, n_- = 2, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$T = 3$ (26 bases)

Base: 121

$$\begin{aligned}
121\bar{3}_{[\# = 5]} \quad \Delta = -194, n_+ = 1, n_- = 3, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
121\bar{3}_{[\# = 2]} \quad \Delta = -194, n_+ = 1, n_- = 3, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
121\bar{3}_{[\# = 3]} \quad \Delta = -194, n_+ = 1, n_- = 3, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
121\bar{3}_{[\# = 4]} \quad \Delta = -194, n_+ = 1, n_- = 3, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
121\bar{3}_{[\# = 6]} \quad \Delta = -194, n_+ = 1, n_- = 3, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
121\bar{3}_{[\# = 7]} \quad \Delta = -194, n_+ = 1, n_- = 3, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
121\bar{3}_{[\# = 8]} \quad \Delta = -194, n_+ = 1, n_- = 3, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
121\bar{3}_{[\# = 9]} \quad \Delta = -194, n_+ = 1, n_- = 3, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
\overset{4}{121}_{[\# = 2]} \quad \Delta = -113, n_+ = 1, n_- = 3, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 0 \\ 2 & 0 & 8 \end{pmatrix}
\end{aligned}$$

Base: 212

$$212 \overset{3}{[\# = 4]} \quad \Delta = -194, n_+ = 1, n_- = 3, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$212 \overset{4}{[\# = 2]} \quad \Delta = -113, n_+ = 1, n_- = 3, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 0 \\ 2 & 0 & 8 \end{pmatrix}$$

$$212 \overset{12}{\quad} \Delta = -434, n_+ = 1, n_- = 3, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -10 & -8 \\ 2 & -8 & 8 \end{pmatrix}$$

$$212 \overset{8}{\quad} \Delta = -319, n_+ = 1, n_- = 3, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -6 & -4 \\ 2 & -4 & 8 \end{pmatrix}$$

$$212 \overset{7}{\quad} \Delta = -291, n_+ = 1, n_- = 3, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -5 & -3 \\ 2 & -3 & 8 \end{pmatrix}$$

$$212 \overset{6}{\quad} \Delta = -264, n_+ = 1, n_- = 3, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -4 & -2 \\ 2 & -2 & 8 \end{pmatrix}$$

$$212 \overset{5}{\quad} \Delta = -238, n_+ = 1, n_- = 3, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$212 \overset{4}{\quad} \Delta = -214, n_+ = 1, n_- = 3, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 8 \end{pmatrix}$$

$$212 \overset{2}{\quad} \Delta = -186, n_+ = 1, n_- = 3, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$212 \overset{3}{[\# = 7]} \quad \Delta = -194, n_+ = 1, n_- = 3, n_0 = 0, |\det| = 196 \quad \begin{pmatrix} 0 & 14 & 2 \\ 14 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$212 \overset{3}{[\# = 8]} \quad \Delta = -194, n_+ = 1, n_- = 3, n_0 = 0, |\det| = 256 \quad \begin{pmatrix} 0 & 16 & 2 \\ 16 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$212 \overset{3}{[\# = 9]} \quad \Delta = -194, n_+ = 1, n_- = 3, n_0 = 0, |\det| = 324 \quad \begin{pmatrix} 0 & 18 & 2 \\ 18 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$212 \overset{3}{[\# = 2]} \quad \Delta = -194, n_+ = 1, n_- = 3, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$212 \overset{3}{\quad} \Delta = -194, n_+ = 1, n_- = 3, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$212 \overset{3}{[\# = 6]} \quad \Delta = -194, n_+ = 1, n_- = 3, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$212 \overset{3}{[\# = 3]} \quad \Delta = -194, n_+ = 1, n_- = 3, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$212 \overset{3}{[\# = 5]} \quad \Delta = -194, n_+ = 1, n_- = 3, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$T = 4$ (67 bases)

Base: 1221

$$1221 \overset{3}{[\# = 6]} \quad \Delta = -165, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$\begin{aligned}
1221\mathbf{3}_{[\#=3]} \quad \Delta = -165, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1221\mathbf{3}_{[\#=2]} \quad \Delta = -165, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1221\mathbf{3}_{[\#=4]} \quad \Delta = -165, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1221\mathbf{3}_{[\#=9]} \quad \Delta = -165, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
1221\mathbf{3}_{[\#=8]} \quad \Delta = -165, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1221\mathbf{3}_{[\#=7]} \quad \Delta = -165, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
1221\mathbf{3}_{[\#=5]} \quad \Delta = -165, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{2}{122}$

$$\begin{aligned}
\overset{2}{221}\mathbf{7} \quad \Delta = -262, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -4 & -2 \\ 2 & -2 & 8 \end{pmatrix} \\
\overset{2}{221}\mathbf{12} \quad \Delta = -405, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -9 & -7 \\ 2 & -7 & 8 \end{pmatrix} \\
\overset{2}{221}\mathbf{8} \quad \Delta = -290, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -5 & -3 \\ 2 & -3 & 8 \end{pmatrix} \\
\overset{2}{221}\mathbf{6} \quad \Delta = -235, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
\overset{2}{221}\mathbf{5} \quad \Delta = -209, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 8 \end{pmatrix} \\
\overset{2}{221}\mathbf{4} \quad \Delta = -185, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
\overset{2}{221}\mathbf{3}_{[\#=2]} \quad \Delta = -165, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\overset{2}{221}\mathbf{3}_{[\#=6]} \quad \Delta = -165, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 144 & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
\overset{2}{221}\mathbf{2} \quad \Delta = -157, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\overset{2}{221}\mathbf{3}_{[\#=4]} \quad \Delta = -165, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\overset{2}{221}\mathbf{3}_{[\#=9]} \quad \Delta = -165, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 324 & \begin{pmatrix} 0 & 18 & 2 \\ 18 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
\overset{2}{221}\mathbf{3} \quad \Delta = -165, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
{}^2_{2213} \text{ }_{[\# = 3]} \quad \Delta = -165, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
{}^2_{2213} \text{ }_{[\# = 7]} \quad \Delta = -165, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 196 & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
{}^2_{2213} \text{ }_{[\# = 5]} \quad \Delta = -165, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
{}^2_{2213} \text{ }_{[\# = 8]} \quad \Delta = -165, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 256 & \begin{pmatrix} 0 & 16 & 2 \\ 16 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix}
\end{aligned}$$

Base: 2213

$$\begin{aligned}
{}^3_{2213} \text{ }_{[\# = 3]} \quad \Delta = -173, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
{}^3_{2213} \quad \Delta = -174, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
{}^4_{2213} \quad \Delta = -193, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 9 & \begin{pmatrix} 9 & 5 \\ 5 & 9 \end{pmatrix} \\
{}^5_{2213} \quad \Delta = -217, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & -2 & 0 \\ 2 & 0 & 8 \end{pmatrix} \\
{}^2_{2213} \quad \Delta = -165, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
{}^6_{2213} \quad \Delta = -243, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
{}^3_{2213} \quad \Delta = -173, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
{}^3_{2213} \text{ }_{[\# = 9]} \quad \Delta = -173, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 729 & \begin{pmatrix} 0 & 27 & 2 \\ 27 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
{}^3_{2213} \text{ }_{[\# = 8]} \quad \Delta = -173, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 576 & \begin{pmatrix} 0 & 24 & 2 \\ 24 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
{}^3_{2213} \text{ }_{[\# = 7]} \quad \Delta = -173, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 441 & \begin{pmatrix} 0 & 21 & 2 \\ 21 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
{}^3_{2213} \text{ }_{[\# = 6]} \quad \Delta = -173, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 324 & \begin{pmatrix} 0 & 18 & 2 \\ 18 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
{}^3_{2213} \text{ }_{[\# = 5]} \quad \Delta = -173, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 225 & \begin{pmatrix} 0 & 15 & 2 \\ 15 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
{}^3_{2213} \text{ }_{[\# = 4]} \quad \Delta = -173, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 144 & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
{}^3_{2213} \text{ }_{[\# = 2]} \quad \Delta = -173, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1312

$${}^6_{1312} \quad \Delta = -243, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -4 & -2 \\ 2 & -2 & 8 \end{pmatrix}$$

$$\begin{aligned}
1312^5 \quad \Delta = -217, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
1312^4 \quad \Delta = -193, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 8 \end{pmatrix} \\
1312^3 \quad \Delta = -173, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
21313_{[\# = 7]} \quad \Delta = -173, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
21313_{[\# = 8]} \quad \Delta = -173, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1312^2 \quad \Delta = -165, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
21313_{[\# = 5]} \quad \Delta = -173, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
21313_{[\# = 9]} \quad \Delta = -173, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1312^3_{[\# = 5]} \quad \Delta = -173, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1312^3_{[\# = 8]} \quad \Delta = -173, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 256 & \begin{pmatrix} 0 & 16 & 2 \\ 16 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
21313_{[\# = 4]} \quad \Delta = -173, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1312^3_{[\# = 4]} \quad \Delta = -173, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21313_{[\# = 3]} \quad \Delta = -173, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1312^3_{[\# = 3]} \quad \Delta = -173, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
21313_{[\# = 2]} \quad \Delta = -173, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1312^3_{[\# = 2]} \quad \Delta = -173, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1312^3_{[\# = 6]} \quad \Delta = -173, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 144 & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1312^3_{[\# = 7]} \quad \Delta = -173, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 196 & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
21313_{[\# = 6]} \quad \Delta = -173, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

$$1312^{\textcolor{red}{3}}_{[\#=9]} \quad \Delta = -173, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 324 \quad \begin{pmatrix} 0 & 18 & 2 \\ 18 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

Base: 131^1

$$131^1\textcolor{red}{3}_{[\#=4]} \quad \Delta = -173, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$131^1\textcolor{red}{3}_{[\#=5]} \quad \Delta = -173, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$131^1\textcolor{red}{3}_{[\#=7]} \quad \Delta = -173, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$131^1\textcolor{red}{3}_{[\#=8]} \quad \Delta = -173, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$131^1\textcolor{red}{3}_{[\#=9]} \quad \Delta = -173, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$131^1\textcolor{red}{3}_{[\#=2]} \quad \Delta = -173, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$131^1\textcolor{red}{3}_{[\#=3]} \quad \Delta = -173, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$131^1\textcolor{red}{3}_{[\#=6]} \quad \Delta = -173, n_+ = 1, n_- = 4, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$T = 5$ **(159 bases)**

Base: 12221

$$12221\textcolor{red}{3}_{[\#=2]} \quad \Delta = -136, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$12221\textcolor{red}{3}_{[\#=3]} \quad \Delta = -136, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$12221\textcolor{red}{3}_{[\#=4]} \quad \Delta = -136, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$12221\textcolor{red}{3}_{[\#=5]} \quad \Delta = -136, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$12221\textcolor{red}{3}_{[\#=8]} \quad \Delta = -136, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$12221\textcolor{red}{3}_{[\#=7]} \quad \Delta = -136, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$12221\textcolor{red}{3}_{[\#=9]} \quad \Delta = -136, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$12221\textcolor{red}{3}_{[\#=6]} \quad \Delta = -136, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

Base: 1222^2

$$\begin{aligned}
{}^2_{2221}{}^3 \quad \Delta = -136, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
{}^2_{2221}{}^3_{[\# = 2]} \quad \Delta = -136, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
{}^2_{2221}{}^6 \quad \Delta = -206, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 8 \end{pmatrix} \\
{}^2_{2221}{}^7 \quad \Delta = -233, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
{}^2_{2221}{}^5 \quad \Delta = -180, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
{}^2_{2221}{}^3_{[\# = 4]} \quad \Delta = -136, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
{}^2_{2221}{}^3_{[\# = 5]} \quad \Delta = -136, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
{}^2_{2221}{}^3_{[\# = 6]} \quad \Delta = -136, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 144 & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
{}^2_{2221}{}^3_{[\# = 7]} \quad \Delta = -136, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 196 & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
{}^2_{2221}{}^3_{[\# = 8]} \quad \Delta = -136, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 256 & \begin{pmatrix} 0 & 16 & 2 \\ 16 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
{}^2_{2221}{}^3_{[\# = 9]} \quad \Delta = -136, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 324 & \begin{pmatrix} 0 & 18 & 2 \\ 18 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
{}^2_{2221}{}^2 \quad \Delta = -128, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
{}^2_{2221}{}^3_{[\# = 3]} \quad \Delta = -136, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
{}^2_{2221}{}^8 \quad \Delta = -261, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -4 & -2 \\ 2 & -2 & 8 \end{pmatrix} \\
{}^2_{2221}{}^4 \quad \Delta = -156, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
{}^2_{2221}{}^{12} \quad \Delta = -376, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -8 & -6 \\ 2 & -6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 12312

$$\begin{aligned}
21321{}^3_{[\# = 7]} \quad \Delta = -145, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
21321{}^3_{[\# = 5]} \quad \Delta = -145, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
12312{}^3_{[\# = 3]} \quad \Delta = -145, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
21321\overset{3}{\color{red}3}_{[\# = 2]} \quad \Delta = -145, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
12312\overset{3}{\color{red}3}_{[\# = 2]} \quad \Delta = -145, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
12312\overset{3}{\color{red}3} \quad \Delta = -145, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
12312\overset{4}{\color{red}4} \quad \Delta = -165, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 8 \end{pmatrix} \\
12312\overset{5}{\color{red}5} \quad \Delta = -189, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
21321\overset{3}{\color{red}3}_{[\# = 6]} \quad \Delta = -145, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
21321\overset{3}{\color{red}3}_{[\# = 3]} \quad \Delta = -145, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
12312\overset{3}{\color{red}3}_{[\# = 4]} \quad \Delta = -145, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21321\overset{3}{\color{red}3}_{[\# = 4]} \quad \Delta = -145, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
12312\overset{3}{\color{red}3}_{[\# = 5]} \quad \Delta = -145, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
21321\overset{3}{\color{red}3}_{[\# = 8]} \quad \Delta = -145, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
21321\overset{3}{\color{red}3}_{[\# = 9]} \quad \Delta = -145, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
12312\overset{2}{\color{red}2} \quad \Delta = -137, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\text{Base: } 13131 & \\
13131\overset{4}{\color{red}4} \quad \Delta = -172, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
13131\overset{3}{\color{red}3}_{[\# = 9]} \quad \Delta = -152, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
13131\overset{3}{\color{red}3}_{[\# = 8]} \quad \Delta = -152, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
13131\overset{3}{\color{red}3}_{[\# = 7]} \quad \Delta = -152, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
13131\overset{3}{\color{red}3}_{[\# = 6]} \quad \Delta = -152, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
13131\overset{3}{\color{red}3}_{[\# = 5]} \quad \Delta = -152, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

$$13131\overset{3}{3} [\# = 4] \quad \Delta = -152, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$13131\overset{3}{3} [\# = 3] \quad \Delta = -152, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$13131\overset{3}{3} [\# = 2] \quad \Delta = -152, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$13131\overset{3}{3} [\# = 2] \quad \Delta = -152, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 21 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$13131\overset{3}{3} \quad \Delta = -152, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$13131\overset{3}{3} [\# = 3] \quad \Delta = -152, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 51 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$13131\overset{2}{2} \quad \Delta = -144, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 13213

$$13213\overset{3}{3} [\# = 2] \quad \Delta = -153, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$13213\overset{3}{3} [\# = 5] \quad \Delta = -153, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 225 \quad \begin{pmatrix} 0 & 15 & 2 \\ 15 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$31231\overset{3}{3} [\# = 5] \quad \Delta = -153, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$13213\overset{3}{3} \quad \Delta = -153, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$31231\overset{3}{3} [\# = 2] \quad \Delta = -153, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$13213\overset{3}{3} [\# = 4] \quad \Delta = -153, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$31231\overset{3}{3} [\# = 4] \quad \Delta = -153, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$13213\overset{3}{3} [\# = 3] \quad \Delta = -153, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$31231\overset{3}{3} [\# = 3] \quad \Delta = -153, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$13213\overset{2}{2} \quad \Delta = -145, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$13213\overset{4}{4} \quad \Delta = -173, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 9 & 5 \\ 5 & 9 \end{pmatrix}$$

Base: 31313

$$31313\overset{3}{3} \quad \Delta = -160, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$31313^{(3)}_{[\# = 2]} \quad \Delta = -160, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 21 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$31313^{(3)}_{[\# = 3]} \quad \Delta = -160, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 51 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$31313^{(4)} \quad \Delta = -180, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$31313^{(2)} \quad \Delta = -152, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 23123

$$23123^{(3)}_{[\# = 2]} \quad \Delta = -154, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 37 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$23123^{(4)} \quad \Delta = -174, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 6 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$23123^{(3)} \quad \Delta = -154, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$23123^{(3)}_{[\# = 3]} \quad \Delta = -154, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 225 \quad \begin{pmatrix} 0 & 15 & 2 \\ 15 & 87 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$23123^{(2)} \quad \Delta = -146, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 1321²

$$13213^{(2)}_{[\# = 6]} \quad \Delta = -145, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$13217^{(2)} \quad \Delta = -242, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -4 & -2 \\ 2 & -2 & 8 \end{pmatrix}$$

$$13215^{(2)} \quad \Delta = -189, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 8 \end{pmatrix}$$

$$13214^{(2)} \quad \Delta = -165, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$13218^{(2)} \quad \Delta = -270, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -5 & -3 \\ 2 & -3 & 8 \end{pmatrix}$$

$$13212^{(2)} \quad \Delta = -137, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$13216^{(2)} \quad \Delta = -215, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$13213^{(2)}_{[\# = 9]} \quad \Delta = -145, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 324 \quad \begin{pmatrix} 0 & 18 & 2 \\ 18 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix}$$

$$13213^{(2)}_{[\# = 8]} \quad \Delta = -145, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 256 \quad \begin{pmatrix} 0 & 16 & 2 \\ 16 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$13213^{(2)}_{[\# = 7]} \quad \Delta = -145, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 196 \quad \begin{pmatrix} 0 & 14 & 2 \\ 14 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$\begin{aligned}
& \overset{2}{1231}\mathbf{3}_{[\# = 3]} \quad \Delta = -145, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 13\overset{2}{21}\mathbf{3} \quad \Delta = -145, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& 13\overset{2}{21}\mathbf{3}_{[\# = 5]} \quad \Delta = -145, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& 12\overset{2}{31}\mathbf{3}_{[\# = 5]} \quad \Delta = -145, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& 13\overset{2}{21}\mathbf{3}_{[\# = 4]} \quad \Delta = -145, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 12\overset{2}{31}\mathbf{3}_{[\# = 4]} \quad \Delta = -145, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 13\overset{2}{21}\mathbf{3}_{[\# = 3]} \quad \Delta = -145, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 12\overset{2}{31}\mathbf{3}_{[\# = 2]} \quad \Delta = -145, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 13\overset{2}{21}\mathbf{3}_{[\# = 2]} \quad \Delta = -145, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \text{Base: } \overset{1}{1321} \\
& \overset{1}{1321}\mathbf{3}_{[\# = 9]} \quad \Delta = -145, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
& \overset{1}{1321}\mathbf{3}_{[\# = 4]} \quad \Delta = -145, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{1}{1321}\mathbf{3}_{[\# = 6]} \quad \Delta = -145, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& 12\overset{1}{31}\mathbf{3}_{[\# = 4]} \quad \Delta = -145, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{1}{1321}\mathbf{3}_{[\# = 3]} \quad \Delta = -145, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 12\overset{1}{31}\mathbf{3}_{[\# = 5]} \quad \Delta = -145, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{1}{1321}\mathbf{3}_{[\# = 5]} \quad \Delta = -145, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& 12\overset{1}{31}\mathbf{3}_{[\# = 3]} \quad \Delta = -145, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{1}{1321}\mathbf{3}_{[\# = 2]} \quad \Delta = -145, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{1}{1321}\mathbf{3}_{[\# = 7]} \quad \Delta = -145, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
{}^1_{1321}\mathbf{3}_{[\#=8]} \quad \Delta = -145, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
{}^1_{1231}\mathbf{3}_{[\#=2]} \quad \Delta = -145, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: ${}^1_{2313}$

$$\begin{aligned}
{}^{2,1}_{2313} \quad \Delta = -147, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
{}^{1,2}_{2313} \quad \Delta = -145, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
{}^{4,1}_{2313} \quad \Delta = -173, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
{}^{2,1}_{2313} \quad \Delta = -145, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
{}^{3,1}_{2313} [\#=5] \quad \Delta = -153, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
{}^{3,1}_{2313} [\#=4] \quad \Delta = -153, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
{}^{3,1}_{2313} [\#=3] \quad \Delta = -153, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
{}^{3,1}_{2313} [\#=2] \quad \Delta = -153, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
{}^{1,3}_{2313} \quad \Delta = -153, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
{}^{3,1}_{2313} \quad \Delta = -153, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
{}^{1,3}_{2313} [\#=2] \quad \Delta = -153, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
{}^{5,1}_{2313} \quad \Delta = -197, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 8 \end{pmatrix}
\end{aligned}$$

Base: ${}^{21}_{232}$

$$\begin{aligned}
{}^{21}_{232} [\#=2] \quad \Delta = -147, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 37 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
{}^{21}_{232} \quad \Delta = -167, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 6 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
{}^{21}_{232} \quad \Delta = -139, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
{}^{21}_{232} \quad \Delta = -147, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

$$2\overset{21}{3}2_{[\# = 3]} \quad \Delta = -147, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 87 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$2\overset{321}{3}2 \quad \Delta = -148, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: 41222

$$\overset{4}{4}1222 \quad \Delta = -184, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\overset{2}{4}1222 \quad \Delta = -156, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$22214\overset{12}{2}_{[\# = 2]} \quad \Delta = -55, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$\overset{3}{4}1222_{[\# = 3]} \quad \Delta = -164, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$\overset{3}{4}1222 \quad \Delta = -164, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\overset{3}{4}1222_{[\# = 2]} \quad \Delta = -164, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: $2\overset{2,1}{3}1$

$$2\overset{2,1}{3}1\overset{3}{3}_{[\# = 3]} \quad \Delta = -147, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$2\overset{2,1}{3}1\overset{3}{3}_{[\# = 2]} \quad \Delta = -147, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$2\overset{2,1}{3}1\overset{2}{2} \quad \Delta = -139, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$2\overset{2,1}{3}1\overset{4}{4} \quad \Delta = -167, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

Base: 14122

$$\overset{4}{1}4122 \quad \Delta = -184, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1412\overset{3}{2} \quad \Delta = -165, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$\overset{2}{1}4122_{[\# = 2]} \quad \Delta = -55, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$\overset{3}{1}4122_{[\# = 3]} \quad \Delta = -164, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 51 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$\overset{3}{1}4122_{[\# = 2]} \quad \Delta = -164, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 21 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$22141\overset{3}{3}_{[\# = 2]} \quad \Delta = -164, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$22141\overset{3}{3}_{[\# = 3]} \quad \Delta = -164, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\begin{aligned}
14122 \quad \Delta = -156, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
14122 \quad \Delta = -164, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: 21412

$$\begin{aligned}
21412 \quad \Delta = -55, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
21412 \quad \Delta = -184, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
21412 \quad \Delta = -156, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21412 \quad \Delta = -164, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21412 \quad \Delta = -164, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 21 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21412 \quad \Delta = -164, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 51 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\begin{smallmatrix} 21 \\ 141 \end{smallmatrix}$

$$\begin{aligned}
141 \quad \Delta = -55, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
141 \quad \Delta = -164, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
141 \quad \Delta = -184, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 8 \end{pmatrix} \\
141 \quad \Delta = -156, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
141 \quad \Delta = -164, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
141 \quad \Delta = -164, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
141 \quad \Delta = -164, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
141 \quad \Delta = -164, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\begin{smallmatrix} 1,1 \\ 141 \end{smallmatrix}$

$$\begin{aligned}
141 \quad \Delta = -164, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
141 \quad \Delta = -164, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

$$1 \overset{2,1,1}{4} 1 \text{ } [\# = 2] \quad \Delta = -55, n_+ = 1, n_- = 5, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$T = 6$ (248 bases)

Base: 122221

$$122221\overset{3}{3} \text{ } [\# = 4] \quad \Delta = -107, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$122221\overset{3}{3} \text{ } [\# = 3] \quad \Delta = -107, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$122221\overset{3}{3} \text{ } [\# = 2] \quad \Delta = -107, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$122221\overset{3}{3} \text{ } [\# = 7] \quad \Delta = -107, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix}$$

$$122221\overset{3}{3} \text{ } [\# = 5] \quad \Delta = -107, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix}$$

$$122221\overset{3}{3} \text{ } [\# = 6] \quad \Delta = -107, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

$$122221\overset{3}{3} \text{ } [\# = 8] \quad \Delta = -107, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix}$$

$$122221\overset{3}{3} \text{ } [\# = 9] \quad \Delta = -107, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix}$$

Base: $12\overset{2}{2}22$

$$\overset{2}{2}2221\overset{1}{2} \text{ } \Delta = -347, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -7 & -5 \\ 2 & -5 & 8 \end{pmatrix}$$

$$\overset{2}{2}2221\overset{8}{8} \text{ } \Delta = -232, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$\overset{2}{2}2221\overset{7}{7} \text{ } \Delta = -204, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 8 \end{pmatrix}$$

$$\overset{2}{2}2221\overset{6}{6} \text{ } \Delta = -177, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$\overset{2}{2}2221\overset{5}{5} \text{ } \Delta = -151, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$\overset{2}{2}2221\overset{4}{4} \text{ } \Delta = -127, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\overset{2}{2}2221\overset{3}{3} \text{ } [\# = 9] \quad \Delta = -107, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 324 \quad \begin{pmatrix} 0 & 18 & 2 \\ 18 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix}$$

$$\overset{2}{2}2221\overset{3}{3} \text{ } [\# = 6] \quad \Delta = -107, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

$$\overset{2}{2}2221\overset{3}{3} \text{ } [\# = 5] \quad \Delta = -107, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix}$$

$$\begin{aligned}
& \overset{2}{22221\textcolor{red}{3}}_{[\#=8]} \quad \Delta = -107, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 256 \quad \begin{pmatrix} 0 & 16 & 2 \\ 16 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
& \overset{2}{22221\textcolor{red}{3}}_{[\#=4]} \quad \Delta = -107, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& \overset{2}{22221\textcolor{red}{3}}_{[\#=3]} \quad \Delta = -107, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{2}{22221\textcolor{red}{3}}_{[\#=2]} \quad \Delta = -107, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{2}{22221\textcolor{red}{3}} \quad \Delta = -107, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{2}{22221\textcolor{red}{3}}_{[\#=7]} \quad \Delta = -107, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 196 \quad \begin{pmatrix} 0 & 14 & 2 \\ 14 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
& \overset{2}{22221\textcolor{red}{2}} \quad \Delta = -99, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \textbf{Base: } 122312 \\
& 213221\textcolor{red}{3}_{[\#=3]} \quad \Delta = -116, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 12231\textcolor{red}{2}^5 \quad \Delta = -160, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& 12231\textcolor{red}{2}^4 \quad \Delta = -136, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 8 \end{pmatrix} \\
& 213221\textcolor{red}{3}_{[\#=7]} \quad \Delta = -116, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
& 213221\textcolor{red}{3}_{[\#=9]} \quad \Delta = -116, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
& 213221\textcolor{red}{3}_{[\#=8]} \quad \Delta = -116, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
& 12231\textcolor{red}{2}^3_{[\#=3]} \quad \Delta = -116, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 12231\textcolor{red}{2}^3 \quad \Delta = -116, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
& 213221\textcolor{red}{3}_{[\#=4]} \quad \Delta = -116, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& 213221\textcolor{red}{3}_{[\#=2]} \quad \Delta = -116, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 12231\textcolor{red}{2}^3_{[\#=2]} \quad \Delta = -116, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 12231\textcolor{red}{2}^2 \quad \Delta = -108, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
122312^3_{[\#=5]} \quad \Delta = -116, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
2132213_{[\#=5]} \quad \Delta = -116, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
122312^3_{[\#=4]} \quad \Delta = -116, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
2132213_{[\#=6]} \quad \Delta = -116, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
\text{Base: } 123131 & \\
1231313_{[\#=8]} \quad \Delta = -124, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
1313213_{[\#=7]} \quad \Delta = -124, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1231313_{[\#=4]} \quad \Delta = -124, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
1231313_{[\#=6]} \quad \Delta = -124, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
1231313_{[\#=7]} \quad \Delta = -124, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
1313213_{[\#=8]} \quad \Delta = -124, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1231313_{[\#=9]} \quad \Delta = -124, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
1313213_{[\#=9]} \quad \Delta = -124, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1231313_{[\#=5]} \quad \Delta = -124, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
123131^3 \quad \Delta = -124, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
123131^3_{[\#=2]} \quad \Delta = -124, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 21 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1313213_{[\#=2]} \quad \Delta = -124, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1231313_{[\#=2]} \quad \Delta = -124, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
1313213_{[\#=3]} \quad \Delta = -124, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1313213_{[\#=4]} \quad \Delta = -124, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
131321\bar{3} \text{ } [\# = 5] \quad \Delta = -124, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
131321\bar{3} \text{ } [\# = 6] \quad \Delta = -124, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
123131\bar{3} \text{ } [\# = 3] \quad \Delta = -124, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
1231\bar{3}1 \quad \Delta = -116, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\text{Base: } 123213 \\
12321\bar{3} \quad \Delta = -146, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 9 & 5 \\ 5 & 9 \end{pmatrix} \\
312321\bar{3} \text{ } [\# = 2] \quad \Delta = -126, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
12321\bar{3} \text{ } [\# = 2] \quad \Delta = -126, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
12321\bar{3} \quad \Delta = -126, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
312321\bar{3} \text{ } [\# = 8] \quad \Delta = -126, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
312321\bar{3} \text{ } [\# = 7] \quad \Delta = -126, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
312321\bar{3} \text{ } [\# = 6] \quad \Delta = -126, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
312321\bar{3} \text{ } [\# = 5] \quad \Delta = -126, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
12321\bar{3} \text{ } [\# = 5] \quad \Delta = -126, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 225 \quad & \begin{pmatrix} 0 & 15 & 2 \\ 15 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
312321\bar{3} \text{ } [\# = 4] \quad \Delta = -126, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
12321\bar{3} \text{ } [\# = 4] \quad \Delta = -126, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 144 \quad & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
312321\bar{3} \text{ } [\# = 3] \quad \Delta = -126, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
12321\bar{3} \text{ } [\# = 3] \quad \Delta = -126, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
312321\bar{3} \text{ } [\# = 9] \quad \Delta = -126, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
12321\bar{3} \quad \Delta = -118, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

$$\text{Base: } 1\bar{3}221$$

$$\begin{aligned}
13221\overset{1}{\textcolor{red}{3}}_{[\# = 9]} \quad \Delta = -116, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
12231\overset{1}{\textcolor{red}{3}}_{[\# = 2]} \quad \Delta = -116, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
13221\overset{1}{\textcolor{red}{3}}_{[\# = 2]} \quad \Delta = -116, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
12231\overset{1}{\textcolor{red}{3}}_{[\# = 3]} \quad \Delta = -116, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
13221\overset{1}{\textcolor{red}{3}}_{[\# = 3]} \quad \Delta = -116, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
12231\overset{1}{\textcolor{red}{3}}_{[\# = 4]} \quad \Delta = -116, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
13221\overset{1}{\textcolor{red}{3}}_{[\# = 8]} \quad \Delta = -116, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
13221\overset{1}{\textcolor{red}{3}}_{[\# = 7]} \quad \Delta = -116, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
13221\overset{1}{\textcolor{red}{3}}_{[\# = 6]} \quad \Delta = -116, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
13221\overset{1}{\textcolor{red}{3}}_{[\# = 5]} \quad \Delta = -116, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
12231\overset{1}{\textcolor{red}{3}}_{[\# = 5]} \quad \Delta = -116, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
13221\overset{1}{\textcolor{red}{3}}_{[\# = 4]} \quad \Delta = -116, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

Base: 12321

$$\begin{aligned}
12\overset{31}{\textcolor{red}{3}}21_{[\# = 2]} \quad \Delta = -118, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
12321\overset{1}{\textcolor{red}{3}}_{[\# = 2]} \quad \Delta = -118, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
12\overset{31}{\textcolor{red}{3}}21_{[\# = 3]} \quad \Delta = -118, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
12321\overset{1}{\textcolor{red}{3}}_{[\# = 8]} \quad \Delta = -118, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
12321\overset{1}{\textcolor{red}{3}}_{[\# = 9]} \quad \Delta = -118, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
12321\overset{1}{\textcolor{red}{3}}_{[\# = 7]} \quad \Delta = -118, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
12321\overset{1}{\textcolor{red}{3}}_{[\# = 6]} \quad \Delta = -118, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
12321\overset{1}{3} \text{ }_{[\# = 5]} \quad \Delta = -118, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
12321\overset{1}{3} \text{ }_{[\# = 4]} \quad \Delta = -118, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
12321\overset{1}{3} \text{ }_{[\# = 3]} \quad \Delta = -118, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{2}{23131}$

$$\begin{aligned}
\overset{2}{23131}\overset{2}{6} \quad \Delta = -196, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 8 \end{pmatrix} \\
\overset{2}{23131}\overset{2}{3} \text{ }_{[\# = 4]} \quad \Delta = -126, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
\overset{2}{23131}\overset{2}{4} \quad \Delta = -146, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\overset{2}{23131}\overset{2}{5} \quad \Delta = -170, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
\overset{2}{23131}\overset{2}{3} \text{ }_{[\# = 9]} \quad \Delta = -126, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 324 \quad & \begin{pmatrix} 0 & 18 & 2 \\ 18 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
\overset{2}{23131}\overset{2}{3} \quad \Delta = -126, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\overset{2}{23131}\overset{2}{3} \quad \Delta = -126, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\overset{2}{23131}\overset{2}{3} \text{ }_{[\# = 2]} \quad \Delta = -126, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\overset{2}{23131}\overset{2}{3} \text{ }_{[\# = 3]} \quad \Delta = -126, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\overset{2}{23131}\overset{2}{3} \text{ }_{[\# = 5]} \quad \Delta = -126, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 100 \quad & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
\overset{2}{23131}\overset{2}{2} \quad \Delta = -119, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\overset{2}{23131}\overset{2}{2} \quad \Delta = -118, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\overset{2}{23131}\overset{2}{3} \text{ }_{[\# = 6]} \quad \Delta = -126, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 144 \quad & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
\overset{2}{23131}\overset{2}{2} \quad \Delta = -118, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\overset{2}{23131}\overset{2}{3} \text{ }_{[\# = 7]} \quad \Delta = -126, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 196 \quad & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
\overset{2}{23131}\overset{2}{3} \text{ }_{[\# = 8]} \quad \Delta = -126, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 256 \quad & \begin{pmatrix} 0 & 16 & 2 \\ 16 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}
\end{aligned}$$

Base: $12\bar{3}2$

$$\begin{aligned}
2\bar{3}21\bar{2} \quad \Delta = -118, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2\bar{3}21\bar{4} \quad \Delta = -146, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
12\bar{3}2 \quad \Delta = -118, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2\bar{3}21\bar{3} \text{ [#=7]} \quad \Delta = -126, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 196 & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
2\bar{3}21\bar{5} \quad \Delta = -170, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
2\bar{3}21\bar{3} \text{ [#=9]} \quad \Delta = -126, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 324 & \begin{pmatrix} 0 & 18 & 2 \\ 18 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
2\bar{3}21\bar{6} \quad \Delta = -196, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 8 \end{pmatrix} \\
2\bar{3}21\bar{3} \text{ [#=8]} \quad \Delta = -126, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 256 & \begin{pmatrix} 0 & 16 & 2 \\ 16 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
2\bar{3}21\bar{3} \quad \Delta = -126, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2\bar{3}21\bar{3} \text{ [#=6]} \quad \Delta = -126, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 144 & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
2\bar{3}21\bar{3} \text{ [#=5]} \quad \Delta = -126, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
2\bar{3}21\bar{7} \quad \Delta = -223, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
12\bar{3}2 \quad \Delta = -126, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2\bar{3}21\bar{3} \text{ [#=4]} \quad \Delta = -126, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
2\bar{3}21\bar{3} \text{ [#=2]} \quad \Delta = -126, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
2\bar{3}21\bar{8} \quad \Delta = -251, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -4 & -2 \\ 2 & -2 & 8 \end{pmatrix} \\
2\bar{3}21\bar{3} \text{ [#=3]} \quad \Delta = -126, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

Base: 412231

$$\begin{aligned}
132214\bar{2} \text{ [#=2]} \quad \Delta = -35, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\bar{4}12231 \text{ [#=3]} \quad \Delta = -144, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 144 & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 249 & 29 \\ 2 & 29 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
412231\overset{3}{3} [\# = 2] \quad \Delta = -144, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
412231\overset{3}{3} [\# = 5] \quad \Delta = -144, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
412231\overset{3}{3} [\# = 2] \quad \Delta = -144, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 109 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
412231\overset{2}{2} \quad \Delta = -136, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
412231\overset{4}{4} \quad \Delta = -164, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
412231\overset{3}{3} \quad \Delta = -144, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
412231\overset{3}{3} [\# = 3] \quad \Delta = -144, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
412231\overset{3}{3} [\# = 4] \quad \Delta = -144, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}
\end{aligned}$$

Base: 412313

$$\begin{aligned}
412313\overset{2}{2} \quad \Delta = -144, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
412313\overset{3}{3} \quad \Delta = -152, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
412313\overset{3}{3} \quad \Delta = -152, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
412313\overset{3}{3} [\# = 2] \quad \Delta = -152, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
412313\overset{2}{2} \quad \Delta = -144, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: 413132

$$\begin{aligned}
413132\overset{2}{2} \quad \Delta = -144, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
413132\overset{2}{2} \quad \Delta = -144, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
413132\overset{2}{2} \quad \Delta = -145, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
413132\overset{3}{3} [\# = 2] \quad \Delta = -152, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
413132\overset{3}{3} \quad \Delta = -152, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

$$413132 \quad \Delta = -152, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 41231

$$412313 \quad [\# = 3] \quad \Delta = -146, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$412313 \quad [\# = 2] \quad \Delta = -146, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$41231 \quad \Delta = -146, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$412314 \quad \Delta = -166, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$412312 \quad \Delta = -138, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$41231 \quad \Delta = -138, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 41322

$$41322 \quad [\# = 4] \quad \Delta = -144, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 109 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$41322 \quad \Delta = -144, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$41322 \quad [\# = 5] \quad \Delta = -144, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 225 \quad \begin{pmatrix} 0 & 15 & 2 \\ 15 & 172 & 24 \\ 2 & 24 & 8 \end{pmatrix}$$

$$41322 \quad [\# = 2] \quad \Delta = -144, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$41322 \quad \Delta = -136, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$41322 \quad \Delta = -136, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$41322 \quad \Delta = -188, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$41322 \quad [\# = 3] \quad \Delta = -144, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 60 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$41322 \quad \Delta = -164, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: 141231

$$141231 \quad \Delta = -144, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1412313 \quad [\# = 2] \quad \Delta = -144, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$132141\overset{3}{3} \text{ }_{[\# = 3]} \quad \Delta = -144, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$141231\overset{3}{3} \text{ }_{[\# = 3]} \quad \Delta = -144, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$141231\overset{3}{3} \text{ }_{[\# = 4]} \quad \Delta = -144, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$141231\overset{3}{3} \text{ }_{[\# = 5]} \quad \Delta = -144, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$141\overset{2}{2}31 \quad \Delta = -136, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$132141\overset{3}{3} \text{ }_{[\# = 2]} \quad \Delta = -144, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: 141313

$$14131\overset{3}{3} \text{ }_{[\# = 2]} \quad \Delta = -151, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$14131\overset{4}{3} \quad \Delta = -171, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 9 & 5 \\ 5 & 9 \end{pmatrix}$$

$$14131\overset{2}{3} \quad \Delta = -143, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$14131\overset{2}{3} \quad \Delta = -143, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$14131\overset{3}{3} \quad \Delta = -151, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$313141\overset{3}{3} \text{ }_{[\# = 3]} \quad \Delta = -151, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$14131\overset{3}{3} \text{ }_{[\# = 3]} \quad \Delta = -151, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$14131\overset{3}{3} \quad \Delta = -151, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$313141\overset{3}{3} \text{ }_{[\# = 2]} \quad \Delta = -151, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$14131\overset{2}{3} \quad \Delta = -144, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: 321413

$$32141\overset{3}{3} \quad \Delta = -152, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$32141\overset{3}{3} \quad \Delta = -152, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$32141\overset{2}{3} \quad \Delta = -144, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$321413^2 \quad \Delta = -144, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: 214131

$$2141313^{\#=8} \quad \Delta = -143, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$2141313^{\#=7} \quad \Delta = -143, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$2141313^{\#=6} \quad \Delta = -143, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$2141313^{\#=5} \quad \Delta = -143, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$2141313^{\#=4} \quad \Delta = -143, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$2141313^{\#=3} \quad \Delta = -143, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$214131^3_{\#=3} \quad \Delta = -143, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 51 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$2141313^{\#=2} \quad \Delta = -143, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$214131^3_{\#=2} \quad \Delta = -143, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 21 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$214131^3 \quad \Delta = -143, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$2141313^{\#=9} \quad \Delta = -143, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$214131^3 \quad \Delta = -143, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$214131^4 \quad \Delta = -163, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$214131^2 \quad \Delta = -135, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$214131^2 \quad \Delta = -135, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 14132¹

$$14132^{41} \quad \Delta = -164, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$14132^{51} \quad \Delta = -188, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 8 \end{pmatrix}$$

$$14132^{2,1} \quad \Delta = -138, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$\begin{aligned}
141\overset{21}{3}2 \quad \Delta = -136, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
141\overset{21}{3}2 \quad \Delta = -136, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
141\overset{31}{3}2 \text{ } [\# = 4] \quad \Delta = -144, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
141\overset{31}{3}2 \text{ } [\# = 5] \quad \Delta = -144, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 100 \quad & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
141\overset{31}{3}2 \text{ } [\# = 3] \quad \Delta = -144, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2\overset{3141}{3}1 \text{ } [\# = 3] \quad \Delta = -144, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
141\overset{31}{3}2 \text{ } [\# = 2] \quad \Delta = -144, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2\overset{3141}{3}1 \text{ } [\# = 2] \quad \Delta = -144, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\text{Base: } 1\overset{131}{4}1 & \\
1\overset{131}{4}1\overset{13}{3} \text{ } [\# = 3] \quad \Delta = -143, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1\overset{131}{4}1 \quad \Delta = -135, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1\overset{3131}{4}1 \text{ } [\# = 7] \quad \Delta = -143, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1\overset{3131}{4}1 \text{ } [\# = 8] \quad \Delta = -143, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1\overset{3131}{4}1 \text{ } [\# = 9] \quad \Delta = -143, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1\overset{3131}{4}1 \text{ } [\# = 5] \quad \Delta = -143, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1\overset{3131}{4}1 \text{ } [\# = 4] \quad \Delta = -143, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1\overset{3131}{4}1 \text{ } [\# = 3] \quad \Delta = -143, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1\overset{3131}{4}1 \text{ } [\# = 2] \quad \Delta = -143, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
1\overset{131}{4}1\overset{13}{3} \text{ } [\# = 2] \quad \Delta = -143, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1\overset{131}{4}1 \quad \Delta = -143, n_+ = 1, n_- = 6, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}
\end{aligned}$$

$$\overset{3131}{1 \ 4 \ 1} \text{ [#=6]} \quad \Delta = -143, \ n_+ = 1, \ n_- = 6, \ n_0 = 0, \ |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: $\overset{31}{1413}$

$$\overset{31}{31414} \quad \Delta = -171, \ n_+ = 1, \ n_- = 6, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$\overset{313}{1413} \quad \Delta = -151, \ n_+ = 1, \ n_- = 6, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\overset{31}{31413} \text{ [#=2]} \quad \Delta = -151, \ n_+ = 1, \ n_- = 6, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\overset{31}{31413} \quad \Delta = -151, \ n_+ = 1, \ n_- = 6, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$\overset{31}{31413} \text{ [#=3]} \quad \Delta = -151, \ n_+ = 1, \ n_- = 6, \ n_0 = 0, \ |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$\overset{31}{31412} \quad \Delta = -143, \ n_+ = 1, \ n_- = 6, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\overset{312}{1413} \quad \Delta = -143, \ n_+ = 1, \ n_- = 6, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 222215

$$\overset{2}{222215} \quad \Delta = -151, \ n_+ = 1, \ n_- = 6, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\overset{3}{222215} \quad \Delta = -159, \ n_+ = 1, \ n_- = 6, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 222151

$$\overset{3}{222151} \quad \Delta = -159, \ n_+ = 1, \ n_- = 6, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 17 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\overset{2}{222151} \quad \Delta = -151, \ n_+ = 1, \ n_- = 6, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 215122

$$\overset{2}{215122} \quad \Delta = -151, \ n_+ = 1, \ n_- = 6, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\overset{3}{215122} \quad \Delta = -160, \ n_+ = 1, \ n_- = 6, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$\overset{2}{215122} \quad \Delta = -151, \ n_+ = 1, \ n_- = 6, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\overset{3}{215122} \quad \Delta = -159, \ n_+ = 1, \ n_- = 6, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\overset{3}{215122} \quad \Delta = -159, \ n_+ = 1, \ n_- = 6, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: $\overset{221}{1 \ 5 \ 1}$

$$\overset{221}{1 \ 5 \ 1} \quad \Delta = -151, \ n_+ = 1, \ n_- = 6, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\begin{array}{l} \begin{array}{c} 221 \\ 1 \ 5 \ 1 \end{array} \quad \Delta = -159, \ n_+ = 1, \ n_- = 6, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \end{array}$$

$$\begin{array}{l} \begin{array}{c} 221 \\ 1 \ 5 \ 1 \end{array} \quad \Delta = -160, \ n_+ = 1, \ n_- = 6, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \end{array}$$

Base: $\begin{array}{c} 21 \\ 1 \ 5 \ 12 \end{array}$

$$\begin{array}{l} \begin{array}{c} 21 \textcolor{red}{2} \\ 1 \ 5 \ 12 \end{array} \quad \Delta = -151, \ n_+ = 1, \ n_- = 6, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \end{array}$$

$$\begin{array}{l} \begin{array}{c} 21 \textcolor{red}{3} \\ 1 \ 5 \ 12 \end{array} \quad \Delta = -159, \ n_+ = 1, \ n_- = 6, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \end{array}$$

Base: $\begin{array}{c} 1,1 \\ 1 \ 5 \ 12 \end{array}$

$$\begin{array}{l} \begin{array}{c} 1,1 \textcolor{red}{2} \\ 1 \ 5 \ 12 \end{array} \quad \Delta = -151, \ n_+ = 1, \ n_- = 6, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \end{array}$$

$$\begin{array}{l} \begin{array}{c} 1,1 \textcolor{red}{3} \\ 1 \ 5 \ 12 \end{array} \quad \Delta = -159, \ n_+ = 1, \ n_- = 6, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \end{array}$$

$T = 7$ **(371 bases)**

Base: 1222221

$$1222221 \textcolor{red}{3} \text{ [#=7]} \quad \Delta = -78, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix}$$

$$1222221 \textcolor{red}{3} \text{ [#=8]} \quad \Delta = -78, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix}$$

$$1222221 \textcolor{red}{3} \text{ [#=9]} \quad \Delta = -78, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix}$$

$$1222221 \textcolor{red}{3} \text{ [#=6]} \quad \Delta = -78, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$1222221 \textcolor{red}{3} \text{ [#=5]} \quad \Delta = -78, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

$$1222221 \textcolor{red}{3} \text{ [#=4]} \quad \Delta = -78, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$1222221 \textcolor{red}{3} \text{ [#=3]} \quad \Delta = -78, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$1222221 \textcolor{red}{3} \text{ [#=2]} \quad \Delta = -78, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: $\begin{array}{c} 2 \\ 222222 \end{array}$

$$\begin{array}{l} \begin{array}{c} 2 \\ 222221 \textcolor{red}{5} \end{array} \quad \Delta = -122, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \end{array}$$

$$\begin{array}{l} \begin{array}{c} 2 \\ 222221 \textcolor{red}{8} \end{array} \quad \Delta = -203, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 8 \end{pmatrix} \end{array}$$

$$\begin{array}{l} \begin{array}{c} 2 \\ 222221 \textcolor{red}{6} \end{array} \quad \Delta = -148, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \end{array}$$

$$\begin{aligned}
2^2 2222214 \quad \Delta = -98, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2^2 22222112 \quad \Delta = -318, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -6 & -4 \\ 2 & -4 & 8 \end{pmatrix} \\
2^2 2222217 \quad \Delta = -175, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
2^2 2222213 \text{ [#=2]} \quad \Delta = -78, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
2^2 2222213 \text{ [#=3]} \quad \Delta = -78, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
2^2 2222213 \text{ [#=5]} \quad \Delta = -78, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
2^2 2222213 \quad \Delta = -78, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2^2 2222213 \text{ [#=4]} \quad \Delta = -78, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
2^2 2222213 \text{ [#=6]} \quad \Delta = -78, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 144 & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
2^2 2222213 \text{ [#=8]} \quad \Delta = -78, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 256 & \begin{pmatrix} 0 & 16 & 2 \\ 16 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
2^2 2222212 \quad \Delta = -70, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2^2 2222213 \text{ [#=7]} \quad \Delta = -78, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 196 & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
2^2 2222213 \text{ [#=9]} \quad \Delta = -78, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 324 & \begin{pmatrix} 0 & 18 & 2 \\ 18 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix} \\
\text{Base: } 1223131 & \\
13132213 \text{ [#=6]} \quad \Delta = -95, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
12231313 \text{ [#=3]} \quad \Delta = -95, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
13132213 \text{ [#=4]} \quad \Delta = -95, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
12231313 \text{ [#=4]} \quad \Delta = -95, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
13132213 \text{ [#=5]} \quad \Delta = -95, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
12231313 \text{ [#=5]} \quad \Delta = -95, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1313221\overset{3}{3} \text{ [#=7]} \quad \Delta = -95, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
1223131\overset{3}{3} \text{ [#=7]} \quad \Delta = -95, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
1223131\overset{3}{3} \text{ [#=2]} \quad \Delta = -95, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 21 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1223131\overset{3}{3} \text{ [#=8]} \quad \Delta = -95, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
1313221\overset{3}{3} \text{ [#=9]} \quad \Delta = -95, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
1223131\overset{3}{3} \text{ [#=9]} \quad \Delta = -95, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
1223131\overset{3}{3} \quad \Delta = -95, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1313221\overset{3}{3} \text{ [#=3]} \quad \Delta = -95, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1223131\overset{3}{3} \text{ [#=2]} \quad \Delta = -95, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
1313221\overset{3}{3} \text{ [#=2]} \quad \Delta = -95, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1223131\overset{2}{2} \quad \Delta = -87, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1313221\overset{3}{3} \text{ [#=8]} \quad \Delta = -95, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
1223131\overset{3}{3} \text{ [#=6]} \quad \Delta = -95, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
\text{Base: } 1231321 & \\
1231321\overset{3}{3} \text{ [#=7]} \quad \Delta = -96, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
1231321\overset{2}{2} \quad \Delta = -88, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1231321\overset{3}{3} \text{ [#=8]} \quad \Delta = -96, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
1231321\overset{3}{3} \text{ [#=2]} \quad \Delta = -96, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1231321\overset{3}{3} \text{ [#=4]} \quad \Delta = -96, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1231321\overset{3}{3} \text{ [#=5]} \quad \Delta = -96, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1231321^{\textcolor{red}{3}} \quad \Delta = -96, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1231321^{\textcolor{red}{3}}_{[\# = 6]} \quad \Delta = -96, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
1231321^{\textcolor{red}{3}}_{[\# = 9]} \quad \Delta = -96, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
1231321^{\textcolor{red}{3}}_{[\# = 3]} \quad \Delta = -96, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

Base: $2^2 31321$

$$\begin{aligned}
2^2 31321^{\textcolor{red}{8}} \quad \Delta = -223, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
2^2 31321^{\textcolor{red}{3}}_{[\# = 9]} \quad \Delta = -98, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 324 & \begin{pmatrix} 0 & 18 & 2 \\ 18 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
2^2 31321^{\textcolor{red}{7}} \quad \Delta = -195, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 8 \end{pmatrix} \\
2^2 31321^{\textcolor{red}{6}} \quad \Delta = -168, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
2^2 31321^{\textcolor{red}{5}} \quad \Delta = -142, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2^2 31321^{\textcolor{red}{4}} \quad \Delta = -118, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2^2 31321^{\textcolor{red}{3}}_{[\# = 8]} \quad \Delta = -98, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 256 & \begin{pmatrix} 0 & 16 & 2 \\ 16 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
2^2 31321^{\textcolor{red}{3}}_{[\# = 7]} \quad \Delta = -98, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 196 & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
2^2 31321^{\textcolor{red}{3}}_{[\# = 6]} \quad \Delta = -98, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 144 & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
2^2 31321^{\textcolor{red}{3}}_{[\# = 5]} \quad \Delta = -98, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
2^2 31321^{\textcolor{red}{3}}_{[\# = 4]} \quad \Delta = -98, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
2^2 31321^{\textcolor{red}{3}}_{[\# = 3]} \quad \Delta = -98, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
2^2 31321^{\textcolor{red}{3}}_{[\# = 2]} \quad \Delta = -98, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
2^2 31321^{\textcolor{red}{3}} \quad \Delta = -98, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2^2 31321^{\textcolor{red}{2}} \quad \Delta = -92, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

$$231321^{\textcolor{red}{2}} \quad \Delta = -90, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$231321^{\textcolor{red}{2}} \quad \Delta = -90, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: 4123214

$$4123214^{\textcolor{red}{3}} \quad \Delta = -145, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$4123214^{\textcolor{red}{2}} \quad \Delta = -137, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 1412321

$$1412321^{\textcolor{red}{3}}_{[\# = 9]} \quad \Delta = -117, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$1412321^{\textcolor{red}{3}}_{[\# = 8]} \quad \Delta = -117, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1412321^{\textcolor{red}{3}} \quad \Delta = -117, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1412321^{\textcolor{red}{3}}_{[\# = 7]} \quad \Delta = -117, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$1232141^{\textcolor{red}{3}}_{[\# = 2]} \quad \Delta = -117, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1412321^{\textcolor{red}{3}}_{[\# = 6]} \quad \Delta = -117, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1412321^{\textcolor{red}{3}}_{[\# = 2]} \quad \Delta = -117, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1412321^{\textcolor{red}{2}} \quad \Delta = -109, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1232141^{\textcolor{red}{3}}_{[\# = 3]} \quad \Delta = -117, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1412321^{\textcolor{red}{3}}_{[\# = 3]} \quad \Delta = -117, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1412321^{\textcolor{red}{3}}_{[\# = 4]} \quad \Delta = -117, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1412321^{\textcolor{red}{3}}_{[\# = 5]} \quad \Delta = -117, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

Base: 4123141

$$4123141^{\textcolor{red}{2}} \quad \Delta = -135, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$4123141^{\textcolor{red}{3}}_{[\# = 2]} \quad \Delta = -143, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$4123141\overset{3}{\color{red}}\text{ }_{[\#=3]} \quad \Delta = -143, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$\overset{2}{\color{red}}4123141 \quad \Delta = -135, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\overset{3}{\color{red}}4123141 \quad \Delta = -143, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

Base: 2141321

$$\overset{4}{\color{red}}2141321 \quad \Delta = -135, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$2141321\overset{3}{\color{red}}\text{ }_{[\#=4]} \quad \Delta = -115, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$2141321\overset{3}{\color{red}}\text{ }_{[\#=5]} \quad \Delta = -115, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$\overset{2}{\color{red}}2141321 \quad \Delta = -107, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\overset{2}{\color{red}}2141321 \quad \Delta = -107, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$\overset{3}{\color{red}}2141321 \quad \Delta = -115, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$2141321\overset{3}{\color{red}}\text{ }_{[\#=9]} \quad \Delta = -115, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$2141321\overset{3}{\color{red}}\text{ }_{[\#=8]} \quad \Delta = -115, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$2141321\overset{3}{\color{red}}\text{ }_{[\#=7]} \quad \Delta = -115, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$\overset{3}{\color{red}}2141321\text{ }_{[\#=2]} \quad \Delta = -115, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 21 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\overset{3}{\color{red}}2141321\text{ }_{[\#=3]} \quad \Delta = -115, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 51 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$2141321\overset{3}{\color{red}}\text{ }_{[\#=3]} \quad \Delta = -115, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$2141321\overset{3}{\color{red}}\text{ }_{[\#=2]} \quad \Delta = -115, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$2141321\overset{3}{\color{red}}\text{ }_{[\#=6]} \quad \Delta = -115, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: $\overset{2}{141321}$

$$\overset{2}{141321}\overset{2}{\color{red}} \quad \Delta = -109, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\begin{aligned}
141321\overset{2}{4} \quad \Delta = -137, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
12\overset{3141}{3}2 \text{ [#=2]} \quad \Delta = -117, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
141321\overset{2}{3} \text{ [#=9]} \quad \Delta = -117, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 324 \quad & \begin{pmatrix} 0 & 18 & 2 \\ 18 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
141321\overset{2}{3} \text{ [#=8]} \quad \Delta = -117, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 256 \quad & \begin{pmatrix} 0 & 16 & 2 \\ 16 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
141321\overset{2}{3} \text{ [#=7]} \quad \Delta = -117, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 196 \quad & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
141321\overset{2}{3} \text{ [#=6]} \quad \Delta = -117, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 144 \quad & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
141321\overset{2}{3} \text{ [#=5]} \quad \Delta = -117, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 100 \quad & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
141321\overset{2}{3} \text{ [#=3]} \quad \Delta = -117, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
12\overset{3141}{3}2 \text{ [#=3]} \quad \Delta = -117, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
141321\overset{2}{3} \text{ [#=2]} \quad \Delta = -117, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
141321\overset{2}{8} \quad \Delta = -242, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -4 & -2 \\ 2 & -2 & 8 \end{pmatrix} \\
141321\overset{2}{3} \quad \Delta = -117, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
141321\overset{2}{7} \quad \Delta = -214, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
141321\overset{2}{6} \quad \Delta = -187, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 8 \end{pmatrix} \\
141321\overset{22}{2} \quad \Delta = -109, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
141321\overset{2}{5} \quad \Delta = -161, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
141321\overset{2}{3} \text{ [#=4]} \quad \Delta = -117, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
\text{Base: } 1314131 \\
1314131\overset{2}{3} \text{ [#=4]} \quad \Delta = -122, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1314131\overset{2}{3} \text{ [#=9]} \quad \Delta = -122, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1314131\bar{3} \text{ } [\# = 8] \quad \Delta = -122, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1314131\bar{3} \text{ } [\# = 7] \quad \Delta = -122, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
1314131\bar{3} \text{ } [\# = 6] \quad \Delta = -122, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1314131\bar{3} \text{ } [\# = 5] \quad \Delta = -122, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1314131\bar{3} \text{ } [\# = 3] \quad \Delta = -122, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1314131\bar{3} \text{ } [\# = 2] \quad \Delta = -122, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1314131\bar{3} \quad \Delta = -122, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1314131\bar{2} \quad \Delta = -114, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 4131413

$$\begin{aligned}
4131413\bar{2} \quad \Delta = -142, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
4131413\bar{3} \quad \Delta = -150, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
4131413\bar{3} \quad \Delta = -150, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
4131413\bar{3} \quad \Delta = -150, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
4131413\bar{2} \quad \Delta = -142, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
4131413\bar{2} \quad \Delta = -142, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
4131413\bar{2} \quad \Delta = -143, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1413141

$$\begin{aligned}
1413141\bar{3} \text{ } [\# = 2] \quad \Delta = -142, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1413141\bar{3} \text{ } [\# = 3] \quad \Delta = -142, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1413141\bar{2} \quad \Delta = -135, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

$$1413141 \quad \Delta = -142, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1413141 \quad \Delta = -134, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 2141 $\overset{2}{3}$ 2

$$2141\overset{2}{3}2 \quad \Delta = -109, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$2141\overset{4}{3}2 \quad \Delta = -137, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$2 \overset{3}{3} \overset{2}{2} \quad \Delta = -118, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$2141\overset{3}{3}2 \text{ [#=3]} \quad \Delta = -117, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 123 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$2141\overset{3}{3}2 \quad \Delta = -117, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$2141\overset{3}{3}2 \text{ [#=2]} \quad \Delta = -117, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$2141\overset{2}{3}2 \quad \Delta = -109, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 4141232

$$4141\overset{3}{2}32 \quad \Delta = -145, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 37 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$4141\overset{2}{2}32 \quad \Delta = -137, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$4141\overset{2}{2}32 \quad \Delta = -137, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: 1414123

$$14141\overset{2}{2}3 \quad \Delta = -135, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$3214141\overset{3}{3} \text{ [#=3]} \quad \Delta = -143, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$14141\overset{3}{2}3 \quad \Delta = -143, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$3214141\overset{3}{3} \text{ [#=2]} \quad \Delta = -143, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$14141\overset{2}{2}3 \quad \Delta = -135, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 2141412

$$214141\overset{3}{2} \text{ [#=3]} \quad \Delta = -134, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 87 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$2141412 \quad \Delta = -134, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$2141412 \text{ [#=2]} \quad \Delta = -134, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 37 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$2141412 \quad \Delta = -126, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$2141412 \quad \Delta = -126, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$2141412 \quad \Delta = -154, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{1231}{4} 1$$

$$1 \overset{1231}{4} 1 \quad \Delta = -107, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1 \overset{1231}{4} 13 \text{ [#=2]} \quad \Delta = -115, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{31231}{4} 1 \text{ [#=3]} \quad \Delta = -115, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \overset{31231}{4} 1 \text{ [#=5]} \quad \Delta = -115, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1 \overset{31231}{4} 1 \text{ [#=9]} \quad \Delta = -115, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$1 \overset{31231}{4} 1 \text{ [#=4]} \quad \Delta = -115, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{1231}{4} 13 \text{ [#=3]} \quad \Delta = -115, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \overset{31231}{4} 1 \text{ [#=2]} \quad \Delta = -115, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{31231}{4} 1 \text{ [#=8]} \quad \Delta = -115, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1 \overset{31231}{4} 1 \text{ [#=7]} \quad \Delta = -115, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$1 \overset{31231}{4} 1 \text{ [#=6]} \quad \Delta = -115, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{231}{4} 1$$

$$1 \overset{231}{4} 13 \text{ [#=2]} \quad \Delta = -117, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{231}{4} 1 \quad \Delta = -109, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\begin{aligned}
& \overset{231}{1\ 4\ 1\ 4} \quad \Delta = -137, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{231}{1\ 4\ 1\ 2} \quad \Delta = -109, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{231}{1\ 4\ 1\ 3} \text{ [#=3]} \quad \Delta = -117, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{231}{1\ 4\ 1\ 3} \quad \Delta = -117, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{231}{41\ 4\ 1}$

$$\begin{aligned}
& \overset{231}{41\ 4\ 1\ 4} \quad \Delta = -163, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{231}{41\ 4\ 1\ 3} \text{ [#=3]} \quad \Delta = -143, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 60 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{231}{41\ 4\ 1\ 3} \quad \Delta = -143, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{231}{41\ 4\ 1\ 3} \text{ [#=2]} \quad \Delta = -143, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{231}{41\ 4\ 1\ 2} \quad \Delta = -135, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{2231}{41\ 4\ 1} \quad \Delta = -135, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{231}{41\ 4\ 1} \quad \Delta = -135, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{141412}$

$$\begin{aligned}
& \overset{1}{141412} \text{ }^3 \quad \Delta = -134, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
& \overset{31}{1\ 4\ 1412} \text{ [#=2]} \quad \Delta = -134, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{1}{141412} \text{ }^3 \text{ [#=3]} \quad \Delta = -134, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{1}{141412} \text{ }^3 \text{ [#=2]} \quad \Delta = -134, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{31}{1\ 4\ 1412} \text{ [#=3]} \quad \Delta = -134, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{1}{141412} \text{ }^4 \quad \Delta = -154, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 8 \end{pmatrix} \\
& \overset{1}{141412} \text{ }^2 \quad \Delta = -126, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

$$\overset{12}{141412} \quad \Delta = -126, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: $\overset{31}{14141}$

$$\overset{31}{14141} \quad \Delta = -134, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\overset{31}{141413} \text{ [#=2]} \quad \Delta = -142, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$\overset{31}{314141} \quad \Delta = -142, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\overset{31}{14141} \quad \Delta = -142, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\overset{31}{314141} \text{ [#=3]} \quad \Delta = -142, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 60 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$\overset{31}{141413} \text{ [#=3]} \quad \Delta = -142, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$\overset{31}{314141} \text{ [#=2]} \quad \Delta = -142, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$\overset{312}{14141} \quad \Delta = -134, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\overset{41}{314141} \quad \Delta = -162, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\overset{21}{314141} \quad \Delta = -134, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: $\overset{1}{14141}$

$$\overset{121}{14141} \quad \Delta = -126, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\overset{31}{14141} \text{ [#=2]} \quad \Delta = -134, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\overset{31}{14141} \text{ [#=3]} \quad \Delta = -134, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 3132215

$$\overset{3}{3132215} \quad \Delta = -147, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\overset{3}{3132215} \text{ [#=2]} \quad \Delta = -147, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 21 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\overset{3}{3132215} \quad \Delta = -147, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\overset{2}{3132215} \quad \Delta = -139, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\overset{2}{3132215} \quad \Delta = -139, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 2313215

$$\overset{3}{2313215} \quad \Delta = -148, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\overset{2}{2313215} \quad \Delta = -142, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\overset{2}{2313215} \quad \Delta = -140, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\overset{2}{2313215} \quad \Delta = -140, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: 2231315

$$\overset{2}{2231315} \quad \Delta = -140, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\overset{2}{2231315} \quad \Delta = -139, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\overset{2}{2231315} \quad \Delta = -139, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$\overset{3}{2231315} \quad \Delta = -147, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\overset{3}{2231315} \text{ [#=2]} \quad \Delta = -147, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 196 \quad \begin{pmatrix} 0 & 14 & 2 \\ 14 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

Base: 1322151

$$1512231\overset{3}{3} \text{ [#=5]} \quad \Delta = -139, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1512231\overset{3}{3} \text{ [#=4]} \quad \Delta = -139, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1512231\overset{3}{3} \text{ [#=3]} \quad \Delta = -139, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1512231\overset{3}{3} \text{ [#=2]} \quad \Delta = -139, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1322151 \quad \Delta = -139, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 17 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1322151 \quad \Delta = -131, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 1512313

$$1512313 \text{ [#=2]} \quad \Delta = -147, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\overset{2}{1512313} \quad \Delta = -139, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1512313^2 \quad \Delta = -139, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1512313^3 \quad \Delta = -147, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: 151231^2

$$1512314^2 \quad \Delta = -161, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$1512313^2_{[\#=2]} \quad \Delta = -141, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1512313^2_{[\#=3]} \quad \Delta = -141, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1512313^2 \quad \Delta = -141, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$151231^2 \quad \Delta = -133, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1512312^2 \quad \Delta = -133, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: 2313151

$$2313151^2 \quad \Delta = -141, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$2313151^3 \quad \Delta = -147, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 32 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$2313151^3_{[\#=2]} \quad \Delta = -147, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 137 & 21 \\ 2 & 21 & 8 \end{pmatrix}$$

$$2313151^2 \quad \Delta = -140, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$2313151^2 \quad \Delta = -139, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$2313151^2 \quad \Delta = -139, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 15131^{22}

$$15131^{22} \quad \Delta = -131, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$15131^{222} \quad \Delta = -131, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$15131^{22} \quad \Delta = -159, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$15131^{22} \quad \Delta = -183, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$\begin{aligned}
151\overset{22}{3}1\overset{3}{3} \text{ [#=4]} \quad \Delta = -139, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 144 \quad & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 109 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
151\overset{22}{3}1\overset{3}{3} \text{ [#=3]} \quad \Delta = -139, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 60 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
151\overset{22}{3}1\overset{3}{3} \text{ [#=5]} \quad \Delta = -139, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 225 \quad & \begin{pmatrix} 0 & 15 & 2 \\ 15 & 172 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
151\overset{22}{3}1\overset{3}{3} \text{ [#=2]} \quad \Delta = -139, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
151\overset{22}{3}1\overset{3}{3} \quad \Delta = -139, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\text{Base: } 1321512 \\
132151\overset{3}{2} \quad \Delta = -139, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
2151231\overset{3}{3} \text{ [#=2]} \quad \Delta = -139, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
132151\overset{2}{2} \quad \Delta = -131, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
132151\overset{2}{2} \quad \Delta = -131, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2151231\overset{3}{3} \text{ [#=5]} \quad \Delta = -139, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
2151231\overset{3}{3} \text{ [#=3]} \quad \Delta = -139, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2151231\overset{3}{3} \text{ [#=4]} \quad \Delta = -139, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\text{Base: } 3215123 \\
32151\overset{2}{2} \quad \Delta = -140, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\text{Base: } 3151223 \\
3151\overset{2}{2}223 \quad \Delta = -139, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
3151\overset{2}{2}223 \quad \Delta = -139, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
3151\overset{3}{2}223 \quad \Delta = -147, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\text{Base: } 2151313 \\
2151\overset{3}{3}13 \text{ [#=3]} \quad \Delta = -146, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2151\overset{4}{3}13 \quad \Delta = -166, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 9 & 5 \\ 5 & 9 \end{pmatrix}
\end{aligned}$$

$$2151313^2 \quad \Delta = -138, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$2151313^2 \quad \Delta = -138, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$2151313^3 \quad \Delta = -146, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$2151313^2 \quad \Delta = -138, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$2151313^3_{[\#=2]} \quad \Delta = -146, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$2151313^3 \quad \Delta = -146, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: 1315122

$$1315122^2 \quad \Delta = -130, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$22151313^3_{[\#=7]} \quad \Delta = -138, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$22151313^3_{[\#=9]} \quad \Delta = -138, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix}$$

$$22151313^3_{[\#=8]} \quad \Delta = -138, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$22151313^3_{[\#=6]} \quad \Delta = -138, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$1315122^2 \quad \Delta = -130, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$22151313^3_{[\#=4]} \quad \Delta = -138, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$22151313^3_{[\#=3]} \quad \Delta = -138, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$22151313^3_{[\#=2]} \quad \Delta = -138, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1315122^3 \quad \Delta = -138, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$22151313^3_{[\#=5]} \quad \Delta = -138, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1315122^3 \quad \Delta = -139, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

Base: 215131

$$2151313^2_{[\#=4]} \quad \Delta = -139, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$2151\overset{2}{3}1\overset{3}{3} \text{ }_{[\# = 5]} \quad \Delta = -139, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$2151\overset{2}{3}1\overset{2}{3} \text{ }_{[\# = 2]} \quad \Delta = -139, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$2151\overset{3}{3}1\overset{2}{3} \quad \Delta = -139, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$2151\overset{2}{3}1\overset{4}{4} \quad \Delta = -159, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$2151\overset{2}{3}1\overset{5}{5} \quad \Delta = -183, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 8 \end{pmatrix}$$

$$2151\overset{2}{3}1\overset{3}{3} \quad \Delta = -139, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$2151\overset{2}{3}1\overset{3}{3} \text{ }_{[\# = 3]} \quad \Delta = -139, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$2151\overset{2}{3}1\overset{2}{2} \quad \Delta = -131, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$2151\overset{2}{3}1\overset{2}{2} \quad \Delta = -131, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$2151\overset{2}{3}1\overset{2}{2} \quad \Delta = -131, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: $1\overset{1}{5}1231$

$$1\overset{1}{5}1231\overset{3}{3} \text{ }_{[\# = 5]} \quad \Delta = -139, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1\overset{1}{5}1231\overset{3}{3} \text{ }_{[\# = 2]} \quad \Delta = -139, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1\overset{1}{5}1231\overset{3}{3} \text{ }_{[\# = 3]} \quad \Delta = -139, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1\overset{1}{5}1231\overset{3}{3} \text{ }_{[\# = 4]} \quad \Delta = -139, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1\overset{1}{5}1231\overset{2}{2} \quad \Delta = -131, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: $1\overset{3131}{5}1$

$$1\overset{3131}{5}1 \quad \Delta = -138, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1\overset{3131}{5}1 \quad \Delta = -138, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1\overset{3131}{5}1 \quad \Delta = -166, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 9 & 5 \\ 5 & 9 \end{pmatrix}$$

$$1\overset{3131}{5}1 \quad \Delta = -146, n_+ = 1, n_- = 7, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$\begin{aligned}
& \overset{3131}{1 \ 5 \ 1} \text{ [#=3]} \quad \Delta = -146, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 5 \ 1} \text{ [#=2]} \quad \Delta = -146, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{131}{1 \ 5 \ 1}$

$$\begin{aligned}
& \overset{5131}{1 \ 5 \ 1} \quad \Delta = -183, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 8 \end{pmatrix} \\
& \overset{2131}{1 \ 5 \ 1} \quad \Delta = -131, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 5 \ 1} \text{ [#=3]} \quad \Delta = -139, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{4131}{1 \ 5 \ 1} \quad \Delta = -159, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
& \overset{3131}{1 \ 5 \ 1} \text{ [#=4]} \quad \Delta = -139, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 5 \ 1} \quad \Delta = -139, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 5 \ 1} \quad \Delta = -131, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 5 \ 1} \text{ [#=2]} \quad \Delta = -139, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 5 \ 1} \text{ [#=5]} \quad \Delta = -139, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{321}{31 \ 5 \ 1}$

$$\begin{aligned}
& \overset{321}{31 \ 5 \ 1} \quad \Delta = -139, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{321}{31 \ 5 \ 12} \quad \Delta = -139, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{2321}{31 \ 5 \ 1} \quad \Delta = -139, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{131}{1 \ 5 \ 12}$

$$\begin{aligned}
& \overset{3131}{1 \ 5 \ 12} \text{ [#=3]} \quad \Delta = -138, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{1312}{1 \ 5 \ 12} \quad \Delta = -130, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 5 \ 12} \quad \Delta = -130, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{1313}{1 \ 5 \ 12} \quad \Delta = -138, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{3131}{1 \ 5 \ 12} \ [\# = 4] \quad \Delta = -138, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 5 \ 12} \ [\# = 5] \quad \Delta = -138, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 5 \ 12} \ [\# = 6] \quad \Delta = -138, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 5 \ 12} \ [\# = 7] \quad \Delta = -138, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 5 \ 12} \ [\# = 8] \quad \Delta = -138, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 5 \ 12} \ [\# = 9] \quad \Delta = -138, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 5 \ 12} \ [\# = 2] \quad \Delta = -138, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{21}{31 \ 5 \ 13}$

$$\begin{aligned}
& \overset{321}{31 \ 5 \ 13} \quad \Delta = -147, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{21}{31 \ 5 \ 13} \quad \Delta = -146, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{21}{31 \ 5 \ 13} \quad \Delta = -138, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{221}{31 \ 5 \ 13} \quad \Delta = -138, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1,131}{1 \ 5 \ 1}$

$$\begin{aligned}
& \overset{1,3131}{1 \ 5 \ 1} \ [\# = 9] \quad \Delta = -138, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
& \overset{1,3131}{1 \ 5 \ 1} \ [\# = 7] \quad \Delta = -138, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
& \overset{1,3131}{1 \ 5 \ 1} \ [\# = 6] \quad \Delta = -138, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{1,3131}{1 \ 5 \ 1} \ [\# = 5] \quad \Delta = -138, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{1,3131}{1 \ 5 \ 1} \ [\# = 4] \quad \Delta = -138, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{1,131}{1 \ 5 \ 1} \quad \Delta = -130, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{1,3131}{1 \ 5 \ 1} \ [\# = 2] \quad \Delta = -138, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{1,3131}{1 \ 5 \ 1} \text{ }_{[\# = 3]} \quad \Delta = -138, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{1,3131}{1 \ 5 \ 1} \text{ }_{[\# = 8]} \quad \Delta = -138, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix}
\end{aligned}$$

Base: $31 \overset{1,31}{5} 1$

$$\overset{21,31}{31 \ 5 \ 1} \quad \Delta = -138, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$31 \overset{1,31}{5} \textcolor{red}{12} \quad \Delta = -138, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$31 \overset{1,31}{5} \textcolor{red}{13} \quad \Delta = -146, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: 6122222

$$\overset{3}{6122222} \quad \Delta = -156, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 27 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$\overset{2}{6122222} \quad \Delta = -148, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 1612222

$$161\overset{2}{2222} \quad \Delta = -148, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$161\overset{3}{2222} \quad \Delta = -156, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 17 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 2161222

$$2161\overset{2}{222} \quad \Delta = -148, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$2161\overset{3}{222} \quad \Delta = -156, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$2161\overset{3}{222} \quad \Delta = -156, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$2161\overset{2}{222} \quad \Delta = -148, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 2216122

$$22161\overset{3}{22} \quad \Delta = -156, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$22161\overset{3}{22} \quad \Delta = -157, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$22161\overset{2}{22} \quad \Delta = -148, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: $1 \overset{2221}{6} 1$

$$\overset{2221}{1 \ 6 \ 1} \quad \Delta = -148, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\overset{2221}{1 \ 6 \ 1} \quad \Delta = -156, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: $\overset{221}{1 \ 6 \ 12}$

$$\overset{221}{1 \ 6 \ 12} \quad \Delta = -157, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$\overset{221}{1 \ 6 \ 12} \quad \Delta = -148, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\overset{221\textcolor{red}{2}}{1 \ 6 \ 12} \quad \Delta = -148, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\overset{221\textcolor{red}{3}}{1 \ 6 \ 12} \quad \Delta = -156, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\overset{221}{1 \ 6 \ 12} \quad \Delta = -156, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: $\overset{21}{21 \ 6 \ 12}$

$$\overset{\textcolor{red}{221}}{21 \ 6 \ 12} \quad \Delta = -148, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\overset{\textcolor{red}{321}}{21 \ 6 \ 12} \quad \Delta = -156, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: $\overset{221,1}{1 \ 6 \ 1}$

$$\overset{221,1}{1 \ 6 \ 1} \quad \Delta = -156, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\overset{221,1}{1 \ 6 \ 1} \quad \Delta = -157, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$\overset{221,1}{1 \ 6 \ 1} \quad \Delta = -148, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: $\overset{21,21}{1 \ 6 \ 1}$

$$\overset{21,21}{1 \ 6 \ 1} \quad \Delta = -148, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\overset{21,21}{1 \ 6 \ 1} \quad \Delta = -156, \ n_+ = 1, \ n_- = 7, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$T = 8$ (572 bases)

Base: 12222221

$$12222221\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -49, \ n_+ = 1, \ n_- = 8, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 60 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$12222221\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -49, \ n_+ = 1, \ n_- = 8, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 25 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$12222221\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -49, \ n_+ = 1, \ n_- = 8, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 172 & 34 \\ 2 & 34 & 8 \end{pmatrix}$$

$$\begin{aligned}
12222221\mathbf{3}_{[\# = 4]} \quad \Delta = -49, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 109 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
12222221\mathbf{3}_{[\# = 9]} \quad \Delta = -49, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 564 & 62 \\ 2 & 62 & 8 \end{pmatrix} \\
12222221\mathbf{3}_{[\# = 6]} \quad \Delta = -49, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 249 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
12222221\mathbf{3}_{[\# = 7]} \quad \Delta = -49, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 340 & 48 \\ 2 & 48 & 8 \end{pmatrix} \\
12222221\mathbf{3}_{[\# = 8]} \quad \Delta = -49, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 445 & 55 \\ 2 & 55 & 8 \end{pmatrix} \\
\text{Base: } 1222222^2 \\
2^2222221\mathbf{3}_{[\# = 9]} \quad \Delta = -49, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 324 & \begin{pmatrix} 0 & 18 & 2 \\ 18 & 564 & 62 \\ 2 & 62 & 8 \end{pmatrix} \\
2^2222221\mathbf{7} \quad \Delta = -146, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2^2222221\mathbf{8} \quad \Delta = -174, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
2^2222221\mathbf{12} \quad \Delta = -289, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -5 & -3 \\ 2 & -3 & 8 \end{pmatrix} \\
2^2222221\mathbf{5} \quad \Delta = -93, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2^2222221\mathbf{2} \quad \Delta = -41, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
2^2222221\mathbf{6} \quad \Delta = -119, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2^2222221\mathbf{4} \quad \Delta = -69, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2^2222221\mathbf{3}_{[\# = 3]} \quad \Delta = -49, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 60 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
2^2222221\mathbf{3}_{[\# = 7]} \quad \Delta = -49, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 196 & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 340 & 48 \\ 2 & 48 & 8 \end{pmatrix} \\
2^2222221\mathbf{3}_{[\# = 6]} \quad \Delta = -49, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 144 & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 249 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
2^2222221\mathbf{3}_{[\# = 5]} \quad \Delta = -49, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 172 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
2^2222221\mathbf{3}_{[\# = 4]} \quad \Delta = -49, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 109 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
2^2222221\mathbf{3} \quad \Delta = -49, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{2}{2222221}\mathbf{3} \text{ }_{[\# = 2]} \quad \Delta = -49, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& \overset{2}{2222221}\mathbf{3} \text{ }_{[\# = 8]} \quad \Delta = -49, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 256 \quad \begin{pmatrix} 0 & 16 & 2 \\ 16 & 445 & 55 \\ 2 & 55 & 8 \end{pmatrix} \\
& \text{Base: } 12231321 \\
& 12231321\mathbf{3} \text{ }_{[\# = 7]} \quad \Delta = -67, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& 12313221\mathbf{3} \text{ }_{[\# = 2]} \quad \Delta = -67, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 12231321\mathbf{3} \text{ }_{[\# = 2]} \quad \Delta = -67, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& 12313221\mathbf{3} \text{ }_{[\# = 3]} \quad \Delta = -67, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& 12231321\mathbf{3} \text{ }_{[\# = 3]} \quad \Delta = -67, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& 12313221\mathbf{3} \text{ }_{[\# = 8]} \quad \Delta = -67, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
& 12231321\mathbf{3} \text{ }_{[\# = 8]} \quad \Delta = -67, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& 12313221\mathbf{3} \text{ }_{[\# = 9]} \quad \Delta = -67, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix} \\
& \overset{2}{12231321} \quad \Delta = -59, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 12231321\mathbf{3} \text{ }_{[\# = 9]} \quad \Delta = -67, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& \overset{3}{12231321} \quad \Delta = -67, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 12313221\mathbf{3} \text{ }_{[\# = 7]} \quad \Delta = -67, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
& 12313221\mathbf{3} \text{ }_{[\# = 4]} \quad \Delta = -67, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& 12231321\mathbf{3} \text{ }_{[\# = 4]} \quad \Delta = -67, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& 12313221\mathbf{3} \text{ }_{[\# = 5]} \quad \Delta = -67, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
& 12231321\mathbf{3} \text{ }_{[\# = 5]} \quad \Delta = -67, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& 12313221\mathbf{3} \text{ }_{[\# = 6]} \quad \Delta = -67, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix}
\end{aligned}$$

$$12231321\mathbf{3}_{[\# = 6]} \quad \Delta = -67, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

Base: $2\mathbf{3}13221$

$$2\mathbf{3}13221\mathbf{12} \quad \Delta = -309, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -6 & -4 \\ 2 & -4 & 8 \end{pmatrix}$$

$$2\mathbf{3}13221\mathbf{5} \quad \Delta = -113, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$2\mathbf{3}13221\mathbf{8} \quad \Delta = -194, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 8 \end{pmatrix}$$

$$2\mathbf{3}13221\mathbf{4} \quad \Delta = -89, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$2\mathbf{3}13221\mathbf{7} \quad \Delta = -166, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$2\mathbf{3}13221\mathbf{3}_{[\# = 2]} \quad \Delta = -69, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$2\mathbf{3}13221\mathbf{6} \quad \Delta = -139, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$2\mathbf{3}13221\mathbf{2} \quad \Delta = -61, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$2\mathbf{3}13221\mathbf{3}_{[\# = 5]} \quad \Delta = -69, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

$$2\mathbf{3}13221\mathbf{3}_{[\# = 4]} \quad \Delta = -69, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$2\mathbf{3}13221\mathbf{3} \quad \Delta = -69, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$2\mathbf{3}13221\mathbf{3}_{[\# = 3]} \quad \Delta = -69, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$2\mathbf{3}13221\mathbf{3}_{[\# = 8]} \quad \Delta = -69, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 256 \quad \begin{pmatrix} 0 & 16 & 2 \\ 16 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix}$$

$$2\mathbf{3}13221\mathbf{3}_{[\# = 6]} \quad \Delta = -69, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$2\mathbf{3}13221\mathbf{3}_{[\# = 9]} \quad \Delta = -69, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 324 \quad \begin{pmatrix} 0 & 18 & 2 \\ 18 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix}$$

$$2\mathbf{3}13221\mathbf{3}_{[\# = 7]} \quad \Delta = -69, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 196 \quad \begin{pmatrix} 0 & 14 & 2 \\ 14 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix}$$

$$2\mathbf{3}13221\mathbf{2} \quad \Delta = -61, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 21413221

$$\mathbf{4}21413221 \quad \Delta = -106, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$\begin{aligned}
21413221^2 \quad \Delta = -78, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
21413221^3 \quad \Delta = -86, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21413221^3_{[\#=2]} \quad \Delta = -86, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 21 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21413221^3_{[\#=2]} \quad \Delta = -86, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
21413221^3_{[\#=3]} \quad \Delta = -86, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 51 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21413221^3_{[\#=3]} \quad \Delta = -86, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21413221^3_{[\#=4]} \quad \Delta = -86, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21413221^3_{[\#=7]} \quad \Delta = -86, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
21413221^3_{[\#=5]} \quad \Delta = -86, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
21413221^2 \quad \Delta = -78, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21413221^3_{[\#=6]} \quad \Delta = -86, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
21413221^3_{[\#=9]} \quad \Delta = -86, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
21413221^3_{[\#=8]} \quad \Delta = -86, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
\text{Base: } 13141321 & \\
13141321^3_{[\#=5]} \quad \Delta = -94, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
12314131^3_{[\#=6]} \quad \Delta = -94, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
13141321^3_{[\#=6]} \quad \Delta = -94, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
12314131^3_{[\#=5]} \quad \Delta = -94, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
13141321^3_{[\#=4]} \quad \Delta = -94, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
12314131^3_{[\#=4]} \quad \Delta = -94, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
12314131\overset{\textcolor{red}{3}}{3} \quad \Delta = -94, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
13141321\overset{\textcolor{red}{3}}{3} \quad \Delta = -94, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
13141321\overset{\textcolor{red}{3}}{3} \quad \Delta = -94, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
12314131\overset{\textcolor{red}{3}}{3} \quad \Delta = -94, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
12314131\overset{\textcolor{red}{3}}{3} \quad \Delta = -94, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
12314131\overset{\textcolor{red}{3}}{3} \quad \Delta = -94, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
13141321\overset{\textcolor{red}{3}}{3} \quad \Delta = -94, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
12314131\overset{\textcolor{red}{3}}{3} \quad \Delta = -94, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
13141321\overset{\textcolor{red}{3}}{3} \quad \Delta = -94, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
13141321\overset{\textcolor{red}{3}}{3} \quad \Delta = -94, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
13141321\overset{\textcolor{red}{3}}{3} \quad \Delta = -94, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
13141321\overset{\textcolor{red}{2}}{2} \quad \Delta = -86, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
13141321\overset{\textcolor{red}{2}}{2} \quad \Delta = -86, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\textbf{Base: } 13141\overset{\textcolor{red}{2}}{2} & \\
2\overset{\textcolor{red}{5}}{3}2 \quad \Delta = -140, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2\overset{\textcolor{red}{3}}{3}2 \quad \Delta = -96, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2\overset{\textcolor{red}{3}}{3}2 \quad \Delta = -96, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 324 \quad & \begin{pmatrix} 0 & 18 & 2 \\ 18 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
2\overset{\textcolor{red}{6}}{3}2 \quad \Delta = -166, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
2\overset{\textcolor{red}{4}}{3}2 \quad \Delta = -116, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2\overset{\textcolor{red}{3}}{3}2 \quad \Delta = -96, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
2 \overset{3}{3} 13141 \quad \Delta = -96, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
2 \overset{3}{3} 13141 \quad \Delta = -96, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
2 \overset{3}{3} 13141 \quad \Delta = -96, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 256 & \begin{pmatrix} 0 & 16 & 2 \\ 16 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
2 \overset{3}{3} 13141 \quad \Delta = -96, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 144 & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
2 \overset{3}{3} 13141 \quad \Delta = -96, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 196 & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
13141 \overset{3}{3} 2 \quad \Delta = -96, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
13141 \overset{2}{2} 32 \quad \Delta = -89, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2 \overset{2}{3} 13141 \quad \Delta = -88, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2 \overset{3}{3} 13141 \quad \Delta = -96, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
13141 \overset{2}{2} 32 \quad \Delta = -88, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
13141 \overset{22}{22} 32 \quad \Delta = -88, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\text{Base: } 41314141 & \\
413141 \overset{2}{2} 41 \quad \Delta = -133, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
413141 \overset{3}{3} 41 \quad \Delta = -141, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 37 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
413141 \overset{3}{3} 41 \quad \Delta = -141, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
41314141 \overset{3}{3} [\# = 2] \quad \Delta = -141, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
41314141 \overset{3}{3} [\# = 3] \quad \Delta = -141, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
413141 \overset{2}{2} 41 \quad \Delta = -134, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
413141 \overset{2}{2} 41 \quad \Delta = -133, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
413141 \overset{2}{2} 41 \quad \Delta = -133, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 21414131

$$\begin{aligned}
21414131^{\textcolor{red}{2}} \quad \Delta = -105, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
21414131^{\textcolor{red}{4}} \quad \Delta = -133, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21414131^{\textcolor{red}{2}} \quad \Delta = -105, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21414131^{\textcolor{red}{3}} \text{ [#=2] } \quad \Delta = -113, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 37 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21414131^{\textcolor{red}{3}} \text{ [#=2] } \quad \Delta = -113, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
21414131^{\textcolor{red}{3}} \text{ [#=3] } \quad \Delta = -113, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 87 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
21414131^{\textcolor{red}{3}} \text{ [#=3] } \quad \Delta = -113, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
21414131^{\textcolor{red}{3}} \text{ [#=4] } \quad \Delta = -113, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21414131^{\textcolor{red}{3}} \text{ [#=6] } \quad \Delta = -113, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
21414131^{\textcolor{red}{3}} \text{ [#=7] } \quad \Delta = -113, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21414131^{\textcolor{red}{3}} \text{ [#=8] } \quad \Delta = -113, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21414131^{\textcolor{red}{3}} \text{ [#=9] } \quad \Delta = -113, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21414131^{\textcolor{red}{3}} \quad \Delta = -113, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
21414131^{\textcolor{red}{3}} \text{ [#=5] } \quad \Delta = -113, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21414131^{\textcolor{red}{3}} \quad \Delta = -113, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
21414131^{\textcolor{red}{2}} \quad \Delta = -105, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1^{\textcolor{red}{12231}} 4^{\textcolor{red}{1}} 1$

$$\begin{aligned}
1^{\textcolor{red}{312231}} 4^{\textcolor{red}{1}} 1 \text{ [#=8] } \quad \Delta = -86, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1^{\textcolor{red}{12231}} 4^{\textcolor{red}{1}} 1^{\textcolor{red}{3}} \text{ [#=3] } \quad \Delta = -86, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1^{\textcolor{red}{312231}} 4^{\textcolor{red}{1}} 1 \text{ [#=9] } \quad \Delta = -86, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1^{\overline{312231}} 1^{\overline{4}} 1^{\overline{1}} [\# = 2] \quad \Delta = -86, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1^{\overline{312231}} 1^{\overline{4}} 1^{\overline{1}} [\# = 4] \quad \Delta = -86, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{\overline{12231}} 1^{\overline{4}} 1^{\overline{13}} [\# = 2] \quad \Delta = -86, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1^{\overline{312231}} 1^{\overline{4}} 1^{\overline{1}} [\# = 5] \quad \Delta = -86, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1^{\overline{312231}} 1^{\overline{4}} 1^{\overline{1}} [\# = 6] \quad \Delta = -86, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1^{\overline{312231}} 1^{\overline{4}} 1^{\overline{1}} [\# = 3] \quad \Delta = -86, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1^{\overline{312231}} 1^{\overline{4}} 1^{\overline{1}} [\# = 7] \quad \Delta = -86, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
1^{\overline{12231}} 1^{\overline{4}} 1^{\overline{1}} \quad \Delta = -78, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1414131^{\overline{1}}$

$$\begin{aligned}
1414131^{\overline{1}} 1^{\overline{3}} [\# = 4] \quad \Delta = -113, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1414131^{\overline{12}} \quad \Delta = -105, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1414131^{\overline{1}} 1^{\overline{4}} 14131^{\overline{3}} [\# = 3] \quad \Delta = -113, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1414131^{\overline{1}} 1^{\overline{4}} 14131^{\overline{3}} [\# = 5] \quad \Delta = -113, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1414131^{\overline{1}} 1^{\overline{4}} 14131^{\overline{3}} [\# = 2] \quad \Delta = -113, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
1414131^{\overline{1}} 1^{\overline{4}} 14131^{\overline{3}} [\# = 2] \quad \Delta = -113, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1414131^{\overline{1}} 1^{\overline{3}} \quad \Delta = -113, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
1414131^{\overline{1}} 1^{\overline{4}} 14131^{\overline{3}} [\# = 6] \quad \Delta = -113, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1414131^{\overline{1}} 1^{\overline{4}} 14131^{\overline{3}} [\# = 7] \quad \Delta = -113, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1414131^{\overline{1}} 1^{\overline{4}} 14131^{\overline{3}} [\# = 9] \quad \Delta = -113, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1414131^{\overline{1}} 1^{\overline{4}} 14131^{\overline{3}} [\# = 8] \quad \Delta = -113, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

$$1414131\bar{3} \text{ }_{[\# = 3]} \quad \Delta = -113, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1414131 \quad \Delta = -105, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: 4141413

$$414\bar{4}1413 \text{ }_{[\# = 3]} \quad \Delta = -141, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 60 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$414141\bar{3} \quad \Delta = -141, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$414\bar{4}1413 \text{ }_{[\# = 2]} \quad \Delta = -141, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$414\bar{4}1413 \quad \Delta = -141, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$414\bar{4}1413 \quad \Delta = -161, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$414141\bar{3} \quad \Delta = -133, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$414\bar{4}1413 \quad \Delta = -133, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$414141\bar{3} \quad \Delta = -133, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$414141\bar{3} \quad \Delta = -133, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 1414141

$$1414\bar{4}141 \quad \Delta = -153, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$1414\bar{4}141 \text{ }_{[\# = 3]} \quad \Delta = -133, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1414141\bar{3} \text{ }_{[\# = 3]} \quad \Delta = -133, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1414\bar{4}141 \text{ }_{[\# = 2]} \quad \Delta = -133, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1414141\bar{3} \text{ }_{[\# = 2]} \quad \Delta = -133, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1414\bar{4}141 \quad \Delta = -133, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1414\bar{4}141 \quad \Delta = -125, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1414141 \quad \Delta = -125, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 14132215

$$14132215 \quad \Delta = -130, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$14132215 \quad \Delta = -130, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$512231413 \text{ [#=2]} \quad \Delta = -138, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$512231413 \text{ [#=3]} \quad \Delta = -138, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$14132215 \quad \Delta = -138, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 31413215

$$31413215 \quad \Delta = -140, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$31413215 \quad \Delta = -138, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$31413215 \quad \Delta = -138, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$31413215 \quad \Delta = -138, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$31413215 \quad \Delta = -146, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: 23141315

$$23141315 \quad \Delta = -146, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$23141315 \quad \Delta = -138, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$23141315 \quad \Delta = -138, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$23141315 \quad \Delta = -138, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$23141315 \quad \Delta = -139, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 14132151

$$14132151 \quad \Delta = -132, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$151231413 \text{ [#=2]} \quad \Delta = -138, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$15123141\overset{3}{3} \text{ }_{[\# = 3]} \quad \Delta = -138, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$14132151 \quad \Delta = -130, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$14132151 \quad \Delta = -130, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 41232151

$$41232151 \quad \Delta = -132, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$41232151 \quad \Delta = -132, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$41232151 \quad \Delta = -140, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

Base: 31413151

$$31413151 \quad \Delta = -137, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$31413151 \quad \Delta = -137, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$31413151 \quad \Delta = -137, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$31413151 \quad \Delta = -138, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$31413151 \quad \Delta = -145, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$31413151 \quad \Delta = -145, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: 12321512

$$21512321\overset{3}{3} \text{ }_{[\# = 4]} \quad \Delta = -112, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$12321512 \quad \Delta = -104, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$12321512 \quad \Delta = -104, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$21512321\overset{3}{3} \text{ }_{[\# = 9]} \quad \Delta = -112, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$21512321\overset{3}{3} \text{ }_{[\# = 8]} \quad \Delta = -112, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$12321512 \quad \Delta = -112, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$\begin{aligned}
21512321\bar{3} \text{ } [\# = 2] \quad \Delta = -112, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21512321\bar{3} \text{ } [\# = 3] \quad \Delta = -112, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21512321\bar{3} \text{ } [\# = 7] \quad \Delta = -112, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
21512321\bar{3} \text{ } [\# = 5] \quad \Delta = -112, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
21512321\bar{3} \text{ } [\# = 6] \quad \Delta = -112, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}
\end{aligned}$$

Base: 41321513

$$\begin{aligned}
\bar{2}41321513 \quad \Delta = -138, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\bar{2}41321513 \quad \Delta = -140, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\bar{2}41321513 \quad \Delta = -138, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\bar{2}41321513 \quad \Delta = -138, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 14131512

$$\begin{aligned}
\bar{2}14131512 \quad \Delta = -129, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\bar{2}14131512 \quad \Delta = -130, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
21513141\bar{3} \text{ } [\# = 3] \quad \Delta = -137, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
21513141\bar{3} \text{ } [\# = 2] \quad \Delta = -137, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\bar{3}14131512 \quad \Delta = -137, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
\bar{3}14131512 \quad \Delta = -137, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\bar{2}14131512 \quad \Delta = -129, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\bar{2}14131512 \quad \Delta = -129, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: 41231512

$$\bar{3}41231512 \quad \Delta = -138, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$\begin{aligned}
41231512^2 \quad \Delta = -130, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
41231512^2 \quad \Delta = -130, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
41231512^2 \quad \Delta = -130, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
41231512^3 \quad \Delta = -138, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}
\end{aligned}$$

Base: 13151231

$$\begin{aligned}
131512313^{\#2} \quad \Delta = -118, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
132151313^{\#2} \quad \Delta = -118, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
132151313^{\#6} \quad \Delta = -118, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
132151313^{\#3} \quad \Delta = -118, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
131512313^{\#3} \quad \Delta = -118, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
131512313^{\#4} \quad \Delta = -118, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
132151313^{\#5} \quad \Delta = -118, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
131512313^{\#5} \quad \Delta = -118, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
132151313^{\#7} \quad \Delta = -118, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
132151313^{\#8} \quad \Delta = -118, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
132151313^{\#9} \quad \Delta = -118, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
132151313^{\#4} \quad \Delta = -118, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
13151231^2 \quad \Delta = -110, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
13151231^2 \quad \Delta = -110, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: 12315122

$$12315122^2 \quad \Delta = -102, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\begin{aligned}
12315122^2 \quad \Delta = -102, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
12315122^3 \quad \Delta = -110, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
12315122^3 \quad \Delta = -111, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
221513213^{\# = 2} \quad \Delta = -110, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
221513213^{\# = 9} \quad \Delta = -110, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
221513213^{\# = 4} \quad \Delta = -110, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
221513213^{\# = 8} \quad \Delta = -110, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
221513213^{\# = 3} \quad \Delta = -110, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
221513213^{\# = 5} \quad \Delta = -110, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
221513213^{\# = 7} \quad \Delta = -110, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
221513213^{\# = 6} \quad \Delta = -110, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}
\end{aligned}$$

Base: 41315123

$$\begin{aligned}
41315123^2 \quad \Delta = -138, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
41315123^2 \quad \Delta = -138, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
41315123^2 \quad \Delta = -138, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
41315123^2 \quad \Delta = -139, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
41315123^3 \quad \Delta = -146, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: 13151313

$$\begin{aligned}
13151313^2 \quad \Delta = -117, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
13151313^2 \quad \Delta = -117, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
13151313^4 \quad \Delta = -145, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 & \begin{pmatrix} 9 & 5 \\ 5 & 9 \end{pmatrix}
\end{aligned}$$

$$13151\overset{3}{13} \text{ }_{[\# = 3]} \quad \Delta = -125, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$13151\overset{2}{13} \quad \Delta = -117, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$13151\overset{2}{13} \quad \Delta = -118, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$13151\overset{3}{13} \quad \Delta = -125, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$31315131\overset{3}{3} \text{ }_{[\# = 2]} \quad \Delta = -125, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$13151\overset{3}{13} \text{ }_{[\# = 2]} \quad \Delta = -125, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$31315131\overset{3}{3} \text{ }_{[\# = 6]} \quad \Delta = -125, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$31315131\overset{3}{3} \text{ }_{[\# = 9]} \quad \Delta = -125, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix}$$

$$31315131\overset{3}{3} \text{ }_{[\# = 7]} \quad \Delta = -125, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$31315131\overset{3}{3} \text{ }_{[\# = 8]} \quad \Delta = -125, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$31315131\overset{3}{3} \text{ }_{[\# = 4]} \quad \Delta = -125, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$31315131\overset{3}{3} \text{ }_{[\# = 3]} \quad \Delta = -125, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$31315131\overset{3}{3} \text{ }_{[\# = 5]} \quad \Delta = -125, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

Base: $\overset{2}{13}15131$

$$\overset{2}{13}15131 \quad \Delta = -110, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$13151\overset{2}{13}1\overset{2}{2} \quad \Delta = -110, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\overset{2}{13}15131 \quad \Delta = -110, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$13151\overset{2}{13}1\overset{2}{5} \quad \Delta = -162, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 8 \end{pmatrix}$$

$$\overset{2}{13}15131\overset{3}{3} \text{ }_{[\# = 6]} \quad \Delta = -118, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$\overset{2}{13}15131\overset{3}{3} \text{ }_{[\# = 3]} \quad \Delta = -118, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$\begin{aligned}
13151\overset{2}{131}\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -118, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
13151\overset{2}{131}\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -118, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
13151\overset{2}{131}\textcolor{red}{3} \text{ [#=7]} \quad \Delta = -118, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
13151\overset{2}{131}\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -118, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
13151\overset{2}{131}\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -118, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
13151\overset{2}{131}\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -118, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
13151\overset{2}{131}\textcolor{red}{3} \text{ [#=8]} \quad \Delta = -118, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
13151\overset{2}{131}\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -118, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 100 \quad & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
13151\overset{2}{131}\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -118, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
13151\overset{2}{131}\textcolor{red}{4} \quad \Delta = -138, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
13151\overset{2}{131}\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -118, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\textbf{Base: } 1512321 \quad & \\
1512321\overset{1}{\textcolor{red}{3}} \text{ [#=2]} \quad \Delta = -112, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1512321\overset{1}{\textcolor{red}{3}} \text{ [#=3]} \quad \Delta = -112, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1512321\overset{1}{\textcolor{red}{3}} \text{ [#=5]} \quad \Delta = -112, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1512321\overset{1}{\textcolor{red}{3}} \text{ [#=4]} \quad \Delta = -112, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1512321 \overset{1}{\textcolor{red}{2}} \quad \Delta = -104, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1512321\overset{1}{\textcolor{red}{3}} \text{ [#=9]} \quad \Delta = -112, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1512321\overset{1}{\textcolor{red}{3}} \text{ [#=8]} \quad \Delta = -112, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1512321\overset{1}{\textcolor{red}{3}} \text{ [#=7]} \quad \Delta = -112, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}
\end{aligned}$$

$$1512321\overset{1}{3} \text{ [#=6]} \quad \Delta = -112, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

Base: $14131\overset{1}{5}1$

$$14131\overset{2}{5}1 \quad \Delta = -129, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1\overset{3}{5}14131 \text{ [#=3]} \quad \Delta = -137, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1\overset{3}{5}1\overset{1}{5}1 \text{ [#=2]} \quad \Delta = -137, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$14131\overset{3}{5}1 \quad \Delta = -137, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$14131\overset{2}{5}1 \quad \Delta = -130, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$14131\overset{2}{5}1 \quad \Delta = -129, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: $41231\overset{1}{5}1$

$$41231\overset{2}{5}1 \quad \Delta = -130, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$41231\overset{2}{5}1 \quad \Delta = -130, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$41231\overset{3}{5}1 \quad \Delta = -138, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

Base: $1\overset{1231}{5}12$

$$1\overset{1231}{5}1\overset{2}{2} \quad \Delta = -102, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1\overset{1231}{5}12 \quad \Delta = -102, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1\overset{3}{5}12\overset{1231}{12} \text{ [#=9]} \quad \Delta = -110, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$1\overset{3}{5}12\overset{1231}{12} \text{ [#=2]} \quad \Delta = -110, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1\overset{3}{5}12\overset{1231}{12} \text{ [#=4]} \quad \Delta = -110, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1\overset{3}{5}12\overset{1231}{12} \text{ [#=7]} \quad \Delta = -110, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$1\overset{1231}{5}1\overset{3}{2} \quad \Delta = -110, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$1\overset{3}{5}12\overset{1231}{12} \text{ [#=8]} \quad \Delta = -110, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$\begin{aligned}
1 \overset{3}{5} 12 \text{ } [\# = 5] \quad \Delta = -110, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1 \overset{3}{5} 12 \text{ } [\# = 6] \quad \Delta = -110, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1 \overset{3}{5} 12 \text{ } [\# = 3] \quad \Delta = -110, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

Base: 4131513

$$\begin{aligned}
4131513 \overset{2}{5} \overset{1}{13} \quad \Delta = -137, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
4131513 \overset{3}{5} \overset{1}{13} \quad \Delta = -145, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
4131513 \overset{2}{5} \overset{1}{13} \quad \Delta = -138, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
4131513 \overset{1}{5} \overset{2}{13} \quad \Delta = -137, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
4131513 \overset{2}{5} \overset{1}{13} \quad \Delta = -137, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
4131513 \overset{2}{5} \overset{1}{13} \quad \Delta = -137, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1315131

$$\begin{aligned}
1315131 \overset{1}{5} \overset{3}{13} \text{ } [\# = 6] \quad \Delta = -117, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1315131 \overset{1}{5} \overset{3}{13} \text{ } [\# = 7] \quad \Delta = -117, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
1315131 \overset{1}{5} \overset{3}{13} \text{ } [\# = 9] \quad \Delta = -117, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
1315131 \overset{1}{5} \overset{3}{13} \text{ } [\# = 8] \quad \Delta = -117, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1315131 \overset{1}{5} \overset{3}{13} \text{ } [\# = 5] \quad \Delta = -117, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1315131 \overset{1}{5} \overset{3}{13} \text{ } [\# = 4] \quad \Delta = -117, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1315131 \overset{1}{5} \overset{3}{13} \text{ } [\# = 3] \quad \Delta = -117, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1315131 \overset{1}{5} \overset{3}{13} \text{ } [\# = 2] \quad \Delta = -117, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1315131 \overset{2}{5} \overset{1}{13} \quad \Delta = -109, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: $31 \overset{131}{5} 13$

$$\begin{aligned}
31 \overset{3}{5} 13 \quad \Delta = -125, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
31 \overset{3}{5} 13 \text{ [#=5]} \quad \Delta = -125, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
31 \overset{3}{5} 13 \text{ [#=6]} \quad \Delta = -125, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 144 & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
31 \overset{3}{5} 13 \text{ [#=4]} \quad \Delta = -125, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
31 \overset{1}{5} 13 \quad \Delta = -118, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
31 \overset{2}{5} 13 \quad \Delta = -117, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
31 \overset{2}{5} 13 \quad \Delta = -117, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
31 \overset{1}{5} 13 \quad \Delta = -117, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
31 \overset{5}{5} 13 \quad \Delta = -169, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
31 \overset{4}{5} 13 \quad \Delta = -145, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
31 \overset{3}{5} 13 \text{ [#=7]} \quad \Delta = -125, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 196 & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
31 \overset{6}{5} 13 \quad \Delta = -195, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 8 \end{pmatrix} \\
31 \overset{3}{5} 13 \text{ [#=8]} \quad \Delta = -125, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 256 & \begin{pmatrix} 0 & 16 & 2 \\ 16 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
31 \overset{3}{5} 13 \text{ [#=3]} \quad \Delta = -125, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
31 \overset{3}{5} 13 \text{ [#=2]} \quad \Delta = -125, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
31 \overset{3}{5} 13 \text{ [#=9]} \quad \Delta = -125, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 324 & \begin{pmatrix} 0 & 18 & 2 \\ 18 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{1,1231}{5} 1 & \\
1 \overset{1,31231}{5} 1 \text{ [#=4]} \quad \Delta = -110, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1 \overset{1,31231}{5} 1 \text{ [#=3]} \quad \Delta = -110, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \overset{1,31231}{5} 1 \text{ [#=7]} \quad \Delta = -110, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1^{1,3,1231} 1^{1 \text{ [#}=2]} \quad \Delta = -110, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{1,3,1231} 1^{1 \text{ [#}=5]} \quad \Delta = -110, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1^{1,3,1231} 1^{1 \text{ [#}=9]} \quad \Delta = -110, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1^{1,3,1231} 1^{1 \text{ [#}=8]} \quad \Delta = -110, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1^{1,1231} 1^{1 \text{ [#}=4]} \quad \Delta = -102, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1^{1,3,1231} 1^{1 \text{ [#}=6]} \quad \Delta = -110, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}
\end{aligned}$$

Base: 61223132

$$\begin{aligned}
61223132^{2} \quad \Delta = -137, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
61223132^{3} \quad \Delta = -145, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
61223132^{3} \quad \Delta = -145, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 27 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
61223132^{2} \quad \Delta = -137, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: 61231322

$$\begin{aligned}
61231322^{3} \quad \Delta = -145, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 32 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
61231322^{2} \quad \Delta = -139, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
61231322^{2} \quad \Delta = -137, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
61231322^{2} \quad \Delta = -137, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

Base: 16122313

$$\begin{aligned}
16122313^{3} \quad \Delta = -144, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
16122313^{2} \quad \Delta = -136, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
16122313^{2} \quad \Delta = -136, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
16122313^{3} \text{ [#}=2] \quad \Delta = -144, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

$$16122313^3 \quad \Delta = -144, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 17 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 16123132

$$16123132^2 \quad \Delta = -139, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$16123132^3 \quad \Delta = -145, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$16123132^2 \quad \Delta = -137, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$16123132^2 \quad \Delta = -137, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$16123132^2 \quad \Delta = -139, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 16131322

$$16131322^2 \quad \Delta = -136, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$16131322^2 \quad \Delta = -136, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$16131322^{3 \text{ [#=2]}} \quad \Delta = -144, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 196 \quad \begin{pmatrix} 0 & 14 & 2 \\ 14 & 137 & 21 \\ 2 & 21 & 8 \end{pmatrix}$$

$$16131322^3 \quad \Delta = -144, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 32 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 21612231

$$21612231^3 \quad \Delta = -136, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$216122313^{3 \text{ [#=5]}} \quad \Delta = -136, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$216122313^{3 \text{ [#=4]}} \quad \Delta = -136, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$216122313^{3 \text{ [#=3]}} \quad \Delta = -136, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$216122313^{3 \text{ [#=2]}} \quad \Delta = -136, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$21612231^3 \quad \Delta = -136, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$21612231^2 \quad \Delta = -128, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$21612231^2 \quad \Delta = -128, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 21612313

$$\begin{aligned}
21612313^2 \quad \Delta = -138, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
21612313^2 \quad \Delta = -136, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21612313^2 \quad \Delta = -136, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21612313^2 \quad \Delta = -136, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21612313^{31} [\#=2] \quad \Delta = -144, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
21612313^3 \quad \Delta = -144, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21612313^3 \quad \Delta = -144, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: 2161232¹

$$\begin{aligned}
2161232^{41} \quad \Delta = -158, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
2161232^{21} \quad \Delta = -130, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2161232^{21} \quad \Delta = -130, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2161232^{21} \quad \Delta = -130, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2161232^{31} [\#=2] \quad \Delta = -138, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2161232^{31} [\#=3] \quad \Delta = -138, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2161232^3 \quad \Delta = -138, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: 32216123

$$\begin{aligned}
32216123^3 \quad \Delta = -145, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
32216123^2 \quad \Delta = -137, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
32216123^2 \quad \Delta = -137, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 22161231

$$221612313^{11} [\#=2] \quad \Delta = -136, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$22161231^2 \quad \Delta = -128, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$22161231^3 \quad \Delta = -137, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$22161231^3 \quad \Delta = -136, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$22161231^3 \text{ [#=3]} \quad \Delta = -136, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$22161231^3 \text{ [#=4]} \quad \Delta = -136, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$22161231^3 \text{ [#=5]} \quad \Delta = -136, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$22161231^2 \quad \Delta = -128, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 21613132

$$21613132^2 \quad \Delta = -136, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$21613132^2 \quad \Delta = -138, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$21613132^2 \quad \Delta = -136, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$21613132^2 \quad \Delta = -136, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$21613132^3 \text{ [#=2]} \quad \Delta = -144, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$21613132^3 \quad \Delta = -144, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$21613132^3 \quad \Delta = -144, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 22216131

$$22216131^3 \text{ [#=6]} \quad \Delta = -135, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$22216131^2 \quad \Delta = -127, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$22216131^2 \quad \Delta = -127, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$22216131^3 \quad \Delta = -135, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 17 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\begin{aligned}
22216131\bar{3} \text{ }_{[\# = 2]} \quad \Delta = -135, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
22216131\bar{3} \text{ }_{[\# = 3]} \quad \Delta = -135, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
22216131\bar{3} \text{ }_{[\# = 4]} \quad \Delta = -135, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
22216131\bar{3} \text{ }_{[\# = 5]} \quad \Delta = -135, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
22216131\bar{3} \text{ }_{[\# = 7]} \quad \Delta = -135, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
22216131\bar{3} \text{ }_{[\# = 8]} \quad \Delta = -135, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
22216131\bar{3} \text{ }_{[\# = 9]} \quad \Delta = -135, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 22161313

$$\begin{aligned}
2216131\bar{3}^4 \quad \Delta = -163, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 & \begin{pmatrix} 9 & 5 \\ 5 & 9 \end{pmatrix} \\
2216131\bar{3}^3 \text{ }_{[\# = 3]} \quad \Delta = -143, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2216131\bar{3}^3 \quad \Delta = -143, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2216131\bar{3}^2 \quad \Delta = -135, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2216131\bar{3}^2 \quad \Delta = -135, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2216131\bar{3}^2 \quad \Delta = -135, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2216131\bar{3}^3 \text{ }_{[\# = 2]} \quad \Delta = -143, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2216131\bar{3}^3 \quad \Delta = -144, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2216131\bar{3}^3 \quad \Delta = -143, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{13221}{6} 1$

$$\begin{aligned}
1 \overset{13221}{6} 1 \quad \Delta = -128, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \overset{313221}{6} 1 \text{ }_{[\# = 2]} \quad \Delta = -136, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
1 \overset{13221}{6} 1 \quad \Delta = -136, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{313221}{1 \ 6 \ 1} \text{ }_{[\# = 3]} \quad \Delta = -136, \ n_+ = 1, \ n_- = 8, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{313221}{1 \ 6 \ 1} \text{ }_{[\# = 4]} \quad \Delta = -136, \ n_+ = 1, \ n_- = 8, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{313221}{1 \ 6 \ 1} \text{ }_{[\# = 5]} \quad \Delta = -136, \ n_+ = 1, \ n_- = 8, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{31321}{1 \ 6 \ 1}$

$$\begin{aligned}
& \overset{31321}{1 \ 6 \ 1} \quad \Delta = -144, \ n_+ = 1, \ n_- = 8, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{31321}{1 \ 6 \ 1} \text{ }_{[\# = 2]} \quad \Delta = -144, \ n_+ = 1, \ n_- = 8, \ n_0 = 0, \ |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{31321}{1 \ 6 \ 1} \quad \Delta = -136, \ n_+ = 1, \ n_- = 8, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{31321}{1 \ 6 \ 1} \quad \Delta = -138, \ n_+ = 1, \ n_- = 8, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{31321}{1 \ 6 \ 1} \quad \Delta = -136, \ n_+ = 1, \ n_- = 8, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{2321}{1 \ 6 \ 1}$

$$\begin{aligned}
& \overset{2321}{1 \ 6 \ 1} \text{ }_{[\# = 2]} \quad \Delta = -138, \ n_+ = 1, \ n_- = 8, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{2321}{1 \ 6 \ 1} \quad \Delta = -158, \ n_+ = 1, \ n_- = 8, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
& \overset{2321}{1 \ 6 \ 1} \quad \Delta = -130, \ n_+ = 1, \ n_- = 8, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{2321}{1 \ 6 \ 1} \quad \Delta = -130, \ n_+ = 1, \ n_- = 8, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{2321}{1 \ 6 \ 1} \text{ }_{[\# = 3]} \quad \Delta = -138, \ n_+ = 1, \ n_- = 8, \ n_0 = 0, \ |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{23131}{1 \ 6 \ 1}$

$$\begin{aligned}
& \overset{23131}{1 \ 6 \ 1} \quad \Delta = -144, \ n_+ = 1, \ n_- = 8, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{23131}{1 \ 6 \ 1} \quad \Delta = -136, \ n_+ = 1, \ n_- = 8, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{23131}{1 \ 6 \ 1} \quad \Delta = -136, \ n_+ = 1, \ n_- = 8, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{23131}{1 \ 6 \ 1} \text{ }_{[\# = 2]} \quad \Delta = -144, \ n_+ = 1, \ n_- = 8, \ n_0 = 0, \ |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& \overset{23131}{1 \ 6 \ 1} \quad \Delta = -138, \ n_+ = 1, \ n_- = 8, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{1321}{6} 12$

$$1 \overset{31321}{6} 12 \text{ [\#}=3] \quad \Delta = -136, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1 \overset{31321}{6} 12 \text{ [\#}=2] \quad \Delta = -136, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$1 \overset{13213}{6} 12 \quad \Delta = -136, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{31321}{6} 12 \text{ [\#}=4] \quad \Delta = -136, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{31321}{6} 12 \text{ [\#}=5] \quad \Delta = -136, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{1321}{6} 12 \quad \Delta = -128, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{13212}{6} 12 \quad \Delta = -128, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: $1 \overset{321}{6} 123$

$$321 \overset{321}{6} 1\overset{3}{3} \quad \Delta = -145, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$321 \overset{321}{6} 1\overset{2}{2} \quad \Delta = -137, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{3212}{6} 123 \quad \Delta = -137, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: $1\overset{31}{6} 1223$

$$1\overset{313}{6} 1223 \quad \Delta = -144, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$3221\overset{31}{6} 1\overset{3}{3} \quad \Delta = -144, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1\overset{312}{6} 1223 \quad \Delta = -136, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$3221\overset{31}{6} 1\overset{2}{2} \quad \Delta = -136, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1\overset{31}{6} 1223 \quad \Delta = -136, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: $1 \overset{3131}{6} 12$

$$1 \overset{3131}{6} 12 \text{ [\#}=3] \quad \Delta = -143, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{3131}{6} 12 \quad \Delta = -135, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\begin{aligned}
& \overset{3131}{1 \ 6 \ 12} \quad \Delta = -163, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 9 & 5 \\ 5 & 9 \end{pmatrix} \\
& \overset{3131}{1 \ 6 \ 12} \text{ [#=2]} \quad \Delta = -143, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 6 \ 12} \quad \Delta = -143, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{3131\textcolor{red}{3}}{1 \ 6 \ 12} \quad \Delta = -143, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 6 \ 12} \quad \Delta = -135, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{3131\textcolor{red}{2}}{1 \ 6 \ 12} \quad \Delta = -135, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \textbf{Base: } \overset{131}{1 \ 6 \ 122} \\
& \overset{131}{1 \ 6 \ 122} \quad \Delta = -127, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{131\textcolor{red}{2}}{1 \ 6 \ 122} \quad \Delta = -127, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{131\textcolor{red}{3}}{1 \ 6 \ 122} \quad \Delta = -135, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{\textcolor{red}{3}131}{1 \ 6 \ 122} \text{ [#=3]} \quad \Delta = -135, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{\textcolor{red}{3}131}{1 \ 6 \ 122} \text{ [#=4]} \quad \Delta = -135, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{\textcolor{red}{3}131}{1 \ 6 \ 122} \text{ [#=5]} \quad \Delta = -135, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{\textcolor{red}{3}131}{1 \ 6 \ 122} \text{ [#=6]} \quad \Delta = -135, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{\textcolor{red}{3}131}{1 \ 6 \ 122} \text{ [#=7]} \quad \Delta = -135, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{\textcolor{red}{3}131}{1 \ 6 \ 122} \text{ [#=8]} \quad \Delta = -135, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{\textcolor{red}{3}131}{1 \ 6 \ 122} \text{ [#=9]} \quad \Delta = -135, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{\textcolor{red}{3}131}{1 \ 6 \ 122} \text{ [#=2]} \quad \Delta = -135, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
& \overset{131\textcolor{red}{3}}{1 \ 6 \ 122} \quad \Delta = -136, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

$$\textbf{Base: } \overset{31}{21 \ 6 \ 123}$$

$${}^{331}_{21\bar{6}123} \quad \Delta = -144, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$${}^{231}_{21\bar{6}123} \quad \Delta = -136, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$${}^{312}_{21\bar{6}123} \quad \Delta = -136, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$${}^{31}_{21\bar{6}123} \quad \Delta = -136, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: ${}^{131}_{21\bar{6}12}$

$${}^{2131}_{21\bar{6}12} \quad \Delta = -127, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$${}^{3131}_{21\bar{6}12 [\# = 6]} \quad \Delta = -135, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$${}^{3131}_{21\bar{6}12 [\# = 7]} \quad \Delta = -135, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$${}^{3131}_{21\bar{6}12 [\# = 4]} \quad \Delta = -135, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$${}^{3131}_{21\bar{6}12 [\# = 3]} \quad \Delta = -135, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$${}^{3131}_{21\bar{6}12} \quad \Delta = -135, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$${}^{3131}_{21\bar{6}12 [\# = 2]} \quad \Delta = -135, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$${}^{3131}_{21\bar{6}12 [\# = 8]} \quad \Delta = -135, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$${}^{3131}_{21\bar{6}12 [\# = 9]} \quad \Delta = -135, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$${}^{3131}_{21\bar{6}12 [\# = 5]} \quad \Delta = -135, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$${}^{131}_{21\bar{6}12} \quad \Delta = -127, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: ${}^{31}_{221\bar{6}13}$

$${}^{331}_{221\bar{6}13} \quad \Delta = -143, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$${}^{231}_{221\bar{6}13} \quad \Delta = -135, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$${}^{312}_{221\bar{6}13} \quad \Delta = -135, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\overset{3}{2}21\overset{31}{6}13 \quad \Delta = -144, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{1,1321}{6} 1$$

$$1 \overset{1,31321}{6} 1 \text{ [#=4]} \quad \Delta = -136, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1 \overset{1,1321}{6} 1 \quad \Delta = -128, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{1,31321}{6} 1 \text{ [#=2]} \quad \Delta = -136, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \overset{1,31321}{6} 1 \text{ [#=5]} \quad \Delta = -136, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1 \overset{1,31321}{6} 1 \text{ [#=3]} \quad \Delta = -136, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{1,3131}{6} 1$$

$$1 \overset{1,3131}{6} 1 \quad \Delta = -163, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \overset{1,3131}{6} 1 \quad \Delta = -135, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{1,3131}{6} 1 \text{ [#=3]} \quad \Delta = -143, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 132 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$1 \overset{1,3131}{6} 1 \text{ [#=2]} \quad \Delta = -143, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$1 \overset{1,3131}{6} 1 \quad \Delta = -143, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{1,3131}{6} 1 \quad \Delta = -135, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{1,31}{6} 123$$

$$1 \overset{1,31}{6} 123 \quad \Delta = -136, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$321 \overset{1,31}{6} \overset{13}{13} \quad \Delta = -144, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1 \overset{1,31}{6} \overset{2}{12}3 \quad \Delta = -136, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$321 \overset{1,31}{6} \overset{12}{12} \quad \Delta = -136, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{21,131}{6} 1$$

$$1 \overset{21,131}{6} 1 \quad \Delta = -127, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$\begin{aligned}
& \begin{matrix} 21, \textcolor{red}{3}131 \\ 1 \quad 6 \quad 1 \end{matrix} \quad [\# = 4] \quad \Delta = -135, \quad n_+ = 1, \quad n_- = 8, \quad n_0 = 0, \quad |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \begin{matrix} 21, \textcolor{red}{3}131 \\ 1 \quad 6 \quad 1 \end{matrix} \quad [\# = 5] \quad \Delta = -135, \quad n_+ = 1, \quad n_- = 8, \quad n_0 = 0, \quad |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \begin{matrix} 21, \textcolor{red}{3}131 \\ 1 \quad 6 \quad 1 \end{matrix} \quad [\# = 3] \quad \Delta = -135, \quad n_+ = 1, \quad n_- = 8, \quad n_0 = 0, \quad |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \begin{matrix} 21, \textcolor{red}{3}131 \\ 1 \quad 6 \quad 1 \end{matrix} \quad [\# = 6] \quad \Delta = -135, \quad n_+ = 1, \quad n_- = 8, \quad n_0 = 0, \quad |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \begin{matrix} 21, \textcolor{red}{3}131 \\ 1 \quad 6 \quad 1 \end{matrix} \quad [\# = 7] \quad \Delta = -135, \quad n_+ = 1, \quad n_- = 8, \quad n_0 = 0, \quad |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
& \begin{matrix} 21, \textcolor{red}{3}131 \\ 1 \quad 6 \quad 1 \end{matrix} \quad [\# = 8] \quad \Delta = -135, \quad n_+ = 1, \quad n_- = 8, \quad n_0 = 0, \quad |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& \begin{matrix} 21, \textcolor{red}{3}131 \\ 1 \quad 6 \quad 1 \end{matrix} \quad [\# = 2] \quad \Delta = -135, \quad n_+ = 1, \quad n_- = 8, \quad n_0 = 0, \quad |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \begin{matrix} 21, 131 \\ 1 \quad 6 \quad 1 \end{matrix} \quad \Delta = -135, \quad n_+ = 1, \quad n_- = 8, \quad n_0 = 0, \quad |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
& \begin{matrix} 21, \textcolor{red}{3}131 \\ 1 \quad 6 \quad 1 \end{matrix} \quad [\# = 9] \quad \Delta = -135, \quad n_+ = 1, \quad n_- = 8, \quad n_0 = 0, \quad |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
& \begin{matrix} 21, 131 \\ 1 \quad 6 \quad 1 \end{matrix} \quad \Delta = -127, \quad n_+ = 1, \quad n_- = 8, \quad n_0 = 0, \quad |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \textbf{Base: } \begin{matrix} 21, 31 \\ 1 \quad 6 \quad 13 \end{matrix} \\
& \begin{matrix} 21, 31 \\ 31 \quad 6 \quad 12 \end{matrix} \quad \Delta = -135, \quad n_+ = 1, \quad n_- = 8, \quad n_0 = 0, \quad |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \begin{matrix} 21, \textcolor{red}{3}12 \\ 1 \quad 6 \quad 13 \end{matrix} \quad \Delta = -135, \quad n_+ = 1, \quad n_- = 8, \quad n_0 = 0, \quad |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \begin{matrix} 21, 31 \\ 31 \quad 6 \quad 13 \end{matrix} \quad \Delta = -143, \quad n_+ = 1, \quad n_- = 8, \quad n_0 = 0, \quad |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \begin{matrix} 21, 31 \\ 1 \quad 6 \quad 13 \end{matrix} \quad \Delta = -135, \quad n_+ = 1, \quad n_- = 8, \quad n_0 = 0, \quad |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \begin{matrix} 21, 31 \\ 1 \quad 6 \quad 13 \end{matrix} \quad \Delta = -143, \quad n_+ = 1, \quad n_- = 8, \quad n_0 = 0, \quad |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \textbf{Base: } 71222222 \\
& \begin{matrix} \textcolor{red}{2} \\ 71222222 \end{matrix} \quad \Delta = -146, \quad n_+ = 1, \quad n_- = 8, \quad n_0 = 0, \quad |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \textbf{Base: } 17122222 \\
& \begin{matrix} \textcolor{red}{2} \\ 17122222 \end{matrix} \quad \Delta = -146, \quad n_+ = 1, \quad n_- = 8, \quad n_0 = 0, \quad |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \textbf{Base: } 21712222 \\
& \begin{matrix} \textcolor{red}{2} \\ 21712222 \end{matrix} \quad \Delta = -146, \quad n_+ = 1, \quad n_- = 8, \quad n_0 = 0, \quad |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

$$^{221}1712222 \quad \Delta = -146, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: $^{221}171222$

$$^{3221}171222 \quad \Delta = -155, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$^{221}171222 \quad \Delta = -146, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$^{221}171222 \quad \Delta = -146, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: $^{22221}1 \begin{smallmatrix} 7 \\ 7 \end{smallmatrix} 1$

$$^{22221}1 \begin{smallmatrix} 7 \\ 7 \end{smallmatrix} 1 \quad \Delta = -146, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: $^{2221}1 \begin{smallmatrix} 7 \\ 7 \end{smallmatrix} 12$

$$^{2221}1 \begin{smallmatrix} 7 \\ 7 \end{smallmatrix} 12 \quad \Delta = -146, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$^{22212}1 \begin{smallmatrix} 7 \\ 7 \end{smallmatrix} 12 \quad \Delta = -146, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: $^{221}1 \begin{smallmatrix} 7 \\ 7 \end{smallmatrix} 122$

$$^{2212}1 \begin{smallmatrix} 7 \\ 7 \end{smallmatrix} 122 \quad \Delta = -146, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$^{2213}1 \begin{smallmatrix} 7 \\ 7 \end{smallmatrix} 122 \quad \Delta = -155, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: $^{221}21 \begin{smallmatrix} 7 \\ 7 \end{smallmatrix} 12$

$$^{221}21 \begin{smallmatrix} 7 \\ 7 \end{smallmatrix} 12 \quad \Delta = -155, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$^{2221}21 \begin{smallmatrix} 7 \\ 7 \end{smallmatrix} 12 \quad \Delta = -146, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$^{221}21 \begin{smallmatrix} 7 \\ 7 \end{smallmatrix} 12 \quad \Delta = -146, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: $^{1,2221}1 \begin{smallmatrix} 7 \\ 7 \end{smallmatrix} 1$

$$^{1,2221}1 \begin{smallmatrix} 7 \\ 7 \end{smallmatrix} 1 \quad \Delta = -146, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: $^{221,21}1 \begin{smallmatrix} 7 \\ 7 \end{smallmatrix} 1$

$$^{221,21}1 \begin{smallmatrix} 7 \\ 7 \end{smallmatrix} 1 \quad \Delta = -146, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$^{221,21}1 \begin{smallmatrix} 7 \\ 7 \end{smallmatrix} 1 \quad \Delta = -146, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$^{221,21}1 \begin{smallmatrix} 7 \\ 7 \end{smallmatrix} 1 \quad \Delta = -155, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

Base: $^{21,21}1 \begin{smallmatrix} 7 \\ 7 \end{smallmatrix} 12$

$$1 \overset{21,21}{7} \overset{2}{12} \quad \Delta = -146, n_+ = 1, n_- = 8, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$T = 9$ (875 bases)

Base: 122222221

$$\begin{aligned} 122222221 \overset{3}{\text{red}}_{[\# = 5]} \quad \Delta = -20, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 197 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\ 122222221 \overset{3}{\text{red}}_{[\# = 8]} \quad \Delta = -20, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 509 & 63 \\ 2 & 63 & 8 \end{pmatrix} \\ 122222221 \overset{3}{\text{red}}_{[\# = 9]} \quad \Delta = -20, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 645 & 71 \\ 2 & 71 & 8 \end{pmatrix} \\ 122222221 \overset{3}{\text{red}}_{[\# = 2]} \quad \Delta = -20, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 29 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\ 122222221 \overset{3}{\text{red}}_{[\# = 7]} \quad \Delta = -20, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 389 & 55 \\ 2 & 55 & 8 \end{pmatrix} \\ 122222221 \overset{3}{\text{red}}_{[\# = 4]} \quad \Delta = -20, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 125 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\ 122222221 \overset{3}{\text{red}}_{[\# = 6]} \quad \Delta = -20, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 285 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\ 122222221 \overset{3}{\text{red}}_{[\# = 3]} \quad \Delta = -20, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 69 & 23 \\ 2 & 23 & 8 \end{pmatrix} \end{aligned}$$

Base: 12222222²

$$\begin{aligned} 22222221 \overset{2}{\text{red}} \quad \Delta = -40, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\ 22222221 \overset{2}{\text{red}} \quad \Delta = -90, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\ 22222221 \overset{2}{\text{red}} \quad \Delta = -64, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\ 22222221 \overset{2}{\text{red}} \quad \Delta = -145, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\ 22222221 \overset{2}{\text{red}} \quad \Delta = -117, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\ 22222221 \overset{2}{\text{red}} \quad \Delta = -260, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -4 & -2 \\ 2 & -2 & 8 \end{pmatrix} \\ 22222221 \overset{2}{\text{red}} \quad \Delta = -12, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\ 22222221 \overset{2}{\text{red}} \quad \Delta = -20, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 7 \\ 2 & 7 & 8 \end{pmatrix} \end{aligned}$$

$$\begin{aligned}
2^2 2222221\mathbf{3} \text{ }_{[\# = 3]} \quad \Delta = -20, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
2^2 2222221\mathbf{3} \text{ }_{[\# = 8]} \quad \Delta = -20, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 256 & \begin{pmatrix} 0 & 16 & 2 \\ 16 & 509 & 63 \\ 2 & 63 & 8 \end{pmatrix} \\
2^2 2222221\mathbf{3} \text{ }_{[\# = 9]} \quad \Delta = -20, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 324 & \begin{pmatrix} 0 & 18 & 2 \\ 18 & 645 & 71 \\ 2 & 71 & 8 \end{pmatrix} \\
2^2 2222221\mathbf{3} \text{ }_{[\# = 7]} \quad \Delta = -20, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 196 & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 389 & 55 \\ 2 & 55 & 8 \end{pmatrix} \\
2^2 2222221\mathbf{3} \text{ }_{[\# = 5]} \quad \Delta = -20, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 197 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
2^2 2222221\mathbf{3} \text{ }_{[\# = 6]} \quad \Delta = -20, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 144 & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 285 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
2^2 2222221\mathbf{3} \text{ }_{[\# = 2]} \quad \Delta = -20, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
2^2 2222221\mathbf{3} \text{ }_{[\# = 4]} \quad \Delta = -20, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 31 \\ 2 & 31 & 8 \end{pmatrix}
\end{aligned}$$

Base: 122313221

$$\begin{aligned}
122313221\mathbf{3} \text{ }_{[\# = 2]} \quad \Delta = -38, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 25 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
122313221\mathbf{3} \text{ }_{[\# = 8]} \quad \Delta = -38, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 445 & 55 \\ 2 & 55 & 8 \end{pmatrix} \\
122313221\mathbf{3} \text{ }_{[\# = 9]} \quad \Delta = -38, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 564 & 62 \\ 2 & 62 & 8 \end{pmatrix} \\
1223\mathbf{3} 2221 \quad \Delta = -38, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
122313221\mathbf{3} \text{ }_{[\# = 4]} \quad \Delta = -38, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 109 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
122313221\mathbf{3} \text{ }_{[\# = 5]} \quad \Delta = -38, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 172 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
122313221\mathbf{3} \text{ }_{[\# = 6]} \quad \Delta = -38, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 249 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
122313221\mathbf{3} \text{ }_{[\# = 7]} \quad \Delta = -38, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 340 & 48 \\ 2 & 48 & 8 \end{pmatrix} \\
1223\mathbf{2} 2221 \quad \Delta = -30, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
122313221\mathbf{3} \text{ }_{[\# = 3]} \quad \Delta = -38, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 60 & 20 \\ 2 & 20 & 8 \end{pmatrix}
\end{aligned}$$

Base: 131413221

$$\begin{array}{ll}
131413221\overset{3}{3}_{[\# = 5]} & \Delta = -65, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
122314131\overset{3}{3}_{[\# = 7]} & \Delta = -65, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
122314131\overset{3}{3}_{[\# = 5]} & \Delta = -65, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
122314131\overset{3}{3}_{[\# = 4]} & \Delta = -65, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
131413221\overset{3}{3} & \Delta = -65, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
122314131\overset{3}{3}_{[\# = 2]} & \Delta = -65, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
131413221\overset{3}{3}_{[\# = 2]} & \Delta = -65, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
122314131\overset{3}{3}_{[\# = 3]} & \Delta = -65, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
131413221\overset{3}{3}_{[\# = 3]} & \Delta = -65, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
131413221\overset{3}{3}_{[\# = 4]} & \Delta = -65, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
131413221\overset{3}{3}_{[\# = 6]} & \Delta = -65, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
131413221\overset{3}{3}_{[\# = 7]} & \Delta = -65, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
131413221\overset{3}{3}_{[\# = 8]} & \Delta = -65, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
122314131\overset{3}{3}_{[\# = 9]} & \Delta = -65, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
131413221\overset{3}{3}_{[\# = 9]} & \Delta = -65, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
122314131\overset{3}{3}_{[\# = 8]} & \Delta = -65, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
131413221\overset{2}{2} & \Delta = -57, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
131413221\overset{2}{2} & \Delta = -57, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
122314131\overset{3}{3}_{[\# = 6]} & \Delta = -65, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix}
\end{array}$$

Base: 123141321

$$\begin{aligned}
123141321\bar{3} \text{ }_{[\# = 4]} \quad \Delta = -66, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
123141321\bar{2} \quad \Delta = -58, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
123141321\bar{3} \text{ }_{[\# = 9]} \quad \Delta = -66, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
123141321\bar{3} \text{ }_{[\# = 8]} \quad \Delta = -66, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
123141321\bar{3} \text{ }_{[\# = 7]} \quad \Delta = -66, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
123141321\bar{3} \text{ }_{[\# = 6]} \quad \Delta = -66, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
123141321\bar{3} \text{ }_{[\# = 5]} \quad \Delta = -66, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
123141321\bar{3} \text{ }_{[\# = 3]} \quad \Delta = -66, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
123141321\bar{3} \text{ }_{[\# = 2]} \quad \Delta = -66, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: $123141\bar{3}2$

$$\begin{aligned}
2\bar{6}^{123141} \quad 2 \quad \Delta = -138, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2\bar{5}^{123141} \quad 2 \quad \Delta = -112, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2\bar{7}^{123141} \quad 2 \quad \Delta = -165, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
2\bar{4}^{123141} \quad 2 \quad \Delta = -88, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2\bar{8}^{123141} \quad 2 \quad \Delta = -193, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 8 \end{pmatrix} \\
2\bar{3}^{123141} \text{ }_{[\# = 2]} \quad \Delta = -68, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
2\bar{3}^{123141} \text{ }_{[\# = 9]} \quad \Delta = -68, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 324 \quad & \begin{pmatrix} 0 & 18 & 2 \\ 18 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix} \\
2\bar{3}^{123141} \text{ }_{[\# = 3]} \quad \Delta = -68, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
2\bar{3}^{123141} \text{ }_{[\# = 8]} \quad \Delta = -68, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 256 \quad & \begin{pmatrix} 0 & 16 & 2 \\ 16 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
2\bar{3}^{123141} \text{ }_{[\# = 7]} \quad \Delta = -68, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 196 \quad & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix}
\end{aligned}$$

$$2 \overset{3}{1}23141 \quad \Delta = -68, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$2 \overset{3}{1}23141 \quad \Delta = -68, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$2 \overset{3}{1}23141 \quad \Delta = -68, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

$$2 \overset{3}{1}23141 \quad \Delta = -68, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$123141\overset{2}{3}2 \quad \Delta = -62, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$2 \overset{2}{1}23141 \quad \Delta = -60, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$123141\overset{22}{3}2 \quad \Delta = -60, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$123141\overset{2}{3}2 \quad \Delta = -60, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 214141321

$$214141321\overset{3}{3} \quad \Delta = -85, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$214141321\overset{3}{3} \quad \Delta = -85, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$214141321\overset{3}{3} \quad \Delta = -85, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$214141321\overset{3}{3} \quad \Delta = -85, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$214141321\overset{3}{3} \quad \Delta = -85, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$214141321\overset{3}{3} \quad \Delta = -85, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$214141321\overset{3}{3} \quad \Delta = -85, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$214141321 \quad \Delta = -85, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 87 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$214141321 \quad \Delta = -85, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 37 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$214141321\overset{3}{3} \quad \Delta = -85, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$214141321 \quad \Delta = -105, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$214141321 \quad \Delta = -85, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$214141321 \quad \Delta = -77, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$214141321 \quad \Delta = -77, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$214141321 \quad \Delta = -77, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 131414131

$$131414131 \quad \Delta = -92, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$131414131 \quad \Delta = -84, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$131414131 \quad \Delta = -84, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$131414131 \quad \Delta = -92, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$131414131 \quad \Delta = -92, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$131414131 \quad \Delta = -92, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$131414131 \quad \Delta = -92, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$131414131 \quad \Delta = -92, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$131414131 \quad \Delta = -92, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$131414131 \quad \Delta = -92, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$131414131 \quad \Delta = -92, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

Base: 21414132

$$2 \quad \begin{matrix} 3 \\ 214141 \\ 3 \end{matrix} \quad 2 \quad \Delta = -88, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$21414132 \quad \Delta = -79, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$21414132 \quad \Delta = -79, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$\begin{aligned}
\overset{2}{2}14141\overset{2}{3}2 \quad \Delta = -79, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\overset{3}{2}14141\overset{2}{3}2 \quad \Delta = -87, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 15 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\overset{3}{2}14141\overset{2}{3}2 \text{ [#=2]} \quad \Delta = -87, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 69 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
\overset{3}{2}14141\overset{2}{3}2 \text{ [#=3]} \quad \Delta = -87, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 144 & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 159 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
\overset{4}{2}14141\overset{2}{3}2 \quad \Delta = -107, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 14 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 214141412

$$\begin{aligned}
\overset{3}{2}14141412 \text{ [#=2]} \quad \Delta = -104, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
\overset{3}{2}14141412 \text{ [#=3]} \quad \Delta = -104, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 123 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
\overset{3}{2}14141412 \quad \Delta = -104, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\overset{4}{2}14141412 \quad \Delta = -124, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\overset{2}{2}14141412 \quad \Delta = -96, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\overset{2}{2}14141412 \quad \Delta = -96, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{1}4141321$

$$\begin{aligned}
\overset{1}{1}4141321\overset{1}{3} \text{ [#=7]} \quad \Delta = -85, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
\overset{1}{1}4141321\overset{1}{3} \text{ [#=8]} \quad \Delta = -85, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
\overset{1}{1}4141321\overset{1}{3} \text{ [#=9]} \quad \Delta = -85, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
\overset{1}{1}4141321\overset{1}{3} \text{ [#=2]} \quad \Delta = -85, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\overset{31}{1}4141321 \text{ [#=3]} \quad \Delta = -85, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\overset{31}{1}4141321 \text{ [#=2]} \quad \Delta = -85, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\overset{1}{1}4141321\overset{2}{3} \quad \Delta = -77, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
14141321 \quad \Delta = -77, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
141413213 \quad [\# = 5] \quad \Delta = -85, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
141413213 \quad [\# = 4] \quad \Delta = -85, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
141413213 \quad [\# = 3] \quad \Delta = -85, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
141413213 \quad [\# = 6] \quad \Delta = -85, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1414132

$$\begin{aligned}
1414132 \quad \Delta = -107, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1414132 \quad \Delta = -79, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1414132 \quad \Delta = -79, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1414132 \quad \Delta = -79, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1414132 \quad \Delta = -87, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1414132 \quad [\# = 2] \quad \Delta = -87, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1414132 \quad [\# = 3] \quad \Delta = -87, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

Base: 14141412

$$\begin{aligned}
14141412 \quad \Delta = -104, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
14141412 \quad [\# = 2] \quad \Delta = -104, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
14141412 \quad [\# = 2] \quad \Delta = -104, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
14141412 \quad [\# = 3] \quad \Delta = -104, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
14141412 \quad [\# = 3] \quad \Delta = -104, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
14141412 \quad \Delta = -124, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
14141412 \quad \Delta = -96, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
14141412 \quad \Delta = -96, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
14141412 \quad \Delta = -96, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 41414141

$$\begin{aligned}
41414141 \quad \Delta = -152, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
41414141 \quad \Delta = -124, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
414141413 \quad [\# = 3] \quad \Delta = -132, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
41414141 \quad \Delta = -124, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
41414141 \quad [\# = 2] \quad \Delta = -132, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
41414141 \quad \Delta = -124, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
414141413 \quad [\# = 2] \quad \Delta = -132, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
41414141 \quad \Delta = -132, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
41414141 \quad [\# = 3] \quad \Delta = -132, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
41414141 \quad \Delta = -124, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 14141411

$$\begin{aligned}
14141411 \quad [\# = 3] \quad \Delta = -104, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
14141411 \quad [\# = 2] \quad \Delta = -104, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
14141411 \quad \Delta = -96, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 141413215

$$\begin{aligned}
141413215 \quad \Delta = -129, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
141413215 \quad \Delta = -129, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

$$512314141\overset{2}{3} \text{ }_{[\# = 3]} \quad \Delta = -137, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$141413215 \quad \Delta = -129, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$14141\overset{2}{3}215 \quad \Delta = -131, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$512314141\overset{2}{3} \text{ }_{[\# = 2]} \quad \Delta = -137, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: 314141315

$$314141\overset{2}{3}15 \quad \Delta = -137, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$314141\overset{2}{3}15 \quad \Delta = -136, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$314141\overset{2}{3}15 \quad \Delta = -136, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$314141\overset{2}{3}15 \quad \Delta = -136, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$314141\overset{3}{3}15 \quad \Delta = -144, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$314141\overset{3}{3}15 \quad \Delta = -144, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$314141\overset{2}{3}15 \quad \Delta = -136, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: 151232151

$$1512321\overset{2}{5}1 \quad \Delta = -127, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 141413151

$$151314141\overset{2}{3} \text{ }_{[\# = 2]} \quad \Delta = -136, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1414131\overset{2}{5}1 \quad \Delta = -128, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1414131\overset{2}{5}1 \quad \Delta = -128, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1414131\overset{2}{5}1 \quad \Delta = -128, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1414131\overset{2}{5}1 \quad \Delta = -129, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$151314141\overset{2}{3} \text{ }_{[\# = 3]} \quad \Delta = -136, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$141413151^3 \quad \Delta = -136, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: 315123215

$$315123215^2 \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$315123215^2 \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$315123215^2 \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

Base: 215132215

$$215132215^2 \quad \Delta = -125, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$215132215^3 \quad \Delta = -133, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$215132215^3 \quad \Delta = -133, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$215132215^2 \quad \Delta = -125, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$215132215^2 \quad \Delta = -125, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 123215131

$$1315123213_{[\# = 7]} \quad \Delta = -91, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix}$$

$$1232151313_{[\# = 6]} \quad \Delta = -91, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$1315123213_{[\# = 6]} \quad \Delta = -91, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

$$1232151313_{[\# = 5]} \quad \Delta = -91, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$1232151313_{[\# = 4]} \quad \Delta = -91, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$1315123213_{[\# = 4]} \quad \Delta = -91, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$1232151313_{[\# = 3]} \quad \Delta = -91, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$1315123213_{[\# = 3]} \quad \Delta = -91, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1232151313_{[\# = 2]} \quad \Delta = -91, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$\begin{aligned}
131512321\bar{3} \text{ }_{[\# = 2]} \quad \Delta = -91, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
123215131\bar{3} \text{ }_{[\# = 9]} \quad \Delta = -91, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
131512321\bar{3} \text{ }_{[\# = 9]} \quad \Delta = -91, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
123215131\bar{3} \text{ }_{[\# = 8]} \quad \Delta = -91, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
131512321\bar{3} \text{ }_{[\# = 5]} \quad \Delta = -91, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
131512321\bar{3} \text{ }_{[\# = 8]} \quad \Delta = -91, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
123215131\bar{3} \text{ }_{[\# = 7]} \quad \Delta = -91, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
123215131\bar{2} \quad \Delta = -83, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
123215131\bar{2} \quad \Delta = -83, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 122315122

$$\begin{aligned}
221513221\bar{3} \text{ }_{[\# = 4]} \quad \Delta = -81, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
122315122\bar{2} \quad \Delta = -73, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
122315122\bar{2} \quad \Delta = -73, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
221513221\bar{3} \text{ }_{[\# = 9]} \quad \Delta = -81, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
122315122\bar{3} \quad \Delta = -82, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
221513221\bar{3} \text{ }_{[\# = 8]} \quad \Delta = -81, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
221513221\bar{3} \text{ }_{[\# = 6]} \quad \Delta = -81, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
221513221\bar{3} \text{ }_{[\# = 5]} \quad \Delta = -81, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
221513221\bar{3} \text{ }_{[\# = 3]} \quad \Delta = -81, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
221513221\bar{3} \text{ }_{[\# = 7]} \quad \Delta = -81, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix}
\end{aligned}$$

$$221513221\overset{3}{3}_{[\# = 2]} \quad \Delta = -81, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$122315122\overset{3}{3} \quad \Delta = -81, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: 123151231

$$123151231\overset{3}{3}_{[\# = 5]} \quad \Delta = -90, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$132151321\overset{3}{3}_{[\# = 2]} \quad \Delta = -90, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$123151231\overset{3}{3}_{[\# = 2]} \quad \Delta = -90, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$123151231\overset{2}{3} \quad \Delta = -82, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$132151321\overset{3}{3}_{[\# = 3]} \quad \Delta = -90, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$123151231\overset{3}{3}_{[\# = 3]} \quad \Delta = -90, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$123151231\overset{3}{3}_{[\# = 4]} \quad \Delta = -90, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$132151321\overset{3}{3}_{[\# = 4]} \quad \Delta = -90, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$132151321\overset{3}{3}_{[\# = 5]} \quad \Delta = -90, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix}$$

$$123151231\overset{2}{3} \quad \Delta = -82, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$132151321\overset{3}{3}_{[\# = 6]} \quad \Delta = -90, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

$$132151321\overset{3}{3}_{[\# = 7]} \quad \Delta = -90, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix}$$

$$132151321\overset{3}{3}_{[\# = 8]} \quad \Delta = -90, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix}$$

$$132151321\overset{3}{3}_{[\# = 9]} \quad \Delta = -90, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix}$$

Base: 321513215

$$321513215\overset{2}{3} \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$321513215\overset{2}{3} \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$321513215 \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 231512315

$$231512315 \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$231512315 \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$231512315 \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: 215132151

$$215132151 \quad \Delta = -125, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$215132151 \quad \Delta = -125, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$215132151 \quad \Delta = -125, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$215132151 \quad \Delta = -133, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: 315123151

$$315123151 \quad \Delta = -133, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$315123151 \quad \Delta = -133, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$315123151 \quad \Delta = -133, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 232151315

$$232151315 \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$232151315 \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$232151315 \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$232151315 \quad \Delta = -136, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 414131513

$$414131513 \quad \Delta = -136, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$414131513 \quad \Delta = -136, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\begin{aligned}
414131513 \quad \Delta = -144, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 37 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
414131513 \quad \Delta = -137, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
414131513 \quad \Delta = -136, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
414131513 \quad \Delta = -136, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: 141315131

$$\begin{aligned}
1413151313_{[\# = 9]} \quad \Delta = -116, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
141315131 \quad \Delta = -108, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
141315131 \quad \Delta = -109, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
141315131 \quad \Delta = -108, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1413151313_{[\# = 3]} \quad \Delta = -116, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1413151313_{[\# = 7]} \quad \Delta = -116, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1413151313_{[\# = 8]} \quad \Delta = -116, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1413151313_{[\# = 4]} \quad \Delta = -116, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1413151313_{[\# = 2]} \quad \Delta = -116, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1315131413_{[\# = 2]} \quad \Delta = -116, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1413151313_{[\# = 6]} \quad \Delta = -116, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
141315131 \quad \Delta = -108, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
141315131 \quad \Delta = -116, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1413151313_{[\# = 5]} \quad \Delta = -116, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1315131413_{[\# = 3]} \quad \Delta = -116, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: 413151314

$$413151314^2 \quad \Delta = -137, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$413151314^2 \quad \Delta = -136, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$413151314^2 \quad \Delta = -136, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$413151314^3 \quad \Delta = -144, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: 412315131

$$412315131^3_{[\# = 9]} \quad \Delta = -117, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$412315131^3_{[\# = 2]} \quad \Delta = -117, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$412315131^3_{[\# = 4]} \quad \Delta = -117, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$412315131^3_{[\# = 3]} \quad \Delta = -117, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$412315131^2 \quad \Delta = -109, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$412315131^2 \quad \Delta = -109, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$412315131^2 \quad \Delta = -109, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$412315131^3_{[\# = 8]} \quad \Delta = -117, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$412315131^3 \quad \Delta = -117, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 41 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$412315131^3_{[\# = 6]} \quad \Delta = -117, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$412315131^3_{[\# = 5]} \quad \Delta = -117, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$412315131^3_{[\# = 7]} \quad \Delta = -117, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

Base: 123151313

$$313151321^3_{[\# = 9]} \quad \Delta = -97, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$313151321^3_{[\# = 8]} \quad \Delta = -97, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$313151321\overset{3}{3} \text{ }_{[\# = 4]} \quad \Delta = -97, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$123151313\overset{3}{3} \text{ }_{[\# = 2]} \quad \Delta = -97, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$123151313\overset{4}{3} \quad \Delta = -117, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 9 & 5 \\ 5 & 9 \end{pmatrix}$$

$$313151321\overset{3}{3} \text{ }_{[\# = 2]} \quad \Delta = -97, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$123151313\overset{2}{3} \quad \Delta = -89, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$123151313\overset{2}{3} \quad \Delta = -89, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$123151313\overset{2}{3} \quad \Delta = -89, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$313151321\overset{3}{3} \text{ }_{[\# = 7]} \quad \Delta = -97, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$313151321\overset{3}{3} \text{ }_{[\# = 6]} \quad \Delta = -97, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$123151313\overset{2}{3} \quad \Delta = -90, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$123151313\overset{3}{3} \text{ }_{[\# = 3]} \quad \Delta = -97, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$313151321\overset{3}{3} \text{ }_{[\# = 5]} \quad \Delta = -97, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$123151313\overset{3}{3} \quad \Delta = -97, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$313151321\overset{3}{3} \text{ }_{[\# = 3]} \quad \Delta = -97, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: $\overset{2}{13151321}$

$$123151\overset{2}{3}1\overset{5}{5} \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 8 \end{pmatrix}$$

$$\overset{2}{13151321}\overset{2}{2} \quad \Delta = -82, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$123151\overset{2}{3}1\overset{2}{2} \quad \Delta = -82, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\overset{2}{13151321}\overset{2}{2} \quad \Delta = -82, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$123151\overset{2}{3}1\overset{4}{4} \quad \Delta = -110, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$\begin{aligned}
13151321\overset{2}{3} \text{ [#=3]} \quad \Delta = -90, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
12315131\overset{2}{3} \text{ [#=2]} \quad \Delta = -90, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
12315131\overset{2}{3} \text{ [#=4]} \quad \Delta = -90, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
12315131\overset{2}{3} \text{ [#=3]} \quad \Delta = -90, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
13151321\overset{2}{3} \text{ [#=7]} \quad \Delta = -90, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
13151321\overset{2}{3} \text{ [#=6]} \quad \Delta = -90, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
13151321\overset{2}{3} \text{ [#=5]} \quad \Delta = -90, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
12315131\overset{2}{3} \text{ [#=5]} \quad \Delta = -90, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
13151321\overset{2}{3} \text{ [#=4]} \quad \Delta = -90, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
13151321\overset{2}{3} \text{ [#=2]} \quad \Delta = -90, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
13151321\overset{2}{3} \text{ [#=8]} \quad \Delta = -90, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
13151321\overset{2}{3} \text{ [#=9]} \quad \Delta = -90, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}
\end{aligned}$$

Base: 321513151

$$\begin{aligned}
321513151\overset{2}{1} \quad \Delta = -133, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
321513151\overset{2}{1} \quad \Delta = -133, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
321513151\overset{2}{1} \quad \Delta = -133, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
321513151\overset{2}{1} \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 215131512

$$\begin{aligned}
215131512\overset{3}{1} \quad \Delta = -132, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
215131512\overset{2}{1} \quad \Delta = -124, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

$$215131512 \quad \Delta = -124, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$215131512 \quad \Delta = -125, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \begin{smallmatrix} 512231 \\ 5 \end{smallmatrix} 1$$

$$1 \begin{smallmatrix} 512231 \\ 5 \end{smallmatrix} 1 \quad \Delta = -125, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \begin{smallmatrix} 512231 \\ 5 \end{smallmatrix} 1 \quad \Delta = -125, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \begin{smallmatrix} 512231 \\ 5 \end{smallmatrix} 1 \quad \Delta = -133, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\text{Base: } 15132151$$

$$15132151 \quad \Delta = -125, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$15132151 \quad \Delta = -125, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \begin{smallmatrix} 215131 \\ 5 \end{smallmatrix} 1$$

$$1 \begin{smallmatrix} 215131 \\ 5 \end{smallmatrix} 1 \quad \Delta = -125, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \begin{smallmatrix} 215131 \\ 5 \end{smallmatrix} 1 \quad \Delta = -124, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \begin{smallmatrix} 215131 \\ 5 \end{smallmatrix} 1 \quad \Delta = -124, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1 \begin{smallmatrix} 215131 \\ 5 \end{smallmatrix} 1 \quad \Delta = -124, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \begin{smallmatrix} 215131 \\ 5 \end{smallmatrix} 1 \quad \Delta = -132, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$\text{Base: } 1 \begin{smallmatrix} 12231 \\ 5 \end{smallmatrix} 12$$

$$1 \begin{smallmatrix} 12231 \\ 5 \end{smallmatrix} 12 \quad \Delta = -73, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1 \begin{smallmatrix} 312231 \\ 5 \end{smallmatrix} 12 \text{ [#=7]} \quad \Delta = -81, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix}$$

$$1 \begin{smallmatrix} 12231 \\ 5 \end{smallmatrix} 12 \quad \Delta = -73, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \begin{smallmatrix} 312231 \\ 5 \end{smallmatrix} 12 \text{ [#=9]} \quad \Delta = -81, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix}$$

$$1 \begin{smallmatrix} 312231 \\ 5 \end{smallmatrix} 12 \text{ [#=6]} \quad \Delta = -81, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

$$1 \overset{3}{1} 5^{12231} 12_{[\# = 5]} \quad \Delta = -81, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix}$$

$$1 \overset{3}{1} 5^{12231} 12_{[\# = 4]} \quad \Delta = -81, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$1 \overset{3}{1} 5^{12231} 12_{[\# = 3]} \quad \Delta = -81, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1 \overset{3}{1} 5^{12231} 12_{[\# = 2]} \quad \Delta = -81, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1 \overset{12231}{5} \overset{3}{1} 12 \quad \Delta = -81, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$1 \overset{3}{1} 5^{12231} 12_{[\# = 8]} \quad \Delta = -81, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix}$$

Base: $31 \overset{51231}{5} 1$

$$31 \overset{51231}{5} \overset{12}{1} \quad \Delta = -133, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$31 \overset{51231}{5} 1 \quad \Delta = -133, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$31 \overset{51231}{5} 1 \quad \Delta = -133, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$31 \overset{251231}{5} 1 \quad \Delta = -133, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: $1 \overset{15131}{5} 13$

$$1 \overset{15131}{5} 13 \quad \Delta = -133, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$15131 \overset{31}{5} 13 \quad \Delta = -140, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1 \overset{15131}{5} 13 \quad \Delta = -132, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{15131}{5} 13 \quad \Delta = -132, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$15131 \overset{21}{5} 13 \quad \Delta = -132, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{15131}{5} \overset{12}{1} 13 \quad \Delta = -132, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: $1 \overset{15131}{5} 1$

$$1 \overset{15131}{5} 1 \quad \Delta = -125, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{15131}{5} 1 \quad \Delta = -124, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: $131\bar{5}1321$

$$131\bar{5}1321\bar{3} \text{ [#=7]} \quad \Delta = -89, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$131\bar{5}1321\bar{3} \text{ [#=2]} \quad \Delta = -89, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1231\bar{5}131\bar{3} \text{ [#=2]} \quad \Delta = -89, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$131\bar{5}1321\bar{3} \text{ [#=9]} \quad \Delta = -89, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$1231\bar{5}131\bar{3} \text{ [#=9]} \quad \Delta = -89, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix}$$

$$131\bar{5}1321\bar{3} \text{ [#=8]} \quad \Delta = -89, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$131\bar{5}1321\bar{3} \text{ [#=5]} \quad \Delta = -89, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$1231\bar{5}131\bar{3} \text{ [#=8]} \quad \Delta = -89, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$1231\bar{5}131\bar{3} \text{ [#=7]} \quad \Delta = -89, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$131\bar{5}1321\bar{3} \text{ [#=6]} \quad \Delta = -89, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$1231\bar{5}131\bar{3} \text{ [#=6]} \quad \Delta = -89, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$1231\bar{5}131\bar{3} \text{ [#=5]} \quad \Delta = -89, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$131\bar{5}1321\bar{3} \text{ [#=4]} \quad \Delta = -89, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1231\bar{5}131\bar{3} \text{ [#=4]} \quad \Delta = -89, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1231\bar{5}131\bar{3} \text{ [#=3]} \quad \Delta = -89, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$131\bar{5}1321 \quad \Delta = -81, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$131\bar{5}1321 \quad \Delta = -81, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$131\bar{5}1321\bar{3} \text{ [#=3]} \quad \Delta = -89, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: $231 \begin{smallmatrix} 5131 \\ 5 \end{smallmatrix} 1$

$$231 \overset{5131}{5} 1 \quad \Delta = -133, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$231 \overset{5131}{5} \textcolor{red}{12} \quad \Delta = -133, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$231 \overset{5131}{5} 1 \quad \Delta = -133, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$231 \overset{\textcolor{red}{2}5131}{5} 1 \quad \Delta = -133, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$231 \overset{5131}{5} 1 \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$231 \overset{5131}{5} \textcolor{red}{13} \quad \Delta = -141, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\textbf{Base: } 31 \overset{1231}{5} 13$$

$$31 \overset{1231}{5} 13 \quad \Delta = -89, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$31 \overset{\textcolor{red}{3}1231}{5} 13 \quad \Delta = -97, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$31 \overset{\textcolor{red}{3}1231}{5} 13 \text{ [#=4]} \quad \Delta = -97, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$31 \overset{\textcolor{red}{2}1231}{5} 13 \quad \Delta = -89, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$31 \overset{\textcolor{red}{4}1231}{5} 13 \quad \Delta = -117, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$31 \overset{\textcolor{red}{3}1231}{5} 13 \text{ [#=3]} \quad \Delta = -97, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$31 \overset{\textcolor{red}{3}1231}{5} 13 \text{ [#=9]} \quad \Delta = -97, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 324 \quad \begin{pmatrix} 0 & 18 & 2 \\ 18 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix}$$

$$31 \overset{\textcolor{red}{3}1231}{5} 13 \text{ [#=2]} \quad \Delta = -97, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$31 \overset{\textcolor{red}{2}1231}{5} 13 \quad \Delta = -89, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$31 \overset{\textcolor{red}{8}1231}{5} 13 \quad \Delta = -222, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$31 \overset{\textcolor{red}{3}1231}{5} 13 \text{ [#=6]} \quad \Delta = -97, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

$$31 \overset{\textcolor{red}{3}1231}{5} 13 \text{ [#=7]} \quad \Delta = -97, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 196 \quad \begin{pmatrix} 0 & 14 & 2 \\ 14 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix}$$

$$31 \overset{\textcolor{red}{3}1231}{5} 13 \text{ [#=8]} \quad \Delta = -97, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 256 \quad \begin{pmatrix} 0 & 16 & 2 \\ 16 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix}$$

$$\begin{aligned}
31 \overset{7}{5} 13 \quad \Delta = -194, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 8 \end{pmatrix} \\
31 \overset{5}{5} 13 \quad \Delta = -141, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
31 \overset{3}{5} 13 \text{ [#=5]} \quad \Delta = -97, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
31 \overset{6}{5} 13 \quad \Delta = -167, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{1,12231}{5} 1$

$$\begin{aligned}
1 \overset{1,312231}{5} 1 \text{ [#=6]} \quad \Delta = -81, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
1 \overset{1,12231}{5} 1 \quad \Delta = -73, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1 \overset{1,312231}{5} 1 \text{ [#=8]} \quad \Delta = -81, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
1 \overset{1,312231}{5} 1 \text{ [#=7]} \quad \Delta = -81, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
1 \overset{1,312231}{5} 1 \text{ [#=2]} \quad \Delta = -81, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1 \overset{1,312231}{5} 1 \text{ [#=5]} \quad \Delta = -81, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
1 \overset{1,312231}{5} 1 \text{ [#=9]} \quad \Delta = -81, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
1 \overset{1,312231}{5} 1 \text{ [#=4]} \quad \Delta = -81, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1 \overset{1,312231}{5} 1 \text{ [#=3]} \quad \Delta = -81, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

Base: 612231413

$$\begin{aligned}
612231413 \overset{3}{} \quad \Delta = -143, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 27 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
612231413 \overset{2}{} \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
612231413 \overset{2}{} \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
612231413 \overset{2}{} \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
612231413 \overset{3}{} \quad \Delta = -143, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: 612314132

$$612314132^2 \quad \Delta = -136, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$612314132^2 \quad \Delta = -136, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$612314132^2 \quad \Delta = -136, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$612314132^2 \quad \Delta = -138, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 613141322

$$613141322^2 \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$613141322^2 \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$613141322^2 \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$613141322^3 \quad \Delta = -143, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 60 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

Base: 141322161

$$141322161^3 \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 27 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$141322161^2 \quad \Delta = -127, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$141322161^2 \quad \Delta = -127, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1612231413_{[\# = 2]} \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1612231413_{[\# = 3]} \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 314132161

$$314132161^2 \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$314132161^2 \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$314132161^2 \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$314132161^3 \quad \Delta = -143, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$314132161^2 \quad \Delta = -137, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 231413161

$$\begin{aligned}
231413161^2 \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
231413161^2 \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
231413161^2 \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
231413161^3 \quad \Delta = -143, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 60 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
231413161^2 \quad \Delta = -137, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 141321612

$$\begin{aligned}
2161231413^{\#=2} \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
141321612^3 \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
141321612^2 \quad \Delta = -127, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2161231413^{\#=3} \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
141321612^2 \quad \Delta = -127, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
141321612^2 \quad \Delta = -127, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
141321612^2 \quad \Delta = -129, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 412321612

$$\begin{aligned}
412321612^2 \quad \Delta = -129, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
412321612^2 \quad \Delta = -129, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
412321612^2 \quad \Delta = -129, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
412321612^3 \quad \Delta = -137, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
412321612^3 \quad \Delta = -137, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

Base: 221612321

$$\begin{aligned}
2216123213^{\#=9} \quad \Delta = -109, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
221612321^2 \quad \Delta = -101, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
221612321\textcolor{red}{3} \text{ }_{[\# = 3]} \quad \Delta = -109, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
221612321\textcolor{red}{3} \text{ }_{[\# = 4]} \quad \Delta = -109, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
221612321\textcolor{red}{3} \text{ }_{[\# = 5]} \quad \Delta = -109, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
221612321\textcolor{red}{3} \text{ }_{[\# = 6]} \quad \Delta = -109, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\textcolor{red}{3} 221612321 \quad \Delta = -109, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
221612321\textcolor{red}{3} \text{ }_{[\# = 7]} \quad \Delta = -109, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
221612321\textcolor{red}{3} \text{ }_{[\# = 8]} \quad \Delta = -109, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
\textcolor{red}{3} 221612321 \quad \Delta = -110, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\textcolor{red}{2} 221612321 \quad \Delta = -101, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
221612321\textcolor{red}{3} \text{ }_{[\# = 2]} \quad \Delta = -109, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: 413216123

$$\begin{aligned}
413216123 \textcolor{red}{2} \quad \Delta = -136, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
413216123 \textcolor{red}{2} \quad \Delta = -136, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
413216123 \textcolor{red}{2} \quad \Delta = -136, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
413216123 \textcolor{red}{2} \quad \Delta = -138, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 132161231

$$\begin{aligned}
132161231\textcolor{red}{3} \text{ }_{[\# = 3]} \quad \Delta = -116, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
132161231\textcolor{red}{3} \text{ }_{[\# = 4]} \quad \Delta = -116, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
132161231\textcolor{red}{3} \text{ }_{[\# = 5]} \quad \Delta = -116, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
132161231\textcolor{red}{3} \text{ }_{[\# = 2]} \quad \Delta = -116, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

$$132161231^2 \quad \Delta = -108, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 413221613

$$413221613^3 \quad \Delta = -143, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 42 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$413221613^2 \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$413221613^2 \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$413221613^2 \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

Base: 314131612

$$314131612^3 \quad \Delta = -142, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 32 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$314131612^3 \quad \Delta = -142, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$314131612^3 \quad \Delta = -142, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$314131612^2 \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$314131612^2 \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$314131612^2 \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$314131612^2 \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: 131612231

$$131612231^3_{[\#=3]} \quad \Delta = -115, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$132216131^3_{[\#=3]} \quad \Delta = -115, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$132216131^3_{[\#=5]} \quad \Delta = -115, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$131612231^3_{[\#=5]} \quad \Delta = -115, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$132216131^3_{[\#=6]} \quad \Delta = -115, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$132216131^3_{[\#=9]} \quad \Delta = -115, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix}$$

$$132216131\overset{3}{\color{red}}_{[\# = 8]} \quad \Delta = -115, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$132216131\overset{3}{\color{red}}_{[\# = 7]} \quad \Delta = -115, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$131612231\overset{2}{\color{red}} \quad \Delta = -107, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$131612231\overset{2}{\color{red}} \quad \Delta = -107, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$131612231\overset{3}{\color{red}}_{[\# = 4]} \quad \Delta = -115, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$132216131\overset{3}{\color{red}}_{[\# = 4]} \quad \Delta = -115, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$131612231\overset{3}{\color{red}}_{[\# = 2]} \quad \Delta = -115, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$132216131\overset{3}{\color{red}}_{[\# = 2]} \quad \Delta = -115, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$131612231\overset{3}{\color{red}} \quad \Delta = -115, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: 141316122

$$141316122\overset{3}{\color{red}} \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$141316122\overset{2}{\color{red}} \quad \Delta = -126, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$141316122\overset{2}{\color{red}} \quad \Delta = -126, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$141316122\overset{2}{\color{red}} \quad \Delta = -126, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$141316122\overset{3}{\color{red}} \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$221613141\overset{3}{\color{red}}_{[\# = 3]} \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$221613141\overset{3}{\color{red}}_{[\# = 2]} \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$141316122\overset{3}{\color{red}} \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: 413161223

$$413161223\overset{3}{\color{red}} \quad \Delta = -143, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$413161223\overset{2}{\color{red}} \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$\begin{aligned}
413161223 \quad \Delta = -143, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
413161223 \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
413161223 \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 131612313

$$\begin{aligned}
131612313 \quad [\# = 2] \quad \Delta = -123, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
131612313 \quad \Delta = -117, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
131612313 \quad \Delta = -115, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
131612313 \quad \Delta = -115, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
131612313 \quad \Delta = -115, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
131612313 \quad \Delta = -123, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
3132161313 \quad [\# = 7] \quad \Delta = -123, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
3132161313 \quad [\# = 6] \quad \Delta = -123, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
3132161313 \quad [\# = 5] \quad \Delta = -123, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
3132161313 \quad [\# = 4] \quad \Delta = -123, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
3132161313 \quad [\# = 8] \quad \Delta = -123, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
3132161313 \quad [\# = 9] \quad \Delta = -123, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
3132161313 \quad [\# = 3] \quad \Delta = -123, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
3132161313 \quad [\# = 2] \quad \Delta = -123, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: 132161313

$$\begin{aligned}
132161313 \quad \Delta = -143, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 9 & 5 \\ 5 & 9 \end{pmatrix} \\
3131612313 \quad [\# = 4] \quad \Delta = -123, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
132161313 \quad \Delta = -115, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
132161313 \quad \Delta = -115, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
132161313 \quad \Delta = -115, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
132161313_{[\# = 3]} \quad \Delta = -123, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
3131612313_{[\# = 3]} \quad \Delta = -123, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
132161313_{[\# = 2]} \quad \Delta = -123, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
3131612313_{[\# = 2]} \quad \Delta = -123, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
132161313 \quad \Delta = -123, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
3131612313_{[\# = 5]} \quad \Delta = -123, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

Base: 13161232¹

$$\begin{aligned}
13161232 \quad \Delta = -109, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
13161232 \quad \Delta = -109, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
13161232 \quad \Delta = -109, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
13161232_{[\# = 3]} \quad \Delta = -117, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2 \quad 3^{\frac{3131612}{1}}_{[\# = 3]} \quad \Delta = -117, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
13161232_{[\# = 2]} \quad \Delta = -117, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2 \quad 3^{\frac{3131612}{1}}_{[\# = 2]} \quad \Delta = -117, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2 \quad 3^{\frac{3131612}{1}}_{[\# = 5]} \quad \Delta = -117, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
2 \quad 3^{\frac{3131612}{1}}_{[\# = 6]} \quad \Delta = -117, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
2 \quad 3^{\frac{3131612}{1}}_{[\# = 7]} \quad \Delta = -117, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}
\end{aligned}$$

$$2 \overset{3}{3} 1 \text{ }_{[\# = 8]} \quad \Delta = -117, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$2 \overset{3}{3} 1 \text{ }_{[\# = 9]} \quad \Delta = -117, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix}$$

$$2 \overset{3}{3} 1 \text{ }_{[\# = 4]} \quad \Delta = -117, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$13161232 \overset{41}{\quad} \quad \Delta = -137, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

Base: 131613132

$$131613132 \overset{3}{\quad} \quad \Delta = -123, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$131613132 \overset{2}{\quad} \quad \Delta = -117, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$131613132 \overset{2}{\quad} \quad \Delta = -115, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$131613132 \overset{2}{\quad} \quad \Delta = -115, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$131613132 \overset{2}{\quad} \quad \Delta = -115, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$231316131\overset{3}{\quad} \text{ }_{[\# = 9]} \quad \Delta = -123, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix}$$

$$231316131\overset{3}{\quad} \text{ }_{[\# = 8]} \quad \Delta = -123, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$231316131\overset{3}{\quad} \text{ }_{[\# = 7]} \quad \Delta = -123, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$231316131\overset{3}{\quad} \text{ }_{[\# = 6]} \quad \Delta = -123, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$231316131\overset{3}{\quad} \text{ }_{[\# = 5]} \quad \Delta = -123, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$231316131\overset{3}{\quad} \text{ }_{[\# = 4]} \quad \Delta = -123, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$231316131\overset{3}{\quad} \text{ }_{[\# = 3]} \quad \Delta = -123, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$131613132 \overset{3}{\quad} \text{ }_{[\# = 2]} \quad \Delta = -123, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$231316131\overset{3}{\quad} \text{ }_{[\# = 2]} \quad \Delta = -123, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: 313161313

$$313161313 \overset{3}{\quad} \text{ }_{[\# = 3]} \quad \Delta = -130, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 132 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$\begin{aligned}
& \overset{3}{3}13161313 \quad \Delta = -130, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{3}{3}13161313 \text{ [#=2]} \quad \Delta = -130, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& \overset{4}{3}13161313 \quad \Delta = -150, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{2}{3}13161313 \quad \Delta = -122, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{2}{3}13161313 \quad \Delta = -122, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \text{Base: } \overset{1}{1}6123141 \\
& \overset{1}{1}6123141\overset{3}{3} \text{ [#=2]} \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& \overset{1}{1}6123141\overset{3}{3} \text{ [#=3]} \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& \overset{1}{1}\overset{2}{2}6123141 \quad \Delta = -127, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{1}{1}6123141\overset{2}{2} \quad \Delta = -127, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{1}{1}6123141\overset{2}{2} \quad \Delta = -129, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \text{Base: } \overset{1}{1}41232161 \\
& \overset{3}{3}\overset{1}{1}41232161 \quad \Delta = -137, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& 41232161\overset{2}{2}\overset{1}{1} \quad \Delta = -129, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{2}{2}\overset{1}{1}41232161 \quad \Delta = -129, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \text{Base: } \overset{1}{1}16131413 \\
& \overset{1}{1}\overset{3}{3}16131413 \quad \Delta = -142, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{1}{1}16131413\overset{2}{2} \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{1}{1}\overset{2}{2}16131413 \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{1}{1}16131413\overset{2}{2} \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{1}{1}16131413\overset{3}{3} \quad \Delta = -142, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{12321}{6} 12$

$$1 \overset{12321}{6} 12 \quad \Delta = -101, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{312321}{6} 12 \text{ [#=4]} \quad \Delta = -109, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{12321}{6} \overset{2}{12} \quad \Delta = -101, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{312321}{6} 12 \text{ [#=7]} \quad \Delta = -109, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$1 \overset{312321}{6} 12 \text{ [#=6]} \quad \Delta = -109, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1 \overset{312321}{6} 12 \text{ [#=8]} \quad \Delta = -109, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1 \overset{312321}{6} 12 \text{ [#=9]} \quad \Delta = -109, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$1 \overset{312321}{6} 12 \text{ [#=3]} \quad \Delta = -109, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \overset{312321}{6} 12 \text{ [#=2]} \quad \Delta = -109, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{12321}{6} \overset{3}{12} \quad \Delta = -109, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{312321}{6} 12 \text{ [#=5]} \quad \Delta = -109, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

Base: $41321 \overset{31}{6} 1$

$$41321 \overset{31}{6} 1 \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$41321 \overset{2}{6} \overset{31}{1} \quad \Delta = -137, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$41321 \overset{31}{6} \overset{12}{1} \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$41321 \overset{2}{6} \overset{31}{1} \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$41321 \overset{231}{6} 1 \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: $1613141 \overset{21}{1}$

$$1613141 \overset{21}{1} \quad \Delta = -126, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1613141 \overset{21}{6} \overset{2}{1} \quad \Delta = -126, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\begin{aligned}
& \overset{21}{1} \overset{2}{6} 13141 \quad \Delta = -126, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{21}{1} \overset{2}{6} 13141 \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{21}{1} \overset{2}{6} 13141 \overset{3}{3} [\#=3] \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& \overset{21}{1} \overset{2}{6} 13141 \overset{3}{3} [\#=2] \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& \overset{21}{1} \overset{3}{6} 13141 \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \text{Base: } \overset{1321}{1} \overset{6}{6} 131 \\
& \overset{1321}{1} \overset{6}{6} 131 \quad \Delta = -107, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{3}{1} \overset{31321}{6} 131 [\#=5] \quad \Delta = -115, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{1321}{1} \overset{2}{6} 131 \quad \Delta = -107, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{1321}{1} \overset{6}{6} 131 \overset{3}{3} [\#=6] \quad \Delta = -115, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{1321}{1} \overset{6}{6} 131 \overset{3}{3} [\#=7] \quad \Delta = -115, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
& \overset{1321}{1} \overset{6}{6} 131 \overset{3}{3} [\#=8] \quad \Delta = -115, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& \overset{1321}{1} \overset{6}{6} 131 \overset{3}{3} [\#=9] \quad \Delta = -115, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
& \overset{1321}{1} \overset{6}{6} 131 \overset{3}{3} [\#=5] \quad \Delta = -115, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{1321}{1} \overset{6}{6} 131 \overset{3}{3} [\#=2] \quad \Delta = -115, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{3}{1} \overset{31321}{6} 131 [\#=2] \quad \Delta = -115, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
& \overset{1321}{1} \overset{6}{6} 131 \overset{3}{3} [\#=3] \quad \Delta = -115, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{3}{1} \overset{31321}{6} 131 [\#=3] \quad \Delta = -115, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{1321}{1} \overset{6}{6} 131 \overset{3}{3} [\#=4] \quad \Delta = -115, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{3}{1} \overset{31321}{6} 131 [\#=4] \quad \Delta = -115, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: $4131 \overset{321}{6} 1$

$$4131 \overset{321}{6} 1 \overset{2}{} \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$4131 \overset{321}{6} 1 \overset{3}{} \quad \Delta = -143, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$4131 \overset{2321}{6} 1 \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$4131 \overset{321}{6} 1 \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$4131 \overset{2}{6} \overset{321}{1} \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: $1 \overset{3131}{6} 131$

$$1 \overset{3131}{6} 131 \overset{3}{} [\# = 5] \quad \Delta = -122, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1 \overset{3131}{6} 131 \overset{3}{} [\# = 4] \quad \Delta = -122, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1 \overset{3131}{6} 131 \overset{3}{} [\# = 8] \quad \Delta = -122, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$1 \overset{3131}{6} 131 \quad \Delta = -114, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{3131}{6} 131 [\# = 2] \quad \Delta = -122, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \overset{3131}{6} 131 \overset{3}{} [\# = 3] \quad \Delta = -122, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{3131}{6} 131 \overset{3}{} [\# = 9] \quad \Delta = -122, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix}$$

$$1 \overset{3131}{6} 131 [\# = 3] \quad \Delta = -122, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{3131}{6} 131 \quad \Delta = -142, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 9 & 5 \\ 5 & 9 \end{pmatrix}$$

$$1 \overset{3131}{6} 131 \overset{2}{} \quad \Delta = -114, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{3131}{6} 131 \quad \Delta = -114, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{3131}{6} 131 \quad \Delta = -122, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1 \overset{3131}{6} 131 \overset{3}{} [\# = 6] \quad \Delta = -122, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$\begin{aligned}
& \overset{3131}{1 \ 6 \ 131\textcolor{red}{3}}_{[\#=7]} \quad \Delta = -122, \ n_+ = 1, \ n_- = 9, \ n_0 = 0, \ |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 6 \ 131\textcolor{red}{3}}_{[\#=2]} \quad \Delta = -122, \ n_+ = 1, \ n_- = 9, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{31}{1321\ 6\ 13}$

$$\begin{aligned}
& \overset{31}{31\ 6\ 1231\textcolor{red}{3}} \quad \Delta = -123, \ n_+ = 1, \ n_- = 9, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{31}{31\ 6\ 1231\textcolor{red}{3}}_{[\#=3]} \quad \Delta = -123, \ n_+ = 1, \ n_- = 9, \ n_0 = 0, \ |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{31}{31\ 6\ 1231\textcolor{red}{3}}_{[\#=5]} \quad \Delta = -123, \ n_+ = 1, \ n_- = 9, \ n_0 = 0, \ |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& \overset{\textcolor{red}{2}}{1321\ 6\ 13} \overset{31}{} \quad \Delta = -117, \ n_+ = 1, \ n_- = 9, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{31}{31\ 6\ 1231\textcolor{red}{3}}_{[\#=4]} \quad \Delta = -123, \ n_+ = 1, \ n_- = 9, \ n_0 = 0, \ |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{31}{31\ 6\ 1231\textcolor{red}{5}} \quad \Delta = -167, \ n_+ = 1, \ n_- = 9, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
& \overset{31}{31\ 6\ 1231\textcolor{red}{3}}_{[\#=2]} \quad \Delta = -123, \ n_+ = 1, \ n_- = 9, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{31}{31\ 6\ 1231\textcolor{red}{4}} \quad \Delta = -143, \ n_+ = 1, \ n_- = 9, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{31}{31\ 6\ 1231\textcolor{red}{2}} \quad \Delta = -115, \ n_+ = 1, \ n_- = 9, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{\textcolor{red}{2}31}{1321\ 6\ 13} \quad \Delta = -115, \ n_+ = 1, \ n_- = 9, \ n_0 = 0, \ |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{31\textcolor{red}{2}}{1321\ 6\ 13} \quad \Delta = -115, \ n_+ = 1, \ n_- = 9, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{31}{4131\ 6\ 12}$

$$\begin{aligned}
& \overset{\textcolor{red}{2}31}{4131\ 6\ 12} \quad \Delta = -134, \ n_+ = 1, \ n_- = 9, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{\textcolor{red}{2}}{4131\ 6\ 12} \overset{31}{} \quad \Delta = -134, \ n_+ = 1, \ n_- = 9, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{31\textcolor{red}{2}}{4131\ 6\ 12} \quad \Delta = -134, \ n_+ = 1, \ n_- = 9, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{31}{4131\ 6\ 12} \quad \Delta = -134, \ n_+ = 1, \ n_- = 9, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{\textcolor{red}{3}}{4131\ 6\ 12} \overset{31}{} \quad \Delta = -142, \ n_+ = 1, \ n_- = 9, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

$$4131\overset{31}{6}12 \quad \Delta = -142, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: $131\overset{31}{6}123$

$$131\overset{31}{6}123 \quad \Delta = -115, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$131\overset{231}{6}123 \quad \Delta = -115, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$321\overset{31}{6}131\overset{2}{2} \quad \Delta = -115, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$131\overset{31}{6}123 \quad \Delta = -115, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$321\overset{31}{6}131\overset{3}{3} \quad \Delta = -123, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$321\overset{31}{6}131\overset{4}{4} \quad \Delta = -143, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$321\overset{31}{6}131\overset{6}{6} \quad \Delta = -193, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 8 \end{pmatrix}$$

$$321\overset{31}{6}131\overset{3}{3}_{[\# = 2]} \quad \Delta = -123, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$321\overset{31}{6}131\overset{5}{5} \quad \Delta = -167, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$321\overset{31}{6}131\overset{3}{3}_{[\# = 6]} \quad \Delta = -123, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$321\overset{31}{6}131\overset{3}{3}_{[\# = 7]} \quad \Delta = -123, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 196 \quad \begin{pmatrix} 0 & 14 & 2 \\ 14 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$321\overset{31}{6}131\overset{3}{3}_{[\# = 8]} \quad \Delta = -123, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 256 \quad \begin{pmatrix} 0 & 16 & 2 \\ 16 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$321\overset{31}{6}131\overset{3}{3}_{[\# = 5]} \quad \Delta = -123, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$321\overset{31}{6}131\overset{3}{3}_{[\# = 4]} \quad \Delta = -123, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$321\overset{31}{6}131\overset{3}{3}_{[\# = 3]} \quad \Delta = -123, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$321\overset{31}{6}131\overset{3}{3}_{[\# = 9]} \quad \Delta = -123, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 324 \quad \begin{pmatrix} 0 & 18 & 2 \\ 18 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

Base: $21\overset{131}{6}131$

$$21\overset{131}{6}131\overset{3}{3}_{[\# = 2]} \quad \Delta = -114, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\begin{aligned}
& \overset{3131}{21 \ 6 \ 131} \quad \Delta = -114, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{131}{21 \ 6 \ 131} \overset{3}{\# = 5} \quad \Delta = -114, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{2131}{21 \ 6 \ 131} \quad \Delta = -106, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{131}{21 \ 6 \ 131} \overset{3}{\# = 8} \quad \Delta = -114, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& \overset{131}{21 \ 6 \ 131} \overset{2}{\# = 4} \quad \Delta = -106, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{131}{21 \ 6 \ 131} \overset{3}{\# = 7} \quad \Delta = -114, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
& \overset{131}{21 \ 6 \ 131} \overset{3}{\# = 9} \quad \Delta = -114, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
& \overset{131}{21 \ 6 \ 131} \overset{3}{\# = 4} \quad \Delta = -114, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{131}{21 \ 6 \ 131} \overset{3}{\# = 3} \quad \Delta = -114, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{131}{21 \ 6 \ 131} \overset{3}{\# = 6} \quad \Delta = -114, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \text{Base: } \overset{3131}{31 \ 6 \ 13} \\
& \overset{3131}{31 \ 6 \ 13} \overset{3}{\# = 2} \quad \Delta = -130, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 93 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& \overset{23131}{31 \ 6 \ 13} \quad \Delta = -123, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{3131}{31 \ 6 \ 13} \quad \Delta = -150, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 20 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{3131}{31 \ 6 \ 13} \quad \Delta = -122, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{3131}{31 \ 6 \ 13} \quad \Delta = -122, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{3131}{31 \ 6 \ 13} \overset{3}{\# = 3} \quad \Delta = -130, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 324 \quad \begin{pmatrix} 0 & 18 & 2 \\ 18 & 213 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
& \overset{3131}{31 \ 6 \ 13} \quad \Delta = -130, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 21 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{23131}{31 \ 6 \ 13} \quad \Delta = -122, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \text{Base: } \overset{1,12321}{1 \ 6 \ 1}
\end{aligned}$$

$$\begin{array}{ll}
1 \overset{1, \textcolor{red}{3} 12321}{6} 1 \text{ [#}=9] & \Delta = -109, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1 \overset{1, \textcolor{red}{3} 12321}{6} 1 \text{ [#}=6] & \Delta = -109, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1 \overset{1, \textcolor{red}{3} 12321}{6} 1 \text{ [#}=7] & \Delta = -109, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1 \overset{1, \textcolor{red}{3} 12321}{6} 1 \text{ [#}=8] & \Delta = -109, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1 \overset{1, \textcolor{red}{3} 12321}{6} 1 \text{ [#}=5] & \Delta = -109, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1 \overset{1, \textcolor{red}{3} 12321}{6} 1 \text{ [#}=4] & \Delta = -109, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1 \overset{1, \textcolor{red}{3} 12321}{6} 1 \text{ [#}=3] & \Delta = -109, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \overset{1, \textcolor{red}{3} 12321}{6} 1 \text{ [#}=2] & \Delta = -109, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \overset{1, 12321}{6} 1 & \Delta = -101, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{array}$$

Base: $1 \overset{1, 131}{6} 131$

$$\begin{array}{ll}
1 \overset{1, 131}{6} 131 \textcolor{red}{3} \text{ [#}=7] & \Delta = -114, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
1 \overset{1, 131}{6} 131 \textcolor{red}{3} \text{ [#}=8] & \Delta = -114, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1 \overset{1, 131}{6} 131 \textcolor{red}{3} \text{ [#}=4] & \Delta = -114, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \overset{1, 131}{6} 131 \textcolor{red}{3} \text{ [#}=9] & \Delta = -114, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
1 \overset{1, 131}{6} 131 \textcolor{red}{3} \text{ [#}=5] & \Delta = -114, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1 \overset{1, 131}{6} 131 \textcolor{red}{3} \text{ [#}=2] & \Delta = -114, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1 \overset{1, 131}{6} 131 \textcolor{red}{3} \text{ [#}=3] & \Delta = -114, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{1, 131}{6} 131 \textcolor{red}{3} \text{ [#}=6] & \Delta = -114, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1 \overset{1, 131 \textcolor{red}{2}}{6} 131 & \Delta = -106, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{array}$$

Base: $131 \overset{1, 31}{6} 13$

$$\begin{aligned}
31 \overset{1,31}{6} 131\textcolor{red}{3} \quad \Delta = -122, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
31 \overset{1,31}{6} 131\textcolor{red}{3} [\# = 7] \quad \Delta = -122, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 196 \quad & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
31 \overset{1,31}{6} 131\textcolor{red}{3} [\# = 4] \quad \Delta = -122, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
131 \overset{\textcolor{red}{3},31}{6} 13 \quad \Delta = -122, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
31 \overset{1,31}{6} 131\textcolor{red}{3} [\# = 9] \quad \Delta = -122, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 324 \quad & \begin{pmatrix} 0 & 18 & 2 \\ 18 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
31 \overset{1,31}{6} 131\textcolor{red}{3} [\# = 8] \quad \Delta = -122, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 256 \quad & \begin{pmatrix} 0 & 16 & 2 \\ 16 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
31 \overset{1,31}{6} 131\textcolor{red}{5} \quad \Delta = -166, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
131 \overset{\textcolor{red}{2},31}{6} 13 \quad \Delta = -114, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
31 \overset{1,31}{6} 131\textcolor{red}{4} \quad \Delta = -142, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
131 \overset{1,31\textcolor{red}{2}}{6} 13 \quad \Delta = -114, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
131 \overset{\textcolor{red}{2},31}{6} 13 \quad \Delta = -114, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
31 \overset{1,31}{6} 131\textcolor{red}{2} \quad \Delta = -114, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
31 \overset{1,31}{6} 131\textcolor{red}{3} [\# = 2] \quad \Delta = -122, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
31 \overset{1,31}{6} 131\textcolor{red}{3} [\# = 6] \quad \Delta = -122, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 144 \quad & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
31 \overset{1,31}{6} 131\textcolor{red}{3} [\# = 3] \quad \Delta = -122, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
31 \overset{1,31}{6} 131\textcolor{red}{3} [\# = 5] \quad \Delta = -122, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 100 \quad & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
31 \overset{1,31}{6} 131\textcolor{red}{6} \quad \Delta = -192, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 8 \end{pmatrix}
\end{aligned}$$

Base: 712231322

$$\begin{aligned}
712231322 \overset{\textcolor{red}{2}}{} \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
712231322 \overset{\textcolor{red}{3}}{} \quad \Delta = -143, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 32 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

$$712231322 \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: 171223132

$$171223132 \quad \Delta = -143, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$171223132 \quad \Delta = -137, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$171223132 \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$171223132 \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 171231322

$$171231322 \quad \Delta = -137, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$171231322 \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$171231322 \quad \Delta = -143, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 32 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$171231322 \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 217122313

$$217122313 \quad \Delta = -142, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$217122313 \text{ }_{[\# = 2]} \quad \Delta = -142, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$217122313 \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$217122313 \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$217122313 \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: 217123132

$$217123132 \quad \Delta = -137, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$217123132 \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$217123132 \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$217123132 \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$217123132^3 \quad \Delta = -143, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$217123132^2 \quad \Delta = -137, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 221712231

$$221712231^2 \quad \Delta = -126, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$221712231^2 \quad \Delta = -126, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$221712231^3 \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$2217122313_{[\# = 5]} \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$2217122313_{[\# = 3]} \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$2217122313_{[\# = 2]} \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$2217122313_{[\# = 4]} \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: 322171223

$$322171223^2 \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 221712313

$$221712313^3_{[\# = 2]} \quad \Delta = -142, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$221712313^2 \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$221712313^2 \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$221712313^2 \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$221712313^3 \quad \Delta = -142, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$221712313^3 \quad \Delta = -143, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$221712313^2 \quad \Delta = -136, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: 222171231

$$2221712313_{[\# = 3]} \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$222171231^2 \quad \Delta = -126, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$222171231^3_{[\#=5]} \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$222171231^3_{[\#=4]} \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$222171231^3_{[\#=2]} \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$222171231^2 \quad \Delta = -126, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 22171232^1

$$22171232^{41} \quad \Delta = -156, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$22171232^{21} \quad \Delta = -128, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$22171232^{21} \quad \Delta = -128, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$22171232^{21} \quad \Delta = -128, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$22171232^{31}_{[\#=3]} \quad \Delta = -136, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$22171232^{31} \quad \Delta = -137, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$22171232^{31}_{[\#=2]} \quad \Delta = -136, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: $1^{313221} \begin{smallmatrix} 7 \\ 1 \end{smallmatrix}$

$$1^{313221} \begin{smallmatrix} 7 \\ 1 \end{smallmatrix} \quad \Delta = -142, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1^{313221} \begin{smallmatrix} 7 \\ 1 \end{smallmatrix} \quad \Delta = -142, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1^{313221} \begin{smallmatrix} 7 \\ 1 \end{smallmatrix} \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1^{313221} \begin{smallmatrix} 7 \\ 1 \end{smallmatrix} \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: $1^{231321} \begin{smallmatrix} 7 \\ 1 \end{smallmatrix}$

$$1^{231321} \begin{smallmatrix} 7 \\ 1 \end{smallmatrix} \quad \Delta = -143, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1^{231321} \begin{smallmatrix} 7 \\ 1 \end{smallmatrix} \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{231321}{7} 1 \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{231321}{7} 1 \quad \Delta = -137, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{231321}{7} 1 \quad \Delta = -137, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: $1 \overset{13221}{7} 12$

$$1 \overset{313221}{7} 12 \text{ [#=5]} \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{313221}{7} 12 \text{ [#=4]} \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{313221}{7} 12 \text{ [#=3]} \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1 \overset{313221}{7} 12 \text{ [#=2]} \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$1 \overset{13221}{7} 12 \quad \Delta = -126, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{13221}{7} 12 \quad \Delta = -126, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: $1 \overset{31321}{7} 12$

$$1 \overset{31321}{7} 12 \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{31321}{7} 12 \text{ [#=2]} \quad \Delta = -142, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \overset{31321}{7} 12 \quad \Delta = -142, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1 \overset{31321}{7} 12 \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{31321}{7} 12 \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{31321}{7} 12 \quad \Delta = -136, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: $1 \overset{2321}{7} 12$

$$1 \overset{2321}{7} 12 \quad \Delta = -156, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$1 \overset{2321}{7} 12 \text{ [#=3]} \quad \Delta = -136, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{2321}{7} 12 \text{ [#=2]} \quad \Delta = -136, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\begin{aligned}
1 \overset{2321}{7} 12 \quad \Delta = -128, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1 \overset{2321}{7} 12 \quad \Delta = -128, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \overset{2321}{7} 12 \quad \Delta = -128, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{321}{7} 1223$

$$\begin{aligned}
3221 \overset{321}{7} 12 \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1 \overset{321}{7} 1223 \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1 \overset{321}{7} 1223 \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{1321}{7} 122$

$$\begin{aligned}
1 \overset{31321}{7} 122 \text{ [#=5]} \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1 \overset{31321}{7} 122 \text{ [#=3]} \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1 \overset{1321}{7} 122 \quad \Delta = -126, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1 \overset{31321}{7} 122 \text{ [#=2]} \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
1 \overset{31321}{7} 122 \text{ [#=4]} \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1 \overset{1321}{7} \overset{3}{122} \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1 \overset{1321}{7} \overset{2}{122} \quad \Delta = -126, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: $21 \overset{1321}{7} 12$

$$\begin{aligned}
21 \overset{21321}{7} 12 \quad \Delta = -126, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21 \overset{31321}{7} 12 \text{ [#=2]} \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
21 \overset{31321}{7} 12 \text{ [#=5]} \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21 \overset{31321}{7} 12 \text{ [#=4]} \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21 \overset{31321}{7} 12 \text{ [#=3]} \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

$$21 \overset{1321}{7} 12 \quad \Delta = -126, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: $21 \overset{321}{7} 123$

$$21 \overset{321}{7} 12\overset{2}{3} \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\overset{2}{2} 1 \overset{321}{7} 123 \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$321 \overset{321}{7} 12\overset{3}{3} \quad \Delta = -144, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: $1 \overset{1,13221}{7} 1$

$$1 \overset{1,13221}{7} 1 \quad \Delta = -126, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{1,313221}{7} 1 \text{ [#=4]} \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1 \overset{1,313221}{7} 1 \text{ [#=3]} \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{1,313221}{7} 1 \text{ [#=2]} \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \overset{1,313221}{7} 1 \text{ [#=5]} \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

Base: $1 \overset{1,31321}{7} 1$

$$1 \overset{1,31321}{7} 1 \quad \Delta = -142, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{1,31321}{7} 1 \quad \Delta = -136, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{1,31321}{7} 1 \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{1,31321}{7} 1 \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{1,31321}{7} 1 \text{ [#=2]} \quad \Delta = -142, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

Base: $1 \overset{1,2321}{7} 1$

$$1 \overset{1,2321}{7} 1 \text{ [#=2]} \quad \Delta = -136, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1 \overset{1,2321}{7} 1 \quad \Delta = -128, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \overset{1,2321}{7} 1 \quad \Delta = -128, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1^{1,2321} \begin{smallmatrix} 1 \\ 7 \\ 1 \end{smallmatrix} [\# = 3] \quad \Delta = -136, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 60 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1^{1,2321} \begin{smallmatrix} 1 \\ 7 \\ 1 \end{smallmatrix} \quad \Delta = -156, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\text{Base: } 1^{21,1321} \begin{smallmatrix} 1 \\ 7 \\ 1 \end{smallmatrix}$$

$$1^{21,31321} \begin{smallmatrix} 1 \\ 7 \\ 1 \end{smallmatrix} [\# = 2] \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1^{21,1321} \begin{smallmatrix} 1 \\ 7 \\ 1 \end{smallmatrix} \quad \Delta = -126, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1^{21,1321} \begin{smallmatrix} 1 \\ 7 \\ 1 \end{smallmatrix} \quad \Delta = -126, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1^{21,31321} \begin{smallmatrix} 1 \\ 7 \\ 1 \end{smallmatrix} [\# = 3] \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1^{21,31321} \begin{smallmatrix} 1 \\ 7 \\ 1 \end{smallmatrix} [\# = 4] \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1^{21,31321} \begin{smallmatrix} 1 \\ 7 \\ 1 \end{smallmatrix} [\# = 5] \quad \Delta = -134, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$\text{Base: } 1^{1,321} \begin{smallmatrix} 1 \\ 7 \\ 123 \end{smallmatrix}$$

$$1^{1,3212} \begin{smallmatrix} 1 \\ 7 \\ 123 \end{smallmatrix} \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$321^{1,321} \begin{smallmatrix} 1 \\ 7 \\ 12 \end{smallmatrix} \quad \Delta = -135, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\text{Base: } 812222222$$

$$812222222^2 \quad \Delta = -145, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$\text{Base: } 181222222$$

$$181222222^2 \quad \Delta = -145, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$\text{Base: } 218122222$$

$$218122222^2 \quad \Delta = -145, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$218122222^2 \quad \Delta = -145, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\text{Base: } 221812222$$

$$221812222^2 \quad \Delta = -145, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$221812222^2 \quad \Delta = -145, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$221812222^3 \quad \Delta = -154, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: 222181222

$$222181222 \quad \Delta = -145, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: $1 \overset{222221}{8} 1$

$$1 \overset{222221}{8} 1 \quad \Delta = -145, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: $1 \overset{22221}{8} 12$

$$1 \overset{22221}{8} 12 \quad \Delta = -145, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{22221}{8} 12 \quad \Delta = -145, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: $1 \overset{2221}{8} 122$

$$1 \overset{2221}{8} 122 \quad \Delta = -145, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{2221}{8} 122 \quad \Delta = -145, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{2221}{8} 122 \quad \Delta = -154, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: $21 \overset{2221}{8} 12$

$$21 \overset{2221}{8} 12 \quad \Delta = -145, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$21 \overset{2221}{8} 12 \quad \Delta = -145, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: $21 \overset{221}{8} 122$

$$21 \overset{221}{8} 122 \quad \Delta = -145, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$21 \overset{221}{8} 122 \quad \Delta = -154, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$21 \overset{221}{8} 122 \quad \Delta = -145, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: $1 \overset{1,22221}{8} 1$

$$1 \overset{1,22221}{8} 1 \quad \Delta = -145, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: $1 \overset{21,2221}{8} 1$

$$1 \overset{21,2221}{8} 1 \quad \Delta = -145, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1 \overset{21,2221}{8} 1 \quad \Delta = -145, n_+ = 1, n_- = 9, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: $1 \overset{221,221}{8} 1$

$$\begin{aligned}
& \overset{221,221}{1 \ 8 \ 1} \quad \Delta = -154, \ n_+ = 1, \ n_- = 9, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{221,221}{1 \ 8 \ 1} \quad \Delta = -145, \ n_+ = 1, \ n_- = 9, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{221,21}{1 \ 8 \ 12}$

$$\begin{aligned}
& \overset{221,21}{1 \ 8 \ 12} \quad \Delta = -154, \ n_+ = 1, \ n_- = 9, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
& \overset{221,21}{1 \ 8 \ 12} \quad \Delta = -145, \ n_+ = 1, \ n_- = 9, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{221,21}{1 \ 8 \ 12} \quad \Delta = -145, \ n_+ = 1, \ n_- = 9, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{21,21}{21 \ 8 \ 12}$

$$\overset{221,21}{21 \ 8 \ 12} \quad \Delta = -145, \ n_+ = 1, \ n_- = 9, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$T = 10$ (1084 bases)

Base: $\overset{2}{12222222}$

$$\begin{aligned}
& \overset{2}{22222221} \textcolor{red}{8} \quad \Delta = -116, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{2}{22222221} \textcolor{red}{7} \quad \Delta = -88, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{2}{22222221} \textcolor{red}{12} \quad \Delta = -231, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& \overset{2}{22222221} \textcolor{red}{6} \quad \Delta = -61, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{2}{22222221} \textcolor{red}{5} \quad \Delta = -35, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{2}{22222221} \textcolor{red}{4} \quad \Delta = -11, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1231413221

$$\begin{aligned}
& 1231413221 \textcolor{red}{3} \text{ [#=9]} \quad \Delta = -37, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
& 1231413221 \textcolor{red}{2} \quad \Delta = -29, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 1231413221 \textcolor{red}{2} \quad \Delta = -29, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& 1223141321 \textcolor{red}{3} \text{ [#=2]} \quad \Delta = -37, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 1231413221 \textcolor{red}{3} \text{ [#=2]} \quad \Delta = -37, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1223141321\mathbf{3}_{[\# = 3]} \quad \Delta = -37, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1223141321\mathbf{3}_{[\# = 6]} \quad \Delta = -37, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1223141321\mathbf{3}_{[\# = 9]} \quad \Delta = -37, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1231413221\mathbf{3}_{[\# = 7]} \quad \Delta = -37, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
1223141321\mathbf{3}_{[\# = 7]} \quad \Delta = -37, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1231413221\mathbf{3}_{[\# = 3]} \quad \Delta = -37, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1223141321\mathbf{3}_{[\# = 8]} \quad \Delta = -37, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1223141321\mathbf{3}_{[\# = 4]} \quad \Delta = -37, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1231413221\mathbf{3}_{[\# = 4]} \quad \Delta = -37, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1223141321\mathbf{3}_{[\# = 5]} \quad \Delta = -37, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1231413221\mathbf{3}_{[\# = 5]} \quad \Delta = -37, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1231413221\mathbf{3}_{[\# = 8]} \quad \Delta = -37, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1231413221\mathbf{3}_{[\# = 6]} \quad \Delta = -37, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1223141\overset{2}{3}2$

$$\begin{aligned}
2\overset{4}{1}223141\overset{2}{3}2 \quad \Delta = -59, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2\overset{2}{1}223141\overset{2}{3}2 \quad \Delta = -31, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
2\overset{12}{1}223141\overset{2}{3}2 \quad \Delta = -279, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -5 & -3 \\ 2 & -3 & 8 \end{pmatrix} \\
2\overset{5}{1}223141\overset{2}{3}2 \quad \Delta = -83, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2\overset{8}{1}223141\overset{2}{3}2 \quad \Delta = -164, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
2\overset{7}{1}223141\overset{2}{3}2 \quad \Delta = -136, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
2^{\overset{6}{1}223141} 3^{\overset{2}{2}} 2 \quad \Delta = -109, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
122314132^{\overset{2}{2}} \quad \Delta = -31, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
122314132^{\overset{2}{2}} \quad \Delta = -31, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2^{\overset{3}{1}223141} 3^{\overset{2}{2}} 2_{[\# = 4]} \quad \Delta = -39, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 109 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
2^{\overset{3}{1}223141} 3^{\overset{2}{2}} 2_{[\# = 9]} \quad \Delta = -39, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 324 & \begin{pmatrix} 0 & 18 & 2 \\ 18 & 564 & 62 \\ 2 & 62 & 8 \end{pmatrix} \\
2^{\overset{3}{1}223141} 3^{\overset{2}{2}} 2_{[\# = 2]} \quad \Delta = -39, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
2^{\overset{3}{1}223141} 3^{\overset{2}{2}} 2_{[\# = 8]} \quad \Delta = -39, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 256 & \begin{pmatrix} 0 & 16 & 2 \\ 16 & 445 & 55 \\ 2 & 55 & 8 \end{pmatrix} \\
2^{\overset{3}{1}223141} 3^{\overset{2}{2}} 2_{[\# = 3]} \quad \Delta = -39, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 60 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
2^{\overset{3}{1}223141} 3^{\overset{2}{2}} 2_{[\# = 7]} \quad \Delta = -39, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 196 & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 340 & 48 \\ 2 & 48 & 8 \end{pmatrix} \\
2^{\overset{3}{1}223141} 3^{\overset{2}{2}} 2_{[\# = 6]} \quad \Delta = -39, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 144 & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 249 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
2^{\overset{3}{1}223141} 3^{\overset{2}{2}} 2_{[\# = 5]} \quad \Delta = -39, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 172 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
2^{\overset{3}{1}223141} 3^{\overset{2}{2}} 2 \quad \Delta = -39, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 2141413221

$$\begin{aligned}
2141413221^{\overset{3}{3}} \quad \Delta = -56, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2141413221^{\overset{4}{4}} \quad \Delta = -76, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2141413221^{\overset{2}{2}} \quad \Delta = -48, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2141413221^{\overset{3}{3}}_{[\# = 3]} \quad \Delta = -56, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2141413221^{\overset{3}{3}}_{[\# = 4]} \quad \Delta = -56, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
2141413221^{\overset{3}{3}}_{[\# = 2]} \quad \Delta = -56, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 37 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
2141413221^{\overset{3}{3}}_{[\# = 5]} \quad \Delta = -56, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
2141413221\textcolor{red}{3}_{[\# = 6]} \quad \Delta = -56, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
2141413221\textcolor{red}{3}_{[\# = 7]} \quad \Delta = -56, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
2141413221\textcolor{red}{3}_{[\# = 8]} \quad \Delta = -56, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
2141413221\textcolor{red}{3}_{[\# = 9]} \quad \Delta = -56, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
\textcolor{red}{3}2141413221_{[\# = 3]} \quad \Delta = -56, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 87 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
2141413221\textcolor{red}{3}_{[\# = 2]} \quad \Delta = -56, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\textcolor{red}{2}2141413221 \quad \Delta = -48, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
214141\textcolor{red}{2}3221 \quad \Delta = -48, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1314141321

$$\begin{aligned}
1314141321\textcolor{red}{3}_{[\# = 6]} \quad \Delta = -64, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\textcolor{red}{3}1314141321 \quad \Delta = -64, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1231414131\textcolor{red}{3}_{[\# = 6]} \quad \Delta = -64, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1231414131\textcolor{red}{3}_{[\# = 7]} \quad \Delta = -64, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1314141321\textcolor{red}{3}_{[\# = 3]} \quad \Delta = -64, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1314141321\textcolor{red}{3}_{[\# = 5]} \quad \Delta = -64, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1314141321\textcolor{red}{3}_{[\# = 9]} \quad \Delta = -64, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1231414131\textcolor{red}{3}_{[\# = 5]} \quad \Delta = -64, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1314141321\textcolor{red}{3}_{[\# = 4]} \quad \Delta = -64, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1231414131\textcolor{red}{3}_{[\# = 4]} \quad \Delta = -64, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1231414131\textcolor{red}{3}_{[\# = 3]} \quad \Delta = -64, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1314141321\overset{2}{3} \text{ [#=2]} \quad \Delta = -64, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1231414131\overset{2}{3} \text{ [#=2]} \quad \Delta = -64, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1314141321\overset{2}{3} \text{ [#=7]} \quad \Delta = -64, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1231414131\overset{2}{3} \text{ [#=8]} \quad \Delta = -64, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1231414131\overset{2}{3} \text{ [#=9]} \quad \Delta = -64, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1314141321\overset{2}{3} \text{ [#=8]} \quad \Delta = -64, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1314141321\overset{2}{3} \quad \Delta = -56, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1314141321\overset{2}{3} \quad \Delta = -56, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1314141321\overset{2}{3} \quad \Delta = -56, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\text{Base: } 1314141\overset{2}{3}2 & \\
1314141\overset{2}{3}2 \quad \Delta = -58, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2\overset{5}{3}1314141\overset{2}{3}2 \quad \Delta = -110, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2\overset{6}{3}1314141\overset{2}{3}2 \quad \Delta = -136, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2\overset{4}{3}1314141\overset{2}{3}2 \quad \Delta = -86, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2\overset{3}{3}1314141\overset{2}{3}2 \text{ [#=5]} \quad \Delta = -66, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 100 \quad & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
2\overset{3}{3}1314141\overset{2}{3}2 \text{ [#=4]} \quad \Delta = -66, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
2\overset{3}{3}1314141\overset{2}{3}2 \text{ [#=3]} \quad \Delta = -66, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
2\overset{3}{3}1314141\overset{2}{3}2 \text{ [#=2]} \quad \Delta = -66, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1314141\overset{3}{3}2 \quad \Delta = -66, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 15 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
2\overset{3}{3}1314141\overset{2}{3}2 \text{ [#=6]} \quad \Delta = -66, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 144 \quad & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
2^{\overset{3}{1}314141} 3^{\overset{2}{2}} 2 \quad \Delta = -66, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2^{\overset{3}{1}314141} 3^{\overset{2}{2}} 2 \text{ [#=8]} \quad \Delta = -66, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 256 \quad & \begin{pmatrix} 0 & 16 & 2 \\ 16 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
2^{\overset{3}{1}314141} 3^{\overset{2}{2}} 2 \text{ [#=9]} \quad \Delta = -66, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 324 \quad & \begin{pmatrix} 0 & 18 & 2 \\ 18 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix} \\
2^{\overset{3}{1}314141} 3^{\overset{2}{2}} 2 \text{ [#=7]} \quad \Delta = -66, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 196 \quad & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
2^{\overset{2}{1}314141} 3^{\overset{2}{2}} 2 \quad \Delta = -58, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
131414132^{\overset{2}{2}} \quad \Delta = -59, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
131414132^{\overset{2}{2}} \quad \Delta = -58, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
131414132^{\overset{2}{2}} \quad \Delta = -58, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\textbf{Base: } 2141414131 & \\
2141414131\overset{3}{3} \text{ [#=3]} \quad \Delta = -83, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2141414131^{\overset{4}{4}} \quad \Delta = -103, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2141414131 \text{ [#=3]}^{\overset{3}{3}} \quad \Delta = -83, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 123 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
2141414131\overset{3}{3} \text{ [#=2]} \quad \Delta = -83, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
2141414131 \text{ [#=2]}^{\overset{3}{3}} \quad \Delta = -83, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
2141414131^{\overset{3}{3}} \quad \Delta = -83, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
2141414131^{\overset{3}{3}} \quad \Delta = -83, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
2141414131\overset{3}{3} \text{ [#=9]} \quad \Delta = -83, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2141414131\overset{3}{3} \text{ [#=8]} \quad \Delta = -83, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
2141414131\overset{3}{3} \text{ [#=7]} \quad \Delta = -83, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2141414131\overset{3}{3} \text{ [#=6]} \quad \Delta = -83, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
2141414131\overset{2}{3}_{[\#=5]} \quad \Delta = -83, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2141414131\overset{2}{3}_{[\#=4]} \quad \Delta = -83, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2141414131\overset{2}{3} \quad \Delta = -75, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2141414131\overset{2}{3} \quad \Delta = -75, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2141414131\overset{2}{3} \quad \Delta = -75, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2141414131\overset{2}{3} \quad \Delta = -75, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{141413221}$

$$\begin{aligned}
\overset{1}{141413221}\overset{2}{3} \quad \Delta = -48, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\overset{1}{141413221}\overset{2}{3}_{[\#=9]} \quad \Delta = -56, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
\overset{31}{141413221}_{[\#=2]} \quad \Delta = -56, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\overset{31}{141413221}_{[\#=3]} \quad \Delta = -56, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\overset{1}{141413221}\overset{2}{3}_{[\#=3]} \quad \Delta = -56, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\overset{1}{141413221}\overset{2}{3}_{[\#=8]} \quad \Delta = -56, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
\overset{1}{141413221}\overset{2}{3}_{[\#=7]} \quad \Delta = -56, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
\overset{1}{141413221}\overset{2}{3}_{[\#=6]} \quad \Delta = -56, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
\overset{1}{141413221}\overset{2}{3}_{[\#=5]} \quad \Delta = -56, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\overset{1}{141413221}\overset{2}{3}_{[\#=4]} \quad \Delta = -56, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\overset{1}{141413221}\overset{2}{3} \quad \Delta = -48, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\overset{1}{141413221}\overset{2}{3}_{[\#=2]} \quad \Delta = -56, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{141414131}$

$$\begin{aligned}
141414131\overset{1}{\color{red}3} \text{ }_{[\# = 9]} \quad \Delta = -83, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
141414131\overset{1}{\color{red}3} \text{ }_{[\# = 8]} \quad \Delta = -83, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
141414131\overset{1}{\color{red}3} \text{ }_{[\# = 7]} \quad \Delta = -83, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
141414131\overset{1}{\color{red}3} \text{ }_{[\# = 6]} \quad \Delta = -83, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
141414131\overset{1}{\color{red}3} \text{ }_{[\# = 5]} \quad \Delta = -83, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
141414131\overset{1}{\color{red}3} \text{ }_{[\# = 2]} \quad \Delta = -83, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
141414131\overset{1}{\color{red}3} \text{ }_{[\# = 4]} \quad \Delta = -83, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
141414131\overset{1}{\color{red}3} \text{ }_{[\# = 3]} \quad \Delta = -83, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
141414131\overset{31}{\color{red}3} \text{ }_{[\# = 2]} \quad \Delta = -83, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
141414131\overset{1}{\color{red}3} \quad \Delta = -83, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
141414131\overset{31}{\color{red}3} \text{ }_{[\# = 3]} \quad \Delta = -83, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
141414131\overset{1}{\color{red}2} \quad \Delta = -75, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
141414131\overset{12}{\color{red}2} \quad \Delta = -75, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
141414131\overset{1}{\color{red}2} \quad \Delta = -75, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1414141315

$$\begin{aligned}
5131414141\overset{1}{\color{red}3} \text{ }_{[\# = 3]} \quad \Delta = -135, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1414141315\overset{2}{\color{red}2} \quad \Delta = -128, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1414141315\overset{2}{\color{red}2} \quad \Delta = -127, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1414141315\overset{2}{\color{red}2} \quad \Delta = -127, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1414141315\overset{2}{\color{red}2} \quad \Delta = -127, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

$$1414141315^2 \quad \Delta = -127, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$51314141413^{\#2} \quad \Delta = -135, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1414141315^3 \quad \Delta = -135, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 1223151231

$$13215132213^{\#4} \quad \Delta = -61, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$1223151231^2 \quad \Delta = -53, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1223151231^2 \quad \Delta = -53, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$13215132213^{\#2} \quad \Delta = -61, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$12231512313^{\#2} \quad \Delta = -61, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$13215132213^{\#3} \quad \Delta = -61, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$12231512313^{\#3} \quad \Delta = -61, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$13215132213^{\#5} \quad \Delta = -61, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

$$12231512313^{\#5} \quad \Delta = -61, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$13215132213^{\#6} \quad \Delta = -61, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$12231512313^{\#4} \quad \Delta = -61, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$13215132213^{\#8} \quad \Delta = -61, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix}$$

$$13215132213^{\#7} \quad \Delta = -61, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix}$$

$$13215132213^{\#9} \quad \Delta = -61, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix}$$

Base: 1231512321

$$12321513213^{\#8} \quad \Delta = -63, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix}$$

$$\begin{aligned}
1231512321^2 \quad \Delta = -55, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1231512321^3_{[\# = 5]} \quad \Delta = -63, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
1231512321^3_{[\# = 4]} \quad \Delta = -63, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
1231512321^2 \quad \Delta = -55, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1232151321^3_{[\# = 5]} \quad \Delta = -63, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
1231512321^3_{[\# = 3]} \quad \Delta = -63, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
1232151321^3_{[\# = 6]} \quad \Delta = -63, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1231512321^3_{[\# = 6]} \quad \Delta = -63, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
1232151321^3_{[\# = 7]} \quad \Delta = -63, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
1231512321^3_{[\# = 7]} \quad \Delta = -63, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
1232151321^3_{[\# = 3]} \quad \Delta = -63, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1231512321^3_{[\# = 9]} \quad \Delta = -63, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
1232151321^3_{[\# = 9]} \quad \Delta = -63, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix} \\
1231512321^3_{[\# = 8]} \quad \Delta = -63, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
1231512321^3_{[\# = 2]} \quad \Delta = -63, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
1232151321^3_{[\# = 2]} \quad \Delta = -63, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1232151321^3_{[\# = 4]} \quad \Delta = -63, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
\text{Base: } 1315132215 \\
5122315131^3_{[\# = 9]} \quad \Delta = -112, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
1315132215^3 \quad \Delta = -112, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
5122315131\overset{2}{3}_{[\# = 2]} \quad \Delta = -112, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
5122315131\overset{2}{3}_{[\# = 3]} \quad \Delta = -112, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
5122315131\overset{2}{3}_{[\# = 4]} \quad \Delta = -112, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
5122315131\overset{2}{3}_{[\# = 5]} \quad \Delta = -112, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
5122315131\overset{2}{3}_{[\# = 6]} \quad \Delta = -112, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
131\overset{2}{5}132215 \quad \Delta = -104, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
131\overset{2}{5}132215 \quad \Delta = -104, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
131\overset{2}{5}132215 \quad \Delta = -104, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
5122315131\overset{2}{3}_{[\# = 7]} \quad \Delta = -112, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
5122315131\overset{2}{3}_{[\# = 8]} \quad \Delta = -112, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1223151313

$$\begin{aligned}
122315131\overset{2}{3} \quad \Delta = -60, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
122315131\overset{2}{3} \quad \Delta = -60, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
122315131\overset{2}{3} \quad \Delta = -60, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
122315131\overset{4}{3} \quad \Delta = -88, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 9 & 5 \\ 5 & 9 \end{pmatrix} \\
3131513221\overset{2}{3}_{[\# = 3]} \quad \Delta = -68, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
3131513221\overset{2}{3}_{[\# = 6]} \quad \Delta = -68, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
122315131\overset{3}{3} \quad \Delta = -68, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
3131513221\overset{2}{3}_{[\# = 4]} \quad \Delta = -68, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
3131513221\overset{2}{3}_{[\# = 9]} \quad \Delta = -68, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1223151313^{3}_{[\# = 3]} \quad \Delta = -68, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
31315132213^{3}_{[\# = 2]} \quad \Delta = -68, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1223151313^{3}_{[\# = 2]} \quad \Delta = -68, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
31315132213^{3}_{[\# = 8]} \quad \Delta = -68, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
31315132213^{3}_{[\# = 7]} \quad \Delta = -68, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
1223151313^{2} \quad \Delta = -61, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
31315132213^{3}_{[\# = 5]} \quad \Delta = -68, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1413151321

$$\begin{aligned}
1413151321^{2} \quad \Delta = -80, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1413151321^{2} \quad \Delta = -80, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1413151321^{2} \quad \Delta = -80, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
14131513213^{3}_{[\# = 9]} \quad \Delta = -88, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
1413151321^{2} \quad \Delta = -81, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
14131513213^{3}_{[\# = 8]} \quad \Delta = -88, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
14131513213^{3}_{[\# = 3]} \quad \Delta = -88, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
12315131413^{3}_{[\# = 2]} \quad \Delta = -88, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
12315131413^{3}_{[\# = 3]} \quad \Delta = -88, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1413151321^{3} \quad \Delta = -88, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
14131513213^{3}_{[\# = 2]} \quad \Delta = -88, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
14131513213^{3}_{[\# = 7]} \quad \Delta = -88, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1413151321\overset{3}{\color{red}}_{[\# = 6]} \quad \Delta = -88, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1413151321\overset{3}{\color{red}}_{[\# = 5]} \quad \Delta = -88, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1413151321\overset{3}{\color{red}}_{[\# = 4]} \quad \Delta = -88, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

Base: 4131513215

$$\begin{aligned}
\overset{2}{\color{red}}{4131513215} \quad \Delta = -132, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\overset{2}{\color{red}}{4131513215} \quad \Delta = -132, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\overset{2}{\color{red}}{4131513215} \quad \Delta = -132, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\overset{2}{\color{red}}{4131513215} \quad \Delta = -132, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\overset{2}{\color{red}}{4131513215} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\overset{3}{\color{red}}{4131513215} \quad \Delta = -140, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: 4123151321

$$\begin{aligned}
4123151321\overset{3}{\color{red}}_{[\# = 2]} \quad \Delta = -89, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
4123151321\overset{2}{\color{red}} \quad \Delta = -81, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
4123151321\overset{2}{\color{red}} \quad \Delta = -81, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
4123151321\overset{3}{\color{red}} \quad \Delta = -89, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 57 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
4123151321\overset{3}{\color{red}}_{[\# = 9]} \quad \Delta = -89, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
4123151321\overset{3}{\color{red}}_{[\# = 8]} \quad \Delta = -89, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
4123151321\overset{2}{\color{red}} \quad \Delta = -81, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 58 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
4123151321\overset{3}{\color{red}}_{[\# = 3]} \quad \Delta = -89, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
4123151321\overset{3}{\color{red}}_{[\# = 4]} \quad \Delta = -89, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
4123151321\overset{2}{3}_{[\# = 5]} \quad \Delta = -89, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
4123151321\overset{2}{3}_{[\# = 6]} \quad \Delta = -89, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
4123151321\overset{2}{3}_{[\# = 7]} \quad \Delta = -89, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1315132151

$$\begin{aligned}
1512315131\overset{2}{3}_{[\# = 3]} \quad \Delta = -112, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1512315131\overset{2}{3}_{[\# = 2]} \quad \Delta = -112, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1512315131\overset{2}{3}_{[\# = 8]} \quad \Delta = -112, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1512315131\overset{2}{3}_{[\# = 7]} \quad \Delta = -112, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1512315131\overset{2}{3}_{[\# = 9]} \quad \Delta = -112, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1512315131\overset{2}{3}_{[\# = 6]} \quad \Delta = -112, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1512315131\overset{2}{3}_{[\# = 5]} \quad \Delta = -112, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1512315131\overset{2}{3}_{[\# = 4]} \quad \Delta = -112, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1315132151\overset{2}{2} \quad \Delta = -104, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1315132151\overset{2}{2} \quad \Delta = -104, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1315132151\overset{2}{2} \quad \Delta = -104, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{2}{131513221}$

$$\begin{aligned}
\overset{2}{131513221}\overset{2}{3}_{[\# = 7]} \quad \Delta = -61, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
1223151\overset{2}{3}1\overset{2}{5} \quad \Delta = -105, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 8 \end{pmatrix} \\
\overset{2}{131513221}\overset{2}{2} \quad \Delta = -53, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\overset{2}{131513221}\overset{2}{2} \quad \Delta = -53, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{2}{131513221}\overset{2}{3} \text{ }_{[\# = 8]} \quad \Delta = -61, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
& 1223151\overset{2}{31}\overset{2}{4} \quad \Delta = -81, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
& 1223151\overset{2}{31}\overset{2}{2} \quad \Delta = -53, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 1223151\overset{2}{31}\overset{2}{3} \text{ }_{[\# = 4]} \quad \Delta = -61, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{2}{131513221}\overset{2}{3} \text{ }_{[\# = 3]} \quad \Delta = -61, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{2}{131513221}\overset{2}{3} \text{ }_{[\# = 9]} \quad \Delta = -61, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
& 1223151\overset{2}{31}\overset{2}{3} \text{ }_{[\# = 2]} \quad \Delta = -61, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{2}{131513221}\overset{2}{3} \text{ }_{[\# = 2]} \quad \Delta = -61, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& 1223151\overset{2}{31}\overset{2}{3} \text{ }_{[\# = 3]} \quad \Delta = -61, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{2}{131513221}\overset{2}{3} \text{ }_{[\# = 4]} \quad \Delta = -61, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& 1223151\overset{2}{31}\overset{2}{3} \text{ }_{[\# = 5]} \quad \Delta = -61, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{2}{131513221}\overset{2}{3} \text{ }_{[\# = 5]} \quad \Delta = -61, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
& \overset{2}{131513221}\overset{2}{3} \text{ }_{[\# = 6]} \quad \Delta = -61, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
& \text{Base: } 4131513151 \\
& 413151\overset{2}{31}51 \quad \Delta = -132, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 413151\overset{2}{31}51 \quad \Delta = -131, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 413151\overset{2}{31}51 \quad \Delta = -131, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 413151\overset{2}{31}51 \quad \Delta = -131, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 413151\overset{2}{31}51 \quad \Delta = -132, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{3}{41315131}51 \quad \Delta = -139, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

$$4131513151^2 \quad \Delta = -131, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 1315131512

$$1315131512^3 \quad \Delta = -111, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$1315131512^2 \quad \Delta = -103, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1315131512^2 \quad \Delta = -103, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1315131512^2 \quad \Delta = -103, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1315131512^2 \quad \Delta = -103, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1315131512^2 \quad \Delta = -104, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$2151315131^3_{[\# = 3]} \quad \Delta = -111, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$2151315131^3_{[\# = 4]} \quad \Delta = -111, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$2151315131^3_{[\# = 5]} \quad \Delta = -111, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$2151315131^3_{[\# = 6]} \quad \Delta = -111, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$2151315131^3_{[\# = 7]} \quad \Delta = -111, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$2151315131^3_{[\# = 2]} \quad \Delta = -111, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$2151315131^3_{[\# = 8]} \quad \Delta = -111, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$2151315131^3_{[\# = 9]} \quad \Delta = -111, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

Base: 151315131^1

$$151315131^1_{[\# = 9]} \quad \Delta = -111, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$151315131^1^2 \quad \Delta = -103, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$151315131^1^2 \quad \Delta = -103, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$151315131^1^2 \quad \Delta = -103, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\begin{aligned}
151315131\bar{3} \text{ }_{[\# = 6]} \quad \Delta = -111, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
151315131\bar{3} \text{ }_{[\# = 7]} \quad \Delta = -111, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
151315131\bar{3} \text{ }_{[\# = 2]} \quad \Delta = -111, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
151315131\bar{3} \text{ }_{[\# = 8]} \quad \Delta = -111, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
151315131\bar{3} \text{ }_{[\# = 3]} \quad \Delta = -111, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
151315131\bar{2} \quad \Delta = -104, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
151315131\bar{3} \text{ }_{[\# = 5]} \quad \Delta = -111, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
151315131\bar{3} \text{ }_{[\# = 4]} \quad \Delta = -111, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

Base: 131513221

$$\begin{aligned}
122315131\bar{3} \text{ }_{[\# = 2]} \quad \Delta = -60, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
131513221\bar{3} \text{ }_{[\# = 4]} \quad \Delta = -60, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
122315131\bar{3} \text{ }_{[\# = 4]} \quad \Delta = -60, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
122315131\bar{3} \text{ }_{[\# = 8]} \quad \Delta = -60, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
131513221\bar{3} \text{ }_{[\# = 8]} \quad \Delta = -60, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
122315131\bar{3} \text{ }_{[\# = 9]} \quad \Delta = -60, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
131513221\bar{2} \quad \Delta = -52, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
131513221\bar{2}^1 \quad \Delta = -52, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
131513221\bar{3} \text{ }_{[\# = 7]} \quad \Delta = -60, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
122315131\bar{3} \text{ }_{[\# = 5]} \quad \Delta = -60, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
122315131\bar{3} \text{ }_{[\# = 7]} \quad \Delta = -60, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
12231\overset{1}{5}131\overset{1}{3} \text{ }_{[\# = 3]} \quad \Delta = -60, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
131\overset{1}{5}13221\overset{1}{3} \text{ }_{[\# = 3]} \quad \Delta = -60, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
131\overset{1}{5}13221\overset{1}{3} \text{ }_{[\# = 9]} \quad \Delta = -60, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
131\overset{1}{5}13221\overset{1}{3} \text{ }_{[\# = 2]} \quad \Delta = -60, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
12231\overset{1}{5}131\overset{1}{3} \text{ }_{[\# = 6]} \quad \Delta = -60, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
131\overset{1}{5}13221\overset{1}{3} \text{ }_{[\# = 5]} \quad \Delta = -60, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
131\overset{1}{5}13221\overset{1}{3} \text{ }_{[\# = 6]} \quad \Delta = -60, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1231\overset{1}{5}1321$

$$\begin{aligned}
1231\overset{1}{5}1321\overset{1}{3} \text{ }_{[\# = 3]} \quad \Delta = -61, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1231\overset{1}{5}1321\overset{1}{3} \text{ }_{[\# = 9]} \quad \Delta = -61, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1231\overset{1}{5}1321\overset{1}{3} \text{ }_{[\# = 8]} \quad \Delta = -61, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1231\overset{1}{5}1321\overset{1}{3} \text{ }_{[\# = 4]} \quad \Delta = -61, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1231\overset{1}{5}1321\overset{1}{3} \text{ }_{[\# = 7]} \quad \Delta = -61, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1231\overset{21}{5}1321 \quad \Delta = -53, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1231\overset{1}{5}1321\overset{1}{3} \text{ }_{[\# = 6]} \quad \Delta = -61, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1231\overset{1}{5}1321\overset{1}{3} \text{ }_{[\# = 2]} \quad \Delta = -61, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1231\overset{1}{5}1321\overset{1}{3} \text{ }_{[\# = 5]} \quad \Delta = -61, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}
\end{aligned}$$

Base: $31\overset{12231}{5}13$

$$\begin{aligned}
31\overset{312231}{5}13 \text{ }_{[\# = 5]} \quad \Delta = -68, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 100 \quad & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
31\overset{312231}{5}13 \text{ }_{[\# = 6]} \quad \Delta = -68, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 144 \quad & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
31 \overset{5}{1} 2231 \quad \Delta = -112, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
31 \overset{12231}{5} 13 \quad \Delta = -60, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
31 \overset{312231}{5} 13 [\# = 7] \quad \Delta = -68, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 196 & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
31 \overset{312231}{5} 13 [\# = 8] \quad \Delta = -68, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 256 & \begin{pmatrix} 0 & 16 & 2 \\ 16 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
31 \overset{312231}{5} 13 [\# = 4] \quad \Delta = -68, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
31 \overset{312231}{5} 13 [\# = 3] \quad \Delta = -68, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
31 \overset{312231}{5} 13 [\# = 2] \quad \Delta = -68, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
31 \overset{312231}{5} 13 \quad \Delta = -68, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
31 \overset{712231}{5} 13 \quad \Delta = -165, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
31 \overset{212231}{5} 13 \quad \Delta = -60, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
31 \overset{212231}{5} 13 \quad \Delta = -60, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
31 \overset{412231}{5} 13 \quad \Delta = -88, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
31 \overset{812231}{5} 13 \quad \Delta = -193, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 8 \end{pmatrix} \\
31 \overset{1212231}{5} 13 \quad \Delta = -308, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -6 & -4 \\ 2 & -4 & 8 \end{pmatrix} \\
31 \overset{612231}{5} 13 \quad \Delta = -138, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
31 \overset{312231}{5} 13 [\# = 9] \quad \Delta = -68, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 324 & \begin{pmatrix} 0 & 18 & 2 \\ 18 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix}
\end{aligned}$$

Base: 6122314141

$$\begin{aligned}
6122314141 \overset{3}{5} [\# = 2] \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
6122314141 \overset{3}{5} \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 63 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
6122314141 \overset{3}{5} [\# = 3] \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
6122314141 \quad \Delta = -126, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
6122314141 \quad \Delta = -126, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
6122314141 \quad \Delta = -126, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 6122315123

$$\begin{aligned}
6122315123 \quad \Delta = -139, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 27 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
6122315123 \quad \Delta = -131, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
6122315123 \quad \Delta = -131, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
6122315123 \quad \Delta = -131, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 612231513

$$\begin{aligned}
612231513 \quad \Delta = -138, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 27 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
612231513 \quad \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
612231513 \quad \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
612231513 \quad \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
612231513 \quad \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: 6123141413

$$\begin{aligned}
6123141413 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
6123141413 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
6123141413 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
6123141413 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
6123141413 \quad \Delta = -136, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
6123141413 \quad \Delta = -142, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: 6123151232

$$\begin{aligned}
6123151232 \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
6123151232 \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
6123151232 \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: 6123215132

$$\begin{aligned}
6123215132 \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
6123215132 \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
6123215132 \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{aligned}$$

Base: 612315¹132

$$\begin{aligned}
612315132 \quad \Delta = -131, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
612315132 \quad \Delta = -131, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
612315132 \quad \Delta = -139, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
612315132 \quad \Delta = -131, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
612315132 \quad \Delta = -131, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: 6131414132

$$\begin{aligned}
6131414132 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
6131414132 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
6131414132 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
6131414132 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
6131414132 \quad \Delta = -142, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 60 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

Base: 613151¹322

$$613151322 \quad \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$6131\overset{2}{1}51322 \quad \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$6131\overset{2}{5}1322 \quad \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$6131\overset{21}{5}1322 \quad \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$6131\overset{31}{5}1322 \quad \Delta = -138, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 1612231512

$$1612231\overset{2}{5}12 \quad \Delta = -122, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1612231\overset{2}{5}12 \quad \Delta = -122, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1612231\overset{2}{5}12 \quad \Delta = -122, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1612231\overset{3}{5}12 \quad \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$1612231\overset{3}{5}12 \quad \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 17 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 1612231¹51

$$1612231\overset{21}{5}1 \quad \Delta = -122, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1612231\overset{3}{5}1 \quad \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 17 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1612231\overset{2}{5}1 \quad \Delta = -122, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 1414132161

$$14141321\overset{2}{6}1 \quad \Delta = -126, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1612314141\overset{3}{3} \text{ [#=3]} \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1612314141\overset{3}{3} \text{ [#=2]} \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$14141321\overset{2}{6}1 \quad \Delta = -126, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$14141321\overset{2}{6}1 \quad \Delta = -128, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$14141321\overset{2}{6}1 \quad \Delta = -126, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: 1612315123

$$\begin{aligned}
1612315123 \quad \Delta = -131, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1612315123 \quad \Delta = -131, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1612315123 \quad \Delta = -131, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1612315¹13

$$\begin{aligned}
161231513 \quad \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
161231513 \quad \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
161231513 \quad \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
161231513 \quad \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
161231513 \quad \Delta = -138, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1612321513

$$\begin{aligned}
1612321513 \quad \Delta = -132, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1612321513 \quad \Delta = -132, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1612321513 \quad \Delta = -132, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: 3141413161

$$\begin{aligned}
3141413161 \quad \Delta = -141, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
3141413161 \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
3141413161 \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
3141413161 \quad \Delta = -141, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 60 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
3141413161 \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
3141413161 \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1613151232

$$1613151232 \quad \Delta = -132, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$\begin{aligned}
1613151232^2 \quad \Delta = -132, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1613151232^2 \quad \Delta = -132, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

Base: 161315132^1

$$\begin{aligned}
161315132^{21} \quad \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
161315132^{12} \quad \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
161315132^{21} \quad \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
161315132^{21} \quad \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
161315132^{31} \quad \Delta = -138, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 2161231512

$$\begin{aligned}
2161231512^3 \quad \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2161231512^3 \quad \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
2161231512^2 \quad \Delta = -122, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2161231512^2 \quad \Delta = -122, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2161231512^2 \quad \Delta = -122, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2161231512^2 \quad \Delta = -122, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: 216123151^1

$$\begin{aligned}
216123151^3 \quad \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
216123151^{21} \quad \Delta = -122, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
216123151^{21} \quad \Delta = -122, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
216123151^{21} \quad \Delta = -122, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: 2161232151

$$2161232151^3 \quad \Delta = -132, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$2161232151^2 \quad \Delta = -124, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$2161232151^2 \quad \Delta = -124, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$2161232151^2 \quad \Delta = -124, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 3216123151

$$3216123151^2 \quad \Delta = -131, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$3216123151^2 \quad \Delta = -131, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$3216123151^2 \quad \Delta = -131, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 3216123215

$$3216123215^2 \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$3216123215^2 \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$3216123215^2 \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

Base: 1321612321

$$1321612321^3_{[\# = 2]} \quad \Delta = -89, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1321612321^3_{[\# = 5]} \quad \Delta = -89, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1321612321^2 \quad \Delta = -81, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1321612321^2 \quad \Delta = -81, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1232161231^3_{[\# = 5]} \quad \Delta = -89, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1321612321^3_{[\# = 6]} \quad \Delta = -89, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1321612321^3_{[\# = 7]} \quad \Delta = -89, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$1321612321^3_{[\# = 4]} \quad \Delta = -89, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1321612321^3_{[\# = 8]} \quad \Delta = -89, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1321612321\overset{2}{3} \text{ [#=9]} \quad \Delta = -89, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$1232161231\overset{2}{3} \text{ [#=2]} \quad \Delta = -89, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1232161231\overset{2}{3} \text{ [#=3]} \quad \Delta = -89, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1321612321\overset{2}{3} \text{ [#=3]} \quad \Delta = -89, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1232161231\overset{2}{3} \text{ [#=4]} \quad \Delta = -89, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: 2321612315

$$232161231\overset{2}{5} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$232161231\overset{2}{5} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$232161231\overset{2}{5} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 3161223151

$$31612231\overset{2}{5} \quad \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$31612231\overset{2}{5} \quad \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$31612231\overset{2}{5} \quad \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$31612231\overset{3}{5} \quad \Delta = -138, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: 4141321613

$$4141321\overset{2}{6}13 \quad \Delta = -136, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$4141321\overset{2}{6}13 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$4141321\overset{2}{6}13 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$4141321\overset{2}{6}13 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$4141321\overset{2}{6}13 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: 1414131612

$$1414131\overset{2}{6}12 \quad \Delta = -125, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\begin{aligned}
1414131612^2 \quad \Delta = -125, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1414131612^2 \quad \Delta = -125, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1414131612^2 \quad \Delta = -125, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21613141413^{\#=2} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
21613141413^{\#=3} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1414131612^3 \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
1414131612^3 \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 32 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 2161315123

$$\begin{aligned}
2161315123^2 \quad \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2161315123^2 \quad \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
2161315123^2 \quad \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2161315123^3 \quad \Delta = -138, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2161315123^2 \quad \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 216131513¹

$$\begin{aligned}
216131513^2 \quad \Delta = -129, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
216131513^{21} \quad \Delta = -129, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
216131513^{21} \quad \Delta = -129, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
216131513^{12} \quad \Delta = -129, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
216131513^{31} \quad \Delta = -137, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
216131513^3 \quad \Delta = -137, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
216131513^{21} \quad \Delta = -129, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: 2216131512

$$\begin{aligned}
2216131512^2 \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2216131512^3 \quad \Delta = -129, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
2216131512^2 \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2216131512^2 \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2216131512^2 \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2216131512^3 \quad \Delta = -129, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2216131512^3 \quad \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: 221613151¹

$$\begin{aligned}
221613151^{21} \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
221613151^{21} \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
221613151^{21} \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
221613151^{31} \quad \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
221613151^{31} \quad \Delta = -129, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: 3221613151

$$\begin{aligned}
3221613151^2 \quad \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
3221613151^2 \quad \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
3221613151^3 \quad \Delta = -138, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
3221613151^2 \quad \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1413216131

$$\begin{aligned}
1413216131^2 \quad \Delta = -106, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1413216131^2 \quad \Delta = -106, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1413216131^2 \quad \Delta = -106, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1413216131^3_{[\# = 5]} \quad \Delta = -114, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1413216131^3_{[\# = 2]} \quad \Delta = -114, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
1316123141^3_{[\# = 2]} \quad \Delta = -114, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1316123141^3_{[\# = 3]} \quad \Delta = -114, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1413216131^3_{[\# = 3]} \quad \Delta = -114, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1413216131^3_{[\# = 9]} \quad \Delta = -114, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1413216131^3_{[\# = 8]} \quad \Delta = -114, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1413216131^3_{[\# = 7]} \quad \Delta = -114, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1413216131^3_{[\# = 6]} \quad \Delta = -114, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1413216131^2 \quad \Delta = -108, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1413216131^3_{[\# = 4]} \quad \Delta = -114, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\textbf{Base: } 4123216131 & \\
4123216131^3_{[\# = 7]} \quad \Delta = -116, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
4123216131^2 \quad \Delta = -108, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
4123216131^2 \quad \Delta = -108, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
4123216131^2 \quad \Delta = -108, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
4123216131^3_{[\# = 8]} \quad \Delta = -116, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
4123216131^3 \quad \Delta = -116, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
4123216131^3_{[\# = 2]} \quad \Delta = -116, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
4123216131\overset{3}{3} \text{ }_{[\# = 9]} \quad \Delta = -116, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
4123216131\overset{3}{3} \text{ }_{[\# = 6]} \quad \Delta = -116, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
4123216131\overset{3}{3} \text{ }_{[\# = 5]} \quad \Delta = -116, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
4123216131\overset{3}{3} \text{ }_{[\# = 4]} \quad \Delta = -116, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
4123216131\overset{3}{3} \text{ }_{[\# = 3]} \quad \Delta = -116, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: 4141316123

$$\begin{aligned}
4141\overset{3}{3}16123 \quad \Delta = -142, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 37 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
4141\overset{2}{2}16123 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
4141\overset{2}{2}16123 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
4141\overset{2}{2}16123 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
4141\overset{2}{2}16123 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1232161313

$$\begin{aligned}
3131612321\overset{3}{3} \text{ }_{[\# = 9]} \quad \Delta = -96, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
3131612321\overset{3}{3} \text{ }_{[\# = 7]} \quad \Delta = -96, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
3131612321\overset{3}{3} \text{ }_{[\# = 6]} \quad \Delta = -96, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
123216131\overset{2}{3} \quad \Delta = -88, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
123216131\overset{2}{3} \quad \Delta = -88, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
3131612321\overset{3}{3} \text{ }_{[\# = 5]} \quad \Delta = -96, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
3131612321\overset{3}{3} \text{ }_{[\# = 4]} \quad \Delta = -96, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
123216131\overset{3}{3} \text{ }_{[\# = 3]} \quad \Delta = -96, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
3131612321\overset{3}{3} \text{ }_{[\# = 3]} \quad \Delta = -96, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1232161313\overset{3}{3} \text{ }_{[\# = 2]} \quad \Delta = -96, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1232161313\overset{2}{2} \quad \Delta = -88, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1232161313\overset{4}{4} \quad \Delta = -116, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 9 & 5 \\ 5 & 9 \end{pmatrix} \\
3131612321\overset{3}{3} \text{ }_{[\# = 2]} \quad \Delta = -96, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
3131612321\overset{3}{3} \text{ }_{[\# = 8]} \quad \Delta = -96, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1232161313\overset{3}{3} \quad \Delta = -96, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\text{Base: } 4131612314 \\
4131612314\overset{2}{2} \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
4131612314\overset{2}{2} \quad \Delta = -136, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
4131612314\overset{2}{2} \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
4131612314\overset{2}{2} \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
4131612314\overset{2}{2} \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
4131612314\overset{3}{3} \quad \Delta = -142, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\text{Base: } 1413161231 \\
1413161231\overset{3}{3} \text{ }_{[\# = 4]} \quad \Delta = -114, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1413161231\overset{2}{2} \quad \Delta = -106, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1413161231\overset{2}{2} \quad \Delta = -106, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1413161231\overset{2}{2} \quad \Delta = -106, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1413161231\overset{3}{3} \quad \Delta = -114, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1321613141\overset{3}{3} \text{ }_{[\# = 2]} \quad \Delta = -114, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1413161231\overset{\textcolor{red}{3}}{3}_{[\#=2]} \quad \Delta = -114, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
1413161231\overset{\textcolor{red}{3}}{3}_{[\#=5]} \quad \Delta = -114, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1413161231\overset{\textcolor{red}{3}}{3}_{[\#=3]} \quad \Delta = -114, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1321613141\overset{\textcolor{red}{3}}{3}_{[\#=3]} \quad \Delta = -114, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 3141316131

$$\begin{aligned}
\overset{\textcolor{red}{3}}{3}3141316131 \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
3141316131\overset{\textcolor{red}{3}}{3} \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 32 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
3141316131\overset{\textcolor{red}{2}}{2} \quad \Delta = -113, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
3141316131\overset{\textcolor{red}{2}}{2} \quad \Delta = -113, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
3141316131\overset{\textcolor{red}{2}}{2} \quad \Delta = -113, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
3141316131\overset{\textcolor{red}{2}}{2} \quad \Delta = -113, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
3141316131\overset{\textcolor{red}{3}}{3}_{[\#=2]} \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
3141316131\overset{\textcolor{red}{3}}{3}_{[\#=3]} \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
3141316131\overset{\textcolor{red}{3}}{3}_{[\#=9]} \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
3141316131\overset{\textcolor{red}{3}}{3}_{[\#=8]} \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
3141316131\overset{\textcolor{red}{3}}{3}_{[\#=7]} \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
3141316131\overset{\textcolor{red}{3}}{3}_{[\#=6]} \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
3141316131\overset{\textcolor{red}{3}}{3}_{[\#=5]} \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
3141316131\overset{\textcolor{red}{3}}{3}_{[\#=4]} \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1413161313

$$1413161313^3_{[\# = 2]} \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1413161313^2 \quad \Delta = -113, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1413161313^2 \quad \Delta = -113, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1413161313^2 \quad \Delta = -113, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1413161313^2 \quad \Delta = -113, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1413161313^3 \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$31316131413^3_{[\# = 2]} \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$31316131413^3_{[\# = 3]} \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1413161313^3_{[\# = 3]} \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1413161313^4 \quad \Delta = -141, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 9 & 5 \\ 5 & 9 \end{pmatrix}$$

$$1413161313^3 \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$\mathbf{Base:} \quad 1^{\begin{smallmatrix} 2151321 \\ 6 \end{smallmatrix}} 1$$

$$1^{\begin{smallmatrix} 2151321 \\ 6 \end{smallmatrix}} 1 \quad \Delta = -122, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1^{\begin{smallmatrix} 2151321 \\ 6 \end{smallmatrix}} 1 \quad \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$1^{\begin{smallmatrix} 2151321 \\ 6 \end{smallmatrix}} 1 \quad \Delta = -122, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1^{\begin{smallmatrix} 2151321 \\ 6 \end{smallmatrix}} 1 \quad \Delta = -122, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\mathbf{Base:} \quad 1^{\begin{smallmatrix} 151321 \\ 6 \end{smallmatrix}} 1$$

$$1^{\begin{smallmatrix} 151321 \\ 6 \end{smallmatrix}} 1 \quad \Delta = -122, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1^{\begin{smallmatrix} 151321 \\ 6 \end{smallmatrix}} 1 \quad \Delta = -122, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\mathbf{Base:} \quad 1^{\begin{smallmatrix} 1512321 \\ 6 \end{smallmatrix}} 1$$

$$1^{\begin{smallmatrix} 1512321 \\ 6 \end{smallmatrix}} 1 \quad \Delta = -124, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{1512321}{6} 1 \quad \Delta = -124, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: $1 \overset{1}{161314141}$

$$1 \overset{1}{161314141} \overset{3}{1} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 37 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1 \overset{1}{161314141} \overset{3}{1} \text{ [#=2]} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$1 \overset{1}{161314141} \overset{3}{1} \text{ [#=3]} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$1 \overset{1}{161314141} \overset{2}{1} \quad \Delta = -125, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{1}{161314141} \overset{2}{1} \quad \Delta = -125, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{1}{161314141} \overset{2}{1} \quad \Delta = -125, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: $1 \overset{3215131}{6} 1$

$$1 \overset{3215131}{6} 1 \quad \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{3215131}{6} 1 \quad \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{3215131}{6} 1 \quad \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: $1 \overset{315131}{6} 1$

$$1 \overset{315131}{6} 1 \quad \Delta = -129, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{315131}{6} 1 \quad \Delta = -129, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{315131}{6} 1 \quad \Delta = -129, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{315131}{6} 1 \quad \Delta = -129, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{315131}{6} 1 \quad \Delta = -137, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: $1 \overset{151321}{6} 13$

$$1 \overset{151321}{6} 13 \quad \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$31 \overset{151321}{6} \overset{13}{1} \quad \Delta = -138, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$31 \overset{151321}{6} \overset{12}{12} \quad \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{151321}{6} \overset{2}{13} \quad \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{151321}{6} 13 \quad \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: $1 \overset{512321}{6} 13$

$$31 \overset{512321}{6} \overset{12}{12} \quad \Delta = -132, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{512321}{6} \overset{2}{13} \quad \Delta = -132, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{512321}{6} 13 \quad \Delta = -132, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \overset{512321}{6} 13 \quad \Delta = -132, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$31 \overset{512321}{6} \overset{13}{13} \quad \Delta = -140, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: $1 \overset{215131}{6} 12$

$$1 \overset{215131}{6} \overset{2}{12} \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{215131}{6} 12 \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1 \overset{215131}{6} 12 \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{215131}{6} 12 \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{215131}{6} 12 \quad \Delta = -129, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$1 \overset{215131}{6} \overset{3}{12} \quad \Delta = -129, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: $1 \overset{15131}{6} 12$

$$1 \overset{15131}{6} \overset{3}{12} \quad \Delta = -129, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{15131}{6} 12 \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{15131}{6} 12 \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{15131}{6} \overset{2}{12} \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: $1 \overset{12321}{6} 131$

$$1 \overset{12321}{6} 131 \overset{\color{red}3}{\#}=4] \quad \Delta = -88, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1 \overset{\color{red}3}{12321} 131 [\#]=3] \quad \Delta = -88, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \overset{12321}{6} 131 \overset{\color{red}3}{\#}=2] \quad \Delta = -88, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \overset{12321}{6} 131 \overset{\color{red}3}{\#}=3] \quad \Delta = -88, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{\color{red}3}{12321} 131 [\#]=2] \quad \Delta = -88, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{12321}{6} 131 \overset{\color{red}3}{\#}=8] \quad \Delta = -88, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$1 \overset{12321}{6} 131 \overset{\color{red}2}{\#}=1] \quad \Delta = -80, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{12321}{6} 131 \overset{\color{red}3}{\#}=5] \quad \Delta = -88, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1 \overset{12321}{6} 131 \overset{\color{red}3}{\#}=6] \quad \Delta = -88, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$1 \overset{12321}{6} 131 \overset{\color{red}3}{\#}=7] \quad \Delta = -88, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$1 \overset{\color{red}3}{12321} 131 [\#]=5] \quad \Delta = -88, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1 \overset{\color{red}3}{12321} 131 [\#]=7] \quad \Delta = -88, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$1 \overset{\color{red}3}{12321} 131 [\#]=8] \quad \Delta = -88, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1 \overset{12321}{6} 131 \overset{\color{red}3}{\#}=9] \quad \Delta = -88, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix}$$

$$1 \overset{\color{red}3}{12321} 131 [\#]=9] \quad \Delta = -88, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$1 \overset{12321}{6} 131 \quad \Delta = -80, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{\color{red}3}{12321} 131 [\#]=6] \quad \Delta = -88, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1 \overset{\color{red}3}{12321} 131 [\#]=4] \quad \Delta = -88, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: $1 \overset{15131}{6} 123$

$$\begin{array}{ll}
321 \overset{15131}{6} \overset{2}{12} & \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
321 \overset{15131}{6} \overset{13}{12} & \Delta = -138, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1 \overset{15131}{6} \overset{2}{123} & \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \overset{15131}{6} \overset{123}{123} & \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1 \overset{15131}{6} \overset{123}{123} & \Delta = -130, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{5131}{6} 1232 & \\
1 \overset{5131}{6} \overset{2}{1232} & \Delta = -132, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
2321 \overset{5131}{6} \overset{2}{12} & \Delta = -132, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2321 \overset{5131}{6} \overset{13}{12} & \Delta = -140, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1 \overset{5131}{6} 1232 & \Delta = -132, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1 \overset{5131}{6} 1232 & \Delta = -132, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\text{Base: } 414131 \overset{31}{6} 1 & \\
414131 \overset{31}{6} 1 & \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
414131 \overset{2}{6} \overset{31}{1} & \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
414131 \overset{2}{6} \overset{31}{1} & \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
414131 \overset{31}{6} \overset{12}{12} & \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
414131 \overset{231}{6} 1 & \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
414131 \overset{3}{6} \overset{31}{1} & \Delta = -141, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 37 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{131}{6} 13141 & \\
1 \overset{131}{6} \overset{2}{13141} & \Delta = -105, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \overset{131}{6} \overset{2}{13141} & \Delta = -105, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
& \overset{131}{1\ 6\ 13141} \quad \Delta = -105, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{3131}{1\ 6\ 13141} \text{ [#=6]} \quad \Delta = -113, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& \overset{3131}{1\ 6\ 13141} \text{ [#=5]} \quad \Delta = -113, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& \overset{131}{1\ 6\ 13141} \overset{3}{} \quad \Delta = -113, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{3131}{1\ 6\ 13141} \text{ [#=3]} \quad \Delta = -113, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{3131}{1\ 6\ 13141} \text{ [#=2]} \quad \Delta = -113, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{3131}{1\ 6\ 13141} \text{ [#=9]} \quad \Delta = -113, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
& \overset{131}{1\ 6\ 13141} \overset{3}{} \text{ [#=2]} \quad \Delta = -113, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& \overset{3131}{1\ 6\ 13141} \text{ [#=8]} \quad \Delta = -113, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
& \overset{3131}{1\ 6\ 13141} \text{ [#=4]} \quad \Delta = -113, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{131}{1\ 6\ 13141} \overset{3}{} \text{ [#=3]} \quad \Delta = -113, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& \overset{3131}{1\ 6\ 13141} \text{ [#=7]} \quad \Delta = -113, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{413161314}$

$$\begin{aligned}
& \overset{3}{} \overset{1}{} \quad \Delta = -141, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 4131\overset{21}{6\ 1314} \quad \Delta = -133, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 4131\overset{21}{6\ 1314} \quad \Delta = -133, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 4131\overset{2}{} \overset{1}{} \quad \Delta = -133, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{31}{12321613}$

$$\begin{aligned}
& \overset{31}{31\ 6\ 12321} \overset{8}{} \quad \Delta = -221, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& \overset{31}{31\ 6\ 12321} \overset{3}{} \text{ [#=8]} \quad \Delta = -96, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 256 \quad \begin{pmatrix} 0 & 16 & 2 \\ 16 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
31 \overset{31}{6} 12321 \overset{31}{3} [\# = 3] \quad \Delta = -96, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
31 \overset{31}{6} 12321 \overset{31}{3} [\# = 4] \quad \Delta = -96, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
31 \overset{31}{6} 12321 \overset{31}{3} [\# = 2] \quad \Delta = -96, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
31 \overset{31}{6} 12321 \overset{31}{3} \quad \Delta = -96, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
31 \overset{31}{6} 12321 \overset{31}{3} [\# = 9] \quad \Delta = -96, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 324 \quad & \begin{pmatrix} 0 & 18 & 2 \\ 18 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
31 \overset{31}{6} 12321 \overset{31}{6} \quad \Delta = -166, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
31 \overset{31}{6} 12321 \overset{31}{3} [\# = 5] \quad \Delta = -96, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 100 \quad & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
31 \overset{31}{6} 12321 \overset{31}{3} [\# = 7] \quad \Delta = -96, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 196 \quad & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
31 \overset{31}{6} 12321 \overset{31}{3} [\# = 6] \quad \Delta = -96, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 144 \quad & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
31 \overset{31}{6} 12321 \overset{31}{4} \quad \Delta = -116, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
31 \overset{31}{6} 12321 \overset{31}{7} \quad \Delta = -193, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 8 \end{pmatrix} \\
12321 \overset{231}{6} 13 \quad \Delta = -88, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
31 \overset{31}{6} 12321 \overset{31}{2} \quad \Delta = -88, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
12321 \overset{312}{6} 13 \quad \Delta = -88, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\text{Base: } 21 \overset{15131}{6} 13 & \\
\overset{315131}{21} \overset{315131}{6} 13 \quad \Delta = -137, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
21 \overset{151312}{6} 13 \quad \Delta = -129, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\overset{215131}{21} \overset{215131}{6} 13 \quad \Delta = -129, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21 \overset{15131}{6} 13 \quad \Delta = -129, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21 \overset{15131}{6} 13 \quad \Delta = -129, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: $^{31}31613141$

$$^{31}31613141^2 \quad \Delta = -113, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$^{31}316131414 \quad \Delta = -141, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$^{31}316131412 \quad \Delta = -113, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$^{31}31613141^3 \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 21 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$^{31}316131413_{[\# = 2]} \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$^{31}316131413_{[\# = 3]} \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$^{231}31613141 \quad \Delta = -113, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$^{31}316131413 \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$^{312}31613141 \quad \Delta = -113, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

Base: $^{31}41316131$

$$^{31}413161313_{[\# = 7]} \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 196 \quad \begin{pmatrix} 0 & 14 & 2 \\ 14 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$^{31}41316131 \quad \Delta = -113, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$^2_{31}41316131 \quad \Delta = -113, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$^{231}41316131 \quad \Delta = -113, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$^{31}413161312 \quad \Delta = -113, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$^{312}41316131 \quad \Delta = -113, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$^{31}413161316 \quad \Delta = -191, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 8 \end{pmatrix}$$

$$^3_{31}41316131 \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$^{31}413161315 \quad \Delta = -165, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$\begin{aligned}
& 4131\overset{31}{6}131\mathbf{3}_{[\#=2]} \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 4131\overset{31}{6}131\mathbf{4} \quad \Delta = -141, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& 4131\overset{31}{6}131\mathbf{3} \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 4131\overset{31}{6}131\mathbf{3}_{[\#=8]} \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 256 \quad \begin{pmatrix} 0 & 16 & 2 \\ 16 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
& 4131\overset{31}{6}131\mathbf{3}_{[\#=9]} \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 324 \quad \begin{pmatrix} 0 & 18 & 2 \\ 18 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
& 4131\overset{31}{6}131\mathbf{3}_{[\#=6]} \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& 4131\overset{31}{6}131\mathbf{3}_{[\#=5]} \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& 4131\overset{31}{6}131\mathbf{3}_{[\#=3]} \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 4131\overset{31}{6}131\mathbf{3}_{[\#=4]} \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \text{Base: } 131\overset{131}{6}131 \\
& 131\overset{131}{6}131\mathbf{3}_{[\#=7]} \quad \Delta = -93, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
& 131\overset{131}{6}131\mathbf{3}_{[\#=4]} \quad \Delta = -93, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 131\overset{131}{6}131\mathbf{3}_{[\#=6]} \quad \Delta = -93, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& 131\overset{131}{6}131\mathbf{3}_{[\#=5]} \quad \Delta = -93, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& 131\overset{131}{6}131\mathbf{3}_{[\#=9]} \quad \Delta = -93, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
& 131\overset{2131}{6}131 \quad \Delta = -85, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 131\overset{131}{6}131\mathbf{3}_{[\#=8]} \quad \Delta = -93, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& 131\overset{131}{6}131\mathbf{3}_{[\#=2]} \quad \Delta = -93, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 131\overset{131}{6}131\mathbf{3}_{[\#=3]} \quad \Delta = -93, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \text{Base: } 1\overset{1,215131}{6}1
\end{aligned}$$

$$1 \overset{1,215131}{6} 1 \quad \Delta = -129, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \overset{1,215131}{6} 1 \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{1,215131}{6} 1 \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{1,215131}{6} 1 \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: $1 \overset{1,15131}{6} 1$

$$1 \overset{1,15131}{6} 1 \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{1,15131}{6} 1 \quad \Delta = -121, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: $1 \overset{1,15131}{6} 13$

$$1 \overset{1,15131}{6} 13 \quad \Delta = -129, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$31 \overset{1,15131}{6} 13 \quad \Delta = -137, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$31 \overset{1,15131}{6} 12 \quad \Delta = -129, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{1,15131}{6} 13 \quad \Delta = -129, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{1,15131}{6} 13 \quad \Delta = -129, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 7122314132

$$7122314132 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$7122314132 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$7122314132 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: 7123141322

$$7123141322 \quad \Delta = -136, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$7123141322 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$7123141322 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$7123141322 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

Base: 3141322171

$$3141322171 \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$3141322171 \quad \Delta = -141, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$3141322171 \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$3141322171 \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: 2314132171

$$2314132171 \quad \Delta = -136, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$2314132171 \quad \Delta = -136, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$2314132171 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$2314132171 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$2314132171 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: 1413221712

$$1413221712 \quad \Delta = -125, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1413221712 \quad \Delta = -125, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1413221712 \quad \Delta = -125, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$21712231413_{[\# = 2]} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$21712231413_{[\# = 3]} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 3141321712

$$3141321712 \quad \Delta = -135, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$3141321712 \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$3141321712 \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$3141321712^2 \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$3141321712^2 \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$3141321712^3 \quad \Delta = -141, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 4132217123

$$4132217123^2 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$4132217123^2 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$4132217123^2 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 1413217122

$$1413217122^2 \quad \Delta = -125, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1413217122^2 \quad \Delta = -125, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1413217122^2 \quad \Delta = -125, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1413217122^3 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$1413217122^2 \quad \Delta = -127, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$22171231413_{[\# = 2]} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$22171231413_{[\# = 3]} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 4123217122

$$4123217122^2 \quad \Delta = -127, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$4123217122^2 \quad \Delta = -127, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$4123217122^3 \quad \Delta = -135, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$4123217122^3 \quad \Delta = -136, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$4123217122^2 \quad \Delta = -127, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 1321712231

$$\begin{aligned}
1321712231\overset{2}{3}_{[\# = 3]} \quad \Delta = -114, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1321712231\overset{2}{3} \quad \Delta = -106, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1321712231\overset{2}{3} \quad \Delta = -106, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1321712231\overset{2}{3}_{[\# = 5]} \quad \Delta = -114, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1322171231\overset{2}{3}_{[\# = 5]} \quad \Delta = -114, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1321712231\overset{2}{3}_{[\# = 4]} \quad \Delta = -114, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1322171231\overset{2}{3}_{[\# = 4]} \quad \Delta = -114, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1322171231\overset{2}{3}_{[\# = 3]} \quad \Delta = -114, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1321712231\overset{2}{3}_{[\# = 2]} \quad \Delta = -114, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
1322171231\overset{2}{3}_{[\# = 2]} \quad \Delta = -114, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\text{Base: } 2221712321 & \\
2221712321\overset{2}{3}_{[\# = 3]} \quad \Delta = -107, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2221712321\overset{2}{3}_{[\# = 7]} \quad \Delta = -107, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
2221712321\overset{2}{3}_{[\# = 8]} \quad \Delta = -107, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
2221712321\overset{2}{3}_{[\# = 9]} \quad \Delta = -107, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
2221712321\overset{2}{3}_{[\# = 2]} \quad \Delta = -107, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2221712321\overset{2}{3}_{[\# = 6]} \quad \Delta = -107, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
2221712321\overset{2}{3}_{[\# = 5]} \quad \Delta = -107, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
2221712321\overset{2}{3} \quad \Delta = -99, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2221712321\overset{2}{3} \quad \Delta = -99, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

$$2221712321\overset{3}{\color{red}}_{[\# = 4]} \quad \Delta = -107, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 4132171223

$$4132171223 \overset{2}{\color{red}} \quad \Delta = -136, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$4132171223 \overset{2}{\color{red}} \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$4132171223 \overset{2}{\color{red}} \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$4132171223 \overset{2}{\color{red}} \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 1321712313

$$1321712313 \overset{3}{\color{red}} \quad \Delta = -122, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1321712313 \overset{2}{\color{red}} \quad \Delta = -114, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1321712313 \overset{2}{\color{red}} \quad \Delta = -114, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1321712313 \overset{2}{\color{red}} \quad \Delta = -114, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$3132171231\overset{3}{\color{red}}_{[\# = 2]} \quad \Delta = -122, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$3132171231\overset{3}{\color{red}}_{[\# = 5]} \quad \Delta = -122, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1321712313 \overset{3}{\color{red}}_{[\# = 2]} \quad \Delta = -122, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$3132171231\overset{3}{\color{red}}_{[\# = 3]} \quad \Delta = -122, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$3132171231\overset{3}{\color{red}}_{[\# = 4]} \quad \Delta = -122, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1321712313 \overset{2}{\color{red}} \quad \Delta = -116, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: 132171232¹

$$132171232 \overset{41}{\color{red}} \quad \Delta = -136, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$132171232 \overset{31}{\color{red}}_{[\# = 3]} \quad \Delta = -116, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$2 \overset{31321712}{\color{red}} 3 \overset{1}{\color{red}}_{[\# = 2]} \quad \Delta = -116, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1321712\overset{31}{3}2_{[\# = 2]} \quad \Delta = -116, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$2\overset{31321712}{3}1_{[\# = 3]} \quad \Delta = -116, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$2\overset{31321712}{3}1_{[\# = 4]} \quad \Delta = -116, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$2\overset{31321712}{3}1_{[\# = 5]} \quad \Delta = -116, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1321712\overset{2}{3}\overset{1}{2}2 \quad \Delta = -108, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1321712\overset{21}{3}2 \quad \Delta = -108, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1321712\overset{2}{3}\overset{1}{2}2 \quad \Delta = -108, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: $\overset{1}{171223141}$

$$\overset{1}{171223141}\overset{2}{41} \quad \Delta = -125, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\overset{12}{171223141} \quad \Delta = -125, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$\overset{1}{171223141}\overset{3}{3}_{[\# = 3]} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$\overset{1}{171223141}\overset{3}{3}_{[\# = 2]} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

Base: $\overset{1}{171231413}$

$$\overset{1}{171231413}\overset{2}{413} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\overset{1}{171231413}\overset{2}{413} \quad \Delta = -135, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\overset{12}{171231413} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$\overset{1}{171231413}\overset{3}{413} \quad \Delta = -141, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$\overset{1}{171231413}\overset{2}{413} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: $\overset{21}{17123141}$

$$\overset{21}{17123141} \quad \Delta = -125, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\overset{21}{17123141}\overset{2}{41} \quad \Delta = -127, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\begin{aligned}
1 \overset{21}{7} 123141 \overset{2}{} \quad \Delta = -125, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \overset{21}{7} 123141 \overset{2}{} \quad \Delta = -125, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1 \overset{21}{7} 123141 \overset{3}{}_{[\# = 3]} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
1 \overset{21}{7} 123141 \overset{3}{}_{[\# = 2]} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}
\end{aligned}$$

Base: $412321 \overset{21}{7} 1$

$$\begin{aligned}
412321 \overset{21}{7} 1 \overset{2}{} \quad \Delta = -127, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
412321 \overset{21}{7} 1 \quad \Delta = -127, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
412321 \overset{21}{7} 1 \overset{2}{} \quad \Delta = -127, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
412321 \overset{21}{7} 1 \overset{3}{} \quad \Delta = -135, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{12321}{7} 122$

$$\begin{aligned}
1 \overset{312321}{7} 122_{[\# = 6]} \quad \Delta = -107, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \overset{12321}{7} 122 \overset{2}{} \quad \Delta = -99, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \overset{12321}{7} 122 \quad \Delta = -99, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1 \overset{12321}{7} 122 \overset{3}{} \quad \Delta = -108, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1 \overset{312321}{7} 122_{[\# = 4]} \quad \Delta = -107, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \overset{312321}{7} 122_{[\# = 8]} \quad \Delta = -107, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1 \overset{312321}{7} 122_{[\# = 9]} \quad \Delta = -107, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1 \overset{312321}{7} 122_{[\# = 7]} \quad \Delta = -107, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1 \overset{312321}{7} 122_{[\# = 3]} \quad \Delta = -107, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1 \overset{312321}{7} 122_{[\# = 2]} \quad \Delta = -107, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

$$1 \overset{312321}{7} 122 \text{ [#}=5] \quad \Delta = -107, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

Base: $41321 \overset{321}{7} 1$

$$41321 \overset{321}{7} 1 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$41321 \overset{321}{7} 1 \overset{2}{2} \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$41321 \overset{2}{7} \overset{321}{1} \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$41321 \overset{2321}{7} 1 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$41321 \overset{2}{7} \overset{321}{1} \quad \Delta = -136, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: $1 \overset{1321}{7} 1231$

$$1 \overset{1321}{7} 1231 \overset{3}{3} \text{ [#}=5] \quad \Delta = -114, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1 \overset{1321}{7} 1231 \overset{2}{2} \quad \Delta = -106, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{1321}{7} 1231 \overset{3}{3} \text{ [#}=4] \quad \Delta = -114, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1 \overset{1321}{7} 1231 \overset{3}{3} \text{ [#}=3] \quad \Delta = -114, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{1321}{7} 1231 \overset{3}{3} \text{ [#}=2] \quad \Delta = -114, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: $21 \overset{12321}{7} 12$

$$21 \overset{212321}{7} 12 \quad \Delta = -99, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$21 \overset{12321}{7} 12 \quad \Delta = -99, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$21 \overset{312321}{7} 12 \text{ [#}=5] \quad \Delta = -107, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$21 \overset{312321}{7} 12 \text{ [#}=7] \quad \Delta = -107, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$21 \overset{312321}{7} 12 \text{ [#}=6] \quad \Delta = -107, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$21 \overset{312321}{7} 12 \text{ [#}=8] \quad \Delta = -107, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$21 \overset{312321}{7} 12 \text{ [#}=3] \quad \Delta = -107, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\begin{aligned}
21 \overset{3}{7} 12 \text{ }_{[\# = 4]} \quad \Delta = -107, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21 \overset{3}{7} 12 \text{ }_{[\# = 2]} \quad \Delta = -107, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21 \overset{3}{7} 12 \text{ }_{[\# = 9]} \quad \Delta = -107, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}
\end{aligned}$$

Base: $412321 \overset{1,1}{7} 1$

$$\begin{aligned}
412321 \overset{2,1,1}{7} 1 \quad \Delta = -127, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
412321 \overset{3}{7} 1 \text{ }_{[\# = 1]} \quad \Delta = -135, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
412321 \overset{2}{7} 1 \text{ }_{[\# = 1]} \quad \Delta = -127, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{21,12321}{7} 1$

$$\begin{aligned}
1 \overset{21,12321}{7} 1 \quad \Delta = -99, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \overset{21,12321}{7} 1 \quad \Delta = -99, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1 \overset{21,3}{7} 12321 \text{ }_{[\# = 3]} \quad \Delta = -107, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \overset{21,3}{7} 12321 \text{ }_{[\# = 2]} \quad \Delta = -107, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \overset{21,3}{7} 12321 \text{ }_{[\# = 4]} \quad \Delta = -107, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1 \overset{21,3}{7} 12321 \text{ }_{[\# = 5]} \quad \Delta = -107, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1 \overset{21,3}{7} 12321 \text{ }_{[\# = 6]} \quad \Delta = -107, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1 \overset{21,3}{7} 12321 \text{ }_{[\# = 9]} \quad \Delta = -107, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1 \overset{21,3}{7} 12321 \text{ }_{[\# = 7]} \quad \Delta = -107, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1 \overset{21,3}{7} 12321 \text{ }_{[\# = 8]} \quad \Delta = -107, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1812231322

$$\begin{aligned}
1812231322 \overset{3}{7} \quad \Delta = -142, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 32 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1812231322 \overset{2}{7} \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

$$1812231\overset{2}{3}22 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: 2181223132

$$21812231\overset{3}{2} \quad \Delta = -142, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$21812231\overset{2}{2} \quad \Delta = -136, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$2181\overset{2}{2}23132 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$21812231\overset{2}{2} \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$2181\overset{2}{2}23132 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 2181231322

$$2181\overset{2}{2}31322 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$2181231\overset{2}{3}22 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$2181\overset{2}{2}31322 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$2181231\overset{3}{2}22 \quad \Delta = -142, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 32 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$2181\overset{2}{2}31322 \quad \Delta = -136, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 2218122313

$$221812231\overset{3}{3} \quad \Delta = -141, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$221812231\overset{3}{3} \quad \Delta = -141, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$221812231\overset{2}{3} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$22181\overset{2}{2}2313 \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$22181\overset{3}{2}2313 \quad \Delta = -142, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$22181\overset{2}{2}2313 \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 2221812231

$$222181\overset{2}{2}231 \quad \Delta = -125, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\begin{aligned}
2221812231^2 \quad \Delta = -125, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2221812231^3_{[\# = 5]} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2221812231^3_{[\# = 4]} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2221812231^3_{[\# = 3]} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2221812231^3_{[\# = 2]} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}
\end{aligned}$$

Base: 2218123132

$$\begin{aligned}
2218123132^2 \quad \Delta = -136, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2218123132^3 \quad \Delta = -143, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2218123132^2 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2218123132^2 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
2218123132^2 \quad \Delta = -136, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2218123132^2 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2218123132^3 \quad \Delta = -142, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 222181231

$$\begin{aligned}
2222181231^2 \quad \Delta = -125, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2222181231^3_{[\# = 2]} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
2222181231^3_{[\# = 3]} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2222181231^3_{[\# = 5]} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2222181231^3_{[\# = 4]} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2222181231^2 \quad \Delta = -125, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 2221812313

$$\begin{aligned}
2221812\overset{2}{3}13 \quad \Delta = -135, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2221812\overset{2}{3}13 \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2221812\overset{2}{3}13 \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2221812\overset{2}{3}13 \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2221812\overset{3}{3}13 \quad \Delta = -141, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2221812\overset{3}{3}13_{[\# = 2]} \quad \Delta = -141, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: $2221812\overset{1}{3}2$

$$\begin{aligned}
2221812\overset{2}{3}\overset{1}{2} \quad \Delta = -127, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2221812\overset{2}{3}\overset{1}{2} \quad \Delta = -127, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2221812\overset{41}{3}2 \quad \Delta = -155, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
2221812\overset{31}{3}2_{[\# = 2]} \quad \Delta = -135, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2221812\overset{31}{3}2_{[\# = 3]} \quad \Delta = -135, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2221812\overset{21}{3}2 \quad \Delta = -127, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1^{\overset{2313221}{8}}1$

$$\begin{aligned}
1^{\overset{2313221}{8}}1 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1^{\overset{2313221}{8}}1 \quad \Delta = -142, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1^{\overset{2313221}{8}}1 \quad \Delta = -136, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1^{\overset{2313221}{8}}1 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1^{\overset{2231321}{8}}1$

$$\begin{aligned}
1^{\overset{2231321}{8}}1 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1^{\overset{2231321}{8}}1 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{2231321}{1 \ 8 \ 1} \quad \Delta = -142, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 32 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{2231321}{1 \ 8 \ 1} \quad \Delta = -136, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{313221}{1 \ 8 \ 12}$

$$\begin{aligned}
& \overset{313221}{1 \ 8 \ 12} \text{ [#=2]} \quad \Delta = -141, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{313221}{1 \ 8 \ 12} \quad \Delta = -141, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{313221}{1 \ 8 \ 12} \quad \Delta = -133, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{313221 \textcolor{red}{2}}{1 \ 8 \ 12} \quad \Delta = -133, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{313221}{1 \ 8 \ 12} \quad \Delta = -133, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{231321}{1 \ 8 \ 12}$

$$\begin{aligned}
& \overset{231321}{1 \ 8 \ 12} \quad \Delta = -136, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{231321}{1 \ 8 \ 12} \quad \Delta = -142, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{231321}{1 \ 8 \ 12} \quad \Delta = -136, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{231321}{1 \ 8 \ 12} \quad \Delta = -134, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{231321}{1 \ 8 \ 12} \quad \Delta = -134, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{231321 \textcolor{red}{2}}{1 \ 8 \ 12} \quad \Delta = -134, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{13221}{1 \ 8 \ 122}$

$$\begin{aligned}
& \overset{13221}{1 \ 8 \ 122} \quad \Delta = -125, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{13221 \textcolor{red}{2}}{1 \ 8 \ 122} \quad \Delta = -125, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{13221 \textcolor{red}{3}}{1 \ 8 \ 122} \quad \Delta = -134, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{\textcolor{red}{3}13221}{1 \ 8 \ 122} \text{ [#=2]} \quad \Delta = -133, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
& \overset{\textcolor{red}{3}13221}{1 \ 8 \ 122} \text{ [#=3]} \quad \Delta = -133, \ n_+ = 1, \ n_- = 10, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

$$1 \overset{313221}{8} 122 \text{ [#}=4] \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{313221}{8} 122 \text{ [#}=5] \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{3221}{8} 1223$$

$$3221 \overset{3221}{8} 1 \overset{2}{2} \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{3221}{8} 1223 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{31321}{8} 122$$

$$1 \overset{31321}{8} 122 \quad \Delta = -135, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1 \overset{31321}{8} 122 \text{ [#}=2] \quad \Delta = -141, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \overset{31321}{8} 122 \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{31321}{8} 122 \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{31321}{8} 122 \quad \Delta = -141, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1 \overset{31321}{8} 122 \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{31321}{8} 122 \overset{3}{2} \quad \Delta = -142, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{1321}{8} 1222$$

$$1 \overset{31321}{8} 1222 \text{ [#}=3] \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1 \overset{31321}{8} 1222 \text{ [#}=4] \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{31321}{8} 1222 \text{ [#}=5] \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{1321}{8} 1222 \quad \Delta = -125, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{1321}{8} 1222 \quad \Delta = -125, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{31321}{8} 1222 \text{ [#}=2] \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{2321}{8} 122$$

$$\begin{aligned}
& \overset{2321}{1} \overset{2}{8} 122 \quad \Delta = -127, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{2321}{1} \overset{2321}{8} 122 \text{ [#=3]} \quad \Delta = -135, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{2321}{1} \overset{2321}{8} 122 \text{ [#=2]} \quad \Delta = -135, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{2321}{1} \overset{2321}{8} 122 \quad \Delta = -127, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{2321}{1} \overset{2321}{8} 122 \quad \Delta = -127, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{2321}{1} \overset{2321}{8} 122 \quad \Delta = -155, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
& \overset{2321}{1} \overset{2321}{8} \overset{3}{122} \quad \Delta = -136, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{13221}{21} \overset{13221}{8} 12$

$$\begin{aligned}
& \overset{313221}{21} \overset{313221}{8} 12 \text{ [#=3]} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{13221}{21} \overset{13221}{8} 12 \quad \Delta = -125, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{313221}{21} \overset{313221}{8} 12 \text{ [#=2]} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
& \overset{213221}{21} \overset{213221}{8} 12 \quad \Delta = -125, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{313221}{21} \overset{313221}{8} 12 \text{ [#=4]} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{313221}{21} \overset{313221}{8} 12 \text{ [#=5]} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{31321}{21} \overset{31321}{8} 12$

$$\begin{aligned}
& \overset{31321}{21} \overset{31321}{8} 12 \quad \Delta = -141, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{31321}{21} \overset{31321}{8} 12 \quad \Delta = -135, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{31321}{21} \overset{31321}{8} 12 \text{ [#=2]} \quad \Delta = -141, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{231321}{21} \overset{231321}{8} 12 \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{31321}{21} \overset{31321}{8} 12 \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

$$21 \overset{31321}{8} 12 \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: $21 \overset{2321}{8} 12$

$$21 \overset{2321}{8} 12 \text{ [#=3]} \quad \Delta = -135, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$21 \overset{2321}{8} 12 \quad \Delta = -127, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$21 \overset{2321}{8} 12 \quad \Delta = -155, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$21 \overset{2321}{8} 12 \quad \Delta = -127, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$21 \overset{2321}{8} 12 \quad \Delta = -127, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$21 \overset{2321}{8} 12 \text{ [#=2]} \quad \Delta = -135, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: $21 \overset{321}{8} 1223$

$$3221 \overset{321}{8} 12\textcolor{red}{3} \quad \Delta = -143, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$21 \overset{321}{8} 1223 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$21 \overset{2321}{8} 1223 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$21 \overset{321\textcolor{red}{2}}{8} 1223 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: $21 \overset{1321}{8} 122$

$$21 \overset{31321}{8} 122 \text{ [#=3]} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$21 \overset{1321\textcolor{red}{2}}{8} 122 \quad \Delta = -125, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$21 \overset{31321}{8} 122 \text{ [#=2]} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$21 \overset{31321}{8} 122 \text{ [#=5]} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$21 \overset{31321}{8} 122 \text{ [#=4]} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$21 \overset{21321}{8} 122 \quad \Delta = -125, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$21 \overset{1321}{8} 122 \quad \Delta = -125, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$${}^{1321}_{21} \overset{3}{8} 122 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: ${}^{321}_{221} 8 123$

$$\overset{3}{221} \overset{321}{8} 123 \quad \Delta = -143, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$${}^{2321}_{221} 8 123 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$${}^{3212}_{221} 8 123 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: ${}^{1,313221}_{1^8} 1$

$${}^{1,313221}_{1^8} 1 \text{ [#=2]} \quad \Delta = -141, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$${}^{1,313221}_{1^8} 1 \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$${}^{1,313221}_{1^8} 1 \quad \Delta = -141, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$${}^{1,313221}_{1^8} 1 \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: ${}^{1,231321}_{1^8} 1$

$${}^{1,231321}_{1^8} 1 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$${}^{1,231321}_{1^8} 1 \quad \Delta = -142, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 32 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$${}^{1,231321}_{1^8} 1 \quad \Delta = -136, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$${}^{1,231321}_{1^8} 1 \quad \Delta = -136, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$${}^{1,231321}_{1^8} 1 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: ${}^{21,13221}_{1^8} 1$

$${}^{21,13221}_{1^8} 1 \quad \Delta = -125, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$${}^{21,13221}_{1^8} 1 \quad \Delta = -125, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$${}^{21,313221}_{1^8} 1 \text{ [#=4]} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$${}^{21,313221}_{1^8} 1 \text{ [#=5]} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1 \overset{21, \textcolor{red}{3} 13221}{8} 1 \text{ } [\# = 3] \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{21, \textcolor{red}{3} 13221}{8} 1 \text{ } [\# = 2] \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{21, 31321}{8} 1$$

$$1 \overset{21, 31321}{8} 1 \quad \Delta = -135, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{21, 31321}{8} 1 \quad \Delta = -141, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{21, 31321}{8} 1 \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1 \overset{21, 31321}{8} 1 \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{21, 31321}{8} 1 \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{21, 31321}{8} 1 \text{ } [\# = 2] \quad \Delta = -141, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{21, 2321}{8} 1$$

$$1 \overset{21, 2321}{8} 1 \quad \Delta = -127, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{21, 2321}{8} 1 \quad \Delta = -155, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{21, 2321}{8} 1 \text{ } [\# = 2] \quad \Delta = -135, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1 \overset{21, 2321}{8} 1 \text{ } [\# = 3] \quad \Delta = -135, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 60 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1 \overset{21, 2321}{8} 1 \quad \Delta = -127, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \overset{21, 2321}{8} 1 \quad \Delta = -127, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{1, 321}{8} 1223$$

$$1 \overset{1, 321}{8} 1223 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$3221 \overset{1, 321}{8} 1 \textcolor{red}{2} \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{1, 321 \textcolor{red}{2}}{8} 1223 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{221, 1321}{8} 1$$

$$\begin{aligned}
1 \overset{221,1321}{8} 1 \quad \Delta = -125, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \overset{221,1321}{8} 1 \quad \Delta = -125, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1 \overset{221,1321}{8} 1 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
1 \overset{221,\textcolor{red}{3}1321}{8} 1 \text{ [#=2]} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1 \overset{221,\textcolor{red}{3}1321}{8} 1 \text{ [#=3]} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{221,\textcolor{red}{3}1321}{8} 1 \text{ [#=4]} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \overset{221,\textcolor{red}{3}1321}{8} 1 \text{ [#=5]} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{21,1321}{8} 12$

$$\begin{aligned}
1 \overset{21,1321}{8} 12 \quad \Delta = -125, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \overset{21,\textcolor{red}{3}1321}{8} 12 \text{ [#=2]} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1 \overset{21,\textcolor{red}{3}1321}{8} 12 \text{ [#=4]} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \overset{21,\textcolor{red}{3}1321}{8} 12 \text{ [#=5]} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1 \overset{21,\textcolor{red}{3}1321}{8} 12 \text{ [#=3]} \quad \Delta = -133, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{21,1321\textcolor{red}{2}}{8} 12 \quad \Delta = -125, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{21,321}{8} 123$

$$\begin{aligned}
1 \overset{21,321}{8} 123 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
321 \overset{21,321}{8} \textcolor{red}{12} \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1 \overset{\textcolor{red}{3}21,321}{8} 123 \quad \Delta = -143, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1 \overset{21,321\textcolor{red}{2}}{8} 123 \quad \Delta = -134, n_+ = 1, n_- = 10, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

$T = 11$ (1218 bases)

Base: $12222222\overset{2}{2}$

$${}^2_{2222222221\textcolor{red}{12}} \quad \Delta = -202, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 8 \end{pmatrix}$$

$${}^2_{2222222221\textcolor{red}{5}} \quad \Delta = -6, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$${}^2_{2222222221\textcolor{red}{8}} \quad \Delta = -87, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$${}^2_{2222222221\textcolor{red}{7}} \quad \Delta = -59, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$${}^2_{2222222221\textcolor{red}{6}} \quad \Delta = -32, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 12231413221

$$12231413221\textcolor{red}{3} \text{ [#=7]} \quad \Delta = -8, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix}$$

$$12231413221\textcolor{red}{3} \text{ [#=8]} \quad \Delta = -8, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix}$$

$$12231413221\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -8, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$12231413221\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -8, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix}$$

$$12231413221\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -8, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix}$$

$$12231413221\textcolor{red}{3} \text{ [#=6]} \quad \Delta = -8, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

$$12231413221\textcolor{red}{2} \quad \Delta = 0, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$12231413221\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -8, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$12231413221\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -8, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

Base: 13141413221

$$13141413221\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -35, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix}$$

$$13141413221\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -35, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$13141413221\textcolor{red}{3} \quad \Delta = -35, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$12231414131\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -35, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$\begin{aligned}
13141413221\overset{2}{3} \text{ }_{[\# = 8]} \quad \Delta = -35, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
12231414131\overset{2}{3} \text{ }_{[\# = 8]} \quad \Delta = -35, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
13141413221\overset{2}{3} \text{ }_{[\# = 7]} \quad \Delta = -35, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
12231414131\overset{2}{3} \text{ }_{[\# = 7]} \quad \Delta = -35, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
13141413221\overset{2}{3} \text{ }_{[\# = 6]} \quad \Delta = -35, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
12231414131\overset{2}{3} \text{ }_{[\# = 6]} \quad \Delta = -35, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
13141413221\overset{2}{3} \text{ }_{[\# = 5]} \quad \Delta = -35, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
12231414131\overset{2}{3} \text{ }_{[\# = 5]} \quad \Delta = -35, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
13141413221\overset{2}{3} \text{ }_{[\# = 4]} \quad \Delta = -35, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
12231414131\overset{2}{3} \text{ }_{[\# = 4]} \quad \Delta = -35, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
13141413221\overset{2}{3} \text{ }_{[\# = 3]} \quad \Delta = -35, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
12231414131\overset{2}{3} \text{ }_{[\# = 3]} \quad \Delta = -35, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
12231414131\overset{2}{3} \text{ }_{[\# = 2]} \quad \Delta = -35, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
13141413221\overset{2}{2} \quad \Delta = -27, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
13141413221\overset{2}{2} \quad \Delta = -27, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
13141413221\overset{2}{2} \quad \Delta = -27, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 12314141321

$$\begin{aligned}
12314141321\overset{2}{3} \text{ }_{[\# = 7]} \quad \Delta = -36, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
12314141321\overset{2}{3} \text{ }_{[\# = 6]} \quad \Delta = -36, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
12314141321\overset{2}{3} \text{ }_{[\# = 8]} \quad \Delta = -36, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
12314141321\overset{1}{3} \text{ [#=5]} \quad \Delta = -36, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
12314141321\overset{1}{3} \text{ [#=4]} \quad \Delta = -36, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
12314141321\overset{1}{3} \text{ [#=9]} \quad \Delta = -36, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
12314141321\overset{1}{3} \text{ [#=3]} \quad \Delta = -36, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
12314141321\overset{1}{3} \text{ [#=2]} \quad \Delta = -36, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
12314141321\overset{2}{2} \quad \Delta = -28, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
12314141321\overset{2}{2} \quad \Delta = -28, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\text{Base: } 12314141\overset{2}{3}2 & \\
2\overset{5}{3}2 \quad \Delta = -82, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2\overset{3}{3}2 \text{ [#=3]} \quad \Delta = -38, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 60 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
2\overset{6}{3}2 \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2\overset{7}{3}2 \quad \Delta = -135, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2\overset{8}{3}2 \quad \Delta = -163, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
2\overset{4}{3}2 \quad \Delta = -58, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2\overset{3}{3}2 \text{ [#=2]} \quad \Delta = -38, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
2\overset{3}{3}2 \text{ [#=6]} \quad \Delta = -38, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 144 \quad & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 249 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
2\overset{3}{3}2 \text{ [#=7]} \quad \Delta = -38, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 196 \quad & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 340 & 48 \\ 2 & 48 & 8 \end{pmatrix} \\
2\overset{3}{3}2 \text{ [#=9]} \quad \Delta = -38, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 324 \quad & \begin{pmatrix} 0 & 18 & 2 \\ 18 & 564 & 62 \\ 2 & 62 & 8 \end{pmatrix} \\
2\overset{3}{3}2 \text{ [#=5]} \quad \Delta = -38, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 100 \quad & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 172 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
2\overset{3}{3}2 \text{ [#=8]} \quad \Delta = -38, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 256 \quad & \begin{pmatrix} 0 & 16 & 2 \\ 16 & 445 & 55 \\ 2 & 55 & 8 \end{pmatrix}
\end{aligned}$$

$$2^{\overset{3}{1}2314141} \quad 2^{\overset{2}{3}} \quad 2 \quad [\#=4] \quad \Delta = -38, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 109 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$2^{\overset{3}{1}2314141} \quad 2^{\overset{2}{3}} \quad 2 \quad \Delta = -38, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$12314141\overset{2}{3}2 \quad \Delta = -30, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$2^{\overset{2}{1}2314141} \quad 2^{\overset{2}{3}} \quad 2 \quad \Delta = -30, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$12314141\overset{2}{3}2 \quad \Delta = -30, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$12314141\overset{2}{3}2 \quad \Delta = -30, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$12314141\overset{2}{3}2 \quad \Delta = -32, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: 21414141321

$$21414141321\overset{3}{3} \quad [\#=3] \quad \Delta = -55, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$21414141321\overset{3}{3} \quad [\#=2] \quad \Delta = -55, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$21414141321\overset{3}{3} \quad [\#=7] \quad \Delta = -55, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$21414141321\overset{3}{3} \quad [\#=6] \quad \Delta = -55, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$21414141321\overset{3}{3} \quad [\#=5] \quad \Delta = -55, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$21414141321\overset{3}{3} \quad [\#=4] \quad \Delta = -55, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$21414141321 \quad [\#=3] \quad \Delta = -55, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 123 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$21414141321\overset{3}{3} \quad [\#=8] \quad \Delta = -55, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$21414141321\overset{3}{3} \quad [\#=9] \quad \Delta = -55, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$21414141321 \quad \Delta = -55, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$21414141321 \quad \Delta = -75, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$21414141321 \quad [\#=2] \quad \Delta = -55, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$21414141321 \quad \Delta = -47, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$21414141321 \quad \Delta = -47, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$21414141321 \quad \Delta = -47, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$21414141321 \quad \Delta = -47, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: 13141414131

$$13141414131 \quad \Delta = -54, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$13141414131\mathbf{3} \text{ [#=5]} \quad \Delta = -62, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix}$$

$$13141414131\mathbf{3} \text{ [#=9]} \quad \Delta = -62, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix}$$

$$13141414131\mathbf{3} \text{ [#=8]} \quad \Delta = -62, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix}$$

$$13141414131\mathbf{3} \text{ [#=6]} \quad \Delta = -62, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

$$13141414131\mathbf{3} \text{ [#=3]} \quad \Delta = -62, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$13141414131\mathbf{3} \text{ [#=7]} \quad \Delta = -62, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix}$$

$$13141414131 \quad \Delta = -62, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$13141414131\mathbf{3} \text{ [#=2]} \quad \Delta = -62, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$13141414131\mathbf{3} \text{ [#=4]} \quad \Delta = -62, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$13141414131 \quad \Delta = -54, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 2141414132

$$2^{\mathbf{3}21414141} \quad \Delta = -58, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$2141414132 \quad \Delta = -49, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$2141414132 \quad \Delta = -49, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\begin{aligned}
\overset{4}{2}141414132^2 \quad \Delta = -77, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\overset{3}{2}141414132^2 \quad \Delta = -57, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\overset{3}{2}141414132^2_{[\# = 2]} \quad \Delta = -57, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 85 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
\overset{3}{2}141414132^2_{[\# = 3]} \quad \Delta = -57, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 144 & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 195 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
\overset{2}{2}141414132^2 \quad \Delta = -49, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 20 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
2141414132^{\overset{2}{2}} \quad \Delta = -49, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 21414141412

$$\begin{aligned}
\overset{4}{2}1414141412 \quad \Delta = -94, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 14 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\overset{3}{2}1414141412_{[\# = 2]} \quad \Delta = -74, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 69 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
\overset{3}{2}1414141412_{[\# = 3]} \quad \Delta = -74, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 159 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
\overset{3}{2}1414141412 \quad \Delta = -74, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 15 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\overset{2}{2}1414141412 \quad \Delta = -66, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\overset{2}{2}1414141412 \quad \Delta = -66, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21414141412^{\overset{2}{2}} \quad \Delta = -66, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{1}414141321$

$$\begin{aligned}
\overset{1}{1}414141321\overset{3}{3}_{[\# = 4]} \quad \Delta = -55, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\overset{1}{1}414141321\overset{3}{3}_{[\# = 3]} \quad \Delta = -55, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\overset{31}{1}414141321_{[\# = 3]} \quad \Delta = -55, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\overset{1}{1}414141321\overset{3}{3}_{[\# = 6]} \quad \Delta = -55, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\overset{1}{1}414141321\overset{3}{3}_{[\# = 2]} \quad \Delta = -55, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{1}{1414141321} \overset{3}{\text{red}}_{[\# = 7]} \quad \Delta = -55, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& \overset{31}{1414141321}_{[\# = 2]} \quad \Delta = -55, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{1}{1414141321} \overset{3}{\text{red}}_{[\# = 5]} \quad \Delta = -55, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{1}{1414141321} \overset{3}{\text{red}}_{[\# = 8]} \quad \Delta = -55, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{1}{1414141321} \overset{3}{\text{red}}_{[\# = 9]} \quad \Delta = -55, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{1}{1414141321} \overset{2}{\text{red}} \quad \Delta = -47, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{12}{1414141321} \quad \Delta = -47, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{1}{1414141321} \overset{2}{\text{red}} \quad \Delta = -47, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \text{Base: } \overset{1}{141414132} \overset{2}{\text{red}} \\
& \overset{1}{141414132} \overset{22}{\text{red}} \quad \Delta = -49, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{41}{141414132} \overset{2}{\text{red}} \quad \Delta = -77, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{31}{141414132} \overset{2}{\text{red}} \quad \Delta = -57, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{31}{141414132} \overset{2}{\text{red}}_{[\# = 2]} \quad \Delta = -57, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{31}{141414132} \overset{2}{\text{red}}_{[\# = 3]} \quad \Delta = -57, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{21}{141414132} \overset{2}{\text{red}} \quad \Delta = -49, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{1}{141414132} \overset{2}{\text{red}} \quad \Delta = -49, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{12}{141414132} \overset{2}{\text{red}} \quad \Delta = -49, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \text{Base: } \overset{1}{1414141412} \\
& \overset{1}{1414141412} \overset{2}{\text{red}} \quad \Delta = -66, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{1}{1414141412} \overset{4}{\text{red}} \quad \Delta = -94, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
{}^1_{1414141412} \quad \Delta = -66, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
{}^1_{1414141412} \quad \Delta = -66, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
{}^1_{1414141412} \quad \Delta = -74, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
{}^{31}_{1414141412} \quad [\# = 2] \quad \Delta = -74, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
{}^1_{1414141412} \quad \Delta = -74, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
{}^{31}_{1414141412} \quad [\# = 3] \quad \Delta = -74, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
{}^1_{1414141412} \quad \Delta = -74, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
{}^{12}_{1414141412} \quad \Delta = -66, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: ${}^1_{141414141}$

$$\begin{aligned}
{}^1_{141414141} \quad \Delta = -66, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
{}^{12}_{141414141} \quad \Delta = -66, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
{}^{31}_{141414141} \quad [\# = 2] \quad \Delta = -74, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
{}^{31}_{141414141} \quad [\# = 3] \quad \Delta = -74, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

Base: 12231512321

$$\begin{aligned}
{}^{2}_{12231512321} \quad \Delta = -26, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
{}^{2}_{12231512321} \quad \Delta = -26, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^{122315123213} \quad [\# = 6] \quad \Delta = -34, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
{}^{122315123213} \quad [\# = 5] \quad \Delta = -34, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
{}^{122315123213} \quad [\# = 8] \quad \Delta = -34, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
{}^{123215132213} \quad [\# = 8] \quad \Delta = -34, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 445 & 55 \\ 2 & 55 & 8 \end{pmatrix} \\
{}^{122315123213} \quad [\# = 3] \quad \Delta = -34, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
12321513221\bar{3} \text{ }_{[\# = 4]} \quad \Delta = -34, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 109 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
12231512321\bar{3} \text{ }_{[\# = 7]} \quad \Delta = -34, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
12321513221\bar{3} \text{ }_{[\# = 2]} \quad \Delta = -34, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 25 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
12321513221\bar{3} \text{ }_{[\# = 3]} \quad \Delta = -34, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 60 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
12321513221\bar{3} \text{ }_{[\# = 7]} \quad \Delta = -34, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 340 & 48 \\ 2 & 48 & 8 \end{pmatrix} \\
12231512321\bar{3} \text{ }_{[\# = 9]} \quad \Delta = -34, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
12231512321\bar{3} \text{ }_{[\# = 4]} \quad \Delta = -34, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
12321513221\bar{3} \text{ }_{[\# = 5]} \quad \Delta = -34, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 172 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
12321513221\bar{3} \text{ }_{[\# = 6]} \quad \Delta = -34, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 249 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
12321513221\bar{3} \text{ }_{[\# = 9]} \quad \Delta = -34, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 564 & 62 \\ 2 & 62 & 8 \end{pmatrix} \\
12231512321\bar{3} \text{ }_{[\# = 2]} \quad \Delta = -34, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}
\end{aligned}$$

Base: 14131513221

$$\begin{aligned}
14131513221\bar{3} \text{ }_{[\# = 5]} \quad \Delta = -59, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
12231513141\bar{3} \text{ }_{[\# = 3]} \quad \Delta = -59, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
14131513221\bar{3} \text{ }_{[\# = 6]} \quad \Delta = -59, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
14131513221\bar{3} \text{ }_{[\# = 7]} \quad \Delta = -59, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
141\bar{3}1513221 \quad \Delta = -52, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
141\bar{3}1513221 \quad \Delta = -51, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
14131513221\bar{3} \text{ }_{[\# = 8]} \quad \Delta = -59, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
14131513221\bar{3} \text{ }_{[\# = 9]} \quad \Delta = -59, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
14131513221^2 \quad \Delta = -51, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
14131513221^2 \quad \Delta = -51, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
14131513221^3_{[\# = 4]} \quad \Delta = -59, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
14131513221^3_{[\# = 3]} \quad \Delta = -59, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
14131513221^3_{[\# = 2]} \quad \Delta = -59, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
12231513141^3_{[\# = 2]} \quad \Delta = -59, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
14131513221^3 \quad \Delta = -59, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: 12315132215

$$\begin{aligned}
51223151321^3_{[\# = 4]} \quad \Delta = -84, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
12315132215^2 \quad \Delta = -76, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
12315132215^2 \quad \Delta = -76, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
12315132215^2 \quad \Delta = -76, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
51223151321^3_{[\# = 3]} \quad \Delta = -84, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
51223151321^3_{[\# = 2]} \quad \Delta = -84, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
51223151321^3_{[\# = 5]} \quad \Delta = -84, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
51223151321^3_{[\# = 6]} \quad \Delta = -84, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
51223151321^3_{[\# = 7]} \quad \Delta = -84, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
51223151321^3_{[\# = 9]} \quad \Delta = -84, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
12315132215^3 \quad \Delta = -84, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
51223151321^3_{[\# = 8]} \quad \Delta = -84, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}
\end{aligned}$$

Base: 41231513221

$$\begin{aligned}
41231513221^2 \quad \Delta = -52, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 37 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
41231513221^3_{[\# = 9]} \quad \Delta = -60, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
41231513221^2 \quad \Delta = -52, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 74 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
41231513221^2 \quad \Delta = -52, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
41231513221^3_{[\# = 5]} \quad \Delta = -60, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
41231513221^3_{[\# = 4]} \quad \Delta = -60, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
41231513221^3_{[\# = 8]} \quad \Delta = -60, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
41231513221^3_{[\# = 7]} \quad \Delta = -60, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
41231513221^3 \quad \Delta = -60, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 73 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
41231513221^3_{[\# = 2]} \quad \Delta = -60, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
41231513221^3_{[\# = 3]} \quad \Delta = -60, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
41231513221^3_{[\# = 6]} \quad \Delta = -60, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}
\end{aligned}$$

Base: 51231513215

$$\begin{aligned}
51231513215^2 \quad \Delta = -128, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
51231513215^2 \quad \Delta = -128, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: 12315132151

$$\begin{aligned}
12315132151^2 \quad \Delta = -76, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
12315132151^2 \quad \Delta = -76, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
12315132151^2 \quad \Delta = -76, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
15123151321^3_{[\# = 2]} \quad \Delta = -84, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
15123151321\bar{3} \text{ }_{[\# = 4]} \quad \Delta = -84, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
15123151321\bar{3} \text{ }_{[\# = 5]} \quad \Delta = -84, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
15123151321\bar{3} \text{ }_{[\# = 6]} \quad \Delta = -84, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
15123151321\bar{3} \text{ }_{[\# = 7]} \quad \Delta = -84, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
15123151321\bar{3} \text{ }_{[\# = 8]} \quad \Delta = -84, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
15123151321\bar{3} \text{ }_{[\# = 9]} \quad \Delta = -84, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
15123151321\bar{3} \text{ }_{[\# = 3]} \quad \Delta = -84, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

Base: 15131513215

$$\begin{aligned}
151\bar{3}1513215 \quad \Delta = -128, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
151\bar{3}1513215 \quad \Delta = -127, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
151315\bar{3}215 \quad \Delta = -127, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
151315\bar{3}215 \quad \Delta = -127, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
151315132\bar{5} \quad \Delta = -127, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: 12315131512

$$\begin{aligned}
21513151321\bar{3} \text{ }_{[\# = 3]} \quad \Delta = -83, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
21513151321\bar{3} \text{ }_{[\# = 9]} \quad \Delta = -83, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
21513151321\bar{3} \text{ }_{[\# = 8]} \quad \Delta = -83, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
21513151321\bar{3} \text{ }_{[\# = 7]} \quad \Delta = -83, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
21513151321\bar{3} \text{ }_{[\# = 6]} \quad \Delta = -83, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
21513151321\bar{3} \text{ }_{[\# = 5]} \quad \Delta = -83, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix}
\end{aligned}$$

$$21513151321\overset{2}{3} \text{ [#=4]} \quad \Delta = -83, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$12315131512 \quad \Delta = -75, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$12315131512 \quad \Delta = -75, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$12315131512 \quad \Delta = -75, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$12315131512 \quad \Delta = -83, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$21513151321\overset{2}{3} \text{ [#=2]} \quad \Delta = -83, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$12315131512 \quad \Delta = -75, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$12315131512 \quad \Delta = -76, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 13151315131

$$13151315131 \quad \Delta = -83, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$13151315131\overset{2}{3} \text{ [#=2]} \quad \Delta = -90, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$13151315131\overset{2}{3} \text{ [#=4]} \quad \Delta = -90, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$13151315131\overset{2}{3} \text{ [#=8]} \quad \Delta = -90, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix}$$

$$13151315131\overset{2}{3} \text{ [#=3]} \quad \Delta = -90, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$13151315131 \quad \Delta = -82, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$13151315131\overset{2}{3} \text{ [#=9]} \quad \Delta = -90, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix}$$

$$13151315131\overset{2}{3} \text{ [#=7]} \quad \Delta = -90, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix}$$

$$13151315131\overset{2}{3} \text{ [#=5]} \quad \Delta = -90, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix}$$

$$13151315131\overset{2}{3} \text{ [#=6]} \quad \Delta = -90, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

$$13151315131 \quad \Delta = -82, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 31513151315

$$\begin{aligned}
31513151315^2 \quad \Delta = -134, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
31513151315^2 \quad \Delta = -134, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
31513151315^2 \quad \Delta = -134, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
31513151315^2 \quad \Delta = -134, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
31513151315^2 \quad \Delta = -135, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
31513151315^2 \quad \Delta = -135, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 15131513151

$$\begin{aligned}
15131513151^2 \quad \Delta = -127, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
15131513151^2 \quad \Delta = -126, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
15131513151^2 \quad \Delta = -126, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1513151321

$$\begin{aligned}
1513151321^1 3_{[\# = 4]} \quad \Delta = -83, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1513151321^1 3_{[\# = 3]} \quad \Delta = -83, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1513151321^1 2 \quad \Delta = -75, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1513151321^1 2 \quad \Delta = -75, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1513151321^{12} \quad \Delta = -75, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1513151321^1 2 \quad \Delta = -76, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1513151321^1 3_{[\# = 2]} \quad \Delta = -83, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1513151321^1 3_{[\# = 8]} \quad \Delta = -83, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
1513151321^1 3_{[\# = 5]} \quad \Delta = -83, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{1}{1513151321}\mathbf{3}_{[\# = 6]} \quad \Delta = -83, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
& \overset{1}{1513151321}\mathbf{3}_{[\# = 7]} \quad \Delta = -83, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
& \overset{1}{1513151321}\mathbf{3}_{[\# = 9]} \quad \Delta = -83, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
& \text{Base: } \overset{1}{1231513221} \\
& \overset{1}{1223151321}\mathbf{3}_{[\# = 4]} \quad \Delta = -32, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{21}{1231513221} \quad \Delta = -24, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{1}{1223151321}\mathbf{3}_{[\# = 7]} \quad \Delta = -32, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
& \overset{1}{1231513221}\mathbf{3}_{[\# = 7]} \quad \Delta = -32, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
& \overset{1}{1231513221}\mathbf{3}_{[\# = 6]} \quad \Delta = -32, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
& \overset{1}{1223151321}\mathbf{3}_{[\# = 2]} \quad \Delta = -32, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{1}{1231513221}\mathbf{3}_{[\# = 2]} \quad \Delta = -32, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{1}{1223151321}\mathbf{3}_{[\# = 3]} \quad \Delta = -32, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{1}{1231513221}\mathbf{3}_{[\# = 3]} \quad \Delta = -32, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{12}{1231513221} \quad \Delta = -24, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{1}{1223151321}\mathbf{3}_{[\# = 9]} \quad \Delta = -32, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
& \overset{1}{1231513221}\mathbf{3}_{[\# = 4]} \quad \Delta = -32, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& \overset{1}{1223151321}\mathbf{3}_{[\# = 5]} \quad \Delta = -32, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& \overset{1}{1231513221}\mathbf{3}_{[\# = 8]} \quad \Delta = -32, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
& \overset{1}{1223151321}\mathbf{3}_{[\# = 6]} \quad \Delta = -32, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& \overset{1}{1231513221}\mathbf{3}_{[\# = 5]} \quad \Delta = -32, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1223151321\overset{1}{3} \text{ [#=8]} \quad \Delta = -32, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1231513221\overset{1}{3} \text{ [#=9]} \quad \Delta = -32, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix}
\end{aligned}$$

Base: 41315132216

$$\begin{aligned}
4131513221\overset{2}{6} \quad \Delta = -130, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
4131513221\overset{2}{6} \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
4131513221\overset{2}{6} \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
4131513221\overset{2}{6} \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
4131513221\overset{2}{6} \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
4131513221\overset{3}{6} \quad \Delta = -137, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
4131513221\overset{3}{6} \quad \Delta = -137, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: 61231414141

$$\begin{aligned}
612314141\overset{2}{4}1 \quad \Delta = -125, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
61231414141\overset{1}{3} \text{ [#=2]} \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
61231414141\overset{1}{3} \text{ [#=3]} \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
612314141\overset{2}{4}1 \quad \Delta = -127, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
612314141\overset{2}{4}1 \quad \Delta = -125, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
612314141\overset{2}{4}1 \quad \Delta = -125, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
612314141\overset{2}{4}1 \quad \Delta = -125, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{aligned}$$

Base: 61314141413

$$\begin{aligned}
61314141\overset{3}{4}13 \quad \Delta = -140, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 60 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
61314141\overset{3}{4}13 \quad \Delta = -140, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
61314141413 \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
61314141413 \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
61314141413 \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
61314141413 \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
61314141413 \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 16122315131

$$\begin{aligned}
16122315131 \textcolor{red}{3} \text{ [#=2]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
16122315131 \textcolor{red}{3} \text{ [#=5]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
16122315131 \textcolor{red}{3} \text{ [#=9]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
16122315131 \textcolor{red}{3} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 17 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
16122315131 \textcolor{red}{3} \text{ [#=3]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
16122315131 \textcolor{red}{3} \text{ [#=4]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
16122315131 \textcolor{red}{3} \text{ [#=8]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
16122315131 \textcolor{red}{3} \text{ [#=7]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
16122315131 \textcolor{red}{3} \text{ [#=6]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
16122315131 \textcolor{red}{2} \quad \Delta = -101, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
16122315131 \textcolor{red}{2} \quad \Delta = -101, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
16122315131 \textcolor{red}{2} \quad \Delta = -101, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 41315132161

$$41315132161 \textcolor{red}{2} \quad \Delta = -130, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\begin{aligned}
41315132161^2 \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
41315132161^2 \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
41315132161^3 \quad \Delta = -137, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
41315132161^2 \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
41315132161^2 \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 14141413161

$$\begin{aligned}
161314141413^{\text{red}}_{[\# = 2]} \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
14141413161^2 \quad \Delta = -124, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
14141413161^2 \quad \Delta = -124, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
14141413161^2 \quad \Delta = -124, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
14141413161^2 \quad \Delta = -124, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
161314141413^{\text{red}}_{[\# = 3]} \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
14141413161^3 \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 60 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

Base: 21612315131

$$\begin{aligned}
21612315131^{\text{red}}_{[\# = 4]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21612315131^{\text{red}}_{[\# = 9]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
21612315131^{\text{red}}_{[\# = 8]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
21612315131^{\text{red}}_{[\# = 7]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
21612315131^{\text{red}}_{[\# = 6]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21612315131^2 \quad \Delta = -101, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
21612315131^2 \quad \Delta = -101, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
21612315131^2 \quad \Delta = -101, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21612315131^2 \quad \Delta = -101, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21612315131^3_{[\#=5]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
21612315131^3_{[\#=3]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
21612315131^3_{[\#=2]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21612315131^3 \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: 12321612321

$$\begin{aligned}
12321612321^3_{[\#=5]} \quad \Delta = -62, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
12321612321^3_{[\#=8]} \quad \Delta = -62, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
12321612321^3_{[\#=7]} \quad \Delta = -62, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
12321612321^3_{[\#=6]} \quad \Delta = -62, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
12321612321^3_{[\#=9]} \quad \Delta = -62, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
12321612321^3_{[\#=2]} \quad \Delta = -62, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
12321612321^2 \quad \Delta = -54, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
12321612321^3_{[\#=4]} \quad \Delta = -62, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
12321612321^3_{[\#=3]} \quad \Delta = -62, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

Base: 61232161232

$$\begin{aligned}
61232161232^2 \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
61232161232^2 \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

$${}^2 61232161232 \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 16123216123

$$16123216123 \quad \Delta = -130, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$16123216123 \quad \Delta = -130, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$${}^2 16123216123 \quad \Delta = -130, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 21612321612

$${}^3 21612321612 \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$${}^2 21612321612 \quad \Delta = -121, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$${}^2 21612321612 \quad \Delta = -121, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 41315131612

$$41315131612 \quad \Delta = -136, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$41315131612 \quad \Delta = -128, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$41315131612 \quad \Delta = -128, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$${}^2 41315131612 \quad \Delta = -128, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$41315131612 \quad \Delta = -128, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$${}^3 41315131612 \quad \Delta = -136, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$${}^2 41315131612 \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$${}^2 41315131612 \quad \Delta = -128, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: 22161315131

$${}^3 22161315131 \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$22161315131 \quad \Delta = -100, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$${}^3 22161315131 \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\begin{aligned}
22161315131\textcolor{red}{3}_{[\# = 5]} \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
22161315131\textcolor{red}{3}_{[\# = 4]} \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
22161315131\textcolor{red}{2} \quad \Delta = -100, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
22161315131\textcolor{red}{2} \quad \Delta = -100, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
22161315131\textcolor{red}{2} \quad \Delta = -100, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
22161315131\textcolor{red}{3}_{[\# = 3]} \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
22161315131\textcolor{red}{3}_{[\# = 9]} \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
22161315131\textcolor{red}{3}_{[\# = 2]} \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
22161315131\textcolor{red}{3}_{[\# = 8]} \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
22161315131\textcolor{red}{3}_{[\# = 7]} \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
22161315131\textcolor{red}{3}_{[\# = 6]} \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\textbf{Base: } 13161231512 & \\
21513216131\textcolor{red}{3}_{[\# = 4]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
13161231512\textcolor{red}{3} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
21513216131\textcolor{red}{3}_{[\# = 3]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21513216131\textcolor{red}{3}_{[\# = 6]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
21513216131\textcolor{red}{3}_{[\# = 7]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
21513216131\textcolor{red}{3}_{[\# = 9]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
21513216131\textcolor{red}{3}_{[\# = 8]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
21513216131\textcolor{red}{3}_{[\# = 5]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

$$21513216131\bar{3} \text{ }_{[\# = 2]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1316123151\bar{2} \quad \Delta = -101, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1316123151\bar{2} \quad \Delta = -101, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1316123151\bar{2} \quad \Delta = -101, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1316123151\bar{2} \quad \Delta = -101, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 13161231¹51

$$1 \text{ }^{\bar{3}13161231} \text{ }_5 \text{ }^1 \text{ }_{[\# = 9]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix}$$

$$1 \text{ }^{\bar{3}13161231} \text{ }_5 \text{ }^1 \text{ }_{[\# = 7]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$1 \text{ }^{\bar{3}13161231} \text{ }_5 \text{ }^1 \text{ }_{[\# = 6]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$1 \text{ }^{\bar{3}13161231} \text{ }_5 \text{ }^1 \text{ }_{[\# = 5]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1 \text{ }^{\bar{3}13161231} \text{ }_5 \text{ }^1 \text{ }_{[\# = 4]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1 \text{ }^{\bar{3}13161231} \text{ }_5 \text{ }^1 \text{ }_{[\# = 8]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$1316123151\bar{2} \quad \Delta = -101, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \text{ }^{\bar{3}13161231} \text{ }_5 \text{ }^1 \text{ }_{[\# = 2]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \text{ }^{\bar{3}13161231} \text{ }_5 \text{ }^1 \text{ }_{[\# = 3]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1316123151\bar{2} \quad \Delta = -101, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1316123151\bar{2}^1 \quad \Delta = -101, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: 13161232151

$$15123216131\bar{3} \text{ }_{[\# = 7]} \quad \Delta = -111, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$15123216131\bar{3} \text{ }_{[\# = 5]} \quad \Delta = -111, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$\begin{array}{ll}
15123216131\overset{2}{3} \text{ [#=6]} & \Delta = -111, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
15123216131\overset{2}{3} \text{ [#=4]} & \Delta = -111, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
15123216131\overset{2}{3} \text{ [#=8]} & \Delta = -111, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
15123216131\overset{2}{3} \text{ [#=9]} & \Delta = -111, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
15123216131\overset{2}{3} \text{ [#=2]} & \Delta = -111, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
15123216131\overset{2}{3} \text{ [#=3]} & \Delta = -111, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
13161232151\overset{2}{2} & \Delta = -103, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
13161232151\overset{2}{2} & \Delta = -103, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
13161232151\overset{2}{2} & \Delta = -103, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\text{Base: } 41316123151 & \\
41316123151\overset{3}{3} & \Delta = -137, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
41316123151\overset{2}{2} & \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
41316123151\overset{2}{2} & \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
41316123151\overset{2}{2} & \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
41316123151\overset{2}{2} & \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\text{Base: } 41316123215 & \\
41316123215\overset{3}{3} & \Delta = -139, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
41316123215\overset{2}{2} & \Delta = -131, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
41316123215\overset{2}{2} & \Delta = -131, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
41316123215\overset{2}{2} & \Delta = -131, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{array}$$

$$41316123215 \quad \Delta = -131, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 14131612321

$$141316123213_{[\# = 6]} \quad \Delta = -87, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$14131612321 \quad \Delta = -79, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$14131612321 \quad \Delta = -79, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$141316123213_{[\# = 7]} \quad \Delta = -87, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$141316123213_{[\# = 8]} \quad \Delta = -87, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$123216131413_{[\# = 2]} \quad \Delta = -87, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$141316123213_{[\# = 2]} \quad \Delta = -87, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$14131612321 \quad \Delta = -87, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$141316123213_{[\# = 9]} \quad \Delta = -87, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$123216131413_{[\# = 3]} \quad \Delta = -87, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$141316123213_{[\# = 3]} \quad \Delta = -87, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$141316123213_{[\# = 4]} \quad \Delta = -87, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$141316123213_{[\# = 5]} \quad \Delta = -87, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$14131612321 \quad \Delta = -79, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 13216131512

$$13216131512 \quad \Delta = -101, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$13216131512 \quad \Delta = -101, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$13216131512 \quad \Delta = -101, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\begin{aligned}
13216131512^{\quad 2} \quad \Delta = -101, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
215131612313^{\quad 3} \text{ [#=3]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
215131612313^{\quad 3} \text{ [#=5]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
13216131512^{\quad 3} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
215131612313^{\quad 3} \text{ [#=2]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
215131612313^{\quad 3} \text{ [#=4]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

Base: 41321613151

$$\begin{aligned}
41321613151^{\quad 2} \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 & \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
41321613151^{\quad 2} \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
41321613151^{\quad 2} \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
41321613151^{\quad 2} \quad \Delta = -131, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
41321613151^{\quad 2} \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1321613151¹

$$\begin{aligned}
1321613151^{\quad 2} \quad 1^{\quad 1} \quad \Delta = -101, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1^{\quad 3} 13216131^{\quad 1} \quad 5^{\quad 1} \quad 1 \text{ [#=5]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1^{\quad 3} 13216131^{\quad 1} \quad 5^{\quad 1} \quad 1 \text{ [#=4]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{\quad 3} 13216131^{\quad 1} \quad 5^{\quad 1} \quad 1 \text{ [#=3]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1^{\quad 3} 13216131^{\quad 1} \quad 5^{\quad 1} \quad 1 \text{ [#=2]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1321613151^{\quad 21} \quad \Delta = -101, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1321613151^{\quad 2} \quad 1^{\quad 1} \quad \Delta = -101, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 41414131613

$$\begin{aligned}
41414131613^2 \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
41414131613^2 \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
41414131613^2 \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
41414131613^2 \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
41414131613^2 \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 144 & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
41414131613^3 \quad \Delta = -140, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 144 & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 81 & 17 \\ 2 & 17 & 8 \end{pmatrix}
\end{aligned}$$

Base: 14141316131

$$\begin{aligned}
14141316131^3 \quad \Delta = -112, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 32 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
14141316131^3_{[\# = 6]} \quad \Delta = -112, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
14141316131^2 \quad \Delta = -104, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
14141316131^2 \quad \Delta = -104, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
14141316131^2 \quad \Delta = -104, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
14141316131^2 \quad \Delta = -104, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
14141316131^3_{[\# = 2]} \quad \Delta = -112, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
13161314141^3_{[\# = 2]} \quad \Delta = -112, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
13161314141^3_{[\# = 3]} \quad \Delta = -112, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
14141316131^3_{[\# = 3]} \quad \Delta = -112, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
14141316131^3_{[\# = 4]} \quad \Delta = -112, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
14141316131^3_{[\# = 9]} \quad \Delta = -112, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
14141316131^3_{[\# = 5]} \quad \Delta = -112, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

$$14141316131\overset{3}{3} \text{ }_{[\# = 8]} \quad \Delta = -112, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$14141316131\overset{3}{3} \text{ }_{[\# = 7]} \quad \Delta = -112, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 41413161314

$$4141316131\overset{3}{4} \quad \Delta = -140, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 37 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$4141316131\overset{2}{4} \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$4141316131\overset{2}{4} \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$4141316131\overset{2}{4} \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$4141316131\overset{2}{4} \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$4141316131\overset{2}{4} \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$4141316131\overset{3}{4} \quad \Delta = -140, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: 14131613141

$$1413161314\overset{2}{1} \quad \Delta = -104, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1413161314\overset{2}{1} \quad \Delta = -104, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1413161314\overset{3}{1} \quad \Delta = -112, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 21 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1413161314\overset{3}{1}\overset{3}{3} \text{ }_{[\# = 2]} \quad \Delta = -112, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1413161314\overset{3}{1}\overset{3}{3} \text{ }_{[\# = 3]} \quad \Delta = -112, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: 13161315123

$$32151316131\overset{3}{3} \text{ }_{[\# = 8]} \quad \Delta = -117, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$1316131512\overset{2}{3} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$32151316131\overset{3}{3} \text{ }_{[\# = 7]} \quad \Delta = -117, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$1316131512\overset{2}{3} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\begin{aligned}
13161315123^2 \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
321513161313^{\#3} \quad \Delta = -117, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
321513161313^{\#2} \quad \Delta = -117, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
321513161313^{\#9} \quad \Delta = -117, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
13161315123^2 \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
321513161313^{\#6} \quad \Delta = -117, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
321513161313^{\#5} \quad \Delta = -117, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
321513161313^{\#4} \quad \Delta = -117, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1316131513^1

$$\begin{aligned}
31^{31316131} 5^1 1^{\#4} \quad \Delta = -116, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
31^{31316131} 5^1 1^{\#3} \quad \Delta = -116, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1316131513^{31} \quad \Delta = -116, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
31^{31316131} 5^1 1^{\#2} \quad \Delta = -116, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
31^{31316131} 5^1 1^{\#8} \quad \Delta = -116, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
31^{31316131} 5^1 1^{\#5} \quad \Delta = -116, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
31^{31316131} 5^1 1^{\#6} \quad \Delta = -116, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
31^{31316131} 5^1 1^{\#7} \quad \Delta = -116, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
31^{31316131} 5^1 1^{\#9} \quad \Delta = -116, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
1316131513^{12} \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1316131513^{21} \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

$$1316131\overset{2}{1}\overset{1}{1}3 \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1316131\overset{2}{1}\overset{1}{1}3 \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1316131\overset{2}{1}\overset{1}{1}3 \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 31316131512

$$3131613151\overset{3}{2} \quad \Delta = -116, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$3131613151\overset{3}{2} \quad \Delta = -116, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$3131613151\overset{3}{2} \quad \Delta = -116, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 132 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$3131613151\overset{3}{2} \quad \Delta = -116, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$3131613151\overset{4}{2} \quad \Delta = -136, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$3131613151\overset{2}{2} \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$3131613151\overset{2}{2} \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$3131613151\overset{2}{2} \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$3131613151\overset{2}{2} \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$3131613151\overset{2}{2} \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: 3131613151¹

$$3131613151\overset{2}{1} \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$3131613151\overset{3}{1} \quad \Delta = -116, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$3131613151\overset{3}{1} \quad \Delta = -116, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 132 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$3131613151\overset{4}{1} \quad \Delta = -136, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$3131613151\overset{3}{1} \quad \Delta = -116, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$3131613151\overset{2}{1} \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\begin{aligned}
3131613151^{\mathbf{2}1} \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
3131613151^{\mathbf{2}1} \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: 21613161223

$$\begin{aligned}
21613161223^{\mathbf{3}} \quad \Delta = -135, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21613161223^{\mathbf{2}} \quad \Delta = -127, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
21613161223^{\mathbf{2}} \quad \Delta = -127, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21613161223^{\mathbf{2}} \quad \Delta = -127, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21613161223^{\mathbf{2}} \quad \Delta = -127, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21613161223^{\mathbf{3}} \quad \Delta = -135, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: 22161316122

$$\begin{aligned}
22161316122^{\mathbf{2}} \quad \Delta = -118, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
22161316122^{\mathbf{2}} \quad \Delta = -118, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
22161316122^{\mathbf{3}} \quad \Delta = -126, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
22161316122^{\mathbf{3}} \quad \Delta = -127, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1^{\mathbf{3}13151321} 6^{\mathbf{1}} 1$

$$\begin{aligned}
1^{\mathbf{3}13151321} 6^{\mathbf{1}} 1 \text{ [#=9]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1^{\mathbf{3}13151321} 6^{\mathbf{1}} 1 \text{ [#=8]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1^{\mathbf{3}13151321} 6^{\mathbf{1}} 1 \text{ [#=6]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{\mathbf{3}13151321} 6^{\mathbf{1}} 1 \text{ [#=5]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1^{\mathbf{3}13151321} 6^{\mathbf{1}} 1 \text{ [#=4]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{\mathbf{3}13151321} 6^{\mathbf{1}} 1 \text{ [#=3]} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

$$1^{\overline{3}13151321} \quad 1^{\overline{6}} \quad 1 \quad [\# = 2] \quad \Delta = -109, \quad n_+ = 1, \quad n_- = 11, \quad n_0 = 0, \quad |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1^{\overline{3}13151321} \quad 1^{\overline{6}} \quad 1 \quad [\# = 7] \quad \Delta = -109, \quad n_+ = 1, \quad n_- = 11, \quad n_0 = 0, \quad |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$1^{\overline{13}151321} \quad 1^{\overline{6}} \quad 1 \quad \Delta = -101, \quad n_+ = 1, \quad n_- = 11, \quad n_0 = 0, \quad |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1^{\overline{13}151321} \quad 1^{\overline{6}} \quad 1 \quad \Delta = -101, \quad n_+ = 1, \quad n_- = 11, \quad n_0 = 0, \quad |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1^{\overline{13}151321} \quad 1^{\overline{6}} \quad 1 \quad \Delta = -101, \quad n_+ = 1, \quad n_- = 11, \quad n_0 = 0, \quad |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 1612321612

$$1612321612^{\overline{2}} \quad \Delta = -121, \quad n_+ = 1, \quad n_- = 11, \quad n_0 = 0, \quad |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1612321612^{\overline{2}} \quad \Delta = -121, \quad n_+ = 1, \quad n_- = 11, \quad n_0 = 0, \quad |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1612321612^{\overline{1}2} \quad \Delta = -121, \quad n_+ = 1, \quad n_- = 11, \quad n_0 = 0, \quad |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1612321612^{\overline{1}3} \quad \Delta = -129, \quad n_+ = 1, \quad n_- = 11, \quad n_0 = 0, \quad |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

Base: 161232161

$$161232161^{\overline{1}2} \quad \Delta = -121, \quad n_+ = 1, \quad n_- = 11, \quad n_0 = 0, \quad |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 4131513161

$$4131513161^{\overline{2}} \quad \Delta = -129, \quad n_+ = 1, \quad n_- = 11, \quad n_0 = 0, \quad |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$4131513161^{\overline{3}} \quad \Delta = -136, \quad n_+ = 1, \quad n_- = 11, \quad n_0 = 0, \quad |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$4131513161^{\overline{2}} \quad \Delta = -128, \quad n_+ = 1, \quad n_- = 11, \quad n_0 = 0, \quad |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$4131513161^{\overline{2}1} \quad \Delta = -128, \quad n_+ = 1, \quad n_- = 11, \quad n_0 = 0, \quad |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$4131513161^{\overline{2}1} \quad \Delta = -128, \quad n_+ = 1, \quad n_- = 11, \quad n_0 = 0, \quad |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$4131513161^{\overline{2}} \quad \Delta = -128, \quad n_+ = 1, \quad n_- = 11, \quad n_0 = 0, \quad |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: 1613161223

$$1613161223^{\overline{1}2} \quad \Delta = -127, \quad n_+ = 1, \quad n_- = 11, \quad n_0 = 0, \quad |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\begin{aligned}
1^{13}161223 \quad \Delta = -135, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1^{12}161223 \quad \Delta = -127, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1^{12}161223 \quad \Delta = -127, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1^{31}61232161$

$$\begin{aligned}
1^{31}61232161 \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1^{312}61232161 \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
1^{31}61232161 \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
31^{21}61232161 \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
31^{31}61232161 \quad \Delta = -137, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1^{1315131}6^{12}$

$$\begin{aligned}
1^{1315131}6^{12} \quad \Delta = -100, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1^{1315131}6^{12} \quad \Delta = -100, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1^{31315131}6^{12} [\# = 3] \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1^{1315131}6^{12} \quad \Delta = -100, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1^{31315131}6^{12} [\# = 5] \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1^{31315131}6^{12} [\# = 7] \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1^{31315131}6^{12} [\# = 2] \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1^{13151313}6^{12} \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1^{31315131}6^{12} [\# = 9] \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1^{13151312}6^{12} \quad \Delta = -100, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1 \overset{31315131}{6} 12 \text{ [#}=4] \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \overset{31315131}{6} 12 \text{ [#}=6] \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \overset{31315131}{6} 12 \text{ [#}=8] \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{21}{16}1316122$

$$\begin{aligned}
\overset{21}{16}1316122 \overset{3}{6} \quad \Delta = -127, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
\overset{21}{16}1316122 \overset{2}{6} \quad \Delta = -118, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\overset{21}{16}1316122 \overset{2}{6} \quad \Delta = -118, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\overset{21}{16}1316122 \quad \Delta = -118, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\overset{21}{16}1316122 \quad \Delta = -126, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\overset{21}{16}1316122 \overset{3}{6} \quad \Delta = -126, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\overset{21}{16}1316122 \overset{2}{6} \quad \Delta = -118, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{321}{16}131612$

$$\begin{aligned}
\overset{321}{16}131612 \overset{3}{6} \quad \Delta = -135, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
\overset{321}{16}131612 \overset{2}{6} \quad \Delta = -127, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\overset{31}{321}6131612 \quad \Delta = -135, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\overset{321}{16}131612 \overset{2}{6} \quad \Delta = -127, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\overset{321}{16}131612 \quad \Delta = -127, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\overset{21}{321}6131612 \quad \Delta = -127, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\overset{321}{16}131612 \overset{2}{6} \quad \Delta = -127, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{2321}{16}13161$

$$\overset{31}{2321}613161 \quad \Delta = -137, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\begin{aligned}
1 \overset{2321}{6} 13161 \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1 \overset{2321}{6} 13161 \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
2321 \overset{21}{6} 13161 \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \overset{2321}{6} \overset{2}{13161} \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{215131}{6} 131$

$$\begin{aligned}
1 \overset{215131}{6} 131 \overset{3}{\text{[#}=6]} \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1 \overset{215131}{6} 131 \overset{3}{\text{[#}=2]} \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1 \overset{215131}{6} 131 \overset{3}{\text{[#}=9]} \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
1 \overset{215131}{6} 131 \overset{3}{\text{[#}=7]} \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
1 \overset{215131}{6} 131 \overset{3}{\text{[#}=5]} \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1 \overset{215131}{6} 131 \overset{3}{\text{[#}=4]} \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \overset{215131}{6} 131 \overset{3}{\text{[#}=3]} \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{215131}{6} 131 \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
1 \overset{215131}{6} 131 \overset{3}{\text{[#}=8]} \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1 \overset{215131}{6} 131 \quad \Delta = -100, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1 \overset{215131}{6} 131 \quad \Delta = -100, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1 \overset{215131}{6} \overset{2}{131} \quad \Delta = -100, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1 \overset{215131}{6} 131 \quad \Delta = -100, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{15131}{6} 131$

$$1 \overset{15131}{6} 131 \overset{3}{\text{[#}=9]} \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix}$$

$$\begin{array}{ll}
1^{15131} \bar{6} \text{ 131}\bar{\textcolor{red}{3}}_{[\# = 2]} \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1^{15131} \bar{6} \text{ 131}\bar{\textcolor{red}{3}}_{[\# = 4]} \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{15131} \bar{6} \text{ 131}\bar{\textcolor{red}{3}}_{[\# = 5]} \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1^{15131} \bar{6} \text{ 131}\bar{\textcolor{red}{3}}_{[\# = 6]} \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1^{15131} \bar{6} \text{ 131}\bar{\textcolor{red}{3}}_{[\# = 7]} \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
1^{15131} \bar{6} \text{ 131}\bar{\textcolor{red}{3}}_{[\# = 8]} \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1^{15131} \bar{6} \text{ 131} \quad \Delta = -100, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1^{15131} \bar{6} \text{ 131}\bar{\textcolor{red}{3}}_{[\# = 3]} \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1^{15131} \bar{6} \text{ 131} \quad \Delta = -100, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1^{15131} \bar{6} \text{ 131} \quad \Delta = -100, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\text{Base: } 4131^{15131} \bar{6} \text{ 1} & \\
\bar{\textcolor{red}{3}}^{15131} 4131 \bar{6} \text{ 1} \quad \Delta = -136, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\bar{\textcolor{red}{2}}^{15131} 4131 \bar{6} \text{ 1} \quad \Delta = -128, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
4131^{15131} \bar{6} \text{ 1} \quad \Delta = -128, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
4131^{\bar{\textcolor{red}{2}}15131} \bar{6} \text{ 1} \quad \Delta = -128, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
4131^{15131} \bar{6} \text{ 1}\bar{\textcolor{red}{2}} \quad \Delta = -128, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
4131^{15131} \bar{6} \text{ 1} \quad \Delta = -128, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\text{Base: } 161316123^1 & \\
161316123^{\bar{\textcolor{red}{1}}\bar{\textcolor{red}{2}}1} \quad \Delta = -127, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
161316123^1 \bar{\textcolor{red}{2}}1 \quad \Delta = -127, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{array}$$

$${}^1_{16}131{}^{31}_6123 \quad \Delta = -135, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$${}^1_{16}131{}^{21}_6123 \quad \Delta = -127, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$${}^1_{16}131{}^{12}_6123 \quad \Delta = -127, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: ${}^{31}_{21}6131612$

$${}^{331}_{21}6131612 \quad \Delta = -134, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 23 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$${}^{231}_{21}6131612 \quad \Delta = -126, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 24 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$${}^{312}_{21}6131612 \quad \Delta = -126, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$${}^{312}_{21}6131612 \quad \Delta = -126, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$${}^{31}_{21}6131612 \quad \Delta = -126, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$${}^{31}_{21}6131612 \quad \Delta = -126, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$${}^{31}_{21}6131612 \quad \Delta = -134, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

Base: ${}^{21}_{16}131{}^{21}_61$

$${}^{21}_{16}131{}^{21}_61 \quad \Delta = -126, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$${}^{21}_{16}131{}^{21}_61 \quad \Delta = -118, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$${}^{212}_{16}131{}^{21}_61 \quad \Delta = -118, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: ${}^{215131}_{31}613$

$${}^{215131}_{31}613 \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$${}^{3215131}_{31}613 \quad \Delta = -117, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$${}^{215131}_{31}613 \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$${}^{215131}_{31}613 \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$${}^{2215131}_{31}613 \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$31 \overset{215131}{6} 13 \quad \Delta = -116, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: $31 \overset{15131}{6} 13$

$$31 \overset{15131}{6} 13 \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$31 \overset{315131}{6} 13 \quad \Delta = -116, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$31 \overset{15131}{6} 13 \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$31 \overset{215131}{6} 13 \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$31 \overset{215131}{6} 13 \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: $131 \overset{15131}{6} 13$

$$31 \overset{15131}{6} 131 \overset{15131}{5} \quad \Delta = -160, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$31 \overset{15131}{6} 131 \overset{15131}{3} [\# = 7] \quad \Delta = -116, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 196 \quad \begin{pmatrix} 0 & 14 & 2 \\ 14 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$31 \overset{15131}{6} 131 \overset{15131}{3} [\# = 6] \quad \Delta = -116, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$31 \overset{15131}{6} 131 \overset{15131}{3} [\# = 9] \quad \Delta = -116, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 324 \quad \begin{pmatrix} 0 & 18 & 2 \\ 18 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$31 \overset{15131}{6} 131 \overset{15131}{3} [\# = 2] \quad \Delta = -116, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$31 \overset{15131}{6} 131 \overset{15131}{3} [\# = 3] \quad \Delta = -116, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$31 \overset{15131}{6} 131 \overset{15131}{3} [\# = 4] \quad \Delta = -116, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$31 \overset{15131}{6} 131 \overset{15131}{3} [\# = 5] \quad \Delta = -116, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$31 \overset{15131}{6} 131 \overset{15131}{3} \quad \Delta = -116, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$31 \overset{15131}{6} 131 \overset{15131}{3} [\# = 8] \quad \Delta = -116, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 256 \quad \begin{pmatrix} 0 & 16 & 2 \\ 16 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$31 \overset{15131}{6} 131 \overset{15131}{6} \quad \Delta = -186, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 8 \end{pmatrix}$$

$$131 \overset{15131}{6} 13 \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$131 \overset{15131}{6} 13 \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$131 \overset{15131}{6} \overset{2}{13} \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\overset{2}{131} \overset{15131}{6} 13 \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$31 \overset{15131}{6} 131 \overset{2}{2} \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$31 \overset{15131}{6} 131 \overset{4}{2} \quad \Delta = -136, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: $\overset{1}{16} \overset{21}{131} \overset{21}{6} 13$

$$\overset{1}{16} \overset{21}{131} \overset{21}{6} 13 \quad \Delta = -126, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\overset{1}{16} \overset{2}{131} \overset{21}{6} 13 \quad \Delta = -126, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\overset{1}{16} \overset{21}{131} \overset{2}{6} 13 \quad \Delta = -126, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\overset{1}{16} \overset{2}{131} \overset{21}{6} 13 \quad \Delta = -126, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\overset{1}{16} \overset{21}{131} \overset{21}{6} 13 \quad \Delta = -134, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: $\overset{1,1315131}{1} \overset{21}{6} \overset{21}{1}$

$$\overset{1,1315131}{1} \overset{21}{6} \overset{21}{1} \quad \Delta = -100, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\overset{1,3}{1} \overset{1315131}{6} \overset{21}{1} [\# = 9] \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$\overset{1,3}{1} \overset{1315131}{6} \overset{21}{1} [\# = 2] \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\overset{1,3}{1} \overset{1315131}{6} \overset{21}{1} [\# = 3] \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$\overset{1,3}{1} \overset{1315131}{6} \overset{21}{1} [\# = 4] \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$\overset{1,3}{1} \overset{1315131}{6} \overset{21}{1} [\# = 5] \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$\overset{1,3}{1} \overset{1315131}{6} \overset{21}{1} [\# = 6] \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$\overset{1,3}{1} \overset{1315131}{6} \overset{21}{1} [\# = 7] \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$1^{1,1315131} \quad 1 \quad 6 \quad 1 \quad \Delta = -100, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1^{1,31315131} \quad 1 \quad 6 \quad 1 \quad [\# = 8] \quad \Delta = -108, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$1^{1,1315131} \quad 1 \quad 6 \quad 1 \quad \Delta = -100, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 71223141413

$$71223141413^3 \quad \Delta = -140, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$71223141413^2 \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$71223141413^2 \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$71223141413^2 \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$71223141413^2 \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 71223151232

$$71223151232^2 \quad \Delta = -131, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$71223151232^2 \quad \Delta = -131, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$71223151232^2 \quad \Delta = -131, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: 7122315132

$$7122315132^1 \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$7122315132^{12} \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$7122315132^{21} \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$7122315132^{21} \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$7122315132^{31} \quad \Delta = -137, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 71231414132

$$71231414132^2 \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$71231414132^2 \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$71231414132^2 \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$71231414132^2 \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$71231414132^2 \quad \Delta = -135, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 7123151322¹

$$7123151322^{31} \quad \Delta = -137, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$7123151322^{21} \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$7123151322^{21} \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$7123151322^{12} \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$7123151322^{21} \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 71232151322

$$71232151322^2 \quad \Delta = -131, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$71232151322^2 \quad \Delta = -131, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$71232151322^2 \quad \Delta = -131, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix}$$

Base: 14141322171

$$171223141413^{21} \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$14141322171^2 \quad \Delta = -124, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$14141322171^2 \quad \Delta = -124, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$14141322171^2 \quad \Delta = -124, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$171223141413^{21} \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 17122315123

$$17122315123^2 \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$17122315123^2 \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1712231\overset{2}{5}123 \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 1712231¹513

$$1712231\overset{2}{5}\overset{1}{1}3 \quad \Delta = -128, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1712231\overset{2}{5}\overset{1}{1}3 \quad \Delta = -128, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1712231\overset{3}{5}\overset{1}{1}3 \quad \Delta = -136, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1712231\overset{2}{5}\overset{1}{1}3 \quad \Delta = -128, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1712231\overset{1}{5}\overset{2}{1}3 \quad \Delta = -128, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: 31414132171

$$314141\overset{2}{3}2171 \quad \Delta = -134, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$314141\overset{2}{3}2171 \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$314141\overset{3}{3}2171 \quad \Delta = -140, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$314141\overset{2}{3}2171 \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$314141\overset{2}{3}2171 \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$314141\overset{2}{3}2171 \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: 17123151232

$$171231\overset{2}{5}1232 \quad \Delta = -131, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$171231\overset{2}{5}1232 \quad \Delta = -131, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$171231\overset{2}{5}1232 \quad \Delta = -131, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: 1712315¹132

$$171231\overset{2}{5}\overset{1}{1}32 \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$171231\overset{2}{5}\overset{1}{1}32 \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$171231\overset{31}{5}132 \quad \Delta = -137, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$171231\overset{2}{5}\overset{1}{1}32 \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$171231\overset{12}{5}132 \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 17123215132

$$171232\overset{2}{1}5132 \quad \Delta = -131, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$1712321\overset{2}{5}132 \quad \Delta = -131, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1712321\overset{2}{5}132 \quad \Delta = -131, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 21712231512

$$2171223\overset{2}{1}512 \quad \Delta = -120, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$21712231\overset{3}{5}12 \quad \Delta = -128, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$21712231\overset{2}{5}12 \quad \Delta = -120, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$21712231\overset{2}{5}12 \quad \Delta = -120, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$21712231\overset{2}{5}12 \quad \Delta = -120, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: 21712231¹51

$$21712231\overset{2}{5}\overset{1}{1} \quad \Delta = -120, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$21712231\overset{21}{5}1 \quad \Delta = -120, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$21712231\overset{2}{5}\overset{1}{1} \quad \Delta = -120, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 14141321712

$$1414132\overset{2}{1}712 \quad \Delta = -124, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$14141321\overset{2}{7}12 \quad \Delta = -124, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$14141321\overset{2}{7}12 \quad \Delta = -124, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$21712314141\overset{3}{\#}=2] \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\begin{aligned}
14141\overset{2}{3}21712 \quad \Delta = -126, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21712314141\overset{3}{3} \text{ [#=3]} \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
14141\overset{2}{3}21712 \quad \Delta = -124, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\text{Base: } 21712315123 \\
\overset{2}{2}1712315123 \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2171231\overset{2}{5}123 \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2171231\overset{2}{5}123 \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
2171231\overset{2}{5}123 \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\text{Base: } 2171231\overset{1}{5}13 \\
\overset{2}{2}171231\overset{1}{5}13 \quad \Delta = -128, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2171231\overset{21}{5}13 \quad \Delta = -128, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2171231\overset{21}{5}13 \quad \Delta = -128, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
2171231\overset{12}{5}13 \quad \Delta = -128, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2171231\overset{31}{5}13 \quad \Delta = -136, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2171231\overset{2}{5}13 \quad \Delta = -128, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\text{Base: } 21712321513 \\
21712321\overset{2}{5}13 \quad \Delta = -130, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21712321\overset{2}{5}13 \quad \Delta = -130, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21712321\overset{2}{5}13 \quad \Delta = -130, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21712321\overset{2}{5}13 \quad \Delta = -130, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\text{Base: } 32171223151 \\
\overset{2}{3}2171223151 \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

$$32171223151^2 \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$32171223151^2 \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: 22171231512

$$22171231512^2 \quad \Delta = -120, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$22171231512^3 \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$22171231512^3 \quad \Delta = -128, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$22171231512^2 \quad \Delta = -120, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$22171231512^2 \quad \Delta = -120, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$22171231512^2 \quad \Delta = -120, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 2217123151¹

$$2217123151^{21} \quad \Delta = -120, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$2217123151^{21} \quad \Delta = -120, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$2217123151^3 \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$2217123151^{21} \quad \Delta = -120, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 22171232151

$$22171232151^2 \quad \Delta = -122, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$22171232151^2 \quad \Delta = -122, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$22171232151^2 \quad \Delta = -122, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$22171232151^3 \quad \Delta = -131, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: 32217123151

$$32217123151^2 \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$32217123151^2 \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$3221712315^2 \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 32217123215

$$32217123215^2 \quad \Delta = -131, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$32217123215^2 \quad \Delta = -131, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$32217123215^2 \quad \Delta = -131, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 23217122315

$$23217122315^2 \quad \Delta = -131, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 50 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$23217122315^2 \quad \Delta = -131, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$23217122315^2 \quad \Delta = -131, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: 12321712231

$$12321712231^3_{[\#=5]} \quad \Delta = -87, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$13221712321^3_{[\#=3]} \quad \Delta = -87, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$12321712231^3_{[\#=2]} \quad \Delta = -87, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$13221712321^3_{[\#=2]} \quad \Delta = -87, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$13221712321^3_{[\#=4]} \quad \Delta = -87, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$12321712231^3_{[\#=3]} \quad \Delta = -87, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$12321712231^3_{[\#=4]} \quad \Delta = -87, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$13221712321^3_{[\#=6]} \quad \Delta = -87, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$13221712321^3_{[\#=7]} \quad \Delta = -87, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$13221712321^3_{[\#=8]} \quad \Delta = -87, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$12321712231^2 \quad \Delta = -79, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\begin{aligned}
12321712231 \quad \Delta = -79, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
132217123213_{[\# = 9]} \quad \Delta = -87, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
132217123213_{[\# = 5]} \quad \Delta = -87, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}
\end{aligned}$$

Base: 41413217123

$$\begin{aligned}
41413217123 \quad \Delta = -135, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
41413217123 \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
41413217123 \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
41413217123 \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
41413217123 \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{aligned}$$

Base: 41321712314

$$\begin{aligned}
41321712314 \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
41321712314 \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
41321712314 \quad \Delta = -135, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 14132171231

$$\begin{aligned}
132171231413_{[\# = 3]} \quad \Delta = -113, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
132171231413_{[\# = 2]} \quad \Delta = -113, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
14132171231 \quad \Delta = -105, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
141321712313_{[\# = 2]} \quad \Delta = -113, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
141321712313_{[\# = 3]} \quad \Delta = -113, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
141321712313_{[\# = 4]} \quad \Delta = -113, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
141321712313_{[\# = 5]} \quad \Delta = -113, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

$$14132171231^2 \quad \Delta = -107, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$14132171231^2 \quad \Delta = -105, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$14132171231^2 \quad \Delta = -105, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 41232171231

$$412321712313_{[\#=3]} \quad \Delta = -115, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$41232171231^2 \quad \Delta = -107, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$41232171231^2 \quad \Delta = -107, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$41232171231^2 \quad \Delta = -107, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$412321712313_{[\#=2]} \quad \Delta = -115, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$412321712313_{[\#=4]} \quad \Delta = -115, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$41232171231^3 \quad \Delta = -115, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$412321712313_{[\#=5]} \quad \Delta = -115, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 12321712313

$$12321712313^3 \quad \Delta = -95, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$313217123213_{[\#=6]} \quad \Delta = -95, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$313217123213_{[\#=5]} \quad \Delta = -95, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$12321712313^2 \quad \Delta = -89, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$313217123213_{[\#=4]} \quad \Delta = -95, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$313217123213_{[\#=3]} \quad \Delta = -95, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$12321712313^3_{[\#=2]} \quad \Delta = -95, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\begin{aligned}
12321712313 \quad \Delta = -87, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
12321712313 \quad \Delta = -87, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
313217123213 \quad \Delta = -95, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
313217123213 \quad \Delta = -95, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
12321712313 \quad \Delta = -87, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
313217123213 \quad \Delta = -95, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
313217123213 \quad \Delta = -95, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1232171232¹

$$\begin{aligned}
1232171232 \quad \Delta = -81, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2 \quad 3 \quad 1 \quad \Delta = -89, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
2 \quad 3 \quad 1 \quad \Delta = -89, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1232171232 \quad \Delta = -89, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1232171232 \quad \Delta = -81, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1232171232 \quad \Delta = -81, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2 \quad 3 \quad 1 \quad \Delta = -89, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1232171232 \quad \Delta = -89, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2 \quad 3 \quad 1 \quad \Delta = -89, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
2 \quad 3 \quad 1 \quad \Delta = -89, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
2 \quad 3 \quad 1 \quad \Delta = -89, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
2 \quad 3 \quad 1 \quad \Delta = -89, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}
\end{aligned}$$

$$12321712\overset{41}{3}2 \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$2\overset{3}{12321712}3\overset{1}{1} [\# = 4] \quad \Delta = -89, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$\text{Base: } 1\overset{21513221}{7}1$$

$$1\overset{21513221}{7}1 \quad \Delta = -128, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$1\overset{21513221}{7}1 \quad \Delta = -120, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1\overset{21513221}{7}1 \quad \Delta = -120, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1\overset{21513221}{7}1 \quad \Delta = -120, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\text{Base: } 1\overset{1513221}{7}1$$

$$1\overset{1513221}{7}1 \quad \Delta = -120, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1\overset{1513221}{7}1 \quad \Delta = -120, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\text{Base: } 1\overset{1}{712314141}$$

$$1\overset{1}{712314141}\overset{2}{3} \quad \Delta = -124, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$1\overset{1}{712314141}\overset{2}{3} \quad \Delta = -126, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1\overset{1}{712314141}\overset{2}{3} \quad \Delta = -124, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1\overset{1}{712314141}\overset{2}{3} \quad \Delta = -124, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1\overset{1}{712314141}\overset{3}{3} [\# = 2] \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$1\overset{1}{712314141}\overset{3}{3} [\# = 3] \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$\text{Base: } 1\overset{32151321}{7}1$$

$$1\overset{32151321}{7}1 \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1\overset{32151321}{7}1 \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1\overset{32151321}{7}1 \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\text{Base: } 1\overset{3151321}{7}1$$

$$1 \begin{smallmatrix} 3151321 \\ 7 \end{smallmatrix} 1 \quad \Delta = -136, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1 \overset{3151321}{7} 1 \quad \Delta = -128, \quad n_+ = 1, \quad n_- = 11, \quad n_0 = 0, \quad |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{3151321}{7} 1 \quad \Delta = -128, \quad n_+ = 1, \quad n_- = 11, \quad n_0 = 0, \quad |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \begin{smallmatrix} & 3151321 \\ & 7 \\ & 1 \end{smallmatrix} \Delta = -128, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \begin{smallmatrix} 3151321 \\ 7 \end{smallmatrix} 1 \quad \Delta = -128, \, n_+ = 1, \, n_- = 11, \, n_0 = 0, \, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: $1 \overset{31512321}{7} 1$

$$1 \begin{smallmatrix} 31512321 \\ 7 \end{smallmatrix} 1 \quad \Delta = -130, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{31512321}{7} 1 \quad \Delta = -130, \, n_+ = 1, \, n_- = 11, \, n_0 = 0, \, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \begin{smallmatrix} 31512321 \\ 7 \end{smallmatrix} 1 \quad \Delta = -130, \, n_+ = 1, \, n_- = 11, \, n_0 = 0, \, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: $1 \overset{2151321}{7} 12$

$$1^{2151321}{}^{\textcolor{red}{2}}{}_{712} \quad \Delta = -120, \quad n_+ = 1, \quad n_- = 11, \quad n_0 = 0, \quad |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1^{2151321} 7^{12} \quad \Delta = -120, \quad n_+ = 1, \quad n_- = 11, \quad n_0 = 0, \quad |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1 \begin{smallmatrix} 2151321 \\ 7 \end{smallmatrix} 12 \quad \Delta = -120, \ n_+ = 1, \ n_- = 11, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \quad \begin{smallmatrix} 2151321 \\ 7 \quad 12 \end{smallmatrix} \quad \Delta = -128, \, n_+ = 1, \, n_- = 11, \, n_0 = 0, \, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$1 \begin{smallmatrix} 2151321 \\ 7 \end{smallmatrix} 12 \quad \Delta = -120, \, n_+ = 1, \, n_- = 11, \, n_0 = 0, \, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: $1^{151321} 7^{12}$

$$1 \overset{151321}{7} 12 \quad \Delta = -120, \quad n_+ = 1, \quad n_- = 11, \quad n_0 = 0, \quad |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1^{151321} \overset{\text{red}}{7} 12 \quad \Delta = -120, \, n_+ = 1, \, n_- = 11, \, n_0 = 0, \, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \begin{smallmatrix} 151321 \\ 7 \end{smallmatrix} 12 \quad \Delta = -120, \, n_+ = 1, \, n_- = 11, \, n_0 = 0, \, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: $1^{15} 7^{12} 12^{21}$

$$1 \overset{1512321}{7} 12 \quad \Delta = -122, \quad n_+ = 1, \quad n_- = 11, \quad n_0 = 0, \quad |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \begin{smallmatrix} 1512321 \\ 7 \end{smallmatrix} 12 \quad \Delta = -122, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \begin{smallmatrix} 1512321 \\ 7 \end{smallmatrix} 12 \quad \Delta = -122, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: $1 \begin{smallmatrix} 151321 \\ 7 \end{smallmatrix} 123$

$$1 \begin{smallmatrix} 151321 \\ 7 \end{smallmatrix} 123 \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \begin{smallmatrix} 151321 \\ 7 \end{smallmatrix} 123 \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1 \begin{smallmatrix} 151321 \\ 7 \end{smallmatrix} 123 \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$321 \begin{smallmatrix} 151321 \\ 7 \end{smallmatrix} 12 \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: $1 \begin{smallmatrix} 512321 \\ 7 \end{smallmatrix} 123$

$$321 \begin{smallmatrix} 512321 \\ 7 \end{smallmatrix} 12 \quad \Delta = -131, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \begin{smallmatrix} 512321 \\ 7 \end{smallmatrix} 123 \quad \Delta = -131, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1 \begin{smallmatrix} 512321 \\ 7 \end{smallmatrix} 123 \quad \Delta = -131, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \begin{smallmatrix} 512321 \\ 7 \end{smallmatrix} 123 \quad \Delta = -131, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: $1 \begin{smallmatrix} 12321 \\ 7 \end{smallmatrix} 1231$

$$1 \begin{smallmatrix} 312321 \\ 7 \end{smallmatrix} 1231 \quad [\# = 7] \quad \Delta = -87, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$1 \begin{smallmatrix} 12321 \\ 7 \end{smallmatrix} 1231 \quad \Delta = -79, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \begin{smallmatrix} 12321 \\ 7 \end{smallmatrix} 1231 \quad [\# = 2] \quad \Delta = -87, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \begin{smallmatrix} 312321 \\ 7 \end{smallmatrix} 1231 \quad [\# = 2] \quad \Delta = -87, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \begin{smallmatrix} 312321 \\ 7 \end{smallmatrix} 1231 \quad [\# = 6] \quad \Delta = -87, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1 \begin{smallmatrix} 312321 \\ 7 \end{smallmatrix} 1231 \quad [\# = 5] \quad \Delta = -87, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1 \begin{smallmatrix} 312321 \\ 7 \end{smallmatrix} 1231 \quad [\# = 3] \quad \Delta = -87, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \begin{smallmatrix} 12321 \\ 7 \end{smallmatrix} 1231 \quad [\# = 4] \quad \Delta = -87, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$\begin{aligned}
1 \overset{312321}{7} 1231 \text{ }_{[\# = 4]} \quad \Delta = -87, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \overset{12321}{7} 1231 \overset{3}{\text{ }}_{[\# = 5]} \quad \Delta = -87, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1 \overset{312321}{7} 1231 \text{ }_{[\# = 9]} \quad \Delta = -87, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1 \overset{12321}{7} 1231 \overset{2}{\text{ }} \quad \Delta = -79, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \overset{12321}{7} 1231 \overset{3}{\text{ }}_{[\# = 3]} \quad \Delta = -87, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{312321}{7} 1231 \text{ }_{[\# = 8]} \quad \Delta = -87, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{51321}{7} 1232$

$$\begin{aligned}
1 \overset{51321}{7} 1232 \quad \Delta = -131, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 100 \quad & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1 \overset{51321}{7} 1232 \quad \Delta = -131, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2321 \overset{51321}{7} \overset{2}{\text{ }} \quad \Delta = -131, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1 \overset{51321}{7} 1232 \overset{2}{\text{ }} \quad \Delta = -131, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{1,2151321}{7} 1$

$$\begin{aligned}
1 \overset{1,2151321}{7} 1 \quad \Delta = -128, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1 \overset{1,2151321}{7} 1 \quad \Delta = -120, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{1,2151321}{7} 1 \quad \Delta = -120, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \overset{1,2151321}{7} 1 \quad \Delta = -120, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{1,151321}{7} 1$

$$\begin{aligned}
1 \overset{1,151321}{7} 1 \quad \Delta = -120, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{1,151321}{7} 1 \quad \Delta = -120, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{1,1512321}{7} 1$

$$1 \overset{1,1512321}{7} 1 \quad \Delta = -122, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1^{1,1512321} \quad 1 \quad 7 \quad 1 \quad \Delta = -122, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 81223141322

$$81223141322^2 \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$81223141322^2 \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$81223141322^2 \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

Base: 23141322181

$$23141322181^2 \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$23141322181^2 \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$23141322181^2 \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$23141322181^2 \quad \Delta = -135, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 22314132181

$$22314132181^2 \quad \Delta = -135, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$22314132181^2 \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$22314132181^2 \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$22314132181^2 \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

Base: 31413221812

$$31413221812^3 \quad \Delta = -140, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$31413221812^2 \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$31413221812^2 \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$31413221812^2 \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$31413221812^2 \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 23141321812

$$23141321812^2 \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$23141321812^2 \quad \Delta = -135, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$23141321812^2 \quad \Delta = -135, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$23141321812^2 \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$23141321812^2 \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$23141321812^2 \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: 14132218122

$$14132218122^3 \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$221812231413_{[\# = 2]} \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$221812231413_{[\# = 3]} \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$14132218122^2 \quad \Delta = -124, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$14132218122^2 \quad \Delta = -124, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$14132218122^2 \quad \Delta = -124, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 13221812231

$$132218122313_{[\# = 2]} \quad \Delta = -113, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$132218122313_{[\# = 3]} \quad \Delta = -113, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$132218122313_{[\# = 5]} \quad \Delta = -113, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$132218122313_{[\# = 4]} \quad \Delta = -113, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$13221812231^2 \quad \Delta = -105, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 41322181223

$$41322181223^2 \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$\begin{aligned}
41322181223 \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
41322181223 \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

Base: 31413218122

$$\begin{aligned}
31413218122 \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
31413218122 \quad \Delta = -134, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
31413218122 \quad \Delta = -140, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
31413218122 \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
31413218122 \quad \Delta = -141, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
31413218122 \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
31413218122 \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 13218122313

$$\begin{aligned}
13218122313 \quad [\# = 2] \quad \Delta = -121, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
13218122313 \quad \Delta = -113, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
13218122313 \quad \Delta = -113, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
13218122313 \quad \Delta = -113, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
313221812313 \quad [\# = 4] \quad \Delta = -121, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
313221812313 \quad [\# = 3] \quad \Delta = -121, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
313221812313 \quad [\# = 5] \quad \Delta = -121, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
13218122313 \quad \Delta = -121, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
313221812313 \quad [\# = 2] \quad \Delta = -121, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: 14132181222

$$\begin{aligned}
22218123141\overset{2}{3} \text{ }_{[\# = 3]} \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
22218123141\overset{2}{3} \text{ }_{[\# = 2]} \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
14132181222\overset{2}{} \quad \Delta = -124, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
14132181222\overset{2}{} \quad \Delta = -124, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
14132181222\overset{2}{} \quad \Delta = -126, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
14132181222\overset{2}{} \quad \Delta = -124, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\text{Base: } 13221812313 \\
13221812313\overset{2}{} \quad \Delta = -113, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
31321812231\overset{2}{3} \text{ }_{[\# = 4]} \quad \Delta = -121, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
31321812231\overset{2}{3} \text{ }_{[\# = 5]} \quad \Delta = -121, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
13221812313\overset{3}{} \quad \Delta = -121, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
31321812231\overset{2}{3} \text{ }_{[\# = 3]} \quad \Delta = -121, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
13221812313\overset{3}{} \text{ }_{[\# = 2]} \quad \Delta = -121, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
31321812231\overset{2}{3} \text{ }_{[\# = 2]} \quad \Delta = -121, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
13221812313\overset{2}{} \quad \Delta = -115, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
13221812313\overset{2}{} \quad \Delta = -113, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
13221812313\overset{2}{} \quad \Delta = -113, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\text{Base: } 22221812321 \\
22221812321\overset{2}{3} \text{ }_{[\# = 2]} \quad \Delta = -106, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
22221812321\overset{2}{3} \text{ }_{[\# = 3]} \quad \Delta = -106, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
22221812321\overset{3}{\text{[}\#\text{=4}\text{]}} \quad \Delta = -106, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
22221812321\overset{3}{\text{[}\#\text{=5}\text{]}} \quad \Delta = -106, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
22221812321\overset{3}{\text{[}\#\text{=9}\text{]}} \quad \Delta = -106, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
22221812321\overset{3}{\text{[}\#\text{=8}\text{]}} \quad \Delta = -106, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
22221812321\overset{3}{\text{[}\#\text{=7}\text{]}} \quad \Delta = -106, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
22221812321\overset{3}{\text{[}\#\text{=6}\text{]}} \quad \Delta = -106, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
22221812321\overset{2}{\text{[}\#\text{=5}\text{]}} \quad \Delta = -98, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
22221812321\overset{2}{\text{[}\#\text{=4}\text{]}} \quad \Delta = -98, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 41232181222

$$\begin{aligned}
41232181222\overset{2}{\text{[}\#\text{=4}\text{]}} \quad \Delta = -126, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
41232181222\overset{3}{\text{[}\#\text{=3}\text{]}} \quad \Delta = -134, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
41232181222\overset{2}{\text{[}\#\text{=2}\text{]}} \quad \Delta = -126, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
41232181222\overset{2}{\text{[}\#\text{=1}\text{]}} \quad \Delta = -126, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1322181232¹

$$\begin{aligned}
1322181232\overset{2}{\text{[}\#\text{=3}\text{]}}\overset{1}{\text{[}\#\text{=2}\text{]}} \quad \Delta = -107, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2\overset{3}{\text{[}\#\text{=2}\text{]}}\overset{1}{\text{[}\#\text{=1}\text{]}}\overset{1}{\text{[}\#\text{=0}\text{]}} \quad \Delta = -115, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1322181232\overset{3}{\text{[}\#\text{=2}\text{]}}\overset{1}{\text{[}\#\text{=1}\text{]}} \quad \Delta = -115, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2\overset{3}{\text{[}\#\text{=1}\text{]}}\overset{1}{\text{[}\#\text{=0}\text{]}}\overset{1}{\text{[}\#\text{=0}\text{]}} \quad \Delta = -115, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1322181232\overset{3}{\text{[}\#\text{=1}\text{]}}\overset{1}{\text{[}\#\text{=0}\text{]}} \quad \Delta = -115, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2\overset{3}{\text{[}\#\text{=0}\text{]}}\overset{1}{\text{[}\#\text{=0}\text{]}}\overset{1}{\text{[}\#\text{=0}\text{]}} \quad \Delta = -115, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1322181232\overset{4}{\text{[}\#\text{=0}\text{]}} \quad \Delta = -135, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}
\end{aligned}$$

$$2 \overset{3}{1} 3 \overset{2}{1} 1 \text{ } [\# = 5] \quad \Delta = -115, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1322181232 \overset{2}{1} \quad \Delta = -107, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1322181232 \overset{21}{1} \quad \Delta = -107, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: 31321812313

$$3 \overset{3}{1} 321812313 \text{ } [\# = 2] \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$3 \overset{3}{1} 321812313 \quad \Delta = -129, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$3 \overset{2}{1} 321812313 \quad \Delta = -123, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$3 \overset{2}{1} 321812313 \quad \Delta = -121, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$3 \overset{2}{1} 321812313 \quad \Delta = -121, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 13218123132

$$231321812313 \overset{3}{1} \text{ } [\# = 3] \quad \Delta = -122, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$231321812313 \overset{3}{1} \text{ } [\# = 4] \quad \Delta = -122, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$231321812313 \overset{3}{1} \text{ } [\# = 5] \quad \Delta = -122, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$231321812313 \overset{3}{1} \text{ } [\# = 2] \quad \Delta = -122, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$13218123132 \overset{3}{1} \quad \Delta = -122, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$13218123132 \overset{2}{1} \quad \Delta = -114, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$13218123132 \overset{2}{1} \quad \Delta = -114, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$13218123132 \overset{2}{1} \quad \Delta = -114, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$13218123132 \overset{2}{1} \quad \Delta = -116, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$13218123132 \overset{2}{1} \quad \Delta = -116, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: $\overset{31321812}{2 \text{ } 3 \text{ } 1}$

$$\begin{aligned}
& 2^{\overline{31321812}} \quad 3 \quad 1 \quad \Delta = -117, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& 2^{\overline{31321812}} \quad 3 \quad 1 \quad \textcolor{red}{3} \text{ [#=2]} \quad \Delta = -123, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& 2^{\overline{31321812}} \quad 3 \quad 1 \text{ [#=2]} \quad \Delta = -123, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 2^{\overline{31321812}} \quad 3 \quad 1 \quad \Delta = -123, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& 2^{\overline{31321812}} \quad 3 \quad 1 \quad \Delta = -115, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 2^{\overline{31321812}} \quad 3 \quad 1 \quad \Delta = -115, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 2^{\overline{31321812}} \quad 3 \quad 1 \quad \Delta = -115, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 2^{\overline{31321812}} \quad 3 \quad 1 \quad \textcolor{red}{2} \quad \Delta = -115, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 2^{\overline{31321812}} \quad 3 \quad 1 \quad \textcolor{red}{3} \text{ [#=3]} \quad \Delta = -123, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 60 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& 2^{\overline{31321812}} \quad 3 \quad 1 \quad \textcolor{red}{4} \quad \Delta = -143, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \textbf{Base: } 2^{\overline{2321812}} \quad 3 \quad 1 \\
& 2^{\overline{2321812}} \quad 3 \quad 1 \quad \textcolor{red}{3} \text{ [#=3]} \quad \Delta = -117, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 60 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& 2^{\overline{2321812}} \quad 3 \quad 1 \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 2^{\overline{2321812}} \quad 3 \quad 1 \quad \textcolor{red}{2} \quad \Delta = -109, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 2^{\overline{2321812}} \quad 3 \quad 1 \quad \textcolor{red}{4} \quad \Delta = -137, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 2^{\overline{2321812}} \quad 3 \quad 1 \quad \textcolor{red}{3} \text{ [#=2]} \quad \Delta = -117, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \textbf{Base: } 1812231413 \\
& 1812231413 \quad \textcolor{red}{3} \quad \Delta = -140, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& 1812231413 \quad \textcolor{red}{12} \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 1812231413 \quad \textcolor{red}{2} \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

$${}^1_{1812231413}{}^2 \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: ${}^1_{1812314132}$

$${}^1{}^2_{1812314132} \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$${}^1{}^2_{1812314132} \quad \Delta = -135, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$${}^1_{1812314132}{}^2 \quad \Delta = -135, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$${}^1_{1812314132}{}^2 \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$${}^1_{1812314132}{}^2 \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: ${}^{21}_{181223141}$

$${}^{21}_{181223141} \quad \Delta = -124, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$${}^{21}{}^2_{181223141} \quad \Delta = -124, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$${}^{21}_{181223141}{}^2 \quad \Delta = -124, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$${}^{21}_{181223141}{}^3 \text{ [#=3]} \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$${}^{21}_{181223141}{}^3 \text{ [#=2]} \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

Base: ${}^{21}_{181231413}$

$${}^{21}_{181231413}{}^2 \quad \Delta = -134, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$${}^{21}{}^2_{181231413} \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$${}^{21}_{181231413} \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$${}^{21}_{181231413}{}^3 \quad \Delta = -140, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$${}^{21}_{181231413}{}^2 \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$${}^{21}_{181231413}{}^2 \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: ${}^{221}_{412321} 8 \ 1$

$$412321 \overset{3}{8} 1 \overset{221}{} \quad \Delta = -134, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$412321 \overset{2}{8} 1 \overset{221}{} \quad \Delta = -126, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$412321 \overset{221}{} 8 1 \quad \Delta = -126, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$412321 \overset{221}{} 8 1 \quad \Delta = -135, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$412321 \overset{2221}{} 8 1 \quad \Delta = -126, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: $413221 \overset{321}{} 8 1$

$$413221 \overset{2321}{} 8 1 \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$413221 \overset{2}{8} 1 \overset{321}{} \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$413221 \overset{321}{} 8 1 \overset{2}{} \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$413221 \overset{321}{} 8 1 \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: $1 \overset{221}{} 8 123141$

$$1 \overset{221}{} 8 123141 \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \overset{221}{} 8 123141 \overset{3}{\text{red}} \text{ [#=2]} \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$1 \overset{221}{} 8 123141 \overset{3}{\text{red}} \text{ [#=3]} \quad \Delta = -132, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$1 \overset{221}{} 8 123141 \overset{2}{\text{red}} \quad \Delta = -126, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{221}{\text{red}} 8 123141 \quad \Delta = -124, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{221}{} 8 123141 \overset{2}{\text{red}} \quad \Delta = -124, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{221}{} 8 123141 \quad \Delta = -124, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

Base: $1 \overset{13221}{} 8 1231$

$$1 \overset{13221}{} 8 1231 \overset{3}{\text{red}} \text{ [#=2]} \quad \Delta = -113, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \overset{13221}{} 8 1231 \overset{3}{\text{red}} \text{ [#=5]} \quad \Delta = -113, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$\begin{array}{ll}
1 \overset{313221}{8} 1231_{[\# = 5]} \quad \Delta = -113, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1 \overset{313221}{8} 1231_{[\# = 2]} \quad \Delta = -113, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
1 \overset{13221}{8} 1231\overset{3}{\color{red}}_{[\# = 3]} \quad \Delta = -113, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{313221}{8} 1231_{[\# = 4]} \quad \Delta = -113, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1 \overset{13221}{8} 1231\overset{3}{\color{red}}_{[\# = 4]} \quad \Delta = -113, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \overset{313221}{8} 1231_{[\# = 3]} \quad \Delta = -113, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1 \overset{13221\overset{2}{\color{red}}}{8} 1231 \quad \Delta = -105, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \overset{13221}{8} 1231 \quad \Delta = -105, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{12321}{8} 1222 & \\
1 \overset{312321}{8} 1222_{[\# = 2]} \quad \Delta = -106, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1 \overset{312321}{8} 1222_{[\# = 4]} \quad \Delta = -106, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \overset{312321}{8} 1222_{[\# = 3]} \quad \Delta = -106, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1 \overset{12321\overset{2}{\color{red}}}{8} 1222 \quad \Delta = -98, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{312321}{8} 1222_{[\# = 5]} \quad \Delta = -106, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1 \overset{312321}{8} 1222_{[\# = 6]} \quad \Delta = -106, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \overset{312321}{8} 1222_{[\# = 7]} \quad \Delta = -106, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1 \overset{312321}{8} 1222_{[\# = 8]} \quad \Delta = -106, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1 \overset{12321}{8} 1222 \quad \Delta = -98, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1 \overset{312321}{8} 1222_{[\# = 9]} \quad \Delta = -106, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
\text{Base: } 41321 \overset{3221}{8} 1 &
\end{array}$$

$$41321 \overset{2}{8} \overset{3221}{1} \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$41321 \overset{2}{8} \overset{3221}{1} \quad \Delta = -135, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$41321 \overset{2}{8} \overset{3221}{1} \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$41321 \overset{3221}{8} \overset{12}{1} \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$41321 \overset{3221}{8} \overset{1}{1} \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$\text{Base: } \overset{31321}{1} \overset{8}{8} \overset{1231}{1}$$

$$\overset{31321}{1} \overset{8}{8} \overset{1231}{1} \overset{2}{1} \quad \Delta = -113, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\overset{31321}{1} \overset{8}{8} \overset{1231}{1} \quad \Delta = -121, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$\overset{31321}{1} \overset{8}{8} \overset{1231}{1} \overset{3}{3} \text{ [#=5]} \quad \Delta = -121, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$\overset{31321}{1} \overset{8}{8} \overset{1231}{1} \overset{3}{3} \text{ [#=2]} \quad \Delta = -121, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\overset{31321}{1} \overset{8}{8} \overset{1231}{1} \text{ [#=2]} \quad \Delta = -121, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\overset{31321}{1} \overset{8}{8} \overset{1231}{1} \overset{3}{3} \text{ [#=3]} \quad \Delta = -121, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$\overset{31321}{1} \overset{8}{8} \overset{1231}{1} \overset{3}{3} \text{ [#=4]} \quad \Delta = -121, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$\overset{31321}{1} \overset{8}{8} \overset{1231}{1} \quad \Delta = -115, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$\overset{31321}{1} \overset{8}{8} \overset{1231}{1} \quad \Delta = -113, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\overset{31321}{1} \overset{8}{8} \overset{1231}{1} \quad \Delta = -113, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\text{Base: } \overset{2321}{1} \overset{8}{8} \overset{1231}{1}$$

$$\overset{2321}{1} \overset{8}{8} \overset{1231}{1} \overset{3}{3} \text{ [#=4]} \quad \Delta = -115, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$\overset{2321}{1} \overset{8}{8} \overset{1231}{1} \text{ [#=3]} \quad \Delta = -115, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$\overset{2321}{1} \overset{8}{8} \overset{1231}{1} \overset{2}{1} \quad \Delta = -107, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\begin{aligned}
& \overset{2321}{1 \ 8 \ 1231} \quad \Delta = -107, \ n_+ = 1, \ n_- = 11, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{2321}{1 \ 8 \ 1231} \quad \Delta = -107, \ n_+ = 1, \ n_- = 11, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{2321}{1 \ 8 \ 1231\textcolor{red}{3} \ [\# = 3]} \quad \Delta = -115, \ n_+ = 1, \ n_- = 11, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{2321}{1 \ 8 \ 1231 \ [\# = 2]} \quad \Delta = -115, \ n_+ = 1, \ n_- = 11, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{2321}{1 \ 8 \ 1231\textcolor{red}{3} \ [\# = 5]} \quad \Delta = -115, \ n_+ = 1, \ n_- = 11, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{2321}{1 \ 8 \ 1231\textcolor{red}{3} \ [\# = 2]} \quad \Delta = -115, \ n_+ = 1, \ n_- = 11, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{2321}{1 \ 8 \ 1231} \quad \Delta = -135, \ n_+ = 1, \ n_- = 11, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}
\end{aligned}$$

Base: $\overset{21}{412321812}$

$$\begin{aligned}
& \overset{\textcolor{red}{2}}{412321\overset{21}{812}} \quad \Delta = -126, \ n_+ = 1, \ n_- = 11, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 412321\overset{21\textcolor{red}{2}}{812} \quad \Delta = -126, \ n_+ = 1, \ n_- = 11, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 412321\overset{\textcolor{red}{2}21}{812} \quad \Delta = -126, \ n_+ = 1, \ n_- = 11, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{\textcolor{red}{3}}{412321\overset{21}{812}} \quad \Delta = -134, \ n_+ = 1, \ n_- = 11, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{21}{218123141}$

$$\begin{aligned}
& \overset{21\textcolor{red}{2}}{218123141} \quad \Delta = -124, \ n_+ = 1, \ n_- = 11, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{21}{218123141\textcolor{red}{3} \ [\# = 3]} \quad \Delta = -132, \ n_+ = 1, \ n_- = 11, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& \overset{21}{218123141\textcolor{red}{3} \ [\# = 2]} \quad \Delta = -132, \ n_+ = 1, \ n_- = 11, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& \overset{\textcolor{red}{2}21}{218123141} \quad \Delta = -124, \ n_+ = 1, \ n_- = 11, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{21}{218123141\textcolor{red}{2}} \quad \Delta = -124, \ n_+ = 1, \ n_- = 11, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{21}{218123141\textcolor{red}{2}} \quad \Delta = -126, \ n_+ = 1, \ n_- = 11, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{12321}{21 \ 8 \ 122}$

$$\overset{\textcolor{red}{3}12321}{21 \ 8 \ 122 \ [\# = 4]} \quad \Delta = -106, \ n_+ = 1, \ n_- = 11, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\begin{aligned}
21 \overset{3}{1}2321 \quad 22 \text{ [#}=6] \quad \Delta = -106, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21 \overset{3}{1}2321 \quad 22 \text{ [#}=5] \quad \Delta = -106, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
21 \overset{1}{2}321 \overset{3}{1} \quad 22 \quad \Delta = -107, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21 \overset{3}{1}2321 \quad 22 \text{ [#}=7] \quad \Delta = -106, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
21 \overset{3}{1}2321 \quad 22 \text{ [#}=8] \quad \Delta = -106, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
21 \overset{3}{1}2321 \quad 22 \text{ [#}=9] \quad \Delta = -106, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
21 \overset{2}{1}2321 \quad 22 \quad \Delta = -98, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21 \overset{1}{2}321 \overset{2}{1} \quad 22 \quad \Delta = -98, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21 \overset{3}{1}2321 \quad 22 \text{ [#}=3] \quad \Delta = -106, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
21 \overset{1}{2}321 \quad 22 \quad \Delta = -98, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21 \overset{3}{1}2321 \quad 22 \text{ [#}=2] \quad \Delta = -106, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\text{Base: } 41321 \overset{321}{8} 12 \\
41321 \overset{321}{8} 12 \overset{2}{1} \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
41321 \overset{2}{8} 12 \overset{321}{1} \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
41321 \overset{321}{8} 12 \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
41321 \overset{2}{8} 12 \overset{321}{1} \quad \Delta = -133, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
41321 \overset{2}{8} 12 \overset{321}{1} \quad \Delta = -135, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\text{Base: } 21 \overset{1321}{8} 1231 \\
21 \overset{1321}{8} 1231 \overset{3}{1} \text{ [#}=5] \quad \Delta = -113, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
21 \overset{2}{1}321 \quad 21 \quad \Delta = -105, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

$$21 \overset{1321}{8} 1231 \quad \Delta = -105, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$21 \overset{1321}{8} 1231\overset{\color{red}{3}}{[\# = 4]} \quad \Delta = -113, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$21 \overset{1321}{8} 1231\overset{\color{red}{3}}{[\# = 3]} \quad \Delta = -113, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$21 \overset{1321}{8} 1231\overset{\color{red}{3}}{[\# = 2]} \quad \Delta = -113, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: $1321 \overset{321}{8} 123$

$$321 \overset{321}{8} 1231\overset{\color{red}{3}}{[\# = 2]} \quad \Delta = -122, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1321 \overset{\color{red}{2}321}{8} 123 \quad \Delta = -114, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$321 \overset{321}{8} 1231\overset{\color{red}{2}}{} \quad \Delta = -114, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1321 \overset{321\color{red}{2}}{8} 123 \quad \Delta = -114, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$321 \overset{321}{8} 1231\overset{\color{red}{3}}{[\# = 3]} \quad \Delta = -122, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$321 \overset{321}{8} 1231\overset{\color{red}{3}}{[\# = 4]} \quad \Delta = -122, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$321 \overset{321}{8} 1231\overset{\color{red}{3}}{[\# = 5]} \quad \Delta = -122, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$321 \overset{321}{8} 1231\overset{\color{red}{3}}{} \quad \Delta = -122, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1321 \overset{\color{red}{2}}{8} \overset{321}{123} \quad \Delta = -116, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$321 \overset{321}{8} 1231\overset{\color{red}{5}}{} \quad \Delta = -166, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$321 \overset{321}{8} 1231\overset{\color{red}{4}}{} \quad \Delta = -142, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: $1 \overset{221,12321}{8} 1$

$$1 \overset{221,12321}{8} 1 \quad \Delta = -107, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$1 \overset{221,\color{red}{3}12321}{8} 1 [\# = 7] \quad \Delta = -106, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$1 \overset{221,12321}{8} 1 \quad \Delta = -98, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{221,12321}{8} 1 \quad \Delta = -98, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\begin{aligned}
1 \overset{221, \textcolor{red}{3}12321}{8} 1 \text{ [#}=2] \quad \Delta = -106, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \overset{221, \textcolor{red}{3}12321}{8} 1 \text{ [#}=3] \quad \Delta = -106, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \overset{221, \textcolor{red}{3}12321}{8} 1 \text{ [#}=4] \quad \Delta = -106, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1 \overset{221, \textcolor{red}{3}12321}{8} 1 \text{ [#}=5] \quad \Delta = -106, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1 \overset{221, \textcolor{red}{3}12321}{8} 1 \text{ [#}=6] \quad \Delta = -106, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1 \overset{221, \textcolor{red}{3}12321}{8} 1 \text{ [#}=9] \quad \Delta = -106, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1 \overset{221, \textcolor{red}{3}12321}{8} 1 \text{ [#}=8] \quad \Delta = -106, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{1,1321}{8} 1231$

$$\begin{aligned}
1 \overset{1,1321 \textcolor{red}{2}}{8} 1231 \quad \Delta = -105, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \overset{1,1321}{8} 1231 \textcolor{red}{3} \text{ [#}=5] \quad \Delta = -113, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1 \overset{1,1321}{8} 1231 \textcolor{red}{3} \text{ [#}=3] \quad \Delta = -113, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{1,1321}{8} 1231 \textcolor{red}{3} \text{ [#}=2] \quad \Delta = -113, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1 \overset{1,1321}{8} 1231 \textcolor{red}{3} \text{ [#}=4] \quad \Delta = -113, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{21,12321}{8} 12$

$$\begin{aligned}
1 \overset{21, \textcolor{red}{3}12321}{8} 12 \text{ [#}=6] \quad \Delta = -106, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1 \overset{21,12321 \textcolor{red}{2}}{8} 12 \quad \Delta = -98, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1 \overset{21,12321}{8} 12 \quad \Delta = -98, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \overset{21, \textcolor{red}{3}12321}{8} 12 \text{ [#}=3] \quad \Delta = -106, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \overset{21, \textcolor{red}{3}12321}{8} 12 \text{ [#}=4] \quad \Delta = -106, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1 \overset{21, \textcolor{red}{3}12321}{8} 12 \text{ [#}=5] \quad \Delta = -106, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1^{21, \textcolor{red}{3}12321} 8^{12} [\# = 7] \quad \Delta = -106, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1^{21, \textcolor{red}{3}12321} 8^{12} [\# = 8] \quad \Delta = -106, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1^{21, \textcolor{red}{3}12321} 8^{12} [\# = 9] \quad \Delta = -106, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1^{21, \textcolor{red}{3}12321} 8^{12} [\# = 2] \quad \Delta = -106, n_+ = 1, n_- = 11, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

$T = 12$ (1502 bases)

Base: 1222222222²

$$\begin{aligned}
2^{22222222221\textcolor{red}{6}} \quad \Delta = -3, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
2^{22222222221\textcolor{red}{12}} \quad \Delta = -173, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
2^{22222222221\textcolor{red}{7}} \quad \Delta = -30, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2^{22222222221\textcolor{red}{8}} \quad \Delta = -58, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: 123141413221

$$\begin{aligned}
122314141321\textcolor{red}{3} [\# = 3] \quad \Delta = -7, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
123141413221\textcolor{red}{3} [\# = 2] \quad \Delta = -7, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
122314141321\textcolor{red}{3} [\# = 8] \quad \Delta = -7, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
123141413221\textcolor{red}{3} [\# = 4] \quad \Delta = -7, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
123141413221\textcolor{red}{3} [\# = 3] \quad \Delta = -7, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
123141413221\textcolor{red}{3} [\# = 8] \quad \Delta = -7, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
122314141321\textcolor{red}{3} [\# = 9] \quad \Delta = -7, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
122314141321\textcolor{red}{3} [\# = 2] \quad \Delta = -7, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
122314141321\textcolor{red}{3} [\# = 4] \quad \Delta = -7, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
123141413221\textcolor{red}{3} [\# = 7] \quad \Delta = -7, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
122314141321\bar{3} \text{ }_{[\# = 7]} \quad \Delta = -7, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
123141413221\bar{3} \text{ }_{[\# = 6]} \quad \Delta = -7, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
123141413221\bar{3} \text{ }_{[\# = 5]} \quad \Delta = -7, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
122314141321\bar{3} \text{ }_{[\# = 6]} \quad \Delta = -7, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
123141413221\bar{3} \text{ }_{[\# = 9]} \quad \Delta = -7, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
122314141321\bar{3} \text{ }_{[\# = 5]} \quad \Delta = -7, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix}
\end{aligned}$$

Base: $122314141\bar{3}2$

$$\begin{aligned}
2 \overset{\bar{3}}{122314141} 2 \text{ }_{[\# = 4]} \quad \Delta = -9, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
2 \overset{5}{122314141} 2 \quad \Delta = -53, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1223141\overset{2}{4}1\bar{3}2 \quad \Delta = -1, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2 \overset{\bar{3}}{122314141} 2 \text{ }_{[\# = 7]} \quad \Delta = -9, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 196 \quad & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 389 & 55 \\ 2 & 55 & 8 \end{pmatrix} \\
2 \overset{\bar{3}}{122314141} 2 \text{ }_{[\# = 2]} \quad \Delta = -9, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
2 \overset{\bar{3}}{122314141} 2 \text{ }_{[\# = 6]} \quad \Delta = -9, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 144 \quad & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 285 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
2 \overset{\bar{3}}{122314141} 2 \text{ }_{[\# = 5]} \quad \Delta = -9, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 100 \quad & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 197 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
2 \overset{\bar{3}}{122314141} 2 \quad \Delta = -9, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
2 \overset{\bar{3}}{122314141} 2 \text{ }_{[\# = 8]} \quad \Delta = -9, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 256 \quad & \begin{pmatrix} 0 & 16 & 2 \\ 16 & 509 & 63 \\ 2 & 63 & 8 \end{pmatrix} \\
2 \overset{\bar{3}}{122314141} 2 \text{ }_{[\# = 3]} \quad \Delta = -9, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
2 \overset{2}{122314141} 2 \quad \Delta = -1, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1223141\overset{2}{4}1\bar{3}2 \quad \Delta = -1, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1223141\overset{2}{4}1\bar{3}2 \quad \Delta = -1, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$2^{\overset{7}{1}22314141} 3^{\quad} 2^{\quad} \quad \Delta = -106, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$2^{\overset{8}{1}22314141} 3^{\quad} 2^{\quad} \quad \Delta = -134, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$2^{\overset{4}{1}22314141} 3^{\quad} 2^{\quad} \quad \Delta = -29, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$2^{\overset{3}{1}22314141} 3^{\quad} 2^{\quad} [\# = 9] \quad \Delta = -9, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 324 \quad \begin{pmatrix} 0 & 18 & 2 \\ 18 & 645 & 71 \\ 2 & 71 & 8 \end{pmatrix}$$

$$2^{\overset{6}{1}22314141} 3^{\quad} 2^{\quad} \quad \Delta = -79, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$2^{\overset{12}{1}22314141} 3^{\quad} 2^{\quad} \quad \Delta = -249, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -4 & -2 \\ 2 & -2 & 8 \end{pmatrix}$$

Base: 214141413221

$$214141413221\overset{3}{3} [\# = 4] \quad \Delta = -26, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$214141413221\overset{3}{3} [\# = 3] \quad \Delta = -26, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$214141413221\overset{3}{3} [\# = 5] \quad \Delta = -26, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$214141413221\overset{3}{3} [\# = 6] \quad \Delta = -26, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$214141413221\overset{3}{3} [\# = 7] \quad \Delta = -26, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$214141413221\overset{3}{3} [\# = 2] \quad \Delta = -26, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$214141413221\overset{3}{3} [\# = 8] \quad \Delta = -26, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$214141413221\overset{3}{3} [\# = 9] \quad \Delta = -26, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix}$$

$$214141413221\overset{3}{3} \quad \Delta = -26, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$214141413221\overset{3}{3} [\# = 2] \quad \Delta = -26, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$214141413221\overset{3}{3} [\# = 3] \quad \Delta = -26, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 123 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$214141413221\overset{4}{4} \quad \Delta = -46, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$214141413221\overset{2}{2} \quad \Delta = -18, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\begin{aligned}
214141413221 \quad \Delta = -18, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
214141413221 \quad \Delta = -18, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
214141413221 \quad \Delta = -18, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: 131414141321

$$\begin{aligned}
123141414131\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -34, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
123141414131\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -34, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
131414141321\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -34, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
131414141321\textcolor{red}{3} \text{ [#=6]} \quad \Delta = -34, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
131414141321\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -34, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
123141414131\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -34, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
131414141321\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -34, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
123141414131\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -34, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
131414141321\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -34, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
123141414131\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -34, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
131414141321\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -34, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
131414141321 \quad \Delta = -34, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
123141414131\textcolor{red}{3} \text{ [#=6]} \quad \Delta = -34, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
131414141321\textcolor{red}{3} \text{ [#=8]} \quad \Delta = -34, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
123141414131\textcolor{red}{3} \text{ [#=8]} \quad \Delta = -34, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
131414141321\textcolor{red}{3} \text{ [#=7]} \quad \Delta = -34, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
123141414131\overset{\textcolor{red}{3}}{3} \text{ [#=7]} \quad \Delta = -34, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
131414141321 \quad \Delta = -26, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
131414141321 \quad \Delta = -26, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
131414141321 \quad \Delta = -26, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
131414141321 \quad \Delta = -26, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\textbf{Base: } 13141414132^2 \\
2^{\overset{\textcolor{red}{3}}{3}131414141} 2 \text{ [#=5]} \quad \Delta = -36, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 100 \quad & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 172 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
2^{\overset{\textcolor{red}{3}}{3}131414141} 2 \text{ [#=6]} \quad \Delta = -36, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 144 \quad & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 249 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
2^{\overset{\textcolor{red}{3}}{3}131414141} 2 \text{ [#=7]} \quad \Delta = -36, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 196 \quad & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 340 & 48 \\ 2 & 48 & 8 \end{pmatrix} \\
2^{\overset{\textcolor{red}{3}}{3}131414141} 2 \quad \Delta = -36, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2^{\overset{\textcolor{red}{5}}{5}131414141} 2 \quad \Delta = -80, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2^{\overset{\textcolor{red}{4}}{4}131414141} 2 \quad \Delta = -56, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
13141414132^{\overset{\textcolor{red}{3}}{3}} \quad \Delta = -36, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
2^{\overset{\textcolor{red}{3}}{3}131414141} 2 \text{ [#=9]} \quad \Delta = -36, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 324 \quad & \begin{pmatrix} 0 & 18 & 2 \\ 18 & 564 & 62 \\ 2 & 62 & 8 \end{pmatrix} \\
2^{\overset{\textcolor{red}{3}}{3}131414141} 2 \text{ [#=8]} \quad \Delta = -36, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 256 \quad & \begin{pmatrix} 0 & 16 & 2 \\ 16 & 445 & 55 \\ 2 & 55 & 8 \end{pmatrix} \\
2^{\overset{\textcolor{red}{3}}{3}131414141} 2 \text{ [#=3]} \quad \Delta = -36, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 60 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
2^{\overset{\textcolor{red}{3}}{3}131414141} 2 \text{ [#=4]} \quad \Delta = -36, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 109 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
2^{\overset{\textcolor{red}{6}}{6}131414141} 2 \quad \Delta = -106, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2^{\overset{\textcolor{red}{2}}{2}131414141} 2 \quad \Delta = -28, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
13141414132^{\overset{\textcolor{red}{2}}{2}} \quad \Delta = -28, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 20 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

$$13141414132^{\overset{2}{2}} \quad \Delta = -28, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$13141414132^{\overset{2}{2}} \quad \Delta = -28, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$13141414132^{\overset{2}{2}} \quad \Delta = -28, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$2^{\overset{3}{3}} 131414141^{\overset{2}{2}} \quad \Delta = -36, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$13141414132^{\overset{2}{2}} \quad \Delta = -29, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 214141414131

$$214141414131^{\overset{2}{2}} \quad \Delta = -45, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$214141414131^{\overset{2}{2}} \quad \Delta = -45, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$214141414131^{\overset{2}{2}} \quad \Delta = -45, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$214141414131\overset{3}{3} \quad \Delta = -53, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$214141414131^{\overset{3}{3}} \quad \Delta = -53, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 69 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$214141414131\overset{3}{3} \quad \Delta = -53, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$214141414131\overset{3}{3} \quad \Delta = -53, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$214141414131\overset{3}{3} \quad \Delta = -53, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$214141414131\overset{3}{3} \quad \Delta = -53, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$214141414131\overset{3}{3} \quad \Delta = -53, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$214141414131^{\overset{3}{3}} \quad \Delta = -53, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 159 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

$$214141414131^{\overset{3}{3}} \quad \Delta = -53, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$214141414131\overset{3}{3} \quad \Delta = -53, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$214141414131\overset{3}{3} \quad \Delta = -53, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$\begin{aligned}
\overset{4}{214141414131} \quad \Delta = -73, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 14 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\overset{3}{214141414131} \quad \Delta = -53, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 15 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
214141414131 \overset{2}{} \quad \Delta = -45, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
214141414131 \overset{2}{} \quad \Delta = -45, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{14141413221}$

$$\begin{aligned}
\overset{1}{14141413221} \overset{3}{}_{[\#=7]} \quad \Delta = -26, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
\overset{1}{14141413221} \overset{3}{}_{[\#=9]} \quad \Delta = -26, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
\overset{1}{14141413221} \overset{3}{}_{[\#=3]} \quad \Delta = -26, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\overset{1}{14141413221} \overset{3}{}_{[\#=8]} \quad \Delta = -26, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
\overset{1}{14141413221} \overset{3}{}_{[\#=6]} \quad \Delta = -26, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
\overset{1}{14141413221} \overset{3}{}_{[\#=5]} \quad \Delta = -26, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\overset{1}{14141413221} \overset{3}{}_{[\#=4]} \quad \Delta = -26, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\overset{31}{14141413221} [\#=3] \quad \Delta = -26, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\overset{1}{14141413221} \overset{3}{}_{[\#=2]} \quad \Delta = -26, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\overset{1}{14141413221} \overset{2}{} \quad \Delta = -18, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\overset{1}{14141413221} \overset{2}{} \quad \Delta = -18, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\overset{12}{14141413221} \quad \Delta = -18, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\overset{31}{14141413221} [\#=2] \quad \Delta = -26, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{14141414131}$

$$\overset{1}{14141414131} \overset{3}{}_{[\#=7]} \quad \Delta = -53, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\begin{aligned}
14141414131^{12} \quad \Delta = -45, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
14141414131^1 \quad \Delta = -53, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
14141414131^1_{\# = 9} \quad \Delta = -53, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
14141414131^{31}_{\# = 2} \quad \Delta = -53, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
14141414131^1_{\# = 8} \quad \Delta = -53, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
14141414131^1_{\# = 6} \quad \Delta = -53, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
14141414131^1_{\# = 5} \quad \Delta = -53, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
14141414131^1_{\# = 2} \quad \Delta = -53, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
14141414131^1_{\# = 3} \quad \Delta = -53, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
14141414131^{31}_{\# = 3} \quad \Delta = -53, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
14141414131^1_{\# = 4} \quad \Delta = -53, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
14141414131^1 \quad \Delta = -45, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
14141414131^1 \quad \Delta = -45, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
14141414131^1 \quad \Delta = -45, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 122315132215

$$\begin{aligned}
5122315132213_{\# = 6} \quad \Delta = -55, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
5122315132213_{\# = 5} \quad \Delta = -55, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
5122315132213_{\# = 9} \quad \Delta = -55, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
5122315132213_{\# = 4} \quad \Delta = -55, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
122315132215^3 \quad \Delta = -55, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
512231513221\overset{2}{3} \text{ [#=3]} \quad \Delta = -55, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
512231513221\overset{2}{3} \text{ [#=2]} \quad \Delta = -55, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
512231513221\overset{2}{3} \text{ [#=7]} \quad \Delta = -55, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
512231513221\overset{2}{3} \text{ [#=8]} \quad \Delta = -55, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
12231513221\overset{2}{5} \quad \Delta = -47, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
12231513221\overset{2}{5} \quad \Delta = -47, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
12231513221\overset{2}{5} \quad \Delta = -47, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 122315132151

$$\begin{aligned}
151231513221\overset{2}{3} \text{ [#=2]} \quad \Delta = -55, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
151231513221\overset{2}{3} \text{ [#=8]} \quad \Delta = -55, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
151231513221\overset{2}{3} \text{ [#=3]} \quad \Delta = -55, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
151231513221\overset{2}{3} \text{ [#=4]} \quad \Delta = -55, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
151231513221\overset{2}{3} \text{ [#=9]} \quad \Delta = -55, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix} \\
151231513221\overset{2}{3} \text{ [#=5]} \quad \Delta = -55, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
151231513221\overset{2}{3} \text{ [#=6]} \quad \Delta = -55, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1223151321\overset{2}{5} \quad \Delta = -47, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1223151321\overset{2}{5} \quad \Delta = -47, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1223151321\overset{2}{5} \quad \Delta = -47, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
151231513221\overset{2}{3} \text{ [#=7]} \quad \Delta = -55, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix}
\end{aligned}$$

Base: 122315131512

$$\begin{aligned}
215131513221\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -54, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
215131513221\textcolor{red}{3} \text{ [#=8]} \quad \Delta = -54, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
122315131512 \quad \Delta = -46, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
122315131512 \quad \Delta = -46, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
122315131512 \quad \Delta = -46, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
122315131512 \quad \Delta = -46, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
122315131512 \quad \Delta = -47, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
215131513221\textcolor{red}{3} \text{ [#=7]} \quad \Delta = -54, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
215131513221\textcolor{red}{3} \text{ [#=6]} \quad \Delta = -54, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
215131513221\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -54, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
215131513221\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -54, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix} \\
215131513221\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -54, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
215131513221\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -54, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
122315131512 \quad \Delta = -54, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}
\end{aligned}$$

Base: 123151315131

$$\begin{aligned}
123151315131\textcolor{red}{3} \text{ [#=6]} \quad \Delta = -62, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
131513151321\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -62, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
131513151321\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -62, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
131513151321\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -62, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
123151315131\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -62, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
123151315131\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -62, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
131513151321\textcolor{red}{3} \text{ [#=6]} \quad \Delta = -62, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
123151315131\textcolor{red}{2} \quad \Delta = -55, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
123151315131\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -62, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
123151315131\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -62, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
123151315131\textcolor{red}{3} \text{ [#=8]} \quad \Delta = -62, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
131513151321\textcolor{red}{3} \text{ [#=8]} \quad \Delta = -62, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
123151315131\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -62, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
131513151321\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -62, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix} \\
131513151321\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -62, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
131513151321\textcolor{red}{3} \text{ [#=7]} \quad \Delta = -62, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
123151315131\textcolor{red}{2} \quad \Delta = -54, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
123151315131\textcolor{red}{2} \quad \Delta = -54, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
123151315131\textcolor{red}{2} \quad \Delta = -54, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
123151315131\textcolor{red}{2} \quad \Delta = -54, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
123151315131\textcolor{red}{3} \text{ [#=7]} \quad \Delta = -62, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
\textbf{Base: } 15131513221\textcolor{red}{1} \\
15131513221\textcolor{red}{2} \quad \Delta = -46, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
15131513221\textcolor{red}{12} \quad \Delta = -46, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
15131513221\textcolor{red}{12} \quad \Delta = -46, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

$$15131513221\overset{1}{\mathbf{3}}_{[\# = 8]} \quad \Delta = -54, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix}$$

$$15131513221\overset{1}{\mathbf{2}} \quad \Delta = -47, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$15131513221\overset{1}{\mathbf{3}}_{[\# = 9]} \quad \Delta = -54, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix}$$

$$15131513221\overset{1}{\mathbf{3}}_{[\# = 2]} \quad \Delta = -54, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$15131513221\overset{1}{\mathbf{3}}_{[\# = 3]} \quad \Delta = -54, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$15131513221\overset{1}{\mathbf{3}}_{[\# = 4]} \quad \Delta = -54, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$15131513221\overset{1}{\mathbf{3}}_{[\# = 5]} \quad \Delta = -54, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

$$15131513221\overset{1}{\mathbf{3}}_{[\# = 6]} \quad \Delta = -54, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$15131513221\overset{1}{\mathbf{3}}_{[\# = 7]} \quad \Delta = -54, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix}$$

Base: 12231513221

$$12231513221\overset{1}{\mathbf{3}}_{[\# = 2]} \quad \Delta = -3, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$12231513221\overset{1}{\mathbf{3}}_{[\# = 9]} \quad \Delta = -3, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix}$$

$$12231513221\overset{1}{\mathbf{3}}_{[\# = 7]} \quad \Delta = -3, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix}$$

$$12231513221\overset{1}{\mathbf{3}}_{[\# = 6]} \quad \Delta = -3, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

$$12231513221\overset{1}{\mathbf{3}}_{[\# = 8]} \quad \Delta = -3, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix}$$

$$12231513221\overset{1}{\mathbf{3}}_{[\# = 4]} \quad \Delta = -3, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$12231513221\overset{1}{\mathbf{3}}_{[\# = 3]} \quad \Delta = -3, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$12231513221\overset{1}{\mathbf{3}}_{[\# = 5]} \quad \Delta = -3, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix}$$

Base: 612231513215

$$612231513215\overset{2}{\mathbf{3}} \quad \Delta = -125, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\begin{aligned}
612231513215^{\textcolor{red}{3}} \quad \Delta = -133, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 27 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
612231513215^{\textcolor{red}{2}} \quad \Delta = -125, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
612231513215^{\textcolor{red}{2}} \quad \Delta = -125, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
612231513215^{\textcolor{red}{2}} \quad \Delta = -125, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 612231513151

$$\begin{aligned}
612231513151^{\textcolor{red}{2}} \quad \Delta = -124, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
612231513151^{\textcolor{red}{2}} \quad \Delta = -124, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
612231513151^{\textcolor{red}{2}} \quad \Delta = -124, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
612231513151^{\textcolor{red}{3}} \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 63 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
612231513151^{\textcolor{red}{2}} \quad \Delta = -124, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
612231513151^{\textcolor{red}{2}} \quad \Delta = -125, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 612315131513

$$\begin{aligned}
612315131513^{\textcolor{red}{2}} \quad \Delta = -133, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
612315131513^{\textcolor{red}{2}} \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
612315131513^{\textcolor{red}{2}} \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
612315131513^{\textcolor{red}{2}} \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
612315131513^{\textcolor{red}{2}} \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
612315131513^{\textcolor{red}{2}} \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 613141414141

$$\begin{aligned}
613141414141^{\textcolor{red}{2}} \quad \Delta = -123, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
613141414141^{\textcolor{red}{3}} \text{ [#=3] } \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
613141414141 \quad \Delta = -123, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
613141414141 \quad \Delta = -123, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 142 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
613141414141 \quad \Delta = -123, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
613141414141 \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
613141414141 \textcolor{red}{3} \text{ [#=2]} \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
613141414141 \quad \Delta = -123, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: 613151315132

$$\begin{aligned}
613151315132 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
613151315132 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
613151315132 \quad \Delta = -133, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
613151315132 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
613151315132 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
613151315132 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 161223151321

$$\begin{aligned}
161223151321 \textcolor{red}{3} \text{ [#=4]} \quad \Delta = -81, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
161223151321 \textcolor{red}{3} \text{ [#=3]} \quad \Delta = -81, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
161223151321 \textcolor{red}{3} \text{ [#=9]} \quad \Delta = -81, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
161223151321 \textcolor{red}{3} \text{ [#=8]} \quad \Delta = -81, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
161223151321 \textcolor{red}{3} \text{ [#=7]} \quad \Delta = -81, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
161223151321 \textcolor{red}{3} \text{ [#=6]} \quad \Delta = -81, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix}
\end{aligned}$$

$$161223151321\overset{2}{3} \text{ [#=5]} \quad \Delta = -81, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$161223151321\overset{2}{2} \quad \Delta = -73, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$161223151321\overset{2}{1} \quad \Delta = -73, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$161223151321\overset{3}{3} \text{ [#=2]} \quad \Delta = -81, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$161223151321\overset{3}{2} \quad \Delta = -81, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 17 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$161223151321\overset{2}{1} \quad \Delta = -73, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: 161231513215

$$161231513215\overset{2}{5} \quad \Delta = -125, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$161231513215\overset{2}{4} \quad \Delta = -125, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$161231513215\overset{2}{3} \quad \Delta = -125, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$161231513215\overset{2}{2} \quad \Delta = -125, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 161231513151

$$161231513151\overset{2}{5} \quad \Delta = -124, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$161231513151\overset{2}{4} \quad \Delta = -125, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$161231513151\overset{2}{3} \quad \Delta = -124, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$161231513151\overset{2}{2} \quad \Delta = -124, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$161231513151\overset{2}{1} \quad \Delta = -124, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 161315131513

$$161315131513\overset{2}{3} \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$161315131513\overset{2}{2} \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$161315131513\overset{2}{1} \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$\begin{aligned}
161315131513^2 \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
161315131513^2 \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
161315131513^2 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: 216123151321

$$\begin{aligned}
2161231513213^{\#2} \quad \Delta = -81, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2161231513213^{\#7} \quad \Delta = -81, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
2161231513213^{\#8} \quad \Delta = -81, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
2161231513213^{\#9} \quad \Delta = -81, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
2161231513213^{\#6} \quad \Delta = -81, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
2161231513213^{\#5} \quad \Delta = -81, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
2161231513213^{\#4} \quad \Delta = -81, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
2161231513213^{\#3} \quad \Delta = -81, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
216123151321^3 \quad \Delta = -81, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
216123151321^2 \quad \Delta = -73, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
216123151321^2 \quad \Delta = -73, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
216123151321^2 \quad \Delta = -73, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
216123151321^2 \quad \Delta = -73, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 316123151315

$$\begin{aligned}
316123151315^2 \quad \Delta = -133, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
316123151315^2 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

$$316123151315 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$316123151315 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$316123151315 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$316123151315 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 216131513215

$$216131513215 \quad \Delta = -124, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$216131513215 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$216131513215 \quad \Delta = -124, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$216131513215 \quad \Delta = -124, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$216131513215 \quad \Delta = -124, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$216131513215 \quad \Delta = -124, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 216131513151

$$216131513151 \quad \Delta = -124, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$216131513151 \quad \Delta = -123, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$216131513151 \quad \Delta = -123, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$216131513151 \quad \Delta = -123, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$216131513151 \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$216131513151 \quad \Delta = -123, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$216131513151 \quad \Delta = -123, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 221613151321

$$221613151321 \quad \Delta = -72, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$\begin{aligned}
221613151321^2 \quad \Delta = -72, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
221613151321^3_{\# = 3} \quad \Delta = -80, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
221613151321^3_{\# = 4} \quad \Delta = -80, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
221613151321^3_{\# = 5} \quad \Delta = -80, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
221613151321^3_{\# = 6} \quad \Delta = -80, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
221613151321^3_{\# = 8} \quad \Delta = -80, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
221613151321^3 \quad \Delta = -81, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
221613151321^3_{\# = 2} \quad \Delta = -80, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
221613151321^2 \quad \Delta = -72, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
221613151321^2 \quad \Delta = -72, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
221613151321^3 \quad \Delta = -80, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
221613151321^3_{\# = 9} \quad \Delta = -80, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
221613151321^3_{\# = 7} \quad \Delta = -80, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
\text{Base: } 131612315131 & \\
131513216131^3_{\# = 9} \quad \Delta = -88, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
131612315131^3_{\# = 6} \quad \Delta = -88, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
131513216131^3_{\# = 7} \quad \Delta = -88, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
131612315131^3_{\# = 7} \quad \Delta = -88, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
131513216131^3_{\# = 8} \quad \Delta = -88, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
131612315131^3_{\# = 8} \quad \Delta = -88, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
131612315131\overset{2}{3} \text{ [#=9]} \quad \Delta = -88, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
131513216131\overset{2}{3} \text{ [#=6]} \quad \Delta = -88, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
131513216131\overset{2}{3} \text{ [#=3]} \quad \Delta = -88, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
131612315131\overset{2}{3} \quad \Delta = -80, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
131612315131\overset{2}{3} \text{ [#=3]} \quad \Delta = -88, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
131513216131\overset{2}{3} \text{ [#=4]} \quad \Delta = -88, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
131612315131\overset{2}{3} \text{ [#=5]} \quad \Delta = -88, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
131612315131\overset{2}{3} \text{ [#=2]} \quad \Delta = -88, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
131513216131\overset{2}{3} \text{ [#=5]} \quad \Delta = -88, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
131612315131\overset{2}{3} \text{ [#=4]} \quad \Delta = -88, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
131513216131\overset{2}{3} \text{ [#=2]} \quad \Delta = -88, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
131612315131\overset{2}{3} \quad \Delta = -80, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
131612315131\overset{2}{3} \quad \Delta = -80, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
131612315131\overset{2}{3} \quad \Delta = -80, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 321613151315

$$\begin{aligned}
321613151315\overset{2}{3} \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
321613151315\overset{2}{3} \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
321613151315\overset{2}{3} \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
321613151315\overset{2}{3} \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
321613151315\overset{2}{3} \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

$$321613151315^2 \quad \Delta = -133, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 413161232161

$$413161232161^2 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$413161232161^2 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$413161232161^2 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$413161232161^2 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$413161232161^3 \quad \Delta = -136, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

Base: 315132161315

$$315132161315^2 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$315132161315^2 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$315132161315^2 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$315132161315^2 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$315132161315^2 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: 132161315131

$$132161315131^3 \text{ [#=9]} \quad \Delta = -88, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$132161315131^3 \text{ [#=4]} \quad \Delta = -88, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$131513161231^3 \text{ [#=4]} \quad \Delta = -88, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$132161315131^3 \text{ [#=7]} \quad \Delta = -88, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$132161315131^3 \text{ [#=6]} \quad \Delta = -88, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$132161315131^3 \text{ [#=2]} \quad \Delta = -88, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$131513161231^3 \text{ [#=3]} \quad \Delta = -88, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$\begin{aligned}
132161315131^2 \quad \Delta = -80, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
132161315131^2 \quad \Delta = -80, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
132161315131^2 \quad \Delta = -80, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1315131612313_{\# = 2} \quad \Delta = -88, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1321613151313_{\# = 3} \quad \Delta = -88, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
132161315131^2 \quad \Delta = -80, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1315131612313_{\# = 5} \quad \Delta = -88, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1321613151313_{\# = 5} \quad \Delta = -88, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1321613151313_{\# = 8} \quad \Delta = -88, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

Base: 151321613151

$$\begin{aligned}
151321613151^2 \quad \Delta = -124, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
151321613151^2 \quad \Delta = -124, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
151321613151^2 \quad \Delta = -124, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
151321613151^2 \quad \Delta = -124, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 513216131513

$$\begin{aligned}
513216131513^2 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
513216131513^2 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 100 \quad & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
513216131513^2 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
513216131513^2 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
513216131513^2 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: 512321613151

$$\begin{aligned}
512321613151^2 \quad \Delta = -126, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
512321613151^2 \quad \Delta = -126, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
512321613151^2 \quad \Delta = -126, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
512321613151^2 \quad \Delta = -126, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 123216131512

$$\begin{aligned}
2151316123213_{\#3} \quad \Delta = -82, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
2151316123213_{\#4} \quad \Delta = -82, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
123216131512^2 \quad \Delta = -74, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2151316123213_{\#8} \quad \Delta = -82, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
2151316123213_{\#7} \quad \Delta = -82, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
2151316123213_{\#6} \quad \Delta = -82, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
123216131512^3 \quad \Delta = -82, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
2151316123213_{\#2} \quad \Delta = -82, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
123216131512^2 \quad \Delta = -74, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
123216131512^2 \quad \Delta = -74, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
123216131512^2 \quad \Delta = -74, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2151316123213_{\#9} \quad \Delta = -82, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
2151316123213_{\#5} \quad \Delta = -82, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
\text{Base: } 216123216131 \\
216123216131_{\#8} \quad \Delta = -108, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
216123216131^3 \quad \Delta = -108, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
216123216131\overset{2}{3} \text{ }_{[\#=7]} \quad \Delta = -108, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
216123216131\overset{2}{3} \text{ }_{[\#=6]} \quad \Delta = -108, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
216123216131\overset{2}{3} \text{ }_{[\#=5]} \quad \Delta = -108, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
216123216131\overset{2}{3} \text{ }_{[\#=4]} \quad \Delta = -108, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
216123216131\overset{2}{3} \text{ }_{[\#=3]} \quad \Delta = -108, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
216123216131\overset{2}{3} \text{ }_{[\#=2]} \quad \Delta = -108, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
216123216131\overset{2}{5} \quad \Delta = -100, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
216123216131\overset{2}{5} \quad \Delta = -100, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
216123216131\overset{2}{5} \quad \Delta = -100, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
216123216131\overset{2}{5} \quad \Delta = -100, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
216123216131\overset{2}{3} \text{ }_{[\#=9]} \quad \Delta = -108, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 123216131¹₅₁

$$\begin{aligned}
1\overset{3}{1}23216131\overset{1}{5} \text{ }_1 \text{ }_{[\#=7]} \quad \Delta = -82, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1\overset{3}{1}23216131\overset{1}{5} \text{ }_1 \text{ }_{[\#=4]} \quad \Delta = -82, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1\overset{3}{1}23216131\overset{1}{5} \text{ }_1 \text{ }_{[\#=8]} \quad \Delta = -82, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1\overset{3}{1}23216131\overset{1}{5} \text{ }_1 \text{ }_{[\#=6]} \quad \Delta = -82, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1\overset{3}{1}23216131\overset{1}{5} \text{ }_1 \text{ }_{[\#=5]} \quad \Delta = -82, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1\overset{3}{1}23216131\overset{1}{5} \text{ }_1 \text{ }_{[\#=2]} \quad \Delta = -82, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1\overset{3}{1}23216131\overset{1}{5} \text{ }_1 \text{ }_{[\#=3]} \quad \Delta = -82, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
123216131\overset{2}{5}\overset{1}{1} \quad \Delta = -74, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$1^{\overline{3}123216131} \quad \Delta = -82, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$123216131\overline{2}1 \quad \Delta = -74, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$123216131\overline{2}1 \quad \Delta = -74, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 413151316131

$$413151316131\overline{2} \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$413151316131\overline{3} \quad [\# = 7] \quad \Delta = -115, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$413151316131\overline{3} \quad [\# = 3] \quad \Delta = -115, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$413151316131\overline{3} \quad [\# = 4] \quad \Delta = -115, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$413151316131\overline{3} \quad [\# = 5] \quad \Delta = -115, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$413151316131\overline{3} \quad [\# = 6] \quad \Delta = -115, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$413151316131\overline{3} \quad [\# = 8] \quad \Delta = -115, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$413151316131\overline{3} \quad [\# = 9] \quad \Delta = -115, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$413151316131\overline{3} \quad [\# = 2] \quad \Delta = -115, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$413151316131\overline{3} \quad \Delta = -115, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$413151316131\overline{2} \quad \Delta = -108, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$413151316131\overline{2} \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$413151316131\overline{2} \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$413151316131\overline{2} \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$413151316131\overline{2} \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: 313161315131

$$\begin{aligned}
313161315131^2 \quad \Delta = -87, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
313161315131^3_{\# = 2} \quad \Delta = -95, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
313161315131^4 \quad \Delta = -115, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
313161315131^3_{\# = 3} \quad \Delta = -95, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 132 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
313161315131^3_{\# = 2} \quad \Delta = -95, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
313161315131^3_{\# = 7} \quad \Delta = -95, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
313161315131^3 \quad \Delta = -95, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
313161315131^3_{\# = 8} \quad \Delta = -95, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
313161315131^3_{\# = 9} \quad \Delta = -95, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
313161315131^3_{\# = 4} \quad \Delta = -95, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
313161315131^2 \quad \Delta = -87, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
313161315131^2 \quad \Delta = -87, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
313161315131^3_{\# = 5} \quad \Delta = -95, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
313161315131^3_{\# = 6} \quad \Delta = -95, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
313161315131^2 \quad \Delta = -87, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
313161315131^3_{\# = 3} \quad \Delta = -95, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
313161315131^2 \quad \Delta = -87, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\text{Base: } 414131613151 \\
414131613151^2 \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
414131613151^2 \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

$$414131613151^2 \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$414131613151^2 \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$414131613151^3 \quad \Delta = -135, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 37 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$414131613151^2 \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 141316131512

$$141316131512^3 \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$141316131512^2 \quad \Delta = -99, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$141316131512^2 \quad \Delta = -99, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$141316131512^2 \quad \Delta = -99, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$141316131512^2 \quad \Delta = -99, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$2151316131413^{\#=3} \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$141316131512^3 \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$2151316131413^{\#=2} \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$141316131512^2 \quad \Delta = -99, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 14131613151¹

$$14131613151^3 \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$14131613151^{21} \quad \Delta = -99, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$14131613151^{21} \quad \Delta = -99, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$14131613151^{21} \quad \Delta = -99, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1^{3141316131} 5^1 \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$14131613151^{21} \quad \Delta = -99, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1^{\overline{3141316131}} 5^{\overline{1}} 1_{[\# = 2]} \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: 413216131612

$$413216131612^{\overline{3}} \quad \Delta = -134, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$413216131612^{\overline{2}} \quad \Delta = -126, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$413216131612^{\overline{2}} \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$413216131612^{\overline{2}} \quad \Delta = -126, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$413216131612^{\overline{2}} \quad \Delta = -126, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$413216131612^{\overline{2}} \quad \Delta = -126, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$413216131612^{\overline{2}} \quad \Delta = -126, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: 221613161231

$$221613161231^{\overline{3}} \quad \Delta = -106, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$221613161231^{\overline{3}} \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$221613161231^{\overline{3}}_{[\# = 2]} \quad \Delta = -106, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$221613161231^{\overline{3}}_{[\# = 5]} \quad \Delta = -106, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$221613161231^{\overline{3}}_{[\# = 3]} \quad \Delta = -106, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$221613161231^{\overline{3}}_{[\# = 4]} \quad \Delta = -106, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$221613161231^{\overline{2}} \quad \Delta = -98, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$221613161231^{\overline{2}} \quad \Delta = -98, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$221613161231^{\overline{2}} \quad \Delta = -98, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$221613161231^{\overline{2}} \quad \Delta = -98, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

Base: 131613161223

$$\begin{aligned}
322161316131\overset{3}{3} \text{ }_{[\# = 2]} \quad \Delta = -114, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
322161316131\overset{3}{3} \text{ }_{[\# = 3]} \quad \Delta = -114, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
322161316131\overset{3}{3} \text{ }_{[\# = 4]} \quad \Delta = -114, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
322161316131\overset{3}{3} \text{ }_{[\# = 5]} \quad \Delta = -114, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
131613161223\overset{3}{3} \quad \Delta = -114, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
131613161223\overset{2}{2} \quad \Delta = -106, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
131613161223\overset{2}{2} \quad \Delta = -106, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
131613161223\overset{2}{2} \quad \Delta = -106, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
131613161223\overset{2}{2} \quad \Delta = -106, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
322161316131\overset{3}{3} \text{ }_{[\# = 7]} \quad \Delta = -114, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
322161316131\overset{3}{3} \text{ }_{[\# = 6]} \quad \Delta = -114, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
322161316131\overset{3}{3} \text{ }_{[\# = 8]} \quad \Delta = -114, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
322161316131\overset{3}{3} \text{ }_{[\# = 9]} \quad \Delta = -114, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix}
\end{aligned}$$

Base: 221613161313

$$\begin{aligned}
221613161313\overset{3}{3} \quad \Delta = -113, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
221613161313\overset{3}{3} \quad \Delta = -113, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
221613161313\overset{3}{3} \quad \Delta = -114, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
221613161313\overset{4}{4} \quad \Delta = -133, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 9 & 5 \\ 5 & 9 \end{pmatrix} \\
221613161313\overset{2}{2} \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
221613161313\overset{2}{2} \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
221613161313 \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
221613161313 \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
221613161313 \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
221613161313 \quad \Delta = -113, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
221613161313 \quad \Delta = -113, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{123151321}{6} 1 & \\
1 \overset{3123151321}{6} 1 \quad \Delta = -81, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
1 \overset{3123151321}{6} 1 \quad \Delta = -81, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1 \overset{3123151321}{6} 1 \quad \Delta = -81, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1 \overset{3123151321}{6} 1 \quad \Delta = -81, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \overset{3123151321}{6} 1 \quad \Delta = -81, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{123151321}{6} 1 \quad \Delta = -73, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1 \overset{3123151321}{6} 1 \quad \Delta = -81, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1 \overset{3123151321}{6} 1 \quad \Delta = -81, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1 \overset{3123151321}{6} 1 \quad \Delta = -81, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
1 \overset{123151321}{6} 1 \quad \Delta = -73, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1 \overset{123151321}{6} 1 \quad \Delta = -73, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\text{Base: } 16123216131 \overset{1}{1} & \\
16123216131 \overset{1}{1} \quad \Delta = -108, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
16123216131 \overset{1}{1} \quad \Delta = -108, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

$$16123216131\bar{3} \text{ }_{[\# = 8]} \quad \Delta = -108, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$16123216131\bar{3} \text{ }_{[\# = 2]} \quad \Delta = -108, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$16123216131\bar{3} \text{ }_{[\# = 3]} \quad \Delta = -108, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$16123216131\bar{3} \text{ }_{[\# = 4]} \quad \Delta = -108, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$16123216131\bar{2} \quad \Delta = -100, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$16123216131\bar{2} \quad \Delta = -100, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$16123216131\bar{12} \quad \Delta = -100, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$16123216131\bar{3} \text{ }_{[\# = 7]} \quad \Delta = -108, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$16123216131\bar{3} \text{ }_{[\# = 9]} \quad \Delta = -108, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \begin{smallmatrix} 512315131 \\ 6 \end{smallmatrix} 1$$

$$1 \begin{smallmatrix} 512315131 \\ 6 \end{smallmatrix} 1 \quad \Delta = -124, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \begin{smallmatrix} 512315131 \\ 6 \end{smallmatrix} 1 \quad \Delta = -124, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \begin{smallmatrix} 512315131 \\ 6 \end{smallmatrix} 1 \quad \Delta = -124, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \begin{smallmatrix} 512315131 \\ 6 \end{smallmatrix} 1 \quad \Delta = -124, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \begin{smallmatrix} 151315131 \\ 6 \end{smallmatrix} 1$$

$$1 \begin{smallmatrix} 151315131 \\ 6 \end{smallmatrix} 1 \quad \Delta = -124, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \begin{smallmatrix} 151315131 \\ 6 \end{smallmatrix} 1 \quad \Delta = -123, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \begin{smallmatrix} 151315131 \\ 6 \end{smallmatrix} 1 \quad \Delta = -123, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \begin{smallmatrix} 151315131 \\ 6 \end{smallmatrix} 1 \quad \Delta = -123, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \begin{smallmatrix} 151315131 \\ 6 \end{smallmatrix} 1 \quad \Delta = -123, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\text{Base: } 413216131\bar{1}$$

$$\begin{aligned}
41321613161^{\overset{2}{1}} \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
41321613161^{\overset{2}{1}} \quad \Delta = -126, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
41321613161^{\overset{2}{1}} \quad \Delta = -126, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
41321613161^{\overset{2}{1}} \quad \Delta = -126, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
41321613161^{\overset{2}{1}} \quad \Delta = -126, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1^{\overset{12315131}{6}} 12$

$$\begin{aligned}
1^{\overset{312315131}{6}} 12 \text{ [#=7]} \quad \Delta = -80, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
1^{\overset{312315131}{6}} 12 \text{ [#=2]} \quad \Delta = -80, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1^{\overset{12315131}{6}} 12 \quad \Delta = -72, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1^{\overset{12315131}{6}} 12^{\overset{2}{1}} \quad \Delta = -72, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1^{\overset{12315131}{6}} 12 \quad \Delta = -72, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1^{\overset{12315131}{6}} 12 \quad \Delta = -72, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1^{\overset{312315131}{6}} 12 \text{ [#=3]} \quad \Delta = -80, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1^{\overset{312315131}{6}} 12 \text{ [#=4]} \quad \Delta = -80, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{\overset{312315131}{6}} 12 \text{ [#=9]} \quad \Delta = -80, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
1^{\overset{12315131}{6}} 12^{\overset{3}{1}} \quad \Delta = -80, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1^{\overset{312315131}{6}} 12 \text{ [#=5]} \quad \Delta = -80, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1^{\overset{312315131}{6}} 12 \text{ [#=6]} \quad \Delta = -80, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1^{\overset{312315131}{6}} 12 \text{ [#=8]} \quad \Delta = -80, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix}
\end{aligned}$$

Base: $16^{\overset{21}{1}} 13161231$

$$\begin{aligned}
16^{21\bar{2}}13161231 \quad \Delta = -98, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
16^{21}13161231 \quad \Delta = -98, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
16^{21\bar{2}}13161231 \quad \Delta = -98, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
16^{21\bar{2}}13161231 \quad \Delta = -98, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
16^{21}13161231\bar{3} \text{ [#=4]} \quad \Delta = -106, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
16^{21}13161231\bar{3} \text{ [#=3]} \quad \Delta = -106, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
16^{21}13161231 \quad \Delta = -106, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
16^{21}13161231\bar{3} \text{ [#=5]} \quad \Delta = -106, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
16^{21}13161231\bar{3} \text{ [#=2]} \quad \Delta = -106, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}
\end{aligned}$$

Base: $16^{21}13161313$

$$\begin{aligned}
16^{21\bar{2}}13161313 \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
16^{21\bar{4}}13161313 \quad \Delta = -133, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 9 & 5 \\ 5 & 9 \end{pmatrix} \\
16^{21\bar{2}}13161313 \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
16^{21\bar{3}}13161313 \text{ [#=2]} \quad \Delta = -113, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
16^{21}13161313 \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
16^{21\bar{2}}13161313 \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
16^{21\bar{2}}13161313 \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
16^{21\bar{3}}13161313 \text{ [#=3]} \quad \Delta = -113, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
16^{21}13161313 \quad \Delta = -113, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
16^{21\bar{3}}13161313 \quad \Delta = -113, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{321}{6} 1316131$

$$1 \overset{321}{6} 1316131 \quad \Delta = -106, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1 \overset{321}{6} 1316131\overset{3}{3} [\# = 7] \quad \Delta = -114, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$321\overset{31}{6} 1316131 \quad \Delta = -114, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{321}{6} 1316131\overset{3}{3} [\# = 2] \quad \Delta = -114, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$1 \overset{321}{6} 1316131\overset{3}{3} [\# = 3] \quad \Delta = -114, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1 \overset{321}{6} 1316131\overset{3}{3} [\# = 4] \quad \Delta = -114, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{321}{6} 1316131\overset{3}{3} [\# = 9] \quad \Delta = -114, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{321}{6} 1316131\overset{3}{3} [\# = 5] \quad \Delta = -114, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{321}{6} 1316131\overset{3}{3} [\# = 6] \quad \Delta = -114, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \overset{321}{6} 1316131\overset{2}{3} \quad \Delta = -106, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1 \overset{321}{6} 1316131\overset{2}{3} \quad \Delta = -106, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1 \overset{321}{6} 1316131\overset{2}{3} \quad \Delta = -106, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{321}{6} 1316131\overset{3}{3} [\# = 8] \quad \Delta = -114, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$321\overset{21}{6} 1316131 \quad \Delta = -106, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: $1 \overset{51315131}{6} 13$

$$1 \overset{51315131}{6} 13 \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{51315131}{6} 13 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$31 \overset{51315131}{6} \overset{13}{3} \quad \Delta = -139, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$31 \overset{51315131}{6} \overset{12}{3} \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{51315131}{6} 13 \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1 \overset{51315131}{6} 13 \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{51315131}{6} \overset{2}{13} \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{51315131}{6} 13 \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{1315131}{6} 131$$

$$1 \overset{\overset{3}{1315131}}{6} 131 \text{ [#=8]} \quad \Delta = -87, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1 \overset{1315131}{6} 131 \overset{3}{3} \text{ [#=5]} \quad \Delta = -87, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1 \overset{1315131}{6} 131 \overset{3}{3} \text{ [#=3]} \quad \Delta = -87, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{1315131}{6} 131 \overset{3}{3} \text{ [#=6]} \quad \Delta = -87, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$1 \overset{\overset{3}{1315131}}{6} 131 \text{ [#=6]} \quad \Delta = -87, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1 \overset{\overset{3}{1315131}}{6} 131 \text{ [#=4]} \quad \Delta = -87, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{1315131}{6} 131 \overset{3}{3} \text{ [#=4]} \quad \Delta = -87, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1 \overset{\overset{3}{1315131}}{6} 131 \text{ [#=3]} \quad \Delta = -87, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \overset{\overset{3}{1315131}}{6} 131 \text{ [#=2]} \quad \Delta = -87, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{1315131}{6} 131 \overset{3}{3} \text{ [#=7]} \quad \Delta = -87, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$1 \overset{\overset{3}{1315131}}{6} 131 \text{ [#=7]} \quad \Delta = -87, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$1 \overset{1315131}{6} 131 \overset{3}{3} \text{ [#=8]} \quad \Delta = -87, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$1 \overset{1315131}{6} 131 \overset{3}{3} \text{ [#=9]} \quad \Delta = -87, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix}$$

$$1 \overset{1315131}{6} 131 \quad \Delta = -79, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1 \overset{1315131}{6} 131 \quad \Delta = -79, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{1315131}{6} 131\overset{\color{red}{3}}{[\# = 2]} \quad \Delta = -87, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \overset{1315131}{6} 131 \quad \Delta = -79, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{\color{red}{3}1315131}{6} 131 [\# = 5] \quad \Delta = -87, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1 \overset{1315131\color{red}{2}}{6} 131 \quad \Delta = -79, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{\color{red}{3}1315131}{6} 131 [\# = 9] \quad \Delta = -87, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{315131}{6} 1315$$

$$1 \overset{315131}{6} 131\overset{\color{red}{2}}{5} \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{315131\color{red}{2}}{6} 1315 \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$5131 \overset{315131}{6} 1\overset{\color{red}{3}}{3} \quad \Delta = -139, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$5131 \overset{315131}{6} 1\overset{\color{red}{2}}{2} \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \overset{315131}{6} 1315 \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1 \overset{315131}{6} 1315 \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{315131}{6} 1315 \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{15131}{6} 13151$$

$$1 \overset{15131\color{red}{2}}{6} 13151 \quad \Delta = -123, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$15131 \overset{15131}{6} 1\overset{\color{red}{3}}{3} \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$15131 \overset{15131}{6} 1\overset{\color{red}{2}}{2} \quad \Delta = -123, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{15131}{6} 131\overset{\color{red}{2}}{51} \quad \Delta = -123, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{131}{6} 1316122$$

$$1 \overset{131}{6} 131\overset{\color{red}{2}}{6}122 \quad \Delta = -97, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{131}{6} 1316122 \quad \Delta = -97, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\begin{aligned}
& \overset{131}{1} \overset{2}{6} 1316122 \quad \Delta = -97, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{131}{1} \overset{3}{6} 1316122 \quad \Delta = -106, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
& \overset{3131}{1} \overset{3}{6} 1316122 \text{ [#=3]} \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{3131}{1} \overset{3}{6} 1316122 \text{ [#=4]} \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{3131}{1} \overset{3}{6} 1316122 \text{ [#=5]} \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& \overset{131}{1} \overset{3}{6} 1316122 \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{3131}{1} \overset{3}{6} 1316122 \text{ [#=6]} \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& \overset{3131}{1} \overset{3}{6} 1316122 \text{ [#=7]} \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
& \overset{3131}{1} \overset{3}{6} 1316122 \text{ [#=8]} \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
& \overset{3131}{1} \overset{3}{6} 1316122 \text{ [#=9]} \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
& \overset{3131}{1} \overset{3}{6} 1316122 \text{ [#=2]} \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{1312}{1} \overset{2}{6} 1316122 \quad \Delta = -97, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{41316131612}$

$$\begin{aligned}
& \overset{1}{41316131612} \overset{2}{6} \quad \Delta = -125, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{3}{41316131612} \overset{1}{6} \quad \Delta = -133, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{21}{41316131612} \quad \Delta = -125, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{1}{41316131612} \overset{3}{6} \quad \Delta = -133, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{21}{41316131612} \quad \Delta = -125, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{12}{41316131612} \quad \Delta = -125, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{2}{41316131612} \overset{1}{6} \quad \Delta = -125, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

$$4131\overset{1}{6}131\overset{2}{6}12 \quad \Delta = -125, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: $4131\overset{1}{6}131\overset{1}{6}1$

$$4131\overset{3}{6}1\overset{1}{3}1\overset{1}{6}1 \quad \Delta = -133, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$4131\overset{21}{6}131\overset{1}{6}1 \quad \Delta = -125, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$4131\overset{2}{6}1\overset{1}{3}1\overset{1}{6}1 \quad \Delta = -125, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$4131\overset{1}{6}131\overset{21}{6}1 \quad \Delta = -125, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$4131\overset{12}{6}1\overset{1}{3}1\overset{1}{6}1 \quad \Delta = -125, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$4131\overset{21}{6}1\overset{1}{3}1\overset{1}{6}1 \quad \Delta = -125, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: $21\overset{31}{6}1316131$

$$21\overset{31}{6}1316131\overset{3}{1} \text{ [#=3]} \quad \Delta = -113, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$21\overset{31}{6}131\overset{2}{6}131 \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$21\overset{312}{6}1316131 \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$21\overset{231}{6}1316131 \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 24 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$21\overset{31}{6}1316131 \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$21\overset{31}{6}131\overset{2}{6}131 \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$21\overset{31}{6}1316131\overset{3}{1} \text{ [#=2]} \quad \Delta = -113, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$21\overset{31}{6}1316131\overset{3}{1} \text{ [#=5]} \quad \Delta = -113, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$21\overset{31}{6}1316131\overset{3}{1} \text{ [#=6]} \quad \Delta = -113, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$21\overset{31}{6}1316131\overset{3}{1} \text{ [#=4]} \quad \Delta = -113, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$21\overset{31}{6}1316131\overset{3}{1} \text{ [#=7]} \quad \Delta = -113, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\begin{aligned}
21\overset{31}{6}1316131\overset{\color{red}{3}}{1} \text{ }_{[\# = 8]} \quad \Delta = -113, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21\overset{31}{6}1316131\overset{\color{red}{3}}{1} \text{ }_{[\# = 9]} \quad \Delta = -113, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21\overset{\color{red}{3}31}{6}1316131 \quad \Delta = -113, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 23 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

Base: $16\overset{21}{1}316\overset{1}{1}31$

$$\begin{aligned}
16\overset{21}{1}316\overset{1}{1}31\overset{\color{red}{3}}{1} \text{ }_{[\# = 2]} \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
16\overset{21}{1}316\overset{1}{1}31 \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
16\overset{21}{1}316\overset{1}{1}31\overset{\color{red}{3}}{1} \text{ }_{[\# = 9]} \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
16\overset{21}{1}316\overset{1}{1}31\overset{\color{red}{3}}{1} \text{ }_{[\# = 3]} \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
16\overset{21}{1}316\overset{1}{1}31\overset{\color{red}{3}}{1} \text{ }_{[\# = 4]} \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
16\overset{21}{1}316\overset{1}{1}31\overset{\color{red}{3}}{1} \text{ }_{[\# = 8]} \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
16\overset{21}{1}316\overset{1}{1}31\overset{\color{red}{3}}{1} \text{ }_{[\# = 7]} \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
16\overset{21}{1}316\overset{1}{1}31\overset{\color{red}{3}}{1} \text{ }_{[\# = 5]} \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
16\overset{21}{1}316\overset{1}{1}31\overset{\color{red}{3}}{1} \text{ }_{[\# = 6]} \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
16\overset{21}{1}316\overset{\color{red}{2}1}{1}31 \quad \Delta = -97, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
16\overset{\color{red}{2}1}{1}316\overset{1}{1}31 \quad \Delta = -97, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
16\overset{21}{1}316\overset{1}{1}31 \quad \Delta = -97, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
16\overset{21}{1}316\overset{\color{red}{1}2}{1}31 \quad \Delta = -97, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: $31\overset{1315131}{6}13$

$$\begin{aligned}
31\overset{1315131}{6}13 \quad \Delta = -88, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
31\overset{\color{red}{6}1315131}{6}13 \quad \Delta = -165, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
31^{\overline{41315131}} 6^{13} \quad \Delta = -115, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
31^{\overline{1315131}} 6^{13} \quad \Delta = -87, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
31^{\overline{1315131}} 6^{13} \quad \Delta = -87, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
31^{\overline{1315131}} 6^{13} \quad \Delta = -87, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
31^{\overline{21315131}} 6^{13} \quad \Delta = -87, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
31^{\overline{21315131}} 6^{13} \quad \Delta = -87, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
31^{\overline{31315131}} 6^{13} [\# = 5] \quad \Delta = -95, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
31^{\overline{31315131}} 6^{13} [\# = 7] \quad \Delta = -95, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 196 & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
31^{\overline{31315131}} 6^{13} [\# = 9] \quad \Delta = -95, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 324 & \begin{pmatrix} 0 & 18 & 2 \\ 18 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
31^{\overline{31315131}} 6^{13} [\# = 8] \quad \Delta = -95, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 256 & \begin{pmatrix} 0 & 16 & 2 \\ 16 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
31^{\overline{31315131}} 6^{13} [\# = 6] \quad \Delta = -95, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 144 & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
31^{\overline{31315131}} 6^{13} [\# = 2] \quad \Delta = -95, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
31^{\overline{31315131}} 6^{13} [\# = 3] \quad \Delta = -95, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
31^{\overline{31315131}} 6^{13} \quad \Delta = -95, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
31^{\overline{31315131}} 6^{13} [\# = 4] \quad \Delta = -95, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix}
\end{aligned}$$

Base: $31^{\overline{31}} 6^{13} 1316122$

$$\begin{aligned}
31^{\overline{31}} 6^{13} 1316122^{\overline{3}} \quad \Delta = -114, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
31^{\overline{312}} 6^{13} 1316122 \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
31^{\overline{31}} 6^{13} 1316122^{\overline{3}} \quad \Delta = -114, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
31^{\overline{31}} 6^{13} 1316122^{\overline{2}} \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{231}{3161316122} \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 31\overset{31}{6}1316122 \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 31\overset{31}{6}1316122 \quad \Delta = -113, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 27 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \text{Base: } 131\overset{31}{6}131612 \\
& 31\overset{3131}{6}131612 \quad \Delta = -113, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 31\overset{3131}{6}131612 \text{ [#=4]} \quad \Delta = -113, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& 31\overset{3131}{6}131612 \text{ [#=7]} \quad \Delta = -113, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 196 \quad \begin{pmatrix} 0 & 14 & 2 \\ 14 & 389 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
& 31\overset{3131}{6}131612 \text{ [#=8]} \quad \Delta = -113, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 256 \quad \begin{pmatrix} 0 & 16 & 2 \\ 16 & 509 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
& 131\overset{312}{6}131612 \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 31\overset{3131}{6}131612 \text{ [#=3]} \quad \Delta = -113, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& 31\overset{3131}{6}131612 \text{ [#=2]} \quad \Delta = -113, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 131\overset{31}{6}131612 \quad \Delta = -113, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
& 131\overset{231}{6}131612 \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 24 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 31\overset{3131}{6}131612 \text{ [#=6]} \quad \Delta = -113, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 285 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
& 31\overset{3131}{6}131612 \text{ [#=5]} \quad \Delta = -113, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 197 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
& 31\overset{3131}{6}131612 \text{ [#=9]} \quad \Delta = -113, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 324 \quad \begin{pmatrix} 0 & 18 & 2 \\ 18 & 645 & 53 \\ 2 & 53 & 8 \end{pmatrix} \\
& 131\overset{31}{6}131612 \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 131\overset{31}{6}131612 \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& 31\overset{2131}{6}131612 \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 131\overset{31}{6}131612 \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
31 \overset{4}{6} 131612 \quad \Delta = -133, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
31 \overset{5}{6} 131612 \quad \Delta = -157, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
31 \overset{6}{6} 131612 \quad \Delta = -183, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: $131\overset{31}{6}131\overset{1}{6}1$

$$\begin{aligned}
31 \overset{4}{6} 131\overset{1}{6}1 \quad \Delta = -133, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
31 \overset{3}{6} 131\overset{1}{6}1 \quad \Delta = -113, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
31 \overset{3}{6} 131\overset{1}{6}1 \text{ [#=6] } \quad \Delta = -113, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 144 & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 285 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
31 \overset{3}{6} 131\overset{1}{6}1 \text{ [#=7] } \quad \Delta = -113, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 196 & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 389 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
31 \overset{3}{6} 131\overset{1}{6}1 \text{ [#=8] } \quad \Delta = -113, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 256 & \begin{pmatrix} 0 & 16 & 2 \\ 16 & 509 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
31 \overset{3}{6} 131\overset{1}{6}1 \text{ [#=9] } \quad \Delta = -113, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 324 & \begin{pmatrix} 0 & 18 & 2 \\ 18 & 645 & 53 \\ 2 & 53 & 8 \end{pmatrix} \\
31 \overset{3}{6} 131\overset{1}{6}1 \text{ [#=5] } \quad \Delta = -113, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 197 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
31 \overset{3}{6} 131\overset{1}{6}1 \text{ [#=4] } \quad \Delta = -113, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
31 \overset{3}{6} 131\overset{1}{6}1 \text{ [#=3] } \quad \Delta = -113, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
31 \overset{3}{6} 131\overset{1}{6}1 \text{ [#=2] } \quad \Delta = -113, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
131\overset{31}{6}131\overset{1}{6}1 \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
131\overset{2}{6}131\overset{1}{6}1 \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 24 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
131\overset{3}{6}131\overset{1}{6}1 \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
31 \overset{6}{6} 131\overset{1}{6}1 \quad \Delta = -183, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
31 \overset{5}{6} 131\overset{1}{6}1 \quad \Delta = -157, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
131\overset{31}{6}131\overset{2}{6}1 \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

$$31 \overset{2131}{6} \overset{1}{13161} \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: $31 \overset{31}{6} \overset{21}{13161}$

$$31 \overset{31}{6} \overset{21}{13161} \quad \Delta = -113, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$31 \overset{31}{6} \overset{21}{13161} \overset{13}{6} \quad \Delta = -113, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$31 \overset{31}{6} \overset{21}{13161} \overset{12}{6} \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\overset{231}{31} \overset{21}{6} \overset{21}{13161} \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$31 \overset{31}{6} \overset{221}{13161} \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$31 \overset{31}{6} \overset{21}{13161} \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$31 \overset{312}{6} \overset{21}{13161} \quad \Delta = -105, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$31 \overset{31}{6} \overset{321}{13161} \quad \Delta = -114, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: $1 \overset{1,12315131}{6} 1$

$$1 \overset{1,312315131}{6} 1 \text{ [#=4]} \quad \Delta = -80, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$1 \overset{1,12315131}{6} 1 \quad \Delta = -72, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{1,12315131}{6} 1 \quad \Delta = -72, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{1,12315131}{6} 1 \quad \Delta = -72, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{1,312315131}{6} 1 \text{ [#=5]} \quad \Delta = -80, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix}$$

$$1 \overset{1,312315131}{6} 1 \text{ [#=9]} \quad \Delta = -80, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix}$$

$$1 \overset{1,312315131}{6} 1 \text{ [#=8]} \quad \Delta = -80, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix}$$

$$1 \overset{1,312315131}{6} 1 \text{ [#=7]} \quad \Delta = -80, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix}$$

$$1 \overset{1,312315131}{6} 1 \text{ [#=6]} \quad \Delta = -80, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

$$1^{1,3^{12315131}} 6^1 1 \text{ [#}=3] \quad \Delta = -80, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1^{1,3^{12315131}} 6^1 1 \text{ [#}=2] \quad \Delta = -80, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

Base: 712231414141

$$712231414141^2 \quad \Delta = -123, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$712231414141^2 \quad \Delta = -123, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$712231414141^2 \quad \Delta = -123, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

$$712231414141^2 \quad \Delta = -123, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 89 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$712231414141^3 \text{ [#}=3] \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$712231414141^3 \text{ [#}=2] \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

Base: 712314141413

$$712314141413^3 \quad \Delta = -139, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$712314141413^2 \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$712314141413^2 \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$712314141413^2 \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$712314141413^2 \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$712314141413^2 \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$712314141413^2 \quad \Delta = -133, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 413151322171

$$413151322171^2 \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$413151322171^3 \quad \Delta = -135, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$413151322171^2 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\begin{aligned}
413151322171^2 \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
413151322171^2 \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
413151322171^2 \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: 141414132171

$$\begin{aligned}
141414132171^2 \quad \Delta = -125, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1712314141413^{\#=3} \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
141414132171^2 \quad \Delta = -123, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
141414132171^2 \quad \Delta = -123, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
141414132171^2 \quad \Delta = -123, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
141414132171^2 \quad \Delta = -123, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1712314141413^{\#=2} \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: 232161232171

$$\begin{aligned}
232161232171^2 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
232161232171^2 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
232161232171^2 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix}
\end{aligned}$$

Base: 217122315131

$$\begin{aligned}
217122315131^{\#=9} \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
217122315131^2 \quad \Delta = -99, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
217122315131^2 \quad \Delta = -99, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
217122315131^2 \quad \Delta = -99, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
217122315131^2 \quad \Delta = -99, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
217122315131\overset{3}{3} \text{ [#}=6] \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
217122315131\overset{3}{3} \text{ [#}=2] \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
217122315131\overset{3}{3} \text{ [#}=3] \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
217122315131\overset{3}{3} \text{ [#}=4] \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
217122315131\overset{3}{3} \text{ [#}=5] \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
217122315131\overset{3}{3} \text{ [#}=7] \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
217122315131\overset{3}{3} \text{ [#}=8] \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

Base: 413151321712

$$\begin{aligned}
\overset{3}{4}13151321712 \quad \Delta = -135, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\overset{2}{4}13151321712 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\overset{2}{4}13151321712 \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\overset{2}{4}13151321712 \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\overset{2}{4}13151321712 \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\overset{2}{4}13151321712 \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\overset{2}{4}13151321712 \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: 321612321712

$$\begin{aligned}
\overset{2}{3}21612321712 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
321612321712 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\overset{2}{3}21612321712 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
\overset{2}{3}21612321712 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: ³¹1612321712

$$1\overset{31}{6}12321712 \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$31\overset{21}{6}12321712 \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1\overset{31}{6}12321712 \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1\overset{31}{6}12321\overset{2}{7}12 \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1\overset{31}{6}12321\overset{2}{7}12 \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$31\overset{31}{6}12321712 \quad \Delta = -135, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 221712315131

$$22171231\overset{2}{5}131 \quad \Delta = -99, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$22171231\overset{2}{5}131 \quad \Delta = -99, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$22171231\overset{2}{5}131 \quad \Delta = -99, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$221712315131\overset{3}{3} \text{ [#=2]} \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$221712315131\overset{3}{3} \text{ [#=3]} \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$221712315131\overset{3}{3} \text{ [#=9]} \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$221712315131\overset{3}{3} \text{ [#=4]} \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$221712315131\overset{3}{3} \text{ [#=8]} \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$221712315131\overset{3}{3} \text{ [#=7]} \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$221712315131\overset{3}{3} \text{ [#=5]} \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$221712315131\overset{3}{3} \text{ [#=6]} \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$22171\overset{2}{2}315131 \quad \Delta = -99, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$221712315131\overset{3}{3} \quad \Delta = -108, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: 216123217122

$$216123217122^2 \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$216123217122^2 \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$216123217122^3 \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$216123217122^2 \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$216123217122^2 \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$216123217122^3 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

Base: 16123217122

$$16123217122^1 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$16123217122^{12} \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$16123217122^1 \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$16123217122^1 \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 161232171223

$$161232171223^2 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$161232171223^2 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$161232171223^2 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

Base: 132171231512

$$215132171231^3 \quad [\# = 5] \quad \Delta = -108, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$132171231512^2 \quad \Delta = -100, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$132171231512^2 \quad \Delta = -100, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$132171231512^3 \quad \Delta = -108, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$132171231512^2 \quad \Delta = -100, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$\begin{aligned}
132171231512 \quad \Delta = -100, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2151321712313 \quad [\# = 2] \quad \Delta = -108, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2151321712313 \quad [\# = 3] \quad \Delta = -108, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2151321712313 \quad [\# = 4] \quad \Delta = -108, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

Base: 13217123151

$$\begin{aligned}
13217123151 \quad \Delta = -100, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
13217123151 \quad \Delta = -100, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
13217123151 \quad \Delta = -100, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \quad 5 \quad 1 \quad [\# = 4] \quad \Delta = -108, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \quad 5 \quad 1 \quad [\# = 5] \quad \Delta = -108, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1 \quad 5 \quad 1 \quad [\# = 2] \quad \Delta = -108, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1 \quad 5 \quad 1 \quad [\# = 3] \quad \Delta = -108, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 132171232151

$$\begin{aligned}
1512321712313 \quad [\# = 5] \quad \Delta = -110, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1512321712313 \quad [\# = 4] \quad \Delta = -110, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1512321712313 \quad [\# = 3] \quad \Delta = -110, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1512321712313 \quad [\# = 2] \quad \Delta = -110, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
132171232151 \quad \Delta = -102, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
132171232151 \quad \Delta = -102, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
132171232151 \quad \Delta = -102, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 413217123151

$$\begin{aligned}
413217123151^2 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
413217123151^2 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
413217123151^2 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
413217123151^2 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
413217123151^2 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

Base: 141321712321

$$\begin{aligned}
1413217123213_{[\#=5]} \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
141321712321^2 \quad \Delta = -78, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1413217123213_{[\#=3]} \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1413217123213_{[\#=4]} \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1232171231413_{[\#=3]} \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
141321712321^2 \quad \Delta = -80, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1413217123213_{[\#=6]} \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1413217123213_{[\#=2]} \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1232171231413_{[\#=2]} \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
141321712321^2 \quad \Delta = -78, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1413217123213_{[\#=9]} \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
141321712321^2 \quad \Delta = -78, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1413217123213_{[\#=8]} \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1413217123213_{[\#=7]} \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{aligned}$$

Base: 413217123215

$$413217123215^2 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$413217123215^2 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$413217123215^2 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$413217123215^2 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$413217123215^2 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 412321712321

$$412321712321^3_{\# = 8} \quad \Delta = -88, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$412321712321^3_{\# = 7} \quad \Delta = -88, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$412321712321^3_{\# = 5} \quad \Delta = -88, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$412321712321^3_{\# = 2} \quad \Delta = -88, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$412321712321^2 \quad \Delta = -80, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$412321712321^3_{\# = 3} \quad \Delta = -88, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$412321712321^3_{\# = 9} \quad \Delta = -88, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$412321712321^3_{\# = 4} \quad \Delta = -88, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$412321712321^3 \quad \Delta = -88, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$412321712321^2 \quad \Delta = -80, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$412321712321^2 \quad \Delta = -80, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$412321712321^3_{\# = 6} \quad \Delta = -88, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: $1^{131513221}_7 1$

$$1^{3131513221}_7 1_{\# = 4} \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\begin{aligned}
1^{\overline{131513221}} 7^{\overline{1}} 1 \quad \Delta = -99, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1^{\overline{131513221}} 7^{\overline{1}} 1 \quad \Delta = -99, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1^{\overline{3131513221}} 7^{\overline{1}} 1_{[\# = 9]} \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1^{\overline{3131513221}} 7^{\overline{1}} 1_{[\# = 6]} \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{\overline{3131513221}} 7^{\overline{1}} 1_{[\# = 3]} \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1^{\overline{3131513221}} 7^{\overline{1}} 1_{[\# = 7]} \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1^{\overline{3131513221}} 7^{\overline{1}} 1_{[\# = 8]} \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1^{\overline{3131513221}} 7^{\overline{1}} 1_{[\# = 5]} \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1^{\overline{3131513221}} 7^{\overline{1}} 1_{[\# = 2]} \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1^{\overline{131513221}} 7^{\overline{1}} 1 \quad \Delta = -99, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: $413151321\overline{71}$

$$\begin{aligned}
413151321\overline{71}^{\overline{2}} \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
413151321\overline{71}^{\overline{2}} \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
413151321\overline{71}^{\overline{21}} \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
413151321\overline{71}^{\overline{2}} \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
413151321\overline{71}^{\overline{2}} \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
413151321\overline{71}^{\overline{3}} \quad \Delta = -135, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

Base: $17123216123^{\overline{1}}$

$$\begin{aligned}
17123216123^{\overline{12}} \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
17123216123^{\overline{1}} \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$1\overset{1}{7}12321\overset{2}{6}123 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: $1\overset{31}{6}12321\overset{1}{7}1$

$$1\overset{31}{6}12321\overset{21}{7}1 \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$31\overset{21}{6}12321\overset{1}{7}1 \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1\overset{31}{6}12321\overset{1}{7}1 \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$1\overset{31}{6}12321\overset{1}{7}1 \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$31\overset{31}{6}12321\overset{1}{7}1 \quad \Delta = -135, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: $1\overset{13151321}{7}12$

$$1\overset{313151321}{7}12 \text{ [#=8]} \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1\overset{313151321}{7}12 \text{ [#=7]} \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$1\overset{313151321}{7}12 \text{ [#=2]} \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1\overset{13151321}{7}12 \quad \Delta = -99, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1\overset{13151321}{7}12 \quad \Delta = -99, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1\overset{13151321}{7}12 \quad \Delta = -99, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1\overset{313151321}{7}12 \text{ [#=3]} \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1\overset{313151321}{7}12 \text{ [#=4]} \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1\overset{13151321}{7}12 \quad \Delta = -99, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1\overset{313151321}{7}12 \text{ [#=5]} \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1\overset{313151321}{7}12 \text{ [#=6]} \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1\overset{313151321}{7}12 \text{ [#=9]} \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

Base: $1\overset{21}{7}12321612$

$$\begin{aligned}
1 \overset{21}{7} 12321612 \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \overset{21}{7} 12321612 \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1 \overset{21}{7} 12321612 \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{21}{7} 12321612 \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1 \overset{21}{7} 12321612 \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{1}{6} 12321 \overset{21}{7} 1$

$$\begin{aligned}
1 \overset{1}{6} 12321 \overset{21}{7} 1 \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1 \overset{1}{6} 12321 \overset{21}{7} 1 \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{1}{6} 12321 \overset{21}{7} 1 \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{321}{7} 1232161$

$$\begin{aligned}
1 \overset{321}{7} 1232161 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
321 \overset{21}{7} 1232161 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1 \overset{321}{7} 1232161 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1 \overset{321}{7} 1232161 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{12321}{7} 12321$

$$\begin{aligned}
1 \overset{12321}{7} 12321 \overset{3}{\text{[#}=5]} \quad \Delta = -60, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1 \overset{12321}{7} 12321 \overset{3}{\text{[#}=4]} \quad \Delta = -60, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1 \overset{12321}{7} 12321 \overset{2}{\text{[#}=9]} \quad \Delta = -52, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \overset{12321}{7} 12321 \overset{3}{\text{[#}=9]} \quad \Delta = -60, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1 \overset{12321}{7} 12321 \overset{3}{\text{[#}=3]} \quad \Delta = -60, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \overset{12321}{7} 12321 \overset{3}{\text{[#}=6]} \quad \Delta = -60, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \begin{matrix} 12321 \\ 1 \quad 7 \quad 12321 \end{matrix} \begin{matrix} \textcolor{red}{3} \\ \text{[#}=7 \end{matrix} \quad \Delta = -60, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
& \begin{matrix} 12321 \\ 1 \quad 7 \quad 12321 \end{matrix} \begin{matrix} \textcolor{red}{3} \\ \text{[#}=8 \end{matrix} \quad \Delta = -60, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
& \begin{matrix} 12321 \\ 1 \quad 7 \quad 12321 \end{matrix} \begin{matrix} \textcolor{red}{3} \\ \text{[#}=2 \end{matrix} \quad \Delta = -60, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\begin{matrix} 2321 \\ 612321 \end{matrix} \begin{matrix} 7 \\ 7 \end{matrix} 1$

$$\begin{aligned}
& \begin{matrix} 2321 \\ 612321 \end{matrix} \begin{matrix} 7 \\ 7 \end{matrix} 1 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \begin{matrix} \textcolor{red}{2} \\ 612321 \end{matrix} \begin{matrix} 2321 \\ 7 \end{matrix} 1 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \begin{matrix} \textcolor{red}{2} \\ 612321 \end{matrix} \begin{matrix} 2321 \\ 7 \end{matrix} 1 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& \begin{matrix} 2321 \\ 612321 \end{matrix} \begin{matrix} 7 \\ 7 \end{matrix} \textcolor{red}{12} \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\begin{matrix} 1,13151321 \\ 1 \end{matrix} \begin{matrix} 7 \\ 7 \end{matrix} 1$

$$\begin{aligned}
& \begin{matrix} 1, \textcolor{red}{3}13151321 \\ 1 \end{matrix} \begin{matrix} 7 \\ 7 \end{matrix} 1 \text{ [#}=7 \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
& \begin{matrix} 1, \textcolor{red}{3}13151321 \\ 1 \end{matrix} \begin{matrix} 7 \\ 7 \end{matrix} 1 \text{ [#}=9 \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
& \begin{matrix} 1,13151321 \\ 1 \end{matrix} \begin{matrix} 7 \\ 7 \end{matrix} 1 \quad \Delta = -99, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \begin{matrix} 1,13151321 \\ 1 \end{matrix} \begin{matrix} 7 \\ 7 \end{matrix} 1 \quad \Delta = -99, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \begin{matrix} 1,13151321 \\ 1 \end{matrix} \begin{matrix} 7 \\ 7 \end{matrix} 1 \quad \Delta = -99, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \begin{matrix} 1, \textcolor{red}{3}13151321 \\ 1 \end{matrix} \begin{matrix} 7 \\ 7 \end{matrix} 1 \text{ [#}=2 \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \begin{matrix} 1, \textcolor{red}{3}13151321 \\ 1 \end{matrix} \begin{matrix} 7 \\ 7 \end{matrix} 1 \text{ [#}=3 \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \begin{matrix} 1, \textcolor{red}{3}13151321 \\ 1 \end{matrix} \begin{matrix} 7 \\ 7 \end{matrix} 1 \text{ [#}=5 \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& \begin{matrix} 1, \textcolor{red}{3}13151321 \\ 1 \end{matrix} \begin{matrix} 7 \\ 7 \end{matrix} 1 \text{ [#}=4 \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \begin{matrix} 1, \textcolor{red}{3}13151321 \\ 1 \end{matrix} \begin{matrix} 7 \\ 7 \end{matrix} 1 \text{ [#}=8 \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
& \begin{matrix} 1, \textcolor{red}{3}13151321 \\ 1 \end{matrix} \begin{matrix} 7 \\ 7 \end{matrix} 1 \text{ [#}=6 \quad \Delta = -107, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}
\end{aligned}$$

Base: 812231414132

$$\begin{aligned}
812231414132^2 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
812231414132^2 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
812231414132^2 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
812231414132^2 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

Base: 8122315¹1322

$$\begin{aligned}
8122315^{21}1322 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
8122315^{12}1322 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
8122315^11322^2 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
8122315^{31}1322 \quad \Delta = -136, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
8122315^{21}1322 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

Base: 812314141322

$$\begin{aligned}
812314141322^2 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
812314141322^2 \quad \Delta = -134, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
812314141322^2 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
812314141322^2 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
812314141322^2 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

Base: 314141322181

$$\begin{aligned}
314141322181^2 \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
314141322181^2 \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
314141322181^2 \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
314141322181^2 \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

$$314141322181 \quad \Delta = -139, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 181223151232

$$181223151232 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$181223151232 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$181223151232 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: 18122315132

$$18122315132 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$18122315132 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$18122315132 \quad \Delta = -136, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$18122315132 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$18122315132 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: 231414132181

$$231414132181 \quad \Delta = -134, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$231414132181 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$231414132181 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$231414132181 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$231414132181 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$231414132181 \quad \Delta = -134, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 181232151322

$$181232151322 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$181232151322 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix}$$

$$181232151322 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: 1812315¹1322

$$1812315^{21}1322 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1812315^{21}1322 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1812315^{12}1322 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1812315^{21}1322 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1812315^{31}1322 \quad \Delta = -136, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 141413221812

$$141413221812^{21} \quad \Delta = -123, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$141413221812^{21} \quad \Delta = -123, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$141413221812^{21} \quad \Delta = -123, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$218122314141^{31} \text{ [#=3]} \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$218122314141^{31} \text{ [#=2]} \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$141413221812^{21} \quad \Delta = -123, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: 218122315123

$$218122315123^{21} \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$218122315123^{21} \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$218122315123^{21} \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$218122315123^{21} \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 218122315¹13

$$218122315^{21}13 \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$218122315^{31}13 \quad \Delta = -135, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$218122315^{12}13 \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\begin{aligned}
21812231\overset{21}{5}13 \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21812231\overset{21}{5}13 \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21812231\overset{2}{5}\overset{1}{1}3 \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 314141321812

$$\begin{aligned}
314141321812 \quad \Delta = -139, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
314141\overset{2}{3}21812 \quad \Delta = -133, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
314141321812 \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
314141321812 \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
314141\overset{2}{3}21812 \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
314141321812 \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
314141321812 \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: 218123215132

$$\begin{aligned}
21812321\overset{2}{5}132 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21812321\overset{2}{5}132 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21812321\overset{2}{5}132 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
21812321\overset{2}{5}132 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 218123151232

$$\begin{aligned}
218123151\overset{2}{2}32 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
218123151\overset{2}{2}32 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
218123151\overset{2}{2}32 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
218123151\overset{2}{2}32 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

Base: 2181231¹5132

$$\begin{aligned}
2181231\overset{31}{5}132 \quad \Delta = -136, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2181231\overset{2}{5}\overset{1}{1}32 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2181231\overset{21}{5}132 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
2181231\overset{12}{5}132 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
2181231\overset{21}{5}132 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2181231\overset{2}{5}\overset{1}{1}32 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 221812231512

$$\begin{aligned}
221812231\overset{3}{5}1512 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
221812231\overset{2}{5}\overset{1}{1}512 \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
221812231\overset{2}{5}1512 \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
221812231\overset{2}{5}1512 \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
221812231\overset{2}{5}1512 \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
221812231\overset{3}{5}1512 \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}
\end{aligned}$$

Base: 221812231¹51

$$\begin{aligned}
221812231\overset{21}{5}1 \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
221812231\overset{2}{5}\overset{1}{1}1 \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
221812231\overset{3}{5}\overset{1}{1}1 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
221812231\overset{2}{5}\overset{1}{1}1 \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 322181223151

$$\begin{aligned}
3221812231\overset{2}{5}1 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
3221812231\overset{2}{5}1 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

$$322181223151 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 414132218123

$$414132218123 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$414132218123 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$414132218123 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$414132218123 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

Base: 141413218122

$$141413218122 \quad \Delta = -123, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$141413218122 \quad \Delta = -123, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$141413218122 \quad \Delta = -123, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$141413218122 \quad \Delta = -123, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$141413218122 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$141413218122 \quad \Delta = -125, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$2218123141413_{[\# = 2]} \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$2218123141413_{[\# = 3]} \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 221812321513

$$221812321513 \quad \Delta = -129, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$221812321513 \quad \Delta = -138, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$221812321513 \quad \Delta = -129, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$221812321513 \quad \Delta = -129, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$221812321513 \quad \Delta = -129, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: 221812315123

$$\begin{aligned}
\overset{3}{221812315123} \quad \Delta = -137, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\overset{2}{221812315123} \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\overset{2}{221812315123} \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\overset{2}{221812315123} \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\overset{2}{221812315123} \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 22181231¹513

$$\begin{aligned}
\overset{12}{22181231513} \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\overset{3}{22181231513} \quad \Delta = -136, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\overset{21}{22181231513} \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\overset{2}{22181231513} \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\overset{21}{22181231513} \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\overset{2}{22181231513} \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\overset{31}{22181231513} \quad \Delta = -135, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: 141322181231

$$\begin{aligned}
141322181231\overset{3}{\text{red}} \text{ [#=2]} \quad \Delta = -112, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
141322181231\overset{3}{\text{red}} \text{ [#=5]} \quad \Delta = -112, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
141322181231\overset{3}{\text{red}} \text{ [#=4]} \quad \Delta = -112, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
141322181231\overset{3}{\text{red}} \text{ [#=3]} \quad \Delta = -112, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
132181223141\overset{3}{\text{red}} \text{ [#=2]} \quad \Delta = -112, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
132181223141\overset{3}{\text{red}} \text{ [#=3]} \quad \Delta = -112, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
141322181231^2 \quad \Delta = -104, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
141322181231^2 \quad \Delta = -104, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
141322181231^2 \quad \Delta = -104, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 222181231512

$$\begin{aligned}
222181231512^2 \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
222181231512^2 \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
222181231512^2 \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
222181231512^2 \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
222181231512^3 \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}
\end{aligned}$$

Base: 22218123151¹

$$\begin{aligned}
22218123151^{21} \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
22218123151^{21} \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
22218123151^{21} \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 123218122313

$$\begin{aligned}
123218122313^2 \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
3132218123213^{\#6} \quad \Delta = -94, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
3132218123213^{\#7} \quad \Delta = -94, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
3132218123213^{\#5} \quad \Delta = -94, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
3132218123213^{\#4} \quad \Delta = -94, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
3132218123213^{\#3} \quad \Delta = -94, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
123218122313^3 \quad \Delta = -94, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
313221812321\overset{2}{3} \text{ [#=2]} \quad \Delta = -94, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
12321812231\overset{2}{3} \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
12321812231\overset{3}{3} \quad \Delta = -94, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
313221812321\overset{2}{3} \text{ [#=9]} \quad \Delta = -94, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
12321812231\overset{2}{3} \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
313221812321\overset{2}{3} \text{ [#=8]} \quad \Delta = -94, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
\text{Base: } 412321812231 \\
412321812231\overset{2}{3} \text{ [#=2]} \quad \Delta = -114, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
412321812231\overset{2}{3} \text{ [#=5]} \quad \Delta = -114, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
412321812231\overset{2}{3} \quad \Delta = -106, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
412321812231\overset{2}{3} \quad \Delta = -106, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
412321812231\overset{2}{3} \quad \Delta = -106, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
412321812231\overset{2}{3} \text{ [#=4]} \quad \Delta = -114, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
412321812231\overset{2}{3} \text{ [#=3]} \quad \Delta = -114, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
412321812231\overset{3}{3} \quad \Delta = -114, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\text{Base: } 413218122314 \\
413218122314\overset{2}{3} \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
413218122314\overset{2}{3} \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
413218122314\overset{2}{3} \quad \Delta = -134, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
413218122314\overset{2}{3} \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

$$413218122314 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: 141321812231

$$141321812231 \quad \Delta = -106, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1413218122313 \text{ [#=5]} \quad \Delta = -112, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1413218122313 \text{ [#=4]} \quad \Delta = -112, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1413218122313 \text{ [#=3]} \quad \Delta = -112, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1322181231413 \text{ [#=3]} \quad \Delta = -112, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1413218122313 \text{ [#=2]} \quad \Delta = -112, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$1322181231413 \text{ [#=2]} \quad \Delta = -112, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$141321812231 \quad \Delta = -104, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$141321812231 \quad \Delta = -104, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$141321812231 \quad \Delta = -104, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 414132181223

$$414132181223 \quad \Delta = -134, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$414132181223 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$414132181223 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$414132181223 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$414132181223 \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

Base: 222181232151

$$222181232151 \quad \Delta = -121, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$222181232151 \quad \Delta = -121, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$222181232151 \quad \Delta = -121, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 123218123132

$$2313218123213 \quad [\# = 7] \quad \Delta = -95, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$123218123132 \quad \Delta = -89, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$123218123132 \quad \Delta = -89, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$123218123132 \quad \Delta = -87, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$123218123132 \quad \Delta = -95, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$2313218123213 \quad [\# = 2] \quad \Delta = -95, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$2313218123213 \quad [\# = 3] \quad \Delta = -95, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$2313218123213 \quad [\# = 4] \quad \Delta = -95, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$2313218123213 \quad [\# = 5] \quad \Delta = -95, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$2313218123213 \quad [\# = 8] \quad \Delta = -95, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$2313218123213 \quad [\# = 9] \quad \Delta = -95, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$123218123132 \quad \Delta = -87, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$123218123132 \quad \Delta = -87, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$2313218123213 \quad [\# = 6] \quad \Delta = -95, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

Base: 412321812313

$$412321812313 \quad \Delta = -122, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$412321812313 \quad \Delta = -116, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$412321812313 \quad \Delta = -114, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\begin{aligned}
412321812313^2 \quad \Delta = -114, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
412321812313^2 \quad \Delta = -114, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
412321812313^2 \quad \Delta = -114, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
412321812313^3_{[\#=2]} \quad \Delta = -122, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
412321812313^3 \quad \Delta = -122, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: 141321812313

$$\begin{aligned}
141321812313^2 \quad \Delta = -114, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
141321812313^2 \quad \Delta = -112, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
141321812313^2 \quad \Delta = -112, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
141321812313^2 \quad \Delta = -112, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
141321812313^2 \quad \Delta = -112, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
141321812313^3 \quad \Delta = -120, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
141321812313^3_{[\#=2]} \quad \Delta = -120, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
3132181231413_{[\#=2]} \quad \Delta = -120, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
3132181231413_{[\#=3]} \quad \Delta = -120, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
141321812313^2 \quad \Delta = -114, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 314132181231

$$\begin{aligned}
3141321812313_{[\#=4]} \quad \Delta = -120, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
3141321812313_{[\#=3]} \quad \Delta = -120, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
3141321812313_{[\#=5]} \quad \Delta = -120, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
314132181231 \quad \Delta = -114, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
314132181231 \quad \Delta = -120, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
314132181231 \textcolor{red}{3} \text{ [#=2]} \quad \Delta = -120, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
314132181231 \quad \Delta = -112, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
314132181231 \quad \Delta = -112, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
314132181231 \quad \Delta = -112, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
314132181231 \quad \Delta = -112, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 14132181232¹

$$\begin{aligned}
14132181232 \textcolor{red}{31} \text{ [#=3]} \quad \Delta = -114, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
14132181232 \quad \Delta = -108, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
14132181232 \quad \Delta = -106, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
14132181232 \quad \Delta = -106, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
14132181232 \quad \Delta = -106, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
14132181232 \quad \Delta = -106, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2 \textcolor{red}{3141321812} \textcolor{red}{3} \textcolor{red}{1} \text{ [#=3]} \quad \Delta = -114, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
14132181232 \textcolor{red}{31} \text{ [#=2]} \quad \Delta = -114, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2 \textcolor{red}{3141321812} \textcolor{red}{3} \textcolor{red}{1} \text{ [#=2]} \quad \Delta = -114, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
14132181232 \quad \Delta = -134, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}
\end{aligned}$$

Base: 41232181232¹

$$41232181232 \quad \Delta = -116, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$412321812\overset{31}{3}2 \text{ }_{[\# = 2]} \quad \Delta = -116, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$412321812\overset{2}{3}\overset{1}{2} \quad \Delta = -108, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$412321812\overset{31}{3}2 \text{ }_{[\# = 3]} \quad \Delta = -116, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$412321812\overset{2}{3}\overset{1}{2} \quad \Delta = -108, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$412321812\overset{21}{3}2 \quad \Delta = -108, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$412321812\overset{2}{3}\overset{1}{2} \quad \Delta = -108, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$412321812\overset{41}{3}2 \quad \Delta = -136, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

Base: 18122314141

$$18122314141 \overset{12}{\quad} \quad \Delta = -123, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$18122314141 \overset{1}{\quad} \overset{2}{\quad} \quad \Delta = -123, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$18122314141 \overset{1}{\quad} \overset{2}{\quad} \quad \Delta = -123, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$18122314141\overset{1}{\quad}\overset{3}{\quad} \text{ }_{[\# = 3]} \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$18122314141\overset{1}{\quad}\overset{3}{\quad} \text{ }_{[\# = 2]} \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

Base: $1 \overset{321513221}{8} 1$

$$1 \overset{321513221}{8} 1 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{321513221}{8} 1 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{321513221}{8} 1 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: $1 \overset{31513221}{8} 1$

$$1 \overset{31513221}{8} 1 \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{31513221}{8} 1 \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{31513221}{8} 1 \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\begin{aligned}
& \text{Base: } 1 \overset{31513221}{8} 1 \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 1 \overset{31513221}{8} 1 \quad \Delta = -135, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{1}{18123141413}$

$$\begin{aligned}
& 1 \overset{1}{18123141413} \overset{2}{8} \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 1 \overset{1}{18123141413} \overset{2}{8} \quad \Delta = -133, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 1 \overset{1}{18123141413} \overset{3}{8} \quad \Delta = -139, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& 1 \overset{12}{18123141413} \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 1 \overset{1}{18123141413} \overset{2}{8} \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 1 \overset{1}{18123141413} \overset{2}{8} \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{231512321}{8} 1$

$$\begin{aligned}
& 1 \overset{231512321}{8} 1 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 1 \overset{231512321}{8} 1 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& 1 \overset{231512321}{8} 1 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{232151321}{8} 1$

$$\begin{aligned}
& 1 \overset{232151321}{8} 1 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 1 \overset{232151321}{8} 1 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 1 \overset{232151321}{8} 1 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{23151321}{8} 1$

$$\begin{aligned}
& 1 \overset{23151321}{8} 1 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 1 \overset{23151321}{8} 1 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 1 \overset{23151321}{8} 1 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

$$1 \overset{23151321}{8} 1 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{23151321}{8} 1 \quad \Delta = -136, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{21513221}{8} 12$$

$$1 \overset{21513221}{8} 12 \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{21513221}{8} 12 \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1 \overset{21513221}{8} 12 \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$1 \overset{21513221}{8} 12 \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{21513221}{8} 12 \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{1513221}{8} 12$$

$$1 \overset{1513221}{8} 12 \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{1513221}{8} 12 \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{1513221}{8} 12 \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$\text{Base: } 18 \overset{21}{12314141}$$

$$18 \overset{21}{12314141} \quad \Delta = -123, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$18 \overset{21}{12314141} \quad \Delta = -123, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$18 \overset{21}{12314141} \quad \Delta = -123, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$18 \overset{21}{12314141} \quad \Delta = -123, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$18 \overset{21}{12314141} \text{ [#=2]} \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$18 \overset{21}{12314141} \text{ [#=3]} \quad \Delta = -131, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$18 \overset{21}{12314141} \quad \Delta = -125, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{31512321}{8} 12$$

$$\begin{array}{ll}
1 \overset{31512321}{8} 12 & \Delta = -129, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{31512321}{8} 12 & \Delta = -129, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1 \overset{31512321}{8} 12 & \Delta = -129, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1 \overset{31512321}{8} 12 & \Delta = -129, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{32151321}{8} 12 & \\
1 \overset{32151321}{8} 12 & \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1 \overset{32151321}{8} 12 & \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{32151321}{8} 12 & \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \overset{32151321}{8} 12 & \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{3151321}{8} 12 & \\
1 \overset{3151321}{8} 12 & \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{3151321}{8} 12 & \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1 \overset{3151321}{8} 12 & \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1 \overset{3151321}{8} 12 & \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1 \overset{3151321}{8} 12 & \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \overset{3151321}{8} 12 & \Delta = -135, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{1512321}{8} 122 & \\
1 \overset{1512321}{8} 122 & \Delta = -121, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \overset{1512321}{8} 122 & \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1 \overset{1512321}{8} 122 & \Delta = -121, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \overset{1512321}{8} 122 & \Delta = -121, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{array}$$

Base: $1 \overset{1513221}{8} 123$

$$1 \overset{1513221}{8} 123 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1 \overset{1513221}{8} 123 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{1513221}{8} \overset{2}{123} \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$321 \overset{1513221}{8} \overset{1}{2} \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: $1 \overset{2151321}{8} 122$

$$1 \overset{2151321}{8} 122 \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1 \overset{2151321}{8} 122 \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{2151321}{8} \overset{3}{122} \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{2151321}{8} 122 \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{2151321}{8} 122 \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$1 \overset{2151321}{8} \overset{2}{122} \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: $1 \overset{151321}{8} 122$

$$1 \overset{151321}{8} 122 \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{151321}{8} 122 \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{151321}{8} \overset{2}{122} \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{151321}{8} \overset{3}{122} \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: $1 \overset{13221}{8} 12321$

$$\overset{3}{1} \overset{13221}{8} 12321 \text{ [#=2]} \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$1 \overset{13221}{8} \overset{2}{12321} \quad \Delta = -78, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{13221}{8} 12321 \overset{3}{\text{[#=2]}} \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\begin{array}{ll}
1 \overset{313221}{8} 12321_{[\# = 4]} \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1 \overset{13221}{8} 12321\overset{3}{\# = 5} \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1 \overset{313221}{8} 12321_{[\# = 5]} \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1 \overset{13221}{8} 12321\overset{3}{\# = 6} \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1 \overset{13221}{8} 12321\overset{3}{\# = 7} \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1 \overset{13221}{8} 12321\overset{3}{\# = 9} \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1 \overset{13221}{8} 12321\overset{3}{\# = 4} \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1 \overset{13221}{8} 12321 \quad \Delta = -78, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \overset{313221}{8} 12321_{[\# = 3]} \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1 \overset{13221}{8} 12321\overset{3}{\# = 8} \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1 \overset{13221}{8} 12321\overset{3}{\# = 3} \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{512321}{8} 1223 & \\
1 \overset{512321}{8} 1223 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1 \overset{512321}{8} 1223 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1 \overset{512321}{8} 1223 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
3221 \overset{512321}{8} 1\overset{2}{\# = 3} \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{513221}{8} 1232 & \\
2321 \overset{513221}{8} 1\overset{2}{\# = 3} \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1 \overset{513221}{8} 1232 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1 \overset{513221}{8} 1232 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{array}$$

$$1 \overset{513221}{8} 1232 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 50 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

Base: $1 \overset{151321}{8} 1223$

$$1 \overset{151321}{8} 1223 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1 \overset{151321}{8} 1223 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{151321}{8} \overset{2}{1223} \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$3221 \overset{151321}{8} \overset{1}{2} \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: $4141321 \overset{321}{8} 1$

$$4141321 \overset{2}{8} \overset{321}{1} \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$4141321 \overset{2}{8} \overset{321}{1} \quad \Delta = -134, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$4141321 \overset{2}{8} \overset{321}{1} \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$4141321 \overset{321}{8} \overset{1}{1} \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$4141321 \overset{321}{8} \overset{1}{2} \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$4141321 \overset{2}{8} \overset{321}{1} \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: $412321 \overset{1321}{8} 1$

$$412321 \overset{31321}{8} \overset{1}{1} [\# = 3] \quad \Delta = -114, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$412321 \overset{31321}{8} \overset{1}{1} [\# = 5] \quad \Delta = -114, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$412321 \overset{31321}{8} \overset{1}{1} [\# = 2] \quad \Delta = -114, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$412321 \overset{3}{8} \overset{1321}{1} \quad \Delta = -114, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$412321 \overset{1321}{8} \overset{1}{1} \quad \Delta = -106, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$412321 \overset{2}{8} \overset{1321}{1} \quad \Delta = -106, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$412321 \overset{2}{8} \overset{1321}{1} \quad \Delta = -106, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$412321 \overset{31321}{8 \ 1} \text{ [#=4]} \quad \Delta = -114, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: 41321812314

$$41321812314 \overset{2}{1} \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$41321812314 \overset{21}{1} \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$41321812314 \overset{21}{1} \quad \Delta = -132, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$41321812314 \overset{2}{1} \quad \Delta = -134, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: $1 \overset{1321}{8} 123141$

$$1 \overset{1321}{8} 123141 \overset{2}{1} \quad \Delta = -104, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{1321}{8} 123141 \overset{2}{1} \quad \Delta = -106, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{1321}{8} 123141 \overset{3}{1} \text{ [#=3]} \quad \Delta = -112, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$1 \overset{31321}{8} 123141 \text{ [#=5]} \quad \Delta = -112, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$1 \overset{31321}{8} 123141 \text{ [#=4]} \quad \Delta = -112, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1 \overset{13212}{8} 123141 \quad \Delta = -104, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{1321}{8} 123141 \quad \Delta = -104, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1 \overset{31321}{8} 123141 \text{ [#=2]} \quad \Delta = -112, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{1321}{8} 123141 \overset{3}{1} \text{ [#=2]} \quad \Delta = -112, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$1 \overset{31321}{8} 123141 \text{ [#=3]} \quad \Delta = -112, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: $1 \overset{31321}{8} 12321$

$$1 \overset{31321}{8} 12321 \overset{3}{1} \text{ [#=3]} \quad \Delta = -94, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1 \overset{31321}{8} 12321 \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{31321}{8} 12321 \quad \Delta = -88, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$\begin{aligned}
& \overset{31321}{1 \ 8 \ 12321\mathbf{3}}_{[\#=6]} \quad \Delta = -94, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& \overset{31321}{1 \ 8 \ 12321} \quad \Delta = -94, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{31321}{1 \ 8 \ 12321\mathbf{3}}_{[\#=2]} \quad \Delta = -94, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{31321}{1 \ 8 \ 12321}_{[\#=2]} \quad \Delta = -94, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{31321}{1 \ 8 \ 12321\mathbf{3}}_{[\#=8]} \quad \Delta = -94, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
& \overset{31321}{1 \ 8 \ 12321\mathbf{3}}_{[\#=4]} \quad \Delta = -94, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{31321}{1 \ 8 \ 12321\mathbf{3}}_{[\#=5]} \quad \Delta = -94, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& \overset{31321}{1 \ 8 \ 12321\mathbf{3}}_{[\#=7]} \quad \Delta = -94, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
& \overset{31321}{1 \ 8 \ 12321} \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{31321\mathbf{2}}{1 \ 8 \ 12321} \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{31321}{1 \ 8 \ 12321\mathbf{3}}_{[\#=9]} \quad \Delta = -94, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
& \text{Base: } \overset{2321}{1 \ 8 \ 12321} \\
& \overset{2321}{1 \ 8 \ 12321} \quad \Delta = -80, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{2321\mathbf{2}}{1 \ 8 \ 12321} \quad \Delta = -80, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{2321}{1 \ 8 \ 12321\mathbf{3}}_{[\#=9]} \quad \Delta = -88, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
& \overset{2321}{1 \ 8 \ 12321\mathbf{3}}_{[\#=5]} \quad \Delta = -88, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& \overset{2321}{1 \ 8 \ 12321} \quad \Delta = -108, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
& \overset{2321}{1 \ 8 \ 12321} \quad \Delta = -80, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{2321}{1 \ 8 \ 12321\mathbf{3}}_{[\#=8]} \quad \Delta = -88, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
& \overset{2321}{1 \ 8 \ 12321\mathbf{3}}_{[\#=7]} \quad \Delta = -88, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{2321}{1 \ 8 \ 12321} \overset{\color{red}{3}}{[\# = 6]} \quad \Delta = -88, \ n_+ = 1, \ n_- = 12, \ n_0 = 0, \ |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& \overset{2321}{1 \ 8 \ 12321} \overset{\color{red}{3}}{[\# = 2]} \quad \Delta = -88, \ n_+ = 1, \ n_- = 12, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{2321}{1 \ 8 \ 12321} [\# = 2] \quad \Delta = -88, \ n_+ = 1, \ n_- = 12, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{2321}{1 \ 8 \ 12321} \overset{\color{red}{3}}{[\# = 3]} \quad \Delta = -88, \ n_+ = 1, \ n_- = 12, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{2321}{1 \ 8 \ 12321} [\# = 3] \quad \Delta = -88, \ n_+ = 1, \ n_- = 12, \ n_0 = 0, \ |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{2321}{1 \ 8 \ 12321} \overset{\color{red}{3}}{[\# = 4]} \quad \Delta = -88, \ n_+ = 1, \ n_- = 12, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

Base: $21 \overset{1512321}{8} 12$

$$\begin{aligned}
& \overset{\color{red}{2}1512321}{21 \ 8 \ 12} \quad \Delta = -121, \ n_+ = 1, \ n_- = 12, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{1512321}{21 \ 8 \ 12} \quad \Delta = -121, \ n_+ = 1, \ n_- = 12, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{1512321}{21 \ 8 \ 12} \quad \Delta = -121, \ n_+ = 1, \ n_- = 12, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: $21 \overset{2151321}{8} 12$

$$\begin{aligned}
& \overset{2151321}{21 \ 8 \ 12} \quad \Delta = -127, \ n_+ = 1, \ n_- = 12, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
& \overset{2151321}{21 \ 8 \ 12} \quad \Delta = -119, \ n_+ = 1, \ n_- = 12, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{\color{red}{2}2151321}{21 \ 8 \ 12} \quad \Delta = -119, \ n_+ = 1, \ n_- = 12, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{2151321}{21 \ 8 \ 12} \quad \Delta = -119, \ n_+ = 1, \ n_- = 12, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{2151321}{21 \ 8 \ 12} \quad \Delta = -119, \ n_+ = 1, \ n_- = 12, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: $21 \overset{151321}{8} 12$

$$\begin{aligned}
& \overset{151321}{21 \ 8 \ 12} \quad \Delta = -119, \ n_+ = 1, \ n_- = 12, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{\color{red}{2}151321}{21 \ 8 \ 12} \quad \Delta = -119, \ n_+ = 1, \ n_- = 12, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{151321}{21 \ 8 \ 12} \quad \Delta = -119, \ n_+ = 1, \ n_- = 12, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: $21 \overset{512321}{8} 123$

$$\begin{aligned}
21 \overset{2}{8} 123 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21 \overset{512321}{8} 123 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21 \overset{512321}{8} 123 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
21 \overset{512321}{8} 123 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

Base: $21 \overset{151321}{8} 123$

$$\begin{aligned}
21 \overset{151321}{8} 123 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
21 \overset{151321}{8} 123 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21 \overset{151321}{8} 123 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21 \overset{2}{8} 123 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
321 \overset{151321}{8} 123 \quad \Delta = -137, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: $21 \overset{12321}{8} 1231$

$$\begin{aligned}
21 \overset{12321}{8} 1231 \overset{3}{\# = 5} \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
21 \overset{12321}{8} 1231 \overset{3}{\# = 2} \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
21 \overset{312321}{8} 1231 \overset{3}{\# = 2} \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21 \overset{312321}{8} 1231 \overset{3}{\# = 9} \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
21 \overset{12321}{8} 1231 \overset{3}{\# = 4} \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21 \overset{312321}{8} 1231 \overset{3}{\# = 8} \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
21 \overset{312321}{8} 1231 \overset{3}{\# = 4} \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21 \overset{312321}{8} 1231 \overset{3}{\# = 7} \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
21 \overset{312321}{8} 1231 \overset{3}{\# = 3} \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
21 \overset{12321}{8} 1231 \textcolor{red}{3} \text{ [#=3]} \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21 \overset{12321}{8} 1231 \quad \Delta = -78, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21 \overset{12321\textcolor{red}{2}}{8} 1231 \quad \Delta = -78, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21 \overset{\textcolor{red}{2}12321}{8} 1231 \quad \Delta = -78, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21 \overset{\textcolor{red}{3}12321}{8} 1231 \text{ [#=5]} \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
21 \overset{\textcolor{red}{3}12321}{8} 1231 \text{ [#=6]} \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\textbf{Base: } 21 \overset{51321}{8} 1232 \\
21 \overset{51321}{8} 12\textcolor{red}{3} \quad \Delta = -139, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21 \overset{\textcolor{red}{2}51321}{8} 1232 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21 \overset{51321\textcolor{red}{2}}{8} 1232 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21 \overset{51321}{8} 1232 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
21 \overset{51321}{8} 1232 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 100 \quad & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\textbf{Base: } 12321 \overset{321}{8} 123 \\
321 \overset{321}{8} 12321\textcolor{red}{3} \quad \Delta = -95, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
321 \overset{321}{8} 12321\textcolor{red}{8} \quad \Delta = -220, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
321 \overset{321}{8} 12321\textcolor{red}{7} \quad \Delta = -192, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 8 \end{pmatrix} \\
321 \overset{321}{8} 12321\textcolor{red}{5} \quad \Delta = -139, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
321 \overset{321}{8} 12321\textcolor{red}{6} \quad \Delta = -165, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
321 \overset{321}{8} 12321\textcolor{red}{2} \quad \Delta = -87, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
321 \overset{321}{8} 12321\textcolor{red}{4} \quad \Delta = -115, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{ll}
12321 \overset{321}{8} 123 \quad \Delta = -87, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
321 \overset{321}{8} 12321\mathbf{3} \text{ [#=7]} \quad \Delta = -95, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 196 & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
321 \overset{321}{8} 12321\mathbf{3} \text{ [#=8]} \quad \Delta = -95, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 256 & \begin{pmatrix} 0 & 16 & 2 \\ 16 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
321 \overset{321}{8} 12321\mathbf{3} \text{ [#=5]} \quad \Delta = -95, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
321 \overset{321}{8} 12321\mathbf{3} \text{ [#=9]} \quad \Delta = -95, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 324 & \begin{pmatrix} 0 & 18 & 2 \\ 18 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
321 \overset{321}{8} 12321\mathbf{3} \text{ [#=6]} \quad \Delta = -95, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 144 & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
321 \overset{321}{8} 12321\mathbf{3} \text{ [#=2]} \quad \Delta = -95, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
321 \overset{321}{8} 12321\mathbf{3} \text{ [#=3]} \quad \Delta = -95, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
321 \overset{321}{8} 12321\mathbf{3} \text{ [#=4]} \quad \Delta = -95, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
12321 \overset{2321}{8} 123 \quad \Delta = -87, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{1,21513221}{8} 1 & \\
1 \overset{1,21513221}{8} 1 \quad \Delta = -127, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1 \overset{1,21513221}{8} 1 \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \overset{1,21513221}{8} 1 \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \overset{1,21513221}{8} 1 \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{1,1513221}{8} 1 & \\
1 \overset{1,1513221}{8} 1 \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1 \overset{1,1513221}{8} 1 \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{1,31512321}{8} 1 & \\
1 \overset{1,31512321}{8} 1 \quad \Delta = -129, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1 \overset{1,31512321}{8} 1 \quad \Delta = -129, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{array}$$

$$1 \overset{1,31512321}{8} 1 \quad \Delta = -129, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: $1 \overset{1,32151321}{8} 1$

$$1 \overset{1,32151321}{8} 1 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1 \overset{1,32151321}{8} 1 \quad \Delta = -128, \quad n_+ = 1, \quad n_- = 12, \quad n_0 = 0, \quad |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1 \overset{1,32151321}{8} 1 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: $1 \overset{1,3151321}{8} 1$

$$1^{1,3151321}_8 1 \quad \Delta = -135, \quad n_+ = 1, \quad n_- = 12, \quad n_0 = 0, \quad |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{1,3151321}{8} 1 \quad \Delta = -127, \quad n_+ = 1, \quad n_- = 12, \quad n_0 = 0, \quad |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1^{1,3151321}_8 1 \quad \Delta = -127, \quad n_+ = 1, \quad n_- = 12, \quad n_0 = 0, \quad |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1^{1,3151321}_8 1 \quad \Delta = -127, \quad n_+ = 1, \quad n_- = 12, \quad n_0 = 0, \quad |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1^{1,3151321}_8 1 \quad \Delta = -127, \quad n_+ = 1, \quad n_- = 12, \quad n_0 = 0, \quad |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: $1 \overset{21,2151321}{8} 1$

$$1^{21,2151321} 8^1 1^1 \quad \Delta = -119, \quad n_+ = 1, \quad n_- = 12, \quad n_0 = 0, \quad |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1^{21,2151321} 8^{1} 1 \quad \Delta = -127, \quad n_+ = 1, \quad n_- = 12, \quad n_0 = 0, \quad |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1^{21,2151321} 8^1 1^1 \quad \Delta = -119, \quad n_+ = 1, \quad n_- = 12, \quad n_0 = 0, \quad |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1^{21, 2151321} 8 \quad 1 \quad \Delta = -119, \quad n_+ = 1, \quad n_- = 12, \quad n_0 = 0, \quad |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1^{21,2151321} 8^1 1^1 \quad \Delta = -119, \quad n_+ = 1, \quad n_- = 12, \quad n_0 = 0, \quad |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: $1 \overset{21,151321}{8} 1$

$$1^{21,151321} 8^1 1 \quad \Delta = -119, \quad n_+ = 1, \quad n_- = 12, \quad n_0 = 0, \quad |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1^{21,151321} 8^1 1^1 \quad \Delta = -119, \quad n_+ = 1, \quad n_- = 12, \quad n_0 = 0, \quad |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1^{21,151321} 8^1 1^1 \quad \Delta = -119, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: $1 \overset{21,1512321}{8} 1$

$$1 \overset{21,1512321}{8} 1 \quad \Delta = -121, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{21,1512321}{8} 1 \quad \Delta = -121, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{21,1512321}{8} 1 \quad \Delta = -121, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{1,512321}{8} 123$$

$$1 \overset{1,512321}{8} 123 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \overset{1,512321}{8} 123 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$321 \overset{1,512321}{8} \textcolor{red}{12} \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{1,512321\textcolor{red}{2}}{8} 123 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{1,151321}{8} 123$$

$$1 \overset{1,151321}{8} 123 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{1,151321}{8} 123 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1 \overset{1,151321\textcolor{red}{2}}{8} 123 \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$321 \overset{1,151321}{8} \textcolor{red}{12} \quad \Delta = -128, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{1,12321}{8} 1231$$

$$1 \overset{1,12321}{8} 1231 \quad \Delta = -78, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{1,12321\textcolor{red}{2}}{8} 1231 \quad \Delta = -78, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{1,312321}{8} 1231 \text{ [#=9]} \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$1 \overset{1,312321}{8} 1231 \text{ [#=7]} \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$1 \overset{1,12321}{8} 1231\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1 \overset{1,312321}{8} 1231 \text{ [#=6]} \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$1 \overset{1,312321}{8} 1231 \text{ [#=5]} \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$\begin{aligned}
1 \text{ } ^{1,312321} \text{ } 8 \text{ } 1231 \text{ } [\# = 4] \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1 \text{ } ^{1,312321} \text{ } 8 \text{ } 1231 \text{ } [\# = 8] \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1 \text{ } ^{1,12321} \text{ } 8 \text{ } 1231 \text{ } ^{1231} \text{ } [\# = 4] \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \text{ } ^{1,12321} \text{ } 8 \text{ } 1231 \text{ } ^{1231} \text{ } [\# = 2] \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1 \text{ } ^{1,312321} \text{ } 8 \text{ } 1231 \text{ } [\# = 2] \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \text{ } ^{1,12321} \text{ } 8 \text{ } 1231 \text{ } ^{1231} \text{ } [\# = 3] \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \text{ } ^{1,312321} \text{ } 8 \text{ } 1231 \text{ } [\# = 3] \quad \Delta = -86, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \text{ } ^{1,51321} \text{ } 8 \text{ } 1232$

$$\begin{aligned}
1 \text{ } ^{1,51321} \text{ } 8 \text{ } 1232 \text{ } ^{1232} \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
2321 \text{ } ^{1,51321} \text{ } 8 \text{ } ^{1232} \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1 \text{ } ^{1,51321} \text{ } 8 \text{ } 1232 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1 \text{ } ^{1,51321} \text{ } 8 \text{ } 1232 \quad \Delta = -130, n_+ = 1, n_- = 12, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

$T = 13$ (1666 bases)

Base: 12122222222222

$$121 \text{ } ^{12} \text{ } 2222222222 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 130 & 22 \\ 2 & 22 & 8 \end{pmatrix}$$

Base: 11212222222222

$$1121 \text{ } ^{12} \text{ } 2222222222 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 108 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

Base: 21121222222222

$$\begin{aligned}
21121 \text{ } ^{12} \text{ } 2222222222 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21121 \text{ } ^{12} \text{ } 2222222222 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 88 & 18 \\ 2 & 18 & 8 \end{pmatrix}
\end{aligned}$$

Base: 22222112122222

$$222221 \text{ } ^{12} \text{ } 12122222 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: 22221121222222

$$\begin{aligned}
22221121^222222 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
22221121^222222 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\text{Base: } 22211212222222 & \\
2221121^22222222 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
2221121^22222222 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\text{Base: } 22112122222222 & \\
221121^222222222 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 70 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
221121^222222222 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\text{Base: } 1^{2222222221}12 & \\
1^{2222222221}12 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 100 \quad & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 88 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
\text{Base: } 1^{222222221}12 & \\
1^{222222221}12 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 70 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
1^{222222221}12^2 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\text{Base: } 1^{222221}121222 & \\
1^{222221}121222 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1^{222221}121222^2 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\text{Base: } 1^{2222221}1222 & \\
1^{2222221}1222 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1^{2222221}1222 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\text{Base: } 1^{22222221}122 & \\
1^{22222221}122 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1^{22222221}122 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\text{Base: } 21^{22222221}12 &
\end{aligned}$$

$$\begin{array}{ll}
\text{Base: } 21 \overset{222222221}{12} 12 & \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21 \overset{222222221}{12} 12 & \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\text{Base: } 21 \overset{22221}{12} 12222 & \\
21 \overset{222212}{12} 12222 & \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21 \overset{222221}{12} 12222 & \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\text{Base: } 21 \overset{222221}{12} 1222 & \\
21 \overset{222221}{12} 1222 & \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21 \overset{2222221}{12} 1222 & \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21 \overset{2222212}{12} 1222 & \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\text{Base: } 21 \overset{2222221}{12} 122 & \\
21 \overset{2222221}{12} 122 & \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
21 \overset{22222212}{12} 122 & \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21 \overset{22222221}{12} 122 & \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\text{Base: } 2221 \overset{2221}{12} 1222 & \\
2221 \overset{2221}{12} 1222 & \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\text{Base: } 221 \overset{22221}{12} 1222 & \\
221 \overset{22221}{12} 1222 & \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
221 \overset{22221}{12} 1222 & \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
221 \overset{222212}{12} 1222 & \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\text{Base: } 221 \overset{222221}{12} 122 & \\
221 \overset{222221}{12} 122 & \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
221 \overset{222221}{12} 122 & \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{1,222222221}{12} 1 &
\end{array}$$

$$\begin{array}{l}
1 \overset{1,222222221}{12} 1 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 88 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{21,22222221}{12} 1 \\
1 \overset{21,22222221}{12} 1 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1 \overset{21,22222221}{12} 1 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 70 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{1,22221}{12} 1222 \\
1 \overset{1,22221}{12} 1222 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{1,22221}{12} 1222 \\
1 \overset{1,22221}{12} 1222 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1 \overset{1,22221}{12} 1222 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{221,2222221}{12} 1 \\
1 \overset{221,2222221}{12} 1 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1 \overset{221,2222221}{12} 1 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{21,2222221}{12} 12 \\
1 \overset{21,2222221}{12} 12 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1 \overset{21,2222221}{12} 12 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{21,22221}{12} 1222 \\
1 \overset{21,22221}{12} 1222 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{21,22221}{12} 1222 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1 \overset{21,22221}{12} 1222 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{221,222221}{12} 12 \\
1 \overset{221,222221}{12} 12 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1 \overset{221,222221}{12} 12 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1 \overset{221,222221}{12} 12 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{array}$$

Base: $1 \overset{221,2221}{12} 1222$

$$1 \overset{221,2221}{12} 1222 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{221,2221\textcolor{red}{2}}{12} 1222 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: $1 \overset{221,22221}{12} 122$

$$1 \overset{221,22221\textcolor{red}{2}}{12} 122 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{221,22221}{12} 122 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: $21 \overset{21,222221}{12} 12$

$$21 \overset{21,222221}{12} 12 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$21 \overset{\textcolor{red}{2}21,222221}{12} 12 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: $21 \overset{21,2221}{12} 1222$

$$21 \overset{21,2221\textcolor{red}{2}}{12} 1222 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$21 \overset{\textcolor{red}{2}21,2221}{12} 1222 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: $21 \overset{221,22221}{12} 12$

$$21 \overset{221,22221}{12} 12 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$21 \overset{221,22221}{12} 12 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$21 \overset{\textcolor{red}{2}221,22221}{12} 12 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: $21 \overset{221,2221}{12} 122$

$$21 \overset{221,2221}{12} 122 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$21 \overset{221,2221\textcolor{red}{2}}{12} 122 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$21 \overset{\textcolor{red}{2}221,2221}{12} 122 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: $221 \overset{221,221}{12} 122$

$$221 \overset{\textcolor{red}{2}221,221}{12} 122 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 122222222222^2

$${}^2_{222222222221}12 \quad \Delta = -144, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$${}^2_{222222222221}8 \quad \Delta = -29, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$${}^2_{222222222221}7 \quad \Delta = -1, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 1314141413221

$$12231414141313 \text{ [#=3]} \quad \Delta = -5, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$13141414132213 \text{ [#=9]} \quad \Delta = -5, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix}$$

$${}^3_{1314141413221} \quad \Delta = -5, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$12231414141313 \text{ [#=2]} \quad \Delta = -5, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$12231414141313 \text{ [#=6]} \quad \Delta = -5, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

$$13141414132213 \text{ [#=3]} \quad \Delta = -5, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$12231414141313 \text{ [#=4]} \quad \Delta = -5, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$13141414132213 \text{ [#=4]} \quad \Delta = -5, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$12231414141313 \text{ [#=7]} \quad \Delta = -5, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix}$$

$$12231414141313 \text{ [#=5]} \quad \Delta = -5, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix}$$

$$13141414132213 \text{ [#=7]} \quad \Delta = -5, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$12231414141313 \text{ [#=8]} \quad \Delta = -5, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix}$$

$$13141414132213 \text{ [#=8]} \quad \Delta = -5, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$13141414132213 \text{ [#=6]} \quad \Delta = -5, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$13141414132213 \text{ [#=5]} \quad \Delta = -5, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$13141414132213 \text{ [#=2]} \quad \Delta = -5, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1223141414131\bar{3} \text{ }_{[\# = 9]} \quad \Delta = -5, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix}$$

Base: 1231414141321

$$1231414141321\bar{3} \text{ }_{[\# = 6]} \quad \Delta = -6, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$1231414141321\bar{3} \text{ }_{[\# = 3]} \quad \Delta = -6, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$1231414141321\bar{3} \text{ }_{[\# = 2]} \quad \Delta = -6, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1231414141321\bar{3} \text{ }_{[\# = 4]} \quad \Delta = -6, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$1231414141321\bar{3} \text{ }_{[\# = 8]} \quad \Delta = -6, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix}$$

$$1231414141321\bar{3} \text{ }_{[\# = 7]} \quad \Delta = -6, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix}$$

$$1231414141321\bar{3} \text{ }_{[\# = 9]} \quad \Delta = -6, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix}$$

$$1231414141321\bar{3} \text{ }_{[\# = 5]} \quad \Delta = -6, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

Base: 123141414132²

$$2^{\bar{4}1231414141} \bar{3}^2 \quad \Delta = -28, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$123141414132^{\bar{2}} \quad \Delta = 0, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$123141414132^{\bar{2}} \quad \Delta = 0, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 20 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$2^{\bar{3}1231414141} \bar{3}^2 \quad \Delta = -8, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$2^{\bar{3}1231414141} \bar{3}^2 \text{ }_{[\# = 2]} \quad \Delta = -8, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$2^{\bar{3}1231414141} \bar{3}^2 \text{ }_{[\# = 3]} \quad \Delta = -8, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$2^{\bar{3}1231414141} \bar{3}^2 \text{ }_{[\# = 4]} \quad \Delta = -8, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$2^{\bar{3}1231414141} \bar{3}^2 \text{ }_{[\# = 5]} \quad \Delta = -8, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 197 & 39 \\ 2 & 39 & 8 \end{pmatrix}$$

$$2^{\bar{3}1231414141} \bar{3}^2 \text{ }_{[\# = 6]} \quad \Delta = -8, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 285 & 47 \\ 2 & 47 & 8 \end{pmatrix}$$

$$2^{\overset{3}{1}231414141} 3^{\overset{2}{2}} 2_{[\# = 7]} \quad \Delta = -8, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 196 \quad \begin{pmatrix} 0 & 14 & 2 \\ 14 & 389 & 55 \\ 2 & 55 & 8 \end{pmatrix}$$

$$2^{\overset{3}{1}231414141} 3^{\overset{2}{2}} 2_{[\# = 8]} \quad \Delta = -8, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 256 \quad \begin{pmatrix} 0 & 16 & 2 \\ 16 & 509 & 63 \\ 2 & 63 & 8 \end{pmatrix}$$

$$2^{\overset{3}{1}231414141} 3^{\overset{2}{2}} 2_{[\# = 9]} \quad \Delta = -8, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 324 \quad \begin{pmatrix} 0 & 18 & 2 \\ 18 & 645 & 71 \\ 2 & 71 & 8 \end{pmatrix}$$

$$2^{\overset{5}{1}231414141} 3^{\overset{2}{2}} 2 \quad \Delta = -52, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$2^{\overset{6}{1}231414141} 3^{\overset{2}{2}} 2 \quad \Delta = -78, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$2^{\overset{7}{1}231414141} 3^{\overset{2}{2}} 2 \quad \Delta = -105, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$2^{\overset{8}{1}231414141} 3^{\overset{2}{2}} 2 \quad \Delta = -133, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$123141414132^{\overset{2}{2}} \quad \Delta = -2, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$123141414132^{\overset{2}{2}} \quad \Delta = 0, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$2^{\overset{2}{1}231414141} 3^{\overset{2}{2}} 2 \quad \Delta = 0, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$123141414132^{\overset{2}{2}} \quad \Delta = 0, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 2141414141321

$$2141414141321^{\overset{4}{4}} \quad \Delta = -45, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 14 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$2141414141321^{\overset{2}{2}} \quad \Delta = -17, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$2141414141321\overset{3}{3}_{[\# = 9]} \quad \Delta = -25, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$2141414141321\overset{3}{3}_{[\# = 4]} \quad \Delta = -25, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$2141414141321\overset{3}{3}_{[\# = 3]} \quad \Delta = -25, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$2141414141321^{\overset{3}{3}}_{[\# = 3]} \quad \Delta = -25, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 159 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

$$2141414141321\overset{3}{3}_{[\# = 2]} \quad \Delta = -25, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$2141414141321^{\overset{3}{3}} \quad \Delta = -25, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 15 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$\begin{aligned}
& \overset{3}{2141414141321} \text{ [#=2]} \quad \Delta = -25, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 69 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& 2141414141321\overset{3}{\text{}} \text{ [#=8]} \quad \Delta = -25, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& 2141414141321\overset{3}{\text{}} \text{ [#=7]} \quad \Delta = -25, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& 2141414141321\overset{3}{\text{}} \text{ [#=6]} \quad \Delta = -25, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 2141414141321\overset{3}{\text{}} \text{ [#=5]} \quad \Delta = -25, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{2}{2141414141321} \quad \Delta = -17, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 2141414141321\overset{2}{\text{}} \quad \Delta = -17, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 2141414141321\overset{2}{\text{}} \quad \Delta = -17, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& 2141414141321\overset{2}{\text{}} \quad \Delta = -17, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \text{Base: } 1314141414131 \\
& \overset{3}{1314141414131} \quad \Delta = -32, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 15 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& 1314141414131\overset{2}{\text{}} \quad \Delta = -24, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 1314141414131\overset{2}{\text{}} \quad \Delta = -24, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 1314141414131\overset{2}{\text{}} \quad \Delta = -24, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 1314141414131\overset{3}{\text{}} \text{ [#=5]} \quad \Delta = -32, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
& 1314141414131\overset{3}{\text{}} \text{ [#=2]} \quad \Delta = -32, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 1314141414131\overset{3}{\text{}} \text{ [#=8]} \quad \Delta = -32, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
& 1314141414131\overset{3}{\text{}} \text{ [#=6]} \quad \Delta = -32, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
& 1314141414131\overset{3}{\text{}} \text{ [#=7]} \quad \Delta = -32, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
& 1314141414131\overset{3}{\text{}} \text{ [#=4]} \quad \Delta = -32, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix}
\end{aligned}$$

$$1314141414131\overset{3}{3} \text{ }_{[\# = 3]} \quad \Delta = -32, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$1314141414131\overset{3}{3} \text{ }_{[\# = 9]} \quad \Delta = -32, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix}$$

Base: 2141414141 $\overset{2}{3}2$

$$\overset{3}{2}141414141\overset{2}{3}2 \quad \Delta = -27, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 23 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$\overset{4}{2}141414141\overset{2}{3}2 \quad \Delta = -47, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 22 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$21414141\overset{2}{4}1\overset{2}{3}2 \quad \Delta = -19, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$21414141\overset{2}{4}1\overset{2}{3}2 \quad \Delta = -19, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 20 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$21414141\overset{2}{4}1\overset{2}{3}2 \quad \Delta = -19, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\overset{2}{2}141414141\overset{2}{3}2 \quad \Delta = -19, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 24 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$21414141\overset{2}{4}1\overset{2}{3}2 \quad \Delta = -19, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\overset{3}{2} \overset{2141414141}{3} \overset{2}{2} \quad \Delta = -28, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\overset{3}{2}141414141\overset{2}{3}2 \text{ }_{[\# = 2]} \quad \Delta = -27, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 101 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$\overset{3}{2}141414141\overset{2}{3}2 \text{ }_{[\# = 3]} \quad \Delta = -27, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 231 & 41 \\ 2 & 41 & 8 \end{pmatrix}$$

Base: 2141414141412

$$\overset{4}{2}141414141412 \quad \Delta = -64, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 18 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\overset{3}{2}141414141412 \text{ }_{[\# = 2]} \quad \Delta = -44, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 85 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$\overset{3}{2}141414141412 \quad \Delta = -44, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 19 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$\overset{3}{2}141414141412 \text{ }_{[\# = 3]} \quad \Delta = -44, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 195 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$\overset{2}{2}141414141412 \quad \Delta = -36, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 20 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$\overset{2}{2}141414141412 \quad \Delta = -36, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$2141414141412 \quad \Delta = -36, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 141414141321

$$14141414141321\overset{1}{3} \quad \Delta = -25, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$14141414141321\overset{1}{3} \quad \Delta = -25, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1\overset{31}{4}1414141321 \quad \Delta = -25, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$14141414141321\overset{1}{3} \quad \Delta = -25, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$14141414141321\overset{1}{3} \quad \Delta = -25, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$14141414141321\overset{1}{3} \quad \Delta = -25, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$14141414141321\overset{1}{3} \quad \Delta = -25, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$14141414141321 \quad \Delta = -17, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$14141414141321 \quad \Delta = -17, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$14141414141321 \quad \Delta = -17, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$14141414141321 \quad \Delta = -17, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1\overset{31}{4}1414141321 \quad \Delta = -25, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$14141414141321\overset{1}{3} \quad \Delta = -25, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$14141414141321\overset{1}{3} \quad \Delta = -25, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: 14141414132

$$1414141414132 \quad \Delta = -19, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1414141414132 \quad \Delta = -19, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1\overset{21}{4}141414132 \quad \Delta = -19, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\begin{aligned}
& \overset{1}{14}141414\overset{22}{132} \quad \Delta = -19, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{12}{14}141414\overset{2}{132} \quad \Delta = -19, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 20 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{31}{14}141414\overset{2}{132} \text{ [#=2]} \quad \Delta = -27, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& \overset{31}{14}141414\overset{2}{132} \text{ [#=3]} \quad \Delta = -27, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 60 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
& \overset{31}{14}141414\overset{2}{132} \quad \Delta = -27, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{41}{14}141414\overset{2}{132} \quad \Delta = -47, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{14}1414141412$

$$\begin{aligned}
& \overset{31}{14}1414141412 \text{ [#=3]} \quad \Delta = -44, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{1}{14}14141414\overset{4}{12} \quad \Delta = -64, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 8 \end{pmatrix} \\
& \overset{1}{14}14141414\overset{2}{12} \quad \Delta = -36, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{12}{14}1414141412 \quad \Delta = -36, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{1}{14}14\overset{2}{14}141412 \quad \Delta = -36, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{31}{14}1414141412 \text{ [#=2]} \quad \Delta = -44, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{1}{14}14141414\overset{3}{12} \text{ [#=2]} \quad \Delta = -44, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{1}{14}14141414\overset{3}{12} \text{ [#=3]} \quad \Delta = -44, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{1}{14}1414\overset{2}{14}1412 \quad \Delta = -36, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{1}{14}141414\overset{2}{14}12 \quad \Delta = -36, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{1}{14}141414\overset{3}{14}12 \quad \Delta = -44, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{14}141414\overset{1}{14}1$

$$\overset{31}{14}141414\overset{1}{14}1 \text{ [#=2]} \quad \Delta = -44, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$\begin{aligned}
& \overset{1}{1} \overset{2}{4} 1 4 1 4 1 4 1 \overset{1}{1} \quad \Delta = -36, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{1}{1} 4 1 \overset{2}{4} 1 4 1 4 1 \overset{1}{1} \quad \Delta = -36, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{3}{1} 4 1 4 1 4 1 4 1 \overset{1}{1} \text{ [#=3]} \quad \Delta = -44, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1223151315131

$$\begin{aligned}
1223151315131 \text{3} \text{ [#=8]} \quad \Delta = -33, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
1315131513221 \text{3} \text{ [#=4]} \quad \Delta = -33, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 109 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1223151315131 \text{3} \text{ [#=4]} \quad \Delta = -33, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
1223151315131 \text{3} \text{ [#=5]} \quad \Delta = -33, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
1315131513221 \text{3} \text{ [#=6]} \quad \Delta = -33, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 249 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
1223151315131 \text{3} \text{ [#=9]} \quad \Delta = -33, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
1315131513221 \text{3} \text{ [#=3]} \quad \Delta = -33, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 60 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
1223151315131 \text{3} \text{ [#=3]} \quad \Delta = -33, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
1223151315131 \text{3} \text{ [#=6]} \quad \Delta = -33, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
1315131513221 \text{3} \text{ [#=7]} \quad \Delta = -33, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 340 & 48 \\ 2 & 48 & 8 \end{pmatrix} \\
1223151315131 \text{3} \text{ [#=7]} \quad \Delta = -33, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
1315131513221 \text{3} \text{ [#=9]} \quad \Delta = -33, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 564 & 62 \\ 2 & 62 & 8 \end{pmatrix} \\
1315131513221 \text{3} \text{ [#=8]} \quad \Delta = -33, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 445 & 55 \\ 2 & 55 & 8 \end{pmatrix} \\
1315131513221 \text{3} \text{ [#=2]} \quad \Delta = -33, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 25 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1315131513221 \text{3} \text{ [#=5]} \quad \Delta = -33, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 172 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
1223151315131 \text{3} \text{ [#=2]} \quad \Delta = -33, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1223151\overset{2}{3}15131 \quad \Delta = -26, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1223151\overset{2}{3}15131 \quad \Delta = -25, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1223151\overset{2}{3}15131 \quad \Delta = -25, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1223151\overset{2}{3}15131 \quad \Delta = -25, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1223151\overset{2}{3}15131 \quad \Delta = -25, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1231513151321

$$\begin{aligned}
1231513151\overset{2}{3}21 \quad \Delta = -26, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1231513151\overset{2}{3}21 \quad \Delta = -27, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1231513151\overset{2}{3}21 \quad \Delta = -26, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1231513151321\overset{2}{3} \text{ [#=7]} \quad \Delta = -34, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 340 & 48 \\ 2 & 48 & 8 \end{pmatrix} \\
1231513151321\overset{2}{3} \text{ [#=3]} \quad \Delta = -34, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 60 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
1231513151321\overset{2}{3} \text{ [#=2]} \quad \Delta = -34, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 25 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1231513151321\overset{2}{3} \text{ [#=9]} \quad \Delta = -34, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 564 & 62 \\ 2 & 62 & 8 \end{pmatrix} \\
1231513151321\overset{2}{3} \text{ [#=8]} \quad \Delta = -34, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 445 & 55 \\ 2 & 55 & 8 \end{pmatrix} \\
1231513151321\overset{2}{3} \text{ [#=4]} \quad \Delta = -34, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 109 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1231513151321\overset{2}{3} \text{ [#=6]} \quad \Delta = -34, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 249 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
1231513151321\overset{2}{3} \text{ [#=5]} \quad \Delta = -34, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 172 & 34 \\ 2 & 34 & 8 \end{pmatrix}
\end{aligned}$$

Base: 6122315132216

$$\begin{aligned}
612231513221\overset{2}{6} \quad \Delta = -122, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
612231513221\overset{2}{6} \quad \Delta = -122, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

$$6122315132216 \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 27 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

Base: 1612231513221

$$1612231513221 \quad \Delta = -44, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$16122315132213 \text{ [#=5]} \quad \Delta = -52, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$1612231513221 \quad \Delta = -44, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1612231513221 \quad \Delta = -44, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$16122315132213 \text{ [#=2]} \quad \Delta = -52, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$16122315132213 \text{ [#=6]} \quad \Delta = -52, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$16122315132213 \text{ [#=7]} \quad \Delta = -52, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$1612231513221 \quad \Delta = -52, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 17 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$16122315132213 \text{ [#=8]} \quad \Delta = -52, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$16122315132213 \text{ [#=9]} \quad \Delta = -52, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$16122315132213 \text{ [#=3]} \quad \Delta = -52, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$16122315132213 \text{ [#=4]} \quad \Delta = -52, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

Base: 6122315132161

$$6122315132161 \quad \Delta = -122, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$6122315132161 \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 63 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$6122315132161 \quad \Delta = -122, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$6122315132161 \quad \Delta = -122, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$6122315132161 \quad \Delta = -122, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: 1612315132161

$$\begin{aligned}
1612315132161 \quad \Delta = -122, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1612315132161 \quad \Delta = -122, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{aligned}$$

Base: 2161231513221

$$\begin{aligned}
2161231513221 \quad \Delta = -44, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2161231513221\textcolor{red}{3} \text{ [#=7]} \quad \Delta = -52, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
2161231513221 \quad \Delta = -44, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2161231513221 \quad \Delta = -44, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2161231513221 \quad \Delta = -44, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2161231513221\textcolor{red}{3} \text{ [#=8]} \quad \Delta = -52, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
2161231513221 \quad \Delta = -52, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2161231513221\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -52, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
2161231513221\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -52, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
2161231513221\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -52, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
2161231513221\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -52, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
2161231513221\textcolor{red}{3} \text{ [#=6]} \quad \Delta = -52, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
2161231513221\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -52, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}
\end{aligned}$$

Base: 6123151321613

$$\begin{aligned}
6123151321613 \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
6123151321613 \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
6123151321613 \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

6123151321613 ²	$\Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, \det = 81$	$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$
6123151321613 ²	$\Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$
Base: 3161231513161		
3161231513161 ²	$\Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, \det = 64$	$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$
3161231513161 ²	$\Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$
3161231513161 ²	$\Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$
3161231513161 ²	$\Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$
3161231513161 ²	$\Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, \det = 81$	$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix}$
Base: 6122315131612		
6122315131612 ²	$\Delta = -121, n_+ = 1, n_- = 13, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$
6122315131612 ³	$\Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 63 & 15 \\ 2 & 15 & 8 \end{pmatrix}$
6122315131612 ³	$\Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, \det = 4$	$\begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$
6122315131612 ²	$\Delta = -121, n_+ = 1, n_- = 13, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$
6122315131612 ²	$\Delta = -121, n_+ = 1, n_- = 13, n_0 = 0, \det = 4$	$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$
6122315131612 ²	$\Delta = -121, n_+ = 1, n_- = 13, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$
6122315131612 ²	$\Delta = -121, n_+ = 1, n_- = 13, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
Base: 1612315131612		
1612315131612 ²	$\Delta = -121, n_+ = 1, n_- = 13, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$
1612315131612 ²	$\Delta = -121, n_+ = 1, n_- = 13, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
1612315131612 ²	$\Delta = -121, n_+ = 1, n_- = 13, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$
1612315131612 ²	$\Delta = -121, n_+ = 1, n_- = 13, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$

$$1612315131612^2 \quad \Delta = -121, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1612315131612^3 \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

Base: 2161315131612

$$2161315131612^3 \quad \Delta = -128, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$2161315131612^2 \quad \Delta = -120, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$2161315131612^2 \quad \Delta = -120, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$2161315131612^2 \quad \Delta = -120, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 2216131513221

$$2216131513221^2 \quad \Delta = -43, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$2216131513221^3 \text{ [#=4]} \quad \Delta = -51, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$2216131513221^2 \quad \Delta = -43, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$2216131513221^3 \text{ [#=8]} \quad \Delta = -51, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$2216131513221^3 \text{ [#=2]} \quad \Delta = -51, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$2216131513221^3 \text{ [#=3]} \quad \Delta = -51, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$2216131513221^3 \text{ [#=5]} \quad \Delta = -51, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$2216131513221^3 \text{ [#=6]} \quad \Delta = -51, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$2216131513221^3 \text{ [#=7]} \quad \Delta = -51, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$2216131513221^3 \text{ [#=9]} \quad \Delta = -51, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$2216131513221^3 \quad \Delta = -51, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$2216131513221^3 \quad \Delta = -52, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$2216131513221^2 \quad \Delta = -43, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$2216131513221 \quad \Delta = -43, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: 1316123151321

$$1231513216131\bar{3} \text{ [#=2]} \quad \Delta = -60, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1231513216131\bar{3} \text{ [#=8]} \quad \Delta = -60, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$1316123151321\bar{3} \text{ [#=8]} \quad \Delta = -60, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$1231513216131\bar{3} \text{ [#=9]} \quad \Delta = -60, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix}$$

$$1316123151321 \quad \Delta = -52, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1316123151321\bar{3} \text{ [#=6]} \quad \Delta = -60, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$1316123151321 \quad \Delta = -52, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1316123151321 \quad \Delta = -52, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1316123151321 \quad \Delta = -52, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1316123151321\bar{3} \text{ [#=5]} \quad \Delta = -60, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1231513216131\bar{3} \text{ [#=7]} \quad \Delta = -60, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$1316123151321\bar{3} \text{ [#=9]} \quad \Delta = -60, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix}$$

$$1316123151321\bar{3} \text{ [#=7]} \quad \Delta = -60, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$1231513216131\bar{3} \text{ [#=6]} \quad \Delta = -60, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$1231513216131\bar{3} \text{ [#=5]} \quad \Delta = -60, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1316123151321\bar{3} \text{ [#=4]} \quad \Delta = -60, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1316123151321\bar{3} \text{ [#=2]} \quad \Delta = -60, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1231513216131\bar{3} \text{ [#=4]} \quad \Delta = -60, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1231513216131\overset{2}{3} \text{ }_{[\# = 3]} \quad \Delta = -60, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1316123151321\overset{2}{3} \text{ }_{[\# = 3]} \quad \Delta = -60, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 6123151316123

$$61231513161\overset{2}{2}3 \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$61231513161\overset{2}{2}3 \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$61231513161\overset{2}{2}3 \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$61231513161\overset{2}{2}3 \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$61231513161\overset{2}{2}3 \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: 1613151316123

$$16131513161\overset{2}{2}3 \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$16131513161\overset{2}{2}3 \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$16131513161\overset{2}{2}3 \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$16131513161\overset{2}{2}3 \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$16131513161\overset{2}{2}3 \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 1321613151321

$$1321613151321\overset{2}{3} \quad \Delta = -52, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1231513161231\overset{2}{3} \text{ }_{[\# = 2]} \quad \Delta = -60, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1321613151321\overset{2}{3} \text{ }_{[\# = 2]} \quad \Delta = -60, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1321613151321\overset{2}{3} \quad \Delta = -52, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1321613151321\overset{2}{3} \quad \Delta = -52, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1321613151321\overset{2}{3} \quad \Delta = -52, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\begin{aligned}
1321613151321\bar{3} \text{ } [\# = 9] \quad \Delta = -60, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
1231513161231\bar{3} \text{ } [\# = 3] \quad \Delta = -60, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1231513161231\bar{3} \text{ } [\# = 4] \quad \Delta = -60, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1321613151321\bar{3} \text{ } [\# = 4] \quad \Delta = -60, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1231513161231\bar{3} \text{ } [\# = 5] \quad \Delta = -60, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1321613151321\bar{3} \text{ } [\# = 3] \quad \Delta = -60, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1321613151321\bar{3} \text{ } [\# = 8] \quad \Delta = -60, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1321613151321\bar{3} \text{ } [\# = 7] \quad \Delta = -60, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
1321613151321\bar{3} \text{ } [\# = 6] \quad \Delta = -60, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1321613151321\bar{3} \text{ } [\# = 5] \quad \Delta = -60, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

Base: 6131513161232

$$\begin{aligned}
6131513161232 \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
6131513161232 \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
6131513161232 \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
6131513161232 \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
6131513161232 \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1232161315131

$$\begin{aligned}
1315131612321\bar{3} \text{ } [\# = 3] \quad \Delta = -61, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1232161315131 \quad \Delta = -53, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1232161315131\bar{3} \text{ } [\# = 3] \quad \Delta = -61, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1232161315131\bar{3} \text{ } [\# = 2] \quad \Delta = -61, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1315131612321\bar{3} \text{ } [\# = 2] \quad \Delta = -61, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1232161315131\bar{3} \text{ } [\# = 9] \quad \Delta = -61, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1315131612321\bar{3} \text{ } [\# = 7] \quad \Delta = -61, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1232161315131\bar{3} \text{ } [\# = 7] \quad \Delta = -61, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1315131612321\bar{3} \text{ } [\# = 8] \quad \Delta = -61, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1232161315131\bar{3} \text{ } [\# = 8] \quad \Delta = -61, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1315131612321\bar{3} \text{ } [\# = 9] \quad \Delta = -61, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1232161315131\bar{2} \quad \Delta = -53, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1315131612321\bar{3} \text{ } [\# = 4] \quad \Delta = -61, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1232161315131\bar{2} \quad \Delta = -53, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1232161315131\bar{2} \quad \Delta = -53, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1232161315131\bar{3} \text{ } [\# = 4] \quad \Delta = -61, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1315131612321\bar{3} \text{ } [\# = 5] \quad \Delta = -61, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1232161315131\bar{3} \text{ } [\# = 5] \quad \Delta = -61, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1232161315131\bar{3} \text{ } [\# = 6] \quad \Delta = -61, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1315131612321\bar{3} \text{ } [\# = 6] \quad \Delta = -61, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}
\end{aligned}$$

Base: 6123216131513

$$\begin{aligned}
6123216131513\bar{2} \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 100 \quad & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
6123216131513\bar{2} \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

$$6123216131513 \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$6123216131513 \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$6123216131513 \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 1612321613151

$$1612321613151 \quad \Delta = -123, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1612321613151 \quad \Delta = -123, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1612321613151 \quad \Delta = -123, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1612321613151 \quad \Delta = -123, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 3161232161315

$$3161232161315 \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$3161232161315 \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$3161232161315 \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$3161232161315 \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$3161232161315 \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: 6131612315132

$$6131612315132 \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$6131612315132 \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$6131612315132 \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 225 \quad \begin{pmatrix} 0 & 15 & 2 \\ 15 & 163 & 25 \\ 2 & 25 & 8 \end{pmatrix}$$

$$6131612315132 \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$6131612315132 \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

Base: 1316123216131

$$1316123216131 \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\begin{aligned}
1316123216131\bar{3} \text{ } [\# = 7] \quad \Delta = -87, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1316123216131\bar{3} \text{ } [\# = 6] \quad \Delta = -87, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1316123216131\bar{3} \text{ } [\# = 2] \quad \Delta = -87, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1316123216131\bar{3} \text{ } [\# = 8] \quad \Delta = -87, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1316123216131\bar{3} \text{ } [\# = 9] \quad \Delta = -87, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1316123216131\bar{3} \text{ } [\# = 5] \quad \Delta = -87, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1316123216131\bar{3} \text{ } [\# = 3] \quad \Delta = -87, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1316123216131\bar{3} \text{ } [\# = 4] \quad \Delta = -87, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
13161\bar{2}3216131 \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1316131513215

$$\begin{aligned}
13161\bar{2}31513215 \quad \Delta = -103, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
13161\bar{2}31513215 \quad \Delta = -103, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
13161315\bar{2}3215 \quad \Delta = -103, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
5123151316131\bar{3} \text{ } [\# = 4] \quad \Delta = -111, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
5123151316131\bar{3} \text{ } [\# = 5] \quad \Delta = -111, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
13161315\bar{2}3215 \quad \Delta = -103, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
5123151316131\bar{3} \text{ } [\# = 2] \quad \Delta = -111, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
13161\bar{2}31513215 \quad \Delta = -103, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
5123151316131\bar{3} \text{ } [\# = 8] \quad \Delta = -111, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
5123151316131\bar{3} \text{ } [\# = 7] \quad \Delta = -111, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
5123151316131\bar{3} \text{ } [\# = 9] \quad \Delta = -111, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
5123151316131\bar{3} \text{ } [\# = 6] \quad \Delta = -111, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
5123151316131\bar{3} \text{ } [\# = 3] \quad \Delta = -111, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1316131513151

$$\begin{aligned}
1316131513151\bar{2} \quad \Delta = -103, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1316131513151\bar{2} \quad \Delta = -102, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1513151316131\bar{3} \text{ } [\# = 2] \quad \Delta = -110, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1513151316131\bar{3} \text{ } [\# = 7] \quad \Delta = -110, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
1316131513151\bar{2} \quad \Delta = -102, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1316131513151\bar{2} \quad \Delta = -102, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1513151316131\bar{3} \text{ } [\# = 4] \quad \Delta = -110, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1513151316131\bar{3} \text{ } [\# = 5] \quad \Delta = -110, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1513151316131\bar{3} \text{ } [\# = 6] \quad \Delta = -110, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1513151316131\bar{3} \text{ } [\# = 3] \quad \Delta = -110, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1513151316131\bar{3} \text{ } [\# = 8] \quad \Delta = -110, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1316131513151\bar{2} \quad \Delta = -102, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1316131513151\bar{2} \quad \Delta = -102, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1513151316131\bar{3} \text{ } [\# = 9] \quad \Delta = -110, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix}
\end{aligned}$$

Base: 3131613151321

$$3131613151321\bar{2} \quad \Delta = -59, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\begin{aligned}
3131613151321\overset{\textcolor{red}{3}}{\text{ }}_{[\# = 4]} \quad \Delta = -67, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
3131613151321\overset{\textcolor{red}{3}}{\text{ }}_{[\# = 3]} \quad \Delta = -67, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
3131613151321\overset{\textcolor{red}{3}}{\text{ }}_{[\# = 9]} \quad \Delta = -67, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
3131613151321\overset{\textcolor{red}{2}}{\text{ }} \quad \Delta = -59, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
3131613151321\overset{\textcolor{red}{3}}{\text{ }}_{[\# = 5]} \quad \Delta = -67, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
3131613151321\overset{\textcolor{red}{3}}{\text{ }}_{[\# = 6]} \quad \Delta = -67, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
3131613151321\overset{\textcolor{red}{2}}{\text{ }} \quad \Delta = -59, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
3131613151321\overset{\textcolor{red}{2}}{\text{ }} \quad \Delta = -59, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
3131613151321\overset{\textcolor{red}{3}}{\text{ }}_{[\# = 2]} \quad \Delta = -67, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
3131613151321\overset{\textcolor{red}{3}}{\text{ }}_{[\# = 7]} \quad \Delta = -67, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
3131613151321\overset{\textcolor{red}{3}}{\text{ }}_{[\# = 8]} \quad \Delta = -67, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
3131613151321\overset{\textcolor{red}{4}}{\text{ }} \quad \Delta = -87, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
3131613151321\overset{\textcolor{red}{2}}{\text{ }} \quad \Delta = -59, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
3131613151321\overset{\textcolor{red}{3}}{\text{ }}_{[\# = 2]} \quad \Delta = -67, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
3131613151321\overset{\textcolor{red}{3}}{\text{ }} \quad \Delta = -67, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
3131613151321\overset{\textcolor{red}{3}}{\text{ }}_{[\# = 3]} \quad \Delta = -67, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 132 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
\textbf{Base: } 4131613151315 & \\
4131613151315\overset{\textcolor{red}{2}}{\text{ }} \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
4131613151315\overset{\textcolor{red}{2}}{\text{ }} \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
4131613151315\overset{\textcolor{red}{2}}{\text{ }} \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
4131613151315^2 \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
4131613151315^2 \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
4131613151315^2 \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
4131613151315^2 \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
4131613151315^3 \quad \Delta = -138, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1413161315131

$$\begin{aligned}
14131613151313^{\#=6} \quad \Delta = -86, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1413161315131^2 \quad \Delta = -78, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1413161315131^2 \quad \Delta = -78, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1413161315131^2 \quad \Delta = -78, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1413161315131^2 \quad \Delta = -78, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1413161315131^2 \quad \Delta = -78, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
14131613151313^{\#=9} \quad \Delta = -86, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
13151316131413^{\#=3} \quad \Delta = -86, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
14131613151313^{\#=7} \quad \Delta = -86, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
14131613151313^{\#=2} \quad \Delta = -86, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
13151316131413^{\#=2} \quad \Delta = -86, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
14131613151313^{\#=5} \quad \Delta = -86, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
14131613151313^{\#=3} \quad \Delta = -86, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
14131613151313^{\#=4} \quad \Delta = -86, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

$$1413161315131\overset{3}{3} \text{ [#=8]} \quad \Delta = -86, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1413161315131\overset{3}{3} \quad \Delta = -86, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 2151316131512

$$2151316131512\overset{3}{3} \quad \Delta = -102, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$2151316131512\overset{2}{2} \quad \Delta = -94, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$2151316131512\overset{2}{2} \quad \Delta = -94, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$2151316131512\overset{2}{2} \quad \Delta = -94, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: $1 \overset{2151316131}{5} 1$

$$1 \overset{2151316131}{5} 1 \quad \Delta = -94, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{2151316131}{5} 1 \quad \Delta = -102, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$1 \overset{2151316131}{5} 1 \quad \Delta = -94, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{2151316131}{5} 1 \quad \Delta = -94, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1 \overset{2151316131}{5} 1 \quad \Delta = -94, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{2151316131}{5} 1 \quad \Delta = -94, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: $1 \overset{151316131}{5} 1$

$$1 \overset{151316131}{5} 1 \quad \Delta = -94, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{151316131}{5} 1 \quad \Delta = -94, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 1613161231513

$$1613161231513\overset{2}{2} \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$1613161231513\overset{2}{2} \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1613161231513\overset{2}{2} \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1613161231513\overset{2}{2} \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$161316123151^2 \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: 2161316123151

$$2161316123151^2 \quad \Delta = -121, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$2161316123151^2 \quad \Delta = -121, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$2161316123151^2 \quad \Delta = -121, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$2161316123151^2 \quad \Delta = -121, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$2161316123151^2 \quad \Delta = -121, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$2161316123151^3 \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: 2161316123215

$$2161316123215^2 \quad \Delta = -123, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$2161316123215^2 \quad \Delta = -123, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$2161316123215^2 \quad \Delta = -123, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$2161316123215^2 \quad \Delta = -123, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$2161316123215^2 \quad \Delta = -123, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$2161316123215^3 \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: 2216131612321

$$2216131612321^3 \quad \Delta = -80, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$2216131612321^3 \text{ [#=2]} \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$2216131612321^2 \quad \Delta = -71, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$2216131612321^3 \text{ [#=9]} \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$2216131612321^3 \text{ [#=4]} \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\begin{aligned}
2216131612321\bar{3} \text{ } [\# = 5] \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
2216131612321\bar{2} \quad \Delta = -71, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
2216131612321\bar{2} \quad \Delta = -71, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2216131612321\bar{2} \quad \Delta = -71, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
2216131612321\bar{3} \text{ } [\# = 6] \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
2216131612321\bar{3} \text{ } [\# = 7] \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
2216131612321\bar{3} \text{ } [\# = 3] \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2216131612321\bar{3} \text{ } [\# = 8] \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
2216131612321\bar{3} \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\text{Base: } 3216131612315 \\
3216131612315\bar{2} \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 100 \quad & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
3216131612315\bar{2} \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
3216131612315\bar{2} \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
3216131612315\bar{2} \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
3216131612315\bar{2} \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\text{Base: } 1321613161231 \\
1321613161231\bar{3} \text{ } [\# = 3] \quad \Delta = -86, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1321613161231\bar{3} \text{ } [\# = 2] \quad \Delta = -86, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1321613161231\bar{3} \text{ } [\# = 5] \quad \Delta = -86, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1321613161231\bar{3} \text{ } [\# = 4] \quad \Delta = -86, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1321613161231 \quad \Delta = -78, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1321613161231 \quad \Delta = -78, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: 4141316131612

$$\begin{aligned}
4141316131612 \quad \Delta = -132, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 101 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
4141316131612 \quad \Delta = -124, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
4141316131612 \quad \Delta = -124, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
4141316131612 \quad \Delta = -132, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
4141316131612 \quad \Delta = -124, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
4141316131612 \quad \Delta = -124, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
4141316131612 \quad \Delta = -124, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
4141316131612 \quad \Delta = -124, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1413161316122

$$\begin{aligned}
1413161316122 \quad \Delta = -104, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
22161316131413 \quad [\#=2] \quad \Delta = -104, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1413161316122 \quad \Delta = -105, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
1413161316122 \quad \Delta = -96, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1413161316122 \quad \Delta = -96, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1413161316122 \quad \Delta = -96, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1413161316122 \quad \Delta = -96, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1413161316122 \quad \Delta = -104, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 21 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
22161316131413 \quad [\#=3] \quad \Delta = -104, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

$$1413161316122 \quad \Delta = -96, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 4132161316131

$$4132161316131\textcolor{red}{3} \text{ [#=7]} \quad \Delta = -113, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$4132161316131 \quad \Delta = -105, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$4132161316131 \quad \Delta = -105, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$4132161316131 \quad \Delta = -105, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$4132161316131 \quad \Delta = -105, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$4132161316131 \quad \Delta = -105, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$4132161316131 \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$4132161316131\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -113, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$4132161316131\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -113, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$4132161316131\textcolor{red}{3} \text{ [#=8]} \quad \Delta = -113, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$4132161316131\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -113, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$4132161316131\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -113, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$4132161316131\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -113, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$4132161316131\textcolor{red}{3} \text{ [#=6]} \quad \Delta = -113, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: 1321613161313

$$3131613161231\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -93, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1321613161313 \quad \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1321613161313 \quad \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\begin{aligned}
1321613161313^2 \quad \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1321613161313^2 \quad \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1321613161313^2 \quad \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1321613161313^3 \quad \Delta = -93, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
31316131612313^{3\#2} \quad \Delta = -93, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1321613161313^3 \#2 \quad \Delta = -93, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
31316131612313^{3\#5} \quad \Delta = -93, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
31316131612313^{3\#3} \quad \Delta = -93, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1321613161313^3 \#3 \quad \Delta = -93, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1321613161313^4 \quad \Delta = -113, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 9 & 5 \\ 5 & 9 \end{pmatrix}
\end{aligned}$$

Base: 3131613161313

$$\begin{aligned}
3131613161313^2 \quad \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
3131613161313^4 \quad \Delta = -120, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
3131613161313^2 \quad \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
3131613161313^2 \quad \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
3131613161313^3 \#2 \quad \Delta = -100, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 93 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
3131613161313^3 \quad \Delta = -100, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 21 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
3131613161313^3 \#3 \quad \Delta = -100, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 213 & 29 \\ 2 & 29 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \begin{smallmatrix} 1223151321 \\ 6 \end{smallmatrix} 1$

$$\begin{aligned}
1 \begin{smallmatrix} 31223151321 \\ 6 \end{smallmatrix} 1 \#7 \quad \Delta = -52, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1 \begin{smallmatrix} 31223151321 \\ 6 \end{smallmatrix} 1 \#6 \quad \Delta = -52, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{3}{1} \overset{12223151321}{6} 1 \text{ } [\# = 5] \quad \Delta = -52, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& \overset{3}{1} \overset{12223151321}{6} 1 \text{ } [\# = 2] \quad \Delta = -52, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{3}{1} \overset{12223151321}{6} 1 \text{ } [\# = 3] \quad \Delta = -52, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{3}{1} \overset{12223151321}{6} 1 \text{ } [\# = 8] \quad \Delta = -52, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
& \overset{3}{1} \overset{12223151321}{6} 1 \text{ } [\# = 9] \quad \Delta = -52, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
& \overset{3}{1} \overset{12223151321}{6} 1 \text{ } [\# = 4] \quad \Delta = -52, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{12223151321}{1} \overset{12223151321}{6} 1 \quad \Delta = -44, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{12223151321}{1} \overset{12223151321}{6} 1 \quad \Delta = -44, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{12223151321}{1} \overset{12223151321}{6} 1 \quad \Delta = -44, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{6}12231513161$

$$\begin{aligned}
& \overset{3}{6}12231513161 \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 63 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& 612231513161 \overset{2}{1} \quad \Delta = -121, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 612231513161 \overset{2}{1} \quad \Delta = -121, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 612231513161 \overset{2}{1} \quad \Delta = -121, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 612231513161 \overset{2}{1} \quad \Delta = -121, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{16}1315132161$

$$\begin{aligned}
& \overset{1}{16}1315132161 \overset{2}{1} \quad \Delta = -121, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{1}{16}1315132161 \overset{2}{1} \quad \Delta = -121, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{1}{16}1315132161 \overset{2}{1} \quad \Delta = -121, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{1}{16}1315132161 \overset{2}{1} \quad \Delta = -121, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{16}1315131612$

$$\begin{array}{ll}
{}^1\overset{2}{161315131612} & \Delta = -120, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
{}^{12}161315131612 & \Delta = -120, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^1\overset{2}{161315131612} & \Delta = -120, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
{}^1161315131612 & \Delta = -120, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
{}^1161315131612 & \Delta = -120, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
{}^1161315131612 & \Delta = -128, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
\text{Base: } {}^116131513161 & \\
{}^{12}16131513161 & \Delta = -120, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^1\overset{2}{16131513161} & \Delta = -120, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\text{Base: } {}^1161316123151 & \\
{}^1\overset{2}{161316123151} & \Delta = -121, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
{}^{12}161316123151 & \Delta = -121, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
{}^1\overset{2}{161316123151} & \Delta = -121, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
{}^1161316123151 & \Delta = -121, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\text{Base: } {}^1161316123215 & \\
{}^1\overset{2}{161316123215} & \Delta = -123, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
{}^{12}161316123215 & \Delta = -123, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
{}^1161316123215 & \Delta = -123, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
{}^1161316123215 & \Delta = -123, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\text{Base: } 414131613161 & \\
414131613161 & \Delta = -124, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{array}$$

$$414131613161^{\overset{3}{1}} \quad \Delta = -132, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 101 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$414131613161^{\overset{2}{1}} \quad \Delta = -124, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$414131613161^{\overset{2}{1}} \quad \Delta = -124, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$414131613161^{\overset{2}{1}} \quad \Delta = -124, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$414131613161^{\overset{2}{1}} \quad \Delta = -124, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$\text{Base: } 1^{\overset{122315131}{6}} 12$$

$$1^{\overset{122315131}{6}} 12 \quad \Delta = -43, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1^{\overset{3122315131}{6}} 12 \text{ [#=4]} \quad \Delta = -51, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1^{\overset{122315131}{6}} 12 \quad \Delta = -43, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1^{\overset{3122315131}{6}} 12 \text{ [#=9]} \quad \Delta = -51, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$1^{\overset{3122315131}{6}} 12 \text{ [#=5]} \quad \Delta = -51, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$1^{\overset{3122315131}{6}} 12 \text{ [#=6]} \quad \Delta = -51, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$1^{\overset{122315131}{6}} 12^{\overset{2}{1}} \quad \Delta = -43, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1^{\overset{122315131}{6}} 12 \quad \Delta = -43, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1^{\overset{3122315131}{6}} 12 \text{ [#=3]} \quad \Delta = -51, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1^{\overset{3122315131}{6}} 12 \text{ [#=2]} \quad \Delta = -51, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1^{\overset{3122315131}{6}} 12 \text{ [#=7]} \quad \Delta = -51, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$1^{\overset{3122315131}{6}} 12 \text{ [#=8]} \quad \Delta = -51, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$1^{\overset{122315131}{6}} 12^{\overset{3}{1}} \quad \Delta = -51, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\text{Base: } 16^{\overset{21}{1}} 131612321$$

$$\begin{aligned}
& \overset{21}{16} \overset{2}{131612321} \quad \Delta = -71, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{21}{16} 131612321\overset{3}{3} \text{ [#=3]} \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{21}{16} 131612321\overset{3}{3} \text{ [#=4]} \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{21}{16} 131612321\overset{3}{3} \text{ [#=5]} \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{21}{16} 131612321\overset{3}{3} \text{ [#=6]} \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{21}{16} 131612321\overset{3}{3} \text{ [#=7]} \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& \overset{21}{16} 131612321\overset{3}{3} \text{ [#=8]} \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{21}{16} 131612321 \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{21}{16} 131612321\overset{3}{3} \text{ [#=9]} \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{21}{16} 131612321\overset{3}{3} \text{ [#=2]} \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{21}{16} \overset{2}{131612321} \quad \Delta = -71, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{21}{16} 131612321 \quad \Delta = -71, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{21}{16} \overset{2}{131612321} \quad \Delta = -71, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \text{Base: } \overset{21}{16} 131613141 \\
& \overset{21}{16} 131613141\overset{3}{3} \text{ [#=3]} \quad \Delta = -104, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& \overset{21}{16} 131613141\overset{3}{3} \text{ [#=2]} \quad \Delta = -104, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& \overset{21}{16} \overset{2}{131613141} \quad \Delta = -96, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{21}{16} 131613141\overset{3}{3} \quad \Delta = -104, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{21}{16} 131613141 \quad \Delta = -104, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{21}{16} \overset{2}{131613141} \quad \Delta = -96, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{ll}
{}^{21}16131613141 & \Delta = -96, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
{}^{21}16131613141 & \Delta = -96, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
{}^{21}16131613141 & \Delta = -96, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{array}$$

Base: $612315131\overset{31}{6}1$

$$\begin{array}{ll}
612315131\overset{2}{6}1 & \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
612315131\overset{2}{6}1 & \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
612315131\overset{2}{6}1 & \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
612315131\overset{31}{6}1 & \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
612315131\overset{31}{6}12 & \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
612315131\overset{231}{6}1 & \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{array}$$

Base: $16131513161\overset{31}{6}$

$$\begin{array}{ll}
16131513161\overset{31}{6} & \Delta = -128, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
16131513161\overset{31}{6} & \Delta = -128, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
16131513161\overset{31}{6} & \Delta = -128, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
16131513161\overset{31}{6} & \Delta = -128, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
316131513161 & \Delta = -128, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
16131513161\overset{31}{6} & \Delta = -128, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
316131513161 & \Delta = -136, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{array}$$

Base: $16131612315\overset{31}{6}$

$$\begin{array}{ll}
316131612315 & \Delta = -137, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
16131612315\overset{31}{6} & \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{ll}
\begin{array}{c} 31 \\ 16131612315 \end{array} \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\begin{array}{c} 31 \\ 16131612315 \end{array} \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\begin{array}{c} 21 \\ 316131612315 \end{array} \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\begin{array}{c} 31 \\ 16131612315 \end{array} \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\begin{array}{c} 31 \\ 16131612315 \end{array} \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\text{Base: } 1 \begin{array}{c} 12315131 \\ 6 \end{array} 131 & \\
\begin{array}{c} 312315131 \\ 1 \quad 6 \quad 131 \end{array} [\# = 2] \quad \Delta = -59, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\begin{array}{c} 312315131 \\ 1 \quad 6 \quad 131 \end{array} [\# = 8] \quad \Delta = -59, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
\begin{array}{c} 12315131 \\ 1 \quad 6 \quad 131 \end{array} 3 [\# = 9] \quad \Delta = -59, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
\begin{array}{c} 312315131 \\ 1 \quad 6 \quad 131 \end{array} [\# = 4] \quad \Delta = -59, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\begin{array}{c} 12315131 \\ 1 \quad 6 \quad 131 \end{array} 3 [\# = 8] \quad \Delta = -59, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
\begin{array}{c} 312315131 \\ 1 \quad 6 \quad 131 \end{array} [\# = 3] \quad \Delta = -59, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\begin{array}{c} 12315131 \\ 1 \quad 6 \quad 131 \end{array} 3 [\# = 3] \quad \Delta = -59, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\begin{array}{c} 312315131 \\ 1 \quad 6 \quad 131 \end{array} [\# = 9] \quad \Delta = -59, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
\begin{array}{c} 312315131 \\ 1 \quad 6 \quad 131 \end{array} [\# = 5] \quad \Delta = -59, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\begin{array}{c} 12315131 \\ 1 \quad 6 \quad 131 \end{array} 3 [\# = 2] \quad \Delta = -59, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\begin{array}{c} 312315131 \\ 1 \quad 6 \quad 131 \end{array} [\# = 7] \quad \Delta = -59, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
\begin{array}{c} 12315131 \\ 1 \quad 6 \quad 131 \end{array} \quad \Delta = -51, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\begin{array}{c} 12315131 \\ 1 \quad 6 \quad 131 \end{array} \quad \Delta = -51, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\begin{array}{c} 12315131 \\ 1 \quad 6 \quad 131 \end{array} \quad \Delta = -51, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
1^{12315131} 6^{131\mathbf{3}}_{[\#=7]} \quad \Delta = -59, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
1^{12315131} 6^{131\mathbf{3}}_{[\#=5]} \quad \Delta = -59, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1^{\mathbf{3}12315131} 6^{131}_{[\#=6]} \quad \Delta = -59, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1^{12315131} 6^{131\mathbf{3}}_{[\#=6]} \quad \Delta = -59, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1^{12315131} 6^{131\mathbf{3}}_{[\#=4]} \quad \Delta = -59, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{12315131} 6^{131} \quad \Delta = -51, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: $6131^{2315131} 6^1$

$$\begin{aligned}
6131^{2315131} 6^{1\mathbf{3}} \quad \Delta = -137, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
6131^{2315131} 6^1 \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 121 \quad & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
6131^{2315131} 6^1 \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
6131^{\mathbf{2}2315131} 6^1 \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
6131^{2315131} 6^{1\mathbf{2}} \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
6131^{2315131} 6^1 \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 100 \quad & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
6131^{\mathbf{2}2315131} 6^1 \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1^{131} 6^{13161231}$

$$\begin{aligned}
1^{\mathbf{3}131} 6^{13161231}_{[\#=7]} \quad \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1^{\mathbf{3}131} 6^{13161231}_{[\#=4]} \quad \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1^{\mathbf{3}131} 6^{13161231}_{[\#=5]} \quad \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1^{131} 6^{13161231\mathbf{3}}_{[\#=5]} \quad \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1^{\mathbf{3}131} 6^{13161231}_{[\#=3]} \quad \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{3131}{1 \ 6 \ 13161231} \text{ }_{[\# = 6]} \quad \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 13161231} \overset{3}{} \text{ }_{[\# = 2]} \quad \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 6 \ 13161231} \text{ }_{[\# = 9]} \quad \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 6 \ 13161231} \text{ }_{[\# = 2]} \quad \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 13161231} \overset{3}{} \text{ }_{[\# = 3]} \quad \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 6 \ 13161231} \text{ }_{[\# = 8]} \quad \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 13161231} \overset{2}{} \quad \Delta = -77, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 13161231} \overset{3}{} \text{ }_{[\# = 4]} \quad \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 13161231} \overset{2}{} \quad \Delta = -77, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 13161231} \quad \Delta = -77, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 13161231} \overset{2}{} \quad \Delta = -77, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \text{Base: } \overset{131}{1 \ 6 \ 13161313} \\
& \overset{131}{1 \ 6 \ 13161313} \overset{3}{} \quad \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 13161313} \overset{2}{} \quad \Delta = -84, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 13161313} \overset{2}{} \quad \Delta = -84, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 13161313} \quad \Delta = -84, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 13161313} \overset{2}{} \quad \Delta = -84, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 13161313} \overset{2}{} \quad \Delta = -84, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 6 \ 13161313} \text{ }_{[\# = 6]} \quad \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 6 \ 13161313} \text{ }_{[\# = 8]} \quad \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{ll}
\overset{3131}{1 \ 6 \ 13161313} \text{ }_{[\# = 7]} \quad \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
\overset{3131}{1 \ 6 \ 13161313} \text{ }_{[\# = 5]} \quad \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
\overset{3131}{1 \ 6 \ 13161313} \text{ }_{[\# = 4]} \quad \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
\overset{131}{1 \ 6 \ 13161313} \overset{4}{} \quad \Delta = -112, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 9 & 5 \\ 5 & 9 \end{pmatrix} \\
\overset{131}{1 \ 6 \ 13161313} \overset{3}{} \text{ }_{[\# = 3]} \quad \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\overset{3131}{1 \ 6 \ 13161313} \text{ }_{[\# = 9]} \quad \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
\overset{3131}{1 \ 6 \ 13161313} \text{ }_{[\# = 2]} \quad \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\overset{131}{1 \ 6 \ 13161313} \overset{3}{} \text{ }_{[\# = 2]} \quad \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\overset{3131}{1 \ 6 \ 13161313} \text{ }_{[\# = 3]} \quad \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\text{Base: } 413161316131 \overset{1}{} & \\
413161316131 \overset{1}{} \overset{2}{} \quad \Delta = -104, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
413161316131 \overset{1}{} \text{ }_{[\# = 4]} \quad \Delta = -112, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
413161316131 \overset{1}{} \text{ }_{[\# = 5]} \quad \Delta = -112, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
413161316131 \overset{1}{} \text{ }_{[\# = 9]} \quad \Delta = -112, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
413161316131 \overset{1}{} \text{ }_{[\# = 8]} \quad \Delta = -112, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
413161316131 \overset{21}{} \quad \Delta = -104, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
413161316131 \overset{21}{} \quad \Delta = -104, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
413161316131 \overset{12}{} \quad \Delta = -104, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
413161316131 \overset{3}{} \overset{1}{} \quad \Delta = -112, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
413161316131 \overset{1}{} \text{ }_{[\# = 2]} \quad \Delta = -112, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
413161316131 \quad \Delta = -104, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
413161316131 \quad \Delta = -104, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
4131613161313_{[\#=3]} \quad \Delta = -112, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
4131613161313_{[\#=6]} \quad \Delta = -112, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
4131613161313_{[\#=7]} \quad \Delta = -112, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{315131}{6} 13161$

$$\begin{aligned}
315131613161 \quad \Delta = -136, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \overset{315131}{6} 13161 \quad \Delta = -128, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
1 \overset{315131}{6} 13161 \quad \Delta = -128, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1 \overset{315131}{6} 13161 \quad \Delta = -128, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
315131613161 \quad \Delta = -128, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \overset{315131}{6} 13161 \quad \Delta = -128, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1 \overset{315131}{6} 13161 \quad \Delta = -128, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{5131}{6} 1316123$

$$\begin{aligned}
1 \overset{5131}{6} 1316123 \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{5131}{6} 1316123 \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \overset{5131}{6} 1316123 \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
513161316123 \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1 \overset{5131}{6} 1316123 \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1 \overset{5131}{6} 1316123 \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

$$5131\overset{31}{6}1316123 \quad \Delta = -137, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: $1\overset{15131}{6}131612$

$$1\overset{15131}{6}\overset{3}{131612} \quad \Delta = -128, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$1\overset{15131}{6}131612 \quad \Delta = -120, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$15131\overset{21}{6}131612 \quad \Delta = -120, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1\overset{15131}{6}\overset{2}{131612} \quad \Delta = -120, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1\overset{15131}{6}\overset{2}{131612} \quad \Delta = -120, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$15131\overset{31}{6}131612 \quad \Delta = -128, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1\overset{15131}{6}131612 \quad \Delta = -120, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1\overset{15131}{6}\overset{2}{131612} \quad \Delta = -120, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: $1\overset{1}{6}1\overset{1}{3}1613151$

$$1\overset{1}{6}1\overset{1}{3}\overset{2}{1613151} \quad \Delta = -120, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1\overset{1}{6}1\overset{31}{3}151 \quad \Delta = -128, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1\overset{1}{6}1\overset{1}{3}\overset{2}{1613151} \quad \Delta = -120, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1\overset{1}{6}1\overset{21}{3}151 \quad \Delta = -120, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1\overset{1}{6}1\overset{21}{3}151 \quad \Delta = -120, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1\overset{1}{6}1\overset{2}{3}\overset{1}{1613151} \quad \Delta = -120, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: $31\overset{12315131}{6}13$

$$31\overset{412315131}{6}13 \quad \Delta = -87, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$31\overset{12315131}{6}13 \quad \Delta = -59, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$31\overset{512315131}{6}13 \quad \Delta = -111, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\begin{aligned}
31^{12315131} 6^{13} \quad \Delta = -59, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
31^{212315131} 6^{13} \quad \Delta = -59, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
31^{212315131} 6^{13} \quad \Delta = -59, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
31^{312315131} 6^{13} [\# = 4] \quad \Delta = -67, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
31^{612315131} 6^{13} \quad \Delta = -137, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
31^{312315131} 6^{13} [\# = 6] \quad \Delta = -67, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 144 & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
31^{312315131} 6^{13} [\# = 9] \quad \Delta = -67, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 324 & \begin{pmatrix} 0 & 18 & 2 \\ 18 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix} \\
31^{312315131} 6^{13} [\# = 8] \quad \Delta = -67, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 256 & \begin{pmatrix} 0 & 16 & 2 \\ 16 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
31^{312315131} 6^{13} [\# = 3] \quad \Delta = -67, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
31^{312315131} 6^{13} [\# = 2] \quad \Delta = -67, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
31^{312315131} 6^{13} [\# = 7] \quad \Delta = -67, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 196 & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
31^{312315131} 6^{13} \quad \Delta = -67, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
31^{712315131} 6^{13} \quad \Delta = -164, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
31^{312315131} 6^{13} [\# = 5] \quad \Delta = -67, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
31^{12315131} 6^{13} \quad \Delta = -59, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
31^{812315131} 6^{13} \quad \Delta = -192, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 8 \end{pmatrix} \\
\text{Base: } 31^{31} 6^{13} 13161231 & \\
31^{31} 6^{13} 13161231^2 \quad \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
31^{31^2} 6^{13} 13161231 \quad \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
31^{31} 6^{13} 13161231^2 \quad \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
31 \bar{6} 13161231\bar{3} \text{ }_{[\# = 2]} \quad \Delta = -93, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
31 \bar{6} 13161231\bar{3} \text{ }_{[\# = 5]} \quad \Delta = -93, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 100 \quad & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
31 \bar{6} 13161231\bar{2} \quad \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\bar{2}31 \bar{6} 13161231 \quad \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
31 \bar{6} 13161231\bar{2} \quad \Delta = -87, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
31 \bar{6} 13161231\bar{3} \text{ }_{[\# = 4]} \quad \Delta = -93, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
31 \bar{6} 13161231\bar{4} \quad \Delta = -113, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
31 \bar{6} 13161231\bar{3} \text{ }_{[\# = 3]} \quad \Delta = -93, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
31 \bar{6} 13161231\bar{3} \quad \Delta = -93, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\text{Base: } 31 \bar{6} 13161313 & \\
31 \bar{6} 13161313\bar{3} \quad \Delta = -100, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 30 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
31 \bar{6} 13161313\bar{2} \quad \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
31 \bar{6} 13161313\bar{2} \quad \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
\bar{2}31 \bar{6} 13161313 \quad \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
31 \bar{6} 13161313\bar{2} \quad \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
31 \bar{6} 13161313\bar{3} \text{ }_{[\# = 2]} \quad \Delta = -100, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 144 \quad & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 129 & 25 \\ 2 & 25 & 8 \end{pmatrix} \\
31 \bar{6} 13161313\bar{2} \quad \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
31 \bar{6} 13161313\bar{3} \text{ }_{[\# = 3]} \quad \Delta = -100, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 324 \quad & \begin{pmatrix} 0 & 18 & 2 \\ 18 & 294 & 38 \\ 2 & 38 & 8 \end{pmatrix} \\
31 \bar{6} 13161313\bar{4} \quad \Delta = -120, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
31 \bar{6} 13161313\bar{2} \quad \Delta = -93, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: $16^{31}13161315$

$$31\overset{21}{6}13161315 \quad \Delta = -128, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$16^{31}131\overset{21}{6}1315 \quad \Delta = -128, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$16^{31}13161\overset{12}{3}15 \quad \Delta = -128, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$16^{31}131613\overset{1}{2}15 \quad \Delta = -128, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$16^{31}13161315 \quad \Delta = -128, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$16^{31}\overset{12}{6}13161315 \quad \Delta = -128, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$16^{31}1316131\overset{21}{5} \quad \Delta = -128, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$31\overset{31}{6}13161315 \quad \Delta = -136, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$16^{31}131\overset{31}{6}1315 \quad \Delta = -136, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: $1316^{31}1316131$

$$31\overset{6131}{6}1316131 \quad \Delta = -162, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1316^{31}13161\overset{2}{3}1 \quad \Delta = -84, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1316^{312}1316131 \quad \Delta = -84, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$131\overset{231}{6}1316131 \quad \Delta = -84, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 24 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$31\overset{4131}{6}1316131 \quad \Delta = -112, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1316^{31}1316131 \quad \Delta = -84, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1316^{31}131613\overset{2}{1} \quad \Delta = -84, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$31\overset{2131}{6}1316131 \quad \Delta = -84, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1316^{31}1316131\overset{3}{3} \text{ [#=8]} \quad \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\begin{array}{ll}
{}^{31}131\bar{6}1316131\bar{3} \text{ }_{[\# = 3]} & \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
{}^{3131}31\bar{6}1316131 \text{ }_{[\# = 7]} & \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 196 \quad \begin{pmatrix} 0 & 14 & 2 \\ 14 & 389 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
{}^{31}131\bar{6}1316131\bar{3} \text{ }_{[\# = 6]} & \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
{}^{3131}31\bar{6}1316131 \text{ }_{[\# = 6]} & \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 285 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
{}^{31}131\bar{6}1316131\bar{3} \text{ }_{[\# = 9]} & \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
{}^{31}131\bar{6}1316131\bar{3} \text{ }_{[\# = 7]} & \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
{}^{5131}31\bar{6}1316131 & \Delta = -136, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
{}^{3131}31\bar{6}1316131 & \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
{}^{3131}31\bar{6}1316131 \text{ }_{[\# = 9]} & \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 324 \quad \begin{pmatrix} 0 & 18 & 2 \\ 18 & 645 & 53 \\ 2 & 53 & 8 \end{pmatrix} \\
{}^{3131}31\bar{6}1316131 \text{ }_{[\# = 2]} & \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
{}^{31}131\bar{6}1316131\bar{3} \text{ }_{[\# = 2]} & \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
{}^{3131}31\bar{6}1316131 \text{ }_{[\# = 3]} & \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
{}^{31}131\bar{6}1316131\bar{3} \text{ }_{[\# = 5]} & \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
{}^{3131}31\bar{6}1316131 \text{ }_{[\# = 8]} & \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 256 \quad \begin{pmatrix} 0 & 16 & 2 \\ 16 & 509 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
{}^{3131}31\bar{6}1316131 \text{ }_{[\# = 4]} & \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
{}^{31}131\bar{6}1316131\bar{3} \text{ }_{[\# = 4]} & \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
{}^{3131}31\bar{6}1316131 \text{ }_{[\# = 5]} & \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 197 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
\text{Base: } {}^{131}1\bar{6}1316131 & \\
{}^{3131}1\bar{6}1316131 \text{ }_{[\# = 2]} & \Delta = -84, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
{}^{3131}1\bar{6}1316131 \text{ }_{[\# = 9]} & \Delta = -84, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
& \overset{3131}{1\ 6\ 1316131} \overset{1}{[\# = 8]} \quad \Delta = -84, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
& \overset{3131}{1\ 6\ 1316131} \overset{1}{[\# = 3]} \quad \Delta = -84, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{3131}{1\ 6\ 1316131} \overset{1}{[\# = 6]} \quad \Delta = -84, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& \overset{3131}{1\ 6\ 1316131} \overset{1}{[\# = 7]} \quad \Delta = -84, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
& \overset{3131}{1\ 6\ 1316131} \overset{1}{[\# = 5]} \quad \Delta = -84, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& \overset{131}{1\ 6\ 1316131} \overset{1}{} \quad \Delta = -76, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{3131}{1\ 6\ 1316131} \overset{1}{[\# = 4]} \quad \Delta = -84, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{1312}{1\ 6\ 1316131} \overset{1}{} \quad \Delta = -76, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \text{Base: } \overset{31}{31\ 6\ 1316131} \overset{1}{} \\
& \overset{31}{31\ 6\ 1316131\ 3} \overset{1}{} \quad \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{312}{31\ 6\ 1316131} \overset{1}{} \quad \Delta = -84, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{31}{31\ 6\ 131\ 6\ 131} \overset{21}{} \quad \Delta = -84, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{31}{31\ 6\ 1316131\ 3} \overset{1}{[\# = 8]} \quad \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 256 \quad \begin{pmatrix} 0 & 16 & 2 \\ 16 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
& \overset{31}{31\ 6\ 1316131} \overset{12}{} \quad \Delta = -84, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{31}{31\ 6\ 1316131} \overset{21}{} \quad \Delta = -84, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{231}{31\ 6\ 1316131} \overset{1}{} \quad \Delta = -84, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{31}{31\ 6\ 1316131\ 3} \overset{1}{[\# = 9]} \quad \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 324 \quad \begin{pmatrix} 0 & 18 & 2 \\ 18 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
& \overset{31}{31\ 6\ 1316131\ 3} \overset{1}{[\# = 2]} \quad \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{31}{31\ 6\ 1316131\ 3} \overset{1}{[\# = 3]} \quad \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{31}{31\ 6\ 1316131\ 6} \overset{1}{} \quad \Delta = -162, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& 31 \overset{1}{6} 1316131\overset{1}{2} \quad \Delta = -84, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 31 \overset{1}{6} 1316131\overset{1}{3} \text{ [#=4]} \quad \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& 31 \overset{1}{6} 131\overset{31}{6} 131 \quad \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 31 \overset{1}{6} 1316131\overset{1}{4} \quad \Delta = -112, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 31 \overset{1}{6} 1316131\overset{1}{3} \text{ [#=5]} \quad \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
& 31 \overset{1}{6} 1316131\overset{1}{3} \text{ [#=6]} \quad \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
& 31 \overset{1}{6} 1316131\overset{1}{3} \text{ [#=7]} \quad \Delta = -92, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 196 \quad \begin{pmatrix} 0 & 14 & 2 \\ 14 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
& \text{Base: } 1 \overset{1,122315131}{6} 1 \\
& 1 \overset{1,3122315131}{6} 1 \text{ [#=6]} \quad \Delta = -51, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
& 1 \overset{1,122315131}{6} 1 \quad \Delta = -43, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 1 \overset{1,3122315131}{6} 1 \text{ [#=5]} \quad \Delta = -51, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
& 1 \overset{1,3122315131}{6} 1 \text{ [#=7]} \quad \Delta = -51, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
& 1 \overset{1,3122315131}{6} 1 \text{ [#=4]} \quad \Delta = -51, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& 1 \overset{1,3122315131}{6} 1 \text{ [#=8]} \quad \Delta = -51, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
& 1 \overset{1,3122315131}{6} 1 \text{ [#=2]} \quad \Delta = -51, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 1 \overset{1,3122315131}{6} 1 \text{ [#=9]} \quad \Delta = -51, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix} \\
& 1 \overset{1,3122315131}{6} 1 \text{ [#=3]} \quad \Delta = -51, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& 1 \overset{1,122315131}{6} 1 \quad \Delta = -43, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 1 \overset{1,122315131}{6} 1 \quad \Delta = -43, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \text{Base: } 7122315131513
\end{aligned}$$

$$\begin{aligned}
7122315131513^2 \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
7122315131513^2 \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
7122315131513^2 \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
7122315131513^2 \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
7122315131513^2 \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
7122315131513^2 \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: 7123141414141

$$\begin{aligned}
71231414141413^{\#2} \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
7123141414141^2 \quad \Delta = -122, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
7123141414141^2 \quad \Delta = -122, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 142 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
7123141414141^2 \quad \Delta = -122, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 89 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
7123141414141^2 \quad \Delta = -122, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
7123141414141^2 \quad \Delta = -122, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
71231414141413^{\#3} \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
7123141414141^2 \quad \Delta = -124, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{aligned}$$

Base: 7123151315132

$$\begin{aligned}
7123151315132^2 \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
7123151315132^2 \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
7123151315132^2 \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
7123151315132^2 \quad \Delta = -132, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

$$7123151\overset{2}{3}15132 \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

$$71231513151\overset{2}{3}2 \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: 1712231513215

$$171223151321\overset{2}{5} \quad \Delta = -123, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1712231\overset{2}{5}13215 \quad \Delta = -123, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$171223151321\overset{2}{5} \quad \Delta = -123, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$171223151321\overset{2}{5} \quad \Delta = -123, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: 1712231513151

$$17122315131\overset{2}{5}1 \quad \Delta = -123, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$17122315131\overset{2}{5}1 \quad \Delta = -122, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1712231\overset{2}{5}13151 \quad \Delta = -122, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$17122315131\overset{2}{5}1 \quad \Delta = -122, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$17122315131\overset{2}{5}1 \quad \Delta = -122, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 1712315131513

$$1712315131\overset{2}{5}13 \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1712315131\overset{2}{5}13 \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1712315131\overset{2}{5}13 \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1712315131\overset{2}{5}13 \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1712315131\overset{2}{5}13 \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1712315131\overset{2}{5}13 \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 2171223151321

$$2171223151321\overset{2}{3} \text{ [#=4]} \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$\begin{aligned}
2171223151321\overset{\textcolor{red}{3}}{3} \text{ [#=7]} \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
2171223151321\overset{\textcolor{red}{3}}{3} \text{ [#=8]} \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
2171223151321\overset{\textcolor{red}{3}}{3} \text{ [#=6]} \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
2171223151321\overset{\textcolor{red}{3}}{3} \text{ [#=9]} \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
2171223151321\overset{\textcolor{red}{3}}{3} \text{ [#=3]} \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2171223151321\overset{\textcolor{red}{3}}{3} \text{ [#=5]} \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
2171223151321\overset{\textcolor{red}{2}}{2} \quad \Delta = -71, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2171223151321\overset{\textcolor{red}{2}}{2} \quad \Delta = -71, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2171223151321\overset{\textcolor{red}{2}}{2} \quad \Delta = -71, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2171223151321\overset{\textcolor{red}{2}}{2} \quad \Delta = -71, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2171223151321\overset{\textcolor{red}{3}}{3} \text{ [#=2]} \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: 2171231513215

$$\begin{aligned}
2171231513215\overset{\textcolor{red}{2}}{2} \quad \Delta = -123, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2171231513215\overset{\textcolor{red}{2}}{2} \quad \Delta = -123, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2171231513215\overset{\textcolor{red}{2}}{2} \quad \Delta = -123, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2171231513215\overset{\textcolor{red}{2}}{2} \quad \Delta = -123, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2171231513215\overset{\textcolor{red}{2}}{2} \quad \Delta = -123, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: 2171231513151

$$\begin{aligned}
2171231513151\overset{\textcolor{red}{2}}{2} \quad \Delta = -122, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2171231513151\overset{\textcolor{red}{2}}{2} \quad \Delta = -122, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

$$217123151\overset{2}{3}151 \quad \Delta = -122, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$217123151\overset{2}{3}151 \quad \Delta = -123, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$217123151\overset{2}{3}151 \quad \Delta = -122, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$217123151\overset{2}{3}151 \quad \Delta = -122, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 4131612321712

$$4131612321712 \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$4131612321712 \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$4131612321712 \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$4131612321712 \quad \Delta = -134, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$4131612321712 \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$4131612321712 \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 2217123151321

$$2217123151\overset{2}{3}21 \quad \Delta = -71, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$2217123151321\overset{2}{3} \text{ [#=5]} \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$2217123151321\overset{2}{3} \text{ [#=2]} \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$2217123151321\overset{2}{3} \text{ [#=3]} \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$2217123151321\overset{2}{3} \text{ [#=4]} \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$2217123151\overset{2}{3}21 \quad \Delta = -71, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$2217123151\overset{2}{3}21 \quad \Delta = -71, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$2217123151\overset{2}{3}21 \quad \Delta = -71, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\begin{aligned}
2217123151321\bar{3} \text{ } [\# = 6] \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
2217123151321\bar{3} \text{ } [\# = 7] \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
2217123151321\bar{3} \text{ } [\# = 8] \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
2217123151321\bar{3} \text{ } [\# = 9] \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
\bar{3} 2217123151321 \quad \Delta = -80, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1316123217122

$$\begin{aligned}
2217123216131\bar{3} \text{ } [\# = 8] \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
131612321\bar{2} 7122 \quad \Delta = -98, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
13161\bar{2} 23217122 \quad \Delta = -98, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
131612321\bar{2} 7122 \quad \Delta = -98, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
13161\bar{2} 23217122 \quad \Delta = -98, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
131612321\bar{3} 7122 \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
2217123216131\bar{3} \text{ } [\# = 5] \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
2217123216131\bar{3} \text{ } [\# = 2] \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
2217123216131\bar{3} \text{ } [\# = 3] \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
2217123216131\bar{3} \text{ } [\# = 4] \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
2217123216131\bar{3} \text{ } [\# = 9] \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
2217123216131\bar{3} \text{ } [\# = 6] \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
2217123216131\bar{3} \text{ } [\# = 7] \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}
\end{aligned}$$

Base: 3217123151315

$$3217123151\bar{2} 315 \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$\begin{aligned}
3217123151315^2 \quad \Delta = -132, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
3217123151315^2 \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
3217123151315^2 \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
3217123151315^2 \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
3217123151315^2 \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1321712315131

$$\begin{aligned}
13151321712313^{\#=3} \quad \Delta = -87, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1321712315131^2 \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1321712315131^2 \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1321712315131^2 \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
13217123151313^{\#=3} \quad \Delta = -87, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
13217123151313^{\#=9} \quad \Delta = -87, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
13217123151313^{\#=8} \quad \Delta = -87, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
13217123151313^{\#=7} \quad \Delta = -87, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
13217123151313^{\#=6} \quad \Delta = -87, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1321712315131^2 \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
13151321712313^{\#=2} \quad \Delta = -87, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
13217123151313^{\#=2} \quad \Delta = -87, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
13217123151313^{\#=5} \quad \Delta = -87, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
13151321712313^{\#=5} \quad \Delta = -87, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

$$1321712315131\bar{3} \text{ } [\# = 4] \quad \Delta = -87, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1315132171231\bar{3} \text{ } [\# = 4] \quad \Delta = -87, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: 1513217123151

$$151\bar{3}217123151 \quad \Delta = -123, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$151321\bar{7}123151 \quad \Delta = -123, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 5132171231513

$$51321712315\bar{1}3 \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$513217123151\bar{3} \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$51321\bar{7}1231513 \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$51321\bar{7}123151\bar{3} \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$513217123151\bar{3} \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 2161232171231

$$2161232171231\bar{3} \text{ } [\# = 5] \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$2161232171231\bar{3} \text{ } [\# = 4] \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$2161232171231\bar{3} \text{ } [\# = 3] \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$2161232171231\bar{3} \text{ } [\# = 2] \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$2161\bar{2}32171231 \quad \Delta = -99, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$21612321\bar{7}1231 \quad \Delta = -99, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$2161\bar{2}32171231 \quad \Delta = -99, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$2161232\bar{1}71231 \quad \Delta = -99, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$2161232171231\bar{3} \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: 1513217123215

$$\begin{aligned}
1513217123215^2 \quad \Delta = -125, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1513217123215^2 \quad \Delta = -125, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1513217123215^2 \quad \Delta = -125, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1513217123215^2 \quad \Delta = -125, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1232171231512

$$\begin{aligned}
1232171231512^2 \quad \Delta = -73, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21513217123213^{\#4} \quad \Delta = -81, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
21513217123213^{\#5} \quad \Delta = -81, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1232171231512^2 \quad \Delta = -73, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1232171231512^2 \quad \Delta = -73, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1232171231512^2 \quad \Delta = -73, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
21513217123213^{\#6} \quad \Delta = -81, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
21513217123213^{\#7} \quad \Delta = -81, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
21513217123213^{\#8} \quad \Delta = -81, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
21513217123213^{\#9} \quad \Delta = -81, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
21513217123213^{\#3} \quad \Delta = -81, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1232171231512^3 \quad \Delta = -81, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
21513217123213^{\#2} \quad \Delta = -81, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: 123217123151¹

$$\begin{aligned}
1^{31232171231} 5^1 \quad \Delta = -81, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
123217123151^{21} \quad \Delta = -73, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
123217123151^1 \quad \Delta = -73, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
123217123151^{21} \quad \Delta = -73, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1^{31232171231} 5^1 1_{[\# = 9]} \quad \Delta = -81, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1^{31232171231} 5^1 1_{[\# = 7]} \quad \Delta = -81, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1^{31232171231} 5^1 1_{[\# = 2]} \quad \Delta = -81, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{31232171231} 5^1 1_{[\# = 3]} \quad \Delta = -81, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{31232171231} 5^1 1_{[\# = 6]} \quad \Delta = -81, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1^{31232171231} 5^1 1_{[\# = 5]} \quad \Delta = -81, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1^{31232171231} 5^1 1_{[\# = 8]} \quad \Delta = -81, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}
\end{aligned}$$

Base: 4132171232161

$$\begin{aligned}
4132171232161^2 \quad \Delta = -127, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
4132171232161^2 \quad \Delta = -127, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
4132171232161^2 \quad \Delta = -127, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
4132171232161^2 \quad \Delta = -127, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
4132171232161^2 \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: 161232171231

$$\begin{aligned}
161232171231^1 \quad \Delta = -99, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
161232171231^{12} \quad \Delta = -99, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
161232171231^1 \quad \Delta = -99, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
161232171231^1 3_{[\# = 2]} \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}
\end{aligned}$$

$${}^1_{161232171231\bar{3}} \text{ [#=3]} \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$${}^1_{161232171231\bar{3}} \text{ [#=5]} \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$${}^1_{161232171231\bar{3}} \text{ [#=4]} \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: 5123217123215

$${}^2_{5123217123215} \quad \Delta = -127, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$${}^2_{5123217123215} \quad \Delta = -127, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 1232171232151

$${}^1_{1512321712321\bar{3}} \text{ [#=5]} \quad \Delta = -83, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$${}^1_{1512321712321\bar{3}} \text{ [#=7]} \quad \Delta = -83, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$${}^1_{1512321712321\bar{3}} \text{ [#=9]} \quad \Delta = -83, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$${}^2_{1232171232151} \quad \Delta = -75, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$${}^2_{1232171232151} \quad \Delta = -75, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$${}^2_{1232171232151} \quad \Delta = -75, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$${}^1_{1512321712321\bar{3}} \text{ [#=2]} \quad \Delta = -83, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$${}^1_{1512321712321\bar{3}} \text{ [#=3]} \quad \Delta = -83, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$${}^1_{1512321712321\bar{3}} \text{ [#=8]} \quad \Delta = -83, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$${}^1_{1512321712321\bar{3}} \text{ [#=6]} \quad \Delta = -83, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$${}^1_{1512321712321\bar{3}} \text{ [#=4]} \quad \Delta = -83, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

Base: 2171232171223

$${}^2_{2171232171223} \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$${}^2_{2171232171223} \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$\begin{aligned}
2171232171223 \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
2171232171223 \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 2217123217122

$$\begin{aligned}
2217123217122 \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2217123217122 \quad \Delta = -117, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
2217123217122 \quad \Delta = -117, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{1231513221}{7} 1$

$$\begin{aligned}
1 \overset{31231513221}{7} 1 \text{ [#=6]} \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1 \overset{31231513221}{7} 1 \text{ [#=9]} \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
1 \overset{31231513221}{7} 1 \text{ [#=4]} \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \overset{31231513221}{7} 1 \text{ [#=3]} \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{31231513221}{7} 1 \text{ [#=2]} \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1 \overset{31231513221}{7} 1 \text{ [#=7]} \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
1 \overset{1231513221}{7} 1 \quad \Delta = -71, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{1231513221}{7} 1 \quad \Delta = -71, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1 \overset{1231513221}{7} 1 \quad \Delta = -71, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1 \overset{31231513221}{7} 1 \text{ [#=8]} \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1 \overset{31231513221}{7} 1 \text{ [#=5]} \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{5123151321}{7} 1$

$$\begin{aligned}
1 \overset{5123151321}{7} 1 \quad \Delta = -123, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{5123151321}{7} 1 \quad \Delta = -123, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$1 \overset{5123151321}{7} 1 \quad \Delta = -123, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \overset{5123151321}{7} 1 \quad \Delta = -123, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{1513151321}{7} 1$$

$$1 \overset{1513151321}{7} 1 \quad \Delta = -122, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{1513151321}{7} 1 \quad \Delta = -122, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{1513151321}{7} 1 \quad \Delta = -122, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{1513151321}{7} 1 \quad \Delta = -123, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{1513151321}{7} 1 \quad \Delta = -122, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\text{Base: } 4131612321\overset{1}{7}1$$

$$413161\overset{2}{2}321\overset{1}{7}1 \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$4131612321\overset{3}{7}1 \quad \Delta = -134, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$4131612321\overset{2}{7}1 \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$413161\overset{2}{2}321\overset{1}{7}1 \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$4131612321\overset{21}{7}1 \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\text{Base: } 171232171223\overset{1}{7}$$

$$171232171223\overset{2}{7} \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$171232171223\overset{2}{7} \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$171232171223\overset{12}{7} \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{123151321}{7} 12$$

$$1 \overset{123151321}{7} 12 \quad \Delta = -71, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1 \overset{123151321}{7} \overset{2}{12} \quad \Delta = -71, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\begin{aligned}
1^{123151321} 7^{12} \quad \Delta = -71, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1^{123151321} 7^{12} \quad \Delta = -71, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1^{3123151321} 7^{12} [\# = 4] \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{3123151321} 7^{12} [\# = 7] \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
1^{3123151321} 7^{12} [\# = 9] \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
1^{3123151321} 7^{12} [\# = 6] \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1^{3123151321} 7^{12} [\# = 8] \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1^{3123151321} 7^{12} [\# = 2] \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1^{3123151321} 7^{12} [\# = 3] \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1^{3123151321} 7^{12} [\# = 5] \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

Base: $17^{21}123216131$

$$\begin{aligned}
17^{21}1232161313 [\# = 4] \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
17^{21}123216131 \quad \Delta = -98, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
17^{21}123216131^2 \quad \Delta = -98, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
17^{21}123216131^2 \quad \Delta = -98, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
17^{21}1232161313 [\# = 8] \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
17^{21}1232161313 [\# = 7] \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
17^{21}1232161313 [\# = 6] \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
17^{21}1232161313 [\# = 9] \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
17^{21}1232161313 [\# = 2] \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}
\end{aligned}$$

$$^{21}_{17}123216131\bar{3}_{[\# = 3]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$^{21}_{17}123216131\bar{2} \quad \Delta = -98, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$^{21}_{17}123216131\bar{3}_{[\# = 5]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: $^{21}_{17}123217122$

$$^{21}_{17}123217122\bar{2} \quad \Delta = -117, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$^{21}_{17}123217122 \quad \Delta = -117, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$^{21}_{17}123217122\bar{2} \quad \Delta = -117, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$^{21}_{17}123217122\bar{2} \quad \Delta = -117, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$^{21}_{17}123217122\bar{3} \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

Base: $^{321}_{17}12321712$

$$^{321}_{17}12321712\bar{21} \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$^{321}_{17}12321712 \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$^{321}_{17}12321712\bar{2} \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$^{321}_{17}12321712\bar{2} \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$^{321}_{17}12321712\bar{2} \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

Base: $^{2321}_{17}1232171$

$$^{2321}_{17}1232171 \quad \Delta = -128, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

$$^{2321}_{17}1232171\bar{21} \quad \Delta = -128, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$^{2321}_{17}1232171\bar{2} \quad \Delta = -128, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$^{2321}_{17}1232171\bar{2} \quad \Delta = -128, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix}$$

Base: $^{1}_{17}123217123$

$$\begin{aligned}
& \overset{1}{1}712321\overset{21}{7}123 \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{12}{1}712321\overset{1}{7}123 \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{1}{1}712321\overset{12}{7}123 \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{1}{1}712321\overset{21}{7}123 \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{21}{1}7\overset{21}{1}2321\overset{21}{7}1$

$$\begin{aligned}
& \overset{212}{1}7\overset{21}{1}2321\overset{21}{7}1 \quad \Delta = -117, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{21}{1}7\overset{21}{1}2321\overset{21}{7}1 \quad \Delta = -117, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1,123151321}{1}7\overset{1,123151321}{1}$

$$\begin{aligned}
& \overset{1,3123151321}{1}7\overset{1,123151321}{1} \quad [\# = 4] \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& \overset{1,123151321}{1}7\overset{1,123151321}{1} \quad \Delta = -71, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{1,123151321}{1}7\overset{1,123151321}{1} \quad \Delta = -71, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{1,123151321}{1}7\overset{1,123151321}{1} \quad \Delta = -71, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{1,3123151321}{1}7\overset{1,123151321}{1} \quad [\# = 3] \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{1,3123151321}{1}7\overset{1,123151321}{1} \quad [\# = 2] \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{1,3123151321}{1}7\overset{1,123151321}{1} \quad [\# = 5] \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
& \overset{1,3123151321}{1}7\overset{1,123151321}{1} \quad [\# = 6] \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
& \overset{1,3123151321}{1}7\overset{1,123151321}{1} \quad [\# = 9] \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
& \overset{1,3123151321}{1}7\overset{1,123151321}{1} \quad [\# = 8] \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
& \overset{1,3123151321}{1}7\overset{1,123151321}{1} \quad [\# = 7] \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix}
\end{aligned}$$

Base: 8122314141413

$$812231414141\overset{2}{3} \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$8122314141413^2 \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$8122314141413^2 \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$8122314141413^2 \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$8122314141413^3 \quad \Delta = -138, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$8122314141413^2 \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 8123141414132

$$8123141414132^2 \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$8123141414132^2 \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$8123141414132^2 \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$8123141414132^2 \quad \Delta = -133, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$8123141414132^2 \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$8123141414132^2 \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: 1414141322181

$$18122314141413^{\#=3} \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1414141322181^2 \quad \Delta = -122, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1414141322181^2 \quad \Delta = -122, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1414141322181^2 \quad \Delta = -122, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1414141322181^2 \quad \Delta = -122, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$18122314141413^{\#=2} \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: 3141414132181

$$3141414132181^2 \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\begin{array}{ll}
3141414132181 & \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
3141414132181 & \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
3141414132181 & \Delta = -132, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
3141414132181 & \Delta = -138, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
3141414132181 & \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
3141414132181 & \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
\text{Base: } 4131513221812 & \\
4131513221812 & \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
4131513221812 & \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
4131513221812 & \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
4131513221812 & \Delta = -127, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
4131513221812 & \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
4131513221812 & \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
4131513221812 & \Delta = -134, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\text{Base: } 1414141321812 & \\
1414141321812 & \Delta = -122, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1414141321812 & \Delta = -124, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1414141321812 & \Delta = -122, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
21812314141413 \text{ [#=2]} & \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
21812314141413 \text{ [#=3]} & \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{array}$$

$$1414141321812^2 \quad \Delta = -122, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1414141321812^2 \quad \Delta = -122, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1414141321812^2 \quad \Delta = -122, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 2321612321812

$$2321612321812^2 \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$2321612321812^2 \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

$$2321612321812^2 \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix}$$

$$2321612321812^2 \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: 2218122315131

$$2218122315131^3 \text{ [#=3]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$2218122315131^2 \quad \Delta = -98, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$2218122315131^2 \quad \Delta = -98, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$2218122315131^2 \quad \Delta = -98, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$2218122315131^3 \text{ [#=9]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$2218122315131^3 \text{ [#=8]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$2218122315131^3 \text{ [#=7]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$2218122315131^3 \text{ [#=6]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$2218122315131^3 \text{ [#=2]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$2218122315131^2 \quad \Delta = -98, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$2218122315131^3 \text{ [#=4]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$2218122315131\overset{3}{3} \text{ }_{[\# = 5]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$2218122315131 \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: 4131513218122

$$4131513218122 \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$4131513218122 \quad \Delta = -127, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$4131513218122 \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$4131513218122 \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$4131513218122 \quad \Delta = -134, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$4131513218122 \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$4131513218122 \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$4131513218122 \quad \Delta = -135, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

Base: 3216123218122

$$3216123218122 \quad \Delta = -127, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$3216123218122 \quad \Delta = -136, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$3216123218122 \quad \Delta = -127, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$3216123218122 \quad \Delta = -127, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$3216123218122 \quad \Delta = -127, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: $\overset{31}{16}123218122$

$$\overset{31}{16}123218122 \quad \Delta = -135, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$\overset{31}{16}123218122 \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\overset{21}{31}6123218122 \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\overset{31}{16}123218122 \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\begin{aligned}
& \overset{31}{1} \overset{2}{6} 123218122 \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{31}{3} \overset{1}{6} 123218122 \quad \Delta = -134, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{31}{1} \overset{2}{6} 123218122 \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1321812231512

$$\begin{aligned}
& 1321812231 \overset{2}{5} 12 \quad \Delta = -99, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& 2151322181231 \overset{3}{3} \text{ [#=2]} \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 2151322181231 \overset{3}{3} \text{ [#=5]} \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& 2151322181231 \overset{3}{3} \text{ [#=4]} \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 1321812231 \overset{2}{5} 12 \quad \Delta = -99, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 1321812231 \overset{2}{5} 12 \quad \Delta = -99, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 1321812231 \overset{2}{5} 12 \quad \Delta = -99, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 2151322181231 \overset{3}{3} \text{ [#=3]} \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 1321812231 \overset{3}{5} 12 \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}
\end{aligned}$$

Base: 2221812315131

$$\begin{aligned}
& 2221812315131 \overset{3}{3} \text{ [#=6]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 2221812315131 \overset{3}{3} \text{ [#=9]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& 2221812315131 \overset{3}{3} \text{ [#=3]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 2221812315131 \overset{2}{5} 12 \quad \Delta = -98, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 2221812315131 \overset{3}{3} \text{ [#=7]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& 2221812315131 \overset{3}{3} \text{ [#=8]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& 2221812315131 \overset{3}{3} \text{ [#=5]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
2221812315131\bar{3} \text{ } [\# = 2] \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2221812315131\bar{2} \quad \Delta = -98, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2221812315131\bar{2} \quad \Delta = -98, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2221812315131\bar{2} \quad \Delta = -98, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2221812315131\bar{3} \text{ } [\# = 4] \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\text{Base: } 132181223151\bar{1} \\
1\bar{3}1321812231\bar{1} \text{ } [\# = 3] \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1\bar{3}1321812231\bar{1} \text{ } [\# = 4] \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1\bar{3}1321812231\bar{1} \text{ } [\# = 5] \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
132181223151\bar{2} \quad \Delta = -99, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1\bar{3}1321812231\bar{1} \text{ } [\# = 2] \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
132181223151\bar{2}^1 \quad \Delta = -99, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
132181223151\bar{2}^1 \quad \Delta = -99, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\text{Base: } 1413221812321 \\
1413221812321\bar{3} \text{ } [\# = 5] \quad \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1413221812321\bar{3} \text{ } [\# = 8] \quad \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1413221812321\bar{3} \text{ } [\# = 4] \quad \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1413221812321\bar{3} \text{ } [\# = 6] \quad \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1232181223141\bar{3} \text{ } [\# = 2] \quad \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1413221812321\bar{3} \text{ } [\# = 3] \quad \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1413221812321\bar{3} \text{ } [\# = 2] \quad \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1413221812321\bar{3} \text{ } [\# = 9] \quad \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1413221812321\bar{3} \text{ } [\# = 7] \quad \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1413221812321^2 \quad \Delta = -77, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1413221812321^2 \quad \Delta = -77, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1413221812321^2 \quad \Delta = -77, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1232181223141\bar{3} \text{ } [\# = 3] \quad \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 4132218123215

$$\begin{aligned}
4132218123215^2 \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
4132218123215^2 \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
4132218123215^2 \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
4132218123215^2 \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1322181232151

$$\begin{aligned}
1512321812231\bar{3} \text{ } [\# = 2] \quad \Delta = -109, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1322181232151^2 \quad \Delta = -101, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1322181232151^2 \quad \Delta = -101, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1322181232151^2 \quad \Delta = -101, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1512321812231\bar{3} \text{ } [\# = 3] \quad \Delta = -109, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1512321812231\bar{3} \text{ } [\# = 5] \quad \Delta = -109, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1512321812231\bar{3} \text{ } [\# = 4] \quad \Delta = -109, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

Base: 4132218123151

$$\begin{aligned}
4132218123151^2 \quad \Delta = -127, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
4132218123151^2 \quad \Delta = -127, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
4132218123151^2 \quad \Delta = -127, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
4132218123151^2 \quad \Delta = -127, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

Base: 4132181223151

$$\begin{aligned}
4132181223151^2 \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
4132181223151^2 \quad \Delta = -127, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
4132181223151^2 \quad \Delta = -127, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
4132181223151^2 \quad \Delta = -127, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
4132181223151^2 \quad \Delta = -127, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1322181231512

$$\begin{aligned}
21513218122313^{\#=5} \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
21513218122313^{\#=4} \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21513218122313^{\#=3} \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21513218122313^{\#=2} \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1322181231512^3 \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
1322181231512^2 \quad \Delta = -99, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1322181231512^2 \quad \Delta = -99, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1322181231512^2 \quad \Delta = -99, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1322181231512^2 \quad \Delta = -99, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: 132218123151¹

$$\begin{aligned}
132218123151^1 \quad \Delta = -99, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
132218123151^2 \quad \Delta = -99, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
132218123151^{21} \quad \Delta = -99, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1^{31322181231} 5^1 1_{[\# = 2]} \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1^{31322181231} 5^1 1_{[\# = 3]} \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1^{31322181231} 5^1 1_{[\# = 4]} \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{31322181231} 5^1 1_{[\# = 5]} \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

Base: 2161232181222

$$\begin{aligned}
2161232181222^2 \quad \Delta = -118, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2161232181222^2 \quad \Delta = -118, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2161232181222^2 \quad \Delta = -118, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
2161232181222^3 \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2161232181222^2 \quad \Delta = -118, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 161232181222

$$\begin{aligned}
161232181222^{12} \quad \Delta = -118, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
161232181222^1 \quad \Delta = -118, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
161232181222^1 \quad \Delta = -118, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 4141413218123

$$\begin{aligned}
4141413218123^2 \quad \Delta = -133, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
4141413218123^2 \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
4141413218123^2 \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
4141413218123 \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
4141413218123 \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 144 & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
4141413218123 \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
\text{Base: } 4123218123141 & \\
4123218123141\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -113, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
4123218123141 \quad \Delta = -105, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
4123218123141 \quad \Delta = -105, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
4123218123141 \quad \Delta = -105, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
4123218123141 \quad \Delta = -105, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
4123218123141\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -113, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
4123218123141 \quad \Delta = -113, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 41 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
4123218123141 \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\text{Base: } 1413218123141 & \\
1413218123141\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -111, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1413218123141 \quad \Delta = -105, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1413218123141 \quad \Delta = -103, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1413218123141 \quad \Delta = -103, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1413218123141\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -111, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\text{Base: } 3141321812321 & \\
3141321812321\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -93, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
3141321812321 \quad \Delta = -87, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{ll}
3141321812321\overset{3}{\color{red}}\text{ }_{[\#=7]} & \Delta = -93, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
3141321812321\overset{3}{\color{red}}\text{ }_{[\#=8]} & \Delta = -93, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
3141321812321\overset{3}{\color{red}}\text{ }_{[\#=3]} & \Delta = -93, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
3141321812321\overset{3}{\color{red}}\text{ }_{[\#=6]} & \Delta = -93, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
3141321812321\overset{3}{\color{red}}\text{ }_{[\#=9]} & \Delta = -93, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
3141321812321\overset{3}{\color{red}}\text{ }_{[\#=4]} & \Delta = -93, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
3141321812321\overset{3}{\color{red}}\text{ }_{[\#=5]} & \Delta = -93, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\overset{3}{\color{red}}{3141321812321} & \Delta = -93, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
314132181\overset{2}{\color{red}}{2321} & \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
314132181\overset{2}{\color{red}}{2321} & \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\overset{2}{\color{red}}{3141321812321} & \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
314132181\overset{2}{\color{red}}{2321} & \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\textbf{Base: } 4141321812314 & \\
414132181\overset{2}{\color{red}}{2314} & \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
414132181\overset{2}{\color{red}}{2314} & \Delta = -133, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
414132181\overset{2}{\color{red}}{2314} & \Delta = -133, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
414132181\overset{2}{\color{red}}{2314} & \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
414132181\overset{2}{\color{red}}{2314} & \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\overset{2}{\color{red}}{4141321812314} & \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
414132181\overset{2}{\color{red}}{2314} & \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{array}$$

Base: 1414132181231

$$\begin{aligned}
1414132181231^2 \quad \Delta = -103, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
13218123141413^{\#2} \quad \Delta = -111, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
14141321812313^{\#3} \quad \Delta = -111, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
13218123141413^{\#3} \quad \Delta = -111, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
14141321812313^{\#2} \quad \Delta = -111, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
14141321812313^{\#4} \quad \Delta = -111, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
14141321812313^{\#5} \quad \Delta = -111, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1414132181231^2 \quad \Delta = -103, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1414132181231^2 \quad \Delta = -103, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1414132181231^2 \quad \Delta = -103, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1414132181231^2 \quad \Delta = -105, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 3132181232151

$$\begin{aligned}
3132181232151^2 \quad \Delta = -111, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
3132181232151^3 \quad \Delta = -117, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
3132181232151^2 \quad \Delta = -109, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
3132181232151^2 \quad \Delta = -109, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
3132181232151^3 \quad \Delta = -117, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
3132181232151^2 \quad \Delta = -109, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
3132181232151^2 \quad \Delta = -109, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1321812321513

$$\begin{aligned}
1321812321513^2 \quad \Delta = -109, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1321812321513^2 \quad \Delta = -109, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1321812321513^2 \quad \Delta = -109, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1321812321513^2 \quad \Delta = -109, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
31512321812313^{\#5} \quad \Delta = -117, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
31512321812313^{\#3} \quad \Delta = -117, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
31512321812313^{\#2} \quad \Delta = -117, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
31512321812313^{\#4} \quad \Delta = -117, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

Base: 3132181231512

$$\begin{aligned}
3132181231512^2 \quad \Delta = -109, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
3132181231512^3 \quad \Delta = -115, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
3132181231512^2 \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
3132181231512^2 \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
3132181231512^2 \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
3132181231512^3_{\#2} \quad \Delta = -115, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
3132181231512^3 \quad \Delta = -115, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
3132181231512^2 \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
3132181231512^2 \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1321812315123

$$\begin{aligned}
1321812315123^2 \quad \Delta = -108, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1321812315123^2 \quad \Delta = -108, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
13218123151\overset{2}{3} \quad \Delta = -108, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
13218123151\overset{2}{3} \quad \Delta = -108, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
3215132181231\overset{1}{3} \text{ [#=2]} \quad \Delta = -116, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
3215132181231\overset{1}{3} \text{ [#=5]} \quad \Delta = -116, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
3215132181231\overset{1}{3} \text{ [#=4]} \quad \Delta = -116, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
3215132181231\overset{1}{3} \text{ [#=3]} \quad \Delta = -116, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 31321812315¹1

$$\begin{aligned}
31321812315\overset{1}{1}\overset{2}{3} \quad \Delta = -109, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
3132181\overset{1}{2}315\overset{1}{1}\overset{2}{3} \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
31321812315\overset{1}{1}\overset{2}{3} \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
31321812315\overset{1}{1}\overset{2}{3} \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
31321812315\overset{1}{1}\overset{2}{3} \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
31321812315\overset{1}{1}\overset{2}{3} \text{ [#=2]} \quad \Delta = -115, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
31321812315\overset{1}{1}\overset{2}{3} \quad \Delta = -115, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1321812315¹13

$$\begin{aligned}
1321812315\overset{1}{1}\overset{2}{3} \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
132181231\overset{1}{5}13\overset{2}{3} \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1321812315\overset{1}{1}\overset{2}{3} \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1321812315\overset{1}{1}\overset{2}{3} \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1321812315\overset{1}{1}\overset{2}{3} \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
31 \overset{3}{5} 1 \text{ [#=2]} \quad \Delta = -115, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
31 \overset{3}{5} 1 \text{ [#=5]} \quad \Delta = -115, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
31 \overset{3}{5} 1 \text{ [#=4]} \quad \Delta = -115, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
31 \overset{3}{5} 1 \text{ [#=3]} \quad \Delta = -115, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
132181231 \overset{31}{5} 13 \quad \Delta = -115, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: 4123218123214

$$\begin{aligned}
4123218123214 \overset{3}{} \quad \Delta = -115, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
4123218123214 \overset{2}{} \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
4123218123214 \overset{2}{} \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: $2 \overset{2151321812}{3} 1$

$$\begin{aligned}
2 \overset{2151321812}{3} 1 \quad \Delta = -101, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2 \overset{2151321812}{3} 1 \quad \Delta = -101, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2 \overset{2151321812}{3} 1 \overset{12}{} \quad \Delta = -101, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2 \overset{2151321812}{3} 1 \quad \Delta = -101, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2 \overset{2151321812}{3} 1 \quad \Delta = -101, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2 \overset{2151321812}{3} 1 \quad \Delta = -109, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
2 \overset{2151321812}{3} 1 \overset{14}{} \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2 \overset{2151321812}{3} 1 \overset{13}{} \text{ [#=2]} \quad \Delta = -109, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
2 \overset{2151321812}{3} 1 \overset{13}{} \text{ [#=3]} \quad \Delta = -109, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 60 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

Base: $2 \overset{151321812}{3} 1$

$$2 \overset{151321812}{3} 1 \overset{14}{} \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\begin{aligned}
& 2^{151321812} 3^{13} [\# = 3] \quad \Delta = -109, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 60 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& 2^{151321812} 3^1 \quad \Delta = -101, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 2^{151321812} 3^1 \quad \Delta = -101, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 2^{151321812} 3^1 \quad \Delta = -101, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 2^{151321812} 3^{12} \quad \Delta = -101, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 2^{151321812} 3^{13} [\# = 2] \quad \Delta = -109, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \text{Base: } 2^{1512321812} 3^1 \\
& 2^{1512321812} 3^{14} \quad \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 2^{1512321812} 3^{13} [\# = 2] \quad \Delta = -111, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& 2^{1512321812} 3^{12} \quad \Delta = -103, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 2^{1512321812} 3^{13} [\# = 3] \quad \Delta = -111, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 60 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& 2^{1512321812} 3^1 \quad \Delta = -103, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 2^{1512321812} 3^1 \quad \Delta = -103, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 2^{1512321812} 3^1 \quad \Delta = -103, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \text{Base: } 413151322181^1 \\
& 413151322181^2 \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 413151322181^1 \quad \Delta = -127, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 413151322181^2 \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 413151322181^1 \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 413151322181^{21} \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

$$413151322181 \quad \Delta = -134, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

Base: 181231414141

$$181231414141 \quad \Delta = -122, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1812314141413 \text{ [#=2]} \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$181231414141 \quad \Delta = -122, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

$$181231414141 \quad \Delta = -122, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$1812314141413 \text{ [#=3]} \quad \Delta = -130, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$181231414141 \quad \Delta = -124, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$181231414141 \quad \Delta = -122, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 181232161232

$$181232161232 \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$181232161232 \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$181232161232 \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: $1^{131513221} 8^{12}$

$$1^{131513221} 8^{12} \text{ [#=5]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1^{131513221} 8^{12} \text{ [#=7]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$1^{131513221} 8^{12} \text{ [#=6]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1^{131513221} 8^{12} \quad \Delta = -98, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1^{131513221} 8^{12} \text{ [#=4]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1^{131513221} 8^{12} \text{ [#=2]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1^{131513221} 8^{12} \quad \Delta = -98, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\begin{aligned}
& \overset{131513221}{1} \quad 8 \quad 12 \quad \Delta = -98, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{131513221}{1} \quad 8 \quad 12 \quad \Delta = -98, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{3131513221}{1} \quad 8 \quad 12 \text{ [#=8]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{3131513221}{1} \quad 8 \quad 12 \text{ [#=9]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{3131513221}{1} \quad 8 \quad 12 \text{ [#=3]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{21}{41315132181}$

$$\begin{aligned}
& \overset{3}{41315132181} \quad \Delta = -134, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& 413151321 \overset{221}{81} \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 413151321 \overset{21}{81} \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{2}{41315132181} \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{2}{41315132181} \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{2}{41315132181} \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{2}{41315132181} \quad \Delta = -127, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{21}{18123216123}$

$$\begin{aligned}
& \overset{212}{18123216123} \quad \Delta = -127, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{21}{18123216123} \quad \Delta = -127, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{21}{18123216123} \quad \Delta = -127, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{21}{18123216123} \quad \Delta = -127, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{31}{161232181}$

$$\begin{aligned}
& \overset{31}{161232181} \quad \Delta = -134, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{31}{161232181} \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

$$1 \overset{31}{6} 12321 \overset{21}{8} 1 \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$31 \overset{21}{6} 12321 \overset{21}{8} 1 \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \overset{31}{6} 12321 \overset{21}{8} 1 \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1 \overset{31}{6} 12321 \overset{21}{8} 1 \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

Base: $1 \overset{13151321}{8} 122$

$$1 \overset{313151321}{8} 122 \text{ [#=8]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1 \overset{13151321}{8} \overset{2}{122} \quad \Delta = -98, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{13151321}{8} 122 \quad \Delta = -98, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1 \overset{13151321}{8} 122 \quad \Delta = -98, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{13151321}{8} \overset{3}{122} \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{313151321}{8} 122 \text{ [#=3]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \overset{313151321}{8} 122 \text{ [#=9]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$1 \overset{313151321}{8} 122 \text{ [#=7]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$1 \overset{13151321}{8} 122 \quad \Delta = -98, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{313151321}{8} 122 \text{ [#=6]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1 \overset{313151321}{8} 122 \text{ [#=5]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1 \overset{313151321}{8} 122 \text{ [#=4]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{313151321}{8} 122 \text{ [#=2]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: $1 \overset{221}{8} 12321612$

$$1 \overset{221}{8} \overset{2}{12321612} \quad \Delta = -118, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$\begin{aligned}
& \overset{221}{1 \ 8 \ 12321612} \quad \Delta = -118, \ n_+ = 1, \ n_- = 13, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{221}{1 \ 8 \ 12321612} \overset{2}{} \quad \Delta = -118, \ n_+ = 1, \ n_- = 13, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{221}{1 \ 8 \ 12321612} \overset{3}{} \quad \Delta = -126, \ n_+ = 1, \ n_- = 13, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
& \overset{221}{1 \ 8 \ 12321612} \overset{2}{} \quad \Delta = -118, \ n_+ = 1, \ n_- = 13, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{221}{1 \ 8 \ 12321612} \quad \Delta = -127, \ n_+ = 1, \ n_- = 13, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{1612321} \overset{221}{8 \ 1}$

$$\begin{aligned}
& \overset{1}{1612321} \overset{221}{8 \ 1} \quad \Delta = -118, \ n_+ = 1, \ n_- = 13, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{1}{1612321} \overset{221}{8 \ 1} \overset{2}{} \quad \Delta = -118, \ n_+ = 1, \ n_- = 13, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{1}{1612321} \overset{221}{8 \ 1} \overset{2}{} \quad \Delta = -118, \ n_+ = 1, \ n_- = 13, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{1}{1612321} \overset{221}{8 \ 1} \quad \Delta = -127, \ n_+ = 1, \ n_- = 13, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}
\end{aligned}$$

Base: $\overset{3221}{1 \ 8 \ 1232161}$

$$\begin{aligned}
& \overset{3221}{1 \ 8 \ 1232161} \quad \Delta = -127, \ n_+ = 1, \ n_- = 13, \ n_0 = 0, \ |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{3221}{1 \ 8 \ 1232161} \overset{21}{} \quad \Delta = -127, \ n_+ = 1, \ n_- = 13, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{3221}{1 \ 8 \ 1232161} \overset{2}{} \quad \Delta = -127, \ n_+ = 1, \ n_- = 13, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{3221}{1 \ 8 \ 1232161} \overset{2}{} \quad \Delta = -127, \ n_+ = 1, \ n_- = 13, \ n_0 = 0, \ |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1512321}{1 \ 8 \ 1231}$

$$\begin{aligned}
& \overset{1512321}{1 \ 8 \ 1231} \quad \Delta = -101, \ n_+ = 1, \ n_- = 13, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{1512321}{1 \ 8 \ 1231} \overset{3}{} \text{ [#=5]} \quad \Delta = -109, \ n_+ = 1, \ n_- = 13, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{1512321}{1 \ 8 \ 1231} \overset{3}{} \text{ [#=4]} \quad \Delta = -109, \ n_+ = 1, \ n_- = 13, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{1512321}{1 \ 8 \ 1231} \overset{2}{} \quad \Delta = -101, \ n_+ = 1, \ n_- = 13, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{1512321}{1 \ 8 \ 1231} \overset{3}{} \text{ [#=3]} \quad \Delta = -109, \ n_+ = 1, \ n_- = 13, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{1512321}{1} \overset{8}{8} 1231 \quad \Delta = -101, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{1512321}{1} \overset{8}{8} 1231 \textcolor{red}{3} [\#=2] \quad \Delta = -109, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{151321}{41321} \overset{8}{8} 1$

$$\begin{aligned}
& \overset{\textcolor{red}{2}}{41321} \overset{151321}{8} 1 \quad \Delta = -127, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{\textcolor{red}{2}}{41321} \overset{151321}{8} 1 \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 41321 \overset{151321}{8} 1 \textcolor{red}{2} \quad \Delta = -127, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 41321 \overset{\textcolor{red}{2}151321}{8} 1 \quad \Delta = -127, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 41321 \overset{151321}{8} 1 \quad \Delta = -127, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& 41321 \overset{151321}{8} 1 \quad \Delta = -127, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{2151321}{1} \overset{8}{8} 1231$

$$\begin{aligned}
& \overset{2151321}{1} \overset{8}{8} 1231 \textcolor{red}{3} [\#=2] \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{2151321}{1} \overset{8}{8} 1231 \textcolor{red}{2} \quad \Delta = -99, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{2151321}{1} \overset{8}{8} 1231 \textcolor{red}{3} [\#=5] \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{2151321}{1} \overset{8}{8} 1231 \quad \Delta = -99, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{2151321}{1} \overset{8}{8} 1231 \quad \Delta = -99, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{2151321}{1} \overset{8}{8} 1231 \textcolor{red}{3} [\#=3] \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{2151321}{1} \overset{8}{8} 1231 \quad \Delta = -99, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{2151321}{1} \overset{8}{8} 1231 \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
& \overset{2151321}{1} \overset{8}{8} 1231 \textcolor{red}{3} [\#=4] \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{151321}{1} \overset{8}{8} 1231$

$$\overset{151321}{1} \overset{8}{8} 1231 \quad \Delta = -99, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\begin{aligned}
& \overset{151321}{1 \ 8 \ 1231} \quad \Delta = -99, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{151321}{1 \ 8 \ 1231} \overset{2}{} \quad \Delta = -99, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{151321}{1 \ 8 \ 1231} \overset{3}{}_{[\#=5]} \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{151321}{1 \ 8 \ 1231} \overset{3}{}_{[\#=2]} \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{151321}{1 \ 8 \ 1231} \overset{3}{}_{[\#=3]} \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{151321}{1 \ 8 \ 1231} \overset{3}{}_{[\#=4]} \quad \Delta = -107, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \text{Base: } \overset{12321}{412321} \overset{1}{8 \ 1} \\
& \overset{312321}{412321} \overset{8 \ 1}{}_{[\#=5]} \quad \Delta = -87, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{312321}{412321} \overset{8 \ 1}{}_{[\#=2]} \quad \Delta = -87, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{12321}{412321} \overset{8 \ 1}{} \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{312321}{412321} \overset{8 \ 1}{}_{[\#=6]} \quad \Delta = -87, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{312321}{412321} \overset{8 \ 1}{}_{[\#=7]} \quad \Delta = -87, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& \overset{312321}{412321} \overset{8 \ 1}{}_{[\#=8]} \quad \Delta = -87, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{312321}{412321} \overset{8 \ 1}{}_{[\#=9]} \quad \Delta = -87, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{3}{412321} \overset{12321}{8 \ 1} \quad \Delta = -87, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{312321}{412321} \overset{8 \ 1}{}_{[\#=3]} \quad \Delta = -87, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{312321}{412321} \overset{8 \ 1}{}_{[\#=4]} \quad \Delta = -87, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{2}{412321} \overset{12321}{8 \ 1} \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{212321}{412321} \overset{8 \ 1}{} \quad \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \text{Base: } \overset{12321}{1 \ 8 \ 123141}
\end{aligned}$$

$$\begin{array}{ll}
1 \overset{12321}{8} \overset{2}{123141} & \Delta = -77, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1 \overset{12321}{8} \overset{2}{123141} & \Delta = -79, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1 \overset{312321}{8} \overset{2}{123141} [\# = 7] & \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
1 \overset{12321}{8} \overset{2}{123141} \overset{3}{123141} [\# = 2] & \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
1 \overset{312321}{8} \overset{2}{123141} [\# = 3] & \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1 \overset{312321}{8} \overset{2}{123141} [\# = 2] & \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1 \overset{12321}{8} \overset{2}{123141} \overset{3}{123141} [\# = 3] & \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
1 \overset{312321}{8} \overset{2}{123141} [\# = 4] & \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1 \overset{312321}{8} \overset{2}{123141} [\# = 5] & \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
1 \overset{312321}{8} \overset{2}{123141} [\# = 6] & \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
1 \overset{312321}{8} \overset{2}{123141} [\# = 9] & \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
1 \overset{12321}{8} \overset{2}{123141} & \Delta = -77, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1 \overset{12321}{8} \overset{2}{123141} & \Delta = -77, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \overset{312321}{8} \overset{2}{123141} [\# = 8] & \Delta = -85, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
\text{Base: } 41321 \overset{512321}{8} 1 & \\
41321 \overset{512321}{8} 1 & \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
41321 \overset{2}{512321} \overset{2}{8} 1 & \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
41321 \overset{512321}{8} 1 & \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
41321 \overset{2}{512321} \overset{2}{8} 1 & \Delta = -131, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
41321 \overset{2}{512321} \overset{2}{8} 1 & \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{array}$$

$$41321 \overset{512321}{8} \overset{12}{12} \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\text{Base: } 21 \overset{13151321}{8} 12$$

$$21 \overset{313151321}{8} 12 \text{ [#=8]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$21 \overset{313151321}{8} 12 \text{ [#=9]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$21 \overset{313151321}{8} 12 \text{ [#=4]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$21 \overset{313151321}{8} 12 \text{ [#=7]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$21 \overset{313151321}{8} 12 \text{ [#=6]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$21 \overset{313151321}{8} 12 \text{ [#=5]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$21 \overset{313151321}{8} 12 \text{ [#=3]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$21 \overset{313151321}{8} 12 \text{ [#=2]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$21 \overset{213151321}{8} 12 \quad \Delta = -98, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$21 \overset{13151321}{8} 12 \quad \Delta = -98, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$21 \overset{13151321}{8} 12 \quad \Delta = -98, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$21 \overset{13151321}{8} 12 \quad \Delta = -98, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\text{Base: } 21 \overset{21}{8} 12321612$$

$$21 \overset{21}{8} 12321 \overset{3}{6} 12 \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$21 \overset{221}{8} 12321612 \quad \Delta = -118, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$21 \overset{212}{8} 12321612 \quad \Delta = -118, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$21 \overset{21}{8} 12321 \overset{2}{6} 12 \quad \Delta = -118, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$21 \overset{21}{8} 12321 \overset{2}{6} 12 \quad \Delta = -118, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$\text{Base: } 1612321 \overset{1}{8} \overset{21}{12}$$

$$\begin{aligned}
& \overset{1}{16}12321\overset{21}{8}12 \quad \Delta = -118, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{1}{16}12321\overset{21}{8}12 \quad \Delta = -118, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{1}{16}12321\overset{21}{8}12 \quad \Delta = -118, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{321}{21}81232161$

$$\begin{aligned}
& \overset{321}{321}81232161 \quad \Delta = -136, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{321}{21}81232161 \quad \Delta = -127, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{2321}{21}81232161 \quad \Delta = -127, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 24 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{321}{21}81232161 \quad \Delta = -127, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& \overset{321}{21}81232161 \quad \Delta = -127, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{12321}{21}812321$

$$\begin{aligned}
& \overset{12321}{21}812321\overset{3}{\text{red}}_{[\# = 9]} \quad \Delta = -59, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
& \overset{12321}{21}812321\overset{3}{\text{red}}_{[\# = 8]} \quad \Delta = -59, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
& \overset{12321}{21}812321\overset{3}{\text{red}}_{[\# = 7]} \quad \Delta = -59, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
& \overset{12321}{21}812321\overset{3}{\text{red}}_{[\# = 6]} \quad \Delta = -59, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& \overset{12321}{21}812321\overset{3}{\text{red}}_{[\# = 5]} \quad \Delta = -59, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& \overset{12321}{21}812321\overset{3}{\text{red}}_{[\# = 4]} \quad \Delta = -59, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{12321}{21}812321\overset{3}{\text{red}}_{[\# = 3]} \quad \Delta = -59, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{12321}{21}812321\overset{3}{\text{red}}_{[\# = 2]} \quad \Delta = -59, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{2}{\text{red}}12321 \quad \Delta = -51, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{12321}{21}812321\overset{2}{\text{red}} \quad \Delta = -51, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{2321}{61}2321812$

$$612321 \overset{2}{8} 12 \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$612321 \overset{2321}{8} 12 \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$612321 \overset{2}{8} \overset{2321}{12} \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$612321 \overset{2321}{8} \overset{2}{12} \quad \Delta = -129, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$612321 \overset{2321}{8} 12 \overset{3}{3} \quad \Delta = -138, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{1,131513221}{8} 1$$

$$1 \overset{1,131513221}{8} 1 \quad \Delta = -98, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{1,131513221}{8} 1 \quad \Delta = -98, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{1,3131513221}{8} 1 \text{ [#=7]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$1 \overset{1,131513221}{8} 1 \quad \Delta = -98, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{1,3131513221}{8} 1 \text{ [#=8]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$1 \overset{1,3131513221}{8} 1 \text{ [#=9]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$1 \overset{1,3131513221}{8} 1 \text{ [#=6]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$1 \overset{1,3131513221}{8} 1 \text{ [#=4]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1 \overset{1,3131513221}{8} 1 \text{ [#=5]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$1 \overset{1,3131513221}{8} 1 \text{ [#=3]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1 \overset{1,3131513221}{8} 1 \text{ [#=2]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\text{Base: } 413151321 \overset{1,1}{8} 1$$

$$413151321 \overset{2}{8} \overset{1,1}{1} \quad \Delta = -127, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$413151321 \overset{2,1,1}{8} 1 \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$413151321 \overset{2}{8} \overset{1,1}{1} \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$413151321 \overset{2}{8} \overset{1,1}{1} \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$413151321 \overset{2}{8} \overset{1,1}{1} \quad \Delta = -126, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$413151321 \overset{3}{8} \overset{1,1}{1} \quad \Delta = -134, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$\text{Base: } 1^{\overset{21,13151321}{8}} 1$$

$$1^{\overset{21,313151321}{8}} 1_{[\# = 7]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$1^{\overset{21,13151321}{8}} 1 \quad \Delta = -98, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1^{\overset{21,13151321}{8}} 1 \quad \Delta = -98, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1^{\overset{21,13151321}{8}} 1 \quad \Delta = -98, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1^{\overset{21,13151321}{8}} 1 \quad \Delta = -98, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1^{\overset{21,313151321}{8}} 1_{[\# = 6]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$1^{\overset{21,313151321}{8}} 1_{[\# = 4]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1^{\overset{21,313151321}{8}} 1_{[\# = 8]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$1^{\overset{21,313151321}{8}} 1_{[\# = 2]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1^{\overset{21,313151321}{8}} 1_{[\# = 5]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$1^{\overset{21,313151321}{8}} 1_{[\# = 9]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$1^{\overset{21,313151321}{8}} 1_{[\# = 3]} \quad \Delta = -106, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$\text{Base: } 1^{\overset{1,12321}{8}} 12321$$

$$1^{\overset{1,12321}{8}} 12321 \overset{3}{1}_{[\# = 4]} \quad \Delta = -59, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1^{\overset{1,12321}{8}} 12321 \overset{3}{1}_{[\# = 2]} \quad \Delta = -59, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\begin{aligned}
1^{1,12321} \text{ }^2 \text{ } 8 \text{ } 12321 \quad \Delta = -51, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1^{1,12321} \text{ } 8 \text{ } 12321 \text{ }^3 \text{ }_{[\# = 8]} \quad \Delta = -59, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1^{1,12321} \text{ } 8 \text{ } 12321 \text{ }^3 \text{ }_{[\# = 9]} \quad \Delta = -59, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1^{1,12321} \text{ } 8 \text{ } 12321 \text{ }^3 \text{ }_{[\# = 7]} \quad \Delta = -59, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1^{1,12321} \text{ } 8 \text{ } 12321 \text{ }^3 \text{ }_{[\# = 6]} \quad \Delta = -59, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1^{1,12321} \text{ } 8 \text{ } 12321 \text{ }^3 \text{ }_{[\# = 5]} \quad \Delta = -59, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1^{1,12321} \text{ } 8 \text{ } 12321 \text{ }^3 \text{ }_{[\# = 3]} \quad \Delta = -59, n_+ = 1, n_- = 13, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

$T = 14$ (1842 bases)

Base: 222222211212231

$$\begin{aligned}
222222211212231 \text{ }^3 \text{ }_{[\# = 4]} \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
222222211212231 \text{ }^3 \text{ }_{[\# = 3]} \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
222222211212231 \text{ }^3 \text{ }_{[\# = 2]} \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
222222211212231 \text{ }^3 \text{ }_{[\# = 5]} \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
22222221121 \text{ }^2 \text{ } 2231 \quad \Delta = -124, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
22222221121 \text{ }^2 \text{ } 2231 \quad \Delta = -124, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 70 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{aligned}$$

Base: 222222112122313

$$\begin{aligned}
22222211212231 \text{ }^3 \text{ } 3 \quad \Delta = -140, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
22222211212231 \text{ }^3 \text{ }_{[\# = 2]} \quad \Delta = -140, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
22222211212231 \text{ }^2 \text{ } 3 \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
22222211212231 \text{ }^2 \text{ } 3 \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
22222211212231 \text{ }^2 \text{ } 3 \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

Base: 222221121223132

$$222221121223132^3 \quad \Delta = -141, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$222221121223132^2 \quad \Delta = -135, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$222221121223132^2 \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$222221121223132^2 \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$222221121223132^2 \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: 222211212231322

$$222211212231322^3 \quad \Delta = -141, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 32 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$222211212231322^2 \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$222211212231322^2 \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$222211212231322^2 \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: $1 \overset{13221}{12} 122222$

$$1 \overset{313221}{12} 122222 \text{ [#=2]} \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$1 \overset{13221}{12} 122222 \quad \Delta = -124, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{132212}{12} 122222 \quad \Delta = -124, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1 \overset{313221}{12} 122222 \text{ [#=3]} \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1 \overset{313221}{12} 122222 \text{ [#=4]} \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{313221}{12} 122222 \text{ [#=5]} \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: $1 \overset{313221}{12} 122222$

$$1 \overset{3132212}{12} 122222 \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1 \overset{313221}{12} 122222 \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\begin{aligned}
& \overset{313221}{1 \ 12 \ 122222} \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{313221}{1 \ 12 \ 122222} \quad \Delta = -140, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{313221}{1 \ 12 \ 122222} \text{ [#=2]} \quad \Delta = -140, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{2313221}{1 \ 12 \ 12222}$

$$\begin{aligned}
& \overset{2313221}{1 \ 12 \ 12222} \quad \Delta = -135, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{2313221}{1 \ 12 \ 12222} \text{ 2} \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{2313221}{1 \ 12 \ 12222} \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{2313221}{1 \ 12 \ 12222} \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{2313221}{1 \ 12 \ 12222} \quad \Delta = -141, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{22313221}{1 \ 12 \ 1222}$

$$\begin{aligned}
& \overset{22313221}{1 \ 12 \ 1222} \text{ 2} \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{22313221}{1 \ 12 \ 1222} \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{22313221}{1 \ 12 \ 1222} \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{22313221}{1 \ 12 \ 1222} \quad \Delta = -141, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 32 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{13221}{21 \ 12 \ 122222}$

$$\begin{aligned}
& \overset{13221}{21 \ 12 \ 122222} \quad \Delta = -124, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{13221}{21 \ 12 \ 122222} \text{ 2} \quad \Delta = -124, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{213221}{21 \ 12 \ 122222} \quad \Delta = -124, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{313221}{21 \ 12 \ 122222} \text{ [#=5]} \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{313221}{21 \ 12 \ 122222} \text{ [#=2]} \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
& \overset{313221}{21 \ 12 \ 122222} \text{ [#=3]} \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

$$21 \overset{313221}{12} 12222 \text{ }_{[\#=4]} \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: $21 \overset{313221}{12} 12222$

$$21 \overset{2313221}{12} 12222 \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$21 \overset{313221}{12} \overset{2}{12222} \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$21 \overset{313221}{12} 12222 \quad \Delta = -140, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$21 \overset{313221}{12} 12222 \text{ }_{[\#=2]} \quad \Delta = -140, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$21 \overset{313221}{12} 12222 \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$21 \overset{313221}{12} 12222 \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: $21 \overset{2313221}{12} 1222$

$$21 \overset{2313221}{12} \overset{2}{1222} \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$21 \overset{2313221}{12} 1222 \quad \Delta = -141, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$21 \overset{2313221}{12} 1222 \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$21 \overset{22313221}{12} 1222 \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$21 \overset{2313221}{12} 1222 \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$21 \overset{2313221}{12} 1222 \quad \Delta = -135, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: $21 \overset{22313221}{12} 122$

$$21 \overset{222313221}{12} 122 \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$21 \overset{22313221}{12} \overset{2}{122} \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$21 \overset{22313221}{12} 122 \quad \Delta = -141, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 32 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$21 \overset{22313221}{12} 122 \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$21 \overset{22313221}{12} 122 \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: $22221 \overset{3221}{12} 1223$

$$22221 \overset{23221}{12} 1223 \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$22221 \overset{32212}{12} 1223 \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: $2221 \overset{13221}{12} 1222$

$$2221 \overset{13221}{12} 1222 \quad \Delta = -124, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$2221 \overset{313221}{12} 1222 \text{ [#=2]} \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$2221 \overset{313221}{12} 1222 \text{ [#=4]} \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$2221 \overset{213221}{12} 1222 \quad \Delta = -124, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$2221 \overset{313221}{12} 1222 \text{ [#=3]} \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$2221 \overset{313221}{12} 1222 \text{ [#=5]} \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: $221 \overset{13221}{12} 12222$

$$221 \overset{213221}{12} 12222 \quad \Delta = -124, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$221 \overset{13221}{12} 12222 \quad \Delta = -124, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$221 \overset{132212}{12} 12222 \quad \Delta = -124, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$221 \overset{313221}{12} 12222 \text{ [#=2]} \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$221 \overset{313221}{12} 12222 \text{ [#=3]} \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$221 \overset{313221}{12} 12222 \text{ [#=4]} \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$221 \overset{313221}{12} 12222 \text{ [#=5]} \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: $221 \overset{313221}{12} 1222$

$$221 \overset{313221}{12} 1222 \quad \Delta = -140, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$221 \overset{313221}{12} 1222 \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$221 \overset{313221}{12} 1222 \text{ }_{[\#=2]} \quad \Delta = -140, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$221 \overset{313221}{12} 1222 \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$221 \overset{313221}{12} \overset{2}{1222} \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$221 \overset{2}{313221} 1222 \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\text{Base: } 221 \overset{2313221}{12} 122$$

$$221 \overset{2313221}{12} 122 \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$221 \overset{2313221}{12} 122 \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$221 \overset{2313221}{12} 122 \quad \Delta = -141, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$221 \overset{2313221}{12} 122 \quad \Delta = -135, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$221 \overset{2}{2313221} 122 \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{1,13221}{12} 122222$$

$$1 \overset{1,13221}{12} 122222 \quad \Delta = -124, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{1,13221}{12} \overset{2}{122222} \quad \Delta = -124, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1 \overset{1,313221}{12} 122222 \text{ }_{[\#=5]} \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1 \overset{1,313221}{12} 122222 \text{ }_{[\#=4]} \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1 \overset{1,313221}{12} 122222 \text{ }_{[\#=3]} \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{1,313221}{12} 122222 \text{ }_{[\#=2]} \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{1,313221}{12} 12222$$

$$1 \overset{1,313221}{12} 12222 \quad \Delta = -140, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{1,313221}{12} 12222 \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{1,313221}{12} \overset{2}{12222} \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\begin{array}{l}
\begin{array}{l} 1^{1,313221} \\ 1 \quad 12 \quad 12222 \end{array} \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\\
\begin{array}{l} 1^{1,313221} \\ 1 \quad 12 \quad 12222 \end{array} \quad [\#=2] \quad \Delta = -140, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
\\
\text{Base: } 1^{1,2313221} \quad 1222 \\
\begin{array}{l} 1^{1,2313221} \\ 1 \quad 12 \quad 1222 \end{array} \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\\
\begin{array}{l} 1^{1,2313221} \\ 1 \quad 12 \quad 1222 \end{array} \quad \Delta = -135, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\\
\begin{array}{l} 1^{1,2313221} \\ 1 \quad 12 \quad 1222 \end{array} \quad \Delta = -141, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 32 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\\
\begin{array}{l} 1^{1,2313221\textcolor{red}{2}} \\ 1 \quad 12 \quad 1222 \end{array} \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\\
\begin{array}{l} 1^{1,2313221} \\ 1 \quad 12 \quad 1222 \end{array} \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\\
\text{Base: } 1^{221,22313221} \quad 1 \\
\begin{array}{l} 1^{221,22313221} \\ 1 \quad 12 \quad 1 \end{array} \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
\\
\begin{array}{l} 1^{221,22313221} \\ 1 \quad 12 \quad 1 \end{array} \quad \Delta = -141, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 60 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\\
\begin{array}{l} 1^{221,22313221} \\ 1 \quad 12 \quad 1 \end{array} \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\\
\begin{array}{l} 1^{221,22313221} \\ 1 \quad 12 \quad 1 \end{array} \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\\
\text{Base: } 1^{21,22313221} \quad 12 \\
\begin{array}{l} 1^{21,22313221} \\ 1 \quad 12 \quad 12 \end{array} \quad \Delta = -141, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 60 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\\
\begin{array}{l} 1^{21,22313221} \\ 1 \quad 12 \quad 12 \end{array} \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\\
\begin{array}{l} 1^{21,22313221} \\ 1 \quad 12 \quad 12 \end{array} \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
\\
\begin{array}{l} 1^{21,22313221\textcolor{red}{2}} \\ 1 \quad 12 \quad 12 \end{array} \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\\
\text{Base: } 1^{21,13221} \quad 12222 \\
\begin{array}{l} 1^{21,13221\textcolor{red}{2}} \\ 1 \quad 12 \quad 12222 \end{array} \quad \Delta = -124, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\\
\begin{array}{l} 1^{21,\textcolor{red}{3}13221} \\ 1 \quad 12 \quad 12222 \end{array} \quad [\#=5] \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
& \text{Base: } 1^{21, \textcolor{red}{3}13221} 12^{12222} [\#=4] \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 1^{21, \textcolor{red}{3}13221} 12^{12222} [\#=3] \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 1^{21, \textcolor{red}{3}13221} 12^{12222} [\#=2] \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 1^{21, 13221} 12^{12222} \quad \Delta = -124, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& 1^{21, 13221} 12^{12222} \quad \Delta = -124, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \text{Base: } 1^{21, 313221} 12^{1222} \\
& 1^{21, 313221} 12^{1222} \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& 1^{21, 313221} 12^{1222} \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 1^{21, 313221} 12^{1222} \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 1^{21, 313221} 12^{1222} \quad \Delta = -140, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 1^{21, 313221} 12^{1222} [\#=2] \quad \Delta = -140, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& 1^{21, 313221 \textcolor{red}{2}} 12^{1222} \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \text{Base: } 1^{221, 2313221} 12^{12} \\
& 1^{221, 2313221 \textcolor{red}{2}} 12^{12} \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 1^{221, 2313221} 12^{12} \quad \Delta = -141, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 32 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 1^{221, 2313221} 12^{12} \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 1^{221, 2313221} 12^{12} \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 1^{221, 2313221} 12^{12} \quad \Delta = -135, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 1^{221, 2313221} 12^{12} \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \text{Base: } 2221^{1, 3221} 12^{1223} \\
& 2221^{1, 3221 \textcolor{red}{2}} 12^{1223} \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{21,3221}{2221} \ 12 \ 1223 \quad \Delta = -133, \ n_+ = 1, \ n_- = 14, \ n_0 = 0, \ |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{21,3221}{2221} \ 12 \ 1223 \quad \Delta = -133, \ n_+ = 1, \ n_- = 14, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{221,13221}{12} 1222$

$$\begin{aligned}
& \overset{221,313221}{1} \ 12 \ 1222 \ [\# = 5] \quad \Delta = -132, \ n_+ = 1, \ n_- = 14, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{221,313221}{1} \ 12 \ 1222 \ [\# = 4] \quad \Delta = -132, \ n_+ = 1, \ n_- = 14, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{221,313221}{1} \ 12 \ 1222 \ [\# = 3] \quad \Delta = -132, \ n_+ = 1, \ n_- = 14, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{221,313221}{1} \ 12 \ 1222 \ [\# = 2] \quad \Delta = -132, \ n_+ = 1, \ n_- = 14, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

$$\overset{221,13221}{1} \ 12 \ 1222 \quad \Delta = -124, \ n_+ = 1, \ n_- = 14, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\overset{221,13221}{1} \ 12 \ 1222 \quad \Delta = -124, \ n_+ = 1, \ n_- = 14, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$\overset{221,13221}{1} \ 12 \ 1222 \quad \Delta = -124, \ n_+ = 1, \ n_- = 14, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: $1 \overset{221,313221}{12} 122$

$$\overset{221,313221}{1} \ 12 \ 122 \quad \Delta = -132, \ n_+ = 1, \ n_- = 14, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\overset{221,313221}{1} \ 12 \ 122 \quad \Delta = -132, \ n_+ = 1, \ n_- = 14, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\overset{221,313221}{1} \ 12 \ 122 \quad \Delta = -132, \ n_+ = 1, \ n_- = 14, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\overset{221,313221}{1} \ 12 \ 122 \ [\# = 2] \quad \Delta = -140, \ n_+ = 1, \ n_- = 14, \ n_0 = 0, \ |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$\overset{221,313221}{1} \ 12 \ 122 \quad \Delta = -140, \ n_+ = 1, \ n_- = 14, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: $21 \overset{21,2313221}{12} 12$

$$\overset{21,2313221}{21} \ 12 \ 12 \quad \Delta = -141, \ n_+ = 1, \ n_- = 14, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 32 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\overset{221,2313221}{21} \ 12 \ 12 \quad \Delta = -133, \ n_+ = 1, \ n_- = 14, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\overset{21,2313221}{21} \ 12 \ 12 \quad \Delta = -135, \ n_+ = 1, \ n_- = 14, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\overset{21,2313221}{21} \ 12 \ 12 \quad \Delta = -133, \ n_+ = 1, \ n_- = 14, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$21^{21,2313221} 12 \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: $21^{21,13221} 12 1222$

$$21^{21,313221} 12 1222 \text{ [#=4]} \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$21^{21,13221} 12 1222 \quad \Delta = -124, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$21^{21,13221\textcolor{red}{2}} 12 1222 \quad \Delta = -124, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$21^{21,313221} 12 1222 \text{ [#=5]} \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$21^{21,313221} 12 1222 \text{ [#=2]} \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$21^{\textcolor{red}{2}21,13221} 12 1222 \quad \Delta = -124, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$21^{21,313221} 12 1222 \text{ [#=3]} \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: $21^{221,313221} 12 12$

$$21^{221,313221} 12 12 \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$21^{221,313221} 12 12 \quad \Delta = -140, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$21^{221,313221} 12 12 \text{ [#=2]} \quad \Delta = -140, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$21^{221,313221} 12 12 \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$21^{221,313221} 12 12 \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$21^{\textcolor{red}{2}221,313221} 12 12 \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: $21^{221,3221} 12 1223$

$$21^{\textcolor{red}{3}221,3221} 12 1223 \quad \Delta = -142, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$21^{\textcolor{red}{2}221,3221} 12 1223 \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$21^{221,3221\textcolor{red}{2}} 12 1223 \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$21^{221,3221} 12 1223 \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

Base: $21^{221,13221} 12^{221} 122$

$$21^{221,13221} 12^{221} 122 \quad \Delta = -124, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$21^{221,313221} 12^{221} 122 \text{ [#=3]} \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$21^{221,313221} 12^{221} 122 \text{ [#=4]} \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$21^{221,313221} 12^{221} 122 \text{ [#=5]} \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$21^{221,313221} 12^{221} 122 \text{ [#=2]} \quad \Delta = -132, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$21^{221,13221} 12^{221} 122 \quad \Delta = -124, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$21^{221,13221} 12^{221} 122 \quad \Delta = -124, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 12222222222222^2

$$22222222222221^2 \quad \Delta = 0, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$22222222222221^2 \quad \Delta = -115, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: 12231414141413^2

$$2^{712231414141} 3^2 \quad \Delta = -76, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$2^{512231414141} 3^2 \quad \Delta = -23, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$2^{612231414141} 3^2 \quad \Delta = -49, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$2^{812231414141} 3^2 \quad \Delta = -104, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$2^{1212231414141} 3^2 \quad \Delta = -219, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

Base: 21414141413221

$$21414141413221 \quad \Delta = -16, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 14 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 13141414141321

$$13141414141321^3 \text{ [#=3]} \quad \Delta = -4, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$12314141414131^3 \text{ [#=9]} \quad \Delta = -4, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix}$$

$$\begin{array}{ll}
13141414141321\overset{3}{3} \text{ [#=2]} & \Delta = -4, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
12314141414131\overset{3}{3} \text{ [#=2]} & \Delta = -4, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
12314141414131\overset{3}{3} \text{ [#=5]} & \Delta = -4, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
13141414141321\overset{3}{3} \text{ [#=5]} & \Delta = -4, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
12314141414131\overset{3}{3} \text{ [#=3]} & \Delta = -4, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
12314141414131\overset{3}{3} \text{ [#=6]} & \Delta = -4, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
13141414141321\overset{3}{3} \text{ [#=6]} & \Delta = -4, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
12314141414131\overset{3}{3} \text{ [#=7]} & \Delta = -4, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
13141414141321\overset{3}{3} \text{ [#=7]} & \Delta = -4, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
12314141414131\overset{3}{3} \text{ [#=8]} & \Delta = -4, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
13141414141321\overset{3}{3} \text{ [#=8]} & \Delta = -4, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
12314141414131\overset{3}{3} \text{ [#=4]} & \Delta = -4, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
13141414141321\overset{3}{3} \text{ [#=9]} & \Delta = -4, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
13141414141321\overset{3}{3} & \Delta = -4, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 15 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
13141414141321\overset{3}{3} \text{ [#=4]} & \Delta = -4, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\text{Base: } 13141414141321^2 & \\
2^{\overset{3}{3}13141414141} 2 \text{ [#=6]} & \Delta = -6, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 285 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
2^{\overset{3}{3}13141414141} 2 \text{ [#=9]} & \Delta = -6, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 324 \quad \begin{pmatrix} 0 & 18 & 2 \\ 18 & 645 & 71 \\ 2 & 71 & 8 \end{pmatrix} \\
2^{\overset{3}{3}13141414141} 2 \text{ [#=5]} & \Delta = -6, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 197 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
2^{\overset{3}{3}13141414141} 2 \text{ [#=3]} & \Delta = -6, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 23 \\ 2 & 23 & 8 \end{pmatrix}
\end{array}$$

$$2^{\overset{3}{1}3141414141} 3^{\quad} 2_{[\# = 2]} \quad \Delta = -6, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$2^{\overset{3}{1}3141414141} 3^{\quad} 2 \quad \Delta = -6, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$2^{\overset{3}{1}3141414141} 3^{\quad} 2_{[\# = 8]} \quad \Delta = -6, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 256 \quad \begin{pmatrix} 0 & 16 & 2 \\ 16 & 509 & 63 \\ 2 & 63 & 8 \end{pmatrix}$$

$$2^{\overset{5}{1}3141414141} 3^{\quad} 2 \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$2^{\overset{3}{1}3141414141} 3^{\quad} 2_{[\# = 7]} \quad \Delta = -6, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 196 \quad \begin{pmatrix} 0 & 14 & 2 \\ 14 & 389 & 55 \\ 2 & 55 & 8 \end{pmatrix}$$

$$2^{\overset{3}{1}3141414141} 3^{\quad} 2_{[\# = 4]} \quad \Delta = -6, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$2^{\overset{6}{1}3141414141} 3^{\quad} 2 \quad \Delta = -76, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$2^{\overset{4}{1}3141414141} 3^{\quad} 2 \quad \Delta = -26, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$131414141414132^{\overset{2}{}} \quad \Delta = -6, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 23 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

Base: 21414141414131

$$21414141414131\overset{3}{3}_{[\# = 3]} \quad \Delta = -23, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$21414141414131\overset{3}{3}_{[\# = 4]} \quad \Delta = -23, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$21414141414131^{\overset{2}{}} \quad \Delta = -15, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$21414141414131^{\overset{2}{}} \quad \Delta = -15, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 20 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$21414141414131^{\overset{3}{}}_{[\# = 3]} \quad \Delta = -23, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 195 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$21414141414131\overset{3}{3}_{[\# = 2]} \quad \Delta = -23, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$21414141414131^{\overset{2}{}} \quad \Delta = -15, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$21414141414131^{\overset{2}{}} \quad \Delta = -15, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$21414141414131^{\overset{2}{}} \quad \Delta = -15, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$21414141414131^{\overset{2}{}} \quad \Delta = -15, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\begin{aligned}
21414141414131\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -23, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21414141414131\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -23, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 85 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
21414141414131\textcolor{red}{3} \quad \Delta = -23, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 19 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21414141414131\textcolor{red}{3} \quad \Delta = -23, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
21414141414131\textcolor{red}{3} \text{ [#=6]} \quad \Delta = -23, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
21414141414131\textcolor{red}{3} \text{ [#=7]} \quad \Delta = -23, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21414141414131\textcolor{red}{3} \text{ [#=8]} \quad \Delta = -23, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21414141414131\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -23, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21414141414131\textcolor{red}{4} \quad \Delta = -43, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 18 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\textbf{Base: } 1414141414131\textcolor{red}{1} \\
1414141414131\textcolor{red}{2} \quad \Delta = -15, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1414141414131\textcolor{red}{2} \quad \Delta = -15, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1414141414131\textcolor{red}{2} \quad \Delta = -15, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1414141414131\textcolor{red}{2} \quad \Delta = -15, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1414141414131\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -23, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1414141414131\textcolor{red}{12} \quad \Delta = -15, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1414141414131\textcolor{red}{3} \text{ [#=7]} \quad \Delta = -23, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1414141414131\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -23, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1414141414131\textcolor{red}{3} \quad \Delta = -23, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
1414141414131\textcolor{red}{3} \text{ [#=8]} \quad \Delta = -23, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{31}{1} 4 14141414131 \text{ } [\# = 2] \quad \Delta = -23, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{1}{1} 414141414131 \overset{3}{3} \text{ } [\# = 4] \quad \Delta = -23, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{1}{1} 414141414131 \overset{3}{3} \text{ } [\# = 2] \quad \Delta = -23, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
& \overset{31}{1} 4 14141414131 \text{ } [\# = 3] \quad \Delta = -23, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{1}{1} 414141414131 \overset{3}{3} \text{ } [\# = 9] \quad \Delta = -23, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{1}{1} 414141414131 \overset{3}{3} \text{ } [\# = 6] \quad \Delta = -23, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: 12231513151321

$$\begin{aligned}
& 12231513151321 \overset{3}{3} \text{ } [\# = 2] \quad \Delta = -5, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& 12315131513221 \overset{3}{3} \text{ } [\# = 6] \quad \Delta = -5, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 285 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
& 12231513151321 \overset{3}{3} \text{ } [\# = 5] \quad \Delta = -5, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& 12315131513221 \overset{3}{3} \text{ } [\# = 8] \quad \Delta = -5, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 509 & 63 \\ 2 & 63 & 8 \end{pmatrix} \\
& 12315131513221 \overset{3}{3} \text{ } [\# = 2] \quad \Delta = -5, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 29 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& 12315131513221 \overset{3}{3} \text{ } [\# = 3] \quad \Delta = -5, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 69 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& 12231513151321 \overset{3}{3} \text{ } [\# = 3] \quad \Delta = -5, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& 12315131513221 \overset{3}{3} \text{ } [\# = 4] \quad \Delta = -5, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 125 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
& 12231513151321 \overset{3}{3} \text{ } [\# = 4] \quad \Delta = -5, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& 12315131513221 \overset{3}{3} \text{ } [\# = 5] \quad \Delta = -5, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 197 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
& 12231513151321 \overset{3}{3} \text{ } [\# = 8] \quad \Delta = -5, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& 12315131513221 \overset{3}{3} \text{ } [\# = 7] \quad \Delta = -5, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 389 & 55 \\ 2 & 55 & 8 \end{pmatrix} \\
& 12315131513221 \overset{3}{3} \text{ } [\# = 9] \quad \Delta = -5, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 645 & 71 \\ 2 & 71 & 8 \end{pmatrix}
\end{aligned}$$

12231513151321 $\bar{3}$ [#=9]	$\Delta = -5, n_+ = 1, n_- = 14, n_0 = 0, \det = 81$	$\begin{pmatrix} 0 & 9 & 2 \\ 9 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$
12231513151321 $\bar{3}$ [#=7]	$\Delta = -5, n_+ = 1, n_- = 14, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$
12231513151321 $\bar{3}$ [#=6]	$\Delta = -5, n_+ = 1, n_- = 14, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$
Base: 13161231513221		
12231513216131 $\bar{3}$ [#=6]	$\Delta = -31, n_+ = 1, n_- = 14, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix}$
13161231513221 $\bar{3}$ [#=7]	$\Delta = -31, n_+ = 1, n_- = 14, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$
13161231513221 $\bar{3}$ [#=5]	$\Delta = -31, n_+ = 1, n_- = 14, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$
13161231513221 $\bar{3}$ [#=4]	$\Delta = -31, n_+ = 1, n_- = 14, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$
12231513216131 $\bar{3}$ [#=4]	$\Delta = -31, n_+ = 1, n_- = 14, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$
13161231513221 $\bar{3}$ [#=3]	$\Delta = -31, n_+ = 1, n_- = 14, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$
12231513216131 $\bar{3}$ [#=3]	$\Delta = -31, n_+ = 1, n_- = 14, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$
12231513216131 $\bar{3}$ [#=5]	$\Delta = -31, n_+ = 1, n_- = 14, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$
12231513216131 $\bar{3}$ [#=2]	$\Delta = -31, n_+ = 1, n_- = 14, n_0 = 0, \det = 4$	$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$
13161231513221 $\bar{3}$ [#=2]	$\Delta = -31, n_+ = 1, n_- = 14, n_0 = 0, \det = 4$	$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$
13161231513221 $\bar{3}$ [#=6]	$\Delta = -31, n_+ = 1, n_- = 14, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$
12231513216131 $\bar{3}$ [#=7]	$\Delta = -31, n_+ = 1, n_- = 14, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}$
12231513216131 $\bar{3}$ [#=8]	$\Delta = -31, n_+ = 1, n_- = 14, n_0 = 0, \det = 64$	$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix}$
13161231513221 $\bar{3}$ [#=8]	$\Delta = -31, n_+ = 1, n_- = 14, n_0 = 0, \det = 64$	$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$
12231513216131 $\bar{3}$ [#=9]	$\Delta = -31, n_+ = 1, n_- = 14, n_0 = 0, \det = 81$	$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix}$
13161231513221 $\bar{3}$ [#=9]	$\Delta = -31, n_+ = 1, n_- = 14, n_0 = 0, \det = 81$	$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$

$$\begin{aligned}
13161231513221^2 \quad \Delta = -23, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
13161231513221^2 \quad \Delta = -23, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
13161231513221^2 \quad \Delta = -23, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
13161231513221^2 \quad \Delta = -23, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 13216131513221

$$\begin{aligned}
13216131513221^2 \quad \Delta = -23, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
13216131513221^2 \quad \Delta = -23, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
13216131513221^2 \quad \Delta = -23, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
13216131513221^2 \quad \Delta = -23, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
132161315132213_{[\#=6]} \quad \Delta = -31, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
122315131612313_{[\#=4]} \quad \Delta = -31, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
122315131612313_{[\#=2]} \quad \Delta = -31, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
132161315132213_{[\#=3]} \quad \Delta = -31, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
132161315132213_{[\#=7]} \quad \Delta = -31, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
132161315132213_{[\#=8]} \quad \Delta = -31, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
132161315132213_{[\#=9]} \quad \Delta = -31, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
132161315132213_{[\#=4]} \quad \Delta = -31, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
122315131612313_{[\#=3]} \quad \Delta = -31, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
132161315132213_{[\#=2]} \quad \Delta = -31, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
122315131612313_{[\#=5]} \quad \Delta = -31, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
13216131513221\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -31, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
\textbf{Base: } 12321613151321 & \\
12315131612321\textcolor{red}{3} \text{ [#=7]} \quad \Delta = -33, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
12321613151321\textcolor{red}{3} \text{ [#=7]} \quad \Delta = -33, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
12321613151321\textcolor{red}{3} \text{ [#=6]} \quad \Delta = -33, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
12315131612321\textcolor{red}{3} \text{ [#=8]} \quad \Delta = -33, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
12321613151321\textcolor{red}{3} \text{ [#=8]} \quad \Delta = -33, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
12315131612321\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -33, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
12315131612321\textcolor{red}{3} \text{ [#=6]} \quad \Delta = -33, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
12321613151321\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -33, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
12321613151321\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -33, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
12315131612321\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -33, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
12315131612321\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -33, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
12321613151321\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -33, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
12321613151321\textcolor{red}{2} \quad \Delta = -25, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
12315131612321\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -33, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
12321613151321\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -33, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
12315131612321\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -33, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
12321613151321\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -33, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
12321613151321\textcolor{red}{2} \quad \Delta = -25, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

$$12321613151321 \quad \Delta = -25, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$12321613151321 \quad \Delta = -25, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: 61223151316131

$$61223151316131\bar{3} \text{ [#=4]} \quad \Delta = -108, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$61223151316131\bar{3} \text{ [#=8]} \quad \Delta = -108, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$61223151316131\bar{3} \text{ [#=9]} \quad \Delta = -108, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$61223151316131\bar{3} \text{ [#=7]} \quad \Delta = -108, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$61223151316131\bar{3} \text{ [#=6]} \quad \Delta = -108, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$61223151316131\bar{3} \text{ [#=5]} \quad \Delta = -108, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$61223151316131 \quad \Delta = -100, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$61223151316131\bar{3} \text{ [#=3]} \quad \Delta = -108, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$61223151316131\bar{3} \text{ [#=2]} \quad \Delta = -108, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$61223151316131 \quad \Delta = -100, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$61223151316131 \quad \Delta = -100, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$61223151316131 \quad \Delta = -100, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$61223151316131 \quad \Delta = -108, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 63 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$61223151316131 \quad \Delta = -100, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: 16123151316131

$$16123151316131 \quad \Delta = -100, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$16123151316131 \quad \Delta = -100, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$\begin{array}{ll}
16123151316131^2 & \Delta = -100, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
16123151316131^2 & \Delta = -100, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
16123151316131^2 & \Delta = -100, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
161231513161313_{\#8} & \Delta = -108, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
161231513161313_{\#2} & \Delta = -108, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
161231513161313_{\#3} & \Delta = -108, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
161231513161313_{\#9} & \Delta = -108, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
161231513161313_{\#7} & \Delta = -108, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
161231513161313_{\#6} & \Delta = -108, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
161231513161313_{\#5} & \Delta = -108, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
161231513161313_{\#4} & \Delta = -108, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\text{Base: } 21613151316131 & \\
216131513161313_{\#6} & \Delta = -107, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
21613151316131^2 & \Delta = -99, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
216131513161313_{\#5} & \Delta = -107, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
216131513161313_{\#4} & \Delta = -107, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
216131513161313_{\#3} & \Delta = -107, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
216131513161313_{\#2} & \Delta = -107, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
21613151316131^3 & \Delta = -107, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
216131513161313_{\#9} & \Delta = -107, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
21613151316131\bar{3} \text{ }_{[\# = 7]} \quad \Delta = -107, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21613151316131\bar{3} \text{ }_{[\# = 8]} \quad \Delta = -107, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21613151316131\bar{2} \quad \Delta = -99, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21613151316131\bar{2} \quad \Delta = -99, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21613151316131\bar{2} \quad \Delta = -99, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
21613151316131\bar{2} \quad \Delta = -99, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21613151316131\bar{2} \quad \Delta = -99, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\text{Base: } 31316131513221 \\
31316131513221\bar{2} \quad \Delta = -30, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
31316131513221\bar{2} \quad \Delta = -30, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
31316131513221\bar{2} \quad \Delta = -30, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
31316131513221\bar{2} \quad \Delta = -30, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
31316131513221\bar{2} \quad \Delta = -30, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
31316131513221\bar{3} \text{ }_{[\# = 3]} \quad \Delta = -38, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
31316131513221\bar{3} \text{ }_{[\# = 3]} \quad \Delta = -38, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 132 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
31316131513221\bar{3} \quad \Delta = -38, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
31316131513221\bar{3} \text{ }_{[\# = 2]} \quad \Delta = -38, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
31316131513221\bar{3} \text{ }_{[\# = 4]} \quad \Delta = -38, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
31316131513221\bar{3} \text{ }_{[\# = 5]} \quad \Delta = -38, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
31316131513221\bar{3} \text{ }_{[\# = 2]} \quad \Delta = -38, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
31316131513221\overset{\textcolor{red}{3}}{3} \text{ [#=6]} \quad \Delta = -38, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
31316131513221\overset{\textcolor{red}{3}}{3} \text{ [#=8]} \quad \Delta = -38, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
31316131513221\overset{\textcolor{red}{3}}{3} \text{ [#=7]} \quad \Delta = -38, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
31316131513221\overset{\textcolor{red}{3}}{3} \text{ [#=9]} \quad \Delta = -38, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}
\end{aligned}$$

$$\overset{\textcolor{red}{4}}{3} 31316131513221 \quad \Delta = -58, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: 41316131513216

$$\begin{aligned}
\overset{\textcolor{red}{3}}{4} 41316131513216 \quad \Delta = -136, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
413161\overset{\textcolor{red}{2}}{3} 1513216 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
413161\overset{\textcolor{red}{2}}{3} 1513216 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
413161\overset{\textcolor{red}{2}}{3} 1513216 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
413161\overset{\textcolor{red}{2}}{3} 1513216 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
413161\overset{\textcolor{red}{2}}{3} 1513216 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
413161\overset{\textcolor{red}{2}}{3} 1513216 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 41316131513161

$$\begin{aligned}
\overset{\textcolor{red}{3}}{4} 41316131513161 \quad \Delta = -135, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
413161\overset{\textcolor{red}{2}}{3} 1513161 \quad \Delta = -127, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
413161\overset{\textcolor{red}{2}}{3} 1513161 \quad \Delta = -127, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
413161\overset{\textcolor{red}{2}}{3} 1513161 \quad \Delta = -127, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
413161\overset{\textcolor{red}{2}}{3} 1513161 \quad \Delta = -127, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
413161\overset{\textcolor{red}{2}}{3} 1513161 \quad \Delta = -127, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

$$41316131513161 \quad \Delta = -127, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 14131613151321

$$14131613151321\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -58, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$14131613151321 \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$14131613151321 \quad \Delta = -58, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$12315131613141\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -58, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$14131613151321\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -58, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$14131613151321\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -58, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix}$$

$$14131613151321\textcolor{red}{3} \text{ [#=8]} \quad \Delta = -58, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$12315131613141\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -58, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$14131613151321 \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$14131613151321\textcolor{red}{3} \text{ [#=7]} \quad \Delta = -58, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$14131613151321\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -58, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$14131613151321\textcolor{red}{3} \text{ [#=6]} \quad \Delta = -58, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$14131613151321\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -58, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$14131613151321 \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$14131613151321 \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$14131613151321 \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: 51315131613151

$$51315131613151 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$\begin{aligned}
51315131613151^2 \quad \Delta = -126, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
51315131613151^2 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
51315131613151^2 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
51315131613151^2 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
51315131613151^2 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
51315131613151^2 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 13151316131512

$$\begin{aligned}
13151316131512^2 \quad \Delta = -73, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
13151316131512^3 \quad \Delta = -81, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
215131613151313^{\text{[#}=2]} \quad \Delta = -81, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
215131613151313^{\text{[#}=3]} \quad \Delta = -81, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
215131613151313^{\text{[#}=4]} \quad \Delta = -81, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
215131613151313^{\text{[#}=5]} \quad \Delta = -81, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
13151316131512^2 \quad \Delta = -73, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
215131613151313^{\text{[#}=6]} \quad \Delta = -81, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
215131613151313^{\text{[#}=7]} \quad \Delta = -81, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
215131613151313^{\text{[#}=8]} \quad \Delta = -81, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
215131613151313^{\text{[#}=9]} \quad \Delta = -81, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
13151316131512^2 \quad \Delta = -73, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
13151316131512^2 \quad \Delta = -73, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$13151316131512 \quad \Delta = -73, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$13151316131512 \quad \Delta = -73, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: 1315131613151¹

$$1 \overset{3}{1}3151316131 \overset{5}{5} 1 \text{ [#=7]} \quad \Delta = -81, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$1 \overset{3}{1}3151316131 \overset{5}{5} 1 \text{ [#=9]} \quad \Delta = -81, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$1 \overset{3}{1}3151316131 \overset{5}{5} 1 \text{ [#=8]} \quad \Delta = -81, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$1 \overset{3}{1}3151316131 \overset{5}{5} 1 \text{ [#=6]} \quad \Delta = -81, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$1 \overset{3}{1}3151316131 \overset{5}{5} 1 \text{ [#=5]} \quad \Delta = -81, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$1 \overset{3}{1}3151316131 \overset{5}{5} 1 \text{ [#=3]} \quad \Delta = -81, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1315131613151 \overset{2}{2} \quad \Delta = -73, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1315131613151 \overset{2}{2} \quad \Delta = -73, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1315131613151 \overset{2}{2} \quad \Delta = -73, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1315131613151 \overset{2}{2} \quad \Delta = -73, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1315131613151 \overset{21}{21} \quad \Delta = -73, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{3}{1}3151316131 \overset{5}{5} 1 \text{ [#=4]} \quad \Delta = -81, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1 \overset{3}{1}3151316131 \overset{5}{5} 1 \text{ [#=2]} \quad \Delta = -81, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 16131612321613

$$16131612321613 \overset{2}{2} \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$16131612321613 \overset{2}{2} \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$16131612321613 \overset{2}{2} \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$16131612321613^2 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$16131612321613^2 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: 21613161232161

$$21613161232161^3 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$21613161232161^2 \quad \Delta = -120, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$21613161232161^2 \quad \Delta = -120, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$21613161232161^2 \quad \Delta = -120, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$21613161232161^2 \quad \Delta = -120, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$21613161232161^2 \quad \Delta = -120, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 61232161316123

$$61232161316123^2 \quad \Delta = -129, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$61232161316123^2 \quad \Delta = -129, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$61232161316123^2 \quad \Delta = -129, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$61232161316123^2 \quad \Delta = -129, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$61232161316123^2 \quad \Delta = -129, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

Base: 13216131612321

$$13216131612321^2 \quad \Delta = -51, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$132161316123213^{\text{[#}=7]} \quad \Delta = -59, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$132161316123213^{\text{[#}=8]} \quad \Delta = -59, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$132161316123213^{\text{[#}=9]} \quad \Delta = -59, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$123216131612313^{\text{[#}=3]} \quad \Delta = -59, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$\begin{aligned}
13216131612321\bar{3} \text{ }_{[\# = 6]} \quad \Delta = -59, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
13216131612321\bar{3} \text{ }_{[\# = 5]} \quad \Delta = -59, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
12321613161231\bar{3} \text{ }_{[\# = 5]} \quad \Delta = -59, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
13216131612321\bar{3} \text{ }_{[\# = 4]} \quad \Delta = -59, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
13216131612321 \quad \Delta = -51, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
13216131612321 \quad \Delta = -51, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
13216131612321 \quad \Delta = -51, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
12321613161231\bar{3} \text{ }_{[\# = 4]} \quad \Delta = -59, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
13216131612321\bar{3} \text{ }_{[\# = 3]} \quad \Delta = -59, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
13216131612321\bar{3} \text{ }_{[\# = 2]} \quad \Delta = -59, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
12321613161231\bar{3} \text{ }_{[\# = 2]} \quad \Delta = -59, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: 22161316131512

$$\begin{aligned}
22161316131512 \quad \Delta = -91, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
22161316131512 \quad \Delta = -91, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
22161316131512 \quad \Delta = -91, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
22161316131512 \quad \Delta = -99, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
22161316131512 \quad \Delta = -100, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
22161316131512 \quad \Delta = -91, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
22161316131512 \quad \Delta = -91, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
22161316131512 \quad \Delta = -91, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

$$22161316131512 \quad \Delta = -99, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 13161316123151

$$15132161316131\bar{3} \quad [\# = 5] \quad \Delta = -108, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$15132161316131\bar{3} \quad [\# = 6] \quad \Delta = -108, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$15132161316131\bar{3} \quad [\# = 3] \quad \Delta = -108, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$13161316123151 \quad \Delta = -100, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$15132161316131\bar{3} \quad [\# = 9] \quad \Delta = -108, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix}$$

$$13161316123151 \quad \Delta = -100, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$13161316123151 \quad \Delta = -100, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$13161316123151 \quad \Delta = -100, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$15132161316131\bar{3} \quad [\# = 4] \quad \Delta = -108, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$13161316123151 \quad \Delta = -100, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$15132161316131\bar{3} \quad [\# = 2] \quad \Delta = -108, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$15132161316131\bar{3} \quad [\# = 7] \quad \Delta = -108, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$15132161316131\bar{3} \quad [\# = 8] \quad \Delta = -108, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

Base: 2216131613151¹

$$2216131613151 \quad \Delta = -91, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$2216131613151 \quad \Delta = -99, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$2216131613151 \quad \Delta = -91, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$2216131613151 \quad \Delta = -91, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\begin{aligned}
2216131613151^1 \quad \Delta = -91, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2216131613151^2 \quad \Delta = -91, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
2216131613151^3 \quad \Delta = -100, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: 13161316123215

$$\begin{aligned}
512321613161313^{\#8} \quad \Delta = -110, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
512321613161313^{\#7} \quad \Delta = -110, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
512321613161313^{\#9} \quad \Delta = -110, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
13161316123215^2 \quad \Delta = -102, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
512321613161313^{\#6} \quad \Delta = -110, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
512321613161313^{\#2} \quad \Delta = -110, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
512321613161313^{\#4} \quad \Delta = -110, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
512321613161313^{\#3} \quad \Delta = -110, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
512321613161313^{\#5} \quad \Delta = -110, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
13161316123215^2 \quad \Delta = -102, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
13161316123215^2 \quad \Delta = -102, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
13161316123215^2 \quad \Delta = -102, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
13161316123215^2 \quad \Delta = -102, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 12321613161313

$$\begin{aligned}
12321613161313^4 \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 9 & 5 \\ 5 & 9 \end{pmatrix} \\
12321613161313^2 \quad \Delta = -58, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
12321613161313^2 \quad \Delta = -58, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
31316131612321\overset{2}{3} \text{ }_{[\# = 5]} \quad \Delta = -66, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
31316131612321\overset{3}{3} \text{ }_{[\# = 9]} \quad \Delta = -66, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
31316131612321\overset{3}{3} \text{ }_{[\# = 8]} \quad \Delta = -66, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
31316131612321\overset{3}{3} \text{ }_{[\# = 7]} \quad \Delta = -66, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
31316131612321\overset{3}{3} \text{ }_{[\# = 6]} \quad \Delta = -66, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
1232161316131\overset{2}{3} \quad \Delta = -58, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1232161316131\overset{3}{3} \quad \Delta = -66, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
31316131612321\overset{3}{3} \text{ }_{[\# = 2]} \quad \Delta = -66, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1232161316131\overset{3}{3} \text{ }_{[\# = 2]} \quad \Delta = -66, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
31316131612321\overset{3}{3} \text{ }_{[\# = 3]} \quad \Delta = -66, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1232161316131\overset{3}{3} \text{ }_{[\# = 3]} \quad \Delta = -66, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
31316131612321\overset{3}{3} \text{ }_{[\# = 4]} \quad \Delta = -66, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1232161316131\overset{2}{3} \quad \Delta = -58, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1232161316131\overset{2}{3} \quad \Delta = -58, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 14131613161231

$$\begin{aligned}
14131613161231\overset{3}{3} \text{ }_{[\# = 2]} \quad \Delta = -84, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
14131613161231\overset{3}{3} \text{ }_{[\# = 5]} \quad \Delta = -84, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
14131613161231\overset{3}{3} \text{ }_{[\# = 4]} \quad \Delta = -84, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
13216131613141\overset{3}{3} \text{ }_{[\# = 2]} \quad \Delta = -84, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
14131613161231\overset{3}{3} \quad \Delta = -84, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 21 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
14131613161231^2 \quad \Delta = -76, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
14131613161231^2 \quad \Delta = -76, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
14131613161231^2 \quad \Delta = -76, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
14131613161231^2 \quad \Delta = -76, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
14131613161231^2 \quad \Delta = -76, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
132161316131413^{\#3} \quad \Delta = -84, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
141316131612313^{\#3} \quad \Delta = -84, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: 41316131612315

$$\begin{aligned}
41316131612315^2 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
41316131612315^3 \quad \Delta = -136, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
41316131612315^2 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
41316131612315^2 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
41316131612315^2 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
41316131612315^2 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
41316131612315^2 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: 41413161316131

$$\begin{aligned}
414131613161313^{\#2} \quad \Delta = -111, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
41413161316131^3 \quad \Delta = -111, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 101 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
414131613161313^{\#8} \quad \Delta = -111, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
414131613161313^{\#7} \quad \Delta = -111, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
41413161316131\bar{3} \text{ }_{[\# = 6]} \quad \Delta = -111, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
41413161316131\bar{3} \text{ }_{[\# = 5]} \quad \Delta = -111, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
41413161316131\bar{2} \quad \Delta = -103, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
41413161316131\bar{2} \quad \Delta = -103, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
41413161316131\bar{2} \quad \Delta = -103, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
41413161316131\bar{2} \quad \Delta = -103, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
41413161316131\bar{3} \text{ }_{[\# = 9]} \quad \Delta = -111, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
41413161316131\bar{3} \text{ }_{[\# = 3]} \quad \Delta = -111, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
41413161316131\bar{3} \text{ }_{[\# = 4]} \quad \Delta = -111, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
41413161316131\bar{2} \quad \Delta = -103, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
41413161316131\bar{2} \quad \Delta = -103, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
\text{Base: } 14131613161313 & \\
31316131613141\bar{3} \text{ }_{[\# = 2]} \quad \Delta = -91, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
14131613161313\bar{2} \quad \Delta = -83, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
14131613161313\bar{2} \quad \Delta = -83, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
14131613161313\bar{3} \quad \Delta = -91, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
14131613161313\bar{3} \text{ }_{[\# = 3]} \quad \Delta = -91, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
14131613161313\bar{2} \quad \Delta = -83, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
14131613161313\bar{2} \quad \Delta = -83, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
31316131613141\bar{3} \text{ }_{[\# = 3]} \quad \Delta = -91, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

$$1413161316131\overset{3}{3} \text{ }_{[\# = 2]} \quad \Delta = -91, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1413161316131\overset{2}{3} \quad \Delta = -83, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1413161316131\overset{2}{3} \quad \Delta = -83, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1413161316131\overset{3}{3} \quad \Delta = -91, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 21 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1413161316131\overset{4}{3} \quad \Delta = -111, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 9 & 5 \\ 5 & 9 \end{pmatrix}$$

Base: $\overset{1}{1613151316131}$

$$\overset{1}{1613151316131}\overset{2}{3} \quad \Delta = -99, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$\overset{1}{1613151316131}\overset{3}{3} \text{ }_{[\# = 5]} \quad \Delta = -107, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\overset{1}{1613151316131}\overset{3}{3} \text{ }_{[\# = 2]} \quad \Delta = -107, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$\overset{1}{1613151316131}\overset{3}{3} \text{ }_{[\# = 9]} \quad \Delta = -107, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$\overset{1}{1613151316131}\overset{2}{3} \quad \Delta = -99, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$\overset{1}{1613151316131}\overset{2}{3} \quad \Delta = -99, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\overset{1}{1613151316131}\overset{3}{3} \text{ }_{[\# = 3]} \quad \Delta = -107, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$\overset{1}{1613151316131}\overset{2}{3} \quad \Delta = -99, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$\overset{1}{1613151316131}\overset{3}{3} \text{ }_{[\# = 6]} \quad \Delta = -107, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\overset{1}{1613151316131}\overset{3}{3} \text{ }_{[\# = 4]} \quad \Delta = -107, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\overset{1}{1613151316131}\overset{3}{3} \text{ }_{[\# = 7]} \quad \Delta = -107, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\overset{1}{1613151316131}\overset{3}{3} \text{ }_{[\# = 8]} \quad \Delta = -107, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\overset{1}{1613151316131}\overset{2}{3} \quad \Delta = -99, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: $\overset{1}{1613161232161}$

$$\begin{array}{ll}
1\overset{1}{6}1\overset{2}{3}161232161 & \Delta = -120, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1\overset{1}{6}1\overset{2}{3}161232161 & \Delta = -120, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
1\overset{1}{6}1\overset{2}{3}161232161 & \Delta = -120, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1\overset{1}{6}1\overset{2}{3}161232161 & \Delta = -120, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{array}$$

Base: $1\overset{21}{6}1316131512$

$$\begin{array}{ll}
1\overset{21}{6}1\overset{2}{3}16131512 & \Delta = -91, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1\overset{21}{6}1\overset{2}{3}16131512 & \Delta = -91, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1\overset{21}{6}1\overset{2}{3}16131512 & \Delta = -91, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1\overset{21}{6}1\overset{2}{3}16131512 & \Delta = -91, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1\overset{21}{6}1\overset{2}{3}16131512 & \Delta = -91, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1\overset{21}{6}1\overset{2}{3}16131512 & \Delta = -91, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1\overset{21}{6}1\overset{2}{3}16131512 & \Delta = -99, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1\overset{21}{6}1\overset{2}{3}16131512 & \Delta = -99, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}
\end{array}$$

Base: $1\overset{21}{6}1\overset{1}{3}1613151$

$$\begin{array}{ll}
1\overset{21}{6}1\overset{1}{3}1613151 & \Delta = -99, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1\overset{21}{6}1\overset{2}{3}1\overset{1}{3}1613151 & \Delta = -91, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1\overset{21}{6}1\overset{2}{3}1\overset{2}{1}1613151 & \Delta = -91, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1\overset{21}{6}1\overset{1}{3}1613151 & \Delta = -91, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1\overset{21}{6}1\overset{2}{3}1\overset{1}{3}1613151 & \Delta = -91, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1\overset{21}{6}1\overset{2}{3}1\overset{1}{3}1613151 & \Delta = -91, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{array}$$

Base: $6123216131\overset{31}{6}1$

$$\begin{aligned}
6123216131\overset{31}{6}1 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
6123216131\overset{2}{6}1\overset{31}{6}1 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
6123216131\overset{2}{6}1\overset{31}{6}1 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
6123216131\overset{2}{6}1\overset{31}{6}1 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
6123216131\overset{231}{6}1 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
6123216131\overset{31}{6}1\overset{2}{6}1 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1\overset{122315131}{6}131$

$$\begin{aligned}
1\overset{122315131}{6}131 \quad \Delta = -22, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1\overset{122315131}{6}131 \quad \Delta = -22, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1\overset{122315131}{6}131 \quad \Delta = -22, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1\overset{122315131}{6}131\overset{2}{6}1 \quad \Delta = -22, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1\overset{122315131}{6}131\overset{3}{6}1\text{ [#=4]} \quad \Delta = -30, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1\overset{3122315131}{6}131\text{ [#=8]} \quad \Delta = -30, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1\overset{122315131}{6}131\overset{3}{6}1\text{ [#=7]} \quad \Delta = -30, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
1\overset{3122315131}{6}131\text{ [#=7]} \quad \Delta = -30, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1\overset{122315131}{6}131\overset{3}{6}1\text{ [#=8]} \quad \Delta = -30, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1\overset{3122315131}{6}131\text{ [#=6]} \quad \Delta = -30, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1\overset{122315131}{6}131\overset{3}{6}1\text{ [#=9]} \quad \Delta = -30, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
1\overset{122315131}{6}131\overset{3}{6}1\text{ [#=6]} \quad \Delta = -30, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1\overset{3122315131}{6}131\text{ [#=5]} \quad \Delta = -30, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1^{122315131} 6^{131} 131 \textcolor{red}{3} [\# = 5] \quad \Delta = -30, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1^{3122315131} 6^{131} 131 [\# = 2] \quad \Delta = -30, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{122315131} 6^{131} 131 \textcolor{red}{3} [\# = 2] \quad \Delta = -30, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1^{3122315131} 6^{131} 131 [\# = 9] \quad \Delta = -30, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1^{3122315131} 6^{131} 131 [\# = 3] \quad \Delta = -30, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{122315131} 6^{131} 131 \textcolor{red}{3} [\# = 3] \quad \Delta = -30, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1^{3122315131} 6^{131} 131 [\# = 4] \quad \Delta = -30, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1^{131} 6^{131} 131612321$

$$\begin{aligned}
1^{131} 6^{131} 131612321 \textcolor{red}{3} [\# = 8] \quad \Delta = -58, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1^{131} 6^{131} 131612321 \textcolor{red}{3} [\# = 4] \quad \Delta = -58, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{131} 6^{131} 131612321 \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1^{131} 6^{131} 131612321 \textcolor{red}{3} [\# = 9] \quad \Delta = -58, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1^{3131} 6^{131} 131612321 [\# = 7] \quad \Delta = -58, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1^{131} 6^{131} 131612321 \textcolor{red}{3} [\# = 6] \quad \Delta = -58, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{3131} 6^{131} 131612321 [\# = 6] \quad \Delta = -58, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1^{131} 6^{131} 131612321 \textcolor{red}{3} [\# = 5] \quad \Delta = -58, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1^{131} 6^{131} 131612321 \textcolor{red}{3} [\# = 7] \quad \Delta = -58, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1^{3131} 6^{131} 131612321 [\# = 5] \quad \Delta = -58, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1^{131} 6^{131} 131612321 \textcolor{red}{2} \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1^{3131} 6^{131} 131612321 [\# = 2] \quad \Delta = -58, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{3131}{1 \ 6 \ 131612321} \text{ }_{[\# = 8]} \quad \Delta = -58, \ n_+ = 1, \ n_- = 14, \ n_0 = 0, \ |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 6 \ 131612321} \text{ }_{[\# = 9]} \quad \Delta = -58, \ n_+ = 1, \ n_- = 14, \ n_0 = 0, \ |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 131612321} \overset{3}{\text{ }}_{[\# = 3]} \quad \Delta = -58, \ n_+ = 1, \ n_- = 14, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 6 \ 131612321} \text{ }_{[\# = 3]} \quad \Delta = -58, \ n_+ = 1, \ n_- = 14, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 6 \ 131612321} \text{ }_{[\# = 4]} \quad \Delta = -58, \ n_+ = 1, \ n_- = 14, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 131612321} \overset{3}{\text{ }}_{[\# = 2]} \quad \Delta = -58, \ n_+ = 1, \ n_- = 14, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 131612321} \overset{2}{\text{ }} \quad \Delta = -50, \ n_+ = 1, \ n_- = 14, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 131612321} \overset{2}{\text{ }} \quad \Delta = -50, \ n_+ = 1, \ n_- = 14, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{131}{1 \ 6 \ 131613141}$

$$\begin{aligned}
& \overset{3131}{1 \ 6 \ 131613141} \text{ }_{[\# = 6]} \quad \Delta = -83, \ n_+ = 1, \ n_- = 14, \ n_0 = 0, \ |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 6 \ 131613141} \text{ }_{[\# = 8]} \quad \Delta = -83, \ n_+ = 1, \ n_- = 14, \ n_0 = 0, \ |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 6 \ 131613141} \text{ }_{[\# = 7]} \quad \Delta = -83, \ n_+ = 1, \ n_- = 14, \ n_0 = 0, \ |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 6 \ 131613141} \text{ }_{[\# = 2]} \quad \Delta = -83, \ n_+ = 1, \ n_- = 14, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 131613141} \overset{3}{\text{ }} \quad \Delta = -83, \ n_+ = 1, \ n_- = 14, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 131613141} \overset{3}{\text{ }}_{[\# = 2]} \quad \Delta = -83, \ n_+ = 1, \ n_- = 14, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 6 \ 131613141} \text{ }_{[\# = 3]} \quad \Delta = -83, \ n_+ = 1, \ n_- = 14, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 6 \ 131613141} \text{ }_{[\# = 4]} \quad \Delta = -83, \ n_+ = 1, \ n_- = 14, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 6 \ 131613141} \text{ }_{[\# = 5]} \quad \Delta = -83, \ n_+ = 1, \ n_- = 14, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 131613141} \overset{3}{\text{ }}_{[\# = 3]} \quad \Delta = -83, \ n_+ = 1, \ n_- = 14, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 131613141} \quad \Delta = -75, \ n_+ = 1, \ n_- = 14, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{lcl}
\begin{array}{c} 131 \\ 1 \ 6 \end{array} \begin{array}{c} \textcolor{red}{2} \\ 131613141 \end{array} & \Delta = -75, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\begin{array}{c} 131 \\ 1 \ 6 \end{array} \begin{array}{c} \textcolor{red}{2} \\ 131613141 \end{array} & \Delta = -75, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\begin{array}{c} 131 \\ 1 \ 6 \end{array} \begin{array}{c} \textcolor{red}{2} \\ 131613141 \end{array} & \Delta = -75, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\begin{array}{c} 131\textcolor{red}{2} \\ 1 \ 6 \end{array} \begin{array}{c} 131613141 \end{array} & \Delta = -75, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\begin{array}{c} \textcolor{red}{3}131 \\ 1 \ 6 \end{array} \begin{array}{c} 131613141 \end{array} [\#=9] & \Delta = -83, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix}
\end{array}$$

Base: $\begin{array}{c} 5131 \\ 41316131 \end{array} \begin{array}{c} 6 \ 1 \end{array}$

$$\begin{array}{lcl}
\begin{array}{c} \textcolor{red}{2} \\ 41316131 \end{array} \begin{array}{c} 5131 \\ 6 \ 1 \end{array} & \Delta = -127, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\begin{array}{c} \textcolor{red}{2} \\ 41316131 \end{array} \begin{array}{c} 5131 \\ 6 \ 1 \end{array} & \Delta = -127, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\begin{array}{c} \textcolor{red}{2}5131 \\ 41316131 \end{array} \begin{array}{c} 6 \ 1 \end{array} & \Delta = -127, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\begin{array}{c} \textcolor{red}{2} \\ 41316131 \end{array} \begin{array}{c} 5131 \\ 6 \ 1 \end{array} & \Delta = -127, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\begin{array}{c} 5131 \\ 41316131 \end{array} \begin{array}{c} 6 \ 1 \end{array} & \Delta = -127, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\begin{array}{c} \textcolor{red}{3} \\ 41316131 \end{array} \begin{array}{c} 5131 \\ 6 \ 1 \end{array} & \Delta = -135, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\begin{array}{c} 5131 \\ 41316131 \end{array} \begin{array}{c} 6 \ 1 \end{array} & \Delta = -127, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\begin{array}{c} 5131 \\ 41316131 \end{array} \begin{array}{c} 6 \ \textcolor{red}{12} \end{array} & \Delta = -127, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\begin{array}{c} 5131 \\ 41316131 \end{array} \begin{array}{c} 6 \ \textcolor{red}{13} \end{array} & \Delta = -135, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{array}$$

Base: $\begin{array}{c} 15131 \\ 1 \ 6 \end{array} \begin{array}{c} 1316131 \end{array}$

$$\begin{array}{lcl}
\begin{array}{c} 15131 \\ 1 \ 6 \end{array} \begin{array}{c} 1316131\textcolor{red}{3} \end{array} [\#=4] & \Delta = -107, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\begin{array}{c} \textcolor{red}{3}1 \\ 15131 \end{array} \begin{array}{c} 6 \ 1316131 \end{array} & \Delta = -107, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\begin{array}{c} 15131 \\ 1 \ 6 \end{array} \begin{array}{c} 1316131\textcolor{red}{3} \end{array} [\#=2] & \Delta = -107, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
\begin{array}{c} 15131 \\ 1 \ 6 \end{array} \begin{array}{c} 1316131\textcolor{red}{3} \end{array} [\#=3] & \Delta = -107, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
1^{15131} \bar{6} \ 1316131\bar{3} \ [\#=5] \quad \Delta = -107, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1^{15131} \bar{6} \ 1316131\bar{3} \ [\#=6] \quad \Delta = -107, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1^{15131} \bar{6} \ 1316131\bar{3} \ [\#=7] \quad \Delta = -107, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1^{15131} \bar{6} \ 1316131\bar{3} \ [\#=8] \quad \Delta = -107, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{15131} \bar{6} \ 1316131\bar{3} \ [\#=9] \quad \Delta = -107, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1^{15131} \bar{6} \ 1316131\bar{2} \quad \Delta = -99, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1^{15131} \bar{6} \ 1316131 \quad \Delta = -99, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1^{15131} \bar{6} \ 1316131 \quad \Delta = -99, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
15131\bar{6}^{21}1316131 \quad \Delta = -99, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1^{15131} \bar{6} \ 1316131\bar{2} \quad \Delta = -99, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1^{15131} \bar{6} \ 1316131\bar{2} \quad \Delta = -99, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

Base: 6131613161232

$$\begin{aligned}
6131\bar{6}^{31}13161232 \quad \Delta = -136, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
6131613161232 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
6131613161232 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
6131613161232 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
6131613161232 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
6131613161232 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
6131\bar{6}^{21}13161232 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1613161316123

$$\begin{array}{ll}
1613161316123 & \Delta = -126, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1613161316123 & \Delta = -134, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1613161316123 & \Delta = -126, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1613161316123 & \Delta = -126, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1613161316123 & \Delta = -126, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1613161316123 & \Delta = -126, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1613161316123 & \Delta = -126, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{array}$$

Base: 2161316131612

$$\begin{array}{ll}
2161316131612 & \Delta = -117, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2161316131612 & \Delta = -117, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2161316131612 & \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2161316131612 & \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2161316131612 & \Delta = -117, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
2161316131612 & \Delta = -117, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{array}$$

Base: 161316131612

$$\begin{array}{ll}
161316131612 & \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
161316131612 & \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
161316131612 & \Delta = -117, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
161316131612 & \Delta = -117, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
161316131612 & \Delta = -117, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{array}$$

$$161316131612 \quad \Delta = -117, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$161316131612 \quad \Delta = -117, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$161316131612 \quad \Delta = -117, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\text{Base: } 31 \overset{122315131}{6} 13$$

$$31 \overset{5122315131}{6} 13 \quad \Delta = -82, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$31 \overset{12122315131}{6} 13 \quad \Delta = -278, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -5 & -3 \\ 2 & -3 & 8 \end{pmatrix}$$

$$31 \overset{7122315131}{6} 13 \quad \Delta = -135, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$31 \overset{122315131}{6} 13 \quad \Delta = -30, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$31 \overset{2122315131}{6} 13 \quad \Delta = -30, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$31 \overset{8122315131}{6} 13 \quad \Delta = -163, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$31 \overset{3122315131}{6} 13 \text{ [#=2]} \quad \Delta = -38, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$31 \overset{3122315131}{6} 13 \text{ [#=3]} \quad \Delta = -38, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 60 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$31 \overset{3122315131}{6} 13 \text{ [#=4]} \quad \Delta = -38, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 109 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$31 \overset{3122315131}{6} 13 \text{ [#=5]} \quad \Delta = -38, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 172 & 34 \\ 2 & 34 & 8 \end{pmatrix}$$

$$31 \overset{3122315131}{6} 13 \quad \Delta = -38, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$31 \overset{3122315131}{6} 13 \text{ [#=9]} \quad \Delta = -38, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 324 \quad \begin{pmatrix} 0 & 18 & 2 \\ 18 & 564 & 62 \\ 2 & 62 & 8 \end{pmatrix}$$

$$31 \overset{3122315131}{6} 13 \text{ [#=6]} \quad \Delta = -38, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 249 & 41 \\ 2 & 41 & 8 \end{pmatrix}$$

$$31 \overset{3122315131}{6} 13 \text{ [#=8]} \quad \Delta = -38, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 256 \quad \begin{pmatrix} 0 & 16 & 2 \\ 16 & 445 & 55 \\ 2 & 55 & 8 \end{pmatrix}$$

$$31 \overset{3122315131}{6} 13 \text{ [#=7]} \quad \Delta = -38, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 196 \quad \begin{pmatrix} 0 & 14 & 2 \\ 14 & 340 & 48 \\ 2 & 48 & 8 \end{pmatrix}$$

$$31 \overset{6122315131}{6} 13 \quad \Delta = -108, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$31 \overset{4}{1} 2 2 3 1 5 1 3 1 \quad \Delta = -58, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$31 \overset{1}{1} 2 2 3 1 5 1 3 1 \quad \Delta = -30, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$31 \overset{1}{1} 2 2 3 1 5 1 3 1 \quad \Delta = -30, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$31 \overset{2}{1} 2 2 3 1 5 1 3 1 \quad \Delta = -30, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: $31 \overset{31}{6} 1 3 1 6 1 2 3 2 1$

$$31 \overset{31}{6} 1 3 1 6 1 2 3 2 1 \overset{3}{3} \quad [\# = 7] \quad \Delta = -66, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 196 \quad \begin{pmatrix} 0 & 14 & 2 \\ 14 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix}$$

$$31 \overset{31}{6} 1 3 1 6 1 2 3 2 1 \overset{3}{3} \quad [\# = 8] \quad \Delta = -66, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 256 \quad \begin{pmatrix} 0 & 16 & 2 \\ 16 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix}$$

$$31 \overset{31}{6} 1 3 1 6 1 2 3 2 1 \overset{3}{3} \quad [\# = 9] \quad \Delta = -66, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 324 \quad \begin{pmatrix} 0 & 18 & 2 \\ 18 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix}$$

$$31 \overset{31}{6} 1 3 1 6 1 2 3 2 1 \overset{6}{6} \quad \Delta = -136, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$31 \overset{31}{6} 1 3 1 6 1 2 3 2 1 \overset{3}{3} \quad [\# = 6] \quad \Delta = -66, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$31 \overset{31}{6} 1 3 1 6 1 2 3 2 1 \overset{3}{3} \quad [\# = 2] \quad \Delta = -66, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$31 \overset{31}{6} 1 3 1 6 1 2 3 2 1 \overset{3}{3} \quad [\# = 4] \quad \Delta = -66, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$31 \overset{31}{6} 1 3 1 6 1 2 3 2 1 \overset{3}{3} \quad [\# = 3] \quad \Delta = -66, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$31 \overset{31}{6} 1 3 1 6 1 2 3 2 1 \overset{3}{3} \quad \Delta = -66, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$31 \overset{31}{6} 1 3 1 6 1 2 3 2 1 \overset{8}{8} \quad \Delta = -191, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 8 \end{pmatrix}$$

$$31 \overset{31}{6} 1 3 1 6 1 2 3 2 1 \overset{7}{7} \quad \Delta = -163, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$31 \overset{31}{6} 1 3 1 6 1 2 3 2 1 \overset{4}{4} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$31 \overset{31}{6} 1 3 1 6 1 2 3 2 1 \overset{2}{2} \quad \Delta = -58, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$31 \overset{31}{6} 1 3 1 6 1 2 3 2 1 \overset{2}{2} \quad \Delta = -58, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\overset{2}{3} 3 1 \overset{31}{6} 1 3 1 6 1 2 3 2 1 \quad \Delta = -58, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\begin{aligned}
^{31} 316131612321^2 \quad \Delta = -58, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
^{31} 316131612321^5 \quad \Delta = -110, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
^{312} 316131612321 \quad \Delta = -58, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
^{31} 316131612321^3_{[\# = 5]} \quad \Delta = -66, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix}
\end{aligned}$$

Base: $^{31} 316131613141$

$$\begin{aligned}
^{31} 316131613141^4 \quad \Delta = -111, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
^{31} 316131613141^3 \quad \Delta = -91, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 30 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
^{31} 316131613141^3 \quad \Delta = -91, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
^{31} 316131613141^3_{[\# = 2]} \quad \Delta = -91, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
^{31} 316131613141^3_{[\# = 3]} \quad \Delta = -91, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
^{231} 316131613141 \quad \Delta = -83, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
^{31} 316131613141^2 \quad \Delta = -83, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
^{31} 316131613141^2 \quad \Delta = -83, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
^{31} 316131613141^2 \quad \Delta = -83, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
^{31} 316131613141^2 \quad \Delta = -83, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
^{312} 316131613141 \quad \Delta = -83, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

Base: $^{31} 16131613161^1$

$$\begin{aligned}
^{31} 16131613161^1 \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
^{31} 16131613161^1 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
^{31} 16131613161^1 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

$$1\overset{31}{6}1\overset{21}{3}1613161 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$31\overset{21}{6}1\overset{1}{3}1613161 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1\overset{31}{6}131\overset{21}{6}13161 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1\overset{31}{6}1\overset{2}{3}1\overset{1}{6}13161 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1\overset{31}{6}131\overset{31}{6}13161 \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: $1\overset{1}{6}131\overset{1}{6}1\overset{1}{3}161$

$$1\overset{1}{6}1\overset{2}{3}1\overset{1}{6}1\overset{1}{3}161 \quad \Delta = -117, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1\overset{1}{6}131\overset{21}{6}1\overset{1}{3}161 \quad \Delta = -117, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1\overset{1}{6}131\overset{31}{6}1\overset{1}{3}161 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1\overset{1}{6}131\overset{21}{6}1\overset{1}{3}161 \quad \Delta = -117, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: 71223151321613

$$71223151321\overset{2}{6}13 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$71223151321\overset{2}{6}13 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$71223151321\overset{2}{6}13 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$712231\overset{2}{5}1321613 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$71223151321\overset{2}{6}13 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: 71223151316123

$$7122315131\overset{2}{6}123 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$7122315131\overset{2}{6}123 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$712231\overset{2}{5}1316123 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$712231\overset{2}{5}1316123 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$71223151316123 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: 712231513161³¹

$$712231513161 \quad \Delta = -127, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$7122315131612 \quad \Delta = -127, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$712231513161 \quad \Delta = -127, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$712231513161 \quad \Delta = -127, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$712231513161 \quad \Delta = -127, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$712231513161 \quad \Delta = -127, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: 71231513161232

$$71231513161232 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$71231513161232 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$71231513161232 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$71231513161232 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$71231513161232 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

Base: 71232161315132

$$71232161315132 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$71232161315132 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

$$71232161315132 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$71232161315132 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$71232161315132 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix}$$

Base: 61223151322171

$$61223151322171 \quad \Delta = -120, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

³ 61223151322171	$\Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 63 & 15 \\ 2 & 15 & 8 \end{pmatrix}$
61223151322 ² 171	$\Delta = -120, n_+ = 1, n_- = 14, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
61223151322171 ²	$\Delta = -120, n_+ = 1, n_- = 14, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
² 61223151322171	$\Delta = -120, n_+ = 1, n_- = 14, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$
Base: 16123151322171		
16123151322171 ²	$\Delta = -120, n_+ = 1, n_- = 14, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
16123151322171 ²	$\Delta = -120, n_+ = 1, n_- = 14, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$
16123151322171 ²	$\Delta = -120, n_+ = 1, n_- = 14, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
16123151322171 ²	$\Delta = -120, n_+ = 1, n_- = 14, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$
Base: 21613151322171		
³ 21613151322171	$\Delta = -127, n_+ = 1, n_- = 14, n_0 = 0, \det = 4$	$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$
² 21613151322171	$\Delta = -119, n_+ = 1, n_- = 14, n_0 = 0, \det = 4$	$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$
21613151322171 ²	$\Delta = -119, n_+ = 1, n_- = 14, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$
21613151322171 ²	$\Delta = -119, n_+ = 1, n_- = 14, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
21613151322171 ²	$\Delta = -119, n_+ = 1, n_- = 14, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$
21613151322171 ²	$\Delta = -119, n_+ = 1, n_- = 14, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
Base: ¹ 1613151322171		
¹ 1613151322171 ²	$\Delta = -119, n_+ = 1, n_- = 14, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
¹² 1613151322171	$\Delta = -119, n_+ = 1, n_- = 14, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$
¹ 1613151322171 ²	$\Delta = -119, n_+ = 1, n_- = 14, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$
¹ 1613151322171 ²	$\Delta = -119, n_+ = 1, n_- = 14, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
Base: 31612315132171		

$$\begin{array}{ll}
31612315132171 & \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
31612315132171 & \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
31612315132171 & \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
31612315132171 & \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
31612315132171 & \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\text{Base: } 32161315132171 & \\
32161315132171 & \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
32161315132171 & \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
32161315132171 & \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
32161315132171 & \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
32161315132171 & \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
\text{Base: } 161315132171 & \\
161315132171 & \Delta = -127, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
161315132171 & \Delta = -127, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
161315132171 & \Delta = -135, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
161315132171 & \Delta = -127, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
161315132171 & \Delta = -127, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
161315132171 & \Delta = -127, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
161315132171 & \Delta = -127, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
\text{Base: } 31513161232171 & \\
31513161232171 & \Delta = -129, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
31513161232171 \quad \Delta = -129, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
31513161232171 \quad \Delta = -129, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
31513161232171 \quad \Delta = -129, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
31513161232171 \quad \Delta = -129, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix}
\end{aligned}$$

Base: 21712231513221

$$\begin{aligned}
21712231513221\mathbf{3} \text{ [#=8]} \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
21712231513221\mathbf{3} \text{ [#=6]} \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
21712231513221\mathbf{3} \text{ [#=9]} \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
21712231513221\mathbf{3} \text{ [#=4]} \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
21712231513221\mathbf{3} \text{ [#=5]} \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
21712231513221\mathbf{3} \text{ [#=7]} \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
21712231513221\mathbf{3} \text{ [#=3]} \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21712231513221\mathbf{3} \text{ [#=2]} \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21712231513221 \quad \Delta = -42, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21712231513221 \quad \Delta = -42, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21712231513221 \quad \Delta = -42, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
21712231513221 \quad \Delta = -42, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 61223151321712

$$\begin{aligned}
61223151321712 \quad \Delta = -120, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
61223151321712 \quad \Delta = -120, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{ll}
61223151321712^2 & \Delta = -120, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
61223151321712^2 & \Delta = -120, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
61223151321712^3 & \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 63 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
61223151321712^2 & \Delta = -120, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
\text{Base: } 16123151321712 & \\
16123151321712^2 & \Delta = -120, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
16123151321712^2 & \Delta = -120, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
16123151321712^2 & \Delta = -120, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
16123151321712^2 & \Delta = -120, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
16123151321712^2 & \Delta = -120, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\text{Base: } 21613151321712 & \\
21613151321712^2 & \Delta = -119, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21613151321712^3 & \Delta = -127, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
21613151321712^2 & \Delta = -119, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
21613151321712^2 & \Delta = -119, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
21613151321712^2 & \Delta = -119, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21613151321712^2 & \Delta = -119, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21613151321712^2 & \Delta = -119, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\text{Base: } 1613151321712^1 & \\
1613151321712^1 & \Delta = -119, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1613151321712^1 & \Delta = -119, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{array}$$

$$1613151321712 \quad \Delta = -119, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1613151321712 \quad \Delta = -119, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1613151321712 \quad \Delta = -119, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: 15131612321712

$$15131612321712 \quad \Delta = -121, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$15131612321712 \quad \Delta = -121, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$15131612321712 \quad \Delta = -121, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$15131612321712 \quad \Delta = -121, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$15131612321712 \quad \Delta = -121, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: 22171231513221

$$22171231513221\textcolor{red}{3} \text{ [#=7]} \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$22171231513221\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$22171231513221\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$22171231513221\textcolor{red}{3} \text{ [#=6]} \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$22171231513221\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$22171231513221\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$22171231513221\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$22171231513221\textcolor{red}{3} \text{ [#=8]} \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$22171231513221 \quad \Delta = -42, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$22171231513221 \quad \Delta = -51, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\begin{aligned}
22171231513221 \quad \Delta = -42, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
22171231513221 \quad \Delta = -42, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
22171231513221 \quad \Delta = -42, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 61231513217123

$$\begin{aligned}
61231513217123 \quad \Delta = -129, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
61231513217123 \quad \Delta = -129, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
61231513217123 \quad \Delta = -129, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
61231513217123 \quad \Delta = -129, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
61231513217123 \quad \Delta = -129, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: 16131513217123

$$\begin{aligned}
16131513217123 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
16131513217123 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
16131513217123 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
16131513217123 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
16131513217123 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 51316123217123

$$\begin{aligned}
51316123217123 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
51316123217123 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
51316123217123 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
51316123217123 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
51316123217123 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

Base: 13217123151321

$$\begin{aligned}
13217123151321\mathbf{\bar{3}}_{[\# = 5]} \quad \Delta = -59, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
13217123151321\mathbf{\bar{2}} \quad \Delta = -51, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
13217123151321\mathbf{\bar{2}} \quad \Delta = -51, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
13217123151321\mathbf{\bar{2}} \quad \Delta = -51, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
13217123151321\mathbf{\bar{2}} \quad \Delta = -51, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
13217123151321\mathbf{\bar{3}}_{[\# = 8]} \quad \Delta = -59, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
12315132171231\mathbf{\bar{3}}_{[\# = 5]} \quad \Delta = -59, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
13217123151321\mathbf{\bar{3}}_{[\# = 6]} \quad \Delta = -59, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
13217123151321\mathbf{\bar{3}}_{[\# = 4]} \quad \Delta = -59, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
12315132171231\mathbf{\bar{3}}_{[\# = 4]} \quad \Delta = -59, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
13217123151321\mathbf{\bar{3}}_{[\# = 3]} \quad \Delta = -59, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
12315132171231\mathbf{\bar{3}}_{[\# = 3]} \quad \Delta = -59, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
13217123151321\mathbf{\bar{3}}_{[\# = 2]} \quad \Delta = -59, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
12315132171231\mathbf{\bar{3}}_{[\# = 2]} \quad \Delta = -59, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
13217123151321\mathbf{\bar{3}}_{[\# = 7]} \quad \Delta = -59, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
13217123151321\mathbf{\bar{3}}_{[\# = 9]} \quad \Delta = -59, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
\mathbf{Base:} \quad 13161232171231 \\
13217123216131\mathbf{\bar{3}}_{[\# = 3]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
13161232171231\mathbf{\bar{3}}_{[\# = 3]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
13217123216131\mathbf{\bar{3}}_{[\# = 4]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
13161232171231\bar{3} \text{ }_{[\# = 4]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
13217123216131\bar{3} \text{ }_{[\# = 9]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
13217123216131\bar{3} \text{ }_{[\# = 8]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
13217123216131\bar{3} \text{ }_{[\# = 7]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
13217123216131\bar{3} \text{ }_{[\# = 6]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
13217123216131\bar{3} \text{ }_{[\# = 5]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
13161232171231\bar{3} \text{ }_{[\# = 5]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
13161232171231\bar{3} \text{ }_{[\# = 2]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
13161\bar{2}232171231 \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
13161232171231\bar{2} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
13217123216131\bar{3} \text{ }_{[\# = 2]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
13161\bar{2}232171231 \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
13161232171231\bar{2} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\text{Base: } 12321712315131 \\
12321712315131\bar{2} \quad \Delta = -52, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
12321712315131\bar{3} \text{ }_{[\# = 9]} \quad \Delta = -60, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
13151321712321\bar{3} \text{ }_{[\# = 9]} \quad \Delta = -60, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
12321712315131\bar{3} \text{ }_{[\# = 8]} \quad \Delta = -60, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
13151321712321\bar{3} \text{ }_{[\# = 8]} \quad \Delta = -60, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
12321712315131\bar{3} \text{ }_{[\# = 7]} \quad \Delta = -60, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
13151321712321\bar{3} \text{ }_{[\# = 7]} \quad \Delta = -60, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
13151321712321\bar{3} \text{ }_{[\# = 6]} \quad \Delta = -60, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
13151321712321\bar{3} \text{ }_{[\# = 5]} \quad \Delta = -60, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
12321712315131 \quad \Delta = -52, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
12321712315131 \quad \Delta = -52, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
12321712315131 \quad \Delta = -52, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
13151321712321\bar{3} \text{ }_{[\# = 4]} \quad \Delta = -60, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
12321712315131\bar{3} \text{ }_{[\# = 6]} \quad \Delta = -60, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
12321712315131\bar{3} \text{ }_{[\# = 3]} \quad \Delta = -60, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
13151321712321\bar{3} \text{ }_{[\# = 3]} \quad \Delta = -60, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
12321712315131\bar{3} \text{ }_{[\# = 2]} \quad \Delta = -60, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
13151321712321\bar{3} \text{ }_{[\# = 2]} \quad \Delta = -60, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
12321712315131\bar{3} \text{ }_{[\# = 4]} \quad \Delta = -60, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
12321712315131\bar{3} \text{ }_{[\# = 5]} \quad \Delta = -60, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

Base: 61232171231513

$$\begin{aligned}
61232171231513 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 100 \quad & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
61232171231513 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
61232171231513 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
61232171231513 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 121 \quad & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
61232171231513 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 61315132171232

$$\begin{aligned}
61315132171232^2 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
61315132171232^2 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
61315132171232^2 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
61315132171232^2 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
61315132171232^2 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 16123217123151

$$\begin{aligned}
16123217123151^2 \quad \Delta = -122, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
16123217123151^2 \quad \Delta = -122, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
16123217123151^2 \quad \Delta = -122, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
16123217123151^2 \quad \Delta = -122, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 31612321712315

$$\begin{aligned}
31612321712315^2 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
31612321712315^2 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
31612321712315^2 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
31612321712315^2 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
31612321712315^2 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: 21612321712321

$$\begin{aligned}
21612321712321^3_{[\# = 8]} \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
21612321712321^2 \quad \Delta = -72, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21612321712321^2 \quad \Delta = -72, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21612321712321^2 \quad \Delta = -72, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
21612321712321^2 \quad \Delta = -72, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
216123217123213_{[\# = 9]} \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
216123217123213_{[\# = 6]} \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
216123217123213_{[\# = 4]} \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
216123217123213_{[\# = 7]} \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
21612321712321^3 \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
216123217123213_{[\# = 2]} \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
216123217123213_{[\# = 5]} \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
216123217123213_{[\# = 3]} \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: 16123217123215

$$\begin{aligned}
16123217123215^2 \quad \Delta = -124, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
16123217123215^2 \quad \Delta = -124, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
16123217123215^2 \quad \Delta = -124, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
16123217123215^2 \quad \Delta = -124, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1612321712321^1

$$\begin{aligned}
1612321712321^1 \quad \Delta = -72, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
16123217123213_{[\# = 4]}^1 \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1612321712321^1 \quad \Delta = -72, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1612321712321^1 \quad \Delta = -72, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
16123217123213_{[\# = 5]}^1 \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{1}{1612321712321\textcolor{red}{3}}_{[\#=7]} \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& \overset{1}{1612321712321\textcolor{red}{3}}_{[\#=6]} \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{1}{1612321712321\textcolor{red}{3}}_{[\#=2]} \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{1}{1612321712321\textcolor{red}{3}}_{[\#=3]} \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{1}{1612321712321\textcolor{red}{3}}_{[\#=8]} \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{1}{1612321712321\textcolor{red}{3}}_{[\#=9]} \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}
\end{aligned}$$

Base: 41321712321712

$$\begin{aligned}
& \overset{\textcolor{red}{2}}{41321712321712} \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 41321712321\overset{\textcolor{red}{2}}{712} \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 41321\overset{\textcolor{red}{2}}{712321712} \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& 41321712321\overset{\textcolor{red}{2}}{712} \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& 41321712321\overset{\textcolor{red}{2}}{712} \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& 41321\overset{\textcolor{red}{2}}{712321712} \quad \Delta = -127, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: 22171232171231

$$\begin{aligned}
& 22171232171231\textcolor{red}{3}_{[\#=5]} \quad \Delta = -105, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 22171232171231\overset{\textcolor{red}{3}}{} \quad \Delta = -106, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 22171232171231\textcolor{red}{3}_{[\#=4]} \quad \Delta = -105, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 22171232171231\textcolor{red}{3}_{[\#=2]} \quad \Delta = -105, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
& 22171232171231\overset{\textcolor{red}{2}}{} \quad \Delta = -97, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 22171232171231\overset{\textcolor{red}{2}}{} \quad \Delta = -97, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
22171232171231^2 \quad \Delta = -97, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
22171232171231^2 \quad \Delta = -97, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
22171232171231^3_{[\# = 3]} \quad \Delta = -105, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\text{Base: } 1^{12231513221}_7 1 & \\
1^{312231513221}_7 1_{[\# = 2]} \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{312231513221}_7 1_{[\# = 3]} \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{312231513221}_7 1_{[\# = 9]} \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1^{312231513221}_7 1_{[\# = 7]} \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1^{312231513221}_7 1_{[\# = 8]} \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1^{312231513221}_7 1_{[\# = 6]} \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1^{312231513221}_7 1_{[\# = 5]} \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1^{312231513221}_7 1_{[\# = 4]} \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1^{12231513221}_7 1 \quad \Delta = -42, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1^{12231513221}_7 1 \quad \Delta = -42, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1^{12231513221}_7 1 \quad \Delta = -42, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\text{Base: } 6122315132171^1 & \\
6122315132171^{21} \quad \Delta = -120, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
6122315132171^2 \quad \Delta = -120, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
6122315132171^1 \quad \Delta = -120, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
6122315132171^3 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 63 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

$${}^2_6 122315132171 \quad \Delta = -120, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

Base: ${}^1_{17} 12315132161$

$${}^1_{17} {}^2_{12} 315132161 \quad \Delta = -120, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$${}^1_{17} 12315132161 \quad \Delta = -120, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$${}^{12}_{17} 12315132161 \quad \Delta = -120, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$${}^1_{17} {}^2_{12} 315132161 \quad \Delta = -120, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: ${}^1_{17} 12315131612$

$${}^1_{17} 12315131612 \quad \Delta = -127, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$${}^1_{17} {}^2_{12} 315131612 \quad \Delta = -119, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$${}^1_{17} {}^2_{12} 315131612 \quad \Delta = -119, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$${}^{12}_{17} 315131612 \quad \Delta = -119, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$${}^1_{17} 12315131612 \quad \Delta = -119, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$${}^1_{17} {}^2_{12} 315131612 \quad \Delta = -119, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: ${}^1_{16} 1315132171$

$${}^1_{16} {}^2_{12} 315132171 \quad \Delta = -119, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$${}^{12}_{16} 315132171 \quad \Delta = -119, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$${}^1_{16} {}^2_{12} 315132171 \quad \Delta = -119, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$${}^1_{16} {}^{21}_{12} 315132171 \quad \Delta = -119, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: ${}^1_{17} 12321613151$

$${}^1_{17} {}^2_{12} 321613151 \quad \Delta = -121, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$${}^{12}_{17} 321613151 \quad \Delta = -121, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$\begin{aligned}
1712321613151 \quad \Delta = -121, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1712321613151 \quad \Delta = -121, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 4132171232171

$$\begin{aligned}
4132171232171 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
4132171232171 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
4132171232171 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
4132171232171 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
4132171232171 \quad \Delta = -127, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1^{1223151321}_7 12$

$$\begin{aligned}
1^{1223151321}_7 12 \quad \Delta = -42, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1^{1223151321}_7 12 \quad \Delta = -42, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1^{1223151321}_7 12 \quad \Delta = -42, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1^{1223151321}_7 12 \quad \Delta = -42, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1^{31223151321}_7 12 \text{ [#=3]} \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{31223151321}_7 12 \text{ [#=9]} \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1^{31223151321}_7 12 \text{ [#=7]} \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1^{31223151321}_7 12 \text{ [#=5]} \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1^{31223151321}_7 12 \text{ [#=2]} \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{31223151321}_7 12 \text{ [#=8]} \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1^{31223151321}_7 12 \text{ [#=4]} \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

$$1 \overset{31223151321}{7} 12 \text{ [#}=6] \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

Base: $1 \overset{21}{7} 1232171231$

$$1 \overset{21}{7} 1232171231 \overset{3}{\text{ [#}=2]} \quad \Delta = -105, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$1 \overset{21}{7} 1232171231 \overset{2}{\text{ [#}=5]} \quad \Delta = -97, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{21}{7} 1232171231 \overset{3}{\text{ [#}=5]} \quad \Delta = -105, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{21}{7} 1232171231 \overset{2}{\text{ [#}=3]} \quad \Delta = -97, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{21}{7} 1232171231 \overset{2}{\text{ [#}=4]} \quad \Delta = -97, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{21}{7} 1232171231 \quad \Delta = -97, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{21}{7} 1232171231 \overset{3}{\text{ [#}=3]} \quad \Delta = -105, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1 \overset{21}{7} 1232171231 \overset{3}{\text{ [#}=4]} \quad \Delta = -105, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: $1 \overset{1,1223151321}{7} 1$

$$1 \overset{1,31223151321}{7} 1 \text{ [#}=8] \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix}$$

$$1 \overset{1,1223151321}{7} 1 \quad \Delta = -42, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{1,31223151321}{7} 1 \text{ [#}=9] \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix}$$

$$1 \overset{1,31223151321}{7} 1 \text{ [#}=6] \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$1 \overset{1,31223151321}{7} 1 \text{ [#}=3] \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$1 \overset{1,31223151321}{7} 1 \text{ [#}=7] \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix}$$

$$1 \overset{1,31223151321}{7} 1 \text{ [#}=2] \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1 \overset{1,31223151321}{7} 1 \text{ [#}=4] \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$1 \overset{1,31223151321}{7} 1 \text{ [#}=5] \quad \Delta = -50, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

$$1^{1,1223151321} \quad 1 \quad 7 \quad 1 \quad \Delta = -42, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1^{1,1223151321} \quad 1 \quad 7 \quad 1 \quad \Delta = -42, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 81223141414141

$$81223141414141\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -129, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$81223141414141\textcolor{red}{2} \quad \Delta = -121, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

$$81223141414141\textcolor{red}{2} \quad \Delta = -121, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$81223141414141\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -129, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$81223141414141\textcolor{red}{2} \quad \Delta = -121, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 142 & 24 \\ 2 & 24 & 8 \end{pmatrix}$$

$$81223141414141\textcolor{red}{2} \quad \Delta = -121, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 118 & 22 \\ 2 & 22 & 8 \end{pmatrix}$$

$$81223141414141\textcolor{red}{2} \quad \Delta = -121, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: 81223151315132

$$81223151315132\textcolor{red}{2} \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$81223151315132\textcolor{red}{2} \quad \Delta = -131, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$81223151315132\textcolor{red}{2} \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$81223151315132\textcolor{red}{2} \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$81223151315132\textcolor{red}{2} \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$81223151315132\textcolor{red}{2} \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

Base: 81231414141413

$$81231414141413\textcolor{red}{2} \quad \Delta = -129, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$81231414141413\textcolor{red}{2} \quad \Delta = -129, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$81231414141413\textcolor{red}{3} \quad \Delta = -137, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$81231414141413^2 \quad \Delta = -129, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$81231414141413^2 \quad \Delta = -131, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$81231414141413^2 \quad \Delta = -129, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$81231414141413^2 \quad \Delta = -129, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$81231414141413^2 \quad \Delta = -129, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

Base: 81231513151322

$$81231513151322^2 \quad \Delta = -131, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$81231513151322^2 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 196 \quad \begin{pmatrix} 0 & 14 & 2 \\ 14 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix}$$

$$81231513151322^2 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$81231513151322^2 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$81231513151322^2 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$81231513151322^2 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix}$$

Base: 18122315131513

$$18122315131513^2 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$18122315131513^2 \quad \Delta = -129, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$18122315131513^2 \quad \Delta = -129, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$18122315131513^2 \quad \Delta = -129, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$18122315131513^2 \quad \Delta = -129, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$18122315131513^2 \quad \Delta = -129, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: 14141414132181

$$14141414132181^2 \quad \Delta = -121, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$\begin{aligned}
18123141414141\overset{2}{3} \text{ [#=2]} \quad \Delta = -129, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
141414141321\overset{2}{81} \quad \Delta = -121, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
141414141321\overset{2}{81} \quad \Delta = -123, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
141414141321\overset{2}{81} \quad \Delta = -121, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
141414141321\overset{2}{81} \quad \Delta = -121, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
141414141321\overset{2}{81} \quad \Delta = -121, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
18123141414141\overset{2}{3} \text{ [#=3]} \quad \Delta = -129, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 18123151315132

$$\begin{aligned}
181231513151\overset{2}{32} \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
181231513151\overset{2}{32} \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
181231513151\overset{2}{32} \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
181231513151\overset{2}{32} \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
181231513151\overset{2}{32} \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
181231513151\overset{2}{32} \quad \Delta = -131, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

Base: 21812231513215

$$\begin{aligned}
2181223151321\overset{2}{5} \quad \Delta = -122, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2181223151321\overset{2}{5} \quad \Delta = -122, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
2181223151321\overset{2}{5} \quad \Delta = -122, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
2181223151321\overset{2}{5} \quad \Delta = -122, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2181223151321\overset{2}{5} \quad \Delta = -122, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 21812231513151

$$\begin{aligned}
2181223151\overset{2}{3}151 \quad \Delta = -122, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2181223151\overset{2}{3}151 \quad \Delta = -121, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2181223151\overset{2}{3}151 \quad \Delta = -121, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
2181223151\overset{2}{3}151 \quad \Delta = -121, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2181223151\overset{2}{3}151 \quad \Delta = -121, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
2181223151\overset{2}{3}151 \quad \Delta = -121, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 21812315131513

$$\begin{aligned}
2181231513151\overset{2}{3} \quad \Delta = -129, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2181231513151\overset{2}{3} \quad \Delta = -129, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
2181231513151\overset{2}{3} \quad \Delta = -129, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
2181231513151\overset{2}{3} \quad \Delta = -129, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2181231513151\overset{2}{3} \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
2181231513151\overset{2}{3} \quad \Delta = -129, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
2181231513151\overset{2}{3} \quad \Delta = -129, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{2321}{1 \ 7 \ 12321812}$

$$\begin{aligned}
\overset{2321}{1 \ 7 \ 12321812}\overset{2} \quad \Delta = -127, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\overset{2321}{1 \ 7 \ 12321812} \quad \Delta = -127, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
\overset{2321}{1 \ 7 \ 12321812}\overset{2} \quad \Delta = -127, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\overset{2321}{1 \ 7 \ 12321812}\overset{2} \quad \Delta = -127, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
2321\overset{21}{7 \ 12321812} \quad \Delta = -127, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: 22181223151321

$$\begin{aligned}
22181223151321^3 \quad \Delta = -79, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
22181223151321^3_{[\# = 6]} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
22181223151321^3_{[\# = 4]} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
22181223151321^3_{[\# = 7]} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
22181223151321^3_{[\# = 9]} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
22181223151321^2 \quad \Delta = -70, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
22181223151321^2 \quad \Delta = -70, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
22181223151321^2 \quad \Delta = -70, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
22181223151321^2 \quad \Delta = -70, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
22181223151321^3_{[\# = 8]} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
22181223151321^3_{[\# = 2]} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
22181223151321^3_{[\# = 5]} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
22181223151321^3_{[\# = 3]} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 22181231513215

$$\begin{aligned}
22181231513215^2 \quad \Delta = -122, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
22181231513215^2 \quad \Delta = -122, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
22181231513215^2 \quad \Delta = -122, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
22181231513215^2 \quad \Delta = -122, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
22181231513215^3 \quad \Delta = -131, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
22181231513215^2 \quad \Delta = -122, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: 32181223151315

$$\begin{aligned}
32181223151\overset{2}{3}15 \quad \Delta = -131, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
32181223151\overset{2}{3}15 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
32181223151\overset{2}{3}15 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
32181223151\overset{2}{3}15 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
32181223151\overset{2}{3}15 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
32181223151\overset{2}{3}15 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 22181231513151

$$\begin{aligned}
2218123151\overset{2}{3}151 \quad \Delta = -121, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2218123151\overset{2}{3}151 \quad \Delta = -121, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2218123151\overset{2}{3}151 \quad \Delta = -121, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
2218123151\overset{2}{3}151 \quad \Delta = -121, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2218123151\overset{3}{3}151 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2218123151\overset{2}{3}151 \quad \Delta = -121, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2218123151\overset{2}{3}151 \quad \Delta = -122, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: 41316123218122

$$\begin{aligned}
413161232181\overset{3}{2}2 \quad \Delta = -134, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
413161232181\overset{3}{2}2 \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
413161232181\overset{2}{2}2 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
413161232181\overset{2}{2}2 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
413161232181\overset{2}{2}2 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

$$41316123218122 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$41316123218122 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: 32217123218122

$$32217123218122 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$32217123218122 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$32217123218122 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$32217123218122 \quad \Delta = -134, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$32217123218122 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

Base: $1 \overset{321}{7} 123218122$

$$1 \overset{321}{7} 123218122 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{321}{7} 123218122 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$1 \overset{321}{7} 123218122 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{321}{7} 123218122 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$321 \overset{21}{7} 123218122 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \overset{321}{7} 123218122 \quad \Delta = -134, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

Base: 13218122315131

$$13151322181231\overset{3}{\text{ }}_{[\# = 5]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$13151322181231\overset{3}{\text{ }}_{[\# = 2]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$13218122315131 \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$13218122315131\overset{3}{\text{ }}_{[\# = 4]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$13151322181231\overset{3}{\text{ }}_{[\# = 4]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$13218122315131\overset{3}{\text{ }}_{[\# = 3]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\begin{aligned}
13151322181231\mathbf{3}_{[\# = 3]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
13218122315131\mathbf{2} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
13218122315131\mathbf{3}_{[\# = 7]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
13218122315131\mathbf{2} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
13218122315131\mathbf{3}_{[\# = 5]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
13218122315131\mathbf{3}_{[\# = 9]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
13218122315131\mathbf{3}_{[\# = 6]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
13218122315131\mathbf{3}_{[\# = 2]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
13218122315131\mathbf{3}_{[\# = 8]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
13218122315131\mathbf{2} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: 22218123151321

$$\begin{aligned}
22218123151321\mathbf{3}_{[\# = 7]} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
22218123151321\mathbf{3}_{[\# = 3]} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
22218123151321\mathbf{3}_{[\# = 8]} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
22218123151321\mathbf{3}_{[\# = 2]} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
22218123151321\mathbf{3}_{[\# = 9]} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
22218123151321\mathbf{3}_{[\# = 4]} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
22218123151321\mathbf{3}_{[\# = 5]} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
22218123151321\mathbf{3}_{[\# = 6]} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
22218123151321\mathbf{2} \quad \Delta = -70, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

$$22218123151321 \quad \Delta = -70, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$22218123151321 \quad \Delta = -70, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$22218123151321 \quad \Delta = -70, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 15132218123215

$$15132218123215 \quad \Delta = -124, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$15132218123215 \quad \Delta = -124, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$15132218123215 \quad \Delta = -124, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$15132218123215 \quad \Delta = -124, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

Base: 15132218123151

$$15132218123151 \quad \Delta = -122, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$15132218123151 \quad \Delta = -122, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$15132218123151 \quad \Delta = -122, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$15132218123151 \quad \Delta = -122, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

Base: 12321812231512

$$21513221812321\mathbf{3} \text{ [#=5]} \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$12321812231512 \quad \Delta = -72, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$12321812231512 \quad \Delta = -72, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$12321812231512 \quad \Delta = -72, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$12321812231512 \quad \Delta = -72, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$21513221812321\mathbf{3} \text{ [#=3]} \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$21513221812321\mathbf{3} \text{ [#=7]} \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$\begin{aligned}
21513221812321\bar{3} \text{ }_{[\# = 8]} \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
21513221812321\bar{3} \text{ }_{[\# = 6]} \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
21513221812321\bar{3} \text{ }_{[\# = 4]} \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
21513221812321\bar{3} \text{ }_{[\# = 9]} \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1232181223151\bar{2} \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
21513221812321\bar{3} \text{ }_{[\# = 2]} \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: 51322181231513

$$\begin{aligned}
5132218123151\bar{3} \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
5132218123151\bar{3} \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
5132218123151\bar{3} \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 169 \quad & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
5132218123151\bar{3} \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
5132218123151\bar{3} \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 121 \quad & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}
\end{aligned}$$

Base: 31513221812315

$$\begin{aligned}
31513221812315 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
31513221812315 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 196 \quad & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 68 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
31513221812315 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
31513221812315 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 100 \quad & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
31513221812315 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 13221812315131

$$\begin{aligned}
13151321812231\bar{3} \text{ }_{[\# = 3]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
13151321812231\bar{3} \text{ }_{[\# = 2]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
13221812315131\overset{2}{3} \text{ }_{[\# = 2]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
13151321812231\overset{2}{3} \text{ }_{[\# = 5]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
13221812315131\overset{2}{3} \text{ }_{[\# = 5]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
13221812315131\overset{2}{3} \text{ }_{[\# = 6]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
13221812315131\overset{2}{3} \text{ }_{[\# = 7]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
13221812315131\overset{2}{3} \text{ }_{[\# = 8]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
13221812315131\overset{2}{3} \text{ }_{[\# = 9]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
13151321812231\overset{2}{3} \text{ }_{[\# = 4]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
13221812315131\overset{2}{3} \text{ }_{[\# = 4]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
13221812315131\overset{2}{3} \text{ }_{[\# = 3]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
13221812315131\overset{2}{1} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1322181\overset{2}{2}2315131 \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
13221812315131\overset{2}{1} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
13221812315131\overset{2}{1} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\text{Base: } 32218123151315 & \\
32218123151315\overset{2}{1} \quad \Delta = -131, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
32218123151315\overset{2}{1} \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
32218123151315\overset{2}{1} \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
322181\overset{2}{2}23151315 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
32218123151315\overset{2}{1} \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

$$32218123151315 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 1232181223151

$$1232181223151 \quad \Delta = -72, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1^{312321812231} 5^1 1 \text{ [#=2]} \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1^{312321812231} 5^1 1 \text{ [#=8]} \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$1232181223151 \quad \Delta = -72, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1^{312321812231} 5^1 1 \text{ [#=7]} \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$1232181223151 \quad \Delta = -72, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1^{312321812231} 5^1 1 \text{ [#=9]} \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$1^{312321812231} 5^1 1 \text{ [#=3]} \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1^{312321812231} 5^1 1 \text{ [#=4]} \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1^{312321812231} 5^1 1 \text{ [#=6]} \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$1^{312321812231} 5^1 1 \text{ [#=5]} \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

Base: 41322181232161

$$41322181232161 \quad \Delta = -126, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$41322181232161 \quad \Delta = -126, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$41322181232161 \quad \Delta = -126, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$41322181232161 \quad \Delta = -126, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix}$$

Base: 21612321812231

$$216123218122313 \text{ [#=2]} \quad \Delta = -106, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$216123218122313 \text{ [#=5]} \quad \Delta = -106, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\begin{aligned}
& \overset{3}{21612321812231} \quad \Delta = -106, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 21612321812231\overset{3}{\text{ } [\# = 4]} \quad \Delta = -106, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 21612321812231\overset{3}{\text{ } [\# = 3]} \quad \Delta = -106, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& 21612321812231\overset{2}{\text{ }} \quad \Delta = -98, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 21612321812231\overset{2}{\text{ }} \quad \Delta = -98, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 21612321812231\overset{2}{\text{ }} \quad \Delta = -98, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 21612321812231\overset{2}{\text{ }} \quad \Delta = -98, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \text{Base: } 13161232181222 \\
& 13161232181222\overset{2}{\text{ }} \quad \Delta = -97, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 13161232181222\overset{2}{\text{ }} \quad \Delta = -97, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 22218123216131\overset{3}{\text{ } [\# = 4]} \quad \Delta = -105, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& 22218123216131\overset{3}{\text{ } [\# = 8]} \quad \Delta = -105, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
& 22218123216131\overset{3}{\text{ } [\# = 7]} \quad \Delta = -105, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
& 22218123216131\overset{3}{\text{ } [\# = 6]} \quad \Delta = -105, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& 22218123216131\overset{3}{\text{ } [\# = 5]} \quad \Delta = -105, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& 22218123216131\overset{3}{\text{ } [\# = 9]} \quad \Delta = -105, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
& 22218123216131\overset{3}{\text{ } [\# = 3]} \quad \Delta = -105, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 13161232181222\overset{2}{\text{ }} \quad \Delta = -97, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 22218123216131\overset{3}{\text{ } [\# = 2]} \quad \Delta = -105, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 13161232181222\overset{2}{\text{ }} \quad \Delta = -97, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1612321812231

$$1612321812231 \quad \Delta = -98, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1612321812231 \text{ }^1\text{ }^2\text{ }^3 \text{ }_{[\# = 5]} \quad \Delta = -106, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1612321812231 \text{ }^1\text{ }^2\text{ }^3 \text{ }_{[\# = 4]} \quad \Delta = -106, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1612321812231 \text{ }^1\text{ }^2\text{ }^3 \text{ }_{[\# = 2]} \quad \Delta = -106, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$1612321812231 \text{ }^1\text{ }^2\text{ }^3 \text{ }_{[\# = 3]} \quad \Delta = -106, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1612321812231 \text{ }^1\text{ }^2\text{ }^3 \quad \Delta = -98, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1612321812231 \text{ }^1\text{ }^2\text{ }^3 \quad \Delta = -98, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 22171232181222

$$22171232181222 \text{ }^3 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$22171232181222 \text{ }^2 \quad \Delta = -116, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$22171232181222 \text{ }^2 \quad \Delta = -116, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$22171232181222 \text{ }^2 \quad \Delta = -116, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$22171232181222 \text{ }^2 \quad \Delta = -116, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

Base: 171232181222

$$171232181222 \text{ }^{21} \quad \Delta = -116, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$171232181222 \text{ }^{21} \text{ }^2 \quad \Delta = -116, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$171232181222 \text{ }^{21} \text{ }^2 \quad \Delta = -116, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$171232181222 \text{ }^{21} \text{ }^2 \quad \Delta = -116, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 14141321812321

$$14141321812321 \text{ }^1 \text{ }^2 \text{ }^3 \text{ }_{[\# = 4]} \quad \Delta = -84, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$14141321812321 \text{ }^1 \text{ }^2 \quad \Delta = -76, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\begin{aligned}
14141321812321^2 \quad \Delta = -76, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
14141321812321^2 \quad \Delta = -76, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
14141321812321^2 \quad \Delta = -76, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
141413218123213_{[\# = 9]} \quad \Delta = -84, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
141413218123213_{[\# = 8]} \quad \Delta = -84, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
141413218123213_{[\# = 7]} \quad \Delta = -84, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
141413218123213_{[\# = 6]} \quad \Delta = -84, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
141413218123213_{[\# = 5]} \quad \Delta = -84, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
141413218123213_{[\# = 3]} \quad \Delta = -84, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
123218123141413_{[\# = 2]} \quad \Delta = -84, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
123218123141413_{[\# = 3]} \quad \Delta = -84, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
14141321812321^2 \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
141413218123213_{[\# = 2]} \quad \Delta = -84, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: 41413218123215

$$\begin{aligned}
41413218123215^2 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
41413218123215^2 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
41413218123215^2 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
41413218123215^2 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
41413218123215^2 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
41413218123215^2 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

Base: 14132181232151

$$\begin{aligned}
14132181232151 \quad \Delta = -102, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
14132181232151 \quad \Delta = -100, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
14132181232151 \quad \Delta = -100, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
14132181232151 \quad \Delta = -100, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
14132181232151 \quad \Delta = -100, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
15123218123141\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -108, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
15123218123141\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -108, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: 41413218123151

$$\begin{aligned}
41413218123151 \quad \Delta = -126, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
41413218123151 \quad \Delta = -126, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
41413218123151 \quad \Delta = -126, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
41413218123151 \quad \Delta = -126, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
41413218123151 \quad \Delta = -126, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
41413218123151 \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: 41232181231512

$$\begin{aligned}
41232181231512 \quad \Delta = -108, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
41232181231512 \quad \Delta = -108, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
41232181231512 \quad \Delta = -100, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
41232181231512 \quad \Delta = -100, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
41232181231512 \quad \Delta = -100, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
41232181231512 \quad \Delta = -100, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
41232181231512 \quad \Delta = -100, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 14132181231512

$$\begin{aligned}
14132181231512 \quad \Delta = -100, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
215132181231413 \quad [\# = 3] \quad \Delta = -106, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
215132181231413 \quad [\# = 2] \quad \Delta = -106, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
14132181231512 \quad \Delta = -98, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
14132181231512 \quad \Delta = -98, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
14132181231512 \quad \Delta = -98, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
14132181231512 \quad \Delta = -106, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
14132181231512 \quad \Delta = -98, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
14132181231512 \quad \Delta = -98, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: 12321812315123

$$\begin{aligned}
321513218123213 \quad [\# = 2] \quad \Delta = -89, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
321513218123213 \quad [\# = 9] \quad \Delta = -89, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
321513218123213 \quad [\# = 6] \quad \Delta = -89, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
321513218123213 \quad [\# = 4] \quad \Delta = -89, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
321513218123213 \quad [\# = 3] \quad \Delta = -89, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
321513218123213 \quad [\# = 5] \quad \Delta = -89, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
321513218123213 \quad [\# = 7] \quad \Delta = -89, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
12321812315123 \quad \Delta = -81, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
32151321812321\overset{2}{3} \text{ [#=8]} \quad \Delta = -89, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
12321812315123 \quad \Delta = -81, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1232181\overset{2}{2}315123 \quad \Delta = -81, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1232181\overset{2}{2}315123 \quad \Delta = -81, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\text{Base: } 31321812315131 \\
3132181\overset{2}{2}315131 \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
3132181\overset{2}{2}315131 \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
3132181\overset{2}{2}315131 \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
313218123151\overset{2}{3}1 \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
313218123151\overset{2}{3}1 \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
3132181\overset{2}{2}315131 \quad \Delta = -88, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
31321812315131\overset{2}{3} \text{ [#=8]} \quad \Delta = -94, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
31321812315131 \quad \Delta = -94, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
31321812315131\overset{2}{3} \text{ [#=7]} \quad \Delta = -94, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
31321812315131\overset{2}{3} \text{ [#=9]} \quad \Delta = -94, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
31321812315131 \text{ [#=2]} \quad \Delta = -94, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
31321812315131\overset{2}{3} \text{ [#=2]} \quad \Delta = -94, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
31321812315131\overset{2}{3} \text{ [#=3]} \quad \Delta = -94, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
31321812315131\overset{2}{3} \text{ [#=4]} \quad \Delta = -94, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
31321812315131\overset{2}{3} \text{ [#=5]} \quad \Delta = -94, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

$$31321812315131\overset{3}{\color{red}}_{[\# = 6]} \quad \Delta = -94, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: 41315132181231

$$41315132181231\overset{2}{\color{red}} \quad \Delta = -106, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$41315132181231\overset{2}{\color{red}} \quad \Delta = -106, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$41315132181231\overset{2}{\color{red}} \quad \Delta = -106, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$41315132181231\overset{3}{\color{red}} \quad \Delta = -114, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$41315132181231\overset{3}{\color{red}}_{[\# = 2]} \quad \Delta = -114, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$41315132181231\overset{3}{\color{red}}_{[\# = 3]} \quad \Delta = -114, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$41315132181231\overset{3}{\color{red}}_{[\# = 4]} \quad \Delta = -114, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$41315132181231\overset{3}{\color{red}}_{[\# = 5]} \quad \Delta = -114, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$41315132181231\overset{2}{\color{red}} \quad \Delta = -107, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$41315132181231\overset{2}{\color{red}} \quad \Delta = -106, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$41315132181231\overset{2}{\color{red}} \quad \Delta = -106, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: 41232181231¹₅₁

$$41232181231\overset{3}{\color{red}}\overset{1}{\substack{51}} \quad \Delta = -108, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$41232181231\overset{2}{\color{red}}\overset{1}{\substack{51}} \quad \Delta = -100, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$41232181231\overset{21}{\color{red}}\overset{1}{\substack{51}} \quad \Delta = -100, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$41232181231\overset{2}{\color{red}}\overset{1}{\substack{51}} \quad \Delta = -100, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$41232181231\overset{2}{\color{red}}\overset{1}{\substack{51}} \quad \Delta = -100, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 14132181231¹₅₁

$$14132181231\overset{2}{\color{red}}\overset{1}{\substack{51}} \quad \Delta = -98, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$14132181231\overset{2}{5}1 \quad \Delta = -100, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$14132181231\overset{2}{5}1 \quad \Delta = -98, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$14132181231\overset{21}{5}1 \quad \Delta = -98, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$14132181231\overset{2}{5}1 \quad \Delta = -98, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1\overset{314132181231}{5}1 \text{ [#=2]} \quad \Delta = -106, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1\overset{314132181231}{5}1 \text{ [#=3]} \quad \Delta = -106, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 1232181231¹513

$$1232181231\overset{31}{5}13 \quad \Delta = -88, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$31\overset{31232181231}{5}1 \text{ [#=6]} \quad \Delta = -88, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$31\overset{31232181231}{5}1 \text{ [#=7]} \quad \Delta = -88, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$31\overset{31232181231}{5}1 \text{ [#=8]} \quad \Delta = -88, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$31\overset{31232181231}{5}1 \text{ [#=3]} \quad \Delta = -88, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$31\overset{31232181231}{5}1 \text{ [#=4]} \quad \Delta = -88, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$31\overset{31232181231}{5}1 \text{ [#=5]} \quad \Delta = -88, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$31\overset{31232181231}{5}1 \text{ [#=2]} \quad \Delta = -88, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1232181231\overset{21}{5}13 \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1232181231\overset{2}{5}13 \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1232181231\overset{12}{5}13 \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1232181231\overset{2}{5}13 \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$31\overset{31232181231}{5}1 \text{ [#=9]} \quad \Delta = -88, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$1232181231\overset{2}{1}513 \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 21612321812313

$$21612321812313 \quad \Delta = -114, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$2161232181231\overset{2}{3}3 \quad \Delta = -108, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$2161232181231\overset{3}{3} \quad \Delta = -114, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$2161232181231\overset{3}{3} \text{ [#=2]} \quad \Delta = -114, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$2161232181231\overset{2}{3}3 \quad \Delta = -106, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$21\overset{2}{6}12321812313 \quad \Delta = -106, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$2161232181231\overset{2}{3}3 \quad \Delta = -106, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$2161232181231\overset{2}{3}3 \quad \Delta = -106, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$2161232181231\overset{2}{3}3 \quad \Delta = -106, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 32161232181231

$$32161232181231\overset{2}{3} \quad \Delta = -107, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$321\overset{2}{6}1232181231 \quad \Delta = -107, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$32161232181231\overset{3}{3} \text{ [#=2]} \quad \Delta = -115, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$32161232181231\overset{3}{3} \text{ [#=3]} \quad \Delta = -115, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$32161232181231\overset{2}{3} \quad \Delta = -107, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$32161232181231\overset{3}{3} \text{ [#=4]} \quad \Delta = -115, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$32161232181231\overset{3}{3} \text{ [#=5]} \quad \Delta = -115, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$321\overset{2}{6}1232181231 \quad \Delta = -107, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

Base: ¹1612321812313

$$\begin{aligned}
& \overset{1}{16}1232181231\overset{3}{3} \quad \Delta = -114, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{1}{16}1232181231\overset{3}{3} \text{ [#=2]} \quad \Delta = -114, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{1}{16}1232181231\overset{2}{2} \quad \Delta = -106, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{1}{16}1232181231\overset{2}{2} \quad \Delta = -106, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{1}{16}1232181231\overset{2}{2} \quad \Delta = -106, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{1}{16}1232181231\overset{2}{2} \quad \Delta = -106, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{1}{16}1232181231\overset{2}{2} \quad \Delta = -108, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \text{Base: } \overset{31}{16}1232181231 \\
& \overset{31}{16}1232181231\overset{3}{3} \text{ [#=3]} \quad \Delta = -114, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{31}{31}61232181231 \quad \Delta = -114, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{31}{16}1232181231\overset{3}{3} \text{ [#=5]} \quad \Delta = -114, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{31}{16}1232181231\overset{3}{3} \text{ [#=4]} \quad \Delta = -114, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{31}{16}1232181231\overset{2}{2} \quad \Delta = -106, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{31}{16}1232181231\overset{2}{2} \quad \Delta = -106, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& \overset{31}{16}1232181231\overset{2}{2} \quad \Delta = -106, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{31}{16}1232181231 \quad \Delta = -106, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{21}{31}61232181231 \quad \Delta = -106, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{31}{16}1232181231\overset{3}{3} \text{ [#=2]} \quad \Delta = -114, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
& \text{Base: } 41232181232151 \\
& 412321812321\overset{2}{5}1 \quad \Delta = -102, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
41232181232151 \quad \Delta = -110, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
41232181232151 \quad \Delta = -102, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
41232181232151 \quad \Delta = -102, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
41232181232151 \quad \Delta = -102, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 12321812321513

$$\begin{aligned}
31512321812321\mathbf{3} \text{ } [\# = 9] \quad \Delta = -90, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
31512321812321\mathbf{3} \text{ } [\# = 8] \quad \Delta = -90, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
31512321812321\mathbf{3} \text{ } [\# = 4] \quad \Delta = -90, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
31512321812321\mathbf{3} \text{ } [\# = 7] \quad \Delta = -90, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
31512321812321\mathbf{3} \text{ } [\# = 6] \quad \Delta = -90, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
31512321812321\mathbf{3} \text{ } [\# = 5] \quad \Delta = -90, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
31512321812321\mathbf{3} \text{ } [\# = 3] \quad \Delta = -90, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
31512321812321\mathbf{3} \text{ } [\# = 2] \quad \Delta = -90, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
12321812321513 \quad \Delta = -82, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
12321812321513 \quad \Delta = -82, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
12321812321513 \quad \Delta = -82, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
12321812321513 \quad \Delta = -82, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: $2^{\frac{13151321812}{3}} 1$

$$\begin{aligned}
2^{\frac{13151321812}{3}} 1 \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2^{\frac{313151321812}{3}} 1 \text{ } [\# = 4] \quad \Delta = -88, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{ll}
2^{\overline{313151321812}} 3^{\overline{1}} 1_{[\# = 8]} & \Delta = -88, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
2^{\overline{313151321812}} 3^{\overline{1}} 1_{[\# = 7]} & \Delta = -88, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
2^{\overline{313151321812}} 3^{\overline{1}} 1_{[\# = 6]} & \Delta = -88, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
2^{\overline{313151321812}} 3^{\overline{1}} 1_{[\# = 5]} & \Delta = -88, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
2^{\overline{313151321812}} 3^{\overline{1}} 1_{[\# = 3]} & \Delta = -88, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2^{\overline{13151321812}} 3^{\overline{13}} 1_{[\# = 3]} & \Delta = -88, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 60 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
2^{\overline{313151321812}} 3^{\overline{1}} 1_{[\# = 2]} & \Delta = -88, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2^{\overline{13151321812}} 3^{\overline{13}} 1_{[\# = 2]} & \Delta = -88, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
2^{\overline{313151321812}} 3^{\overline{1}} 1_{[\# = 9]} & \Delta = -88, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
2^{\overline{13151321812}} 3^{\overline{1}} 1 & \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2^{\overline{13151321812}} 3^{\overline{12}} 1 & \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2^{\overline{13151321812}} 3^{\overline{1}} 1 & \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2^{\overline{13151321812}} 3^{\overline{1}} 1 & \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2^{\overline{13151321812}} 3^{\overline{14}} 1 & \Delta = -108, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\text{Base: } 2161232181232^{\overline{1}} & \\
2161232181232^{\overline{21}} & \Delta = -100, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2161232181232^{\overline{3}} 1 & \Delta = -108, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2161232181232^{\overline{2}} 1 & \Delta = -100, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2161232181232^{\overline{41}} & \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
2161232181232^{\overline{2}} 1 & \Delta = -100, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
2161232181232^{\overset{2}{}}{}^{\underset{1}{}} \quad \Delta = -100, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2161232181232^{\overset{2}{}}{}^{\underset{1}{}} \quad \Delta = -100, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
2161232181232^{\overset{31}{}}{}^{\underset{2}{}}_{[\# = 2]} \quad \Delta = -108, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2161232181232^{\overset{31}{}}{}^{\underset{2}{}}_{[\# = 3]} \quad \Delta = -108, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: $161232181232^{\overset{1}{}}{}^{\underset{1}{}}$

$$\begin{aligned}
161232181232^{\overset{1}{}}{}^{\overset{2}{}}{}^{\underset{1}{}} \quad \Delta = -100, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
161232181232^{\overset{1}{}}{}^{\overset{31}{}}{}^{\underset{2}{}}_{[\# = 2]} \quad \Delta = -108, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
161232181232^{\overset{1}{}}{}^{\overset{31}{}}{}^{\underset{2}{}}_{[\# = 3]} \quad \Delta = -108, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
161232181232^{\overset{12}{}}{}^{\underset{1}{}} \quad \Delta = -100, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
161232181232^{\overset{1}{}}{}^{\overset{41}{}}{}^{\underset{2}{}} \quad \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
161232181232^{\overset{1}{}}{}^{\overset{2}{}}{}^{\underset{1}{}} \quad \Delta = -100, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
161232181232^{\overset{1}{}}{}^{\overset{21}{}}{}^{\underset{2}{}} \quad \Delta = -100, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1^{\overset{51231513221}{}}{}^{\underset{8}{}}{}^{\underset{1}{}}$

$$\begin{aligned}
1^{\overset{51231513221}{}}{}^{\underset{8}{}}{}^{\underset{1}{}} \quad \Delta = -122, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1^{\overset{51231513221}{}}{}^{\underset{8}{}}{}^{\underset{1}{}} \quad \Delta = -122, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1^{\overset{51231513221}{}}{}^{\underset{8}{}}{}^{\underset{1}{}} \quad \Delta = -122, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1^{\overset{51231513221}{}}{}^{\underset{8}{}}{}^{\underset{1}{}} \quad \Delta = -122, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1^{\overset{15131513221}{}}{}^{\underset{8}{}}{}^{\underset{1}{}}$

$$\begin{aligned}
1^{\overset{15131513221}{}}{}^{\underset{8}{}}{}^{\underset{1}{}} \quad \Delta = -122, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1^{\overset{15131513221}{}}{}^{\underset{8}{}}{}^{\underset{1}{}} \quad \Delta = -121, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1^{\overset{15131513221}{}}{}^{\underset{8}{}}{}^{\underset{1}{}} \quad \Delta = -121, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

$$1 \overset{15131513221}{8} 1 \quad \Delta = -121, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{15131513221}{8} 1 \quad \Delta = -121, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{31513151321}{8} 1$$

$$1 \overset{31513151321}{8} 1 \quad \Delta = -129, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{31513151321}{8} 1 \quad \Delta = -129, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1 \overset{31513151321}{8} 1 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{31513151321}{8} 1 \quad \Delta = -129, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1 \overset{31513151321}{8} 1 \quad \Delta = -129, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{31513151321}{8} 1 \quad \Delta = -129, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{2321}{7} \overset{1}{1232181}$$

$$1 \overset{2321}{7} \overset{1}{1232181} \quad \Delta = -127, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix}$$

$$1 \overset{2321}{7} \overset{21}{1232181} \quad \Delta = -127, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1 \overset{2321}{7} \overset{1}{1232181} \quad \Delta = -127, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

$$2321 \overset{21}{7} \overset{1}{1232181} \quad \Delta = -127, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{1231513221}{8} 12$$

$$1 \overset{31231513221}{8} 12 \text{ [#=3]} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{31231513221}{8} 12 \text{ [#=5]} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1 \overset{31231513221}{8} 12 \text{ [#=4]} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1 \overset{31231513221}{8} 12 \text{ [#=7]} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$1 \overset{31231513221}{8} 12 \text{ [#=6]} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$1 \overset{31231513221}{8} 12 \text{ [#=2]} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1^{\overline{31231513221}} 8^{12} \text{ [#}=8] \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$1^{\overline{31231513221}} 8^{12} \text{ [#}=9] \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix}$$

$$1^{\overline{1231513221}} 8^{12} \quad \Delta = -70, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1^{\overline{1231513221}} 8^{12} \quad \Delta = -70, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1^{\overline{1231513221}2} 8^{12} \quad \Delta = -70, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1^{\overline{1231513221}} 8^{12} \quad \Delta = -70, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: $1^{\overline{5123151321}} 8^{12}$

$$1^{\overline{5123151321}} 8^{12} \quad \Delta = -122, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1^{\overline{5123151321}2} 8^{12} \quad \Delta = -122, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1^{\overline{5123151321}} 8^{12} \quad \Delta = -122, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1^{\overline{5123151321}} 8^{12} \quad \Delta = -122, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1^{\overline{5123151321}} 8^{12} \quad \Delta = -122, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: $1^{\overline{1513151321}} 8^{12}$

$$1^{\overline{1513151321}} 8^{12} \quad \Delta = -122, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1^{\overline{1513151321}} 8^{12} \quad \Delta = -121, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1^{\overline{1513151321}} 8^{12} \quad \Delta = -121, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1^{\overline{1513151321}} 8^{12} \quad \Delta = -121, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1^{\overline{1513151321}} 8^{12} \quad \Delta = -121, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1^{\overline{1513151321}2} 8^{12} \quad \Delta = -121, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: $41316123218^{21}1$

$$\overline{3} 41316123218^{21}1 \quad \Delta = -133, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$4131612321\overset{2}{8}1^{21} \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$4131612321\overset{2}{8}1^{21} \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$4131612321\overset{21}{8}1 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$4131612321\overset{2}{8}1^{21} \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$4131612321\overset{221}{8}1 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: $18^{21}1232171223$

$$18^{21}1232171223 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$18^{21}1232171223 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$18^{21}1232171223 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$18^{21}1232171223 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: $1\overset{321}{7}12321\overset{21}{8}1$

$$1\overset{321}{7}12321\overset{21}{8}1 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$321\overset{21}{7}12321\overset{21}{8}1 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1\overset{321}{7}12321\overset{21}{8}1 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$1\overset{321}{7}12321\overset{221}{8}1 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1\overset{321}{7}12321\overset{21}{8}1 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: $1\overset{123151321}{8}122$

$$1\overset{3123151321}{8}122 \text{ [#=6]} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$1\overset{123151321}{8}122 \overset{3}{} \quad \Delta = -79, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1\overset{3123151321}{8}122 \text{ [#=4]} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1\overset{3123151321}{8}122 \text{ [#=3]} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$\begin{aligned}
& 1^{\overline{3123151321}} 8^{221} 122_{[\# = 5]} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& 1^{\overline{3123151321}} 8^{221} 122_{[\# = 2]} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 1^{\overline{123151321}} 8^{221} 122 \quad \Delta = -70, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 1^{\overline{123151321}} 8^{221} 122 \quad \Delta = -70, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& 1^{\overline{123151321}} 8^{221} 122 \quad \Delta = -70, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 1^{\overline{3123151321}} 8^{221} 122_{[\# = 7]} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
& 1^{\overline{3123151321}} 8^{221} 122_{[\# = 8]} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& 1^{\overline{123151321}} 8^{221} 122 \quad \Delta = -70, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 1^{\overline{3123151321}} 8^{221} 122_{[\# = 9]} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
& \text{Base: } 1^{\overline{221}} 8^{221} 123216131 \\
& 1^{\overline{221}} 8^{221} 123216131 \quad \Delta = -97, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 1^{\overline{221}} 8^{221} 123216131 \quad \Delta = -105, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& 1^{\overline{221}} 8^{221} 123216131 \quad \Delta = -105, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
& 1^{\overline{221}} 8^{221} 123216131 \quad \Delta = -105, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 1^{\overline{221}} 8^{221} 123216131 \quad \Delta = -106, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 1^{\overline{221}} 8^{221} 123216131 \quad \Delta = -105, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 1^{\overline{221}} 8^{221} 123216131 \quad \Delta = -105, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 1^{\overline{221}} 8^{221} 123216131 \quad \Delta = -105, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 1^{\overline{221}} 8^{221} 123216131 \quad \Delta = -105, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 1^{\overline{221}} 8^{221} 123216131 \quad \Delta = -97, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{221}{1} \overset{2}{8} 123216131 \quad \Delta = -97, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{221}{1} \overset{221}{8} 123216131 \quad \Delta = -97, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{221}{1} \overset{221}{8} 123216131 \overset{3}{3} [\# = 9] \quad \Delta = -105, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{221}{1} \overset{221}{8} 123217122$

$$\begin{aligned}
& \overset{221}{1} \overset{2}{8} 123217122 \quad \Delta = -116, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{221}{1} \overset{221}{8} 123217122 \quad \Delta = -116, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{221}{1} \overset{221}{8} 12321 \overset{2}{7} 122 \quad \Delta = -116, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{221}{1} \overset{221}{8} 123217122 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{221}{1} \overset{221}{8} 12321 \overset{3}{7} 122 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
& \overset{221}{1} \overset{2}{8} 123217122 \quad \Delta = -116, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{21}{1} \overset{221}{7} 12321 \overset{221}{8} 1$

$$\begin{aligned}
& \overset{21}{1} \overset{221}{7} 12321 \overset{221}{8} 1 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
& \overset{21}{1} \overset{2}{7} 12321 \overset{221}{8} 1 \quad \Delta = -116, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{21}{1} \overset{221}{7} 12321 \overset{221}{8} 1 \quad \Delta = -116, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{21}{1} \overset{221}{7} 12321 \overset{221}{8} 1 \quad \Delta = -116, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{21}{1} \overset{2}{7} 12321 \overset{221}{8} 1 \quad \Delta = -116, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{3221}{1} \overset{3221}{8} 12321712$

$$\begin{aligned}
& \overset{3221}{1} \overset{3221}{8} 12321 \overset{2}{7} 12 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{3221}{1} \overset{21}{8} 12321712 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{3221}{1} \overset{2}{8} 12321712 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& \overset{3221}{1} \overset{3221}{8} 12321712 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}
\end{aligned}$$

$$1 \overset{3221}{8} \overset{2}{12321712} \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 171232181223

$$171232181223 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$171232181223 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$171232181223 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$171232181223 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: $1 \overset{513151321}{8} 123$

$$1 \overset{513151321}{8} 123 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \overset{513151321}{8} 123 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{513151321}{8} \overset{2}{123} \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$321 \overset{513151321}{8} \overset{2}{12} \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{513151321}{8} 123 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{513151321}{8} 123 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1 \overset{513151321}{8} 123 \quad \Delta = -131, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: $1 \overset{151321}{8} 123151$

$$151321 \overset{151321}{8} \overset{2}{12} \quad \Delta = -122, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{151321}{8} \overset{2}{123151} \quad \Delta = -122, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{151321}{8} \overset{2}{123151} \quad \Delta = -122, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: $1 \overset{3151321}{8} 12315$

$$1 \overset{3151321}{8} 12315 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{3151321}{8} 12315 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\begin{aligned}
1 \overset{3151321}{8} 12315 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1 \overset{3151321}{8} 12315 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
1 \overset{3151321}{8} 12315 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
51321 \overset{3151321}{8} 12 \quad \Delta = -130, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{13151321}{8} 1231$

$$\begin{aligned}
1 \overset{313151321}{8} 1231 \text{ [#=3]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1 \overset{13151321}{8} 1231 \overset{3}{3} \text{ [#=3]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{13151321}{8} 1231 \overset{3}{3} \text{ [#=2]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1 \overset{313151321}{8} 1231 \text{ [#=2]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1 \overset{313151321}{8} 1231 \text{ [#=5]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1 \overset{13151321}{8} 1231 \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1 \overset{13151321}{8} 1231 \overset{3}{3} \text{ [#=4]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \overset{313151321}{8} 1231 \text{ [#=4]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \overset{13151321}{8} 1231 \overset{3}{3} \text{ [#=5]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1 \overset{313151321}{8} 1231 \text{ [#=9]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1 \overset{313151321}{8} 1231 \text{ [#=8]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1 \overset{313151321}{8} 1231 \text{ [#=7]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1 \overset{313151321}{8} 1231 \text{ [#=6]} \quad \Delta = -86, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \overset{13151321}{8} 1231 \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1 \overset{13151321}{8} 1231 \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$1 \overset{13151321}{8} 1231 \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: $1 \overset{1321}{8} 12321612$

$$1 \overset{31321}{8} 12321612 \text{ [#=2]} \quad \Delta = -106, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{31321}{8} 12321612 \text{ [#=5]} \quad \Delta = -106, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$1 \overset{31321}{8} 12321612 \text{ [#=4]} \quad \Delta = -106, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1 \overset{31321}{8} 12321612 \text{ [#=3]} \quad \Delta = -106, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1 \overset{1321}{8} 12321612 \overset{3}{} \quad \Delta = -106, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$1 \overset{1321}{8} 12321612 \quad \Delta = -98, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1 \overset{1321}{8} 12321612 \overset{2}{} \quad \Delta = -98, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1 \overset{1321}{8} 12321612 \overset{2}{} \quad \Delta = -98, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{1321}{8} 12321612 \overset{2}{} \quad \Delta = -98, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: $1 \overset{1}{6} 1232181231$

$$1 \overset{1}{6} 1232181231 \overset{1}{3} \text{ [#=4]} \quad \Delta = -106, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1 \overset{1}{6} 1232181231 \overset{1}{3} \text{ [#=5]} \quad \Delta = -106, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1 \overset{1}{6} 1232181231 \overset{1}{3} \text{ [#=3]} \quad \Delta = -106, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{1}{6} 1232181231 \overset{1}{3} \text{ [#=2]} \quad \Delta = -106, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \overset{1}{6} 1232181231 \overset{1}{2} \quad \Delta = -98, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{1}{6} 1232181231 \overset{2}{1} \quad \Delta = -98, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{1}{6} 1232181231 \overset{2}{1} \quad \Delta = -98, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: $1 \overset{512321}{8} 123215$

$$1 \overset{512321}{8} 123215 \overset{2}{} \quad \Delta = -126, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$\begin{aligned}
& 512321 \overset{512321}{8} \textcolor{red}{12} \quad \Delta = -126, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 1 \overset{512321}{8} \textcolor{red}{2} 123215 \quad \Delta = -126, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{1512321}{8} 12321$

$$\begin{aligned}
& 1 \overset{1512321}{8} 12321 \quad \Delta = -74, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 1 \overset{1512321}{8} 12321 \quad \Delta = -74, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 1 \overset{1512321}{8} \textcolor{red}{2} 12321 \quad \Delta = -74, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 1 \overset{1512321}{8} 12321 \textcolor{red}{3} [\# = 8] \quad \Delta = -82, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
& 1 \overset{1512321}{8} 12321 \textcolor{red}{3} [\# = 7] \quad \Delta = -82, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
& 1 \overset{1512321}{8} 12321 \textcolor{red}{3} [\# = 3] \quad \Delta = -82, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 1 \overset{1512321}{8} 12321 \textcolor{red}{3} [\# = 4] \quad \Delta = -82, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& 1 \overset{1512321}{8} 12321 \textcolor{red}{3} [\# = 5] \quad \Delta = -82, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& 1 \overset{1512321}{8} 12321 \textcolor{red}{3} [\# = 9] \quad \Delta = -82, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
& 1 \overset{1512321}{8} 12321 \textcolor{red}{3} [\# = 6] \quad \Delta = -82, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& 1 \overset{1512321}{8} 12321 \textcolor{red}{3} [\# = 2] \quad \Delta = -82, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{512321}{8} 123151$

$$\begin{aligned}
& 151321 \overset{512321}{8} \textcolor{red}{12} \quad \Delta = -124, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 1 \overset{512321}{8} \textcolor{red}{2} 123151 \quad \Delta = -124, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 1 \overset{512321}{8} 123151 \quad \Delta = -124, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& 1 \overset{512321}{8} \textcolor{red}{2} 123151 \quad \Delta = -124, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 1 \overset{512321}{8} 123151 \quad \Delta = -124, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{2151321}{8} 12321$

$$\begin{array}{l}
1 \quad 8 \quad 12321 \textcolor{red}{3} \text{ [#=6]} \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1 \quad 8 \quad 12321 \textcolor{red}{3} \text{ [#=4]} \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1 \quad 8 \quad 12321 \textcolor{red}{3} \text{ [#=7]} \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1 \quad 8 \quad 12321 \textcolor{red}{2} \quad \Delta = -72, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \quad 8 \quad 12321 \quad \Delta = -72, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1 \quad 8 \quad 12321 \textcolor{red}{3} \text{ [#=5]} \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1 \quad 8 \quad 12321 \quad \Delta = -72, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1 \quad 8 \quad 12321 \quad \Delta = -72, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \quad 8 \quad 12321 \textcolor{red}{3} \text{ [#=8]} \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1 \quad 8 \quad 12321 \textcolor{red}{3} \text{ [#=2]} \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \quad 8 \quad 12321 \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
1 \quad 8 \quad 12321 \textcolor{red}{3} \text{ [#=3]} \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \quad 8 \quad 12321 \textcolor{red}{3} \text{ [#=9]} \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
\textbf{Base: } 1 \quad 8 \quad 12321 \\
1 \quad 8 \quad 12321 \quad \Delta = -72, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1 \quad 8 \quad 12321 \textcolor{red}{2} \quad \Delta = -72, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \quad 8 \quad 12321 \quad \Delta = -72, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \quad 8 \quad 12321 \textcolor{red}{3} \text{ [#=2]} \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \quad 8 \quad 12321 \textcolor{red}{3} \text{ [#=3]} \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \quad 8 \quad 12321 \textcolor{red}{3} \text{ [#=4]} \quad \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{lcl}
1^{151321} \begin{array}{c} 8 \\ 12321 \end{array} \begin{array}{c} \textcolor{red}{3} \\ \text{[#}=5 \end{array} & \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1^{151321} \begin{array}{c} 8 \\ 12321 \end{array} \begin{array}{c} \textcolor{red}{3} \\ \text{[#}=9 \end{array} & \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1^{151321} \begin{array}{c} 8 \\ 12321 \end{array} \begin{array}{c} \textcolor{red}{3} \\ \text{[#}=8 \end{array} & \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1^{151321} \begin{array}{c} 8 \\ 12321 \end{array} \begin{array}{c} \textcolor{red}{3} \\ \text{[#}=7 \end{array} & \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1^{151321} \begin{array}{c} 8 \\ 12321 \end{array} \begin{array}{c} \textcolor{red}{3} \\ \text{[#}=6 \end{array} & \Delta = -80, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}
\end{array}$$

Base: 4132181232161

$$\begin{array}{lcl}
41321 \begin{array}{c} \textcolor{red}{21} \\ 81232161 \end{array} & \Delta = -126, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
41321 \begin{array}{c} \textcolor{red}{2} \\ 181232161 \end{array} & \Delta = -128, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
41321 \begin{array}{c} \textcolor{red}{21} \\ 81232161 \end{array} & \Delta = -126, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
41321 \begin{array}{c} \textcolor{red}{2} \\ 181232161 \end{array} & \Delta = -126, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
41321 \begin{array}{c} \textcolor{red}{12} \\ 81232161 \end{array} & \Delta = -126, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
41321 \begin{array}{c} \textcolor{red}{1} \\ 181232161 \end{array} & \Delta = -126, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{array}$$

Base: $21^{123151321} \begin{array}{c} 8 \\ 12 \end{array}$

$$\begin{array}{lcl}
21^{123151321} \begin{array}{c} \textcolor{red}{2} \\ 8 \\ 12 \end{array} & \Delta = -70, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21^{123151321} \begin{array}{c} 8 \\ 12 \end{array} & \Delta = -70, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21^{123151321} \begin{array}{c} 8 \\ 12 \end{array} & \Delta = -70, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
21^{123151321} \begin{array}{c} 8 \\ 12 \end{array} & \Delta = -70, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21^{123151321} \begin{array}{c} \textcolor{red}{3} \\ 8 \\ 12 \end{array} \text{[#}=9 & \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
21^{123151321} \begin{array}{c} \textcolor{red}{3} \\ 8 \\ 12 \end{array} \text{[#}=3 & \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21^{123151321} \begin{array}{c} \textcolor{red}{3} \\ 8 \\ 12 \end{array} \text{[#}=8 & \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
21^{3123151321} 8^{12} [\# = 7] \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
21^{3123151321} 8^{12} [\# = 6] \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
21^{3123151321} 8^{12} [\# = 5] \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
21^{3123151321} 8^{12} [\# = 4] \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21^{3123151321} 8^{12} [\# = 2] \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: $21^{21} 8^{123216131}$

$$\begin{aligned}
21^{21} 8^{123216131} 3 [\# = 7] \quad \Delta = -105, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21^{212} 8^{123216131} \quad \Delta = -97, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21^{21} 8^{123216131} 2 \quad \Delta = -97, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21^{21} 8^{123216131} 3 [\# = 4] \quad \Delta = -105, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21^{21} 8^{123216131} 3 [\# = 9] \quad \Delta = -105, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21^{21} 8^{123216131} 3 [\# = 6] \quad \Delta = -105, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
21^{21} 8^{123216131} 3 [\# = 5] \quad \Delta = -105, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21^{21} 8^{123216131} 3 [\# = 2] \quad \Delta = -105, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
21^{21} 8^{123216131} 2 \quad \Delta = -97, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
21^{221} 8^{123216131} \quad \Delta = -97, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21^{21} 8^{123216131} 3 [\# = 8] \quad \Delta = -105, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21^{21} 8^{123216131} 3 [\# = 3] \quad \Delta = -105, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: $21^{21} 8^{123217122}$

$$21^{212} 8^{123217122} \quad \Delta = -116, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\begin{aligned}
{}^{21}_{21}8123217122 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
{}^{221}_{21}8123217122 \quad \Delta = -116, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
{}^{21}_{21}8123217122 \quad \Delta = -116, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
{}^{21}_{21}8123217122 \quad \Delta = -116, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: ${}^{21}_{17}12321812$

$$\begin{aligned}
{}^{21}_{17}12321812 \quad \Delta = -116, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
{}^{212}_{17}12321812 \quad \Delta = -116, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^{21}_{17}12321812 \quad \Delta = -116, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
{}^{21}_{17}12321812 \quad \Delta = -116, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: ${}^{321}_{21}812321712$

$$\begin{aligned}
{}^{321}_{21}812321712 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^{3212}_{21}812321712 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
{}^{2321}_{21}812321712 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 24 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^{321}_{21}812321712 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
{}^{321}_{21}812321712 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
{}^{321}_{321}812321712 \quad \Delta = -134, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: ${}^1_{17}123218123$

$$\begin{aligned}
{}^1_{17}123218123 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^1_{17}123218123 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
{}^1_{17}123218123 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
{}^{12}_{17}123218123 \quad \Delta = -125, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

$$1712321 \overset{1}{8} \overset{321}{123} \quad \Delta = -134, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: $21 \overset{2321}{8} 1232171$

$$21 \overset{2321}{8} 1232171 \quad \Delta = -127, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

$$2321 \overset{321}{8} 1232171 \quad \Delta = -136, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$21 \overset{2321}{8} 1232171 \quad \Delta = -127, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 48 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$21 \overset{2321}{8} 1232171 \quad \Delta = -127, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix}$$

$$21 \overset{2321}{8} 1232171 \quad \Delta = -127, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: $1 \overset{1,1231513221}{8} 1$

$$1 \overset{1,1231513221}{8} 1 \quad \Delta = -70, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{1,31231513221}{8} 1 \text{ [#=5]} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix}$$

$$1 \overset{1,31231513221}{8} 1 \text{ [#=6]} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

$$1 \overset{1,31231513221}{8} 1 \text{ [#=7]} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix}$$

$$1 \overset{1,31231513221}{8} 1 \text{ [#=8]} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix}$$

$$1 \overset{1,31231513221}{8} 1 \text{ [#=9]} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix}$$

$$1 \overset{1,31231513221}{8} 1 \text{ [#=2]} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1 \overset{1,31231513221}{8} 1 \text{ [#=3]} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1 \overset{1,31231513221}{8} 1 \text{ [#=4]} \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$1 \overset{1,1231513221}{8} 1 \quad \Delta = -70, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{1,1231513221}{8} 1 \quad \Delta = -70, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: $1 \overset{1,5123151321}{8} 1$

$$1 \overset{1,5123151321}{8} 1 \quad \Delta = -122, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\begin{array}{l}
1^{1,5123151321} \quad 1 \quad 8 \quad 1 \quad \Delta = -122, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{1,5123151321} \quad 1 \quad 8 \quad 1 \quad \Delta = -122, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1^{1,5123151321} \quad 1 \quad 8 \quad 1 \quad \Delta = -122, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{array}$$

Base: $1^{1,1513151321} \quad 1 \quad 8 \quad 1$

$$\begin{array}{l}
1^{1,1513151321} \quad 1 \quad 8 \quad 1 \quad \Delta = -122, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{1,1513151321} \quad 1 \quad 8 \quad 1 \quad \Delta = -121, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1^{1,1513151321} \quad 1 \quad 8 \quad 1 \quad \Delta = -121, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1^{1,1513151321} \quad 1 \quad 8 \quad 1 \quad \Delta = -121, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{1,1513151321} \quad 1 \quad 8 \quad 1 \quad \Delta = -121, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{array}$$

Base: $1^{21,123151321} \quad 1 \quad 8 \quad 1$

$$\begin{array}{l}
1^{21,\textcolor{red}{3}123151321} \quad 1 \quad 8 \quad 1 \quad [\# = 8] \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
1^{21,\textcolor{red}{3}123151321} \quad 1 \quad 8 \quad 1 \quad [\# = 7] \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
1^{21,\textcolor{red}{3}123151321} \quad 1 \quad 8 \quad 1 \quad [\# = 5] \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
1^{21,\textcolor{red}{3}123151321} \quad 1 \quad 8 \quad 1 \quad [\# = 4] \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1^{21,\textcolor{red}{3}123151321} \quad 1 \quad 8 \quad 1 \quad [\# = 3] \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1^{21,\textcolor{red}{3}123151321} \quad 1 \quad 8 \quad 1 \quad [\# = 2] \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1^{21,\textcolor{red}{3}123151321} \quad 1 \quad 8 \quad 1 \quad [\# = 6] \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
1^{21,\textcolor{red}{3}123151321} \quad 1 \quad 8 \quad 1 \quad [\# = 9] \quad \Delta = -78, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
1^{21,123151321} \quad 1 \quad 8 \quad 1 \quad \Delta = -70, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1^{21,123151321} \quad 1 \quad 8 \quad 1 \quad \Delta = -70, n_+ = 1, n_- = 14, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
& \overset{21,123151321}{1 \quad 8 \quad 1} \quad \Delta = -70, \, n_+ = 1, \, n_- = 14, \, n_0 = 0, \, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{21,123151321}{1 \quad 8 \quad 1} \quad \Delta = -70, \, n_+ = 1, \, n_- = 14, \, n_0 = 0, \, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

$T = 15$ (1810 bases)

Base: 2313221121223132

$$\begin{aligned}
& \overset{3}{2313221121223132} \quad \Delta = -130, \, n_+ = 1, \, n_- = 15, \, n_0 = 0, \, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 32 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{2}{2313221121223132} \quad \Delta = -124, \, n_+ = 1, \, n_- = 15, \, n_0 = 0, \, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{2}{2313221121223132} \quad \Delta = -122, \, n_+ = 1, \, n_- = 15, \, n_0 = 0, \, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{2}{2313221121223132} \quad \Delta = -122, \, n_+ = 1, \, n_- = 15, \, n_0 = 0, \, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

Base: 2231322112122313

$$\begin{aligned}
& \overset{3}{2231322112122313} \text{ [#=2]} \quad \Delta = -129, \, n_+ = 1, \, n_- = 15, \, n_0 = 0, \, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{3}{2231322112122313} \quad \Delta = -129, \, n_+ = 1, \, n_- = 15, \, n_0 = 0, \, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{3}{2231322112122313} \quad \Delta = -129, \, n_+ = 1, \, n_- = 15, \, n_0 = 0, \, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 60 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{2}{2231322112122313} \quad \Delta = -121, \, n_+ = 1, \, n_- = 15, \, n_0 = 0, \, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{2}{2231322112122313} \quad \Delta = -121, \, n_+ = 1, \, n_- = 15, \, n_0 = 0, \, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{2}{2231322112122313} \quad \Delta = -121, \, n_+ = 1, \, n_- = 15, \, n_0 = 0, \, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{2}{2231322112122313} \quad \Delta = -121, \, n_+ = 1, \, n_- = 15, \, n_0 = 0, \, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1413221121222222

$$\begin{aligned}
& 2222221121223141\overset{3}{3} \text{ [#=2]} \quad \Delta = -131, \, n_+ = 1, \, n_- = 15, \, n_0 = 0, \, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 2222221121223141\overset{3}{3} \text{ [#=3]} \quad \Delta = -131, \, n_+ = 1, \, n_- = 15, \, n_0 = 0, \, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{2}{1413221121222222} \quad \Delta = -123, \, n_+ = 1, \, n_- = 15, \, n_0 = 0, \, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{2}{1413221121222222} \quad \Delta = -123, \, n_+ = 1, \, n_- = 15, \, n_0 = 0, \, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

$$1413221121222222 \quad \Delta = -123, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 3141322112122222

$$3141322112122222 \quad \Delta = -131, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$3141322112122222 \quad \Delta = -139, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$3141322112122222 \quad \Delta = -131, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$3141322112122222 \quad \Delta = -131, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$3141322112122222 \quad \Delta = -131, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 2314132211212222

$$2314132211212222 \quad \Delta = -132, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$2314132211212222 \quad \Delta = -132, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$2314132211212222 \quad \Delta = -134, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$2314132211212222 \quad \Delta = -132, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$2314132211212222 \quad \Delta = -132, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 2231413221121222

$$2231413221121222 \quad \Delta = -132, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$2231413221121222 \quad \Delta = -132, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$2231413221121222 \quad \Delta = -132, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 70 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$2231413221121222 \quad \Delta = -132, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: $1 \overset{313221}{12} 1223132$

$$1 \overset{313221}{12} 1223132 \quad \Delta = -121, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1 \overset{313221}{12} 1223132 \quad \Delta = -121, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1 \overset{313221}{12} 1223132 \quad \Delta = -123, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\begin{aligned}
& \overset{313221}{1 \ 12 \ 1223132} \quad \Delta = -121, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{313221}{1 \ 12 \ 1223132} \quad \Delta = -121, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{313221}{1 \ 12 \ 1223132} \text{ [#=2]} \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{313221}{1 \ 12 \ 1223132} \text{ }^{\textcolor{red}{3}} \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 32 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{313221}{1 \ 12 \ 1223132} \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{22313221}{1 \ 12 \ 12231}$

$$\begin{aligned}
& \overset{22313221}{1 \ 12 \ 12231} \quad \Delta = -113, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{22313221}{1 \ 12 \ 12231} \text{ }^{\textcolor{red}{2}} \quad \Delta = -113, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{22313221}{1 \ 12 \ 12231} \textcolor{red}{3} \text{ [#=5]} \quad \Delta = -121, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{22313221}{1 \ 12 \ 12231} \textcolor{red}{3} \text{ [#=4]} \quad \Delta = -121, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{22313221}{1 \ 12 \ 12231} \textcolor{red}{3} \text{ [#=3]} \quad \Delta = -121, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{22313221}{1 \ 12 \ 12231} \textcolor{red}{3} \text{ [#=2]} \quad \Delta = -121, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{22313221}{1 \ 12 \ 12231} \quad \Delta = -113, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{22313221}{1 \ 12 \ 12231} \quad \Delta = -121, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 32 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{222221}{1 \ 12 \ 1223141}$

$$\begin{aligned}
& \overset{222221}{1 \ 12 \ 1223141} \textcolor{red}{3} \text{ [#=3]} \quad \Delta = -131, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& \overset{222221}{1 \ 12 \ 1223141} \textcolor{red}{3} \text{ [#=2]} \quad \Delta = -131, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& \overset{222221}{1 \ 12 \ 1223141} \quad \Delta = -123, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 76 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
& \overset{222221}{1 \ 12 \ 1223141} \text{ }^{\textcolor{red}{2}} \quad \Delta = -123, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& \overset{222221}{1 \ 12 \ 1223141} \text{ }^{\textcolor{red}{2}} \quad \Delta = -123, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{22221}{1 \ 12 \ 12231413}$

$$\begin{array}{ll}
\begin{array}{l} 22221 \\ 1 \ 12 \ 12231413 \end{array} \quad \Delta = -131, \ n_+ = 1, \ n_- = 15, \ n_0 = 0, \ |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\begin{array}{l} 22221 \\ 1 \ 12 \ 12231413 \end{array} \quad \Delta = -131, \ n_+ = 1, \ n_- = 15, \ n_0 = 0, \ |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\begin{array}{l} 22221 \\ 1 \ 12 \ 12231413 \end{array} \quad \Delta = -139, \ n_+ = 1, \ n_- = 15, \ n_0 = 0, \ |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\begin{array}{l} 22221 \\ 1 \ 12 \ 12231413 \end{array} \quad \Delta = -131, \ n_+ = 1, \ n_- = 15, \ n_0 = 0, \ |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\begin{array}{l} 22221 \\ 1 \ 12 \ 12231413 \end{array} \quad \Delta = -131, \ n_+ = 1, \ n_- = 15, \ n_0 = 0, \ |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\text{Base: } \begin{array}{l} 2221 \\ 1 \ 12 \ 122314132 \end{array} & \\
\begin{array}{l} 2221 \\ 1 \ 12 \ 122314132 \end{array} \quad \Delta = -134, \ n_+ = 1, \ n_- = 15, \ n_0 = 0, \ |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\begin{array}{l} 2221 \\ 1 \ 12 \ 122314132 \end{array} \quad \Delta = -132, \ n_+ = 1, \ n_- = 15, \ n_0 = 0, \ |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\begin{array}{l} 2221 \\ 1 \ 12 \ 122314132 \end{array} \quad \Delta = -132, \ n_+ = 1, \ n_- = 15, \ n_0 = 0, \ |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\begin{array}{l} 2221 \\ 1 \ 12 \ 122314132 \end{array} \quad \Delta = -132, \ n_+ = 1, \ n_- = 15, \ n_0 = 0, \ |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\begin{array}{l} 2221 \\ 1 \ 12 \ 122314132 \end{array} \quad \Delta = -132, \ n_+ = 1, \ n_- = 15, \ n_0 = 0, \ |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\text{Base: } \begin{array}{l} 221 \\ 1 \ 12 \ 1223141322 \end{array} & \\
\begin{array}{l} 221 \\ 1 \ 12 \ 1223141322 \end{array} \quad \Delta = -132, \ n_+ = 1, \ n_- = 15, \ n_0 = 0, \ |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\begin{array}{l} 221 \\ 1 \ 12 \ 1223141322 \end{array} \quad \Delta = -132, \ n_+ = 1, \ n_- = 15, \ n_0 = 0, \ |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\begin{array}{l} 221 \\ 1 \ 12 \ 1223141322 \end{array} \quad \Delta = -132, \ n_+ = 1, \ n_- = 15, \ n_0 = 0, \ |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
\begin{array}{l} 221 \\ 1 \ 12 \ 1223141322 \end{array} \quad \Delta = -132, \ n_+ = 1, \ n_- = 15, \ n_0 = 0, \ |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\text{Base: } \begin{array}{l} 21 \\ 21121223141322 \end{array} & \\
\begin{array}{l} 21 \\ 21121223141322 \end{array} \quad \Delta = -132, \ n_+ = 1, \ n_- = 15, \ n_0 = 0, \ |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
\begin{array}{l} 21 \\ 21121223141322 \end{array} \quad \Delta = -132, \ n_+ = 1, \ n_- = 15, \ n_0 = 0, \ |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\begin{array}{l} 21 \\ 21121223141322 \end{array} \quad \Delta = -132, \ n_+ = 1, \ n_- = 15, \ n_0 = 0, \ |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\begin{array}{l} 21 \\ 21121223141322 \end{array} \quad \Delta = -132, \ n_+ = 1, \ n_- = 15, \ n_0 = 0, \ |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{array}$$

Base: $21 \overset{313221}{12} 122313$

$$21 \overset{2313221}{12} 122313 \quad \Delta = -120, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$21 \overset{313221}{12} 122313 \overset{2}{} \quad \Delta = -120, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$21 \overset{313221}{12} 122313 \overset{3}{}_{[\#=2]} \quad \Delta = -128, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$21 \overset{313221}{12} 122313 \overset{3}{} \quad \Delta = -128, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$21 \overset{313221}{12} 122313 \overset{2}{} \quad \Delta = -120, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: $21 \overset{2313221}{12} 12231$

$$21 \overset{2313221}{12} 12231 \overset{3}{}_{[\#=3]} \quad \Delta = -121, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$21 \overset{2313221}{12} 12231 \quad \Delta = -121, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$21 \overset{2313221}{12} 12231 \overset{3}{}_{[\#=4]} \quad \Delta = -121, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$21 \overset{2313221}{12} 12231 \quad \Delta = -113, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$21 \overset{2313221}{12} 12231 \quad \Delta = -113, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$21 \overset{22313221}{12} 12231 \quad \Delta = -113, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$21 \overset{2313221}{12} 12231 \overset{3}{}_{[\#=2]} \quad \Delta = -121, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$21 \overset{2313221}{12} 12231 \overset{2}{} \quad \Delta = -113, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$21 \overset{2313221}{12} 12231 \quad \Delta = -115, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$21 \overset{2313221}{12} 12231 \overset{3}{}_{[\#=5]} \quad \Delta = -121, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

Base: $21 \overset{22221}{12} 1223141$

$$21 \overset{22221}{12} 1223141 \quad \Delta = -123, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$21 \overset{22221}{12} 1223141 \overset{3}{}_{[\#=2]} \quad \Delta = -131, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$21 \overset{22221}{12} 1223141 \overset{3}{}_{[\#=3]} \quad \Delta = -131, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$\begin{aligned}
21 \overset{22221}{12} 1223141 \quad \Delta = -123, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
21 \overset{22221}{12} \overset{2}{1223141} \quad \Delta = -123, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21 \overset{22221}{12} 1223141 \quad \Delta = -123, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: $21 \overset{2221}{12} 12231413$

$$\begin{aligned}
21 \overset{2221}{12} \overset{2}{12231413} \quad \Delta = -131, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21 \overset{2221}{12} 12231413 \quad \Delta = -131, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21 \overset{2221}{12} 12231413 \quad \Delta = -131, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21 \overset{2221}{12} \overset{2}{12231413} \quad \Delta = -131, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21 \overset{2221}{12} \overset{3}{12231413} \quad \Delta = -139, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
21 \overset{2221}{12} 12231413 \quad \Delta = -131, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: $21 \overset{221}{12} 122314132$

$$\begin{aligned}
21 \overset{221}{12} \overset{2}{122314132} \quad \Delta = -134, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
21 \overset{221}{12} 122314132 \quad \Delta = -132, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21 \overset{221}{12} \overset{2}{122314132} \quad \Delta = -132, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21 \overset{221}{12} 122314132 \quad \Delta = -132, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21 \overset{221}{12} \overset{2}{122314132} \quad \Delta = -132, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21 \overset{221}{12} \overset{2}{122314132} \quad \Delta = -132, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: $2221 \overset{13221}{12} 12231$

$$\begin{aligned}
2221 \overset{13221}{12} \overset{2}{12231} \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2221 \overset{13221}{12} 12231 \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2221 \overset{13221}{12} 12231 \overset{3}{[\# = 3]} \quad \Delta = -112, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{lcl}
221 \overset{13221}{12} 12231 \textcolor{red}{3} [\#=2] & \Delta = -112, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
221 \overset{13221}{12} 12231 \textcolor{red}{3} [\#=4] & \Delta = -112, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
221 \overset{13221}{12} 12231 \textcolor{red}{3} [\#=5] & \Delta = -112, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\textbf{Base: } 3221 \overset{313221}{12} 1223 & & \\
3221 \overset{313221}{12} 1223 & \Delta = -121, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
3221 \overset{313221}{12} 1223 & \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 21 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
3221 \overset{313221}{12} 1223 [\#=2] & \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 144 & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 93 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
3221 \overset{\textcolor{red}{2}313221}{12} 1223 & \Delta = -122, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
3221 \overset{\textcolor{red}{2}313221}{12} 1223 & \Delta = -121, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
3221 \overset{313221}{12} 1223 & \Delta = -121, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
\textbf{Base: } 221 \overset{313221}{12} 12231 & & \\
221 \overset{313221}{12} 12231 & \Delta = -112, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
221 \overset{313221}{12} 12231 & \Delta = -120, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
221 \overset{313221}{12} 12231 \textcolor{red}{3} [\#=2] & \Delta = -120, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
221 \overset{313221}{12} 12231 [\#=2] & \Delta = -120, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
221 \overset{313221}{12} 12231 \textcolor{red}{3} [\#=5] & \Delta = -120, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
221 \overset{313221}{12} 12231 \textcolor{red}{3} [\#=4] & \Delta = -120, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
221 \overset{313221}{12} 12231 \textcolor{red}{3} [\#=3] & \Delta = -120, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
221 \overset{\textcolor{red}{2}313221}{12} 12231 & \Delta = -112, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
221 \overset{313221}{12} 12231 & \Delta = -112, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{array}$$

$$221 \overset{313221}{12} \overset{2}{12231} \quad \Delta = -112, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: $413221 \overset{3221}{12} 1222$

$$413221 \overset{3221}{12} 1222 \quad \Delta = -132, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$413221 \overset{3221}{12} \overset{2}{1222} \quad \Delta = -132, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$\overset{2}{413221} \overset{3221}{12} 1222 \quad \Delta = -132, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$413221 \overset{2}{3221} 12 1222 \quad \Delta = -132, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

Base: $221 \overset{2221}{12} 1223141$

$$\overset{2}{221} \overset{2221}{12} 1223141 \quad \Delta = -123, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$221 \overset{2221}{12} \overset{2}{1223141} \quad \Delta = -123, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$221 \overset{2221}{12} \overset{2}{1223141} \quad \Delta = -123, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$221 \overset{2221}{12} 1223141 \quad \Delta = -123, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$221 \overset{2221}{12} 1223141 \overset{3}{\text{[#}=3]} \quad \Delta = -131, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$221 \overset{2221}{12} 1223141 \overset{3}{\text{[#}=2]} \quad \Delta = -131, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

Base: $221 \overset{221}{12} 12231413$

$$221 \overset{221}{12} \overset{2}{12231413} \quad \Delta = -131, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\overset{2}{221} 12 12231413 \quad \Delta = -131, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$221 \overset{221}{12} \overset{2}{12231413} \quad \Delta = -131, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$221 \overset{221}{12} \overset{2}{12231413} \quad \Delta = -131, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$221 \overset{221}{12} \overset{3}{12231413} \quad \Delta = -139, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: $1 \overset{1,313221}{12} 122313$

$$1 \overset{1,313221}{12} \overset{2}{122313} \quad \Delta = -120, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1^{1,313221} \overset{3}{12} 122313 \text{ [#=2]} \quad \Delta = -128, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$1^{1,313221} \overset{3}{12} 122313 \quad \Delta = -128, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1^{1,313221} \overset{2}{12} 122313 \quad \Delta = -120, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\text{Base: } 1^{1,2313221} 12231$$

$$1^{1,2313221} 12231 \quad \Delta = -115, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1^{1,2313221} \overset{2}{12} 12231 \quad \Delta = -113, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1^{1,2313221} 12231 \quad \Delta = -113, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1^{1,2313221} 12231 \quad \Delta = -113, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1^{1,2313221} 12231 \overset{3}{13} \text{ [#=3]} \quad \Delta = -121, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1^{1,2313221} 12231 \overset{3}{13} \text{ [#=5]} \quad \Delta = -121, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1^{1,2313221} 12231 \overset{3}{13} \text{ [#=2]} \quad \Delta = -121, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1^{1,2313221} 12231 \quad \Delta = -121, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 32 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1^{1,2313221} 12231 \overset{3}{13} \text{ [#=4]} \quad \Delta = -121, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$\text{Base: } 1^{21,313221} 12231$$

$$1^{21,313221} 12231 \quad \Delta = -112, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1^{21,313221} \overset{2}{12} 12231 \quad \Delta = -112, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1^{21,313221} 12231 \overset{3}{13} \text{ [#=3]} \quad \Delta = -120, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1^{21,313221} 12231 \text{ [#=2]} \quad \Delta = -120, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$1^{21,313221} 12231 \overset{3}{13} \text{ [#=5]} \quad \Delta = -120, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1^{21,313221} 12231 \quad \Delta = -120, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{21,313221}{12} 12231\textcolor{red}{3} \text{ [#}=2] \quad \Delta = -120, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \overset{21,313221}{12} 12231\textcolor{red}{3} \text{ [#}=4] \quad \Delta = -120, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1 \overset{21,313221}{12} 12231 \quad \Delta = -112, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1 \overset{21,313221}{12} 12231 \quad \Delta = -112, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: $1 \overset{1,3221}{12} 1223$

$$3221 \overset{1,3221}{12} 12231\textcolor{red}{3} \text{ [#}=4] \quad \Delta = -121, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$13221 \overset{\textcolor{red}{2},3221}{12} 1223 \quad \Delta = -113, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$3221 \overset{1,3221}{12} 12231\textcolor{red}{2} \quad \Delta = -113, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$13221 \overset{\textcolor{red}{2},3221}{12} 1223 \quad \Delta = -113, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$13221 \overset{1,3221\textcolor{red}{2}}{12} 1223 \quad \Delta = -113, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$3221 \overset{1,3221}{12} 12231\textcolor{red}{4} \quad \Delta = -141, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$3221 \overset{1,3221}{12} 12231\textcolor{red}{3} \quad \Delta = -121, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$3221 \overset{1,3221}{12} 12231\textcolor{red}{3} \text{ [#}=3] \quad \Delta = -121, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$3221 \overset{1,3221}{12} 12231\textcolor{red}{3} \text{ [#}=2] \quad \Delta = -121, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$3221 \overset{1,3221}{12} 12231\textcolor{red}{3} \text{ [#}=5] \quad \Delta = -121, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$3221 \overset{1,3221}{12} 12231\textcolor{red}{5} \quad \Delta = -165, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

Base: $1 \overset{221,13221}{12} 12231$

$$1 \overset{221,13221}{12} 12231\textcolor{red}{3} \text{ [#}=5] \quad \Delta = -112, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1 \overset{221,13221}{12} 12231\textcolor{red}{3} \text{ [#}=3] \quad \Delta = -112, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{221,13221}{12} 12231\textcolor{red}{3} \text{ [#}=2] \quad \Delta = -112, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1^{221,13221} \quad 12 \quad 12231\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -112, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1^{221,13221} \quad 12 \quad 12231 \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1^{221,13221\textcolor{red}{2}} \quad 12 \quad 12231 \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: $21^{21,13221} \quad 12 \quad 12231$

$$21^{21,13221} \quad 12 \quad 12231\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -112, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$21^{\textcolor{red}{2}21,13221} \quad 12 \quad 12231 \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$21^{21,13221\textcolor{red}{2}} \quad 12 \quad 12231 \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$21^{21,13221} \quad 12 \quad 12231\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -112, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$21^{21,13221} \quad 12 \quad 12231\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -112, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$21^{21,13221} \quad 12 \quad 12231\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -112, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: 122222222222^2

$$2^2222222222221\textcolor{red}{12} \quad \Delta = -86, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 123141414141^2

$$2^{\textcolor{red}{6}123141414141} \quad 3 \quad 2 \quad \Delta = -48, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$2^{\textcolor{red}{7}123141414141} \quad 3 \quad 2 \quad \Delta = -75, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$2^{\textcolor{red}{8}123141414141} \quad 3 \quad 2 \quad \Delta = -103, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$2^{\textcolor{red}{5}123141414141} \quad 3 \quad 2 \quad \Delta = -22, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 214141414141321

$$214141414141321^{\textcolor{red}{4}} \quad \Delta = -15, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 18 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 131414141414131

$$131414141414131\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -2, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 109 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$131414141414131\textcolor{red}{3} \text{ [#=7]} \quad \Delta = -2, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 340 & 48 \\ 2 & 48 & 8 \end{pmatrix}$$

$$\begin{aligned}
131414141414131^{\color{red}3} \quad \Delta = -2, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 19 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
131414141414131^{\color{red}3} \text{ [#=6]} \quad \Delta = -2, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 249 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
131414141414131^{\color{red}3} \text{ [#=9]} \quad \Delta = -2, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 564 & 62 \\ 2 & 62 & 8 \end{pmatrix} \\
131414141414131^{\color{red}3} \text{ [#=5]} \quad \Delta = -2, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 172 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
131414141414131^{\color{red}3} \text{ [#=2]} \quad \Delta = -2, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 25 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
131414141414131^{\color{red}3} \text{ [#=8]} \quad \Delta = -2, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 445 & 55 \\ 2 & 55 & 8 \end{pmatrix} \\
131414141414131^{\color{red}3} \text{ [#=3]} \quad \Delta = -2, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 60 & 20 \\ 2 & 20 & 8 \end{pmatrix}
\end{aligned}$$

Base: $21414141414132^{\color{red}2}$

$$21414141414132^{\color{red}4} \quad \Delta = -17, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

Base: 214141414141412

$$\begin{aligned}
214141414141412^{\color{red}2} \quad \Delta = -6, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 20 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
214141414141412^{\color{red}3} \quad \Delta = -14, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 23 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
214141414141412^{\color{red}3} \text{ [#=3]} \quad \Delta = -14, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 231 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
214141414141412^{\color{red}3} \text{ [#=2]} \quad \Delta = -14, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 101 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
214141414141412^{\color{red}2} \quad \Delta = -6, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 24 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
214141414141412^{\color{red}4} \quad \Delta = -34, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 22 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
214141414141412^{\color{red}2} \quad \Delta = -6, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
214141414141412^{\color{red}2} \quad \Delta = -6, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1414141414132^{\color{red}1}$

$$1414141414132^{\color{red}41} \quad \Delta = -17, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: $14141414141412^{\color{red}1}$

$$1\overset{31}{4}141414141412 \text{ }_{[\# = 3]} \quad \Delta = -14, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 60 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$1\overset{1}{4}141414141412 \overset{3}{12} \text{ }_{[\# = 3]} \quad \Delta = -14, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1\overset{1}{4}141414141412 \overset{2}{12} \quad \Delta = -6, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1\overset{1}{4}141414141412 \overset{2}{12} \quad \Delta = -6, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1\overset{1}{4}141414141412 \overset{3}{12} \text{ }_{[\# = 2]} \quad \Delta = -14, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1\overset{31}{4}141414141412 \text{ }_{[\# = 2]} \quad \Delta = -14, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 25 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$1\overset{12}{4}141414141412 \quad \Delta = -6, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 20 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1\overset{1}{4}141414141412 \overset{2}{12} \quad \Delta = -6, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1\overset{1}{4}141414141412 \overset{4}{12} \quad \Delta = -34, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 8 \end{pmatrix}$$

$$1\overset{1}{4}141414141412 \overset{3}{12} \quad \Delta = -14, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$1\overset{1}{4}141414141412 \overset{2}{12} \quad \Delta = -6, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1\overset{1}{4}141414141412 \overset{2}{12} \quad \Delta = -6, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: $1\overset{1}{4}14141414141\overset{1}{1}$

$$1\overset{31}{4}14141414141\overset{1}{1} \text{ }_{[\# = 3]} \quad \Delta = -14, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 60 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$1\overset{31}{4}14141414141\overset{1}{1} \text{ }_{[\# = 2]} \quad \Delta = -14, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 25 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$1\overset{1}{4}14141414141\overset{1}{1} \overset{2}{12} \quad \Delta = -6, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1\overset{12}{4}14141414141\overset{1}{1} \quad \Delta = -6, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 20 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1\overset{1}{4}14141414141\overset{1}{1} \overset{2}{12} \quad \Delta = -6, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 123216131513221

$$122315131612321\overset{3}{1} \text{ }_{[\# = 2]} \quad \Delta = -4, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$123216131513221\overset{3}{1} \text{ }_{[\# = 4]} \quad \Delta = -4, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$\begin{aligned}
122315131612321\overset{3}{\color{red}}_{[\# = 9]} \quad \Delta = -4, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
123216131513221\overset{3}{\color{red}}_{[\# = 9]} \quad \Delta = -4, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
122315131612321\overset{3}{\color{red}}_{[\# = 4]} \quad \Delta = -4, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
123216131513221\overset{3}{\color{red}}_{[\# = 5]} \quad \Delta = -4, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
123216131513221\overset{3}{\color{red}}_{[\# = 8]} \quad \Delta = -4, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
122315131612321\overset{3}{\color{red}}_{[\# = 5]} \quad \Delta = -4, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
123216131513221\overset{3}{\color{red}}_{[\# = 3]} \quad \Delta = -4, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
123216131513221\overset{3}{\color{red}}_{[\# = 2]} \quad \Delta = -4, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
122315131612321\overset{3}{\color{red}}_{[\# = 6]} \quad \Delta = -4, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
123216131513221\overset{3}{\color{red}}_{[\# = 6]} \quad \Delta = -4, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
122315131612321\overset{3}{\color{red}}_{[\# = 3]} \quad \Delta = -4, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
122315131612321\overset{3}{\color{red}}_{[\# = 8]} \quad \Delta = -4, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
122315131612321\overset{3}{\color{red}}_{[\# = 7]} \quad \Delta = -4, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
123216131513221\overset{3}{\color{red}}_{[\# = 7]} \quad \Delta = -4, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}
\end{aligned}$$

Base: 141316131513221

$$\begin{aligned}
141316131513221\overset{2}{\color{red}} \quad \Delta = -21, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
141316131513221\overset{2}{\color{red}} \quad \Delta = -21, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
141316131513221\overset{3}{\color{red}} \quad \Delta = -29, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
122315131613141\overset{3}{\color{red}}_{[\# = 2]} \quad \Delta = -29, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
141316131513221\overset{3}{\color{red}}_{[\# = 2]} \quad \Delta = -29, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
122315131613141\overset{2}{3} \quad [\# = 3] \quad \Delta = -29, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
141316131513221\overset{2}{3} \quad [\# = 3] \quad \Delta = -29, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
141316131513221\overset{2}{3} \quad [\# = 4] \quad \Delta = -29, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
141316131513221\overset{2}{3} \quad [\# = 5] \quad \Delta = -29, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
141316131513221\overset{2}{3} \quad [\# = 6] \quad \Delta = -29, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
141316131513221\overset{2}{3} \quad [\# = 7] \quad \Delta = -29, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
141316131513221\overset{2}{3} \quad [\# = 8] \quad \Delta = -29, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
141316131513221\overset{2}{3} \quad [\# = 9] \quad \Delta = -29, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1413161\overset{2}{3}1513221 \quad \Delta = -21, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1413161\overset{2}{3}1513221 \quad \Delta = -21, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1413161\overset{2}{3}1513221 \quad \Delta = -21, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 612315131613151

$$\begin{aligned}
6123151\overset{2}{3}1613151 \quad \Delta = -123, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
61231513161\overset{2}{3}151 \quad \Delta = -123, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
6123151\overset{2}{3}1613151 \quad \Delta = -123, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
6123151\overset{2}{3}1613151 \quad \Delta = -123, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
61231513161\overset{2}{3}151 \quad \Delta = -123, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
61231513161\overset{2}{3}151 \quad \Delta = -123, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 123151316131512

$$1231513161\overset{2}{3}1512 \quad \Delta = -45, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\begin{aligned}
123151316131512^2 \quad \Delta = -45, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
123151316131512^2 \quad \Delta = -45, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
215131613151321^3_{\# = 6} \quad \Delta = -53, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
215131613151321^3_{\# = 9} \quad \Delta = -53, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
215131613151321^3_{\# = 8} \quad \Delta = -53, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
215131613151321^3_{\# = 7} \quad \Delta = -53, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
215131613151321^3_{\# = 5} \quad \Delta = -53, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
215131613151321^3_{\# = 2} \quad \Delta = -53, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
215131613151321^3_{\# = 4} \quad \Delta = -53, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
215131613151321^3_{\# = 3} \quad \Delta = -53, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
123151316131512^3 \quad \Delta = -53, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
123151316131512^2 \quad \Delta = -45, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
123151316131512^2 \quad \Delta = -45, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
123151316131512^2 \quad \Delta = -45, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: 131513161315131

$$\begin{aligned}
131513161315131^3_{\# = 2} \quad \Delta = -60, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
131513161315131^3_{\# = 3} \quad \Delta = -60, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
131513161315131^3_{\# = 4} \quad \Delta = -60, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
131513161315131^3_{\# = 5} \quad \Delta = -60, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
131513161315131^3_{\# = 6} \quad \Delta = -60, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
131513161315131^2 \quad \Delta = -52, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1315131613151313^{\#7} \quad \Delta = -60, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1315131613151313^{\#8} \quad \Delta = -60, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1315131613151313^{\#9} \quad \Delta = -60, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
131513161315131^2 \quad \Delta = -52, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
131513161315131^2 \quad \Delta = -52, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\text{Base: } 613151316131513 \\
613151316131513^2 \quad \Delta = -130, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
613151316131513^2 \quad \Delta = -130, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
613151316131513^2 \quad \Delta = -130, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
613151316131513^2 \quad \Delta = -130, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
613151316131513^2 \quad \Delta = -130, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
613151316131513^2 \quad \Delta = -130, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
613151316131513^2 \quad \Delta = -130, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
\text{Base: } 161315131613151 \\
161315131613151^2 \quad \Delta = -122, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
161315131613151^2 \quad \Delta = -122, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
161315131613151^2 \quad \Delta = -122, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
161315131613151^2 \quad \Delta = -122, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
161315131613151^2 \quad \Delta = -122, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

$$16131513161315^2 \quad \Delta = -122, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 316131513161315

$$316131513161315^2 \quad \Delta = -130, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$316131513161315^2 \quad \Delta = -130, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 58 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$316131513161315^2 \quad \Delta = -130, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 63 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$316131513161315^2 \quad \Delta = -130, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$316131513161315^2 \quad \Delta = -130, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$316131513161315^2 \quad \Delta = -130, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$316131513161315^2 \quad \Delta = -130, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

Base: 131613151316131

$$131613151316131^2 \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$131613151316131^2 \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$131613151316131^2 \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$131613151316131^3_{[\# = 6]} \quad \Delta = -86, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$131613151316131^3_{[\# = 5]} \quad \Delta = -86, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$131613151316131^3_{[\# = 9]} \quad \Delta = -86, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$131613151316131^3_{[\# = 7]} \quad \Delta = -86, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$131613151316131^3_{[\# = 8]} \quad \Delta = -86, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$131613151316131^3_{[\# = 4]} \quad \Delta = -86, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$131613151316131^3_{[\# = 2]} \quad \Delta = -86, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$131613151316131\overset{2}{3} \text{ }_{[\# = 3]} \quad \Delta = -86, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: 12315131613151

$$123151\overset{2}{3}16131\overset{1}{5}1 \quad \Delta = -45, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1231513\overset{2}{1}6131\overset{1}{5}1 \quad \Delta = -45, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$123151316131\overset{2}{1}51 \quad \Delta = -45, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$123151316131\overset{2}{1}51 \quad \Delta = -45, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$123151\overset{2}{3}16131\overset{1}{5}1 \quad \Delta = -45, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1\overset{3}{1}23151316131\overset{1}{5}1 \text{ }_{[\# = 8]} \quad \Delta = -53, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix}$$

$$1\overset{3}{1}23151316131\overset{1}{5}1 \text{ }_{[\# = 7]} \quad \Delta = -53, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix}$$

$$1\overset{3}{1}23151316131\overset{1}{5}1 \text{ }_{[\# = 6]} \quad \Delta = -53, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

$$1\overset{3}{1}23151316131\overset{1}{5}1 \text{ }_{[\# = 5]} \quad \Delta = -53, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix}$$

$$1\overset{3}{1}23151316131\overset{1}{5}1 \text{ }_{[\# = 4]} \quad \Delta = -53, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$1\overset{3}{1}23151316131\overset{1}{5}1 \text{ }_{[\# = 2]} \quad \Delta = -53, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1\overset{3}{1}23151316131\overset{1}{5}1 \text{ }_{[\# = 9]} \quad \Delta = -53, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix}$$

$$1\overset{3}{1}23151316131\overset{1}{5}1 \text{ }_{[\# = 3]} \quad \Delta = -53, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

Base: 123216131612321

$$123216131612321\overset{2}{3} \text{ }_{[\# = 9]} \quad \Delta = -32, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix}$$

$$123216131612321\overset{2}{3} \text{ }_{[\# = 8]} \quad \Delta = -32, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix}$$

$$1232161\overset{2}{3}1612321 \quad \Delta = -24, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1232161\overset{2}{3}1612321 \quad \Delta = -24, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$\begin{aligned}
123216131612321\overset{\textcolor{red}{3}}{3}_{[\# = 2]} \quad \Delta = -32, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
123216131612321\overset{\textcolor{red}{3}}{3}_{[\# = 4]} \quad \Delta = -32, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
123216131612321\overset{\textcolor{red}{3}}{3}_{[\# = 5]} \quad \Delta = -32, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
123216131612321\overset{\textcolor{red}{3}}{3}_{[\# = 6]} \quad \Delta = -32, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
123216131612321\overset{\textcolor{red}{3}}{3}_{[\# = 3]} \quad \Delta = -32, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
123216131612321\overset{\textcolor{red}{3}}{3}_{[\# = 7]} \quad \Delta = -32, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix}
\end{aligned}$$

Base: 216131613151315

$$\begin{aligned}
2161\overset{\textcolor{red}{2}}{3}1613151315 \quad \Delta = -122, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
216131613151\overset{\textcolor{red}{2}}{3}15 \quad \Delta = -123, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21613161315131\overset{\textcolor{red}{3}}{5} \quad \Delta = -130, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21613161\overset{\textcolor{red}{2}}{3}151315 \quad \Delta = -122, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
216131613151\overset{\textcolor{red}{2}}{3}15 \quad \Delta = -122, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
216131613151\overset{\textcolor{red}{2}}{3}15 \quad \Delta = -122, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
216131613151\overset{\textcolor{red}{2}}{3}15 \quad \Delta = -122, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
216131613151\overset{\textcolor{red}{2}}{3}15 \quad \Delta = -122, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21613161\overset{\textcolor{red}{2}}{3}151315 \quad \Delta = -122, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: 221613161315131

$$\begin{aligned}
22161\overset{\textcolor{red}{2}}{3}161315131 \quad \Delta = -70, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
2216131613151\overset{\textcolor{red}{2}}{3}1 \quad \Delta = -70, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
221613161315131\overset{\textcolor{red}{3}}{3}_{[\# = 5]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
221613161315131\overset{2}{3}_{[\# = 3]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
221613161315131\overset{2}{3} \quad \Delta = -70, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
221613161315131\overset{2}{3} \quad \Delta = -70, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
221613161315131\overset{2}{3} \quad \Delta = -70, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
221613161315131\overset{2}{3} \quad \Delta = -70, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
221613161315131\overset{2}{3}_{[\# = 6]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
221613161315131\overset{2}{3}_{[\# = 4]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
221613161315131\overset{2}{3}_{[\# = 2]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
221613161315131\overset{2}{3}_{[\# = 7]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
221613161315131\overset{2}{3}_{[\# = 8]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
221613161315131\overset{2}{3}_{[\# = 9]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
221613161315131\overset{3}{3} \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
221613161315131\overset{3}{3} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\text{Base: } 131613161232161 \\
161232161316131\overset{2}{3}_{[\# = 8]} \quad \Delta = -107, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
161232161316131\overset{2}{3}_{[\# = 6]} \quad \Delta = -107, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
161232161316131\overset{2}{3}_{[\# = 9]} \quad \Delta = -107, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
131613161232161\overset{2}{3} \quad \Delta = -99, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
131613161232161\overset{2}{3} \quad \Delta = -99, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
131613161232161\overset{2}{3} \quad \Delta = -99, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
131613161232161 \quad \Delta = -99, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
131613161232161 \quad \Delta = -99, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1612321613161313 \quad [\# = 5] \quad \Delta = -107, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1612321613161313 \quad [\# = 4] \quad \Delta = -107, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1612321613161313 \quad [\# = 3] \quad \Delta = -107, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1612321613161313 \quad [\# = 2] \quad \Delta = -107, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1612321613161313 \quad [\# = 7] \quad \Delta = -107, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}
\end{aligned}$$

Base: 413161316123216

$$\begin{aligned}
413161316123216 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
413161316123216 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
413161316123216 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
413161316123216 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
413161316123216 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
413161316123216 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
413161316123216 \quad \Delta = -135, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: 141316131612321

$$\begin{aligned}
1232161316131413 \quad [\# = 2] \quad \Delta = -57, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
141316131612321 \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
141316131612321 \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
141316131612321 \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
141316131612321^2 \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
141316131612321^3_{[\# = 5]} \quad \Delta = -57, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
141316131612321^3_{[\# = 6]} \quad \Delta = -57, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
141316131612321^3_{[\# = 7]} \quad \Delta = -57, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
141316131612321^3 \quad \Delta = -57, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 21 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
141316131612321^2 \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
141316131612321^3_{[\# = 2]} \quad \Delta = -57, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
123216131613141^3_{[\# = 3]} \quad \Delta = -57, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
141316131612321^3_{[\# = 9]} \quad \Delta = -57, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
141316131612321^3_{[\# = 8]} \quad \Delta = -57, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
141316131612321^3_{[\# = 3]} \quad \Delta = -57, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
141316131612321^3_{[\# = 4]} \quad \Delta = -57, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: 132161316131512

$$\begin{aligned}
132161316131512^2 \quad \Delta = -71, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
132161316131512^2 \quad \Delta = -71, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
132161316131512^3 \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
215131613161231^3_{[\# = 2]} \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
215131613161231^3_{[\# = 3]} \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
215131613161231^3_{[\# = 5]} \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
215131613161231^3_{[\# = 4]} \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
132161316131512^2 \quad \Delta = -71, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
132161316131512^2 \quad \Delta = -71, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
132161316131512^2 \quad \Delta = -71, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
132161316131512^2 \quad \Delta = -71, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: 513216131613151

$$\begin{aligned}
513216131613151^2 \quad \Delta = -123, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
513216131613151^2 \quad \Delta = -123, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
513216131613151^2 \quad \Delta = -123, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
513216131613151^2 \quad \Delta = -123, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
513216131613151^2 \quad \Delta = -123, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
513216131613151^2 \quad \Delta = -123, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 13216131613151¹

$$\begin{aligned}
1^{3132161316131} 5^1 1_{[\# = 3]} \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{3132161316131} 5^1 1_{[\# = 5]} \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1^{3132161316131} 5^1 1_{[\# = 2]} \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
13216131613151^2 1^1 \quad \Delta = -71, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
13216131613151^2 1^1 \quad \Delta = -71, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
13216131613151^2 1^1 \quad \Delta = -71, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
13216131613151^2 1^1 \quad \Delta = -71, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
13216131613151^2 1^1 \quad \Delta = -71, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

$$1^{\overline{3132161316131}}_5 1_{[\# = 4]} \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

Base: 141316131613141

$$141316131613141\overline{3}_{[\# = 2]} \quad \Delta = -82, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$141316131613141^{\overline{2}} \quad \Delta = -74, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$141316131613141^{\overline{2}} \quad \Delta = -74, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$141316131613141^{\overline{2}} \quad \Delta = -74, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$141316131613141^{\overline{3}} \quad \Delta = -82, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 30 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$141316131613141\overline{3}_{[\# = 3]} \quad \Delta = -82, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

Base: 313161316131512

$$313161316131512^{\overline{3}}_{[\# = 2]} \quad \Delta = -86, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 93 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$313161316131512^{\overline{3}}_{[\# = 3]} \quad \Delta = -86, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 213 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

$$313161316131512^{\overline{4}} \quad \Delta = -106, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$313161316131512^{\overline{2}} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$313161316131512^{\overline{2}} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$313161316131512^{\overline{2}} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$313161316131512^{\overline{2}} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$313161316131512^{\overline{2}} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$313161316131512^{\overline{2}} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$313161316131512^{\overline{2}} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$313161316131512^{\overline{3}} \quad \Delta = -86, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 21 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$313161316131512^{\overline{3}} \quad \Delta = -86, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

Base: 31316131613151¹

$$31316131613151^{\overset{4}{1}} \quad \Delta = -106, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$31316131613151^{\overset{3}{1}} \quad \Delta = -86, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 21 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$31316131613151^{\overset{3}{1}}_{[\#=2]} \quad \Delta = -86, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 93 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$31316131613151^{\overset{3}{1}}_{[\#=3]} \quad \Delta = -86, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 213 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

$$31316131613151^{\overset{2}{1}} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$31316131613151^{\overset{2}{1}} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$31316131613151^{\overset{2}{1}} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$31316131613151^{\overset{2}{1}} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$31316131613151^{\overset{2}{1}} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$31316131613151^{\overset{2}{1}} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 221613161316122

$$221613161316122^{\overset{2}{1}} \quad \Delta = -88, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$221613161316122^{\overset{2}{1}} \quad \Delta = -88, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$221613161316122^{\overset{2}{1}} \quad \Delta = -88, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$221613161316122^{\overset{3}{1}} \quad \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 27 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$221613161316122^{\overset{3}{1}} \quad \Delta = -97, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 16131613151315¹

$$16131613151315^{\overset{1}{2}} \quad \Delta = -122, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$16131613151315^{\overset{1}{2}} \quad \Delta = -122, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$16131613151315^{\overset{1}{2}} \quad \Delta = -122, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\begin{aligned}
{}^1_{16131613151315}{}^2 \quad \Delta = -122, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
{}^1_{16131613151315}{}^2 \quad \Delta = -123, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
{}^1_{16131613151315}{}^2 \quad \Delta = -122, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
{}^1_{16131613151315}{}^2 \quad \Delta = -122, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: ${}^{21}_{1613161315131}$

$$\begin{aligned}
{}^{21}_{1613161315131}{}^3_{[\# = 5]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
{}^{21}_{1613161315131}{}^2 \quad \Delta = -70, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
{}^{21}_{1613161315131}{}^2 \quad \Delta = -70, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
{}^{21}_{1613161315131} \quad \Delta = -70, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
{}^{21}_{1613161315131}{}^2 \quad \Delta = -70, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
{}^{21}_{1613161315131}{}^3_{[\# = 9]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
{}^{21}_{1613161315131}{}^3_{[\# = 8]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
{}^{21}_{1613161315131} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
{}^{21}_{1613161315131}{}^3_{[\# = 2]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
{}^{21}_{1613161315131}{}^3_{[\# = 3]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
{}^{21}_{1613161315131}{}^3_{[\# = 4]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
{}^{21}_{1613161315131}{}^3_{[\# = 6]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
{}^{21}_{1613161315131}{}^3_{[\# = 7]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
{}^{21}_{1613161315131}{}^2 \quad \Delta = -70, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
{}^{21}_{1613161315131}{}^2 \quad \Delta = -70, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: $16^{21}13161316122$

$$\begin{aligned}
16^{21}13161316122^2 \quad \Delta = -88, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
16^{21}13161316122 \quad \Delta = -88, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
16^{21}13161316122^2 \quad \Delta = -88, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
16^{21}13161316122^3 \quad \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
16^{21}13161316122 \quad \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
16^{21}13161316122^2 \quad \Delta = -88, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
16^{21}13161316122^3 \quad \Delta = -97, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
16^{21}13161316122^2 \quad \Delta = -88, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
16^{21}13161316122^2 \quad \Delta = -88, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: $16^{21}131613161^{21}$

$$\begin{aligned}
16^{21}131613161^{21} \quad \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
16^{21}131613161^2 \quad \Delta = -88, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
16^{21}131613161^2 \quad \Delta = -88, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
16^{21}131613161^{21} \quad \Delta = -88, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1^{131}61316131512$

$$\begin{aligned}
1^{131}61316131512^2 \quad \Delta = -70, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1^{131}61316131512^2 \quad \Delta = -70, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1^{131}61316131512^2 \quad \Delta = -70, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1^{131}61316131512^2 \quad \Delta = -70, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1^{3131}61316131512 \text{ [#=8]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{ll}
1 \overset{3131}{6} 1316131512 \text{ }_{[\#=9]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1 \overset{3131}{6} 1316131512 \text{ }_{[\#=7]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1 \overset{3131}{6} 1316131512 \text{ }_{[\#=6]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1 \overset{3131}{6} 1316131512 \text{ }_{[\#=5]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1 \overset{3131}{6} 1316131512 \text{ }_{[\#=4]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1 \overset{3131}{6} 1316131512 \text{ }_{[\#=3]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \overset{3131}{6} 1316131512 \text{ }_{[\#=2]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \overset{131}{6} 1316131512 \overset{3}{} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
1 \overset{1312}{6} 1316131512 \quad \Delta = -70, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \overset{131}{6} 1316131512 \quad \Delta = -70, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{131}{6} 131613151 \overset{1}{} & \\
1 \overset{3131}{6} 131613151 \overset{1}{} \text{ }_{[\#=7]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1 \overset{3131}{6} 131613151 \overset{1}{} \text{ }_{[\#=4]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1 \overset{131}{6} 131613151 \overset{21}{} \quad \Delta = -70, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1 \overset{3131}{6} 131613151 \overset{1}{} \text{ }_{[\#=5]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1 \overset{3131}{6} 131613151 \overset{1}{} \text{ }_{[\#=6]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1 \overset{131}{6} 131613151 \overset{2}{6} \overset{1}{} \quad \Delta = -70, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1 \overset{131}{6} 131613151 \overset{1}{} \quad \Delta = -70, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \overset{131}{6} 131613151 \overset{2}{6} \overset{1}{} \quad \Delta = -70, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \overset{3131}{6} 131613151 \overset{1}{} \text{ }_{[\#=8]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}
\end{array}$$

$$1 \overset{3131}{6} 131613151 \overset{1}{[\# = 9]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$1 \overset{3131}{6} 131613151 \overset{1}{[\# = 2]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{3131}{6} 131613151 \overset{1}{[\# = 3]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1 \overset{1312}{6} 131613151 \overset{1}{} \quad \Delta = -70, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: $1 \overset{5131}{6} 131613151$

$$1 \overset{5131}{6} 131613151 \quad \Delta = -122, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{51312}{6} 131613151 \quad \Delta = -122, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1 \overset{5131}{6} 131613151 \overset{2}{} \quad \Delta = -122, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{5131}{6} 131613151 \overset{2}{} \quad \Delta = -122, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$5131 \overset{31}{6} 131613151 \quad \Delta = -130, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{5131}{6} 131613151 \quad \Delta = -122, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$5131 \overset{21}{6} 131613151 \quad \Delta = -122, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \overset{5131}{6} 131613151 \overset{2}{} \quad \Delta = -122, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: $41316131613161 \overset{1}{} \quad \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9$

$$41316131 \overset{31}{6} 13161 \quad \Delta = -132, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$41316131613161 \overset{1}{\overset{3}{} \quad \Delta = -132, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16} \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$41316131 \overset{21}{6} 13161 \quad \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$41316131613161 \overset{1}{\overset{2}{} \quad \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25} \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$41316131613161 \overset{1}{\overset{2}{} \quad \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16} \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$41316131613161 \overset{1}{\overset{2}{} \quad \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49} \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\begin{aligned}
41316131613161 \quad \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
41316131613161 \quad \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
41316131613161 \quad \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 21613161316131

$$\begin{aligned}
21613161316131 \quad \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21613161316131 \quad \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21613161316131 \quad \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21613161316131 \quad \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21613161316131 \quad \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21613161316131 \quad \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21613161316131 \quad \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
21613161316131\textcolor{red}{3} \text{ [#=6]} \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21613161316131\textcolor{red}{3} \text{ [#=8]} \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
21613161316131\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
21613161316131\textcolor{red}{3} \text{ [#=7]} \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
21613161316131 \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
21613161316131\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21613161316131\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
21613161316131\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
21613161316131\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

$$21613161316131 \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: 1613161316131

$$1613161316131\textcolor{red}{3} \text{ [#=7]} \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$1613161316131\textcolor{red}{2} \quad \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1613161316131\textcolor{red}{12} \quad \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1613161316131\textcolor{red}{12} \quad \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1613161316131\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1613161316131\textcolor{red}{3} \text{ [#=6]} \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1613161316131\textcolor{red}{31} \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1613161316131\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1613161316131\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1613161316131\textcolor{red}{2} \quad \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1613161316131\textcolor{red}{21} \quad \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1613161316131\textcolor{red}{21} \quad \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1613161316131\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$1613161316131\textcolor{red}{3} \text{ [#=8]} \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1613161316131\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: 3161316131512

$$3161316131512\textcolor{red}{231} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$3161316131512\textcolor{red}{3} \quad \Delta = -86, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 15 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$\begin{aligned}
31 \overline{6} 1316131512\overline{3} \quad \Delta = -87, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
31 \overline{6} 1316131512 \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
31 \overline{6} 1316131512 \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
31 \overline{6} 1316131512 \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
31 \overline{6} 1316131512 \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
31 \overline{6} 1316131512 \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: $31 \overline{6} 131613151$

$$\begin{aligned}
31 \overline{6} 131613151 \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
31 \overline{6} 131613151 \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
31 \overline{6} 131613151\overline{2} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\overline{2}31 \overline{6} 131613151 \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
31 \overline{6} 131613151 \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
31 \overline{6} 131613151 \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
31 \overline{6} 131613151\overline{3} \quad \Delta = -86, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 413161315132217

$$\begin{aligned}
413161315132217 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 5 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
413161315132217 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
413161315132217 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
413161315132217 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
413161315132217 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{ll}
413161315132217^2 & \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
413161315132217^3 & \Delta = -134, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\text{Base: } 712321613161232 & \\
712321613161232^2 & \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
712321613161232^2 & \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
712321613161232^2 & \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
712321613161232^2 & \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
712321613161232^2 & \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
\text{Base: } 131613151322171 & \\
1712231513161313^{\text{red}}_{[\# = 8]} & \Delta = -106, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
131613151322171^2 & \Delta = -98, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
131613151322171^2 & \Delta = -98, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
131613151322171^2 & \Delta = -98, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
131613151322171^2 & \Delta = -98, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
131613151322171^2 & \Delta = -98, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1712231513161313^{\text{red}}_{[\# = 7]} & \Delta = -106, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1712231513161313^{\text{red}}_{[\# = 6]} & \Delta = -106, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1712231513161313^{\text{red}}_{[\# = 5]} & \Delta = -106, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1712231513161313^{\text{red}}_{[\# = 3]} & \Delta = -106, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1712231513161313^{\text{red}}_{[\# = 2]} & \Delta = -106, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{array}$$

$$171223151316131\overset{2}{3}_{[\# = 4]} \quad \Delta = -106, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$171223151316131\overset{2}{3}_{[\# = 9]} \quad \Delta = -106, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

Base: 171223151322171

$$171223151322171 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$171223151322171 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: 413161315132171

$$413161315132171 \quad \Delta = -134, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$413161315132171 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$413161315132171 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$413161315132171 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

$$413161315132171 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$413161315132171 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$413161315132171 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 321613161232171

$$321613161232171 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

$$321613161232171 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$321613161232171 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$321613161232171 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$321613161232171 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix}$$

Base: ³¹1613161232171

$$\overset{31}{1}613161232171 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$\overset{31}{1}613161232171 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

$$\begin{aligned}
31 \text{ } 6 \text{ } 13161232171 \quad \Delta = -134, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
31 \text{ } 16 \text{ } 13161232171 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
31 \text{ } 16 \text{ } 13161232171 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
21 \text{ } 6 \text{ } 13161232171 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
31 \text{ } 16 \text{ } 13161232171 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: 131613151321712

$$\begin{aligned}
131613151321712 \quad \Delta = -98, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
2 \text{ } 131613151321712 \quad \Delta = -98, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2 \text{ } 131613151321712 \quad \Delta = -98, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
2 \text{ } 131613151321712 \quad \Delta = -98, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2 \text{ } 131613151321712 \quad \Delta = -98, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
2 \text{ } 131613151321712 \quad \Delta = -98, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
2171231513161313_{[\# = 3]} \quad \Delta = -106, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
2171231513161313_{[\# = 2]} \quad \Delta = -106, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
2171231513161313_{[\# = 9]} \quad \Delta = -106, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
2171231513161313_{[\# = 8]} \quad \Delta = -106, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
2171231513161313_{[\# = 7]} \quad \Delta = -106, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
2171231513161313_{[\# = 6]} \quad \Delta = -106, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
2171231513161313_{[\# = 5]} \quad \Delta = -106, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
2171231513161313_{[\# = 4]} \quad \Delta = -106, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

Base: 171223151321712

$$171223151321712^2 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$171223151321712^2 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$171223151321712^2 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$171223151321712^2 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$171223151321712^2 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: 216131612321712

$$216131612321712^2 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$216131612321712^3 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$216131612321712^2 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$216131612321712^2 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$216131612321712^2 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$216131612321712^2 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$216131612321712^2 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: ¹16131612321712

$$16131612321712^1 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$16131612321712^{1,2} \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$16131612321712^{1,2} \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$16131612321712^1 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$16131612321712^1 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

Base: 217123151321712

$$217123151321712^2 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$217123151321712 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$217123151321712 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 712231513217123

$$712231513217123 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$712231513217123 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$712231513217123 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$712231513217123 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$712231513217123 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

Base: 171231513217123

$$171231513217123 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$171231513217123 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$171231513217123 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$171231513217123 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$171231513217123 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 161316123217123

$$161316123217123 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$161316123217123 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$161316123217123 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$161316123217123 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$161316123217123 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 132171231513221

$$132171231513221 \quad \Delta = -22, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\begin{array}{ll}
122315132171231\mathbf{3}_{[\# = 4]} & \Delta = -30, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
132171231513221\mathbf{3}_{[\# = 2]} & \Delta = -30, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
132171231513221\mathbf{3}_{[\# = 3]} & \Delta = -30, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
13217123151\mathbf{3}^2221 & \Delta = -22, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
132171231513221\mathbf{3}^2 & \Delta = -22, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
132171231513221\mathbf{3}^2 & \Delta = -22, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
132171231513221\mathbf{3}_{[\# = 4]} & \Delta = -30, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
122315132171231\mathbf{3}_{[\# = 5]} & \Delta = -30, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
132171231513221\mathbf{3}_{[\# = 5]} & \Delta = -30, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
132171231513221\mathbf{3}_{[\# = 9]} & \Delta = -30, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
132171231513221\mathbf{3}_{[\# = 6]} & \Delta = -30, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
132171231513221\mathbf{3}_{[\# = 8]} & \Delta = -30, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
122315132171231\mathbf{3}_{[\# = 2]} & \Delta = -30, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
122315132171231\mathbf{3}_{[\# = 3]} & \Delta = -30, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
132171231513221\mathbf{3}_{[\# = 7]} & \Delta = -30, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
\text{Base: } 123217123151321 & \\
123217123151321\mathbf{3}_{[\# = 4]} & \Delta = -32, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
123151321712321\mathbf{3}_{[\# = 4]} & \Delta = -32, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
12321\mathbf{3}^27123151321 & \Delta = -24, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
123217123151\mathbf{3}^221 & \Delta = -24, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
123217123151321^2 \quad \Delta = -24, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
123217123151321^2 \quad \Delta = -24, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
123217123151321^3_{[\# = 3]} \quad \Delta = -32, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
123217123151321^3_{[\# = 8]} \quad \Delta = -32, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
123151321712321^3_{[\# = 8]} \quad \Delta = -32, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
123151321712321^3_{[\# = 5]} \quad \Delta = -32, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
123217123151321^3_{[\# = 7]} \quad \Delta = -32, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
123217123151321^3_{[\# = 5]} \quad \Delta = -32, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
123217123151321^3_{[\# = 2]} \quad \Delta = -32, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
123151321712321^3_{[\# = 7]} \quad \Delta = -32, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
123217123151321^3_{[\# = 6]} \quad \Delta = -32, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
123151321712321^3_{[\# = 2]} \quad \Delta = -32, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
123151321712321^3_{[\# = 6]} \quad \Delta = -32, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
123151321712321^3_{[\# = 3]} \quad \Delta = -32, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
123151321712321^3_{[\# = 9]} \quad \Delta = -32, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
123217123151321^3_{[\# = 9]} \quad \Delta = -32, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix}
\end{aligned}$$

Base: 712315132171232

$$\begin{aligned}
712315132171232^2 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
712315132171232^2 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
712315132171232^2 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

$$712315132171232 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$712315132171232 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

Base: 712321712315132

$$712321712315132 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix}$$

$$712321712315132 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$712321712315132 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 225 \quad \begin{pmatrix} 0 & 15 & 2 \\ 15 & 163 & 25 \\ 2 & 25 & 8 \end{pmatrix}$$

$$712321712315132 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$712321712315132 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

Base: 131612321712321

$$123217123216131\mathbf{3}_{[\# = 9]} \quad \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$123217123216131\mathbf{3}_{[\# = 2]} \quad \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$131612321712321\mathbf{3}_{[\# = 6]} \quad \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$123217123216131\mathbf{3}_{[\# = 7]} \quad \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$123217123216131\mathbf{3}_{[\# = 5]} \quad \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$131612321712321\mathbf{3}_{[\# = 5]} \quad \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$123217123216131\mathbf{3}_{[\# = 6]} \quad \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$131612321712321\mathbf{3}_{[\# = 2]} \quad \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$131612321712321\mathbf{3}_{[\# = 4]} \quad \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$123217123216131\mathbf{3}_{[\# = 3]} \quad \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$131612321712321\mathbf{3}_{[\# = 3]} \quad \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$123217123216131\overset{2}{3}_{[\# = 4]} \quad \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$131612321712321\overset{2}{3} \quad \Delta = -51, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$131612321712321\overset{2}{3}_{[\# = 7]} \quad \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$123217123216131\overset{2}{3}_{[\# = 8]} \quad \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$131612321712321\overset{2}{3}_{[\# = 8]} \quad \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$131612321712321\overset{2}{3}_{[\# = 9]} \quad \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$131612321712321\overset{2}{3} \quad \Delta = -51, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$131612321712321\overset{2}{3} \quad \Delta = -51, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$131612321712321\overset{2}{3} \quad \Delta = -51, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 613161232171232

$$613161232171232\overset{2}{3} \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix}$$

$$613161232171232\overset{2}{3} \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 225 \quad \begin{pmatrix} 0 & 15 & 2 \\ 15 & 163 & 25 \\ 2 & 25 & 8 \end{pmatrix}$$

$$613161232171232\overset{2}{3} \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$613161232171232\overset{2}{3} \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$613161232171232\overset{2}{3} \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: 161232171232161

$$161232171232161\overset{2}{3} \quad \Delta = -121, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$161232171232161\overset{2}{3} \quad \Delta = -121, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

Base: 612321712321613

$$612321712321613\overset{2}{3} \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$612321712321613\overset{2}{3} \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

612321712321613	$\Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, \det = 100$	$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$
612321712321613	$\Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, \det = 121$	$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$
612321712321613	$\Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, \det = 81$	$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$
Base: 171232171231513		
171232171231513	$\Delta = -128, n_+ = 1, n_- = 15, n_0 = 0, \det = 121$	$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$
171232171231513	$\Delta = -128, n_+ = 1, n_- = 15, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$
171232171231513	$\Delta = -128, n_+ = 1, n_- = 15, n_0 = 0, \det = 100$	$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$
171232171231513	$\Delta = -128, n_+ = 1, n_- = 15, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$
171232171231513	$\Delta = -128, n_+ = 1, n_- = 15, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$
Base: 217123217123151		
217123217123151	$\Delta = -120, n_+ = 1, n_- = 15, n_0 = 0, \det = 4$	$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$
217123217123151	$\Delta = -120, n_+ = 1, n_- = 15, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$
217123217123151	$\Delta = -120, n_+ = 1, n_- = 15, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$
217123217123151	$\Delta = -120, n_+ = 1, n_- = 15, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$
217123217123151	$\Delta = -120, n_+ = 1, n_- = 15, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$
Base: 217123217123215		
217123217123215	$\Delta = -122, n_+ = 1, n_- = 15, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$
217123217123215	$\Delta = -122, n_+ = 1, n_- = 15, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$
217123217123215	$\Delta = -122, n_+ = 1, n_- = 15, n_0 = 0, \det = 4$	$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$
217123217123215	$\Delta = -122, n_+ = 1, n_- = 15, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$
217123217123215	$\Delta = -122, n_+ = 1, n_- = 15, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$
Base: 221712321712321		

$$221712321712321\overset{3}{3}_{[\# = 7]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$221712321712321\overset{3}{3} \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$221712321712321\overset{2}{2} \quad \Delta = -70, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$221712321712321\overset{2}{2} \quad \Delta = -70, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$221712321712321\overset{2}{2} \quad \Delta = -70, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$221712321712321\overset{2}{2} \quad \Delta = -70, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$221712321712321\overset{3}{3}_{[\# = 5]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$221712321712321\overset{3}{3}_{[\# = 4]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$221712321712321\overset{3}{3}_{[\# = 9]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$221712321712321\overset{3}{3}_{[\# = 3]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$221712321712321\overset{3}{3}_{[\# = 2]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$221712321712321\overset{3}{3}_{[\# = 6]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$221712321712321\overset{3}{3}_{[\# = 8]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

Base: 321712321712315

$$321712321712315\overset{2}{2} \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$321712321712315\overset{2}{2} \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$321712321712315\overset{2}{2} \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$321712321712315\overset{2}{2} \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$321712321712315\overset{2}{2} \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 132171232171231

$$\begin{aligned}
132171232171231\overset{2}{3} \text{ }_{[\# = 5]} \quad \Delta = -85, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
132171232171231\overset{2}{3} \text{ }_{[\# = 2]} \quad \Delta = -85, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
132171232171231\overset{2}{3} \text{ }_{[\# = 4]} \quad \Delta = -85, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
132171232171231\overset{2}{3} \text{ }_{[\# = 3]} \quad \Delta = -85, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
132171232171231\overset{2}{3} \quad \Delta = -77, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
132171232171231\overset{2}{3} \quad \Delta = -77, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

Base: 17123151316131

$$\begin{aligned}
\overset{1}{17123151316131}\overset{2}{3} \text{ }_{[\# = 8]} \quad \Delta = -106, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\overset{1}{17123151316131}\overset{2}{3} \text{ }_{[\# = 9]} \quad \Delta = -106, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\overset{1}{17123151316131}\overset{2}{3} \text{ }_{[\# = 6]} \quad \Delta = -106, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\overset{1}{17123151316131}\overset{2}{3} \quad \Delta = -98, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
\overset{1}{17123151316131}\overset{2}{3} \quad \Delta = -98, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\overset{1}{17123151316131}\overset{2}{3} \quad \Delta = -98, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\overset{1}{17123151316131}\overset{2}{3} \quad \Delta = -98, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\overset{1}{17123151316131}\overset{2}{3} \quad \Delta = -98, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\overset{1}{17123151316131}\overset{2}{3} \text{ }_{[\# = 5]} \quad \Delta = -106, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\overset{1}{17123151316131}\overset{2}{3} \text{ }_{[\# = 4]} \quad \Delta = -106, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\overset{1}{17123151316131}\overset{2}{3} \text{ }_{[\# = 2]} \quad \Delta = -106, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
\overset{1}{17123151316131}\overset{2}{3} \text{ }_{[\# = 3]} \quad \Delta = -106, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\overset{1}{17123151316131}\overset{2}{3} \text{ }_{[\# = 7]} \quad \Delta = -106, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 17123151322171

$$\begin{aligned}
17123151322171 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
17123151322171 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
17123151322171 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
17123151322171 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

Base: 17123151321712

$$\begin{aligned}
17123151321712 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
17123151321712 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
17123151321712 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
17123151321712 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
17123151321712 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: 17123216131612

$$\begin{aligned}
17123216131612 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
17123216131612 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
17123216131612 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
17123216131612 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
17123216131612 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
17123216131612 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1613161232171

$$\begin{aligned}
1613161232171 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
1613161232171 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{1}{16} \overset{2}{13} 16 12 32 1 \overset{1}{7} 1 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{1}{16} 13 16 12 32 1 \overset{2}{7} \overset{1}{1} \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{17} 12 32 17 12 31 51$

$$\begin{aligned}
& \overset{1}{17} 12 32 17 12 31 \overset{2}{5} 1 \quad \Delta = -120, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{1}{17} 12 32 17 12 31 51 \quad \Delta = -120, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{1}{17} 12 32 \overset{2}{7} 12 31 51 \quad \Delta = -120, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{1}{17} 12 32 17 12 31 \overset{2}{5} 1 \quad \Delta = -120, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{17} 12 32 17 12 32 15$

$$\begin{aligned}
& \overset{1}{17} 12 32 17 12 32 \overset{2}{1} 5 \quad \Delta = -122, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{1}{17} 12 32 17 12 32 \overset{2}{1} 5 \quad \Delta = -122, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{1}{17} 12 32 17 12 32 \overset{2}{1} 5 \quad \Delta = -122, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{1}{17} 12 32 \overset{2}{7} 12 32 15 \quad \Delta = -122, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{17} 12 31 51 32 1 \overset{1}{7} 1$

$$\begin{aligned}
& \overset{1}{17} 12 31 \overset{2}{5} 1 32 1 \overset{1}{7} 1 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{1}{17} 12 31 51 32 1 \overset{2}{7} \overset{1}{1} \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{21}{17} 12 32 17 12 32 1$

$$\begin{aligned}
& \overset{21}{17} \overset{2}{1} 2 32 17 12 32 1 \quad \Delta = -70, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{21}{17} 12 32 17 12 32 1 \overset{3}{\text{[#}=6]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{21}{17} 12 32 17 12 32 1 \overset{3}{\text{[#}=9]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{21}{17} 12 32 17 12 32 1 \overset{3}{\text{[#}=8]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{21}{17} 12 32 17 12 32 1 \overset{3}{\text{[#}=7]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{21}{17}12321712321\overset{2}{3} \text{ }_{[\# = 5]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{21}{17}12321712321\overset{2}{3} \text{ }_{[\# = 2]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{21}{17}12321712321\overset{2}{3} \text{ }_{[\# = 4]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{21}{17}12321712321\overset{2}{3} \text{ }_{[\# = 3]} \quad \Delta = -78, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{21}{17}12321712321\overset{2}{2} \quad \Delta = -70, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{21}{17}12321712321\overset{2}{2} \quad \Delta = -70, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{21}{17}12321712321\overset{2}{2} \quad \Delta = -70, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \text{Base: } 812231513161232 \\
& 812231513161232\overset{2}{2} \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& 812231513161232\overset{2}{2} \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
& 812231513161232\overset{2}{2} \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 812231513161232\overset{2}{2} \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& 812231513161232\overset{2}{2} \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& \text{Base: } 812314141414141 \\
& 812314141414141\overset{2}{2} \quad \Delta = -120, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 118 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
& 812314141414141\overset{2}{2} \quad \Delta = -120, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 812314141414141\overset{2}{3} \text{ }_{[\# = 2]} \quad \Delta = -128, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& 812314141414141\overset{2}{3} \text{ }_{[\# = 3]} \quad \Delta = -128, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& 812314141414141\overset{2}{2} \quad \Delta = -122, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& 812314141414141\overset{2}{2} \quad \Delta = -120, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
812314141414141 \quad \Delta = -120, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
812314141414141 \quad \Delta = -120, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 142 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
812314141414141 \quad \Delta = -120, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 218 & 30 \\ 2 & 30 & 8 \end{pmatrix}
\end{aligned}$$

Base: 812321613151322

$$\begin{aligned}
812321613151322 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
812321613151322 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 196 & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
812321613151322 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
812321613151322 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 256 & \begin{pmatrix} 0 & 16 & 2 \\ 16 & 206 & 28 \\ 2 & 28 & 8 \end{pmatrix} \\
812321613151322 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

Base: 316123151322181

$$\begin{aligned}
316123151322181 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
316123151322181 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
316123151322181 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
316123151322181 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
316123151322181 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}
\end{aligned}$$

Base: 321613151322181

$$\begin{aligned}
321613151322181 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
321613151322181 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
321613151322181 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
321613151322181 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
321613151322181 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}
\end{aligned}$$

³¹
Base: 1613151322181

$$\begin{aligned}
^{21}31613151322181 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
^{312}1613151322181 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
^{31}1613151322181 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
^{31}1613151322181^2 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
^{31}1613151322181^2 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
^{31}1613151322181^2 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
^{31}31613151322181 \quad \Delta = -134, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 232161315132181

$$\begin{aligned}
232161315132181^2 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
232161315132181^2 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
232161315132181^2 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
232161315132181^2 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
232161315132181^2 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

Base: 231513161232181

$$\begin{aligned}
231513161232181^2 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
231513161232181^2 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 206 & 28 \\ 2 & 28 & 8 \end{pmatrix} \\
^{2}231513161232181 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
^{2}231513161232181 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
^{2}231513161232181 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

Base: 612231513221812

$$612231513221812^2 \quad \Delta = -119, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

³ 612231513221812	$\Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 63 & 15 \\ 2 & 15 & 8 \end{pmatrix}$
² 612231513221812	$\Delta = -119, n_+ = 1, n_- = 15, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$
612231513221812 ²	$\Delta = -119, n_+ = 1, n_- = 15, n_0 = 0, \det = 4$	$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$
612231513221812 ²	$\Delta = -119, n_+ = 1, n_- = 15, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
612231513221812 ²	$\Delta = -119, n_+ = 1, n_- = 15, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$
Base: 161231513221812		
161231513221812 ²	$\Delta = -119, n_+ = 1, n_- = 15, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
161231513221812 ²	$\Delta = -119, n_+ = 1, n_- = 15, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$
161231513221812 ²	$\Delta = -119, n_+ = 1, n_- = 15, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$
161231513221812 ²	$\Delta = -119, n_+ = 1, n_- = 15, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
161231513221812 ²	$\Delta = -119, n_+ = 1, n_- = 15, n_0 = 0, \det = 4$	$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$
Base: 216131513221812		
216131513221812 ²	$\Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, \det = 4$	$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$
216131513221812 ²	$\Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, \det = 4$	$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$
216131513221812 ²	$\Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
216131513221812 ²	$\Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$
216131513221812 ³	$\Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, \det = 4$	$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$
216131513221812 ²	$\Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
216131513221812 ²	$\Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$
Base: ¹ 16131513221812		
¹ 16131513221812 ²	$\Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$

$$\begin{aligned}
& \overset{1}{16}131513221812 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{1}{16}131513221812 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{1}{16}131513221812 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{1}{16}131513221812 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: 316123151321812

$$\begin{aligned}
& \overset{2}{31}6123151321812 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 316123151321812 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& 316123151321812 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 316123151321812 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& 316123151321812 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
& 316123151321812 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix}
\end{aligned}$$

Base: 321613151321812

$$\begin{aligned}
& \overset{2}{32}1613151321812 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& 321613151321812 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& 321613151321812 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{2}{32}1613151321812 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 321613151321812 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& 321613151321812 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}
\end{aligned}$$

Base: 315131612321812

$$\begin{aligned}
& 315131612321812 \quad \Delta = -128, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
& \overset{2}{31}5131612321812 \quad \Delta = -128, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
315131612321812^2 \quad \Delta = -128, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
315131612321812^2 \quad \Delta = -128, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
315131612321812^2 \quad \Delta = -128, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
315131612321812^2 \quad \Delta = -128, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: $16^{31}13151321812$

$$\begin{aligned}
16^{31}13151321812^2 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
16^{31}13151321812^2 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
316^{21}13151321812 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
16^{31}13151321812^2 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
16^{31}13151321812^2 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
316^{31}13151321812 \quad \Delta = -134, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
16^{31}13151321812^2 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
16^{31}13151321812 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: 221812231513221

$$\begin{aligned}
221812231513221^2 \quad \Delta = -41, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
221812231513221^2 \quad \Delta = -41, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
221812231513221^2 \quad \Delta = -41, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
221812231513221^2 \quad \Delta = -41, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
221812231513221^3_{[\#=2]} \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
221812231513221^3_{[\#=3]} \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
221812231513221\overset{3}{\color{red}}\substack{[\#\!=4]} \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
221812231513221\overset{3}{\color{red}}\substack{[\#\!=5]} \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
221812231513221\overset{3}{\color{red}}\substack{[\#\!=6]} \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
221812231513221\overset{3}{\color{red}}\substack{[\#\!=7]} \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
221812231513221\overset{3}{\color{red}}\substack{[\#\!=9]} \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
221812231513221\overset{3}{\color{red}}\substack{[\#\!=8]} \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
\overset{3}{\color{red}}221812231513221 \quad \Delta = -50, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: 612231513218122

$$\begin{aligned}
612231513218122\overset{2}{\color{red}} \quad \Delta = -119, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
612231513218122\overset{3}{\color{red}} \quad \Delta = -128, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
612231513218122\overset{2}{\color{red}} \quad \Delta = -119, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
612231513218122\overset{2}{\color{red}} \quad \Delta = -119, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
612231513218122\overset{2}{\color{red}} \quad \Delta = -119, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
612231513218122\overset{3}{\color{red}} \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 63 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
612231513218122\overset{2}{\color{red}} \quad \Delta = -119, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 612315132218123

$$\begin{aligned}
612315132218123\overset{2}{\color{red}} \quad \Delta = -128, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 121 \quad & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
612315132218123\overset{2}{\color{red}} \quad \Delta = -128, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
612315132218123\overset{2}{\color{red}} \quad \Delta = -128, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
612315132218123\overset{2}{\color{red}} \quad \Delta = -128, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
612315132218123\overset{2}{\color{red}} \quad \Delta = -128, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: 161231513218122

$$161231513218122^2 \quad \Delta = -119, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$161231513218122^2 \quad \Delta = -119, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$161231513218122^2 \quad \Delta = -119, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$161231513218122^2 \quad \Delta = -119, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$161231513218122^2 \quad \Delta = -119, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$161231513218122^3 \quad \Delta = -128, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

Base: 161315132218123

$$161315132218123^2 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$161315132218123^2 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$161315132218123^2 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$161315132218123^2 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$161315132218123^2 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 216131513218122

$$216131513218122^3 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$216131513218122^3 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$216131513218122^2 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$216131513218122^2 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$216131513218122^2 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$216131513218122^2 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$216131513218122^2 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$216131513218122^2 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 151316123218122

$$\begin{aligned}
151316123218122^3 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
151316123218122^2 \quad \Delta = -120, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
151316123218122^2 \quad \Delta = -120, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
151316123218122^2 \quad \Delta = -120, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
151316123218122^2 \quad \Delta = -120, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
151316123218122^2 \quad \Delta = -120, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: 16131513218122

$$\begin{aligned}
16131513218122^3 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
16131513218122^2 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
16131513218122^2 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
16131513218122^{12} \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
16131513218122^{12} \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
16131513218122^{12} \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 413217123218122

$$\begin{aligned}
413217123218122^2 \quad \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
413217123218122^2 \quad \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
413217123218122^2 \quad \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
413217123218122^2 \quad \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
413217123218122^2 \quad \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
413217123218122^3 \quad \Delta = -133, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
413217123218122^2 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: 222181231513221

$$222181231513221\textcolor{red}{3} \text{ [#=6]} \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$222181231513221\textcolor{red}{2} \quad \Delta = -41, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$222181231513221\textcolor{red}{2} \quad \Delta = -41, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$222181231513221\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$222181231513221\textcolor{red}{3} \text{ [#=7]} \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$222181231513221\textcolor{red}{2} \quad \Delta = -41, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$222181231513221\textcolor{red}{2} \quad \Delta = -41, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$222181231513221\textcolor{red}{3} \text{ [#=8]} \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$222181231513221\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$222181231513221\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$222181231513221\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$222181231513221\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

Base: 132181223151321

$$132181223151321\textcolor{red}{2} \quad \Delta = -50, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$132181223151321\textcolor{red}{2} \quad \Delta = -50, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$132181223151321\textcolor{red}{2} \quad \Delta = -50, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$132181223151321\textcolor{red}{2} \quad \Delta = -50, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$132181223151321\textcolor{red}{3} \text{ [#=7]} \quad \Delta = -58, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$132181223151321\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -58, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix}$$

$$\begin{array}{ll}
123151322181231\mathbf{3}_{[\# = 2]} & \Delta = -58, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
132181223151321\mathbf{3}_{[\# = 5]} & \Delta = -58, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
123151322181231\mathbf{3}_{[\# = 5]} & \Delta = -58, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
132181223151321\mathbf{3}_{[\# = 4]} & \Delta = -58, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
123151322181231\mathbf{3}_{[\# = 3]} & \Delta = -58, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
132181223151321\mathbf{3}_{[\# = 3]} & \Delta = -58, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
123151322181231\mathbf{3}_{[\# = 4]} & \Delta = -58, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
132181223151321\mathbf{3}_{[\# = 2]} & \Delta = -58, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
132181223151321\mathbf{3}_{[\# = 8]} & \Delta = -58, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
132181223151321\mathbf{3}_{[\# = 6]} & \Delta = -58, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
\text{Base: } 123218122315131 & \\
131513221812321\mathbf{3}_{[\# = 3]} & \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
123218122315131\mathbf{3}_{[\# = 3]} & \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
123218122315131\mathbf{3}_{[\# = 2]} & \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
131513221812321\mathbf{3}_{[\# = 2]} & \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
131513221812321\mathbf{3}_{[\# = 9]} & \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
123218122315131\mathbf{3}_{[\# = 9]} & \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
131513221812321\mathbf{3}_{[\# = 4]} & \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1232181223151\mathbf{3}_{[\# = 2]}^2 & \Delta = -51, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
131513221812321\mathbf{3}_{[\# = 7]} & \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{ll}
123218122315131\mathbf{3}_{[\# = 8]} & \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
123218122315131\mathbf{3}_{[\# = 4]} & \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
131513221812321\mathbf{3}_{[\# = 5]} & \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
123218122315131\mathbf{3}_{[\# = 5]} & \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
131513221812321\mathbf{3}_{[\# = 6]} & \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
12321812231\overset{2}{5}131 & \Delta = -51, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
12321812231\overset{2}{5}131 & \Delta = -51, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
12321\overset{2}{8}122315131 & \Delta = -51, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
123218122315131\mathbf{3}_{[\# = 7]} & \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
131513221812321\mathbf{3}_{[\# = 8]} & \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
123218122315131\mathbf{3}_{[\# = 6]} & \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\text{Base: } 132218123151321 & \\
132218123151321\mathbf{3}_{[\# = 3]} & \Delta = -58, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
132218123151321\mathbf{3}_{[\# = 9]} & \Delta = -58, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
132218123151321\mathbf{3}_{[\# = 8]} & \Delta = -58, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
132218123151321\mathbf{3}_{[\# = 7]} & \Delta = -58, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
132218123151321\mathbf{3}_{[\# = 6]} & \Delta = -58, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
132218123151321\mathbf{3}_{[\# = 5]} & \Delta = -58, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
123151321812231\mathbf{3}_{[\# = 5]} & \Delta = -58, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
123151321812231\mathbf{3}_{[\# = 2]} & \Delta = -58, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
132218123151321\overset{2}{3} \text{ }_{[\# = 2]} \quad \Delta = -58, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
123151321812231\overset{2}{3} \text{ }_{[\# = 3]} \quad \Delta = -58, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
132218123151321\overset{2}{2} \quad \Delta = -50, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
132218123151321\overset{2}{2} \quad \Delta = -50, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
132218123151321\overset{2}{2} \quad \Delta = -50, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
123151321812231\overset{2}{3} \text{ }_{[\# = 4]} \quad \Delta = -58, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
132218123151321\overset{2}{3} \text{ }_{[\# = 4]} \quad \Delta = -58, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
132218123151321\overset{2}{2} \quad \Delta = -50, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: 612315132181223

$$\begin{aligned}
612315132181223\overset{2}{2} \quad \Delta = -128, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
612315132181223\overset{2}{2} \quad \Delta = -128, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
612315132181223\overset{2}{2} \quad \Delta = -128, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
612315132181223\overset{2}{2} \quad \Delta = -128, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
612315132181223\overset{2}{2} \quad \Delta = -128, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: 612321812231513

$$\begin{aligned}
612321812231513\overset{2}{2} \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
612321812231513\overset{2}{2} \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
612321812231513\overset{2}{2} \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 196 \quad & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
612321812231513\overset{2}{2} \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 100 \quad & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
612321812231513\overset{2}{2} \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 613151322181232

² 613151322181232	$\Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$
6131513221 ² 81232	$\Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, \det = 256$	$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 174 & 26 \\ 2 & 26 & 8 \end{pmatrix}$
613151 ² 322181232	$\Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, \det = 121$	$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$
613151322181232	$\Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, \det = 81$	$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$
613151322181 ² 232	$\Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, \det = 64$	$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
Base: 161232181223151		
1612321812231 ² 51	$\Delta = -121, n_+ = 1, n_- = 15, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$
161232181 ² 223151	$\Delta = -121, n_+ = 1, n_- = 15, n_0 = 0, \det = 81$	$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
1612321812231 ² 51	$\Delta = -121, n_+ = 1, n_- = 15, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$
161 ² 232181223151	$\Delta = -121, n_+ = 1, n_- = 15, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$
Base: 161315132181223		
161 ² 315132181223	$\Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$
161315132181 ² 223	$\Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$
1613151321 ² 81223	$\Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, \det = 81$	$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$
1613151 ² 32181223	$\Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, \det = 64$	$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
161315132181 ² 223	$\Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$
Base: 316123218122315		
31612321812231 ² 5	$\Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$
3161 ² 23218122315	$\Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$
31612321 ² 8122315	$\Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, \det = 121$	$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$
3161 ² 23218122315	$\Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, \det = 81$	$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$

$$316123218122315 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 50 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

Base: 131612321812231

$$131612321812231 \quad \Delta = -77, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$132218123216131\mathbf{3}_{[\# = 5]} \quad \Delta = -85, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$132218123216131\mathbf{3}_{[\# = 4]} \quad \Delta = -85, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$131612321812231\mathbf{3}_{[\# = 3]} \quad \Delta = -85, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$132218123216131\mathbf{3}_{[\# = 3]} \quad \Delta = -85, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$131612321812231\mathbf{3}_{[\# = 2]} \quad \Delta = -85, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$132218123216131\mathbf{3}_{[\# = 2]} \quad \Delta = -85, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$132218123216131\mathbf{3}_{[\# = 6]} \quad \Delta = -85, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$132218123216131\mathbf{3}_{[\# = 7]} \quad \Delta = -85, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$132218123216131\mathbf{3}_{[\# = 9]} \quad \Delta = -85, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$131612321812231\mathbf{3}_{[\# = 5]} \quad \Delta = -85, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$132218123216131\mathbf{3}_{[\# = 8]} \quad \Delta = -85, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$131612321812231\mathbf{3}_{[\# = 4]} \quad \Delta = -85, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$131612321812231 \quad \Delta = -77, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$131612321812231 \quad \Delta = -77, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$131612321812231 \quad \Delta = -77, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 513161232181223

$$513161232181223 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$513161232181223 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$513161232181223 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$513161232181223 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$513161232181223 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

Base: 413221812321712

$$413221812321712 \quad \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$413221812321712 \quad \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$413221812321712 \quad \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$413221812321712 \quad \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$413221812321712 \quad \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix}$$

Base: 221712321812231

$$221712321812231 \quad \Delta = -105, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$221712321812231 \text{ }_{[\# = 2]} \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$221712321812231 \text{ }_{[\# = 3]} \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$221712321812231 \text{ }_{[\# = 4]} \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$221712321812231 \quad \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$221712321812231 \quad \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$221712321812231 \quad \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$221712321812231 \text{ }_{[\# = 5]} \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$221712321812231 \quad \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 132171232181222

$$\begin{aligned}
222181232171231\overset{\textcolor{red}{3}}{3}_{[\# = 4]} \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
222181232171231\overset{\textcolor{red}{3}}{3}_{[\# = 3]} \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
222181232171231\overset{\textcolor{red}{3}}{3}_{[\# = 2]} \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
132171232181222\overset{\textcolor{red}{2}}{2} \quad \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
132171232181222\overset{\textcolor{red}{2}}{2} \quad \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
132171232181222\overset{\textcolor{red}{2}}{2} \quad \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
132171232181222\overset{\textcolor{red}{2}}{2} \quad \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
222181232171231\overset{\textcolor{red}{3}}{3}_{[\# = 5]} \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
\textbf{Base: } 413221812321\overset{1}{71} \\
413221812321\overset{1}{71}\overset{\textcolor{red}{2}}{2} \quad \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
413221812321\overset{1}{71}\overset{\textcolor{red}{2}}{2} \quad \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
413221812321\overset{1}{71}\overset{\textcolor{red}{21}}{21} \quad \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
413221812321\overset{1}{71}\overset{\textcolor{red}{2}}{2} \quad \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
\textbf{Base: } 1\overset{21}{7}12321812231 \\
1\overset{21}{7}12321812231\overset{\textcolor{red}{3}}{3}_{[\# = 3]} \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1\overset{21}{7}12321812231\overset{\textcolor{red}{3}}{3}_{[\# = 2]} \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
1\overset{21}{7}12321812231\overset{\textcolor{red}{3}}{3}_{[\# = 4]} \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1\overset{21}{7}12321812231\overset{\textcolor{red}{3}}{3}_{[\# = 5]} \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1\overset{21}{7}12321812231\overset{\textcolor{red}{2}}{2} \quad \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1\overset{21}{7}12321812231\overset{\textcolor{red}{2}}{2} \quad \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

$$1\overset{21}{7}12321812231 \quad \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1\overset{21}{7}12321\overset{2}{8}12231 \quad \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 151232181231512

$$151232181\overset{2}{2}31512 \quad \Delta = -95, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$151232181\overset{2}{2}31512 \quad \Delta = -95, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$151232181231\overset{2}{5}12 \quad \Delta = -95, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$151\overset{2}{2}32181231512 \quad \Delta = -95, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$15123\overset{2}{2}181231512 \quad \Delta = -95, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$151232181231\overset{3}{5}12 \quad \Delta = -103, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

Base: 215132181231512

$$\overset{2}{2}15132181231512 \quad \Delta = -93, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\overset{3}{2}15132181231512 \quad \Delta = -101, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$2151\overset{2}{3}2181231512 \quad \Delta = -93, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$2151\overset{2}{3}2181231512 \quad \Delta = -93, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 412321812315131

$$412321812315131\overset{3}{3}_{[\# = 3]} \quad \Delta = -87, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\overset{2}{4}12321812315131 \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$4123\overset{2}{2}1812315131 \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$41232181231\overset{2}{5}131 \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$41232181231\overset{2}{5}131 \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$41232181231\overset{2}{5}131 \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\begin{aligned}
412321812315131\textcolor{red}{3} \text{ }_{[\# = 2]} \quad \Delta = -87, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
412321812315131\textcolor{red}{3} \text{ }_{[\# = 8]} \quad \Delta = -87, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
412321812315131\textcolor{red}{3} \text{ }_{[\# = 6]} \quad \Delta = -87, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
412321812315131\textcolor{red}{3} \text{ }_{[\# = 7]} \quad \Delta = -87, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
412321812315131\textcolor{red}{3} \text{ }_{[\# = 9]} \quad \Delta = -87, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
\textcolor{red}{3} 412321812315131 \quad \Delta = -87, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
412321812315131\textcolor{red}{3} \text{ }_{[\# = 5]} \quad \Delta = -87, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
412321812315131\textcolor{red}{3} \text{ }_{[\# = 4]} \quad \Delta = -87, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: 141321812315131

$$\begin{aligned}
131513218123141\textcolor{red}{3} \text{ }_{[\# = 2]} \quad \Delta = -85, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
141321812315131 \textcolor{red}{2} \quad \Delta = -77, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
141321812315131 \textcolor{red}{2} \quad \Delta = -77, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
141321812315131 \textcolor{red}{2} \quad \Delta = -77, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
141321812315131 \textcolor{red}{2} \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
141321812315131\textcolor{red}{3} \text{ }_{[\# = 8]} \quad \Delta = -85, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
141321812315131\textcolor{red}{3} \text{ }_{[\# = 7]} \quad \Delta = -85, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
141321812315131\textcolor{red}{3} \text{ }_{[\# = 6]} \quad \Delta = -85, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
141321812315131\textcolor{red}{3} \text{ }_{[\# = 5]} \quad \Delta = -85, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
141321812315131 \textcolor{red}{2} \quad \Delta = -77, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
141321812315131\textcolor{red}{3} \text{ }_{[\# = 2]} \quad \Delta = -85, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
141321812315131\bar{3} \text{ }_{[\# = 9]} \quad \Delta = -85, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
131513218123141\bar{3} \text{ }_{[\# = 3]} \quad \Delta = -85, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
141321812315131\bar{3} \text{ }_{[\# = 4]} \quad \Delta = -85, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
141321812315131\bar{2} \quad \Delta = -77, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
141321812315131\bar{3} \text{ }_{[\# = 3]} \quad \Delta = -85, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\text{Base: } 413151321812321 & \\
413151321812321\bar{3} \text{ }_{[\# = 9]} \quad \Delta = -87, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
413151321812321\bar{2} \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
413151321812321\bar{2} \quad \Delta = -80, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
413151321812321\bar{3} \text{ }_{[\# = 4]} \quad \Delta = -87, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
413151321812321\bar{3} \text{ }_{[\# = 6]} \quad \Delta = -87, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
413151321812321\bar{3} \text{ }_{[\# = 8]} \quad \Delta = -87, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
413151321812321\bar{3} \text{ }_{[\# = 3]} \quad \Delta = -87, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
413151321812321\bar{3} \text{ }_{[\# = 2]} \quad \Delta = -87, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
413151321812321\bar{3} \text{ }_{[\# = 5]} \quad \Delta = -87, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
413151321812321\bar{3} \quad \Delta = -87, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
413151321812321\bar{3} \text{ }_{[\# = 7]} \quad \Delta = -87, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
413151321812321\bar{2} \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
413151321812321\bar{2} \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
413151321812321\bar{2} \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

$$413151321812321 \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: 313218123151321

$$313218123151321 \quad \Delta = -58, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$313218123151321\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -66, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix}$$

$$313218123151321 \quad \Delta = -66, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$313218123151321 \text{ [#=2]} \quad \Delta = -66, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$313218123151321\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -66, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$313218123151321\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -66, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$313218123151321\textcolor{red}{3} \text{ [#=8]} \quad \Delta = -66, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$313218123151321\textcolor{red}{3} \text{ [#=7]} \quad \Delta = -66, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$313218123151321\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -66, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$313218123151321\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -66, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$313218123151321\textcolor{red}{3} \text{ [#=6]} \quad \Delta = -66, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$313218123151321 \quad \Delta = -58, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$313218123151321 \quad \Delta = -58, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$313218123151321 \quad \Delta = -58, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$313218123151321 \quad \Delta = -58, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$313218123151321 \quad \Delta = -60, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 132181231513215

$$132181231513215 \quad \Delta = -102, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$\begin{aligned}
132181231513215^2 \quad \Delta = -102, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
132181231513215^2 \quad \Delta = -102, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
132181231513215^2 \quad \Delta = -102, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
132181231513215^2 \quad \Delta = -102, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
5123151321812313_{[\# = 2]} \quad \Delta = -110, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
5123151321812313_{[\# = 5]} \quad \Delta = -110, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
5123151321812313_{[\# = 4]} \quad \Delta = -110, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
5123151321812313_{[\# = 3]} \quad \Delta = -110, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 413218123151315

$$\begin{aligned}
413218123151315^2 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
413218123151315^2 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
413218123151315^2 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
413218123151315^2 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
413218123151315^2 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
413218123151315^2 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
413218123151315^2 \quad \Delta = -131, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
413218123151315^2 \quad \Delta = -130, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 132181231513151

$$\begin{aligned}
1513151321812313_{[\# = 2]} \quad \Delta = -109, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1513151321812313_{[\# = 4]} \quad \Delta = -109, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

$$151315132181231\overset{\textcolor{red}{3}}{3}_{[\# = 5]} \quad \Delta = -109, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$151315132181231\overset{\textcolor{red}{3}}{3}_{[\# = 3]} \quad \Delta = -109, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1321812315131\overset{\textcolor{red}{2}}{5} \quad \Delta = -101, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1321812315131\overset{\textcolor{red}{2}}{5} \quad \Delta = -101, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1321812315131\overset{\textcolor{red}{2}}{5} \quad \Delta = -102, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1321812315131\overset{\textcolor{red}{2}}{5} \quad \Delta = -101, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1321812315131\overset{\textcolor{red}{2}}{5} \quad \Delta = -101, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1321812315131\overset{\textcolor{red}{2}}{5} \quad \Delta = -101, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{151232181231}{5} 1$$

$$1 \overset{151232181231}{5} 1 \quad \Delta = -95, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{151232181231}{5} 1 \quad \Delta = -95, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{151232181231}{5} 1 \quad \Delta = -95, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{151232181231}{5} 1 \quad \Delta = -95, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{215132181231}{5} 1$$

$$1 \overset{215132181231}{5} 1 \quad \Delta = -93, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{215132181231}{5} 1 \quad \Delta = -93, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{215132181231}{5} 1 \quad \Delta = -93, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{215132181231}{5} 1 \quad \Delta = -101, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$1 \overset{215132181231}{5} 1 \quad \Delta = -93, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{215132181231}{5} 1 \quad \Delta = -93, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{15132181231}{5} 1$$

$$\begin{array}{l}
\begin{array}{c} 15132181231 \\ 1 \quad 5 \quad 1 \end{array} \quad \Delta = -93, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\begin{array}{c} 15132181231 \\ 1 \quad 5 \quad 1 \end{array} \quad \Delta = -93, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{array}$$

Base: 414132181232161

$$\begin{array}{l}
414132181232161^2 \quad \Delta = -125, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
414132181232161^2 \quad \Delta = -125, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
414132181232161^2 \quad \Delta = -125, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
414132181232161^2 \quad \Delta = -125, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
414132181232161^2 \quad \Delta = -125, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
414132181232161^2 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{array}$$

Base: 141321812321612

$$\begin{array}{l}
141321812321612^2 \quad \Delta = -99, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
141321812321612^2 \quad \Delta = -97, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
141321812321612^2 \quad \Delta = -97, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
141321812321612^2 \quad \Delta = -97, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
141321812321612^2 \quad \Delta = -97, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
141321812321612^2 \quad \Delta = -97, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
2161232181231413^{\#3} \quad \Delta = -105, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
2161232181231413^{\#2} \quad \Delta = -105, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
141321812321612^3 \quad \Delta = -105, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}
\end{array}$$

Base: 131612321812313

$$131612321812313^2 \quad \Delta = -87, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$\begin{aligned}
313218123216131\overset{\textcolor{red}{3}}{3} \text{ [#=8]} \quad \Delta = -93, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
313218123216131\overset{\textcolor{red}{3}}{3} \text{ [#=4]} \quad \Delta = -93, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
313218123216131\overset{\textcolor{red}{3}}{3} \text{ [#=3]} \quad \Delta = -93, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
313218123216131\overset{\textcolor{red}{3}}{3} \text{ [#=7]} \quad \Delta = -93, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
313218123216131\overset{\textcolor{red}{3}}{3} \text{ [#=6]} \quad \Delta = -93, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
313218123216131\overset{\textcolor{red}{3}}{3} \text{ [#=2]} \quad \Delta = -93, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
13161232181231\overset{\textcolor{red}{3}}{3} \text{ [#=2]} \quad \Delta = -93, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
313218123216131\overset{\textcolor{red}{3}}{3} \text{ [#=9]} \quad \Delta = -93, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
313218123216131\overset{\textcolor{red}{3}}{3} \text{ [#=5]} \quad \Delta = -93, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
13161232181231\overset{\textcolor{red}{3}}{3} \quad \Delta = -93, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
13161232181231\overset{\textcolor{red}{2}}{2} \quad \Delta = -85, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
13161232181231\overset{\textcolor{red}{2}}{2} \quad \Delta = -85, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
13161232181231\overset{\textcolor{red}{2}}{2} \quad \Delta = -85, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
13161232181231\overset{\textcolor{red}{2}}{2} \quad \Delta = -85, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
13161232181231\overset{\textcolor{red}{2}}{2} \quad \Delta = -85, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\textbf{Base: } 413161232181231 & \\
413161232181231\overset{\textcolor{red}{3}}{3} \text{ [#=4]} \quad \Delta = -113, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
413161232181231\overset{\textcolor{red}{2}}{2} \quad \Delta = -105, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
413161232181231\overset{\textcolor{red}{2}}{2} \quad \Delta = -105, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
413161232181231\overset{\textcolor{red}{3}}{3} \quad \Delta = -113, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
413161232181231\overset{2}{3}_{[\# = 2]} \quad \Delta = -113, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
413161232181231\overset{2}{3} \quad \Delta = -105, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
413161232181231\overset{2}{3} \quad \Delta = -105, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
413161232181231\overset{2}{3} \quad \Delta = -105, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
413161232181231\overset{2}{3}_{[\# = 3]} \quad \Delta = -113, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
413161232181231\overset{2}{3}_{[\# = 5]} \quad \Delta = -113, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 16123218123141

$$\begin{aligned}
16123218123141\overset{1}{\overset{2}{3}} \quad \Delta = -99, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
16123218123141\overset{1}{\overset{2}{3}}_{[\# = 2]} \quad \Delta = -105, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
16123218123141\overset{1}{\overset{2}{3}}_{[\# = 3]} \quad \Delta = -105, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
16123218123141\overset{1}{\overset{2}{3}} \quad \Delta = -97, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
16123218123141\overset{1}{\overset{2}{3}} \quad \Delta = -97, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
16123218123141\overset{1}{\overset{2}{3}} \quad \Delta = -97, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
16123218123141\overset{1}{\overset{2}{3}} \quad \Delta = -97, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

Base: 221712321812313

$$\begin{aligned}
221712321812313\overset{2}{3} \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
221712321812313\overset{2}{3} \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
221712321812313\overset{3}{[\# = 2]} \quad \Delta = -112, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
221712321812313\overset{2}{3} \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
221712321812313\overset{2}{3} \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

$$221712321812313^2 \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$221712321812313^3 \quad \Delta = -113, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$221712321812313^2 \quad \Delta = -106, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$221712321812313^3 \quad \Delta = -112, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: 322171232181231

$$3221712321812313^{\# = 3} \quad \Delta = -113, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$3221712321812313^{\# = 5} \quad \Delta = -113, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$322171232181231^2 \quad \Delta = -105, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$3221712321812313^{\# = 4} \quad \Delta = -113, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$322171232181231^2 \quad \Delta = -105, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$3221712321812313^{\# = 2} \quad \Delta = -113, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$322171232181231^2 \quad \Delta = -105, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$322171232181231^2 \quad \Delta = -105, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: $^{21}1712321812313$

$$^{21}1712321812313^2 \quad \Delta = -106, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$^{21}1712321812313^3 \quad \Delta = -112, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$^{21}1712321812313^{\# = 2} \quad \Delta = -112, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$^{21}1712321812313^2 \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$^{21}1712321812313^2 \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$^{21}1712321812313^2 \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{21}{7} 12321812313 \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{21}{7} 12321812313 \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: $1 \overset{321}{7} 1232181231$

$$1 \overset{321}{7} 1232181231 \quad \Delta = -105, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1 \overset{321}{7} 1232181231 \quad \Delta = -105, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$1 \overset{321}{7} 1232181231 \overset{2}{3} [\#=5] \quad \Delta = -113, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{321}{7} 1232181231 \quad \Delta = -105, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{321}{7} 1232181231 \quad \Delta = -105, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{321}{7} 1232181231 \overset{2}{3} [\#=2] \quad \Delta = -113, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$1 \overset{321}{7} 1232181231 \overset{2}{3} [\#=3] \quad \Delta = -113, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1 \overset{321}{7} 1232181231 \overset{2}{3} [\#=4] \quad \Delta = -113, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$321 \overset{21}{7} 1232181231 \quad \Delta = -105, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: 151232181232151

$$151 \overset{2}{232181232151} \quad \Delta = -97, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1512321 \overset{2}{81232151} \quad \Delta = -97, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 412321812321612

$$412321812321612 \overset{3}{2} \quad \Delta = -107, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$412321812321612 \overset{2}{2} \quad \Delta = -99, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$412321812321612 \overset{2}{2} \quad \Delta = -99, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$412321812321612 \overset{2}{2} \quad \Delta = -99, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$412321812321612 \overset{2}{2} \quad \Delta = -99, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$412321812321612 \quad \Delta = -99, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$412321812321612 \quad \Delta = -107, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 41 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

Base: 321612321812321

$$321612321812321\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -88, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$321612321812321 \quad \Delta = -80, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$321612321812321 \quad \Delta = -80, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$321612321812321 \quad \Delta = -80, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$321612321812321 \quad \Delta = -80, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$321612321812321\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -88, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$321612321812321\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -88, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$321612321812321\textcolor{red}{3} \text{ [#=6]} \quad \Delta = -88, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$321612321812321\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -88, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$321612321812321\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -88, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$321612321812321\textcolor{red}{3} \text{ [#=8]} \quad \Delta = -88, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$321612321812321\textcolor{red}{3} \text{ [#=7]} \quad \Delta = -88, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

Base: 412321812321¹61

$$412321812321\textcolor{red}{3} \text{ }^161 \quad \Delta = -107, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 41 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$412321812321\textcolor{red}{2} \text{ }^161 \quad \Delta = -99, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$412321812321\textcolor{red}{2} \text{ }^161 \quad \Delta = -99, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$412321812321\textcolor{red}{2} \text{ }^161 \quad \Delta = -99, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$41232181232161^{\mathbf{21}} \quad \Delta = -99, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: $16^{31}12321812321$

$$316^{21}12321812321 \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$16^{31}12321812321\mathbf{3}_{[\# = 7]} \quad \Delta = -87, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$16^{31}12321812321\mathbf{3}_{[\# = 8]} \quad \Delta = -87, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$16^{31}12321812321\mathbf{3}_{[\# = 9]} \quad \Delta = -87, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$16^{31}12321812321\mathbf{3}_{[\# = 6]} \quad \Delta = -87, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$16^{31}12321812321\mathbf{3}_{[\# = 5]} \quad \Delta = -87, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$16^{31}12321812321\mathbf{3}_{[\# = 4]} \quad \Delta = -87, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$16^{31}12321812321\mathbf{3}_{[\# = 3]} \quad \Delta = -87, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$16^{31}12321812321\mathbf{3}_{[\# = 2]} \quad \Delta = -87, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$316^{31}12321812321 \quad \Delta = -87, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$16^{31}12321812321^{\mathbf{2}} \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$16^{31}12321812321 \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$16^{31\mathbf{2}}12321812321 \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$16^{31}12321812321^{\mathbf{2}} \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: $2^{123151321812}3^1$

$$2^{3123151321812}3^1_{[\# = 4]} \quad \Delta = -60, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$2^{123151321812}3^1 \quad \Delta = -52, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$2^{123151321812}3^1\mathbf{13}_{[\# = 2]} \quad \Delta = -60, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$\begin{aligned}
2^{\overline{3123151321812}}_3 1_{[\# = 8]} \quad \Delta = -60, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
2^{\overline{123151321812}}_3 1 \quad \Delta = -52, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2^{\overline{123151321812}}_3 1 \quad \Delta = -52, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2^{\overline{3123151321812}}_3 1_{[\# = 2]} \quad \Delta = -60, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2^{\overline{123151321812}}_3 1\overline{3}_{[\# = 3]} \quad \Delta = -60, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 60 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
2^{\overline{3123151321812}}_3 1_{[\# = 3]} \quad \Delta = -60, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2^{\overline{3123151321812}}_3 1_{[\# = 5]} \quad \Delta = -60, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
2^{\overline{3123151321812}}_3 1_{[\# = 6]} \quad \Delta = -60, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
2^{\overline{3123151321812}}_3 1_{[\# = 7]} \quad \Delta = -60, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
2^{\overline{3123151321812}}_3 1_{[\# = 9]} \quad \Delta = -60, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
2^{\overline{123151321812}}_3 1\overline{2} \quad \Delta = -52, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2^{\overline{123151321812}}_3 1 \quad \Delta = -52, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2^{\overline{123151321812}}_3 1\overline{4} \quad \Delta = -80, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\text{Base: } 131612321812\overline{3}^1_2 \\
2^{\overline{3131612321812}}_3 1_{[\# = 8]} \quad \Delta = -87, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
131612321812\overline{3}^{\overline{41}}_2 \quad \Delta = -107, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
131612321812\overline{3}^{\overline{2}}_1 \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
131612321812\overline{3}^{\overline{2}}_1 \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
131612321812\overline{3}^{\overline{2}}_1 \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
131612321812\overline{3}^{\overline{21}}_2 \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
13161232181232^1 \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
13161232181232^{31} \quad [\# = 3] \quad \Delta = -87, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2^{3131612321812} 3^1 \quad [\# = 3] \quad \Delta = -87, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
13161232181232^{31} \quad [\# = 2] \quad \Delta = -87, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2^{3131612321812} 3^1 \quad [\# = 2] \quad \Delta = -87, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
2^{3131612321812} 3^1 \quad [\# = 9] \quad \Delta = -87, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
2^{3131612321812} 3^1 \quad [\# = 4] \quad \Delta = -87, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
2^{3131612321812} 3^1 \quad [\# = 7] \quad \Delta = -87, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
2^{3131612321812} 3^1 \quad [\# = 6] \quad \Delta = -87, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
2^{3131612321812} 3^1 \quad [\# = 5] \quad \Delta = -87, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
\text{Base: } 22171232181232^1 & \\
22171232181232^{41} \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
22171232181232^2 \quad \Delta = -98, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
22171232181232^2 \quad \Delta = -98, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
22171232181232^{21} \quad \Delta = -98, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
22171232181232^2 \quad \Delta = -98, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
22171232181232^2 \quad \Delta = -98, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
22171232181232^3 \quad \Delta = -107, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
22171232181232^{31} \quad [\# = 3] \quad \Delta = -106, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
22171232181232^{31} \quad [\# = 2] \quad \Delta = -106, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: $17^{21}1232181232^1$

$$17^{21}1232181232^1 \quad \Delta = -98, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$17^{21}1232181232^1 \quad \Delta = -98, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$17^{21}1232181232^1 \quad \Delta = -98, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$17^{21}1232181232^1 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$17^{21}1232181232^1 \quad [\# = 3] \quad \Delta = -106, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$17^{21}1232181232^1 \quad [\# = 2] \quad \Delta = -106, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$17^{21}1232181232^1 \quad \Delta = -98, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$17^{21}1232181232^1 \quad \Delta = -98, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 222181232181222

$$222181232181222^1 \quad \Delta = -115, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$222181232181222^1 \quad \Delta = -115, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: 61223151322181

$$61223151322181^1 \quad \Delta = -119, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$61223151322181^1 \quad \Delta = -119, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$61223151322181^1 \quad \Delta = -119, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$61223151322181^1 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 63 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$61223151322181^1 \quad \Delta = -119, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: 18122315132161

$$18122315132161^1 \quad \Delta = -119, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$18122315132161^1 \quad \Delta = -119, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$18122315132161^1 \quad \Delta = -119, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$${}^1_{18122315132161} \quad \Delta = -119, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: ${}^1_{18122315131612}$

$${}^1_{18122315131612} \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$${}^1_{18122315131612} \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$${}^1_{18122315131612} \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$${}^1_{18122315131612} \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$${}^{12}_{18122315131612} \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$${}^1_{18122315131612} \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: ${}^1_{1613151322181}$

$${}^1_{1613151322181} \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$${}^1_{1613151322181} \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$${}^1_{1613151322181} \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$${}^{12}_{1613151322181} \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: ${}^1_{18123151321613}$

$${}^{12}_{18123151321613} \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$${}^1_{18123151321613} \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$${}^1_{18123151321613} \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$${}^1_{18123151321613} \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$${}^1_{18123151321613} \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: ${}^1_{18123151316123}$

$${}^1_{18123151316123} \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\begin{aligned}
{}^1_{18123151316123} \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^1_{18123151316123} \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
{}^{12}_{18123151316123} \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^1_{18123151316123} \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

Base: ${}^1_{18123216131513}$

$$\begin{aligned}
{}^1_{18123216131513} \quad \Delta = -128, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
{}^{12}_{18123216131513} \quad \Delta = -128, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^1_{18123216131513} \quad \Delta = -128, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
{}^1_{18123216131513} \quad \Delta = -128, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^1_{18123216131513} \quad \Delta = -128, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: ${}^{31}_{161315132181}$

$$\begin{aligned}
{}^{31}_{161315132181} \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
{}^{31}_{3161315132181} \quad \Delta = -134, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
{}^{31}_{161315132181} \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
{}^{31}_{161315132181} \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
{}^{21}_{3161315132181} \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
{}^{312}_{161315132181} \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
{}^{31}_{161315132181} \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}
\end{aligned}$$

Base: ${}^{12231513221}_{1^8 12}$

$$\begin{aligned}
{}^{312231513221}_{1^8 12} \quad [\# = 3] \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
{}^{312231513221}_{1^8 12} \quad [\# = 2] \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

$$1^{\overline{312231513221}} 8^{12} [\# = 9] \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$1^{\overline{12231513221}} 8^{12} \quad \Delta = -41, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1^{\overline{312231513221}} 8^{12} [\# = 4] \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1^{\overline{312231513221}} 8^{12} [\# = 7] \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$1^{\overline{312231513221}} 8^{12} [\# = 5] \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$1^{\overline{312231513221}} 8^{12} [\# = 6] \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$1^{\overline{12231513221}} 8^{12} \quad \Delta = -41, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1^{\overline{12231513221}} 8^{12} \quad \Delta = -41, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1^{\overline{312231513221}} 8^{12} [\# = 8] \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$1^{\overline{12231513221}} 8^{12} \quad \Delta = -41, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: $61223151321\overline{81}^{21}$

$$61223151321\overline{81}^{21} \quad \Delta = -119, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$61223151321\overline{81}^{21} \quad \Delta = -119, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$61223151321\overline{81}^{21} \quad \Delta = -119, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$61223151321\overline{81}^{21} \quad \Delta = -119, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$61223151321\overline{81}^{21} \quad \Delta = -119, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$61223151321\overline{81}^{21} \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 63 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

Base: $18123151321\overline{61}^{21}$

$$18123151321\overline{61}^{21} \quad \Delta = -119, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$18123151321\overline{61}^{21} \quad \Delta = -119, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$\begin{array}{ll}
\overset{21}{1}812315132161^{\overset{2}{2}} & \Delta = -119, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\overset{21}{1}812315132161^{\overset{2}{2}} & \Delta = -119, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
\overset{21}{1}812315132161 & \Delta = -119, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\text{Base: } \overset{21}{1}812315131612 & \\
\overset{21}{1}812315131612^{\overset{2}{2}} & \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\overset{21}{1}812315131612^{\overset{2}{2}} & \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\overset{21}{1}812315131612 & \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\overset{21}{1}812315131612^{\overset{2}{2}} & \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\overset{21}{1}812315131612^{\overset{2}{2}} & \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\overset{21}{1}812315131612^{\overset{3}{3}} & \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
\overset{21}{1}812315131612^{\overset{2}{2}} & \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
\text{Base: } \overset{21}{1}812321613151 & \\
\overset{21}{1}812321613151^{\overset{2}{2}} & \Delta = -120, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\overset{21}{1}812321613151^{\overset{2}{2}} & \Delta = -120, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\overset{21}{1}812321613151^{\overset{2}{2}} & \Delta = -120, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\overset{21}{1}812321613151 & \Delta = -120, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\overset{21}{1}812321613151^{\overset{2}{2}} & \Delta = -120, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\text{Base: } \overset{1}{1}61315132181^{\overset{21}{21}} & \\
\overset{1}{1}61315132181^{\overset{21}{21}} & \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\overset{1}{1}61315132181^{\overset{21}{21}} & \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\overset{1}{1}61315132181^{\overset{221}{221}} & \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
& \overset{1}{1} \overset{2}{6} 13151321 \overset{21}{8} 1 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{1}{1} \overset{2}{6} 13151321 \overset{21}{8} 1 \quad \Delta = -118, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

Base: $41321712321 \overset{21}{8} 1$

$$\begin{aligned}
& 41321712321 \overset{21}{8} 1 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 41321712321 \overset{21}{8} 1 \quad \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 41321712321 \overset{21}{8} 1 \quad \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& 41321712321 \overset{21}{8} 1 \quad \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& 41321712321 \overset{21}{8} 1 \quad \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 41321712321 \overset{21}{8} 1 \quad \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{1223151321}{8} 122$

$$\begin{aligned}
& 1 \overset{3}{1} 223151321 \overset{21}{8} 122 \text{ [#=7]} \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
& 1 \overset{3}{1} 223151321 \overset{21}{8} 122 \text{ [#=3]} \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 1 \overset{1223151321}{8} 122 \overset{2}{2} \quad \Delta = -41, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 1 \overset{1223151321}{8} 122 \overset{3}{2} \quad \Delta = -50, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 1 \overset{3}{1} 223151321 \overset{21}{8} 122 \text{ [#=5]} \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& 1 \overset{3}{1} 223151321 \overset{21}{8} 122 \text{ [#=9]} \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
& 1 \overset{3}{1} 223151321 \overset{21}{8} 122 \text{ [#=8]} \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
& 1 \overset{3}{1} 223151321 \overset{21}{8} 122 \text{ [#=2]} \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 1 \overset{1223151321}{8} 122 \quad \Delta = -41, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& 1 \overset{3}{1} 223151321 \overset{21}{8} 122 \text{ [#=6]} \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}
\end{aligned}$$

$$1 \overset{31223151321}{8} 122 \text{ [#}=4] \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1 \overset{1223151321}{8} 122 \quad \Delta = -41, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{1223151321}{8} 122 \quad \Delta = -41, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: $1 \overset{221}{8} 1232171231$

$$1 \overset{221}{8} 1232171231 \overset{3}{\text{[#}=4]} \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{221}{8} 1232171231 \overset{3}{\text{[#}=3]} \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1 \overset{221}{8} 1232171231 \overset{3}{\text{[#}=2]} \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$1 \overset{221}{8} 1232171231 \overset{3}{\text{[#}=5]} \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{221}{8} 1232171231 \overset{2}{\text{[#}=4]} \quad \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{221}{8} 1232171231 \overset{2}{\text{[#}=3]} \quad \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{221}{8} 1232171231 \overset{2}{\text{[#}=2]} \quad \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1 \overset{221}{8} 1232171231 \overset{2}{\text{[#}=1]} \quad \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{221}{8} 1232171231 \overset{2}{\text{[#}=0]} \quad \Delta = -105, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: $1 \overset{221}{8} 1232181222$

$$1 \overset{221}{8} 1232181222 \overset{2}{\text{[#}=4]} \quad \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \overset{221}{8} 1232181222 \overset{2}{\text{[#}=3]} \quad \Delta = -115, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{221}{8} 1232181222 \overset{2}{\text{[#}=2]} \quad \Delta = -115, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{221}{8} 1232181222 \overset{2}{\text{[#}=1]} \quad \Delta = -115, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1 \overset{221}{8} 1232181222 \overset{2}{\text{[#}=0]} \quad \Delta = -115, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: $1 \overset{3221}{8} 123218122$

$$1 \overset{3221}{8} 123218122 \overset{2}{\text{[#}=4]} \quad \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\begin{array}{ll}
1 \overset{3221}{8} 123218122 \overset{3}{\Delta} = -133, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
1 \overset{3221}{8} 123218122 \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
3221 \overset{21}{8} 123218122 \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1 \overset{3221}{8} 123218122 \overset{2}{\Delta} = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1 \overset{3221}{8} 123218122 \overset{2}{\Delta} = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
\text{Base: } 6123151321 \overset{321}{8} 1 & \\
6123151321 \overset{321}{8} 1 \Delta = -128, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
6123151321 \overset{2}{8} 1 \overset{321}{\Delta} = -128, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
6123151321 \overset{2}{8} 1 \overset{321}{\Delta} = -128, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
6123151321 \overset{321}{8} 1 \overset{2}{\Delta} = -128, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
6123151321 \overset{2}{8} 1 \overset{2321}{\Delta} = -128, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
6123151321 \overset{2}{8} 1 \overset{321}{\Delta} = -128, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{321}{8} 1231513161 & \\
1 \overset{321}{8} 1231513161 \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
321 \overset{21}{8} 1231513161 \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1 \overset{321}{8} 1231513161 \overset{2}{\Delta} = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1 \overset{321}{8} 1231513161 \overset{2}{\Delta} = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1 \overset{321}{8} 1231513161 \overset{2}{\Delta} = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
1 \overset{321}{8} 1231513161 \overset{2}{\Delta} = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{321}{8} 1232161315 & \\
321 \overset{21}{8} 1232161315 \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{ll}
1 \overset{321}{8} 1232161315 & \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \overset{321}{8} 12321\overset{2}{6}1315 & \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1 \overset{321}{8} 12321\overset{2}{6}1315 & \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{321}{8} 12321\overset{2}{6}1315 & \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1 \overset{321}{8} 12321\overset{2}{6}1315 & \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{123151321}{8} 1231 & \\
1 \overset{3123151321}{8} 1231 \text{ [#=6]} & \Delta = -58, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1 \overset{3123151321}{8} 1231 \text{ [#=5]} & \Delta = -58, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1 \overset{123151321}{8} 1231\overset{3}{3} \text{ [#=2]} & \Delta = -58, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1 \overset{123151321}{8} 1231 & \Delta = -50, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \overset{3123151321}{8} 1231 \text{ [#=7]} & \Delta = -58, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
1 \overset{3123151321}{8} 1231 \text{ [#=4]} & \Delta = -58, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \overset{123151321}{8} 1231\overset{3}{3} \text{ [#=4]} & \Delta = -58, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \overset{3123151321}{8} 1231 \text{ [#=3]} & \Delta = -58, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{123151321}{8} 1231\overset{3}{3} \text{ [#=3]} & \Delta = -58, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{3123151321}{8} 1231 \text{ [#=2]} & \Delta = -58, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1 \overset{123151321}{8} 1231\overset{3}{3} \text{ [#=5]} & \Delta = -58, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1 \overset{123151321}{8} 1231 & \Delta = -50, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1 \overset{123151321}{8} 1231\overset{2}{2} & \Delta = -50, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \overset{3123151321}{8} 1231 \text{ [#=8]} & \Delta = -58, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{l}
\begin{array}{c} 123151321 \\ 1 \quad 8 \quad 1231 \end{array} \quad \Delta = -50, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\begin{array}{c} 3123151321 \\ 1 \quad 8 \quad 1231 \end{array} \quad [\# = 9] \quad \Delta = -58, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
\text{Base: } \begin{array}{c} 1321 \\ 1 \quad 8 \quad 123216131 \end{array} \\
\begin{array}{c} 1321 \\ 1 \quad 8 \quad 123216131 \end{array} \begin{array}{c} 3 \\ \end{array} [\# = 3] \quad \Delta = -85, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\begin{array}{c} 1321 \\ 1 \quad 8 \quad 123216131 \end{array} \begin{array}{c} 3 \\ \end{array} [\# = 2] \quad \Delta = -85, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
\begin{array}{c} 31321 \\ 1 \quad 8 \quad 123216131 \end{array} [\# = 4] \quad \Delta = -85, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
\begin{array}{c} 31321 \\ 1 \quad 8 \quad 123216131 \end{array} [\# = 2] \quad \Delta = -85, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\begin{array}{c} 1321 \\ 1 \quad 8 \quad 123216131 \end{array} \begin{array}{c} 3 \\ \end{array} [\# = 6] \quad \Delta = -85, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\begin{array}{c} 1321 \\ 1 \quad 8 \quad 123216131 \end{array} \begin{array}{c} 2 \\ \end{array} \quad \Delta = -77, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\begin{array}{c} 1321 \\ 1 \quad 8 \quad 123216131 \end{array} \begin{array}{c} 3 \\ \end{array} [\# = 9] \quad \Delta = -85, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\begin{array}{c} 31321 \\ 1 \quad 8 \quad 123216131 \end{array} [\# = 5] \quad \Delta = -85, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
\begin{array}{c} 1321 \\ 1 \quad 8 \quad 123216131 \end{array} \begin{array}{c} 3 \\ \end{array} [\# = 5] \quad \Delta = -85, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\begin{array}{c} 1321 \\ 1 \quad 8 \quad 123216131 \end{array} \begin{array}{c} 3 \\ \end{array} [\# = 4] \quad \Delta = -85, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\begin{array}{c} 31321 \\ 1 \quad 8 \quad 123216131 \end{array} [\# = 3] \quad \Delta = -85, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\begin{array}{c} 1321 \\ 1 \quad 8 \quad 123216131 \end{array} \begin{array}{c} 2 \\ \end{array} \quad \Delta = -77, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\begin{array}{c} 1321 \\ 1 \quad 8 \quad 123216131 \end{array} \quad \Delta = -77, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\begin{array}{c} 1321 \\ 1 \quad 8 \quad 123216131 \end{array} \begin{array}{c} 2 \\ \end{array} \quad \Delta = -77, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\begin{array}{c} 1321 \\ 1 \quad 8 \quad 123216131 \end{array} \begin{array}{c} 3 \\ \end{array} [\# = 7] \quad \Delta = -85, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\begin{array}{c} 1321 \\ 1 \quad 8 \quad 123216131 \end{array} \begin{array}{c} 3 \\ \end{array} [\# = 8] \quad \Delta = -85, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\text{Base: } \begin{array}{c} 1321 \\ 1 \quad 8 \quad 123217122 \end{array}
\end{array}$$

$$\begin{array}{ll}
1 \overset{1321}{8} \overset{2}{123217122} & \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1 \overset{1321}{8} \overset{2}{123217122} & \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{1321}{8} \overset{3}{123217122} & \Delta = -105, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
1 \overset{1321}{8} 123217122 & \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1 \overset{1321}{8} \overset{2}{123217122} & \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1 \overset{31321}{8} 123217122 \text{ [#=3]} & \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \overset{31321}{8} 123217122 \text{ [#=2]} & \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \overset{31321}{8} 123217122 \text{ [#=4]} & \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1 \overset{31321}{8} 123217122 \text{ [#=5]} & \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{21}{7} \overset{1}{1232181231} & \\
1 \overset{21}{7} \overset{1}{1232181231} \overset{3}{\text{ [#=2]}} & \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1 \overset{21}{7} \overset{1}{1232181231} \overset{3}{\text{ [#=4]}} & \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \overset{21}{7} \overset{1}{1232181231} \overset{3}{\text{ [#=3]}} & \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{21}{7} \overset{1}{1232181231} \overset{3}{\text{ [#=5]}} & \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1 \overset{21}{7} \overset{1}{1232181231} \overset{2}{\text{ [#=2]}} & \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \overset{21}{7} \overset{2}{1232181231} \overset{1}{\text{ [#=1]}} & \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1 \overset{21}{7} \overset{2}{1232181231} \overset{1}{\text{ [#=1]}} & \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{21}{7} \overset{1}{1232181231} & \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{13151321}{8} 12321 & \\
1 \overset{313151321}{8} 12321 \text{ [#=4]} & \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{ll}
1^{13151321} 8^{12321} \quad \Delta = -51, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1^{13151321} 8^{12321} \textcolor{red}{3} [\# = 4] \quad \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1^{\textcolor{red}{3}13151321} 8^{12321} [\# = 3] \quad \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1^{13151321} 8^{12321} \textcolor{red}{3} [\# = 3] \quad \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{13151321} 8^{12321} \textcolor{red}{3} [\# = 8] \quad \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1^{\textcolor{red}{3}13151321} 8^{12321} [\# = 7] \quad \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1^{13151321} 8^{12321} \textcolor{red}{3} [\# = 5] \quad \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1^{\textcolor{red}{3}13151321} 8^{12321} [\# = 5] \quad \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1^{13151321} 8^{12321} \textcolor{red}{3} [\# = 6] \quad \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1^{\textcolor{red}{3}13151321} 8^{12321} [\# = 6] \quad \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{13151321} 8^{12321} \textcolor{red}{3} [\# = 7] \quad \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1^{13151321} 8^{12321} \textcolor{red}{3} [\# = 2] \quad \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{\textcolor{red}{3}13151321} 8^{12321} [\# = 2] \quad \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1^{\textcolor{red}{3}13151321} 8^{12321} [\# = 9] \quad \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1^{13151321} 8^{12321} \textcolor{red}{3} [\# = 9] \quad \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1^{\textcolor{red}{3}13151321} 8^{12321} [\# = 8] \quad \Delta = -59, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1^{13151321} 8^{12321} \textcolor{red}{2} \quad \Delta = -51, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1^{13151321} 8^{12321} \quad \Delta = -51, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1^{13151321} 8^{12321} \quad \Delta = -51, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{array}$$

Base: $612321 \textcolor{red}{3} 8^{3151321} 1$

$$\begin{aligned}
612321 \overset{3151321}{8} 1 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
612321 \overset{23151321}{8} 1 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
612321 \overset{2}{8} \overset{3151321}{1} \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
612321 \overset{3151321}{8} 1 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
612321 \overset{3151321}{8} 1 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
612321 \overset{3151321}{8} 12 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\text{Base: } 613151321 \overset{2321}{8} 1 & \\
613151321 \overset{2321}{8} 12 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
613151321 \overset{2321}{8} 1 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
613151321 \overset{2}{8} \overset{2321}{1} \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
613151321 \overset{22321}{8} 1 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
613151321 \overset{2}{8} \overset{2321}{1} \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
613151321 \overset{2}{8} \overset{2321}{1} \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{512321}{8} 1232161 & \\
1 \overset{512321}{8} \overset{2}{1232161} \quad \Delta = -123, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1 \overset{512321}{8} 1232161 \quad \Delta = -123, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1 \overset{512321}{8} \overset{2}{1232161} \quad \Delta = -123, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
1 \overset{512321}{8} 1232161 \quad \Delta = -123, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
512321 \overset{21}{8} 1232161 \quad \Delta = -123, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{151321}{8} 1232161 & \\
1 \overset{151321}{8} \overset{2}{1232161} \quad \Delta = -121, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{151321}{1 \ 8 \ 1232161} \quad \Delta = -121, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{151321}{1 \ 8 \ 1232161} \quad \Delta = -121, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 151321\overset{21}{8}1232161 \quad \Delta = -121, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{151321}{1 \ 8 \ 1232161} \overset{2}{} \quad \Delta = -121, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{12321}{1 \ 8 \ 12321612}$

$$\begin{aligned}
& \overset{312321}{1 \ 8 \ 12321612} \text{ [#=6]} \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
& \overset{312321}{1 \ 8 \ 12321612} \text{ [#=2]} \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{312321}{1 \ 8 \ 12321612} \text{ [#=8]} \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
& \overset{312321}{1 \ 8 \ 12321612} \text{ [#=5]} \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
& \overset{312321}{1 \ 8 \ 12321612} \text{ [#=4]} \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& \overset{312321}{1 \ 8 \ 12321612} \text{ [#=7]} \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
& \overset{12321}{1 \ 8 \ 12321612} \overset{3}{} \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
& \overset{312321}{1 \ 8 \ 12321612} \text{ [#=9]} \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
& \overset{12321}{1 \ 8 \ 12321612} \overset{2}{} \quad \Delta = -71, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{12321}{1 \ 8 \ 12321612} \overset{2}{} \quad \Delta = -71, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{12321}{1 \ 8 \ 12321612} \overset{2}{} \quad \Delta = -71, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{12321}{1 \ 8 \ 12321612} \quad \Delta = -71, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{312321}{1 \ 8 \ 12321612} \text{ [#=3]} \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{51321}{1 \ 8 \ 12321613}$

$$\overset{51321}{1 \ 8 \ 12321613} \overset{2}{} \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$1 \overset{51321}{8} \overset{2}{12321613} \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$1 \overset{51321}{8} \overset{2}{12321613} \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$51321 \overset{21}{8} 12321613 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \overset{51321}{8} 12321613 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{51321}{8} 12321613 \quad \Delta = -129, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

Base: $1612321\overset{1}{8}12321$

$$1612321\overset{1}{8}12321\overset{1}{3} \text{ [#=3]} \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1612321\overset{1}{8}1\overset{2}{2321} \quad \Delta = -71, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1612321\overset{21}{8}12321 \quad \Delta = -71, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$161\overset{12}{2321}\overset{1}{8}12321 \quad \Delta = -71, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1612321\overset{1}{8}12321\overset{1}{3} \text{ [#=9]} \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$1612321\overset{1}{8}12321\overset{1}{3} \text{ [#=7]} \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$1612321\overset{1}{8}12321\overset{1}{3} \text{ [#=6]} \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$1612321\overset{1}{8}12321\overset{1}{3} \text{ [#=5]} \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$1612321\overset{1}{8}12321\overset{1}{3} \text{ [#=4]} \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1612321\overset{1}{8}12321\overset{1}{3} \text{ [#=8]} \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$1612321\overset{1}{8}12321\overset{1}{3} \text{ [#=2]} \quad \Delta = -79, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: $41321\overset{1}{8}12321712$

$$41321\overset{1}{8}1\overset{2}{2321}712 \quad \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$41321\overset{21}{8}12321712 \quad \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\begin{array}{ll}
41321\overset{1}{8}12321\overset{2}{7}12 & \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
41321\overset{2}{8}12321712 & \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
41321\overset{21}{8}12321712 & \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
41321\overset{12}{8}12321712 & \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
41321\overset{2}{8}12321712 & \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\text{Base: } 41321\overset{1}{8}12321\overset{1}{7}1 & \\
41321\overset{1}{8}12321\overset{21}{7}1 & \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
41321\overset{12}{8}12321\overset{1}{7}1 & \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
41321\overset{21}{8}12321\overset{1}{7}1 & \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
41321\overset{2}{8}12321\overset{1}{7}1 & \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
41321\overset{21}{8}12321\overset{1}{7}1 & \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
41321\overset{2}{8}12321\overset{1}{7}1 & \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\text{Base: } 1\overset{21}{8}12321\overset{1}{8}1223 & \\
1\overset{212}{8}12321\overset{1}{8}1223 & \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1\overset{21}{8}12321\overset{21}{8}1223 & \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1\overset{21}{8}12321\overset{1}{8}1223 & \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1\overset{21}{8}12321\overset{12}{8}1223 & \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1\overset{21}{8}12321\overset{21}{8}1223 & \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\text{Base: } 1\overset{221}{8}12321\overset{221}{8}1 & \\
1\overset{221}{8}12321\overset{221}{8}1 & \Delta = -115, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1\overset{221}{8}12321\overset{221}{8}1 & \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{array}$$

$$^{221\bar{2}}_{1\ 8\ 12321}\ ^{221}_{8\ 1}\quad \Delta = -115, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: $^{1223151321}_{21\ 8\ 12}$

$$^{1223151321}_{21\ 8\ 12}\quad \Delta = -41, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$^{1223151321}_{21\ 8\ 12}\quad \Delta = -41, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$^{21223151321}_{21\ 8\ 12}\quad \Delta = -41, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$^{1223151321}_{21\ 8\ 12}\quad \Delta = -41, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$^{31223151321}_{21\ 8\ 12\ [\# = 6]}\quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$^{31223151321}_{21\ 8\ 12\ [\# = 8]}\quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$^{31223151321}_{21\ 8\ 12\ [\# = 9]}\quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$^{31223151321}_{21\ 8\ 12\ [\# = 7]}\quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$^{31223151321}_{21\ 8\ 12\ [\# = 3]}\quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$^{31223151321}_{21\ 8\ 12\ [\# = 2]}\quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$^{31223151321}_{21\ 8\ 12\ [\# = 4]}\quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$^{31223151321}_{21\ 8\ 12\ [\# = 5]}\quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

Base: $^{21}_{21\ 8\ 1232171231}$

$$^{21\bar{2}}_{21\ 8\ 1232171231}\quad \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$^{21}_{21\ 8\ 1232171231}\ ^{21\bar{2}}_{21\ 8\ 1232171231}\quad \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$^{221}_{21\ 8\ 1232171231}\quad \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$^{21}_{21\ 8\ 1232171231}\ ^{21\bar{2}}_{21\ 8\ 1232171231}\quad \Delta = -96, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$^{21}_{21\ 8\ 1232171231}\ ^{21\bar{2}}_{21\ 8\ 1232171231}\ ^{21\bar{2}}_{21\ 8\ 1232171231}\ [\# = 4]\quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$^{21}_{21} 8 1232171231 \textcolor{red}{3} \text{ [#}=5] \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$^{21}_{21} 8 1232171231 \textcolor{red}{3} \text{ [#}=2] \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$^{21}_{21} 8 1232171231 \textcolor{red}{3} \text{ [#}=3] \quad \Delta = -104, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: $^{21}_{21} 8 1232181222$

$$^{21\textcolor{red}{2}}_{21} 8 1232181222 \quad \Delta = -115, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$^{21}_{21} 8 1232181222 \textcolor{red}{2} \quad \Delta = -115, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$\textcolor{red}{2}^{21}_{21} 8 1232181222 \quad \Delta = -115, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$^{21}_{21} 8 1232181222 \textcolor{red}{2} \quad \Delta = -115, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: $^{221}_{1} \textcolor{red}{8} 12321 \textcolor{red}{8}^{21}_{12}$

$$^{221}_{1} \textcolor{red}{8} 12321 \textcolor{red}{8}^{21}_{12} \quad \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$^{221}_{1} \textcolor{red}{8} 12321 \textcolor{red}{8}^{21}_{12} \quad \Delta = -115, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$^{221}_{1} \textcolor{red}{8} 12321 \textcolor{red}{8}^{21\textcolor{red}{2}}_{12} \quad \Delta = -115, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\textcolor{red}{2}^{221\textcolor{red}{2}}_{1} \textcolor{red}{8} 12321 \textcolor{red}{8}^{21}_{12} \quad \Delta = -115, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$^{221}_{1} \textcolor{red}{8} 12321 \textcolor{red}{8}^{221\textcolor{red}{2}}_{12} \quad \Delta = -115, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: $^{321}_{21} \textcolor{red}{8} 123218122$

$$^{321\textcolor{red}{2}}_{21} \textcolor{red}{8} 123218122 \quad \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$\textcolor{red}{3}^{321}_{21} \textcolor{red}{8} 123218122 \quad \Delta = -133, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$^{321}_{21} \textcolor{red}{8} 123218122 \textcolor{red}{3} \quad \Delta = -133, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$^{321}_{21} \textcolor{red}{8} 123218122 \textcolor{red}{2} \quad \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$^{321}_{21} \textcolor{red}{8} 123218122 \quad \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$\textcolor{red}{2}^{321}_{21} \textcolor{red}{8} 123218122 \quad \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 24 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$^{321}_{21} 8^{21} 123218122 \quad \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: $^{2321}_{21} 8^{21} 12321812$

$$^{321}_{2321} 8^{21} 12321812 \quad \Delta = -135, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$^{2321}_{21} 8^{21} 12321812 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix}$$

$$^{2321}_{21} 8^{21} 12321812 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$^{22321}_{21} 8^{21} 12321812 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 48 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$^{2321}_{21} 8^{21} 12321812 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

$$^{2321}_{21} 8^{21} 12321812 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: $^{21}_{18} 12321^{21}_{8} 123$

$$^{21}_{18} 12321^{21}_{8} 123 \quad \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$^{21}_{18} 12321^{221}_{8} 123 \quad \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$^{21}_{18} 12321^{21}_{8} 123 \quad \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$^{21}_{18} 12321^{21}_{8} 123 \quad \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$^{21}_{18} 12321^{21}_{8} 123 \quad \Delta = -124, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$^{21}_{18} 12321^{321}_{8} 123 \quad \Delta = -133, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: $^{21}_{21} 8^{21} 12321^{21}_{8} 12$

$$^{21}_{21} 8^{21} 12321^{21}_{8} 12 \quad \Delta = -115, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$^{221}_{21} 8^{21} 12321^{21}_{8} 12 \quad \Delta = -115, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: $^1_{18} 12321^{21}_{8} 1232$

$$^1_{18} 12321^{221}_{8} 1232 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$^1_{18} 12321^{321}_{8} 1232 \quad \Delta = -135, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\begin{aligned}
& \overset{1}{1} \overset{2}{8} 12321 \overset{21}{8} 1232 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{1}{1} 12321 \overset{21}{8} 1232 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{1}{1} 12321 \overset{21}{8} \overset{2}{12} 32 \quad \Delta = -126, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{1,12231513221}{8} 1$

$$\begin{aligned}
& \overset{1,12231513221}{1} \overset{1}{8} 1 \quad \Delta = -41, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{1,12231513221}{1} \overset{1}{8} 1 \quad \Delta = -41, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{1,12231513221}{1} \overset{1}{8} 1 \quad \Delta = -41, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{1,3}{1} \overset{12231513221}{8} 1 \text{ [#=5]} \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
& \overset{1,3}{1} \overset{12231513221}{8} 1 \text{ [#=8]} \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
& \overset{1,3}{1} \overset{12231513221}{8} 1 \text{ [#=4]} \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& \overset{1,3}{1} \overset{12231513221}{8} 1 \text{ [#=3]} \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{1,3}{1} \overset{12231513221}{8} 1 \text{ [#=9]} \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix} \\
& \overset{1,3}{1} \overset{12231513221}{8} 1 \text{ [#=7]} \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
& \overset{1,3}{1} \overset{12231513221}{8} 1 \text{ [#=6]} \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
& \overset{1,3}{1} \overset{12231513221}{8} 1 \text{ [#=2]} \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

Base: $61223151321 \overset{1,1}{8} 1$

$$\begin{aligned}
& \overset{2}{6} 1223151321 \overset{1,1}{8} 1 \quad \Delta = -119, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& \overset{3}{6} 1223151321 \overset{1,1}{8} 1 \quad \Delta = -127, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 63 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& 61223151321 \overset{2}{8} \overset{1,1}{1} \quad \Delta = -119, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 61223151321 \overset{21,1}{8} 1 \quad \Delta = -119, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

$$61223151321 \overset{2}{8} \overset{1,1}{1} \quad \Delta = -119, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: $1 \overset{21,1223151321}{8} 1$

$$1 \overset{21,1223151321}{8} 1 \quad \Delta = -41, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1 \overset{21,1223151321}{8} 1 \quad \Delta = -41, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{21,1223151321}{8} 1 \quad \Delta = -41, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{21,\textcolor{red}{3}1223151321}{8} 1 \text{ [#=5]} \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

$$1 \overset{21,1223151321}{8} 1 \quad \Delta = -41, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{21,\textcolor{red}{3}1223151321}{8} 1 \text{ [#=2]} \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1 \overset{21,\textcolor{red}{3}1223151321}{8} 1 \text{ [#=3]} \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$1 \overset{21,\textcolor{red}{3}1223151321}{8} 1 \text{ [#=9]} \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix}$$

$$1 \overset{21,\textcolor{red}{3}1223151321}{8} 1 \text{ [#=8]} \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix}$$

$$1 \overset{21,\textcolor{red}{3}1223151321}{8} 1 \text{ [#=7]} \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix}$$

$$1 \overset{21,\textcolor{red}{3}1223151321}{8} 1 \text{ [#=6]} \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$1 \overset{21,\textcolor{red}{3}1223151321}{8} 1 \text{ [#=4]} \quad \Delta = -49, n_+ = 1, n_- = 15, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$T = 16$ (2131 bases)

Base: 14132211212231322

$$14132211212231322 \overset{2}{8} \quad \Delta = -112, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$22313221121223141\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$22313221121223141\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$14132211212231322 \overset{3}{8} \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 32 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\begin{aligned}
14132211212231322^2 \quad \Delta = -112, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
14132211212231322^2 \quad \Delta = -112, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
14132211212231322^2 \quad \Delta = -112, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: 31413221121223132

$$\begin{aligned}
31413221121223132^3 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
31413221121223132^2 \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
31413221121223132^2 \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
31413221121223132^2 \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
31413221121223132^2 \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
31413221121223132^2 \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
31413221121223132^2 \quad \Delta = -122, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
31413221121223132^3 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 23141322112122313

$$\begin{aligned}
23141322112122313^2 \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
23141322112122313^2 \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
23141322112122313^2 \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
23141322112122313^2 \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
23141322112122313^2 \quad \Delta = -122, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
23141322112122313^3 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
23141322112122313^3_{[\# = 2]} \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

$$23141322112122313 \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: 22314132211212231

$$22314132211212231 \quad \Delta = -112, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$22314132211212231 \quad \Delta = -112, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$223141322112122313_{[\# = 2]} \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$223141322112122313_{[\# = 3]} \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$223141322112122313_{[\# = 4]} \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$22314132211212231 \quad \Delta = -112, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 70 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$223141322112122313_{[\# = 5]} \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$22314132211212231 \quad \Delta = -112, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 1414132211212222

$$222221121223141413_{[\# = 2]} \quad \Delta = -130, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$222221121223141413_{[\# = 3]} \quad \Delta = -130, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1414132211212222 \quad \Delta = -122, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1414132211212222 \quad \Delta = -122, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1414132211212222 \quad \Delta = -122, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1414132211212222 \quad \Delta = -122, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 31414132211212222

$$31414132211212222 \quad \Delta = -130, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$31414132211212222 \quad \Delta = -130, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$31414132211212222 \quad \Delta = -130, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$\begin{array}{lll}
3141413221121222 & \Delta = -130, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
3141413221121222 & \Delta = -130, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
3141413221121222 & \Delta = -138, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\text{Base: } 23141413221121222 & & \\
23141413221121222 & \Delta = -131, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
23141413221121222 & \Delta = -131, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
23141413221121222 & \Delta = -131, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 70 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
23141413221121222 & \Delta = -131, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
23141413221121222 & \Delta = -133, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
23141413221121222 & \Delta = -131, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
\text{Base: } 22314141322112122 & & \\
22314141322112122 & \Delta = -131, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
22314141322112122 & \Delta = -131, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
22314141322112122 & \Delta = -131, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
22314141322112122 & \Delta = -131, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 88 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
22314141322112122 & \Delta = -131, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\text{Base: } 22222211212231512 & & \\
22222211212231512 & \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
22222211212231512 & \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
22222211212231512 & \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
22222211212231512 & \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{ll}
22222211212231512^2 & \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\text{Base: } 22222112122315123 & \\
22222112122315123^2 & \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
22222112122315123^2 & \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
22222112122315123^2 & \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
22222112122315123^2 & \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\text{Base: } 22221121223151232 & \\
22221121223151232^2 & \Delta = -129, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
22221121223151232^2 & \Delta = -129, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
22221121223151232^2 & \Delta = -129, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
22221121223151232^2 & \Delta = -129, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\text{Base: } 2222221121223151^1 & \\
2222221121223151^2 & \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2222221121223151^{21} & \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2222221121223151^2 & \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\text{Base: } 2222211212231513^1 & \\
2222211212231513^2 & \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
2222211212231513^{21} & \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
2222211212231513^{31} & \Delta = -134, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2222211212231513^{21} & \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2222211212231513^2 & \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{array}$$

$$2222211212231\overset{1}{5}\overset{2}{1}3 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: 222211212231¹5132

$$222211212231\overset{1}{5}\overset{2}{1}32 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$222211212231\overset{2}{5}\overset{21}{1}32 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$222211212231\overset{21}{5}\overset{2}{1}32 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$22221121\overset{2}{2}231\overset{1}{5}132 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$222211212231\overset{31}{5}132 \quad \Delta = -135, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$222211212231\overset{2}{5}\overset{1}{1}32 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 22211212231¹51322

$$22211212231\overset{31}{5}1322 \quad \Delta = -135, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$22211212231\overset{21}{5}1322 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$22211212231\overset{2}{5}\overset{1}{1}322 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$22211212231\overset{12}{5}1322 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$2221121\overset{2}{2}231\overset{1}{5}1322 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$22211212231\overset{21}{5}1322 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 11²¹2122314141322

$$11\overset{21}{2}122314141322 \quad \Delta = -131, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$11\overset{21}{2}1223141\overset{2}{4}1322 \quad \Delta = -131, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$11\overset{21}{2}1223141\overset{2}{4}1322 \quad \Delta = -131, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$11\overset{21}{2}12231\overset{2}{4}141322 \quad \Delta = -131, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$11\overset{21}{2}1223141\overset{2}{4}1322 \quad \Delta = -131, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 70 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

Base: $1 \overset{2313221}{12} 1223141$

$$1 \overset{2313221}{12} 1223141\overset{3}{\text{}}_{[\#=3]} \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$1 \overset{2313221}{12} 1223141 \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 57 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1 \overset{2313221}{12} 1223141\overset{3}{\text{}}_{[\#=2]} \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$1 \overset{2313221}{12} 1223141\overset{2}{\text{}} \quad \Delta = -112, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{2313221}{12} 1223141 \quad \Delta = -112, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 76 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

$$1 \overset{2313221}{12} 1223141 \quad \Delta = -112, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 58 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$1 \overset{2313221}{12} 1223141\overset{2}{\text{}} \quad \Delta = -112, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$1 \overset{2313221}{12} 1223141 \quad \Delta = -114, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: $1 \overset{313221}{12} 12231413$

$$1 \overset{313221}{12} 12231413_{[\#=2]} \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$1 \overset{313221}{12} 12231413\overset{2}{\text{}} \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{313221}{12} 12231413\overset{2}{\text{}} \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{313221}{12} 12231413 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{313221}{12} 12231413 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{313221}{12} 12231413\overset{2}{\text{}} \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{313221}{12} 12231413 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{313221}{12} 12231413\overset{3}{\text{}} \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: $1 \overset{13221}{12} 122314132$

$$1 \overset{13221}{12} 122314132\overset{2}{\text{}} \quad \Delta = -112, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{313221}{12} 122314132_{[\#=5]} \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$\begin{array}{l}
\begin{array}{l} \textcolor{red}{3}13221 \\ 1 \ 12 \ 122314132 \end{array} \begin{array}{l} [\# = 4] \\ \Delta = -120, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 16 \end{array} \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\begin{array}{l} \textcolor{red}{3}13221 \\ 1 \ 12 \ 122314132 \end{array} \begin{array}{l} [\# = 3] \\ \Delta = -120, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 9 \end{array} \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\begin{array}{l} \textcolor{red}{3}13221 \\ 1 \ 12 \ 122314132 \end{array} \begin{array}{l} [\# = 2] \\ \Delta = -120, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 4 \end{array} \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\begin{array}{l} 13221 \\ 1 \ 12 \ 122314132 \end{array} \Delta = -112, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 16 \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\begin{array}{l} 13221 \textcolor{red}{2} \\ 1 \ 12 \ 122314132 \end{array} \Delta = -112, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 49 \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\begin{array}{l} 13221 \textcolor{red}{2} \\ 1 \ 12 \ 122314132 \end{array} \Delta = -112, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 49 \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\begin{array}{l} 13221 \textcolor{red}{2} \\ 1 \ 12 \ 122314132 \end{array} \Delta = -114, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 4 \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\textbf{Base: } \begin{array}{l} 22221 \\ 1 \ 12 \ 122314141 \end{array} \\
\begin{array}{l} 22221 \\ 1 \ 12 \ 122314141 \end{array} \begin{array}{l} \textcolor{red}{3} [\# = 3] \\ \Delta = -130, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 9 \end{array} \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
\begin{array}{l} 22221 \\ 1 \ 12 \ 122314141 \end{array} \Delta = -122, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 25 \begin{pmatrix} 0 & 5 & 2 \\ 5 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
\begin{array}{l} 22221 \textcolor{red}{2} \\ 1 \ 12 \ 122314141 \end{array} \Delta = -122, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 36 \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
\begin{array}{l} 22221 \textcolor{red}{2} \\ 1 \ 12 \ 122314141 \end{array} \Delta = -122, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 25 \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\begin{array}{l} 22221 \textcolor{red}{2} \\ 1 \ 12 \ 122314141 \end{array} \Delta = -122, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 9 \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\begin{array}{l} 22221 \\ 1 \ 12 \ 122314141 \end{array} \begin{array}{l} \textcolor{red}{3} [\# = 2] \\ \Delta = -130, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 4 \end{array} \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
\textbf{Base: } \begin{array}{l} 2221 \\ 1 \ 12 \ 1223141413 \end{array} \\
\begin{array}{l} 2221 \textcolor{red}{2} \\ 1 \ 12 \ 1223141413 \end{array} \Delta = -130, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 9 \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\begin{array}{l} 2221 \\ 1 \ 12 \ 1223141413 \end{array} \Delta = -130, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 16 \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\begin{array}{l} 2221 \textcolor{red}{2} \\ 1 \ 12 \ 1223141413 \end{array} \Delta = -130, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 25 \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\begin{array}{l} 2221 \textcolor{red}{2} \\ 1 \ 12 \ 1223141413 \end{array} \Delta = -130, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 49 \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\begin{array}{l} 2221 \textcolor{red}{2} \\ 1 \ 12 \ 1223141413 \end{array} \Delta = -130, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 49 \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{array}$$

$$1 \overset{2221}{12} 1223141413^{\color{red}3} \quad \Delta = -138, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: $1 \overset{221}{12} 12231414132$

$$1 \overset{221}{12} 12231414132 \quad \Delta = -131, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{221}{12} 12231414132^{\color{red}2} \quad \Delta = -131, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{221}{12} 12231414132^{\color{red}2} \quad \Delta = -131, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1 \overset{221}{12} 12231414132^{\color{red}2} \quad \Delta = -131, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1 \overset{221}{12} 12231414132^{\color{red}2} \quad \Delta = -131, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1 \overset{221}{12} 12231414132^{\color{red}2} \quad \Delta = -133, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: $1 \overset{21513221}{12} 122222$

$$1 \overset{21513221}{12} 122222 \quad \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{21513221}{12} 122222 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$1 \overset{21513221}{12} 122222^{\color{red}2} \quad \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1 \overset{21513221}{12} 122222 \quad \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{21513221}{12} 122222 \quad \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: $1 \overset{321513221}{12} 12222$

$$1 \overset{321513221}{12} 12222 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{321513221}{12} 12222 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{321513221}{12} 12222 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{321513221}{12} 12222^{\color{red}2} \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: $1 \overset{2321513221}{12} 1222$

$$1 \overset{2321513221}{12} 1222 \quad \Delta = -129, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$\begin{array}{ll}
1 \overset{2321513221}{12} 1222 \quad \Delta = -129, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{2321513221}{12} 1222 \quad \Delta = -129, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1 \overset{2321513221}{12} 1222 \quad \Delta = -129, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{1513221}{12} 122222 & \\
1 \overset{1513221}{12} 122222 \quad \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{1513221}{12} 122222 \quad \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1 \overset{1513221}{12} 122222 \quad \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{31513221}{12} 12222 & \\
1 \overset{31513221}{12} 12222 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1 \overset{31513221}{12} 12222 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1 \overset{31513221}{12} 12222 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1 \overset{31513221}{12} 12222 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \overset{31513221}{12} 12222 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1 \overset{31513221}{12} 12222 \quad \Delta = -134, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{231513221}{12} 1222 & \\
1 \overset{231513221}{12} 1222 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \overset{231513221}{12} 1222 \quad \Delta = -135, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1 \overset{231513221}{12} 1222 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1 \overset{231513221}{12} 1222 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \overset{231513221}{12} 1222 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1 \overset{231513221}{12} 1222 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{array}$$

Base: $1^{2231513221} 12^{12} 122$

$$\begin{aligned}
1^{2231513221} 12^{12} 122 \quad \Delta = -135, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1^{2231513221} 12^{12} 122 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1^{2231513221} 12^{12} 122 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{2231513221} 12^{12} 122 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{2231513221} 12^{12} 122 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1^{2231513221} 12^{12} 122 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

Base: $21^{21} 12^{12} 2231414132$

$$\begin{aligned}
21^{21} 12^{12} 2231414132 \quad \Delta = -131, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21^{21} 12^{12} 2231414132 \quad \Delta = -131, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
21^{21} 12^{12} 2231414132 \quad \Delta = -131, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21^{21} 12^{12} 2231414132 \quad \Delta = -133, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
21^{21} 12^{12} 2231414132 \quad \Delta = -131, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
21^{21} 12^{12} 2231414132 \quad \Delta = -131, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: $21^{2231513221} 12^{12} 12$

$$\begin{aligned}
21^{2231513221} 12^{12} 12 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
21^{2231513221} 12^{12} 12 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21^{2231513221} 12^{12} 12 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21^{2231513221} 12^{12} 12 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
21^{2231513221} 12^{12} 12 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21^{2231513221} 12^{12} 12 \quad \Delta = -135, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: $21 \overset{313221}{12} 1223141$

$$\begin{aligned}
21 \overset{313221}{12} 1223141 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 21 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
21 \overset{313221}{12} 1223141 \text{ [#=2]} \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 93 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
21 \overset{313221}{12} 1223141 \textcolor{red}{3} \text{ [#=3]} \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
21 \overset{313221}{12} 1223141 \textcolor{red}{3} \text{ [#=2]} \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
21 \overset{313221}{12} 12231\textcolor{red}{4}1 \quad \Delta = -111, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21 \textcolor{red}{2} \overset{313221}{12} 1223141 \quad \Delta = -111, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21 \overset{313221}{12} 1223141 \quad \Delta = -111, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21 \overset{313221}{12} 12231\textcolor{red}{4} \quad \Delta = -111, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
21 \overset{313221}{12} 1223141 \quad \Delta = -111, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

Base: $21 \overset{13221}{12} 12231413$

$$\begin{aligned}
21 \overset{13221}{12} 12231\textcolor{red}{4}13 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
21 \overset{313221}{12} 12231413 \text{ [#=5]} \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
21 \overset{313221}{12} 12231413 \text{ [#=4]} \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21 \overset{313221}{12} 12231413 \text{ [#=2]} \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
21 \overset{313221}{12} 12231413 \text{ [#=3]} \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21 \overset{13221}{12} 12231\textcolor{red}{4}13 \quad \Delta = -111, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21 \overset{13221}{12} 12231\textcolor{red}{4}13 \quad \Delta = -111, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21 \textcolor{red}{2} \overset{13221}{12} 12231413 \quad \Delta = -111, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21 \overset{13221}{12} 12231413 \quad \Delta = -111, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

$$21 \overset{13221}{12} \overset{2}{12231413} \quad \Delta = -111, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: $21 \overset{2221}{12} 122314141$

$$21 \overset{2221}{12} 122314141 \overset{3}{3} \text{ [#=3]} \quad \Delta = -130, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$21 \overset{2221}{12} 122314141 \overset{3}{3} \text{ [#=2]} \quad \Delta = -130, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$21 \overset{2221}{12} 122314141 \quad \Delta = -122, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$21 \overset{2221}{12} 122314141 \quad \Delta = -122, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$21 \overset{2221}{12} 122314141 \overset{2}{2} \quad \Delta = -122, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$21 \overset{2221}{12} 122314141 \overset{2}{2} \quad \Delta = -122, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$21 \overset{2221}{12} 122314141 \overset{2}{2} \quad \Delta = -122, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: $21 \overset{221}{12} 1223141413$

$$21 \overset{221}{12} 1223141413 \overset{3}{3} \quad \Delta = -138, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$21 \overset{221}{12} 1223141413 \overset{2}{2} \quad \Delta = -130, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$21 \overset{2221}{12} 1223141413 \quad \Delta = -130, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$21 \overset{221}{12} 1223141413 \overset{2}{2} \quad \Delta = -130, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$21 \overset{221}{12} 1223141413 \overset{2}{2} \quad \Delta = -130, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$21 \overset{221}{12} 1223141413 \quad \Delta = -130, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$21 \overset{221}{12} 1223141413 \overset{2}{2} \quad \Delta = -130, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: $21 \overset{21513221}{12} 12222$

$$21 \overset{21513221}{12} 12222 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$21 \overset{21513221}{12} 12222 \quad \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$21 \overset{21513221}{12} 12222 \quad \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$\begin{array}{ll}
21^{21513221} 12^{12222} & \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
21^{21513221\textcolor{red}{2}} 12^{12222} & \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21^{\textcolor{red}{2}21513221} 12^{12222} & \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\textbf{Base: } 21^{321513221} 12^{1222} & \\
21^{321513221} 12^{1222} & \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21^{321513221} 12^{1222} & \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21^{\textcolor{red}{2}321513221} 12^{1222} & \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21^{321513221\textcolor{red}{2}} 12^{1222} & \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21^{321513221} 12^{1222} & \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\textbf{Base: } 21^{2321513221} 12^{122} & \\
21^{\textcolor{red}{2}2321513221} 12^{122} & \Delta = -129, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21^{2321513221\textcolor{red}{2}} 12^{122} & \Delta = -129, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21^{2321513221} 12^{122} & \Delta = -129, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
21^{2321513221} 12^{122} & \Delta = -129, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21^{2321513221} 12^{122} & \Delta = -129, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\textbf{Base: } 21^{1513221} 12^{12222} & \\
21^{1513221\textcolor{red}{2}} 12^{12222} & \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21^{\textcolor{red}{2}1513221} 12^{12222} & \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21^{1513221} 12^{12222} & \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21^{1513221} 12^{12222} & \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\textbf{Base: } 21^{31513221} 12^{1222} &
\end{array}$$

$$\begin{aligned}
21^{31513221} 12^{1222} \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21^{31513221} 12^{1222} \quad \Delta = -134, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
21^{31513221} 12^{1222} \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21^{31513221} 12^{1222} \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21^{231513221} 12^{1222} \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21^{315132212} 12^{1222} \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21^{31513221} 12^{1222} \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: $21^{231513221} 12^{1222}$

$$\begin{aligned}
21^{231513221} 12^{122} \quad \Delta = -135, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21^{2231513221} 12^{122} \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21^{2315132212} 12^{122} \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21^{231513221} 12^{122} \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21^{231513221} 12^{122} \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21^{231513221} 12^{122} \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
21^{231513221} 12^{122} \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: $13221^{13221} 12^{12231}$

$$\begin{aligned}
13221^{13221} 12^{12231} \mathbf{3} \text{ [#=3]} \quad \Delta = -92, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
13221^{13221} 12^{12231} \mathbf{3} \text{ [#=2]} \quad \Delta = -92, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
13221^{13221} 12^{12231} \mathbf{3} \text{ [#=4]} \quad \Delta = -92, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
13221^{13221} 12^{12231} \mathbf{3} \text{ [#=5]} \quad \Delta = -92, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

$$13221 \overset{2}{1}3221 \quad \Delta = -84, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: $413221 \overset{3221}{12} 12231$

$$413221 \overset{2}{1}3221 \quad \Delta = -112, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$413221 \overset{3221}{12} 12231 \overset{4}{1} \quad \Delta = -140, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$413221 \overset{3221}{12} 12231 \overset{5}{1} \quad \Delta = -164, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$413221 \overset{3221}{12} 12231 \overset{3}{1} \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$413221 \overset{3221}{12} 12231 \overset{3}{1} [\# = 2] \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$413221 \overset{3221}{12} 12231 \overset{3}{1} [\# = 3] \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$413221 \overset{3221}{12} 12231 \overset{2}{1} \quad \Delta = -112, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$413221 \overset{3221}{12} 12231 \quad \Delta = -112, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$413221 \overset{3221}{12} 12231 \overset{3}{1} [\# = 4] \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$413221 \overset{3221}{12} 12231 \overset{3}{1} [\# = 5] \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$413221 \overset{3221}{12} 12231 \overset{2}{1} \quad \Delta = -112, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$413221 \overset{2}{1}3221 \quad \Delta = -112, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

Base: $413221 \overset{221}{121} 122314$

$$413221 \overset{2}{1}3221 \quad \Delta = -131, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$413221 \overset{221}{121} 122314 \quad \Delta = -131, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$413221 \overset{2}{1}3221 \quad \Delta = -131, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: $3221 \overset{3221}{12} 1223141$

$$3221 \overset{3221}{12} 1223141 \overset{3}{1} \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$3221 \overset{3221}{12} 1223141 \overset{2}{1} \quad \Delta = -112, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\begin{aligned}
& \overset{3221}{3221\ 12\ 1223141\textcolor{red}{3}}_{[\#=3]} \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{3221}{3221\ 12\ 1223141\textcolor{red}{3}}_{[\#=2]} \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{3221\textcolor{red}{2}}{3221\ 12\ 1223141} \quad \Delta = -112, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{\textcolor{red}{2}3221}{3221\ 12\ 1223141} \quad \Delta = -112, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{3221}{3221\ 12\ 1223141\textcolor{red}{2}} \quad \Delta = -112, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{3221}{3221\ 12\ 1223141\textcolor{red}{4}} \quad \Delta = -140, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \textbf{Base: } \overset{13221}{221\ 12\ 1223141} \\
& \overset{13221}{221\ 12\ 1223141} \quad \Delta = -103, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{\textcolor{red}{3}13221}{221\ 12\ 1223141}_{[\#=3]} \quad \Delta = -111, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{13221}{221\ 12\ 1223141\textcolor{red}{3}}_{[\#=3]} \quad \Delta = -111, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& \overset{\textcolor{red}{3}13221}{221\ 12\ 1223141}_{[\#=4]} \quad \Delta = -111, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{\textcolor{red}{3}13221}{221\ 12\ 1223141}_{[\#=2]} \quad \Delta = -111, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{13221}{221\ 12\ 1223141\textcolor{red}{3}}_{[\#=2]} \quad \Delta = -111, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& \overset{13221\textcolor{red}{2}}{221\ 12\ 1223141} \quad \Delta = -103, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& \overset{13221}{221\ 12\ 1223141\textcolor{red}{2}} \quad \Delta = -103, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{\textcolor{red}{2}13221}{221\ 12\ 1223141} \quad \Delta = -103, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{\textcolor{red}{3}13221}{221\ 12\ 1223141}_{[\#=5]} \quad \Delta = -111, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& \textbf{Base: } \overset{3221}{41413221\ 12\ 122} \\
& \overset{\textcolor{red}{2}3221}{41413221\ 12\ 122} \quad \Delta = -131, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& \overset{3221\textcolor{red}{2}}{41413221\ 12\ 122} \quad \Delta = -131, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{2}{4}1413221 \overset{3221}{12} 122 \quad \Delta = -131, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 41413221 \overset{3221}{12} 122 \quad \Delta = -131, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& 41413221 \overset{2}{12} \overset{3221}{12} 122 \quad \Delta = -131, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: $221 \overset{221}{12} 122314141$

$$\begin{aligned}
& 221 \overset{221}{12} 122314141 \overset{3}{\#=3} \quad \Delta = -130, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& 221 \overset{221}{12} 122314141 \overset{3}{\#=2} \quad \Delta = -130, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& 221 \overset{221}{12} 122314141 \overset{2}{\#} \quad \Delta = -122, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 221 \overset{221}{12} 122314141 \overset{2}{\#} \quad \Delta = -122, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& 221 \overset{221}{12} 122314141 \overset{2}{\#} \quad \Delta = -122, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& 221 \overset{221}{12} 122314141 \overset{2}{\#} \quad \Delta = -122, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: $2221 \overset{1513221}{12} 1223$

$$\begin{aligned}
& 2221 \overset{1513221}{12} 1223 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 2221 \overset{1513221}{12} 1223 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 2221 \overset{1513221}{12} \overset{2}{12} 223 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 2221 \overset{2}{1513221} \overset{2}{12} 1223 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: $221 \overset{21513221}{12} 1222$

$$\begin{aligned}
& 221 \overset{21513221}{12} 1222 \quad \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 221 \overset{21513221}{12} 1222 \quad \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 221 \overset{21513221}{12} 1222 \quad \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& 221 \overset{2}{21513221} \overset{2}{12} 1222 \quad \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 221 \overset{21513221}{12} \overset{2}{12} 222 \quad \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

$$221 \overset{21513221}{12} 1222 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$\text{Base: } 221 \overset{321513221}{12} 122$$

$$221 \overset{321513221}{12} 122 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$221 \overset{321513221}{12} 122 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$221 \overset{321513221}{12} 122 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$221 \overset{2321513221}{12} 122 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\text{Base: } 221 \overset{1513221}{12} 1222$$

$$221 \overset{1513221}{12} 1222 \quad \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$221 \overset{1513221}{12} 1222 \quad \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$221 \overset{1513221}{12} 1222 \quad \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$221 \overset{21513221}{12} 1222 \quad \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\text{Base: } 221 \overset{31513221}{12} 122$$

$$221 \overset{31513221}{12} 122 \quad \Delta = -134, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$221 \overset{31513221}{12} 122 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$221 \overset{31513221}{12} 122 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$221 \overset{31513221}{12} 122 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$221 \overset{31513221}{12} 122 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$221 \overset{231513221}{12} 122 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{21,2231513221}{12} 1$$

$$1 \overset{21,2231513221}{12} 1 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1 \overset{21,2231513221}{12} 1 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1 \overset{21,2231513221}{12} 1 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$1 \overset{21,2231513221}{12} 1 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1 \overset{21,2231513221}{12} 1 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 70 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$1 \overset{21,2231513221}{12} 1 \quad \Delta = -135, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{1,21513221}{12} 12222$$

$$1 \overset{1,21513221}{12} 12222 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \overset{1,21513221}{12} 12222 \quad \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{1,21513221}{12} 12222 \quad \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{1,21513221}{12} 12222 \quad \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{1,21513221}{12} 12222 \quad \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{1,321513221}{12} 1222$$

$$1 \overset{1,321513221}{12} 1222 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{1,321513221}{12} 1222 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1 \overset{1,321513221}{12} 1222 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1 \overset{1,321513221}{12} 1222 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{221,2321513221}{12} 1$$

$$1 \overset{221,2321513221}{12} 1 \quad \Delta = -129, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{221,2321513221}{12} 1 \quad \Delta = -129, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1 \overset{221,2321513221}{12} 1 \quad \Delta = -129, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1 \overset{221,2321513221}{12} 1 \quad \Delta = -129, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{1,1513221}{12} 12222$$

$$\begin{array}{lcl}
1^{1,1513221} \begin{array}{c} 12 \\ 12222 \end{array} & \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1^{1,1513221} \begin{array}{c} 12 \\ 12222 \end{array} & \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{1,1513221\textcolor{red}{2}} \begin{array}{c} 12 \\ 12222 \end{array} & \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\textbf{Base: } 1^{1,31513221} \begin{array}{c} 12 \\ 1222 \end{array} & & \\
1^{1,31513221} \begin{array}{c} 12 \\ 1222 \end{array} & \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1^{1,31513221} \begin{array}{c} 12 \\ 1222 \end{array} & \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{1,31513221} \begin{array}{c} 12 \\ 1222 \end{array} & \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1^{1,31513221} \begin{array}{c} 12 \\ 1222 \end{array} & \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{1,31513221\textcolor{red}{2}} \begin{array}{c} 12 \\ 1222 \end{array} & \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1^{1,31513221} \begin{array}{c} 12 \\ 1222 \end{array} & \Delta = -134, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\textbf{Base: } 1^{221,231513221} \begin{array}{c} 12 \\ 1 \end{array} & & \\
1^{221,231513221} \begin{array}{c} 12 \\ 1 \end{array} & \Delta = -135, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1^{221,231513221} \begin{array}{c} 12 \\ 1 \end{array} & \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{221,231513221} \begin{array}{c} 12 \\ 1 \end{array} & \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{221,231513221} \begin{array}{c} 12 \\ 1 \end{array} & \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1^{221,231513221} \begin{array}{c} 12 \\ 1 \end{array} & \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1^{221,231513221} \begin{array}{c} 12 \\ 1 \end{array} & \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\textbf{Base: } 1^{21,2321513221} \begin{array}{c} 12 \\ 12 \end{array} & & \\
1^{21,2321513221\textcolor{red}{2}} \begin{array}{c} 12 \\ 12 \end{array} & \Delta = -129, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1^{21,2321513221} \begin{array}{c} 12 \\ 12 \end{array} & \Delta = -129, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
1^{21,2321513221} \begin{array}{c} 12 \\ 12 \end{array} & \Delta = -129, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{array}$$

$$1^{21,2321513221} \begin{smallmatrix} 12 \\ 12 \end{smallmatrix} \quad \Delta = -129, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

Base: $1^{21,231513221} \begin{smallmatrix} 12 \\ 12 \end{smallmatrix}$

$$1^{21,231513221} \begin{smallmatrix} 12 \\ 12 \end{smallmatrix} \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1^{21,231513221\color{red}2} \begin{smallmatrix} 12 \\ 12 \end{smallmatrix} \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1^{21,231513221} \begin{smallmatrix} 12 \\ 12 \end{smallmatrix} \quad \Delta = -135, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1^{21,231513221} \begin{smallmatrix} 12 \\ 12 \end{smallmatrix} \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1^{21,231513221} \begin{smallmatrix} 12 \\ 12 \end{smallmatrix} \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1^{21,231513221} \begin{smallmatrix} 12 \\ 12 \end{smallmatrix} \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: $1^{21,21513221} \begin{smallmatrix} 12 \\ 1222 \end{smallmatrix}$

$$1^{21,21513221} \begin{smallmatrix} 12 \\ 1222 \end{smallmatrix} \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1^{21,21513221\color{red}2} \begin{smallmatrix} 12 \\ 1222 \end{smallmatrix} \quad \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1^{21,21513221} \begin{smallmatrix} 12 \\ 1222 \end{smallmatrix} \quad \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1^{21,21513221} \begin{smallmatrix} 12 \\ 1222 \end{smallmatrix} \quad \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1^{21,21513221} \begin{smallmatrix} 12 \\ 1222 \end{smallmatrix} \quad \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1^{21,21513221} \begin{smallmatrix} 12 \\ 1222 \end{smallmatrix} \quad \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: $1^{221,321513221} \begin{smallmatrix} 12 \\ 12 \end{smallmatrix}$

$$1^{221,321513221} \begin{smallmatrix} 12 \\ 12 \end{smallmatrix} \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1^{221,321513221} \begin{smallmatrix} 12 \\ 12 \end{smallmatrix} \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1^{221,321513221} \begin{smallmatrix} 12 \\ 12 \end{smallmatrix} \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1^{221,321513221} \begin{smallmatrix} 12 \\ 12 \end{smallmatrix} \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1^{221,321513221\color{red}2} \begin{smallmatrix} 12 \\ 12 \end{smallmatrix} \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: $1 \overset{21,1513221}{12} 1222$

$$1 \overset{21,1513221}{12} \overset{2}{1222} \quad \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{21,1513221}{12} 1222 \quad \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{21,1513221}{12} 1222 \quad \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{21,1513221}{12} 1222 \quad \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: $1 \overset{221,31513221}{12} 12$

$$1 \overset{221,31513221}{12} 12 \quad \Delta = -134, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{221,31513221}{12} 12 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{221,31513221}{12} 12 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1 \overset{221,31513221}{12} 12 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1 \overset{221,31513221}{12} 12 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \overset{221,31513221}{12} 12 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{221,31513221}{12} \overset{2}{12} \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: $1 \overset{221,1513221}{12} 1223$

$$1 \overset{3}{221,1513221} 12 1223 \quad \Delta = -136, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$3221 \overset{221,1513221}{12} \overset{2}{1} \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{221,1513221}{12} 1223 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{221,1513221}{12} 1223 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{221,1513221}{12} 1223 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1 \overset{221,1513221}{12} 1223 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

Base: $1 \overset{221,21513221}{12} 122$

Base: $21 \overset{21,1513221}{12} 1223$

$$\begin{aligned}
21 \overset{21,1513221}{12} 1223 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21 \overset{21,1513221}{12} 1223 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21 \overset{21,1513221}{12} 1223 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
21 \overset{21,1513221}{12} 1223 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: $21 \overset{221,21513221}{12} 12$

$$\begin{aligned}
21 \overset{221,21513221}{12} 12 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
21 \overset{221,21513221}{12} 12 \quad \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21 \overset{221,21513221}{12} 12 \quad \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21 \overset{221,21513221}{12} 12 \quad \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21 \overset{221,21513221}{12} 12 \quad \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21 \overset{221,21513221}{12} 12 \quad \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: $21 \overset{221,1513221}{12} 12$

$$\begin{aligned}
21 \overset{221,1513221}{12} 12 \quad \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21 \overset{221,1513221}{12} 12 \quad \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21 \overset{221,1513221}{12} 12 \quad \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21 \overset{221,1513221}{12} 12 \quad \Delta = -118, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1222222222222222^2

$$2222222222222221 \overset{2}{12} \quad \Delta = -57, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: 122314141414132^2

$$\begin{aligned}
2 \overset{8}{1223141414141} 2 \quad \Delta = -74, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2 \overset{7}{1223141414141} 2 \quad \Delta = -46, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

$$2^{\overline{6}1223141414141}3^{\overline{2}}2 \quad \Delta = -19, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$2^{\overline{12}1223141414141}3^{\overline{2}}2 \quad \Delta = -189, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 8 \end{pmatrix}$$

Base: 131414141414132²

$$2^{\overline{6}1314141414141}3^{\overline{2}}2 \quad \Delta = -46, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$2^{\overline{5}1314141414141}3^{\overline{2}}2 \quad \Delta = -20, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 2141414141414131

$$2^{\overline{4}1414141414141}31 \quad \Delta = -13, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 22 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: 1223151316131512

$$1223151316131512^{\overline{3}} \quad \Delta = -24, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$2151316131513221^{\overline{3}}_{[\# = 2]} \quad \Delta = -24, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$2151316131513221^{\overline{3}}_{[\# = 3]} \quad \Delta = -24, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$2151316131513221^{\overline{3}}_{[\# = 4]} \quad \Delta = -24, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$2151316131513221^{\overline{3}}_{[\# = 5]} \quad \Delta = -24, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

$$1223151316131512^{\overline{2}} \quad \Delta = -16, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1223151316131512^{\overline{2}} \quad \Delta = -16, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$2151316131513221^{\overline{3}}_{[\# = 6]} \quad \Delta = -24, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$2151316131513221^{\overline{3}}_{[\# = 7]} \quad \Delta = -24, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix}$$

$$2151316131513221^{\overline{3}}_{[\# = 8]} \quad \Delta = -24, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix}$$

$$2151316131513221^{\overline{3}}_{[\# = 9]} \quad \Delta = -24, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix}$$

$$1223151316131512^{\overline{2}} \quad \Delta = -16, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1223151316131512^{\overline{2}} \quad \Delta = -16, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1223151316131512^2 \quad \Delta = -16, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1223151316131512^2 \quad \Delta = -16, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 1315131613151321

$$1315131613151321^2 \quad \Delta = -24, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$12315131613151313_{[\# = 5]} \quad \Delta = -32, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$13151316131513213_{[\# = 6]} \quad \Delta = -32, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$12315131613151313_{[\# = 7]} \quad \Delta = -32, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$1315131613151321^2 \quad \Delta = -24, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$12315131613151313_{[\# = 4]} \quad \Delta = -32, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$13151316131513213_{[\# = 7]} \quad \Delta = -32, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$12315131613151313_{[\# = 8]} \quad \Delta = -32, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$13151316131513213_{[\# = 8]} \quad \Delta = -32, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$12315131613151313_{[\# = 2]} \quad \Delta = -32, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$13151316131513213_{[\# = 9]} \quad \Delta = -32, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix}$$

$$13151316131513213_{[\# = 2]} \quad \Delta = -32, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$12315131613151313_{[\# = 3]} \quad \Delta = -32, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1315131613151321^2 \quad \Delta = -24, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1315131613151321^2 \quad \Delta = -24, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1315131613151321^2 \quad \Delta = -24, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1315131613151321^2 \quad \Delta = -24, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\begin{aligned}
1231513161315131\bar{3} \text{ }_{[\# = 6]} \quad \Delta = -32, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1315131613151321\bar{3} \text{ }_{[\# = 5]} \quad \Delta = -32, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1315131613151321\bar{3} \text{ }_{[\# = 4]} \quad \Delta = -32, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1315131613151321\bar{3} \text{ }_{[\# = 3]} \quad \Delta = -32, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1231513161315131\bar{3} \text{ }_{[\# = 9]} \quad \Delta = -32, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1223151316131¹51

$$\begin{aligned}
1\bar{3}1223151316131\bar{1} \text{ }_{[\# = 9]} \quad \Delta = -24, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix} \\
1223151316131\bar{1}^2 \quad \Delta = -16, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1223151316131\bar{1}^2 \quad \Delta = -16, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1\bar{3}1223151316131\bar{1} \text{ }_{[\# = 2]} \quad \Delta = -24, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1223151316131\bar{1}^{21} \quad \Delta = -16, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1\bar{3}1223151316131\bar{1} \text{ }_{[\# = 5]} \quad \Delta = -24, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
1\bar{3}1223151316131\bar{1} \text{ }_{[\# = 4]} \quad \Delta = -24, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1223151316131\bar{1}^2 \quad \Delta = -16, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1223151316131\bar{1}^2 \quad \Delta = -16, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1\bar{3}1223151316131\bar{1} \text{ }_{[\# = 8]} \quad \Delta = -24, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
1\bar{3}1223151316131\bar{1} \text{ }_{[\# = 7]} \quad \Delta = -24, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
1\bar{3}1223151316131\bar{1} \text{ }_{[\# = 6]} \quad \Delta = -24, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1\bar{3}1223151316131\bar{1} \text{ }_{[\# = 3]} \quad \Delta = -24, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1613161315131613

$$\begin{aligned}
1613161315131613^2 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1613161315131613^2 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
1613161315131613^2 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 144 & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
1613161315131613^2 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1613161315131613^2 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1613161315131613^2 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1613161315131613^2 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 6123151316131612

$$\begin{aligned}
6123151316131612^3 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
6123151316131612^2 \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
6123151316131612^2 \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
6123151316131612^2 \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
6123151316131612^2 \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
6123151316131612^2 \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
6123151316131612^2 \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
6123151316131612^2 \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix}
\end{aligned}$$

Base: 2161316131513161

$$\begin{aligned}
2161316131513161^2 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
2161316131513161^2 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
2161316131513161^2 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
2161316131513161^2 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
2161316131513161 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
2161316131513161 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2161316131513161 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
2161316131513161 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: 2216131613151321

$$\begin{aligned}
2216131613151321 \textcolor{red}{3} \text{ [#=8]} \quad \Delta = -50, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
2216131613151321 \quad \Delta = -42, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2216131613151321 \quad \Delta = -42, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2216131613151321 \quad \Delta = -42, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
2216131613151321 \quad \Delta = -42, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2216131613151321 \textcolor{red}{3} \text{ [#=9]} \quad \Delta = -50, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
2216131613151321 \quad \Delta = -42, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2216131613151321 \quad \Delta = -42, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
2216131613151321 \quad \Delta = -51, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2216131613151321 \textcolor{red}{3} \text{ [#=2]} \quad \Delta = -50, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2216131613151321 \textcolor{red}{3} \text{ [#=6]} \quad \Delta = -50, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
2216131613151321 \quad \Delta = -50, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2216131613151321 \textcolor{red}{3} \text{ [#=4]} \quad \Delta = -50, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
2216131613151321 \textcolor{red}{3} \text{ [#=5]} \quad \Delta = -50, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
2216131613151321 \textcolor{red}{3} \text{ [#=3]} \quad \Delta = -50, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

$$2216131613151321\overset{2}{3} \text{ }_{[\# = 7]} \quad \Delta = -50, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

Base: 6131513161316123

$$6131513161316123 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$6131513161316123 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$6131513161316123 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

$$6131513161316123 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$6131513161316123 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$6131513161316123 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$6131513161316123 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 1321613161315131

$$1321613161315131\overset{2}{3} \text{ }_{[\# = 7]} \quad \Delta = -58, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$1321613161315131\overset{2}{3} \text{ }_{[\# = 4]} \quad \Delta = -58, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1321613161315131\overset{2}{3} \text{ }_{[\# = 8]} \quad \Delta = -58, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1315131613161231\overset{2}{3} \text{ }_{[\# = 5]} \quad \Delta = -58, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$1321613161315131\overset{2}{3} \text{ }_{[\# = 9]} \quad \Delta = -58, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$1321613161315131 \quad \Delta = -50, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1321613161315131\overset{2}{3} \text{ }_{[\# = 5]} \quad \Delta = -58, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1321613161315131\overset{2}{3} \text{ }_{[\# = 6]} \quad \Delta = -58, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1315131613161231\overset{2}{3} \text{ }_{[\# = 4]} \quad \Delta = -58, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1321613161315131 \quad \Delta = -50, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\begin{aligned}
1321613161315131 \quad \Delta = -50, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1321613161315131 \quad \Delta = -50, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1321613161315131 \quad \Delta = -50, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1321613161315131 \quad \Delta = -50, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
13151316131612313_{[\# = 3]} \quad \Delta = -58, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
13151316131612313_{[\# = 2]} \quad \Delta = -58, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
13216131613151313_{[\# = 2]} \quad \Delta = -58, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
13216131613151313_{[\# = 3]} \quad \Delta = -58, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\text{Base: } 1232161316131512 & \\
1232161316131512 \quad \Delta = -44, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1232161316131512 \quad \Delta = -44, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1232161316131512 \quad \Delta = -44, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1232161316131512 \quad \Delta = -44, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1232161316131512 \quad \Delta = -44, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1232161316131512 \quad \Delta = -44, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21513161316123213_{[\# = 8]} \quad \Delta = -52, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
21513161316123213_{[\# = 9]} \quad \Delta = -52, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
21513161316123213_{[\# = 6]} \quad \Delta = -52, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
21513161316123213_{[\# = 5]} \quad \Delta = -52, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
21513161316123213_{[\# = 4]} \quad \Delta = -52, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
2151316131612321\overset{3}{3} \text{ }_{[\# = 3]} \quad \Delta = -52, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
2151316131612321\overset{3}{3} \text{ }_{[\# = 2]} \quad \Delta = -52, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
123216131613151\overset{3}{2} \quad \Delta = -52, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
2151316131612321\overset{3}{3} \text{ }_{[\# = 7]} \quad \Delta = -52, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix}
\end{aligned}$$

Base: 6123216131613151

$$\begin{aligned}
6123216131613151\overset{2}{1} \quad \Delta = -122, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
6123216131613151\overset{2}{1} \quad \Delta = -122, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
6123216131613151\overset{2}{1} \quad \Delta = -122, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
6123216131613151\overset{2}{1} \quad \Delta = -122, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
6123216131613151\overset{2}{1} \quad \Delta = -122, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
6123216131613151\overset{2}{1} \quad \Delta = -122, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 100 \quad & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{aligned}$$

Base: 123216131613151¹

$$\begin{aligned}
1\overset{3}{1}232161316131\overset{1}{5}1 \text{ }_{[\# = 2]} \quad \Delta = -52, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1\overset{3}{1}232161316131\overset{1}{5}1 \text{ }_{[\# = 4]} \quad \Delta = -52, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1\overset{3}{1}232161316131\overset{1}{5}1 \text{ }_{[\# = 3]} \quad \Delta = -52, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1\overset{3}{1}232161316131\overset{1}{5}1 \text{ }_{[\# = 7]} \quad \Delta = -52, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
123216131613151\overset{2}{1} \quad \Delta = -44, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
123216131613151\overset{2}{1} \quad \Delta = -44, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
123216131613151\overset{2}{1} \quad \Delta = -44, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1\overset{3}{1}232161316131\overset{1}{5}1 \text{ }_{[\# = 9]} \quad \Delta = -52, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
123216131613151\overset{2}{1} \quad \Delta = -44, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

$$1^{\overline{31232161316131}}_5 1_{[\# = 8]} \quad \Delta = -52, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix}$$

$$1^{\overline{31232161316131}}_5 1_{[\# = 6]} \quad \Delta = -52, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

$$1^{\overline{31232161316131}}_5 1_{[\# = 5]} \quad \Delta = -52, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix}$$

$$123216131613151^{\overline{2}}_1 \quad \Delta = -44, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 1316131613151315

$$1316131613151315^{\overline{2}} \quad \Delta = -102, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1316131613151315^{\overline{2}} \quad \Delta = -101, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$51315131613161313^{\overline{2}}_{[\# = 2]} \quad \Delta = -109, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$51315131613161313^{\overline{2}}_{[\# = 3]} \quad \Delta = -109, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$51315131613161313^{\overline{2}}_{[\# = 5]} \quad \Delta = -109, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$51315131613161313^{\overline{2}}_{[\# = 9]} \quad \Delta = -109, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix}$$

$$51315131613161313^{\overline{2}}_{[\# = 6]} \quad \Delta = -109, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$51315131613161313^{\overline{2}}_{[\# = 8]} \quad \Delta = -109, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$51315131613161313^{\overline{2}}_{[\# = 4]} \quad \Delta = -109, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$51315131613161313^{\overline{2}}_{[\# = 7]} \quad \Delta = -109, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$1316131613151315^{\overline{2}} \quad \Delta = -101, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1316131613151315^{\overline{2}} \quad \Delta = -101, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1316131613151315^{\overline{2}} \quad \Delta = -101, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1316131613151315^{\overline{2}} \quad \Delta = -101, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1316131613151315^{\overline{2}} \quad \Delta = -101, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1316131613151315 \quad \Delta = -101, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 3131613161315131

$$3131613161315131 \text{ }_{[\# = 3]} \quad \Delta = -65, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 213 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

$$3131613161315131 \quad \Delta = -57, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$3131613161315131 \quad \Delta = -57, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$3131613161315131 \quad \Delta = -57, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$3131613161315131 \quad \Delta = -57, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$3131613161315131 \quad \Delta = -57, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$3131613161315131 \quad \Delta = -57, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$3131613161315131 \quad \Delta = -57, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$3131613161315131 \quad \Delta = -85, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$3131613161315131 \quad \Delta = -65, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 21 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$31316131613151313 \text{ }_{[\# = 9]} \quad \Delta = -65, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$3131613161315131 \text{ }_{[\# = 2]} \quad \Delta = -65, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 93 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$31316131613151313 \text{ }_{[\# = 2]} \quad \Delta = -65, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$31316131613151313 \text{ }_{[\# = 7]} \quad \Delta = -65, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$31316131613151313 \text{ }_{[\# = 6]} \quad \Delta = -65, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$31316131613151313 \text{ }_{[\# = 3]} \quad \Delta = -65, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$31316131613151313 \text{ }_{[\# = 8]} \quad \Delta = -65, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$31316131613151313 \text{ }_{[\# = 5]} \quad \Delta = -65, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$3131613161315131\overset{3}{3} \text{ }_{[\# = 4]} \quad \Delta = -65, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 1413161316131512

$$1413161316131512 \quad \Delta = -69, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1413161316131512 \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$1413161316131512 \quad \Delta = -69, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1413161316131512 \quad \Delta = -69, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1413161316131512 \quad \Delta = -69, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1413161316131512 \quad \Delta = -69, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1413161316131512 \quad \Delta = -69, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1413161316131512 \quad \Delta = -69, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$2151316131613141\overset{3}{3} \text{ }_{[\# = 2]} \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1413161316131512 \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 21 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$2151316131613141\overset{3}{3} \text{ }_{[\# = 3]} \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: 141316131613151¹

$$1\overset{3}{1413161316131}5\overset{1}{1} \text{ }_{[\# = 3]} \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$141316131613151\overset{2}{1} \quad \Delta = -69, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$141316131613151\overset{2}{1} \quad \Delta = -69, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$141316131613151\overset{2}{1} \quad \Delta = -69, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$141316131613151\overset{2}{1} \quad \Delta = -69, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$141316131613151\overset{2}{1} \quad \Delta = -69, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$141316131613151\overset{2}{1} \quad \Delta = -69, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1^{\overline{31413161316131}}_5 1_{[\# = 2]} \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$141316131613151^{\overline{3}}_1 \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 21 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

Base: 2161316131612315

$$2161316131612315^{\overline{2}} \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$2161316131612315^{\overline{2}} \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$2161316131612315^{\overline{2}} \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$2161316131612315^{\overline{2}} \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$2161316131612315^{\overline{2}} \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$2161316131612315^{\overline{2}} \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$2161316131612315^{\overline{2}} \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$2161316131612315^{\overline{3}} \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: 2216131613161231

$$2216131613161231\overline{3}_{[\# = 5]} \quad \Delta = -76, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$2216131613161231^{\overline{3}} \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$2216131613161231^{\overline{3}} \quad \Delta = -76, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 27 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$2216131613161231\overline{3}_{[\# = 2]} \quad \Delta = -76, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$2216131613161231\overline{3}_{[\# = 3]} \quad \Delta = -76, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$2216131613161231\overline{3}_{[\# = 4]} \quad \Delta = -76, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$2216131613161231^{\overline{2}} \quad \Delta = -68, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$2216131613161231^{\overline{2}} \quad \Delta = -68, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$2216131613161231^2 \quad \Delta = -68, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$2216131613161231^2 \quad \Delta = -68, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$2216131613161231^2 \quad \Delta = -68, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$2216131613161231^2 \quad \Delta = -68, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 2216131613161313

$$2216131613161313^3 \quad \Delta = -83, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 27 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$2216131613161313^2 \quad \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$2216131613161313^3 \quad \Delta = -84, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$2216131613161313^3_{[\# = 3]} \quad \Delta = -83, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$2216131613161313^3_{[\# = 2]} \quad \Delta = -83, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$2216131613161313^3 \quad \Delta = -83, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$2216131613161313^2 \quad \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$2216131613161313^2 \quad \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$2216131613161313^2 \quad \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$2216131613161313^2 \quad \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$2216131613161313^2 \quad \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$2216131613161313^4 \quad \Delta = -103, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 9 & 5 \\ 5 & 9 \end{pmatrix}$$

$$2216131613161313^2 \quad \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 612315131613161¹

$$612315131613161^2 \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$612315131613161^2 \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$\begin{array}{ll}
61231513161\overset{2}{1}\overset{1}{} & \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
61231513161\overset{2}{1}\overset{1}{} & \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
61231513161\overset{2}{1}\overset{1}{} & \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
61231513161\overset{2}{1}\overset{1}{} & \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix}
\end{array}$$

Base: $\overset{1}{}161316131513161$

$$\begin{array}{ll}
\overset{1}{}161316131513161\overset{2}{} & \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
\overset{1}{}161316131513161\overset{2}{} & \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
\overset{1}{}161316131513161\overset{2}{} & \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
\overset{1}{}161316131513161\overset{2}{} & \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\overset{1}{2}161316131513161 & \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\overset{1}{2}161316131513161 & \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{array}$$

Base: $\overset{1}{}161316131612315$

$$\begin{array}{ll}
\overset{1}{}161316131612315\overset{2}{} & \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\overset{1}{}161316131612315\overset{2}{} & \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\overset{1}{2}161316131612315 & \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\overset{1}{}161316131612315\overset{2}{} & \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\overset{1}{2}161316131612315 & \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\overset{1}{2}161316131612315 & \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{array}$$

Base: $\overset{21}{}16131613151321$

$$\begin{array}{ll}
\overset{21}{}16131613151321\overset{2}{3} [\# = 2] & \Delta = -50, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\overset{21}{}16131613151321\overset{2}{} & \Delta = -42, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{ll}
{}^{21}16131613151321\mathbf{3}_{[\#=7]} & \Delta = -50, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
{}^{21}16131613151321\mathbf{3}_{[\#=8]} & \Delta = -50, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
{}^{21}16131613151321\mathbf{3}_{[\#=9]} & \Delta = -50, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
{}^{21}16131613151321\mathbf{3}_{[\#=3]} & \Delta = -50, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
{}^{21}\mathbf{2}16131613151321 & \Delta = -42, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
{}^{21}16131613151321\mathbf{2} & \Delta = -42, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
{}^{21}\mathbf{2}16131613151321 & \Delta = -42, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
{}^{21}16131613151321 & \Delta = -42, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
{}^{21}16131613151321\mathbf{3}_{[\#=6]} & \Delta = -50, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
{}^{21}16131613151321\mathbf{3}_{[\#=5]} & \Delta = -50, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
{}^{21}16131613151321\mathbf{3}_{[\#=4]} & \Delta = -50, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
{}^{21}\mathbf{2}16131613151321 & \Delta = -42, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
{}^{21}16131613151321 & \Delta = -50, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\text{Base: } {}^{21}16131613161231 & \\
{}^{21}16131613161231\mathbf{3}_{[\#=2]} & \Delta = -76, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
{}^{21}\mathbf{2}16131613161231 & \Delta = -68, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^{21}16131613161231\mathbf{2} & \Delta = -68, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
{}^{21}16131613161231 & \Delta = -68, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
{}^{21}\mathbf{2}16131613161231 & \Delta = -68, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
{}^{21}16131613161231 & \Delta = -76, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{ll}
{}^{21}16131613161231\overset{3}{\color{red}}_{[\#=3]} & \Delta = -76, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
{}^{21}16131613161231\overset{3}{\color{red}}_{[\#=4]} & \Delta = -76, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
{}^{21}16131613161231\overset{3}{\color{red}}_{[\#=5]} & \Delta = -76, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
{}^{21}16131613161231\overset{2}{\color{red}} & \Delta = -68, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
{}^{21}16131613161231\overset{2}{\color{red}} & \Delta = -68, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\text{Base: } {}^{21}16131613161313 & \\
{}^{21}16131613161313\overset{2}{\color{red}} & \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
{}^{21}16131613161313\overset{2}{\color{red}} & \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
{}^{21}16131613161313\overset{2}{\color{red}} & \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
{}^{21}16131613161313 & \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
{}^{21}16131613161313\overset{2}{\color{red}} & \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
{}^{21}16131613161313\overset{4}{\color{red}} & \Delta = -103, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 9 & 5 \\ 5 & 9 \end{pmatrix} \\
{}^{21}16131613161313\overset{2}{\color{red}} & \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
{}^{21}16131613161313\overset{2}{\color{red}} & \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^{21}16131613161313\overset{3}{\color{red}}_{[\#=3]} & \Delta = -83, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
{}^{21}16131613161313\overset{3}{\color{red}}_{[\#=2]} & \Delta = -83, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
{}^{21}16131613161313\overset{3}{\color{red}} & \Delta = -83, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
{}^{21}16131613161313 & \Delta = -83, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\text{Base: } {}^{31}61315131613161 & \\
{}^{31}61315131613161\overset{2}{\color{red}} & \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{ll}
\overset{2}{6}13151316131\overset{31}{6}1 & \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
613151316131\overset{231}{6}1 & \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
613151316131\overset{2}{6}1\overset{31}{6}1 & \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
613151316131\overset{31}{6}1 & \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
613151316131\overset{2}{6}1\overset{31}{6}1 & \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
613151316131\overset{2}{6}1\overset{31}{6}1 & \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
613151316131\overset{31}{6}1\overset{2}{6}1 & \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\text{Base: } 1\overset{131}{6}13161315131 & \\
1\overset{131}{6}13161315131\overset{131}{3} [\#=8] & \Delta = -57, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1\overset{1312}{6}13161315131 & \Delta = -49, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1\overset{131}{6}13161315131\overset{2}{6}1 & \Delta = -49, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1\overset{3131}{6}13161315131 [\#=5] & \Delta = -57, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1\overset{131}{6}13161315131\overset{131}{3} [\#=5] & \Delta = -57, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1\overset{3131}{6}13161315131 [\#=8] & \Delta = -57, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1\overset{131}{6}13161315131\overset{131}{3} [\#=9] & \Delta = -57, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1\overset{3131}{6}13161315131 [\#=9] & \Delta = -57, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1\overset{3131}{6}13161315131 [\#=6] & \Delta = -57, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1\overset{131}{6}13161315131\overset{131}{3} [\#=6] & \Delta = -57, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1\overset{3131}{6}13161315131 [\#=7] & \Delta = -57, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1\overset{3131}{6}13161315131 [\#=2] & \Delta = -57, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{ll}
\begin{array}{c} 131 \\ 1 \ 6 \ 13161315131\textcolor{red}{3} \end{array} \begin{array}{l} [\# = 2] \\ \Delta = -57, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \end{array} & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\begin{array}{c} 131 \\ 1 \ 6 \ 13161315131\textcolor{red}{3} \end{array} \begin{array}{l} [\# = 7] \\ \Delta = -57, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \end{array} & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
\begin{array}{c} \textcolor{red}{3}131 \\ 1 \ 6 \ 13161315131 \end{array} \begin{array}{l} [\# = 3] \\ \Delta = -57, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \end{array} & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\begin{array}{c} 131 \\ 1 \ 6 \ 13161315131\textcolor{red}{3} \end{array} \begin{array}{l} [\# = 4] \\ \Delta = -57, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \end{array} & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\begin{array}{c} \textcolor{red}{3}131 \\ 1 \ 6 \ 13161315131 \end{array} \begin{array}{l} [\# = 4] \\ \Delta = -57, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \end{array} & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
\begin{array}{c} 131 \\ 1 \ 6 \ 13161315131\textcolor{red}{3} \end{array} \begin{array}{l} [\# = 3] \\ \Delta = -57, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \end{array} & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\begin{array}{c} 131 \\ 1 \ 6 \ 13161315131\textcolor{red}{2} \end{array} & \begin{array}{l} \Delta = -49, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \end{array} \\
\begin{array}{c} 131 \\ 1 \ 6 \ 13161315131 \end{array} & \begin{array}{l} \Delta = -49, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \end{array} \\
\begin{array}{c} 131 \\ 1 \ 6 \ 13161315131\textcolor{red}{2} \end{array} & \begin{array}{l} \Delta = -49, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \\ \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \end{array} \\
\begin{array}{c} 131 \ \textcolor{red}{2} \\ 1 \ 6 \ 13161315131 \end{array} & \begin{array}{l} \Delta = -49, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \\ \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \end{array} \\
\text{Base: } \begin{array}{c} 131 \\ 1 \ 6 \ 13161316122 \end{array} & \\
\begin{array}{c} 131 \\ 1 \ 6 \ 13161316122\textcolor{red}{2} \end{array} & \begin{array}{l} \Delta = -67, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \\ \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \end{array} \\
\begin{array}{c} 131 \\ 1 \ 6 \ 13161316122\textcolor{red}{3} \end{array} & \begin{array}{l} \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \\ \begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \end{array} \\
\begin{array}{c} \textcolor{red}{3}131 \\ 1 \ 6 \ 13161316122 \end{array} \begin{array}{l} [\# = 9] \\ \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \end{array} & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
\begin{array}{c} \textcolor{red}{3}131 \\ 1 \ 6 \ 13161316122 \end{array} \begin{array}{l} [\# = 8] \\ \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \end{array} & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
\begin{array}{c} \textcolor{red}{3}131 \\ 1 \ 6 \ 13161316122 \end{array} \begin{array}{l} [\# = 2] \\ \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \end{array} & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\begin{array}{c} 131 \\ 1 \ 6 \ 13161316122 \end{array} & \begin{array}{l} \Delta = -67, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \end{array} \\
\begin{array}{c} 131 \\ 1 \ 6 \ 13161316122\textcolor{red}{2} \end{array} & \begin{array}{l} \Delta = -67, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \\ \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \end{array} \\
\begin{array}{c} 131\textcolor{red}{2} \\ 1 \ 6 \ 13161316122 \end{array} & \begin{array}{l} \Delta = -67, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \\ \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \end{array} \\
\begin{array}{c} 131 \ \textcolor{red}{2} \\ 1 \ 6 \ 13161316122 \end{array} & \begin{array}{l} \Delta = -67, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \\ \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \end{array}
\end{array}$$

$$\begin{aligned}
& \overset{131}{1} \overset{2}{6} 13161316122 \quad \Delta = -67, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{3131}{1} \overset{2}{6} 13161316122 \text{ [#}=7] \quad \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
& \overset{3131}{1} \overset{2}{6} 13161316122 \text{ [#}=6] \quad \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
& \overset{3131}{1} \overset{2}{6} 13161316122 \text{ [#}=5] \quad \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
& \overset{3131}{1} \overset{2}{6} 13161316122 \text{ [#}=4] \quad \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& \overset{3131}{1} \overset{2}{6} 13161316122 \text{ [#}=3] \quad \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{131}{1} \overset{3}{6} 13161316122 \quad \Delta = -76, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}
\end{aligned}$$

Base: $\overset{5131}{1} \overset{2}{6} 1316131612$

$$\begin{aligned}
& \overset{5131}{1} \overset{2}{6} 1316131612 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{5131}{1} \overset{2}{6} 1316131612 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{5131}{1} \overset{2}{6} 1316131612 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
& \overset{5131}{1} \overset{2}{6} 1316131612 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{31}{5131} \overset{2}{6} 1316131612 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{5131}{1} \overset{2}{6} 1316131612 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{5131}{1} \overset{2}{6} 1316131612 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
& \overset{5131}{1} \overset{3}{6} 1316131612 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
& \overset{21}{5131} \overset{2}{6} 1316131612 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{5131}{1} \overset{2}{6} 1316131612 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{16} \overset{1}{131613161315}$

$$\begin{aligned}
& \overset{1}{16} \overset{2}{131613161315} \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{1}{16} \overset{2}{131613161315} \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1\overset{1}{6}131\overset{2}{6}131\overset{1}{6}1315 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1\overset{1}{6}131\overset{1}{6}131\overset{1}{6}1315 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1\overset{1}{6}131\overset{1}{6}131\overset{1}{6}1315 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1\overset{1}{6}131\overset{1}{6}131\overset{31}{6}1315 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1\overset{1}{6}131\overset{1}{6}131\overset{21}{6}1315 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1\overset{1}{6}131\overset{1}{6}131\overset{21}{6}1315 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1\overset{21}{6}131\overset{1}{6}131\overset{1}{6}131$

$$\begin{aligned}
1\overset{21}{6}131\overset{1}{6}131\overset{1}{6}131 \quad \Delta = -67, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1\overset{21}{6}131\overset{1}{6}131\overset{1}{6}131\overset{3}{3} \text{ [#=9]} \quad \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
1\overset{21}{6}131\overset{1}{6}131\overset{1}{6}131 \quad \Delta = -67, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1\overset{21}{6}131\overset{1}{6}131\overset{1}{6}131\overset{3}{3} \text{ [#=7]} \quad \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
1\overset{21}{6}131\overset{1}{6}131\overset{1}{6}131 \quad \Delta = -67, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1\overset{21}{6}131\overset{1}{6}131\overset{1}{6}131 \quad \Delta = -67, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1\overset{21}{6}131\overset{1}{6}131\overset{1}{6}131 \quad \Delta = -67, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1\overset{21}{6}131\overset{1}{6}131\overset{1}{6}131\overset{3}{3} \text{ [#=8]} \quad \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1\overset{21}{6}131\overset{1}{6}131\overset{1}{6}131 \quad \Delta = -67, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1\overset{21}{6}131\overset{1}{6}131\overset{1}{6}131\overset{3}{3} \text{ [#=4]} \quad \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1\overset{21}{6}131\overset{1}{6}131\overset{1}{6}131 \quad \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1\overset{21}{6}131\overset{1}{6}131\overset{1}{6}131\overset{3}{3} \text{ [#=2]} \quad \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1\overset{21}{6}131\overset{1}{6}131\overset{1}{6}131\overset{3}{3} \text{ [#=3]} \quad \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{l}
\begin{array}{c} 21 \\ 1613161316131\mathbf{\bar{3}} \end{array} \begin{array}{c} 1 \\ \end{array} \quad \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\begin{array}{c} 21 \\ 1613161316131\mathbf{\bar{3}} \end{array} \begin{array}{c} 1 \\ \end{array} \quad \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix}
\end{array}$$

Base: 613161316131513

$$\begin{array}{l}
\begin{array}{c} 1 \\ 61316131613151\mathbf{\bar{3}} \end{array} \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\begin{array}{c} 31 \\ 613161316131513 \end{array} \quad \Delta = -135, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\begin{array}{c} 1 \\ 61316131613151\mathbf{\bar{3}} \end{array} \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\begin{array}{c} 1 \\ 61316131613151\mathbf{\bar{3}} \end{array} \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\begin{array}{c} 21 \\ 613161316131513 \end{array} \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\begin{array}{c} 21 \\ 613161316131513 \end{array} \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
\begin{array}{c} 12 \\ 613161316131513 \end{array} \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
\begin{array}{c} 1 \\ 61316131613151\mathbf{\bar{3}} \end{array} \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
\begin{array}{c} 2 \\ 61316131613151\mathbf{\bar{3}} \end{array} \begin{array}{c} 1 \\ \end{array} \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{array}$$

Base: 161316131613151

$$\begin{array}{l}
\begin{array}{c} 1 \\ 161316131613151\mathbf{\bar{3}} \end{array} \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\begin{array}{c} 21 \\ 161316131613151 \end{array} \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\begin{array}{c} 1 \\ 161316131613151\mathbf{\bar{3}} \end{array} \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\begin{array}{c} 1 \\ 161316131613151\mathbf{\bar{3}} \end{array} \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\begin{array}{c} 12 \\ 161316131613151 \end{array} \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\begin{array}{c} 2 \\ 161316131613151\mathbf{\bar{3}} \end{array} \begin{array}{c} 1 \\ \end{array} \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\begin{array}{c} 21 \\ 161316131613151 \end{array} \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{array}$$

$$16131\overset{31}{6}131613151 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: $316131\overset{1}{6}13161315$

$$316131\overset{1}{6}13161\overset{2}{3}15 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$316131\overset{21}{6}13161315 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 58 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$316131\overset{1}{6}1\overset{2}{3}161315 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$\overset{2}{3}16131\overset{1}{6}13161315 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$316131\overset{21}{6}13161315 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$316131\overset{1}{6}1\overset{2}{3}161315 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$316131\overset{31}{6}13161315 \quad \Delta = -135, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\overset{2}{3}16131\overset{1}{6}13161315 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$316131\overset{12}{6}13161315 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 63 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

Base: $1316131\overset{1}{6}1316131$

$$1316131\overset{1}{6}1316131\overset{3}{3} \text{ [#=5]} \quad \Delta = -83, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1316131\overset{31}{6}1316131 \quad \Delta = -83, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1316131\overset{1}{6}1316131\overset{3}{3} \text{ [#=4]} \quad \Delta = -83, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1316131\overset{1}{6}1316131\overset{3}{3} \text{ [#=6]} \quad \Delta = -83, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$1316131\overset{1}{6}1316131\overset{3}{3} \text{ [#=7]} \quad \Delta = -83, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$1316131\overset{1}{6}1316131\overset{3}{3} \text{ [#=8]} \quad \Delta = -83, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$1316131\overset{1}{6}1316131\overset{3}{3} \text{ [#=9]} \quad \Delta = -83, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix}$$

$$1316131\overset{1}{6}1316131\overset{3}{3} \text{ [#=3]} \quad \Delta = -83, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$\begin{aligned}
131613161316131\overset{1}{3} \text{ }_{[\# = 2]} \quad \Delta = -83, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
131613161316131\overset{2}{1} \quad \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
131613161316131\overset{2}{1} \quad \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
131613161316131\overset{2}{1} \quad \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1316131\overset{2}{6}1316131 \quad \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\text{Base: } 31613161315131 \quad & \\
31\overset{31}{6}13161315131\overset{2}{1} \quad \Delta = -57, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
31\overset{31}{6}13161315131\overset{2}{1} \quad \Delta = -57, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
31\overset{231}{6}13161315131 \quad \Delta = -57, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
31\overset{31}{6}13161315131\overset{2}{1} \quad \Delta = -57, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
31\overset{31}{6}13161315131\overset{2}{1} \quad \Delta = -57, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
31\overset{31}{6}1316131\overset{2}{5}131 \quad \Delta = -57, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
31\overset{31}{6}13161315131\overset{2}{1} \quad \Delta = -57, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
31\overset{31}{6}13161315131\overset{2}{1} \quad \Delta = -58, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
31\overset{31}{6}13161315131\overset{2}{3} \text{ }_{[\# = 5]} \quad \Delta = -65, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 100 \quad & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
31\overset{31}{6}13161315131\overset{2}{5} \quad \Delta = -109, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
31\overset{31}{6}13161315131\overset{2}{3} \text{ }_{[\# = 7]} \quad \Delta = -65, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 196 \quad & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
31\overset{31}{6}13161315131\overset{2}{3} \text{ }_{[\# = 8]} \quad \Delta = -65, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 256 \quad & \begin{pmatrix} 0 & 16 & 2 \\ 16 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
31\overset{31}{6}13161315131\overset{2}{3} \text{ }_{[\# = 6]} \quad \Delta = -65, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 144 \quad & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
31\overset{31}{6}13161315131\overset{2}{3} \text{ }_{[\# = 9]} \quad \Delta = -65, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 324 \quad & \begin{pmatrix} 0 & 18 & 2 \\ 18 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
^{31}31613161315131\mathbf{4} \quad \Delta = -85, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
^{31}31613161315131\mathbf{3} \quad \Delta = -65, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
^{31}31613161315131\mathbf{3}_{[\# = 2]} \quad \Delta = -65, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
^{31}31613161315131\mathbf{6} \quad \Delta = -135, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
^{31}31613161315131\mathbf{3}_{[\# = 3]} \quad \Delta = -65, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
^{31}31613161315131\mathbf{3}_{[\# = 4]} \quad \Delta = -65, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix}
\end{aligned}$$

Base: $^{31}31613161316122$

$$\begin{aligned}
^{31}31613161316122\mathbf{3} \quad \Delta = -83, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 36 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
^{31}\mathbf{2}31613161316122 \quad \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
^{31}31613161316122\mathbf{2} \quad \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
^{31}31613161316122\mathbf{2} \quad \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
^{31}31613161316122\mathbf{2} \quad \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
^{31}31613161316122\mathbf{2} \quad \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 37 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
^{31}31613161316122\mathbf{3} \quad \Delta = -84, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
^{31}31613161316122\mathbf{3} \quad \Delta = -84, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 15 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
^{\mathbf{2}31}31613161316122 \quad \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: $^{31}316131613161\mathbf{21}$

$$\begin{aligned}
^{31}316131613161\mathbf{21} \quad \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
^{31}\mathbf{2}316131613161\mathbf{21} \quad \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
^{31}316131613161\mathbf{21}\mathbf{3} \quad \Delta = -83, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{ll}
{}^{31} 1 \overset{2}{6} 1316131 \overset{21}{6} 1 & \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^{31} 1 \overset{2}{6} 1316131 \overset{21}{6} 1 & \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
{}^{231} 1 \overset{2}{6} 1316131 \overset{21}{6} 1 & \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
{}^{31} 1 \overset{2}{6} 1316131 \overset{21}{6} 1 \overset{2}{6} 1 & \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
{}^{31} 1 \overset{2}{6} 1316131 \overset{21}{6} 1 & \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^{31} 1 \overset{2}{6} 1316131 \overset{321}{6} 1 & \Delta = -84, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
{}^{31} 1 \overset{2}{6} 1316131 \overset{21}{6} 1 & \Delta = -83, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 15 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{array}$$

Base: 7122315131613151

$$\begin{array}{ll}
71223151316131 \overset{2}{5} 1 & \Delta = -121, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
71223151316131 \overset{2}{5} 1 & \Delta = -121, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
71223151316131 \overset{2}{5} 1 & \Delta = -121, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
71223151316131 \overset{2}{5} 1 & \Delta = -121, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
71223151316131 \overset{2}{5} 1 & \Delta = -121, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
71223151316131 \overset{2}{5} 1 & \Delta = -121, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{array}$$

Base: 7123151316131513

$$\begin{array}{ll}
7123151316131 \overset{2}{5} 13 & \Delta = -129, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
7123151316131 \overset{2}{5} 13 & \Delta = -129, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
7123151316131 \overset{2}{5} 13 & \Delta = -129, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
7123151316131 \overset{2}{5} 13 & \Delta = -129, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
7123151316131 \overset{2}{5} 13 & \Delta = -129, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
7123151316131513^2 \quad \Delta = -129, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
7123151316131513^2 \quad \Delta = -129, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1513161315132171

$$\begin{aligned}
1513161315132171^2 \quad \Delta = -121, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
1513161315132171^2 \quad \Delta = -121, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1513161315132171^2 \quad \Delta = -121, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1513161315132171^2 \quad \Delta = -121, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1513161315132171^2 \quad \Delta = -121, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
1513161315132171^2 \quad \Delta = -121, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

Base: 4131613161232171

$$\begin{aligned}
4131613161232171^3 \quad \Delta = -133, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
4131613161232171^2 \quad \Delta = -125, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
4131613161232171^2 \quad \Delta = -125, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
4131613161232171^2 \quad \Delta = -125, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
4131613161232171^2 \quad \Delta = -125, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
4131613161232171^2 \quad \Delta = -125, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
4131613161232171^2 \quad \Delta = -125, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1316131612321712

$$\begin{aligned}
21712321613161313^3_{[\# = 6]} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
21712321613161313^3_{[\# = 7]} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
21712321613161313^3_{[\# = 8]} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}
\end{aligned}$$

$$2171232161316131\overset{2}{3} \text{ }_{[\# = 9]} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$1316131612321712 \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$2171232161316131\overset{2}{3} \text{ }_{[\# = 4]} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$2171232161316131\overset{2}{3} \text{ }_{[\# = 2]} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$2171232161316131\overset{2}{3} \text{ }_{[\# = 3]} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$2171232161316131\overset{2}{3} \text{ }_{[\# = 5]} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$1316131612321712 \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1316131612321712 \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1316131612321712 \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1316131612321712 \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1316131612321712 \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 1232171231513221

$$1223151321712321\overset{2}{3} \text{ }_{[\# = 3]} \quad \Delta = -3, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1232171231513221\overset{2}{3} \text{ }_{[\# = 3]} \quad \Delta = -3, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1232171231513221\overset{2}{3} \text{ }_{[\# = 7]} \quad \Delta = -3, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$1232171231513221\overset{2}{3} \text{ }_{[\# = 9]} \quad \Delta = -3, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$1223151321712321\overset{2}{3} \text{ }_{[\# = 9]} \quad \Delta = -3, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$1223151321712321\overset{2}{3} \text{ }_{[\# = 5]} \quad \Delta = -3, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$1232171231513221\overset{2}{3} \text{ }_{[\# = 8]} \quad \Delta = -3, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$1232171231513221\overset{2}{3} \text{ }_{[\# = 4]} \quad \Delta = -3, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$\begin{aligned}
1223151321712321\overset{2}{3} \text{ [#=4]} \quad \Delta = -3, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1223151321712321\overset{2}{3} \text{ [#=7]} \quad \Delta = -3, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1232171231513221\overset{2}{3} \text{ [#=6]} \quad \Delta = -3, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1223151321712321\overset{2}{3} \text{ [#=2]} \quad \Delta = -3, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1232171231513221\overset{2}{3} \text{ [#=2]} \quad \Delta = -3, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1232171231513221\overset{2}{3} \text{ [#=5]} \quad \Delta = -3, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1223151321712321\overset{2}{3} \text{ [#=8]} \quad \Delta = -3, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1223151321712321\overset{2}{3} \text{ [#=6]} \quad \Delta = -3, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}
\end{aligned}$$

Base: 3161232171232171

$$\begin{aligned}
3161\overset{2}{2}32171232171 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
3161\overset{2}{2}32171232171 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
31612321712321\overset{2}{7}1 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
31612321\overset{2}{7}71232171 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 163 & 25 \\ 2 & 25 & 8 \end{pmatrix} \\
31612321\overset{2}{7}71232171 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1612321712321712

$$\begin{aligned}
1612321\overset{2}{7}712321712 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1612321712321\overset{2}{7}12 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1612321\overset{2}{7}712321712 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
161\overset{2}{2}321712321712 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1612321712321\overset{2}{7}12 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: 6123217123217123

$$\begin{aligned}
6123217123217123^2 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
6123217123217123^2 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
6123217123217123^2 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
6123217123217123^2 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
6123217123217123^2 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1321712321712321

$$\begin{aligned}
1321712321712321^3_{[\# = 2]} \quad \Delta = -58, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1321712321712321^3_{[\# = 7]} \quad \Delta = -58, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1321712321712321^3_{[\# = 8]} \quad \Delta = -58, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1321712321712321^2 \quad \Delta = -50, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1321712321712321^2 \quad \Delta = -50, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1321712321712321^2 \quad \Delta = -50, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1321712321712321^2 \quad \Delta = -50, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1232171232171231^3_{[\# = 3]} \quad \Delta = -58, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1321712321712321^3_{[\# = 5]} \quad \Delta = -58, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1321712321712321^3_{[\# = 9]} \quad \Delta = -58, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1321712321712321^3_{[\# = 4]} \quad \Delta = -58, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1321712321712321^3_{[\# = 6]} \quad \Delta = -58, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1232171232171231^3_{[\# = 4]} \quad \Delta = -58, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1321712321712321^3_{[\# = 3]} \quad \Delta = -58, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

$$1232171232171231\overset{1}{\textcolor{red}{3}}_{[\# = 2]} \quad \Delta = -58, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1232171232171231\overset{1}{\textcolor{red}{3}}_{[\# = 5]} \quad \Delta = -58, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

Base: $\overset{1}{171232161316131}$

$$\overset{1}{171232161316131}\overset{1}{\textcolor{red}{3}}_{[\# = 9]} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$\overset{1}{171232161316131}\overset{2}{\textcolor{red}{3}} \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$\overset{1}{171232161316131}\overset{2}{\textcolor{red}{3}} \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$\overset{1}{171232161316131}\overset{2}{\textcolor{red}{3}} \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$\overset{1}{171232161316131}\overset{2}{\textcolor{red}{3}} \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$\overset{1}{171232161316131}\overset{1}{\textcolor{red}{3}}_{[\# = 5]} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\overset{1}{171232161316131}\overset{1}{\textcolor{red}{3}}_{[\# = 3]} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$\overset{1}{171232161316131}\overset{1}{\textcolor{red}{3}}_{[\# = 7]} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\overset{1}{171232161316131}\overset{1}{\textcolor{red}{3}}_{[\# = 8]} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\overset{1}{171232161316131}\overset{1}{\textcolor{red}{3}}_{[\# = 4]} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\overset{1}{171232161316131}\overset{1}{\textcolor{red}{3}}_{[\# = 6]} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\overset{1}{171232161316131}\overset{1}{\textcolor{red}{3}}_{[\# = 2]} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$\overset{1}{171232161316131}\overset{2}{\textcolor{red}{3}} \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: $\overset{1}{171232171232161}$

$$\overset{1}{171232171232161}\overset{2}{\textcolor{red}{3}} \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\overset{1}{171232171232161}\overset{2}{\textcolor{red}{3}} \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$\overset{1}{171232171232161}\overset{2}{\textcolor{red}{3}} \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$${}^1 171232171232161 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

Base: 8122315132171232

$$8122315132171232 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$8122315132171232 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

$${}^2 8122315132171232 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$8122315132171232 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$8122315132171232 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix}$$

Base: 8123217123151322

$$8123217123151322 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 256 \quad \begin{pmatrix} 0 & 16 & 2 \\ 16 & 206 & 28 \\ 2 & 28 & 8 \end{pmatrix}$$

$$8123217123151322 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 361 \quad \begin{pmatrix} 0 & 19 & 2 \\ 19 & 283 & 33 \\ 2 & 33 & 8 \end{pmatrix}$$

$${}^2 8123217123151322 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$8123217123151322 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix}$$

$$8123217123151322 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: 4131613151322181

$$4131613151322181 \quad \Delta = -125, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

$${}^3 4131613151322181 \quad \Delta = -133, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$4131613151322181 \quad \Delta = -125, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$4131613151322181 \quad \Delta = -125, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$${}^2 4131613151322181 \quad \Delta = -125, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$4131613151322181 \quad \Delta = -125, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$${}^2 4131613151322181 \quad \Delta = -125, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: 3217123151322181

$$\begin{aligned}
3217123151322181^2 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
3217123151322181^2 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
3217123151322181^2 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
3217123151322181^2 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
3217123151322181^2 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

Base: 2321613161232181

$$\begin{aligned}
2321613161232181^2 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
2321613161232181^2 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
2321613161232181^2 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
2321613161232181^2 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
2321613161232181^2 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 206 & 28 \\ 2 & 28 & 8 \end{pmatrix}
\end{aligned}$$

Base: 2315132171232181

$$\begin{aligned}
2315132171232181^2 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
2315132171232181^2 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
2315132171232181^2 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 206 & 28 \\ 2 & 28 & 8 \end{pmatrix} \\
2315132171232181^2 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
2315132171232181^2 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 225 & \begin{pmatrix} 0 & 15 & 2 \\ 15 & 283 & 33 \\ 2 & 33 & 8 \end{pmatrix}
\end{aligned}$$

Base: 2321712315132181

$$\begin{aligned}
2321712315132181^2 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
2321712315132181^2 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
2321712315132181^2 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
2321712315132181 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
2321712315132181 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 219 & 29 \\ 2 & 29 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1316131513221812

$$\begin{aligned}
2181223151316131\overset{2}{3} \text{ [#=4]} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
2181223151316131\overset{2}{3} \text{ [#=5]} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
2181223151316131\overset{2}{3} \text{ [#=6]} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
2181223151316131\overset{2}{3} \text{ [#=7]} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
2181223151316131\overset{2}{3} \text{ [#=8]} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
2181223151316131\overset{2}{3} \text{ [#=9]} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
2181223151316131\overset{2}{3} \text{ [#=3]} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
2181223151316131\overset{2}{3} \text{ [#=2]} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

$$1316131513221812 \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1316131513221812 \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1316131513221812 \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1316131513221812 \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1316131513221812 \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1316131513221812 \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: 1712231513221812

$$1712231513221812 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$1712231513221812 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$\begin{array}{ll}
1712231513221812^2 & \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1712231513221812^2 & \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1712231513221812^2 & \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\text{Base: } 2171231513221812 & \\
2171231513221812^2 & \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
2171231513221812^2 & \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
2171231513221812^2 & \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
2171231513221812^2 & \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
2171231513221812^2 & \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2171231513221812^2 & \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\text{Base: } 171231513221812^1 & \\
171231513221812^1 2 & \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
171231513221812^1 2 & \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
171231513221812^1 2 & \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
171231513221812^1 2 & \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
171231513221812^{12} & \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
\text{Base: } 4131613151321812 & \\
4131613151321812^2 & \Delta = -125, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
4131613151321812^2 & \Delta = -125, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
4131613151321812^3 & \Delta = -133, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
4131613151321812^2 & \Delta = -125, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
4131613151321812 \quad \Delta = -125, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
4131613151321812 \quad \Delta = -125, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
4131613151321812 \quad \Delta = -125, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
4131613151321812 \quad \Delta = -125, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 3216131612321812

$$\begin{aligned}
3216131612321812 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
3216131612321812 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
3216131612321812 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
3216131612321812 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
3216131612321812 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
3216131612321812 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix}
\end{aligned}$$

Base: ³¹16131612321812

$$\begin{aligned}
³¹16131612321812 \quad \Delta = -125, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
³¹16131612321812 \quad \Delta = -125, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
³¹²16131612321812 \quad \Delta = -125, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
³¹16131612321812 \quad \Delta = -125, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
²¹316131612321812 \quad \Delta = -125, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
³¹16131612321812 \quad \Delta = -125, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
³¹16131612321812 \quad \Delta = -125, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
³¹316131612321812 \quad \Delta = -133, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 3217123151321812

$$\begin{aligned}
3217123151321812^2 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
3217123151321812^2 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
3217123151321812^2 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
3217123151321812^2 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
3217123151321812^2 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
3217123151321812^2 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: 3151321712321812

$$\begin{aligned}
3151321712321812^2 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
3151321712321812^2 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
3151321712321812^2 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
3151321712321812^2 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
3151321712321812^2 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 163 & 25 \\ 2 & 25 & 8 \end{pmatrix} \\
3151321712321812^2 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1316131513218122

$$\begin{aligned}
22181231513161313_{[\# = 5]}^2 \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1316131513218122^2 \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
22181231513161313_{[\# = 9]}^2 \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
22181231513161313_{[\# = 8]}^2 \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1316131513218122^2 \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
22181231513161313_{[\# = 7]}^2 \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
2218123151316131\overset{2}{3} \text{ [#=6]} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1316131513218122 \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1316131513218122 \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1316131513218122 \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
2218123151316131\overset{2}{3} \text{ [#=4]} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
2218123151316131\overset{2}{3} \text{ [#=3]} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
2218123151316131\overset{2}{3} \text{ [#=2]} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1316131513218122 \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1316131513218122 \quad \Delta = -106, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
\text{Base: } 2161316123218122 \\
2161316123218122 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
2161316123218122 \quad \Delta = -125, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2161316123218122 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
2161316123218122 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
2161316123218122 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
2161316123218122 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2161316123218122 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2161316123218122 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
\text{Base: } \overset{1}{16}1316123218122 \\
\overset{1}{16}1316123218122 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
\overset{1}{16}1316123218122 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1^1 161316123218122^2 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1^1 161316123218122^2 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
1^1 161316123218122^2 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1^1 161316123218122^2 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 7122315132218123

$$\begin{aligned}
7122315132218123^2 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
7122315132218123^2 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
7122315132218123^2 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
7122315132218123^2 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
7122315132218123^2 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1712231513218122

$$\begin{aligned}
1712231513218122^2 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1712231513218122^2 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1712231513218122^2 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
1712231513218122^2 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1712231513218122^3 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
1712231513218122^2 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1712315132218123

$$\begin{aligned}
1712315132218123^2 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1712315132218123^2 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
1712315132218123^2 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$1712315132218123 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1712315132218123 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: 2171231513218122

$$2171231513218122 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$2171231513218122 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$2171231513218122 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$2171231513218122 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$2171231513218122 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$2171231513218122 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$2171231513218122 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

Base: 1513217123218122

$$1513217123218122 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$1513217123218122 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$1513217123218122 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1513217123218122 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1513217123218122 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1513217123218122 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

Base: 5123217123218122

$$5123217123218122 \quad \Delta = -121, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$5123217123218122 \quad \Delta = -121, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$5123217123218122 \quad \Delta = -121, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$5123217123218122 \quad \Delta = -121, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$5123217123218122^3 \quad \Delta = -130, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$5123217123218122^2 \quad \Delta = -121, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

Base: 171231513218122^1

$$171231513218122^1 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$171231513218122^1 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$171231513218122^1 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$171231513218122^{12} \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$171231513218122^1 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$171231513218122^1 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

Base: 1321812231513221

$$1321812231513221^3_{[\# = 2]} \quad \Delta = -29, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1321812231513221^3_{[\# = 9]} \quad \Delta = -29, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$1321812231513221^3_{[\# = 8]} \quad \Delta = -29, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$1321812231513221^2 \quad \Delta = -21, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1321812231513221^2 \quad \Delta = -21, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1223151322181231^3_{[\# = 5]} \quad \Delta = -29, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1321812231513221^3_{[\# = 7]} \quad \Delta = -29, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$1321812231513221^3_{[\# = 6]} \quad \Delta = -29, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$1321812231513221^3_{[\# = 5]} \quad \Delta = -29, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$1321812231513221^3_{[\# = 4]} \quad \Delta = -29, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1321812231513221^2 \quad \Delta = -21, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$\begin{aligned}
1321812231513221^2 \quad \Delta = -21, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1223151322181231^3_{[\# = 4]} \quad \Delta = -29, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1321812231513221^3_{[\# = 3]} \quad \Delta = -29, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1223151322181231^3_{[\# = 3]} \quad \Delta = -29, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1223151322181231^3_{[\# = 2]} \quad \Delta = -29, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\text{Base: } 1322181231513221 & \\
1322181231513221^3_{[\# = 6]} \quad \Delta = -29, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1322181231513221^3_{[\# = 5]} \quad \Delta = -29, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1322181231513221^3_{[\# = 7]} \quad \Delta = -29, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1322181231513221^3_{[\# = 8]} \quad \Delta = -29, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1322181231513221^3_{[\# = 9]} \quad \Delta = -29, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1223151321812231^3_{[\# = 3]} \quad \Delta = -29, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1322181231513221^2 \quad \Delta = -21, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1223151321812231^3_{[\# = 5]} \quad \Delta = -29, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1322181231513221^3_{[\# = 4]} \quad \Delta = -29, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1223151321812231^3_{[\# = 4]} \quad \Delta = -29, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1223151321812231^3_{[\# = 2]} \quad \Delta = -29, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1322181231513221^3_{[\# = 3]} \quad \Delta = -29, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1322181231513221^2 \quad \Delta = -21, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1322181231513221^2 \quad \Delta = -21, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1322181231513221 \quad \Delta = -21, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
13221812315132213_{[\# = 2]} \quad \Delta = -29, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\text{Base: } 1232181223151321 & \\
12315132218123213_{[\# = 7]} \quad \Delta = -31, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
12315132218123213_{[\# = 5]} \quad \Delta = -31, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1232181223151321 \quad \Delta = -23, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1232181223151321 \quad \Delta = -23, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
12315132218123213_{[\# = 2]} \quad \Delta = -31, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
12321812231513213_{[\# = 2]} \quad \Delta = -31, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
12315132218123213_{[\# = 3]} \quad \Delta = -31, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
12321812231513213_{[\# = 3]} \quad \Delta = -31, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
12315132218123213_{[\# = 4]} \quad \Delta = -31, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
12321812231513213_{[\# = 4]} \quad \Delta = -31, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
12321812231513213_{[\# = 5]} \quad \Delta = -31, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
12321812231513213_{[\# = 7]} \quad \Delta = -31, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
12315132218123213_{[\# = 6]} \quad \Delta = -31, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
12321812231513213_{[\# = 6]} \quad \Delta = -31, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1232181223151321 \quad \Delta = -23, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
12315132218123213_{[\# = 8]} \quad \Delta = -31, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
12321812231513213_{[\# = 8]} \quad \Delta = -31, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix}
\end{aligned}$$

$$1231513221812321\overset{2}{3}_{[\# = 9]} \quad \Delta = -31, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$1232181223151321\overset{2}{3}_{[\# = 9]} \quad \Delta = -31, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix}$$

$$1232181\overset{2}{2}223151321 \quad \Delta = -23, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 1613161232181223

$$16131612321\overset{2}{8}1223 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$16131612321\overset{2}{8}1223 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$16131612321\overset{2}{8}1223 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$16131612321\overset{2}{8}1223 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$16131612321\overset{2}{8}1223 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

Base: 7122315132181223

$$71223151321\overset{2}{8}1223 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$71223151321\overset{2}{8}1223 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$71223151321\overset{2}{8}1223 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$71223151321\overset{2}{8}1223 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$71223151321\overset{2}{8}1223 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: 7123151322181232

$$71231513221\overset{2}{8}1232 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$71231513221\overset{2}{8}1232 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$71231513221\overset{2}{8}1232 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 256 \quad \begin{pmatrix} 0 & 16 & 2 \\ 16 & 174 & 26 \\ 2 & 26 & 8 \end{pmatrix}$$

$$71231513221\overset{2}{8}1232 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$71231513221\overset{2}{8}1232 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

Base: 7123218122315132

$$\begin{array}{ll}
7123218122315132^2 & \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
7123218122315132^2 & \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 361 \quad \begin{pmatrix} 0 & 19 & 2 \\ 19 & 264 & 32 \\ 2 & 32 & 8 \end{pmatrix} \\
7123218122315132^2 & \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
7123218122315132^2 & \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
7123218122315132^2 & \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\text{Base: } 1712315132181223 & \\
1712315132181223^2 & \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1712315132181223^2 & \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1712315132181223^2 & \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1712315132181223^2 & \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1712315132181223^2 & \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\text{Base: } 1712321812231513 & \\
1712321812231513^2 & \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1712321812231513^2 & \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 196 \quad \begin{pmatrix} 0 & 14 & 2 \\ 14 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
1712321812231513^2 & \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
1712321812231513^2 & \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1712321812231513^2 & \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\text{Base: } 2171232181223151 & \\
2171232181223151^2 & \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
2171232181223151^2 & \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2171232181223151^2 & \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
2171232181223151 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
2171232181223151 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 3217123218122315

$$\begin{aligned}
3217123218122315 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 50 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
3217123218122315 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
3217123218122315 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
3217123218122315 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
3217123218122315 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1322181232171231

$$\begin{aligned}
1322181232171231 \quad \Delta = -84, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1321712321812231 \quad \Delta = -76, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1321712321812231 \quad \Delta = -76, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1321712321812231 \quad \Delta = -76, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1321712321812231 \quad \Delta = -76, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1321712321812231 \quad \Delta = -84, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
1322181232171231 \quad \Delta = -84, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1322181232171231 \quad \Delta = -84, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1321712321812231 \quad \Delta = -84, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1321712321812231 \quad \Delta = -84, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1321712321812231 \quad \Delta = -84, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

$$1322181232171231\overset{2}{3} \text{ }_{[\# = 4]} \quad \Delta = -84, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

Base: 5132171232181223

$$5132171232181\overset{2}{2}23 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$51\overset{2}{3}2171232181223 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$51321\overset{2}{7}1232181223 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$5132171232181\overset{2}{2}23 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$51\overset{2}{3}2171232181223 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

Base: 1232171232181222

$$2221812321712321\overset{2}{3} \text{ }_{[\# = 7]} \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix}$$

$$1232171232181\overset{2}{2}22 \quad \Delta = -69, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$2221812321712321\overset{2}{3} \text{ }_{[\# = 9]} \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix}$$

$$2221812321712321\overset{2}{3} \text{ }_{[\# = 3]} \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$2221812321712321\overset{2}{3} \text{ }_{[\# = 4]} \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$2221812321712321\overset{2}{3} \text{ }_{[\# = 2]} \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$2221812321712321\overset{2}{3} \text{ }_{[\# = 8]} \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix}$$

$$2221812321712321\overset{2}{3} \text{ }_{[\# = 5]} \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix}$$

$$2221812321712321\overset{2}{3} \text{ }_{[\# = 6]} \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

$$1232171232181\overset{2}{2}22 \quad \Delta = -69, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$12321\overset{2}{7}1232181222 \quad \Delta = -69, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$123\overset{2}{2}171232181222 \quad \Delta = -69, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

Base: ¹171232181223151

$$\begin{array}{ll}
171232181223151^{\overset{1}{2}} & \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
171232181223151^{\overset{1}{2}} & \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
171232181223151^{\overset{1}{2}} & \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
171232181223151^{\overset{1}{2}} & \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\text{Base: } 1315132181232151 & \\
1315132181232151^{\overset{2}{2}} & \Delta = -74, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1315132181232151^{\overset{2}{2}} & \Delta = -74, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1315132181232151^{\overset{2}{2}} & \Delta = -74, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1315132181232151^{\overset{2}{2}} & \Delta = -74, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
15123218123151313_{\text{[#}=8]} & \Delta = -82, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1315132181232151^{\overset{2}{2}} & \Delta = -74, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
15123218123151313_{\text{[#}=6]} & \Delta = -82, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
15123218123151313_{\text{[#}=7]} & \Delta = -82, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
15123218123151313_{\text{[#}=2]} & \Delta = -82, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
15123218123151313_{\text{[#}=3]} & \Delta = -82, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
15123218123151313_{\text{[#}=4]} & \Delta = -82, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
15123218123151313_{\text{[#}=9]} & \Delta = -82, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
15123218123151313_{\text{[#}=5]} & \Delta = -82, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
\text{Base: } 1315132181231512 & \\
1315132181231512^{\overset{2}{2}} & \Delta = -72, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
1315132181231512^2 \quad \Delta = -72, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1315132181231512^2 \quad \Delta = -72, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1315132181231512^2 \quad \Delta = -72, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1315132181231512^2 \quad \Delta = -72, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1315132181231512^2 \quad \Delta = -72, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1315132181231512^3 \quad \Delta = -80, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
21513218123151313_{\#8} \quad \Delta = -80, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
21513218123151313_{\#3} \quad \Delta = -80, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21513218123151313_{\#2} \quad \Delta = -80, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21513218123151313_{\#9} \quad \Delta = -80, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
21513218123151313_{\#6} \quad \Delta = -80, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
21513218123151313_{\#5} \quad \Delta = -80, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
21513218123151313_{\#4} \quad \Delta = -80, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
21513218123151313_{\#7} \quad \Delta = -80, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
\text{Base: } 4123218123151321 & \\
41232181231513213_{\#5} \quad \Delta = -59, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
4123218123151321^2 \quad \Delta = -51, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
4123218123151321^2 \quad \Delta = -51, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
4123218123151321^3 \quad \Delta = -59, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
41232181231513213_{\#3} \quad \Delta = -59, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
4123218123151321\bar{3} \text{ }_{[\# = 7]} \quad \Delta = -59, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
4123218123151321\bar{3} \text{ }_{[\# = 6]} \quad \Delta = -59, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
4123218123151321\bar{3} \text{ }_{[\# = 8]} \quad \Delta = -59, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
4123218123151321\bar{2} \quad \Delta = -51, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
4123218123151321\bar{2} \quad \Delta = -51, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
4123218123151321\bar{3} \text{ }_{[\# = 2]} \quad \Delta = -59, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
4123218123151321\bar{3} \text{ }_{[\# = 9]} \quad \Delta = -59, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
4123218123151321\bar{3} \text{ }_{[\# = 4]} \quad \Delta = -59, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
4123218123151321\bar{2} \quad \Delta = -51, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1413218123151321

$$\begin{aligned}
1413218123151321\bar{3} \text{ }_{[\# = 8]} \quad \Delta = -57, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1413218123151321\bar{3} \text{ }_{[\# = 6]} \quad \Delta = -57, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1231513218123141\bar{3} \text{ }_{[\# = 3]} \quad \Delta = -57, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1413218123151321\bar{2} \quad \Delta = -49, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1413218123151321\bar{3} \text{ }_{[\# = 2]} \quad \Delta = -57, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1413218123151321\bar{3} \text{ }_{[\# = 7]} \quad \Delta = -57, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
1231513218123141\bar{3} \text{ }_{[\# = 2]} \quad \Delta = -57, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1413218123151321\bar{2} \quad \Delta = -49, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1413218123151321\bar{2} \quad \Delta = -49, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1413218123151321\bar{2} \quad \Delta = -49, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1413218123151321\overset{2}{3} \text{ [#=3]} \quad \Delta = -57, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1413218123151321\overset{2}{3} \text{ [#=4]} \quad \Delta = -57, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1413218123151321\overset{2}{2} \quad \Delta = -51, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1413218123151321\overset{2}{3} \text{ [#=5]} \quad \Delta = -57, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1413218123151321\overset{2}{2} \quad \Delta = -49, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1413218123151321\overset{2}{3} \text{ [#=9]} \quad \Delta = -57, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
\text{Base: } 3132181231513221 & \\
3132181231513221\overset{2}{3} \text{ [#=3]} \quad \Delta = -37, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
3132181231513221\overset{2}{3} \text{ [#=5]} \quad \Delta = -37, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
3132181231513221\overset{2}{3} \text{ [#=9]} \quad \Delta = -37, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
3132181231513221\overset{2}{3} \text{ [#=8]} \quad \Delta = -37, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
3132181231513221\overset{2}{3} \text{ [#=6]} \quad \Delta = -37, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
3132181231513221\overset{2}{2} \quad \Delta = -29, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
3132181231513221\overset{2}{2} \quad \Delta = -29, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
3132181231513221\overset{2}{2} \quad \Delta = -29, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
3132181231513221\overset{2}{3} \text{ [#=7]} \quad \Delta = -37, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
3132181231513221\overset{2}{3} \text{ [#=2]} \quad \Delta = -37, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
3132181231513221\overset{3}{3} \text{ [#=2]} \quad \Delta = -37, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
3132181231513221\overset{3}{2} \quad \Delta = -37, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
3132181231513221\overset{3}{3} \text{ [#=4]} \quad \Delta = -37, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

$$3132181231513221 \quad \Delta = -31, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$3132181231513221 \quad \Delta = -29, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$3132181231513221 \quad \Delta = -29, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 1232181231513215

$$5123151321812321\mathbf{3} \text{ [#=2]} \quad \Delta = -83, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$5123151321812321\mathbf{3} \text{ [#=4]} \quad \Delta = -83, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$5123151321812321\mathbf{3} \text{ [#=8]} \quad \Delta = -83, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$5123151321812321\mathbf{3} \text{ [#=9]} \quad \Delta = -83, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$5123151321812321\mathbf{3} \text{ [#=3]} \quad \Delta = -83, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$5123151321812321\mathbf{3} \text{ [#=5]} \quad \Delta = -83, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$5123151321812321\mathbf{3} \text{ [#=7]} \quad \Delta = -83, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$1232181231513215 \quad \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1232181231513215 \quad \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1232181231513215 \quad \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1232181231513215 \quad \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1232181231513215 \quad \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$5123151321812321\mathbf{3} \text{ [#=6]} \quad \Delta = -83, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

Base: 5123218123151315

$$5123218123151315 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$5123218123151315 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$\begin{aligned}
5123218123151315^2 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
5123218123151315^2 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
5123218123151315^2 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
5123218123151315^2 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
5123218123151315^2 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1513218123151315

$$\begin{aligned}
1513218123151315^2 \quad \Delta = -124, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1513218123151315^2 \quad \Delta = -124, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1513218123151315^2 \quad \Delta = -124, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1513218123151315^2 \quad \Delta = -125, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1513218123151315^2 \quad \Delta = -124, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1513218123151315^2 \quad \Delta = -124, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1513218123151315^2 \quad \Delta = -124, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1232181231513151

$$\begin{aligned}
1232181231513151^2 \quad \Delta = -74, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1513151321812321^3_{[\# = 4]} \quad \Delta = -82, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1513151321812321^3_{[\# = 5]} \quad \Delta = -82, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1513151321812321^3_{[\# = 6]} \quad \Delta = -82, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1513151321812321^3_{[\# = 2]} \quad \Delta = -82, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1513151321812321^3_{[\# = 3]} \quad \Delta = -82, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1513151321812321\overset{2}{3} \text{ }_{[\# = 7]} \quad \Delta = -82, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1513151321812321\overset{2}{3} \text{ }_{[\# = 8]} \quad \Delta = -82, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1513151321812321\overset{2}{3} \text{ }_{[\# = 9]} \quad \Delta = -82, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
12321812315131\overset{2}{51} \quad \Delta = -74, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
12321812315131\overset{2}{51} \quad \Delta = -75, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
12321812315131\overset{2}{51} \quad \Delta = -74, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
12321812315131\overset{2}{51} \quad \Delta = -74, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
12321812315131\overset{2}{51} \quad \Delta = -74, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\text{Base: } 1315132181231\overset{1}{51} & \\
1315132181231\overset{2}{51} \text{ }_{[\# = 4]} \quad \Delta = -72, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1\overset{3}{1}315132181231\overset{1}{51} \text{ }_{[\# = 5]} \quad \Delta = -80, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1\overset{3}{1}315132181231\overset{1}{51} \text{ }_{[\# = 5]} \quad \Delta = -80, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1\overset{3}{1}315132181231\overset{1}{51} \text{ }_{[\# = 2]} \quad \Delta = -80, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1\overset{3}{1}315132181231\overset{1}{51} \text{ }_{[\# = 7]} \quad \Delta = -80, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1315132181231\overset{2}{51} \text{ }_{[\# = 3]} \quad \Delta = -72, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1315132181231\overset{2}{51} \text{ }_{[\# = 3]} \quad \Delta = -72, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1315132181231\overset{2}{51} \text{ }_{[\# = 3]} \quad \Delta = -72, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1\overset{3}{1}315132181231\overset{1}{51} \text{ }_{[\# = 3]} \quad \Delta = -80, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1\overset{3}{1}315132181231\overset{1}{51} \text{ }_{[\# = 9]} \quad \Delta = -80, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1\overset{3}{1}315132181231\overset{1}{51} \text{ }_{[\# = 8]} \quad \Delta = -80, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}
\end{aligned}$$

$$131513218123151^1 \quad \Delta = -72, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1^{31315132181231} 5^1 1_{[\# = 6]} \quad \Delta = -80, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

Base: 6122315132181231

$$61223151321812313_{[\# = 4]} \quad \Delta = -107, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$61223151321812313_{[\# = 2]} \quad \Delta = -107, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$61223151321812313_{[\# = 3]} \quad \Delta = -107, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$6122315132181231^3 \quad \Delta = -107, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 63 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$61223151321812313_{[\# = 5]} \quad \Delta = -107, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$6122315132181231^2 \quad \Delta = -99, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$6122315132181231^2 \quad \Delta = -99, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$6122315132181231^2 \quad \Delta = -99, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$6122315132181231^2 \quad \Delta = -99, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$6122315132181231^2 \quad \Delta = -99, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 4132181231513216

$$4132181231513216^2 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$4132181231513216^2 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$4132181231513216^2 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$4132181231513216^2 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$4132181231513216^2 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$4132181231513216^2 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$4132181231513216 \quad \Delta = -129, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 1612315132181231

$$1612315132181231\overset{2}{3} \text{ [#=4]} \quad \Delta = -107, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1612315132181231\overset{2}{3} \text{ [#=3]} \quad \Delta = -107, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1612315132181231\overset{2}{3} \text{ [#=2]} \quad \Delta = -107, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$1612315132181231\overset{2}{3} \text{ [#=5]} \quad \Delta = -107, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1612315132181231 \quad \Delta = -99, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1612315132181231 \quad \Delta = -99, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1612315132181231 \quad \Delta = -99, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1612315132181231 \quad \Delta = -99, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1612315132181231 \quad \Delta = -99, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

Base: 4132181231513161

$$4132181231513161 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$4132181231513161 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$4132181231513161 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

$$4132181231513161 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$4132181231513161 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$4132181231513161 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix}$$

$$4132181231513161 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 2161232181231512

$$2161232181231512 \quad \Delta = -92, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\begin{aligned}
2161232181231512^2 \quad \Delta = -92, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2161232181231512^2 \quad \Delta = -92, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2161232181231512^2 \quad \Delta = -92, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2161232181231512^2 \quad \Delta = -92, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2161232181231512^2 \quad \Delta = -92, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2161232181231512^3 \quad \Delta = -100, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
2161232181231512^3 \quad \Delta = -100, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: 216123218123151¹

$$\begin{aligned}
216123218123151^{21} \quad \Delta = -92, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
216123218123151^{21} \quad \Delta = -92, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
216123218123151^{31} \quad \Delta = -100, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
216123218123151^{21} \quad \Delta = -92, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
216123218123151^{21} \quad \Delta = -92, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
216123218123151^{21} \quad \Delta = -92, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 2161315132181231

$$\begin{aligned}
2161315132181231^3_{[\# = 3]} \quad \Delta = -106, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2161315132181231^3_{[\# = 5]} \quad \Delta = -106, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2161315132181231^3_{[\# = 4]} \quad \Delta = -106, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2161315132181231^3_{[\# = 2]} \quad \Delta = -106, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
2161315132181231^2 \quad \Delta = -98, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2161315132181231^3 \quad \Delta = -106, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{2}{2161315132181231} \quad \Delta = -98, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 2161315132181231 \quad \Delta = -98, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 2161315132181231 \quad \Delta = -98, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 2161315132181231 \quad \Delta = -98, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 2161315132181231 \quad \Delta = -98, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \text{Base: } 1413218123216131 \\
& 1413218123216131 \overset{2}{3} \text{ [#=3]} \quad \Delta = -84, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& 1413218123216131 \quad \Delta = -76, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 1413218123216131 \quad \Delta = -76, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 1413218123216131 \quad \Delta = -76, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& 1413218123216131 \quad \Delta = -76, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 1413218123216131 \overset{2}{3} \text{ [#=9]} \quad \Delta = -84, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 1413218123216131 \overset{2}{3} \text{ [#=4]} \quad \Delta = -84, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 1413218123216131 \overset{2}{3} \text{ [#=5]} \quad \Delta = -84, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 1413218123216131 \quad \Delta = -78, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 1413218123216131 \overset{2}{3} \text{ [#=6]} \quad \Delta = -84, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 1413218123216131 \overset{2}{3} \text{ [#=7]} \quad \Delta = -84, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 1316123218123141 \overset{2}{3} \text{ [#=2]} \quad \Delta = -84, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 1413218123216131 \quad \Delta = -76, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 1316123218123141 \overset{2}{3} \text{ [#=3]} \quad \Delta = -84, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

$$1413218123216131\overset{2}{3} \text{ [#}=8] \quad \Delta = -84, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1413218123216131\overset{2}{3} \text{ [#}=2] \quad \Delta = -84, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

Base: 4132181232161315

$$413218123216131\overset{2}{5} \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$413218123216131\overset{2}{5} \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$413218123216131\overset{2}{5} \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$413218123216131\overset{2}{5} \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$413218123216131\overset{2}{5} \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$413218123216131\overset{2}{5} \quad \Delta = -130, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$413218123216131\overset{2}{5} \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 1513161232181231

$$15131\overset{2}{6}1232181231 \quad \Delta = -100, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1513161232181\overset{2}{2}31 \quad \Delta = -100, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1513161232181231 \quad \Delta = -100, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1513161232181\overset{2}{2}31 \quad \Delta = -100, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1513161232181231\overset{2}{3} \text{ [#}=5] \quad \Delta = -108, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1513161232181231\overset{2}{3} \text{ [#}=2] \quad \Delta = -108, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$1513161232181231\overset{2}{3} \text{ [#}=3] \quad \Delta = -108, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1513161232181231\overset{2}{3} \text{ [#}=4] \quad \Delta = -108, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1513161232181231 \quad \Delta = -100, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

Base: ¹161232181231512

$$\begin{aligned}
{}^1 161232181231512^2 \quad \Delta = -92, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
{}^1 161232181231512^{12} \quad \Delta = -92, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^1 161232181231512^2 \quad \Delta = -92, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
{}^1 161232181231512^2 \quad \Delta = -92, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
{}^1 161232181231512^3 \quad \Delta = -100, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
{}^1 161232181231512^2 \quad \Delta = -92, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: ${}^1 16123218123151^1$

$$\begin{aligned}
{}^1 16123218123151^2 \quad \Delta = -92, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
{}^1 16123218123151^{12} \quad \Delta = -92, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^1 16123218123151^{21} \quad \Delta = -92, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
{}^1 16123218123151^2 \quad \Delta = -92, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: ${}^1 161315132181231$

$$\begin{aligned}
{}^1 161315132181231^3_{[\# = 5]} \quad \Delta = -106, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
{}^1 161315132181231^3_{[\# = 4]} \quad \Delta = -106, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
{}^1 161315132181231^3_{[\# = 3]} \quad \Delta = -106, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
{}^1 161315132181231^3_{[\# = 2]} \quad \Delta = -106, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
{}^1 161315132181231^2 \quad \Delta = -98, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
{}^1 161315132181231^2 \quad \Delta = -98, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^1 161315132181231^2 \quad \Delta = -98, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
{}^1 161315132181231^2 \quad \Delta = -98, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

$$1\overset{2}{6}1315132181231 \quad \Delta = -98, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 4141321812321712

$$4141321812321\overset{2}{7}12 \quad \Delta = -123, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$4141\overset{2}{3}21812321712 \quad \Delta = -123, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$4141321812321\overset{2}{7}12 \quad \Delta = -123, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix}$$

$$4141321812321\overset{2}{7}12 \quad \Delta = -123, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$4141321812321\overset{2}{7}12 \quad \Delta = -123, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$41\overset{2}{4}1321812321712 \quad \Delta = -123, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$4141\overset{2}{3}21812321712 \quad \Delta = -125, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: 1413218123217122

$$141321812321\overset{2}{7}122 \quad \Delta = -95, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$2217123218123141\overset{3}{3}_{[\# = 3]} \quad \Delta = -103, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$2217123218123141\overset{3}{3}_{[\# = 2]} \quad \Delta = -103, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$141321812321\overset{2}{7}122 \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$141321812321\overset{3}{7}122 \quad \Delta = -104, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$141\overset{2}{3}218123217122 \quad \Delta = -95, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$141321812321\overset{2}{7}122 \quad \Delta = -95, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$14132181\overset{2}{2}3217122 \quad \Delta = -95, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$141321812321\overset{2}{7}122 \quad \Delta = -95, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 1321712321812313

$$132171232181231\overset{2}{3} \quad \Delta = -86, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$\begin{aligned}
3132181232171231\overset{2}{3}_{[\neq 5]} \quad \Delta = -92, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
3132181232171231\overset{2}{3}_{[\neq 4]} \quad \Delta = -92, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
132171232181231\overset{2}{3} \quad \Delta = -84, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
132171232181231\overset{2}{3} \quad \Delta = -84, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
132171232181231\overset{2}{3} \quad \Delta = -84, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
132171232181231\overset{2}{3} \quad \Delta = -84, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
132171232181231\overset{2}{3} \quad \Delta = -84, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
3132181232171231\overset{2}{3}_{[\neq 3]} \quad \Delta = -92, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
132171232181231\overset{3}{3}_{[\neq 2]} \quad \Delta = -92, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
3132181232171231\overset{2}{3}_{[\neq 2]} \quad \Delta = -92, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
132171232181231\overset{3}{3} \quad \Delta = -92, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\text{Base: } 4132171232181231 \\
4132171232181231\overset{2}{3}_{[\neq 4]} \quad \Delta = -112, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
4132171232181231\overset{2}{3}_{[\neq 5]} \quad \Delta = -112, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
4132171232181231\overset{2}{3}_{[\neq 2]} \quad \Delta = -112, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
4132171232181231\overset{2}{3}_{[\neq 3]} \quad \Delta = -112, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
4132171232181231\overset{2}{3} \quad \Delta = -106, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
4132171232181231\overset{2}{3} \quad \Delta = -104, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
4132171232181231\overset{2}{3} \quad \Delta = -104, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
4132171232181231\overset{2}{3} \quad \Delta = -104, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

$$4132171232181231 \quad \Delta = -104, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$4132171232181231 \quad \Delta = -104, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 414132181232171¹

$$414132181232171 \quad \Delta = -125, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$414132181232171 \quad \Delta = -123, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$414132181232171 \quad \Delta = -123, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$414132181232171 \quad \Delta = -123, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$414132181232171 \quad \Delta = -123, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix}$$

$$414132181232171 \quad \Delta = -123, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 17²¹123218123141

$$17^{21}123218123141 \quad \Delta = -95, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$17^{21}123218123141\mathbf{3}_{[\# = 3]} \quad \Delta = -103, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$17^{21}123218123141\mathbf{3}_{[\# = 2]} \quad \Delta = -103, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$17^{21}123218123141 \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$17^{21}123218123141 \quad \Delta = -95, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$17^{21}123218123141 \quad \Delta = -95, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$17^{21}123218123141 \quad \Delta = -95, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$17^{21}\mathbf{2}123218123141 \quad \Delta = -95, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

Base: 2161232181232151

$$2161232181232151 \quad \Delta = -94, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$2161232181232151 \quad \Delta = -94, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\begin{aligned}
2161232181232151 \quad \Delta = -94, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2161232181232151 \quad \Delta = -102, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2161232181232151 \quad \Delta = -94, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
2161232181232151 \quad \Delta = -94, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 4123218123216131

$$\begin{aligned}
4123218123216131 \textcolor{red}{3} \text{ [#=9]} \quad \Delta = -86, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
4123218123216131 \textcolor{red}{3} \text{ [#=8]} \quad \Delta = -86, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
4123218123216131 \textcolor{red}{3} \text{ [#=3]} \quad \Delta = -86, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
4123218123216131 \quad \Delta = -78, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
4123218123216131 \textcolor{red}{3} \text{ [#=6]} \quad \Delta = -86, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
4123218123216131 \textcolor{red}{3} \text{ [#=5]} \quad \Delta = -86, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
4123218123216131 \textcolor{red}{3} \text{ [#=4]} \quad \Delta = -86, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
4123218123216131 \textcolor{red}{3} \text{ [#=7]} \quad \Delta = -86, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
4123218123216131 \textcolor{red}{3} \text{ [#=2]} \quad \Delta = -86, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
4123218123216131 \quad \Delta = -86, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 41 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
4123218123216131 \quad \Delta = -78, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
4123218123216131 \quad \Delta = -78, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
4123218123216131 \quad \Delta = -78, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
4123218123216131 \quad \Delta = -78, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: 4131612321812321

$$\begin{aligned}
4131612321812321\overset{2}{3} \text{ }_{[\# = 9]} \quad \Delta = -86, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
4131612321812321\overset{2}{} \quad \Delta = -78, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
4131612321812321\overset{2}{} \quad \Delta = -78, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
4131612321812321\overset{2}{} \quad \Delta = -78, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
4131612321812321\overset{2}{} \quad \Delta = -78, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
4131612321812321\overset{2}{} \quad \Delta = -78, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
4131612321812321\overset{2}{3} \text{ }_{[\# = 2]} \quad \Delta = -86, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
4131612321812321\overset{2}{3} \text{ }_{[\# = 3]} \quad \Delta = -86, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
4131612321812321\overset{2}{3} \text{ }_{[\# = 5]} \quad \Delta = -86, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
4131612321812321\overset{2}{3} \text{ }_{[\# = 4]} \quad \Delta = -86, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
4131612321812321\overset{2}{3} \text{ }_{[\# = 6]} \quad \Delta = -86, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
4131612321812321\overset{2}{3} \text{ }_{[\# = 7]} \quad \Delta = -86, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
4131612321812321\overset{2}{3} \text{ }_{[\# = 8]} \quad \Delta = -86, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
4131612321812321\overset{3}{} \quad \Delta = -86, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\text{Base: } 161232181232151\overset{1}{} & \\
161232181232151\overset{1}{\overset{2}{} } \quad \Delta = -94, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
161232181232151\overset{1}{\overset{2}{} } \quad \Delta = -94, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
161232181232151\overset{1}{\overset{2}{} } \quad \Delta = -94, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
161232181232151\overset{1}{\overset{2}{} } \quad \Delta = -94, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\text{Base: } 4123218123217122 &
\end{aligned}$$

$$\begin{aligned}
4123218123217122^3 \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 41 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
4123218123217122^3 \quad \Delta = -106, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
4123218123217122^2 \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
4123218123217122^2 \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
4123218123217122^2 \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
4123218123217122^2 \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
4123218123217122^2 \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 3221712321812321

$$\begin{aligned}
3221712321812321^3_{[\# = 9]} \quad \Delta = -86, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
3221712321812321^2 \quad \Delta = -78, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
3221712321812321^2 \quad \Delta = -78, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
3221712321812321^2 \quad \Delta = -78, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
3221712321812321^2 \quad \Delta = -78, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
3221712321812321^3_{[\# = 6]} \quad \Delta = -86, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
3221712321812321^3_{[\# = 8]} \quad \Delta = -86, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
3221712321812321^3_{[\# = 2]} \quad \Delta = -86, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
3221712321812321^3_{[\# = 7]} \quad \Delta = -86, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
3221712321812321^3_{[\# = 5]} \quad \Delta = -86, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
3221712321812321^3_{[\# = 4]} \quad \Delta = -86, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
3221712321812321^3_{[\# = 3]} \quad \Delta = -86, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: 412321812321²¹71

$$\begin{aligned}
412321812321 \overset{21}{7}1 \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
412321812321 \overset{21}{7}1 \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 41 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
412321812321 \overset{21}{7}1 \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
412321812321 \overset{21}{7}1 \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
412321812321 \overset{21}{7}1 \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
412321812321 \overset{21}{7}1 \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{321}{7} 12321812321$

$$\begin{aligned}
321 \overset{21}{7} 12321812321 \quad \Delta = -78, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1 \overset{321}{7} 12321812321 \quad \Delta = -78, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1 \overset{321}{7} 12321812321 \quad \Delta = -78, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
1 \overset{321}{7} 12321812321 \quad \Delta = -78, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1 \overset{321}{7} 12321812321 \quad \Delta = -78, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1 \overset{321}{7} 12321812321 \overset{2}{3} [\#=5] \quad \Delta = -86, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1 \overset{321}{7} 12321812321 \overset{2}{3} [\#=2] \quad \Delta = -86, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1 \overset{321}{7} 12321812321 \overset{2}{3} [\#=9] \quad \Delta = -86, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1 \overset{321}{7} 12321812321 \overset{2}{3} [\#=8] \quad \Delta = -86, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1 \overset{321}{7} 12321812321 \overset{2}{3} [\#=7] \quad \Delta = -86, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1 \overset{321}{7} 12321812321 \overset{2}{3} [\#=6] \quad \Delta = -86, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \overset{321}{7} 12321812321 \overset{2}{3} [\#=4] \quad \Delta = -86, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \overset{321}{7} 12321812321 \overset{2}{3} [\#=3] \quad \Delta = -86, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: $2^{1223151321812} 3^1 1$

$$2^{31223151321812} 3^1 1 \text{ [#}=7] \quad \Delta = -31, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$2^{31223151321812} 3^1 1 \text{ [#}=6] \quad \Delta = -31, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$2^{31223151321812} 3^1 1 \text{ [#}=5] \quad \Delta = -31, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$2^{1223151321812} 3^{13} \text{ [#}=2] \quad \Delta = -31, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$2^{1223151321812} 3^{14} \quad \Delta = -51, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$2^{31223151321812} 3^1 1 \text{ [#}=2] \quad \Delta = -31, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$2^{1223151321812} 3^{13} \text{ [#}=3] \quad \Delta = -31, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 60 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$2^{31223151321812} 3^1 1 \text{ [#}=4] \quad \Delta = -31, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$2^{31223151321812} 3^1 1 \text{ [#}=3] \quad \Delta = -31, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$2^{1223151321812} 3^1 \quad \Delta = -23, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$2^{1223151321812} 3^1 \quad \Delta = -23, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$2^{1223151321812} 3^1 \quad \Delta = -23, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$2^{1223151321812} 3^{12} \quad \Delta = -23, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$2^{31223151321812} 3^1 1 \text{ [#}=8] \quad \Delta = -31, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$2^{1223151321812} 3^1 \quad \Delta = -23, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$2^{31223151321812} 3^1 1 \text{ [#}=9] \quad \Delta = -31, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

Base: 132171232181232^1

$$2^{31321712321812} 3^1 1 \text{ [#}=5] \quad \Delta = -86, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$132171232181232^{41} \quad \Delta = -106, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$1321712321812\overset{31}{3}2 \text{ }_{[\# = 2]} \quad \Delta = -86, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$2 \overset{31321712321812}{3} 1 \text{ }_{[\# = 2]} \quad \Delta = -86, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1321712321812\overset{31}{3}2 \text{ }_{[\# = 3]} \quad \Delta = -86, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1321712321812\overset{2}{3}1 \quad \Delta = -78, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1321712321812\overset{21}{3}2 \quad \Delta = -78, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1321712321812\overset{2}{3}1 \quad \Delta = -78, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1321712321812\overset{2}{3}1 \quad \Delta = -78, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1321712321812\overset{2}{3}1 \quad \Delta = -78, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$2 \overset{31321712321812}{3} 1 \text{ }_{[\# = 3]} \quad \Delta = -86, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$2 \overset{31321712321812}{3} 1 \text{ }_{[\# = 4]} \quad \Delta = -86, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

Base: 4132218123218122

$$413221812321812\overset{2}{2} \quad \Delta = -123, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix}$$

$$413221812321812\overset{2}{2} \quad \Delta = -123, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$413221812321812\overset{2}{2} \quad \Delta = -123, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$413221812321812\overset{2}{2} \quad \Delta = -123, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$413221812321812\overset{2}{2} \quad \Delta = -123, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$413221812321812\overset{3}{2} \quad \Delta = -132, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

Base: 2221812321812231

$$2221812321812231\overset{2}{2} \quad \Delta = -95, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$2221812321812231\overset{2}{2} \quad \Delta = -95, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$2221812321812231\overset{3}{2} \text{ }_{[\# = 4]} \quad \Delta = -103, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\begin{aligned}
2221812321812231^2 \quad \Delta = -95, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2221812321812231^3_{[\# = 5]} \quad \Delta = -103, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2221812321812231^3_{[\# = 3]} \quad \Delta = -103, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2221812321812231^2 \quad \Delta = -95, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
2221812321812231^3_{[\# = 2]} \quad \Delta = -103, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
\text{Base: } 2221812321812313 \\
2221812321812313^2 \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2221812321812313^2 \quad \Delta = -103, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
2221812321812313^3_{[\# = 2]} \quad \Delta = -111, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2221812321812313^2 \quad \Delta = -103, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2221812321812313^3 \quad \Delta = -111, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2221812321812313^2 \quad \Delta = -103, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
2221812321812313^2 \quad \Delta = -103, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2221812321812313^2 \quad \Delta = -103, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\text{Base: } 222181232181232^1 \\
222181232181232^{2 \ 1} \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
222181232181232^{3 \ 1}_{[\# = 3]} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
222181232181232^{3 \ 1}_{[\# = 2]} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
222181232181232^{2 \ 1} \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
222181232181232^{2 \ 1} \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
222181232181232^{\overset{2}{1}} \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
222181232181232^{\overset{2}{1}} \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
222181232181232^{\overset{41}{1}} \quad \Delta = -125, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}
\end{aligned}$$

Base: $181223151316131^{\overset{1}{1}}$

$$\begin{aligned}
181223151316131^{\overset{1}{2}} \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
181223151316131^{\overset{1}{2}} \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
181223151316131^{\overset{1}{2}} \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
181223151316131^{\overset{1}{2}} \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
181223151316131^{\overset{12}{2}} \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
1812231513161313^{\overset{1}{\text{[#}=3]}} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1812231513161313^{\overset{1}{\text{[#}=9]}} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1812231513161313^{\overset{1}{\text{[#}=2]}} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
1812231513161313^{\overset{1}{\text{[#}=6]}} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1812231513161313^{\overset{1}{\text{[#}=4]}} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1812231513161313^{\overset{1}{\text{[#}=5]}} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1812231513161313^{\overset{1}{\text{[#}=8]}} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1812231513161313^{\overset{1}{\text{[#}=7]}} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: $181223151322171^{\overset{1}{1}}$

$$\begin{aligned}
181223151322171^{\overset{1}{2}} \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
181223151322171^{\overset{1}{2}} \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{ll}
\overset{1}{18}1223151322171 & \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
\overset{1}{18}1223151322171 & \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\text{Base: } \overset{1}{18}1223151321712 & \\
\overset{1}{18}1223151321712 & \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\overset{1}{18}1223151321712 & \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\overset{1}{18}1223151321712 & \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\overset{1}{18}1223151321712 & \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
\overset{1}{18}1223151321712 & \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\text{Base: } \overset{1}{17}123151322181 & \\
\overset{1}{17}123151322181 & \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\overset{1}{17}123151322181 & \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\overset{1}{17}123151322181 & \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\overset{1}{17}123151322181 & \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
\text{Base: } \overset{1}{4}13161315132181 & \\
\overset{3}{4}13161315132181 & \Delta = -133, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\overset{2}{4}13161315132181 & \Delta = -125, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\overset{2}{4}13161315132181 & \Delta = -125, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\overset{2}{4}13161315132181 & \Delta = -125, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
\overset{2}{4}13161315132181 & \Delta = -125, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\overset{2}{4}13161315132181 & \Delta = -125, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
\overset{2}{4}13161315132181 & \Delta = -125, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}
\end{array}$$

Base: $\overset{1}{181232161316123}$

$$\begin{aligned}
\overset{1}{181232161316123} \quad \overset{2}{\Delta} &= -126, \quad n_+ = 1, \quad n_- = 16, \quad n_0 = 0, \quad |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\overset{1}{181232161316123} \quad \overset{2}{\Delta} &= -126, \quad n_+ = 1, \quad n_- = 16, \quad n_0 = 0, \quad |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\overset{1}{181232161316123} \quad \overset{2}{\Delta} &= -126, \quad n_+ = 1, \quad n_- = 16, \quad n_0 = 0, \quad |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
\overset{1}{181232161316123} \quad \overset{2}{\Delta} &= -126, \quad n_+ = 1, \quad n_- = 16, \quad n_0 = 0, \quad |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\overset{1}{181232161316123} \quad \overset{2}{\Delta} &= -126, \quad n_+ = 1, \quad n_- = 16, \quad n_0 = 0, \quad |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{31}{1613161232181}$

$$\begin{aligned}
\overset{31}{1613161232181} \quad \overset{21}{\Delta} &= -125, \quad n_+ = 1, \quad n_- = 16, \quad n_0 = 0, \quad |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\overset{31}{1613161232181} \quad \overset{21}{\Delta} &= -125, \quad n_+ = 1, \quad n_- = 16, \quad n_0 = 0, \quad |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
\overset{31}{1613161232181} \quad \overset{21}{\Delta} &= -133, \quad n_+ = 1, \quad n_- = 16, \quad n_0 = 0, \quad |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\overset{31}{1613161232181} \quad \overset{21}{\Delta} &= -125, \quad n_+ = 1, \quad n_- = 16, \quad n_0 = 0, \quad |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\overset{31}{1613161232181} \quad \overset{21}{\Delta} &= -125, \quad n_+ = 1, \quad n_- = 16, \quad n_0 = 0, \quad |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\overset{31}{1613161232181} \quad \overset{21}{\Delta} &= -125, \quad n_+ = 1, \quad n_- = 16, \quad n_0 = 0, \quad |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
\overset{31}{1613161232181} \quad \overset{21}{\Delta} &= -125, \quad n_+ = 1, \quad n_- = 16, \quad n_0 = 0, \quad |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{181231513217123}$

$$\begin{aligned}
\overset{1}{181231513217123} \quad \overset{2}{\Delta} &= -126, \quad n_+ = 1, \quad n_- = 16, \quad n_0 = 0, \quad |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
\overset{1}{181231513217123} \quad \overset{2}{\Delta} &= -126, \quad n_+ = 1, \quad n_- = 16, \quad n_0 = 0, \quad |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\overset{1}{181231513217123} \quad \overset{2}{\Delta} &= -126, \quad n_+ = 1, \quad n_- = 16, \quad n_0 = 0, \quad |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\overset{1}{181231513217123} \quad \overset{2}{\Delta} &= -126, \quad n_+ = 1, \quad n_- = 16, \quad n_0 = 0, \quad |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\overset{1}{181231513217123} \quad \overset{2}{\Delta} &= -126, \quad n_+ = 1, \quad n_- = 16, \quad n_0 = 0, \quad |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{181232171231513}$

$$\begin{aligned}
& \overset{1}{1}81232171231513 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{1}{1}81232171231513 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{1}{1}\overset{2}{8}1232171231513 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{1}{1}81232171231513 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& \overset{1}{1}81232171231513 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{21}{1}8123151316131$

$$\begin{aligned}
& \overset{21}{1}8123151316131 \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{21}{1}8123151316131 \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& \overset{21}{1}8123151316131 \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{21}{1}8123151316131 \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{21}{1}8123151316131\overset{3}{3} \text{ [#=5]} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{21}{1}8123151316131\overset{3}{3} \text{ [#=9]} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{21}{1}8123151316131\overset{3}{3} \text{ [#=8]} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{21}{1}8123151316131\overset{3}{3} \text{ [#=4]} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{21}{1}8123151316131\overset{3}{3} \text{ [#=7]} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{21}{1}8123151316131 \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{21}{1}8123151316131 \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{21}{1}8123151316131\overset{3}{3} \text{ [#=2]} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
& \overset{21}{1}8123151316131\overset{3}{3} \text{ [#=3]} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{21}{1}8123151316131\overset{3}{3} \text{ [#=6]} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: $^{21}_{18}123216131612$

$$\begin{aligned}
^{21}_{18}123216131612 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
^{21}_{18}123216131612 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
^{21}_{18}123216131612 \quad \Delta = -125, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
^{21}_{18}123216131612 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
^{21}_{18}123216131612 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
^{21}_{18}123216131612 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
^{21}_{18}123216131612 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

Base: $^1_{16}13161232181$

$$\begin{aligned}
^{12}_{16}13161232181 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
^1_{16}13161232181 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
^1_{16}13161232181 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
^1_{16}13161232181 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
^1_{16}13161232181 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: $^{21}_{18}123151322171$

$$\begin{aligned}
^{21}_{18}123151322171 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
^{21}_{18}123151322171 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
^{21}_{18}123151322171 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
^{21}_{18}123151322171 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
^{21}_{18}123151322171 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: $^{21}_{18}123151321712$

$$\begin{array}{ll}
{}^{21}18123151321712 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
{}^{21}18123151321712 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
{}^{21}18123151321712 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
{}^{21}18123151321712 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
{}^{21}18123151321712 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^{21}18123151321712 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\text{Base: } {}^{21}18123217123151 & \\
{}^{21}18123217123151 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
{}^{21}18123217123151 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
{}^{21}18123217123151 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
{}^{21}18123217123151 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
{}^{21}18123217123151 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\text{Base: } {}^{21}18123217123215 & \\
{}^{21}18123217123215 \quad \Delta = -121, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
{}^{21}18123217123215 \quad \Delta = -121, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^{21}18123217123215 \quad \Delta = -121, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
{}^{21}18123217123215 \quad \Delta = -121, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
{}^{21}18123217123215 \quad \Delta = -121, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\text{Base: } {}^11712315132181 & \\
{}^11712315132181 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
{}^11712315132181 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
& \overset{1}{1}7123151321\overset{221}{8}1 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{1}{1}7123151321\overset{21}{8}1 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{1}{1}7123151321\overset{21}{8}1 \quad \Delta = -117, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: $413221812321\overset{21}{8}1$

$$\begin{aligned}
& 413221812321\overset{221}{8}1 \quad \Delta = -123, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 413221812321\overset{21}{8}1 \quad \Delta = -123, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& 413221812321\overset{21}{8}1 \quad \Delta = -123, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
& \overset{2}{4}13221812321\overset{21}{8}1 \quad \Delta = -123, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 413221812321\overset{21}{8}1 \quad \Delta = -123, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1\overset{221}{8}12321712321$

$$\begin{aligned}
& 1\overset{221}{8}12321712321\overset{221}{3} \text{ [#=9]} \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& 1\overset{221}{8}12321712321 \quad \Delta = -78, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 1\overset{221}{8}12321712321 \quad \Delta = -69, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 1\overset{221}{8}12321712321 \quad \Delta = -69, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 1\overset{221}{8}12321712321 \quad \Delta = -69, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 1\overset{221}{8}12321712321 \quad \Delta = -69, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& 1\overset{221}{8}12321712321\overset{221}{3} \text{ [#=3]} \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 1\overset{221}{8}12321712321\overset{221}{3} \text{ [#=8]} \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& 1\overset{221}{8}12321712321\overset{221}{3} \text{ [#=2]} \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 1\overset{221}{8}12321712321\overset{221}{3} \text{ [#=4]} \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{221}{1 \ 8 \ 12321712321\textcolor{red}{3}}_{[\#=5]} \quad \Delta = -77, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{221}{1 \ 8 \ 12321712321\textcolor{red}{3}}_{[\#=6]} \quad \Delta = -77, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{221}{1 \ 8 \ 12321712321\textcolor{red}{3}}_{[\#=7]} \quad \Delta = -77, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{221}{1 \ 8 \ 12321812231}$

$$\begin{aligned}
& \overset{221\textcolor{red}{2}}{1 \ 8 \ 12321812231} \quad \Delta = -95, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{221}{1 \ 8 \ 12321812231} \quad \Delta = -104, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{221}{1 \ 8 \ 12321812231\textcolor{red}{3}}_{[\#=2]} \quad \Delta = -103, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
& \overset{221}{1 \ 8 \ 12321812231\textcolor{red}{3}}_{[\#=4]} \quad \Delta = -103, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{221}{1 \ 8 \ 12321812231} \quad \Delta = -95, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{221}{1 \ 8 \ 12321812231\textcolor{red}{2}} \quad \Delta = -95, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{221}{1 \ 8 \ 12321812231\textcolor{red}{2}} \quad \Delta = -95, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{221}{1 \ 8 \ 12321812231\textcolor{red}{3}}_{[\#=5]} \quad \Delta = -103, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{221}{1 \ 8 \ 12321812231\textcolor{red}{3}}_{[\#=3]} \quad \Delta = -103, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{221}{1 \ 8 \ 12321812313}$

$$\begin{aligned}
& \overset{221}{1 \ 8 \ 12321812313\textcolor{red}{2}} \quad \Delta = -103, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{221}{1 \ 8 \ 12321812313\textcolor{red}{2}} \quad \Delta = -105, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{221}{1 \ 8 \ 12321812313} \quad \Delta = -103, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{221\textcolor{red}{2}}{1 \ 8 \ 12321812313} \quad \Delta = -103, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{221}{1 \ 8 \ 12321812313\textcolor{red}{2}} \quad \Delta = -103, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{221}{1 \ 8 \ 12321812313\textcolor{red}{2}} \quad \Delta = -103, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{ll}
1 \overset{221}{8} 12321812313 & \Delta = -112, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1 \overset{221}{8} 1232181\overset{3}{2}313 & \Delta = -111, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1 \overset{221}{8} 1232181\overset{3}{2}313_{[\# = 2]} & \Delta = -111, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{array}$$

Base: $1 \overset{221}{8} 1232181232 \overset{1}{}{}$

$$\begin{array}{llll}
\begin{array}{c} 221 \\ 1 \ 8 \ 1232181232 \end{array} & \begin{array}{c} 1 \\ \Delta = -106, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 4 \end{array} & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\begin{array}{c} 221 \\ 1 \ 8 \ 1232181232 \end{array} & \begin{array}{c} 31 \\ \text{[#}=2] \end{array} & \begin{array}{c} \Delta = -105, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 16 \end{array} & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\begin{array}{c} 221 \\ 1 \ 8 \ 1232181232 \end{array} & \begin{array}{c} 41 \\ \Delta = -125, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 4 \end{array} & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
\begin{array}{c} 221 \\ 1 \ 8 \ 1232181232 \end{array} & \begin{array}{c} 1 \\ \Delta = -97, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 9 \end{array} & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\begin{array}{c} 221 \\ 1 \ 8 \ 1232181232 \end{array} & \begin{array}{c} 21 \\ \Delta = -97, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 4 \end{array} & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\begin{array}{c} 221 \\ 1 \ 8 \ 1232181232 \end{array} & \begin{array}{c} 2 \ 1 \\ \Delta = -97, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 16 \end{array} & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\begin{array}{c} 221 \\ 1 \ 8 \ 1232181232 \end{array} & \begin{array}{c} 1 \\ \Delta = -97, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 16 \end{array} & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\begin{array}{c} 221 \\ 1 \ 8 \ 1232181232 \end{array} & \begin{array}{c} 2 \ 1 \\ \Delta = -97, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 16 \end{array} & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\begin{array}{c} 221 \\ 1 \ 8 \ 1232181232 \end{array} & \begin{array}{c} 31 \\ \text{[#}=3] \end{array} & \begin{array}{c} \Delta = -105, \ n_+ = 1, \ n_- = 16, \ n_0 = 0, \ |\det| = 36 \end{array} & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{array}$$

Base: $1 \overset{3221}{8} 1232181231$

$$\begin{array}{lll}
1 \overset{3221}{8} \overset{2}{1232181231} & \Delta = -104, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1 \overset{3221}{8} \overset{2}{1232181231} \textcolor{red}{1} \text{ \#}=4 & \Delta = -112, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1 \overset{3221}{8} \overset{2}{1232181231} \textcolor{red}{1} \text{ \#}=5 & \Delta = -112, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1 \overset{3221}{8} \overset{2}{1232181231} \textcolor{red}{1} \text{ \#}=2 & \Delta = -112, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
1 \overset{3221}{8} \overset{2}{1232181231} \textcolor{red}{1} \text{ \#}=3 & \Delta = -112, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1 \overset{3221}{8} \overset{2}{1232181231} \textcolor{red}{2} & \Delta = -104, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{ll}
3221\overset{21}{8}1232181231 & \Delta = -104, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1\overset{32212}{8}1232181231 & \Delta = -104, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
1\overset{3221}{8}1232181231 & \Delta = -104, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
\text{Base: } 1\overset{321}{8}12321613161 & \\
1\overset{321}{8}12321613161 & \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 163 & 25 \\ 2 & 25 & 8 \end{pmatrix} \\
1\overset{321}{8}12321613161 & \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1\overset{321}{8}12321613161 & \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1\overset{321}{8}12321613161 & \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1\overset{3212}{8}12321613161 & \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
321\overset{21}{8}12321613161 & \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\text{Base: } 71223151321\overset{321}{8}1 & \\
71223151321\overset{2}{8}1 & \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
71223151321\overset{2}{8}1 & \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
71223151321\overset{2}{8}1 & \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
71223151321\overset{321}{8}1 & \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
71223151321\overset{2321}{8}1 & \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
71223151321\overset{321}{8}1\overset{2}{2} & \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\text{Base: } 1\overset{321}{8}12315132171 & \\
1\overset{3212}{8}12315132171 & \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
321\overset{21}{8}12315132171 & \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1\overset{321}{8}12315132171 & \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
& \overset{321}{1} \overset{2}{8} 12315132171 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{321}{1} \overset{2}{8} 12315132171 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{321}{1} \overset{2}{8} 12315132171 \quad \Delta = -126, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{321}{1} \overset{2}{8} 12321712315$

$$\begin{aligned}
& \overset{321}{1} \overset{2}{8} 12321712315 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{321}{1} \overset{2}{8} 12321712315 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{321}{1} \overset{2}{8} 12321712315 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{321}{1} \overset{2}{8} 12321712315 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{321}{1} \overset{2}{8} 12321712315 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{21}{321} \overset{2}{8} 12321712315 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1223151321}{1} \overset{2}{8} 1231$

$$\begin{aligned}
& \overset{1223151321}{1} \overset{2}{8} 1231 \quad \Delta = -21, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{31223151321}{1} \overset{2}{8} 1231 \text{ [#=9]} \quad \Delta = -29, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
& \overset{1223151321}{1} \overset{2}{8} 1231 \overset{3}{\text{[#=5]}} \quad \Delta = -29, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{1223151321}{1} \overset{2}{8} 1231 \overset{3}{\text{[#=2]}} \quad \Delta = -29, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{31223151321}{1} \overset{2}{8} 1231 \text{ [#=5]} \quad \Delta = -29, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& \overset{1223151321}{1} \overset{2}{8} 1231 \quad \Delta = -21, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{31223151321}{1} \overset{2}{8} 1231 \text{ [#=8]} \quad \Delta = -29, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
& \overset{1223151321}{1} \overset{2}{8} 1231 \overset{2}{\text{[#=6]}} \quad \Delta = -21, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{31223151321}{1} \overset{2}{8} 1231 \text{ [#=6]} \quad \Delta = -29, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{1223151321}{1 \quad 8 \quad 1231} \quad \Delta = -21, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{\textcolor{red}{3}1223151321}{1 \quad 8 \quad 1231} \text{ [#=2]} \quad \Delta = -29, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{1223151321}{1 \quad 8 \quad 1231} \textcolor{red}{3} \text{ [#=3]} \quad \Delta = -29, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{\textcolor{red}{3}1223151321}{1 \quad 8 \quad 1231} \text{ [#=3]} \quad \Delta = -29, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{1223151321}{1 \quad 8 \quad 1231} \textcolor{red}{3} \text{ [#=4]} \quad \Delta = -29, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{\textcolor{red}{3}1223151321}{1 \quad 8 \quad 1231} \text{ [#=4]} \quad \Delta = -29, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{\textcolor{red}{3}1223151321}{1 \quad 8 \quad 1231} \text{ [#=7]} \quad \Delta = -29, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1321}{1 \quad 8 \quad 1232171231}$

$$\begin{aligned}
& \overset{1321}{1 \quad 8 \quad 1232171231} \textcolor{red}{3} \text{ [#=5]} \quad \Delta = -84, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{1321}{1 \quad 8 \quad 1232171231} \textcolor{red}{3} \text{ [#=3]} \quad \Delta = -84, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{\textcolor{red}{3}1321}{1 \quad 8 \quad 1232171231} \text{ [#=3]} \quad \Delta = -84, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{1321}{1 \quad 8 \quad 1232171231} \textcolor{red}{3} \text{ [#=2]} \quad \Delta = -84, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
& \overset{1321}{1 \quad 8 \quad 1232171231} \overset{\textcolor{red}{2}}{\quad} \quad \Delta = -76, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{\textcolor{red}{3}1321}{1 \quad 8 \quad 1232171231} \text{ [#=2]} \quad \Delta = -84, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{\textcolor{red}{3}1321}{1 \quad 8 \quad 1232171231} \text{ [#=4]} \quad \Delta = -84, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{1321}{1 \quad 8 \quad 1232171231} \quad \Delta = -76, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{1321}{1 \quad 8 \quad 1232171231} \overset{\textcolor{red}{2}}{\quad} \quad \Delta = -76, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{1321}{1 \quad 8 \quad 1232171231} \overset{\textcolor{red}{2}}{\quad} \quad \Delta = -76, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{1321}{1 \quad 8 \quad 1232171231} \textcolor{red}{3} \text{ [#=4]} \quad \Delta = -84, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{\textcolor{red}{3}1321}{1 \quad 8 \quad 1232171231} \text{ [#=5]} \quad \Delta = -84, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{123151321}{8} 12321$

$$\begin{aligned}
1 \overset{3123151321}{8} 12321 \text{ } [\# = 7] \quad \Delta = -31, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
1 \overset{3123151321}{8} 12321 \text{ } [\# = 3] \quad \Delta = -31, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{123151321}{8} 12321 \textcolor{red}{3} [\# = 8] \quad \Delta = -31, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1 \overset{3123151321}{8} 12321 \text{ } [\# = 8] \quad \Delta = -31, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1 \overset{123151321}{8} 12321 \textcolor{red}{2} \quad \Delta = -23, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \overset{3123151321}{8} 12321 \text{ } [\# = 4] \quad \Delta = -31, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \overset{123151321}{8} 12321 \textcolor{red}{3} [\# = 3] \quad \Delta = -31, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \overset{123151321}{8} 12321 \textcolor{red}{3} [\# = 4] \quad \Delta = -31, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1 \overset{3123151321}{8} 12321 \text{ } [\# = 5] \quad \Delta = -31, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1 \overset{123151321}{8} 12321 \quad \Delta = -23, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \overset{123151321}{8} 12321 \quad \Delta = -23, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1 \overset{123151321}{8} 12321 \quad \Delta = -23, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1 \overset{123151321}{8} 12321 \textcolor{red}{3} [\# = 9] \quad \Delta = -31, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1 \overset{123151321}{8} 12321 \textcolor{red}{3} [\# = 7] \quad \Delta = -31, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1 \overset{3123151321}{8} 12321 \text{ } [\# = 6] \quad \Delta = -31, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1 \overset{123151321}{8} 12321 \textcolor{red}{3} [\# = 6] \quad \Delta = -31, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1 \overset{123151321}{8} 12321 \textcolor{red}{3} [\# = 2] \quad \Delta = -31, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \overset{3123151321}{8} 12321 \text{ } [\# = 2] \quad \Delta = -31, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1 \overset{3123151321}{8} 12321 \text{ } [\# = 9] \quad \Delta = -31, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix}
\end{aligned}$$

$$1^{123151321} 8^{12321} 12321\bar{3} \text{ [#}=5] \quad \Delta = -31, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

Base: 161232181232161

$$161232181232161 \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$161232181232161 \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$161232181232161 \quad \Delta = -120, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: 612321812321613

$$612321812321613 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$612321812321613 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$612321812321613 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$612321812321613 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$612321812321613 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$612321812321613 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: $1^{12321} 8^{12321} 123216131$

$$1^{312321} 8^{12321} 123216131 \text{ [#}=9] \quad \Delta = -58, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix}$$

$$1^{312321} 8^{12321} 123216131 \text{ [#}=5] \quad \Delta = -58, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix}$$

$$1^{312321} 8^{12321} 123216131 \text{ [#}=4] \quad \Delta = -58, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$1^{12321} 8^{12321} 123216131\bar{3} \text{ [#}=3] \quad \Delta = -58, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1^{12321} 8^{12321} 123216131\bar{3} \text{ [#}=4] \quad \Delta = -58, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1^{12321} 8^{12321} 123216131\bar{3} \text{ [#}=9] \quad \Delta = -58, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1^{312321} 8^{12321} 123216131 \text{ [#}=3] \quad \Delta = -58, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1^{12321} 8^{12321} 123216131\bar{3} \text{ [#}=2] \quad \Delta = -58, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$\begin{aligned}
& \overset{3}{1} \overset{12321}{8} 123216131 \text{ }_{[\# = 2]} \quad \Delta = -58, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{12321}{1} \overset{12321}{8} 123216131 \overset{3}{3} \text{ }_{[\# = 7]} \quad \Delta = -58, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{3}{1} \overset{12321}{8} 123216131 \text{ }_{[\# = 6]} \quad \Delta = -58, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
& \overset{12321}{1} \overset{12321}{8} 123216131 \overset{3}{3} \text{ }_{[\# = 6]} \quad \Delta = -58, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{12321}{1} \overset{12321}{8} 123216131 \overset{2}{2} \quad \Delta = -50, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{12321}{1} \overset{12321}{8} 123216131 \quad \Delta = -50, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{3}{1} \overset{12321}{8} 123216131 \text{ }_{[\# = 8]} \quad \Delta = -58, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
& \overset{12321}{1} \overset{12321}{8} 123216131 \overset{2}{2} \quad \Delta = -50, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{12321}{1} \overset{12321}{8} 123216131 \overset{3}{3} \text{ }_{[\# = 5]} \quad \Delta = -58, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{3}{1} \overset{12321}{8} 123216131 \text{ }_{[\# = 7]} \quad \Delta = -58, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
& \overset{12321}{1} \overset{12321}{8} 123216131 \overset{3}{3} \text{ }_{[\# = 8]} \quad \Delta = -58, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{12321}{1} \overset{12321}{8} 123216131 \overset{2}{2} \quad \Delta = -50, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \text{Base: } 6131612321 \overset{2321}{8} 1 \\
& \overset{2}{6} 131612321 \overset{2321}{8} 1 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& 6131612321 \overset{2321}{8} 1 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 6131612321 \overset{2321}{8} \overset{2}{12} \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{2}{6} 131612321 \overset{2321}{8} 1 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 6131612321 \overset{2}{8} \overset{2321}{1} \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 225 \quad \begin{pmatrix} 0 & 15 & 2 \\ 15 & 163 & 25 \\ 2 & 25 & 8 \end{pmatrix} \\
& 6131612321 \overset{2}{8} \overset{2321}{1} \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
& \text{Base: } 7123151321 \overset{2321}{8} 1
\end{aligned}$$

$$\begin{aligned}
7123151321 \overset{2321}{8} 1 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
7123151321 \overset{2}{8} 1 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
7123151321 \overset{2}{8} 1 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
7123151321 \overset{2321}{8} 1 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
7123151321 \overset{2}{8} 1 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
7123151321 \overset{2}{8} 1 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}
\end{aligned}$$

Base: $712321 \overset{23151321}{8} 1$

$$\begin{aligned}
712321 \overset{23151321}{8} 1 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
712321 \overset{23151321}{8} 1 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
712321 \overset{2}{8} 1 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
712321 \overset{2}{8} 1 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
712321 \overset{23151321}{8} 1 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 225 & \begin{pmatrix} 0 & 15 & 2 \\ 15 & 163 & 25 \\ 2 & 25 & 8 \end{pmatrix} \\
712321 \overset{23151321}{8} 1 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{3151321}{8} 1232171$

$$\begin{aligned}
3151321 \overset{21}{8} 1232171 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \overset{3151321}{8} 1232171 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
1 \overset{3151321}{8} 1232171 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 163 & 25 \\ 2 & 25 & 8 \end{pmatrix} \\
1 \overset{3151321}{8} 1232171 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
1 \overset{3151321}{8} 1232171 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \overset{3151321}{8} 1232171 \quad \Delta = -127, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{512321}{8} 12321712$

$$\begin{aligned}
& \overset{512321}{1} \overset{2}{8} 12321712 \quad \Delta = -121, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{512321}{1} \overset{2}{8} 12321712 \quad \Delta = -121, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{512321}{1} \overset{2}{8} 12321712 \quad \Delta = -121, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& \overset{512321}{1} \overset{2}{8} 12321712 \quad \Delta = -121, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 512321 \overset{21}{8} 12321712 \quad \Delta = -121, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{512321}{1} \overset{2}{8} 12321712 \quad \Delta = -121, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{151321}{1} \overset{2}{8} 12321712$

$$\begin{aligned}
& 151321 \overset{21}{8} 12321712 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{151321}{1} \overset{2}{8} 12321712 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{151321}{1} \overset{2}{8} 12321712 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{151321}{1} \overset{2}{8} 12321712 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{151321}{1} \overset{2}{8} 12321712 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{151321}{1} \overset{2}{8} 12321712 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{12321}{1} \overset{2}{8} 123217122$

$$\begin{aligned}
& \overset{12321}{1} \overset{2}{8} 123217122 \quad \Delta = -69, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{312321}{1} \overset{2}{8} 123217122 \text{ [#=3]} \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{312321}{1} \overset{2}{8} 123217122 \text{ [#=2]} \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{312321}{1} \overset{2}{8} 123217122 \text{ [#=4]} \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& \overset{312321}{1} \overset{2}{8} 123217122 \text{ [#=5]} \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
& \overset{312321}{1} \overset{2}{8} 123217122 \text{ [#=6]} \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{l}
1 \overset{3}{8} 123217122_{[\# = 7]} \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
1 \overset{3}{8} 123217122_{[\# = 8]} \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
1 \overset{3}{8} 123217122_{[\# = 9]} \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
1 \overset{12321}{8} 123217122 \overset{3}{} \quad \Delta = -78, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
1 \overset{12321}{8} 123217122 \overset{2}{} \quad \Delta = -69, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1 \overset{12321}{8} 123217122 \overset{2}{} \quad \Delta = -69, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1 \overset{12321}{8} 123217122 \quad \Delta = -69, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{51321}{8} 123217123 \\
1 \overset{51321}{8} 123217123 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
1 \overset{51321}{8} 123217123 \overset{2}{} \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \overset{51321}{8} 123217123 \overset{2}{} \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1 \overset{51321}{8} 123217123 \overset{2}{} \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
51321 \overset{21}{8} 123217123 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1 \overset{51321}{8} 123217123 \quad \Delta = -128, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\text{Base: } 17123218123215 \\
1712321 \overset{21}{8} 123215 \quad \Delta = -121, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1712321 \overset{12}{8} 123215 \overset{1}{} \quad \Delta = -121, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1712321 \overset{1}{8} 123215 \overset{1}{8} \overset{2}{} \quad \Delta = -121, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1712321 \overset{1}{8} 123215 \overset{12}{8} \quad \Delta = -121, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1712321 \overset{1}{8} 123215 \overset{21}{8} \quad \Delta = -121, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\text{Base: } 17123218123151
\end{array}$$

$$1\overset{1}{7}1\overset{2}{2}321\overset{1}{8}123151 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1\overset{1}{7}12321\overset{2}{8}123151 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1\overset{1}{7}12321\overset{1}{8}1\overset{2}{2}3151 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1\overset{1}{7}12321\overset{2}{8}123151 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1\overset{1}{7}12321\overset{1}{8}1\overset{2}{2}3151 \quad \Delta = -119, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: $1\overset{21}{7}12321\overset{1}{8}12321$

$$1\overset{21}{7}12321\overset{1}{8}12321\overset{1}{3} \text{ [#=7]} \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$1\overset{21}{7}12321\overset{1}{8}12321\overset{1}{3} \text{ [#=6]} \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$1\overset{21}{7}12321\overset{1}{8}12321\overset{1}{3} \text{ [#=5]} \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$1\overset{21}{7}12321\overset{1}{8}12321\overset{1}{3} \text{ [#=4]} \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1\overset{21}{7}12321\overset{1}{8}12321\overset{1}{3} \text{ [#=3]} \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1\overset{21}{7}1\overset{2}{2}321\overset{1}{8}12321 \quad \Delta = -69, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1\overset{21}{7}12321\overset{1}{8}12321\overset{1}{3} \text{ [#=8]} \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$1\overset{21}{7}12321\overset{1}{8}12321\overset{1}{3} \text{ [#=9]} \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$1\overset{21}{7}12321\overset{1}{8}1\overset{2}{2}321 \quad \Delta = -69, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1\overset{21}{7}12321\overset{2}{8}12321 \quad \Delta = -69, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1\overset{21}{7}12321\overset{1}{8}12321 \quad \Delta = -69, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1\overset{21}{7}12321\overset{1}{8}12321\overset{1}{3} \text{ [#=2]} \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: $41321\overset{1}{8}123218122$

$$41321\overset{21}{8}123218122 \quad \Delta = -123, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\begin{aligned}
41321\overset{1}{8}123218122 \quad \Delta = -123, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
41321\overset{1}{8}12321\overset{2}{8}122 \quad \Delta = -123, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
41321\overset{2}{8}123218122 \quad \Delta = -123, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
41321\overset{2}{8}123218122 \quad \Delta = -123, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
41321\overset{1}{8}12321\overset{2}{8}122 \quad \Delta = -123, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
41321\overset{1}{8}12321\overset{3}{8}122 \quad \Delta = -132, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
41321\overset{2}{8}123218122 \quad \Delta = -125, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{1321}{8} 1232181222$

$$\begin{aligned}
1 \overset{31321}{8} 1232181222 \text{ [#=2]} \quad \Delta = -103, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \overset{1321}{8} 12321\overset{2}{8}1222 \quad \Delta = -95, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \overset{31321}{8} 1232181222 \text{ [#=5]} \quad \Delta = -103, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1 \overset{1321}{8} 12321\overset{2}{8}1222 \quad \Delta = -95, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{1321}{8} 12321\overset{2}{8}1222 \quad \Delta = -95, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1 \overset{1321}{8} 1232181222 \quad \Delta = -95, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1 \overset{31321}{8} 1232181222 \text{ [#=4]} \quad \Delta = -103, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1 \overset{31321}{8} 1232181222 \text{ [#=3]} \quad \Delta = -103, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

Base: $41321\overset{1}{8}12321\overset{21}{8}1$

$$\begin{aligned}
41321\overset{2}{8}12321\overset{21}{8}1 \quad \Delta = -125, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
41321\overset{2}{8}12321\overset{21}{8}1 \quad \Delta = -123, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
41321\overset{1}{8}12321\overset{21}{8}1 \quad \Delta = -123, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
41321\overset{21}{8}12321\overset{21}{8}1 \quad \Delta = -123, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
41321\overset{21}{8}12321\overset{21}{8}1 \quad \Delta = -123, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
41321\overset{12}{8}12321\overset{21}{8}1 \quad \Delta = -123, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
41321\overset{1}{8}12321\overset{221}{8}1 \quad \Delta = -123, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1\overset{221}{8}12321\overset{1}{8}1231$

$$\begin{aligned}
1\overset{221}{8}12321\overset{12}{8}1231 \quad \Delta = -95, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1\overset{221}{8}12321\overset{1}{8}1231 \quad \Delta = -95, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1\overset{221}{8}12321\overset{21}{8}1231 \quad \Delta = -95, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1\overset{2212}{8}12321\overset{1}{8}1231 \quad \Delta = -95, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1\overset{221}{8}12321\overset{1}{8}1231\overset{3}{\text{[#}=5]} \quad \Delta = -103, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1\overset{221}{8}12321\overset{1}{8}1231\overset{3}{\text{[#}=4]} \quad \Delta = -103, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1\overset{221}{8}12321\overset{1}{8}1231\overset{3}{\text{[#}=3]} \quad \Delta = -103, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1\overset{221}{8}12321\overset{1}{8}1231\overset{3}{\text{[#}=2]} \quad \Delta = -103, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1\overset{221}{8}12321\overset{1}{8}1231 \quad \Delta = -104, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: $21\overset{21}{8}12321712321$

$$\begin{aligned}
21\overset{221}{8}12321712321 \quad \Delta = -69, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21\overset{21}{8}12321712321\overset{3}{\text{[#}=4]} \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21\overset{21}{8}12321712321\overset{3}{\text{[#}=3]} \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
21\overset{21}{8}12321712321\overset{3}{\text{[#}=2]} \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21\overset{21}{8}12321712321\overset{3}{\text{[#}=8]} \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
21 \bar{8} 12321712321 \bar{3} \text{ [#=5]} \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
21 \bar{8} 12321712321 \bar{3} \text{ [#=9]} \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
21 \bar{8} 12321712321 \bar{3} \text{ [#=7]} \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
21 \bar{8} 12321712321 \quad \Delta = -69, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21 \bar{8} 12321712321 \quad \Delta = -69, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21 \bar{8} 12321712321 \bar{3} \text{ [#=6]} \quad \Delta = -77, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21 \bar{8} 12321712321 \quad \Delta = -69, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: $21 \bar{8} 12321812231$

$$\begin{aligned}
21 \bar{8} 12321812231 \bar{3} \text{ [#=4]} \quad \Delta = -103, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21 \bar{8} 12321812231 \bar{3} \text{ [#=3]} \quad \Delta = -103, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
21 \bar{8} 12321812231 \bar{3} \text{ [#=5]} \quad \Delta = -103, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21 \bar{8} 12321812231 \bar{3} \text{ [#=2]} \quad \Delta = -103, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
21 \bar{8} 12321812231 \quad \Delta = -95, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21 \bar{8} 12321812231 \quad \Delta = -95, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21 \bar{8} 12321812231 \quad \Delta = -95, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21 \bar{8} 12321812231 \quad \Delta = -95, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: $21 \bar{8} 12321812313$

$$\begin{aligned}
21 \bar{8} 12321812313 \quad \Delta = -103, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21 \bar{8} 12321812313 \quad \Delta = -103, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21 \bar{8} 12321812313 \quad \Delta = -111, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \begin{matrix} 21 \\ 21\ 8\ 12321812313 \end{matrix} \begin{matrix} 3 \\ \text{[#}=2 \end{matrix} \quad \Delta = -111, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \begin{matrix} 21\ 2 \\ 21\ 8\ 12321812313 \end{matrix} \quad \Delta = -103, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \begin{matrix} 221 \\ 21\ 8\ 12321812313 \end{matrix} \quad \Delta = -103, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \begin{matrix} 21 \\ 21\ 8\ 12321812313 \end{matrix} \begin{matrix} 2 \\ \end{matrix} \quad \Delta = -103, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \begin{matrix} 21 \\ 21\ 8\ 12321812313 \end{matrix} \begin{matrix} 2 \\ \end{matrix} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \text{Base: } \begin{matrix} 21 \\ 21\ 8\ 1232181232 \end{matrix} \begin{matrix} 1 \\ \end{matrix} \\
& \begin{matrix} 21 \\ 21\ 8\ 1232181232 \end{matrix} \begin{matrix} 41 \\ \end{matrix} \quad \Delta = -125, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
& \begin{matrix} 221 \\ 21\ 8\ 1232181232 \end{matrix} \begin{matrix} 1 \\ \end{matrix} \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \begin{matrix} 21 \\ 21\ 8\ 1232181232 \end{matrix} \begin{matrix} 21 \\ \end{matrix} \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \begin{matrix} 21 \\ 21\ 8\ 1232181232 \end{matrix} \begin{matrix} 2 \\ 1 \end{matrix} \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \begin{matrix} 21 \\ 21\ 8\ 1232181232 \end{matrix} \begin{matrix} 31 \\ \text{[#}=2 \end{matrix} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \begin{matrix} 21 \\ 21\ 8\ 1232181232 \end{matrix} \begin{matrix} 2 \\ 1 \end{matrix} \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \begin{matrix} 21\ 2 \\ 21\ 8\ 1232181232 \end{matrix} \begin{matrix} 1 \\ \end{matrix} \quad \Delta = -97, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \begin{matrix} 21 \\ 21\ 8\ 1232181232 \end{matrix} \begin{matrix} 31 \\ \text{[#}=3 \end{matrix} \quad \Delta = -105, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \text{Base: } \begin{matrix} 1321 \\ 1\ 8\ 12321812 \end{matrix} \begin{matrix} 21 \\ \end{matrix} \\
& \begin{matrix} 31321 \\ 1\ 8\ 12321812 \end{matrix} \begin{matrix} 21 \\ \text{[#}=2 \end{matrix} \quad \Delta = -103, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \begin{matrix} 31321 \\ 1\ 8\ 12321812 \end{matrix} \begin{matrix} 21 \\ \text{[#}=4 \end{matrix} \quad \Delta = -103, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \begin{matrix} 1321\ 2 \\ 1\ 8\ 12321812 \end{matrix} \begin{matrix} 21 \\ \end{matrix} \quad \Delta = -95, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \begin{matrix} 1321 \\ 1\ 8\ 12321812 \end{matrix} \begin{matrix} 221 \\ \end{matrix} \quad \Delta = -95, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \begin{matrix} 1321 \\ 1\ 8\ 12321812 \end{matrix} \begin{matrix} 21\ 2 \\ \end{matrix} \quad \Delta = -95, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

$$1 \overset{1321}{8} \overset{21}{12321812} \quad \Delta = -95, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1 \overset{31321}{8} \overset{21}{12321812} [\# = 5] \quad \Delta = -103, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$1 \overset{31321}{8} \overset{21}{12321812} [\# = 3] \quad \Delta = -103, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: $21 \overset{321}{8} 1232181231$

$$21 \overset{321}{8} \overset{2}{1232181231} \quad \Delta = -104, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$321 \overset{321}{8} 1232181231 \quad \Delta = -113, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$21 \overset{321}{8} 1232181231 \overset{3}{[\# = 5]} \quad \Delta = -112, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$21 \overset{321}{8} 1232181231 \overset{3}{[\# = 4]} \quad \Delta = -112, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$21 \overset{321}{8} 1232181231 \overset{3}{[\# = 3]} \quad \Delta = -112, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$21 \overset{321}{8} 1232181231 \overset{3}{[\# = 2]} \quad \Delta = -112, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$21 \overset{321}{8} 1232181231 \overset{2}{\quad} \quad \Delta = -104, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$21 \overset{321}{8} 1232181231 \quad \Delta = -104, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$21 \overset{2321}{8} 1232181231 \quad \Delta = -104, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 24 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$21 \overset{321}{8} \overset{2}{1232181231} \quad \Delta = -104, n_+ = 1, n_- = 16, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$T = 17$ (2318 bases)

Base: 223141414132211212

$$223141414132211212 \overset{2}{\quad} \quad \Delta = -130, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$223141414132211212 \overset{2}{\quad} \quad \Delta = -130, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$223141414132211212 \overset{2}{\quad} \quad \Delta = -130, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$223141414132211212 \overset{2}{\quad} \quad \Delta = -130, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 108 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$223141414132211212 \quad \Delta = -130, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$223141414132211212 \quad \Delta = -130, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: 314132211212231413

$$314132211212231413 \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$314132211212231413 \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$314132211212231413 \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$314132211212231413 \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 141322112122314132

$$141322112122314132 \quad \Delta = -113, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$141322112122314132 \quad \Delta = -111, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$141322112122314132 \quad \Delta = -111, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$141322112122314132 \quad \Delta = -111, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$141322112122314132 \quad \Delta = -111, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$2314132211212231413 \text{ [#=2]} \quad \Delta = -119, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$2314132211212231413 \text{ [#=3]} \quad \Delta = -119, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$141322112122314132 \quad \Delta = -111, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 141413221121223132

$$141413221121223132 \quad \Delta = -111, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$141413221121223132 \quad \Delta = -113, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$141413221121223132 \quad \Delta = -111, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$141413221121223132 \quad \Delta = -111, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$\begin{aligned}
141413221121223132^2 \quad \Delta = -111, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
141413221121223132^2 \quad \Delta = -111, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
2313221121223141413^{\#=3} \quad \Delta = -119, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
141413221121223132^3 \quad \Delta = -119, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2313221121223141413^{\#=2} \quad \Delta = -119, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\text{Base: } 314141322112122313 \\
314141322112122313^3 \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
314141322112122313^3 \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
314141322112122313^2 \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
314141322112122313^3 \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
314141322112122313^2 \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
314141322112122313^2 \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
314141322112122313^2 \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
314141322112122313^2 \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
314141322112122313^2 \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\text{Base: } 231414132211212231 \\
2314141322112122313^{\#=4} \quad \Delta = -119, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
231414132211212231^2 \quad \Delta = -113, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
231414132211212231^2 \quad \Delta = -111, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
231414132211212231^2 \quad \Delta = -111, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
231414132211212231^2 \quad \Delta = -111, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 70 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
231414132211212231^3_{\# = 3} \quad \Delta = -119, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
231414132211212231^2 \quad \Delta = -111, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
231414132211212231^2 \quad \Delta = -111, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
231414132211212231^3_{\# = 5} \quad \Delta = -119, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
231414132211212231^3_{\# = 2} \quad \Delta = -119, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}
\end{aligned}$$

Base: 141414132211212222

$$\begin{aligned}
141414132211212222^2 \quad \Delta = -121, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
141414132211212222^2 \quad \Delta = -121, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
141414132211212222^2 \quad \Delta = -121, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
141414132211212222^2 \quad \Delta = -121, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
141414132211212222^2 \quad \Delta = -121, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
222211212231414141^3_{\# = 3} \quad \Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
222211212231414141^3_{\# = 2} \quad \Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: 314141413221121222

$$\begin{aligned}
314141413221121222^2 \quad \Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
314141413221121222^2 \quad \Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
314141413221121222^2 \quad \Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 70 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
314141413221121222^2 \quad \Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
314141413221121222^2 \quad \Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
314141413221121222^2 \quad \Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
314141413221121222^3 \quad \Delta = -137, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 231414141322112122

$$\begin{aligned}
231414141322112122^2 \quad \Delta = -130, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
231414141322112122^2 \quad \Delta = -130, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 88 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
231414141322112122^2 \quad \Delta = -130, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
231414141322112122^2 \quad \Delta = -130, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
231414141322112122^2 \quad \Delta = -130, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
231414141322112122^2 \quad \Delta = -130, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 121 \quad & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
231414141322112122^2 \quad \Delta = -132, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 223132211212231512

$$\begin{aligned}
223132211212231512^2 \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
223132211212231512^2 \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
223132211212231512^2 \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
223132211212231512^2 \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
223132211212231512^2 \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
223132211212231512^3 \quad \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
223132211212231512^3 \quad \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 60 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

Base: 231322112122315123

$$\begin{aligned}
231322112122315123^2 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
231322112122315123^2 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
231322\overset{2}{1}12122315123 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
2313221121223\overset{2}{1}5123 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
231322112122315\overset{2}{1}23 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
231322\overset{3}{1}12122315123 \quad \Delta = -124, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 32 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
231322\overset{2}{1}12122315123 \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 313221121223151232

$$\begin{aligned}
31322\overset{3}{1}121223151232 \text{ [#=2]} \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
31322\overset{3}{1}121223151232 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
31322\overset{2}{1}121223151232 \quad \Delta = -117, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
313221121223\overset{2}{1}51232 \quad \Delta = -117, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
31322\overset{2}{1}121223151232 \quad \Delta = -117, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
313221121223\overset{2}{1}51232 \quad \Delta = -117, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
31322\overset{2}{1}121223151232 \quad \Delta = -117, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 22222112122315131

$$\begin{aligned}
22222\overset{2}{1}12122315131 \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
22222112122315131\overset{3}{3} \text{ [#=2]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
22222112122315131\overset{3}{3} \text{ [#=4]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
22222112122315131\overset{3}{3} \text{ [#=6]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
22222112122315131\overset{3}{3} \text{ [#=7]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
22222112122315131\overset{3}{3} \text{ [#=8]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

$$222222112122315131^{\color{red}3} \text{ [#=9]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$222222112122315131^{\color{red}3} \text{ [#=5]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$222222112122315131^{\color{red}3} \text{ [#=3]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$222222112122315131^{\color{red}2} \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$222222112122315131^{\color{red}2} \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$222222112122315131^{\color{red}2} \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: 413151322112122222

$$413151322112122222^{\color{red}2} \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$413151322112122222^{\color{red}2} \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$413151322112122222^{\color{red}2} \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$413151322112122222^{\color{red}2} \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$413151322112122222^{\color{red}2} \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$413151322112122222^{\color{red}2} \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$413151322112122222^{\color{red}3} \quad \Delta = -133, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

Base: 22313221121223151¹

$$22313221121223151^{\color{red}2} \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$22313221121223151^{\color{red}21} \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$22313221121223151^{\color{red}2} \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$22313221121223151^{\color{red}3} \quad \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 60 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$22313221121223151^{\color{red}2} \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

Base: 23132211212231513¹

$$\begin{array}{ll}
\overset{2}{2}3132211212231\overset{1}{5}13 & \Delta = -117, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
23132211212231\overset{21}{5}13 & \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
23132211212231\overset{12}{5}13 & \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
23132211212231\overset{2}{5}13 & \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
23132211212231\overset{21}{5}13 & \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
23132211212231\overset{2}{5}13 & \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
23132211212231\overset{31}{5}13 & \Delta = -123, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\overset{3}{2}3132211212231\overset{1}{5}13 & \Delta = -123, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 32 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\overset{2}{2}3132211212231\overset{1}{5}13 & \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{array}$$

Base: 3132211212231¹5132

$$\begin{array}{ll}
3132211212231\overset{21}{5}132 & \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\overset{2}{3}132211212231\overset{1}{5}132 & \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
3132211212231\overset{2}{5}132 & \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
3132211212231\overset{21}{5}132 & \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\overset{3}{3}132211212231\overset{1}{5}132 & \Delta = -123, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
3132211212231\overset{2}{5}132 & \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
3132211212231\overset{12}{5}132 & \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
3132211212231\overset{31}{5}132 & \Delta = -123, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\overset{3}{3}132211212231\overset{1}{5}132 \text{ }_{[\# = 2]} & \Delta = -123, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{array}$$

Base: 132211212231¹51322

$$\begin{array}{ll}
1322112122315^1 1322 & \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1322112122315^{12} 1322 & \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1322112122315^{21} 1322 & \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1322112122315^1 1322 & \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
2231^{3132211212231} 5^1 1 [\# = 5] & \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
2231^{3132211212231} 5^1 1 [\# = 4] & \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
2231^{3132211212231} 5^1 1 [\# = 3] & \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1322112122315^{21} 1322 & \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
2231^{3132211212231} 5^1 1 [\# = 2] & \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1322112122315^{31} 1322 & \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\text{Base: } 11212231414141322 & \\
11212231414141322^{12} & \Delta = -130, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 88 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
11212231414141322^1 & \Delta = -130, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
11212231414141322^1 & \Delta = -130, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
11212231414141322^1 & \Delta = -130, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
11212231414141322^1 & \Delta = -130, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\text{Base: } 1121223141414132^{21} & \\
1121223141414132^{21} & \Delta = -132, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1121223141414132^{21} & \Delta = -130, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1121223141414132^{21} & \Delta = -130, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
^{21} \overset{2}{1121223141414132} \quad \Delta = -130, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 70 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
^{21} \overset{2}{1121223141414132} \quad \Delta = -130, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
^{21} \overset{2}{1121223141414132} \quad \Delta = -130, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
^{21} \overset{2}{1121223141414132} \quad \Delta = -130, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: $14132211\overset{1}{2}12231413$

$$\begin{aligned}
\overset{2}{14132211}\overset{1}{2}12231413 \quad \Delta = -110, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\overset{2}{14132211}\overset{1}{2}12231413 \quad \Delta = -110, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\overset{1}{14132211}\overset{2}{2}12231413 \quad \Delta = -110, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\overset{1}{14132211}\overset{2}{2}12231413 \quad \Delta = -110, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
314132211\overset{1}{2}1223141\overset{3}{3} [\# = 3] \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
314132211\overset{1}{2}1223141\overset{3}{3} [\# = 2] \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
14132211\overset{1}{2}1223141\overset{3}{3} \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\overset{1}{14132211}\overset{2}{2}12231413 \quad \Delta = -110, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1\overset{313221}{12}122314141$

$$\begin{aligned}
\overset{313221}{1}\overset{2}{12}122314141 \quad \Delta = -110, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\overset{313221}{1}\overset{2}{12}122314141 \quad \Delta = -110, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
\overset{313221}{1}\overset{2}{12}122314141 \quad \Delta = -110, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\overset{313221}{1}\overset{2}{12}122314141 \quad \Delta = -110, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\overset{313221}{1}\overset{2}{12}122314141 \quad \Delta = -110, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
\overset{313221}{1}\overset{2}{12}122314141 \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 21 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

$1 \overset{313221}{12} 122314141\overset{3}{3} \text{ }_{[\# = 3]}$	$\Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$
$1 \overset{313221}{12} 122314141\overset{3}{3} \text{ }_{[\# = 2]}$	$\Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, \det = 4$	$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$
$1 \overset{313221}{12} 122314141 \text{ }_{[\# = 2]}$	$\Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 93 & 19 \\ 2 & 19 & 8 \end{pmatrix}$
Base: $1 \overset{13221}{12} 1223141413$		
$1 \overset{313221}{12} 1223141413 \text{ }_{[\# = 2]}$	$\Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, \det = 4$	$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$
$1 \overset{313221}{12} 1223141413 \text{ }_{[\# = 4]}$	$\Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$
$1 \overset{13221}{12} 122314141\overset{3}{3}$	$\Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$
$1 \overset{313221}{12} 1223141413 \text{ }_{[\# = 3]}$	$\Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$
$1 \overset{313221}{12} 1223141413 \text{ }_{[\# = 5]}$	$\Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$
$1 \overset{13221}{12} 122314141\overset{2}{2}$	$\Delta = -110, n_+ = 1, n_- = 17, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$
$1 \overset{13221}{12} 122314141\overset{2}{2}$	$\Delta = -110, n_+ = 1, n_- = 17, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
$1 \overset{13221}{12} 122314141\overset{2}{2}$	$\Delta = -110, n_+ = 1, n_- = 17, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$
$1 \overset{13221}{12} 122314141\overset{2}{2}$	$\Delta = -110, n_+ = 1, n_- = 17, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$
$1 \overset{13221}{12} 1223141413$	$\Delta = -110, n_+ = 1, n_- = 17, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$
Base: $1 \overset{2221}{12} 12231414141$		
$1 \overset{2221}{12} 12231414141\overset{2}{2}$	$\Delta = -121, n_+ = 1, n_- = 17, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix}$
$1 \overset{2221}{12} 12231414141\overset{2}{2}$	$\Delta = -121, n_+ = 1, n_- = 17, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 89 & 19 \\ 2 & 19 & 8 \end{pmatrix}$
$1 \overset{2221}{12} 12231414141$	$\Delta = -121, n_+ = 1, n_- = 17, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
$1 \overset{2221}{12} 12231414141\overset{2}{2}$	$\Delta = -121, n_+ = 1, n_- = 17, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
$1 \overset{2221}{12} 12231414141\overset{2}{2}$	$\Delta = -121, n_+ = 1, n_- = 17, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$

$$\begin{aligned}
& \overset{2221}{1 \ 12 \ 122314141413} \text{ [#=2]} \quad \Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& \overset{2221}{1 \ 12 \ 122314141413} \text{ [#=3]} \quad \Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{221}{112122314141413}$

$$\begin{aligned}
& \overset{221}{112122314141413} \text{ }^3 \quad \Delta = -137, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{221}{112122314141413} \text{ }^2 \quad \Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{221}{112122314141413} \text{ }^2 \quad \Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{221}{112122314141413} \text{ }^2 \quad \Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{221}{112122314141413} \text{ }^2 \quad \Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{221}{112122314141413} \quad \Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{221}{112122314141413} \text{ }^2 \quad \Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{21513221}{1 \ 12 \ 1223132}$

$$\begin{aligned}
& \overset{21513221}{1 \ 12 \ 1223132} \quad \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
& \overset{21513221}{1 \ 12 \ 1223132} \text{ }^3 \quad \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 32 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{21513221}{1 \ 12 \ 1223132} \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{21513221}{1 \ 12 \ 1223132} \text{ }^2 \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{21513221}{1 \ 12 \ 1223132} \text{ }^2 \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{21513221}{1 \ 12 \ 1223132} \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{21513221}{1 \ 12 \ 1223132} \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{21513221}{1 \ 12 \ 1223132} \text{ }^2 \quad \Delta = -109, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{321513221}{1 \ 12 \ 122313}$

$$\overset{321513221}{1 \ 12 \ 122313} \text{ }^3 \quad \Delta = -123, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\begin{aligned}
& \begin{matrix} 321513221 \\ 1 \quad 12 \quad 122313 \end{matrix} \begin{matrix} \textcolor{red}{3} \\ [\# = 2] \end{matrix} \quad \Delta = -123, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& \begin{matrix} 321513221 \\ 1 \quad 12 \quad 122313 \end{matrix} \quad \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \begin{matrix} 321513221 \\ 1 \quad 12 \quad 122313 \end{matrix} \begin{matrix} \textcolor{red}{2} \\ \end{matrix} \quad \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \begin{matrix} 321513221 \\ 1 \quad 12 \quad 122313 \end{matrix} \quad \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \begin{matrix} 321513221 \\ 1 \quad 12 \quad 122313 \end{matrix} \quad \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \begin{matrix} 321513221 \textcolor{red}{2} \\ 1 \quad 12 \quad 122313 \end{matrix} \quad \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \textbf{Base: } \begin{matrix} 2321513221 \\ 1 \quad 12 \quad 12231 \end{matrix} \\
& \begin{matrix} 2321513221 \\ 1 \quad 12 \quad 12231 \end{matrix} \quad \Delta = -109, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \begin{matrix} 2321513221 \textcolor{red}{2} \\ 1 \quad 12 \quad 12231 \end{matrix} \quad \Delta = -109, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \begin{matrix} 2321513221 \\ 1 \quad 12 \quad 12231 \end{matrix} \begin{matrix} \textcolor{red}{3} \\ [\# = 3] \end{matrix} \quad \Delta = -117, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \begin{matrix} 2321513221 \\ 1 \quad 12 \quad 12231 \end{matrix} \begin{matrix} \textcolor{red}{3} \\ [\# = 2] \end{matrix} \quad \Delta = -117, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \begin{matrix} 2321513221 \\ 1 \quad 12 \quad 12231 \end{matrix} \begin{matrix} \textcolor{red}{3} \\ [\# = 4] \end{matrix} \quad \Delta = -117, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \begin{matrix} 2321513221 \\ 1 \quad 12 \quad 12231 \end{matrix} \begin{matrix} \textcolor{red}{3} \\ [\# = 5] \end{matrix} \quad \Delta = -117, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \begin{matrix} 2321513221 \\ 1 \quad 12 \quad 12231 \end{matrix} \quad \Delta = -109, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \begin{matrix} 2321513221 \\ 1 \quad 12 \quad 12231 \end{matrix} \quad \Delta = -109, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \textbf{Base: } \begin{matrix} 131513221 \\ 1 \quad 12 \quad 12222 \end{matrix} \\
& \begin{matrix} \textcolor{red}{3}131513221 \\ 1 \quad 12 \quad 12222 \end{matrix} \begin{matrix} \textcolor{red}{2} \\ [\# = 5] \end{matrix} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \begin{matrix} \textcolor{red}{3}131513221 \\ 1 \quad 12 \quad 12222 \end{matrix} \begin{matrix} \textcolor{red}{2} \\ [\# = 4] \end{matrix} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \begin{matrix} \textcolor{red}{3}131513221 \\ 1 \quad 12 \quad 12222 \end{matrix} \begin{matrix} \textcolor{red}{2} \\ [\# = 3] \end{matrix} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \begin{matrix} \textcolor{red}{3}131513221 \\ 1 \quad 12 \quad 12222 \end{matrix} \begin{matrix} \textcolor{red}{2} \\ [\# = 2] \end{matrix} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{l}
1 \overset{3131513221}{12} 122222 \text{ [#=6]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \overset{3131513221}{12} 122222 \text{ [#=7]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1 \overset{3131513221}{12} 122222 \text{ [#=9]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1 \overset{3131513221}{12} 122222 \text{ [#=8]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1 \overset{131513221}{12} 122222 \overset{2}{} \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1 \overset{131513221}{12} 122222 \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{131513221}{12} 122222 \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1 \overset{131513221}{12} 122222 \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\text{Base: } 4131513221 \overset{22221}{12} 1 \\
4131513221 \overset{2}{12} \overset{22221}{1} \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
4131513221 \overset{2}{12} \overset{22221}{1} \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
4131513221 \overset{22221}{12} 1 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
4131513221 \overset{2}{12} \overset{22221}{1} \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
4131513221 \overset{3}{12} \overset{22221}{1} \quad \Delta = -133, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
4131513221 \overset{2}{12} \overset{22221}{1} \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
4131513221 \overset{22221}{12} \overset{2}{1} \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{1513221}{12} 1223132 \\
1 \overset{1513221}{12} 1223132 \overset{2}{} \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1 \overset{1513221}{12} 1223132 \overset{2}{} \quad \Delta = -109, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1 \overset{1513221}{12} 1223132 \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{array}$$

$$1^{1513221} 12^{1223132} \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1^{1513221} 12^{1223132} \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1^{1513221} 12^{1223132} \quad \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 32 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\text{Base: } 1^{31513221} 12^{122313}$$

$$1^{31513221} 12^{122313} \quad \Delta = -122, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1^{31513221} 12^{122313} \quad \Delta = -122, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1^{31513221} 12^{122313} [\# = 2] \quad \Delta = -122, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$1^{31513221} 12^{122313} \quad \Delta = -114, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1^{31513221} 12^{122313} \quad \Delta = -114, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1^{31513221} 12^{122313} \quad \Delta = -114, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1^{31513221} 12^{122313} \quad \Delta = -114, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1^{31513221} 12^{122313} \quad \Delta = -114, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1^{31513221} 12^{122313} \quad \Delta = -114, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\text{Base: } 1^{231513221} 12^{12231}$$

$$1^{231513221} 12^{12231} \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1^{231513221} 12^{12231} \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1^{231513221} 12^{12231} \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1^{231513221} 12^{12231} \quad \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1^{231513221} 12^{12231} \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1^{231513221} 12^{122313} [\# = 2] \quad \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\begin{aligned}
& \overset{231513221}{1 \quad 12 \quad 12231\textcolor{red}{3}}_{[\#=3]} \quad \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{231513221\textcolor{red}{2}}{1 \quad 12 \quad 12231} \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{231513221}{1 \quad 12 \quad 12231\textcolor{red}{3}}_{[\#=4]} \quad \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{231513221}{1 \quad 12 \quad 12231\textcolor{red}{3}}_{[\#=5]} \quad \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \textbf{Base: } \overset{21}{2112122314141413} \\
& \overset{21}{2112122314141413\textcolor{red}{3}} \quad \Delta = -137, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{21}{2112122314141413\textcolor{red}{2}} \quad \Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{\textcolor{red}{2}21}{2112122314141413} \quad \Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{21}{2112122314141413\textcolor{red}{2}} \quad \Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{21}{2112122314141413\textcolor{red}{2}} \quad \Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{21}{2112122314141413\textcolor{red}{2}} \quad \Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{21\textcolor{red}{2}}{2112122314141413} \quad \Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \textbf{Base: } \overset{21}{1413221121223141} \\
& \overset{21}{1413221121223141\textcolor{red}{3}}_{[\#=3]} \quad \Delta = -110, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{21}{1413221121223141\textcolor{red}{3}}_{[\#=2]} \quad \Delta = -110, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{21}{1413221121223141} \quad \Delta = -102, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{\textcolor{red}{2}21}{1413221121223141} \quad \Delta = -102, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{\textcolor{red}{2}}{1413221121223141\textcolor{red}{2}} \quad \Delta = -102, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \textbf{Base: } \overset{21}{4132211212231414} \\
& \overset{21\textcolor{red}{2}}{4132211212231414} \quad \Delta = -130, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{21}{4132211212231414\textcolor{red}{2}} \quad \Delta = -130, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
413221\overset{21}{1}212231414 \quad \Delta = -130, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
413221\overset{21}{1}212231\overset{2}{4}14 \quad \Delta = -130, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
413221\overset{2}{1}212231414 \quad \Delta = -130, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
\overset{2}{4}13221\overset{21}{1}212231414 \quad \Delta = -130, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: $21 \overset{13221}{12} 122314141$

$$\begin{aligned}
21 \overset{313221}{12} 122314141 \text{ } [\# = 5] \quad \Delta = -110, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
21 \overset{313221}{12} 122314141 \text{ } [\# = 2] \quad \Delta = -110, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21 \overset{13221}{12} 122314141\overset{3}{1} \text{ } [\# = 2] \quad \Delta = -110, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
21 \overset{313221}{12} 122314141 \text{ } [\# = 3] \quad \Delta = -110, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21 \overset{13221}{12} 122314141\overset{3}{1} \text{ } [\# = 3] \quad \Delta = -110, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
21 \overset{313221}{12} 122314141 \text{ } [\# = 4] \quad \Delta = -110, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
21 \overset{13221}{12} 122314\overset{2}{1}41 \quad \Delta = -102, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21 \overset{13221}{12} 122314141 \quad \Delta = -102, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
21 \overset{13221}{12} 122314141 \quad \Delta = -102, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
21 \overset{13221}{12} 122314141 \quad \Delta = -102, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
21 \overset{213221}{12} 122314141 \quad \Delta = -102, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: $4141413221 \overset{3221}{12} 12$

$$\begin{aligned}
4141413221 \overset{23221}{12} 12 \quad \Delta = -130, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 89 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
4141413221 \overset{2}{12} 12 \quad \Delta = -130, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
4141413221 \overset{3221}{12} 12 \quad \Delta = -130, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& 414141\overset{2}{3}221\overset{3221}{12\ 12} \quad \Delta = -130, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
& 414141\overset{3221}{12\ 12} \quad \Delta = -130, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& 414141\overset{2}{3}221\overset{3221}{12\ 12} \quad \Delta = -130, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \text{Base: } 21\overset{221}{12}12231414141 \\
& 21\overset{221}{12}12231414141 \quad \Delta = -121, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& 21\overset{221}{12}12231414141\overset{3}{\color{red}3} \text{ [#=3]} \quad \Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& 21\overset{221}{12}12231414141\overset{3}{\color{red}3} \text{ [#=2]} \quad \Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& 21\overset{221}{12}12231414141 \quad \Delta = -121, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 21\overset{221}{12}12231414141 \quad \Delta = -121, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 21\overset{221}{12}1223\overset{2}{\color{red}4}1414141 \quad \Delta = -121, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
& 21\overset{221}{12}1223\overset{2}{\color{red}4}1414141 \quad \Delta = -121, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 89 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& 21\overset{221}{12}1223\overset{2}{\color{red}4}1414141 \quad \Delta = -121, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \text{Base: } 21\overset{21513221}{12}122313 \\
& 21\overset{21513221}{12}122313\overset{3}{\color{red}3} \quad \Delta = -114, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 21\overset{21513221}{12}122313 \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& 21\overset{21513221}{12}122313 \quad \Delta = -114, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
& 21\overset{21513221}{12}122313\overset{3}{\color{red}3} \text{ [#=2]} \quad \Delta = -114, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& 21\overset{21513221}{12}122313 \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 21\overset{21513221}{12}122313 \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 21\overset{21513221}{12}122313\overset{2}{\color{red}3} \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

$$21^{21513221} 12^{122313} \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$21^{21513221\textcolor{red}{2}} 12^{122313} \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\text{Base: } 21^{321513221} 12^{12231}$$

$$21^{321513221} 12^{12231\textcolor{red}{3}} [\#=4] \quad \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$21^{321513221} 12^{12231\textcolor{red}{3}} [\#=3] \quad \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$21^{321513221} 12^{12231\textcolor{red}{3}} [\#=2] \quad \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$21^{321513221} 12^{12231\textcolor{red}{3}} [\#=5] \quad \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$21^{321513221\textcolor{red}{2}} 12^{12231} \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$21^{321513221} 12^{12231} \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$21^{321513221} 12^{12231} \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$21^{\textcolor{red}{2}321513221} 12^{12231} \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$21^{321513221} 12^{12231} \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\text{Base: } 21^{131513221} 12^{12222}$$

$$21^{131513221\textcolor{red}{2}} 12^{12222} \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$21^{\textcolor{red}{3}131513221} 12^{12222} [\#=7] \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$21^{\textcolor{red}{2}131513221} 12^{12222} \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$21^{\textcolor{red}{3}131513221} 12^{12222} [\#=5] \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$21^{131513221} 12^{12222} \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$21^{\textcolor{red}{3}131513221} 12^{12222} [\#=9] \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$21^{\textcolor{red}{3}131513221} 12^{12222} [\#=8] \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$\begin{aligned}
21^{3131513221} 12^{2222} [\#=6] \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21^{3131513221} 12^{2222} [\#=4] \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21^{3131513221} 12^{2222} [\#=3] \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
21^{131513221} 12^{2222} \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21^{131513221} 12^{2222} \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
21^{3131513221} 12^{2222} [\#=2] \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\text{Base: } 4131513221^{2221} 12^{12} \\
4131513221^{2221} 12^{12} \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
4131513221^{2221} 12^{12} \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
4131513221^{2221} 12^{12} \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
4131513221^{2221} 12^{12} \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
4131513221^{2221} 12^{12} \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
4131513221^{2221} 12^{12} \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
4131513221^{2221} 12^{12} \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
4131513221^{2221} 12^{12} \quad \Delta = -133, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\text{Base: } 21^{1513221} 12^{12} 122313 \\
21^{1513221} 12^{12} 122313 [\#=2] \quad \Delta = -114, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
21^{1513221} 12^{12} 122313 \quad \Delta = -114, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21^{1513221} 12^{12} 122313 \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21^{1513221} 12^{12} 122313 \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
21^{1513221} 12^{\textcolor{red}{2}} 122313 \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21^{\textcolor{red}{2}} 1513221 122313 \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21^{1513221} 12 122313 \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: $21^{31513221} 12 12231$

$$\begin{aligned}
21^{31513221} 12 12231\textcolor{red}{3} [\# = 4] \quad \Delta = -114, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21^{31513221} 12 12231\textcolor{red}{3} [\# = 5] \quad \Delta = -114, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
21^{31513221} 12 12231\textcolor{red}{3} [\# = 3] \quad \Delta = -114, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21^{31513221} 12 12231\textcolor{red}{3} [\# = 2] \quad \Delta = -114, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
21^{\textcolor{red}{2}} 31513221 12231 \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21^{31513221} 12 12231 \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21^{31513221} 12 12231 \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21^{31513221} 12 12231 \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21^{31513221} 12 12231 \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21^{31513221} 12 12231 \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21^{31513221} 12 12231 \quad \Delta = -114, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: $13221^{1513221} 12 1223$

$$\begin{aligned}
13221^{1513221} 12 1223 \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
3221^{1513221} 12 12231\textcolor{red}{3} [\# = 2] \quad \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
3221^{1513221} 12 12231\textcolor{red}{3} [\# = 3] \quad \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
3221^{1513221} 12 12231\textcolor{red}{3} [\# = 4] \quad \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{ll}
3221 \overset{1513221}{12} 12231\textcolor{red}{3} & \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
3221 \overset{1513221}{12} 12231\textcolor{red}{4} & \Delta = -135, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
13221 \overset{1513221}{12} 1223 & \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
13221 \overset{1513221}{12} 1223\textcolor{red}{2} & \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
3221 \overset{1513221}{12} 12231\textcolor{red}{3} [\# = 5] & \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
3221 \overset{1513221}{12} 12231\textcolor{red}{5} & \Delta = -159, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
13221 \overset{\textcolor{red}{2}1513221}{12} 1223 & \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
3221 \overset{1513221}{12} 12231\textcolor{red}{2} & \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\textbf{Base: } 413221 \overset{1513221}{12} 122 & \\
\textcolor{red}{2}413221 \overset{1513221}{12} 122 & \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
413221 \overset{1513221}{12} 122\textcolor{red}{2} & \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
413221 \overset{1513221}{12} 122 & \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
413221 \overset{\textcolor{red}{2}1513221}{12} 122 & \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
413221 \overset{1513221}{12} 122 & \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\textbf{Base: } 3221 \overset{21513221}{12} 1223 & \\
3221 \overset{21513221}{12} 1223 & \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
3221 \overset{\textcolor{red}{3}21513221}{12} 1223 & \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
3221 \overset{21513221}{12} 1223 & \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
3221 \overset{21513221}{12} 1223 & \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
3221 \overset{21513221}{12} 1223 & \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{array}$$

$$3221 \overset{2}{1}21513221 \overset{1}{1}2 \overset{1}{1}223 \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: $221 \overset{21513221}{1}2 \overset{1}{1}2231$

$$221 \overset{21513221}{1}2 \overset{2}{1}2231 \quad \Delta = -98, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$221 \overset{21513221}{1}2 \overset{1}{1}2231 \quad \Delta = -98, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$221 \overset{21513221}{1}2 \overset{1}{1}2231 \overset{3}{1} [\# = 3] \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$221 \overset{21513221}{1}2 \overset{1}{1}2231 \quad \Delta = -98, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$221 \overset{21513221}{1}2 \overset{1}{1}2231 \quad \Delta = -98, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$221 \overset{21513221}{1}2 \overset{1}{1}2231 \quad \Delta = -98, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$221 \overset{21513221}{1}2 \overset{1}{1}2231 \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$221 \overset{21513221}{1}2 \overset{1}{1}2231 \overset{3}{1} [\# = 2] \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$221 \overset{21513221}{1}2 \overset{1}{1}2231 \overset{3}{1} [\# = 4] \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$221 \overset{21513221}{1}2 \overset{1}{1}2231 \overset{3}{1} [\# = 5] \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

Base: $221 \overset{131513221}{1}2 \overset{1}{1}222$

$$221 \overset{2131513221}{1}2 \overset{1}{1}222 \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$221 \overset{131513221}{1}2 \overset{2}{1}222 \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$221 \overset{131513221}{1}2 \overset{1}{1}222 \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$221 \overset{131513221}{1}2 \overset{1}{1}222 \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$221 \overset{131513221}{1}2 \overset{1}{1}222 \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$221 \overset{3131513221}{1}2 \overset{1}{1}222 [\# = 8] \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$221 \overset{3131513221}{1}2 \overset{1}{1}222 [\# = 6] \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$\begin{aligned}
221 \overset{3}{1}31513221 \quad 1222 \text{ [#}=5] \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
221 \overset{3}{1}31513221 \quad 1222 \text{ [#}=4] \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
221 \overset{3}{1}31513221 \quad 1222 \text{ [#}=7] \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
221 \overset{3}{1}31513221 \quad 1222 \text{ [#}=9] \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
221 \overset{3}{1}31513221 \quad 1222 \text{ [#}=3] \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
221 \overset{3}{1}31513221 \quad 1222 \text{ [#}=2] \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: $4131513221 \overset{221}{1}2122$

$$\begin{aligned}
4131513221 \overset{2}{1}2122 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
4131513221 \overset{3}{1}2122 \quad \Delta = -133, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
4131513221 \overset{2}{1}2122 \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
4131513221 \overset{2}{1}2122 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
4131513221 \overset{221}{1}2122 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
4131513221 \overset{2}{1}2122 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
4131513221 \overset{221}{1}2122 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: $3221 \overset{1513221}{1}2 \quad 1223$

$$\begin{aligned}
3221 \overset{1513221}{1}2 \quad 1223 \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
3221 \overset{3}{1}513221 \quad 1223 \quad \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
3221 \overset{2}{1}513221 \quad 1223 \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
3221 \overset{1513221}{1}2 \quad 1223 \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
3221 \overset{2}{1}513221 \quad 1223 \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: $221 \overset{1513221}{1}2 \quad 12231$

$$\begin{aligned}
221 \overset{2}{15}13221 \quad 12 \quad 12231 \quad \Delta = -98, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
221 \overset{15}{13}221 \overset{2}{2} \quad 12 \quad 12231 \quad \Delta = -98, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
221 \overset{15}{13}221 \quad 12 \quad 12231 \quad \Delta = -98, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
221 \overset{15}{13}221 \quad 12 \quad 12231 \quad \Delta = -98, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
221 \overset{15}{13}221 \quad 12 \quad 12231 \overset{3}{3} \text{ [#=2]} \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
221 \overset{15}{13}221 \quad 12 \quad 12231 \overset{3}{3} \text{ [#=3]} \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
221 \overset{15}{13}221 \quad 12 \quad 12231 \overset{3}{3} \text{ [#=4]} \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
221 \overset{15}{13}221 \quad 12 \quad 12231 \overset{3}{3} \text{ [#=5]} \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{1,215}{13}221 \quad 12 \quad 122313 & \\
1 \overset{1,215}{13}221 \quad 12 \quad 12231 \overset{3}{3} \quad \Delta = -114, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \overset{1,215}{13}221 \quad 12 \quad 12231 \overset{2}{2} \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \overset{1,215}{13}221 \quad 12 \quad 12231 \overset{3}{3} \text{ [#=2]} \quad \Delta = -114, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1 \overset{1,215}{13}221 \quad 12 \quad 122313 \quad \Delta = -114, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1 \overset{1,215}{13}221 \quad 12 \quad 122313 \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1 \overset{1,215}{13}221 \quad 12 \quad 122313 \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \overset{1,215}{13}221 \quad 12 \quad 122313 \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \overset{1,215}{13}221 \overset{2}{2} \quad 12 \quad 122313 \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{1,321}{15}13221 \quad 12 \quad 12231 & \\
1 \overset{1,321}{15}13221 \quad 12 \quad 12231 \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1 \overset{1,321}{15}13221 \quad 12 \quad 12231 \overset{3}{3} \text{ [#=3]} \quad \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{l}
1^{1,321513221} \quad 1 \quad 12 \quad 12231\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1^{1,321513221} \quad 1 \quad 12 \quad 12231\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{1,321513221} \quad 1 \quad 12 \quad 12231\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -115, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1^{1,321513221\textcolor{red}{2}} \quad 1 \quad 12 \quad 12231 \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1^{1,321513221} \quad 1 \quad 12 \quad 12231 \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{1,321513221} \quad 1 \quad 12 \quad 12231 \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\text{Base: } 1^{1,131513221} \quad 1 \quad 12 \quad 12222 \\
1^{1,\textcolor{red}{3}131513221} \quad 1 \quad 12 \quad 12222 \text{ [#=7]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1^{1,131513221\textcolor{red}{2}} \quad 1 \quad 12 \quad 12222 \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1^{1,131513221} \quad 1 \quad 12 \quad 12222 \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1^{1,131513221} \quad 1 \quad 12 \quad 12222 \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1^{1,131513221} \quad 1 \quad 12 \quad 12222 \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{1,\textcolor{red}{3}131513221} \quad 1 \quad 12 \quad 12222 \text{ [#=2]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{1,\textcolor{red}{3}131513221} \quad 1 \quad 12 \quad 12222 \text{ [#=3]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{1,\textcolor{red}{3}131513221} \quad 1 \quad 12 \quad 12222 \text{ [#=4]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1^{1,\textcolor{red}{3}131513221} \quad 1 \quad 12 \quad 12222 \text{ [#=5]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1^{1,\textcolor{red}{3}131513221} \quad 1 \quad 12 \quad 12222 \text{ [#=6]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1^{1,\textcolor{red}{3}131513221} \quad 1 \quad 12 \quad 12222 \text{ [#=9]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1^{1,\textcolor{red}{3}131513221} \quad 1 \quad 12 \quad 12222 \text{ [#=8]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
\text{Base: } 1^{1,1513221} \quad 1 \quad 12 \quad 122313
\end{array}$$

$$\begin{array}{l}
1^{1,1513221\textcolor{red}{2}} 12^{122313} \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1^{1,1513221} 12^{12231\textcolor{red}{2}3} \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{1,1513221} 12^{12231\textcolor{red}{3}3} \quad \Delta = -114, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1^{1,1513221} 12^{122313} \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{1,1513221} 12^{122313} \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1^{1,1513221} 12^{12231\textcolor{red}{3}}_{[\#=2]} \quad \Delta = -114, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
\text{Base: } 1^{1,31513221} 12^{12231} \\
1^{1,31513221} 12^{12231} \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{1,31513221} 12^{12231} \quad \Delta = -114, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1^{1,31513221} 12^{12231\textcolor{red}{3}}_{[\#=4]} \quad \Delta = -114, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{1,31513221} 12^{12231\textcolor{red}{3}}_{[\#=3]} \quad \Delta = -114, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1^{1,31513221} 12^{12231\textcolor{red}{3}}_{[\#=2]} \quad \Delta = -114, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1^{1,31513221} 12^{12231\textcolor{red}{3}}_{[\#=5]} \quad \Delta = -114, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1^{1,31513221} 12^{12231} \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1^{1,31513221} 12^{12231} \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1^{1,31513221} 12^{12231} \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{1,31513221\textcolor{red}{2}} 12^{12231} \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\text{Base: } 1^{21,21513221} 12^{12231} \\
1^{21,21513221} 12^{12231} \quad \Delta = -98, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1^{21,21513221} 12^{12231\textcolor{red}{3}}_{[\#=4]} \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
& \begin{matrix} 21,21513221 \\ 1 \quad 12 \quad 12231 \end{matrix} \textcolor{red}{3} \text{ [#=3]} \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \begin{matrix} 21,21513221 \\ 1 \quad 12 \quad 12231 \end{matrix} \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \begin{matrix} 21,21513221 \\ 1 \quad 12 \quad 12231 \end{matrix} \textcolor{red}{3} \text{ [#=2]} \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \begin{matrix} 21,21513221 \\ 1 \quad 12 \quad 12231 \end{matrix} \textcolor{red}{3} \text{ [#=5]} \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \begin{matrix} 21,21513221 \\ 1 \quad 12 \quad 12231 \end{matrix} \quad \Delta = -98, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \begin{matrix} 21,21513221 \\ 1 \quad 12 \quad 12231 \end{matrix} \quad \Delta = -98, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \begin{matrix} 21,21513221 \textcolor{red}{2} \\ 1 \quad 12 \quad 12231 \end{matrix} \quad \Delta = -98, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \begin{matrix} 21,21513221 \\ 1 \quad 12 \quad 12231 \end{matrix} \quad \Delta = -98, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \textbf{Base: } \begin{matrix} 21,131513221 \\ 1 \quad 12 \quad 1222 \end{matrix} \\
& \begin{matrix} 21, \textcolor{red}{3} 131513221 \\ 1 \quad 12 \quad 1222 \end{matrix} \text{ [#=8]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
& \begin{matrix} 21, \textcolor{red}{3} 131513221 \\ 1 \quad 12 \quad 1222 \end{matrix} \text{ [#=9]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
& \begin{matrix} 21, \textcolor{red}{3} 131513221 \\ 1 \quad 12 \quad 1222 \end{matrix} \text{ [#=5]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& \begin{matrix} 21, \textcolor{red}{3} 131513221 \\ 1 \quad 12 \quad 1222 \end{matrix} \text{ [#=6]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& \begin{matrix} 21, \textcolor{red}{3} 131513221 \\ 1 \quad 12 \quad 1222 \end{matrix} \text{ [#=7]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
& \begin{matrix} 21,131513221 \\ 1 \quad 12 \quad 1222 \end{matrix} \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \begin{matrix} 21, \textcolor{red}{3} 131513221 \\ 1 \quad 12 \quad 1222 \end{matrix} \text{ [#=2]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \begin{matrix} 21, \textcolor{red}{3} 131513221 \\ 1 \quad 12 \quad 1222 \end{matrix} \text{ [#=4]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \begin{matrix} 21, \textcolor{red}{3} 131513221 \\ 1 \quad 12 \quad 1222 \end{matrix} \text{ [#=3]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \begin{matrix} 21,131513221 \textcolor{red}{2} \\ 1 \quad 12 \quad 1222 \end{matrix} \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \begin{matrix} 21,131513221 \\ 1 \quad 12 \quad 1222 \end{matrix} \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$1 \overset{21,131513221}{12} 1222 \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1 \overset{21,131513221}{12} 1222 \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: $1 \overset{21,1513221}{12} 12231$

$$1 \overset{21,1513221}{12} 12231 \quad \Delta = -98, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1 \overset{21,1513221}{12} 12231\textcolor{red}{3} [\# = 2] \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \overset{21,1513221}{12} 12231 \quad \Delta = -98, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{21,1513221}{12} 12231 \quad \Delta = -98, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{21,1513221\textcolor{red}{2}}{12} 12231 \quad \Delta = -98, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{21,1513221}{12} 12231\textcolor{red}{3} [\# = 4] \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1 \overset{21,1513221}{12} 12231\textcolor{red}{3} [\# = 5] \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1 \overset{21,1513221}{12} 12231\textcolor{red}{3} [\# = 3] \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: $1 \overset{221,131513221}{12} 122$

$$1 \overset{221,131513221\textcolor{red}{2}}{12} 122 \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{221,\textcolor{red}{3}131513221}{12} 122 [\# = 5] \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$1 \overset{221,\textcolor{red}{3}131513221}{12} 122 [\# = 9] \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$1 \overset{221,\textcolor{red}{3}131513221}{12} 122 [\# = 4] \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1 \overset{221,\textcolor{red}{3}131513221}{12} 122 [\# = 6] \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$1 \overset{221,\textcolor{red}{3}131513221}{12} 122 [\# = 7] \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$1 \overset{221,\textcolor{red}{3}131513221}{12} 122 [\# = 2] \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{221,131513221}{12} 122 \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\begin{aligned}
& 1^{221,131513221} 12 122 \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 1^{221,3131513221} 12 122 \text{ [#=3]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 1^{221,131513221} 12 122 \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 1^{221,3131513221} 12 122 \text{ [#=8]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
& \text{Base: } 21^{221,131513221} 12 12 \\
& 21^{221,3131513221} 12 12 \text{ [#=4]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& 21^{221,131513221} 12 12 \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 21^{221,131513221} 12 12 \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 21^{221,131513221} 12 12 \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 21^{221,131513221} 12 12 \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 21^{221,131513221} 12 12 \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 21^{221,3131513221} 12 12 \text{ [#=9]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
& 21^{221,3131513221} 12 12 \text{ [#=8]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
& 21^{221,3131513221} 12 12 \text{ [#=7]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
& 21^{221,3131513221} 12 12 \text{ [#=6]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& 21^{221,3131513221} 12 12 \text{ [#=5]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& 21^{221,3131513221} 12 12 \text{ [#=3]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 21^{221,3131513221} 12 12 \text{ [#=2]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \text{Base: } 1222222222222222^2 \\
& 2222222222222221^2 \text{ 12} \quad \Delta = -28, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \text{Base: } 1231414141414132^2
\end{aligned}$$

$$2^{\overset{8}{12314141414141}} 3^{12314141414141} 2^{12314141414141} \quad \Delta = -73, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$2^{\overset{7}{12314141414141}} 3^{12314141414141} 2^{12314141414141} \quad \Delta = -45, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$2^{\overset{6}{12314141414141}} 3^{12314141414141} 2^{12314141414141} \quad \Delta = -18, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 21414141414141412

$$2^{\overset{4}{1414141414141412}} \quad \Delta = -4, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 26 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

Base: $\overset{1}{1414141414141412}$

$$1^{414141414141412} \overset{4}{14141414141412} \quad \Delta = -4, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 8 \end{pmatrix}$$

Base: 13151316131513221

$$13151316131513221\overset{3}{\text{[}\#\text{=8]}} \quad \Delta = -3, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$13151316131513221\overset{3}{\text{[}\#\text{=9]}} \quad \Delta = -3, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$12231513161315131\overset{3}{\text{[}\#\text{=8]}} \quad \Delta = -3, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$12231513161315131\overset{3}{\text{[}\#\text{=9]}} \quad \Delta = -3, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$12231513161315131\overset{3}{\text{[}\#\text{=2]}} \quad \Delta = -3, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$13151316131513221\overset{3}{\text{[}\#\text{=2]}} \quad \Delta = -3, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$12231513161315131\overset{3}{\text{[}\#\text{=3]}} \quad \Delta = -3, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$13151316131513221\overset{3}{\text{[}\#\text{=4]}} \quad \Delta = -3, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$13151316131513221\overset{3}{\text{[}\#\text{=3]}} \quad \Delta = -3, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$12231513161315131\overset{3}{\text{[}\#\text{=4]}} \quad \Delta = -3, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$13151316131513221\overset{3}{\text{[}\#\text{=5]}} \quad \Delta = -3, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$13151316131513221\overset{3}{\text{[}\#\text{=6]}} \quad \Delta = -3, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$12231513161315131\overset{3}{\text{[}\#\text{=6]}} \quad \Delta = -3, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$\begin{aligned}
12231513161315131\overset{2}{3} \text{ [#=7]} \quad \Delta = -3, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
13151316131513221\overset{2}{3} \text{ [#=7]} \quad \Delta = -3, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
12231513161315131\overset{2}{3} \text{ [#=5]} \quad \Delta = -3, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}
\end{aligned}$$

Base: 12315131613151321

$$\begin{aligned}
12315131613151321\overset{2}{3} \text{ [#=4]} \quad \Delta = -4, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
12315131613151321\overset{2}{3} \text{ [#=2]} \quad \Delta = -4, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
12315131613151321\overset{2}{3} \text{ [#=3]} \quad \Delta = -4, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
12315131613151321\overset{2}{3} \text{ [#=5]} \quad \Delta = -4, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
12315131613151321\overset{2}{3} \text{ [#=6]} \quad \Delta = -4, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
12315131613151321\overset{2}{3} \text{ [#=7]} \quad \Delta = -4, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
12315131613151321\overset{2}{3} \text{ [#=8]} \quad \Delta = -4, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
12315131613151321\overset{2}{3} \text{ [#=9]} \quad \Delta = -4, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix}
\end{aligned}$$

Base: 22161316131513221

$$\begin{aligned}
22161316131513221\overset{2}{3} \quad \Delta = -13, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
22161316131513221\overset{2}{3} \quad \Delta = -13, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
22161316131513221\overset{2}{3} \quad \Delta = -13, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
22161316131513221\overset{3}{3} \quad \Delta = -22, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
22161316131513221\overset{2}{3} \text{ [#=8]} \quad \Delta = -21, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
22161316131513221\overset{2}{3} \text{ [#=7]} \quad \Delta = -21, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
22161316131513221\overset{2}{3} \text{ [#=6]} \quad \Delta = -21, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
22161316131513221^3 \quad \Delta = -21, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
22161316131513221^3_{[\# = 2]} \quad \Delta = -21, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
22161316131513221^3_{[\# = 5]} \quad \Delta = -21, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
22161316131513221^3_{[\# = 4]} \quad \Delta = -21, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
22161316131513221^3_{[\# = 3]} \quad \Delta = -21, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
22161316131513221^2 \quad \Delta = -13, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
22161316131513221^2 \quad \Delta = -13, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
22161316131513221^3_{[\# = 9]} \quad \Delta = -21, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
22161316131513221^2 \quad \Delta = -13, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\text{Base: } 13216131613151321 & \\
13216131613151321^3_{[\# = 4]} \quad \Delta = -30, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
13216131613151321^2 \quad \Delta = -22, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
13216131613151321^2 \quad \Delta = -22, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
13216131613151321^2 \quad \Delta = -22, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
13216131613151321^2 \quad \Delta = -22, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
13216131613151321^2 \quad \Delta = -22, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
13216131613151321^3_{[\# = 2]} \quad \Delta = -30, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
12315131613161231^3_{[\# = 3]} \quad \Delta = -30, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
13216131613151321^2 \quad \Delta = -22, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
13216131613151321^3_{[\# = 3]} \quad \Delta = -30, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
13216131613151321\mathbf{3}_{[\# = 9]} \quad \Delta = -30, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
12315131613161231\mathbf{3}_{[\# = 2]} \quad \Delta = -30, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
13216131613151321\mathbf{3}_{[\# = 8]} \quad \Delta = -30, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
13216131613151321\mathbf{3}_{[\# = 7]} \quad \Delta = -30, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
13216131613151321\mathbf{3}_{[\# = 6]} \quad \Delta = -30, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
13216131613151321\mathbf{3}_{[\# = 5]} \quad \Delta = -30, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
12315131613161231\mathbf{3}_{[\# = 5]} \quad \Delta = -30, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
12315131613161231\mathbf{3}_{[\# = 4]} \quad \Delta = -30, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

Base: 12321613161315131

$$\begin{aligned}
13151316131612321\mathbf{3}_{[\# = 4]} \quad \Delta = -31, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
12321613161315131\mathbf{3}_{[\# = 7]} \quad \Delta = -31, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
13151316131612321\mathbf{3}_{[\# = 8]} \quad \Delta = -31, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
123216131613151\mathbf{3}_{[\# = 2]}^2 \quad \Delta = -23, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
12321613161315131\mathbf{3}_{[\# = 2]} \quad \Delta = -31, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
13151316131612321\mathbf{3}_{[\# = 7]} \quad \Delta = -31, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
13151316131612321\mathbf{3}_{[\# = 2]} \quad \Delta = -31, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
12321613161315131\mathbf{3}_{[\# = 3]} \quad \Delta = -31, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
13151316131612321\mathbf{3}_{[\# = 3]} \quad \Delta = -31, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
12321613161315131\mathbf{3}_{[\# = 8]} \quad \Delta = -31, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
13151316131612321\mathbf{3}_{[\# = 9]} \quad \Delta = -31, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
12321613161315131\overset{2}{3} \text{ }_{[\# = 9]} \quad \Delta = -31, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
12321613161315131\overset{2}{3} \text{ }_{[\# = 6]} \quad \Delta = -31, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
12321613161315131\overset{2}{3} \quad \Delta = -23, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
12321613161315131\overset{2}{3} \quad \Delta = -23, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
13151316131612321\overset{2}{3} \text{ }_{[\# = 6]} \quad \Delta = -31, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
12321613161315131\overset{2}{3} \text{ }_{[\# = 5]} \quad \Delta = -31, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
12321613161315131\overset{2}{3} \quad \Delta = -23, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
13151316131612321\overset{2}{3} \text{ }_{[\# = 5]} \quad \Delta = -31, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
12321613161315131\overset{2}{3} \quad \Delta = -23, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
12321613161315131\overset{2}{3} \text{ }_{[\# = 4]} \quad \Delta = -31, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
12321613161315131\overset{2}{3} \quad \Delta = -23, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\text{Base: } 61231513161316131 & \\
61231513161316131\overset{2}{3} \text{ }_{[\# = 6]} \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
61231513161316131\overset{2}{3} \text{ }_{[\# = 7]} \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
61231513161316131\overset{2}{3} \quad \Delta = -99, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
61231513161316131\overset{2}{3} \quad \Delta = -99, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
61231513161316131\overset{2}{3} \quad \Delta = -99, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
61231513161316131\overset{2}{3} \quad \Delta = -99, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
61231513161316131\overset{2}{3} \quad \Delta = -99, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
61231513161316131\overset{2}{3} \quad \Delta = -99, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
61231513161316131^2 \quad \Delta = -99, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
61231513161316131^2_{[\# = 4]} \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
61231513161316131^2_{[\# = 3]} \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
61231513161316131^2_{[\# = 2]} \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
61231513161316131^2_{[\# = 9]} \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
61231513161316131^2_{[\# = 5]} \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
61231513161316131^2_{[\# = 8]} \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\text{Base: } 13161316131513161 & \\
16131513161316131^2_{[\# = 6]} \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
16131513161316131^2_{[\# = 4]} \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
13161316131513161^2 \quad \Delta = -98, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
13161316131513161^2 \quad \Delta = -98, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
13161316131513161^2 \quad \Delta = -98, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
13161316131513161^2 \quad \Delta = -98, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
16131513161316131^2_{[\# = 2]} \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
16131513161316131^2_{[\# = 3]} \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
13161316131513161^2 \quad \Delta = -98, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
13161316131513161^2 \quad \Delta = -98, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
13161316131513161^2 \quad \Delta = -98, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
16131513161316131^2_{[\# = 5]} \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
16131513161316131\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
16131513161316131\textcolor{red}{3} \text{ [#=8]} \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
16131513161316131\textcolor{red}{3} \text{ [#=7]} \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
\textbf{Base: } 31316131613151321 \\
\textcolor{red}{3} 31316131613151321 \text{ [#=2]} \quad \Delta = -37, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 93 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
\textcolor{red}{3} 31316131613151321 \quad \Delta = -37, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 21 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
31316131613151321\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -37, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
31316131613151321\textcolor{red}{3} \text{ [#=8]} \quad \Delta = -37, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
31316131613151321\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -37, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
313161\textcolor{red}{2}31613151321 \quad \Delta = -29, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
31316131613151\textcolor{red}{2}321 \quad \Delta = -29, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
3131613161\textcolor{red}{2}3151321 \quad \Delta = -29, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
3131613161\textcolor{red}{2}3151321 \quad \Delta = -29, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
31316131613151321\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -37, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\textcolor{red}{3} 31316131613151321 \text{ [#=3]} \quad \Delta = -37, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 213 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
31316131613151321\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -37, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
31316131613151321\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -37, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\textcolor{red}{2} 31316131613151321 \quad \Delta = -29, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
31316131613151321\textcolor{red}{3} \text{ [#=6]} \quad \Delta = -37, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
31316131613151321\textcolor{red}{3} \text{ [#=7]} \quad \Delta = -37, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
31316131613151321 \quad \Delta = -29, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
31316131613151321 \quad \Delta = -29, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
31316131613151321 \quad \Delta = -57, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 41316131613151316

$$\begin{aligned}
41316131613151316 \quad \Delta = -134, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
41316131613151316 \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
41316131613151316 \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
41316131613151316 \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
41316131613151316 \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
41316131613151316 \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
41316131613151316 \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
41316131613151316 \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
41316131613151316 \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: 14131613161315131

$$\begin{aligned}
14131613161315131 \quad \Delta = -48, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
14131613161315131 \quad \Delta = -48, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
14131613161315131 \quad \Delta = -48, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
14131613161315131 \quad \Delta = -48, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
14131613161315131 \quad \Delta = -48, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
14131613161315131 \quad \Delta = -48, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
14131613161315131^2 \quad \Delta = -48, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
131513161316131413_{\# = 2} \quad \Delta = -56, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
141316131613151313_{\# = 9} \quad \Delta = -56, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
141316131613151313_{\# = 5} \quad \Delta = -56, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
141316131613151313_{\# = 8} \quad \Delta = -56, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
141316131613151313_{\# = 7} \quad \Delta = -56, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
141316131613151313_{\# = 6} \quad \Delta = -56, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
141316131613151313_{\# = 4} \quad \Delta = -56, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
141316131613151313_{\# = 3} \quad \Delta = -56, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
131513161316131413_{\# = 3} \quad \Delta = -56, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
141316131613151313_{\# = 2} \quad \Delta = -56, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

$$14131613161315131^3 \quad \Delta = -56, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 21 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

Base: 21513161316131512

$$\begin{aligned}
21513161316131512^2 \quad \Delta = -64, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
21513161316131512^3 \quad \Delta = -72, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21513161316131512^2 \quad \Delta = -64, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21513161316131512^2 \quad \Delta = -64, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21513161316131512^2 \quad \Delta = -64, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \begin{smallmatrix} 21513161316131 \\ 5 \end{smallmatrix} 1$

$$1 \begin{smallmatrix} 21513161316131 \\ 5 \end{smallmatrix} 1 \quad \Delta = -64, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\begin{aligned}
& \begin{matrix} 21513161316131 \\ 1 \quad \quad 5 \quad \quad 1 \end{matrix} \quad \Delta = -64, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \begin{matrix} 21513161316131 \\ 1 \quad \quad 5 \quad \quad 1 \end{matrix} \quad \Delta = -64, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \begin{matrix} 21513161316131 \\ 1 \quad \quad 5 \quad \quad 1 \end{matrix} \quad \Delta = -64, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \begin{matrix} 21513161316131 \\ 1 \quad \quad 5 \quad \quad 1 \end{matrix} \quad \Delta = -64, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \begin{matrix} 21513161316131 \\ 1 \quad \quad 5 \quad \quad 1 \end{matrix} \quad \Delta = -64, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \begin{matrix} 21513161316131 \\ 1 \quad \quad 5 \quad \quad 1 \end{matrix} \quad \Delta = -64, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \begin{matrix} 21513161316131 \\ 1 \quad \quad 5 \quad \quad 1 \end{matrix} \quad \Delta = -72, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}
\end{aligned}$$

Base: $\begin{matrix} 1513161316131 \\ 1 \quad \quad 5 \quad \quad 1 \end{matrix}$

$$\begin{aligned}
& \begin{matrix} 1513161316131 \\ 1 \quad \quad 5 \quad \quad 1 \end{matrix} \quad \Delta = -64, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \begin{matrix} 1513161316131 \\ 1 \quad \quad 5 \quad \quad 1 \end{matrix} \quad \Delta = -64, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \begin{matrix} 1513161316131 \\ 1 \quad \quad 5 \quad \quad 1 \end{matrix} \quad \Delta = -64, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: 61232161316131612

$$\begin{aligned}
& 61232161316131612 \quad \Delta = -119, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
& 61232161316131612 \quad \Delta = -119, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& 61232161316131612 \quad \Delta = -119, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 61232161316131612 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
& 61232161316131612 \quad \Delta = -119, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 61232161316131612 \quad \Delta = -119, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& 61232161316131612 \quad \Delta = -119, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& 61232161316131612 \quad \Delta = -119, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 168 & 26 \\ 2 & 26 & 8 \end{pmatrix}
\end{aligned}$$

Base: 22161316131612321

$$22161316131612321 \quad \Delta = -49, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\begin{aligned}
22161316131612321^2 \quad \Delta = -41, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
22161316131612321^3 \quad \Delta = -49, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 27 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
22161316131612321^3_{[\# = 2]} \quad \Delta = -49, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
22161316131612321^3_{[\# = 3]} \quad \Delta = -49, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
22161316131612321^3_{[\# = 5]} \quad \Delta = -49, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
22161316131612321^3 \quad \Delta = -50, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
22161316131612321^3_{[\# = 9]} \quad \Delta = -49, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
22161316131612321^3_{[\# = 8]} \quad \Delta = -49, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
22161316131612321^2 \quad \Delta = -41, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
22161316131612321^2 \quad \Delta = -41, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
22161316131612321^2 \quad \Delta = -41, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
22161316131612321^2 \quad \Delta = -41, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
22161316131612321^2 \quad \Delta = -41, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
22161316131612321^3_{[\# = 7]} \quad \Delta = -49, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
22161316131612321^3_{[\# = 6]} \quad \Delta = -49, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\text{Base: } 13216131613161231 & \\
13216131613161231^2 \quad \Delta = -48, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
13216131613161231^3_{[\# = 3]} \quad \Delta = -56, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
13216131613161231^3_{[\# = 2]} \quad \Delta = -56, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
13216131613161231^3_{[\# = 5]} \quad \Delta = -56, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix}
\end{aligned}$$

$$13216131613161231\overset{2}{3}_{[\# = 4]} \quad \Delta = -56, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$13216131613161231\overset{2}{3} \quad \Delta = -48, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$13216131613161231\overset{2}{3} \quad \Delta = -48, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 14131613161316122

$$14131613161316122\overset{2}{3} \quad \Delta = -66, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$14131613161316122\overset{2}{3} \quad \Delta = -66, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$22161316131613141\overset{2}{3}_{[\# = 3]} \quad \Delta = -74, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$14131613161316122\overset{3}{3} \quad \Delta = -74, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$14131613161316122\overset{2}{3} \quad \Delta = -66, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$22161316131613141\overset{2}{3}_{[\# = 2]} \quad \Delta = -74, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$14131613161316122\overset{3}{3} \quad \Delta = -74, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 30 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$14131613161316122\overset{2}{3} \quad \Delta = -66, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$14131613161316122\overset{2}{3} \quad \Delta = -66, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$14131613161316122\overset{2}{3} \quad \Delta = -66, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$14131613161316122\overset{3}{3} \quad \Delta = -75, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$14131613161316122\overset{2}{3} \quad \Delta = -66, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 13161316131612315

$$51321613161316131\overset{2}{3}_{[\# = 7]} \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$51321613161316131\overset{2}{3}_{[\# = 8]} \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$51321613161316131\overset{2}{3}_{[\# = 2]} \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$51321613161316131\overset{2}{3}_{[\# = 3]} \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$\begin{aligned}
51321613161316131\overset{2}{3} \text{ }_{[\# = 4]} \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
51321613161316131\overset{2}{3} \text{ }_{[\# = 5]} \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1316131613161231\overset{2}{5} \quad \Delta = -99, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1316131613161231\overset{2}{5} \quad \Delta = -99, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 100 \quad & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
131\overset{2}{6}1316131612315 \quad \Delta = -99, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
131\overset{2}{6}1316131612315 \quad \Delta = -99, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
13161\overset{2}{3}16131612315 \quad \Delta = -99, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
131613161\overset{2}{3}1612315 \quad \Delta = -99, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
131613161\overset{2}{3}1612315 \quad \Delta = -99, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
51321613161316131\overset{2}{3} \text{ }_{[\# = 9]} \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
51321613161316131\overset{2}{3} \text{ }_{[\# = 6]} \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
\text{Base: } 13216131613161313 & \\
1321613161316131\overset{2}{3} \quad \Delta = -55, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1321613161316131\overset{2}{3} \quad \Delta = -55, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1321\overset{2}{6}131613161313 \quad \Delta = -55, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1321613161\overset{2}{3}161313 \quad \Delta = -55, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1321613161316131\overset{3}{3} \text{ }_{[\# = 3]} \quad \Delta = -63, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
31316131613161231\overset{2}{3} \text{ }_{[\# = 4]} \quad \Delta = -63, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
31316131613161231\overset{2}{3} \text{ }_{[\# = 5]} \quad \Delta = -63, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
1321613161316131\overset{3}{3} \quad \Delta = -63, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

$$31316131613161231\overset{3}{3} [\# = 2] \quad \Delta = -63, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$13216131613161313 \overset{2}{3} \quad \Delta = -55, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$13216131613161313 \overset{4}{3} \quad \Delta = -83, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 9 & 5 \\ 5 & 9 \end{pmatrix}$$

$$31316131613161231\overset{3}{3} [\# = 3] \quad \Delta = -63, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$13216131613161313 \overset{2}{3} \quad \Delta = -55, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$13216131613161313 \overset{2}{3} \quad \Delta = -55, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$13216131613161313 \overset{3}{3} [\# = 2] \quad \Delta = -63, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: 31316131613161313

$$\overset{3}{3} 31316131613161313 [\# = 3] \quad \Delta = -70, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 294 & 38 \\ 2 & 38 & 8 \end{pmatrix}$$

$$\overset{3}{3} 31316131613161313 [\# = 2] \quad \Delta = -70, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 129 & 25 \\ 2 & 25 & 8 \end{pmatrix}$$

$$\overset{3}{3} 31316131613161313 \quad \Delta = -70, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 30 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$\overset{4}{3} 31316131613161313 \quad \Delta = -90, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 29 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$31316131613161313 \overset{2}{3} \quad \Delta = -62, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$31316131613161313 \overset{2}{3} \quad \Delta = -62, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$31316131613161313 \overset{2}{3} \quad \Delta = -62, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$31316131613161313 \overset{2}{3} \quad \Delta = -62, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

Base: 6123216131613161¹

$$6123216131613161 \overset{2}{3} \overset{1}{1} \quad \Delta = -119, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 168 & 26 \\ 2 & 26 & 8 \end{pmatrix}$$

$$\overset{2}{3} 6123216131613161 \overset{1}{1} \quad \Delta = -119, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$6123216131613161 \overset{2}{3} \overset{1}{1} \quad \Delta = -119, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$6123216131613161 \overset{2}{3} \overset{1}{1} \quad \Delta = -119, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$\begin{aligned}
6123216131613161^{\overset{2}{1}} \quad \Delta = -119, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
6123216131613161^{\overset{21}{1}} \quad \Delta = -119, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: $161316131513221^{\overset{21}{1}}$

$$\begin{aligned}
161316131513221^{\overset{21}{1}} \textcolor{red}{3} \text{ [#=4]} \quad \Delta = -21, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
161316131513221^{\overset{21}{1}} \textcolor{red}{3} \text{ [#=3]} \quad \Delta = -21, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
161316131513221^{\overset{21}{1}} \textcolor{red}{3} \text{ [#=2]} \quad \Delta = -21, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
161316131513221^{\overset{21}{1}} \textcolor{red}{3} \text{ [#=9]} \quad \Delta = -21, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
161316131513221^{\overset{21}{1}} \textcolor{red}{3} \text{ [#=8]} \quad \Delta = -21, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
161316131513221^{\overset{21}{1}} \textcolor{red}{3} \text{ [#=7]} \quad \Delta = -21, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
161316131513221^{\overset{21}{1}} \textcolor{red}{3} \text{ [#=5]} \quad \Delta = -21, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
161316131513221^{\overset{21}{2}} \quad \Delta = -13, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
161316131513221^{\overset{21}{1}} \textcolor{red}{2} \quad \Delta = -13, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
161316131513221^{\overset{21}{1}} \quad \Delta = -21, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
161316131513221^{\overset{21}{1}} \textcolor{red}{2} \quad \Delta = -13, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
161316131513221^{\overset{21}{1}} \textcolor{red}{2} \quad \Delta = -13, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
161316131513221^{\overset{21}{1}} \quad \Delta = -13, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
161316131513221^{\overset{21}{1}} \textcolor{red}{2} \quad \Delta = -13, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
161316131513221^{\overset{21}{1}} \textcolor{red}{3} \text{ [#=6]} \quad \Delta = -21, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}
\end{aligned}$$

Base: $161316131612321^{\overset{21}{1}}$

$$161316131612321^{\overset{21}{1}} \textcolor{red}{3} \text{ [#=9]} \quad \Delta = -49, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$\begin{array}{ll}
{}^{21}_{16}1316131612321^2 & \Delta = -41, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
{}^{21}_{16}1316131612321^2 & \Delta = -41, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^{21}_{16}1316131612321^2 & \Delta = -41, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
{}^{21}_{16}1316131612321^2 & \Delta = -41, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
{}^{21}_{16}1316131612321^2 & \Delta = -41, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
{}^{21}_{16}1316131612321^3_{\#7} & \Delta = -49, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
{}^{21}_{16}1316131612321^3_{\#6} & \Delta = -49, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
{}^{21}_{16}1316131612321^3_{\#5} & \Delta = -49, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
{}^{21}_{16}1316131612321^3_{\#4} & \Delta = -49, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
{}^{21}_{16}1316131612321^3_{\#3} & \Delta = -49, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
{}^{21}_{16}1316131612321^3_{\#2} & \Delta = -49, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
{}^{21}_{16}1316131612321 & \Delta = -49, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
{}^{21}_{16}1316131612321^3_{\#8} & \Delta = -49, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
{}^{21}_{16}1316131612321 & \Delta = -41, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\text{Base: } {}^{21}_{16}1316131613141 & \\
{}^{21}_{16}1316131613141^2 & \Delta = -66, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^{21}_{16}1316131613141^2 & \Delta = -66, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
{}^{21}_{16}1316131613141 & \Delta = -74, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 15 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
{}^{21}_{16}1316131613141^3_{\#2} & \Delta = -74, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
{}^{21}_{16}1316131613141^3_{\#3} & \Delta = -74, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{ll}
{}^{21} 1 \overset{3}{6} 1316131613141 & \Delta = -74, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
{}^{21} 1 \overset{2}{6} 1316131613141 & \Delta = -66, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
{}^{21} 1 \overset{2}{6} 1316131613141 & \Delta = -66, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
{}^{21} 1 \overset{2}{6} 1316131613141 & \Delta = -66, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
{}^{21} 1 \overset{2}{6} 1316131613141 & \Delta = -66, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
{}^{21} 1 \overset{2}{6} 1316131613141 & \Delta = -66, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{array}$$

Base: ${}^{131} 1 \overset{131}{6} 131613151321$

$$\begin{array}{ll}
{}^{131} 1 \overset{2}{6} 131613151321 & \Delta = -21, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
{}^{131} 1 \overset{131}{6} 131613151321 & \Delta = -21, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
{}^{131} 1 \overset{2}{6} 131613151321 & \Delta = -21, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
{}^{131} 1 \overset{2}{6} 131613151321 & \Delta = -21, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
{}^{131} 1 \overset{2}{6} 131613151321 & \Delta = -21, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
{}^{131} 1 \overset{2}{6} 131613151321 & \Delta = -21, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
{}^{131} 1 \overset{131}{6} 131613151321 \overset{3}{\#}=4 & \Delta = -29, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
{}^{3131} 1 \overset{131}{6} 131613151321 \overset{3}{\#}=4 & \Delta = -29, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
{}^{131} 1 \overset{131}{6} 131613151321 \overset{3}{\#}=3 & \Delta = -29, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
{}^{3131} 1 \overset{131}{6} 131613151321 \overset{3}{\#}=3 & \Delta = -29, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
{}^{131} 1 \overset{131}{6} 131613151321 \overset{3}{\#}=2 & \Delta = -29, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
{}^{3131} 1 \overset{131}{6} 131613151321 \overset{3}{\#}=2 & \Delta = -29, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
{}^{3131} 1 \overset{131}{6} 131613151321 \overset{3}{\#}=9 & \Delta = -29, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{ll}
\begin{array}{c} 131 \\ 1 \ 6 \ 131613151321\textcolor{red}{3} \end{array} \begin{array}{l} [\#=9] \end{array} & \Delta = -29, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
\begin{array}{c} 131 \\ 1 \ 6 \ 131613151321\textcolor{red}{3} \end{array} \begin{array}{l} [\#=8] \end{array} & \Delta = -29, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
\begin{array}{c} 131 \\ 1 \ 6 \ 131613151321\textcolor{red}{3} \end{array} \begin{array}{l} [\#=6] \end{array} & \Delta = -29, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
\begin{array}{c} \textcolor{red}{3}131 \\ 1 \ 6 \ 131613151321 \end{array} \begin{array}{l} [\#=7] \end{array} & \Delta = -29, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
\begin{array}{c} \textcolor{red}{3}131 \\ 1 \ 6 \ 131613151321 \end{array} \begin{array}{l} [\#=5] \end{array} & \Delta = -29, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
\begin{array}{c} 131 \\ 1 \ 6 \ 131613151321\textcolor{red}{3} \end{array} \begin{array}{l} [\#=5] \end{array} & \Delta = -29, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\begin{array}{c} \textcolor{red}{3}131 \\ 1 \ 6 \ 131613151321 \end{array} \begin{array}{l} [\#=6] \end{array} & \Delta = -29, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
\begin{array}{c} \textcolor{red}{3}131 \\ 1 \ 6 \ 131613151321 \end{array} \begin{array}{l} [\#=8] \end{array} & \Delta = -29, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
\begin{array}{c} 131 \\ 1 \ 6 \ 131613151321\textcolor{red}{3} \end{array} \begin{array}{l} [\#=7] \end{array} & \Delta = -29, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}
\end{array}$$

Base: $\begin{array}{c} 131 \\ 1 \ 6 \ 131613161231 \end{array}$

$$\begin{array}{ll}
\begin{array}{c} 131 \\ 1 \ 6 \ 131613161231\textcolor{red}{2} \end{array} & \Delta = -47, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\begin{array}{c} 131 \\ 1 \ 6 \ 131613161231 \end{array} & \Delta = -47, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\begin{array}{c} 131 \\ 1 \ 6 \ 131613161231\textcolor{red}{2} \end{array} & \Delta = -47, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\begin{array}{c} 131 \\ 1 \ 6 \ 131613161231\textcolor{red}{2} \end{array} & \Delta = -47, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\begin{array}{c} 131 \\ 1 \ 6 \ 131613161231\textcolor{red}{2} \end{array} & \Delta = -47, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\begin{array}{c} 131\textcolor{red}{2} \\ 1 \ 6 \ 131613161231 \end{array} & \Delta = -47, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\begin{array}{c} \textcolor{red}{3}131 \\ 1 \ 6 \ 131613161231 \end{array} \begin{array}{l} [\#=8] \end{array} & \Delta = -55, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
\begin{array}{c} \textcolor{red}{3}131 \\ 1 \ 6 \ 131613161231 \end{array} \begin{array}{l} [\#=7] \end{array} & \Delta = -55, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
\begin{array}{c} \textcolor{red}{3}131 \\ 1 \ 6 \ 131613161231 \end{array} \begin{array}{l} [\#=2] \end{array} & \Delta = -55, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\begin{array}{c} 131 \\ 1 \ 6 \ 131613161231\textcolor{red}{3} \end{array} \begin{array}{l} [\#=2] \end{array} & \Delta = -55, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{ll}
\overset{3131}{1 \ 6 \ 131613161231} \text{ [#=6]} & \Delta = -55, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
\overset{3131}{1 \ 6 \ 131613161231} \text{ [#=5]} & \Delta = -55, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
\overset{3131}{1 \ 6 \ 131613161231} \text{ [#=9]} & \Delta = -55, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
\overset{3131}{1 \ 6 \ 131613161231} \text{ [#=3]} & \Delta = -55, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\overset{131}{1 \ 6 \ 131613161231\textcolor{red}{3}} \text{ [#=3]} & \Delta = -55, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\overset{131}{1 \ 6 \ 131613161231\textcolor{red}{3}} \text{ [#=5]} & \Delta = -55, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\overset{3131}{1 \ 6 \ 131613161231} \text{ [#=4]} & \Delta = -55, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
\overset{131}{1 \ 6 \ 131613161231\textcolor{red}{3}} \text{ [#=4]} & \Delta = -55, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\textbf{Base: } \overset{131}{1 \ 6 \ 131613161313} & \\
\overset{3131}{1 \ 6 \ 131613161313} \text{ [#=5]} & \Delta = -62, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
\overset{131}{1 \ 6 \ 13161316131\textcolor{red}{3}} & \Delta = -62, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\overset{3131}{1 \ 6 \ 131613161313} \text{ [#=9]} & \Delta = -62, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
\overset{3131}{1 \ 6 \ 131613161313} \text{ [#=8]} & \Delta = -62, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
\overset{3131}{1 \ 6 \ 131613161313} \text{ [#=7]} & \Delta = -62, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
\overset{3131}{1 \ 6 \ 131613161313} \text{ [#=4]} & \Delta = -62, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
\overset{3131}{1 \ 6 \ 131613161313} \text{ [#=3]} & \Delta = -62, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\overset{131}{1 \ 6 \ 131613161313} & \Delta = -54, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\overset{131}{1 \ 6 \ 13161316131\textcolor{red}{3}} & \Delta = -54, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\overset{131}{1 \ 6 \ 13161316131\textcolor{red}{2}} & \Delta = -54, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\overset{131}{1 \ 6 \ 13161316131\textcolor{red}{2}} & \Delta = -54, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{ll}
\begin{array}{c} 131 \\ 1 \ 6 \ 131613161313 \end{array} \quad \Delta = -82, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 9 & \begin{pmatrix} 9 & 5 \\ 5 & 9 \end{pmatrix} \\
\begin{array}{c} 131 \\ 1 \ 6 \ 131613161313 \end{array} \quad \Delta = -54, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\begin{array}{c} 1312 \\ 1 \ 6 \ 131613161313 \end{array} \quad \Delta = -54, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\begin{array}{c} 131 \\ 1 \ 6 \ 131613161313 \end{array} \quad \Delta = -62, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\begin{array}{c} 131 \\ 1 \ 6 \ 131613161313 \end{array} \quad \Delta = -62, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\begin{array}{c} 3131 \\ 1 \ 6 \ 131613161313 \end{array} \quad \Delta = -62, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
\begin{array}{c} 131 \\ 1 \ 6 \ 131613161313 \end{array} \quad \Delta = -54, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\begin{array}{c} 3131 \\ 1 \ 6 \ 131613161313 \end{array} \quad \Delta = -62, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\text{Base: } \begin{array}{c} 5131 \\ 1 \ 6 \ 13161316131 \end{array} & \\
\begin{array}{c} 5131 \\ 5131 \ 6 \ 13161316131 \end{array} \quad \Delta = -98, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\begin{array}{c} 5131 \\ 1 \ 6 \ 131613161313 \end{array} \quad \Delta = -106, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\begin{array}{c} 5131 \\ 1 \ 6 \ 131613161313 \end{array} \quad \Delta = -106, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\begin{array}{c} 5131 \\ 1 \ 6 \ 131613161313 \end{array} \quad \Delta = -106, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\begin{array}{c} 5131 \\ 1 \ 6 \ 13161316131 \end{array} \quad \Delta = -98, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
\begin{array}{c} 5131 \\ 1 \ 6 \ 13161316131 \end{array} \quad \Delta = -98, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
\begin{array}{c} 51312 \\ 1 \ 6 \ 13161316131 \end{array} \quad \Delta = -98, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
\begin{array}{c} 5131 \\ 1 \ 6 \ 13161316131 \end{array} \quad \Delta = -98, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\begin{array}{c} 5131 \\ 1 \ 6 \ 13161316131 \end{array} \quad \Delta = -98, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
\begin{array}{c} 5131 \\ 1 \ 6 \ 13161316131 \end{array} \quad \Delta = -98, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\begin{array}{c} 5131 \\ 1 \ 6 \ 13161316131 \end{array} \quad \Delta = -98, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
& 5131\overset{31}{6}13161316131 \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 1\overset{5131}{6}13161316131\overset{3}{3} \text{ [#=2]} \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
& 1\overset{5131}{6}13161316131\overset{3}{3} \text{ [#=5]} \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 1\overset{5131}{6}13161316131\overset{3}{3} \text{ [#=3]} \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& 1\overset{5131}{6}13161316131\overset{3}{3} \text{ [#=4]} \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 1\overset{5131}{6}13161316131\overset{3}{3} \text{ [#=6]} \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: 6131613161316123

$$\begin{aligned}
& 6131\overset{21}{6}13161316123 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& 613161\overset{1}{3}1\overset{2}{6}1316123 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& 61316131\overset{1}{6}1\overset{2}{6}1316123 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 6131613161\overset{1}{3}1\overset{2}{6}123 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 613161316131\overset{1}{6}1\overset{2}{6}23 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 6131\overset{2}{6}1\overset{1}{3}161316123 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 61316131613161\overset{12}{2}3 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
& 6131\overset{21}{6}13161316123 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 6131\overset{31}{6}13161316123 \quad \Delta = -133, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1613161316131612

$$\begin{aligned}
& 161316131613161\overset{21}{2} \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 161\overset{2}{3}161\overset{1}{3}16131612 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 1613161316131\overset{1}{6}1\overset{2}{6}12 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1613161316131612 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1613161316131612 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1613161316131612 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1613161316131612 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1613161316131612 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1613161316131612 \quad \Delta = -124, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1613161316131612 \quad \Delta = -124, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}
\end{aligned}$$

Base: 3161316131613161

$$\begin{aligned}
3161316131613161 \quad \Delta = -124, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
3161316131613161 \quad \Delta = -124, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
3161316131613161 \quad \Delta = -124, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
3161316131613161 \quad \Delta = -124, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
3161316131613161 \quad \Delta = -124, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
3161316131613161 \quad \Delta = -124, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 144 & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
3161316131613161 \quad \Delta = -132, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
3161316131613161 \quad \Delta = -124, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
3161316131613161 \quad \Delta = -124, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

Base: 161316131613161

$$\begin{aligned}
161316131613161 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
161316131613161 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{ll}
{}^1_{16}1316131613161 & \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
{}^1_{16}1316131613161 & \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^1_{16}1316131613161 & \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
{}^{12}_{16}1316131613161 & \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
{}^1_{16}1316131613161 & \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
{}^1_{16}1316131613161 & \Delta = -124, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\text{Base: } {}^1_{61}316131613161 & \\
{}^{21}_{61}316131613161 & \Delta = -124, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
{}^{31}_{61}316131613161 & \Delta = -132, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
{}^1_{61}316131613161 & \Delta = -124, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^1_{61}3161316131612 & \Delta = -124, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
{}^1_{61}316131613161 & \Delta = -124, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
{}^2_{61}1316131613161 & \Delta = -124, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^{21}_{61}316131613161 & \Delta = -124, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
{}^1_{61}316131613161 & \Delta = -124, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
{}^1_{61}316131613161 & \Delta = -124, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
{}^{12}_{61}316131613161 & \Delta = -124, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
\text{Base: } {}^{131}_{16}13161316131 & \\
{}^{1312}_{16}13161316131 & \Delta = -46, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^{131}_{16}13161316131 & \Delta = -46, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
& \overset{131}{1} \overset{1}{6} 13161316131 \quad \Delta = -46, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{3131}{1} \overset{1}{6} 13161316131 \text{ [#=7]} \quad \Delta = -54, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
& \overset{3131}{1} \overset{1}{6} 13161316131 \text{ [#=9]} \quad \Delta = -54, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
& \overset{3131}{1} \overset{1}{6} 13161316131 \text{ [#=4]} \quad \Delta = -54, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& \overset{3131}{1} \overset{1}{6} 13161316131 \text{ [#=5]} \quad \Delta = -54, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
& \overset{3131}{1} \overset{1}{6} 13161316131 \text{ [#=6]} \quad \Delta = -54, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
& \overset{3131}{1} \overset{1}{6} 13161316131 \text{ [#=3]} \quad \Delta = -54, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{3131}{1} \overset{1}{6} 13161316131 \text{ [#=2]} \quad \Delta = -54, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{3131}{1} \overset{1}{6} 13161316131 \text{ [#=8]} \quad \Delta = -54, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{31}{316131613151321}$

$$\begin{aligned}
& \overset{31}{316131613151321} \text{6} \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{31}{316131613151321} \text{4} \quad \Delta = -57, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{31}{316131613151321} \text{8} \quad \Delta = -162, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
& \overset{31}{316131613151321} \text{3} \text{ [#=5]} \quad \Delta = -37, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 172 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
& \overset{31}{316131613151321} \text{3} \text{ [#=3]} \quad \Delta = -37, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 60 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
& \overset{31}{316131613151321} \text{3} \text{ [#=8]} \quad \Delta = -37, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 256 \quad \begin{pmatrix} 0 & 16 & 2 \\ 16 & 445 & 55 \\ 2 & 55 & 8 \end{pmatrix} \\
& \overset{31}{316131613151321} \text{3} \text{ [#=9]} \quad \Delta = -37, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 324 \quad \begin{pmatrix} 0 & 18 & 2 \\ 18 & 564 & 62 \\ 2 & 62 & 8 \end{pmatrix} \\
& \overset{31}{316131613151321} \text{3} \text{ [#=4]} \quad \Delta = -37, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 109 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
& \overset{31}{316131613151321} \text{2} \quad \Delta = -29, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{31}{316131613151321} \text{3} \text{ [#=7]} \quad \Delta = -37, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 196 \quad \begin{pmatrix} 0 & 14 & 2 \\ 14 & 340 & 48 \\ 2 & 48 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{31}{316131613151321\textcolor{red}{3}}_{[\# = 2]} \quad \Delta = -37, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& \overset{31}{316131613151321\textcolor{red}{3}} \quad \Delta = -37, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{31}{316131613151321\textcolor{red}{3}}_{[\# = 6]} \quad \Delta = -37, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 249 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
& \overset{31}{316131613151321\textcolor{red}{2}} \quad \Delta = -29, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{31}{316131613151321\textcolor{red}{2}} \quad \Delta = -29, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& \overset{31}{316131613151321\textcolor{red}{2}} \quad \Delta = -29, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{31}{316131613151321\textcolor{red}{2}} \quad \Delta = -29, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{\textcolor{red}{2}31}{316131613151321} \quad \Delta = -29, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{31\textcolor{red}{2}}{316131613151321} \quad \Delta = -29, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{31}{316131613151321\textcolor{red}{5}} \quad \Delta = -81, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{31}{316131613151321\textcolor{red}{7}} \quad \Delta = -134, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \textbf{Base: } \overset{31}{316131613161231} \\
& \overset{31}{316131613161231\textcolor{red}{2}} \quad \Delta = -57, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{31}{316131613161231\textcolor{red}{5}} \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{31}{316131613161231\textcolor{red}{3}}_{[\# = 2]} \quad \Delta = -63, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{31}{316131613161231\textcolor{red}{3}} \quad \Delta = -63, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{31}{316131613161231\textcolor{red}{3}}_{[\# = 5]} \quad \Delta = -63, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
& \overset{31}{316131613161231\textcolor{red}{3}}_{[\# = 4]} \quad \Delta = -63, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& \overset{31}{316131613161231\textcolor{red}{3}}_{[\# = 3]} \quad \Delta = -63, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{31}{316131613161231\textcolor{red}{4}} \quad \Delta = -83, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{ll}
{}^{31}\overset{2}{316131613161231} & \Delta = -55, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^{31}\overset{2}{316131613161231} & \Delta = -55, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
{}^{231}\overset{31}{316131613161231} & \Delta = -55, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
{}^{31}\overset{31}{316131613161231}\overset{2}{2} & \Delta = -55, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
{}^{31}\overset{2}{316131613161231} & \Delta = -55, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 37 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
{}^{31}\overset{2}{316131613161231} & \Delta = -55, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
{}^{31}\overset{2}{316131613161231} & \Delta = -55, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\text{Base: } {}^{31}\overset{31}{316131613161313} & \\
{}^{31}\overset{2}{316131613161313} & \Delta = -62, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 37 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
{}^{31}\overset{3}{316131613161313} \text{ [#=3]} & \Delta = -70, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 324 \quad \begin{pmatrix} 0 & 18 & 2 \\ 18 & 375 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
{}^{31}\overset{2}{316131613161313} & \Delta = -62, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
{}^{31}\overset{2}{316131613161313} & \Delta = -62, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^{31}\overset{2}{316131613161313} & \Delta = -62, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
{}^{31}\overset{2}{316131613161313} & \Delta = -62, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
{}^{31}\overset{2}{316131613161313} & \Delta = -62, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
{}^{31}\overset{4}{316131613161313} & \Delta = -90, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 38 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
{}^{31}\overset{3}{316131613161313} & \Delta = -70, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 39 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
{}^{31}\overset{3}{316131613161313} \text{ [#=2]} & \Delta = -70, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 165 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
{}^{31}\overset{2}{316131613161313}\overset{2}{2} & \Delta = -63, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
{}^{231}\overset{31}{316131613161313} & \Delta = -62, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{array}$$

Base: $31\overset{31}{6}1316131\overset{1}{6}131$

$$\begin{aligned}
31\overset{231}{6}1316131\overset{1}{6}131 \quad \Delta = -54, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
31\overset{31}{6}1316131\overset{2}{6}131 \quad \Delta = -54, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
31\overset{31}{6}1316131\overset{1}{6}131\overset{1}{6} \quad \Delta = -132, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
31\overset{31}{6}1316131\overset{21}{6}131 \quad \Delta = -54, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
31\overset{31}{6}1316131\overset{2}{6}131 \quad \Delta = -54, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
31\overset{31}{6}1316131\overset{1}{6}131\overset{1}{3} \quad [\# = 2] \quad \Delta = -62, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
31\overset{31}{6}1316131\overset{31}{6}131 \quad \Delta = -62, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
31\overset{31}{6}1316131\overset{1}{6}131\overset{1}{3} \quad \Delta = -62, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
31\overset{312}{6}1316131\overset{1}{6}131 \quad \Delta = -54, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
31\overset{31}{6}1316131\overset{21}{6}131 \quad \Delta = -54, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
31\overset{31}{6}1316131\overset{1}{6}131\overset{1}{2} \quad \Delta = -54, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
31\overset{31}{6}1316131\overset{12}{6}131 \quad \Delta = -54, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
31\overset{31}{6}1316131\overset{1}{6}131\overset{1}{3} \quad [\# = 7] \quad \Delta = -62, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 196 \quad & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
31\overset{31}{6}1316131\overset{1}{6}131\overset{1}{3} \quad [\# = 3] \quad \Delta = -62, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
31\overset{31}{6}1316131\overset{1}{6}131\overset{1}{3} \quad [\# = 5] \quad \Delta = -62, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 100 \quad & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
31\overset{31}{6}1316131\overset{1}{6}131\overset{1}{3} \quad [\# = 6] \quad \Delta = -62, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 144 \quad & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
31\overset{31}{6}1316131\overset{1}{6}131\overset{1}{3} \quad [\# = 8] \quad \Delta = -62, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 256 \quad & \begin{pmatrix} 0 & 16 & 2 \\ 16 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
31\overset{31}{6}1316131\overset{1}{6}131\overset{1}{3} \quad [\# = 9] \quad \Delta = -62, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 324 \quad & \begin{pmatrix} 0 & 18 & 2 \\ 18 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix} \\
31\overset{31}{6}1316131\overset{1}{6}131\overset{1}{3} \quad [\# = 4] \quad \Delta = -62, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{31}{3} \overset{1}{6} 13161316131\overset{4}{4} \quad \Delta = -82, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{31}{3} \overset{1}{6} 13161316131\overset{5}{5} \quad \Delta = -106, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: 71223151316131612

$$\begin{aligned}
& 7122315131613161\overset{2}{2} \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& 7122315131613161\overset{2}{2} \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& 7122315131613161\overset{2}{2} \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 7122315131613161\overset{3}{3} \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
& \overset{2}{7} 1223151316131612 \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 89 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& 7122315131613161\overset{2}{2} \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
& 7122315131613161\overset{2}{2} \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 7122315131613161\overset{2}{2} \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix}
\end{aligned}$$

Base: 7122315131613161¹

$$\begin{aligned}
& 7122315131613161\overset{2}{2}\overset{1}{1} \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 7122315131613161\overset{2}{2}\overset{1}{1} \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
& \overset{2}{7} 122315131613161\overset{1}{1} \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 89 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& 7122315131613161\overset{2}{2}\overset{1}{1} \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
& 7122315131613161\overset{21}{21} \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 7122315131613161\overset{2}{2}\overset{1}{1} \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}
\end{aligned}$$

Base: 71231513161316123

$$\begin{aligned}
& 7123151316131612\overset{2}{2} \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& 7123151316131612\overset{2}{2} \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{ll}
71231513161316123^2 & \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
71231513161316123^2 & \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
71231513161316123^2 & \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
71231513161316123^2 & \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
71231513161316123^2 & \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\text{Base: } 712315131613161^{31} & \\
712315131613161^{31} & \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
712315131613161^{31} & \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
712315131613161^{31} & \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
712315131613161^{31} & \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
712315131613161^{31} & \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
712315131613161^{31} & \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
712315131613161^{231} & \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
712315131613161^{31} & \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
\text{Base: } 71232161316131513 & \\
71232161316131513^2 & \Delta = -128, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
71232161316131513^2 & \Delta = -128, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
71232161316131513^2 & \Delta = -128, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
71232161316131513^2 & \Delta = -128, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
71232161316131513^2 & \Delta = -128, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{ll}
71232161316131513 & \Delta = -128, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
71232161316131513 & \Delta = -128, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\text{Base: } 21613161315132171 & \\
21613161315132171 & \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
21613161315132171 & \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
21613161315132171 & \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21613161315132171 & \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
21613161315132171 & \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
21613161315132171 & \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
21613161315132171 & \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21613161315132171 & \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\text{Base: } 1613161315132171 & \\
1613161315132171 & \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
1613161315132171 & \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
1613161315132171 & \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1613161315132171 & \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1613161315132171 & \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1613161315132171 & \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\text{Base: } 15131613161232171 & \\
15131613161232171 & \Delta = -120, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
15131613161232171 & \Delta = -120, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
15131613161232171 \quad \Delta = -120, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
15131613161232171 \quad \Delta = -120, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
15131613161232171 \quad \Delta = -120, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
15131613161232171 \quad \Delta = -120, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}
\end{aligned}$$

Base: 12321712321712321

$$\begin{aligned}
12321712321712321 \quad \Delta = -23, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
12321712321712321\textcolor{red}{3} \text{ [#=7]} \quad \Delta = -31, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
12321712321712321\textcolor{red}{3} \text{ [#=6]} \quad \Delta = -31, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
12321712321712321\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -31, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
12321712321712321\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -31, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
12321712321712321\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -31, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
12321712321712321 \quad \Delta = -23, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
12321712321712321\textcolor{red}{3} \text{ [#=8]} \quad \Delta = -31, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
12321712321712321\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -31, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
12321712321712321\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -31, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

Base: 71232171232171232

$$\begin{aligned}
71232171232171232 \quad \Delta = -128, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 225 & \begin{pmatrix} 0 & 15 & 2 \\ 15 & 163 & 25 \\ 2 & 25 & 8 \end{pmatrix} \\
71232171232171232 \quad \Delta = -128, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
71232171232171232 \quad \Delta = -128, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
71232171232171232 \quad \Delta = -128, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix}
\end{aligned}$$

Base: 17123217123217123

$$\begin{array}{ll}
17123217123217123^2 & \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
17123217123217123^2 & \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
17123217123217123^2 & \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
17123217123217123^2 & \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
17123217123217123^2 & \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\text{Base: } 21712321712321712 & \\
21712321712321712^2 & \Delta = -117, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
21712321712321712^2 & \Delta = -117, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
21712321712321712^2 & \Delta = -117, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\text{Base: } 1712321712321712^1 & \\
1712321712321712^1 & \Delta = -117, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1712321712321712^1 & \Delta = -117, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1712321712321712^1 & \Delta = -117, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
1712321712321712^1 & \Delta = -117, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1712321712321712^{12} & \Delta = -117, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
\text{Base: } 171232171232171^1 & \\
171232171232171^{12} & \Delta = -117, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
171232171232171^1 & \Delta = -117, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
\text{Base: } 81223151316131513 & \\
81223151316131513^2 & \Delta = -128, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
81223151316131513^2 & \Delta = -128, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{array}$$

81223151316131513 ²	$\Delta = -128, n_+ = 1, n_- = 17, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$
81223151316131513 ²	$\Delta = -128, n_+ = 1, n_- = 17, n_0 = 0, \det = 64$	$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$
81223151316131513 ²	$\Delta = -128, n_+ = 1, n_- = 17, n_0 = 0, \det = 81$	$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$
81223151316131513 ²	$\Delta = -128, n_+ = 1, n_- = 17, n_0 = 0, \det = 100$	$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$
81223151316131513 ²	$\Delta = -128, n_+ = 1, n_- = 17, n_0 = 0, \det = 121$	$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$
Base: 81231513161315132		
81231513161315132 ²	$\Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, \det = 121$	$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$
81231513161315132 ²	$\Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, \det = 121$	$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$
81231513161315132 ²	$\Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, \det = 100$	$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$
81231513161315132 ²	$\Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, \det = 169$	$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix}$
81231513161315132 ²	$\Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, \det = 64$	$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$
81231513161315132 ²	$\Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$
81231513161315132 ²	$\Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, \det = 196$	$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix}$
Base: 15131613151322181		
15131613151322181 ²	$\Delta = -120, n_+ = 1, n_- = 17, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$
15131613151322181 ²	$\Delta = -120, n_+ = 1, n_- = 17, n_0 = 0, \det = 64$	$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$
15131613151322181 ²	$\Delta = -120, n_+ = 1, n_- = 17, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$
15131613151322181 ²	$\Delta = -120, n_+ = 1, n_- = 17, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
15131613151322181 ²	$\Delta = -120, n_+ = 1, n_- = 17, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$
15131613151322181 ²	$\Delta = -120, n_+ = 1, n_- = 17, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$
Base: 31513161315132181		

315131613 ² 15132181	$\Delta = -128, n_+ = 1, n_- = 17, n_0 = 0, \det = 121$	$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix}$
315131613 ² 15132181	$\Delta = -128, n_+ = 1, n_- = 17, n_0 = 0, \det = 100$	$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix}$
31513161315132181	$\Delta = -128, n_+ = 1, n_- = 17, n_0 = 0, \det = 64$	$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$
31513161315132181	$\Delta = -128, n_+ = 1, n_- = 17, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$
31513161315132181	$\Delta = -128, n_+ = 1, n_- = 17, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$
31513161315132181	$\Delta = -128, n_+ = 1, n_- = 17, n_0 = 0, \det = 81$	$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$
31513161315132181	$\Delta = -128, n_+ = 1, n_- = 17, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$
Base: 21812231513221812		
21812231513221812	$\Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
21812231513221812	$\Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$
21812231513221812	$\Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, \det = 4$	$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$
Base: 15131613151321812		
15131613151321812	$\Delta = -120, n_+ = 1, n_- = 17, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$
15131613151321812	$\Delta = -120, n_+ = 1, n_- = 17, n_0 = 0, \det = 64$	$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$
15131613151321812	$\Delta = -120, n_+ = 1, n_- = 17, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$
15131613151321812	$\Delta = -120, n_+ = 1, n_- = 17, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
15131613151321812	$\Delta = -120, n_+ = 1, n_- = 17, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$
15131613151321812	$\Delta = -120, n_+ = 1, n_- = 17, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
15131613151321812	$\Delta = -120, n_+ = 1, n_- = 17, n_0 = 0, \det = 4$	$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$
Base: 41316131612321812		
41316131612321812	$\Delta = -132, n_+ = 1, n_- = 17, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}$

$$\begin{aligned}
41316131612321812^2 \quad \Delta = -124, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
41316131612321812^2 \quad \Delta = -124, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
41316131612321812^2 \quad \Delta = -124, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
41316131612321812^2 \quad \Delta = -124, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
41316131612321812^2 \quad \Delta = -124, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
41316131612321812^2 \quad \Delta = -124, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
41316131612321812^2 \quad \Delta = -124, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: 31612321712321812

$$\begin{aligned}
31612321712321812^2 \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
31612321712321812^2 \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 163 & 25 \\ 2 & 25 & 8 \end{pmatrix} \\
31612321712321812^2 \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
31612321712321812^2 \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
31612321712321812^2 \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
31612321712321812^2 \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: 13161316123218122

$$\begin{aligned}
13161316123218122^2 \quad \Delta = -96, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
13161316123218122^2 \quad \Delta = -96, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
13161316123218122^2 \quad \Delta = -96, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
13161316123218122^2 \quad \Delta = -96, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
221812321613161313^2_{[\# = 7]} \quad \Delta = -104, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
22181232161316131\overset{3}{\color{red}3}_{[\# = 2]} \quad \Delta = -104, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
22181232161316131\overset{3}{\color{red}3}_{[\# = 3]} \quad \Delta = -104, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
22181232161316131\overset{3}{\color{red}3}_{[\# = 8]} \quad \Delta = -104, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
22181232161316131\overset{3}{\color{red}3}_{[\# = 4]} \quad \Delta = -104, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
22181232161316131\overset{3}{\color{red}3}_{[\# = 5]} \quad \Delta = -104, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
22181232161316131\overset{3}{\color{red}3}_{[\# = 6]} \quad \Delta = -104, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
22181232161316131\overset{3}{\color{red}3}_{[\# = 9]} \quad \Delta = -104, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
13161316123218122\overset{3}{\color{red}2} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
13161316123218122\overset{2}{\color{red}2} \quad \Delta = -96, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
13161316123218122\overset{2}{\color{red}2} \quad \Delta = -96, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 16123217123218122

$$\begin{aligned}
16123217123218122\overset{2}{\color{red}2} \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
16123217123218122\overset{2}{\color{red}2} \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
16123217123218122\overset{2}{\color{red}2} \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
16123217123218122\overset{2}{\color{red}2} \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
16123217123218122\overset{2}{\color{red}2} \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
16123217123218122\overset{3}{\color{red}2} \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}
\end{aligned}$$

Base: 18122315132218123

$$\begin{aligned}
18122315132218123\overset{2}{\color{red}2} \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
18122315132218123\overset{2}{\color{red}2} \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
18122315132218123\overset{2}{\color{red}2} \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 121 \quad & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
18122315132218123 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
18122315132218123 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 21812231513218122

$$\begin{aligned}
21812231513218122 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
21812231513218122 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
21812231513218122 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21812231513218122 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21812231513218122 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21812231513218122 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21812231513218122 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: 32181223151321812

$$\begin{aligned}
32181223151321812 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 174 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
32181223151321812 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
32181223151321812 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
32181223151321812 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
32181223151321812 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
32181223151321812 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: 22181231513218122

$$\begin{aligned}
22181231513218122 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
22181231513218122 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
22181231513218122 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

$$22181231513218122 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

Base: 12321812231513221

$$12321812231513221\mathbf{3}_{[\#=5]} \quad \Delta = -2, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$12231513221812321\mathbf{3}_{[\#=5]} \quad \Delta = -2, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$12321812231513221\mathbf{3}_{[\#=6]} \quad \Delta = -2, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$12231513221812321\mathbf{3}_{[\#=4]} \quad \Delta = -2, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$12321812231513221\mathbf{3}_{[\#=4]} \quad \Delta = -2, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$12231513221812321\mathbf{3}_{[\#=7]} \quad \Delta = -2, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$12321812231513221\mathbf{3}_{[\#=7]} \quad \Delta = -2, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$12231513221812321\mathbf{3}_{[\#=6]} \quad \Delta = -2, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$12231513221812321\mathbf{3}_{[\#=8]} \quad \Delta = -2, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$12321812231513221\mathbf{3}_{[\#=8]} \quad \Delta = -2, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$12321812231513221\mathbf{3}_{[\#=3]} \quad \Delta = -2, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$12231513221812321\mathbf{3}_{[\#=9]} \quad \Delta = -2, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$12321812231513221\mathbf{3}_{[\#=9]} \quad \Delta = -2, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$12231513221812321\mathbf{3}_{[\#=3]} \quad \Delta = -2, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$12321812231513221\mathbf{3}_{[\#=2]} \quad \Delta = -2, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$12231513221812321\mathbf{3}_{[\#=2]} \quad \Delta = -2, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 12321712321812231

$$12321712321812231\mathbf{3}_{[\#=5]} \quad \Delta = -57, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\begin{aligned}
13221812321712321\overset{2}{3} \text{ }_{[\# = 2]} \quad \Delta = -57, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
13221812321712321\overset{2}{3} \text{ }_{[\# = 3]} \quad \Delta = -57, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
13221812321712321\overset{2}{3} \text{ }_{[\# = 6]} \quad \Delta = -57, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
13221812321712321\overset{2}{3} \text{ }_{[\# = 7]} \quad \Delta = -57, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
12321712321812231 \overset{2}{\quad} \Delta = -49, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
12321712321812231 \overset{2}{\quad} \Delta = -49, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
13221812321712321\overset{2}{3} \text{ }_{[\# = 8]} \quad \Delta = -57, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
13221812321712321\overset{2}{3} \text{ }_{[\# = 9]} \quad \Delta = -57, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
13221812321712321\overset{2}{3} \text{ }_{[\# = 4]} \quad \Delta = -57, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
12321712321812231\overset{2}{3} \text{ }_{[\# = 3]} \quad \Delta = -57, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
12321712321812231\overset{2}{3} \text{ }_{[\# = 2]} \quad \Delta = -57, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
12321712321812231 \overset{2}{\quad} \Delta = -49, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
12321712321812231 \overset{2}{\quad} \Delta = -49, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
12321712321812231\overset{2}{3} \text{ }_{[\# = 4]} \quad \Delta = -57, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
13221812321712321\overset{2}{3} \text{ }_{[\# = 5]} \quad \Delta = -57, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix}
\end{aligned}$$

Base: 61232171232181223

$$\begin{aligned}
61232171232181223 \overset{2}{\quad} \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
61232171232181223 \overset{2}{\quad} \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
61232171232181223 \overset{2}{\quad} \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 121 \quad & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
61232171232181223 \overset{2}{\quad} \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 100 \quad & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{ll}
\overset{2}{6}1232171232181223 & \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\text{Base: } 81223151322181232 & \\
81223151322181232 & \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
81223151322181232 & \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 256 \quad \begin{pmatrix} 0 & 16 & 2 \\ 16 & 174 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
\overset{2}{8}1223151322181232 & \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
81223151322181232 & \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
81223151322181232 & \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
\text{Base: } 81232181223151322 & \\
81232181223151322 & \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 256 \quad \begin{pmatrix} 0 & 16 & 2 \\ 16 & 206 & 28 \\ 2 & 28 & 8 \end{pmatrix} \\
\overset{2}{8}1232181223151322 & \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
81232181223151322 & \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
81232181223151322 & \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
81232181223151322 & \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 576 \quad \begin{pmatrix} 0 & 24 & 2 \\ 24 & 454 & 42 \\ 2 & 42 & 8 \end{pmatrix} \\
\text{Base: } 18122315132181223 & \\
18122315132181223 & \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\overset{2}{18}122315132181223 & \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
18122315132181223 & \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
18122315132181223 & \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
18122315132181223 & \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\text{Base: } 23218122315132181 & \\
\overset{2}{23}218122315132181 & \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
23218122315132181 & \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{array}$$

232181223151 ² 32181	$\Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, \det = 81$	$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$
232181223151 ² 32181	$\Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, \det = 121$	$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix}$
232181 ² 22315132181	$\Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, \det = 256$	$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 334 & 36 \\ 2 & 36 & 8 \end{pmatrix}$
Base: 21812315132181223		
21812315132181 ² 223	$\Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$
21812315132181 ² 223	$\Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, \det = 81$	$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$
2181 ² 2315132181223	$\Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$
2181 ² 2315132181223	$\Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$
218123151 ² 32181223	$\Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, \det = 64$	$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
2181 ² 2315132181223	$\Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, \det = 4$	$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$
Base: 18123218122315132		
181232181 ² 22315132	$\Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, \det = 361$	$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 264 & 32 \\ 2 & 32 & 8 \end{pmatrix}$
181 ² 23218122315132	$\Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
1812321 ² 8122315132	$\Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, \det = 169$	$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix}$
1812321812231 ² 5132	$\Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$
1812321812231 ² 5132	$\Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, \det = 100$	$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$
Base: 51316131513218123		
513161315132181 ² 23	$\Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$
513161315132181 ² 23	$\Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$
513161315132181 ² 23	$\Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, \det = 121$	$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$
51316131 ² 513218123	$\Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, \det = 169$	$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix}$

$$\begin{aligned}
51316131513218123 \quad \Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 144 & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
51316131513218123 \quad \Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
51316131513218123 \quad \Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 13151321812315131

$$\begin{aligned}
13151321812315131\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -59, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
13151321812315131 \quad \Delta = -51, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
13151321812315131\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -59, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
13151321812315131\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -59, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
13151321812315131\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -59, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
13151321812315131\textcolor{red}{3} \text{ [#=6]} \quad \Delta = -59, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
13151321812315131\textcolor{red}{3} \text{ [#=7]} \quad \Delta = -59, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
13151321812315131\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -59, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
13151321812315131\textcolor{red}{3} \text{ [#=8]} \quad \Delta = -59, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
13151321812315131 \quad \Delta = -51, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
13151321812315131 \quad \Delta = -51, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: 12315132181232151

$$\begin{aligned}
15123218123151321\textcolor{red}{3} \text{ [#=6]} \quad \Delta = -54, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
15123218123151321\textcolor{red}{3} \text{ [#=8]} \quad \Delta = -54, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
15123218123151321\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -54, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
15123218123151321\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -54, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
15123218123151321\overset{2}{3} \text{ [#=2]} \quad \Delta = -54, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
15123218123151321\overset{2}{3} \text{ [#=5]} \quad \Delta = -54, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
15123218123151321\overset{2}{3} \text{ [#=7]} \quad \Delta = -54, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
15123218123151321\overset{2}{3} \text{ [#=9]} \quad \Delta = -54, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
123151321\overset{2}{8}1232151 \quad \Delta = -46, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
123151321\overset{2}{8}1232151 \quad \Delta = -46, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
123151321\overset{2}{8}1232151 \quad \Delta = -46, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
123151321\overset{2}{8}1232151 \quad \Delta = -46, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
123151321\overset{2}{8}1232151 \quad \Delta = -46, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\text{Base: } 12315132181231512 \\
21513218123151321\overset{2}{3} \text{ [#=3]} \quad \Delta = -52, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
21513218123151321\overset{2}{3} \text{ [#=2]} \quad \Delta = -52, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1231513218123151\overset{3}{2} \quad \Delta = -52, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
21513218123151321\overset{2}{3} \text{ [#=4]} \quad \Delta = -52, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
21513218123151321\overset{2}{3} \text{ [#=5]} \quad \Delta = -52, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
21513218123151321\overset{2}{3} \text{ [#=6]} \quad \Delta = -52, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
123151321\overset{2}{8}1231512 \quad \Delta = -44, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
123151321\overset{2}{8}1231512 \quad \Delta = -44, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
123151321\overset{2}{8}1231512 \quad \Delta = -44, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
123151321\overset{2}{8}1231512 \quad \Delta = -44, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
12315132181231512 \quad \Delta = -44, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
12315132181231512 \quad \Delta = -44, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21513218123151321 \textcolor{red}{3} [\# = 7] \quad \Delta = -52, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
21513218123151321 \textcolor{red}{3} [\# = 8] \quad \Delta = -52, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
21513218123151321 \textcolor{red}{3} [\# = 9] \quad \Delta = -52, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
\textbf{Base: } 41232181231513221 & \\
41232181231513221 \textcolor{red}{3} [\# = 3] \quad \Delta = -30, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
41232181231513221 \textcolor{red}{3} [\# = 2] \quad \Delta = -30, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
41232181231513221 \quad \Delta = -30, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
41232181231513221 \textcolor{red}{3} [\# = 4] \quad \Delta = -30, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
41232181231513221 \quad \Delta = -22, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
41232181231513221 \quad \Delta = -22, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
41232181231513221 \quad \Delta = -22, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
41232181231513221 \textcolor{red}{3} [\# = 9] \quad \Delta = -30, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
41232181231513221 \quad \Delta = -22, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
41232181231513221 \textcolor{red}{3} [\# = 8] \quad \Delta = -30, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
41232181231513221 \textcolor{red}{3} [\# = 7] \quad \Delta = -30, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
41232181231513221 \textcolor{red}{3} [\# = 6] \quad \Delta = -30, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
41232181231513221 \textcolor{red}{3} [\# = 5] \quad \Delta = -30, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
41232181231513221 \quad \Delta = -22, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 14132181231513221

$$14132181231513221\overset{2}{3} \text{ [#}=7] \quad \Delta = -28, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$141321812315\overset{2}{1}3221 \quad \Delta = -20, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$14132181231513221\overset{2}{3} \text{ [#}=9] \quad \Delta = -28, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$14132181231513221\overset{2}{3} \text{ [#}=5] \quad \Delta = -28, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$14132181231513221\overset{2}{3} \text{ [#}=4] \quad \Delta = -28, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$14132181231513221\overset{2}{3} \text{ [#}=3] \quad \Delta = -28, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$14132181231513221\overset{2}{2} \quad \Delta = -22, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$12231513218123141\overset{2}{3} \text{ [#}=3] \quad \Delta = -28, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$14132181231513221\overset{2}{3} \text{ [#}=2] \quad \Delta = -28, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$12231513218123141\overset{2}{3} \text{ [#}=2] \quad \Delta = -28, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$14132181231513221\overset{2}{3} \text{ [#}=8] \quad \Delta = -28, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$14132181231513221\overset{2}{3} \text{ [#}=6] \quad \Delta = -28, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$14132181231513221\overset{2}{2} \quad \Delta = -20, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$14132181231513221\overset{2}{2} \quad \Delta = -20, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$14132181231513221\overset{2}{2} \quad \Delta = -20, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$14132181231513221\overset{2}{2} \quad \Delta = -20, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 12315132181231¹₅1

$$\overset{3}{1} \overset{12315132181231}{5} \overset{1}{1} \text{ [#}=4] \quad \Delta = -52, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$12315132181231\overset{2}{5} \overset{1}{1} \quad \Delta = -44, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\begin{aligned}
12315132181231\overset{2}{1}\overset{1}{5} \quad \Delta = -44, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
12315132181231\overset{2}{1}\overset{1}{5} \quad \Delta = -44, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
12315132181231\overset{2}{1}\overset{1}{5} \quad \Delta = -44, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
12315132181231\overset{2}{1}\overset{1}{5} \quad \Delta = -44, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1\overset{3}{1}2315132181231\overset{1}{5} \quad 1 \text{ } [\# = 2] \quad \Delta = -52, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1\overset{3}{1}2315132181231\overset{1}{5} \quad 1 \text{ } [\# = 3] \quad \Delta = -52, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1\overset{3}{1}2315132181231\overset{1}{5} \quad 1 \text{ } [\# = 9] \quad \Delta = -52, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
1\overset{3}{1}2315132181231\overset{1}{5} \quad 1 \text{ } [\# = 8] \quad \Delta = -52, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
1\overset{3}{1}2315132181231\overset{1}{5} \quad 1 \text{ } [\# = 7] \quad \Delta = -52, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
1\overset{3}{1}2315132181231\overset{1}{5} \quad 1 \text{ } [\# = 6] \quad \Delta = -52, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
1\overset{3}{1}2315132181231\overset{1}{5} \quad 1 \text{ } [\# = 5] \quad \Delta = -52, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
\text{Base: } 61223151321812321 & \\
61223151321812321\overset{3}{1} \text{ } [\# = 8] \quad \Delta = -80, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
61223151321812321\overset{3}{1} \text{ } [\# = 7] \quad \Delta = -80, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
61223151321812321\overset{3}{1} \text{ } [\# = 6] \quad \Delta = -80, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
61223151321812321\overset{3}{1} \text{ } [\# = 5] \quad \Delta = -80, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
61223151321812321\overset{3}{1} \text{ } [\# = 4] \quad \Delta = -80, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
61223151321812321\overset{3}{1} \text{ } [\# = 3] \quad \Delta = -80, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
61223151321812321\overset{3}{1} \text{ } [\# = 9] \quad \Delta = -80, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
61223151321812321\overset{2}{1} \quad \Delta = -72, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
61223151321812321^2 \quad \Delta = -72, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
61223151321812321^2 \quad \Delta = -72, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
61223151321812321^2 \quad \Delta = -72, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
61223151321812321^2 \quad \Delta = -72, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
61223151321812321^3 \quad \Delta = -80, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 63 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
61223151321812321^3_{[\# = 2]} \quad \Delta = -80, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: 61231513218123215

$$\begin{aligned}
61231513218123215^2 \quad \Delta = -124, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
61231513218123215^2 \quad \Delta = -124, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
61231513218123215^2 \quad \Delta = -124, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
61231513218123215^2 \quad \Delta = -124, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
61231513218123215^2 \quad \Delta = -124, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
61231513218123215^2 \quad \Delta = -124, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

Base: 61231513218123151

$$\begin{aligned}
61231513218123151^2 \quad \Delta = -122, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
61231513218123151^2 \quad \Delta = -122, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
61231513218123151^2 \quad \Delta = -122, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
61231513218123151^2 \quad \Delta = -122, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
61231513218123151^2 \quad \Delta = -122, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
61231513218123151^2 \quad \Delta = -122, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: 61315132181231513

$$\begin{aligned}
61315132181231513^2 \quad \Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
61315132181231513^2 \quad \Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
61315132181231513^2 \quad \Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
61315132181231513^2 \quad \Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
61315132181231513^2 \quad \Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
61315132181231513^2 \quad \Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
61315132181231513^2 \quad \Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

Base: 16123151321812321

$$\begin{aligned}
16123151321812321^3_{[\# = 9]} \quad \Delta = -80, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
16123151321812321^3_{[\# = 3]} \quad \Delta = -80, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
16123151321812321^2 \quad \Delta = -72, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
16123151321812321^2 \quad \Delta = -72, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
16123151321812321^2 \quad \Delta = -72, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
16123151321812321^2 \quad \Delta = -72, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
16123151321812321^2 \quad \Delta = -72, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
16123151321812321^3_{[\# = 2]} \quad \Delta = -80, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
16123151321812321^3_{[\# = 6]} \quad \Delta = -80, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
16123151321812321^3_{[\# = 8]} \quad \Delta = -80, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
16123151321812321^3_{[\# = 5]} \quad \Delta = -80, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
16123151321812321^3_{[\# = 4]} \quad \Delta = -80, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

$$16123151321812321\overset{2}{3}_{[\# = 7]} \quad \Delta = -80, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

Base: 16123218123151315

$$1612321812315\overset{2}{1}315 \quad \Delta = -123, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1612321812315\overset{2}{1}315 \quad \Delta = -123, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1612321812315\overset{2}{1}315 \quad \Delta = -123, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1612321812315\overset{2}{1}315 \quad \Delta = -123, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1612321812315\overset{2}{1}315 \quad \Delta = -123, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1612321812315\overset{2}{1}315 \quad \Delta = -123, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1612321812315\overset{2}{1}315 \quad \Delta = -124, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 16131513218123215

$$161315\overset{2}{1}3218123215 \quad \Delta = -123, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$161315\overset{2}{1}3218123215 \quad \Delta = -123, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$161315\overset{2}{1}3218123215 \quad \Delta = -123, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$161315\overset{2}{1}3218123215 \quad \Delta = -123, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$161315\overset{2}{1}3218123215 \quad \Delta = -123, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$161315\overset{2}{1}3218123215 \quad \Delta = -123, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

Base: 16131513218123151

$$161315\overset{2}{1}3218123151 \quad \Delta = -121, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$161315\overset{2}{1}3218123151 \quad \Delta = -121, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$161315\overset{2}{1}3218123151 \quad \Delta = -121, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$161315\overset{2}{1}3218123151 \quad \Delta = -121, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$16131513218123151^2 \quad \Delta = -121, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$16131513218123151^2 \quad \Delta = -121, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 21612321812315131

$$21612321812315131^3_{[\# = 8]} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$21612321812315131^2 \quad \Delta = -71, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$21612321812315131^2 \quad \Delta = -71, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$21612321812315131^3_{[\# = 9]} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$21612321812315131^3_{[\# = 7]} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$21612321812315131^2 \quad \Delta = -71, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$21612321812315131^2 \quad \Delta = -71, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$21612321812315131^2 \quad \Delta = -71, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$21612321812315131^2 \quad \Delta = -71, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$21612321812315131^3_{[\# = 3]} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$21612321812315131^3_{[\# = 2]} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$21612321812315131^3 \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$21612321812315131^3_{[\# = 4]} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$21612321812315131^3_{[\# = 5]} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$21612321812315131^3_{[\# = 6]} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: 21613151321812321

$$21613151321812321^2 \quad \Delta = -71, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$\begin{aligned}
21613151321812321^2 \quad \Delta = -71, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21613151321812321^2 \quad \Delta = -71, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21613151321812321^2 \quad \Delta = -71, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21613151321812321^2 \quad \Delta = -71, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21613151321812321^2 \quad \Delta = -71, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21613151321812321^3_{[\#=5]} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
21613151321812321^3_{[\#=8]} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
21613151321812321^3 \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
21613151321812321^3_{[\#=9]} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
21613151321812321^3_{[\#=7]} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
21613151321812321^3_{[\#=6]} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21613151321812321^3_{[\#=2]} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21613151321812321^3_{[\#=3]} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
21613151321812321^3_{[\#=4]} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\text{Base: } 13161232181231512 & \\
21513218123216131^3_{[\#=9]} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
21513218123216131^3_{[\#=8]} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
13161232181231512^2 \quad \Delta = -71, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
13161232181231512^2 \quad \Delta = -71, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21513218123216131^3_{[\#=7]} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
13161232181231512^2 \quad \Delta = -71, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
215132181232161313_{\#6} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
13161232181231512^2 \quad \Delta = -71, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
13161232181231512^2 \quad \Delta = -71, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
13161232181231512^2 \quad \Delta = -71, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
215132181232161313_{\#5} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
215132181232161313_{\#4} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
215132181232161313_{\#3} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
215132181232161313_{\#2} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
13161232181231512^3 \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
\text{Base: } 1316123218123151^1 & \\
1316123218123151^2 \quad \Delta = -71, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1316123218123151^2 \quad \Delta = -71, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1316123218123151^2 \quad \Delta = -71, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1316123218123151^2 \quad \Delta = -71, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1^{\frac{3}{5}} 13161232181231^1_{\#9} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1^{\frac{3}{5}} 13161232181231^1_{\#8} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1^{\frac{3}{5}} 13161232181231^1_{\#6} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1^{\frac{3}{5}} 13161232181231^1_{\#7} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1^{\frac{3}{5}} 13161232181231^1_{\#5} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}
\end{aligned}$$

$$1 \overset{3}{1} 3 1 6 1 2 3 2 1 8 1 2 3 1 \quad 1 \text{ [#}=4] \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1 \overset{3}{1} 3 1 6 1 2 3 2 1 8 1 2 3 1 \quad 1 \text{ [#}=2] \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{3}{1} 3 1 6 1 2 3 2 1 8 1 2 3 1 \quad 1 \text{ [#}=3] \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$13161232181231 \overset{2}{1} 51 \quad \Delta = -71, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: 51316123218123151

$$513161232181231 \overset{2}{5} 1 \quad \Delta = -123, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$513161232181231 \overset{2}{5} 1 \quad \Delta = -123, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$513161232181231 \overset{2}{5} 1 \quad \Delta = -123, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$513161232181231 \overset{2}{5} 1 \quad \Delta = -123, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$513161232181231 \overset{2}{5} 1 \quad \Delta = -123, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$513161232181231 \overset{2}{5} 1 \quad \Delta = -123, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: 31613151321812315

$$3161315132181231 \overset{2}{5} 1 \quad \Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$3161315132181231 \overset{2}{5} 1 \quad \Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 58 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$3161315132181231 \overset{2}{5} 1 \quad \Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 63 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$3161315132181231 \overset{2}{5} 1 \quad \Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 196 \quad \begin{pmatrix} 0 & 14 & 2 \\ 14 & 68 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

$$3161315132181231 \overset{2}{5} 1 \quad \Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$3161315132181231 \overset{2}{5} 1 \quad \Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$3161315132181231 \overset{2}{5} 1 \quad \Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 13218123151316131

$$13218123151316131 \overset{3}{1} \text{ [#}=9] \quad \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$\begin{array}{ll}
13218123151316131\overset{2}{3} \text{ [#=5]} & \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
13161315132181231\overset{2}{3} \text{ [#=5]} & \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
13218123151316131\overset{2}{3} \text{ [#=4]} & \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
13161315132181231\overset{2}{3} \text{ [#=2]} & \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
13218123151316131\overset{2}{3} \text{ [#=3]} & \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
13161315132181231\overset{2}{3} \text{ [#=4]} & \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
13218123151316131\overset{2}{3} \text{ [#=7]} & \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
13218123151316131\overset{2}{3} \text{ [#=6]} & \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
13218123151316131\overset{2}{3} \text{ [#=8]} & \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
13161315132181231\overset{2}{3} & \Delta = -77, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
13161315132181231\overset{2}{3} & \Delta = -77, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
13161315132181231\overset{2}{3} & \Delta = -77, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
13161315132181231\overset{2}{3} & \Delta = -77, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
13161315132181231\overset{2}{3} & \Delta = -77, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
13161315132181231\overset{2}{3} & \Delta = -77, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
13161315132181231\overset{2}{3} \text{ [#=3]} & \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
13218123151316131\overset{2}{3} \text{ [#=2]} & \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\text{Base: } 41321812321613161 & \\
41321812321613161\overset{2}{3} & \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
41321812321613161\overset{2}{3} & \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
41321812321613161^2 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
41321812321613161^2 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
41321812321613161^2 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 163 & 25 \\ 2 & 25 & 8 \end{pmatrix} \\
41321812321613161^2 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
41321812321613161^2 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 21613161232181231

$$\begin{aligned}
21613161232181231^3_{[\# = 5]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21613161232181231^2 \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21613161232181231^2 \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
21613161232181231^2 \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21613161232181231^2 \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21613161232181231^2 \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21613161232181231^2 \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
21613161232181231^3_{[\# = 4]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21613161232181231^3 \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
21613161232181231^3_{[\# = 2]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
21613161232181231^3_{[\# = 3]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1612321812315131^1

$$\begin{aligned}
1612321812315131^2 \quad \Delta = -71, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1612321812315131^1_{[\# = 3]} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{1}{1612321812315131\textcolor{red}{3}}_{[\# = 9]} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{1}{1612321812315131\textcolor{red}{3}}_{[\# = 7]} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& \overset{1}{1612321812315131\textcolor{red}{3}}_{[\# = 6]} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{1}{1612321812315131\textcolor{red}{3}}_{[\# = 8]} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{1}{1612321812315131\textcolor{red}{3}}_{[\# = 2]} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{1}{1612321812315131\textcolor{red}{3}}_{[\# = 5]} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{1}{1612321812315131\textcolor{red}{3}}_{[\# = 4]} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{1}{16123218123151\textcolor{red}{2}31} \quad \Delta = -71, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{1}{1612321812315131\textcolor{red}{2}} \quad \Delta = -71, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{1\textcolor{red}{2}}{1612321812315131} \quad \Delta = -71, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{1}{1612321812315131\textcolor{red}{2}} \quad \Delta = -71, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \textbf{Base: } \overset{1}{1613151321812321} \\
& \overset{1}{1613151321812321\textcolor{red}{2}} \quad \Delta = -71, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{1}{1613151321812321\textcolor{red}{3}}_{[\# = 9]} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{1}{1613151321812321\textcolor{red}{2}} \quad \Delta = -71, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{1}{1613151321812321\textcolor{red}{3}}_{[\# = 8]} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{1}{1613151321812321\textcolor{red}{3}}_{[\# = 7]} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& \overset{1\textcolor{red}{2}}{1613151321812321} \quad \Delta = -71, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{1}{1613151321812321\textcolor{red}{2}} \quad \Delta = -71, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{1}{1613151321812321\textcolor{red}{3}}_{[\# = 6]} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

$$1613151321812321\overset{1}{3}_{[\# = 5]} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1613151321812321\overset{1}{3}_{[\# = 2]} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1613151321812321\overset{2}{3} \quad \Delta = -71, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1613151321812321\overset{1}{3}_{[\# = 3]} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1613151321812321\overset{1}{3}_{[\# = 4]} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 1613161232181231

$$1613161232181231\overset{1}{2} \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1613161232181231\overset{1}{3}_{[\# = 5]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1613161232181231\overset{1}{2}\overset{2}{3} \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1613161232181231\overset{1}{2}\overset{2}{3} \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$1613161232181231\overset{1}{2}\overset{2}{3} \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1613161232181231\overset{1}{2}\overset{2}{3} \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1613161232181231\overset{1}{3}_{[\# = 2]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$1613161232181231\overset{1}{3}_{[\# = 3]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1613161232181231\overset{1}{3}_{[\# = 4]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: 41321812315132217

$$41321812315132217\overset{2}{3} \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$41321812315132217\overset{2}{3} \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$41321812315132217\overset{2}{3} \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$41321812315132217\overset{2}{3} \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\begin{aligned}
41321812315132217^2 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 5 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
41321812315132217^2 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
41321812315132217^2 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 17122315132181231

$$\begin{aligned}
17122315132181231^2 \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
17122315132181231^2 \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
17122315132181231^2 \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
17122315132181231^3_{[\#=2]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
17122315132181231^3_{[\#=5]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
17122315132181231^3_{[\#=4]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
17122315132181231^3_{[\#=3]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
17122315132181231^2 \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
17122315132181231^2 \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

Base: 41321812315132171

$$\begin{aligned}
41321812315132171^2 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
41321812315132171^2 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
41321812315132171^2 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
41321812315132171^2 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
41321812315132171^2 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
41321812315132171^2 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}
\end{aligned}$$

$$41321812315132171 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 21712315132181231

$$21712315132181231 \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$21712315132181231 \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$21712315132181231 \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$21712315132181231 \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$21712315132181231 \textcolor{red}{3}_{[\# = 5]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$21712315132181231 \textcolor{red}{3}_{[\# = 4]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$21712315132181231 \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$21712315132181231 \textcolor{red}{3}_{[\# = 2]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$21712315132181231 \textcolor{red}{3}_{[\# = 3]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$21712315132181231 \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 22171232181231512

$$22171232181231512 \quad \Delta = -98, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$22171232181231512 \quad \Delta = -90, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$22171232181231512 \quad \Delta = -90, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$22171232181231512 \quad \Delta = -90, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$22171232181231512 \quad \Delta = -90, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$22171232181231512 \quad \Delta = -90, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$22171232181231512 \quad \Delta = -99, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$22171232181231512 \quad \Delta = -90, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 22171232181231¹51

$$22171232181231²51 \quad \Delta = -90, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$22171232181231²¹51 \quad \Delta = -90, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$22171232181231²51 \quad \Delta = -90, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$22171232181231³51 \quad \Delta = -99, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$22171232181231²51 \quad \Delta = -90, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$22171232181231²51 \quad \Delta = -90, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 14132181232171231

$$14132181232171231 \quad \Delta = -75, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$13217123218123141\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -83, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$14132181232171231 \quad \Delta = -75, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$14132181232171231 \quad \Delta = -75, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$14132181232171231 \quad \Delta = -75, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$14132181232171231 \quad \Delta = -75, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$14132181232171231 \quad \Delta = -77, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$14132181232171231\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -83, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$14132181232171231\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -83, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$14132181232171231\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -83, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$13217123218123141\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -83, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$14132181232171231\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -83, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

Base: 41321812321712315

$$\begin{aligned}
41321812321712315^2 \quad \Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
41321812321712315^2 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
41321812321712315^2 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
41321812321712315^2 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
41321812321712315^2 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
41321812321712315^2 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
41321812321712315^2 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

Base: 15132171232181231

$$\begin{aligned}
15132171232181231^3_{[\#=4]} \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
15132171232181231^2 \quad \Delta = -99, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
15132171232181231^2 \quad \Delta = -99, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
15132171232181231^2 \quad \Delta = -99, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
15132171232181231^2 \quad \Delta = -99, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
15132171232181231^3_{[\#=2]} \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
15132171232181231^2 \quad \Delta = -99, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
15132171232181231^3_{[\#=3]} \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
15132171232181231^3_{[\#=5]} \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 12321712321812313

$$\begin{aligned}
31321812321712321^3_{[\#=4]} \quad \Delta = -65, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
31321812321712321^3_{[\#=5]} \quad \Delta = -65, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
31321812321712321\overset{3}{3} \text{ }_{[\# = 7]} \quad \Delta = -65, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
31321812321712321\overset{3}{3} \text{ }_{[\# = 8]} \quad \Delta = -65, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
1232171232181231\overset{3}{3} \text{ }_{[\# = 2]} \quad \Delta = -65, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
31321812321712321\overset{3}{3} \text{ }_{[\# = 6]} \quad \Delta = -65, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
1232171232181231\overset{2}{3} \quad \Delta = -59, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1232171232181231\overset{2}{3} \quad \Delta = -57, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
31321812321712321\overset{3}{3} \text{ }_{[\# = 3]} \quad \Delta = -65, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1232171232181231\overset{2}{3} \quad \Delta = -57, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1232171232181231\overset{2}{3} \quad \Delta = -57, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1232171232181231\overset{2}{3} \quad \Delta = -57, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1232171232181231\overset{2}{3} \quad \Delta = -57, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
31321812321712321\overset{3}{3} \text{ }_{[\# = 2]} \quad \Delta = -65, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
31321812321712321\overset{3}{3} \text{ }_{[\# = 9]} \quad \Delta = -65, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
1232171232181231\overset{3}{3} \quad \Delta = -65, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: 51232171232181231

$$\begin{aligned}
51232171232181231\overset{3}{3} \text{ }_{[\# = 2]} \quad \Delta = -109, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
51232171232181231\overset{2}{3} \quad \Delta = -101, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
51232171232181231\overset{2}{3} \quad \Delta = -101, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
51232171232181231\overset{2}{3} \quad \Delta = -101, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
51232171232181231\overset{2}{3} \quad \Delta = -101, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
51232171232181231^{\mathbf{2}} \quad \Delta = -101, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
51232171232181231^{\mathbf{3}}_{[\# = 3]} \quad \Delta = -109, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
51232171232181231^{\mathbf{3}}_{[\# = 5]} \quad \Delta = -109, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
51232171232181231^{\mathbf{3}}_{[\# = 4]} \quad \Delta = -109, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1712315132181231

$$\begin{aligned}
1712315132181231^{\mathbf{2}} \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1712315132181231^{\mathbf{2}} \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1712315132181231^{\mathbf{2}} \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1712315132181231^{\mathbf{12}} \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1712315132181231^{\mathbf{2}} \quad \Delta = -97, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1712315132181231^{\mathbf{3}}_{[\# = 2]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
1712315132181231^{\mathbf{3}}_{[\# = 3]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1712315132181231^{\mathbf{3}}_{[\# = 4]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1712315132181231^{\mathbf{3}}_{[\# = 5]} \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 171232181231512

$$\begin{aligned}
171232181231512^{\mathbf{21}} \quad \Delta = -90, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
171232181231512^{\mathbf{21}} \quad \Delta = -90, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
171232181231512^{\mathbf{21}} \quad \Delta = -90, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
171232181231512^{\mathbf{21}} \quad \Delta = -90, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
171232181231512^{\mathbf{21}} \quad \Delta = -90, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$1\overset{21}{7}1232181231512 \quad \Delta = -90, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1\overset{21}{7}123218123151\overset{3}{2} \quad \Delta = -98, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

Base: $1\overset{21}{7}1232181231\overset{1}{5}1$

$$1\overset{21}{7}1232181231\overset{1}{5}1 \quad \Delta = -90, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1\overset{21}{7}1232181231\overset{1}{5}1 \quad \Delta = -90, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1\overset{21}{7}1232181231\overset{2}{5}1 \quad \Delta = -90, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1\overset{21}{7}1232181231\overset{2}{5}1 \quad \Delta = -90, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1\overset{21}{7}1232181231\overset{2}{5}1 \quad \Delta = -90, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 21612321812321612

$$21612321812321612 \quad \Delta = -91, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$21612321812321612 \quad \Delta = -91, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$21612321812321612 \quad \Delta = -99, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$21612321812321612 \quad \Delta = -91, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

Base: 13161232181232151

$$15123218123216131\overset{3}{3} \text{ [#=4]} \quad \Delta = -81, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$15123218123216131\overset{3}{3} \text{ [#=5]} \quad \Delta = -81, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$15123218123216131\overset{3}{3} \text{ [#=6]} \quad \Delta = -81, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$15123218123216131\overset{3}{3} \text{ [#=7]} \quad \Delta = -81, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$13161232181232151 \quad \Delta = -73, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$15123218123216131\overset{3}{3} \text{ [#=3]} \quad \Delta = -81, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$15123218123216131\overset{3}{3} \text{ [#=2]} \quad \Delta = -81, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\begin{aligned}
13161232181232151^2 \quad \Delta = -73, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
13161232181232151^2 \quad \Delta = -73, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
13161232181232151^2 \quad \Delta = -73, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
13161232181232151^2 \quad \Delta = -73, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
151232181232161313_{\# = 8} \quad \Delta = -81, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
151232181232161313_{\# = 9} \quad \Delta = -81, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}
\end{aligned}$$

Base: 51316123218123215

$$\begin{aligned}
51316123218123215^2 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
51316123218123215^2 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
51316123218123215^2 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
51316123218123215^2 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
51316123218123215^2 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
51316123218123215^2 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{aligned}$$

Base: 15131612321812321

$$\begin{aligned}
151316123218123213_{\# = 7} \quad \Delta = -81, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
151316123218123213_{\# = 6} \quad \Delta = -81, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
151316123218123213_{\# = 3} \quad \Delta = -81, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
151316123218123213_{\# = 2} \quad \Delta = -81, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
151316123218123213_{\# = 9} \quad \Delta = -81, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
151316123218123213_{\# = 8} \quad \Delta = -81, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
15131612321812321\textcolor{red}{3}_{[\# = 5]} \quad \Delta = -81, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
15131612321812321\textcolor{red}{3}_{[\# = 4]} \quad \Delta = -81, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
15131612321812321\textcolor{red}{2} \quad \Delta = -73, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
15131612321812321\textcolor{red}{2} \quad \Delta = -73, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
15131612321812321\textcolor{red}{2} \quad \Delta = -73, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
15131612321812321\textcolor{red}{2} \quad \Delta = -73, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
15131612321812321\textcolor{red}{2} \quad \Delta = -73, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\textbf{Base: } 1612321812321612 \\
1612321812321612\textcolor{red}{2} \quad \Delta = -91, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1612321812321612\textcolor{red}{12} \quad \Delta = -91, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1612321812321612\textcolor{red}{2} \quad \Delta = -91, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1612321812321612\textcolor{red}{2} \quad \Delta = -91, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1612321812321612\textcolor{red}{2} \quad \Delta = -91, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1612321812321612\textcolor{red}{3} \quad \Delta = -99, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
\textbf{Base: } 161232181232161 \\
161232181232161\textcolor{red}{2} \quad \Delta = -91, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
161232181232161\textcolor{red}{12} \quad \Delta = -91, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\textbf{Base: } 22171232181232151 \\
22171232181232151\textcolor{red}{3} \quad \Delta = -101, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
22171232181232151\textcolor{red}{2} \quad \Delta = -92, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
22171232181232151\textcolor{red}{2} \quad \Delta = -92, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
22171232181232151 \quad \Delta = -92, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
22171232181232151 \quad \Delta = -92, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
22171232181232151 \quad \Delta = -92, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 41232181232171231

$$\begin{aligned}
41232181232171231 \quad \Delta = -77, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
41232181232171231 \textcolor{red}{3} \text{ [#=3]} \quad \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
41232181232171231 \textcolor{red}{3} \text{ [#=2]} \quad \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
41232181232171231 \textcolor{red}{3} \quad \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 41 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
41232181232171231 \textcolor{red}{3} \text{ [#=4]} \quad \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
41232181232171231 \textcolor{red}{3} \text{ [#=5]} \quad \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
41232181232171231 \quad \Delta = -77, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
41232181232171231 \quad \Delta = -77, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
41232181232171231 \quad \Delta = -77, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
41232181232171231 \quad \Delta = -77, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 41321712321812321

$$\begin{aligned}
41321712321812321 \quad \Delta = -77, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
41321712321812321 \textcolor{red}{3} \text{ [#=8]} \quad \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
41321712321812321 \quad \Delta = -77, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
41321712321812321 \quad \Delta = -77, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
41321712321812321 \textcolor{red}{3} \text{ [#=2]} \quad \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
41321712321812321\overset{2}{3} \text{ [#=3]} \quad \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
41321712321812321\overset{2}{3} \text{ [#=4]} \quad \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
41321712321812321\overset{2}{3} \text{ [#=5]} \quad \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
41321712321812321\overset{2}{3} \text{ [#=6]} \quad \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
41321712321812321\overset{2}{3} \quad \Delta = -77, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
41321712321812321\overset{2}{3} \text{ [#=9]} \quad \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
41321712321812321\overset{2}{3} \text{ [#=7]} \quad \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
41321712321812321\overset{2}{3} \quad \Delta = -77, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
41321712321812321\overset{2}{3} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\text{Base: } \overset{21}{17}1232181232151 & \\
\overset{21}{17}1232181232151 \quad \Delta = -92, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\overset{21}{17}1232181232151\overset{2}{3} \quad \Delta = -92, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\overset{21}{17}1232181232151\overset{2}{3} \quad \Delta = -92, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\overset{21}{17}1232181232151\overset{2}{3} \quad \Delta = -92, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\overset{21}{17}1232181232151\overset{2}{3} \quad \Delta = -92, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\text{Base: } 1232171232181232\overset{1}{3} & \\
\overset{3}{2}\overset{12321712321812}{3}1 \text{ [#=7]} \quad \Delta = -59, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
1232171232181232\overset{41}{3} \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
\overset{3}{2}\overset{12321712321812}{3}1 \text{ [#=9]} \quad \Delta = -59, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
1232171232181232\overset{31}{3} \text{ [#=2]} \quad \Delta = -59, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{l}
2^{\overset{3}{1}2321712321812} 3^{1} \text{ [#=2]} \quad \Delta = -59, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
2^{\overset{3}{1}2321712321812} 3^{1} \text{ [#=6]} \quad \Delta = -59, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
12321712321812\overset{31}{3}2 \text{ [#=3]} \quad \Delta = -59, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2^{\overset{3}{1}2321712321812} 3^{1} \text{ [#=8]} \quad \Delta = -59, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
2^{\overset{3}{1}2321712321812} 3^{1} \text{ [#=5]} \quad \Delta = -59, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
2^{\overset{3}{1}2321712321812} 3^{1} \text{ [#=4]} \quad \Delta = -59, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1232171232181\overset{2}{1}232 \quad \Delta = -51, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1232171232181\overset{2}{1}232 \quad \Delta = -51, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1232171232181\overset{2}{1}232 \quad \Delta = -51, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
12321712321812\overset{21}{3}2 \quad \Delta = -51, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1232171232181\overset{2}{1}232 \quad \Delta = -51, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2^{\overset{3}{1}2321712321812} 3^{1} \text{ [#=3]} \quad \Delta = -59, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\text{Base: } 21812321812231513 \\
2181\overset{2}{2}321812231513 \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21812321812231\overset{2}{5}13 \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21812321812231\overset{2}{5}13 \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
2181\overset{2}{2}321812231513 \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21812321812231\overset{2}{5}13 \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
21812321812231\overset{2}{5}13 \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 196 \quad \begin{pmatrix} 0 & 14 & 2 \\ 14 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
\text{Base: } 22181232181223151
\end{array}$$

$$\begin{aligned}
22181232181223151^3 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
22181232181223151^2 \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
22181232181223151^2 \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
22181232181223151^2 \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
22181232181223151^2 \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
22181232181223151^2 \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: 13221812321812231

$$\begin{aligned}
13221812321812231^2 \quad \Delta = -75, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
13221812321812231^3_{[\#=2]} \quad \Delta = -83, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
13221812321812231^3_{[\#=3]} \quad \Delta = -83, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
13221812321812231^3_{[\#=4]} \quad \Delta = -83, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
13221812321812231^2 \quad \Delta = -75, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
13221812321812231^3_{[\#=5]} \quad \Delta = -83, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}
\end{aligned}$$

Base: 32218123218122315

$$\begin{aligned}
32218123218122315^2 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
32218123218122315^2 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
32218123218122315^2 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
32218123218122315^2 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 50 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
32218123218122315^2 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: 41413218123218122

$$41413218123218122^2 \quad \Delta = -122, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$\begin{aligned}
41413218123218122^2 \quad \Delta = -122, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
41413218123218122^2 \quad \Delta = -122, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
41413218123218122^2 \quad \Delta = -122, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
41413218123218122^2 \quad \Delta = -122, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
41413218123218122^2 \quad \Delta = -122, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
41413218123218122^2 \quad \Delta = -124, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
41413218123218122^3 \quad \Delta = -131, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}
\end{aligned}$$

Base: 13221812321812313

$$\begin{aligned}
13221812321812313^2 \quad \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
13221812321812313^2 \quad \Delta = -83, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
13221812321812313^2 \quad \Delta = -83, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
13221812321812313^2 \quad \Delta = -83, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
13221812321812313^2 \quad \Delta = -83, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
13221812321812313^2 \quad \Delta = -83, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
313218123218122313^3_{[\#=3]} \quad \Delta = -91, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
13221812321812313^3_{[\#=2]} \quad \Delta = -91, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
313218123218122313^3_{[\#=2]} \quad \Delta = -91, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
13221812321812313^3 \quad \Delta = -91, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
313218123218122313^3_{[\#=5]} \quad \Delta = -91, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
313218123218122313^3_{[\#=4]} \quad \Delta = -91, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

Base: 41232181232181222

$$\begin{aligned}
41232181232181222^2 \quad \Delta = -96, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
41232181232181222^2 \quad \Delta = -96, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
41232181232181222^2 \quad \Delta = -96, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
41232181232181222^3 \quad \Delta = -104, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 41 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
41232181232181222^2 \quad \Delta = -96, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
41232181232181222^2 \quad \Delta = -96, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 41322181232181231

$$\begin{aligned}
41322181232181231^2 \quad \Delta = -103, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
41322181232181231^3_{[\# = 5]} \quad \Delta = -111, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
41322181232181231^3_{[\# = 2]} \quad \Delta = -111, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
41322181232181231^2 \quad \Delta = -103, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
41322181232181231^2 \quad \Delta = -103, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
41322181232181231^2 \quad \Delta = -103, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
41322181232181231^2 \quad \Delta = -103, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
41322181232181231^3_{[\# = 4]} \quad \Delta = -111, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
41322181232181231^3_{[\# = 3]} \quad \Delta = -111, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: 14132181232181222

$$\begin{aligned}
22218123218123141^3_{[\# = 3]} \quad \Delta = -102, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
22218123218123141^3_{[\# = 2]} \quad \Delta = -102, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
14132181232181222^2 \quad \Delta = -96, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
14132181232181222 \quad \Delta = -94, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
14132181232181222 \quad \Delta = -94, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
14132181232181222 \quad \Delta = -94, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
14132181232181222 \quad \Delta = -94, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
14132181232181222 \quad \Delta = -94, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1322181232181232¹

$$\begin{aligned}
1322181232181232 \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
1322181232181232 \quad \Delta = -77, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1322181232181232 \quad \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2^{313221812321812} 3^1 \quad \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
2^{313221812321812} 3^1 \quad \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1322181232181232 \quad \Delta = -77, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1322181232181232 \quad \Delta = -77, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1322181232181232 \quad \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1322181232181232 \quad \Delta = -77, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2^{313221812321812} 3^1 \quad \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
2^{313221812321812} 3^1 \quad \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1322181232181232 \quad \Delta = -77, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: 31321812321812313

$$\begin{aligned}
31321812321812313 \quad \Delta = -99, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 93 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
31321812321812313 \quad \Delta = -93, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$31321812321812313 \quad \Delta = -91, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$31321812321812313 \quad \Delta = -91, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$31321812321812313 \quad \Delta = -91, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$31321812321812313 \quad \Delta = -99, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 21 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$\text{Base: } 2^{\overset{31321812321812}{3}} 1$$

$$2^{\overset{31321812321812}{3}} 1 \quad \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$2^{\overset{31321812321812}{3}} 1 \quad \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$2^{\overset{31321812321812}{3}} 1 \quad \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$2^{\overset{31321812321812}{3}} 1 \quad \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$2^{\overset{31321812321812}{3}} 1 \quad \Delta = -87, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$2^{\overset{31321812321812}{3}} 1 \quad \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$2^{\overset{31321812321812}{3}} 1 \quad \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$2^{\overset{31321812321812}{3}} 1 \quad \Delta = -93, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 96 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$2^{\overset{31321812321812}{3}} 1 \quad \Delta = -93, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$2^{\overset{31321812321812}{3}} 1 \quad \Delta = -93, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 41 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$2^{\overset{31321812321812}{3}} 1 \quad \Delta = -93, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$2^{\overset{31321812321812}{3}} 1 \quad \Delta = -113, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\text{Base: } 2^{\overset{2321812321812}{3}} 1$$

$$2^{\overset{2321812321812}{3}} 1 \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$2^{\overset{2321812321812}{3}} 1 \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\begin{aligned}
2^{2321812321812} 3^1 \quad \Delta = -79, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
2^{2321812321812} 3^{14} \quad \Delta = -107, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2^{2321812321812} 3^{13} [\# = 3] \quad \Delta = -87, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 96 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
2^{2321812321812} 3^{13} [\# = 2] \quad \Delta = -87, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 41 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1812231513221812

$$\begin{aligned}
1812231513221812^2 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1812231513221812^2 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1812231513221812^2 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1812231513221812^2 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1812231513221812^{12} \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1812231513218122

$$\begin{aligned}
1812231513218122^3 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
1812231513218122^2 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1812231513218122^2 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1812231513218122^2 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1812231513218122^{12} \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
1812231513218122^2 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 181223151322181

$$\begin{aligned}
181223151322181^2 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
181223151322181^{12} \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1812315131613151

$$\begin{array}{ll}
{}^1_{1812315131613151} \quad \Delta = -120, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^1_{1812315131613151} \quad \Delta = -120, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
{}^1_{1812315131613151} \quad \Delta = -120, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
{}^{12}_{1812315131613151} \quad \Delta = -120, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^1_{1812315131613151} \quad \Delta = -120, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
{}^1_{1812315131613151} \quad \Delta = -120, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{array}$$

Base: ${}^1_{4131613161232181}$

$$\begin{array}{ll}
{}^2_{4131613161232181} \quad \Delta = -124, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
{}^2_{4131613161232181} \quad \Delta = -124, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^3_{4131613161232181} \quad \Delta = -132, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
{}^{21}_{4131613161232181} \quad \Delta = -124, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
{}^2_{4131613161232181} \quad \Delta = -124, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
{}^2_{4131613161232181} \quad \Delta = -124, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
{}^2_{4131613161232181} \quad \Delta = -124, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{array}$$

Base: ${}^1_{1812321712321613}$

$$\begin{array}{ll}
{}^1_{1812321712321613} \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
{}^1_{1812321712321613} \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
{}^1_{1812321712321613} \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
{}^1_{1812321712321613} \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
{}^{12}_{1812321712321613} \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{array}$$

Base: ${}^1_{1812315132218123}$

$$\begin{aligned}
& \overset{1}{1812315132218123} \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{1}{1812315132218123} \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& \overset{1}{1812315132218123} \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{1}{1812315132218123} \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{12}{1812315132218123} \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{1812321812231513}$

$$\begin{aligned}
& \overset{12}{1812321812231513} \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{1}{1812321812231513} \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& \overset{1}{1812321812231513} \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{1}{1812321812231513} \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{1}{1812321812231513} \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 196 \quad \begin{pmatrix} 0 & 14 & 2 \\ 14 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{1812315132181223}$

$$\begin{aligned}
& \overset{1}{1812315132181223} \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{12}{1812315132181223} \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{1}{1812315132181223} \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{1}{1812315132181223} \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& \overset{1}{1812315132181223} \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{21}{181232161316131}$

$$\begin{aligned}
& \overset{21}{181232161316131} \text{ \textcolor{red}{3} } [\# = 8] \quad \Delta = -104, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{21}{181232161316131} \text{ \textcolor{red}{3} } [\# = 9] \quad \Delta = -104, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{21}{181232161316131} \text{ \textcolor{red}{3} } [\# = 3] \quad \Delta = -104, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

$$18^{21}1232161316131\bar{3} \text{ [#}=7] \quad \Delta = -104, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$18^{21}\bar{2}1232161316131 \quad \Delta = -96, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$18^{21}1232161316131\bar{2} \quad \Delta = -96, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$18^{21}1232161316131\bar{3} \text{ [#}=6] \quad \Delta = -104, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$18^{21}1232161316131\bar{2} \quad \Delta = -96, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$18^{21}1232161316131\bar{3} \text{ [#}=5] \quad \Delta = -104, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$18^{21}1232161316131\bar{3} \text{ [#}=2] \quad \Delta = -104, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$18^{21}1232161316131\bar{2} \quad \Delta = -96, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$18^{21}1232161316131 \quad \Delta = -96, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$18^{21}1232161316131\bar{2} \quad \Delta = -96, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$18^{21}1232161316131\bar{3} \text{ [#}=4] \quad \Delta = -104, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: $18^{21}1232171232161$

$$18^{21}1232171232161\bar{2} \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$18^{21}1232171232161 \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$18^{21}1232171232161\bar{2} \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$18^{21}1232171232161\bar{2} \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$18^{21}\bar{2}1232171232161 \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

Base: $18^{21}1231513221812$

$$18^{21}1231513221812\bar{2} \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$18^{21}\bar{2}1231513221812 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$\begin{array}{ll}
{}^{21}_{18}1231513221812 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
{}^{21}_{18}1231513221812 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
{}^{21}_{18}1231513221812 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
{}^{21}_{18}1231513221812 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\text{Base: } {}^{21}_{18}1231513218122 & \\
{}^{21}_{18}1231513218122 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
{}^{21}_{18}1231513218122 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^{21}_{18}1231513218122 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
{}^{21}_{18}1231513218122 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
{}^{21}_{18}1231513218122 \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
{}^{21}_{18}1231513218122 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
{}^{21}_{18}1231513218122 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\text{Base: } {}^{21}_{18}1232181223151 & \\
{}^{21}_{18}1232181223151 \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^{21}_{18}1232181223151 \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
{}^{21}_{18}1232181223151 \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
{}^{21}_{18}1232181223151 \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
{}^{21}_{18}1232181223151 \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\text{Base: } 4141321812321{}^{21}_{8}1 & \\
4141321812321{}^{21}_{8}1 \quad \Delta = -122, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
4141321812321{}^{21}_{8}1 \quad \Delta = -122, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
4141321812321\overset{21}{8}1 \quad \Delta = -124, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
4141321812321\overset{21}{8}1 \quad \Delta = -122, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
4141321812321\overset{21}{8}1 \quad \Delta = -122, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
4141321812321\overset{21}{8}1 \quad \Delta = -122, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
4141321812321\overset{21}{8}1 \quad \Delta = -122, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{18}1223151321\overset{21}{8}1$

$$\begin{aligned}
1\overset{12}{8}1223151321\overset{21}{8}1 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
181223151321\overset{21}{8}1 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
181223151321\overset{21}{8}1 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
181223151321\overset{21}{8}1 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
181223151321\overset{21}{8}1 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{21}{18}123151321\overset{21}{8}1$

$$\begin{aligned}
\overset{21}{18}123151321\overset{21}{8}1 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\overset{21}{18}123151321\overset{21}{8}1 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\overset{212}{18}123151321\overset{21}{8}1 \quad \Delta = -116, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{221}{18}123218123141$

$$\begin{aligned}
\overset{221}{18}123218123141\overset{3}{3} \text{ [#=3]} \quad \Delta = -102, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
\overset{2212}{18}123218123141 \quad \Delta = -94, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\overset{221}{18}123218123141 \quad \Delta = -94, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\overset{221}{18}123218123141 \quad \Delta = -94, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\overset{221}{18}123218123141 \quad \Delta = -94, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{221}{1\ 8\ 123218123141} \overset{2}{\Delta} = -94, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{221}{1\ 8\ 123218123141} \overset{2}{\Delta} = -96, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{221}{1\ 8\ 123218123141} \Delta = -103, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{221}{1\ 8\ 123218123141} \overset{3}{\Delta} [\# = 2] = -102, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}
\end{aligned}$$

Base: $412321812321 \overset{221}{8\ 1}$

$$\begin{aligned}
& 412321812321 \overset{221}{8\ 1} \overset{2}{\Delta} = -96, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 412321812321 \overset{2}{\Delta} = -96, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 412321812321 \overset{2}{\Delta} = -96, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 412321812321 \overset{2}{\Delta} = -96, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& 412321812321 \overset{221}{8\ 1} \Delta = -96, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 412321812321 \overset{221}{8\ 1} \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
& 412321812321 \overset{3}{\Delta} = -104, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 41 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{3221}{8\ 12321812321}$

$$\begin{aligned}
& \overset{3221}{1\ 8\ 12321812321} \overset{2}{\Delta} = -77, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{3221}{1\ 8\ 12321812321} \overset{3}{\Delta} [\# = 4] = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{3221}{1\ 8\ 12321812321} \overset{3}{\Delta} [\# = 2] = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{3221}{1\ 8\ 12321812321} \Delta = -77, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& 3221 \overset{21}{8\ 12321812321} \Delta = -77, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{3221}{1\ 8\ 12321812321} \overset{2}{\Delta} = -77, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{3221}{1\ 8\ 12321812321} \overset{2}{\Delta} = -77, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{lcl}
\begin{array}{c} 3221 \\ 1 \ 8 \ 12321812321 \end{array} \begin{array}{c} \textcolor{red}{3} \\ \text{[#}=7] \end{array} & \Delta = -85, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
\begin{array}{c} 3221 \\ 1 \ 8 \ 12321812321 \end{array} \begin{array}{c} \textcolor{red}{3} \\ \text{[#}=5] \end{array} & \Delta = -85, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\begin{array}{c} 3221 \\ 1 \ 8 \ 12321812321 \end{array} \begin{array}{c} \textcolor{red}{3} \\ \text{[#}=9] \end{array} & \Delta = -85, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
\begin{array}{c} 3221 \\ 1 \ 8 \ 12321812321 \end{array} \begin{array}{c} \textcolor{red}{3} \\ \text{[#}=3] \end{array} & \Delta = -85, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\begin{array}{c} 3221 \\ 1 \ 8 \ 12321812321 \end{array} \begin{array}{c} \textcolor{red}{3} \\ \text{[#}=6] \end{array} & \Delta = -85, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\begin{array}{c} 3221 \\ 1 \ 8 \ 12321812321 \end{array} \begin{array}{c} \textcolor{red}{3} \\ \text{[#}=8] \end{array} & \Delta = -85, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{array}$$

Base: $\begin{array}{c} 321 \\ 1 \ 8 \ 12321712321 \end{array}$

$$\begin{array}{lcl}
\begin{array}{c} \textcolor{red}{2} \\ 612321712321 \end{array} \begin{array}{c} 321 \\ 8 \ 1 \end{array} & \Delta = -127, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\begin{array}{c} \textcolor{red}{2} \\ 612321712321 \end{array} \begin{array}{c} 321 \\ 8 \ 1 \end{array} & \Delta = -127, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
\begin{array}{c} \textcolor{red}{2} \\ 612321712321 \end{array} \begin{array}{c} 321 \\ 8 \ 1 \end{array} & \Delta = -127, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
\begin{array}{c} 612321712321 \end{array} \begin{array}{c} 321 \\ 8 \ 1 \end{array} & \Delta = -127, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\begin{array}{c} 612321712321 \end{array} \begin{array}{c} 321 \\ 8 \ 1 \end{array} \begin{array}{c} \textcolor{red}{2} \\ 12 \end{array} & \Delta = -127, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\begin{array}{c} 612321712321 \end{array} \begin{array}{c} \textcolor{red}{2} \\ 321 \end{array} \begin{array}{c} 8 \ 1 \end{array} & \Delta = -127, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{array}$$

Base: $\begin{array}{c} 321 \\ 1 \ 8 \ 123151322181 \end{array}$

$$\begin{array}{lcl}
\begin{array}{c} 321 \\ 1 \ 8 \ 123151322181 \end{array} & \Delta = -125, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\begin{array}{c} \textcolor{red}{21} \\ 321 \ 8 \ 123151322181 \end{array} & \Delta = -125, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\begin{array}{c} 321 \\ 1 \ 8 \ 123151322181 \end{array} \begin{array}{c} \textcolor{red}{2} \end{array} & \Delta = -125, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\begin{array}{c} 321 \\ 1 \ 8 \ 123151322181 \end{array} \begin{array}{c} \textcolor{red}{2} \end{array} & \Delta = -125, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
\begin{array}{c} 321 \textcolor{red}{2} \\ 1 \ 8 \ 123151322181 \end{array} & \Delta = -125, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
\begin{array}{c} 321 \\ 1 \ 8 \ 123151322181 \end{array} \begin{array}{c} \textcolor{red}{2} \end{array} & \Delta = -125, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}
\end{array}$$

Base: $\begin{array}{c} 321 \\ 1 \ 8 \ 123151321812 \end{array}$

$$\begin{array}{lll}
\begin{array}{c} 321 \\ 1 \ 8 \ 123151321812 \end{array} & \Delta = -125, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\begin{array}{c} 321 \textcolor{red}{2} \\ 1 \ 8 \ 123151321812 \end{array} & \Delta = -125, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
\begin{array}{c} 321 \\ 1 \ 8 \ 123151321812 \end{array} & \Delta = -125, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
\begin{array}{c} 321 \\ 1 \ 8 \ 123151321812 \end{array} & \Delta = -125, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
\begin{array}{c} 321 \\ 1 \ 8 \ 123151321812 \end{array} & \Delta = -125, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\begin{array}{c} \textcolor{red}{21} \\ 321 \ 8 \ 123151321812 \end{array} & \Delta = -125, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\begin{array}{c} 321 \\ 1 \ 8 \ 123151321812 \end{array} & \Delta = -125, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\textbf{Base: } \begin{array}{c} 321 \\ 1 \ 8 \ 123218122315 \end{array} & & \\
\begin{array}{c} 321 \\ 1 \ 8 \ 123218122315 \end{array} & \Delta = -127, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\begin{array}{c} 321 \\ 1 \ 8 \ 123218122315 \end{array} & \Delta = -127, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 50 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
\begin{array}{c} 321 \\ 1 \ 8 \ 123218122315 \end{array} & \Delta = -127, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\begin{array}{c} \textcolor{red}{21} \\ 321 \ 8 \ 123218122315 \end{array} & \Delta = -127, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\begin{array}{c} 321 \\ 1 \ 8 \ 123218122315 \end{array} & \Delta = -127, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\begin{array}{c} 321 \textcolor{red}{2} \\ 1 \ 8 \ 123218122315 \end{array} & \Delta = -127, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\textbf{Base: } \begin{array}{c} 1 \\ 181231513218123 \end{array} & & \\
\begin{array}{c} \textcolor{red}{12} \\ 181231513218123 \end{array} & \Delta = -125, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\begin{array}{c} 1 \\ 181231513218123 \end{array} & \Delta = -125, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\begin{array}{c} 1 \\ 181231513218123 \end{array} & \Delta = -125, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
\begin{array}{c} 1 \\ 181231513218123 \end{array} & \Delta = -125, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\begin{array}{c} 1 \\ 181231513218123 \end{array} & \Delta = -125, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{array}$$

$$1^{1321} 8^{12321} 1^{123} \quad \Delta = -125, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: $1^{1321} 8^{12321} 1^{12321}$

$$1^{1321} 8^{12321} 1^{12321} \quad \Delta = -49, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1^{1321} 8^{12321} 1^{12321} \quad \Delta = -49, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1^{31321} 8^{12321} 1^{12321} [\# = 2] \quad \Delta = -57, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1^{1321} 8^{12321} 1^{12321} \quad \Delta = -49, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1^{1321} 8^{12321} 1^{12321} 3 [\# = 7] \quad \Delta = -57, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$1^{31321} 8^{12321} 1^{12321} [\# = 3] \quad \Delta = -57, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1^{1321} 8^{12321} 1^{12321} 3 [\# = 3] \quad \Delta = -57, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1^{1321} 8^{12321} 1^{12321} \quad \Delta = -49, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1^{1321} 8^{12321} 1^{12321} 3 [\# = 9] \quad \Delta = -57, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$1^{1321} 8^{12321} 1^{12321} 3 [\# = 8] \quad \Delta = -57, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1^{1321} 8^{12321} 1^{12321} 3 [\# = 6] \quad \Delta = -57, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1^{1321} 8^{12321} 1^{12321} 3 [\# = 5] \quad \Delta = -57, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1^{1321} 8^{12321} 1^{12321} 3 [\# = 2] \quad \Delta = -57, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1^{1321} 8^{12321} 1^{12321} 3 [\# = 4] \quad \Delta = -57, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1^{31321} 8^{12321} 1^{12321} [\# = 5] \quad \Delta = -57, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$1^{31321} 8^{12321} 1^{12321} [\# = 4] \quad \Delta = -57, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

Base: $1^{1223151321} 8^{12321}$

$$1^{31223151321} 8^{12321} [\# = 2] \quad \Delta = -2, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\begin{array}{ll}
1^{1223151321} 8^{12321\mathbf{3}}_{[\#=5]} & \Delta = -2, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1^{1223151321} 8^{12321\mathbf{3}}_{[\#=8]} & \Delta = -2, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1^{1223151321} 8^{12321\mathbf{3}}_{[\#=3]} & \Delta = -2, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{\mathbf{3}1223151321} 8^{12321}_{[\#=4]} & \Delta = -2, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1^{\mathbf{3}1223151321} 8^{12321}_{[\#=9]} & \Delta = -2, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1^{\mathbf{3}1223151321} 8^{12321}_{[\#=5]} & \Delta = -2, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1^{\mathbf{3}1223151321} 8^{12321}_{[\#=7]} & \Delta = -2, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1^{1223151321} 8^{12321\mathbf{3}}_{[\#=7]} & \Delta = -2, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1^{\mathbf{3}1223151321} 8^{12321}_{[\#=8]} & \Delta = -2, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1^{1223151321} 8^{12321\mathbf{3}}_{[\#=9]} & \Delta = -2, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1^{\mathbf{3}1223151321} 8^{12321}_{[\#=6]} & \Delta = -2, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1^{1223151321} 8^{12321\mathbf{3}}_{[\#=6]} & \Delta = -2, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1^{1223151321} 8^{12321\mathbf{3}}_{[\#=2]} & \Delta = -2, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{1223151321} 8^{12321\mathbf{3}}_{[\#=4]} & \Delta = -2, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1^{\mathbf{3}1223151321} 8^{12321}_{[\#=3]} & \Delta = -2, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{array}$$

Base: 3161232181232171

$$\begin{array}{ll}
316123218^{21}1232171 & \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
3161232181^{1232171} & \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
3161232181^{1232171} & \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
316123218^{12}1232171 & \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{array}$$

$$31612321\overset{2}{1}81232171 \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$31612321\overset{2}{8}1\overset{1}{2}32171 \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

Base: $1612321\overset{1}{8}12321712$

$$1612321\overset{2}{8}1\overset{1}{2}321712 \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1612321\overset{21}{8}12321712 \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1612321\overset{21}{8}12321712 \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1612321\overset{12}{8}12321712 \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1612321\overset{1}{8}12321\overset{2}{7}12 \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1612321\overset{1}{8}12321\overset{2}{7}12 \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: $612321\overset{1}{8}123217123$

$$612321\overset{2}{8}1\overset{1}{2}3217123 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$612321\overset{21}{8}123217123 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$612321\overset{1}{8}12321\overset{2}{7}123 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$612321\overset{21}{8}123217123 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$612321\overset{12}{8}123217123 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$612321\overset{1}{8}12321\overset{2}{7}123 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

Base: $1\overset{12321}{8}1232171231$

$$1\overset{312321}{8}1232171231 \text{ [#=4]} \quad \Delta = -57, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$1\overset{12321}{8}1232171231\overset{3}{3} \text{ [#=4]} \quad \Delta = -57, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1\overset{312321}{8}1232171231 \text{ [#=5]} \quad \Delta = -57, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix}$$

$$1\overset{312321}{8}1232171231 \text{ [#=9]} \quad \Delta = -57, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix}$$

$$\begin{array}{ll}
1^{\overline{312321}} \overline{8}^{12321} 1232171231 \text{ }_{[\# = 7]} \Delta = -57, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
1^{\overline{312321}} \overline{8}^{12321} 1232171231 \text{ }_{[\# = 6]} \Delta = -57, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
1^{\overline{12321}} \overline{8}^{12321} 1232171231 \overline{3}^{12321} \text{ }_{[\# = 5]} \Delta = -57, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1^{\overline{312321}} \overline{8}^{12321} 1232171231 \text{ }_{[\# = 8]} \Delta = -57, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
1^{\overline{312321}} \overline{8}^{12321} 1232171231 \text{ }_{[\# = 2]} \Delta = -57, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1^{\overline{12321}} \overline{8}^{12321} 1232171231 \overline{3}^{12321} \text{ }_{[\# = 2]} \Delta = -57, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
1^{\overline{312321}} \overline{8}^{12321} 1232171231 \text{ }_{[\# = 3]} \Delta = -57, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1^{\overline{12321}} \overline{8}^{12321} 1232171231 \overline{3}^{12321} \text{ }_{[\# = 3]} \Delta = -57, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1^{\overline{12321}} \overline{8}^{12321} 1232171231 \overline{2}^{12321} \Delta = -49, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1^{\overline{12321}} \overline{8}^{12321} 1232171231 \Delta = -49, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1^{\overline{12321}} \overline{8}^{12321} 1232171231 \overline{2}^{12321} \Delta = -49, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1^{\overline{12321}} \overline{8}^{12321} 1232171231 \overline{2}^{12321} \Delta = -49, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\text{Base: } 171232181232161 & \\
171232181232161 \overline{1}^{12} \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
171232181232161 \overline{1}^{21} \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
171232181232161 \overline{1}^{12} \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
171232181232161 \overline{1}^{21} \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
171232181232161 \overline{1}^{12} \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\text{Base: } 81223151321 \overline{8}^{2321} 1 & \\
81223151321 \overline{8}^{2321} 1 \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{ll}
81223151321 \overset{2}{8} 1 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
81223151321 \overset{2321}{8} 12 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
81223151321 \overset{2}{8} 1 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
81223151321 \overset{2}{8} 1 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
81223151321 \overset{2}{8} 1 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
\text{Base: } 812321 \overset{223151321}{8} 1 & \\
812321 \overset{223151321}{8} 1 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
812321 \overset{2223151321}{8} 1 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 256 & \begin{pmatrix} 0 & 16 & 2 \\ 16 & 206 & 28 \\ 2 & 28 & 8 \end{pmatrix} \\
812321 \overset{223151321}{8} 1 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
812321 \overset{223151321}{8} 12 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
812321 \overset{223151321}{8} 1 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
812321 \overset{223151321}{8} 1 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 361 & \begin{pmatrix} 0 & 19 & 2 \\ 19 & 283 & 33 \\ 2 & 33 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{23151321}{8} 1232181 & \\
1 \overset{23151321}{8} 1232181 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
1 \overset{23151321}{8} 1232181 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 225 & \begin{pmatrix} 0 & 15 & 2 \\ 15 & 283 & 33 \\ 2 & 33 & 8 \end{pmatrix} \\
23151321 \overset{21}{8} 1232181 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1 \overset{23151321}{8} 1232181 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 206 & 28 \\ 2 & 28 & 8 \end{pmatrix} \\
1 \overset{23151321}{8} 1232181 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1 \overset{23151321}{8} 1232181 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{2321}{8} 12315132181 & \\
2321 \overset{21}{8} 12315132181 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{ll}
1 \overset{2321}{8} 12315132181 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1 \overset{2321}{8} 12315132181 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1 \overset{2321}{8} 12315132181 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
1 \overset{2321}{8} 12315132181 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 219 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
1 \overset{2321}{8} 12315132181 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{3151321}{8} 12321812 & \\
3151321 \overset{21}{8} 12321812 \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \overset{3151321}{8} 12321812 \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \overset{3151321}{8} 12321812 \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
1 \overset{3151321}{8} 12321812 \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1 \overset{3151321}{8} 12321812 \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 163 & 25 \\ 2 & 25 & 8 \end{pmatrix} \\
1 \overset{3151321}{8} 12321812 \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
1 \overset{3151321}{8} 12321812 \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{512321}{8} 123218122 & \\
1 \overset{512321}{8} 123218122 \quad \Delta = -129, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
1 \overset{512321}{8} 123218122 \quad \Delta = -120, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
1 \overset{512321}{8} 123218122 \quad \Delta = -120, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
512321 \overset{21}{8} 123218122 \quad \Delta = -120, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1 \overset{512321}{8} 123218122 \quad \Delta = -120, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1 \overset{512321}{8} 123218122 \quad \Delta = -120, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{l}
\begin{array}{l} 512321 \\ 1 \quad 8 \end{array} \begin{array}{l} \textcolor{red}{2} \\ 123218122 \end{array} \quad \Delta = -120, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\textbf{Base: } \begin{array}{l} 151321 \\ 1 \quad 8 \end{array} 123218122 \\
\begin{array}{l} 151321 \\ 1 \quad 8 \end{array} \begin{array}{l} \textcolor{red}{3} \\ 123218122 \end{array} \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
\begin{array}{l} 151321\textcolor{red}{2} \\ 1 \quad 8 \end{array} 123218122 \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
\begin{array}{l} 151321 \\ 1 \quad 8 \end{array} \begin{array}{l} \textcolor{red}{2} \\ 123218122 \end{array} \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\begin{array}{l} 151321 \\ 1 \quad 8 \end{array} 123218122 \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
\begin{array}{l} 151321 \\ 1 \quad 8 \end{array} 123218122 \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\begin{array}{l} 151321 \\ 1 \quad 8 \end{array} \begin{array}{l} \textcolor{red}{2} \\ 123218122 \end{array} \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
151321\textcolor{red}{21} 123218122 \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\textbf{Base: } \begin{array}{l} 12321 \\ 1 \quad 8 \end{array} 1232181222 \\
\begin{array}{l} 12321 \\ 1 \quad 8 \end{array} 1232181222 \quad \Delta = -68, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\begin{array}{l} 12321 \\ 1 \quad 8 \end{array} \begin{array}{l} \textcolor{red}{2} \\ 1232181222 \end{array} \quad \Delta = -68, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\begin{array}{l} 12321 \\ 1 \quad 8 \end{array} \begin{array}{l} \textcolor{red}{2} \\ 1232181222 \end{array} \quad \Delta = -68, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\begin{array}{l} 12321\textcolor{red}{2} \\ 1 \quad 8 \end{array} 1232181222 \quad \Delta = -68, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\begin{array}{l} \textcolor{red}{3}12321 \\ 1 \quad 8 \end{array} 1232181222 \text{ [#=8]} \quad \Delta = -76, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
\begin{array}{l} \textcolor{red}{3}12321 \\ 1 \quad 8 \end{array} 1232181222 \text{ [#=7]} \quad \Delta = -76, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
\begin{array}{l} \textcolor{red}{3}12321 \\ 1 \quad 8 \end{array} 1232181222 \text{ [#=5]} \quad \Delta = -76, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
\begin{array}{l} \textcolor{red}{3}12321 \\ 1 \quad 8 \end{array} 1232181222 \text{ [#=6]} \quad \Delta = -76, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
\begin{array}{l} \textcolor{red}{3}12321 \\ 1 \quad 8 \end{array} 1232181222 \text{ [#=9]} \quad \Delta = -76, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
\begin{array}{l} \textcolor{red}{3}12321 \\ 1 \quad 8 \end{array} 1232181222 \text{ [#=2]} \quad \Delta = -76, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{l}
\begin{array}{l} \textcolor{red}{3}12321 \\ 1 \ 8 \ 1232181222 \end{array} \begin{array}{l} [\#=3] \\ \Delta = -76, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \end{array} \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\begin{array}{l} \textcolor{red}{3}12321 \\ 1 \ 8 \ 1232181222 \end{array} \begin{array}{l} [\#=4] \\ \Delta = -76, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \end{array} \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
\text{Base: } \begin{array}{l} \textcolor{red}{1}321 \\ 1 \ 8 \ 12321812231 \end{array} \\
\begin{array}{l} \textcolor{red}{1}321\textcolor{red}{2} \\ 1 \ 8 \ 12321812231 \end{array} \begin{array}{l} \Delta = -75, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \end{array} \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\begin{array}{l} \textcolor{red}{1}321\textcolor{red}{2} \\ 1 \ 8 \ 12321812231 \end{array} \begin{array}{l} \Delta = -75, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \end{array} \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\begin{array}{l} \textcolor{red}{1}321 \\ 1 \ 8 \ 12321812231 \end{array} \begin{array}{l} \Delta = -75, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \end{array} \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\begin{array}{l} \textcolor{red}{1}321\textcolor{red}{2} \\ 1 \ 8 \ 12321812231 \end{array} \begin{array}{l} \Delta = -75, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \end{array} \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\begin{array}{l} \textcolor{red}{1}321 \\ 1 \ 8 \ 12321812231\textcolor{red}{3} \end{array} \begin{array}{l} [\#=5] \\ \Delta = -83, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \end{array} \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\begin{array}{l} \textcolor{red}{3}1321 \\ 1 \ 8 \ 12321812231 \end{array} \begin{array}{l} [\#=5] \\ \Delta = -83, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \end{array} \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
\begin{array}{l} \textcolor{red}{1}321 \\ 1 \ 8 \ 12321812231\textcolor{red}{3} \end{array} \begin{array}{l} [\#=4] \\ \Delta = -83, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \end{array} \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\begin{array}{l} \textcolor{red}{3}1321 \\ 1 \ 8 \ 12321812231 \end{array} \begin{array}{l} [\#=4] \\ \Delta = -83, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \end{array} \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
\begin{array}{l} \textcolor{red}{1}321 \\ 1 \ 8 \ 12321812231\textcolor{red}{3} \end{array} \begin{array}{l} [\#=3] \\ \Delta = -83, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \end{array} \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\begin{array}{l} \textcolor{red}{3}1321 \\ 1 \ 8 \ 12321812231 \end{array} \begin{array}{l} [\#=2] \\ \Delta = -83, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \end{array} \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\begin{array}{l} \textcolor{red}{1}321 \\ 1 \ 8 \ 12321812231\textcolor{red}{3} \end{array} \begin{array}{l} [\#=2] \\ \Delta = -83, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \end{array} \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
\begin{array}{l} \textcolor{red}{3}1321 \\ 1 \ 8 \ 12321812231 \end{array} \begin{array}{l} [\#=3] \\ \Delta = -83, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \end{array} \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\text{Base: } \begin{array}{l} \textcolor{red}{5}1321 \\ 1 \ 8 \ 1232181223 \end{array} \\
\begin{array}{l} \textcolor{red}{5}1321\textcolor{red}{2} \\ 1 \ 8 \ 1232181223 \end{array} \begin{array}{l} \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 121 \end{array} \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
\begin{array}{l} \textcolor{red}{5}1321\textcolor{red}{2} \\ 51321 \ 8 \ 1232181223 \end{array} \begin{array}{l} \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \end{array} \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\begin{array}{l} \textcolor{red}{5}1321\textcolor{red}{2} \\ 1 \ 8 \ 1232181223 \end{array} \begin{array}{l} \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \end{array} \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\begin{array}{l} \textcolor{red}{5}1321\textcolor{red}{2} \\ 1 \ 8 \ 1232181223 \end{array} \begin{array}{l} \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \end{array} \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
& \overset{51321}{1 \ 8 \ 1232181223} \quad \Delta = -127, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& \overset{51321}{1 \ 8 \ 1232181223} \quad \Delta = -127, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1321}{1 \ 8 \ 12321812313}$

$$\begin{aligned}
& \overset{1321}{1 \ 8 \ 12321812313} \quad \Delta = -83, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{1321}{1 \ 8 \ 12321812313} \quad \Delta = -83, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{1321}{1 \ 8 \ 12321812313} \quad \Delta = -83, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{1321}{1 \ 8 \ 12321812313} \quad \Delta = -83, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{1321}{1 \ 8 \ 12321812313} \quad \Delta = -83, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{1321}{1 \ 8 \ 12321812313} \quad \Delta = -91, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{31321}{1 \ 8 \ 12321812313} \text{ [#=4]} \quad \Delta = -91, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{31321}{1 \ 8 \ 12321812313} \text{ [#=5]} \quad \Delta = -91, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& \overset{1321}{1 \ 8 \ 12321812313} \quad \Delta = -85, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{31321}{1 \ 8 \ 12321812313} \text{ [#=2]} \quad \Delta = -91, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{1321}{1 \ 8 \ 12321812313} \text{ [#=2]} \quad \Delta = -91, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{31321}{1 \ 8 \ 12321812313} \text{ [#=3]} \quad \Delta = -91, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{4132181232181231}$

$$\begin{aligned}
& \overset{1}{4132181232181231} \text{ [#=4]} \quad \Delta = -111, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{1}{4132181232181231} \quad \Delta = -103, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{21}{4132181232181231} \quad \Delta = -103, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{2}{4132181232181231} \quad \Delta = -103, \ n_+ = 1, \ n_- = 17, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
4132181232181231 \quad \Delta = -103, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
41321812321812313_{[\# = 5]} \quad \Delta = -111, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
4132181232181231 \quad \Delta = -103, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
4132181232181231 \quad \Delta = -103, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
41321812321812313_{[\# = 3]} \quad \Delta = -111, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
41321812321812313_{[\# = 2]} \quad \Delta = -111, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
4132181232181231 \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{1321}{8} 1232181232$

$$\begin{aligned}
1 \overset{31321}{8} 1232181232_{[\# = 5]} \quad \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1 \overset{1321}{8} 1232181232 \quad \Delta = -77, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1 \overset{1321}{8} 1232181232 \quad \Delta = -77, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1 \overset{1321}{8} 1232181232 \quad \Delta = -77, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{1321}{8} 1232181232 \quad \Delta = -77, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1 \overset{1321}{8} 1232181232_{[\# = 2]} \quad \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1 \overset{31321}{8} 1232181232_{[\# = 2]} \quad \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \overset{31321}{8} 1232181232_{[\# = 3]} \quad \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \overset{1321}{8} 1232181232 \quad \Delta = -77, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \overset{1321}{8} 1232181232_{[\# = 3]} \quad \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{1321}{8} 1232181232 \quad \Delta = -105, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
1 \overset{31321}{8} 1232181232_{[\# = 4]} \quad \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{1}812321\overset{1}{8}1231513$

$$\overset{1}{1}812321\overset{1}{8}\overset{2}{1}231513 \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$\overset{1}{1}\overset{2}{8}12321\overset{1}{8}1231513 \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\overset{1}{1}812321\overset{2}{8}\overset{1}{1}231513 \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$\overset{1}{1}812321\overset{2}{8}1231513 \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\overset{1}{1}812321\overset{1}{8}1231\overset{2}{5}13 \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\overset{1}{1}812321\overset{1}{8}1231\overset{2}{5}13 \quad \Delta = -126, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: $\overset{21}{1}812321\overset{1}{8}123215$

$$\overset{21}{1}812321\overset{2}{8}\overset{1}{1}23215 \quad \Delta = -120, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\overset{21}{1}812321\overset{2}{8}\overset{1}{1}23215 \quad \Delta = -120, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\overset{21}{1}812321\overset{1}{8}12321\overset{2}{5} \quad \Delta = -120, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\overset{21}{1}812321\overset{1}{8}123215 \quad \Delta = -120, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\overset{21}{1}\overset{2}{8}12321\overset{1}{8}123215 \quad \Delta = -120, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$\overset{21}{1}812321\overset{1}{8}\overset{2}{1}23215 \quad \Delta = -120, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

Base: $\overset{21}{1}812321\overset{1}{8}123151$

$$\overset{21}{1}812321\overset{2}{8}\overset{1}{1}23151 \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\overset{21}{1}\overset{2}{8}12321\overset{1}{8}123151 \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$\overset{21}{1}812321\overset{2}{8}\overset{1}{1}23151 \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\overset{21}{1}812321\overset{1}{8}123151 \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\overset{21}{1}812321\overset{1}{8}\overset{2}{1}23151 \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$\overset{21}{1}812321\overset{1}{8}1231\overset{2}{5}1 \quad \Delta = -118, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: $1 \overset{221}{8} 12321812321$

$$\begin{aligned}
1 \overset{221}{8} 12321812321\overset{1}{3} \text{ [#=7]} \quad \Delta = -76, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1 \overset{221}{8} 12321812321\overset{1}{3} \text{ [#=2]} \quad \Delta = -76, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \overset{221}{8} 12321812321\overset{1}{3} \text{ [#=9]} \quad \Delta = -76, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1 \overset{221}{8} 12321812321\overset{1}{3} \text{ [#=8]} \quad \Delta = -76, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1 \overset{221}{8} 12321812321\overset{1}{3} \text{ [#=3]} \quad \Delta = -76, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \overset{221}{8} 12321812321\overset{1}{3} \text{ [#=6]} \quad \Delta = -76, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1 \overset{221}{8} 12321812321\overset{1}{3} \text{ [#=4]} \quad \Delta = -76, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1 \overset{221}{8} 12321812321\overset{1}{3} \text{ [#=5]} \quad \Delta = -76, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1 \overset{221}{8} 12321812321\overset{1}{2} \quad \Delta = -68, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1 \overset{221}{8} 12321812321\overset{2}{1} \quad \Delta = -68, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{221}{8} 12321812321\overset{1}{2} \quad \Delta = -77, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1 \overset{221}{8} 12321812321\overset{1}{2} \quad \Delta = -68, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1 \overset{221}{8} 12321812321\overset{1}{2} \quad \Delta = -68, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{321}{8} 12321812315$

$$\begin{aligned}
1 \overset{321}{8} 12321812315\overset{1}{2} \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \overset{321}{8} 12321812315\overset{1}{2} \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1 \overset{321}{8} 12321812315\overset{1}{2} \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 100 \quad & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1 \overset{321}{8} 12321812315\overset{2}{1} \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1 \overset{321}{8} 12321812315\overset{1}{2} \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

$$1 \overset{321}{8} 12321812315 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$321 \overset{21}{8} 12321812315 \quad \Delta = -127, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{1321}{8} 1232181231$$

$$1 \overset{1321}{8} 1232181231 \quad \Delta = -75, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{31321}{8} 1232181231 \text{ [#=4]} \quad \Delta = -83, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1 \overset{1321}{8} 1232181231 \quad \Delta = -75, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1 \overset{31321}{8} 1232181231 \text{ [#=3]} \quad \Delta = -83, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1 \overset{31321}{8} 1232181231 \text{ [#=2]} \quad \Delta = -83, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{31321}{8} 1232181231 \text{ [#=5]} \quad \Delta = -83, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$\text{Base: } 21 \overset{21}{8} 123218123141$$

$$21 \overset{21}{8} 123218123141 \quad \Delta = -96, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$21 \overset{21}{8} 123218123141 \quad \Delta = -94, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$21 \overset{21}{8} 123218123141 \quad \Delta = -94, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$21 \overset{21}{8} 123218123141 \quad \Delta = -94, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$21 \overset{21}{8} 123218123141 \quad \Delta = -94, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$21 \overset{221}{8} 123218123141 \quad \Delta = -94, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$21 \overset{21}{8} 123218123141 \text{ [#=3]} \quad \Delta = -102, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$21 \overset{21}{8} 123218123141 \text{ [#=2]} \quad \Delta = -102, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$\text{Base: } 412321812321 \overset{21}{8} 12$$

$$412321812321 \overset{21}{8} 12 \quad \Delta = -96, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$412321812321 \overset{221}{8} 12 \quad \Delta = -96, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$412321812321\overset{21}{8}12 \quad \Delta = -96, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\overset{2}{4}12321812321\overset{21}{8}12 \quad \Delta = -96, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$412321812321\overset{21}{8}12 \quad \Delta = -96, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$\overset{3}{4}12321812321\overset{21}{8}12 \quad \Delta = -104, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 41 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

Base: $1\overset{12321}{8}\overset{21}{123218}12$

$$1\overset{312321}{8}\overset{21}{123218}12 \text{ [#=2]} \quad \Delta = -76, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1\overset{12321}{8}\overset{21}{123218}12 \quad \Delta = -68, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1\overset{312321}{8}\overset{21}{123218}12 \text{ [#=3]} \quad \Delta = -76, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1\overset{312321}{8}\overset{21}{123218}12 \text{ [#=4]} \quad \Delta = -76, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$1\overset{312321}{8}\overset{21}{123218}12 \text{ [#=8]} \quad \Delta = -76, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix}$$

$$1\overset{312321}{8}\overset{21}{123218}12 \text{ [#=7]} \quad \Delta = -76, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix}$$

$$1\overset{312321}{8}\overset{21}{123218}12 \text{ [#=6]} \quad \Delta = -76, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

$$1\overset{312321}{8}\overset{21}{123218}12 \text{ [#=5]} \quad \Delta = -76, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix}$$

$$1\overset{12321}{8}\overset{21}{123218}12 \quad \Delta = -68, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1\overset{12321}{8}\overset{221}{123218}12 \quad \Delta = -68, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1\overset{312321}{8}\overset{21}{123218}12 \text{ [#=9]} \quad \Delta = -76, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix}$$

$$1\overset{12321}{8}\overset{21}{123218}12 \quad \Delta = -68, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: $21\overset{321}{8}12321812321$

$$21\overset{321}{8}12321812321\overset{3}{8} \text{ [#=2]} \quad \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$21\overset{321}{8}12321812321\overset{3}{8} \text{ [#=9]} \quad \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$\begin{aligned}
21 \overset{321}{8} 12321812321\overset{\color{red}{3}}{[\# = 8]} \quad \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
21 \overset{321}{8} 12321812321\overset{\color{red}{3}}{[\# = 7]} \quad \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
21 \overset{321}{8} 12321812321\overset{\color{red}{3}}{[\# = 6]} \quad \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21 \overset{321}{8} 1232181\overset{\color{red}{2}}{2321} \quad \Delta = -77, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21 \overset{\color{red}{2}321}{8} 12321812321 \quad \Delta = -77, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 24 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21 \overset{321}{8} 1232181\overset{\color{red}{2}}{2321} \quad \Delta = -77, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21 \overset{321\color{red}{2}}{8} 12321812321 \quad \Delta = -77, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
21 \overset{321}{8} 12321812321\overset{\color{red}{3}}{[\# = 5]} \quad \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
21 \overset{321}{8} 12321812321\overset{\color{red}{3}}{[\# = 4]} \quad \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
321 \overset{\color{red}{3}21}{8} 12321812321 \quad \Delta = -86, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21 \overset{321}{8} 12321812321\overset{\color{red}{3}}{[\# = 3]} \quad \Delta = -85, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
21 \overset{321}{8} 12321812321 \quad \Delta = -77, n_+ = 1, n_- = 17, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

$T = 18$ **(2261 bases)**

Base: 2231414141413221121

$$\begin{aligned}
2231414141413221\overset{\color{red}{2}}{1}121 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 \quad & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 130 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
2231414141413221\overset{\color{red}{2}}{1}121 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 \quad & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
2231414141413221\overset{\color{red}{2}}{1}121 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 169 \quad & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
2231414141413221\overset{\color{red}{2}}{1}121 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 225 \quad & \begin{pmatrix} 0 & 15 & 2 \\ 15 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
2231414141413221\overset{\color{red}{2}}{1}121 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
2231414141413221\overset{\color{red}{2}}{1}121 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

Base: 2314141414132211212

$$\begin{aligned}
2314141414132211212 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
2314141414132211212 \quad \Delta = -131, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2314141414132211212 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 108 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
2314141414132211212 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
2314141414132211212 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
2314141414132211212 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2314141414132211212 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
2314141414132211212 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1414132211212231413

$$\begin{aligned}
1414132211212231413 \quad \Delta = -109, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
31413221121223141413 \text{ [#=3]} \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1414132211212231413 \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
31413221121223141413 \text{ [#=2]} \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1414132211212231413 \quad \Delta = -109, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1414132211212231413 \quad \Delta = -109, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1414132211212231413 \quad \Delta = -109, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1414132211212231413 \quad \Delta = -109, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1414132211212231413 \quad \Delta = -109, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: 3141413221121223141

$$31414132211212231413 \text{ [#=3]} \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$\begin{aligned}
3141413221121223141^3 \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 21 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
3141413221121223141^2 \quad \Delta = -109, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
3141413221121223141^2 \quad \Delta = -109, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
3141413221121223141^2 \quad \Delta = -109, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
3141413221121223141^2 \quad \Delta = -109, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 103 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
3141413221121223141^2 \quad \Delta = -109, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 110 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
3141413221121223141^3_{[\# = 2]} \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
3141413221121223141^2 \quad \Delta = -109, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 58 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1414141322112122313

$$\begin{aligned}
1414141322112122313^3_{[\# = 2]} \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1414141322112122313^2 \quad \Delta = -109, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
3132211212231414141^3_{[\# = 3]} \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
3132211212231414141^3_{[\# = 2]} \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1414141322112122313^2 \quad \Delta = -109, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1414141322112122313^3 \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1414141322112122313^2 \quad \Delta = -109, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1414141322112122313^2 \quad \Delta = -109, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1414141322112122313^2 \quad \Delta = -109, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1414141322112122313^2 \quad \Delta = -109, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: 3141414132211212231

$$\begin{aligned}
3141414132211212231^2 \quad \Delta = -109, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 70 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
3141414132211212231^2 \quad \Delta = -109, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
3141414132211212231^3_{[\#=4]} \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
3141414132211212231^2 \quad \Delta = -109, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
3141414132211212231^3_{[\#=5]} \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
3141414132211212231^3_{[\#=3]} \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
3141414132211212231^3 \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
3141414132211212231^2 \quad \Delta = -109, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
3141414132211212231^3_{[\#=2]} \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
3141414132211212231^2 \quad \Delta = -109, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
3141414132211212231^2 \quad \Delta = -109, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\text{Base: } 1414141413221121222 & \\
1414141413221121222^2 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1414141413221121222^2 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1414141413221121222^2 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1414141413221121222^2 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 70 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
1414141413221121222^2 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1414141413221121222^2 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
2221121223141414141^3_{[\#=2]} \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2221121223141414141^3_{[\#=3]} \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 3141414141322112122

$$\begin{aligned}
3141414141322112122 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
3141414141322112122 \quad \Delta = -136, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
3141414141322112122 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
3141414141322112122 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
3141414141322112122 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
3141414141322112122 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
3141414141322112122 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
3141414141322112122 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 88 & 18 \\ 2 & 18 & 8 \end{pmatrix}
\end{aligned}$$

Base: 2314132211212231512

$$\begin{aligned}
2314132211212231512 \quad \Delta = -108, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2314132211212231512 \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
2314132211212231512 \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2314132211212231512 \quad \Delta = -114, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
2314132211212231512 \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
2314132211212231512 \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2314132211212231512 \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
2314132211212231512 \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: 3141322112122315123

$$\begin{aligned}
3141322112122315123 \quad \Delta = -122, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
3141322112122315123 \quad \Delta = -114, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{ll}
3141322112122315123^2 & \Delta = -114, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
3141322112122315123^2 & \Delta = -114, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
3141322112122315123^2 & \Delta = -114, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
3141322112122315123^2 & \Delta = -114, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
3141322112122315123^2 & \Delta = -114, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\text{Base: } 1413221121223151232 & \\
1413221121223151232^2 & \Delta = -108, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1413221121223151232^2 & \Delta = -108, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1413221121223151232^2 & \Delta = -108, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1413221121223151232^2 & \Delta = -108, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
23215132211212231413^{\#2} & \Delta = -116, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
23215132211212231413^{\#3} & \Delta = -116, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1413221121223151232^2 & \Delta = -108, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\text{Base: } 2231322112122315131 & \\
22313221121223151313^{\#7} & \Delta = -94, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
22313221121223151313^{\#6} & \Delta = -94, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
22313221121223151313^{\#8} & \Delta = -94, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
22313221121223151313^{\#9} & \Delta = -94, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
22313221121223151313^{\#2} & \Delta = -94, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
22313221121223151313^{\#4} & \Delta = -94, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
2231322112122315131^{\color{red}3} \text{ [#=3]} \quad \Delta = -94, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2231322112122315131^{\color{red}3} \quad \Delta = -94, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 60 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
2231322112122315131^{\color{red}3} \text{ [#=5]} \quad \Delta = -94, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
2231322112122315131^{\color{red}2} \quad \Delta = -86, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
2231322112122315131^{\color{red}2} \quad \Delta = -86, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2231322112122315131^{\color{red}2} \quad \Delta = -86, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2231322112122315131^{\color{red}2} \quad \Delta = -86, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2231322112122315131^{\color{red}2} \quad \Delta = -86, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\text{Base: } 4131513221121223132 & \\
4131513221121223132^{\color{red}2} \quad \Delta = -114, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
4131513221121223132^{\color{red}3} \quad \Delta = -122, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
4131513221121223132^{\color{red}3} \quad \Delta = -122, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
4131513221121223132^{\color{red}2} \quad \Delta = -116, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
4131513221121223132^{\color{red}2} \quad \Delta = -115, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
4131513221121223132^{\color{red}2} \quad \Delta = -114, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
4131513221121223132^{\color{red}2} \quad \Delta = -114, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
4131513221121223132^{\color{red}2} \quad \Delta = -114, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
4131513221121223132^{\color{red}2} \quad \Delta = -114, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
4131513221121223132^{\color{red}2} \quad \Delta = -114, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\text{Base: } 2222221121223151321 &
\end{aligned}$$

$$\begin{aligned}
2222221121223151321^2 \quad \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2222221121223151321^2 \quad \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2222221121223151321^3_{\#=[4]} \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
2222221121223151321^3_{\#=[5]} \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
2222221121223151321^3_{\#=[7]} \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
2222221121223151321^3_{\#=[6]} \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
2222221121223151321^3_{\#=[3]} \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2222221121223151321^3_{\#=[8]} \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
2222221121223151321^3_{\#=[9]} \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
2222221121223151321^3_{\#=[2]} \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2222221121223151321^2 \quad \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
2222221121223151321^2 \quad \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\text{Base: } 2222211212231513215 \\
2222211212231513215^2 \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2222211212231513215^2 \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2222211212231513215^2 \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
2222211212231513215^2 \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
2222211212231513215^2 \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\text{Base: } 2222211212231513151 \\
2222211212231513151^2 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
2222211212231513151 & \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2222211212231513151 & \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
2222211212231513151 & \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2222211212231513151 & \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
2222211212231513151 & \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: 2222112122315131513

$$\begin{aligned}
2222112122315131513 & \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
2222112122315131513 & \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
2222112122315131513 & \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
2222112122315131513 & \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
2222112122315131513 & \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
2222112122315131513 & \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2222112122315131513 & \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 2221121223151315132

$$\begin{aligned}
2221121223151315132 & \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
2221121223151315132 & \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
2221121223151315132 & \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
2221121223151315132 & \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
2221121223151315132 & \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
2221121223151315132 & \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

$$2221121223151\overset{2}{3}15132 \quad \Delta = -130, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: 2211212231513151322

$$2211212231513151322 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$2211212231513151322 \quad \Delta = -130, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$2211212231513151322 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 196 \quad \begin{pmatrix} 0 & 14 & 2 \\ 14 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix}$$

$$2211212231513151322 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix}$$

$$2211212231513151322 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$221\overset{2}{1}212231513151322 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$2211212231513151322 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 70 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

Base: 231413221121223151¹

$$23141\overset{2}{3}221121223151^1 \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$231\overset{2}{4}13221121223151^1 \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$231413221121223151^1 \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$231413221121223151^{21} \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$231413221\overset{2}{1}21223151^1 \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$231\overset{2}{4}13221121223151^1 \quad \Delta = -108, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 314132211212231513¹

$$3141\overset{3}{3}221121223151^1 \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$3141\overset{2}{3}221121223151^1 \quad \Delta = -113, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$31413221\overset{2}{1}21223151^1 \quad \Delta = -113, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$31413221121223151^1 \quad \Delta = -113, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\begin{aligned}
314132211212231\overset{1}{5}\overset{2}{1}3 \quad \Delta = -113, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
314132211212231\overset{2}{5}\overset{1}{1}3 \quad \Delta = -113, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
314132211212231\overset{2}{5}\overset{1}{1}3 \quad \Delta = -113, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
314132211212231\overset{2}{5}\overset{1}{1}3 \quad \Delta = -113, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
314132211212231\overset{3}{5}\overset{1}{1}3 \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: 14132211212231¹5132

$$\begin{aligned}
14132211212231\overset{2}{5}\overset{1}{1}32 \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
14132211212231\overset{2}{5}\overset{1}{1}32 \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
14132211212231\overset{1}{5}\overset{2}{1}32 \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
14132211212231\overset{2}{5}\overset{1}{1}32 \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
14132211212231\overset{3}{5}\overset{1}{1}32 \quad \Delta = -114, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
14132211212231\overset{2}{5}\overset{1}{1}32 \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
14132211212231\overset{2}{5}\overset{1}{1}32 \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
231\overset{3}{5}\overset{1}{1}4132211212231 \quad \Delta = -114, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
231\overset{3}{5}\overset{1}{1}4132211212231 \quad \Delta = -114, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: 112122314141414132

$$\begin{aligned}
1121223141414141\overset{2}{1}32 \quad \Delta = -131, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1121223141414141\overset{2}{1}32 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 \quad & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1121223141414141\overset{2}{1}32 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1121223141414141\overset{1}{2}\overset{2}{1}32 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 100 \quad & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 88 & 18 \\ 2 & 18 & 8 \end{pmatrix}
\end{aligned}$$

$$112122314141414132 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$112122314141414132 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$112122314141414132 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 11212231414141413

$$11212231414141413 \quad \Delta = -136, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$11212231414141413 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$11212231414141413 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$11212231414141413 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$11212231414141413 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$11212231414141413 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$11212231414141413 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$11212231414141413 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 70 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

Base: $1^{22315131513221}_{12} 12$

$$1^{22315131513221}_{12} 12 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix}$$

$$1^{22315131513221}_{12} 12 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 196 \quad \begin{pmatrix} 0 & 14 & 2 \\ 14 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix}$$

$$1^{22315131513221}_{12} 12 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$1^{22315131513221}_{12} 12 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 70 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$1^{22315131513221}_{12} 12 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1^{22315131513221}_{12} 12 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1^{22315131513221}_{12} 12 \quad \Delta = -130, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

Base: 414132211212231414

$$\begin{array}{ll}
414132211212231414 & \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
414132211212231414 & \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
414132211212231414 & \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
414132211212231414 & \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\text{Base: } 141322112122314141 & \\
141322112122314141 & \Delta = -101, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
141322112122314141 & \Delta = -101, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
141322112122314141 & \Delta = -101, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
141322112122314141\textcolor{red}{3} \text{ [\#}=3] & \Delta = -109, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
141413221121223141\textcolor{red}{3} \text{ [\#}=2] & \Delta = -109, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
141322112122314141\textcolor{red}{3} \text{ [\#}=2] & \Delta = -109, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
141413221121223141\textcolor{red}{3} \text{ [\#}=3] & \Delta = -109, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
141322112122314141 & \Delta = -101, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
141322112122314141 & \Delta = -101, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\text{Base: } 413221121223141414 & \\
413221121223141414 & \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
413221121223141414 & \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
413221121223141414 & \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 76 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
413221121223141414 & \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 58 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
413221121223141414 & \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
& 4132211^1 21223141414^2 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 4132211^2 21223141414^1 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1^{13221} 12^{12231414141}$

$$\begin{aligned}
& 1^{13221} 12^{12231414141} \quad \Delta = -101, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 89 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& 1^{13221} 12^{12231414141} \quad \Delta = -101, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
& 1^{13221} 12^{12231414141} \quad \Delta = -101, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 1^{13221} 12^{12231414141} \quad \Delta = -101, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 1^{313221} 12^{12231414141} [\# = 5] \quad \Delta = -109, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& 1^{313221} 12^{12231414141} [\# = 4] \quad \Delta = -109, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& 1^{13221} 12^{12231414141} 3 [\# = 3] \quad \Delta = -109, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& 1^{313221} 12^{12231414141} [\# = 3] \quad \Delta = -109, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 1^{13221} 12^{12231414141} \quad \Delta = -101, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 1^{13221} 12^{12231414141} 3 [\# = 2] \quad \Delta = -109, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& 1^{313221} 12^{12231414141} [\# = 2] \quad \Delta = -109, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: $414141413221^{3221} 12^1$

$$\begin{aligned}
& 414141413221^{3221} 12^1 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 414141413221^{3221} 12^1 2 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 414141413221^{3221} 12^1 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
& 414141413221^{3221} 12^1 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 225 \quad \begin{pmatrix} 0 & 15 & 2 \\ 15 & 118 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
& 414141413221^{3221} 12^1 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

$$414141413221 \overset{2}{12} 1 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 256 \quad \begin{pmatrix} 0 & 16 & 2 \\ 16 & 142 & 24 \\ 2 & 24 & 8 \end{pmatrix}$$

$$414141413221 \overset{2}{12} 1 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: $1 \overset{221}{12} 1223141414141$

$$1 \overset{221}{12} 1223141414141 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 142 & 24 \\ 2 & 24 & 8 \end{pmatrix}$$

$$1 \overset{221}{12} 1223141414141 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

$$1 \overset{221}{12} 1223141414141 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1 \overset{221}{12} 1223141414141 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1 \overset{221}{12} 1223141414141 \overset{3}{\# = 3} \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$1 \overset{221}{12} 1223141414141 \overset{3}{\# = 2} \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$1 \overset{221}{12} 1223141414141 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 118 & 22 \\ 2 & 22 & 8 \end{pmatrix}$$

$$1 \overset{221}{12} 1223141414141 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: $1 \overset{21513221}{12} 12231413$

$$1 \overset{21513221}{12} 12231413 \quad \Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{21513221}{12} 12231413 \quad \Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{21513221}{12} 12231413 \quad \Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{21513221}{12} 12231413 \quad \Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{21513221}{12} 12231413 \quad \Delta = -113, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1 \overset{21513221}{12} 12231413 \quad \Delta = -113, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \overset{21513221}{12} 12231413 \quad \Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{21513221}{12} 12231413 \quad \Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: $1 \overset{321513221}{12} 1223141$

$$\begin{array}{ll}
1^{321513221} 12^{1223141} \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 76 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
1^{321513221} 12^{1223141} \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 58 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
1^{321513221} 12^{1223141} \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 63 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1^{321513221} 12^{1223141} \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1^{321513221} 12^{1223141} \textcolor{red}{3} [\#=3] \quad \Delta = -114, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
1^{321513221} 12^{1223141} \textcolor{red}{3} [\#=2] \quad \Delta = -114, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
1^{321513221} 12^{1223141} \textcolor{red}{2} \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
\textbf{Base: } 1^{131513221} 12^{1223132} & \\
1^{131513221} 12^{1223132} \textcolor{red}{2} \quad \Delta = -86, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{131513221} 12^{1223132} \quad \Delta = -86, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1^{131513221} 12^{1223132} \textcolor{red}{2} \quad \Delta = -86, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1^{131513221} 12^{1223132} \quad \Delta = -86, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1^{131513221} 12^{1223132} \quad \Delta = -86, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1^{3131513221} 12^{1223132} [\#=5] \quad \Delta = -94, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1^{3131513221} 12^{1223132} [\#=4] \quad \Delta = -94, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{3131513221} 12^{1223132} [\#=9] \quad \Delta = -94, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1^{131513221} 12^{1223132} \textcolor{red}{3} \quad \Delta = -94, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 32 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1^{3131513221} 12^{1223132} [\#=2] \quad \Delta = -94, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1^{3131513221} 12^{1223132} [\#=6] \quad \Delta = -94, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{131513221} 12^{1223132} \textcolor{red}{2} \quad \Delta = -88, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
& \overset{3131513221}{1} \quad 12 \quad 1223132 \quad [\#=3] \quad \Delta = -94, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{3131513221}{1} \quad 12 \quad 1223132 \quad [\#=7] \quad \Delta = -94, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& \overset{3131513221}{1} \quad 12 \quad 1223132 \quad [\#=8] \quad \Delta = -94, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \text{Base: } 4131513221 \quad \overset{313221}{12} \quad 1 \\
& \overset{2}{4131513221} \quad \overset{313221}{12} \quad 1 \quad \Delta = -114, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 4131513221 \quad \overset{313221}{12} \quad 1 \quad [\#=2] \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 4131513221 \quad \overset{313221}{12} \quad 1 \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& 4131513221 \quad \overset{2313221}{12} \quad 1 \quad \Delta = -113, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 4131513221 \quad \overset{2}{12} \quad \overset{313221}{1} \quad \Delta = -113, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 4131513221 \quad \overset{313221}{12} \quad 1 \quad \Delta = -113, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 4131513221 \quad \overset{313221}{12} \quad 1 \quad \Delta = -113, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{2}{4131513221} \quad \overset{313221}{12} \quad 1 \quad \Delta = -113, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 4131513221 \quad \overset{2}{12} \quad \overset{313221}{1} \quad \Delta = -113, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{3}{4131513221} \quad \overset{313221}{12} \quad 1 \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \text{Base: } 1 \quad \overset{1231513221}{12} \quad 122222 \\
& \overset{31231513221}{1} \quad 12 \quad 122222 \quad [\#=3] \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{31231513221}{1} \quad 12 \quad 122222 \quad [\#=4] \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{1231513221}{1} \quad 12 \quad 122222 \quad \overset{2}{122222} \quad \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{1231513221}{1} \quad 12 \quad 122222 \quad \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{1231513221}{1} \quad 12 \quad 122222 \quad \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{1231513221}{1} \quad 12 \quad 12222 \quad \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{\textcolor{red}{3}1231513221}{1} \quad 12 \quad 12222 \quad [\#=5] \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{\textcolor{red}{3}1231513221}{1} \quad 12 \quad 12222 \quad [\#=9] \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
& \overset{\textcolor{red}{3}1231513221}{1} \quad 12 \quad 12222 \quad [\#=8] \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& \overset{\textcolor{red}{3}1231513221}{1} \quad 12 \quad 12222 \quad [\#=7] \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
& \overset{\textcolor{red}{3}1231513221}{1} \quad 12 \quad 12222 \quad [\#=6] \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{\textcolor{red}{3}1231513221}{1} \quad 12 \quad 12222 \quad [\#=2] \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{51231513221}{1} \quad 12 \quad 12222$

$$\begin{aligned}
& \overset{51231513221}{1} \quad 12 \quad 12222 \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{51231513221}{1} \quad 12 \quad 12222 \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{51231513221}{1} \quad 12 \quad 12222 \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{51231513221}{1} \quad 12 \quad 12222 \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{51231513221}{1} \quad 12 \quad 12222 \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{15131513221}{1} \quad 12 \quad 12222$

$$\begin{aligned}
& \overset{15131513221}{1} \quad 12 \quad 12222 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{15131513221}{1} \quad 12 \quad 12222 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{15131513221}{1} \quad 12 \quad 12222 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{15131513221}{1} \quad 12 \quad 12222 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{15131513221}{1} \quad 12 \quad 12222 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{15131513221}{1} \quad 12 \quad 12222 \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{315131513221}{1} \quad 12 \quad 12222$

$1 \overset{315131513221}{12} \overset{2}{1222}$	$\Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$
$1 \overset{315131513221}{12} 1222$	$\Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, \det = 64$	$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
$1 \overset{315131513221}{12} 1222$	$\Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$
$1 \overset{315131513221}{12} 1222$	$\Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$
$1 \overset{315131513221}{12} 1222$	$\Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$
$1 \overset{315131513221}{12} 1222$	$\Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
$1 \overset{315131513221}{12} 1222$	$\Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$
Base: $1 \overset{2315131513221}{12} 122$		
$1 \overset{2315131513221}{12} 122$	$\Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, \det = 121$	$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$
$1 \overset{2315131513221}{12} 122$	$\Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, \det = 100$	$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$
$1 \overset{2315131513221}{12} \overset{2}{122}$	$\Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$
$1 \overset{2315131513221}{12} 122$	$\Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, \det = 81$	$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$
$1 \overset{2315131513221}{12} 122$	$\Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, \det = 64$	$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$
$1 \overset{2315131513221}{12} 122$	$\Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$
$1 \overset{2315131513221}{12} 122$	$\Delta = -130, n_+ = 1, n_- = 18, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$
Base: $1 \overset{1513221}{12} 12231413$		
$1 \overset{1513221}{12} 12231413$	$\Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$
$1 \overset{1513221}{12} 12231413$	$\Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$
$1 \overset{1513221}{12} \overset{3}{12231413}$	$\Delta = -113, n_+ = 1, n_- = 18, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$
$1 \overset{1513221}{12} \overset{2}{12231413}$	$\Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$

$$1^{1513221\bar{2}} 12^{12231413} \quad \Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1^{1513221} 12^{12231413\bar{2}} \quad \Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\text{Base: } 1^{31513221} 12^{1223141}$$

$$1^{31513221} 12^{1223141} \quad \Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1^{31513221} 12^{1223141} \quad \Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 58 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$1^{31513221} 12^{1223141} \quad \Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 76 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

$$1^{31513221} 12^{1223141\bar{2}} \quad \Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1^{31513221\bar{2}} 12^{1223141} \quad \Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$1^{31513221} 12^{1223141\bar{3} \text{ [#}=3]} \quad \Delta = -113, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$1^{31513221} 12^{1223141\bar{3} \text{ [#}=2]} \quad \Delta = -113, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$1^{31513221} 12^{1223141} \quad \Delta = -113, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1^{31513221} 12^{1223141} \quad \Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\text{Base: } 21^{21} 1223141414141$$

$$21^{2\bar{2}1} 1223141414141 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$21^{21\bar{2}} 1223141414141 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 118 & 22 \\ 2 & 22 & 8 \end{pmatrix}$$

$$21^{21} 1223141414141\bar{2} \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 142 & 24 \\ 2 & 24 & 8 \end{pmatrix}$$

$$21^{21} 1223141414141\bar{2} \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

$$21^{21} 1223141414141\bar{2} \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$21^{21} 1223141414141\bar{2} \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$21^{21} 1223141414141\bar{3} \text{ [#}=2] \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$21^{21}1121223141414141\textcolor{red}{3} [\#=3] \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$\textbf{Base: } 21^{2315131513221}_{12} 12$$

$$21^{2315131513221}_{12} 12 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

$$21^{2315131513221}_{12} 12 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$21^{\textcolor{red}{2}315131513221}_{12} 12 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$21^{2315131513221}_{12} 12 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$21^{2315131513221}_{12} 12 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$21^{2315131513221}_{12} 12 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$21^{2315131513221}_{12} 12 \quad \Delta = -130, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$\textbf{Base: } 41413221^{1513221}_{12} 12$$

$$41413221^{\textcolor{red}{2}}_{1513221} 12 12 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$41413221^{\textcolor{red}{2}}_{1513221} 12 12 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$41413221^{\textcolor{red}{2}}_{1513221} 12 12 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$41413221^{1513221\textcolor{red}{2}}_{12} 12 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$41413221^{1513221}_{12} 12 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$41413221^{1513221}_{12} 12 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$\textbf{Base: } 21^{21513221}_{12} 1223141$$

$$21^{21513221}_{12} 1223141 \quad \Delta = -97, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$21^{21513221\textcolor{red}{2}}_{12} 1223141 \quad \Delta = -97, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$21^{21513221}_{12} 1223141 \quad \Delta = -97, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$21^{21513221}_{12} 1223141 \quad \Delta = -97, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$\begin{aligned}
21^{21513221} 12^{1223141} \quad \Delta = -97, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21^{21513221} 12^{1223141} \quad \Delta = -97, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21^{21513221} 12^{1223141} \text{ } \textcolor{red}{3} \text{ } [\# = 2] \quad \Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
21^{21513221} 12^{1223141} \quad \Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21^{21513221} 12^{1223141} \text{ } \textcolor{red}{3} \text{ } [\# = 3] \quad \Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
\textbf{Base: } 21^{131513221} 12^{122313} \\
21^{3131513221} 12^{122313} [\# = 2] \quad \Delta = -93, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21^{131513221} 12^{122313} \quad \Delta = -85, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
21^{131513221} 12^{122313} \text{ } \textcolor{red}{3} \text{ } [\# = 2] \quad \Delta = -93, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
21^{3131513221} 12^{122313} [\# = 7] \quad \Delta = -93, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
21^{3131513221} 12^{122313} [\# = 5] \quad \Delta = -93, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
21^{131513221} 12^{122313} \text{ } \textcolor{red}{3} \quad \Delta = -93, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21^{3131513221} 12^{122313} [\# = 9] \quad \Delta = -93, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
21^{3131513221} 12^{122313} [\# = 3] \quad \Delta = -93, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
21^{3131513221} 12^{122313} [\# = 4] \quad \Delta = -93, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21^{3131513221} 12^{122313} [\# = 6] \quad \Delta = -93, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21^{3131513221} 12^{122313} [\# = 8] \quad \Delta = -93, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
21^{131513221} 12^{122313} \quad \Delta = -85, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21^{131513221} 12^{122313} \quad \Delta = -85, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21^{2131513221} 12^{122313} \quad \Delta = -85, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

$$21 \overset{131513221}{12} \overset{2}{122313} \quad \Delta = -85, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$21 \overset{131513221}{12} \overset{2}{122313} \quad \Delta = -85, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: $4131513221 \overset{13221}{12} 12$

$$4131513221 \overset{13221}{12} 12 \quad \Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$4131513221 \overset{2}{12} \overset{13221}{12} 12 \quad \Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$4131513221 \overset{2}{12} \overset{13221}{12} 12 \quad \Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$4131513221 \overset{2}{12} \overset{13221}{12} 12 \quad \Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$4131513221 \overset{13221}{12} \overset{2}{12} 12 \quad \Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$4131513221 \overset{2}{12} \overset{13221}{12} 12 \quad \Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$4131513221 \overset{2}{12} \overset{13221}{12} 12 \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$4131513221 \overset{3}{12} \overset{13221}{12} 12 \text{ [#=3]} \quad \Delta = -113, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$4131513221 \overset{3}{12} \overset{13221}{12} 12 \text{ [#=4]} \quad \Delta = -113, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$4131513221 \overset{3}{12} \overset{13221}{12} 12 \text{ [#=2]} \quad \Delta = -113, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$4131513221 \overset{3}{12} \overset{13221}{12} 12 \quad \Delta = -113, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$4131513221 \overset{3}{12} \overset{13221}{12} 12 \text{ [#=5]} \quad \Delta = -113, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: $21 \overset{1231513221}{12} 12222$

$$21 \overset{3}{12} \overset{1231513221}{12222} \text{ [#=3]} \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$21 \overset{3}{12} \overset{1231513221}{12222} \text{ [#=5]} \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$21 \overset{3}{12} \overset{1231513221}{12222} \text{ [#=4]} \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$21 \overset{3}{12} \overset{1231513221}{12222} \text{ [#=6]} \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$\begin{array}{ll}
21^{\overline{31231513221}} 12^{12222} [\#=7] & \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
21^{\overline{31231513221}} 12^{12222} [\#=8] & \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
21^{\overline{31231513221}} 12^{12222} [\#=2] & \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
21^{\overline{31231513221}} 12^{12222} [\#=9] & \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
21^{\overline{1231513221}} 12^{12222} & \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21^{\overline{1231513221}} 12^{12222} & \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21^{\overline{21231513221}} 12^{12222} & \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21^{\overline{12315132212}} 12^{12222} & \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21^{\overline{1231513221}} 12^{12222} & \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\text{Base: } 21^{\overline{51231513221}} 12^{1222} & \\
21^{\overline{251231513221}} 12^{1222} & \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21^{\overline{512315132212}} 12^{1222} & \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21^{\overline{51231513221}} 12^{1222} & \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
21^{\overline{51231513221}} 12^{1222} & \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21^{\overline{51231513221}} 12^{1222} & \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21^{\overline{51231513221}} 12^{1222} & \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\text{Base: } 21^{\overline{15131513221}} 12^{1222} & \\
21^{\overline{15131513221}} 12^{1222} & \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21^{\overline{151315132212}} 12^{1222} & \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21^{\overline{215131513221}} 12^{1222} & \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
21^{15131513221}_{12} 1222 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21^{15131513221}_{12} 1222 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21^{15131513221}_{12} 1222 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21^{15131513221}_{12} 1222 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: $21^{315131513221}_{12} 122$

$$\begin{aligned}
21^{2315131513221}_{12} 122 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21^{3151315132212}_{12} 122 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21^{315131513221}_{12} 122 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21^{315131513221}_{12} 122 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21^{315131513221}_{12} 122 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
21^{315131513221}_{12} 122 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21^{315131513221}_{12} 122 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21^{315131513221}_{12} 122 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: $21^{1513221}_{12} 1223141$

$$\begin{aligned}
21^{1513221}_{12} 1223141\textcolor{red}{3} [\# = 3] \quad \Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
21^{1513221}_{12} 1223141\textcolor{red}{3} [\# = 2] \quad \Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
21^{1513221}_{12} 1223141 \quad \Delta = -97, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21^{15132212}_{12} 1223141 \quad \Delta = -97, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
21^{1513221}_{12} 1223141 \quad \Delta = -97, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
21^{21513221}_{12} 1223141 \quad \Delta = -97, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$21 \overset{1513221}{12} \overset{2}{1223141} \quad \Delta = -97, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: $221 \overset{1513221}{12} 1223151$

$$1513221 \overset{1513221}{12} 122\overset{3}{3} \quad \Delta = -130, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$221 \overset{1513221}{12} 1223\overset{2}{151} \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$221 \overset{1513221}{12} 12231\overset{2}{51} \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$\overset{2}{221} \overset{1513221}{12} 1223151 \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

Base: $221 \overset{31513221}{12} 122315$

$$\overset{2}{221} \overset{31513221}{12} 122315 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 55 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$221 \overset{31513221}{12} 1223\overset{2}{15} \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$221 \overset{31513221}{12} 1223\overset{2}{15} \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix}$$

$$221 \overset{31513221}{12} 122315 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$221 \overset{31513221}{12} 122315 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 196 \quad \begin{pmatrix} 0 & 14 & 2 \\ 14 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix}$$

$$221 \overset{31513221}{12} 122315 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$513221 \overset{31513221}{12} 122\overset{3}{3} \quad \Delta = -138, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: $3221 \overset{131513221}{12} 1223$

$$3221 \overset{5131513221}{12} 1223 \quad \Delta = -138, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$3221 \overset{3131513221}{12} 1223 \text{ [#=3]} \quad \Delta = -94, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$3221 \overset{3131513221}{12} 1223 \text{ [#=8]} \quad \Delta = -94, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 256 \quad \begin{pmatrix} 0 & 16 & 2 \\ 16 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix}$$

$$3221 \overset{3131513221}{12} 1223 \text{ [#=9]} \quad \Delta = -94, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 324 \quad \begin{pmatrix} 0 & 18 & 2 \\ 18 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix}$$

$$3221 \overset{3131513221}{12} 1223 \text{ [#=7]} \quad \Delta = -94, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 196 \quad \begin{pmatrix} 0 & 14 & 2 \\ 14 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix}$$

$$3221 \overset{3131513221}{12} 1223 \text{ [#=6]} \quad \Delta = -94, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

$$\begin{aligned}
3221 \overset{3}{1}31513221 \quad 12 \quad 1223 \text{ }_{[\# = 2]} \quad \Delta = -94, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
3221 \overset{6}{1}31513221 \quad 12 \quad 1223 \quad \Delta = -164, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
3221 \overset{4}{1}31513221 \quad 12 \quad 1223 \quad \Delta = -114, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
3221 \overset{1}{1}31513221 \quad 12 \quad 1223 \quad \Delta = -87, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
3221 \overset{1}{1}31513221 \quad 12 \quad 1223 \quad \Delta = -86, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
3221 \overset{2}{1}31513221 \quad 12 \quad 1223 \quad \Delta = -86, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
3221 \overset{3}{1}31513221 \quad 12 \quad 1223 \text{ }_{[\# = 5]} \quad \Delta = -94, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 100 \quad & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
3221 \overset{3}{1}31513221 \quad 12 \quad 1223 \text{ }_{[\# = 4]} \quad \Delta = -94, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
3221 \overset{1}{1}31513221 \quad 12 \quad 1223 \quad \Delta = -86, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 100 \quad & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
3221 \overset{1}{1}31513221 \quad 12 \quad 1223 \quad \Delta = -86, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
3221 \overset{2}{1}31513221 \quad 12 \quad 1223 \quad \Delta = -86, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
3221 \overset{3}{1}31513221 \quad 12 \quad 1223 \quad \Delta = -94, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\text{Base: } 221 \overset{1}{1}31513221 \quad 12 \quad 12231 & \\
221 \overset{1}{1}31513221 \quad 12 \quad 12231 \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
221 \overset{1}{1}31513221 \quad 12 \quad 12231 \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
221 \overset{1}{1}31513221 \quad 12 \quad 12231 \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
221 \overset{1}{1}31513221 \quad 12 \quad 12231\textcolor{red}{3} \text{ }_{[\# = 5]} \quad \Delta = -85, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
221 \overset{1}{1}31513221 \quad 12 \quad 12231\textcolor{red}{3} \text{ }_{[\# = 3]} \quad \Delta = -85, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
221 \overset{3}{1}31513221 \quad 12 \quad 12231 \text{ }_{[\# = 2]} \quad \Delta = -85, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
221 \overset{1}{1}31513221 \quad 12 \quad 12231\textcolor{red}{3} \text{ }_{[\# = 2]} \quad \Delta = -85, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{ll}
221 \overset{3131513221}{12} 12231 \text{ }_{[\# = 5]} \quad \Delta = -85, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
221 \overset{3131513221}{12} 12231 \text{ }_{[\# = 3]} \quad \Delta = -85, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
221 \overset{131513221}{12} 12231 \overset{3}{\text{ }}_{[\# = 4]} \quad \Delta = -85, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
221 \overset{3131513221}{12} 12231 \text{ }_{[\# = 7]} \quad \Delta = -85, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
221 \overset{3131513221}{12} 12231 \text{ }_{[\# = 9]} \quad \Delta = -85, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
221 \overset{3131513221}{12} 12231 \text{ }_{[\# = 8]} \quad \Delta = -85, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
221 \overset{3131513221}{12} 12231 \text{ }_{[\# = 6]} \quad \Delta = -85, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
221 \overset{3131513221}{12} 12231 \text{ }_{[\# = 4]} \quad \Delta = -85, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
221 \overset{2131513221}{12} 12231 \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
221 \overset{131513221}{12} 12231 \overset{2}{\text{ }} \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\text{Base: } 221 \overset{1231513221}{12} 1222 & \\
221 \overset{31231513221}{12} 1222 \text{ }_{[\# = 9]} \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
221 \overset{31231513221}{12} 1222 \text{ }_{[\# = 8]} \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
221 \overset{31231513221}{12} 1222 \text{ }_{[\# = 7]} \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
221 \overset{31231513221}{12} 1222 \text{ }_{[\# = 6]} \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
221 \overset{31231513221}{12} 1222 \text{ }_{[\# = 5]} \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
221 \overset{31231513221}{12} 1222 \text{ }_{[\# = 4]} \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
221 \overset{31231513221}{12} 1222 \text{ }_{[\# = 3]} \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
221 \overset{21231513221}{12} 1222 \quad \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
221 \overset{1231513221}{12} 1222 \overset{2}{\text{ }} \quad \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
221 \overset{1231513221}{12} 1222 \quad \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
221 \overset{1231513221}{12} 1222 \quad \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
221 \overset{1231513221}{12} 1222 \quad \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
221 \overset{\textcolor{red}{3}1231513221}{12} 1222 \text{ [#=2]} \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: $221 \overset{51231513221}{12} 122$

$$\begin{aligned}
221 \overset{51231513221}{12} 122 \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
221 \overset{51231513221}{12} 122 \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
221 \overset{51231513221}{12} 122 \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
221 \overset{\textcolor{red}{2}51231513221}{12} 122 \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
221 \overset{51231513221}{12} 122 \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: $221 \overset{5131513221}{12} 1223$

$$\begin{aligned}
221 \overset{5131513221\textcolor{red}{2}}{12} 1223 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
221 \overset{\textcolor{red}{2}5131513221}{12} 1223 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
221 \overset{5131513221}{12} 1223 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
221 \overset{5131513221}{12} 1223 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
221 \overset{5131513221}{12} 1223 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 \quad & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
221 \overset{5131513221}{12} 1223 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
221 \overset{5131513221}{12} 1223 \quad \Delta = -130, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: $221 \overset{15131513221}{12} 122$

$$\begin{aligned}
221 \overset{15131513221}{12} 122 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
221 \overset{15131513221}{12} 122 \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
221 \overset{15131513221}{12} 122 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
221 \overset{15131513221}{12} 122 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
221 \overset{15131513221}{12} 122 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
221 \overset{215131513221}{12} 122 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{1,22315131513221}{12} 1$

$$\begin{aligned}
1 \overset{1,22315131513221}{12} 1 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1 \overset{1,22315131513221}{12} 1 \quad \Delta = -130, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1 \overset{1,22315131513221}{12} 1 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1 \overset{1,22315131513221}{12} 1 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 206 & 28 \\ 2 & 28 & 8 \end{pmatrix} \\
1 \overset{1,22315131513221}{12} 1 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 88 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
1 \overset{1,22315131513221}{12} 1 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 196 & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 236 & 30 \\ 2 & 30 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{21,2315131513221}{12} 1$

$$\begin{aligned}
1 \overset{21,2315131513221}{12} 1 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 70 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
1 \overset{21,2315131513221}{12} 1 \quad \Delta = -130, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1 \overset{21,2315131513221}{12} 1 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
1 \overset{21,2315131513221}{12} 1 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \overset{21,2315131513221}{12} 1 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
1 \overset{21,2315131513221}{12} 1 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1 \overset{21,2315131513221}{12} 1 \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{1,131513221}{12} 122313$

$$1 \overset{1,131513221}{12} \overset{3}{122313} \quad \Delta = -93, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\begin{array}{ll}
1^{1, \textcolor{red}{3}131513221} 12^{122313} [\#=4] & \Delta = -93, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1^{1,131513221} 12^{122313} [\#=2] \textcolor{red}{3} & \Delta = -93, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1^{1, \textcolor{red}{3}131513221} 12^{122313} [\#=2] & \Delta = -93, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{1, \textcolor{red}{3}131513221} 12^{122313} [\#=3] & \Delta = -93, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{1, \textcolor{red}{3}131513221} 12^{122313} [\#=5] & \Delta = -93, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1^{1, \textcolor{red}{3}131513221} 12^{122313} [\#=9] & \Delta = -93, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1^{1, \textcolor{red}{3}131513221} 12^{122313} [\#=8] & \Delta = -93, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1^{1,131513221} 12^{122313} \textcolor{red}{2} & \Delta = -85, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1^{1,131513221} 12^{122313} & \Delta = -85, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1^{1,131513221} 12^{122313} \textcolor{red}{2} & \Delta = -85, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{1,131513221} 12^{122313} & \Delta = -85, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{1,131513221} 12^{122313} & \Delta = -85, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1^{1, \textcolor{red}{3}131513221} 12^{122313} [\#=7] & \Delta = -93, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1^{1, \textcolor{red}{3}131513221} 12^{122313} [\#=6] & \Delta = -93, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
\textbf{Base: } 1^{1,1231513221} 12^{12222} & \\
1^{1, \textcolor{red}{3}1231513221} 12^{12222} [\#=2] & \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1^{1,1231513221} 12^{12222} \textcolor{red}{2} & \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1^{1,1231513221} 12^{12222} & \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{1,1231513221} 12^{12222} & \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1^{1, \textcolor{red}{3}1231513221} 12^{12222} [\#=3] & \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
1^{1,31231513221} 12^{12222} [\#=4] \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1^{1,31231513221} 12^{12222} [\#=6] \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
1^{1,31231513221} 12^{12222} [\#=7] \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
1^{1,31231513221} 12^{12222} [\#=8] \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
1^{1,31231513221} 12^{12222} [\#=9] \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
1^{1,31231513221} 12^{12222} [\#=5] \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
1^{1,1231513221} 12^{12222} \quad \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1^{1,51231513221} 12^{1222}$

$$\begin{aligned}
1^{1,51231513221} 12^{1222} \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1^{1,51231513221} 12^{1222} \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1^{1,51231513221} 12^{1222} \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1^{1,51231513221} 12^{1222} \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{1,51231513221} 12^{1222} \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1^{1,15131513221} 12^{1222}$

$$\begin{aligned}
1^{1,15131513221} 12^{1222} \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1^{1,15131513221} 12^{1222} \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{1,15131513221} 12^{1222} \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1^{1,15131513221} 12^{1222} \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1^{1,15131513221} 12^{1222} \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1^{1,15131513221} 12^{1222} \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1^{221,315131513221} 12^1$

$$\begin{array}{l}
\begin{array}{c} 221,315131513221 \\ 1 \quad 12 \quad 1 \end{array} \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
\begin{array}{c} 221,315131513221 \\ 1 \quad 12 \quad 1 \end{array} \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
\begin{array}{c} 221,315131513221 \\ 1 \quad 12 \quad 1 \end{array} \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\begin{array}{c} 221,315131513221 \\ 1 \quad 12 \quad 1 \end{array} \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
\begin{array}{c} 221,315131513221 \\ 1 \quad 12 \quad 1 \end{array} \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\begin{array}{c} 221,315131513221 \\ 1 \quad 12 \quad 1 \end{array} \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\begin{array}{c} 221,315131513221 \\ 1 \quad 12 \quad 1 \end{array} \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\text{Base: } \begin{array}{c} 21,315131513221 \\ 1 \quad 12 \quad 12 \end{array} \\
\begin{array}{c} 21,315131513221 \\ 1 \quad 12 \quad 12 \end{array} \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\begin{array}{c} 21,315131513221 \\ 1 \quad 12 \quad 12 \end{array} \quad \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\begin{array}{c} 21,315131513221 \\ 1 \quad 12 \quad 12 \end{array} \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
\begin{array}{c} 21,315131513221 \\ 1 \quad 12 \quad 12 \end{array} \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
\begin{array}{c} 21,315131513221 \\ 1 \quad 12 \quad 12 \end{array} \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\begin{array}{c} 21,315131513221 \textcolor{red}{2} \\ 1 \quad 12 \quad 12 \end{array} \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\begin{array}{c} 21,315131513221 \\ 1 \quad 12 \quad 12 \end{array} \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
\text{Base: } \begin{array}{c} 21,1513221 \\ 1 \quad 12 \quad 1223151 \end{array} \\
\begin{array}{c} 21,1513221 \textcolor{red}{2} \\ 1 \quad 12 \quad 1223151 \end{array} \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\begin{array}{c} 21,1513221 \textcolor{red}{2} \\ 1 \quad 12 \quad 1223151 \end{array} \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
\begin{array}{c} 21,1513221 \\ 1513221 \quad 12 \quad 1 \textcolor{red}{2} \end{array} \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\begin{array}{c} 21,1513221 \\ 1 \quad 12 \quad 1223151 \end{array} \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\text{Base: } \begin{array}{c} 21,31513221 \\ 1 \quad 12 \quad 122315 \end{array}
\end{array}$$

$$\begin{array}{lcl}
1^{21,31513221} 12^{122315} & \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
513221^{21,31513221} 12^{12} 1^{\text{red}} & \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1^{21,31513221} 12^{122315} 1^{\text{red}} & \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
1^{21,31513221} 12^{122315} & \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1^{21,31513221} 12^{122315} 2^{\text{red}} & \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1^{21,31513221} 12^{122315} & \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1^{21,31513221} 12^{122315} & \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 196 & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
\text{Base: } 1^{21,131513221} 12^{12231} & & \\
1^{21,3131513221} 12^{12231} [\# = 7] & \Delta = -85, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1^{21,3131513221} 12^{12231} [\# = 8] & \Delta = -85, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1^{21,131513221} 12^{12231} & \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1^{21,131513221} 12^{12231} & \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1^{21,131513221} 12^{12231} & \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{21,131513221} 12^{12231} & \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1^{21,131513221} 12^{12231} 2^{\text{red}} & \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1^{21,3131513221} 12^{12231} [\# = 6] & \Delta = -85, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1^{21,3131513221} 12^{12231} [\# = 5] & \Delta = -85, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1^{21,131513221} 12^{12231} 3^{\text{red}} [\# = 5] & \Delta = -85, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1^{21,3131513221} 12^{12231} [\# = 4] & \Delta = -85, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1^{21,131513221} 12^{12231} 3^{\text{red}} [\# = 4] & \Delta = -85, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{l}
1^{21, \textcolor{red}{3}131513221} 12^{12231} [\# = 3] \quad \Delta = -85, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{21, 131513221} 12^{12231 \textcolor{red}{3}} [\# = 3] \quad \Delta = -85, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1^{21, \textcolor{red}{3}131513221} 12^{12231} [\# = 2] \quad \Delta = -85, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{21, 131513221} 12^{12231 \textcolor{red}{3}} [\# = 2] \quad \Delta = -85, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1^{21, \textcolor{red}{3}131513221} 12^{12231} [\# = 9] \quad \Delta = -85, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
\text{Base: } 1^{21, 1231513221} 12^{1222} \\
1^{21, 1231513221} 12^{1222} \quad \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{21, \textcolor{red}{3}1231513221} 12^{1222} [\# = 2] \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1^{21, \textcolor{red}{3}1231513221} 12^{1222} [\# = 3] \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1^{21, \textcolor{red}{3}1231513221} 12^{1222} [\# = 5] \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
1^{21, \textcolor{red}{3}1231513221} 12^{1222} [\# = 8] \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
1^{21, \textcolor{red}{3}1231513221} 12^{1222} [\# = 9] \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
1^{21, \textcolor{red}{3}1231513221} 12^{1222} [\# = 4] \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1^{21, \textcolor{red}{3}1231513221} 12^{1222} [\# = 6] \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
1^{21, 1231513221} 12^{1222} \quad \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1^{21, 1231513221} 12^{1222} \quad \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1^{21, 1231513221 \textcolor{red}{2}} 12^{1222} \quad \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1^{21, \textcolor{red}{3}1231513221} 12^{1222} [\# = 7] \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
1^{21, 1231513221} 12^{1222} \quad \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\text{Base: } 1^{221, 51231513221} 12^{12}
\end{array}$$

$$\begin{array}{lcl}
1^{221,51231513221} & \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1^{221,51231513221} & \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1^{221,51231513221} & \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1^{221,51231513221} & \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{221,51231513221\textcolor{red}{2}} & \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1^{221,51231513221} & \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
\textbf{Base: } 1^{21,5131513221} & 1223 & \\
1^{21,5131513221} & \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1^{21,5131513221} & \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1^{21,5131513221} & \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1^{21,5131513221} & \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1^{21,5131513221} & \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
3221^{21,5131513221} & \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1^{21,5131513221\textcolor{red}{2}} & \Delta = -129, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1^{21,5131513221} & \Delta = -130, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\textbf{Base: } 1^{221,15131513221} & 12 & \\
1^{221,15131513221} & \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{221,15131513221} & \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1^{221,15131513221} & \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1^{221,15131513221} & \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{l}
\begin{array}{c} 221,15131513221 \\ 1 \quad 12 \quad 12 \end{array} \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\begin{array}{c} 221,15131513221 \\ 1 \quad 12 \quad 12 \end{array} \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\begin{array}{c} 221,15131513221\textcolor{red}{2} \\ 1 \quad 12 \quad 12 \end{array} \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\textbf{Base: } \begin{array}{c} 221,1231513221 \\ 1 \quad 12 \quad 122 \end{array} \\
\begin{array}{c} 221,\textcolor{red}{3}1231513221 \\ 1 \quad 12 \quad 122 \end{array} \quad [\# = 8] \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
\begin{array}{c} 221,\textcolor{red}{3}1231513221 \\ 1 \quad 12 \quad 122 \end{array} \quad [\# = 7] \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
\begin{array}{c} 221,1231513221 \\ 1 \quad 12 \quad 122 \end{array} \quad \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\begin{array}{c} 221,1231513221\textcolor{red}{2} \\ 1 \quad 12 \quad 122 \end{array} \quad \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\begin{array}{c} 221,1231513221 \\ 1 \quad 12 \quad 122 \end{array} \quad \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\begin{array}{c} 221,1231513221 \\ 1 \quad 12 \quad 122 \end{array} \quad \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\begin{array}{c} 221,\textcolor{red}{3}1231513221 \\ 1 \quad 12 \quad 122 \end{array} \quad [\# = 6] \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
\begin{array}{c} 221,\textcolor{red}{3}1231513221 \\ 1 \quad 12 \quad 122 \end{array} \quad [\# = 9] \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
\begin{array}{c} 221,\textcolor{red}{3}1231513221 \\ 1 \quad 12 \quad 122 \end{array} \quad [\# = 2] \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\begin{array}{c} 221,\textcolor{red}{3}1231513221 \\ 1 \quad 12 \quad 122 \end{array} \quad [\# = 3] \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\begin{array}{c} 221,\textcolor{red}{3}1231513221 \\ 1 \quad 12 \quad 122 \end{array} \quad [\# = 4] \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
\begin{array}{c} 221,\textcolor{red}{3}1231513221 \\ 1 \quad 12 \quad 122 \end{array} \quad [\# = 5] \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
\textbf{Base: } \begin{array}{c} 21,51231513221 \\ 21 \quad 12 \quad 12 \end{array} \\
\begin{array}{c} \textcolor{red}{2}21,51231513221 \\ 21 \quad 12 \quad 12 \end{array} \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\begin{array}{c} 21,51231513221 \\ 21 \quad 12 \quad 12 \end{array} \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\begin{array}{c} 21,51231513221 \\ 21 \quad 12 \quad 12 \end{array} \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{lcl}
21^{21,51231513221} & \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
21^{21,51231513221} & \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\text{Base: } 21^{21,15131513221} & & \\
21^{21,15131513221} & \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21^{21,15131513221} & \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
21^{21,15131513221} & \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21^{21,15131513221} & \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21^{21,15131513221} & \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
21^{21,15131513221} & \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\text{Base: } 21^{221,1231513221} & & \\
21^{221,1231513221} & \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21^{221,31231513221} & \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
21^{221,31231513221} & \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
21^{221,31231513221} & \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
21^{221,31231513221} & \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
21^{221,31231513221} & \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
21^{221,31231513221} & \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
21^{221,31231513221} & \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
21^{221,1231513221} & \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21^{221,1231513221} & \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{array}$$

$$21^{221, \textcolor{red}{3}1231513221}_{12} 12 \text{ [#}=3] \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$21^{221, 1231513221}_{12} 12 \quad \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$21^{\textcolor{red}{2}221, 1231513221}_{12} 12 \quad \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 12231414141414132^2

$$2^{\textcolor{red}{7}122314141414141}_{3} 2 \quad \Delta = -16, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$2^{\textcolor{red}{12}122314141414141}_{3} 2 \quad \Delta = -159, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$2^{\textcolor{red}{8}122314141414141}_{3} 2 \quad \Delta = -44, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: 13141414141414132^2

$$2^{\textcolor{red}{6}131414141414141}_{3} 2 \quad \Delta = -16, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 132161316131513221

$$132161316131513221\textcolor{red}{3} \text{ [#}=8] \quad \Delta = -1, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$132161316131513221\textcolor{red}{3} \text{ [#}=7] \quad \Delta = -1, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$132161316131513221\textcolor{red}{3} \text{ [#}=6] \quad \Delta = -1, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$132161316131513221\textcolor{red}{3} \text{ [#}=5] \quad \Delta = -1, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$122315131613161231\textcolor{red}{3} \text{ [#}=5] \quad \Delta = -1, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$132161316131513221\textcolor{red}{3} \text{ [#}=9] \quad \Delta = -1, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$132161316131513221\textcolor{red}{3} \text{ [#}=4] \quad \Delta = -1, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$122315131613161231\textcolor{red}{3} \text{ [#}=4] \quad \Delta = -1, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$122315131613161231\textcolor{red}{3} \text{ [#}=3] \quad \Delta = -1, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$122315131613161231\textcolor{red}{3} \text{ [#}=2] \quad \Delta = -1, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$132161316131513221\textcolor{red}{3} \text{ [#}=3] \quad \Delta = -1, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$\begin{array}{ll}
132161316131513221\textcolor{red}{3} \text{ [#=2]} & \Delta = -1, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\textbf{Base: } 123216131613151321 & \\
123216131613151321\textcolor{red}{3} \text{ [#=8]} & \Delta = -3, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
123151316131612321\textcolor{red}{3} \text{ [#=9]} & \Delta = -3, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
123151316131612321\textcolor{red}{3} \text{ [#=7]} & \Delta = -3, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
123216131613151321\textcolor{red}{3} \text{ [#=9]} & \Delta = -3, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
123216131613151321\textcolor{red}{3} \text{ [#=7]} & \Delta = -3, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
123216131613151321\textcolor{red}{3} \text{ [#=6]} & \Delta = -3, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
123151316131612321\textcolor{red}{3} \text{ [#=6]} & \Delta = -3, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
123216131613151321\textcolor{red}{3} \text{ [#=5]} & \Delta = -3, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
123151316131612321\textcolor{red}{3} \text{ [#=5]} & \Delta = -3, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
123216131613151321\textcolor{red}{3} \text{ [#=4]} & \Delta = -3, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
123151316131612321\textcolor{red}{3} \text{ [#=8]} & \Delta = -3, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
123216131613151321\textcolor{red}{3} \text{ [#=3]} & \Delta = -3, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
123151316131612321\textcolor{red}{3} \text{ [#=3]} & \Delta = -3, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
123151316131612321\textcolor{red}{3} \text{ [#=2]} & \Delta = -3, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
123216131613151321\textcolor{red}{3} \text{ [#=2]} & \Delta = -3, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
123151316131612321\textcolor{red}{3} \text{ [#=4]} & \Delta = -3, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
\textbf{Base: } 313161316131513221 & \\
\textcolor{red}{4} 313161316131513221 & \Delta = -28, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
313161316131513221^{\color{red}3} \text{ [#=5]} \quad \Delta = -8, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
313161316131513221^{\color{red}2} \quad \Delta = 0, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
313161316131513221^{\color{red}2} \quad \Delta = 0, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
313161316131513221^{\color{red}2} \quad \Delta = 0, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
313161316131513221^{\color{red}3} \text{ [#=2]} \quad \Delta = -8, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 93 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
313161316131513221^{\color{red}3} \quad \Delta = -8, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 21 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
313161316131513221^{\color{red}3} \text{ [#=2]} \quad \Delta = -8, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
313161316131513221^{\color{red}3} \text{ [#=3]} \quad \Delta = -8, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 213 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
313161316131513221^{\color{red}3} \text{ [#=3]} \quad \Delta = -8, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
313161316131513221^{\color{red}3} \text{ [#=4]} \quad \Delta = -8, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
313161316131513221^{\color{red}2} \quad \Delta = 0, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
313161316131513221^{\color{red}3} \text{ [#=6]} \quad \Delta = -8, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
313161316131513221^{\color{red}2} \quad \Delta = 0, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
313161316131513221^{\color{red}2} \quad \Delta = 0, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
313161316131513221^{\color{red}2} \quad \Delta = 0, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
313161316131513221^{\color{red}3} \text{ [#=9]} \quad \Delta = -8, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
313161316131513221^{\color{red}3} \text{ [#=7]} \quad \Delta = -8, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
313161316131513221^{\color{red}3} \text{ [#=8]} \quad \Delta = -8, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
\text{Base: } 141316131613151321 \\
141316131613151321^{\color{red}3} \text{ [#=7]} \quad \Delta = -28, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
141316131613151321\overset{3}{\color{red}}\text{ }_{[\#=5]} \quad \Delta = -28, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
141316131613151321\overset{3}{\color{red}}\text{ }_{[\#=4]} \quad \Delta = -28, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
141316131613151321\overset{3}{\color{red}}\text{ }_{[\#=3]} \quad \Delta = -28, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
123151316131613141\overset{3}{\color{red}}\text{ }_{[\#=3]} \quad \Delta = -28, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
141316131613151321\overset{3}{\color{red}}\text{ }_{[\#=2]} \quad \Delta = -28, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
123151316131613141\overset{3}{\color{red}}\text{ }_{[\#=2]} \quad \Delta = -28, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
141316131613151321\overset{3}{\color{red}}\text{ } \quad \Delta = -28, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 21 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
141316131613151321\overset{3}{\color{red}}\text{ }_{[\#=8]} \quad \Delta = -28, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
141316131613151321\overset{3}{\color{red}}\text{ }_{[\#=9]} \quad \Delta = -28, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
141316131613151321\overset{3}{\color{red}}\text{ }_{[\#=6]} \quad \Delta = -28, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
141316131613151321\overset{2}{\color{red}}\text{ } \quad \Delta = -20, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
141316131613151321\overset{2}{\color{red}}\text{ } \quad \Delta = -20, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
141316131613151321\overset{2}{\color{red}}\text{ } \quad \Delta = -20, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
141316131613151321\overset{2}{\color{red}}\text{ } \quad \Delta = -20, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
141316131613151321\overset{2}{\color{red}}\text{ } \quad \Delta = -20, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
141316131613151321\overset{2}{\color{red}}\text{ } \quad \Delta = -20, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
141316131613151321\overset{2}{\color{red}}\text{ } \quad \Delta = -20, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\text{Base: } 613151316131613151 & \\
613151316131613151\overset{2}{\color{red}}\text{ } \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
613151316131613151\overset{2}{\color{red}}\text{ } \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
613151316131613151^2 \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
613151316131613151^2 \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
613151316131613151^2 \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
613151316131613151^2 \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
613151316131613151^2 \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
613151316131613151^2 \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 131513161316131512

$$\begin{aligned}
2151316131613151313^{\text{[#}=9]} \quad \Delta = -51, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
2151316131613151313^{\text{[#}=8]} \quad \Delta = -51, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
2151316131613151313^{\text{[#}=2]} \quad \Delta = -51, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
131513161316131512^2 \quad \Delta = -43, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
131513161316131512^2 \quad \Delta = -43, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
2151316131613151313^{\text{[#}=7]} \quad \Delta = -51, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
2151316131613151313^{\text{[#}=6]} \quad \Delta = -51, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
2151316131613151313^{\text{[#}=5]} \quad \Delta = -51, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
2151316131613151313^{\text{[#}=4]} \quad \Delta = -51, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
2151316131613151313^{\text{[#}=3]} \quad \Delta = -51, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
131513161316131512^3 \quad \Delta = -51, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
131513161316131512^2 \quad \Delta = -43, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
131513161316131512^2 \quad \Delta = -43, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
131513161316131512^2 \quad \Delta = -43, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
131513161316131512^2 \quad \Delta = -43, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
131513161316131512^2 \quad \Delta = -43, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
131513161316131512^2 \quad \Delta = -43, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 13151316131613151¹

$$\begin{aligned}
1^{\frac{3}{5}131513161316131} 1_{[\# = 6]} \quad \Delta = -51, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
1^{\frac{3}{5}131513161316131} 1_{[\# = 9]} \quad \Delta = -51, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
13151316131613151^2 \quad \Delta = -43, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
13151316131613151^2 \quad \Delta = -43, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
13151316131613151^{2,1} \quad \Delta = -43, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
13151316131613151^2 \quad \Delta = -43, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
13151316131613151^2 \quad \Delta = -43, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
13151316131613151^2 \quad \Delta = -43, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
13151316131613151^2 \quad \Delta = -43, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1^{\frac{3}{5}131513161316131} 1_{[\# = 7]} \quad \Delta = -51, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
1^{\frac{3}{5}131513161316131} 1_{[\# = 8]} \quad \Delta = -51, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
1^{\frac{3}{5}131513161316131} 1_{[\# = 5]} \quad \Delta = -51, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
1^{\frac{3}{5}131513161316131} 1_{[\# = 4]} \quad \Delta = -51, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1^{\frac{3}{5}131513161316131} 1_{[\# = 3]} \quad \Delta = -51, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1^{\frac{3}{5}131513161316131} 1_{[\# = 2]} \quad \Delta = -51, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

Base: 132161316131612321

$$132161316131612321\overset{2}{3} \text{ [#=6]} \quad \Delta = -29, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$132161316131612321\overset{2}{3} \quad \Delta = -21, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$132161316131612321\overset{2}{3} \text{ [#=9]} \quad \Delta = -29, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$132161316131612321\overset{2}{3} \text{ [#=8]} \quad \Delta = -29, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$132161316131612321\overset{2}{3} \text{ [#=7]} \quad \Delta = -29, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$132161316131612321\overset{2}{3} \text{ [#=5]} \quad \Delta = -29, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$123216131613161231\overset{2}{3} \text{ [#=5]} \quad \Delta = -29, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix}$$

$$132161316131612321\overset{2}{3} \text{ [#=4]} \quad \Delta = -29, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$123216131613161231\overset{2}{3} \text{ [#=4]} \quad \Delta = -29, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$132161316131612321\overset{2}{3} \text{ [#=3]} \quad \Delta = -29, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$123216131613161231\overset{2}{3} \text{ [#=3]} \quad \Delta = -29, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$123216131613161231\overset{2}{3} \text{ [#=2]} \quad \Delta = -29, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$132161316131612321\overset{2}{3} \text{ [#=2]} \quad \Delta = -29, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$132161316131612321\overset{2}{3} \quad \Delta = -21, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$132161316131612321\overset{2}{3} \quad \Delta = -21, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$132161316131612321\overset{2}{3} \quad \Delta = -21, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$132161316131612321\overset{2}{3} \quad \Delta = -21, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$132161316131612321\overset{2}{3} \quad \Delta = -21, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 221613161316131512

$$\begin{aligned}
221613161316131512^3 \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
221613161316131512^2 \quad \Delta = -61, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
221613161316131512^2 \quad \Delta = -61, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
221613161316131512^2 \quad \Delta = -61, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
221613161316131512^2 \quad \Delta = -61, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
221613161316131512^2 \quad \Delta = -61, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
221613161316131512^2 \quad \Delta = -61, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
221613161316131512^3 \quad \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
221613161316131512^2 \quad \Delta = -61, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
221613161316131512^2 \quad \Delta = -61, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
221613161316131512^3 \quad \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 27 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

Base: 612321613161316131

$$\begin{aligned}
612321613161316131^2 \quad \Delta = -98, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
612321613161316131^3 \text{ [#=4]} \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
612321613161316131^3 \text{ [#=5]} \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
612321613161316131^2 \quad \Delta = -98, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
612321613161316131^2 \quad \Delta = -98, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
612321613161316131^2 \quad \Delta = -98, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 168 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
612321613161316131^2 \quad \Delta = -98, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
612321613161316131^2 \quad \Delta = -98, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
612321613161316131^2 & \quad \Delta = -98, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
6123216131613161313^{\#=6} & \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
6123216131613161313^{\#=7} & \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
6123216131613161313^{\#=8} & \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
6123216131613161313^{\#=3} & \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
6123216131613161313^{\#=2} & \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
6123216131613161313^{\#=9} & \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 22161316131613151¹

$$\begin{aligned}
22161316131613151^3 & \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
22161316131613151^{21} & \quad \Delta = -61, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
22161316131613151^{21} & \quad \Delta = -61, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
22161316131613151^2 & \quad \Delta = -61, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
22161316131613151^2 & \quad \Delta = -61, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
22161316131613151^2 & \quad \Delta = -61, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
22161316131613151^2 & \quad \Delta = -61, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
22161316131613151^2 & \quad \Delta = -61, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
22161316131613151^3 & \quad \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 27 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

Base: 123216131613161313

$$\begin{aligned}
3131613161316123213^{\#=3} & \quad \Delta = -36, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
3131613161316123213^{\#=4} & \quad \Delta = -36, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
313161316131612321\overset{3}{3} \text{ [#=9]} \quad \Delta = -36, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix} \\
313161316131612321\overset{3}{3} \text{ [#=8]} \quad \Delta = -36, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
313161316131612321\overset{3}{3} \text{ [#=7]} \quad \Delta = -36, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
313161316131612321\overset{3}{3} \text{ [#=6]} \quad \Delta = -36, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
123216131613161313\overset{3}{3} \quad \Delta = -36, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
313161316131612321\overset{3}{3} \text{ [#=2]} \quad \Delta = -36, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
123216131613161313\overset{3}{3} \text{ [#=2]} \quad \Delta = -36, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
123216131613161313\overset{2}{2} \quad \Delta = -28, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
123216131613161313\overset{3}{3} \text{ [#=3]} \quad \Delta = -36, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
123216131613161313\overset{2}{2} \quad \Delta = -28, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
123216131613161313\overset{2}{2} \quad \Delta = -28, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
123216131613161313\overset{2}{2} \quad \Delta = -28, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
123216131613161313\overset{2}{2} \quad \Delta = -28, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
123216131613161313\overset{2}{2} \quad \Delta = -28, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
123216131613161313\overset{4}{4} \quad \Delta = -56, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 9 & 5 \\ 5 & 9 \end{pmatrix} \\
123216131613161313\overset{2}{2} \quad \Delta = -28, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
313161316131612321\overset{3}{3} \text{ [#=5]} \quad \Delta = -36, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
\text{Base: } 141316131613161231 & \\
141316131613161231\overset{3}{3} \text{ [#=2]} \quad \Delta = -54, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
132161316131613141\overset{3}{3} \text{ [#=2]} \quad \Delta = -54, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
141316131613161231^3 \quad \Delta = -54, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 30 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1321613161316131413^3_{\# = 3} \quad \Delta = -54, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1413161316131612313^3_{\# = 3} \quad \Delta = -54, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1413161316131612313^3_{\# = 4} \quad \Delta = -54, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
141316131613161231^2 \quad \Delta = -46, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
141316131613161231^2 \quad \Delta = -46, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
141316131613161231^2 \quad \Delta = -46, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
141316131613161231^2 \quad \Delta = -46, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
141316131613161231^2 \quad \Delta = -46, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1413161316131612313^3_{\# = 5} \quad \Delta = -54, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
141316131613161231^2 \quad \Delta = -46, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
141316131613161231^2 \quad \Delta = -46, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
\textbf{Base: } 141316131613161313 & \\
141316131613161313^2 \quad \Delta = -53, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
141316131613161313^3 \quad \Delta = -61, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
141316131613161313^3_{\# = 3} \quad \Delta = -61, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
141316131613161313^3 \quad \Delta = -61, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 30 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
3131613161316131413^3_{\# = 3} \quad \Delta = -61, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
141316131613161313^3_{\# = 2} \quad \Delta = -61, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
141316131613161313^4 \quad \Delta = -81, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 9 & 5 \\ 5 & 9 \end{pmatrix}
\end{aligned}$$

$$313161316131613141\overset{3}{3} \text{ [#=2]} \quad \Delta = -61, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$141316131613161313 \quad \Delta = -53, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$141316131613161313 \quad \Delta = -53, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$141316131613161313 \quad \Delta = -53, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$141316131613161313 \quad \Delta = -53, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$141316131613161313 \quad \Delta = -53, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$141316131613161313 \quad \Delta = -53, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$141316131613161313 \quad \Delta = -53, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: $\overset{21}{16}13161316131512$

$$\overset{21}{16}13161316131512 \quad \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$\overset{21}{16}13161316131512 \quad \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\overset{21}{16}13161316131512 \quad \Delta = -61, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\overset{21}{16}13161316131512 \quad \Delta = -61, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\overset{21}{16}13161316131512 \quad \Delta = -61, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$\overset{21}{16}13161316131512 \quad \Delta = -61, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$\overset{21}{16}13161316131512 \quad \Delta = -61, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\overset{21}{16}13161316131512 \quad \Delta = -61, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\overset{21}{16}13161316131512 \quad \Delta = -61, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$\overset{21}{16}13161316131512 \quad \Delta = -61, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: $\overset{21}{16}1316131613151$

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$$\begin{array}{lll}
\begin{array}{c} \textcolor{red}{3}131 \\ 1 \ 6 \end{array} 1316131513221 \text{ [#}=5] & \Delta = 0, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
\begin{array}{c} 131 \\ 1 \ 6 \end{array} 1316131513221\textcolor{red}{3} \text{ [#}=4] & \Delta = 0, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
\begin{array}{c} \textcolor{red}{3}131 \\ 1 \ 6 \end{array} 1316131513221 \text{ [#}=2] & \Delta = 0, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\begin{array}{c} \textcolor{red}{3}131 \\ 1 \ 6 \end{array} 1316131513221 \text{ [#}=8] & \Delta = 0, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
\begin{array}{c} 131 \\ 1 \ 6 \end{array} 1316131513221\textcolor{red}{3} \text{ [#}=8] & \Delta = 0, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
\textbf{Base: } \begin{array}{c} 131 \\ 1 \ 6 \end{array} 1316131612321 & & \\
\begin{array}{c} \textcolor{red}{3}131 \\ 1 \ 6 \end{array} 1316131612321 \text{ [#}=4] & \Delta = -28, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
\begin{array}{c} 131 \\ 1 \ 6 \end{array} 1316131612321\textcolor{red}{3} \text{ [#}=2] & \Delta = -28, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\begin{array}{c} \textcolor{red}{3}131 \\ 1 \ 6 \end{array} 1316131612321 \text{ [#}=6] & \Delta = -28, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
\begin{array}{c} \textcolor{red}{3}131 \\ 1 \ 6 \end{array} 1316131612321 \text{ [#}=7] & \Delta = -28, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
\begin{array}{c} 131 \\ 1 \ 6 \end{array} 1316131612321\textcolor{red}{3} \text{ [#}=6] & \Delta = -28, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\begin{array}{c} \textcolor{red}{3}131 \\ 1 \ 6 \end{array} 1316131612321 \text{ [#}=9] & \Delta = -28, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
\begin{array}{c} 131 \\ 1 \ 6 \end{array} 1316131612321\textcolor{red}{3} \text{ [#}=7] & \Delta = -28, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
\begin{array}{c} \textcolor{red}{3}131 \\ 1 \ 6 \end{array} 1316131612321 \text{ [#}=8] & \Delta = -28, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
\begin{array}{c} 131 \\ 1 \ 6 \end{array} 1316131612321\textcolor{red}{3} \text{ [#}=9] & \Delta = -28, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
\begin{array}{c} 131 \\ 1 \ 6 \end{array} 1316131612321\textcolor{red}{3} \text{ [#}=5] & \Delta = -28, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\begin{array}{c} \textcolor{red}{3}131 \\ 1 \ 6 \end{array} 1316131612321 \text{ [#}=5] & \Delta = -28, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
\begin{array}{c} 131 \\ 1 \ 6 \end{array} 1316131612321\textcolor{red}{3} \text{ [#}=4] & \Delta = -28, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\begin{array}{c} \textcolor{red}{3}131 \\ 1 \ 6 \end{array} 1316131612321 \text{ [#}=2] & \Delta = -28, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\begin{array}{c} 131 \\ 1 \ 6 \end{array} 1316131612321\textcolor{red}{3} \text{ [#}=8] & \Delta = -28, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
& \overset{3131}{1 \ 6 \ 1316131612321} \text{ }_{[\# = 3]} \quad \Delta = -28, \ n_+ = 1, \ n_- = 18, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 1316131612321} \overset{3}{\text{ }} \text{ }_{[\# = 3]} \quad \Delta = -28, \ n_+ = 1, \ n_- = 18, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 1316131612321} \overset{2}{\text{ }} \quad \Delta = -20, \ n_+ = 1, \ n_- = 18, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 1316131612321} \overset{2}{\text{ }} \quad \Delta = -20, \ n_+ = 1, \ n_- = 18, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 1316131612321} \quad \Delta = -20, \ n_+ = 1, \ n_- = 18, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 1316131612321} \overset{2}{\text{ }} \quad \Delta = -20, \ n_+ = 1, \ n_- = 18, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 1316131612321} \overset{2}{\text{ }} \quad \Delta = -20, \ n_+ = 1, \ n_- = 18, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 1316131612321} \overset{2}{\text{ }} \quad \Delta = -20, \ n_+ = 1, \ n_- = 18, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \text{Base: } \overset{131}{1 \ 6 \ 1316131613141} \\
& \overset{3131}{1 \ 6 \ 1316131613141} \text{ }_{[\# = 5]} \quad \Delta = -53, \ n_+ = 1, \ n_- = 18, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 1316131613141} \quad \Delta = -45, \ n_+ = 1, \ n_- = 18, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 1316131613141} \overset{2}{\text{ }} \quad \Delta = -45, \ n_+ = 1, \ n_- = 18, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 6 \ 1316131613141} \text{ }_{[\# = 2]} \quad \Delta = -53, \ n_+ = 1, \ n_- = 18, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 6 \ 1316131613141} \text{ }_{[\# = 6]} \quad \Delta = -53, \ n_+ = 1, \ n_- = 18, \ n_0 = 0, \ |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 6 \ 1316131613141} \text{ }_{[\# = 7]} \quad \Delta = -53, \ n_+ = 1, \ n_- = 18, \ n_0 = 0, \ |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 6 \ 1316131613141} \text{ }_{[\# = 8]} \quad \Delta = -53, \ n_+ = 1, \ n_- = 18, \ n_0 = 0, \ |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 6 \ 1316131613141} \text{ }_{[\# = 4]} \quad \Delta = -53, \ n_+ = 1, \ n_- = 18, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 1316131613141} \overset{2}{\text{ }} \quad \Delta = -45, \ n_+ = 1, \ n_- = 18, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 1316131613141} \overset{2}{\text{ }} \quad \Delta = -45, \ n_+ = 1, \ n_- = 18, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 1316131613141} \overset{3}{\text{ }} \text{ }_{[\# = 3]} \quad \Delta = -53, \ n_+ = 1, \ n_- = 18, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{3131}{1 \ 6 \ 1316131613141} \text{ [#=3]} \quad \Delta = -53, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 1316131613141} \overset{3}{\text{13}} \text{ [#=2]} \quad \Delta = -53, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 1316131613141} \overset{3}{\text{13}} \quad \Delta = -53, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 1316131613141} \overset{2}{\text{13}} \quad \Delta = -45, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 1316131613141} \overset{2}{\text{13}} \quad \Delta = -45, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 1316131613141} \overset{2}{\text{13}} \quad \Delta = -45, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 6 \ 1316131613141} \text{ [#=9]} \quad \Delta = -53, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{61316131613161314}$

$$\begin{aligned}
& \overset{2}{61316131613161314} \overset{1}{\text{13}} \quad \Delta = -123, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 6131 \overset{31}{6 \ 131613161314} \quad \Delta = -131, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 613161 \overset{1}{\text{13}} \overset{2}{\text{13}} 613161314 \quad \Delta = -123, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 613161 \overset{1}{\text{13}} \overset{2}{\text{13}} 613161314 \quad \Delta = -123, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 6131 \overset{21}{6 \ 131613161314} \quad \Delta = -123, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 613161 \overset{1}{\text{13}} \overset{3}{\text{13}} 613161314 \quad \Delta = -131, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 613161 \overset{21}{\text{13}} 613161314 \quad \Delta = -123, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& 613161 \overset{12}{\text{13}} 613161314 \quad \Delta = -123, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
& 613161 \overset{1}{\text{13}} \overset{2}{\text{13}} 613161314 \quad \Delta = -123, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& 613161 \overset{1}{\text{13}} \overset{2}{\text{13}} 613161314 \quad \Delta = -123, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 613161 \overset{1}{\text{13}} \overset{2}{\text{13}} 613161314 \quad \Delta = -123, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{16131613161316131}$

$$\begin{aligned}
16131613161316131\overset{1}{3}_{[\# = 5]} \quad \Delta = -103, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
16131613161316131\overset{1}{3}_{[\# = 6]} \quad \Delta = -103, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
16131613161316131\overset{1}{3}_{[\# = 8]} \quad \Delta = -103, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
16131613161316131\overset{1}{3}_{[\# = 3]} \quad \Delta = -103, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
16131613161316131\overset{2}{1} \quad \Delta = -95, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
16131613161316131\overset{1}{2} \quad \Delta = -95, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
16131613161316131\overset{21}{1} \quad \Delta = -95, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
16131613161316131\overset{21}{1} \quad \Delta = -95, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
16131613161316131\overset{12}{2} \quad \Delta = -95, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
16131613161316131\overset{1}{2} \quad \Delta = -95, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
16131613161316131\overset{1}{2} \quad \Delta = -95, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
16131613161316131\overset{1}{2} \quad \Delta = -95, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
16131613161316131\overset{1}{3}_{[\# = 4]} \quad \Delta = -103, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
16131613161316131\overset{1}{3}_{[\# = 7]} \quad \Delta = -103, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
16131613161316131\overset{31}{1} \quad \Delta = -103, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
16131613161316131\overset{1}{3}_{[\# = 2]} \quad \Delta = -103, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
16131613161316131\overset{1}{3}_{[\# = 9]} \quad \Delta = -103, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
\text{Base: } 31\overset{31}{6}1316131513221 & \\
31\overset{312}{6}1316131513221 \quad \Delta = 0, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
31\overset{31}{6}1316131513221 \quad \Delta = 0, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& 31 \overline{6} 1316131513221\overline{3} \text{ [#=2]} \quad \Delta = -8, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& 31 \overline{6} 1316131513221\overline{2}^{31} \quad \Delta = 0, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 31 \overline{6} 1316131513221\overline{5} \quad \Delta = -52, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 31 \overline{6} 1316131513221\overline{8} \quad \Delta = -133, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& 31 \overline{6} 1316131513221\overline{2} \quad \Delta = 0, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 31 \overline{6} 1316131513221\overline{2} \quad \Delta = 0, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 31 \overline{6} 1316131513221\overline{2} \quad \Delta = 0, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& 31 \overline{6} 1316131513221\overline{2} \quad \Delta = 0, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 31 \overline{6} 1316131513221\overline{12} \quad \Delta = -248, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -4 & -2 \\ 2 & -2 & 8 \end{pmatrix} \\
& 31 \overline{6} 1316131513221\overline{6} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 31 \overline{6} 1316131513221\overline{3} \quad \Delta = -8, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 31 \overline{6} 1316131513221\overline{3} \text{ [#=9]} \quad \Delta = -8, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 324 \quad \begin{pmatrix} 0 & 18 & 2 \\ 18 & 645 & 71 \\ 2 & 71 & 8 \end{pmatrix} \\
& 31 \overline{6} 1316131513221\overline{3} \text{ [#=8]} \quad \Delta = -8, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 256 \quad \begin{pmatrix} 0 & 16 & 2 \\ 16 & 509 & 63 \\ 2 & 63 & 8 \end{pmatrix} \\
& 31 \overline{6} 1316131513221\overline{3} \text{ [#=7]} \quad \Delta = -8, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 196 \quad \begin{pmatrix} 0 & 14 & 2 \\ 14 & 389 & 55 \\ 2 & 55 & 8 \end{pmatrix} \\
& 31 \overline{6} 1316131513221\overline{3} \text{ [#=6]} \quad \Delta = -8, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 285 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
& 31 \overline{6} 1316131513221\overline{7} \quad \Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 31 \overline{6} 1316131513221\overline{4} \quad \Delta = -28, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 31 \overline{6} 1316131513221\overline{3} \text{ [#=5]} \quad \Delta = -8, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 197 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
& 31 \overline{6} 1316131513221\overline{3} \text{ [#=4]} \quad \Delta = -8, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 31 \\ 2 & 31 & 8 \end{pmatrix}
\end{aligned}$$

$$^{31}_{3161316131513221\mathbf{\bar{3}}_{[\#=3]}} \quad \Delta = -8, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

Base: $^{31}_{3161316131612321}$

$$^{231}_{3161316131612321} \quad \Delta = -28, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$^{31}_{3161316131612321\mathbf{\bar{5}}} \quad \Delta = -80, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$^{31}_{3161316131612321\mathbf{\bar{7}}} \quad \Delta = -133, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$^{31}_{3161316131612321\mathbf{\bar{6}}} \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$^{31}_{3161316131612321\mathbf{\bar{4}}} \quad \Delta = -56, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$^{31}_{3161316131612321\mathbf{\bar{8}}} \quad \Delta = -161, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$^{31}_{3161316131612321\mathbf{\bar{3}}_{[\#=7]}} \quad \Delta = -36, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 196 \quad \begin{pmatrix} 0 & 14 & 2 \\ 14 & 340 & 48 \\ 2 & 48 & 8 \end{pmatrix}$$

$$^{31}_{3161316131612321\mathbf{\bar{3}}_{[\#=6]}} \quad \Delta = -36, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 249 & 41 \\ 2 & 41 & 8 \end{pmatrix}$$

$$^{312}_{3161316131612321} \quad \Delta = -28, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$^{31}_{3161316131612321\mathbf{\bar{3}}_{[\#=2]}} \quad \Delta = -36, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$^{31}_{3161316131612321\mathbf{\bar{3}}} \quad \Delta = -36, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$^{31}_{3161316131612321\mathbf{\bar{3}}_{[\#=9]}} \quad \Delta = -36, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 324 \quad \begin{pmatrix} 0 & 18 & 2 \\ 18 & 564 & 62 \\ 2 & 62 & 8 \end{pmatrix}$$

$$^{31}_{3161316131612321\mathbf{\bar{3}}_{[\#=8]}} \quad \Delta = -36, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 256 \quad \begin{pmatrix} 0 & 16 & 2 \\ 16 & 445 & 55 \\ 2 & 55 & 8 \end{pmatrix}$$

$$^{31}_{3161316131612321\mathbf{\bar{2}}} \quad \Delta = -28, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$^{31}_{3161316131612321\mathbf{\bar{2}}} \quad \Delta = -28, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$^{31}_{3161316131612321\mathbf{\bar{3}}_{[\#=3]}} \quad \Delta = -36, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 60 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$^{31}_{3161316131612321\mathbf{\bar{3}}_{[\#=4]}} \quad \Delta = -36, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 109 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$^{31}_{3161316131612321\mathbf{\bar{3}}_{[\#=5]}} \quad \Delta = -36, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 172 & 34 \\ 2 & 34 & 8 \end{pmatrix}$$

$$\begin{aligned}
^{31}3161316131612321^2 \quad \Delta = -28, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 37 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
^{31}3161316131612321^2 \quad \Delta = -28, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
^{31}3161316131612321^2 \quad \Delta = -28, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{aligned}$$

Base: $^{31}3161316131613141$

$$\begin{aligned}
^{31}3161316131613141^3_{[\# = 2]} \quad \Delta = -61, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
^{31}3161316131613141^2 \quad \Delta = -53, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
^{31}3161316131613141^2 \quad \Delta = -53, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
^{31}3161316131613141^2 \quad \Delta = -53, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
^{31}3161316131613141^3 \quad \Delta = -61, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
^{31}3161316131613141^3_{[\# = 3]} \quad \Delta = -61, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
^{231}3161316131613141 \quad \Delta = -53, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
^{31}3161316131613141^2 \quad \Delta = -53, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
^{31}3161316131613141^2 \quad \Delta = -53, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 37 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
^{31}3161316131613141^2 \quad \Delta = -53, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
^{31}3161316131613141^2 \quad \Delta = -53, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
^{31}3161316131613141^3 \quad \Delta = -61, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 39 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
^{31}3161316131613141^4 \quad \Delta = -81, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 712231513161316131

$$\begin{aligned}
712231513161316131^2 \quad \Delta = -97, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
712231513161316131^2 \quad \Delta = -97, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
712231513161316131^2 \quad \Delta = -97, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
712231513161316131^2 \quad \Delta = -97, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
7122315131613161313_{\# = 6} \quad \Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
7122315131613161313_{\# = 5} \quad \Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
7122315131613161313_{\# = 4} \quad \Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
7122315131613161313_{\# = 3} \quad \Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
7122315131613161313_{\# = 8} \quad \Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
7122315131613161313_{\# = 2} \quad \Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
7122315131613161313_{\# = 7} \quad \Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
712231513161316131^2 \quad \Delta = -97, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 89 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
712231513161316131^2 \quad \Delta = -97, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
7122315131613161313_{\# = 9} \quad \Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
712231513161316131^2 \quad \Delta = -97, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
\text{Base: } 413161316131513217 \\
413161316131513217^2 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
413161316131513217^2 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
413161316131513217^2 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 5 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
413161316131513217^3 \quad \Delta = -133, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
413161316131513217^2 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
413161316131513217^2 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
413161316131513217 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
413161316131513217 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
413161316131513217 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 712321613161316123

$$\begin{aligned}
712321613161316123 \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
712321613161316123 \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
712321613161316123 \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
712321613161316123 \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
712321613161316123 \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
712321613161316123 \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
712321613161316123 \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{aligned}$$

Base: 7123216131613161³¹61

$$\begin{aligned}
7123216131613161³¹61 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
7123216131613161³¹61 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
7123216131613161²³¹61 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
7123216131613161³¹61 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
7123216131613161³¹61 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
7123216131613161³¹61 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
7123216131613161³¹61 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
7123216131613161³¹61² \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: 131613161315132171

$$\begin{aligned}
131613161315132171 \quad \Delta = -97, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1712315131613161313 \quad [\# = 6] \quad \Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1712315131613161313 \quad [\# = 4] \quad \Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
131613161315132171 \quad \Delta = -97, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
131613161315132171 \quad \Delta = -97, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
131613161315132171 \quad \Delta = -97, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
131613161315132171 \quad \Delta = -97, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
131613161315132171 \quad \Delta = -97, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
131613161315132171 \quad \Delta = -97, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1712315131613161313 \quad [\# = 3] \quad \Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1712315131613161313 \quad [\# = 7] \quad \Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1712315131613161313 \quad [\# = 8] \quad \Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1712315131613161313 \quad [\# = 9] \quad \Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1712315131613161313 \quad [\# = 2] \quad \Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1712315131613161313 \quad [\# = 5] \quad \Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
\text{Base: } 216131613161232171 & \\
216131613161232171 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
216131613161232171 \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
216131613161232171 \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
216131613161232171 \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{ll}
216131613161232171 & \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
216131613161232171 & \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
216131613161232171 & \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
216131613161232171 & \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
\text{Base: } 16131613161232171 & \\
16131613161232171 & \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
16131613161232171 & \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
16131613161232171 & \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
16131613161232171 & \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
16131613161232171 & \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
16131613161232171 & \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\text{Base: } 812231513161316123 & \\
812231513161316123 & \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
812231513161316123 & \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
812231513161316123 & \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
812231513161316123 & \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
812231513161316123 & \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
812231513161316123 & \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
812231513161316123 & \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\text{Base: } 8122315131613161 & \\
8122315131613161 & \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
81223151316131\overset{2}{6}1^{31} \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
81223151316131\overset{2}{6}1^{31} \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
81223151316131\overset{2}{6}1^{31} \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
81223151316131\overset{31}{6}1 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
81223151316131\overset{31}{6}1^2 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
81223151316131\overset{231}{6}1 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
81223151316131\overset{2}{6}1^{31} \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

Base: 812315131613161232

$$\begin{aligned}
812315131613161232 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
812315131613161232 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
812315131613161232 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 196 & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
812315131613161232 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
812315131613161232 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
812315131613161232 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
812315131613161232 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: 812321613161315132

$$\begin{aligned}
812321613161315132 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
812321613161315132 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 196 & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
812321613161315132 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
812321613161315132 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{aligned}$$

812321613161315132 ²	$\Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, \det = 169$	$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix}$
812321613161315132 ²	$\Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, \det = 256$	$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 206 & 28 \\ 2 & 28 & 8 \end{pmatrix}$
812321613161315132 ²	$\Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$
Base: 216131613151322181		
216131613151322181 ²	$\Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, \det = 4$	$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$
216131613151322181 ³	$\Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, \det = 4$	$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$
216131613151322181 ²	$\Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
216131613151322181 ²	$\Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$
216131613151322181 ²	$\Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$
216131613151322181 ²	$\Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, \det = 64$	$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$
216131613151322181 ²	$\Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$
216131613151322181 ²	$\Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$
Base: 16131613151322181 ¹		
16131613151322181 ¹ ²	$\Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$
16131613151322181 ¹ ²	$\Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
16131613151322181 ¹²	$\Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$
16131613151322181 ¹ ²	$\Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, \det = 64$	$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$
16131613151322181 ¹ ²	$\Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$
16131613151322181 ¹ ²	$\Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$
Base: 321613161315132181		
321613161315132181 ²	$\Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, \det = 81$	$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$

$$\begin{aligned}
321613161315132181 \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
321613161315132181 \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
321613161315132181 \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
321613161315132181 \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
321613161315132181 \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
321613161315132181 \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: 315131613161232181

$$\begin{aligned}
315131613161232181 \quad \Delta = -127, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
315131613161232181 \quad \Delta = -127, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
315131613161232181 \quad \Delta = -127, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
315131613161232181 \quad \Delta = -127, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
315131613161232181 \quad \Delta = -127, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 206 & 28 \\ 2 & 28 & 8 \end{pmatrix} \\
315131613161232181 \quad \Delta = -127, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
315131613161232181 \quad \Delta = -127, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: ³¹1613161315132181

$$\begin{aligned}
³¹1613161315132181 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
³¹1613161315132181 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
³¹1613161315132181 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
³¹1613161315132181 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
³¹1613161315132181 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
316^{31}13161315132181 \quad \Delta = -133, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
316^{21}13161315132181 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
16^{31}13161315132181^2 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
16^{31}13161315132181^2 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}
\end{aligned}$$

Base: 232171232171232181

$$\begin{aligned}
232171232171232181^2 \quad \Delta = -127, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
232171232171232181^2 \quad \Delta = -127, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 225 & \begin{pmatrix} 0 & 15 & 2 \\ 15 & 283 & 33 \\ 2 & 33 & 8 \end{pmatrix} \\
232171232171232181^2 \quad \Delta = -127, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 206 & 28 \\ 2 & 28 & 8 \end{pmatrix} \\
232171232171232181^2 \quad \Delta = -127, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 219 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
232171232171232181^2 \quad \Delta = -127, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

Base: 216131613151321812

$$\begin{aligned}
216131613151321812^2 \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
216131613151321812^3 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
216131613151321812^2 \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
216131613151321812^2 \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
216131613151321812^2 \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
216131613151321812^2 \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
216131613151321812^2 \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
216131613151321812^2 \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
216131613151321812^2 \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: 151316131612321812

$$\begin{aligned}
151316131612321812 \quad \Delta = -119, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
151316131612321812 \quad \Delta = -119, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
151316131612321812 \quad \Delta = -119, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
151316131612321812 \quad \Delta = -119, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
151316131612321812 \quad \Delta = -119, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
151316131612321812 \quad \Delta = -119, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
151316131612321812 \quad \Delta = -119, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix}
\end{aligned}$$

Base: 16131613151321812

$$\begin{aligned}
16131613151321812 \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
16131613151321812 \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
16131613151321812 \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
16131613151321812 \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
16131613151321812 \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
16131613151321812 \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
16131613151321812 \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: 321712321712321812

$$\begin{aligned}
321712321712321812 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
321712321712321812 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
321712321712321812 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
321712321712321812 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{ll}
321712321712321812^2 & \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
321712321712321812^2 & \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 163 & 25 \\ 2 & 25 & 8 \end{pmatrix} \\
\text{Base: } 217123217123218122 & \\
217123217123218122^2 & \Delta = -116, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
217123217123218122^3 & \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
217123217123218122^2 & \Delta = -116, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
217123217123218122^2 & \Delta = -116, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
217123217123218122^2 & \Delta = -116, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
217123217123218122^2 & \Delta = -116, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
217123217123218122^2 & \Delta = -116, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\text{Base: } 17123217123218122^1 & \\
17123217123218122^{12} & \Delta = -116, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
17123217123218122^1 & \Delta = -116, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
17123217123218122^1 & \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
17123217123218122^1 & \Delta = -116, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
17123217123218122^1 & \Delta = -116, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
17123217123218122^1 & \Delta = -116, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
\text{Base: } 171232171232181223 & \\
171232171232181223^2 & \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
171232171232181223^2 & \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
171232171232181223^2 & \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
171232171232181223^2 & \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{array}$$

$$171232171232181223 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

Base: 161316131513218123

$$161316131513218123 \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$161316131513218123 \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

$$161316131513218123 \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$161316131513218123 \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$161316131513218123 \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$161316131513218123 \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$161316131513218123 \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 513161316123218123

$$513161316123218123 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$513161316123218123 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$513161316123218123 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$513161316123218123 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$513161316123218123 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$513161316123218123 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 289 \quad \begin{pmatrix} 0 & 17 & 2 \\ 17 & 168 & 26 \\ 2 & 26 & 8 \end{pmatrix}$$

$$513161316123218123 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

Base: 131513218123151321

$$131513218123151321 \quad \Delta = -23, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$131513218123151321 \text{ }_{\text{[#}=4]} \quad \Delta = -31, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$123151321812315131 \text{ }_{\text{[#}=4]} \quad \Delta = -31, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$\begin{aligned}
131513218123151321\overset{2}{3} \text{ [#=3]} \quad \Delta = -31, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
123151321812315131\overset{2}{3} \text{ [#=3]} \quad \Delta = -31, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
131513218123151321\overset{2}{3} \text{ [#=2]} \quad \Delta = -31, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
123151321812315131\overset{2}{3} \text{ [#=2]} \quad \Delta = -31, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
131513218123151321\overset{2}{2} \quad \Delta = -23, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
131513218123151321\overset{2}{2} \quad \Delta = -23, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
131513218123151321\overset{2}{2} \quad \Delta = -23, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
131513218123151321\overset{2}{3} \text{ [#=5]} \quad \Delta = -31, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
131513218123151321\overset{2}{3} \text{ [#=9]} \quad \Delta = -31, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
123151321812315131\overset{2}{3} \text{ [#=9]} \quad \Delta = -31, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
131513218123151321\overset{2}{3} \text{ [#=8]} \quad \Delta = -31, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
123151321812315131\overset{2}{3} \text{ [#=8]} \quad \Delta = -31, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
123151321812315131\overset{2}{3} \text{ [#=6]} \quad \Delta = -31, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
131513218123151321\overset{2}{2} \quad \Delta = -23, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
131513218123151321\overset{2}{2} \quad \Delta = -23, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
131513218123151321\overset{2}{3} \text{ [#=7]} \quad \Delta = -31, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
123151321812315131\overset{2}{3} \text{ [#=7]} \quad \Delta = -31, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
131513218123151321\overset{2}{3} \text{ [#=6]} \quad \Delta = -31, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
123151321812315131\overset{2}{3} \text{ [#=5]} \quad \Delta = -31, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}
\end{aligned}$$

Base: 122315132181232151

$$\begin{aligned}
151232181231513221\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -25, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
151232181231513221\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -25, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
122315132181232151 \textcolor{red}{2} \quad \Delta = -17, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
151232181231513221\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -25, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
122315132181232151 \textcolor{red}{2} \quad \Delta = -17, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
122315132181232151 \textcolor{red}{2} \quad \Delta = -17, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
122315132181232151 \textcolor{red}{2} \quad \Delta = -17, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
122315132181232151 \textcolor{red}{2} \quad \Delta = -17, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
151232181231513221\textcolor{red}{3} \text{ [#=6]} \quad \Delta = -25, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
151232181231513221\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -25, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
151232181231513221\textcolor{red}{3} \text{ [#=7]} \quad \Delta = -25, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
151232181231513221\textcolor{red}{3} \text{ [#=8]} \quad \Delta = -25, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
151232181231513221\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -25, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix} \\
\textbf{Base: } 122315132181231512 \\
215132181231513221\textcolor{red}{3} \text{ [#=8]} \quad \Delta = -23, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
122315132181231512 \textcolor{red}{3} \quad \Delta = -23, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
122315132181231512 \textcolor{red}{2} \quad \Delta = -15, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
122315132181231512 \textcolor{red}{2} \quad \Delta = -15, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
122315132181231512 \textcolor{red}{2} \quad \Delta = -15, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
215132181231513221\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -23, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
215132181231513221\overset{2}{3} \text{ }_{[\#=3]} \quad \Delta = -23, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
12231513218123151\overset{2}{2} \quad \Delta = -15, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
12231513218123151\overset{2}{2} \quad \Delta = -15, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
215132181231513221\overset{2}{3} \text{ }_{[\#=9]} \quad \Delta = -23, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix} \\
215132181231513221\overset{2}{3} \text{ }_{[\#=7]} \quad \Delta = -23, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
215132181231513221\overset{2}{3} \text{ }_{[\#=4]} \quad \Delta = -23, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
215132181231513221\overset{2}{3} \text{ }_{[\#=6]} \quad \Delta = -23, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
215132181231513221\overset{2}{3} \text{ }_{[\#=5]} \quad \Delta = -23, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
12231513218123151\overset{2}{2} \quad \Delta = -15, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\text{Base: } 12231513218123151\overset{1}{1} \\
1\overset{3}{1}22315132181231\overset{1}{5}1 \text{ }_{[\#=4]} \quad \Delta = -23, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1\overset{3}{1}22315132181231\overset{1}{5}1 \text{ }_{[\#=5]} \quad \Delta = -23, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
1\overset{3}{1}22315132181231\overset{1}{5}1 \text{ }_{[\#=9]} \quad \Delta = -23, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix} \\
1\overset{3}{1}22315132181231\overset{1}{5}1 \text{ }_{[\#=8]} \quad \Delta = -23, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
1\overset{3}{1}22315132181231\overset{1}{5}1 \text{ }_{[\#=2]} \quad \Delta = -23, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1\overset{3}{1}22315132181231\overset{1}{5}1 \text{ }_{[\#=3]} \quad \Delta = -23, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1\overset{3}{1}22315132181231\overset{1}{5}1 \text{ }_{[\#=7]} \quad \Delta = -23, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
1\overset{3}{1}22315132181231\overset{1}{5}1 \text{ }_{[\#=6]} \quad \Delta = -23, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
12231513218123151\overset{2}{2}\overset{1}{1} \quad \Delta = -15, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
12231513218123151\overset{2}{2}\overset{1}{1} \quad \Delta = -15, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
12231513218123151^{\overset{2}{1}} \quad \Delta = -15, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
12231513218123151^{\overset{2}{1}} \quad \Delta = -15, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
12231513218123151^{\overset{2}{1}} \quad \Delta = -15, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 612315132181232161

$$\begin{aligned}
612315132181232161^{\overset{2}{1}} \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
612315132181232161^{\overset{2}{1}} \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
612315132181232161^{\overset{2}{1}} \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
612315132181232161^{\overset{2}{1}} \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
612315132181232161^{\overset{2}{1}} \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
612315132181232161^{\overset{2}{1}} \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{aligned}$$

Base: 161232181231513161

$$\begin{aligned}
161232181231513161^{\overset{2}{1}} \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
161232181231513161^{\overset{2}{1}} \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
161232181231513161^{\overset{2}{1}} \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
161232181231513161^{\overset{2}{1}} \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
161232181231513161^{\overset{2}{1}} \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
161232181231513161^{\overset{2}{1}} \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

Base: 216123218123151321

$$\begin{aligned}
216123218123151321^{\overset{2}{1}} \quad \Delta = -43, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
216123218123151321^{\overset{2}{1}} \quad \Delta = -43, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
216123218123151321^{\overset{2}{1}} \quad \Delta = -43, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
216123218123151321\overset{2}{3} \text{ [#=5]} \quad \Delta = -51, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
216123218123151321\overset{2}{3} \text{ [#=4]} \quad \Delta = -51, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
216123218123151321\overset{2}{3} \quad \Delta = -43, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
216123218123151321\overset{2}{3} \quad \Delta = -43, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
216123218123151321\overset{2}{3} \text{ [#=3]} \quad \Delta = -51, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
216123218123151321\overset{2}{3} \text{ [#=2]} \quad \Delta = -51, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
216123218123151321\overset{2}{3} \text{ [#=8]} \quad \Delta = -51, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
216123218123151321\overset{2}{3} \quad \Delta = -43, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
216123218123151321\overset{2}{3} \quad \Delta = -51, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
216123218123151321\overset{2}{3} \text{ [#=9]} \quad \Delta = -51, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
216123218123151321\overset{2}{3} \text{ [#=6]} \quad \Delta = -51, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
216123218123151321\overset{2}{3} \text{ [#=7]} \quad \Delta = -51, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}
\end{aligned}$$

Base: 613151321812321613

$$\begin{aligned}
613151321812321613\overset{2}{3} \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
613151321812321613\overset{2}{3} \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
613151321812321613\overset{2}{3} \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 \quad & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
613151321812321613\overset{2}{3} \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 169 \quad & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
613151321812321613\overset{2}{3} \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 \quad & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
613151321812321613\overset{2}{3} \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
613151321812321613\overset{2}{3} \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{aligned}$$

Base: 131612321812315131

$$\begin{aligned}
131513218123216131\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
131612321812315131\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
131513218123216131\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
131513218123216131\textcolor{red}{3} \text{ [#=6]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
131612321812315131\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
131612321812315131\textcolor{red}{2} \quad \Delta = -50, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
131612321812315131\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
131612321812315131\textcolor{red}{2} \quad \Delta = -50, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
131612321812315131\textcolor{red}{3} \text{ [#=7]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
131513218123216131\textcolor{red}{3} \text{ [#=7]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
131513218123216131\textcolor{red}{3} \text{ [#=8]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
131612321812315131\textcolor{red}{3} \text{ [#=6]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
131612321812315131\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
131612321812315131\textcolor{red}{3} \text{ [#=8]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
131612321812315131\textcolor{red}{2} \quad \Delta = -50, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
131612321812315131\textcolor{red}{2} \quad \Delta = -50, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
131612321812315131\textcolor{red}{2} \quad \Delta = -50, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
131612321812315131\textcolor{red}{2} \quad \Delta = -50, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
131513218123216131\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
131612321812315131\overset{2}{3} \text{ [#=2]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
131513218123216131\overset{2}{3} \text{ [#=3]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
131513218123216131\overset{2}{3} \text{ [#=5]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}
\end{aligned}$$

Base: 613161232181231513

$$\begin{aligned}
61316123218123151\overset{2}{3} \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 169 \quad & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
61316123218123151\overset{2}{3} \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
61316123218123151\overset{2}{3} \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
61316123218123151\overset{2}{3} \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 \quad & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
61316123218123151\overset{2}{3} \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
61316123218123151\overset{2}{3} \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
61316123218123151\overset{2}{3} \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 225 \quad & \begin{pmatrix} 0 & 15 & 2 \\ 15 & 163 & 25 \\ 2 & 25 & 8 \end{pmatrix}
\end{aligned}$$

Base: 612321812315131613

$$\begin{aligned}
61232181231513161\overset{2}{3} \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
61232181231513161\overset{2}{3} \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
61232181231513161\overset{2}{3} \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
61232181231513161\overset{2}{3} \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 100 \quad & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
61232181231513161\overset{2}{3} \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 169 \quad & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
61232181231513161\overset{2}{3} \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 144 \quad & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
61232181231513161\overset{2}{3} \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 196 \quad & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix}
\end{aligned}$$

Base: 131613151321812321

$$131613151321812321\overset{2}{3} \quad \Delta = -50, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$\begin{aligned}
123218123151316131\overset{2}{3} \text{ [#=2]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
123218123151316131\overset{2}{3} \text{ [#=7]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
13161\overset{2}{3}151321812321 \quad \Delta = -50, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
131613151321812321\overset{2}{3} \text{ [#=5]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
123218123151316131\overset{2}{3} \text{ [#=4]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
131613151321812321\overset{2}{3} \text{ [#=3]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
131613151321812321\overset{2}{3} \text{ [#=7]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
131613151\overset{2}{3}1321812321 \quad \Delta = -50, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1316\overset{2}{3}13151321812321 \quad \Delta = -50, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
131613151\overset{2}{3}1321812321 \quad \Delta = -50, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
13161315132181\overset{2}{3}2321 \quad \Delta = -50, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
123218123151316131\overset{2}{3} \text{ [#=5]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
131613151321812321\overset{2}{3} \text{ [#=6]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
131613151321812321\overset{2}{3} \text{ [#=9]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
123218123151316131\overset{2}{3} \text{ [#=9]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
131613151321812321\overset{2}{3} \text{ [#=8]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
123218123151316131\overset{2}{3} \text{ [#=8]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
123218123151316131\overset{2}{3} \text{ [#=3]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
131613151321812321\overset{2}{3} \text{ [#=4]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

$$123218123151316131\overset{2}{3} \text{ [#=6]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$131613151321812321\overset{2}{3} \text{ [#=2]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: 613161315132181232

$$6131613151\overset{2}{3}2181232 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 289 \quad \begin{pmatrix} 0 & 17 & 2 \\ 17 & 202 & 28 \\ 2 & 28 & 8 \end{pmatrix}$$

$$613161315132181232 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$613161315132181232 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$613161315132181\overset{2}{2}32 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$613161315132181\overset{2}{2}32 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 256 \quad \begin{pmatrix} 0 & 16 & 2 \\ 16 & 174 & 26 \\ 2 & 26 & 8 \end{pmatrix}$$

$$613161315132181232 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 324 \quad \begin{pmatrix} 0 & 18 & 2 \\ 18 & 232 & 30 \\ 2 & 30 & 8 \end{pmatrix}$$

$$613161315132181232 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 225 \quad \begin{pmatrix} 0 & 15 & 2 \\ 15 & 163 & 25 \\ 2 & 25 & 8 \end{pmatrix}$$

Base: 161316123218123151

$$161\overset{2}{3}16123218123151 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1613161232181231\overset{2}{5}1 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1613161232181\overset{2}{2}3151 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1613161\overset{2}{2}3218123151 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$161316123218123151 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1613161232181\overset{2}{2}3151 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

Base: 316131612321812315

$$3161\overset{2}{3}1612321812315 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$31613161\overset{2}{2}321812315 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 324 \quad \begin{pmatrix} 0 & 18 & 2 \\ 18 & 124 & 24 \\ 2 & 24 & 8 \end{pmatrix}$$

$$31613161232181\overset{2}{2}315 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$\begin{aligned}
316131612321812315^2 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 196 \quad & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 68 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
316131612321812315^2 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
316131612321812315^2 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 144 \quad & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 58 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
316131612321812315^2 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: 131613161232181231

$$\begin{aligned}
1321812321613161313^{\#7} \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
131613161232181231^2 \quad \Delta = -76, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
131613161232181231^2 \quad \Delta = -76, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
131613161232181231^2 \quad \Delta = -76, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
131613161232181231^2 \quad \Delta = -76, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
131613161232181231^2 \quad \Delta = -76, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
131613161232181231^2 \quad \Delta = -76, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1316131612321812313^{\#5} \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1321812321613161313^{\#6} \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1321812321613161313^{\#5} \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1321812321613161313^{\#4} \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1316131612321812313^{\#4} \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1321812321613161313^{\#8} \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1321812321613161313^{\#9} \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1316131612321812313^{\#3} \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

$$132181232161316131\bar{3} \text{ }_{[\# = 3]} \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$131613161232181231\bar{3} \text{ }_{[\# = 2]} \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$132181232161316131\bar{3} \text{ }_{[\# = 2]} \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 16123218123151321

$$16123218123151\bar{3}21 \quad \Delta = -43, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$16123218123151\bar{3}21 \quad \Delta = -43, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$16123218123151321\bar{3} \text{ }_{[\# = 8]} \quad \Delta = -51, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$16123218123151\bar{3}21 \quad \Delta = -43, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$16123218123151\bar{3}21 \quad \Delta = -43, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$16123218123151321\bar{3} \text{ }_{[\# = 3]} \quad \Delta = -51, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$16123218123151321\bar{3} \text{ }_{[\# = 2]} \quad \Delta = -51, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$16123218123151321\bar{3} \text{ }_{[\# = 5]} \quad \Delta = -51, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$16123218123151321\bar{3} \text{ }_{[\# = 4]} \quad \Delta = -51, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$16123218123151321\bar{3} \text{ }_{[\# = 6]} \quad \Delta = -51, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$16123218123151\bar{3}21 \quad \Delta = -43, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$16123218123151321\bar{3} \text{ }_{[\# = 7]} \quad \Delta = -51, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$16123218123151321\bar{3} \text{ }_{[\# = 9]} \quad \Delta = -51, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix}$$

Base: 712231513218123215

$$71223151321812321\bar{5} \quad \Delta = -122, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$71223151321812321\bar{5} \quad \Delta = -122, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$\begin{aligned}
712231513218123215^2 \quad \Delta = -122, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
712231513218123215^2 \quad \Delta = -122, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
712231513218123215^2 \quad \Delta = -122, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
712231513218123215^2 \quad \Delta = -122, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{aligned}$$

Base: 712231513218123151

$$\begin{aligned}
712231513218123151^2 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
712231513218123151^2 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
712231513218123151^2 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
712231513218123151^2 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
712231513218123151^2 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
712231513218123151^2 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

Base: 712315132181231513

$$\begin{aligned}
712315132181231513^2 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
712315132181231513^2 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
712315132181231513^2 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
712315132181231513^2 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
712315132181231513^2 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
712315132181231513^2 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
712315132181231513^2 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}
\end{aligned}$$

Base: 171223151321812321

$$171223151321812321^3 \text{ [#=8]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$\begin{aligned}
171223151321812321\overset{2}{3} \text{ [#=4]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
171223151321812321\overset{2}{3} \text{ [#=3]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
171223151321812321\overset{2}{3} \text{ [#=2]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
171223151321812321\overset{2}{3} \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
171223151321812321\overset{2}{3} \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
171223151321812321\overset{2}{3} \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
171223151321812321\overset{2}{3} \text{ [#=5]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
171223151321812321\overset{2}{3} \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
171223151321812321\overset{2}{3} \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
171223151321812321\overset{2}{3} \text{ [#=6]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
171223151321812321\overset{2}{3} \text{ [#=7]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
171223151321812321\overset{2}{3} \text{ [#=9]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}
\end{aligned}$$

Base: 171231513218123215

$$\begin{aligned}
171231513218123215\overset{2}{3} \quad \Delta = -122, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
171231513218123215\overset{2}{3} \quad \Delta = -122, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
171231513218123215\overset{2}{3} \quad \Delta = -122, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
171231513218123215\overset{2}{3} \quad \Delta = -122, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
171231513218123215\overset{2}{3} \quad \Delta = -122, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
171231513218123215\overset{2}{3} \quad \Delta = -122, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

Base: 171231513218123151

$$\begin{aligned}
171231513218123151 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
171231513218123151 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
171231513218123151 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
171231513218123151 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
171231513218123151 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
171231513218123151 \quad \Delta = -120, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 217123151321812321

$$\begin{aligned}
217123151321812321 \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
2171231513218123213 \quad [\#=3] \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
217123151321812321 \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
217123151321812321 \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
217123151321812321 \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2171231513218123213 \quad [\#=8] \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
2171231513218123213 \quad [\#=2] \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2171231513218123213 \quad [\#=9] \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
2171231513218123213 \quad [\#=7] \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
2171231513218123213 \quad [\#=6] \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
2171231513218123213 \quad [\#=5] \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
2171231513218123213 \quad [\#=4] \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
217123151321812321 \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

$$217123151321812321 \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: 217123218123151315

$$217123218123151315 \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$217123218123151315 \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$217123218123151315 \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$217123218123151315 \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$217123218123151315 \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$217123218123151315 \quad \Delta = -122, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$217123218123151315 \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$217123218123151315 \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 221712321812315131

$$221712321812315131 \quad \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$221712321812315131 \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$221712321812315131 \quad \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$221712321812315131 \quad \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$221712321812315131 \quad \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$221712321812315131 \text{ } \#=3 \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$221712321812315131 \text{ } \#=4 \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$221712321812315131 \text{ } \#=2 \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$221712321812315131 \text{ } \#=6 \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$\begin{aligned}
221712321812315131\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
221712321812315131\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
221712321812315131\textcolor{red}{3} \text{ [#=8]} \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
221712321812315131\textcolor{red}{2} \quad \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
221712321812315131\textcolor{red}{2} \quad \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
221712321812315131\textcolor{red}{3} \text{ [#=7]} \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
\textbf{Base: } 132171232181231512 & \\
132171232181231512\textcolor{red}{2} \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
215132181232171231\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
132171232181231512\textcolor{red}{2} \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
215132181232171231\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
215132181232171231\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
132171232181231512\textcolor{red}{3} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
132171232181231512\textcolor{red}{2} \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
132171232181231512\textcolor{red}{2} \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
132171232181231512\textcolor{red}{2} \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
215132181232171231\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
132171232181231512\textcolor{red}{2} \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\textbf{Base: } 13217123218123151\textcolor{red}{1} & \\
13217123218123151\textcolor{red}{21} \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
13217123218123151\textcolor{red}{21} \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$13217123218123151 \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$13217123218123151 \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$13217123218123151 \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1 \overset{3}{132171232181231} \underset{5}{1} \text{ [#=2]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{3}{132171232181231} \underset{5}{1} \text{ [#=4]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1 \overset{3}{132171232181231} \underset{5}{1} \text{ [#=5]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$1 \overset{3}{132171232181231} \underset{5}{1} \text{ [#=3]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: 513217123218123151

$$513217123218123151 \quad \Delta = -122, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$513217123218123151 \quad \Delta = -122, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$513217123218123151 \quad \Delta = -122, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$513217123218123151 \quad \Delta = -122, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$513217123218123151 \quad \Delta = -122, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$513217123218123151 \quad \Delta = -122, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

Base: 141321812321712321

$$141321812321712321 \quad \Delta = -48, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$141321812321712321 \quad \Delta = -48, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$141321812321712321 \quad \Delta = -50, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$141321812321712321 \quad \Delta = -48, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$141321812321712321 \quad \Delta = -48, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\begin{aligned}
141321812321712321^2 \quad \Delta = -48, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
141321812321712321^3_{\# = 8} \quad \Delta = -56, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
141321812321712321^3_{\# = 2} \quad \Delta = -56, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
123217123218123141^3_{\# = 3} \quad \Delta = -56, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
141321812321712321^3_{\# = 3} \quad \Delta = -56, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
141321812321712321^3_{\# = 4} \quad \Delta = -56, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
141321812321712321^3_{\# = 5} \quad \Delta = -56, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
141321812321712321^3_{\# = 6} \quad \Delta = -56, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
123217123218123141^3_{\# = 2} \quad \Delta = -56, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
141321812321712321^3_{\# = 7} \quad \Delta = -56, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
141321812321712321^3_{\# = 9} \quad \Delta = -56, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}
\end{aligned}$$

Base: 413218123217123216

$$\begin{aligned}
413218123217123216^2 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
413218123217123216^2 \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
413218123217123216^2 \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
413218123217123216^2 \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
413218123217123216^2 \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
413218123217123216^2 \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
413218123217123216^2 \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

Base: 161232171232181231

$$\begin{aligned}
161232171232181231\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1612321\textcolor{red}{2}71232181231 \quad \Delta = -98, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
161232171232181231 \quad \Delta = -98, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
1612321712321\textcolor{red}{2}81231 \quad \Delta = -98, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
161232171232181\textcolor{red}{2}231 \quad \Delta = -98, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
161232171232181231\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
161232171232181231 \quad \Delta = -98, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
161232171232181231\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
161232171232181231\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
\text{Base: } 17123151321812321 \\
17123151321812321\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
17123151321812321 \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
17123151321812321 \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
17123151321\textcolor{red}{2}812321 \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
17123151321\textcolor{red}{2}812321 \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
17123151321812321 \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
17123151321812321\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
17123151321812321\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
17123151321812321\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
17123151321812321\textcolor{red}{3} \text{ [#=6]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{1}{17123151321812321\textcolor{red}{3}}_{[\#=7]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& \overset{1}{17123151321812321\textcolor{red}{3}}_{[\#=8]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{1}{17123151321812321\textcolor{red}{3}}_{[\#=9]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{17123218123151315}$

$$\begin{aligned}
& \overset{1\textcolor{red}{2}}{17123218123151315} \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{1}{171232181231\textcolor{red}{2}51315} \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{1}{1712321812315\textcolor{red}{2}315} \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{1}{17123218123151\textcolor{red}{2}315} \quad \Delta = -122, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{1}{171232181231513\textcolor{red}{2}15} \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{1}{1712321812315131\textcolor{red}{2}5} \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{1}{17123218123151315\textcolor{red}{2}} \quad \Delta = -121, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{21}{1712321812315131}$

$$\begin{aligned}
& \overset{21}{1712321812315131\textcolor{red}{2}} \quad \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{21}{1712321812315131\textcolor{red}{3}}_{[\#=9]} \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{21\textcolor{red}{2}}{1712321812315131} \quad \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{21}{1712321812315131} \quad \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{21}{17123218123151\textcolor{red}{2}31} \quad \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{21}{1712321812315131\textcolor{red}{2}} \quad \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{21}{1712321812315131\textcolor{red}{2}} \quad \Delta = -69, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{21}{1712321812315131\textcolor{red}{3}}_{[\#=2]} \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{21}{17}12321812315131\overset{\color{red}{3}}{3} \text{ [#=3]} \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{21}{17}12321812315131\overset{\color{red}{3}}{3} \text{ [#=5]} \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{21}{17}12321812315131\overset{\color{red}{3}}{3} \text{ [#=4]} \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{21}{17}12321812315131\overset{\color{red}{3}}{3} \text{ [#=6]} \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{21}{17}12321812315131\overset{\color{red}{3}}{3} \text{ [#=7]} \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& \overset{21}{17}12321812315131\overset{\color{red}{3}}{3} \text{ [#=8]} \quad \Delta = -77, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

Base: 413218123151322181

$$\begin{aligned}
& \overset{\color{red}{2}}{4}13218123151322181 \quad \Delta = -124, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 41321\overset{\color{red}{2}}{8}123151322181 \quad \Delta = -124, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& 4132181231513221\overset{\color{red}{2}}{8}1 \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 4132181\overset{\color{red}{2}}{2}3151322181 \quad \Delta = -124, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
& 4132181231513221\overset{\color{red}{2}}{8}1 \quad \Delta = -124, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
& 413218123151\overset{\color{red}{2}}{3}22181 \quad \Delta = -124, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& 4132181231513221\overset{\color{red}{2}}{8}1 \quad \Delta = -124, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

Base: 218122315132181231

$$\begin{aligned}
& 218122315132181231\overset{\color{red}{3}}{3} \text{ [#=4]} \quad \Delta = -104, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 218122315132181231\overset{\color{red}{3}}{3} \text{ [#=5]} \quad \Delta = -104, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 218122315132181231\overset{\color{red}{3}}{3} \text{ [#=2]} \quad \Delta = -104, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
& 218122315132181231\overset{\color{red}{3}}{3} \text{ [#=3]} \quad \Delta = -104, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& 218122315132181\overset{\color{red}{2}}{2}31 \quad \Delta = -96, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
218122315132181231 \quad \Delta = -96, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
218122315132181231 \quad \Delta = -96, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
218122315132181231 \quad \Delta = -96, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
218122315132181231 \quad \Delta = -96, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
218122315132181231 \quad \Delta = -96, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: 413218123151321812

$$\begin{aligned}
413218123151321812 \quad \Delta = -124, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
413218123151321812 \quad \Delta = -124, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
413218123151321812 \quad \Delta = -124, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
413218123151321812 \quad \Delta = -124, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
413218123151321812 \quad \Delta = -124, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
413218123151321812 \quad \Delta = -124, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
413218123151321812 \quad \Delta = -124, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
413218123151321812 \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: 221812315132181231

$$\begin{aligned}
221812315132181231 \text{ } \textcolor{red}{3} \text{ } [\# = 4] \quad \Delta = -104, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
221812315132181231 \quad \Delta = -96, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
221812315132181231 \quad \Delta = -96, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
221812315132181231 \quad \Delta = -96, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
221812315132181231 \quad \Delta = -96, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
221812315132181231^2 \quad \Delta = -96, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
221812315132181231^2 \quad \Delta = -96, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
221812315132181231^3 \quad \Delta = -105, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
221812315132181231^3 \text{ [#=3]} \quad \Delta = -104, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
221812315132181231^3 \text{ [#=5]} \quad \Delta = -104, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
221812315132181231^3 \text{ [#=2]} \quad \Delta = -104, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
\text{Base: } 216123218123216131 & \\
216123218123216131^3 \text{ [#=7]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
216123218123216131^3 \text{ [#=6]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
216123218123216131^3 \text{ [#=9]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
216123218123216131^3 \text{ [#=2]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
216123218123216131^3 \text{ [#=4]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
216123218123216131^3 \text{ [#=8]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
216123218123216131^3 \text{ [#=5]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
216123218123216131^2 \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
216123218123216131^2 \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
216123218123216131^2 \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
216123218123216131^2 \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
216123218123216131^2 \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
216123218123216131^2 \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

$$216123218123216131\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$216123218123216131\textcolor{red}{3} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 161232181232161315

$$161232181232161315\textcolor{red}{2} \quad \Delta = -122, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$161232181232161315\textcolor{red}{2} \quad \Delta = -122, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$161232181232161315\textcolor{red}{2} \quad \Delta = -122, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$161232181232161315\textcolor{red}{2} \quad \Delta = -122, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$161232181232161315\textcolor{red}{2} \quad \Delta = -122, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$161232181232161315\textcolor{red}{2} \quad \Delta = -122, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

Base: 161316123218123215

$$161316123218123215\textcolor{red}{2} \quad \Delta = -122, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$161316123218123215\textcolor{red}{2} \quad \Delta = -122, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$161316123218123215\textcolor{red}{2} \quad \Delta = -122, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$161316123218123215\textcolor{red}{2} \quad \Delta = -122, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$161316123218123215\textcolor{red}{2} \quad \Delta = -122, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$161316123218123215\textcolor{red}{2} \quad \Delta = -122, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: 216131612321812321

$$216131612321812321\textcolor{red}{2} \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$216131612321812321\textcolor{red}{2} \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$216131612321812321\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$216131612321812321\textcolor{red}{3} \text{ [#=7]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$\begin{aligned}
216131612321812321\overset{3}{\color{red}1} \text{ [#=6]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
216131612321812321\overset{3}{\color{red}1} \text{ [#=5]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
216131612321812321\overset{3}{\color{red}1} \text{ [#=4]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
216131612321812321\overset{3}{\color{red}1} \text{ [#=3]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
216131612321812321\overset{3}{\color{red}1} \text{ [#=8]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
216131612321812321\overset{3}{\color{red}1} \text{ [#=2]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
216131612321812321\overset{3}{\color{red}1} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
21613\overset{2}{\color{red}1}612321812321 \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
21613\overset{2}{\color{red}1}612321812321 \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
216131612321812321\overset{2}{\color{red}1} \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
216131612321812321\overset{2}{\color{red}1} \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\text{Base: } 16123218123216131\overset{1}{\color{red}1} & \\
16123218123216131\overset{1}{\color{red}1}\overset{2}{\color{red}1} \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
16123218123216131\overset{1}{\color{red}1}\overset{3}{\color{red}1} \text{ [#=4]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
16123218123216131\overset{1}{\color{red}1}\overset{3}{\color{red}1} \text{ [#=5]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
16123218123216131\overset{1}{\color{red}1}\overset{3}{\color{red}1} \text{ [#=6]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
16123218123216131\overset{1}{\color{red}1}\overset{2}{\color{red}1} \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
16123218123216131\overset{1}{\color{red}1}\overset{3}{\color{red}1} \text{ [#=9]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
16123218123216131\overset{1}{\color{red}1}\overset{3}{\color{red}1} \text{ [#=8]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
16123218123216131\overset{1}{\color{red}1}\overset{3}{\color{red}1} \text{ [#=7]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$${}^1_{16123218123216131\textcolor{red}{3}}_{[\# = 3]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$${}^1_{16123218123216131\textcolor{red}{3}}_{[\# = 2]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$${}^1_{16123218123216131\textcolor{red}{2}} \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$${}^1_{16123218123216131\textcolor{red}{2}} \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$${}^1_{16123218123216131\textcolor{red}{2}} \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: ${}^1_{16131612321812321}$

$${}^1_{16131612321812321\textcolor{red}{3}}_{[\# = 2]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$${}^1_{16131612321812321\textcolor{red}{3}}_{[\# = 8]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$${}^1_{16131612321812321\textcolor{red}{3}}_{[\# = 6]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$${}^1_{16131612321812321\textcolor{red}{3}}_{[\# = 7]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$${}^1_{16131612321812321\textcolor{red}{3}}_{[\# = 5]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$${}^1_{16131612321812321\textcolor{red}{2}} \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$${}^1_{16131612321812321\textcolor{red}{2}} \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$${}^1_{16131612321812321\textcolor{red}{2}} \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$${}^{1\textcolor{red}{2}}_{16131612321812321} \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$${}^1_{16131612321812321\textcolor{red}{2}} \quad \Delta = -70, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$${}^1_{16131612321812321\textcolor{red}{3}}_{[\# = 9]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$${}^1_{16131612321812321\textcolor{red}{3}}_{[\# = 3]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$${}^1_{16131612321812321\textcolor{red}{3}}_{[\# = 4]} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: ${}^1_{216123218123217122}$

$$\begin{aligned}
216123218123217122^2 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
216123218123217122^2 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
216123218123217122^3 \quad \Delta = -98, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
216123218123217122^3 \quad \Delta = -97, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
216123218123217122^2 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
216123218123217122^2 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
216123218123217122^2 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
216123218123217122^2 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

Base: 16123218123217122^1

$$\begin{aligned}
16123218123217122^1 \quad \Delta = -98, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
16123218123217122^1 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
16123218123217122^{12} \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
16123218123217122^1 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
16123218123217122^1 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
16123218123217122^1 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 132171232181232151

$$\begin{aligned}
1512321812321712313 \text{ [#=3]} \quad \Delta = -80, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1512321812321712313 \text{ [#=5]} \quad \Delta = -80, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1512321812321712313 \text{ [#=4]} \quad \Delta = -80, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1512321812321712313 \text{ [#=2]} \quad \Delta = -80, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
132171232181232151^2 \quad \Delta = -72, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

$$132171232181232151 \quad \Delta = -72, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$132171232181232151 \quad \Delta = -72, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$132171232181232151 \quad \Delta = -72, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$132171232181232151 \quad \Delta = -72, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 513217123218123215

$$513217123218123215 \quad \Delta = -124, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$513217123218123215 \quad \Delta = -124, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$513217123218123215 \quad \Delta = -124, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$513217123218123215 \quad \Delta = -124, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$513217123218123215 \quad \Delta = -124, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$513217123218123215 \quad \Delta = -124, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: 151321712321812321

$$151321712321812321 \quad \Delta = -72, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$151321712321812321\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -80, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$151321712321812321\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -80, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$151321712321812321\textcolor{red}{3} \text{ [#=6]} \quad \Delta = -80, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$151321712321812321\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -80, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$151321712321812321\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -80, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$151321712321812321 \quad \Delta = -72, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$151321712321812321 \quad \Delta = -72, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\begin{aligned}
151321712321812321^{\color{red}3} \text{ [#=8]} \quad \Delta = -80, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
151321712321812321^{\color{red}3} \text{ [#=7]} \quad \Delta = -80, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
151321712321812321^{\color{red}3} \text{ [#=2]} \quad \Delta = -80, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
151321712321812321^{\color{red}2} \quad \Delta = -72, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
151321712321812321^{\color{red}2} \quad \Delta = -72, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\text{Base: } 412321812321712321 & \\
412321812321712321^{\color{red}2} \quad \Delta = -50, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
412321812321712321^{\color{red}3} \text{ [#=7]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
412321812321712321^{\color{red}2} \quad \Delta = -50, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
412321812321712321^{\color{red}2} \quad \Delta = -50, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
412321812321712321^{\color{red}2} \quad \Delta = -50, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
412321812321712321^{\color{red}3} \text{ [#=4]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
412321812321712321^{\color{red}3} \text{ [#=3]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
412321812321712321^{\color{red}3} \text{ [#=5]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
412321812321712321^{\color{red}2} \quad \Delta = -50, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
412321812321712321^{\color{red}3} \text{ [#=6]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
412321812321712321^{\color{red}3} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 41 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
412321812321712321^{\color{red}3} \text{ [#=2]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
412321812321712321^{\color{red}3} \text{ [#=9]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
412321812321712321^{\color{red}3} \text{ [#=8]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

Base: 512321712321812321

$$\begin{aligned}
512321712321812321^2 \quad \Delta = -74, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
512321712321812321^3_{\# = 4} \quad \Delta = -82, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
512321712321812321^3_{\# = 7} \quad \Delta = -82, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
512321712321812321^3_{\# = 2} \quad \Delta = -82, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
512321712321812321^3_{\# = 3} \quad \Delta = -82, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
512321712321812321^3_{\# = 6} \quad \Delta = -82, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
512321712321812321^3_{\# = 5} \quad \Delta = -82, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
512321712321812321^3_{\# = 9} \quad \Delta = -82, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
512321712321812321^2 \quad \Delta = -74, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
512321712321812321^2 \quad \Delta = -74, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
512321712321812321^2 \quad \Delta = -74, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
512321712321812321^2 \quad \Delta = -74, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
512321712321812321^3_{\# = 8} \quad \Delta = -82, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

Base: $17^{21}12321812321612$

$$\begin{aligned}
17^{21}12321812321612^3 \quad \Delta = -97, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
17^{21}12321812321612^2 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
17^{21}12321812321612^2 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
17^{21}12321812321612^2 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
17^{21}12321812321612^2 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
17^{21}12321812321612^2 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

$$1\overset{21}{7}12321812321612 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: $1612321812321\overset{21}{7}1$

$$1\overset{1}{6}12321812321\overset{21}{7}1 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1\overset{1}{6}12321812321\overset{221}{7}1 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1\overset{1}{6}12321812321\overset{21}{7}1 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1\overset{12}{6}12321812321\overset{21}{7}1 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1\overset{1}{6}12321812321\overset{21}{7}1 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: 141321812321812231

$$141321812321812231 \quad \Delta = -74, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$141321812321812231 \quad \Delta = -74, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$141321812321812231 \quad \Delta = -74, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$141321812321812231 \quad \Delta = -76, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$141321812321812231 \quad \Delta = -74, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$141321812321812231 \quad \Delta = -74, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$132218123218123141\overset{2}{3} \quad [\# = 2] \quad \Delta = -82, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$141321812321812231\overset{2}{3} \quad [\# = 4] \quad \Delta = -82, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$141321812321812231\overset{2}{3} \quad [\# = 5] \quad \Delta = -82, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$132218123218123141\overset{2}{3} \quad [\# = 3] \quad \Delta = -82, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$141321812321812231\overset{2}{3} \quad [\# = 2] \quad \Delta = -82, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$141321812321812231\overset{2}{3} \quad [\# = 3] \quad \Delta = -82, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: 413218123218122315

$$\begin{aligned}
413218123218122315^2 \quad \Delta = -128, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
413218123218122315^2 \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
413218123218122315^2 \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
413218123218122315^2 \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
413218123218122315^2 \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 50 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
413218123218122315^2 \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
413218123218122315^2 \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 222181232181232151

$$\begin{aligned}
222181232181232151^2 \quad \Delta = -91, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
222181232181232151^2 \quad \Delta = -91, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
222181232181232151^2 \quad \Delta = -91, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
222181232181232151^2 \quad \Delta = -91, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
222181232181232151^2 \quad \Delta = -91, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 132181232181223151

$$\begin{aligned}
132181232181223151^2 \quad \Delta = -98, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
132181232181223151^2 \quad \Delta = -98, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
132181232181223151^2 \quad \Delta = -98, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1513221812321812313^{\text{red}}_{[\#=4]} \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1513221812321812313^{\text{red}}_{[\#=5]} \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1513221812321812313^{\text{red}}_{[\#=2]} \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
151322181232181231\overset{3}{\color{red}} \text{ [#=3]} \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
132181\overset{2}{\color{red}}232181223151 \quad \Delta = -98, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
132181232181\overset{2}{\color{red}}223151 \quad \Delta = -98, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\text{Base: } 222181232181231512 \\
222181\overset{2}{\color{red}}232181231512 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
222181232181231\overset{3}{\color{red}}512 \quad \Delta = -97, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
222181\overset{2}{\color{red}}232181231512 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
222181\overset{2}{\color{red}}232181231512 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
222181232181\overset{2}{\color{red}}231512 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
222181232181231\overset{2}{\color{red}}512 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
222181232181231\overset{2}{\color{red}}512 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\text{Base: } 222181232181231\overset{1}{\color{red}}51 \\
222181232181231\overset{2}{\color{red}}\overset{1}{\color{red}}51 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
222181\overset{2}{\color{red}}232181231\overset{1}{\color{red}}51 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
222181\overset{2}{\color{red}}232181231\overset{1}{\color{red}}51 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
222181232181\overset{2}{\color{red}}231\overset{1}{\color{red}}51 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
222181\overset{2}{\color{red}}232181231\overset{1}{\color{red}}51 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\text{Base: } 412321812321812231 \\
412321812321812231\overset{3}{\color{red}} \text{ [#=5]} \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
41232181232181\overset{2}{\color{red}}2231 \quad \Delta = -76, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
412321812321812231\overset{3}{\color{red}} \text{ [#=4]} \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
412321812321812231^{\color{red}3} \text{ [#=2]} \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
412321812321812231^{\color{red}3} \text{ [#=3]} \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
412321812321812231^{\color{red}3} \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 41 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
412321812321812231^{\color{red}2} \quad \Delta = -76, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
412321812321812231^{\color{red}2} \quad \Delta = -76, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
412321812321812231^{\color{red}2} \quad \Delta = -76, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
412321812321812231^{\color{red}2} \quad \Delta = -76, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 413221812321812321

$$\begin{aligned}
413221812321812321^{\color{red}3} \text{ [#=2]} \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
413221812321812321^{\color{red}2} \quad \Delta = -76, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
413221812321812321^{\color{red}2} \quad \Delta = -76, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
413221812321812321^{\color{red}3} \text{ [#=3]} \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
413221812321812321^{\color{red}2} \quad \Delta = -76, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
413221812321812321^{\color{red}2} \quad \Delta = -76, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
413221812321812321^{\color{red}2} \quad \Delta = -76, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
413221812321812321^{\color{red}3} \text{ [#=7]} \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
413221812321812321^{\color{red}3} \text{ [#=9]} \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
413221812321812321^{\color{red}3} \text{ [#=8]} \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
413221812321812321^{\color{red}3} \text{ [#=6]} \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
413221812321812321^{\color{red}3} \text{ [#=5]} \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

$$413221812321812321\overset{3}{3} \text{ [#=4]} \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 141321812321812313

$$313218123218123141\overset{3}{3} \text{ [#=3]} \quad \Delta = -90, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$14132181232181231\overset{3}{3} \text{ [#=2]} \quad \Delta = -90, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$14132181232181231\overset{3}{3} \quad \Delta = -90, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$313218123218123141\overset{3}{3} \text{ [#=2]} \quad \Delta = -90, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$14132181\overset{2}{2}2321812313 \quad \Delta = -82, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$141\overset{2}{3}21812321812313 \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$14132181232181231\overset{2}{3} \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$14132181\overset{2}{2}2321812313 \quad \Delta = -82, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$14132181232181231\overset{2}{3} \quad \Delta = -82, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$14132181232181231\overset{2}{3} \quad \Delta = -82, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$14132181232181231\overset{2}{3} \quad \Delta = -82, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$141\overset{2}{3}21812321812313 \quad \Delta = -82, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 414132181232181231

$$414132181232181231\overset{3}{3} \text{ [#=3]} \quad \Delta = -110, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$414132181\overset{2}{2}232181231 \quad \Delta = -102, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$414132181232181231\overset{3}{3} \text{ [#=4]} \quad \Delta = -110, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$414132181232181231\overset{3}{3} \text{ [#=5]} \quad \Delta = -110, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$414132181\overset{2}{2}232181231 \quad \Delta = -104, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$\begin{aligned}
414132181232181231^2 \quad \Delta = -102, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
414132181232181231^2 \quad \Delta = -102, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
414132181232181231^2 \quad \Delta = -102, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
414132181232181231^2 \quad \Delta = -102, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
414132181232181231^2 \quad \Delta = -102, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
414132181232181231^3_{[\# = 2]} \quad \Delta = -110, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}
\end{aligned}$$

Base: 412321812321812313

$$\begin{aligned}
412321812321812313^2 \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
412321812321812313^2 \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
412321812321812313^2 \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
412321812321812313^2 \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
412321812321812313^2 \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
412321812321812313^2 \quad \Delta = -86, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
412321812321812313^3 \quad \Delta = -92, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 41 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
412321812321812313^3 \quad \Delta = -92, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
412321812321812313^3_{[\# = 2]} \quad \Delta = -92, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
412321812321812313^2 \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 41232181232181232¹

$$\begin{aligned}
41232181232181232^2 \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
41232181232181232^{41} \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
41232181232181232^{21} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
412321812321812\overset{31}{3}2 \text{ } [\# = 3] \quad \Delta = -86, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
412321812321812\overset{3}{3}2 \quad \Delta = -86, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 41 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
412321812321812\overset{31}{3}2 \text{ } [\# = 2] \quad \Delta = -86, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
412321812321812\overset{21}{3}2 \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
412321812321812\overset{2}{3}2 \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
412321812321812\overset{2}{3}2 \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
412321812321812\overset{2}{3}2 \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\text{Base: } 141321812321812\overset{1}{3}2 \\
141321812321812\overset{31}{3}2 \text{ } [\# = 3] \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
141321812321812\overset{2}{3}2 \quad \Delta = -76, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
141321812321812\overset{2}{3}2 \quad \Delta = -76, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
141321812321812\overset{41}{3}2 \quad \Delta = -104, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
141321812321812\overset{2}{3}2 \quad \Delta = -76, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
141321812321812\overset{2}{3}2 \quad \Delta = -76, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
141321812321812\overset{2}{3}2 \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2^{\overset{31}{3}}141321812321812\overset{1}{3}2 \text{ } [\# = 3] \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
2^{\overset{31}{3}}141321812321812\overset{1}{3}2 \text{ } [\# = 2] \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
141321812321812\overset{31}{3}2 \text{ } [\# = 2] \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
141321812321812\overset{21}{3}2 \quad \Delta = -76, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
141321812321812\overset{2}{3}2 \quad \Delta = -76, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{18122315132181231}$

$$\begin{aligned}
\overset{1}{18122315132181231} \overset{2}{} \quad \Delta = -96, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
\overset{1}{18122315132181231} \overset{2}{} \quad \Delta = -96, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\overset{1}{18122315132181231} \overset{2}{} \quad \Delta = -96, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\overset{1}{18122315132181231} \overset{2}{} \quad \Delta = -96, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\overset{1}{18122315132181231} \overset{2}{} \quad \Delta = -96, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\overset{1}{18122315132181231} \overset{3}{}_{[\# = 4]} \quad \Delta = -104, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\overset{1}{18122315132181231} \overset{3}{}_{[\# = 2]} \quad \Delta = -104, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
\overset{1}{18122315132181231} \overset{3}{}_{[\# = 3]} \quad \Delta = -104, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\overset{1}{18122315132181231} \overset{3}{}_{[\# = 5]} \quad \Delta = -104, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{18123151316131612}$

$$\begin{aligned}
\overset{1}{18123151316131612} \overset{2}{} \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\overset{1}{18123151316131612} \overset{2}{} \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
\overset{1}{18123151316131612} \overset{2}{} \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
\overset{1}{18123151316131612} \overset{2}{} \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
\overset{1}{18123151316131612} \overset{2}{} \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
\overset{1}{18123151316131612} \overset{2}{} \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\overset{1}{18123151316131612} \overset{3}{} \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
\overset{1}{18123151316131612} \overset{2}{} \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{18123216131613151}$

$$\overset{1}{18123216131613151} \overset{2}{} \quad \Delta = -119, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\begin{array}{lll}
{}^1_{18123216131613151} & \Delta = -119, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
{}^1_{18123216131613151} & \Delta = -119, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^1_{18123216131613151} & \Delta = -119, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
{}^1_{18123216131613151} & \Delta = -119, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
{}^1_{18123216131613151} & \Delta = -119, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\text{Base: } {}^1_{1613161315132181} & & \\
{}^{12}_{1613161315132181} & \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^1_{1613161315132181} & \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
{}^1_{1613161315132181} & \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
{}^1_{1613161315132181} & \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
{}^1_{1613161315132181} & \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
{}^1_{1613161315132181} & \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\text{Base: } {}^1_{18123217123217123} & & \\
{}^1_{18123217123217123} & \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
{}^1_{18123217123217123} & \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
{}^1_{18123217123217123} & \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
{}^1_{18123217123217123} & \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
{}^{12}_{18123217123217123} & \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\text{Base: } {}^1_{41321812315132181} & & \\
{}^2_{41321812315132181} & \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
{}^2_{41321812315132181} & \Delta = -124, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
4132181^{21}2315132181^1 \quad \Delta = -124, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
4132181^{21}2315132181^1 \quad \Delta = -124, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
4132181^{21}2315132181^1 \quad \Delta = -124, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
4132181^{21}2315132181^1 \quad \Delta = -124, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
4132181^{21}2315132181^1 \quad \Delta = -124, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: $18^{21}12321712321712$

$$\begin{aligned}
18^{21}12321712321712^2 \quad \Delta = -116, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
18^{21}12321712321712^2 \quad \Delta = -116, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
18^{21}12321712321712^2 \quad \Delta = -116, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
18^{21}12321712321712^2 \quad \Delta = -116, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
18^{21}12321712321712^2 \quad \Delta = -116, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
18^{21}12321712321712^2 \quad \Delta = -116, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{aligned}$$

Base: 171232171232181^{21}

$$\begin{aligned}
171232171232181^{21} \quad \Delta = -116, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
171232171232181^{21} \quad \Delta = -116, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
171232171232181^{21} \quad \Delta = -116, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
171232171232181^{21} \quad \Delta = -116, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
171232171232181^{21} \quad \Delta = -116, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: $18^{21}12315132181231$

$$\begin{aligned}
18^{21}12315132181231^2 \quad \Delta = -96, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
18^{21}12315132181231^2 \quad \Delta = -96, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{21}{1} \overset{2}{8} 12315132181231 \quad \Delta = -96, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& \overset{21}{1} \overset{2}{8} 12315132181231 \quad \Delta = -96, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{21}{1} 8 12315132181231 \quad \Delta = -96, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{21}{1} 8 12315132181 \overset{2}{2} 31 \quad \Delta = -96, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{21}{1} 8 12315132181231 \overset{3}{3} [\# = 5] \quad \Delta = -104, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{21}{1} 8 12315132181231 \overset{3}{3} [\# = 4] \quad \Delta = -104, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{21}{1} 8 12315132181231 \overset{3}{3} [\# = 3] \quad \Delta = -104, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{21}{1} 8 12315132181231 \overset{3}{3} [\# = 2] \quad \Delta = -104, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{221}{1} 8 1232181231512$

$$\begin{aligned}
& \overset{221}{1} \overset{2}{8} 1232181231512 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{221}{1} \overset{2}{8} 1232181231512 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{221}{1} 8 1232181231512 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{221}{1} 8 1232181231512 \quad \Delta = -98, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{221}{1} 8 1232181231 \overset{3}{5} 12 \quad \Delta = -97, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
& \overset{221}{1} 8 1232181231 \overset{2}{5} 12 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{221}{1} 8 1232181231 \overset{2}{5} 12 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{221}{1} 8 1232181231 \overset{2}{5} 12 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{221}{1} \overset{1}{8} 123218123151$

$$\begin{aligned}
& \overset{221}{1} \overset{2}{8} 1232181231 \overset{1}{5} 1 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{221}{1} 8 1232181231 \overset{1}{5} 1 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

$${}^{221\bar{2}}_1 1\ 8\ 123218123151 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$${}^{221}_1 1\ 8\ 1232181231\bar{5}1 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$${}^{221}_1 1\ 8\ 1232181231\bar{5}\bar{1} \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$${}^{221}_1 1\ 8\ 1232181231\bar{5}1 \quad \Delta = -98, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: ${}^{221}_1 1\ 8\ 1232181232151$

$${}^{221}_1 1\ 8\ 12321812321\bar{5}1 \quad \Delta = -91, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$${}^{221}_1 1\ 8\ 12321812321\bar{5}1 \quad \Delta = -91, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$${}^{221}_1 1\ 8\ 1232181232151 \quad \Delta = -91, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$${}^{221}_1 1\ 8\ 12321812321\bar{5}1 \quad \Delta = -91, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$${}^{221\bar{2}}_1 1\ 8\ 1232181232151 \quad \Delta = -91, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$${}^{221}_1 1\ 8\ 1232181232151 \quad \Delta = -100, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: ${}^{321}_1 1\ 8\ 1232171232171$

$${}^{321}_1 1\ 8\ 1232171232171 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$${}^{321\bar{2}}_1 1\ 8\ 1232171232171 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix}$$

$${}^{321}_1 1\ 8\ 12321712321\bar{7}1 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 163 & 25 \\ 2 & 25 & 8 \end{pmatrix}$$

$${}^{321}_1 1\ 8\ 12321712321\bar{7}1 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix}$$

$${}^{321}_1 1\ 8\ 12321712321\bar{7}1 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$${}^{21}_1 321\ 8\ 1232171232171 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: ${}^{1}_{21712321812321712}$

$${}^{2}_{21712321812321712} \quad \Delta = -116, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$${}^{21}_{21712321\ 8\ 12321712} \quad \Delta = -116, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\begin{aligned}
& \overset{2}{2}1712321\overset{1}{8}12321712 \quad \Delta = -116, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 21712321\overset{2}{8}\overset{1}{1}2321712 \quad \Delta = -116, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1712321\overset{1}{8}123217123$

$$\begin{aligned}
& 1712321\overset{1}{8}\overset{2}{1}23217123 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& 1712321\overset{1}{8}12321\overset{2}{7}123 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& 1712321\overset{1}{8}12321\overset{2}{7}123 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 1712321\overset{2}{8}\overset{1}{1}23217123 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 1712321\overset{2}{8}\overset{1}{1}23217123 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& 1712321\overset{2}{8}\overset{1}{1}23217123 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1\overset{12321}{8}12321712321$

$$\begin{aligned}
& 1\overset{3}{8}\overset{12321}{1}2321712321 \text{ [#=4]} \quad \Delta = -30, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& 1\overset{3}{8}\overset{12321}{1}2321712321 \text{ [#=5]} \quad \Delta = -30, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
& 1\overset{12321}{8}12321712321\overset{3}{1} \text{ [#=4]} \quad \Delta = -30, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 1\overset{3}{8}\overset{12321}{1}2321712321 \text{ [#=7]} \quad \Delta = -30, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
& 1\overset{12321}{8}12321712321\overset{3}{1} \text{ [#=7]} \quad \Delta = -30, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& 1\overset{3}{8}\overset{12321}{1}2321712321 \text{ [#=2]} \quad \Delta = -30, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& 1\overset{12321}{8}12321712321\overset{3}{1} \text{ [#=6]} \quad \Delta = -30, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 1\overset{3}{8}\overset{12321}{1}2321712321 \text{ [#=6]} \quad \Delta = -30, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
& 1\overset{12321}{8}12321712321\overset{3}{1} \text{ [#=5]} \quad \Delta = -30, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& 1\overset{12321}{8}12321712321\overset{3}{1} \text{ [#=2]} \quad \Delta = -30, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{12321}{1} \overset{2}{8} 12321712321 \quad \Delta = -22, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{12321}{1} \overset{12321}{8} 12321712321 \quad \Delta = -22, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{12321}{1} \overset{2}{8} 12321712321 \quad \Delta = -22, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{12321}{1} \overset{2}{8} 12321712321 \quad \Delta = -22, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{312321}{1} \overset{8}{8} 12321712321 \text{ [#=8]} \quad \Delta = -30, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
& \overset{12321}{1} \overset{8}{8} 12321712321 \overset{3}{3} \text{ [#=9]} \quad \Delta = -30, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{12321}{1} \overset{8}{8} 12321712321 \overset{3}{3} \text{ [#=3]} \quad \Delta = -30, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{312321}{1} \overset{8}{8} 12321712321 \text{ [#=3]} \quad \Delta = -30, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{12321}{1} \overset{8}{8} 12321712321 \overset{3}{3} \text{ [#=8]} \quad \Delta = -30, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{312321}{1} \overset{8}{8} 12321712321 \text{ [#=9]} \quad \Delta = -30, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{71232181232171232}$

$$\begin{aligned}
& 712321 \overset{21}{8} 1232171232 \quad \Delta = -127, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 712321 \overset{1}{81232171232} \overset{2}{2} \quad \Delta = -127, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 712321 \overset{2}{81232171232} \overset{1}{1} \quad \Delta = -127, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 712321 \overset{21}{81232171232} \quad \Delta = -127, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
& 712321 \overset{1}{81232171232} \overset{2}{2} \quad \Delta = -127, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
& 712321 \overset{12}{81232171232} \quad \Delta = -127, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 225 \quad \begin{pmatrix} 0 & 15 & 2 \\ 15 & 163 & 25 \\ 2 & 25 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{2321}{712321712321} \overset{8}{8} 1$

$$\begin{aligned}
& 712321712321 \overset{2321}{8} \overset{12}{12} \quad \Delta = -127, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 712321 \overset{2}{712321} \overset{2321}{8} 1 \quad \Delta = -127, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
712321712321 \overset{2}{8} \overset{2321}{1} \quad \Delta = -127, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 225 & \begin{pmatrix} 0 & 15 & 2 \\ 15 & 163 & 25 \\ 2 & 25 & 8 \end{pmatrix} \\
712321712321 \overset{2321}{8} \overset{2}{1} \quad \Delta = -127, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
712321712321 \overset{2}{8} \overset{2321}{1} \quad \Delta = -127, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
712321712321 \overset{2321}{8} \overset{1}{1} \quad \Delta = -127, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1712321812321712

$$\begin{aligned}
171232181 \overset{12}{2321712} \quad \Delta = -116, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
17123218 \overset{21}{12321712} \quad \Delta = -116, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
171232181 \overset{1}{2321712} \overset{2}{1} \quad \Delta = -116, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
171232181 \overset{1}{2321712} \overset{2}{1} \quad \Delta = -116, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
171 \overset{12}{2321812321712} \quad \Delta = -116, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
171232181 \overset{21}{2321712} \quad \Delta = -116, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 171232181232171

$$\begin{aligned}
171232181 \overset{21}{232171} \quad \Delta = -116, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
171 \overset{12}{232181232171} \quad \Delta = -116, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
17123218 \overset{21}{1232171} \quad \Delta = -116, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: 31612321812321812

$$\begin{aligned}
316123218 \overset{21}{12321812} \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
3161232181 \overset{2}{2321812} \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
3161232181 \overset{21}{2321812} \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
3161232181 \overset{12}{2321812} \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
3161232181 \overset{1}{2321812} \overset{2}{1} \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{2}{3}1612321\overset{1}{8}12321812 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 31612321\overset{1}{8}12321\overset{2}{8}12 \quad \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1612321\overset{1}{8}123218122$

$$\begin{aligned}
& 1612321\overset{1}{8}12321\overset{3}{8}122 \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
& 1612321\overset{1}{8}12321\overset{2}{8}122 \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 1612321\overset{1}{8}12321\overset{2}{8}122 \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 161\overset{2}{2}321\overset{1}{8}123218122 \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 1612321\overset{12}{8}123218122 \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& 1612321\overset{21}{8}123218122 \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 1612321\overset{21}{8}123218122 \quad \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1\overset{12321}{8}12321812231$

$$\begin{aligned}
& 1\overset{12321}{8}12321812231\overset{3}{3} \text{ [#=2]} \quad \Delta = -56, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
& 1\overset{312321}{8}12321812231 \text{ [#=3]} \quad \Delta = -56, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& 1\overset{312321}{8}12321812231 \text{ [#=2]} \quad \Delta = -56, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& 1\overset{312321}{8}12321812231 \text{ [#=7]} \quad \Delta = -56, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
& 1\overset{312321}{8}12321812231 \text{ [#=6]} \quad \Delta = -56, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
& 1\overset{12321}{8}12321\overset{2}{8}12231 \quad \Delta = -48, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 1\overset{312321}{8}12321812231 \text{ [#=8]} \quad \Delta = -56, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
& 1\overset{312321}{8}12321812231 \text{ [#=9]} \quad \Delta = -56, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
& 1\overset{12321}{8}12321812231\overset{3}{3} \text{ [#=5]} \quad \Delta = -56, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{3}{1} \overset{12321}{8} 12321812231_{[\# = 5]} \quad \Delta = -56, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
& \overset{12321}{1} \overset{12321}{8} 12321812231 \quad \Delta = -48, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{12321}{1} \overset{12321}{8} 12321812231 \overset{3}{3}_{[\# = 4]} \quad \Delta = -56, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{3}{1} \overset{12321}{8} 12321812231_{[\# = 4]} \quad \Delta = -56, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& \overset{12321}{1} \overset{12321}{8} 12321812231 \overset{3}{3}_{[\# = 3]} \quad \Delta = -56, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{12321}{1} \overset{12321}{8} 12321812231 \overset{2}{2} \quad \Delta = -48, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{12321}{1} \overset{12321}{8} 12321812231 \overset{2}{2} \quad \Delta = -48, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \text{Base: } \overset{1}{6} 1232181232181223 \\
& \overset{1}{6} 1232181232181223 \overset{2}{2} \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& \overset{1}{6} 1232181232181223 \overset{12}{12} \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{2}{6} \overset{1}{12321} 1232181223 \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{2}{6} 1232181232181223 \overset{1}{1} \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& \overset{21}{6} 1232181232181223 \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{1}{6} 1232181232181223 \overset{2}{2} \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \text{Base: } \overset{12321}{1} \overset{12321}{8} 12321812313 \\
& \overset{12321}{1} \overset{12321}{8} 12321812313 \overset{2}{2} \quad \Delta = -56, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{3}{1} \overset{12321}{8} 12321812313_{[\# = 6]} \quad \Delta = -64, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
& \overset{3}{1} \overset{12321}{8} 12321812313_{[\# = 7]} \quad \Delta = -64, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
& \overset{3}{1} \overset{12321}{8} 12321812313_{[\# = 8]} \quad \Delta = -64, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
& \overset{3}{1} \overset{12321}{8} 12321812313_{[\# = 9]} \quad \Delta = -64, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{ll}
1 \overset{3}{8} \overset{12321}{12321812313} \text{ [#=5]} & \Delta = -64, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
1 \overset{3}{8} \overset{12321}{12321812313} \text{ [#=4]} & \Delta = -64, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1 \overset{12321}{8} \overset{3}{12321812313} \text{ [#=2]} & \Delta = -64, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1 \overset{3}{8} \overset{12321}{12321812313} \text{ [#=2]} & \Delta = -64, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1 \overset{12321}{8} \overset{3}{12321812313} & \Delta = -64, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1 \overset{3}{8} \overset{12321}{12321812313} \text{ [#=3]} & \Delta = -64, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1 \overset{12321}{8} \overset{2}{12321812313} & \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1 \overset{12321}{8} \overset{2}{12321812313} & \Delta = -56, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1 \overset{12321}{8} \overset{2}{12321812313} & \Delta = -56, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{12321}{8} \overset{2}{12321812313} & \Delta = -56, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \overset{12321}{8} \overset{2}{12321812313} & \Delta = -56, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{1321}{8} 123218123141 & \\
1 \overset{3}{8} \overset{1321}{123218123141} \text{ [#=5]} & \Delta = -82, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
1 \overset{3}{8} \overset{1321}{123218123141} \text{ [#=4]} & \Delta = -82, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1 \overset{1321}{8} \overset{2}{123218123141} & \Delta = -74, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1 \overset{1321}{8} \overset{2}{123218123141} & \Delta = -74, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1 \overset{1321}{8} \overset{2}{123218123141} & \Delta = -74, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1 \overset{1321}{8} 123218123141 & \Delta = -74, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1 \overset{1321}{8} \overset{2}{123218123141} & \Delta = -74, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \overset{1321}{8} \overset{2}{123218123141} & \Delta = -76, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{l}
\begin{array}{l}
1^{1321} \ 8 \ 123218123141\textcolor{red}{3} \ [\# = 3] \quad \Delta = -82, \ n_+ = 1, \ n_- = 18, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
1^{31321} \ 8 \ 123218123141 \ [\# = 2] \quad \Delta = -82, \ n_+ = 1, \ n_- = 18, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1^{1321} \ 8 \ 123218123141\textcolor{red}{3} \ [\# = 2] \quad \Delta = -82, \ n_+ = 1, \ n_- = 18, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
1^{31321} \ 8 \ 123218123141 \ [\# = 3] \quad \Delta = -82, \ n_+ = 1, \ n_- = 18, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{array} \\
\\
\text{Base: } 1^{512321} \ 8 \ 1232181231 \\
\begin{array}{l}
1^{512321} \ 8 \ 1232181231 \quad \Delta = -100, \ n_+ = 1, \ n_- = 18, \ n_0 = 0, \ |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1^{512321} \ 8 \ 1232181231\textcolor{red}{2} \quad \Delta = -100, \ n_+ = 1, \ n_- = 18, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1^{512321} \ 8 \ 1232181231\textcolor{red}{2} \quad \Delta = -100, \ n_+ = 1, \ n_- = 18, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1^{512321} \ 8 \ 1232181231\textcolor{red}{2} \quad \Delta = -100, \ n_+ = 1, \ n_- = 18, \ n_0 = 0, \ |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
512321\textcolor{red}{21} \ 8 \ 1232181231 \quad \Delta = -100, \ n_+ = 1, \ n_- = 18, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1^{512321} \ 8 \ 1232181231 \quad \Delta = -100, \ n_+ = 1, \ n_- = 18, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1^{512321} \ 8 \ 1232181231\textcolor{red}{3} \ [\# = 4] \quad \Delta = -108, \ n_+ = 1, \ n_- = 18, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1^{512321} \ 8 \ 1232181231\textcolor{red}{3} \ [\# = 3] \quad \Delta = -108, \ n_+ = 1, \ n_- = 18, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1^{512321} \ 8 \ 1232181231\textcolor{red}{3} \ [\# = 5] \quad \Delta = -108, \ n_+ = 1, \ n_- = 18, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1^{512321} \ 8 \ 1232181231\textcolor{red}{3} \ [\# = 2] \quad \Delta = -108, \ n_+ = 1, \ n_- = 18, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}
\end{array} \\
\\
\text{Base: } 41321812321^{51321} \ 8 \ 1 \\
\begin{array}{l}
41321\textcolor{red}{2} \ 8 \ 12321^{51321} \ 8 \ 1 \quad \Delta = -126, \ n_+ = 1, \ n_- = 18, \ n_0 = 0, \ |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
41321\textcolor{red}{2} \ 8 \ 12321^{51321} \ 8 \ 1 \quad \Delta = -128, \ n_+ = 1, \ n_- = 18, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
41321812321^{51321} \ 8 \ 1\textcolor{red}{2} \quad \Delta = -126, \ n_+ = 1, \ n_- = 18, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
41321\textcolor{red}{2} \ 8 \ 12321^{51321} \ 8 \ 1 \quad \Delta = -126, \ n_+ = 1, \ n_- = 18, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{array}
\end{array}$$

$$41321812321 \overset{51321}{8} 1 \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$41321812321 \overset{51321}{8} 1 \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$41321812321 \overset{2}{8} \overset{51321}{1} \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$41321812321 \overset{2}{8} \overset{51321}{1} \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

Base: $1 \overset{151321}{8} 1232181231$

$$1 \overset{151321}{8} 1232181231 \overset{2}{1} \quad \Delta = -98, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{151321}{8} 1232181231 \overset{3}{1} [\# = 3] \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1 \overset{151321}{8} 1232181231 \overset{3}{1} [\# = 4] \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{151321}{8} 1232181231 \quad \Delta = -98, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$1 \overset{151321}{8} 1232181231 \quad \Delta = -98, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$151321 \overset{21}{8} 1232181231 \quad \Delta = -98, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \overset{151321}{8} 1232181231 \overset{2}{1} \quad \Delta = -98, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{151321}{8} 1232181231 \overset{2}{1} \quad \Delta = -98, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$1 \overset{151321}{8} 1232181231 \overset{3}{1} [\# = 5] \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{151321}{8} 1232181231 \overset{3}{1} [\# = 2] \quad \Delta = -106, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

Base: $412321812321 \overset{1321}{8} 1$

$$\overset{2}{4} 12321812321 \overset{1321}{8} 1 \quad \Delta = -76, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$412321812321 \overset{31321}{8} 1 [\# = 4] \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$412321812321 \overset{31321}{8} 1 [\# = 5] \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$412321812321 \overset{31321}{8} 1 [\# = 2] \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$412321812321 \overset{3}{8} \overset{1321}{1} \text{ [#}=3] \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$412321812321 \overset{1321}{8} \overset{1}{1} \quad \Delta = -76, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$412321812321 \overset{2}{8} \overset{1321}{1} \quad \Delta = -76, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$412321812321 \overset{2}{8} \overset{1321}{1} \quad \Delta = -76, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$412321812321 \overset{3}{8} \overset{1321}{1} \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 41 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$412321812321 \overset{2}{8} \overset{1321}{1} \quad \Delta = -76, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: 41321812321812321

$$41321812321812321 \overset{1}{8} \overset{1321}{1} \overset{3}{1} \text{ [#}=4] \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$41321812321812321 \overset{2}{8} \overset{1}{1} \quad \Delta = -76, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$41321812321812321 \overset{21}{8} \overset{1}{1} \quad \Delta = -76, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$41321812321812321 \overset{12}{8} \overset{1}{1} \quad \Delta = -76, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$41321812321812321 \overset{1}{8} \overset{1321}{1} \overset{3}{1} \text{ [#}=8] \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$41321812321812321 \overset{1}{8} \overset{1321}{1} \overset{3}{1} \text{ [#}=6] \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$41321812321812321 \overset{1}{8} \overset{1321}{1} \overset{3}{1} \text{ [#}=5] \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$41321812321812321 \overset{1}{8} \overset{1321}{1} \overset{3}{1} \text{ [#}=7] \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$41321812321812321 \overset{1}{8} \overset{1321}{1} \overset{3}{1} \text{ [#}=3] \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$41321812321812321 \overset{2}{8} \overset{1}{1} \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$41321812321812321 \overset{1}{8} \overset{2}{1} \quad \Delta = -76, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$41321812321812321 \overset{21}{8} \overset{1}{1} \quad \Delta = -76, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$41321812321812321 \overset{1}{8} \overset{2}{1} \quad \Delta = -76, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$\begin{aligned}
& 41321812321812321\overset{1}{3} \text{ }_{[\# = 2]} \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 41321812321812321\overset{1}{3} \text{ }_{[\# = 9]} \quad \Delta = -84, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
& \text{Base: } 1\overset{12321}{8}1232181232 \\
& 1\overset{12321}{8}1232181232 \quad \Delta = -50, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& 1\overset{12321}{8}12321812\overset{21}{3}2 \quad \Delta = -50, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 1\overset{12321}{8}1232181\overset{2}{3}2 \quad \Delta = -50, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 1\overset{312321}{8}1232181232 \text{ }_{[\# = 4]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& 1\overset{12321}{8}1232181\overset{2}{3}2 \quad \Delta = -50, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 1\overset{12321}{8}12321812\overset{1}{3}2 \quad \Delta = -50, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 1\overset{312321}{8}1232181232 \text{ }_{[\# = 9]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
& 1\overset{312321}{8}1232181232 \text{ }_{[\# = 8]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
& 1\overset{312321}{8}1232181232 \text{ }_{[\# = 7]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
& 1\overset{312321}{8}1232181232 \text{ }_{[\# = 6]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
& 1\overset{312321}{8}1232181232 \text{ }_{[\# = 5]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
& 1\overset{12321}{8}12321812\overset{31}{3}2 \text{ }_{[\# = 3]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 1\overset{312321}{8}1232181232 \text{ }_{[\# = 2]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& 1\overset{12321}{8}12321812\overset{31}{3}2 \text{ }_{[\# = 2]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 1\overset{312321}{8}1232181232 \text{ }_{[\# = 3]} \quad \Delta = -58, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& 1\overset{12321}{8}12321812\overset{41}{3}2 \quad \Delta = -78, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
& \text{Base: } 1\overset{1}{8}12321812321613
\end{aligned}$$

$$\begin{array}{ll}
1\bar{8}12321\bar{8}12321613 & \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1\bar{8}12321\bar{8}^{21}12321613 & \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1\bar{8}12321\bar{8}12321\bar{8}^{21}613 & \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1\bar{8}12321\bar{8}^{12}12321613 & \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1\bar{8}12321\bar{8}^{21}12321613 & \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
1\bar{8}^{12}12321\bar{8}12321613 & \Delta = -125, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\text{Base: } 1\bar{8}12321\bar{8}^{21}1232161 & \\
1\bar{8}^{21}\bar{8}^{21}12321\bar{8}1232161 & \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1\bar{8}^{21}12321\bar{8}^{21}1232161 & \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1\bar{8}^{21}12321\bar{8}^{12}1232161 & \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1\bar{8}^{21}12321\bar{8}^{12}1232161 & \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1\bar{8}^{21}12321\bar{8}^{12}1232161 & \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1\bar{8}^{21}12321\bar{8}^{21}1232161 & \Delta = -117, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\text{Base: } 612321\bar{8}12321\bar{8}^{321} & \\
612321\bar{8}^{12}12321\bar{8}^{321} & \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
612321\bar{8}^{21}12321\bar{8}^{321} & \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
612321\bar{8}^{12}12321\bar{8}^{2321} & \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
612321\bar{8}^{21}12321\bar{8}^{321} & \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
612321\bar{8}^{12}12321\bar{8}^{321} & \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
612321\bar{8}^{21}12321\bar{8}^{321} & \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{array}$$

$$612321812321 \overset{1}{8} \overset{321}{12} \quad \Delta = -126, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: $1 \overset{1321}{8} \overset{1}{12321812321}$

$$1 \overset{1321}{8} \overset{1}{12321812321} \overset{1}{3} [\# = 2] \quad \Delta = -56, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{1321}{8} \overset{1}{12321812321} \overset{1}{3} [\# = 8] \quad \Delta = -56, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$1 \overset{1321}{8} \overset{1}{12321812321} \overset{2}{3} \quad \Delta = -48, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{1321}{8} \overset{2}{12321812321} \overset{1}{3} \quad \Delta = -48, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{1321}{8} \overset{1}{12321812321} \overset{2}{3} \quad \Delta = -48, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{1321}{8} \overset{1}{12321812321} \quad \Delta = -48, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1 \overset{31321}{8} \overset{1}{12321812321} [\# = 3] \quad \Delta = -56, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1 \overset{1321}{8} \overset{1}{12321812321} \overset{1}{3} [\# = 3] \quad \Delta = -56, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1 \overset{31321}{8} \overset{1}{12321812321} [\# = 4] \quad \Delta = -56, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1 \overset{1321}{8} \overset{1}{12321812321} \overset{1}{3} [\# = 4] \quad \Delta = -56, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1 \overset{31321}{8} \overset{1}{12321812321} [\# = 5] \quad \Delta = -56, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$1 \overset{1321}{8} \overset{1}{12321812321} \overset{1}{3} [\# = 5] \quad \Delta = -56, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$1 \overset{1321}{8} \overset{1}{12321812321} \overset{1}{3} [\# = 6] \quad \Delta = -56, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$1 \overset{1321}{8} \overset{1}{12321812321} \overset{1}{3} [\# = 7] \quad \Delta = -56, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$1 \overset{31321}{8} \overset{1}{12321812321} [\# = 2] \quad \Delta = -56, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{1321}{8} \overset{1}{12321812321} \overset{1}{3} [\# = 9] \quad \Delta = -56, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

Base: $21 \overset{21}{8} 1232181231512$

$$21 \overset{21}{8} \overset{21}{1232181231512} \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\begin{aligned}
21\bar{8}1232181231512 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21\bar{8}1232181231512 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21\bar{8}1232181231512 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
21\bar{8}1232181231512 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21\bar{8}1232181231512 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21\bar{8}1232181231512 \quad \Delta = -97, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}
\end{aligned}$$

Base: $21\bar{8}123218123151$

$$\begin{aligned}
21\bar{8}123218123151 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21\bar{8}123218123151 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21\bar{8}123218123151 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21\bar{8}123218123151 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21\bar{8}123218123151 \quad \Delta = -89, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: $21\bar{8}1232181232151$

$$\begin{aligned}
21\bar{8}1232181232151 \quad \Delta = -91, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21\bar{8}1232181232151 \quad \Delta = -91, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21\bar{8}1232181232151 \quad \Delta = -91, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21\bar{8}1232181232151 \quad \Delta = -91, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21\bar{8}1232181232151 \quad \Delta = -91, n_+ = 1, n_- = 18, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

$T = 19$ (2286 bases)

Base: 12122314141414141322

$$12122314141414141322 \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$\begin{aligned}
12122314141414141322^2 \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
12122314141414141322^2 \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 289 & \begin{pmatrix} 0 & 17 & 2 \\ 17 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
12122314141414141322^2 \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 144 & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 130 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
12122314141414141322^2 \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 225 & \begin{pmatrix} 0 & 15 & 2 \\ 15 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
12122314141414141322^2 \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
12122314141414141322^2 \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}
\end{aligned}$$

Base: 23141414141413221121

$$\begin{aligned}
23141414141413221121^2 \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
23141414141413221121^2 \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 130 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
23141414141413221121^2 \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 225 & \begin{pmatrix} 0 & 15 & 2 \\ 15 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
23141414141413221121^2 \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
23141414141413221121^2 \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
23141414141413221121^2 \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
23141414141413221121^2 \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
23141414141413221121^2 \quad \Delta = -130, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 31414141414132211212

$$\begin{aligned}
31414141414132211212^2 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
31414141414132211212^3 \quad \Delta = -135, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
31414141414132211212^2 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
31414141414132211212^2 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
31414141414132211212 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
31414141414132211212 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
31414141414132211212 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
31414141414132211212 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 108 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
31414141414132211212 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: 14141322112122314141

$$\begin{aligned}
14141322112122314141 \quad \Delta = -100, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 58 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
14141322112122314141 \quad \Delta = -100, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 76 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
14141322112122314141 \quad \Delta = -100, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
14141322112122314141\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -108, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
14141322112122314141\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -108, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: 41414132211212231414

$$\begin{aligned}
41414132211212231414 \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 89 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
41414132211212231414 \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
41414132211212231414 \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
41414132211212231414 \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
41414132211212231414 \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 144 & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
41414132211212231414 \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
41414132211212231414 \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 14141413221121223141

$$14141413221121223141 \quad \Delta = -100, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\begin{aligned}
14141413221121223141^2 \quad \Delta = -100, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
14141413221121223141^2 \quad \Delta = -100, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
14141413221121223141^2 \quad \Delta = -100, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 103 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
141322112122314141413^{\#2} \quad \Delta = -108, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
141414132211212231413^{\#2} \quad \Delta = -108, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
141414132211212231413^{\#3} \quad \Delta = -108, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
14141413221121223141^2 \quad \Delta = -100, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 110 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
14141413221121223141^2 \quad \Delta = -100, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 58 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
141322112122314141413^{\#3} \quad \Delta = -108, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

Base: 41414141322112122314

$$\begin{aligned}
41414141322112122314^2 \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 144 & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
41414141322112122314^2 \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
41414141322112122314^2 \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 225 & \begin{pmatrix} 0 & 15 & 2 \\ 15 & 118 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
41414141322112122314^2 \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 256 & \begin{pmatrix} 0 & 16 & 2 \\ 16 & 142 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
41414141322112122314^2 \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
41414141322112122314^2 \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
41414141322112122314^2 \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: 14141414132211212231

$$\begin{aligned}
132211212231414141413^{\#2} \quad \Delta = -108, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
141414141322112122313^{\#4} \quad \Delta = -108, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
14141414132211212231\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -108, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
14141414132211212231\textcolor{red}{2} \quad \Delta = -100, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 70 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
14141414132211212231\textcolor{red}{2} \quad \Delta = -100, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
14141414132211212231\textcolor{red}{2} \quad \Delta = -100, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
14141414132211212231\textcolor{red}{2} \quad \Delta = -100, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
14141414132211212231\textcolor{red}{2} \quad \Delta = -100, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
14141414132211212231\textcolor{red}{2} \quad \Delta = -100, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
14141414132211212231\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -108, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
13221121223141414141\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -108, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
14141414132211212231\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -108, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
\textbf{Base: } 41414141413221121223 \\
41414141413221121223\textcolor{red}{2} \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 289 \quad & \begin{pmatrix} 0 & 17 & 2 \\ 17 & 151 & 25 \\ 2 & 25 & 8 \end{pmatrix} \\
41414141413221121223\textcolor{red}{2} \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 400 \quad & \begin{pmatrix} 0 & 20 & 2 \\ 20 & 218 & 30 \\ 2 & 30 & 8 \end{pmatrix} \\
41414141413221121223\textcolor{red}{2} \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 256 \quad & \begin{pmatrix} 0 & 16 & 2 \\ 16 & 142 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
41414141413221121223\textcolor{red}{2} \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 144 \quad & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
41414141413221121223\textcolor{red}{2} \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
41414141413221121223\textcolor{red}{2} \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
41414141413221121223\textcolor{red}{2} \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\textbf{Base: } 14141414141322112122 \\
14141414141322112122\textcolor{red}{2} \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
14141414141322112122^2 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
14141414141322112122^2 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
14141414141322112122^2 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
14141414141322112122^2 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 88 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
14141414141322112122^2 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
221121223141414141413^{\#2} \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
14141414141322112122^2 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
221121223141414141413^{\#3} \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 31414132211212231512

$$\begin{aligned}
31414132211212231512^2 \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
31414132211212231512^2 \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
31414132211212231512^2 \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
31414132211212231512^2 \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
31414132211212231512^2 \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
31414132211212231512^2 \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
31414132211212231512^2 \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
31414132211212231512^3 \quad \Delta = -112, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
31414132211212231512^3 \quad \Delta = -112, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}
\end{aligned}$$

Base: 14141322112122315123

$$\begin{aligned}
321513221121223141413^{\#3} \quad \Delta = -113, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
14141322112122315123^2 \quad \Delta = -105, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{ll}
14141322112122315123^2 & \Delta = -105, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
321513221121223141413^{\#=2} & \Delta = -113, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
14141322112122315123^2 & \Delta = -105, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
14141322112122315123^2 & \Delta = -105, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
14141322112122315123^2 & \Delta = -105, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
14141322112122315123^2 & \Delta = -105, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\text{Base: } 32151322112122315123 & \\
32151322112122315123^2 & \Delta = -110, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
32151322112122315123^2 & \Delta = -110, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
32151322112122315123^2 & \Delta = -110, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\text{Base: } 23215132211212231512 & \\
23215132211212231512^2 & \Delta = -103, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
23215132211212231512^2 & \Delta = -103, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
23215132211212231512^2 & \Delta = -103, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
23215132211212231512^2 & \Delta = -103, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
23215132211212231512^2 & \Delta = -103, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
23215132211212231512^3 & \Delta = -111, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
23215132211212231512^2 & \Delta = -103, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
\text{Base: } 23141322112122315131 & \\
23141322112122315131^2 & \Delta = -85, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
231413221121223151313^{\#=5} & \Delta = -93, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{ll}
23141322112122315131^2 & \Delta = -85, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
23141322112122315131^2 & \Delta = -85, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
23141322112122315131^2 & \Delta = -85, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
23141322112122315131^2 & \Delta = -87, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
23141322112122315131^2 & \Delta = -85, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
23141322112122315131^2 & \Delta = -85, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
23141322112122315131^3_{[\# = 7]} & \Delta = -93, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
23141322112122315131^3_{[\# = 6]} & \Delta = -93, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
23141322112122315131^3_{[\# = 9]} & \Delta = -93, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
23141322112122315131^3_{[\# = 8]} & \Delta = -93, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
23141322112122315131^3_{[\# = 2]} & \Delta = -93, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
23141322112122315131^3_{[\# = 3]} & \Delta = -93, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
23141322112122315131^3_{[\# = 4]} & \Delta = -93, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\text{Base: } 41315132211212231413 & \\
41315132211212231413^2 & \Delta = -112, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
41315132211212231413^2 & \Delta = -112, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
41315132211212231413^2 & \Delta = -112, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
41315132211212231413^3 & \Delta = -120, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
41315132211212231413^2 & \Delta = -112, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
41315132211212231413^2 & \Delta = -112, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{array}$$

$$41315132211212231413 \quad \Delta = -112, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$41315132211212231413 \quad \Delta = -120, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$41315132211212231413 \quad \Delta = -113, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$41315132211212231413 \quad \Delta = -112, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 22313221121223151321

$$22313221121223151321 \quad \Delta = -58, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$22313221121223151321 \quad \Delta = -58, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$22313221121223151321 \quad \Delta = -58, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$22313221121223151321 \quad \Delta = -58, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$22313221121223151321 \quad \Delta = -58, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$22313221121223151321 \text{ }_{[\# = 4]} \quad \Delta = -66, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$22313221121223151321 \text{ }_{[\# = 2]} \quad \Delta = -66, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$22313221121223151321 \quad \Delta = -66, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 60 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$22313221121223151321 \text{ }_{[\# = 3]} \quad \Delta = -66, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$22313221121223151321 \text{ }_{[\# = 9]} \quad \Delta = -66, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix}$$

$$22313221121223151321 \text{ }_{[\# = 7]} \quad \Delta = -66, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$22313221121223151321 \text{ }_{[\# = 6]} \quad \Delta = -66, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$22313221121223151321 \text{ }_{[\# = 5]} \quad \Delta = -66, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$22313221121223151321 \text{ }_{[\# = 8]} \quad \Delta = -66, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

Base: 2222211212231513221

$$\begin{aligned}
22222211212231513221\overset{2}{3} \text{ [#=9]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
22222211212231513221\overset{2}{3} \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
22222211212231513221\overset{2}{3} \text{ [#=3]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
22222211212231513221\overset{2}{3} \text{ [#=4]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
22222211212231513221\overset{2}{3} \text{ [#=5]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
22222211212231513221\overset{2}{3} \text{ [#=6]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
22222211212231513221\overset{2}{3} \text{ [#=7]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
22222211212231513221\overset{2}{3} \text{ [#=8]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
22222211212231513221\overset{2}{3} \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
22222211212231513221\overset{2}{3} \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
22222211212231513221\overset{2}{3} \text{ [#=2]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
22222211212231513221\overset{2}{3} \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\text{Base: } 23132211212231513215 & \\
23132211212231513215\overset{2}{3} \quad \Delta = -110, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
23132211212231513215\overset{3}{3} \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 32 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
23132211212231513215\overset{2}{3} \quad \Delta = -110, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
23132211212231513215\overset{2}{3} \quad \Delta = -112, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
23132211212231513215\overset{2}{3} \quad \Delta = -110, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
23132211212231513215\overset{2}{3} \quad \Delta = -110, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
23132211212231513215\overset{2}{3} \quad \Delta = -110, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

$$23132211212231513215 \quad \Delta = -110, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 23132211212231513151

$$23132211212231513151 \quad \Delta = -109, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$23132211212231513151 \quad \Delta = -109, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$23132211212231513151 \quad \Delta = -109, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$23132211212231513151 \quad \Delta = -109, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$23132211212231513151 \quad \Delta = -111, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$23132211212231513151 \quad \Delta = -109, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$23132211212231513151 \quad \Delta = -109, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$23132211212231513151 \quad \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 32 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$23132211212231513151 \quad \Delta = -110, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: 31322112122315131513

$$31322112122315131513 \quad \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$31322112122315131513 \quad \Delta = -116, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$31322112122315131513 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$31322112122315131513 \text{ [#=2]} \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$31322112122315131513 \quad \Delta = -116, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$31322112122315131513 \quad \Delta = -116, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$31322112122315131513 \quad \Delta = -116, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$31322112122315131513 \quad \Delta = -116, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$31322112122315131513^2 \quad \Delta = -116, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$31322112122315131513^2 \quad \Delta = -116, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 13221121223151315132

$$13221121223151315132^2 \quad \Delta = -109, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$13221121223151315132^2 \quad \Delta = -109, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

$$13221121223151315132^2 \quad \Delta = -109, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$231513151322112122313^{\#2} \quad \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$231513151322112122313^{\#3} \quad \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$13221121223151315132^2 \quad \Delta = -109, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$231513151322112122313^{\#4} \quad \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$231513151322112122313^{\#5} \quad \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$13221121223151315132^2 \quad \Delta = -109, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$13221121223151315132^2 \quad \Delta = -110, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$13221121223151315132^2 \quad \Delta = -109, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

Base: 3141413221121223151¹

$$3141413221121223151^3 \quad \Delta = -112, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$3141413221121223151^2 \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$3141413221121223151^2 \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$3141413221121223151^2 \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$3141413221121223151^2 \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$31414132211212231\overset{21}{5}1 \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$31414132211212231\overset{2}{5}\overset{1}{1} \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$\text{Base: } 1\overset{23215132211212231}{5}1$$

$$1\overset{23215132211212231}{5}1 \quad \Delta = -103, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1\overset{23215132211212231}{5}1 \quad \Delta = -103, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1\overset{23215132211212231}{5}1 \quad \Delta = -103, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1\overset{23215132211212231}{5}1 \quad \Delta = -103, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1\overset{23215132211212231}{5}1 \quad \Delta = -103, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\text{Base: } 1414132211212231\overset{1}{5}13$$

$$1414132211212231\overset{2}{5}\overset{1}{1}3 \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1414132211212231\overset{21}{5}13 \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1414132211212231\overset{31}{5}13 \quad \Delta = -112, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1414132211212231\overset{21}{5}13 \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1414132211212231\overset{12}{5}\overset{1}{1}3 \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1414132211212231\overset{2}{5}\overset{1}{1}3 \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1414132211212231\overset{2}{5}\overset{1}{1}3 \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1414132211212231\overset{2}{5}\overset{1}{1}3 \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$31\overset{31414132211212231}{5}1 \text{ [#=3]} \quad \Delta = -112, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$31\overset{31414132211212231}{5}1 \text{ [#=2]} \quad \Delta = -112, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\text{Base: } 31\overset{3215132211212231}{5}1$$

$$31\overset{3215132211212231}{5}1 \quad \Delta = -109, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$\begin{array}{ll}
31^{32151322111212231}_5 1 & \Delta = -109, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
31^{32151322111212231}_5 1 & \Delta = -109, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
31^{32151322111212231}_5 1 & \Delta = -109, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
31^{232151322111212231}_5 1 & \Delta = -109, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
31^{32151322111212231}_5 12 & \Delta = -109, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
31^{32151322111212231}_5 1 & \Delta = -109, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
31^{32151322111212231}_5 13 & \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{array}$$

Base: $31^{3151322111212231}_5 1$

$$\begin{array}{ll}
31^{3151322111212231}_5 1 & \Delta = -108, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
31^{3151322111212231}_5 12 & \Delta = -108, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
31^{23151322111212231}_5 1 & \Delta = -108, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
31^{3151322111212231}_5 13 & \Delta = -116, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
31^{3151322111212231}_5 1 & \Delta = -108, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{array}$$

Base: $231^{2151322111212231}_5 1$

$$\begin{array}{ll}
231^{2151322111212231}_5 12 & \Delta = -101, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
231^{2151322111212231}_5 13 & \Delta = -109, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
231^{2151322111212231}_5 1 & \Delta = -101, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
231^{2151322111212231}_5 1 & \Delta = -101, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
231^{2151322111212231}_5 1 & \Delta = -101, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
231^{2151322111212231}_5 1 & \Delta = -101, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{array}$$

$$231 \overset{2}{1} 151322111212231 \quad \Delta = -101, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$231 \overset{2}{1} 151322111212231 \quad \Delta = -101, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$231 \overset{2}{1} 151322111212231 \quad \Delta = -109, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$\text{Base: } 1 \overset{23151322111212231}{5} 1$$

$$1 \overset{23151322111212231}{5} 1 \quad \Delta = -101, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1 \overset{23151322111212231}{5} 1 \quad \Delta = -101, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{23151322111212231}{5} 1 \quad \Delta = -101, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{23151322111212231}{5} 1 \quad \Delta = -101, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{23151322111212231}{5} 1 \quad \Delta = -101, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1 \overset{23151322111212231}{5} 1 \quad \Delta = -101, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \overset{23151322111212231}{5} 1 \quad \Delta = -109, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\text{Base: } 61223151322111212222$$

$$612231 \overset{2}{5} 13221112122222 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$612231513221 \overset{2}{1} 12122222 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$612231513221112122222 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$612231 \overset{3}{5} 13221112122222 \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 63 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$612231 \overset{2}{5} 13221112122222 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$6122315132211121 \overset{2}{2} 22222 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\text{Base: } 16123151322111212222$$

$$161231 \overset{2}{5} 13221112122222 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$161231 \overset{2}{5} 13221112122222 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$\begin{array}{ll}
16123151322112122222 & \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
16123151322112122222 & \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
16123151322112122222 & \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\text{Base: } 31612315132211212222 & \\
31612315132211212222 & \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
31612315132211212222 & \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
31612315132211212222 & \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
31612315132211212222 & \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
31612315132211212222 & \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
31612315132211212222 & \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
\text{Base: } 21613151322112122222 & \\
21613151322112122222 & \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21613151322112122222 & \Delta = -125, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
21613151322112122222 & \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21613151322112122222 & \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
21613151322112122222 & \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
21613151322112122222 & \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21613151322112122222 & \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\text{Base: } 32161315132211212222 & \\
32161315132211212222 & \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
32161315132211212222 & \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{ll}
3216131513221^{2}1212222 & \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
32161315132211212222 & \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
32161315132211212222 & \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
3216131513221121^{2}2222 & \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\text{Base: } 23216131513221121222 & \\
23216131513221121222 & \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 70 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
23216131513221121222 & \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
23216131513221121222 & \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
23216131513221121222 & \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
23216131513221121222 & \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
23216131513221121^{2}2222 & \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\text{Base: } 161315132211212222 & \\
1613151322112122222 & \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1613151322112122222 & \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1613151322112122222 & \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1613151322112122222 & \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
16131513221121^{2}22222 & \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\text{Base: } 16^{31}1315132211212222 & \\
316^{21}1315132211212222 & \Delta = -125, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
16^{31}131513221121^{2}2222 & \Delta = -125, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
16^{31}1315132211212222 & \Delta = -125, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
16^{31}1315132211212222 \quad \Delta = -125, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
16^{31}1315132211212222 \quad \Delta = -125, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
16^{31}1315132211212222 \quad \Delta = -125, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
16^{31}1315132211212222 \quad \Delta = -125, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
316^{31}1315132211212222 \quad \Delta = -133, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1121223141414141413

$$\begin{aligned}
1121223141414141413^3 \quad \Delta = -135, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1121223141414141413^2 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1121223141414141413^{12} \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 88 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
1121223141414141413^2 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1121223141414141413^2 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1121223141414141413^2 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1121223141414141413^2 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1121223141414141413^2 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

Base: 112122314141414141

$$\begin{aligned}
112122314141414141^{21} \text{ [#=2]} \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
112122314141414141^{21} \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 218 & 30 \\ 2 & 30 & 8 \end{pmatrix} \\
112122314141414141^{21} \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 151 & 25 \\ 2 & 25 & 8 \end{pmatrix} \\
112122314141414141^{21} \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
112122314141414141^{21} \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{l}
\begin{array}{l} \overset{21}{112122314141414141} \end{array} \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\begin{array}{l} \overset{21}{112122314141414141} \end{array} \overset{3}{\text{[#}=3]} \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
\begin{array}{l} \overset{21}{112122314141414141} \end{array} \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
\begin{array}{l} \overset{21}{112122314141414141} \end{array} \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 142 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
\text{Base: } 4141413221 \overset{1513221}{12} 1 \\
\begin{array}{l} \overset{2}{4141413221} \end{array} \overset{1513221}{12} 1 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\begin{array}{l} \overset{2}{4141413221} \end{array} \overset{1513221}{12} 1 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\begin{array}{l} \overset{2}{4141413221} \end{array} \overset{1513221}{12} 1 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
\begin{array}{l} \overset{2}{4141413221} \end{array} \overset{1513221}{12} 1 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 89 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
\begin{array}{l} \overset{1513221}{4141413221} \end{array} \overset{1513221}{12} 1 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\begin{array}{l} \overset{1513221}{4141413221} \end{array} \overset{1513221}{12} 1 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\begin{array}{l} \overset{1513221}{4141413221} \end{array} \overset{1513221}{12} \overset{12}{12} \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{21513221}{12} 122314141 \\
\begin{array}{l} \overset{21513221}{1} \end{array} \overset{21513221}{12} 122314141 \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\begin{array}{l} \overset{21513221}{1} \end{array} \overset{21513221}{12} 122314141 \overset{3}{\text{[#}=2]} \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
\begin{array}{l} \overset{21513221}{1} \end{array} \overset{21513221}{12} 122314141 \overset{3}{\text{[#}=3]} \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
\begin{array}{l} \overset{21513221}{1} \end{array} \overset{21513221}{12} 122314141 \overset{2}{\text{[#}=2]} \quad \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\begin{array}{l} \overset{21513221}{1} \end{array} \overset{21513221}{12} 122314141 \overset{2}{\text{[#}=2]} \quad \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\begin{array}{l} \overset{21513221}{1} \end{array} \overset{21513221}{12} 122314141 \overset{2}{\text{[#}=2]} \quad \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
\begin{array}{l} \overset{21513221}{1} \end{array} \overset{21513221}{12} 122314141 \quad \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
& \overset{21513221}{1} \overset{12}{12} 122314141 \quad \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{21513221}{1} \overset{12}{12} 122314141 \quad \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{21513221}{1} \overset{12}{12} 122315123$

$$\begin{aligned}
& \overset{21513221}{1} \overset{12}{12} 122315123 \quad \Delta = -101, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{21513221}{1} \overset{12}{12} 122315123 \quad \Delta = -101, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{21513221}{1} \overset{12}{12} \overset{2}{122315123} \quad \Delta = -101, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{21513221}{1} \overset{12}{12} \overset{2}{122315123} \quad \Delta = -101, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{21513221}{1} \overset{12}{12} \overset{2}{122315123} \quad \Delta = -101, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{21513221}{1} \overset{12}{12} 122315123 \quad \Delta = -109, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
& \overset{21513221}{1} \overset{12}{12} 122315123 \quad \Delta = -101, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{131513221}{1} \overset{12}{12} 12231413$

$$\begin{aligned}
& \overset{3131513221}{1} \overset{12}{12} 12231413 \text{ [#=8]} \quad \Delta = -92, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
& \overset{3131513221}{1} \overset{12}{12} 12231413 \text{ [#=9]} \quad \Delta = -92, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
& \overset{131513221}{1} \overset{12}{12} \overset{2}{12231413} \quad \Delta = -84, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{131513221}{1} \overset{12}{12} \overset{2}{12231413} \quad \Delta = -84, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{131513221}{1} \overset{12}{12} 12231413 \quad \Delta = -84, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{131513221}{1} \overset{12}{12} 12231413 \quad \Delta = -84, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{131513221}{1} \overset{12}{12} \overset{2}{12231413} \quad \Delta = -84, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{131513221}{1} \overset{12}{12} 12231413 \quad \Delta = -84, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{3131513221}{1} \overset{12}{12} 12231413 \text{ [#=7]} \quad \Delta = -92, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{3}{1} \overset{131513221}{12} 12231413 \text{ }_{[\# = 6]} \quad \Delta = -92, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& \overset{3}{1} \overset{131513221}{12} 12231413 \text{ }_{[\# = 2]} \quad \Delta = -92, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{131513221}{1} \overset{3}{12} 12231413 \quad \Delta = -92, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{3}{1} \overset{131513221}{12} 12231413 \text{ }_{[\# = 3]} \quad \Delta = -92, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{3}{1} \overset{131513221}{12} 12231413 \text{ }_{[\# = 4]} \quad \Delta = -92, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{3}{1} \overset{131513221}{12} 12231413 \text{ }_{[\# = 5]} \quad \Delta = -92, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{4131513221121223141}$

$$\begin{aligned}
& 41315132211\overset{1}{2}\overset{2}{2}23141 \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 41315132211\overset{1}{2}\overset{2}{2}23141 \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 41315132211\overset{1}{2}\overset{2}{2}23141 \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 41315132211\overset{1}{2}\overset{2}{2}23141 \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 41315132211\overset{1}{2}\overset{2}{2}23141 \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 41315132211\overset{1}{2}\overset{2}{2}23141 \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 41315132211\overset{1}{2}1223141\overset{3}{3} \text{ }_{[\# = 2]} \quad \Delta = -112, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 41315132211\overset{1}{2}1223141 \quad \Delta = -112, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& 41315132211\overset{1}{2}1223141\overset{3}{3} \text{ }_{[\# = 3]} \quad \Delta = -112, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 41315132211\overset{1}{2}\overset{2}{2}23141 \quad \Delta = -105, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1231513221}{1} \overset{2}{12} 1223132$

$$\begin{aligned}
& \overset{1231513221}{1} \overset{2}{12} 1223132 \quad \Delta = -60, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{1231513221}{1} \overset{2}{12} 1223132 \quad \Delta = -58, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& 1^{1231513221} 12^{223132} \Delta = -58, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 1^{1231513221} 12^{223132} \Delta = -58, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 1^{1231513221} 12^{223132} \Delta = -58, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& 1^{1231513221} 12^{223132} \Delta = -58, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 1^{31231513221} 12^{223132} [\#=2] \Delta = -66, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 1^{31231513221} 12^{223132} [\#=6] \Delta = -66, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& 1^{31231513221} 12^{223132} [\#=8] \Delta = -66, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& 1^{31231513221} 12^{223132} [\#=9] \Delta = -66, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
& 1^{31231513221} 12^{223132} [\#=7] \Delta = -66, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
& 1^{31231513221} 12^{223132} [\#=5] \Delta = -66, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& 1^{31231513221} 12^{223132} [\#=4] \Delta = -66, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 1^{31231513221} 12^{223132} [\#=3] \Delta = -66, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 1^{1231513221} 12^{223132} \Delta = -66, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 32 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \text{Base: } 1^{1231513221} 12^{22222} \\
& 1^{1231513221} 12^{22222} \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 1^{312231513221} 12^{22222} [\#=6] \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& 1^{12231513221} 12^{22222} \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 1^{12231513221} 12^{22222} \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& 1^{12231513221} 12^{22222} \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 1^{312231513221} 12^{22222} [\#=5] \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{ll}
1 \overset{\textcolor{red}{3}}{12231513221} 122222 \text{ [#=3]} & \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \overset{\textcolor{red}{3}}{12231513221} 122222 \text{ [#=2]} & \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \overset{\textcolor{red}{3}}{12231513221} 122222 \text{ [#=4]} & \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1 \overset{\textcolor{red}{3}}{12231513221} 122222 \text{ [#=8]} & \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1 \overset{\textcolor{red}{3}}{12231513221} 122222 \text{ [#=9]} & \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1 \overset{\textcolor{red}{3}}{12231513221} 122222 \text{ [#=7]} & \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{51231513221}{12} 122313 & \\
1 \overset{51231513221}{12} \overset{\textcolor{red}{3}}{122313} \text{ [#=2]} & \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1 \overset{51231513221}{12} \overset{\textcolor{red}{3}}{122313} & \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \overset{51231513221}{12} \overset{\textcolor{red}{2}}{122313} & \Delta = -109, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1 \overset{51231513221}{12} 122313 & \Delta = -109, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \overset{51231513221}{12} \overset{\textcolor{red}{2}}{122313} & \Delta = -109, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \overset{51231513221}{12} 122313 & \Delta = -109, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1 \overset{51231513221}{12} 122313 & \Delta = -109, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \overset{51231513221}{12} 122313 & \Delta = -109, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{15131513221}{12} 122313 & \\
1 \overset{15131513221}{12} 122313 & \Delta = -109, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1 \overset{15131513221}{12} 122313 & \Delta = -108, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \overset{15131513221}{12} 122313 & \Delta = -108, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1 \overset{15131513221}{12} \overset{\textcolor{red}{2}}{122313} & \Delta = -108, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{l}
1^{15131513221} 12^{2} 122313 \quad \Delta = -108, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{15131513221} 12^{3} 122313 \quad \Delta = -116, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1^{15131513221} 12^{3} 122313_{[\# = 2]} \quad \Delta = -116, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1^{15131513221} 12 122313 \quad \Delta = -108, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{15131513221} 12 122313 \quad \Delta = -108, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\text{Base: } 1^{315131513221} 12 12231 \\
1^{315131513221} 12 12231 \quad \Delta = -108, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1^{315131513221} 12 122313_{[\# = 4]} \quad \Delta = -116, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{315131513221} 12 122313_{[\# = 3]} \quad \Delta = -116, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1^{315131513221} 12 122313_{[\# = 5]} \quad \Delta = -116, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1^{315131513221} 12 122313_{[\# = 2]} \quad \Delta = -116, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1^{315131513221} 12 12231 \quad \Delta = -108, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1^{315131513221} 12 12231 \quad \Delta = -109, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{315131513221} 12 12231 \quad \Delta = -108, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1^{315131513221} 12 12231 \quad \Delta = -108, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1^{315131513221} 12 12231 \quad \Delta = -108, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1^{315131513221} 12 12231 \quad \Delta = -108, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\text{Base: } 1^{1513221} 12 122314141 \\
1^{1513221} 12 122314141^{2} \quad \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1^{1513221} 12 1223141413_{[\# = 2]} \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{l}
\begin{array}{l} 1^{1513221} \\ 1 \quad 12 \quad 122314141 \end{array} \begin{array}{l} \textcolor{red}{3} \text{ [#=3]} \\ \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \end{array} \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
\\
\begin{array}{l} 1^{1513221} \\ 1 \quad 12 \quad 122314141 \end{array} \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \begin{pmatrix} 0 & 5 & 2 \\ 5 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
\\
\begin{array}{l} 1^{1513221} \\ 1 \quad 12 \quad 122314141 \end{array} \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\\
\begin{array}{l} 1^{1513221} \\ 1 \quad 12 \quad 122314141 \end{array} \begin{array}{l} \textcolor{red}{2} \\ \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \end{array} \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\\
\begin{array}{l} 1^{1513221} \\ 1 \quad 12 \quad 122314141 \end{array} \begin{array}{l} \textcolor{red}{2} \\ \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \end{array} \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
\\
\textbf{Base: } 1^{321513221} 12^{1223151} \\
\\
\begin{array}{l} 1^{321513221} \\ 1 \quad 12 \quad 1223151 \end{array} \begin{array}{l} 1 \\ \Delta = -101, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \end{array} \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\\
\begin{array}{l} 1^{321513221} \\ 1 \quad 12 \quad 1223151 \end{array} \begin{array}{l} 1 \\ \Delta = -101, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \end{array} \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\\
\begin{array}{l} 1^{321513221} \\ 1 \quad 12 \quad 1223151 \end{array} \begin{array}{l} 1 \\ \Delta = -101, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \end{array} \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\\
\begin{array}{l} 1^{321513221} \\ 1 \quad 12 \quad 1223151 \end{array} \begin{array}{l} \textcolor{red}{2} 1 \\ \Delta = -101, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \end{array} \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\\
\begin{array}{l} 1^{321513221} \\ 1 \quad 12 \quad 1223151 \end{array} \begin{array}{l} \textcolor{red}{2} 1 \\ \Delta = -101, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \end{array} \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\\
\textbf{Base: } 1^{21513221} 12^{12231513} \\
\\
\begin{array}{l} 1^{21513221} \\ 1 \quad 12 \quad 12231513 \end{array} \begin{array}{l} 1 \\ \Delta = -100, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \end{array} \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\\
\begin{array}{l} 1^{21513221} \\ 1 \quad 12 \quad 12231513 \end{array} \begin{array}{l} \textcolor{red}{2} 1 \\ \Delta = -100, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \end{array} \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\\
\begin{array}{l} 1^{21513221} \\ 1 \quad 12 \quad 12231513 \end{array} \begin{array}{l} \textcolor{red}{2} 1 \\ \Delta = -100, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \end{array} \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\\
\begin{array}{l} 1^{21513221} \\ 1 \quad 12 \quad 12231513 \end{array} \begin{array}{l} 1 \\ \Delta = -100, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \end{array} \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\\
\begin{array}{l} 1^{21513221} \\ 1 \quad 12 \quad 12231513 \end{array} \begin{array}{l} \textcolor{red}{2} 1 \\ \Delta = -100, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \end{array} \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\\
\begin{array}{l} 1^{21513221} \\ 1 \quad 12 \quad 12231513 \end{array} \begin{array}{l} 1 \\ \Delta = -100, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \end{array} \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\\
\begin{array}{l} 1^{21513221} \\ 1 \quad 12 \quad 12231513 \end{array} \begin{array}{l} 1 \\ \Delta = -108, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \end{array} \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
\\
\begin{array}{l} 1^{21513221} \\ 1 \quad 12 \quad 12231513 \end{array} \begin{array}{l} \textcolor{red}{3} 1 \\ \Delta = -108, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \end{array} \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{array}$$

$$1 \overset{21513221}{12} \overset{12}{12231513} \quad \Delta = -100, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{31513221}{12} \overset{1}{1223151}$$

$$1 \overset{31513221}{12} \overset{1}{1223151} \quad \Delta = -100, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{31513221}{12} \overset{1}{1223151} \quad \Delta = -100, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{31513221}{12} \overset{1}{1223151} \quad \Delta = -100, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{31513221}{12} \overset{1}{1223151} \quad \Delta = -100, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{31513221}{12} \overset{21}{1223151} \quad \Delta = -100, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{31513221}{12} \overset{1}{1223151} \quad \Delta = -100, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{31513221}{12} \overset{1}{1223151} \quad \Delta = -108, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$\text{Base: } 612231513221 \overset{22221}{12} 1$$

$$612231513221 \overset{2}{12} \overset{22221}{1} \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$612231513221 \overset{2}{12} \overset{22221}{1} \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$612231513221 \overset{2}{12} \overset{22221}{1} \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$612231513221 \overset{2}{12} \overset{22221}{1} \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$612231513221 \overset{22221}{12} 1 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$612231513221 \overset{3}{12} \overset{22221}{1} \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 63 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{22221}{12} 122315132161$$

$$1 \overset{22221}{12} \overset{2}{122315132161} \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$1 \overset{22221}{12} \overset{2}{122315132161} \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1 \overset{22221}{12} \overset{2}{122315132161} \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1 \overset{22221}{12} \overset{2}{122315132161} \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{22221}{12} 122315132161 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

Base: $1 \overset{2221}{12} 1223151321613$

$$1 \overset{2221}{12} 1223151321613 \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{2221}{12} 1223151321613^2 \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$1 \overset{2221}{12} 1223151321613^2 \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1 \overset{2221}{12} 1223151321613^2 \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{2221}{12} 1223151321613^2 \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1 \overset{2221}{12} 1223151321613^2 \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: $1 \overset{22221}{12} 122315131612$

$$1 \overset{22221}{12} 122315131612^2 \quad \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{22221}{12} 122315131612^3 \quad \Delta = -125, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$1 \overset{22221}{12} 122315131612^2 \quad \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1 \overset{22221}{12} 122315131612 \quad \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1 \overset{22221}{12} 122315131612^2 \quad \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$1 \overset{22221}{12} 122315131612^2 \quad \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1 \overset{22221}{12} 122315131612^2 \quad \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: $1 \overset{2221}{12} 1223151316123$

$$1 \overset{2221}{12} 1223151316123^2 \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1 \overset{2221}{12} 1223151316123^2 \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{2221}{12} 1223151316123^2 \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{2221}{12} 1223151316123^2 \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$\begin{array}{ll}
\begin{array}{c} 2221 \\ 1 \ 12 \ 1223151316123 \end{array} \quad \Delta = -126, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\begin{array}{c} 2221 \\ 1 \ 12 \ 1223151316123 \end{array} \quad \Delta = -126, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\text{Base: } \begin{array}{c} 221 \\ 11212231513161232 \end{array} & \\
\begin{array}{c} 221 \\ 11212231513161232 \end{array} \quad \Delta = -128, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
\begin{array}{c} 221 \ 2 \\ 11212231513161232 \end{array} \quad \Delta = -128, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\begin{array}{c} 221 \\ 11212231513161232 \end{array} \quad \Delta = -128, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
\begin{array}{c} 221 \\ 11212231513161232 \end{array} \quad \Delta = -128, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\begin{array}{c} 221 \\ 11212231513161232 \end{array} \quad \Delta = -128, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\begin{array}{c} 221 \\ 11212231513161232 \end{array} \quad \Delta = -128, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
\text{Base: } \begin{array}{c} 1 \\ 16131513221121222 \end{array} & \\
\begin{array}{c} 1 \\ 16131513221121222 \end{array} \quad \Delta = -117, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\begin{array}{c} 1 \ 2 \\ 16131513221121222 \end{array} \quad \Delta = -117, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\begin{array}{c} 1 \\ 16131513221121222 \end{array} \quad \Delta = -117, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\begin{array}{c} 1 \\ 16131513221121222 \end{array} \quad \Delta = -117, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\begin{array}{c} 1 \\ 16131513221121222 \end{array} \quad \Delta = -117, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\text{Base: } \begin{array}{c} 31 \\ 16131513221121222 \end{array} & \\
\begin{array}{c} 31 \\ 16131513221121222 \end{array} \quad \Delta = -125, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
\begin{array}{c} 21 \\ 316131513221121222 \end{array} \quad \Delta = -125, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\begin{array}{c} 31 \\ 316131513221121222 \end{array} \quad \Delta = -133, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\begin{array}{c} 31 \\ 16131513221121222 \end{array} \quad \Delta = -125, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\begin{array}{c} 31 \\ 16131513221121222 \end{array} \quad \Delta = -125, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{lcl}
\begin{array}{c} 31 \\ 16 \end{array} \begin{array}{c} 2 \\ 131513221121222 \end{array} \begin{array}{c} 1 \\ 1222 \end{array} & \Delta = -125, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
\begin{array}{c} 31 \\ 16 \end{array} \begin{array}{c} 2 \\ 131513221121222 \end{array} \begin{array}{c} 1 \\ 1222 \end{array} & \Delta = -125, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\begin{array}{c} 31 \\ 16 \end{array} \begin{array}{c} 2 \\ 131513221121222 \end{array} \begin{array}{c} 1 \\ 1222 \end{array} & \Delta = -125, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{array}$$

Base: $21^{21}1212231513161232$

$$\begin{array}{lcl}
21^{21} \begin{array}{c} 2 \\ 1212231513161232 \end{array} & \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
21^{21} \begin{array}{c} 2 \\ 1212231513161232 \end{array} & \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
\begin{array}{c} 2 \\ 21 \end{array} 1212231513161232 & \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21^{21} \begin{array}{c} 2 \\ 1212231513161232 \end{array} & \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
21^{21} \begin{array}{c} 2 \\ 1212231513161232 \end{array} & \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
21^{21} \begin{array}{c} 2 \\ 1212231513161232 \end{array} & \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{array}$$

Base: $21^{21513221}12^{12}12231512$

$$\begin{array}{lcl}
21^{21513221} \begin{array}{c} 2 \\ 12^{12}12231512 \end{array} & \Delta = -92, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21^{21513221} \begin{array}{c} 2 \\ 12^{12}12231512 \end{array} & \Delta = -92, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21^{21513221} \begin{array}{c} 2 \\ 12^{12}12231512 \end{array} & \Delta = -92, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\begin{array}{c} 2 \\ 21 \end{array} 12^{21513221} \begin{array}{c} 2 \\ 12^{12}12231512 \end{array} & \Delta = -92, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21^{21513221} \begin{array}{c} 3 \\ 12^{12}12231512 \end{array} & \Delta = -100, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{array}$$

Base: $21^{131513221}12^{12}1223141$

$$\begin{array}{lcl}
21^{131513221} \begin{array}{c} 1223141 \end{array} \begin{array}{c} 3 \\ \text{[#}=2] \end{array} & \Delta = -84, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
21^{3131513221} \begin{array}{c} 1223141 \end{array} \begin{array}{c} \text{[#}=8] \end{array} & \Delta = -84, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
21^{131513221} \begin{array}{c} 1223141 \end{array} & \Delta = -76, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\begin{array}{c} 2 \\ 21 \end{array} 12^{131513221} \begin{array}{c} 1223141 \end{array} & \Delta = -76, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
21^{131513221} 12^{21223141} \Delta = -76, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21^{131513221} 12^{21223141} \Delta = -76, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
21^{131513221} 12^{21223141} \Delta = -76, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21^{131513221} 12^{21223141} \Delta = -76, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
21^{3131513221} 12^{21223141} [\# = 9] \Delta = -84, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
21^{3131513221} 12^{21223141} [\# = 5] \Delta = -84, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
21^{3131513221} 12^{21223141} [\# = 4] \Delta = -84, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
21^{131513221} 12^{21223141} 3 [\# = 3] \Delta = -84, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
21^{3131513221} 12^{21223141} [\# = 6] \Delta = -84, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
21^{3131513221} 12^{21223141} [\# = 2] \Delta = -84, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
21^{3131513221} 12^{21223141} [\# = 7] \Delta = -84, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
21^{3131513221} 12^{21223141} [\# = 3] \Delta = -84, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\text{Base: } 21^{1231513221} 12^{2122313} & \\
21^{1231513221} 12^{2122313} \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21^{31231513221} 12^{2122313} [\# = 8] \Delta = -65, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
21^{1231513221} 12^{2122313} \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21^{1231513221} 12^{2122313} \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21^{21231513221} 12^{2122313} \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21^{1231513221} 12^{2122313} \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
21^{1231513221} 12^{2122313} \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
21 \overset{3}{1231513221} 12 122313 \text{ [#=3]} \quad \Delta = -65, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21 \overset{3}{1231513221} 12 122313 \text{ [#=7]} \quad \Delta = -65, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
21 \overset{3}{1231513221} 12 122313 \text{ [#=9]} \quad \Delta = -65, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
21 \overset{3}{1231513221} 12 122313 \text{ [#=2]} \quad \Delta = -65, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
21 \overset{3}{1231513221} 12 122313 \text{ [#=6]} \quad \Delta = -65, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
21 \overset{3}{1231513221} 12 122313 \text{ [#=5]} \quad \Delta = -65, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
21 \overset{1231513221}{12} \overset{3}{122313} \quad \Delta = -65, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21 \overset{3}{1231513221} 12 122313 \text{ [#=4]} \quad \Delta = -65, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21 \overset{1231513221}{12} \overset{3}{122313} \text{ [#=2]} \quad \Delta = -65, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
\text{Base: } 21 \overset{12231513221}{12} 12222 & \\
21 \overset{3}{12231513221} 12 12222 \text{ [#=3]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21 \overset{2}{12231513221} 12 12222 \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21 \overset{12231513221}{12} 12222 \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21 \overset{12231513221}{12} \overset{2}{12222} \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21 \overset{3}{12231513221} 12 12222 \text{ [#=8]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
21 \overset{12231513221}{12} 12222 \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
21 \overset{12231513221}{12} 12222 \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21 \overset{3}{12231513221} 12 12222 \text{ [#=2]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21 \overset{3}{12231513221} 12 12222 \text{ [#=9]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
21 \overset{3}{12231513221} 12 12222 \text{ [#=4]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{ll}
21 \overset{3}{1}2231513221 \quad 12222 \text{ [#}=5] & \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
21 \overset{3}{1}2231513221 \quad 12222 \text{ [#}=6] & \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
21 \overset{3}{1}2231513221 \quad 12222 \text{ [#}=7] & \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
\text{Base: } 21 \overset{51231513221}{12} 12231 & \\
21 \overset{51231513221}{12} 12231 \overset{3}{1} \text{ [#}=2] & \Delta = -109, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
21 \overset{51231513221}{12} 12231 \overset{2}{1} & \Delta = -101, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21 \overset{51231513221}{12} 12231 \overset{3}{1} \text{ [#}=4] & \Delta = -109, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21 \overset{51231513221}{12} 12231 & \Delta = -101, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
21 \overset{51231513221}{12} 12231 \overset{3}{1} \text{ [#}=3] & \Delta = -109, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21 \overset{51231513221}{12} 12231 \overset{3}{1} \text{ [#}=5] & \Delta = -109, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
21 \overset{2}{5}1231513221 \quad 12231 & \Delta = -101, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21 \overset{51231513221}{12} 12231 & \Delta = -101, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21 \overset{51231513221}{12} 12231 & \Delta = -101, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21 \overset{51231513221}{12} 12231 & \Delta = -101, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\text{Base: } 413221 \overset{5131513221}{12} 12 & \\
413221 \overset{5131513221}{12} 12 & \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
413221 \overset{5131513221}{12} \overset{2}{1} 12 & \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
413221 \overset{5131513221}{12} 12 & \Delta = -129, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
413221 \overset{2}{5}131513221 \quad 12 12 & \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
413221 \overset{5131513221}{12} 12 & \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
& 413221 \overset{5131513221}{12} 12 \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{2}{413221} \overset{5131513221}{12} 12 \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 413221 \overset{5131513221}{12} 12 \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \text{Base: } 21 \overset{15131513221}{12} 12231 \\
& 21 \overset{15131513221}{12} 12231 \quad \Delta = -100, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 21 \overset{15131513221}{12} 12231 \overset{3}{\text{[#}=5]} \quad \Delta = -108, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& 21 \overset{15131513221}{12} 12231 \overset{3}{\text{[#}=3]} \quad \Delta = -108, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 21 \overset{15131513221}{12} 12231 \overset{3}{\text{[#}=2]} \quad \Delta = -108, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{2}{21} \overset{15131513221}{12} 12231 \quad \Delta = -100, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 21 \overset{15131513221}{12} 12231 \quad \Delta = -100, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 21 \overset{15131513221}{12} 12231 \quad \Delta = -100, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 21 \overset{15131513221}{12} 12231 \overset{2}{\text{[#}=2]} \quad \Delta = -100, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 21 \overset{15131513221}{12} 12231 \quad \Delta = -100, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 21 \overset{15131513221}{12} 12231 \overset{3}{\text{[#}=4]} \quad \Delta = -108, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 21 \overset{15131513221}{12} 12231 \quad \Delta = -101, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \text{Base: } 21 \overset{21513221}{12} \overset{1}{1223151} \\
& 21 \overset{21513221}{12} 12231 \overset{21}{51} \quad \Delta = -92, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{221513221}{21} \overset{1}{12} 12231 \overset{1}{51} \quad \Delta = -92, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 21 \overset{21513221}{12} \overset{21}{12231} \overset{1}{51} \quad \Delta = -92, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 21 \overset{21513221}{12} \overset{1}{12231} \overset{1}{51} \quad \Delta = -92, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
21^{21513221}12^{11}1223151 \quad \Delta = -92, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21^{21513221}12^{11}1223151 \quad \Delta = -100, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
21^{21513221}12^{11}1223151 \quad \Delta = -92, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: $21^{1513221}12^{11}1223151$

$$\begin{aligned}
21^{1513221\textcolor{red}{2}}12^{11}1223151 \quad \Delta = -92, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21^{1513221}12^{21\textcolor{red}{1}}1223151 \quad \Delta = -92, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21^{\textcolor{red}{2}1513221}12^{11}1223151 \quad \Delta = -92, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: $612231513221^{2221}1212$

$$\begin{aligned}
612231513221^{2221}1212 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
612231513221^{2221}1212 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
612231513221^{2221}1212 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
612231513221^{2221}1212 \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 63 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
612231513221^{2221}1212 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
612231513221^{2221}1212 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
612231513221^{2221\textcolor{red}{2}}1212 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: $21^{2221}121222315132161$

$$\begin{aligned}
21^{2221}12^{2\textcolor{red}{1}}122315132161 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
21^{\textcolor{red}{2}2221}121222315132161 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21^{2221}121222315132161 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
21^{2221}121222315132161 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21^{2221}121222315132161 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

$$21 \overset{2221}{12} \overset{2}{122315132161} \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

Base: $\overset{221}{21121223151321613}$

$$21 \overset{221}{12} \overset{2}{1223151321613} \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$21 \overset{221}{12} \overset{2}{1223151321613} \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$21 \overset{221}{12} \overset{2}{1223151321613} \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$21 \overset{221}{12} \overset{2}{1223151321613} \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$21 \overset{221}{12} \overset{2}{1223151321613} \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$21 \overset{221}{12} \overset{2}{1223151321613} \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$\overset{2221}{21} \overset{2}{12} \overset{2}{1223151321613} \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: $\overset{2221}{2112122315131612}$

$$21 \overset{2221}{12} \overset{2}{122315131612} \quad \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$21 \overset{2221}{12} \overset{2}{122315131612} \quad \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$21 \overset{2221}{12} \overset{2}{122315131612} \quad \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$21 \overset{2221}{12} \overset{2}{122315131612} \quad \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$21 \overset{2221}{12} \overset{2}{122315131612} \quad \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$21 \overset{2221}{12} \overset{2}{122315131612} \quad \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$\overset{22221}{21} \overset{2}{12} \overset{2}{122315131612} \quad \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$21 \overset{2221}{12} \overset{3}{122315131612} \quad \Delta = -125, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

Base: $\overset{221}{21121223151316123}$

$$21 \overset{221}{12} \overset{2}{1223151316123} \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$21 \overset{221}{12} \overset{2}{1223151316123} \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\begin{array}{ll}
\begin{array}{c} 221 \\ 21121223151316123 \end{array} & \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\begin{array}{c} 221 \\ 21121223151316123 \end{array} & \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\begin{array}{c} 221 \\ 21121223151316123 \end{array} & \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\begin{array}{c} 221 \\ 21121223151316123 \end{array} & \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\begin{array}{c} 221 \\ 21121223151316123 \end{array} & \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\text{Base: } \begin{array}{c} 1 \\ 16131513221121222 \end{array} & \\
\begin{array}{c} 1 \\ 16131513221121222 \end{array} & \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\begin{array}{c} 1 \\ 16131513221121222 \end{array} & \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\begin{array}{c} 1 \\ 16131513221121222 \end{array} & \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\begin{array}{c} 1 \\ 16131513221121222 \end{array} & \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\begin{array}{c} 1 \\ 16131513221121222 \end{array} & \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\begin{array}{c} 1 \\ 16131513221121222 \end{array} & \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\text{Base: } \begin{array}{c} 31 \\ 161315132211212 \end{array} & \\
\begin{array}{c} 31 \\ 161315132211212 \end{array} & \Delta = -125, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
\begin{array}{c} 31 \\ 161315132211212 \end{array} & \Delta = -125, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\begin{array}{c} 31 \\ 161315132211212 \end{array} & \Delta = -125, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\begin{array}{c} 31 \\ 161315132211212 \end{array} & \Delta = -125, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
\begin{array}{c} 31 \\ 161315132211212 \end{array} & \Delta = -125, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
\begin{array}{c} 31 \\ 3161315132211212 \end{array} & \Delta = -133, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\begin{array}{c} 31 \\ 161315132211212 \end{array} & \Delta = -125, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
& \overset{21}{3}161315132211212 \quad \Delta = -125, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{31}{1}61315132211212 \quad \Delta = -125, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \text{Base: } 3221 \overset{1231513221}{12} 1223 \\
& \overset{31231513221}{3}221 \overset{12}{12} 1223 \text{ [#=9]} \quad \Delta = -66, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 324 \quad \begin{pmatrix} 0 & 18 & 2 \\ 18 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix} \\
& \overset{1231513221}{3}221 \overset{12}{12} 1223 \quad \Delta = -58, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{1231513221}{3}221 \overset{12}{12} 1223 \quad \Delta = -58, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{41231513221}{3}221 \overset{12}{12} 1223 \quad \Delta = -86, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{51231513221}{3}221 \overset{12}{12} 1223 \quad \Delta = -110, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{1231513221}{3}221 \overset{12}{12} 1223 \quad \Delta = -58, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{21231513221}{3}221 \overset{12}{12} 1223 \quad \Delta = -58, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{21231513221}{3}221 \overset{12}{12} 1223 \quad \Delta = -58, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{31231513221}{3}221 \overset{12}{12} 1223 \text{ [#=7]} \quad \Delta = -66, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 196 \quad \begin{pmatrix} 0 & 14 & 2 \\ 14 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
& \overset{31231513221}{3}221 \overset{12}{12} 1223 \text{ [#=6]} \quad \Delta = -66, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
& \overset{31231513221}{3}221 \overset{12}{12} 1223 \text{ [#=8]} \quad \Delta = -66, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 256 \quad \begin{pmatrix} 0 & 16 & 2 \\ 16 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
& \overset{31231513221}{3}221 \overset{12}{12} 1223 \text{ [#=5]} \quad \Delta = -66, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
& \overset{31231513221}{3}221 \overset{12}{12} 1223 \quad \Delta = -66, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{31231513221}{3}221 \overset{12}{12} 1223 \text{ [#=4]} \quad \Delta = -66, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& \overset{31231513221}{3}221 \overset{12}{12} 1223 \text{ [#=2]} \quad \Delta = -66, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{61231513221}{3}221 \overset{12}{12} 1223 \quad \Delta = -136, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{71231513221}{3}221 \overset{12}{12} 1223 \quad \Delta = -163, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
3221 \overset{8}{1231513221} 12 1223 \quad \Delta = -191, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 8 \end{pmatrix} \\
3221 \overset{3}{1231513221} 12 1223 \text{ [#=3]} \quad \Delta = -66, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
\text{Base: } 221 \overset{1231513221}{12} 12231 \quad & \\
221 \overset{1231513221}{12} 12231 \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
221 \overset{3}{1231513221} 12 12231 \text{ [#=4]} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
221 \overset{1231513221}{12} 12231 \overset{3}{\text{[#=4]}} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
221 \overset{3}{1231513221} 12 12231 \text{ [#=5]} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
221 \overset{3}{1231513221} 12 12231 \text{ [#=3]} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
221 \overset{1231513221}{12} 12231 \overset{3}{\text{[#=3]}} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
221 \overset{3}{1231513221} 12 12231 \text{ [#=2]} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
221 \overset{3}{1231513221} 12 12231 \text{ [#=9]} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
221 \overset{1231513221}{12} 12231 \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
221 \overset{1231513221}{12} 12231 \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
221 \overset{1231513221}{12} 12231 \overset{2}{\text{[#=2]}} \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
221 \overset{2}{1231513221} 12 12231 \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
221 \overset{1231513221}{12} 12231 \overset{3}{\text{[#=2]}} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
221 \overset{3}{1231513221} 12 12231 \text{ [#=6]} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
221 \overset{1231513221}{12} 12231 \overset{3}{\text{[#=5]}} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
221 \overset{3}{1231513221} 12 12231 \text{ [#=8]} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
221 \overset{3}{1231513221} 12 12231 \text{ [#=7]} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}
\end{aligned}$$

Base: $221 \overset{12231513221}{12} 1222$

$$221 \overset{\textcolor{red}{3}12231513221}{12} 1222 \text{ [#}=7] \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$221 \overset{12231513221}{12} 1222 \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$221 \overset{\textcolor{red}{2}12231513221}{12} 1222 \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$221 \overset{12231513221}{12} \overset{\textcolor{red}{2}}{1222} \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$221 \overset{12231513221}{12} 1222 \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$221 \overset{12231513221}{12} 1222 \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$221 \overset{\textcolor{red}{3}12231513221}{12} 1222 \text{ [#}=8] \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$221 \overset{\textcolor{red}{3}12231513221}{12} 1222 \text{ [#}=6] \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$221 \overset{\textcolor{red}{3}12231513221}{12} 1222 \text{ [#}=5] \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$221 \overset{\textcolor{red}{3}12231513221}{12} 1222 \text{ [#}=4] \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$221 \overset{\textcolor{red}{3}12231513221}{12} 1222 \text{ [#}=3] \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$221 \overset{\textcolor{red}{3}12231513221}{12} 1222 \text{ [#}=9] \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$221 \overset{\textcolor{red}{3}12231513221}{12} 1222 \text{ [#}=2] \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: $612231513221 \overset{221}{12} 122$

$$612231513221 \overset{\textcolor{red}{3}}{12} \overset{221}{12} 122 \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 63 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$612231513221 \overset{221}{12} \overset{\textcolor{red}{2}}{12} 122 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$612231513221 \overset{\textcolor{red}{2}}{12} \overset{221}{12} 122 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$612231513221 \overset{\textcolor{red}{2}}{12} \overset{221}{12} 122 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$612231513221 \overset{\textcolor{red}{2}}{12} \overset{221}{12} 122 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$612231513221^{2221}12122 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: $61231513221^{3221}12122$

$$61231513221^{3221}12122 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$61231513221^{3221}12122 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$61231513221^{3221}12122 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$61231513221^{3221}12122 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$61231513221^{23221}12122 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$61231513221^{3221}12122 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

Base: $221^{221}12122315132161$

$$221^{221}12122315132161 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$221^{2221}12122315132161 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$221^{221}12122315132161 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$221^{221}12122315132161 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$221^{221}12122315132161 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: $221^{3221}1212231513161$

$$3221^{3221}1212231513161 \quad \Delta = -135, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$221^{3221}1212231513161 \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 174 & 26 \\ 2 & 26 & 8 \end{pmatrix}$$

$$221^{3221}1212231513161 \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$221^{3221}1212231513161 \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$221^{23221}1212231513161 \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 55 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$221^{3221}1212231513161 \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$${}^{3221}_{221} 12 \overset{2}{12231513161} \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

Base: ${}^{221}_{221} 12122315131612$

$${}^{221}_{221} 12122315131612 \quad \Delta = -125, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$${}^{221}_{221} 12122315131612 \quad \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$${}^{221}_{221} 12122315131612 \quad \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$\overset{221}{221} 12122315131612 \quad \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$${}^{221}_{221} 12122315131612 \quad \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$${}^{221\overset{2}{2}}_{221} 12122315131612 \quad \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$${}^{221}_{221} 12122315131612 \quad \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: ${}^1_{16131513221} {}^{221}_{12122}$

$$\overset{1\overset{2}{2}}{16131513221} {}^{221}_{12122} \quad \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$${}^1_{16131513221} \overset{221\overset{2}{2}}{12122} \quad \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$${}^1_{16131513221} \overset{2}{12122} {}^{221}_{12122} \quad \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$${}^1_{16131513221} \overset{2}{12122} {}^{221}_{12122} \quad \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$${}^1_{16131513221} \overset{221\overset{2}{2}}{12122} \quad \Delta = -117, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: ${}^{1,21513221}_{1 \quad 12} 12231512$

$${}^{1,21513221\overset{2}{2}}_{1 \quad 12} 12231512 \quad \Delta = -92, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$${}^{1,21513221}_{1 \quad 12} \overset{2}{12231512} \quad \Delta = -92, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$${}^{1,21513221}_{1 \quad 12} \overset{2}{12231512} \quad \Delta = -92, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$${}^{1,21513221}_{1 \quad 12} \overset{3}{12231512} \quad \Delta = -100, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: ${}^{1,1231513221}_{1 \quad 12} 122313$

$$\begin{aligned}
& 1^{1, \textcolor{red}{3}1231513221} 12^{122313} [\# = 3] \quad \Delta = -65, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& 1^{1, \textcolor{red}{3}1231513221} 12^{122313} [\# = 7] \quad \Delta = -65, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
& 1^{1, \textcolor{red}{3}1231513221} 12^{122313} [\# = 6] \quad \Delta = -65, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
& 1^{1, 1231513221} 12^{\textcolor{red}{3}122313} \quad \Delta = -65, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 1^{1, 1231513221} 12^{\textcolor{red}{3}122313} [\# = 2] \quad \Delta = -65, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& 1^{1, \textcolor{red}{3}1231513221} 12^{122313} [\# = 2] \quad \Delta = -65, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& 1^{1, \textcolor{red}{3}1231513221} 12^{122313} [\# = 8] \quad \Delta = -65, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
& 1^{1, \textcolor{red}{3}1231513221} 12^{122313} [\# = 4] \quad \Delta = -65, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& 1^{1, \textcolor{red}{3}1231513221} 12^{122313} [\# = 5] \quad \Delta = -65, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
& 1^{1, 1231513221} 12^{\textcolor{red}{2}122313} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 1^{1, 1231513221} 12^{122313} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 1^{1, 1231513221} 12^{\textcolor{red}{2}122313} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 1^{1, 1231513221} 12^{122313} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 1^{1, 1231513221} 12^{122313} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 1^{1, \textcolor{red}{3}1231513221} 12^{122313} [\# = 9] \quad \Delta = -65, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
& \text{Base: } 1^{1, 1231513221} 12^{12222} \\
& 1^{1, 1231513221} 12^{12222} \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 1^{1, \textcolor{red}{3}12231513221} 12^{12222} [\# = 3] \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& 1^{1, 12231513221} 12^{\textcolor{red}{2}12222} \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 1^{1, 12231513221} 12^{12222} \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{l}
1^{1,312231513221} \quad 12 \quad 12222 \text{ [#}=9] \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix} \\
1^{1,312231513221} \quad 12 \quad 12222 \text{ [#}=8] \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
1^{1,312231513221} \quad 12 \quad 12222 \text{ [#}=7] \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
1^{1,312231513221} \quad 12 \quad 12222 \text{ [#}=4] \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1^{1,312231513221} \quad 12 \quad 12222 \text{ [#}=5] \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
1^{1,312231513221} \quad 12 \quad 12222 \text{ [#}=6] \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1^{1,312231513221} \quad 12 \quad 12222 \text{ [#}=2] \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{1,12231513221} \quad 12 \quad 12222 \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\text{Base: } 1^{1,51231513221} \quad 12 \quad 12231 \\
1^{1,51231513221} \quad 12 \quad 12231 \quad \Delta = -101, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1^{1,51231513221} \quad 12 \quad 12231 \quad \Delta = -101, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1^{1,51231513221} \quad 12 \quad 12231 \text{ [#}=5] \quad \Delta = -109, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1^{1,51231513221} \quad 12 \quad 12231 \quad \Delta = -101, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{1,51231513221} \quad 12 \quad 12231 \quad \Delta = -101, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1^{1,51231513221} \quad 12 \quad 12231 \text{ [#}=4] \quad \Delta = -109, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{1,51231513221} \quad 12 \quad 12231 \text{ [#}=3] \quad \Delta = -109, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1^{1,51231513221} \quad 12 \quad 12231 \text{ [#}=2] \quad \Delta = -109, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1^{1,51231513221} \quad 12 \quad 12231 \quad \Delta = -101, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
\text{Base: } 1^{1,15131513221} \quad 12 \quad 12231 \\
1^{1,15131513221} \quad 12 \quad 12231 \quad \Delta = -100, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{lcl}
1^{1,15131513221} & 12 & 12231 \quad \Delta = -100, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1^{1,15131513221} & 12 & 12231 \quad \Delta = -100, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1^{1,15131513221} & 12 & 12231 \quad \Delta = -101, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{1,15131513221} & 12 & 12231 \quad \Delta = -100, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1^{1,15131513221} & 12 & 12231 \quad \Delta = -100, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{1,15131513221} & 12 & 12231\textcolor{red}{3}_{[\#=5]} \quad \Delta = -108, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1^{1,15131513221} & 12 & 12231\textcolor{red}{3}_{[\#=3]} \quad \Delta = -108, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1^{1,15131513221} & 12 & 12231\textcolor{red}{3}_{[\#=4]} \quad \Delta = -108, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{1,15131513221} & 12 & 12231\textcolor{red}{3}_{[\#=2]} \quad \Delta = -108, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{array}$$

Base: $1^{1,21513221} 12^{1,21513221} 1223151$

$$\begin{array}{lcl}
1^{1,21513221} & 12 & 12231\textcolor{red}{5}^1 \quad \Delta = -92, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{1,21513221} & 12 & 12231\textcolor{red}{5}^1 \quad \Delta = -100, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1^{1,21513221} & 12 & 12231\textcolor{red}{5}^1 \quad \Delta = -92, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1^{1,21513221} & 12 & 12231\textcolor{red}{5}^1 \quad \Delta = -92, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{1,21513221} & 12 & 12231\textcolor{red}{5}^1 \quad \Delta = -92, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1^{1,21513221} & 12 & 12231\textcolor{red}{5}^1 \quad \Delta = -92, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{array}$$

Base: $1^{1,1513221} 12^{1,1513221} 1223151$

$$\begin{array}{lcl}
1^{1,1513221} & 12 & 12231\textcolor{red}{5}^1 \quad \Delta = -92, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{1,1513221} & 12 & 12231\textcolor{red}{5}^1 \quad \Delta = -92, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{array}$$

Base: $1^{21,1231513221} 12^{21,1231513221} 12231$

$$1^{21,\textcolor{red}{3}1231513221} 12^{21,1231513221} 12231_{[\#=4]} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$\begin{array}{l}
1^{21,1231513221} \quad 12 \quad 12231 \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1^{21,1231513221} \quad 12 \quad 12231 \textcolor{red}{3} \text{ [#=4]} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{21, \textcolor{red}{3}1231513221} \quad 12 \quad 12231 \text{ [#=3]} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1^{21, \textcolor{red}{3}1231513221} \quad 12 \quad 12231 \text{ [#=8]} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
1^{21,1231513221} \quad 12 \quad 12231 \textcolor{red}{3} \text{ [#=5]} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1^{21, \textcolor{red}{3}1231513221} \quad 12 \quad 12231 \text{ [#=5]} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
1^{21, \textcolor{red}{3}1231513221} \quad 12 \quad 12231 \text{ [#=7]} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
1^{21,1231513221} \quad 12 \quad 12231 \textcolor{red}{3} \text{ [#=3]} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1^{21, \textcolor{red}{3}1231513221} \quad 12 \quad 12231 \text{ [#=2]} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1^{21, \textcolor{red}{3}1231513221} \quad 12 \quad 12231 \text{ [#=9]} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
1^{21,1231513221} \quad 12 \quad 12231 \textcolor{red}{3} \text{ [#=2]} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1^{21, \textcolor{red}{3}1231513221} \quad 12 \quad 12231 \text{ [#=6]} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
1^{21,1231513221} \quad 12 \quad 12231 \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{21,1231513221} \quad 12 \quad 12231 \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1^{21,1231513221} \quad 12 \quad 12231 \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1^{21,1231513221 \textcolor{red}{2}} \quad 12 \quad 12231 \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\textbf{Base: } 1^{21,12231513221} \quad 12 \quad 1222 \\
1^{21, \textcolor{red}{3}12231513221} \quad 12 \quad 1222 \text{ [#=7]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
1^{21, \textcolor{red}{3}12231513221} \quad 12 \quad 1222 \text{ [#=8]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
1^{21, \textcolor{red}{3}12231513221} \quad 12 \quad 1222 \text{ [#=9]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{l}
1^{21, \textcolor{red}{3}12231513221} 12 1222 \text{ [#=3]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1^{21, \textcolor{red}{3}12231513221} 12 1222 \text{ [#=2]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{21, \textcolor{red}{3}12231513221} 12 1222 \text{ [#=5]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
1^{21, \textcolor{red}{3}12231513221} 12 1222 \text{ [#=6]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1^{21, 12231513221 \textcolor{red}{2}} 12 1222 \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1^{21, 12231513221} 12 1222 \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1^{21, 12231513221} 12 1222 \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{21, 12231513221} 12 1222 \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1^{21, 12231513221} 12 1222 \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1^{21, \textcolor{red}{3}12231513221} 12 1222 \text{ [#=4]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
\text{Base: } 1^{221, 12231513221} 12 122 \\
1^{221, 12231513221} 12 122 \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{221, \textcolor{red}{3}12231513221} 12 122 \text{ [#=2]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{221, \textcolor{red}{3}12231513221} 12 122 \text{ [#=9]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix} \\
1^{221, \textcolor{red}{3}12231513221} 12 122 \text{ [#=8]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
1^{221, \textcolor{red}{3}12231513221} 12 122 \text{ [#=6]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1^{221, \textcolor{red}{3}12231513221} 12 122 \text{ [#=5]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
1^{221, \textcolor{red}{3}12231513221} 12 122 \text{ [#=4]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1^{221, \textcolor{red}{3}12231513221} 12 122 \text{ [#=3]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1^{221, \textcolor{red}{3}12231513221} 12 122 \text{ [#=7]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix}
\end{array}$$

$$1 \begin{smallmatrix} 221,12231513221 \\ 12 \end{smallmatrix} 122 \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \begin{smallmatrix} 221,12231513221 \\ 12 \end{smallmatrix} 122 \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \begin{smallmatrix} 221,12231513221 \\ 12 \end{smallmatrix} 122 \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: $21 \begin{smallmatrix} 221,12231513221 \\ 12 \end{smallmatrix} 12$

$$21 \begin{smallmatrix} 221,312231513221 \\ 12 \end{smallmatrix} 12 \text{ [#=2]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$21 \begin{smallmatrix} 221,12231513221 \\ 12 \end{smallmatrix} 12 \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$21 \begin{smallmatrix} 221,12231513221 \\ 12 \end{smallmatrix} 12 \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$21 \begin{smallmatrix} 221,312231513221 \\ 12 \end{smallmatrix} 12 \text{ [#=8]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix}$$

$$21 \begin{smallmatrix} 221,312231513221 \\ 12 \end{smallmatrix} 12 \text{ [#=7]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix}$$

$$21 \begin{smallmatrix} 221,312231513221 \\ 12 \end{smallmatrix} 12 \text{ [#=6]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$21 \begin{smallmatrix} 221,312231513221 \\ 12 \end{smallmatrix} 12 \text{ [#=5]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

$$21 \begin{smallmatrix} 221,312231513221 \\ 12 \end{smallmatrix} 12 \text{ [#=4]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$21 \begin{smallmatrix} 221,312231513221 \\ 12 \end{smallmatrix} 12 \text{ [#=3]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$21 \begin{smallmatrix} 221,312231513221 \\ 12 \end{smallmatrix} 12 \text{ [#=9]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix}$$

$$21 \begin{smallmatrix} 221,12231513221 \\ 12 \end{smallmatrix} 12 \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$21 \begin{smallmatrix} 221,12231513221 \\ 12 \end{smallmatrix} 12 \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$21 \begin{smallmatrix} 221,12231513221 \\ 12 \end{smallmatrix} 12 \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: $1231414141414141 \overset{2}{3}2$

$$2 \begin{smallmatrix} 81231414141414141 \\ 3 \end{smallmatrix} 2 \quad \Delta = -43, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$2 \begin{smallmatrix} 71231414141414141 \\ 3 \end{smallmatrix} 2 \quad \Delta = -15, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 1315131613161315131

$$\begin{array}{ll}
1315131613161315131^2 & \Delta = -22, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1315131613161315131^2 & \Delta = -22, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1315131613161315131^2 & \Delta = -22, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1315131613161315131^2 & \Delta = -22, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
13151316131613151313^{\#5} & \Delta = -30, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
13151316131613151313^{\#8} & \Delta = -30, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
13151316131613151313^{\#9} & \Delta = -30, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
13151316131613151313^{\#4} & \Delta = -30, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
13151316131613151313^{\#3} & \Delta = -30, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
13151316131613151313^{\#2} & \Delta = -30, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
13151316131613151313^{\#6} & \Delta = -30, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
13151316131613151313^{\#7} & \Delta = -30, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
\text{Base: } 1231513161316131512 & \\
21513161316131513213^{\#5} & \Delta = -23, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
21513161316131513213^{\#9} & \Delta = -23, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix} \\
21513161316131513213^{\#7} & \Delta = -23, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
21513161316131513213^{\#6} & \Delta = -23, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1231513161316131512^3 & \Delta = -23, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
21513161316131513213^{\#2} & \Delta = -23, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21513161316131513213^{\#8} & \Delta = -23, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{ll}
2151316131613151321\overset{3}{\color{red}} \text{ [#=3]} & \Delta = -23, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
2151316131613151321\overset{3}{\color{red}} \text{ [#=4]} & \Delta = -23, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1231513161316131512\overset{2}{\color{red}} & \Delta = -15, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1231513161316131512\overset{2}{\color{red}} & \Delta = -15, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1231513161316131512\overset{2}{\color{red}} & \Delta = -15, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1231513161316131512\overset{2}{\color{red}} & \Delta = -15, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1231513161316131512\overset{2}{\color{red}} & \Delta = -15, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1231513161316131512\overset{2}{\color{red}} & \Delta = -15, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1231513161316131512\overset{2}{\color{red}} & \Delta = -15, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1231513161316131512\overset{2}{\color{red}} & \Delta = -15, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\text{Base: } 123151316131613151\overset{1}{\color{red}} & \\
123151316131613151\overset{2}{\color{red}}\overset{1}{\color{red}} & \Delta = -15, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
123151316131613151\overset{2}{\color{red}}\overset{1}{\color{red}} & \Delta = -15, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
123151316131613151\overset{2}{\color{red}}\overset{1}{\color{red}} & \Delta = -15, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
123151316131613151\overset{2}{\color{red}}\overset{1}{\color{red}} & \Delta = -15, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
123151316131613151\overset{2}{\color{red}}\overset{1}{\color{red}} & \Delta = -15, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
123151316131613151\overset{2}{\color{red}}\overset{1}{\color{red}} & \Delta = -15, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
123151316131613151\overset{2}{\color{red}}\overset{1}{\color{red}} & \Delta = -15, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1\overset{3}{\color{red}}1231513161316131\overset{1}{\color{red}} \text{ [#=7]} & \Delta = -23, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
1\overset{3}{\color{red}}1231513161316131\overset{1}{\color{red}} \text{ [#=2]} & \Delta = -23, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
1 \overset{\textcolor{red}{3}}{1} 2 3 1 5 1 3 1 6 1 3 1 6 1 3 1 \quad 1 \text{ }_5 \text{ }_1 \text{ }_{[\# = 3]} \quad \Delta = -23, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1 \overset{\textcolor{red}{3}}{1} 2 3 1 5 1 3 1 6 1 3 1 6 1 3 1 \quad 1 \text{ }_5 \text{ }_1 \text{ }_{[\# = 4]} \quad \Delta = -23, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1 \overset{\textcolor{red}{3}}{1} 2 3 1 5 1 3 1 6 1 3 1 6 1 3 1 \quad 1 \text{ }_5 \text{ }_1 \text{ }_{[\# = 9]} \quad \Delta = -23, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix} \\
1 \overset{\textcolor{red}{3}}{1} 2 3 1 5 1 3 1 6 1 3 1 6 1 3 1 \quad 1 \text{ }_5 \text{ }_1 \text{ }_{[\# = 8]} \quad \Delta = -23, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
1 \overset{\textcolor{red}{3}}{1} 2 3 1 5 1 3 1 6 1 3 1 6 1 3 1 \quad 1 \text{ }_5 \text{ }_1 \text{ }_{[\# = 6]} \quad \Delta = -23, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1 \overset{\textcolor{red}{3}}{1} 2 3 1 5 1 3 1 6 1 3 1 6 1 3 1 \quad 1 \text{ }_5 \text{ }_1 \text{ }_{[\# = 5]} \quad \Delta = -23, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1232161316131612321

$$\begin{aligned}
1232161316131612321 \overset{\textcolor{red}{3}}{3} \text{ }_{[\# = 8]} \quad \Delta = -2, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
1232161316131612321 \overset{\textcolor{red}{3}}{3} \text{ }_{[\# = 9]} \quad \Delta = -2, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix} \\
1232161316131612321 \overset{\textcolor{red}{3}}{3} \text{ }_{[\# = 2]} \quad \Delta = -2, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1232161316131612321 \overset{\textcolor{red}{3}}{3} \text{ }_{[\# = 5]} \quad \Delta = -2, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
1232161316131612321 \overset{\textcolor{red}{3}}{3} \text{ }_{[\# = 6]} \quad \Delta = -2, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1232161316131612321 \overset{\textcolor{red}{3}}{3} \text{ }_{[\# = 7]} \quad \Delta = -2, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
1232161316131612321 \overset{\textcolor{red}{3}}{3} \text{ }_{[\# = 4]} \quad \Delta = -2, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1232161316131612321 \overset{\textcolor{red}{3}}{3} \text{ }_{[\# = 3]} \quad \Delta = -2, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix}
\end{aligned}$$

Base: 6131513161316131612

$$\begin{aligned}
6131513161316131612 \quad \overset{\textcolor{red}{2}}{2} \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
6131513161316131612 \quad \overset{\textcolor{red}{2}}{2} \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
6131513161316131612 \quad \overset{\textcolor{red}{3}}{3} \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
6131513161316131612 \quad \overset{\textcolor{red}{2}}{2} \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
6131513161316131612 \quad \overset{\textcolor{red}{2}}{2} \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
6131513161316131612 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 168 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
6131513161316131612 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 207 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
6131513161316131612 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
6131513161316131612 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
6131513161316131612 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 142 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
\text{Base: } 2216131613161315131 & \\
2216131613161315131 \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
2216131613161315131\textcolor{red}{3} \text{ [#=8]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
2216131613161315131\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2216131613161315131 \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
2216131613161315131 \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2216131613161315131 \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2216131613161315131 \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2216131613161315131 \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
2216131613161315131 \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2216131613161315131 \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
2216131613161315131 \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
2216131613161315131\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
2216131613161315131\textcolor{red}{3} \text{ [#=7]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
2216131613161315131 \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 27 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
2216131613161315131\overset{2}{3} \text{ [#=2]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2216131613161315131\overset{2}{3} \text{ [#=5]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
2216131613161315131\overset{2}{3} \text{ [#=4]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
2216131613161315131\overset{2}{3} \text{ [#=6]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1413161316131612321

$$\begin{aligned}
1413161316131612321 \overset{2}{3} \quad \Delta = -19, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1413161316131612321 \overset{2}{3} \quad \Delta = -19, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1413161316131612321 \overset{3}{3} \quad \Delta = -27, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 30 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1232161316131613141\overset{2}{3} \text{ [#=2]} \quad \Delta = -27, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1413161316131612321\overset{2}{3} \text{ [#=2]} \quad \Delta = -27, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1232161316131613141\overset{2}{3} \text{ [#=3]} \quad \Delta = -27, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1413161316131612321\overset{2}{3} \text{ [#=3]} \quad \Delta = -27, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1413161316131612321\overset{2}{3} \text{ [#=6]} \quad \Delta = -27, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1413161316131612321 \overset{2}{3} \quad \Delta = -19, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1413161316131612321 \overset{2}{3} \quad \Delta = -19, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1413161316131612321 \overset{2}{3} \quad \Delta = -19, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1413161316131612321\overset{2}{3} \text{ [#=5]} \quad \Delta = -27, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1413161316131612321 \overset{2}{3} \quad \Delta = -19, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1413161316131612321 \overset{2}{3} \quad \Delta = -19, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1413161316131612321\overset{2}{3} \text{ [#=9]} \quad \Delta = -27, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1413161316131612321\overset{3}{3} \text{ [#=4]} \quad \Delta = -27, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1413161316131612321\overset{3}{3} \text{ [#=8]} \quad \Delta = -27, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1413161316131612321\overset{3}{3} \text{ [#=7]} \quad \Delta = -27, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
\text{Base: } 1321613161316131512 & \\
2151316131613161231\overset{3}{3} \text{ [#=4]} \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
2151316131613161231\overset{3}{3} \text{ [#=5]} \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
1321613161316131512\overset{3}{3} \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
1321613161316131512\overset{2}{2} \quad \Delta = -41, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1321613161316131512\overset{2}{2} \quad \Delta = -41, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1321613161316131512\overset{2}{2} \quad \Delta = -41, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1321613161316131512\overset{2}{2} \quad \Delta = -41, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1321613161316131512\overset{2}{2} \quad \Delta = -41, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2151316131613161231\overset{3}{3} \text{ [#=3]} \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
2151316131613161231\overset{3}{3} \text{ [#=2]} \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1321613161316131512\overset{2}{2} \quad \Delta = -41, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1321613161316131512\overset{2}{2} \quad \Delta = -41, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1321613161316131512\overset{2}{2} \quad \Delta = -41, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\text{Base: } 132161316131613151\overset{1}{1} & \\
132161316131613151\overset{2}{2}\overset{1}{1} \quad \Delta = -41, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1\overset{3}{3}1321613161316131\overset{1}{1} \text{ [#=4]} \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1\overset{3}{3}1321613161316131\overset{1}{1} \text{ [#=3]} \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{l}
1^{\overline{3}1321613161316131} \quad 5 \quad 1 \text{ [#=2]} \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1^{\overline{3}1321613161316131} \quad 5 \quad 1 \text{ [#=5]} \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
132161316131613151^{\overline{2}} \quad \Delta = -41, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
132161316131613151^{\overline{2}} \quad \Delta = -41, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
132161316131613151^{\overline{2}} \quad \Delta = -41, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
132161316131613151^{\overline{2}} \quad \Delta = -41, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
132161316131613151^{\overline{2}1} \quad \Delta = -41, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
132161316131613151^{\overline{2}} \quad \Delta = -41, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\text{Base: } 1413161316131613141 \\
1413161316131613141^{\overline{2}} \quad \Delta = -44, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1413161316131613141^{\overline{2}} \quad \Delta = -44, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 37 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1413161316131613141^{\overline{2}} \quad \Delta = -44, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 40 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
1413161316131613141^{\overline{2}} \quad \Delta = -44, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1413161316131613141\overline{3} \text{ [#=2]} \quad \Delta = -52, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1413161316131613141\overline{3} \text{ [#=3]} \quad \Delta = -52, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1413161316131613141^{\overline{3}} \quad \Delta = -52, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 39 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
\text{Base: } 3131613161316131512 \\
3131613161316131512^{\overline{2}} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
3131613161316131512^{\overline{4}} \quad \Delta = -76, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 29 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
3131613161316131512^{\overline{3}} \quad \Delta = -56, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 30 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
3131613161316131512^{\overline{3}} \quad \Delta = -56, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
& \overset{3}{3131613161316131512} \text{ }_{[\# = 2]} \quad \Delta = -56, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 129 & 25 \\ 2 & 25 & 8 \end{pmatrix} \\
& \overset{3}{3131613161316131512} \text{ }_{[\# = 3]} \quad \Delta = -56, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 294 & 38 \\ 2 & 38 & 8 \end{pmatrix} \\
& 313161\overset{2}{3161316131512} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& 3131613161\overset{2}{316131512} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{2}{3131613161316131512} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& 3131613161\overset{2}{316131512} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 313161\overset{2}{3161316131512} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 31316131613161\overset{2}{31512} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& 31316131613161\overset{2}{31512} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 3131613161\overset{2}{316131512} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \text{Base: } 3131613161316131\overset{1}{51} \\
& 31316131613161\overset{2}{3151} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 313161\overset{2}{316131613151} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{4}{313161316131613151} \quad \Delta = -76, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 29 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 31316131613161\overset{2}{3151} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 313161\overset{2}{316131613151} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{3}{313161316131613151} \text{ }_{[\# = 2]} \quad \Delta = -56, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 129 & 25 \\ 2 & 25 & 8 \end{pmatrix} \\
& \overset{3}{313161316131613151} \quad \Delta = -56, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 30 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{3}{313161316131613151} \text{ }_{[\# = 3]} \quad \Delta = -56, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 294 & 38 \\ 2 & 38 & 8 \end{pmatrix} \\
& \overset{2}{313161316131613151} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
313161316131613151^1 & \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
313161316131613151^2 & \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
313161316131613151^2 & \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 2216131613161316122

$$\begin{aligned}
2216131613161316122^3 & \quad \Delta = -67, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 15 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
2216131613161316122^3 & \quad \Delta = -66, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 36 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
2216131613161316122^2 & \quad \Delta = -58, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
2216131613161316122^2 & \quad \Delta = -58, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
2216131613161316122^2 & \quad \Delta = -58, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 37 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
2216131613161316122^2 & \quad \Delta = -58, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 613151316131613161¹

$$\begin{aligned}
613151316131613161^2 & \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 142 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
613151316131613161^2 & \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 207 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
613151316131613161^2 & \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 168 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
613151316131613161^2 & \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
613151316131613161^2 & \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
613151316131613161^2 & \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
613151316131613161^2 & \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
613151316131613161^{21} & \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 16²¹131613161315131

$$16^{21}131613161315131^3 \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\begin{array}{ll}
{}^{21}1\bar{6}131613161315131\bar{3} \text{ }_{[\# = 4]} & \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
{}^{21}1\bar{6}131613161315131\bar{3} \text{ }_{[\# = 2]} & \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
{}^{21}1\bar{6}131613161315131\bar{3} \text{ }_{[\# = 5]} & \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
{}^{21}1\bar{6}131613161315131\bar{3} \text{ }_{[\# = 8]} & \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
{}^{21}\bar{2}1\bar{6}131613161315131 & \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^{21}1\bar{6}131613161315131 & \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
{}^{21}1\bar{6}131613161315131\bar{3} \text{ }_{[\# = 6]} & \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
{}^{21}1\bar{6}13161316131\bar{5}131 & \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
{}^{21}1\bar{6}13161316131\bar{5}131 & \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
{}^{21}1\bar{6}131613161315131 & \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
{}^{21}1\bar{6}131613161315131 & \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
{}^{21}1\bar{6}131613161315131 & \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
{}^{21}\bar{2}1\bar{6}131613161315131 & \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
{}^{21}1\bar{6}13161316131\bar{5}131 & \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
{}^{21}1\bar{6}131613161315131\bar{3} \text{ }_{[\# = 7]} & \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
{}^{21}1\bar{6}131613161315131\bar{3} \text{ }_{[\# = 9]} & \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
\text{Base: } {}^{21}1\bar{6}131613161316122 & \\
{}^{21}1\bar{6}1316131613161\bar{2}2 & \Delta = -58, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
{}^{21}1\bar{6}1316131613161\bar{2}2 & \Delta = -67, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
{}^{21}1\bar{6}1316131613161\bar{2}2 & \Delta = -66, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{ll}
{}^{21}16131613161316122 & \Delta = -66, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 15 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
{}^{21}16131613161316122 & \Delta = -58, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
{}^{21}16131613161316122 & \Delta = -58, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^{21}16131613161316122 & \Delta = -58, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
{}^{21}16131613161316122 & \Delta = -58, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
{}^{21}16131613161316122 & \Delta = -58, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^{21}16131613161316122 & \Delta = -58, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
{}^{21}16131613161316122 & \Delta = -58, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{array}$$

Base: ${}^{21}161316131613161$

$$\begin{array}{ll}
{}^{21}161316131613161 & \Delta = -66, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 15 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
{}^{21}161316131613161 & \Delta = -58, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^{21}161316131613161 & \Delta = -58, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^{21}161316131613161 & \Delta = -58, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
{}^{21}161316131613161 & \Delta = -58, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{array}$$

Base: ${}^{131}1613161316131512$

$$\begin{array}{ll}
{}^{3131}1613161316131512 \text{ [#=6]} & \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
{}^{1312}1613161316131512 & \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^{3131}1613161316131512 \text{ [#=9]} & \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
{}^{3131}1613161316131512 \text{ [#=8]} & \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
{}^{3131}1613161316131512 \text{ [#=7]} & \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{l}
^{131} 1 \ 6 \ 13161316131512^{\color{red}3} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
^{3131} 1 \ 6 \ 13161316131512 \text{ [#=5]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
^{3131} 1 \ 6 \ 13161316131512 \text{ [#=4]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
^{3131} 1 \ 6 \ 13161316131512 \text{ [#=3]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
^{3131} 1 \ 6 \ 13161316131512 \text{ [#=2]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\\
^{131} 1 \ 6 \ 13161316131512^{\color{red}2} \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
^{131} 1 \ 6 \ 13161316131512^{\color{red}2} \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
^{131} 1 \ 6 \ 13161316131512^{\color{red}2} \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
^{131} 1 \ 6 \ 13161316131512 \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
^{131} 1 \ 6 \ 13161316131512^{\color{red}2} \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
^{131} 1 \ 6 \ 13161316131512^{\color{red}2} \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
^{131} 1 \ 6 \ 13161316131512^{\color{red}2} \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\\
\text{Base: } ^{131} 1 \ 6 \ 1316131613151^{\color{red}1} \\
^{3131} 1 \ 6 \ 1316131613151^{\color{red}1} \text{ [#=6]} \quad \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
\\
^{131} 1 \ 6 \ 1316131613151^{\color{red}2} \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
^{131} 1 \ 6 \ 1316131613151^{\color{red}2} \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
^{1312} 1 \ 6 \ 1316131613151^{\color{red}1} \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
^{131} 1 \ 6 \ 1316131613151^{\color{red}2} \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
^{131} 1 \ 6 \ 1316131613151^{\color{red}1} \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
^{131} 1 \ 6 \ 1316131613151^{\color{red}21} \quad \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{lcl}
\begin{array}{c} 131 \\ 1 \ 6 \ 1316131613151 \end{array} \begin{array}{c} \textcolor{red}{2} \\ \end{array} \begin{array}{c} 1 \\ \end{array} & \Delta = -40, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\begin{array}{c} \textcolor{red}{3131} \\ 1 \ 6 \ 1316131613151 \end{array} \begin{array}{c} 1 \\ \end{array} \text{[#}=9] & \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
\begin{array}{c} \textcolor{red}{3131} \\ 1 \ 6 \ 1316131613151 \end{array} \begin{array}{c} 1 \\ \end{array} \text{[#}=8] & \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
\begin{array}{c} \textcolor{red}{3131} \\ 1 \ 6 \ 1316131613151 \end{array} \begin{array}{c} 1 \\ \end{array} \text{[#}=7] & \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
\begin{array}{c} \textcolor{red}{3131} \\ 1 \ 6 \ 1316131613151 \end{array} \begin{array}{c} 1 \\ \end{array} \text{[#}=5] & \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
\begin{array}{c} \textcolor{red}{3131} \\ 1 \ 6 \ 1316131613151 \end{array} \begin{array}{c} 1 \\ \end{array} \text{[#}=4] & \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
\begin{array}{c} \textcolor{red}{3131} \\ 1 \ 6 \ 1316131613151 \end{array} \begin{array}{c} 1 \\ \end{array} \text{[#}=3] & \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\begin{array}{c} \textcolor{red}{3131} \\ 1 \ 6 \ 1316131613151 \end{array} \begin{array}{c} 1 \\ \end{array} \text{[#}=2] & \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{array}$$

Base: $\begin{array}{c} 1 \\ 613161316131613151 \end{array}$

$$\begin{array}{lcl}
\begin{array}{c} 1 \\ 613161316131613151 \end{array} \begin{array}{c} \textcolor{red}{2} \\ \end{array} & \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
\begin{array}{c} \textcolor{red}{12} \\ 613161316131613151 \end{array} & \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
\begin{array}{c} \textcolor{red}{21} \\ 613161316131613151 \end{array} & \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
\begin{array}{c} \textcolor{red}{2} \\ 613161316131613151 \end{array} \begin{array}{c} 1 \\ \end{array} & \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\begin{array}{c} 1 \\ 613161316131613151 \end{array} \begin{array}{c} \textcolor{red}{2} \\ \end{array} & \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\begin{array}{c} 1 \\ 613161316131613151 \end{array} \begin{array}{c} \textcolor{red}{2} \\ \end{array} & \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\begin{array}{c} 1 \\ 613161316131613151 \end{array} \begin{array}{c} \textcolor{red}{2} \\ \end{array} & \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\begin{array}{c} 1 \\ 613161316131613151 \end{array} \begin{array}{c} \textcolor{red}{2} \\ \end{array} & \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\begin{array}{c} \textcolor{red}{21} \\ 613161316131613151 \end{array} & \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\begin{array}{c} \textcolor{red}{31} \\ 613161316131613151 \end{array} & \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{array}$$

Base: $\begin{array}{c} 31 \\ 31613161316131512 \end{array}$

$$\begin{array}{ll}
{}^{31}\bar{6}13161316131512 & \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
{}^{31}\bar{6}13161316131512 & \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^{31}\bar{6}13161316131512 & \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
{}^{31}\bar{6}13161316131512 & \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 37 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
{}^{31}\bar{6}13161316131512 & \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
{}^{231}\bar{6}13161316131512 & \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
{}^{31}\bar{6}13161316131512 & \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 20 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
{}^{31}\bar{6}13161316131512 & \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
{}^{31}\bar{6}13161316131512\bar{3} & \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
{}^{31}\bar{6}13161316131512 & \Delta = -56, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\text{Base: } {}^{31}\bar{6}1316131613151 & \\
{}^{31}\bar{6}1316131613151 & \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
{}^{31}\bar{6}1316131613151 & \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
{}^{31}\bar{6}1316131613151 & \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^{31}\bar{6}1316131613151 & \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
{}^{231}\bar{6}1316131613151 & \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
{}^{31}\bar{6}1316131613151\bar{2} & \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
{}^{31}\bar{6}1316131613151 & \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
{}^{31}\bar{6}1316131613151 & \Delta = -48, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 37 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
{}^{31}\bar{6}1316131613151\bar{3} & \Delta = -56, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{array}$$

Base: 7123151316131613151

$$\begin{aligned}
7123151^2316131613151 \quad \Delta = -120, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
7123151316131613151^2 \quad \Delta = -120, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
7123151316131613151^2 \quad \Delta = -120, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
7123151316131613151^2 \quad \Delta = -120, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
7123151316131613151^2 \quad \Delta = -120, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
7123151316131613151^2 \quad \Delta = -120, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
7123151316131613151^2 \quad \Delta = -120, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
7123151316131613151^2 \quad \Delta = -120, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: 4131613161316123217

$$\begin{aligned}
4131613161316123217^3 \quad \Delta = -132, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
4131613161316123217^2 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
4131613161316123217^2 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 5 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
4131613161316123217^2 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
4131613161316123217^2 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
4131613161316123217^2 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
4131613161316123217^2 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
4131613161316123217^2 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
4131613161316123217^2 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 37 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1316131613161232171

$$1712321613161316131^3 \text{ [#=6]} \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$\begin{aligned}
1712321613161316131\overset{2}{3} \text{ [#=7]} \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1712321613161316131\overset{2}{3} \text{ [#=8]} \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
13161316131612321\overset{2}{71} \quad \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
13161316131612321\overset{2}{71} \quad \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1712321613161316131\overset{2}{3} \text{ [#=3]} \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1712321613161316131\overset{2}{3} \text{ [#=4]} \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1712321613161316131\overset{2}{3} \text{ [#=2]} \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1712321613161316131\overset{2}{3} \text{ [#=5]} \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
13161316131612321\overset{2}{71} \quad \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 100 \quad & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
13161316131612321\overset{2}{71} \quad \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
13161316131612321\overset{2}{71} \quad \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
13161316131612321\overset{2}{71} \quad \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
13161316131612321\overset{2}{71} \quad \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1712321613161316131\overset{2}{3} \text{ [#=9]} \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}
\end{aligned}$$

Base: 4131613161315132218

$$\begin{aligned}
413161316131513221\overset{2}{8} \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
413161316131513221\overset{2}{8} \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
413161316131513221\overset{2}{8} \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
413161316131513221\overset{2}{8} \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 6 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
413161316131513221\overset{2}{8} \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
4131613161315132218 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
4131613161315132218 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
4131613161315132218 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
4131613161315132218 \quad \Delta = -132, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: 8123216131613161232

$$\begin{aligned}
8123216131613161232 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
8123216131613161232 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
8123216131613161232 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 256 & \begin{pmatrix} 0 & 16 & 2 \\ 16 & 206 & 28 \\ 2 & 28 & 8 \end{pmatrix} \\
8123216131613161232 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 196 & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
8123216131613161232 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
8123216131613161232 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
8123216131613161232 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1316131613151322181

$$\begin{aligned}
1812231513161316131\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1812231513161316131\textcolor{red}{3} \text{ [#=6]} \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1812231513161316131\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1812231513161316131\textcolor{red}{3} \text{ [#=8]} \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1812231513161316131\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1316131613151322181 \quad \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
1812231513161316131\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1316131613151322181^2 \quad \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1316131613151322181^2 \quad \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1316131613151322181^2 \quad \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1316131613151322181^2 \quad \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1316131613151322181^2 \quad \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1316131613151322181^2 \quad \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
181223151316131613131^3_{[\# = 9]} \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
181223151316131613131^3_{[\# = 7]} \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}
\end{aligned}$$

Base: 4131613161315132181

$$\begin{aligned}
4131613161315132181^2 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
4131613161315132181^2 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
4131613161315132181^2 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
4131613161315132181^2 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
4131613161315132181^3 \quad \Delta = -132, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
4131613161315132181^2 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
4131613161315132181^2 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
4131613161315132181^2 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
4131613161315132181^2 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{aligned}$$

Base: 3216131613161232181

$$3216131613161232181^2 \quad \Delta = -125, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix}$$

$$\begin{aligned}
3216131613161232181 \quad \Delta = -125, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
3216131613161232181 \quad \Delta = -125, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
3216131613161232181 \quad \Delta = -125, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
3216131613161232181 \quad \Delta = -125, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 206 & 28 \\ 2 & 28 & 8 \end{pmatrix} \\
3216131613161232181 \quad \Delta = -125, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
3216131613161232181 \quad \Delta = -125, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: $16^{31}131613161232181$

$$\begin{aligned}
316^{31}131613161232181 \quad \Delta = -132, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
16^{31}131613161232181 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
16^{31}131613161232181 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
16^{31}131613161232181 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
16^{31}131613161232181 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 206 & 28 \\ 2 & 28 & 8 \end{pmatrix} \\
16^{31}131613161232181 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
316^{21}131613161232181 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
16^{31}131613161232181 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
16^{31}131613161232181 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1316131613151321812

$$\begin{aligned}
21812315131613161313^{31} \text{ [#=9]} \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1316131613151321812 \quad \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
1316131613151321812 \quad \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1316131613151321812 \quad \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1316131613151321812 \quad \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1316131613151321812 \quad \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1316131613151321812 \quad \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1316131613151321812 \quad \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1316131613151321812 \quad \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
21812315131613161313 \text{ [#=4]} \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
21812315131613161313 \text{ [#=3]} \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21812315131613161313 \text{ [#=2]} \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21812315131613161313 \text{ [#=5]} \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
21812315131613161313 \text{ [#=6]} \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
21812315131613161313 \text{ [#=7]} \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
21812315131613161313 \text{ [#=8]} \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}
\end{aligned}$$

Base: 2161316131612321812

$$\begin{aligned}
2161316131612321812 \quad \Delta = -116, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2161316131612321812 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2161316131612321812 \quad \Delta = -116, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
2161316131612321812 \quad \Delta = -116, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
2161316131612321812 \quad \Delta = -116, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
2161316131612321812 \quad \Delta = -116, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{ll}
2161316131612321812^2 & \Delta = -116, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2161316131612321812^2 & \Delta = -116, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
2161316131612321812^2 & \Delta = -116, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\text{Base: } 161316131612321812^1 & \\
161316131612321812^1 & \Delta = -116, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
161316131612321812^1 & \Delta = -116, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
161316131612321812^1 & \Delta = -116, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
161316131612321812^1 & \Delta = -116, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
161316131612321812^1 & \Delta = -116, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
161316131612321812^1 & \Delta = -116, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
161316131612321812^{12} & \Delta = -116, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\text{Base: } 2218123217123218122 & \\
2218123217123218122^3 & \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2218123217123218122^2 & \Delta = -115, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
2218123217123218122^2 & \Delta = -115, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
2218123217123218122^2 & \Delta = -115, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
\text{Base: } 2181232171232181223 & \\
2181232171232181223^2 & \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
2181232171232181223^2 & \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2181232171232181223^2 & \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
2181232171232181223^2 & \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{array}$$

$$2181232171232181223 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$2181232171232181223 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

Base: 1613161316123218123

$$1613161316123218123 \quad \Delta = -125, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 289 \quad \begin{pmatrix} 0 & 17 & 2 \\ 17 & 168 & 26 \\ 2 & 26 & 8 \end{pmatrix}$$

$$1613161316123218123 \quad \Delta = -125, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$1613161316123218123 \quad \Delta = -125, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1613161316123218123 \quad \Delta = -125, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$1613161316123218123 \quad \Delta = -125, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$1613161316123218123 \quad \Delta = -125, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1613161316123218123 \quad \Delta = -125, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 1231513218123151321

$$1231513218123151321\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -3, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix}$$

$$1231513218123151321\textcolor{red}{3} \text{ [#=8]} \quad \Delta = -3, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix}$$

$$1231513218123151321\textcolor{red}{3} \text{ [#=7]} \quad \Delta = -3, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix}$$

$$1231513218123151321\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -3, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1231513218123151321\textcolor{red}{3} \text{ [#=6]} \quad \Delta = -3, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

$$1231513218123151321\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -3, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix}$$

$$1231513218123151321\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -3, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1231513218123151321\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -3, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

Base: 1315132181231513221

$$1223151321812315131\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -2, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

1223151321812315131 $\bar{3}$ [#=8]	$\Delta = -2, n_+ = 1, n_- = 19, n_0 = 0, \det = 64$	$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$
1315132181231513221 $\bar{3}$ [#=7]	$\Delta = -2, n_+ = 1, n_- = 19, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$
1223151321812315131 $\bar{3}$ [#=7]	$\Delta = -2, n_+ = 1, n_- = 19, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$
1315132181231513221 $\bar{3}$ [#=6]	$\Delta = -2, n_+ = 1, n_- = 19, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$
1223151321812315131 $\bar{3}$ [#=6]	$\Delta = -2, n_+ = 1, n_- = 19, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$
1315132181231513221 $\bar{3}$ [#=5]	$\Delta = -2, n_+ = 1, n_- = 19, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$
1315132181231513221 $\bar{3}$ [#=8]	$\Delta = -2, n_+ = 1, n_- = 19, n_0 = 0, \det = 64$	$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$
1223151321812315131 $\bar{3}$ [#=9]	$\Delta = -2, n_+ = 1, n_- = 19, n_0 = 0, \det = 81$	$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$
1223151321812315131 $\bar{3}$ [#=2]	$\Delta = -2, n_+ = 1, n_- = 19, n_0 = 0, \det = 4$	$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$
1315132181231513221 $\bar{3}$ [#=2]	$\Delta = -2, n_+ = 1, n_- = 19, n_0 = 0, \det = 4$	$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$
1315132181231513221 $\bar{3}$ [#=9]	$\Delta = -2, n_+ = 1, n_- = 19, n_0 = 0, \det = 81$	$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$
1223151321812315131 $\bar{3}$ [#=3]	$\Delta = -2, n_+ = 1, n_- = 19, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$
1315132181231513221 $\bar{3}$ [#=4]	$\Delta = -2, n_+ = 1, n_- = 19, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$
1223151321812315131 $\bar{3}$ [#=4]	$\Delta = -2, n_+ = 1, n_- = 19, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$
1315132181231513221 $\bar{3}$ [#=3]	$\Delta = -2, n_+ = 1, n_- = 19, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$
Base: 2161232181231513221		
2161232181231513221 $\bar{3}$ [#=2]	$\Delta = -22, n_+ = 1, n_- = 19, n_0 = 0, \det = 4$	$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$
2161232181231513221 $\bar{3}$ [#=3]	$\Delta = -22, n_+ = 1, n_- = 19, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$
2161232181231513221 $\bar{3}$ [#=4]	$\Delta = -22, n_+ = 1, n_- = 19, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$
2161232181231513221 $\bar{3}$ [#=5]	$\Delta = -22, n_+ = 1, n_- = 19, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$

$$\begin{aligned}
2161232181231513221\textcolor{red}{3} \text{ [#=6]} \quad \Delta = -22, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
2161232181231513221\textcolor{red}{2} \quad \Delta = -14, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2161232181231513221\textcolor{red}{2} \quad \Delta = -14, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2161232181231513221\textcolor{red}{3} \text{ [#=7]} \quad \Delta = -22, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
2161232181231513221\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -22, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
2161232181231513221\textcolor{red}{3} \text{ [#=8]} \quad \Delta = -22, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
2161232181231513221\textcolor{red}{2} \quad \Delta = -14, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
2161232181231513221\textcolor{red}{2} \quad \Delta = -14, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2161232181231513221\textcolor{red}{2} \quad \Delta = -14, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2161232181231513221\textcolor{red}{2} \quad \Delta = -14, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2161232181231513221\textcolor{red}{3} \quad \Delta = -22, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\textbf{Base: } 1316123218123151321 \\
1316123218123151321\textcolor{red}{2} \quad \Delta = -22, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1316123218123151321\textcolor{red}{3} \text{ [#=7]} \quad \Delta = -30, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
1231513218123216131\textcolor{red}{3} \text{ [#=7]} \quad \Delta = -30, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1316123218123151321\textcolor{red}{3} \text{ [#=6]} \quad \Delta = -30, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1231513218123216131\textcolor{red}{3} \text{ [#=6]} \quad \Delta = -30, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1316123218123151321\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -30, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1231513218123216131\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -30, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1316123218123151321\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -30, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1231513218123216131\overset{\textcolor{red}{3}}{\text{3}}_{[\# = 3]} \quad \Delta = -30, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1231513218123216131\overset{\textcolor{red}{3}}{\text{3}}_{[\# = 8]} \quad \Delta = -30, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1316123218123151321\overset{\textcolor{red}{3}}{\text{3}}_{[\# = 8]} \quad \Delta = -30, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1316123218123151321\overset{\textcolor{red}{3}}{\text{3}}_{[\# = 3]} \quad \Delta = -30, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
13161232181231\overset{\textcolor{red}{2}}{\text{5}}1321 \quad \Delta = -22, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
13161232181231\overset{\textcolor{red}{2}}{\text{5}}1321 \quad \Delta = -22, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
13161232181231\overset{\textcolor{red}{2}}{\text{5}}1321 \quad \Delta = -22, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
131612321\overset{\textcolor{red}{2}}{\text{8}}123151321 \quad \Delta = -22, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
13161232181231\overset{\textcolor{red}{2}}{\text{5}}1321 \quad \Delta = -22, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1231513218123216131\overset{\textcolor{red}{3}}{\text{3}}_{[\# = 4]} \quad \Delta = -30, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1231513218123216131\overset{\textcolor{red}{3}}{\text{3}}_{[\# = 5]} \quad \Delta = -30, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1231513218123216131\overset{\textcolor{red}{3}}{\text{3}}_{[\# = 9]} \quad \Delta = -30, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1316123218123151321\overset{\textcolor{red}{3}}{\text{3}}_{[\# = 9]} \quad \Delta = -30, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
1316123218123151321\overset{\textcolor{red}{3}}{\text{3}}_{[\# = 4]} \quad \Delta = -30, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\textbf{Base: } \overset{1}{16}1232181231513221 \\
\overset{1}{16}1232181231513221\overset{\textcolor{red}{3}}{\text{3}}_{[\# = 4]} \quad \Delta = -22, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
\overset{1}{16}1232181231\overset{\textcolor{red}{2}}{\text{5}}13221 \quad \Delta = -14, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\overset{1}{16}1232181231\overset{\textcolor{red}{2}}{\text{5}}13221 \quad \Delta = -14, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\overset{1}{16}1232181231\overset{\textcolor{red}{2}}{\text{5}}13221 \quad \Delta = -14, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\overset{1}{16}1232181231\overset{\textcolor{red}{2}}{\text{5}}13221 \quad \Delta = -14, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
&161232181231513221 \quad \Delta = -14, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
&161232181231513221\textcolor{red}{3} \text{ [#=8]} \quad \Delta = -22, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
&161232181231513221\textcolor{red}{3} \text{ [#=7]} \quad \Delta = -22, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
&161232181231513221\textcolor{red}{3} \text{ [#=6]} \quad \Delta = -22, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
&161232181231513221\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -22, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
&161232181231513221\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -22, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
&161232181231513221\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -22, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
&161232181231513221\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -22, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}
\end{aligned}$$

Base: 7122315132181232161

$$\begin{aligned}
&7122315132181232161 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
&7122315132181232161 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
&7122315132181232161 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
&7122315132181232161 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 89 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
&7122315132181232161 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
&7122315132181232161 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 7123151321812321613

$$\begin{aligned}
&7123151321812321613 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
&7123151321812321613 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
&7123151321812321613 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
&7123151321812321613 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{aligned}$$

$$7123151321812321613 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$7123151321812321613 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

$$7123151321812321613 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: 1612321812315132171

$$1612321812315132171 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1612321812315132171 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1612321812315132171 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

$$1612321812315132171 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$1612321812315132171 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$1612321812315132171 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix}$$

Base: 3161315132181232171

$$3161315132181232171 \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$3161315132181232171 \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix}$$

$$3161315132181232171 \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 202 & 28 \\ 2 & 28 & 8 \end{pmatrix}$$

$$3161315132181232171 \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 232 & 30 \\ 2 & 30 & 8 \end{pmatrix}$$

$$3161315132181232171 \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 196 \quad \begin{pmatrix} 0 & 14 & 2 \\ 14 & 264 & 32 \\ 2 & 32 & 8 \end{pmatrix}$$

$$3161315132181232171 \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix}$$

$$3161315132181232171 \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: 6123151321812321712

$$6123151321812321712 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$6123151321812321712 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix}$$

$$\begin{aligned}
6123151321812321712 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
6123151321812321712 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
6123151321812321712 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
6123151321812321712 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
6123151321812321712 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1613151321812321712

$$\begin{aligned}
1613151321812321712 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1613151321812321712 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1613151321812321712 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
1613151321812321712 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1613151321812321712 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
1613151321812321712 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
1613151321812321712 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix}
\end{aligned}$$

Base: 2217123218123151321

$$\begin{aligned}
2217123218123151321 \quad \Delta = -41, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2217123218123151321 \quad \Delta = -41, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2217123218123151321 \quad \Delta = -41, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2217123218123151321 \quad \Delta = -41, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
2217123218123151321 \quad \Delta = -41, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2217123218123151321 \quad \Delta = -41, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
2217123218123151321\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\textcolor{red}{3} 2217123218123151321 \quad \Delta = -50, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2217123218123151321\textcolor{red}{3} \text{ [#=8]} \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
2217123218123151321\textcolor{red}{3} \text{ [#=7]} \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
2217123218123151321\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2217123218123151321\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
2217123218123151321\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
2217123218123151321\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
2217123218123151321\textcolor{red}{3} \text{ [#=6]} \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
\textbf{Base: } 6131513218123217123 \\
6131513218123217123 \textcolor{red}{2} \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
6131513218123217123 \textcolor{red}{2} \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
6131513218123217123 \textcolor{red}{2} \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
6131513218123217123 \textcolor{red}{2} \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 169 \quad & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
6131513218123217123 \textcolor{red}{2} \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 \quad & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
6131513218123217123 \textcolor{red}{2} \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
6131513218123217123 \textcolor{red}{2} \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 \quad & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
\textbf{Base: } 1321712321812315131 \\
1321712321812315131\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1315132181232171231\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1321712321812315131\overset{2}{3} \text{ [#=4]} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1321712321812315131\overset{2}{3} \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1321712321812315131\overset{2}{3} \text{ [#=9]} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1315132181232171231\overset{2}{3} \text{ [#=5]} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1321712321812315131\overset{2}{3} \text{ [#=5]} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1321712321812315131\overset{2}{3} \text{ [#=8]} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1321712321812315131\overset{2}{3} \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1321712321812315131\overset{2}{3} \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1321712321812315131\overset{2}{3} \text{ [#=6]} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1321712321812315131\overset{2}{3} \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1315132181232171231\overset{2}{3} \text{ [#=3]} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1321712321812315131\overset{2}{3} \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1321712321812315131\overset{2}{3} \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1321712321812315131\overset{2}{3} \text{ [#=2]} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1315132181232171231\overset{2}{3} \text{ [#=2]} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1321712321812315131\overset{2}{3} \text{ [#=7]} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
\text{Base: } 1232171232181231512 & \\
2151321812321712321\overset{2}{3} \text{ [#=3]} \quad \Delta = -51, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
2151321812321712321\overset{2}{3} \text{ [#=9]} \quad \Delta = -51, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
1232171232181231512\overset{2}{3} \quad \Delta = -43, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1232171232181231512^2 \quad \Delta = -43, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1232171232181231512^2 \quad \Delta = -43, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1232171232181231512^2 \quad \Delta = -43, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1232171232181231512^2 \quad \Delta = -43, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1232171232181231512^2 \quad \Delta = -43, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21513218123217123213^{\#8} \quad \Delta = -51, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
21513218123217123213^{\#6} \quad \Delta = -51, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
1232171232181231512^3 \quad \Delta = -51, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
21513218123217123213^{\#7} \quad \Delta = -51, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
21513218123217123213^{\#2} \quad \Delta = -51, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
21513218123217123213^{\#5} \quad \Delta = -51, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
21513218123217123213^{\#4} \quad \Delta = -51, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
\text{Base: } 123217123218123151^1 & \\
1^{\#5} \quad \Delta = -51, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
1^{\#4} \quad \Delta = -51, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
123217123218123151^2 \quad \Delta = -43, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1^{\#6} \quad \Delta = -51, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
1^{\#7} \quad \Delta = -51, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
1^{\#3} \quad \Delta = -51, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
123217123218123151^2 \quad \Delta = -43, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
123217123218123151^1 \quad \Delta = -43, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
123217123218123151^{21} \quad \Delta = -43, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
123217123218123151^2 \quad \Delta = -43, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1^{31232171232181231} 5^1 \quad [\# = 2] \quad \Delta = -51, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1^{31232171232181231} 5^1 \quad [\# = 9] \quad \Delta = -51, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
1^{31232171232181231} 5^1 \quad [\# = 8] \quad \Delta = -51, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix}
\end{aligned}$$

Base: 6123217123218123151

$$\begin{aligned}
6123217123218123151^2 \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
6123217123218123151^2 \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
6123217123218123151^2 \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
6123217123218123151^2 \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
6123217123218123151^2 \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
6123217123218123151^2 \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 4132181232171232171

$$\begin{aligned}
4132181232171232171^2 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
4132181232171232171^2 \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
4132181232171232171^2 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
4132181232171232171^2 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 163 & 25 \\ 2 & 25 & 8 \end{pmatrix} \\
4132181232171232171^2 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
4132181232171232171^2 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}
\end{aligned}$$

$$4132181^{2}232171232171 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix}$$

Base: 2171232171232181231

$$2171232171232181^{2}231 \quad \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$2171232171232181^{2}231 \quad \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$2171232171232181^{2}231 \quad \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$2171232171232181^{2}231 \quad \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$2171232171232181^{2}231 \quad \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$2171232171232181^{2}231 \quad \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$2171232171232181231^{3} \text{ [#=4]} \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$2171232171232181231^{3} \text{ [#=5]} \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$2171232171232181231^{3} \text{ [#=3]} \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$2171232171232181231^{3} \text{ [#=2]} \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

Base: 6123151321812321¹71

$$6123151321812321^{1}71 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$6123151^{2}321812321^{1}71 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$6123151321812321^{1}71 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix}$$

$$6123151321812321^{21}71 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$6123151321812321^{1}71 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

$$612315132181^{2}2321^{1}71 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

Base: 171232181231513161

$$171232181231^{2}513161 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

$$\begin{aligned}
171232181231513161 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
171232181231513161 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
171232181231513161 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
171232181231513161 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
171232181231513161 \quad \Delta = -118, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

Base: 171232171232181231

$$\begin{aligned}
171232171232181231\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
171232171232181231\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
171232171232181231\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
171232171232181231\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
171232171232181231 \quad \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
171232171232181231 \quad \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
171232171232181231 \quad \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
171232171232181231 \quad \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
171232171232181231 \quad \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 17123218123151321

$$\begin{aligned}
17123218123151321 \quad \Delta = -41, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
17123218123151321\textcolor{red}{3} \text{ [#=8]} \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
17123218123151321\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
17123218123151321\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{21}{17}123218123151321\overset{\color{red}{3}}{3} \text{ }_{[\# = 3]} \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{21}{17}123218123151321\overset{\color{red}{3}}{3} \text{ }_{[\# = 4]} \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{21}{17}123218123151321\overset{\color{red}{3}}{3} \text{ }_{[\# = 7]} \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
& \overset{21}{17}123218123151321\overset{\color{red}{2}}{2} \quad \Delta = -41, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{21}{17}123218123151321 \quad \Delta = -41, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{21}{17}123218123151321\overset{\color{red}{2}}{2} \quad \Delta = -41, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{21}{17}123218123151321\overset{\color{red}{2}}{2} \quad \Delta = -41, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{21}{17}123218123151321\overset{\color{red}{2}}{2} \quad \Delta = -41, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{21}{17}123218123151321\overset{\color{red}{3}}{3} \text{ }_{[\# = 6]} \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{21}{17}123218123151321\overset{\color{red}{3}}{3} \text{ }_{[\# = 5]} \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

Base: 8122315132181231513

$$\begin{aligned}
& 8122315132181231513\overset{\color{red}{2}}{2} \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& 8122315132181231513\overset{\color{red}{2}}{2} \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 8122315132181231513\overset{\color{red}{2}}{2} \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 8122315132181231513\overset{\color{red}{2}}{2} \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& 8122315132181231513\overset{\color{red}{2}}{2} \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
& 8122315132181231513\overset{\color{red}{2}}{2} \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
& 8122315132181231513\overset{\color{red}{2}}{2} \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

Base: 8123151321812315132

$$8123151321812315132\overset{\color{red}{2}}{2} \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$\begin{aligned}
8123151321812315132^2 \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
8123151321812315132^2 \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 289 & \begin{pmatrix} 0 & 17 & 2 \\ 17 & 219 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
8123151321812315132^2 \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
8123151321812315132^2 \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 196 & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
8123151321812315132^2 \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
8123151321812315132^2 \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 225 & \begin{pmatrix} 0 & 15 & 2 \\ 15 & 163 & 25 \\ 2 & 25 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1812231513218123215

$$\begin{aligned}
1812231513218123215^2 \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1812231513218123215^2 \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1812231513218123215^2 \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1812231513218123215^2 \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1812231513218123215^2 \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1812231513218123215^2 \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1812231513218123151

$$\begin{aligned}
1812231513218123151^2 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1812231513218123151^2 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1812231513218123151^2 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1812231513218123151^2 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1812231513218123151^2 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1812231513218123151^2 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1812315132181231513

$$\begin{aligned}
1812315132181^{23}1513 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1812315132181^{23}1513 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
181^{23}2315132181231513 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
18123151^{23}32181231513 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1812315132181^{23}231513 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
181231513218123151^{23}3 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1812315132181^{23}231513 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 2181223151321812321

$$\begin{aligned}
2181223151321812321^{33} \text{ [#=6]} \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
2181223151321812321^{33} \text{ [#=7]} \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
218122315132181^{23}2321 \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2181223151321812321^{33} \text{ [#=5]} \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
2181223151321812321^{33} \text{ [#=8]} \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
2181223151321812321^{33} \text{ [#=3]} \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2181^{23}223151321812321 \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
218122315132181^{23}2321 \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
2181223151^{23}321812321 \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
2181223151321812321^{23} \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
2181^{23}223151321812321 \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
2181223151321812321^{33} \text{ [#=2]} \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
2181223151321812321\overset{2}{3} \text{ [#=4]} \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
2181223151321812321\overset{2}{3} \text{ [#=9]} \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}
\end{aligned}$$

Base: 2181231513218123215

$$\begin{aligned}
218123151321812321\overset{2}{5} \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
218123151321812321\overset{2}{5} \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
218123151321812321\overset{2}{5} \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
218123151321812321\overset{2}{5} \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
218123151321812321\overset{2}{5} \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
218123151321812321\overset{2}{5} \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
218123151321812321\overset{2}{5} \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{aligned}$$

Base: 2181231513218123151

$$\begin{aligned}
21812315132181231\overset{2}{5}1 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
21812315132181231\overset{2}{5}1 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21812315132181231\overset{2}{5}1 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21812315132181231\overset{2}{5}1 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21812315132181231\overset{2}{5}1 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
21812315132181231\overset{2}{5}1 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21812315132181231\overset{2}{5}1 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{aligned}$$

Base: 2218123151321812321

$$\begin{aligned}
2218123151321812321\overset{2}{2} \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2218123151321812321\overset{2}{2} \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
2218123151321812321^2 \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
2218123151321812321^2 \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
2218123151321812321^3 \quad \Delta = -78, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2218123151321812321^3 \text{ [#=4]} \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
2218123151321812321^3 \text{ [#=2]} \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2218123151321812321^3 \text{ [#=3]} \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2218123151321812321^3 \text{ [#=5]} \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
2218123151321812321^3 \text{ [#=6]} \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
2218123151321812321^3 \text{ [#=7]} \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
2218123151321812321^2 \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
2218123151321812321^2 \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
2218123151321812321^3 \text{ [#=9]} \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
2218123151321812321^3 \text{ [#=8]} \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

Base: 3218123151321812315

$$\begin{aligned}
3218123151321812315^2 \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
3218123151321812315^2 \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 63 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
3218123151321812315^2 \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 196 & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 68 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
3218123151321812315^2 \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
3218123151321812315^2 \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
3218123151321812315^2 \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 144 & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 58 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{aligned}$$

$$3218123151321812315^2 \quad \Delta = -128, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: 1321812315132181231

$$13218123151321812313_{\# = 3} \quad \Delta = -84, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$13218123151321812313_{\# = 2} \quad \Delta = -84, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1321812315132181231^2 \quad \Delta = -76, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$1321812315132181231^2 \quad \Delta = -76, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$13218123151321812313_{\# = 5} \quad \Delta = -84, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$13218123151321812313_{\# = 4} \quad \Delta = -84, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1321812315132181231^2 \quad \Delta = -76, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

Base: 1316123218123216131

$$13161232181232161313_{\# = 9} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix}$$

$$13161232181232161313_{\# = 8} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix}$$

$$13161232181232161313_{\# = 2} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$13161232181232161313_{\# = 5} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix}$$

$$13161232181232161313_{\# = 6} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

$$1316123218123216131^2 \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$13161232181232161313_{\# = 3} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$13161232181232161313_{\# = 7} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix}$$

$$13161232181232161313_{\# = 4} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$1316123218123216131^2 \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1316123218123216131 \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: 6131612321812321613

$$6131612321812321613 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$6131612321812321613 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$6131612321812321613 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix}$$

$$6131612321812321613 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 225 \quad \begin{pmatrix} 0 & 15 & 2 \\ 15 & 163 & 25 \\ 2 & 25 & 8 \end{pmatrix}$$

$$6131612321812321613 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$6131612321812321613 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$6131612321812321613 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: 1613161232181232161

$$1613161232181232161 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1613161232181232161 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$1613161232181232161 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix}$$

$$1613161232181232161 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$1613161232181232161 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

$$1613161232181232161 \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 196 & 28 \\ 2 & 28 & 8 \end{pmatrix}$$

Base: 6123218123216131613

$$6123218123216131613 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$6123218123216131613 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$6123218123216131613 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$6123218123216131613 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 196 \quad \begin{pmatrix} 0 & 14 & 2 \\ 14 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix}$$

$$\begin{aligned}
6123218123216131613 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 324 & \begin{pmatrix} 0 & 18 & 2 \\ 18 & 196 & 28 \\ 2 & 28 & 8 \end{pmatrix} \\
6123218123216131613 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 144 & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
6123218123216131613 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1316131612321812321

$$\begin{aligned}
13161316123218123213 \quad [\# = 3] \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
13161316123218123213 \quad [\# = 5] \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
13161316123218123213 \quad [\# = 8] \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
12321812321613161313 \quad [\# = 8] \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
13161316123218123213 \quad [\# = 7] \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1316131612321812321 \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
12321812321613161313 \quad [\# = 9] \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
12321812321613161313 \quad [\# = 3] \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
12321812321613161313 \quad [\# = 7] \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
12321812321613161313 \quad [\# = 2] \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
12321812321613161313 \quad [\# = 4] \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
13161316123218123213 \quad [\# = 4] \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
13161316123218123213 \quad [\# = 6] \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
12321812321613161313 \quad [\# = 6] \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
12321812321613161313 \quad [\# = 5] \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
13161316123218123213 \quad [\# = 2] \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1316131612321812321\overset{2}{3} \text{ [#=9]} \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1316131612321812321\overset{2}{} \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1316131612321812321\overset{2}{} \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1316131612321812321\overset{2}{} \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
1316131612321812321\overset{2}{} \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1316131612321812321\overset{2}{} \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\text{Base: } 6131613161232181232 \\
6131613161232181232\overset{2}{} \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 324 \quad & \begin{pmatrix} 0 & 18 & 2 \\ 18 & 232 & 30 \\ 2 & 30 & 8 \end{pmatrix} \\
6131613161232181232\overset{2}{} \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
6131613161232181232\overset{2}{} \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
6131613161232181232\overset{2}{} \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
6131613161232181232\overset{2}{} \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 225 \quad & \begin{pmatrix} 0 & 15 & 2 \\ 15 & 163 & 25 \\ 2 & 25 & 8 \end{pmatrix} \\
6131613161232181232\overset{2}{} \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 256 \quad & \begin{pmatrix} 0 & 16 & 2 \\ 16 & 174 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
6131613161232181232\overset{2}{} \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 576 \quad & \begin{pmatrix} 0 & 24 & 2 \\ 24 & 406 & 40 \\ 2 & 40 & 8 \end{pmatrix} \\
\text{Base: } 5131612321812321712 \\
5131612321812321712\overset{2}{} \quad \Delta = -120, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
5131612321812321712\overset{2}{} \quad \Delta = -120, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
5131612321812321712\overset{2}{} \quad \Delta = -120, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
5131612321812321712\overset{2}{} \quad \Delta = -120, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
5131612321812321712\overset{2}{} \quad \Delta = -120, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{aligned}$$

$$5131612321812321712^2 \quad \Delta = -120, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix}$$

$$5131612321812321712^2 \quad \Delta = -120, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 196 & 28 \\ 2 & 28 & 8 \end{pmatrix}$$

Base: 1316123218123217122

$$1316123218123217122^2 \quad \Delta = -68, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1316123218123217122^2 \quad \Delta = -68, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1316123218123217122^2 \quad \Delta = -68, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1316123218123217122^2 \quad \Delta = -68, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$2217123218123216131^3_{\# = 9} \quad \Delta = -76, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix}$$

$$1316123218123217122^2 \quad \Delta = -68, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1316123218123217122^2 \quad \Delta = -68, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$2217123218123216131^3_{\# = 8} \quad \Delta = -76, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix}$$

$$2217123218123216131^3_{\# = 2} \quad \Delta = -76, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$2217123218123216131^3_{\# = 3} \quad \Delta = -76, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$2217123218123216131^3_{\# = 4} \quad \Delta = -76, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$2217123218123216131^3_{\# = 6} \quad \Delta = -76, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

$$2217123218123216131^3_{\# = 7} \quad \Delta = -76, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix}$$

$$1316123218123217122^3 \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$2217123218123216131^3_{\# = 5} \quad \Delta = -76, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix}$$

Base: 2217123218123217122

$$2217123218123217122^3 \quad \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$2217123218123217122^2 \quad \Delta = -87, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$2217123218123217122 \quad \Delta = -87, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$2217123218123217122 \quad \Delta = -87, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: 2161232181232171231

$$2161232181232171231 \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$2161232181232171231 \textcolor{red}{3} \text{ [#=2]} \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$2161232181232171231 \textcolor{red}{3} \text{ [#=4]} \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$2161232181232171231 \textcolor{red}{3} \text{ [#=5]} \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$2161232181232171231 \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$2161232181232171231 \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$2161232181232171231 \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$2161232181232171231 \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$2161232181232171231 \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$2161232181232171231 \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$2161232181232171231 \textcolor{red}{3} \text{ [#=3]} \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: ¹161232181232171231

$$\textcolor{red}{1}161232181232171231 \textcolor{red}{3} \text{ [#=3]} \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$\textcolor{red}{1}161232181232171231 \textcolor{red}{3} \text{ [#=5]} \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\textcolor{red}{1}161232181232171231 \textcolor{red}{3} \text{ [#=2]} \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$\textcolor{red}{1}161232181232171231 \textcolor{red}{3} \text{ [#=4]} \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\textcolor{red}{1}^{\textcolor{red}{2}}161232181232171231 \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$\begin{aligned}
161232181232171231^1 \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
161232181232171231^1 \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
161232181232171231^1 \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
161232181232171231^1 \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1612321812321712315

$$\begin{aligned}
1612321812321712315^2 \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1612321812321712315^2 \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1612321812321712315^2 \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1612321812321712315^2 \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1612321812321712315^2 \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1612321812321712315^2 \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1232171232181232151

$$\begin{aligned}
15123218123217123213^{\#4} \quad \Delta = -53, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1232171232181232151^2 \quad \Delta = -45, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1232171232181232151^2 \quad \Delta = -45, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1232171232181232151^2 \quad \Delta = -45, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
15123218123217123213^{\#2} \quad \Delta = -53, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
15123218123217123213^{\#3} \quad \Delta = -53, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
15123218123217123213^{\#5} \quad \Delta = -53, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
15123218123217123213^{\#6} \quad \Delta = -53, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1512321812321712321\bar{3} \text{ [#=8]} \quad \Delta = -53, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
1512321812321712321\bar{3} \text{ [#=9]} \quad \Delta = -53, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
1512321812321712321\bar{3} \text{ [#=7]} \quad \Delta = -53, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
1232171232181\bar{2}232151 \quad \Delta = -45, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1232171\bar{2}232181232151 \quad \Delta = -45, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\text{Base: } 6123217123218123215 \\
61232171232181\bar{2}23215 \quad \Delta = -123, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
612321712321812321\bar{2} \quad \Delta = -123, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
61232171232181\bar{2}23215 \quad \Delta = -123, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
61\bar{2}23217123218123215 \quad \Delta = -123, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
61232171\bar{2}23218123215 \quad \Delta = -123, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 100 \quad & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
61232171\bar{2}23218123215 \quad \Delta = -123, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 \quad & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
\text{Base: } 1612321712321812321 \\
1612321712321812321\bar{3} \text{ [#=3]} \quad \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1612321712321812321\bar{3} \text{ [#=9]} \quad \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1612321712321812321\bar{3} \text{ [#=4]} \quad \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
161232171232181\bar{2}2321 \quad \Delta = -71, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
161\bar{2}2321712321812321 \quad \Delta = -71, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
161232171232181\bar{2}2321 \quad \Delta = -71, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1612321712321812321\bar{3} \text{ [#=8]} \quad \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1612321712321812321\overset{2}{3} \text{ [#=6]} \quad \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1612321712321812321\overset{2}{3} \text{ [#=5]} \quad \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1612321712321812321\overset{2}{3} \text{ [#=2]} \quad \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1612321712321812321\overset{2}{3} \quad \Delta = -71, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
1612321712321812321\overset{2}{3} \text{ [#=7]} \quad \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1612321712321812321\overset{2}{3} \quad \Delta = -71, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
\text{Base: } 171232181232161315 \quad & \\
171232181232161315\overset{2}{3} \quad \Delta = -120, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
171232181232161315\overset{2}{3} \quad \Delta = -120, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
171232181232161315\overset{2}{3} \quad \Delta = -120, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
171232181232161315\overset{2}{3} \quad \Delta = -120, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
171232181232161315\overset{2}{3} \quad \Delta = -120, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
171232181232161315\overset{2}{3} \quad \Delta = -120, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 \quad & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\text{Base: } 17^{21}123218123216131 \quad & \\
17^{21}123218123216131\overset{2}{3} \quad \Delta = -68, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
17^{21}123218123216131\overset{2}{3} \quad \Delta = -68, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
17^{21}123218123216131\overset{2}{3} \quad \Delta = -68, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
17^{21}123218123216131\overset{2}{3} \quad \Delta = -68, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
17^{21}123218123216131\overset{2}{3} \quad \Delta = -68, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
17^{21}123218123216131\overset{2}{3} \quad \Delta = -68, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{21}{17}123218123216131\overset{\color{red}{3}}{3} \text{ }_{[\# = 5]} \quad \Delta = -76, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{21}{17}123218123216131\overset{\color{red}{3}}{3} \text{ }_{[\# = 4]} \quad \Delta = -76, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{21}{17}123218123216131\overset{\color{red}{3}}{3} \text{ }_{[\# = 3]} \quad \Delta = -76, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{21}{17}123218123216131\overset{\color{red}{3}}{3} \text{ }_{[\# = 2]} \quad \Delta = -76, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
& \overset{21}{17}123218123216131\overset{\color{red}{3}}{3} \text{ }_{[\# = 6]} \quad \Delta = -76, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{21}{17}123218123216131\overset{\color{red}{3}}{3} \text{ }_{[\# = 9]} \quad \Delta = -76, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{21}{17}123218123216131\overset{\color{red}{3}}{3} \text{ }_{[\# = 8]} \quad \Delta = -76, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{21}{17}123218123216131\overset{\color{red}{3}}{3} \text{ }_{[\# = 7]} \quad \Delta = -76, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{21}{17}123218123217122$

$$\begin{aligned}
& \overset{21}{17}123218123217122 \quad \Delta = -87, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{21}{17}12321812321712\overset{\color{red}{2}}{2} \quad \Delta = -87, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{21}{17}12321812321712\overset{\color{red}{2}}{2} \quad \Delta = -87, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{21}{17}12321812321712\overset{\color{red}{2}}{2} \quad \Delta = -87, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{21}{17}12321812321712\overset{\color{red}{2}}{2} \quad \Delta = -87, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{21}{17}\overset{\color{red}{2}}{2}123218123217122 \quad \Delta = -87, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{21}{17}12321812321712\overset{\color{red}{3}}{3} \quad \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}
\end{aligned}$$

Base: $\overset{21}{17}12321812321\overset{21}{7}1$

$$\begin{aligned}
& \overset{21}{17}12321812321\overset{21}{7}1 \quad \Delta = -87, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{21}{17}\overset{\color{red}{2}}{2}12321812321\overset{21}{7}1 \quad \Delta = -87, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{21}{17}12321812321\overset{\color{red}{2}}{2}\overset{21}{7}1 \quad \Delta = -87, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: 2218123218123151315

$$\begin{aligned}
2218123218123151315 \quad \Delta = -120, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2218123218123151315 \quad \Delta = -120, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
2218123218123151315 \quad \Delta = -120, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
2218123218123151315 \quad \Delta = -120, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2218123218123151315 \quad \Delta = -120, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2218123218123151315 \quad \Delta = -120, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2218123218123151315 \quad \Delta = -120, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
2218123218123151315 \quad \Delta = -129, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2218123218123151315 \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1513218123218122315

$$\begin{aligned}
1513218123218122315 \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1513218123218122315 \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 50 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
1513218123218122315 \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1513218123218122315 \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1513218123218122315 \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1513218123218122315 \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1322181232181231512

$$\begin{aligned}
1322181232181231512 \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1322181232181231512 \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1322181232181231512 \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1322181232181231512 \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1322181232181231512 \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1322181232181231512 \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21513218123218122313 \text{ [#=3]} \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21513218123218122313 \text{ [#=4]} \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
21513218123218122313 \text{ [#=5]} \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1322181232181231512 \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
21513218123218122313 \text{ [#=2]} \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: 2221812321812315131

$$\begin{aligned}
2221812321812315131 \quad \Delta = -68, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2221812321812315131 \quad \Delta = -68, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2221812321812315131 \quad \Delta = -68, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
2221812321812315131 \quad \Delta = -68, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2221812321812315131 \quad \Delta = -68, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
2221812321812315131 \quad \Delta = -68, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
22218123218123151313 \text{ [#=2]} \quad \Delta = -76, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
22218123218123151313 \text{ [#=8]} \quad \Delta = -76, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
22218123218123151313 \text{ [#=9]} \quad \Delta = -76, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
22218123218123151313 \text{ [#=3]} \quad \Delta = -76, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
22218123218123151313 \text{ [#=4]} \quad \Delta = -76, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
2221812321812315131\bar{3} \text{ } [\# = 5] \quad \Delta = -76, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
2221812321812315131\bar{3} \text{ } [\# = 7] \quad \Delta = -76, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
2221812321812315131\bar{3} \text{ } [\# = 6] \quad \Delta = -76, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1322181232181231¹51

$$\begin{aligned}
132218123218123151\bar{2} \text{ } [\# = 1] \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1\bar{3}1322181232181231\bar{5} \text{ } [\# = 3] \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1\bar{3}1322181232181231\bar{5} \text{ } [\# = 2] \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1\bar{3}1322181232181231\bar{5} \text{ } [\# = 4] \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1\bar{3}1322181232181231\bar{5} \text{ } [\# = 5] \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
132218123218123151\bar{2} \text{ } [\# = 1] \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
132218123218123151\bar{2} \text{ } [\# = 1] \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
132218123218123151\bar{2} \text{ } [\# = 1] \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
132218123218123151\bar{2} \text{ } [\# = 1] \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: 2161232181232181222

$$\begin{aligned}
2161232181232181222\bar{2} \text{ } [\# = 1] \quad \Delta = -88, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2161232181232181222\bar{3} \text{ } [\# = 1] \quad \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
2161232181232181222\bar{2} \text{ } [\# = 1] \quad \Delta = -88, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
2161232181232181222\bar{2} \text{ } [\# = 1] \quad \Delta = -88, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
2161232181232181222\bar{2} \text{ } [\# = 1] \quad \Delta = -88, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2161232181232181222\bar{2} \text{ } [\# = 1] \quad \Delta = -88, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$2161232181232181222 \quad \Delta = -88, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

Base: 161232181232181222

$$161232181232181222 \quad \Delta = -88, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$161232181232181222 \quad \Delta = -88, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$161232181232181222 \quad \Delta = -88, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$161232181232181222 \quad \Delta = -88, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$161232181232181222 \quad \Delta = -88, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 5132218123218123215

$$5132218123218123215 \quad \Delta = -123, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$5132218123218123215 \quad \Delta = -123, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$5132218123218123215 \quad \Delta = -123, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix}$$

$$5132218123218123215 \quad \Delta = -123, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$5132218123218123215 \quad \Delta = -123, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$5132218123218123215 \quad \Delta = -123, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

Base: 1322181232181232151

$$1322181232181232151 \quad \Delta = -71, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1322181232181232151 \quad \Delta = -71, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1322181232181232151 \quad \Delta = -71, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1322181232181232151 \quad \Delta = -71, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1512321812321812231\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1512321812321812231\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$\begin{aligned}
1512321812321812231\overset{2}{3} \text{ [#=2]} \quad \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1512321812321812231\overset{2}{3} \text{ [#=5]} \quad \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1322181232181\overset{2}{2}32151 \quad \Delta = -71, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\text{Base: } 1232181232181223151 \\
12321812321812231\overset{2}{5}1 \quad \Delta = -71, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1232181\overset{2}{2}32181223151 \quad \Delta = -71, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1232181\overset{2}{2}32181223151 \quad \Delta = -71, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1232181\overset{2}{2}32181223151 \quad \Delta = -71, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1232181232181\overset{2}{2}23151 \quad \Delta = -71, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1513221812321812321\overset{2}{3} \text{ [#=2]} \quad \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1513221812321812321\overset{2}{3} \text{ [#=6]} \quad \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1513221812321812321\overset{2}{3} \text{ [#=7]} \quad \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1513221812321812321\overset{2}{3} \text{ [#=5]} \quad \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1513221812321812321\overset{2}{3} \text{ [#=8]} \quad \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1513221812321812321\overset{2}{3} \text{ [#=9]} \quad \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1513221812321812321\overset{2}{3} \text{ [#=4]} \quad \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1513221812321812321\overset{2}{3} \text{ [#=3]} \quad \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\text{Base: } 1413218123218123141 \\
1413218123218123141 \overset{2}{2} \quad \Delta = -75, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1413218123218123141 \overset{2}{2} \quad \Delta = -73, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1413218123218123141^2 \quad \Delta = -73, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1413218123218123141^2 \quad \Delta = -73, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 50 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
1413218123218123141^3_{[\#=3]} \quad \Delta = -81, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1413218123218123141^3_{[\#=2]} \quad \Delta = -81, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

Base: 3132181232181231512

$$\begin{aligned}
3132181232181231512^2 \quad \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
3132181232181231512^2 \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
3132181232181231512^2 \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
3132181232181231512^2 \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
3132181232181231512^2 \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
3132181232181231512^2 \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
3132181232181231512^2 \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
3132181232181231512^2 \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
3132181232181231512^3 \quad \Delta = -85, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
3132181232181231512^3_{[\#=2]} \quad \Delta = -85, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 93 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
3132181232181231512^3 \quad \Delta = -85, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 21 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

Base: 313218123218123151¹

$$\begin{aligned}
313218123218123151^2 \quad \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
313218123218123151^2 \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
313218123218123151^3 \quad \Delta = -85, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 21 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
313218123218123151^3_{[\#=2]} \quad \Delta = -85, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 93 & 19 \\ 2 & 19 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
313218123218123151^1 & \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
313218123218123151^2 & \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
313218123218123151^2 & \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
313218123218123151^2 & \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
313218123218123151^{21} & \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: 4123218123218123214

$$\begin{aligned}
4123218123218123214^2 & \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
4123218123218123214^2 & \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
4123218123218123214^2 & \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
4123218123218123214^3 & \quad \Delta = -85, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 41 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{aligned}$$

Base: 4123218123218123141

$$\begin{aligned}
4123218123218123141^2 & \quad \Delta = -75, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 58 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
4123218123218123141^2 & \quad \Delta = -75, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
4123218123218123141^3 \text{ [#=3]} & \quad \Delta = -83, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
4123218123218123141^3 \text{ [#=2]} & \quad \Delta = -83, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
4123218123218123141^3 & \quad \Delta = -83, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 57 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
4123218123218123141^2 & \quad \Delta = -75, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
4123218123218123141^2 & \quad \Delta = -75, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
4123218123218123141^2 & \quad \Delta = -75, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
4123218123218123141^2 & \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

$$4123218123218123141 \quad \Delta = -75, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 50 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

Base: 4141321812321812321

$$4141321812321812321\overset{2}{3} \text{ [#=3]} \quad \Delta = -83, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$4141321812321812321\overset{2}{3} \text{ [#=7]} \quad \Delta = -83, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$4141321812321812321\overset{2}{3} \quad \Delta = -75, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$4141321812321812321\overset{2}{3} \quad \Delta = -75, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$4141321812321812321\overset{2}{3} \quad \Delta = -75, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix}$$

$$4141321812321812321\overset{2}{3} \text{ [#=9]} \quad \Delta = -83, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$4141321812321812321\overset{2}{3} \text{ [#=8]} \quad \Delta = -83, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$4141321812321812321\overset{2}{3} \text{ [#=6]} \quad \Delta = -83, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$4141321812321812321\overset{2}{3} \text{ [#=2]} \quad \Delta = -83, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$4141321812321812321\overset{2}{3} \quad \Delta = -75, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$4141321812321812321\overset{2}{3} \quad \Delta = -75, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$4141321812321812321\overset{2}{3} \quad \Delta = -75, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$4141321812321812321\overset{2}{3} \text{ [#=4]} \quad \Delta = -83, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$4141321812321812321\overset{2}{3} \text{ [#=5]} \quad \Delta = -83, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$4141321812321812321\overset{2}{3} \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: 3132181232181232151

$$3132181232181232151\overset{2}{3} \quad \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$3132181232181232151\overset{2}{3} \quad \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$\begin{aligned}
& \overset{2}{3}132181232181232151 \quad \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 3132181232181232151 \quad \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 31321812321812321\overset{2}{5}1 \quad \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 31321812321812321\overset{2}{5}1 \quad \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 31321812321812321\overset{2}{5}1 \quad \Delta = -81, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{3}{3}132181232181232151 \text{ [#=2]} \quad \Delta = -87, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 93 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& \overset{3}{3}132181232181232151 \quad \Delta = -87, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 21 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

Base: $2^{\overset{1512321812321812}{3}} 1$

$$\begin{aligned}
& 2^{\overset{1512321812321812}{3}} 1 \quad \Delta = -73, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 2^{\overset{1512321812321812}{3}} 1 \quad \Delta = -73, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 2^{\overset{1512321812321812}{3}} \overset{13}{1} \text{ [#=2]} \quad \Delta = -81, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 41 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& 2^{\overset{1512321812321812}{3}} \overset{12}{1} \quad \Delta = -73, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 2^{\overset{1512321812321812}{3}} 1 \quad \Delta = -73, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 2^{\overset{1512321812321812}{3}} 1 \quad \Delta = -73, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 2^{\overset{1512321812321812}{3}} 1 \quad \Delta = -73, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 2^{\overset{1512321812321812}{3}} \overset{14}{1} \quad \Delta = -101, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 2^{\overset{1512321812321812}{3}} \overset{13}{1} \text{ [#=3]} \quad \Delta = -81, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 96 & 20 \\ 2 & 20 & 8 \end{pmatrix}
\end{aligned}$$

Base: $2^{\overset{2151321812321812}{3}} 1$

$$\begin{aligned}
& 2^{\overset{2151321812321812}{3}} \overset{13}{1} \text{ [#=2]} \quad \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 41 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& 2^{\overset{2151321812321812}{3}} \overset{13}{1} \text{ [#=3]} \quad \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 96 & 20 \\ 2 & 20 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{ll}
2^{\frac{2151321812321812}{3}} 1 & \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
2^{\frac{2151321812321812}{3}} 1 & \Delta = -71, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
2^{\frac{2151321812321812}{3}} 1 & \Delta = -71, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
2^{\frac{2151321812321812}{3}} 1 & \Delta = -71, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2^{\frac{2151321812321812}{3}} 1^{\textcolor{red}{2}} & \Delta = -71, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
2^{\frac{2151321812321812}{3}} 1 & \Delta = -71, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2^{\frac{2151321812321812}{3}} 1 & \Delta = -71, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2^{\frac{2151321812321812}{3}} 1 & \Delta = -71, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2^{\frac{2151321812321812}{3}} 1^{\textcolor{red}{4}} & \Delta = -99, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\textbf{Base: } 2^{\frac{151321812321812}{3}} 1 & \\
2^{\frac{151321812321812}{3}} 1^{\textcolor{red}{4}} & \Delta = -99, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2^{\frac{151321812321812}{3}} 1^{\textcolor{red}{3}}_{[\# = 2]} & \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 41 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
2^{\frac{151321812321812}{3}} 1^{\textcolor{red}{3}}_{[\# = 3]} & \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 96 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
2^{\frac{151321812321812}{3}} 1^{\textcolor{red}{2}} & \Delta = -71, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
2^{\frac{151321812321812}{3}} 1 & \Delta = -71, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
2^{\frac{151321812321812}{3}} 1 & \Delta = -71, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2^{\frac{151321812321812}{3}} 1 & \Delta = -71, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2^{\frac{151321812321812}{3}} 1 & \Delta = -71, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2^{\frac{151321812321812}{3}} 1 & \Delta = -71, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{array}$$

$$\textbf{Base: } 1^{\frac{1}{181223151321812321}}$$

$$\begin{array}{ll}
{}^1_{181223151321812321}{}^2 & \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^1_{181223151321812321}{}^2 & \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
{}^1_{181223151321812321}{}^2 & \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
{}^{12}_{181223151321812321} & \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
{}^1_{181223151321812321}{}^2 & \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
{}^1_{181223151321812321}{}^3_{\# = 9} & \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
{}^1_{181223151321812321}{}^3_{\# = 8} & \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
{}^1_{181223151321812321}{}^3_{\# = 7} & \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
{}^1_{181223151321812321}{}^3_{\# = 6} & \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
{}^1_{181223151321812321}{}^3_{\# = 5} & \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
{}^1_{181223151321812321}{}^3_{\# = 4} & \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
{}^1_{181223151321812321}{}^3_{\# = 3} & \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
{}^1_{181223151321812321}{}^3_{\# = 2} & \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\text{Base: } {}^1_{181231513161316131} & \\
{}^1_{181231513161316131}{}^3_{\# = 4} & \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
{}^1_{181231513161316131}{}^3_{\# = 3} & \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
{}^1_{181231513161316131}{}^3_{\# = 8} & \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
{}^1_{181231513161316131}{}^3_{\# = 6} & \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
{}^1_{181231513161316131}{}^3_{\# = 5} & \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
{}^1_{181231513161316131}{}^3_{\# = 2} & \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{ll}
{}^1_{181231513161316131} \textcolor{red}{3} \text{ [#=9]} & \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
{}^1_{181231513161316131} \textcolor{red}{3} \text{ [#=7]} & \Delta = -104, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
{}^1_{181231513161316131} \textcolor{red}{2} & \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
{}^1_{181231513161316131} \textcolor{red}{2} & \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
{}^1_{181231513161316131} \textcolor{red}{2} & \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
{}^{1\textcolor{red}{2}}_{181231513161316131} & \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
{}^1_{181231513161316131} \textcolor{red}{2} & \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
{}^1_{181231513161316131} \textcolor{red}{2} & \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
{}^1_{181231513161316131} \textcolor{red}{2} & \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\text{Base: } {}^1_{181232161316131612} & \\
{}^1_{181232161316131612} \textcolor{red}{2} & \Delta = -116, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
{}^1_{181232161316131612} \textcolor{red}{2} & \Delta = -116, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
{}^1_{181232161316131612} \textcolor{red}{2} & \Delta = -116, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
{}^1_{181232161316131612} \textcolor{red}{2} & \Delta = -116, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
{}^1_{181232161316131612} \textcolor{red}{2} & \Delta = -116, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
{}^1_{181232161316131612} \textcolor{red}{3} & \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
{}^1_{181232161316131612} \textcolor{red}{2} & \Delta = -116, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 168 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
{}^{1\textcolor{red}{2}}_{181232161316131612} & \Delta = -116, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
\text{Base: } {}^1_{16131613161232181} & \\
{}^1_{16131613161232181} \textcolor{red}{2} & \Delta = -116, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
16131613161232181 & \quad \Delta = -116, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
16131613161232181 & \quad \Delta = -116, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
16131613161232181 & \quad \Delta = -116, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
16131613161232181 & \quad \Delta = -116, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
16131613161232181 & \quad \Delta = -116, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 181232171232181223

$$\begin{aligned}
181232171232181223 & \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
181232171232181223 & \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
181232171232181223 & \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
181232171232181223 & \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
181232171232181223 & \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{aligned}$$

Base: 181231513218123215

$$\begin{aligned}
181231513218123215 & \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
181231513218123215 & \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
181231513218123215 & \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
181231513218123215 & \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
181231513218123215 & \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
181231513218123215 & \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: 181231513218123151

$$\begin{aligned}
181231513218123151 & \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
181231513218123151 & \quad \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{ll}
{}^1 181231513218123151^2 & \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
{}^{12} 181231513218123151^2 & \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^1 181231513218123151^2 & \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
{}^1 181231513218123151^2 & \Delta = -119, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{array}$$

Base: ${}^{21} 18123217123218122$

$$\begin{array}{ll}
{}^{21} 18123217123218122 & \Delta = -115, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
{}^{21} 18123217123218122^2 & \Delta = -115, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
{}^{21} 18123217123218122^2 & \Delta = -115, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^{21} 18123217123218122^2 & \Delta = -115, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
{}^{212} 18123217123218122^2 & \Delta = -115, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
{}^{21} 18123217123218122^3 & \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
{}^{21} 18123217123218122^2 & \Delta = -115, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{array}$$

Base: ${}^{21} 18123151321812321$

$$\begin{array}{ll}
{}^{21} 18123151321812321^2 & \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
{}^{212} 18123151321812321^2 & \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
{}^{21} 18123151321812321^2 & \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
{}^{21} 18123151321812321^2 & \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
{}^{21} 18123151321812321^2 & \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^{21} 18123151321812321^2 & \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
{}^{21} 18123151321812321^3_{[\# = 3]} & \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
& \overset{21}{18123151321812321\overset{2}{3}}_{[\# = 6]} \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{21}{18123151321812321\overset{2}{3}}_{[\# = 9]} \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{21}{18123151321812321\overset{2}{3}}_{[\# = 8]} \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{21}{18123151321812321\overset{2}{3}}_{[\# = 7]} \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& \overset{21}{18123151321812321\overset{2}{3}}_{[\# = 2]} \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{21}{18123151321812321\overset{2}{3}}_{[\# = 5]} \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{21}{18123151321812321\overset{2}{3}}_{[\# = 4]} \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{21}{18123218123151315}$

$$\begin{aligned}
& \overset{21}{1812321812315131\overset{2}{5}} \quad \Delta = -120, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{21}{1812321812315131\overset{2}{5}} \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{21}{181232181231\overset{2}{5}1315} \quad \Delta = -120, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{21}{181232181231\overset{2}{5}1315} \quad \Delta = -120, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{21}{1812321812315131\overset{2}{5}} \quad \Delta = -120, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{21}{1812321812315131\overset{2}{5}} \quad \Delta = -120, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{21}{1812321812315131\overset{2}{5}} \quad \Delta = -120, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{21\overset{2}{1}}{18123218123151315} \quad \Delta = -120, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{21}{1812321712321\overset{21}{81}}$

$$\begin{aligned}
& \overset{21}{1812321712321\overset{21}{81}} \quad \Delta = -115, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{21}{1812321712321\overset{21}{81}} \quad \Delta = -115, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{21\overset{2}{1}}{1812321712321\overset{21}{81}} \quad \Delta = -115, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{221}{1812321812315131}$

$$\begin{aligned}
& \overset{221}{1 \ 8 \ 12321812315131} \overset{\textcolor{red}{3}}{3} \text{ }_{[\# = 4]} \quad \Delta = -76, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{221}{1 \ 8 \ 12321812315131} \overset{\textcolor{red}{2}}{2} \quad \Delta = -68, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{221}{1 \ 8 \ 12321812315131} \overset{\textcolor{red}{2}}{2} \quad \Delta = -68, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{221}{1 \ 8 \ 12321812315131} \overset{\textcolor{red}{2}}{2} \quad \Delta = -68, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{221}{1 \ 8 \ 12321812315131} \overset{\textcolor{red}{2}}{2} \quad \Delta = -68, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{221}{1 \ 8 \ 12321812315131} \quad \Delta = -68, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{221}{1 \ 8 \ 12321812315131} \overset{\textcolor{red}{3}}{3} \text{ }_{[\# = 7]} \quad \Delta = -76, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& \overset{221}{1 \ 8 \ 12321812315131} \overset{\textcolor{red}{3}}{3} \text{ }_{[\# = 8]} \quad \Delta = -76, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{221}{1 \ 8 \ 12321812315131} \quad \Delta = -77, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{221}{1 \ 8 \ 12321812315131} \overset{\textcolor{red}{3}}{3} \text{ }_{[\# = 2]} \quad \Delta = -76, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{221}{1 \ 8 \ 12321812315131} \overset{\textcolor{red}{3}}{3} \text{ }_{[\# = 3]} \quad \Delta = -76, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{221}{1 \ 8 \ 12321812315131} \overset{\textcolor{red}{2}}{2} \quad \Delta = -68, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{221}{1 \ 8 \ 12321812315131} \overset{\textcolor{red}{3}}{3} \text{ }_{[\# = 6]} \quad \Delta = -76, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{221}{1 \ 8 \ 12321812315131} \overset{\textcolor{red}{3}}{3} \text{ }_{[\# = 5]} \quad \Delta = -76, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{221}{1 \ 8 \ 12321812315131} \overset{\textcolor{red}{3}}{3} \text{ }_{[\# = 9]} \quad \Delta = -76, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \textbf{Base: } \overset{221}{1 \ 8 \ 12321812321612} \\
& \overset{221}{1 \ 8 \ 12321812321612} \overset{\textcolor{red}{3}}{3} \quad \Delta = -96, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
& \overset{221}{1 \ 8 \ 12321812321612} \quad \Delta = -97, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{221}{1 \ 8 \ 12321812321612} \overset{\textcolor{red}{2}}{2} \quad \Delta = -88, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{221}{1 \ 8 \ 12321812321612} \overset{\textcolor{red}{2}}{2} \quad \Delta = -88, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{221}{1\ 8\ 12321812321612} \quad \Delta = -88, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{221}{1\ 8\ 12321812321612} \quad \Delta = -88, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{221}{1\ 8\ 12321812321612} \quad \Delta = -88, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{221}{1\ 8\ 12321812321612} \quad \Delta = -88, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{1612321812321} \overset{221}{8\ 1}$

$$\begin{aligned}
& \overset{1}{1612321812321} \overset{221}{8\ 1} \quad \Delta = -88, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{1}{1612321812321} \overset{221}{8\ 1} \quad \Delta = -97, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
& \overset{1}{1612321812321} \overset{221}{8\ 1} \quad \Delta = -88, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{1}{1612321812321} \overset{221}{8\ 1} \quad \Delta = -88, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{1}{1612321812321} \overset{221}{8\ 1} \quad \Delta = -88, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{1}{1612321812321} \overset{221}{8\ 1} \quad \Delta = -88, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{321}{1\ 8\ 12321712321812}$

$$\begin{aligned}
& \overset{321}{1\ 8\ 12321712321812} \quad \Delta = -124, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{321}{1\ 8\ 12321712321812} \quad \Delta = -124, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{321}{1\ 8\ 12321712321812} \quad \Delta = -124, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{321}{1\ 8\ 12321712321812} \quad \Delta = -124, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
& \overset{321}{1\ 8\ 12321712321812} \quad \Delta = -124, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 163 & 25 \\ 2 & 25 & 8 \end{pmatrix} \\
& \overset{321}{1\ 8\ 12321712321812} \quad \Delta = -124, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
& \overset{21}{321\ 8\ 12321712321812} \quad \Delta = -124, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{1812321712321} \overset{1}{8123}$

$$\overset{1}{1812321712321} \overset{1}{8123} \quad \Delta = -124, \ n_+ = 1, \ n_- = 19, \ n_0 = 0, \ |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$\begin{aligned}
& \overset{1}{1}812321712321\overset{21}{8}123 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{12}{1}812321712321\overset{1}{8}123 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{1}{1}812321712321\overset{2}{8}123 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{1}{1}812321712321\overset{21}{8}123 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& \overset{1}{1}812321712321\overset{12}{8}123 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{232171232181232181}$

$$\begin{aligned}
& \overset{2}{232171232181232181} \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{21}{232171232181232181} \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 225 \quad \begin{pmatrix} 0 & 15 & 2 \\ 15 & 163 & 25 \\ 2 & 25 & 8 \end{pmatrix} \\
& \overset{12}{232171232181232181} \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
& \overset{1}{232171232181232181} \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{21}{232171232181232181} \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{2}{232171232181232181} \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{2321}{1 \ 8 \ 1232171232181}$

$$\begin{aligned}
& \overset{2321}{1 \ 8 \ 1232171232181} \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 206 & 28 \\ 2 & 28 & 8 \end{pmatrix} \\
& \overset{2321}{1 \ 8 \ 1232171232181} \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 225 \quad \begin{pmatrix} 0 & 15 & 2 \\ 15 & 283 & 33 \\ 2 & 33 & 8 \end{pmatrix} \\
& \overset{23212}{1 \ 8 \ 1232171232181} \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 219 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
& \overset{2321}{1 \ 8 \ 1232171232181} \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
& \overset{2321}{1 \ 8 \ 1232171232181} \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{21}{232181232171232181} \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{321712321812321812}$

$$\overset{1}{321712321812321812} \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\begin{aligned}
321712321812321812 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
321712321812321812 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
321712321812321812 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
321712321812321812 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
321712321812321812 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
321712321812321812 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{aligned}$$

Base: 217123218123218122

$$\begin{aligned}
217123218123218122 \quad \Delta = -115, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
217123218123218122 \quad \Delta = -115, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
217123218123218122 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
217123218123218122 \quad \Delta = -115, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
217123218123218122 \quad \Delta = -115, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
217123218123218122 \quad \Delta = -115, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
217123218123218122 \quad \Delta = -115, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
217123218123218122 \quad \Delta = -115, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 17123218123218122

$$\begin{aligned}
17123218123218122 \quad \Delta = -115, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
17123218123218122 \quad \Delta = -115, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
17123218123218122 \quad \Delta = -115, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
17123218123218122 \quad \Delta = -115, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
17123218123218122 \quad \Delta = -115, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
17123218123218122 \quad \Delta = -115, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
17123218123218122 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}
\end{aligned}$$

Base: 171232181232181223

$$\begin{aligned}
171232181232181223 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
171232181232181223 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
171232181232181223 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
171232181232181223 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
171232181232181223 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
171232181232181223 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1^{12321} 8^{12321} 123218123141$

$$\begin{aligned}
1^{12321} 8^{12321} 123218123141 \text{ } \textcolor{red}{3} \text{ } [\# = 2] \quad \Delta = -55, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
1^{12321} 8^{12321} 123218123141 \text{ } \textcolor{red}{3} \text{ } [\# = 3] \quad \Delta = -55, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
1^{12321} 8^{12321} 123218123141 \text{ } \textcolor{red}{3} \text{ } [\# = 3] \quad \Delta = -55, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1^{12321} 8^{12321} 123218123141 \text{ } \textcolor{red}{3} \text{ } [\# = 2] \quad \Delta = -55, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{12321} 8^{12321} 123218123141 \text{ } \textcolor{red}{3} \text{ } [\# = 4] \quad \Delta = -55, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1^{12321} 8^{12321} 123218123141 \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1^{12321} 8^{12321} 123218123141 \quad \Delta = -47, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1^{12321} 8^{12321} 123218123141 \quad \Delta = -47, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1^{12321} 8^{12321} 123218123141 \quad \Delta = -47, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{12321}{1} \overset{2}{8} 123218123141 \quad \Delta = -47, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{12321}{1} \overset{2}{8} 123218123141 \quad \Delta = -47, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{312321}{1} \overset{2}{8} 123218123141 \text{ [#=6]} \quad \Delta = -55, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
& \overset{312321}{1} \overset{2}{8} 123218123141 \text{ [#=7]} \quad \Delta = -55, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
& \overset{312321}{1} \overset{2}{8} 123218123141 \text{ [#=9]} \quad \Delta = -55, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix} \\
& \overset{312321}{1} \overset{2}{8} 123218123141 \text{ [#=8]} \quad \Delta = -55, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
& \overset{312321}{1} \overset{2}{8} 123218123141 \text{ [#=5]} \quad \Delta = -55, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{51321}{1} \overset{2}{8} 123218123151$

$$\begin{aligned}
& \overset{51321}{1} \overset{2}{8} 123218123151 \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{51321}{1} \overset{2}{8} 123218123151 \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& \overset{51321}{1} \overset{2}{8} 123218123151 \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{51321}{1} \overset{2}{8} 123218123151 \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{51321}{1} \overset{2}{8} 123218123151 \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& \overset{51321}{1} \overset{2}{8} 123218123151 \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 51321 \overset{21}{8} 123218123151 \quad \Delta = -121, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1321}{1} \overset{2}{8} 1232181231512$

$$\begin{aligned}
& \overset{31321}{1} \overset{2}{8} 1232181231512 \text{ [#=4]} \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{1321}{1} \overset{3}{8} 1232181231512 \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
& \overset{31321}{1} \overset{2}{8} 1232181231512 \text{ [#=3]} \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{31321}{1} \overset{2}{8} 1232181231512 \text{ [#=2]} \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{3}{1} \overset{1321}{8} 1232181231512 \text{ }_{[\# = 5]} \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& \overset{1321}{1} \overset{1321}{8} 1232181231512 \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{1321}{1} \overset{1321}{8} 1232181231512 \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{1321}{1} \overset{1321}{8} 1232181231512 \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{1321}{1} \overset{1321}{8} 1232181231512 \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{1321}{1} \overset{1321}{8} 1232181231512 \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{1321}{1} \overset{1321}{8} 1232181231512 \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \text{Base: } \overset{1321}{1} \overset{1321}{8} 123218123151 \\
& \overset{1321}{1} \overset{1321}{8} 123218123151 \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{3}{1} \overset{31321}{8} 123218123151 \text{ }_{[\# = 2]} \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{1321}{1} \overset{1321}{8} 123218123151 \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{3}{1} \overset{31321}{8} 123218123151 \text{ }_{[\# = 3]} \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{3}{1} \overset{31321}{8} 123218123151 \text{ }_{[\# = 4]} \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{3}{1} \overset{31321}{8} 123218123151 \text{ }_{[\# = 5]} \quad \Delta = -77, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& \overset{1321}{1} \overset{1321}{8} 123218123151 \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{1321}{1} \overset{1321}{8} 123218123151 \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{1321}{1} \overset{1321}{8} 123218123151 \quad \Delta = -69, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \text{Base: } 612321812321812314 \\
& 612321812321812314 \quad \Delta = -125, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& 612321812321812314 \quad \Delta = -125, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
612321\overset{1}{8}12321812314 \quad \Delta = -127, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
612321\overset{1}{8}12321812314 \quad \Delta = -125, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
612321\overset{1}{8}12321812314 \quad \Delta = -125, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
612321\overset{1}{8}12321812314 \quad \Delta = -125, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
612321\overset{21}{8}12321812314 \quad \Delta = -125, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
612321\overset{12}{8}12321812314 \quad \Delta = -125, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1612321¹81232181231

$$\begin{aligned}
1612321\overset{1}{8}1232181231\overset{3}{\text{[#}=4]} \quad \Delta = -105, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1612321\overset{1}{8}1232181231\overset{3}{\text{[#}=5]} \quad \Delta = -105, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1612321\overset{1}{8}1232181231\overset{3}{\text{[#}=3]} \quad \Delta = -105, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1612321\overset{1}{8}1232181231\overset{3}{\text{[#}=2]} \quad \Delta = -105, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1612321\overset{1}{8}1232181231 \quad \Delta = -97, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1612321\overset{21}{8}1232181231 \quad \Delta = -97, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1612321\overset{2}{8}1232181231 \quad \Delta = -97, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1612321\overset{21}{8}1232181231 \quad \Delta = -97, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1612321\overset{12}{8}1232181231 \quad \Delta = -97, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1612321\overset{1}{8}1232181231 \quad \Delta = -97, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: 412321812321¹²³²¹8 1

$$\begin{aligned}
412321812321\overset{12321}{8} 1 \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
412321812321\overset{2}{8}12321\overset{12321}{8} 1 \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
412321812321 \overset{3}{8} 1 \text{ [#}=8] \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
412321812321 \overset{3}{8} 1 \text{ [#}=7] \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
412321812321 \overset{3}{8} 1 \text{ [#}=6] \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
412321812321 \overset{3}{8} 1 \text{ [#}=9] \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
412321812321 \overset{3}{8} 1 \text{ [#}=4] \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
412321812321 \overset{3}{8} 1 \text{ [#}=3] \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
412321812321 \overset{3}{8} 1 \text{ [#}=2] \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
412321812321 \overset{3}{8} 1 \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 41 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
412321812321 \overset{3}{8} 1 \text{ [#}=5] \quad \Delta = -57, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
412321812321 \overset{2}{8} 1 \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
412321812321 \overset{2}{8} 1 \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
412321812321 \overset{2}{8} 1 \quad \Delta = -49, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{512321}{8} 12321812321 & \\
1 \overset{512321}{8} 12321812321 \overset{3}{8} \text{ [#}=2] \quad \Delta = -81, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1 \overset{512321}{8} 12321812321 \quad \Delta = -73, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1 \overset{512321}{8} 12321812321 \quad \Delta = -73, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1 \overset{512321}{8} 12321812321 \overset{2}{8} \quad \Delta = -73, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1 \overset{512321}{8} 12321812321 \overset{2}{8} \quad \Delta = -73, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
1 \overset{512321}{8} 12321812321 \overset{2}{8} \quad \Delta = -73, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1 \overset{512321}{8} 12321812321 \overset{3}{8} \text{ [#}=7] \quad \Delta = -81, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{512321}{1 \ 8 \ 12321812321\mathbf{3}}_{[\#=3]} \quad \Delta = -81, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{512321}{1 \ 8 \ 12321812321\mathbf{3}}_{[\#=4]} \quad \Delta = -81, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{512321}{1 \ 8 \ 12321812321\mathbf{3}}_{[\#=5]} \quad \Delta = -81, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& 512321\overset{21}{8}12321812321 \quad \Delta = -73, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{512321}{1 \ 8 \ 12321812321\mathbf{3}}_{[\#=6]} \quad \Delta = -81, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{512321}{1 \ 8 \ 12321812321\mathbf{3}}_{[\#=8]} \quad \Delta = -81, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{512321}{1 \ 8 \ 12321812321\mathbf{3}}_{[\#=9]} \quad \Delta = -81, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{51321}{1 \ 8 \ 123218123215}$

$$\begin{aligned}
& \overset{51321\mathbf{2}}{1 \ 8 \ 123218123215} \quad \Delta = -123, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& 51321\overset{21}{8}123218123215 \quad \Delta = -123, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{51321}{1 \ 8 \ 123218123215}\overset{2}{} \quad \Delta = -123, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& \overset{51321}{1 \ 8 \ 123218123215}\overset{2}{} \quad \Delta = -123, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{51321}{1 \ 8 \ 123218123215}\overset{2}{} \quad \Delta = -123, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{51321}{1 \ 8 \ 123218123215} \quad \Delta = -123, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& \overset{51321}{1 \ 8 \ 123218123215} \quad \Delta = -123, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1321}{1 \ 8 \ 1232181232151}$

$$\begin{aligned}
& \overset{1321}{1 \ 8 \ 1232181232151}\overset{2}{} \quad \Delta = -71, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{1321\mathbf{2}}{1 \ 8 \ 1232181232151} \quad \Delta = -71, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{1321}{1 \ 8 \ 1232181232151}\overset{2}{} \quad \Delta = -71, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{1321}{1 \ 8 \ 1232181232151}\overset{2}{} \quad \Delta = -71, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{31321}{1 \ 8 \ 1232181232151} \text{ }_{[\#=3]} \quad \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{31321}{1 \ 8 \ 1232181232151} \text{ }_{[\#=2]} \quad \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{31321}{1 \ 8 \ 1232181232151} \text{ }_{[\#=4]} \quad \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{31321}{1 \ 8 \ 1232181232151} \text{ }_{[\#=5]} \quad \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& \overset{1321}{1 \ 8 \ 1232181232151} \quad \Delta = -71, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \text{Base: } \overset{151321}{1 \ 8 \ 12321812321} \\
& \overset{151321}{1 \ 8 \ 12321812321} \overset{2}{} \quad \Delta = -71, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{151321}{1 \ 8 \ 12321812321} \quad \Delta = -71, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 151321 \overset{21}{8 \ 12321812321} \quad \Delta = -71, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{151321}{1 \ 8 \ 12321812321} \quad \Delta = -71, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{151321}{1 \ 8 \ 12321812321} \overset{3}{} \text{ }_{[\#=3]} \quad \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{151321}{1 \ 8 \ 12321812321} \overset{3}{} \text{ }_{[\#=2]} \quad \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{151321}{1 \ 8 \ 12321812321} \overset{2}{} \quad \Delta = -71, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{151321}{1 \ 8 \ 12321812321} \overset{2}{} \quad \Delta = -71, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& \overset{151321}{1 \ 8 \ 12321812321} \overset{3}{} \text{ }_{[\#=5]} \quad \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{151321}{1 \ 8 \ 12321812321} \overset{3}{} \text{ }_{[\#=6]} \quad \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{151321}{1 \ 8 \ 12321812321} \overset{3}{} \text{ }_{[\#=7]} \quad \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& \overset{151321}{1 \ 8 \ 12321812321} \overset{3}{} \text{ }_{[\#=8]} \quad \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{151321}{1 \ 8 \ 12321812321} \overset{3}{} \text{ }_{[\#=4]} \quad \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{151321}{1 \ 8 \ 12321812321} \overset{3}{} \text{ }_{[\#=9]} \quad \Delta = -79, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{1}812321\overset{1}{8}123217123$

$$\begin{aligned}
\overset{1}{1}812321\overset{21}{8}123217123 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\overset{1}{1}812321\overset{21}{8}123217123 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
\overset{12}{1}812321\overset{1}{8}123217123 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\overset{1}{1}812321\overset{1}{8}12321\overset{2}{7}123 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\overset{1}{1}812321\overset{12}{8}123217123 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
\overset{1}{1}812321\overset{1}{8}12321\overset{2}{7}123 \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{21}{1}812321\overset{1}{8}12321712$

$$\begin{aligned}
\overset{21}{1}812321\overset{1}{8}12321\overset{2}{7}12 \quad \Delta = -115, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\overset{21}{1}812321\overset{1}{8}12321712 \quad \Delta = -115, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\overset{21}{1}812321\overset{21}{8}12321712 \quad \Delta = -115, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\overset{21}{1}812321\overset{12}{8}12321712 \quad \Delta = -115, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\overset{21}{1}812321\overset{21}{8}12321712 \quad \Delta = -115, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\overset{21}{1}812321\overset{1}{8}12321\overset{2}{7}12 \quad \Delta = -115, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\overset{212}{1}812321\overset{1}{8}12321712 \quad \Delta = -115, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{1}712321\overset{1}{8}12321\overset{21}{8}1$

$$\begin{aligned}
\overset{1}{1}712321\overset{21}{8}12321\overset{21}{8}1 \quad \Delta = -115, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\overset{1}{1}712321\overset{21}{8}12321\overset{21}{8}1 \quad \Delta = -115, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
\overset{12}{1}712321\overset{1}{8}12321\overset{21}{8}1 \quad \Delta = -115, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
\overset{1}{1}712321\overset{1}{8}12321\overset{21}{8}1 \quad \Delta = -115, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\overset{1}{1}712321\overset{1}{8}12321\overset{221}{8}1 \quad \Delta = -115, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

$$171232181232181 \quad \Delta = -115, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

Base: $1 \overset{321}{8} \overset{1}{1232181232171}$

$$1 \overset{321}{8} \overset{1}{1232181232171} \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$321 \overset{21}{8} \overset{1}{1232181232171} \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{3212}{8} \overset{1}{1232181232171} \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$1 \overset{321}{8} \overset{1}{1232181232171} \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{321}{8} \overset{12}{1232181232171} \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$1 \overset{321}{8} \overset{21}{1232181232171} \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \overset{321}{8} \overset{21}{1232181232171} \quad \Delta = -124, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

Base: $1 \overset{12321}{8} \overset{1}{12321812321}$

$$1 \overset{312321}{8} \overset{1}{12321812321} [\# = 4] \quad \Delta = -29, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$1 \overset{312321}{8} \overset{1}{12321812321} [\# = 3] \quad \Delta = -29, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1 \overset{312321}{8} \overset{1}{12321812321} [\# = 8] \quad \Delta = -29, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix}$$

$$1 \overset{123212}{8} \overset{1}{12321812321} \quad \Delta = -21, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{12321}{8} \overset{1}{12321812321} \quad \Delta = -21, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1 \overset{312321}{8} \overset{1}{12321812321} [\# = 7] \quad \Delta = -29, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix}$$

$$1 \overset{312321}{8} \overset{1}{12321812321} [\# = 5] \quad \Delta = -29, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix}$$

$$1 \overset{312321}{8} \overset{1}{12321812321} [\# = 2] \quad \Delta = -29, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1 \overset{312321}{8} \overset{1}{12321812321} [\# = 6] \quad \Delta = -29, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

$$1 \overset{312321}{8} \overset{1}{12321812321} [\# = 9] \quad \Delta = -29, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix}$$

Base: $712321812321 \overset{1}{8} \overset{2321}{1}$

$$\begin{aligned}
712321812321 \overset{1}{8} \overset{2321}{1} \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
712321812321 \overset{21}{8} \overset{2321}{1} \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
712321812321 \overset{1}{8} \overset{2321}{1} \overset{2}{1} \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
712321812321 \overset{2}{8} \overset{1}{1} \overset{2321}{1} \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
712321812321 \overset{1}{8} \overset{2321}{1} \overset{2}{1} \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
712321812321 \overset{21}{8} \overset{2321}{1} \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
712321812321 \overset{12}{8} \overset{2321}{1} \quad \Delta = -126, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 225 & \begin{pmatrix} 0 & 15 & 2 \\ 15 & 163 & 25 \\ 2 & 25 & 8 \end{pmatrix}
\end{aligned}$$

Base: 21812321812315131

$$\begin{aligned}
21812321812315131 \overset{21}{3} \text{ [#=7]} \quad \Delta = -76, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
21812321812315131 \overset{212}{1} \quad \Delta = -68, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21812321812315131 \overset{21}{1} \overset{2}{2} \quad \Delta = -68, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21812321812315131 \overset{21}{1} \overset{2}{2} \quad \Delta = -68, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21812321812315131 \overset{221}{1} \quad \Delta = -68, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21812321812315131 \overset{21}{1} \overset{2}{2} \quad \Delta = -68, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
21812321812315131 \overset{21}{1} \overset{2}{2} \quad \Delta = -68, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21812321812315131 \overset{21}{3} \text{ [#=8]} \quad \Delta = -76, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
21812321812315131 \overset{21}{3} \text{ [#=9]} \quad \Delta = -76, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
21812321812315131 \overset{21}{3} \text{ [#=3]} \quad \Delta = -76, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
21812321812315131 \overset{21}{3} \text{ [#=6]} \quad \Delta = -76, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21812321812315131 \overset{21}{3} \text{ [#=5]} \quad \Delta = -76, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
&^{21} 21812321812315131\bar{3} \text{ }_{[\# = 2]} \quad \Delta = -76, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
&^{21} 21812321812315131\bar{3} \text{ }_{[\# = 4]} \quad \Delta = -76, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: $^{21} 21812321812321612$

$$\begin{aligned}
&^{21}\bar{2} 21812321812321612 \quad \Delta = -88, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
&^{21} 2181232181\bar{2}321612 \quad \Delta = -88, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
&^{21} 218123218123\bar{2}1612 \quad \Delta = -88, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
&\bar{2}^{21} 21812321812321612 \quad \Delta = -88, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
&^{21} 218123218123\bar{2}1612 \quad \Delta = -88, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
&^{21} 218123\bar{2}1812321612 \quad \Delta = -88, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
&^{21} 218123218123\bar{2}1612 \quad \Delta = -96, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}
\end{aligned}$$

Base: $^1 1612321812321\bar{8}12$

$$\begin{aligned}
&^1 16123\bar{2}1812321\bar{8}12 \quad \Delta = -88, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
&^1\bar{2} 1612321812321\bar{8}12 \quad \Delta = -88, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
&^1 16123218123\bar{2}1\bar{8}12 \quad \Delta = -88, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
&^1 1612321812321\bar{8}\bar{2}12 \quad \Delta = -88, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
&^1 161232181\bar{2}321\bar{8}12 \quad \Delta = -88, n_+ = 1, n_- = 19, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

$T = 20$ (2277 bases)

Base: 121223141414141414132

$$\begin{aligned}
&12122314141414141\bar{4}132 \quad \Delta = -129, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
&^2 121223141414141414132 \quad \Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 130 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
&^2 121223141414141414132 \quad \Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 289 \quad \begin{pmatrix} 0 & 17 & 2 \\ 17 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1212231414141414132 \quad \Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 225 & \begin{pmatrix} 0 & 15 & 2 \\ 15 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1212231414141414132 \quad \Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1212231414141414132 \quad \Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1212231414141414132 \quad \Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1212231414141414132 \quad \Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1212231414141414132 \quad \Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

Base: 314141414141413221121

$$\begin{aligned}
314141414141413221121 \quad \Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 225 & \begin{pmatrix} 0 & 15 & 2 \\ 15 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
314141414141413221121 \quad \Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
314141414141413221121 \quad \Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 130 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
314141414141413221121 \quad \Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
314141414141413221121 \quad \Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
314141414141413221121 \quad \Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
314141414141413221121 \quad \Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
314141414141413221121 \quad \Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
314141414141413221121 \quad \Delta = -134, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 141414141414132211212

$$\begin{aligned}
141414141414132211212 \quad \Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
141414141414132211212 \quad \Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
141414141414132211212 \quad \Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 108 & 20 \\ 2 & 20 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
141414141414132211212 \quad \Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
141414141414132211212 \quad \Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
141414141414132211212 \quad \Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
141414141414132211212 \quad \Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
141414141414132211212 \quad \Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
2112122314141414141413_{[\# = 3]} \quad \Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2112122314141414141413_{[\# = 2]} \quad \Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\text{Base: } 414141413221121223151 & \\
414141413221121223151 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
414141413221121223151 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
414141413221121223151 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 144 & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
414141413221121223151 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 256 & \begin{pmatrix} 0 & 16 & 2 \\ 16 & 142 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
414141413221121223151 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
414141413221121223151 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 225 & \begin{pmatrix} 0 & 15 & 2 \\ 15 & 118 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
414141413221121223151 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\text{Base: } 141414132211212231512 & \\
2151322112122314141413_{[\# = 2]} \quad \Delta = -103, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2151322112122314141413_{[\# = 3]} \quad \Delta = -103, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
141414132211212231512 \quad \Delta = -103, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
141414132211212231512 \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
141414132211212231512 \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
141414132211212231512^2 \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
141414132211212231512^2 \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
141414132211212231512^2 \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
141414132211212231512^2 \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
141414132211212231512^2 \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 314141322112122315131

$$\begin{aligned}
314141322112122315131^3_{[\# = 9]} \quad \Delta = -91, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
314141322112122315131^3_{[\# = 8]} \quad \Delta = -91, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
314141322112122315131^3_{[\# = 7]} \quad \Delta = -91, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
314141322112122315131^3_{[\# = 6]} \quad \Delta = -91, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
314141322112122315131^3_{[\# = 5]} \quad \Delta = -91, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
314141322112122315131^3 \quad \Delta = -91, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
314141322112122315131^3_{[\# = 2]} \quad \Delta = -91, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
314141322112122315131^3_{[\# = 3]} \quad \Delta = -91, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
314141322112122315131^3_{[\# = 4]} \quad \Delta = -91, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
314141322112122315131^2 \quad \Delta = -83, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
314141322112122315131^2 \quad \Delta = -83, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
314141322112122315131^2 \quad \Delta = -83, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
314141322112122315131^2 \quad \Delta = -83, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
314141322112122315131^2 \quad \Delta = -83, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
314141322112122315131 \quad \Delta = -83, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
314141322112122315131 \quad \Delta = -83, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\text{Base: } 131513221121223151232 & \\
131513221121223151232 \quad \Delta = -82, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
2321513221121223151313_{[\# = 4]} \quad \Delta = -90, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
131513221121223151232 \quad \Delta = -82, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
131513221121223151232 \quad \Delta = -82, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
131513221121223151232 \quad \Delta = -82, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
131513221121223151232 \quad \Delta = -82, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
131513221121223151232 \quad \Delta = -82, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2321513221121223151313_{[\# = 3]} \quad \Delta = -90, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
2321513221121223151313_{[\# = 9]} \quad \Delta = -90, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
2321513221121223151313_{[\# = 8]} \quad \Delta = -90, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
2321513221121223151313_{[\# = 7]} \quad \Delta = -90, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
2321513221121223151313_{[\# = 6]} \quad \Delta = -90, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
2321513221121223151313_{[\# = 5]} \quad \Delta = -90, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
2321513221121223151313_{[\# = 2]} \quad \Delta = -90, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\text{Base: } 413151322112122314141 & \\
413151322112122314141 \quad \Delta = -103, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
413151322112122314141 \quad \Delta = -103, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
413151322112122314141^2 \quad \Delta = -103, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
413151322112122314141^2 \quad \Delta = -103, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 76 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
413151322112122314141^2 \quad \Delta = -103, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 58 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
413151322112122314141^2 \quad \Delta = -103, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 63 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
413151322112122314141^2 \quad \Delta = -103, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
413151322112122314141^2 \quad \Delta = -104, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
413151322112122314141^3_{[\# = 3]} \quad \Delta = -111, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
413151322112122314141^3_{[\# = 2]} \quad \Delta = -111, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
413151322112122314141^3 \quad \Delta = -111, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 41 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{aligned}$$

Base: 413151322112122315123

$$\begin{aligned}
413151322112122315123^2 \quad \Delta = -108, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
413151322112122315123^3 \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
413151322112122315123^2 \quad \Delta = -109, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
413151322112122315123^2 \quad \Delta = -108, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
413151322112122315123^2 \quad \Delta = -108, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
413151322112122315123^2 \quad \Delta = -108, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
413151322112122315123^2 \quad \Delta = -108, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
413151322112122315123^2 \quad \Delta = -108, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
413151322112122315123^2 \quad \Delta = -108, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 231413221121223151321

$$\begin{aligned}
231413221121223151321\overset{\textcolor{red}{3}}{3}_{[\# = 8]} \quad \Delta = -65, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
231413221121223151321\overset{\textcolor{red}{2}}{2} \quad \Delta = -57, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
231413221121223151321\overset{\textcolor{red}{2}}{2} \quad \Delta = -57, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
231413221121223151321\overset{\textcolor{red}{2}}{2} \quad \Delta = -57, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
231413221121223151321\overset{\textcolor{red}{2}}{2} \quad \Delta = -57, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
231413221121223151321\overset{\textcolor{red}{2}}{2} \quad \Delta = -57, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
231413221121223151321\overset{\textcolor{red}{2}}{2} \quad \Delta = -57, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
231413221121223151321\overset{\textcolor{red}{2}}{2} \quad \Delta = -59, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
231413221121223151321\overset{\textcolor{red}{3}}{3}_{[\# = 7]} \quad \Delta = -65, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
231413221121223151321\overset{\textcolor{red}{3}}{3}_{[\# = 6]} \quad \Delta = -65, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
231413221121223151321\overset{\textcolor{red}{3}}{3}_{[\# = 5]} \quad \Delta = -65, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
231413221121223151321\overset{\textcolor{red}{3}}{3}_{[\# = 4]} \quad \Delta = -65, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
231413221121223151321\overset{\textcolor{red}{3}}{3}_{[\# = 3]} \quad \Delta = -65, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
231413221121223151321\overset{\textcolor{red}{3}}{3}_{[\# = 9]} \quad \Delta = -65, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
231413221121223151321\overset{\textcolor{red}{3}}{3}_{[\# = 2]} \quad \Delta = -65, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\textbf{Base: } 223132211212231513221 & \\
223132211212231513221\overset{\textcolor{red}{3}}{3}_{[\# = 5]} \quad \Delta = -37, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
223132211212231513221\overset{\textcolor{red}{2}}{2} \quad \Delta = -29, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
223132211212231513221\overset{\textcolor{red}{3}}{3} \quad \Delta = -37, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 60 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
223132211212231513221\overset{\textcolor{red}{3}}{3}_{[\# = 6]} \quad \Delta = -37, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
223132211212231513221\overset{3}{\color{red}}_{[\# = 7]} \quad \Delta = -37, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
223132211212231513221\overset{3}{\color{red}}_{[\# = 8]} \quad \Delta = -37, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
223132211212231513221\overset{3}{\color{red}}_{[\# = 9]} \quad \Delta = -37, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
22313221121223151\overset{2}{\color{red}}3221 \quad \Delta = -29, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
223132211212231513221\overset{3}{\color{red}}_{[\# = 4]} \quad \Delta = -37, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
223132211212231513221\overset{3}{\color{red}}_{[\# = 3]} \quad \Delta = -37, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
223132211212231513221\overset{3}{\color{red}}_{[\# = 2]} \quad \Delta = -37, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
223132211212231513221\overset{2}{\color{red}} \quad \Delta = -29, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
223132211212231513221\overset{2}{\color{red}} \quad \Delta = -29, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
223132211212231513221\overset{2}{\color{red}} \quad \Delta = -29, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\text{Base: } 314132211212231513215 & \\
314132211212231513215\overset{2}{\color{red}} \quad \Delta = -108, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
314132211212231513215\overset{2}{\color{red}} \quad \Delta = -108, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
314132211212231513215\overset{2}{\color{red}} \quad \Delta = -108, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
314132211212231513215\overset{2}{\color{red}} \quad \Delta = -108, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
314132211212231513215\overset{2}{\color{red}} \quad \Delta = -108, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
314132211212231513215\overset{2}{\color{red}} \quad \Delta = -108, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
314132211212231513215\overset{2}{\color{red}} \quad \Delta = -108, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
314132211212231513215\overset{3}{\color{red}} \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\text{Base: } 314132211212231513151 &
\end{aligned}$$

$$\begin{aligned}
31413221121223151\overset{2}{3}151 \quad \Delta = -108, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
31413221121223151\overset{3}{3}151 \quad \Delta = -115, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
31413221121223151\overset{2}{3}151 \quad \Delta = -107, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
31413221121223151\overset{2}{3}151 \quad \Delta = -107, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
31413221121223151\overset{2}{3}151 \quad \Delta = -107, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
31413221121223151\overset{2}{3}151 \quad \Delta = -107, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
31413221121223151\overset{2}{3}151 \quad \Delta = -107, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
31413221121223151\overset{2}{3}151 \quad \Delta = -107, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
31413221121223151\overset{2}{3}151 \quad \Delta = -107, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: 141322112122315131513

$$\begin{aligned}
14132211212231513151\overset{2}{3} \quad \Delta = -107, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
315131513221121223141\overset{3}{3} \text{ [#=2]} \quad \Delta = -115, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
315131513221121223141\overset{3}{3} \text{ [#=3]} \quad \Delta = -115, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1413221121223151\overset{2}{3}1513 \quad \Delta = -108, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1413221121223151\overset{2}{3}1513 \quad \Delta = -107, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1413221121223151\overset{2}{3}1513 \quad \Delta = -107, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1413221121223151\overset{2}{3}1513 \quad \Delta = -107, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1413221121223151\overset{2}{3}1513 \quad \Delta = -107, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1413221121223151\overset{2}{3}1513 \quad \Delta = -107, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1413221121223151\overset{2}{3}1513 \quad \Delta = -107, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 141414132211212231¹51

$$141414132211212231²51 \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1³141414132211212231¹51 \quad \Delta = -103, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$141414132211212231²51 \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$141414132211212231²51 \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$141414132211212231²51 \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$141414132211212231²51 \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$141414132211212231²¹51 \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1³141414132211212231¹51 \quad \Delta = -103, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 41315132211212231¹513

$$41315132211212231²513 \quad \Delta = -107, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$41315132211212231²513 \quad \Delta = -107, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$41315132211212231³¹513 \quad \Delta = -115, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$41315132211212231¹²513 \quad \Delta = -107, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$41315132211212231²513 \quad \Delta = -107, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$41315132211212231²513 \quad \Delta = -107, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$41315132211212231²¹513 \quad \Delta = -107, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$41315132211212231²513 \quad \Delta = -107, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$41315132211212231²¹513 \quad \Delta = -107, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$41315132211212231³513 \quad \Delta = -115, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$41315132211212231513 \quad \Delta = -108, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 13151322112122315132

$$13151322112122315132 \quad \Delta = -80, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$13151322112122315132 \quad \Delta = -80, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$13151322112122315132 \quad \Delta = -80, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$13151322112122315132 \quad \Delta = -80, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$13151322112122315132 \quad \Delta = -80, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$13151322112122315132 \quad \Delta = -80, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$13151322112122315132 \quad \Delta = -80, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$231^{31315132211212231}_5 1 \text{ [#=5]} \quad \Delta = -88, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$231^{31315132211212231}_5 1 \text{ [#=9]} \quad \Delta = -88, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$231^{31315132211212231}_5 1 \text{ [#=8]} \quad \Delta = -88, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$231^{31315132211212231}_5 1 \text{ [#=7]} \quad \Delta = -88, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$231^{31315132211212231}_5 1 \text{ [#=6]} \quad \Delta = -88, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$13151322112122315132 \quad \Delta = -88, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$231^{31315132211212231}_5 1 \text{ [#=2]} \quad \Delta = -88, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$231^{31315132211212231}_5 1 \text{ [#=3]} \quad \Delta = -88, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$231^{31315132211212231}_5 1 \text{ [#=4]} \quad \Delta = -88, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

Base: 612231513221121223132

$$612231513221121223132 \quad \Delta = -115, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 63 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$\begin{aligned}
612231513221121223132^2 \quad \Delta = -109, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
612231513221121223132^3 \quad \Delta = -115, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
612231513221121223132^2 \quad \Delta = -107, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
612231513221121223132^2 \quad \Delta = -107, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
612231513221121223132^2 \quad \Delta = -107, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
612231513221121223132^2 \quad \Delta = -107, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
612231513221121223132^2 \quad \Delta = -107, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
612231513221121223132^2 \quad \Delta = -107, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: 161231513221121223132

$$\begin{aligned}
161231513221121223132^3 \quad \Delta = -115, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
161231513221121223132^2 \quad \Delta = -107, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
161231513221121223132^2 \quad \Delta = -107, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
161231513221121223132^2 \quad \Delta = -107, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
161231513221121223132^2 \quad \Delta = -107, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
161231513221121223132^2 \quad \Delta = -107, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
161231513221121223132^2 \quad \Delta = -107, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
161231513221121223132^2 \quad \Delta = -109, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: 316123151322112122313

$$\begin{aligned}
316123151322112122313^2 \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
316123151322112122313^2 \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
\overset{2}{316123151322112122313} \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
316123151322\overset{2}{112122313} \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
3161231513221121223\overset{3}{13} \text{ [#=2]} \quad \Delta = -122, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\overset{2}{316123151322112122313} \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
316123\overset{2}{151322112122313} \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
31612315\overset{2}{1322112122313} \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
3161231513221121223\overset{3}{13} \quad \Delta = -122, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: 216131513221121223132

$$\begin{aligned}
\overset{3}{216131513221121223132} \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\overset{2}{216131513221121223132} \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2161315\overset{2}{13221121223132} \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
216131513221121223\overset{2}{132} \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
2161315\overset{2}{13221121223132} \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
216131513221121223\overset{2}{132} \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21613151322112\overset{2}{1223132} \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
216131513221121223\overset{2}{132} \quad \Delta = -108, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
21613\overset{2}{1513221121223132} \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
216131513221121223\overset{3}{132} \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 321613151322112122313

$$\begin{aligned}
3216131513221121223\overset{2}{13} \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
321613151322112122313^2 \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
321613151322112122313^2 \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
321613151322112122313^2 \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
321613151322112122313^2 \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
321613151322112122313^2 \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
321613151322112122313^2 \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
321613151322112122313^3 \quad \Delta = -122, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
321613151322112122313^3_{[\# = 2]} \quad \Delta = -122, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: 232161315132211212231

$$\begin{aligned}
232161315132211212231^2 \quad \Delta = -108, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
232161315132211212231^2 \quad \Delta = -108, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
2321613151322112122313^3_{[\# = 5]} \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2321613151322112122313^3_{[\# = 2]} \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
2321613151322112122313^3_{[\# = 3]} \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2321613151322112122313^3_{[\# = 4]} \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
232161315132211212231^2 \quad \Delta = -108, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 70 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
232161315132211212231^2 \quad \Delta = -108, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
232161315132211212231^2 \quad \Delta = -108, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
232161315132211212231^2 \quad \Delta = -108, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}
\end{aligned}$$

Base: 13161315132211212222

$$\begin{aligned}
131613151322112122222 \quad \Delta = -96, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
131613151322112122222 \quad \Delta = -96, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
2222211212231513161313_{[\# = 6]} \quad \Delta = -104, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
131613151322112122222 \quad \Delta = -96, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
2222211212231513161313_{[\# = 5]} \quad \Delta = -104, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
131613151322112122222 \quad \Delta = -96, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
2222211212231513161313_{[\# = 4]} \quad \Delta = -104, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
2222211212231513161313_{[\# = 3]} \quad \Delta = -104, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
131613151322112122222 \quad \Delta = -96, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
2222211212231513161313_{[\# = 7]} \quad \Delta = -104, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
131613151322112122222 \quad \Delta = -96, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
2222211212231513161313_{[\# = 2]} \quad \Delta = -104, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
2222211212231513161313_{[\# = 8]} \quad \Delta = -104, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
2222211212231513161313_{[\# = 9]} \quad \Delta = -104, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}
\end{aligned}$$

Base: 413161315132211212222

$$\begin{aligned}
413161315132211212222 \quad \Delta = -124, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
413161315132211212222 \quad \Delta = -124, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
413161315132211212222 \quad \Delta = -124, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
413161315132211212222 \quad \Delta = -124, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
413161315132211212222 \quad \Delta = -124, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
413161315132211212222 \quad \Delta = -132, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
413161315132211212222 \quad \Delta = -124, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
413161315132211212222 \quad \Delta = -124, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 16131513221121223132

$$\begin{aligned}
16131513221121223132 \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
16131513221121223132 \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
16131513221121223132 \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
16131513221121223132 \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
16131513221121223132 \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
16131513221121223132 \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
16131513221121223132 \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
16131513221121223132 \quad \Delta = -108, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1613151322112122313

$$\begin{aligned}
1613151322112122313 \quad \Delta = -121, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1613151322112122313 \quad \Delta = -121, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
31613151322112122313 \quad \Delta = -121, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1613151322112122313 \quad \Delta = -113, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
31613151322112122313 \quad \Delta = -113, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1613151322112122313 \quad \Delta = -113, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
1613151322112122313 \quad \Delta = -113, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{31}{1}6\overset{2}{1}3151322112122313 \quad \Delta = -113, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{31}{1}613151322112122313 \quad \Delta = -113, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{31}{1}6\overset{2}{1}3151322112122313 \quad \Delta = -113, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{31}{1}6\overset{2}{1}3151322112122313 \quad \Delta = -113, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 171223151322112122222

$$\begin{aligned}
& 171223151322112122222 \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 171223151322112122222 \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 17\overset{2}{1}223151322112122222 \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& 171223151322112122222 \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 171223151322112122222 \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

Base: 217123151322112122222

$$\begin{aligned}
& 21\overset{2}{7}123151322112122222 \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 217123151322112122222 \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 217123151322112122222 \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 217123151322112122222 \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 217123151322112122222 \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 217\overset{2}{1}23151322112122222 \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{aligned}$$

Base: 321712315132211212222

$$\begin{aligned}
& 3217\overset{2}{1}2315132211212222 \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
& 321712315132211212222 \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& 321712315132211212222 \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{ll}
321712315132211212222 & \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
321712315132211212222 & \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
321712315132211212222 & \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\text{Base: } 232171231513221121222 & \\
232171231513221121222 & \Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
232171231513221121222 & \Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
232171231513221121222 & \Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 70 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
232171231513221121222 & \Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
232171231513221121222 & \Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 219 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
232171231513221121222 & \Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
\text{Base: } 17123151322112122222 & \\
17123151322112122222 & \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
17123151322112122222 & \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
17123151322112122222 & \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
17123151322112122222 & \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
17123151322112122222 & \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\text{Base: } 112122314141414141 & \\
112122314141414141 & \Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 142 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
112122314141414141 & \Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 310 & 36 \\ 2 & 36 & 8 \end{pmatrix} \\
112122314141414141 & \Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
112122314141414141 & \Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 188 & 28 \\ 2 & 28 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
& 11212231414141414141 \quad \Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 218 & 30 \\ 2 & 30 & 8 \end{pmatrix} \\
& 11212231414141414141 \quad \Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 11212231414141414141 \quad \Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 11212231414141414141\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& 11212231414141414141\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& \textbf{Base: } 1 \text{ }^{131513221} 12 \text{ }^{131513221} 122314141 \\
& 1 \text{ }^{\textcolor{red}{3}131513221} 12 \text{ }^{131513221} 122314141 \text{ [#=8]} \quad \Delta = -83, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
& 1 \text{ }^{\textcolor{red}{3}131513221} 12 \text{ }^{131513221} 122314141 \text{ [#=4]} \quad \Delta = -83, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& 1 \text{ }^{\textcolor{red}{3}131513221} 12 \text{ }^{131513221} 122314141 \text{ [#=2]} \quad \Delta = -83, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& 1 \text{ }^{\textcolor{red}{3}131513221} 12 \text{ }^{131513221} 122314141 \text{ [#=5]} \quad \Delta = -83, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
& 1 \text{ }^{\textcolor{red}{3}131513221} 12 \text{ }^{131513221} 122314141 \text{ [#=6]} \quad \Delta = -83, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
& 1 \text{ }^{131513221} 12 \text{ }^{131513221} 122314141\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -83, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& 1 \text{ }^{\textcolor{red}{3}131513221} 12 \text{ }^{131513221} 122314141 \text{ [#=9]} \quad \Delta = -83, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
& 1 \text{ }^{131513221} 12 \text{ }^{\textcolor{red}{2}} 122314141 \quad \Delta = -75, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 1 \text{ }^{\textcolor{red}{3}131513221} 12 \text{ }^{131513221} 122314141 \text{ [#=7]} \quad \Delta = -83, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
& 1 \text{ }^{131513221} 12 \text{ }^{131513221} 122314141\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -83, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& 1 \text{ }^{\textcolor{red}{3}131513221} 12 \text{ }^{131513221} 122314141 \text{ [#=3]} \quad \Delta = -83, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& 1 \text{ }^{131513221} 12 \text{ }^{\textcolor{red}{2}} 122314141 \quad \Delta = -75, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& 1 \text{ }^{131513221} 12 \text{ }^{131513221} 122314141 \quad \Delta = -75, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& 1 \text{ }^{131513221} 12 \text{ }^{131513221} 122314141 \quad \Delta = -75, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

$$1^{131513221} 12^{122314141} \quad \Delta = -75, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1^{131513221} 12^{122314141} \quad \Delta = -75, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$\text{Base: } 1^{321513221} 12^{122315131}$$

$$1^{321513221} 12^{122315131} \quad \Delta = -80, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1^{321513221} 12^{122315131} \textcolor{red}{3}_{[\#=6]} \quad \Delta = -88, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$1^{321513221} 12^{122315131} \textcolor{red}{3}_{[\#=7]} \quad \Delta = -88, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$1^{321513221} 12^{122315131} \textcolor{red}{3}_{[\#=5]} \quad \Delta = -88, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$1^{321513221} 12^{122315131} \textcolor{red}{3}_{[\#=3]} \quad \Delta = -88, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1^{321513221} 12^{122315131} \textcolor{red}{3}_{[\#=8]} \quad \Delta = -88, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$1^{321513221} 12^{122315131} \textcolor{red}{3}_{[\#=2]} \quad \Delta = -88, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1^{321513221} 12^{122315131} \textcolor{red}{2} \quad \Delta = -80, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1^{321513221} 12^{122315131} \textcolor{red}{2} \quad \Delta = -80, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1^{321513221} 12^{122315131} \quad \Delta = -80, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1^{321513221} 12^{122315131} \textcolor{red}{3}_{[\#=4]} \quad \Delta = -88, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1^{321513221} 12^{122315131} \textcolor{red}{3}_{[\#=9]} \quad \Delta = -88, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$1^{321513221} 12^{122315131} \quad \Delta = -80, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1^{321513221} 12^{122315131} \textcolor{red}{2} \quad \Delta = -80, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\text{Base: } 4131513221^{21513221} 12^1$$

$$4131513221^{21513221} 12^1 \quad \Delta = -107, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$\textcolor{red}{3} 4131513221^{21513221} 12^1 \quad \Delta = -107, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$\begin{array}{ll}
4131513221 \overset{21513221}{12} 1 & \Delta = -99, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\overset{2}{4131513221} \overset{21513221}{12} 1 & \Delta = -99, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
4131513221 \overset{21513221}{12} 1 & \Delta = -99, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\overset{2}{4131513221} \overset{21513221}{12} 1 & \Delta = -100, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
4131513221 \overset{21513221}{12} 1 & \Delta = -99, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
4131513221 \overset{21513221}{12} 1 & \Delta = -99, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\overset{2}{4131513221} \overset{21513221}{12} 1 & \Delta = -99, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
4131513221 \overset{21513221}{12} 1 & \Delta = -99, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{1231513221}{12} 12231413 & \\
1 \overset{1231513221}{12} 12231413 & \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \overset{1231513221}{12} 12231413 \overset{2}{} & \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1 \overset{1231513221}{12} 12231413 \overset{2}{} & \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \overset{1231513221}{12} 12231413 & \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \overset{31231513221}{12} 12231413 [\#=8] & \Delta = -64, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
1 \overset{1231513221}{12} 12231413 & \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1 \overset{31231513221}{12} 12231413 [\#=2] & \Delta = -64, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1 \overset{31231513221}{12} 12231413 [\#=9] & \Delta = -64, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
1 \overset{1231513221}{12} 12231413 \overset{2}{} & \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1 \overset{1231513221}{12} 12231413 \overset{3}{} & \Delta = -64, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1 \overset{31231513221}{12} 12231413 [\#=3] & \Delta = -64, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
& 1^{\textcolor{red}{3}1231513221} 12^{\textcolor{red}{2}} 12231413 \text{ [#=4]} \quad \Delta = -64, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& 1^{\textcolor{red}{3}1231513221} 12^{\textcolor{red}{2}} 12231413 \text{ [#=5]} \quad \Delta = -64, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
& 1^{\textcolor{red}{3}1231513221} 12^{\textcolor{red}{2}} 12231413 \text{ [#=6]} \quad \Delta = -64, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
& 1^{\textcolor{red}{3}1231513221} 12^{\textcolor{red}{2}} 12231413 \text{ [#=7]} \quad \Delta = -64, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
& \textbf{Base: } 1^{\textcolor{red}{3}1231513221} 12^{\textcolor{red}{2}} 1223132 \\
& 1^{\textcolor{red}{3}1231513221} 12^{\textcolor{red}{2}} 1223132 \quad \Delta = -29, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 1^{\textcolor{red}{3}12231513221} 12^{\textcolor{red}{2}} 1223132 \text{ [#=2]} \quad \Delta = -37, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 1^{\textcolor{red}{3}12231513221} 12^{\textcolor{red}{2}} 1223132 \quad \Delta = -31, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 1^{\textcolor{red}{3}12231513221} 12^{\textcolor{red}{2}} 1223132 \text{ [#=5]} \quad \Delta = -37, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& 1^{\textcolor{red}{3}12231513221} 12^{\textcolor{red}{2}} 1223132 \quad \Delta = -29, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 1^{\textcolor{red}{3}12231513221} 12^{\textcolor{red}{2}} 1223132 \quad \Delta = -29, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 1^{\textcolor{red}{3}12231513221} 12^{\textcolor{red}{2}} 1223132 \quad \Delta = -29, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& 1^{\textcolor{red}{3}12231513221} 12^{\textcolor{red}{2}} 1223132 \quad \Delta = -29, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 1^{\textcolor{red}{3}12231513221} 12^{\textcolor{red}{2}} 1223132 \text{ [#=6]} \quad \Delta = -37, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& 1^{\textcolor{red}{3}12231513221} 12^{\textcolor{red}{2}} 1223132 \text{ [#=7]} \quad \Delta = -37, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
& 1^{\textcolor{red}{3}12231513221} 12^{\textcolor{red}{2}} 1223132 \text{ [#=8]} \quad \Delta = -37, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
& 1^{\textcolor{red}{3}12231513221} 12^{\textcolor{red}{2}} 1223132 \text{ [#=3]} \quad \Delta = -37, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 1^{\textcolor{red}{3}12231513221} 12^{\textcolor{red}{2}} 1223132 \text{ [#=9]} \quad \Delta = -37, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
& 1^{\textcolor{red}{3}12231513221} 12^{\textcolor{red}{2}} 1223132 \quad \Delta = -37, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 32 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 1^{\textcolor{red}{3}12231513221} 12^{\textcolor{red}{2}} 1223132 \text{ [#=4]} \quad \Delta = -37, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{51231513221}{12} 1223141$

$$1 \overset{51231513221}{12} 1223141 \overset{2}{\Delta} = -100, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$1 \overset{51231513221}{12} 1223141 \overset{3}{\Delta} [\#=3] = -108, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$1 \overset{51231513221}{12} 1223141 \overset{3}{\Delta} [\#=2] = -108, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$1 \overset{51231513221}{12} 1223141 \overset{2}{\Delta} = -100, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{51231513221}{12} 1223141 \Delta = -100, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 68 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

$$1 \overset{51231513221}{12} 1223141 \Delta = -100, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 63 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$1 \overset{51231513221}{12} 1223141 \Delta = -100, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 76 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

$$1 \overset{51231513221}{12} 1223141 \Delta = -100, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 58 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

Base: $41413221 \overset{5131513221}{12} 1$

$$41413221 \overset{5131513221}{12} 1 \Delta = -128, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$41413221 \overset{5131513221}{12} 1 \Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$41413221 \overset{5131513221}{12} 1 \Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$41413221 \overset{5131513221}{12} 1 \Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$41413221 \overset{2}{5131513221} \overset{2}{12} 1 \Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$41413221 \overset{5131513221}{12} \overset{12}{\Delta} = -127, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$41413221 \overset{2}{5131513221} \overset{2}{12} 1 \Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$41413221 \overset{2}{5131513221} \overset{2}{12} 1 \Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$41413221 \overset{5131513221}{12} 1 \Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: $1 \overset{15131513221}{12} 1223141$

$$1 \overset{15131513221}{12} 1223141 \Delta = -100, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\begin{array}{ll}
1^{15131513221} 12^{1223141} \quad \Delta = -99, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 76 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
1^{15131513221} 12^{1223141} \quad \Delta = -99, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 58 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
1^{15131513221} 12^{1223141} \quad \Delta = -99, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 63 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1^{15131513221} 12^{1223141} \textcolor{red}{3} [\#=3] \quad \Delta = -107, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
1^{15131513221} 12^{1223141} \textcolor{red}{3} [\#=2] \quad \Delta = -107, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
1^{15131513221} 12^{1223141} \quad \Delta = -99, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1^{15131513221} 12^{1223141} \textcolor{red}{2} \quad \Delta = -99, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1^{15131513221} 12^{1223141} \textcolor{red}{2} \quad \Delta = -99, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{array}$$

Base: $4131513221^{1513221} 12^{12} 1$

$$\begin{array}{ll}
4131513221^{1513221} 12^{12} 1 \quad \Delta = -99, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
4131513221^{1513221} 12^{12} 1 \quad \Delta = -99, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
4131513221^{1513221} 12^{12} 1 \textcolor{red}{2} \quad \Delta = -99, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
4131513221^{1513221} 12^{12} 1 \textcolor{red}{2} \quad \Delta = -99, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
4131513221^{1513221} 12^{12} 1 \textcolor{red}{2} \quad \Delta = -99, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
4131513221^{1513221} 12^{12} 1 \textcolor{red}{2} \quad \Delta = -99, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
4131513221^{1513221} 12^{12} 1 \textcolor{red}{2} \quad \Delta = -100, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
4131513221^{1513221} 12^{12} 1 \textcolor{red}{3} \quad \Delta = -107, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{array}$$

Base: $1^{31513221} 12^{122315131}$

$$\begin{array}{ll}
1^{31513221} 12^{122315131} \textcolor{red}{3} [\#=7] \quad \Delta = -87, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1^{31513221} 12^{122315131} \textcolor{red}{3} [\#=4] \quad \Delta = -87, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
& 1^{31513221} 12^{122315131} \mathbf{3}_{[\#=5]} \quad \Delta = -87, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& 1^{31513221} 12^{122315131} \mathbf{3}_{[\#=6]} \quad \Delta = -87, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& 1^{31513221} 12^{122315131} \quad \Delta = -87, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& 1^{31513221} 12^{122315131} \mathbf{3}_{[\#=2]} \quad \Delta = -87, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 1^{31513221} 12^{122315131} \mathbf{3}_{[\#=8]} \quad \Delta = -87, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
& 1^{31513221} 12^{122315131} \mathbf{3}_{[\#=9]} \quad \Delta = -87, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
& 1^{31513221} 12^{122315131} \quad \Delta = -79, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 1^{31513221} 12^{122315131} \mathbf{2} \quad \Delta = -79, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 1^{31513221} 12^{122315131} \mathbf{2} \quad \Delta = -79, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 1^{31513221} 12^{122315131} \mathbf{3}_{[\#=3]} \quad \Delta = -87, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 1^{31513221} 12^{122315131} \mathbf{2} \quad \Delta = -79, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 1^{31513221} 12^{122315131} \quad \Delta = -79, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 1^{31513221} 12^{122315131} \quad \Delta = -79, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 1^{31513221} 12^{122315131} \quad \Delta = -79, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \mathbf{Base:} \quad 612231513221^{313221} 12^1 1 \\
& 612231513221^{313221} 12^1 1_{[\#=2]} \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 612231513221^{313221} 12^1 1 \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& 612231513221^{\mathbf{3}} 12^{313221} 1 \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 63 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& 612231513221^{313221} 12^1 1 \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 612231513221^{\mathbf{2}} 12^{313221} 1 \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
612231513221 \overset{313221}{12} 1 \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
612231513221 \overset{313221}{12} 1 \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
612231513221 \overset{313221}{12} 1 \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
612231513221 \overset{313221}{12} 1 \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{313221}{12} 122315132161$

$$\begin{aligned}
1 \overset{313221}{12} 122315132161 \text{ [#=2]} \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 93 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1 \overset{313221}{12} 122315132161 \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1 \overset{313221}{12} 122315132161 \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1 \overset{313221}{12} 122315132161 \overset{2}{} \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1 \overset{313221}{12} 122315132161 \overset{2}{} \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \overset{313221}{12} 122315132161 \overset{2}{} \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
1 \overset{313221}{12} 122315132161 \overset{2}{} \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1 \overset{313221}{12} 122315132161 \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 21 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{13221}{12} 1223151321613$

$$\begin{aligned}
1 \overset{13221}{12} 1223151321613 \overset{2}{} \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1 \overset{313221}{12} 1223151321613 \text{ [#=3]} \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{313221}{12} 1223151321613 \text{ [#=4]} \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \overset{313221}{12} 1223151321613 \text{ [#=5]} \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1 \overset{13221}{12} 1223151321613 \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{13221}{12} 1223151321613 \overset{2}{} \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{13221}{1} \overset{2}{12} 1223151321613 \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{13221}{1} \overset{2}{12} 1223151321613 \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{13221}{1} \overset{2}{12} 1223151321613 \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{313221}{1} \overset{2}{12} 1223151321613 \text{ [#=2]} \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{313221}{1} \overset{2}{12} 122315131612$

$$\begin{aligned}
& \overset{313221}{1} \overset{2}{12} 122315131612 \quad \Delta = -105, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{313221}{1} \overset{2}{12} 122315131612 \quad \Delta = -113, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 21 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{313221}{1} \overset{2}{12} 122315131612 \text{ [#=2]} \quad \Delta = -113, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 93 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& \overset{313221}{1} \overset{2}{12} 122315131612 \overset{3}{12} \quad \Delta = -113, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
& \overset{313221}{1} \overset{2}{12} 122315131612 \overset{2}{12} \quad \Delta = -105, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{313221}{1} \overset{2}{12} 122315131612 \quad \Delta = -105, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{313221}{1} \overset{2}{12} 122315131612 \overset{2}{12} \quad \Delta = -105, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{313221}{1} \overset{2}{12} 122315131612 \overset{2}{12} \quad \Delta = -105, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{313221}{1} \overset{2}{12} 122315131612 \overset{2}{12} \quad \Delta = -105, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{313221}{1} \overset{2}{12} 122315131612 \overset{2}{12} \quad \Delta = -105, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{13221}{1} \overset{2}{12} 1223151316123$

$$\begin{aligned}
& \overset{313221}{1} \overset{2}{12} 1223151316123 \text{ [#=3]} \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{313221}{1} \overset{2}{12} 1223151316123 \text{ [#=4]} \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{313221}{1} \overset{2}{12} 1223151316123 \text{ [#=5]} \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{313221}{1} \overset{2}{12} 1223151316123 \text{ [#=2]} \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{ll}
\begin{array}{c} 13221 \\ 1 \ 12 \ 1223151316123 \end{array} \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\begin{array}{c} 13221 \\ 1 \ 12 \ 1223151316123 \end{array} \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\begin{array}{c} 13221 \\ 1 \ 12 \ 1223151316123 \end{array} \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\begin{array}{c} 13221 \\ 1 \ 12 \ 1223151316123 \end{array} \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\begin{array}{c} 13221 \\ 1 \ 12 \ 1223151316123 \end{array} \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\begin{array}{c} 13221 \\ 1 \ 12 \ 1223151316123 \end{array} \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{array}$$

Base: $\begin{array}{c} 22221 \\ 1 \ 12 \ 1223151316131 \end{array}$

$$\begin{array}{ll}
\begin{array}{c} 22221 \\ 1 \ 12 \ 1223151316131 \end{array} \text{ \textcolor{red}{3} } [\#=2] \quad \Delta = -104, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
\begin{array}{c} 22221 \\ 1 \ 12 \ 1223151316131 \end{array} \text{ \textcolor{red}{3} } [\#=8] \quad \Delta = -104, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\begin{array}{c} 22221 \\ 1 \ 12 \ 1223151316131 \end{array} \text{ \textcolor{red}{3} } [\#=7] \quad \Delta = -104, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\begin{array}{c} 22221 \\ 1 \ 12 \ 1223151316131 \end{array} \text{ \textcolor{red}{3} } [\#=9] \quad \Delta = -104, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\begin{array}{c} 22221 \\ 1 \ 12 \ 1223151316131 \end{array} \text{ \textcolor{red}{3} } [\#=3] \quad \Delta = -104, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\begin{array}{c} 22221 \\ 1 \ 12 \ 1223151316131 \end{array} \text{ \textcolor{red}{3} } [\#=5] \quad \Delta = -104, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\begin{array}{c} 22221 \\ 1 \ 12 \ 1223151316131 \end{array} \quad \Delta = -96, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\begin{array}{c} 22221 \\ 1 \ 12 \ 1223151316131 \end{array} \quad \Delta = -96, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\begin{array}{c} 22221 \\ 1 \ 12 \ 1223151316131 \end{array} \quad \Delta = -96, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
\begin{array}{c} 22221 \\ 1 \ 12 \ 1223151316131 \end{array} \quad \Delta = -96, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\begin{array}{c} 22221 \\ 1 \ 12 \ 1223151316131 \end{array} \quad \Delta = -96, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
\begin{array}{c} 22221 \\ 1 \ 12 \ 1223151316131 \end{array} \quad \Delta = -96, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\begin{array}{c} 22221 \\ 1 \ 12 \ 1223151316131 \end{array} \text{ \textcolor{red}{3} } [\#=4] \quad \Delta = -104, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{array}$$

$$1 \overset{22221}{12} 1223151316131\overset{2221}{3} \text{ [#}=6] \quad \Delta = -104, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: $41316131513221 \overset{2221}{12} 1$

$$41316131513221 \overset{2221}{12} 1 \quad \Delta = -124, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\overset{2}{4}1316131513221 \overset{2221}{12} 1 \quad \Delta = -124, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\overset{3}{4}1316131513221 \overset{2221}{12} 1 \quad \Delta = -132, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$413161315\overset{2}{13}221 \overset{2221}{12} 1 \quad \Delta = -124, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$41316131513221 \overset{2221}{12} 1 \quad \Delta = -124, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$\overset{2}{4}1316131513221 \overset{2221}{12} 1 \quad \Delta = -124, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$41316131513221 \overset{2221}{12} 1 \quad \Delta = -124, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$4131613\overset{2}{15}13221 \overset{2221}{12} 1 \quad \Delta = -124, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

Base: $161315132211\overset{1}{2}122313$

$$\overset{1}{16}1315132211\overset{1}{2}122313 \quad \Delta = -105, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\overset{1}{16}1315\overset{2}{13}2211\overset{1}{2}122313 \quad \Delta = -105, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$\overset{1}{16}1315\overset{2}{13}2211\overset{1}{2}122313 \quad \Delta = -105, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$\overset{1}{16}1315132211\overset{2}{1}2122313 \quad \Delta = -105, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$\overset{1}{16}1315132211\overset{2}{1}2122313 \quad \Delta = -105, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\overset{1}{16}1315132211\overset{3}{1}2122313 \text{ [#}=2] \quad \Delta = -113, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$\overset{1}{16}1315132211\overset{3}{1}2122313 \quad \Delta = -113, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\overset{1}{16}1315132211\overset{2}{1}2122313 \quad \Delta = -105, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: $161315132211\overset{31}{2}12231$

$$\overset{31}{16}1315132211\overset{2}{1}212231 \quad \Delta = -105, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$\begin{array}{ll}
1 \overset{31}{6} 1315132211 \overset{1}{12} 12231 \overset{3}{\text{red}}_{[\# = 5]} & \Delta = -113, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1 \overset{31}{6} 1315132211 \overset{1}{12} 12231 \overset{3}{\text{red}}_{[\# = 4]} & \Delta = -113, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \overset{31}{6} 1315132211 \overset{1}{12} 12231 \overset{3}{\text{red}}_{[\# = 3]} & \Delta = -113, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{31}{6} 1315132211 \overset{1}{12} 12231 \overset{3}{\text{red}}_{[\# = 2]} & \Delta = -113, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
31 \overset{31}{6} 1315132211 \overset{1}{12} 12231 & \Delta = -113, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
31 \overset{21}{6} 1315132211 \overset{1}{12} 12231 & \Delta = -105, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1 \overset{31}{6} 1315132211 \overset{1}{12} 12231 & \Delta = -105, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \overset{31}{6} 1315132211 \overset{2}{1} 12231 & \Delta = -105, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1 \overset{31}{6} 1315132211 \overset{1}{12} \overset{2}{2} 231 & \Delta = -105, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{31}{6} 1315132211 \overset{1}{12} 12231 & \Delta = -105, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
1 \overset{31}{6} 1315132211 \overset{1}{12} 12231 & \Delta = -105, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{22221}{12} 1223151322171 & \\
1 \overset{22221}{12} 1223151322171 & \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \overset{22221}{12} 1223151322171 & \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1 \overset{22221}{12} 1223151322171 & \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1 \overset{22221}{12} 1223151322171 & \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1 \overset{22221}{12} \overset{2}{2} 1223151322171 & \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{22221}{12} 1223151321712 & \\
1 \overset{22221}{12} 1223151321712 & \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1 \overset{22221}{12} 1223151321712 & \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{ll}
\begin{array}{l} 22221 \\ 1 \ 12 \ 1223151321712 \end{array} \quad \Delta = -116, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\begin{array}{l} 22221 \\ 1 \ 12 \ 1223151321712 \end{array} \quad \Delta = -116, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\begin{array}{l} 22221 \ 2 \\ 1 \ 12 \ 1223151321712 \end{array} \quad \Delta = -116, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
\begin{array}{l} 22221 \\ 1 \ 12 \ 1223151321712 \end{array} \quad \Delta = -116, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{array}$$

Base: $\begin{array}{l} 2221 \\ 1 \ 12 \ 12231513217123 \end{array}$

$$\begin{array}{ll}
\begin{array}{l} 2221 \ 2 \\ 1 \ 12 \ 12231513217123 \end{array} \quad \Delta = -125, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\begin{array}{l} 2221 \\ 1 \ 12 \ 12231513217123 \end{array} \quad \Delta = -125, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\begin{array}{l} 2221 \\ 1 \ 12 \ 12231513217123 \end{array} \quad \Delta = -125, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\begin{array}{l} 2221 \\ 1 \ 12 \ 12231513217123 \end{array} \quad \Delta = -125, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
\begin{array}{l} 2221 \\ 1 \ 12 \ 12231513217123 \end{array} \quad \Delta = -125, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\begin{array}{l} 2221 \\ 1 \ 12 \ 12231513217123 \end{array} \quad \Delta = -125, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{array}$$

Base: $\begin{array}{l} 221 \\ 1 \ 12122315132171232 \end{array}$

$$\begin{array}{ll}
\begin{array}{l} 221 \\ 1 \ 12122315132171232 \end{array} \quad \Delta = -127, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\begin{array}{l} 221 \\ 1 \ 12122315132171232 \end{array} \quad \Delta = -127, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
\begin{array}{l} 221 \\ 1 \ 12122315132171232 \end{array} \quad \Delta = -127, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
\begin{array}{l} 221 \\ 1 \ 12122315132171232 \end{array} \quad \Delta = -127, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\begin{array}{l} 221 \ 2 \\ 1 \ 12122315132171232 \end{array} \quad \Delta = -127, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\begin{array}{l} 221 \\ 1 \ 12122315132171232 \end{array} \quad \Delta = -127, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{array}$$

Base: $\begin{array}{l} 1 \\ 1712315132211212222 \end{array}$

$$\begin{array}{ll}
\begin{array}{l} 1 \\ 1712315132211212222 \end{array} \quad \Delta = -116, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\begin{array}{l} 1 \\ 1712315132211212222 \end{array} \quad \Delta = -116, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
& \overset{1}{1}71231513221\overset{1}{1}\overset{2}{2}222 \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{1}{1}71231513221\overset{1}{1}\overset{2}{2}222 \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& \overset{1}{1}71231513221\overset{1}{1}\overset{2}{2}222 \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{21}{21}12122315132171232$

$$\begin{aligned}
& \overset{21}{21}12122315132171232 \quad \Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
& \overset{21}{21}\overset{2}{1}22315132171232 \quad \Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{21}{21}121223151321\overset{2}{7}1232 \quad \Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
& \overset{21}{21}12122315132171232 \quad \Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{2}{2}\overset{21}{1}2122315132171232 \quad \Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{21}{21}12122315132171232 \quad \Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{21}{21}\overset{21513221}{12}122315131$

$$\begin{aligned}
& \overset{21}{21}\overset{21513221}{12}122315131\overset{3}{3} \text{ [#=6]} \quad \Delta = -79, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& \overset{21}{21}\overset{21513221}{12}122315131\overset{3}{3} \text{ [#=8]} \quad \Delta = -79, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
& \overset{21}{21}\overset{21513221}{12}122315131\overset{3}{3} \text{ [#=5]} \quad \Delta = -79, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& \overset{21}{21}\overset{21513221}{12}122315131\overset{3}{3} \text{ [#=7]} \quad \Delta = -79, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
& \overset{21}{21}\overset{21513221}{12}122315131\overset{2}{2} \quad \Delta = -71, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{21}{21}\overset{21513221}{12}122315131\overset{3}{3} \text{ [#=4]} \quad \Delta = -79, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{21}{21}\overset{21513221}{12}122315131\overset{3}{3} \text{ [#=3]} \quad \Delta = -79, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{21}{21}\overset{21513221}{12}122315131\overset{3}{3} \text{ [#=2]} \quad \Delta = -79, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{21}{21}\overset{21513221}{12}122315131\overset{3}{3} \text{ [#=9]} \quad \Delta = -79, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
21^{21513221} 12^{122315131} \quad \Delta = -79, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
21^{21513221} 12^{122315131} \quad \Delta = -71, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21^{21513221} 12^{122315131} \quad \Delta = -71, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21^{21513221} 12^{122315131} \quad \Delta = -71, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21^{21513221} 12^{122315131} \quad \Delta = -71, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
21^{21513221} 12^{122315131} \quad \Delta = -71, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21^{21513221} 12^{122315131} \quad \Delta = -71, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: $21^{1231513221} 12^{1223141}$

$$\begin{aligned}
21^{1231513221} 12^{1223141} \quad \Delta = -48, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21^{31231513221} 12^{1223141} \quad [\# = 2] \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21^{1231513221} 12^{1223141} \quad [\# = 2] \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
21^{31231513221} 12^{1223141} \quad [\# = 3] \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
21^{31231513221} 12^{1223141} \quad [\# = 9] \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix} \\
21^{1231513221} 12^{1223141} \quad [\# = 3] \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
21^{31231513221} 12^{1223141} \quad [\# = 4] \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
21^{1231513221} 12^{1223141} \quad \Delta = -48, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
21^{1231513221} 12^{1223141} \quad \Delta = -48, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21^{21231513221} 12^{1223141} \quad \Delta = -48, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21^{1231513221} 12^{1223141} \quad \Delta = -48, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21^{1231513221} 12^{1223141} \quad \Delta = -48, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
21 \overset{3}{1}2231513221 \quad 12 \quad 1223141 \quad [\# = 8] \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
21 \overset{3}{1}2231513221 \quad 12 \quad 1223141 \quad [\# = 7] \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
21 \overset{3}{1}2231513221 \quad 12 \quad 1223141 \quad [\# = 6] \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
21 \overset{3}{1}2231513221 \quad 12 \quad 1223141 \quad [\# = 5] \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix}
\end{aligned}$$

Base: $21 \overset{12231513221}{12} 122313$

$$\begin{aligned}
21 \overset{12231513221}{12} 122313 \quad \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21 \overset{12231513221}{12} 122313 \quad \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
21 \overset{12231513221}{12} 122313 \overset{2}{\quad} \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21 \overset{12231513221}{12} 122313 \overset{2}{\quad} \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21 \overset{2}{1}2231513221 \quad 12 \quad 122313 \quad \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21 \overset{12231513221}{12} 122313 \overset{3}{\quad} [\# = 2] \quad \Delta = -36, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
21 \overset{3}{1}2231513221 \quad 12 \quad 122313 \quad [\# = 2] \quad \Delta = -36, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21 \overset{3}{1}2231513221 \quad 12 \quad 122313 \quad [\# = 8] \quad \Delta = -36, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
21 \overset{3}{1}2231513221 \quad 12 \quad 122313 \quad [\# = 3] \quad \Delta = -36, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21 \overset{12231513221}{12} 122313 \overset{3}{\quad} \Delta = -36, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21 \overset{3}{1}2231513221 \quad 12 \quad 122313 \quad [\# = 4] \quad \Delta = -36, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
21 \overset{3}{1}2231513221 \quad 12 \quad 122313 \quad [\# = 7] \quad \Delta = -36, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
21 \overset{3}{1}2231513221 \quad 12 \quad 122313 \quad [\# = 9] \quad \Delta = -36, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
21 \overset{3}{1}2231513221 \quad 12 \quad 122313 \quad [\# = 6] \quad \Delta = -36, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
21 \overset{3}{1}2231513221 \quad 12 \quad 122313 \quad [\# = 5] \quad \Delta = -36, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}
\end{aligned}$$

$$21^{12231513221}_{12} 122313 \quad \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: $21^{5131513221}_{12} 1223151$

$$21^{5131513221}_{12} 1223151 \quad \Delta = -124, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$21^{5131513221}_{12} 1223151 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$21^{5131513221}_{12} 1223151 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$21^{5131513221}_{12} 1223151^2 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$21^{5131513221}_{12} 1223151^2 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$21^{5131513221}_{12} 1223151 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$21^{5131513221}_{12} 1223151 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$21^{25131513221}_{12} 1223151 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: $21^{1513221}_{12} 122315131$

$$21^{1513221}_{12} 122315131^3 \text{ [#=7]} \quad \Delta = -79, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$21^{1513221}_{12} 122315131^3 \text{ [#=9]} \quad \Delta = -79, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$21^{1513221}_{12} 122315131^3 \text{ [#=8]} \quad \Delta = -79, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$21^{1513221}_{12} 122315131^3 \text{ [#=5]} \quad \Delta = -79, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$21^{1513221}_{12} 122315131^3 \text{ [#=6]} \quad \Delta = -79, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$21^{1513221}_{12} 122315131 \quad \Delta = -71, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$21^{1513221}_{12} 122315131^2 \quad \Delta = -71, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$21^{1513221}_{12} 122315131 \quad \Delta = -71, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$21^{21513221}_{12} 122315131 \quad \Delta = -71, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\begin{aligned}
21^{1513221\bar{2}}12^{122315131} \quad \Delta = -71, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21^{1513221}12^{122315131\bar{2}} \quad \Delta = -71, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21^{1513221}12^{122315131\bar{3}} [\# = 4] \quad \Delta = -79, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
21^{1513221}12^{122315131\bar{3}} [\# = 2] \quad \Delta = -79, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21^{1513221}12^{122315131\bar{3}} [\# = 3] \quad \Delta = -79, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\text{Base: } 612231513221^{13221}12^{12} \\
612231513221^{13221}12^{12} \quad \Delta = -98, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\bar{3}612231513221^{13221}12^{12} \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 63 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
612231513221^{313221}12^{12} [\# = 2] \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
612231513221^{313221}12^{12} [\# = 4] \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
612231513221^{\bar{2}}12^{13221}12^{12} \quad \Delta = -98, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
612231513221^{13221\bar{2}}12^{12} \quad \Delta = -98, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
612231513221^{313221}12^{12} [\# = 3] \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
612231513221^{313221}12^{12} [\# = 5] \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
612231513221^{\bar{2}}12^{13221}12^{12} \quad \Delta = -98, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\bar{2}612231513221^{13221}12^{12} \quad \Delta = -98, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
612231513221^{213221}12^{12} \quad \Delta = -98, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\text{Base: } 4132211\bar{2}12231513216^{21} \\
4132211\bar{2}12231513216^{21} \quad \Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\bar{2}4132211\bar{2}12231513216^{21} \quad \Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& 413221\overset{21}{1}212231513216 \quad \Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 413221\overset{21}{1}212231513216 \quad \Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& 413221\overset{2}{1}212231513216 \quad \Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& 413221\overset{21}{1}212231513216 \quad \Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 413221\overset{21}{1}212231513216 \quad \Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \text{Base: } 21 \overset{13221}{12} 122315132161 \\
& 21 \overset{313221}{12} 122315132161 \text{ } [\# = 5] \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& 21 \overset{313221}{12} 122315132161 \text{ } [\# = 4] \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& 21 \overset{213221}{12} 122315132161 \quad \Delta = -98, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 21 \overset{313221}{12} 122315132161 \text{ } [\# = 2] \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 21 \overset{313221}{12} 122315132161 \text{ } [\# = 3] \quad \Delta = -106, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 21 \overset{13221}{12} 122315132161 \quad \Delta = -98, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 21 \overset{13221}{12} 122315132161 \quad \Delta = -98, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 21 \overset{13221}{12} 122315132161 \quad \Delta = -98, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 21 \overset{13221}{12} 122315132161 \quad \Delta = -98, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 21 \overset{13221}{12} 122315132161 \quad \Delta = -98, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& \text{Base: } 413221\overset{21}{1}212231513161 \\
& 413221\overset{2}{1}212231513161 \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 413221\overset{21}{1}212231513161 \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 413221\overset{21}{1}212231513161 \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
413221\overset{21}{1}212231513161 \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
413221\overset{21}{1}2\overset{2}{1}2231513161 \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
413221\overset{21}{1}212231513\overset{2}{1}61 \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\overset{2}{4}13221\overset{21}{1}212231513161 \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: $21 \overset{13221}{12} 122315131612$

$$\begin{aligned}
21 \overset{13221}{12} \overset{2}{1}22315131612 \quad \Delta = -97, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
21 \overset{13221}{12} 122315131612 \quad \Delta = -97, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\overset{2}{1}3221\overset{13221}{12} 122315131612 \quad \Delta = -97, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21 \overset{313221}{12} 122315131612 \text{ [#=2]} \quad \Delta = -105, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21 \overset{313221}{12} 122315131612 \text{ [#=5]} \quad \Delta = -105, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
21 \overset{13221}{12} 1223151316\overset{3}{1}2 \quad \Delta = -105, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
21 \overset{13221}{12} 1223151316\overset{2}{1}2 \quad \Delta = -97, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21 \overset{13221}{12} 1223151316\overset{2}{1}2 \quad \Delta = -97, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21 \overset{13221}{12} 1223151316\overset{2}{1}2 \quad \Delta = -97, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
21 \overset{313221}{12} 122315131612 \text{ [#=4]} \quad \Delta = -105, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
21 \overset{13221}{12} 1223151316\overset{2}{1}2 \quad \Delta = -97, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
21 \overset{313221}{12} 122315131612 \text{ [#=3]} \quad \Delta = -105, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

Base: $21 \overset{2221}{12} 1223151316131$

$$\begin{aligned}
21 \overset{2221}{12} \overset{2}{1}223151316131 \quad \Delta = -96, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
21 \overset{2221}{12} \overset{2}{1}223151316131 \quad \Delta = -96, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
21 \overset{2221}{12} 1223151316131 \overset{2}{} \quad \Delta = -96, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21 \overset{2221}{12} 1223151316131 \overset{2}{} \quad \Delta = -96, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21 \overset{2221}{12} 1223151316131 \overset{2}{} \quad \Delta = -96, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21 \overset{2221}{12} 1223151316131 \quad \Delta = -96, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
21 \overset{2221}{12} 1223151316131 \overset{2}{} \quad \Delta = -96, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
21 \overset{2221}{12} 1223151316131 \overset{3}{}_{[\#=3]} \quad \Delta = -104, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
21 \overset{2221}{12} 1223151316131 \overset{3}{}_{[\#=4]} \quad \Delta = -104, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21 \overset{2221}{12} 1223151316131 \overset{3}{}_{[\#=5]} \quad \Delta = -104, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21 \overset{2221}{12} 1223151316131 \overset{3}{}_{[\#=6]} \quad \Delta = -104, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
21 \overset{2221}{12} 1223151316131 \overset{3}{}_{[\#=2]} \quad \Delta = -104, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
21 \overset{2221}{12} 1223151316131 \overset{3}{}_{[\#=7]} \quad \Delta = -104, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21 \overset{2221}{12} 1223151316131 \overset{3}{}_{[\#=8]} \quad \Delta = -104, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21 \overset{2221}{12} 1223151316131 \overset{3}{}_{[\#=9]} \quad \Delta = -104, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 41316131513221²²¹1212

$$\begin{aligned}
41316131513221 \overset{221}{12} \overset{2}{12} \quad \Delta = -124, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
41316131513221 \overset{221}{12} \overset{2}{12} \quad \Delta = -124, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
41316131513221 \overset{221}{12} \overset{2}{12} \quad \Delta = -124, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
41316131513221 \overset{221}{12} \overset{2}{12} \quad \Delta = -124, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
41316131513221 \overset{221}{12} \overset{2}{12} \quad \Delta = -124, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
41316131513221 \overset{221}{12} \overset{2}{12} \quad \Delta = -124, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

$$41316131513221\overset{2221}{12}12 \quad \Delta = -124, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$41316131513221\overset{221}{12}12 \quad \Delta = -124, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$41316131513221\overset{3}{12}\overset{221}{12}12 \quad \Delta = -132, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

Base: $\overset{1}{16}131513221\overset{21}{12}12231$

$$\overset{1}{16}131513221\overset{21}{12}12231\overset{3}{1} \text{ [#=2]} \quad \Delta = -105, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\overset{1}{16}131513221\overset{21}{12}12231\overset{3}{1} \text{ [#=3]} \quad \Delta = -105, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$\overset{1}{16}131513221\overset{1}{12}\overset{21}{12}2231 \quad \Delta = -97, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\overset{1}{16}131513221\overset{1}{12}\overset{2}{12}\overset{21}{12}2231 \quad \Delta = -97, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$\overset{1}{16}131513221\overset{21}{12}\overset{2}{12}2231 \quad \Delta = -97, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$\overset{1}{16}131513221\overset{21}{12}12231\overset{3}{1} \text{ [#=4]} \quad \Delta = -105, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$\overset{1}{16}131513221\overset{21}{12}12231\overset{3}{1} \text{ [#=5]} \quad \Delta = -105, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$\overset{1}{16}131513221\overset{1}{12}\overset{2}{12}\overset{21}{12}2231 \quad \Delta = -97, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$\overset{1}{16}131513221\overset{1}{12}\overset{2}{12}\overset{21}{12}2231 \quad \Delta = -97, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$\overset{1}{16}131513221\overset{21}{12}12231 \quad \Delta = -97, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: $\overset{2221}{21}121223151322171$

$$\overset{2221}{21}1212231513221\overset{2}{7}1 \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$\overset{22221}{21}121223151322171 \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\overset{2221}{21}1212231513221\overset{2}{7}1 \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$\overset{2221}{21}1212231513221\overset{2}{7}1 \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$\overset{2221}{21}1212231513221\overset{2}{7}1 \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$${}^{2221}_{21\ 12\ 1223151322171} \quad \Delta = -116, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: $21^{2221} 12^{1223151321712}$

$$21\overset{2221}{12}\overset{2}{122}3151321712 \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$21\overset{2221}{12}\,1223151321712 \quad \Delta = -116, \, n_+ = 1, \, n_- = 20, \, n_0 = 0, \, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$21\ 12\ 1223151321712 \quad \Delta = -116, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$21 \overset{2221}{12} \overset{2}{1223} 151321712 \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$21 \overset{2221}{12} 122315\overset{2}{13}21712 \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$21 \overset{2221}{12} 1223151321712 \overset{2}{\quad} \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$21\overset{2}{12}1223151321712 \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: $21^{221}1212231513217123$

$$211212231513217123 \quad \Delta = -125, \quad n_+ = 1, \quad n_- = 20, \quad n_0 = 0, \quad |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$211212231513217123 \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$211212231513217123 \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$2112^{221\textcolor{red}{2}}2231513217123 \quad \Delta = -125, \, n_+ = 1, \, n_- = 20, \, n_0 = 0, \, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$211212231513217123^{\textcolor{red}{2}} \quad \Delta = -125, \quad n_+ = 1, \quad n_- = 20, \quad n_0 = 0, \quad |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$2112^{221}122315^{2}13217123 \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$211212231513217123 \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: $\overset{1}{1}7\overset{21}{1}231513221121222$

$${}^1_{171231513221121222}{}^{21\textcolor{red}{2}} \quad \Delta = -116, \, n_+ = 1, \, n_- = 20, \, n_0 = 0, \, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$${}^1_{17}1231513221121222 \quad \Delta = -116, \, n_+ = 1, \, n_- = 20, \, n_0 = 0, \, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$${}^1 171231513221121222 \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$\begin{array}{ll}
{}^1_{17}{}^2_{12}{}^{21}_{15}{}^{21}_{13}{}^{21}_{12}{}^{21}_{12}{}^{21}_{22} & \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
{}^{12}_{17}{}^{21}_{12}{}^{21}_{15}{}^{21}_{13}{}^{21}_{12}{}^{21}_{12}{}^{21}_{22} & \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
{}^1_{17}{}^2_{12}{}^{21}_{15}{}^{21}_{13}{}^{21}_{12}{}^{21}_{12}{}^{21}_{22} & \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\text{Base: } {}^{12231513221}_{3221}{}^{12}_{12}{}^{1223} & \\
{}^{812231513221}_{3221}{}^{12}_{12}{}^{1223} & \Delta = -162, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
{}^{1212231513221}_{3221}{}^{12}_{12}{}^{1223} & \Delta = -277, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -5 & -3 \\ 2 & -3 & 8 \end{pmatrix} \\
{}^{312231513221}_{3221}{}^{12}_{12}{}^{1223} \text{ [#=2]} & \Delta = -37, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
{}^{312231513221}_{3221}{}^{12}_{12}{}^{1223} & \Delta = -37, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
{}^{612231513221}_{3221}{}^{12}_{12}{}^{1223} & \Delta = -107, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
{}^{512231513221}_{3221}{}^{12}_{12}{}^{1223} & \Delta = -81, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
{}^{212231513221}_{3221}{}^{12}_{12}{}^{1223} & \Delta = -29, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
{}^{12231513221}_{3221}{}^{12}_{12}{}^{1223} & \Delta = -29, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
{}^{12231513221}_{3221}{}^{12}_{12}{}^{1223} & \Delta = -29, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^{12231513221}_{3221}{}^{12}_{12}{}^{1223} & \Delta = -29, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
{}^{212231513221}_{3221}{}^{12}_{12}{}^{1223} & \Delta = -29, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^{412231513221}_{3221}{}^{12}_{12}{}^{1223} & \Delta = -57, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
{}^{312231513221}_{3221}{}^{12}_{12}{}^{1223} \text{ [#=3]} & \Delta = -37, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 60 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
{}^{312231513221}_{3221}{}^{12}_{12}{}^{1223} \text{ [#=7]} & \Delta = -37, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 196 \quad \begin{pmatrix} 0 & 14 & 2 \\ 14 & 340 & 48 \\ 2 & 48 & 8 \end{pmatrix} \\
{}^{312231513221}_{3221}{}^{12}_{12}{}^{1223} \text{ [#=8]} & \Delta = -37, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 256 \quad \begin{pmatrix} 0 & 16 & 2 \\ 16 & 445 & 55 \\ 2 & 55 & 8 \end{pmatrix} \\
{}^{312231513221}_{3221}{}^{12}_{12}{}^{1223} \text{ [#=9]} & \Delta = -37, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 324 \quad \begin{pmatrix} 0 & 18 & 2 \\ 18 & 564 & 62 \\ 2 & 62 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
3221 \overset{3}{1}2231513221 \quad 12 \quad 1223 \text{ [#}=6] \quad \Delta = -37, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 144 \quad & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 249 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
3221 \overset{3}{1}2231513221 \quad 12 \quad 1223 \text{ [#}=5] \quad \Delta = -37, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 100 \quad & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 172 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
3221 \overset{3}{1}2231513221 \quad 12 \quad 1223 \text{ [#}=4] \quad \Delta = -37, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 109 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
3221 \overset{7}{1}2231513221 \quad 12 \quad 1223 \quad \Delta = -134, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\text{Base: } 221 \overset{12231513221}{12} \quad 12231 & \\
221 \overset{12231513221}{12} \quad 12231 \overset{3}{1} \text{ [#}=2] \quad \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
221 \overset{12231513221}{12} \quad 12231 \overset{2}{1} \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
221 \overset{3}{1}2231513221 \quad 12 \quad 12231 \text{ [#}=9] \quad \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
221 \overset{3}{1}2231513221 \quad 12 \quad 12231 \text{ [#}=8] \quad \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
221 \overset{3}{1}2231513221 \quad 12 \quad 12231 \text{ [#}=6] \quad \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
221 \overset{3}{1}2231513221 \quad 12 \quad 12231 \text{ [#}=7] \quad \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
221 \overset{3}{1}2231513221 \quad 12 \quad 12231 \text{ [#}=5] \quad \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
221 \overset{12231513221}{12} \quad 12231 \overset{3}{1} \text{ [#}=5] \quad \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
221 \overset{3}{1}2231513221 \quad 12 \quad 12231 \text{ [#}=4] \quad \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
221 \overset{12231513221}{12} \quad 12231 \overset{3}{1} \text{ [#}=4] \quad \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
221 \overset{3}{1}2231513221 \quad 12 \quad 12231 \text{ [#}=3] \quad \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
221 \overset{12231513221}{12} \quad 12231 \overset{3}{1} \text{ [#}=3] \quad \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
221 \overset{3}{1}2231513221 \quad 12 \quad 12231 \text{ [#}=2] \quad \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
221 \overset{12231513221}{12} \quad 12231 \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
221 \overset{12231513221}{12} \quad 12231 \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{2}{1}2231513221 \quad 221 \quad 12 \quad 12231 \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{1}{2}231513221 \quad 221 \quad 12 \quad 12231 \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{221}{221}121223151316131$

$$\overset{221}{221}121223151316131 \overset{3}{\# = 5} \quad \Delta = -104, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\overset{221}{221}121223151316131 \overset{3}{\# = 2} \quad \Delta = -104, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$\overset{221}{221}121223151316131 \overset{3}{\# = 6} \quad \Delta = -104, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\overset{221}{221}121223151316131 \overset{3}{\# = 7} \quad \Delta = -104, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\overset{221}{221}121223151316131 \overset{3}{\# = 8} \quad \Delta = -104, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\overset{221}{221}121223151316131 \overset{3}{\# = 4} \quad \Delta = -104, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\overset{221}{221}121223151316131 \overset{3}{\# = 9} \quad \Delta = -104, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$\overset{221}{221}121223151316131 \overset{3}{\# = 3} \quad \Delta = -104, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$\overset{2}{2}221 \quad \overset{221}{221}121223151316131 \quad \Delta = -96, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$\overset{221}{221} \overset{2}{2} \quad \overset{221}{221}121223151316131 \quad \Delta = -96, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$\overset{221}{221}121223151316131 \overset{2}{2} \quad \Delta = -96, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$\overset{221}{221}121223151316131 \overset{2}{2} \quad \Delta = -96, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$\overset{221}{221}121223151316131 \overset{2}{2} \quad \Delta = -96, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$\overset{221}{221}121223151316131 \overset{2}{2} \quad \Delta = -96, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: $\overset{3221}{712231513221}12122$

$$\overset{3221}{712231513221}12122 \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$\overset{2}{712231513221} \overset{3221}{12122} \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$\begin{aligned}
712231513221 \overset{3221}{12} \overset{2}{122} \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
712231513221 \overset{2}{12} \overset{3221}{122} \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
712231513221 \overset{2}{12} \overset{3221}{122} \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
712231513221 \overset{2}{12} \overset{3221}{122} \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

Base: $221 \overset{221}{12} 1223151322171$

$$\begin{aligned}
221 \overset{2}{12} 1223151322171 \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
221 \overset{221}{12} 12231513221 \overset{2}{71} \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
221 \overset{221}{12} 12231513221 \overset{2}{71} \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
221 \overset{221}{12} 1223151322171 \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
221 \overset{221}{12} 12231513221 \overset{2}{71} \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: $221 \overset{3221}{12} 122315132171$

$$\begin{aligned}
221 \overset{3221}{12} 122315132171 \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
221 \overset{2}{12} 122315132171 \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 55 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
3221 \overset{3221}{12} 122315132171 \quad \Delta = -134, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
221 \overset{3221}{12} 1223151321 \overset{2}{71} \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
221 \overset{3221}{12} 1223151321 \overset{2}{71} \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
221 \overset{3221}{12} 1223151321 \overset{2}{71} \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
221 \overset{3221}{12} 122315132171 \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 174 & 26 \\ 2 & 26 & 8 \end{pmatrix}
\end{aligned}$$

Base: $221 \overset{221}{12} 1223151321712$

$$\begin{aligned}
221 \overset{221}{12} 1223151321 \overset{2}{71} 2 \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
221 \overset{221}{12} 1223151321 \overset{2}{71} 2 \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
221\overset{2}{1}21223151321712 \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
221\overset{221}{1}21223151321\overset{2}{7}12 \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
221\overset{221}{1}2122315132171\overset{2}{2} \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
221\overset{221}{1}21223151321712 \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1\overset{1}{7}1231513221\overset{221}{1}2122$

$$\begin{aligned}
1\overset{1}{7}1231513221\overset{221}{1}2122 \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1\overset{1}{7}1231513221\overset{221}{1}2122 \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1\overset{1}{7}1231513221\overset{221}{1}2122 \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1\overset{1}{7}1231513221\overset{221}{1}2122 \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1\overset{1}{7}1231513221\overset{221}{1}2122 \quad \Delta = -116, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1\overset{1,21513221}{12}122315131$

$$\begin{aligned}
1\overset{1,21513221}{12}122315131 \quad \Delta = -71, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1\overset{1,21513221}{12}122315131\overset{3}{3} \text{ [#=7]} \quad \Delta = -79, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1\overset{1,21513221}{12}122315131\overset{3}{3} \text{ [#=6]} \quad \Delta = -79, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1\overset{1,21513221}{12}122315131\overset{3}{3} \text{ [#=8]} \quad \Delta = -79, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1\overset{1,21513221}{12}122315131\overset{3}{3} \text{ [#=5]} \quad \Delta = -79, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1\overset{1,21513221}{12}122315131\overset{3}{3} \text{ [#=9]} \quad \Delta = -79, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1\overset{1,21513221}{12}122315131 \quad \Delta = -71, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1\overset{1,21513221}{12}122315131 \quad \Delta = -71, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1\overset{1,21513221}{12}122315131\overset{2}{2} \quad \Delta = -71, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& 1^{1,21513221} 12^{2} 122315131 \quad \Delta = -71, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 1^{1,21513221} 12^{2} 122315131 \quad \Delta = -71, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 1^{1,21513221} 12^{2} 122315131 \textcolor{red}{3} [\#=4] \quad \Delta = -79, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& 1^{1,21513221} 12^{2} 122315131 \textcolor{red}{3} [\#=3] \quad \Delta = -79, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 1^{1,21513221} 12^{2} 122315131 \textcolor{red}{3} [\#=2] \quad \Delta = -79, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 1^{1,21513221} 12^{2} 122315131 \quad \Delta = -79, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \text{Base: } 1^{1,12231513221} 12^{2} 122313 \\
& 1^{1,\textcolor{red}{3}12231513221} 12^{2} 122313 [\#=6] \quad \Delta = -36, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
& 1^{1,\textcolor{red}{3}12231513221} 12^{2} 122313 [\#=9] \quad \Delta = -36, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix} \\
& 1^{1,\textcolor{red}{3}12231513221} 12^{2} 122313 [\#=8] \quad \Delta = -36, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
& 1^{1,\textcolor{red}{3}12231513221} 12^{2} 122313 [\#=7] \quad \Delta = -36, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
& 1^{1,12231513221\textcolor{red}{2}} 12^{2} 122313 \quad \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 1^{1,\textcolor{red}{3}12231513221} 12^{2} 122313 [\#=4] \quad \Delta = -36, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& 1^{1,12231513221} 12^{2} 122313 \quad \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 1^{1,12231513221} 12^{2} 122313 \quad \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 1^{1,12231513221} 12^{2} 122313 \quad \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 1^{1,12231513221} 12^{2} 122313 \quad \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 1^{1,12231513221} 12^{2} 122313 \textcolor{red}{3} \quad \Delta = -36, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 1^{1,\textcolor{red}{3}12231513221} 12^{2} 122313 [\#=5] \quad \Delta = -36, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
& 1^{1,\textcolor{red}{3}12231513221} 12^{2} 122313 [\#=3] \quad \Delta = -36, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix}
\end{aligned}$$

$$1 \overset{1,3}{12} \overset{12231513221}{122313} [\#=2] \quad \Delta = -36, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1 \overset{1,12231513221}{12} \overset{3}{122313} [\#=2] \quad \Delta = -36, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{1,5131513221}{12} 1223151$$

$$1 \overset{1,5131513221}{12} \overset{2}{1223151} \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{1,5131513221}{12} 1223151 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{1,5131513221}{12} 1223151 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{1,5131513221}{12} 1223151 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \overset{1,5131513221}{12} 1223151 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1513221 \overset{1,5131513221}{12} \overset{12}{12} \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{1,5131513221}{12} 1223151 \quad \Delta = -124, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{1,5131513221}{12} \overset{2}{1223151} \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{1,1513221}{12} 122315131$$

$$1 \overset{1,1513221}{12} 122315131 \overset{3}{3} [\#=6] \quad \Delta = -79, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$1 \overset{1,1513221}{12} 122315131 \overset{3}{3} [\#=7] \quad \Delta = -79, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$1 \overset{1,1513221}{12} \overset{2}{122315131} \quad \Delta = -71, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{1,1513221}{12} 122315131 \quad \Delta = -71, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{1,1513221}{12} 122315131 \quad \Delta = -71, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{1,1513221}{12} \overset{2}{122315131} \quad \Delta = -71, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{1,1513221}{12} \overset{2}{122315131} \quad \Delta = -71, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{1,1513221}{12} 122315131 \overset{3}{3} [\#=5] \quad \Delta = -79, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$\begin{array}{ll}
1^{1,1513221} \quad 1 \quad 12 \quad 122315131\mathbf{3} \quad [\#=4] & \Delta = -79, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1^{1,1513221} \quad 1 \quad 12 \quad 122315131\mathbf{3} \quad [\#=9] & \Delta = -79, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1^{1,1513221} \quad 1 \quad 12 \quad 122315131\mathbf{3} \quad [\#=8] & \Delta = -79, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1^{1,1513221} \quad 1 \quad 12 \quad 122315131\mathbf{3} \quad [\#=2] & \Delta = -79, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{1,1513221} \quad 1 \quad 12 \quad 122315131\mathbf{3} \quad [\#=3] & \Delta = -79, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\text{Base: } 1^{21,12231513221} \quad 12 \quad 12231 & \\
1^{21,\mathbf{3}12231513221} \quad 1 \quad 12 \quad 12231 \quad [\#=4] & \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1^{21,12231513221} \quad 1 \quad 12 \quad 12231\mathbf{3} \quad [\#=4] & \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{21,12231513221} \quad 1 \quad 12 \quad 12231\mathbf{3} \quad [\#=5] & \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1^{21,12231513221} \quad 1 \quad 12 \quad 12231 & \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1^{21,12231513221} \quad 1 \quad 12 \quad 12231 & \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1^{21,\mathbf{3}12231513221} \quad 1 \quad 12 \quad 12231 \quad [\#=2] & \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{21,12231513221} \quad 1 \quad 12 \quad 12231\mathbf{3} \quad [\#=3] & \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1^{21,\mathbf{3}12231513221} \quad 1 \quad 12 \quad 12231 \quad [\#=3] & \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1^{21,\mathbf{3}12231513221} \quad 1 \quad 12 \quad 12231 \quad [\#=5] & \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
1^{21,\mathbf{3}12231513221} \quad 1 \quad 12 \quad 12231 \quad [\#=6] & \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1^{21,12231513221} \quad 1 \quad 12 \quad 12231 & \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1^{21,12231513221} \quad 1 \quad 12 \quad 12231 & \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{21,12231513221\mathbf{2}} \quad 1 \quad 12 \quad 12231 & \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1^{21,\mathbf{3}12231513221} \quad 1 \quad 12 \quad 12231 \quad [\#=7] & \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
1^{21, \textcolor{red}{3}12231513221}_{12} 12231 \text{ [#}=8] \quad \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
1^{21, \textcolor{red}{3}12231513221}_{12} 12231 \text{ [#}=9] \quad \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix} \\
1^{21, 12231513221}_{12} 12231\textcolor{red}{3} \text{ [#}=2] \quad \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: 122314141414141413²2

$$2^{\textcolor{red}{8}12231414141414141}_{3} 2 \quad \Delta = -14, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 13151316131613151321

$$\begin{aligned}
12315131613161315131\textcolor{red}{3} \text{ [#}=2] \quad \Delta = -2, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
12315131613161315131\textcolor{red}{3} \text{ [#}=6] \quad \Delta = -2, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
13151316131613151321\textcolor{red}{3} \text{ [#}=5] \quad \Delta = -2, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
13151316131613151321\textcolor{red}{3} \text{ [#}=6] \quad \Delta = -2, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
12315131613161315131\textcolor{red}{3} \text{ [#}=5] \quad \Delta = -2, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
12315131613161315131\textcolor{red}{3} \text{ [#}=7] \quad \Delta = -2, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
13151316131613151321\textcolor{red}{3} \text{ [#}=4] \quad \Delta = -2, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
13151316131613151321\textcolor{red}{3} \text{ [#}=2] \quad \Delta = -2, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
12315131613161315131\textcolor{red}{3} \text{ [#}=8] \quad \Delta = -2, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
12315131613161315131\textcolor{red}{3} \text{ [#}=4] \quad \Delta = -2, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
13151316131613151321\textcolor{red}{3} \text{ [#}=7] \quad \Delta = -2, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
13151316131613151321\textcolor{red}{3} \text{ [#}=8] \quad \Delta = -2, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
12315131613161315131\textcolor{red}{3} \text{ [#}=3] \quad \Delta = -2, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
12315131613161315131\textcolor{red}{3} \text{ [#}=9] \quad \Delta = -2, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
13151316131613151321\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -2, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
13151316131613151321\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -2, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\textbf{Base: } 22161316131613151321 \\
22161316131613151321\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
22161\textcolor{red}{2}316131613151321 \quad \Delta = -12, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
221613161\textcolor{red}{2}31613151321 \quad \Delta = -12, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
22161316131613151321\textcolor{red}{3} \quad \Delta = -21, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
22161\textcolor{red}{2}316131613151321 \quad \Delta = -12, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
221613161\textcolor{red}{2}31613151321 \quad \Delta = -12, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
22161316131613151\textcolor{red}{2}321 \quad \Delta = -12, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
22161316131613151321\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
22161316131613151321\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
22161316131613151321\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
22161316131613151321\textcolor{red}{3} \text{ [#=6]} \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
22161316131613151321 \quad \Delta = -12, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
22161316131613151321 \quad \Delta = -12, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
22161316131613151\textcolor{red}{2}321 \quad \Delta = -12, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
22161316131613151321\textcolor{red}{3} \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 27 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
22161316131613151321\textcolor{red}{3} \text{ [#=7]} \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
22161316131613151321\textcolor{red}{3} \text{ [#=8]} \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
22161316131613151321\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
\textbf{Base: } 13216131613161315131 & \\
132161316131613151\textcolor{red}{3}^2 \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
132161316131613151\textcolor{red}{3}^2 \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
13151316131613161231\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
132161316131613151\textcolor{red}{3}^2 \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
132161316131613151\textcolor{red}{3}^2 \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
132161316131613151\textcolor{red}{3}^2 \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
13216131613161315131\textcolor{red}{3} \text{ [#=7]} \quad \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
13216131613161315131\textcolor{red}{3} \text{ [#=8]} \quad \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
13216131613161315131\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
13151316131613161231\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
132161316131613151\textcolor{red}{3}^2 \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
132161316131613151\textcolor{red}{3}^2 \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
132161316131613151\textcolor{red}{3}^2 \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
13216131613161315131\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
13151316131613161231\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
13216131613161315131\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
13151316131613161231\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
13216131613161315131\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
13216131613161315131\overset{2}{3} \text{ [#=5]} \quad \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
13216131613161315131\overset{2}{3} \text{ [#=6]} \quad \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\text{Base: } 12321613161316131512 & \\
21513161316131612321\overset{2}{3} \text{ [#=4]} \quad \Delta = -22, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
12321613161316131512 \quad \Delta = -14, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21513161316131612321\overset{2}{3} \text{ [#=6]} \quad \Delta = -22, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
21513161316131612321\overset{2}{3} \text{ [#=9]} \quad \Delta = -22, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix} \\
21513161316131612321\overset{2}{3} \text{ [#=8]} \quad \Delta = -22, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
21513161316131612321\overset{2}{3} \text{ [#=7]} \quad \Delta = -22, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
21513161316131612321\overset{2}{3} \text{ [#=5]} \quad \Delta = -22, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
21513161316131612321\overset{2}{3} \text{ [#=3]} \quad \Delta = -22, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
12321613161316131512 \quad \Delta = -22, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
21513161316131612321\overset{2}{3} \text{ [#=2]} \quad \Delta = -22, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
12321613161316131512 \quad \Delta = -14, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
12321613161316131512 \quad \Delta = -14, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
12321613161316131512 \quad \Delta = -14, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
12321613161316131512 \quad \Delta = -14, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
12321613161316131512 \quad \Delta = -14, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
12321613161316131512 \quad \Delta = -14, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
12321613161316131512 \quad \Delta = -14, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

$$\text{Base: } 1232161316131613151\overset{1}{1}$$

$$\begin{aligned}
1232161316131613151^1 \quad \Delta = -14, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{312321613161316131} 5^1 \quad 1 \text{ [#=8]} \quad \Delta = -22, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
1232161316131613151^1 \quad \Delta = -14, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1232161316131613151^1 \quad \Delta = -14, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1232161316131613151^1 \quad \Delta = -14, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1232161316131613151^1 \quad \Delta = -14, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1232161316131613151^{21} \quad \Delta = -14, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1232161316131613151^1 \quad \Delta = -14, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1^{312321613161316131} 5^1 \quad 1 \text{ [#=2]} \quad \Delta = -22, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1^{312321613161316131} 5^1 \quad 1 \text{ [#=9]} \quad \Delta = -22, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix} \\
1^{312321613161316131} 5^1 \quad 1 \text{ [#=7]} \quad \Delta = -22, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
1^{312321613161316131} 5^1 \quad 1 \text{ [#=6]} \quad \Delta = -22, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1^{312321613161316131} 5^1 \quad 1 \text{ [#=5]} \quad \Delta = -22, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
1^{312321613161316131} 5^1 \quad 1 \text{ [#=4]} \quad \Delta = -22, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1^{312321613161316131} 5^1 \quad 1 \text{ [#=3]} \quad \Delta = -22, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix}
\end{aligned}$$

Base: 61315131613161316131

$$\begin{aligned}
613151316131613161313 \text{ [#=9]} \quad \Delta = -105, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
61315131613161316131^2 \quad \Delta = -97, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
613151316131613161313 \text{ [#=8]} \quad \Delta = -105, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
613151316131613161313 \text{ [#=7]} \quad \Delta = -105, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
61315131613161316131\overset{\textcolor{red}{3}}{3} \text{ [#=6]} \quad \Delta = -105, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
61315131613161316131\overset{\textcolor{red}{3}}{3} \text{ [#=4]} \quad \Delta = -105, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
61315131613161316131\overset{\textcolor{red}{3}}{3} \text{ [#=3]} \quad \Delta = -105, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
61315131613161316131\overset{\textcolor{red}{3}}{3} \text{ [#=2]} \quad \Delta = -105, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
61315131613161316131\overset{\textcolor{red}{3}}{3} \text{ [#=5]} \quad \Delta = -105, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
61315131613161\overset{\textcolor{red}{2}}{3}16131 \quad \Delta = -97, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
61315131613161\overset{\textcolor{red}{2}}{3}16131 \quad \Delta = -97, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
6131513161\overset{\textcolor{red}{2}}{3}161316131 \quad \Delta = -97, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
6131513161\overset{\textcolor{red}{2}}{3}161316131 \quad \Delta = -97, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 100 \quad & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 168 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
613151\overset{\textcolor{red}{2}}{3}1613161316131 \quad \Delta = -97, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 \quad & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 207 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
613151\overset{\textcolor{red}{2}}{3}1613161316131 \quad \Delta = -97, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 142 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
613151\overset{\textcolor{red}{2}}{3}1613161316131 \quad \Delta = -97, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
61315131613161\overset{\textcolor{red}{2}}{3}16131 \quad \Delta = -97, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\textbf{Base: } 31316131613161315131 & \\
31316131613161\overset{\textcolor{red}{2}}{3}161315131 \quad \Delta = -27, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
313161316131613151\overset{\textcolor{red}{4}}{3}1 \quad \Delta = -55, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 29 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
313161316131613151\overset{\textcolor{red}{3}}{3}1 \text{ [#=3]} \quad \Delta = -35, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 294 & 38 \\ 2 & 38 & 8 \end{pmatrix} \\
313161\overset{\textcolor{red}{2}}{3}1613161315131 \quad \Delta = -27, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
313161\overset{\textcolor{red}{2}}{3}1613161315131 \quad \Delta = -27, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
313161\overset{\textcolor{red}{2}}{3}1613161315131 \quad \Delta = -27, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{ll}
31316131613161315131^2 & \Delta = -27, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
31316131613161315131^2 & \Delta = -27, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
31316131613161315131^2 & \Delta = -27, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
31316131613161315131^2 & \Delta = -27, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
31316131613161315131^2 & \Delta = -27, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
313161316131613151313^{\#3} & \Delta = -35, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
313161316131613151313^{\#9} & \Delta = -35, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
313161316131613151313^{\#8} & \Delta = -35, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
313161316131613151313^{\#7} & \Delta = -35, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
313161316131613151313^{\#6} & \Delta = -35, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
313161316131613151313^{\#4} & \Delta = -35, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
313161316131613151313^{\#5} & \Delta = -35, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
313161316131613151313^{\#2} & \Delta = -35, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
31316131613161315131^3 & \Delta = -35, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 30 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
31316131613161315131^3_{\#2} & \Delta = -35, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 129 & 25 \\ 2 & 25 & 8 \end{pmatrix} \\
\text{Base: } 14131613161316131512 & \\
14131613161316131512^2 & \Delta = -39, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
14131613161316131512^2 & \Delta = -39, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
215131613161316131413^{\#3} & \Delta = -47, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
215131613161316131413^{\#2} & \Delta = -47, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{ll}
14131613161316131512^{\color{red}3} & \Delta = -47, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
14131613161316131512^{\color{red}3} & \Delta = -47, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 30 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
14131613161316131512^{\color{red}2} & \Delta = -39, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
14131613161316131512^{\color{red}2} & \Delta = -39, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
14131613161316131512^{\color{red}2} & \Delta = -39, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
14131613161316131512^{\color{red}2} & \Delta = -39, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
14131613161316131512^{\color{red}2} & \Delta = -39, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
14131613161316131512^{\color{red}2} & \Delta = -39, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
14131613161316131512^{\color{red}2} & \Delta = -39, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\textbf{Base: } 1413161316131613151^{\color{red}1} & \\
1413161316131613151^{\color{red}2} & \Delta = -39, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1413161316131613151^{\color{red}21} & \Delta = -39, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1413161316131613151^{\color{red}2} & \Delta = -39, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1413161316131613151^{\color{red}2} & \Delta = -39, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1^{\color{red}3} 14131613161316131^{\color{red}1} & \Delta = -47, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1413161316131613151^{\color{red}3} & \Delta = -47, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 30 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1^{\color{red}3} 14131613161316131^{\color{red}1} & \Delta = -47, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1413161316131613151^{\color{red}2} & \Delta = -39, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1413161316131613151^{\color{red}2} & \Delta = -39, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1413161316131613151^{\color{red}2} & \Delta = -39, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{array}$$

$$1413161316131613151^1 \quad \Delta = -39, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 22161316131613161231

$$22161316131613161231^2 \quad \Delta = -38, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$22161316131613161231^3 \quad \Delta = -46, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 36 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$22161316131613161231^2 \quad \Delta = -38, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$22161316131613161231^2 \quad \Delta = -38, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$22161316131613161231^2 \quad \Delta = -38, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$22161316131613161231^2 \quad \Delta = -38, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 37 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$221613161316131612313^{\#2} \quad \Delta = -46, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$22161316131613161231^2 \quad \Delta = -38, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$22161316131613161231^2 \quad \Delta = -38, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$22161316131613161231^3 \quad \Delta = -47, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 15 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$221613161316131612313^{\#5} \quad \Delta = -46, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$221613161316131612313^{\#3} \quad \Delta = -46, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$221613161316131612313^{\#4} \quad \Delta = -46, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$22161316131613161231^2 \quad \Delta = -38, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

Base: 22161316131613161313

$$22161316131613161313^2 \quad \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$22161316131613161313^2 \quad \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$22161316131613161313^2 \quad \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$\begin{aligned}
22161316131613161313 \quad \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 37 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
22161316131613161313^2 \quad \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
22161316131613161313^2 \quad \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
22161316131613161313^2 \quad \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
22161316131613161313^3 \quad \Delta = -53, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
22161316131613161313^2 \quad \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
22161316131613161313^2 \quad \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
22161316131613161313^3 \quad \Delta = -53, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 36 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
22161316131613161313^3_{[\#=3]} \quad \Delta = -53, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
22161316131613161313^4 \quad \Delta = -73, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 9 & 5 \\ 5 & 9 \end{pmatrix} \\
22161316131613161313^3_{[\#=2]} \quad \Delta = -53, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
22161316131613161313^3 \quad \Delta = -54, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 15 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

Base: $16^{21}1316131613151321$

$$\begin{aligned}
16^{21}1316131613151321^2 \quad \Delta = -12, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
16^{21}13161316131513213_{[\#=6]} \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
16^{21}13161316131513213_{[\#=2]} \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
16^{21}1316131613151321 \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
16^{21}13161316131513213_{[\#=3]} \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
16^{21}13161316131513213_{[\#=9]} \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
16^{21}13161316131513213_{[\#=5]} \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{21}{16}1316131613151321\overset{2}{3}_{[\# = 4]} \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{21}{16}1316131613151321 \quad \Delta = -12, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{21}{16}1316131613151321 \quad \Delta = -12, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{21}{16}1316131613151321 \quad \Delta = -12, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{21}{16}1316131613151321 \quad \Delta = -12, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{21}{16}1316131613151321 \quad \Delta = -12, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{21}{16}1316131613151321 \quad \Delta = -12, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{21}{16}1316131613151321\overset{2}{3}_{[\# = 8]} \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& \overset{21}{16}1316131613151321\overset{2}{3}_{[\# = 7]} \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
& \overset{21}{16}1316131613151321 \quad \Delta = -12, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \text{Base: } \overset{21}{16}1316131613161231 \\
& \overset{21}{16}1316131613161231 \quad \Delta = -38, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{21}{16}1316131613161231 \quad \Delta = -38, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{21}{16}1316131613161231 \quad \Delta = -38, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{21}{16}1316131613161231 \quad \Delta = -38, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& \overset{21}{16}1316131613161231 \quad \Delta = -38, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{21}{16}1316131613161231 \quad \Delta = -38, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{21}{16}1316131613161231\overset{2}{3}_{[\# = 2]} \quad \Delta = -46, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
& \overset{21}{16}1316131613161231\overset{2}{3}_{[\# = 4]} \quad \Delta = -46, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{21}{16}1316131613161231\overset{2}{3}_{[\# = 5]} \quad \Delta = -46, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{l}
^{21} 1 \bar{6} 1316131613161231 \bar{3} \text{ [#=3]} \quad \Delta = -46, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\\
^{21} 1 \bar{6} 1316131613161231 \quad \Delta = -46, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 15 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\\
^{21} 1 \bar{6} 1316131613161231 \quad \Delta = -38, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\\
^{21} 1 \bar{6} 1316131613161231 \quad \Delta = -38, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\\
\text{Base: } ^{21} 1 \bar{6} 1316131613161313 \\
\\
^{21} 1 \bar{6} 1316131613161313 \bar{3} \quad \Delta = -53, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\\
^{21} 1 \bar{6} 1316131613161313 \quad \Delta = -53, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 15 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\\
^{21} 1 \bar{6} 1316131613161313 \bar{4} \quad \Delta = -73, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 9 & 5 \\ 5 & 9 \end{pmatrix} \\
\\
^{21} 1 \bar{6} 1316131613161313 \bar{2} \quad \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\\
^{21} 1 \bar{6} 1316131613161313 \bar{2} \quad \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
\\
^{21} 1 \bar{6} 1316131613161313 \bar{3} \text{ [#=2]} \quad \Delta = -53, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\\
^{21} 1 \bar{6} 1316131613161313 \bar{3} \text{ [#=3]} \quad \Delta = -53, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\\
^{21} 1 \bar{6} 1316131613161313 \bar{2} \quad \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\\
^{21} 1 \bar{6} 1316131613161313 \bar{2} \quad \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\\
^{21} 1 \bar{6} 1316131613161313 \bar{2} \quad \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\\
^{21} 1 \bar{6} 1316131613161313 \bar{2} \quad \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\\
^{21} 1 \bar{6} 1316131613161313 \bar{2} \quad \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\\
^{21} 1 \bar{6} 1316131613161313 \quad \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\\
^{21} 1 \bar{6} 1316131613161313 \bar{2} \quad \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\\
\text{Base: } ^{131} 1 \bar{6} 131613161315131
\end{array}$$

$1 \overset{131}{6} 131613161315131\overset{3}{\color{red}}$ [#=2]	$\Delta = -27, n_+ = 1, n_- = 20, n_0 = 0, \det = 4$	$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$
$1 \overset{131}{6} 131613161\overset{2}{\color{red}}1315131$	$\Delta = -19, n_+ = 1, n_- = 20, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$
$1 \overset{131}{6} 131\overset{2}{\color{red}}613161315131$	$\Delta = -19, n_+ = 1, n_- = 20, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$
$1 \overset{131}{6} 131613161315131\overset{2}{\color{red}}$	$\Delta = -19, n_+ = 1, n_- = 20, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$
$1 \overset{3131}{\color{red}}6 131613161315131$ [#=4]	$\Delta = -27, n_+ = 1, n_- = 20, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix}$
$1 \overset{131}{6} 131613161315131\overset{3}{\color{red}}$ [#=3]	$\Delta = -27, n_+ = 1, n_- = 20, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$
$1 \overset{131}{6} 13161316131\overset{2}{\color{red}}5131$	$\Delta = -19, n_+ = 1, n_- = 20, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$
$1 \overset{131}{6} 131613161315131\overset{2}{\color{red}}$	$\Delta = -19, n_+ = 1, n_- = 20, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$
$1 \overset{131}{6} 1316131613151\overset{2}{\color{red}}31$	$\Delta = -19, n_+ = 1, n_- = 20, n_0 = 0, \det = 4$	$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$
$1 \overset{131}{6} 131613161315131$	$\Delta = -19, n_+ = 1, n_- = 20, n_0 = 0, \det = 4$	$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$
$1 \overset{131}{6} 131\overset{2}{\color{red}}613161315131$	$\Delta = -19, n_+ = 1, n_- = 20, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$
$1 \overset{131}{6} 131613161315131\overset{3}{\color{red}}$ [#=5]	$\Delta = -27, n_+ = 1, n_- = 20, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$
$1 \overset{3131}{\color{red}}6 131613161315131$ [#=2]	$\Delta = -27, n_+ = 1, n_- = 20, n_0 = 0, \det = 4$	$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix}$
$1 \overset{3131}{\color{red}}6 131613161315131$ [#=3]	$\Delta = -27, n_+ = 1, n_- = 20, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$
$1 \overset{131}{6} 131613161315131\overset{3}{\color{red}}$ [#=6]	$\Delta = -27, n_+ = 1, n_- = 20, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$
$1 \overset{3131}{\color{red}}6 131613161315131$ [#=7]	$\Delta = -27, n_+ = 1, n_- = 20, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix}$
$1 \overset{131}{6} 131613161315131\overset{3}{\color{red}}$ [#=7]	$\Delta = -27, n_+ = 1, n_- = 20, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$
$1 \overset{3131}{\color{red}}6 131613161315131$ [#=8]	$\Delta = -27, n_+ = 1, n_- = 20, n_0 = 0, \det = 64$	$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix}$
$1 \overset{131}{6} 131613161315131\overset{3}{\color{red}}$ [#=8]	$\Delta = -27, n_+ = 1, n_- = 20, n_0 = 0, \det = 64$	$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}$

$$\begin{array}{ll}
\begin{array}{l} \textcolor{red}{3}131 \\ 1 \ 6 \ 131613161315131 \end{array} \begin{array}{l} [\# = 9] \\ \Delta = -27, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \end{array} & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
\begin{array}{l} \textcolor{red}{3}131 \\ 1 \ 6 \ 131613161315131 \end{array} \begin{array}{l} [\# = 5] \\ \Delta = -27, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \end{array} & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
\begin{array}{l} 131 \\ 1 \ 6 \ 131613161315131 \end{array} \begin{array}{l} \textcolor{red}{3} [\# = 9] \\ \Delta = -27, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \end{array} & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
\begin{array}{l} 131 \\ 1 \ 6 \ 131613161315131 \end{array} \begin{array}{l} \textcolor{red}{3} [\# = 4] \\ \Delta = -27, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \end{array} & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\begin{array}{l} \textcolor{red}{3}131 \\ 1 \ 6 \ 131613161315131 \end{array} \begin{array}{l} [\# = 6] \\ \Delta = -27, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \end{array} & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
\textbf{Base: } \begin{array}{l} 131 \\ 1 \ 6 \ 131613161316122 \end{array} & \\
\begin{array}{l} \textcolor{red}{3}131 \\ 1 \ 6 \ 131613161316122 \end{array} \begin{array}{l} [\# = 4] \\ \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \end{array} & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
\begin{array}{l} 131 \textcolor{red}{2} \\ 1 \ 6 \ 131613161316122 \end{array} \Delta = -37, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
\begin{array}{l} 131 \textcolor{red}{2} \\ 1 \ 6 \ 131613161316122 \end{array} \Delta = -37, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\begin{array}{l} 131 \textcolor{red}{2} \\ 1 \ 6 \ 131613161316122 \end{array} \Delta = -37, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\begin{array}{l} 131 \textcolor{red}{2} \\ 1 \ 6 \ 131613161316122 \end{array} \Delta = -37, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\begin{array}{l} 131 \textcolor{red}{2} \\ 1 \ 6 \ 131613161316122 \end{array} \Delta = -37, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\begin{array}{l} 131 \textcolor{red}{2} \\ 1 \ 6 \ 131613161316122 \end{array} \Delta = -37, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\begin{array}{l} 131 \textcolor{red}{2} \\ 1 \ 6 \ 131613161316122 \end{array} \Delta = -37, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\begin{array}{l} 131 \textcolor{red}{2} \\ 1 \ 6 \ 131613161316122 \end{array} \Delta = -37, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\begin{array}{l} 131 \textcolor{red}{3} \\ 1 \ 6 \ 131613161316122 \end{array} \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\begin{array}{l} \textcolor{red}{3}131 \\ 1 \ 6 \ 131613161316122 \end{array} \begin{array}{l} [\# = 2] \\ \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \end{array} & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\begin{array}{l} \textcolor{red}{3}131 \\ 1 \ 6 \ 131613161316122 \end{array} \begin{array}{l} [\# = 8] \\ \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \end{array} & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
\begin{array}{l} \textcolor{red}{3}131 \\ 1 \ 6 \ 131613161316122 \end{array} \begin{array}{l} [\# = 7] \\ \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \end{array} & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
\begin{array}{l} \textcolor{red}{3}131 \\ 1 \ 6 \ 131613161316122 \end{array} \begin{array}{l} [\# = 9] \\ \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \end{array} & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
& \overset{3131}{1 \ 6 \ 131613161316122} \text{ [#=5]} \quad \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 6 \ 131613161316122} \text{ [#=3]} \quad \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 6 \ 131613161316122} \text{ [#=6]} \quad \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 131613161316122} \overset{3}{\Delta} = -46, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
& \text{Base: } \overset{21}{1 \ 6 \ 131613161316131} \\
& \overset{21}{1 \ 6 \ 131613161316131} \overset{1}{3} \text{ [#=4]} \quad \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{21}{1 \ 6 \ 131613161316131} \quad \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 15 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{21}{1 \ 6 \ 131613161316131} \overset{1}{3} \text{ [#=6]} \quad \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{21}{1 \ 6 \ 131613161316131} \overset{1}{3} \text{ [#=9]} \quad \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
& \overset{21}{1 \ 6 \ 131613161316131} \overset{1}{3} \text{ [#=8]} \quad \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& \overset{21}{1 \ 6 \ 131613161316131} \overset{1}{3} \text{ [#=7]} \quad \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
& \overset{21}{1 \ 6 \ 131613161316131} \overset{1}{3} \text{ [#=2]} \quad \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{21}{1 \ 6 \ 131613161316131} \overset{1}{2} \quad \Delta = -37, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{21}{1 \ 6 \ 131613161316131} \overset{1}{2} \quad \Delta = -37, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& \overset{21}{1 \ 6 \ 131613161316131} \overset{1}{3} \text{ [#=5]} \quad \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{21}{1 \ 6 \ 131613161316131} \overset{1}{2} \quad \Delta = -37, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{21}{1 \ 6 \ 131613161316131} \overset{2}{\Delta} = -37, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{21}{1 \ 6 \ 131613161316131} \quad \Delta = -37, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{21}{1 \ 6 \ 131613161316131} \overset{2}{\Delta} = -37, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{21}{1 \ 6 \ 131613161316131} \overset{2}{\Delta} = -37, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{l}
\begin{array}{l} \text{21} \\ 16131613161316131 \end{array} \quad \Delta = -37, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\begin{array}{l} \text{21} \\ 16131613161316131 \end{array} \text{3} \text{ [#=3]} \quad \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\text{Base: } \begin{array}{l} 1 \\ 6131613161316131612 \end{array} \\
\begin{array}{l} 1 \\ 6131613161316131612 \end{array} \quad \Delta = -115, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\begin{array}{l} 1 \\ 6131613161316131612 \end{array} \quad \Delta = -115, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\begin{array}{l} 1 \\ 6131613161316131612 \end{array} \quad \Delta = -115, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\begin{array}{l} \text{21} \\ 6131613161316131612 \end{array} \quad \Delta = -115, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\begin{array}{l} \text{2} \\ 6131613161316131612 \end{array} \quad \Delta = -115, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\begin{array}{l} \text{21} \\ 6131613161316131612 \end{array} \quad \Delta = -115, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\begin{array}{l} \text{12} \\ 6131613161316131612 \end{array} \quad \Delta = -115, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
\begin{array}{l} 1 \\ 6131613161316131612 \end{array} \quad \Delta = -115, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\begin{array}{l} \text{31} \\ 6131613161316131612 \end{array} \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\begin{array}{l} 1 \\ 6131613161316131612 \end{array} \quad \Delta = -115, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\begin{array}{l} 1 \\ 6131613161316131612 \end{array} \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\begin{array}{l} 1 \\ 6131613161316131612 \end{array} \quad \Delta = -115, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\text{Base: } \begin{array}{l} 1 \\ 613161316131613161 \end{array} \\
\begin{array}{l} \text{21} \\ 613161316131613161 \end{array} \quad \Delta = -115, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\begin{array}{l} \text{2} \\ 613161316131613161 \end{array} \quad \Delta = -115, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\begin{array}{l} \text{21} \\ 613161316131613161 \end{array} \quad \Delta = -115, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\begin{array}{l} 1 \\ 613161316131613161 \end{array} \quad \Delta = -115, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{ll}
6131\overset{1}{6}13161\overset{2}{3}16131\overset{1}{6}1 & \Delta = -115, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
6131\overset{1}{6}13161\overset{2}{3}16131\overset{1}{6}1 & \Delta = -115, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
6131\overset{1}{6}13161\overset{2}{3}16131\overset{1}{6}1 & \Delta = -115, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
6131\overset{1}{6}13161\overset{2}{3}16131\overset{1}{6}1 & \Delta = -115, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
6131\overset{1}{6}13161\overset{2}{3}16131\overset{1}{6}1 & \Delta = -115, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
6131\overset{3}{6}13161\overset{1}{3}16131\overset{1}{6}1 & \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{array}$$

Base: $31\overset{31}{6}131613161315131$

$$\begin{array}{ll}
31\overset{31}{6}131613161315131\overset{3}{3} \text{ [#=2]} & \Delta = -35, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
31\overset{31}{6}131613161315131\overset{3}{3} \text{ [#=7]} & \Delta = -35, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 196 \quad \begin{pmatrix} 0 & 14 & 2 \\ 14 & 340 & 48 \\ 2 & 48 & 8 \end{pmatrix} \\
31\overset{31}{6}131613161315131\overset{3}{3} & \Delta = -35, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
31\overset{31}{6}131613161315131\overset{5}{5} & \Delta = -79, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
31\overset{31}{6}131613161315131\overset{3}{3} \text{ [#=8]} & \Delta = -35, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 256 \quad \begin{pmatrix} 0 & 16 & 2 \\ 16 & 445 & 55 \\ 2 & 55 & 8 \end{pmatrix} \\
31\overset{31}{6}131613161315131\overset{3}{3} \text{ [#=3]} & \Delta = -35, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 60 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
31\overset{31}{6}131613161315131\overset{3}{3} \text{ [#=6]} & \Delta = -35, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 249 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
31\overset{31}{6}131613161315131\overset{3}{3} \text{ [#=5]} & \Delta = -35, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 172 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
31\overset{31}{6}1316131613151\overset{2}{3}1 & \Delta = -28, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
31\overset{31}{6}131613161315131\overset{3}{3} \text{ [#=4]} & \Delta = -35, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 109 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
31\overset{31}{6}131613161315131\overset{3}{3} \text{ [#=9]} & \Delta = -35, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 324 \quad \begin{pmatrix} 0 & 18 & 2 \\ 18 & 564 & 62 \\ 2 & 62 & 8 \end{pmatrix} \\
31\overset{31}{6}131613161315131\overset{4}{4} & \Delta = -55, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
31\overset{31}{6}1316131613151\overset{2}{3}1 & \Delta = -27, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{array}$$

$^{31}316131613161315131$	$\Delta = -27, n_+ = 1, n_- = 20, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 37 & 15 \\ 2 & 15 & 8 \end{pmatrix}$
$^{31}316131613161315131$	$\Delta = -27, n_+ = 1, n_- = 20, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix}$
$^{31}316131613161315131$	$\Delta = -27, n_+ = 1, n_- = 20, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
$^{31}316131613161315131$	$\Delta = -27, n_+ = 1, n_- = 20, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 20 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
$^{231}316131613161315131$	$\Delta = -27, n_+ = 1, n_- = 20, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$
$^{31}3161316131613151312$	$\Delta = -27, n_+ = 1, n_- = 20, n_0 = 0, \det = 4$	$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 7 \\ 2 & 7 & 8 \end{pmatrix}$
$^{31}316131613161315131$	$\Delta = -27, n_+ = 1, n_- = 20, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$
$^{31}3161316131613151316$	$\Delta = -105, n_+ = 1, n_- = 20, n_0 = 0, \det = 4$	$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$
$^{312}316131613161315131$	$\Delta = -27, n_+ = 1, n_- = 20, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$
Base: $^{31}316131613161316122$		
$^{31}316131613161316122$	$\Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$
$^{31}3161316131613161223$	$\Delta = -54, n_+ = 1, n_- = 20, n_0 = 0, \det = 4$	$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$
$^{31}316131613161316122$	$\Delta = -54, n_+ = 1, n_- = 20, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 11 \\ 2 & 11 & 8 \end{pmatrix}$
$^{31}316131613161316122$	$\Delta = -53, n_+ = 1, n_- = 20, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 45 & 17 \\ 2 & 17 & 8 \end{pmatrix}$
$^{31}316131613161316122$	$\Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix}$
$^{31}316131613161316122$	$\Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
$^{312}316131613161316122$	$\Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$
$^{31}316131613161316122$	$\Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 37 & 15 \\ 2 & 15 & 8 \end{pmatrix}$
$^{31}316131613161316122$	$\Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 16 \\ 2 & 16 & 8 \end{pmatrix}$
$^{231}316131613161316122$	$\Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$

$$^{31}316131613161316122^{\color{red}2} \quad \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 46 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

Base: $^{31}3161316131613161^{21}$

$$^{\color{red}231}3161316131613161^{21} \quad \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$^{31}3161316131613161^{21} \quad \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 37 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$^{31}3161316131613161^{21} \quad \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$^{31}3161316131613161^{21} \quad \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$^{31}3161316131613161^{21} \quad \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$^{31}3161316131613161^{\color{red}2} \quad \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$^{31}3161316131613161^{21} \quad \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 20 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$^{31}3161316131613161^{\color{red}321} \quad \Delta = -54, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$^{31\color{red}2}3161316131613161^{21} \quad \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$^{31}3161316131613161^{21} \quad \Delta = -53, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$^{31}3161316131613161^{\color{red}13} \quad \Delta = -53, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$^{31}3161316131613161^{\color{red}221} \quad \Delta = -45, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

Base: 71231513161316131612

$$71231513161316131612^{\color{red}2} \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$71231513161316131612^{\color{red}3} \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$71231513161316131612^{\color{red}2} \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 168 & 26 \\ 2 & 26 & 8 \end{pmatrix}$$

$$71231513161316131612^{\color{red}2} \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$71231513161316131612^{\color{red}2} \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$71231513161316131612^{\color{red}2} \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 207 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

$$\begin{aligned}
71231513161316131612 \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 142 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
71231513161316131612 \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
71231513161316131612 \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 89 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
71231513161316131612 \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: 7123151316131613161¹

$$\begin{aligned}
7123151316131613161 \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 168 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
7123151316131613161 \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
7123151316131613161 \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 89 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
7123151316131613161 \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
7123151316131613161 \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 142 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
7123151316131613161 \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
7123151316131613161 \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
7123151316131613161 \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 207 & 29 \\ 2 & 29 & 8 \end{pmatrix}
\end{aligned}$$

Base: 71232161316131613151

$$\begin{aligned}
71232161316131613151 \quad \Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
71232161316131613151 \quad \Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
71232161316131613151 \quad \Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
71232161316131613151 \quad \Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
71232161316131613151 \quad \Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
71232161316131613151 \quad \Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}
\end{aligned}$$

² 71232161316131613151	$\Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
712321613161316131 ² 51	$\Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$
Base: 81223151316131613151		
81223151 ² 316131613151	$\Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, \det = 121$	$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$
812231513161316131 ² 51	$\Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, \det = 81$	$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$
² 81223151316131613151	$\Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, \det = 64$	$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$
812231513161316131 ² 51	$\Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$
81223151316131 ² 613151	$\Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$
812231513161316131 ² 51	$\Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, \det = 64$	$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
8122315131 ² 6131613151	$\Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, \det = 100$	$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$
812231513161316131 ² 51	$\Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$
Base: 81231513161316131513		
8123151316131 ² 6131513	$\Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, \det = 100$	$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$
8123151316131613151 ² 3	$\Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, \det = 169$	$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix}$
8123151 ² 3161316131513	$\Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, \det = 196$	$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix}$
8123151316131613151 ² 3	$\Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, \det = 121$	$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$
² 81231513161316131513	$\Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, \det = 64$	$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$
81231513161316131 ² 513	$\Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$
81231513161316131 ² 513	$\Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$
81231513161316131 ² 513	$\Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, \det = 64$	$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$

$$81231513161316131513 \quad \Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

Base: 15131613161315132181

$$15131613161315132181 \quad \Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix}$$

$$15131613161315132181 \quad \Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix}$$

$$15131613161315132181 \quad \Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$15131613161315132181 \quad \Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

$$15131613161315132181 \quad \Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$15131613161315132181 \quad \Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$15131613161315132181 \quad \Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$15131613161315132181 \quad \Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

Base: 41316131613161232181

$$41316131613161232181 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$41316131613161232181 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$41316131613161232181 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix}$$

$$41316131613161232181 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$41316131613161232181 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

$$41316131613161232181 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix}$$

$$41316131613161232181 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 206 & 28 \\ 2 & 28 & 8 \end{pmatrix}$$

$$41316131613161232181 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$41316131613161232181 \quad \Delta = -131, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

Base: 13161316131612321812

$$\begin{aligned}
21812321613161316131\textcolor{red}{3} \text{ [#=8]} \quad \Delta = -103, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
13161316131612321812 \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
13161316131612321812 \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
13161316131612321812 \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
13161316131612321812 \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
13161316131612321812 \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
13161316131612321812 \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
13161316131612321812 \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 100 \quad & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
13161316131612321812 \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
21812321613161316131\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -103, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
21812321613161316131\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -103, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21812321613161316131\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -103, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21812321613161316131\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -103, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
21812321613161316131\textcolor{red}{3} \text{ [#=5]} \quad \Delta = -103, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
21812321613161316131\textcolor{red}{3} \text{ [#=6]} \quad \Delta = -103, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
21812321613161316131\textcolor{red}{3} \text{ [#=7]} \quad \Delta = -103, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
\textbf{Base: } 13161232181231513221 \\
12231513218123216131\textcolor{red}{3} \text{ [#=6]} \quad \Delta = -1, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
12231513218123216131\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -1, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
13161232181231513221\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -1, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
12231513218123216131\overset{2}{3} \text{ [#=3]} \quad \Delta = -1, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
13161232181231513221\overset{2}{3} \text{ [#=6]} \quad \Delta = -1, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
13161232181231513221\overset{2}{3} \text{ [#=2]} \quad \Delta = -1, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
12231513218123216131\overset{2}{3} \text{ [#=2]} \quad \Delta = -1, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
12231513218123216131\overset{2}{3} \text{ [#=7]} \quad \Delta = -1, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
13161232181231513221\overset{2}{3} \text{ [#=7]} \quad \Delta = -1, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
12231513218123216131\overset{2}{3} \text{ [#=8]} \quad \Delta = -1, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
13161232181231513221\overset{2}{3} \text{ [#=8]} \quad \Delta = -1, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
13161232181231513221\overset{2}{3} \text{ [#=5]} \quad \Delta = -1, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
12231513218123216131\overset{2}{3} \text{ [#=9]} \quad \Delta = -1, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
13161232181231513221\overset{2}{3} \text{ [#=9]} \quad \Delta = -1, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
12231513218123216131\overset{2}{3} \text{ [#=5]} \quad \Delta = -1, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
13161232181231513221\overset{2}{3} \text{ [#=4]} \quad \Delta = -1, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

Base: 71223151321812321712

$$\begin{aligned}
71223151321812321712 \overset{2}{\quad} \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
71223151321812321712 \overset{2}{\quad} \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
71223151321812321712 \overset{2}{\quad} \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
71223151321812321712 \overset{2}{\quad} \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
71223151321812321712 \overset{2}{\quad} \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 89 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
71223151321812321712 \overset{2}{\quad} \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix}
\end{aligned}$$

$$71223151321812321712^2 \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 21712321812315132171

$$21712321812315132171^2 \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

$$21712321812315132171^2 \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$21712321812315132171^2 \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$21712321812315132171^2 \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix}$$

$$21712321812315132171^2 \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$21712321812315132171^2 \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$21712321812315132171^2 \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: 22171232181231513221

$$22171232181231513221^3 \quad \Delta = -21, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$22171232181231513221^2 \quad \Delta = -12, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$22171232181231513221^3 \text{ [#=5]} \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$22171232181231513221^3 \text{ [#=4]} \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$22171232181231513221^3 \text{ [#=3]} \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$22171232181231513221^3 \text{ [#=6]} \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$22171232181231513221^3 \text{ [#=7]} \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$22171232181231513221^3 \text{ [#=8]} \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$22171232181231513221^3 \text{ [#=9]} \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$22171232181231513221^3 \text{ [#=2]} \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\begin{aligned}
22171232181231513221^2 \quad \Delta = -12, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
22171232181231513221^2 \quad \Delta = -12, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
22171232181231513221^2 \quad \Delta = -12, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
22171232181231513221^2 \quad \Delta = -12, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
22171232181231513221^2 \quad \Delta = -12, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: 71231513218123217123

$$\begin{aligned}
71231513218123217123^2 \quad \Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
71231513218123217123^2 \quad \Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
71231513218123217123^2 \quad \Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
71231513218123217123^2 \quad \Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
71231513218123217123^2 \quad \Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
71231513218123217123^2 \quad \Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
71231513218123217123^2 \quad \Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 13217123218123151321

$$\begin{aligned}
13217123218123151321^3 \text{ [#=8]} \quad \Delta = -29, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
13217123218123151321^3 \text{ [#=4]} \quad \Delta = -29, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
13217123218123151321^3 \text{ [#=5]} \quad \Delta = -29, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
13217123218123151321^3 \text{ [#=6]} \quad \Delta = -29, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
13217123218123151321^3 \text{ [#=7]} \quad \Delta = -29, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
12315132181232171231^3 \text{ [#=4]} \quad \Delta = -29, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
13217123218123151321\overset{\textcolor{red}{3}}{3} \text{ [#=3]} \quad \Delta = -29, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
12315132181232171231\overset{\textcolor{red}{3}}{3} \text{ [#=3]} \quad \Delta = -29, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
13217123218123151321\overset{\textcolor{red}{3}}{3} \text{ [#=2]} \quad \Delta = -29, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
12315132181232171231\overset{\textcolor{red}{3}}{3} \text{ [#=5]} \quad \Delta = -29, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
12315132181232171231\overset{\textcolor{red}{3}}{3} \text{ [#=2]} \quad \Delta = -29, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
13217123218123151321\overset{\textcolor{red}{3}}{3} \text{ [#=9]} \quad \Delta = -29, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
13217123218123151321 \quad \Delta = -21, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
13217123218123151321 \quad \Delta = -21, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
13217123218123151321 \quad \Delta = -21, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
13217123218123151321 \quad \Delta = -21, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
13217123218123151321 \quad \Delta = -21, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
13217123218123151321 \quad \Delta = -21, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 12321712321812315131

$$\begin{aligned}
12321712321812315131 \quad \Delta = -22, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
13151321812321712321\overset{\textcolor{red}{3}}{3} \text{ [#=3]} \quad \Delta = -30, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
13151321812321712321\overset{\textcolor{red}{3}}{3} \text{ [#=7]} \quad \Delta = -30, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
12321712321812315131\overset{\textcolor{red}{3}}{3} \text{ [#=3]} \quad \Delta = -30, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
13151321812321712321\overset{\textcolor{red}{3}}{3} \text{ [#=4]} \quad \Delta = -30, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
12321712321812315131\overset{\textcolor{red}{3}}{3} \text{ [#=4]} \quad \Delta = -30, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
13151321812321712321\overset{\textcolor{red}{3}}{3} \text{ [#=5]} \quad \Delta = -30, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
12321712321812315131\overset{2}{3} \text{ [#=5]} \quad \Delta = -30, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
12321712321812315131\overset{2}{} \quad \Delta = -22, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
13151321812321712321\overset{2}{3} \text{ [#=6]} \quad \Delta = -30, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
12321712321812315131\overset{2}{} \quad \Delta = -22, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
12321712321812315131\overset{2}{} \quad \Delta = -22, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
12321712321812315131\overset{2}{3} \text{ [#=2]} \quad \Delta = -30, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
13151321812321712321\overset{2}{3} \text{ [#=2]} \quad \Delta = -30, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
12321712321812315131\overset{2}{3} \text{ [#=6]} \quad \Delta = -30, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
12321712321812315131\overset{2}{3} \text{ [#=7]} \quad \Delta = -30, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
13151321812321712321\overset{2}{3} \text{ [#=8]} \quad \Delta = -30, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
12321712321812315131\overset{2}{3} \text{ [#=8]} \quad \Delta = -30, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
13151321812321712321\overset{2}{3} \text{ [#=9]} \quad \Delta = -30, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
12321712321812315131\overset{2}{3} \text{ [#=9]} \quad \Delta = -30, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
12321712321812315131\overset{2}{} \quad \Delta = -22, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
12321712321812315131\overset{2}{} \quad \Delta = -22, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\text{Base: } 71232171232181231513 & \\
71232171232181231513\overset{2}{} \quad \Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 169 \quad & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
71232171232181231513\overset{2}{} \quad \Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 225 \quad & \begin{pmatrix} 0 & 15 & 2 \\ 15 & 163 & 25 \\ 2 & 25 & 8 \end{pmatrix} \\
71232171232181231513\overset{2}{} \quad \Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 169 \quad & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
71232171232181231513\overset{2}{} \quad \Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{ll}
71232171232181231513^2 & \Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
71232171232181231513^2 & \Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
71232171232181231513^2 & \Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
\text{Base: } 17123217123218123151 & \\
17123217123218123151^2 & \Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
17123217123218123151^2 & \Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
17123217123218123151^2 & \Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
17123217123218123151^2 & \Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
17123217123218123151^2 & \Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
17123217123218123151^2 & \Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\text{Base: } 7122315132181232171^1 & \\
7122315132181232171^2 & \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
7122315132181232171^2 & \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
7122315132181232171^2 & \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 89 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
7122315132181232171^2 & \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
7122315132181232171^2 & \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
7122315132181232171^{21} & \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\text{Base: } 1712321812315132171^1 & \\
1712321812315132171^2 & \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1712321812315132171^{12} & \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1712321812315132171^1 & \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
& \overset{1}{1}712321812315132171 \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
& \overset{1}{1}712321812315132171 \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{1}{1}712321812315132171 \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{21}{1}71232181231513221$

$$\begin{aligned}
& \overset{21}{1}71232181231513221\overset{3}{\color{red}} \text{ [#=9]} \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
& \overset{21}{1}71232181231513221 \quad \Delta = -12, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{21}{1}71232181231513221 \quad \Delta = -12, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{21}{1}71232181231513221 \quad \Delta = -12, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{21}{1}71232181231513221 \quad \Delta = -12, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{21}{1}71232181231513221 \quad \Delta = -12, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{21}{1}71232181231513221 \quad \Delta = -12, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{21}{1}71232181231513221\overset{3}{\color{red}} \text{ [#=6]} \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& \overset{21}{1}71232181231513221\overset{3}{\color{red}} \text{ [#=7]} \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
& \overset{21}{1}71232181231513221\overset{3}{\color{red}} \text{ [#=2]} \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{21}{1}71232181231513221\overset{3}{\color{red}} \text{ [#=3]} \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{21}{1}71232181231513221\overset{3}{\color{red}} \text{ [#=4]} \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{21}{1}71232181231513221\overset{3}{\color{red}} \text{ [#=5]} \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& \overset{21}{1}71232181231513221\overset{3}{\color{red}} \text{ [#=8]} \quad \Delta = -20, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}
\end{aligned}$$

Base: 81223151321812321613

$$81223151321812321613 \quad \Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

81223151321812321613 ²	$\Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, \det = 81$	$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$
81223151321812321613 ²	$\Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, \det = 121$	$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$
81223151321812321613 ²	$\Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, \det = 169$	$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix}$
81223151321812321613 ²	$\Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, \det = 121$	$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$
81223151321812321613 ²	$\Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, \det = 81$	$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$
81223151321812321613 ²	$\Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, \det = 64$	$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$
Base: 16123218123151322181		
16123218123151322181 ²	$\Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$
16123218123151322181 ²	$\Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$
16123218123151322181 ²	$\Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, \det = 64$	$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$
16123218123151322181 ²	$\Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$
16123218123151322181 ²	$\Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, \det = 81$	$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix}$
16123218123151322181 ²	$\Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$
Base: 31612321812315132181		
31612321812315132181 ²	$\Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, \det = 81$	$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$
31612321812315132181 ²	$\Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, \det = 121$	$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix}$
31612321812315132181 ²	$\Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, \det = 169$	$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 219 & 29 \\ 2 & 29 & 8 \end{pmatrix}$
31612321812315132181 ²	$\Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, \det = 121$	$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 163 & 25 \\ 2 & 25 & 8 \end{pmatrix}$
31612321812315132181 ²	$\Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, \det = 81$	$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix}$
31612321812315132181 ²	$\Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$

$$\overset{2}{3}1612321812315132181 \quad \Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: 16123218123151321812

$$16123218123151321812 \quad \Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

$$16123218123151321812 \quad \Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$16123218123151321812 \quad \Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$16123218123151321812 \quad \Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$16123218123151321812 \quad \Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$16123218123151321812 \quad \Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix}$$

$$16123218123151321812 \quad \Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

Base: 41321812321712321812

$$\overset{2}{4}1321812321712321812 \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$41321812321712321812 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$41321812321712321812 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$41321812321712321812 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix}$$

$$41321812321712321812 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 163 & 25 \\ 2 & 25 & 8 \end{pmatrix}$$

$$41321812321712321812 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$\overset{2}{4}1321812321712321812 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$41321812321712321812 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix}$$

Base: 22181232171232181231

$$22181232171232181231 \overset{3}{3} [\# = 5] \quad \Delta = -103, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\overset{3}{2}2181232171232181231 \quad \Delta = -104, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\begin{aligned}
22181232171232181231\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -103, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
22181232171232181231\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -103, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
22181232171232181231\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -103, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
221812321\textcolor{red}{2}71232181231 \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
221812321712321\textcolor{red}{2}81231 \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
22181232171232181231 \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
221812321712321\textcolor{red}{2}81231 \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
22181232171232181231 \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
22181\textcolor{red}{2}232171232181231 \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
\textbf{Base: } 61232181231513218123 \\
61232181231513218123 \quad \Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 100 \quad & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
6123218123151321\textcolor{red}{2}8123 \quad \Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
6123218123151321\textcolor{red}{2}8123 \quad \Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 144 \quad & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
61232181231513218123 \quad \Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 169 \quad & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
61232181\textcolor{red}{2}231513218123 \quad \Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 196 \quad & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
6123218123151321\textcolor{red}{2}8123 \quad \Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 \quad & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
61\textcolor{red}{2}232181231513218123 \quad \Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\textbf{Base: } 13218123151321812321 \\
13218123151321812321\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -57, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1321812315132181\textcolor{red}{2}2321 \quad \Delta = -49, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
13218123151321812321^2 \quad \Delta = -49, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
13218123151321812321^2 \quad \Delta = -49, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
123218123151321812313^{\#5} \quad \Delta = -57, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
123218123151321812313^{\#4} \quad \Delta = -57, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
132181231513218123213^{\#5} \quad \Delta = -57, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
132181231513218123213^{\#6} \quad \Delta = -57, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
132181231513218123213^{\#7} \quad \Delta = -57, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
132181231513218123213^{\#8} \quad \Delta = -57, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
132181231513218123213^{\#3} \quad \Delta = -57, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
13218123151321812321^2 \quad \Delta = -49, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
132181231513218123213^{\#4} \quad \Delta = -57, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
123218123151321812313^{\#3} \quad \Delta = -57, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
13218123151321812321^2 \quad \Delta = -49, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
13218123151321812321^2 \quad \Delta = -49, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
123218123151321812313^{\#2} \quad \Delta = -57, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
132181231513218123213^{\#2} \quad \Delta = -57, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\text{Base: } 31613161232181232171 & \\
31613161232181232171^2 \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 324 & \begin{pmatrix} 0 & 18 & 2 \\ 18 & 448 & 42 \\ 2 & 42 & 8 \end{pmatrix} \\
31613161232181232171^2 \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 144 & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 202 & 28 \\ 2 & 28 & 8 \end{pmatrix} \\
31613161232181232171^2 \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 196 & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 264 & 32 \\ 2 & 32 & 8 \end{pmatrix}
\end{aligned}$$

31613161232181232171 ²	$\Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, \det = 81$	$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix}$
31613161232181232171 ²	$\Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$
31613161232181232171 ²	$\Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, \det = 100$	$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix}$
31613161232181232171 ²	$\Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
Base: 16131612321812321712		
16131612321812321712 ²	$\Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, \det = 121$	$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 196 & 28 \\ 2 & 28 & 8 \end{pmatrix}$
16131612321812321712 ²	$\Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, \det = 4$	$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$
16131612321812321712 ²	$\Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$
16131612321812321712 ²	$\Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$
16131612321812321712 ²	$\Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, \det = 81$	$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix}$
16131612321812321712 ²	$\Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$
16131612321812321712 ²	$\Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix}$
Base: 61316123218123217123		
61316123218123217123 ²	$\Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$
61316123218123217123 ²	$\Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, \det = 81$	$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$
61316123218123217123 ²	$\Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, \det = 121$	$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$
61316123218123217123 ²	$\Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, \det = 225$	$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 163 & 25 \\ 2 & 25 & 8 \end{pmatrix}$
61316123218123217123 ²	$\Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, \det = 169$	$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix}$
61316123218123217123 ²	$\Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$
61316123218123217123 ²	$\Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, \det = 81$	$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$
Base: 13161232181232171231		

$$\begin{aligned}
13161232181232171231\overset{2}{3} \text{ }_{[\# = 2]} \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
13217123218123216131\overset{2}{3} \text{ }_{[\# = 2]} \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
13161232181232171231\overset{2}{3} \quad \Delta = -48, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
13161232181232171231\overset{2}{3} \quad \Delta = -48, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
13217123218123216131\overset{2}{3} \text{ }_{[\# = 7]} \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
13217123218123216131\overset{2}{3} \text{ }_{[\# = 5]} \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
13217123218123216131\overset{2}{3} \text{ }_{[\# = 9]} \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
13217123218123216131\overset{2}{3} \text{ }_{[\# = 8]} \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
13161232181232171231\overset{2}{3} \text{ }_{[\# = 4]} \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
13217123218123216131\overset{2}{3} \text{ }_{[\# = 4]} \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
13161232181232171231\overset{2}{3} \text{ }_{[\# = 3]} \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
13217123218123216131\overset{2}{3} \text{ }_{[\# = 6]} \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
13217123218123216131\overset{2}{3} \text{ }_{[\# = 3]} \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
13161232181232171231\overset{2}{3} \quad \Delta = -48, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
13161232181232171231\overset{2}{3} \quad \Delta = -48, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
13161232181232171231\overset{2}{3} \text{ }_{[\# = 5]} \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
13161232181232171231\overset{2}{3} \quad \Delta = -48, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
13161232181232171231\overset{2}{3} \quad \Delta = -48, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\text{Base: } 22171232181232171231 & \\
22171232181232171231\overset{2}{3} \text{ }_{[\# = 4]} \quad \Delta = -75, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
22171232181232171231\overset{3}{3} \text{ }_{[\# = 5]} \quad \Delta = -75, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
22171232181232171231\overset{3}{3} \quad \Delta = -76, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
22171232181232171231\overset{2}{2} \quad \Delta = -67, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
22171232181232171231\overset{2}{2} \quad \Delta = -67, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
22171232181232171231\overset{2}{2} \quad \Delta = -67, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
22171232181232171231\overset{2}{2} \quad \Delta = -67, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
22171232181232171231\overset{3}{3} \text{ }_{[\# = 2]} \quad \Delta = -75, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
22171232181232171231\overset{3}{3} \text{ }_{[\# = 3]} \quad \Delta = -75, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
22171232181232171231\overset{2}{2} \quad \Delta = -67, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
22171232181232171231\overset{2}{2} \quad \Delta = -67, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\text{Base: } 21712321812321712315 \\
21712321812321712315\overset{2}{2} \quad \Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21712321812321712315\overset{2}{2} \quad \Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
21712321812321712315\overset{2}{2} \quad \Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 100 \quad & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
21712321812321712315\overset{2}{2} \quad \Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 \quad & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
21712321812321712315\overset{2}{2} \quad \Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
21712321812321712315\overset{2}{2} \quad \Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21712321812321712315\overset{2}{2} \quad \Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\text{Base: } 61232171232181232161 \\
61232171232181232161\overset{2}{2} \quad \Delta = -120, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 \quad & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 196 & 28 \\ 2 & 28 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
61232171232181232161^2 \quad \Delta = -120, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
61232171232181232161^2 \quad \Delta = -120, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
61232171232181232161^2 \quad \Delta = -120, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
61232171232181232161^2 \quad \Delta = -120, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 168 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
61232171232181232161^2 \quad \Delta = -120, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{aligned}$$

Base: 21612321812321712321

$$\begin{aligned}
216123218123217123213^{[\#=2]} \quad \Delta = -50, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
216123218123217123213^{[\#=3]} \quad \Delta = -50, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
216123218123217123213^{[\#=4]} \quad \Delta = -50, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
216123218123217123213^{[\#=5]} \quad \Delta = -50, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
216123218123217123213^{[\#=6]} \quad \Delta = -50, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
216123218123217123213^{[\#=7]} \quad \Delta = -50, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
216123218123217123213^{[\#=8]} \quad \Delta = -50, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
216123218123217123213^{[\#=9]} \quad \Delta = -50, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
21612321812321712321^3 \quad \Delta = -50, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21612321812321712321^2 \quad \Delta = -42, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21612321812321712321^2 \quad \Delta = -42, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21612321812321712321^2 \quad \Delta = -42, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21612321812321712321^2 \quad \Delta = -42, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
21612321812321712321^2 \quad \Delta = -42, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

$$21612321812321\overset{2}{7}12321 \quad \Delta = -42, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: 1612321812321712321

$$1612321812321712321\overset{1}{3} \text{ [#=3]} \quad \Delta = -50, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1612321812321712321\overset{1}{3} \text{ [#=6]} \quad \Delta = -50, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1612321812321712321\overset{1}{3} \text{ [#=2]} \quad \Delta = -50, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1612321812321712321\overset{1}{3} \text{ [#=9]} \quad \Delta = -50, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$1612321812321712321\overset{1}{3} \text{ [#=5]} \quad \Delta = -50, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1612321812321712321\overset{1}{3} \text{ [#=4]} \quad \Delta = -50, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1612321812321712321\overset{1}{2} \quad \Delta = -42, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1612321812321712321\overset{2}{7} \quad \Delta = -42, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1612321812321712321\overset{2}{7} \quad \Delta = -42, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1612321812321712321\overset{1}{3} \text{ [#=7]} \quad \Delta = -50, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$1612321812321712321\overset{2}{7} \quad \Delta = -42, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1612321812321712321\overset{2}{7} \quad \Delta = -42, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1612321812321712321\overset{1}{3} \text{ [#=8]} \quad \Delta = -50, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

Base: 17123217123218123215

$$1712321712321812321\overset{2}{5} \quad \Delta = -121, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1712321712321812321\overset{2}{5} \quad \Delta = -121, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1712321712321812321\overset{2}{5} \quad \Delta = -121, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1712321712321812321\overset{2}{5} \quad \Delta = -121, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$17123217123218123215 \quad \Delta = -121, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$17123217123218123215 \quad \Delta = -121, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

Base: 21712321712321812321

$$217123217123218123213 \text{ [#=8]} \quad \Delta = -77, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$21712321712321812321 \quad \Delta = -69, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$21712321712321812321 \quad \Delta = -69, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$21712321712321812321 \quad \Delta = -69, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$21712321712321812321 \quad \Delta = -69, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$217123217123218123213 \text{ [#=4]} \quad \Delta = -77, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$217123217123218123213 \text{ [#=3]} \quad \Delta = -77, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$217123217123218123213 \text{ [#=9]} \quad \Delta = -77, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$217123217123218123213 \text{ [#=7]} \quad \Delta = -77, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$217123217123218123213 \text{ [#=6]} \quad \Delta = -77, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$217123217123218123213 \text{ [#=5]} \quad \Delta = -77, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$21712321712321812321 \quad \Delta = -69, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$217123217123218123213 \text{ [#=2]} \quad \Delta = -77, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$21712321712321812321 \quad \Delta = -69, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 1712321812321613161

$$1712321812321613161 \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$1712321812321613161 \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$\begin{aligned}
1712321812321613161 \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
1712321812321613161 \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 163 & 25 \\ 2 & 25 & 8 \end{pmatrix} \\
1712321812321613161 \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1712321812321613161 \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1712321812321712315

$$\begin{aligned}
1712321812321712315 \quad \Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1712321812321712315 \quad \Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1712321812321712315 \quad \Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1712321812321712315 \quad \Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1712321812321712315 \quad \Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1712321812321712315 \quad \Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1712321712321812321

$$\begin{aligned}
1712321712321812321\textcolor{red}{3} \text{ [#=7]} \quad \Delta = -77, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1712321712321812321 \quad \Delta = -69, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1712321712321812321 \quad \Delta = -69, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1712321712321812321 \quad \Delta = -69, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1712321712321812321\textcolor{red}{3} \text{ [#=2]} \quad \Delta = -77, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1712321712321812321\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -77, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1712321712321812321\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -77, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1712321712321812321\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -77, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{1}{17}123217\overset{2}{12}321812321 \quad \Delta = -69, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& \overset{1}{17}12321712321812321\overset{3}{3} \text{ [#=6]} \quad \Delta = -77, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{1}{17}12321712321812321\overset{3}{3} \text{ [#=8]} \quad \Delta = -77, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{1}{17}123217\overset{2}{12}321812321 \quad \Delta = -69, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{1}{17}12321712321812321\overset{3}{3} \text{ [#=5]} \quad \Delta = -77, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \text{Base: } \overset{21}{17}1232181232171231 \\
& \overset{21}{17}123218123217\overset{2}{12}31 \quad \Delta = -67, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{21}{17}1232181232171231\overset{3}{3} \text{ [#=5]} \quad \Delta = -75, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{21}{17}1232181232171231\overset{3}{3} \text{ [#=2]} \quad \Delta = -75, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
& \overset{21}{17}1232181232171231\overset{3}{3} \text{ [#=3]} \quad \Delta = -75, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{21}{17}1232181232171231\overset{3}{3} \text{ [#=4]} \quad \Delta = -75, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{21}{17}1232181232171231 \quad \Delta = -67, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{21}{17}\overset{2}{12}32181232171231 \quad \Delta = -67, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{21}{17}123218123217\overset{2}{12}31 \quad \Delta = -67, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{21}{17}1232181\overset{2}{23}2171231 \quad \Delta = -67, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{21}{17}123218123217\overset{2}{12}31 \quad \Delta = -67, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \text{Base: } 31613151321812321812 \\
& 316131513218123218\overset{2}{12} \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& 3161\overset{2}{31}51321812321812 \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
& 31613151321812321812 \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 202 & 28 \\ 2 & 28 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
31613151321812321812 \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 196 & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 264 & 32 \\ 2 & 32 & 8 \end{pmatrix} \\
31613151321812321812 \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 232 & 30 \\ 2 & 30 & 8 \end{pmatrix} \\
31613151321812321812 \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
31613151321812321812 \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
31613151321812321812 \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: 61231513218123218122

$$\begin{aligned}
61231513218123218122 \quad \Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
61231513218123218122 \quad \Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
61231513218123218122 \quad \Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
61231513218123218122 \quad \Delta = -127, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
61231513218123218122 \quad \Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
61231513218123218122 \quad \Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
61231513218123218122 \quad \Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
61231513218123218122 \quad \Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix}
\end{aligned}$$

Base: 16131513218123218122

$$\begin{aligned}
16131513218123218122 \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
16131513218123218122 \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
16131513218123218122 \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
16131513218123218122 \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
16131513218123218122 \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
16131513218123218122 \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{aligned}$$

$$16131513218123218122^2 \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$16131513218123218122^3 \quad \Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

Base: 51316123218123218122

$$51316123218123218122^2 \quad \Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$51316123218123218122^2 \quad \Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$51316123218123218122^3 \quad \Delta = -128, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$51316123218123218122^2 \quad \Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$51316123218123218122^2 \quad \Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 196 & 28 \\ 2 & 28 & 8 \end{pmatrix}$$

$$51316123218123218122^2 \quad \Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix}$$

$$51316123218123218122^2 \quad \Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$51316123218123218122^2 \quad \Delta = -119, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

Base: 13221812321812315131

$$131513218123218122313^{\text{[#}=5]} \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$13221812321812315131^2 \quad \Delta = -48, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$13221812321812315131^2 \quad \Delta = -48, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$13221812321812315131^2 \quad \Delta = -48, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$131513218123218122313^{\text{[#}=4]} \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$132218123218123151313^{\text{[#}=9]} \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$132218123218123151313^{\text{[#}=4]} \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$132218123218123151313^{\text{[#}=5]} \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$132218123218123151313^{\text{[#}=6]} \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$\begin{aligned}
13221812321812315131\bar{3} \text{ }_{[\# = 7]} \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
13221812321812315131\bar{3} \text{ }_{[\# = 8]} \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
132218123218123151\bar{3}^2 \quad \Delta = -48, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
132218123218123151\bar{3}^2 \quad \Delta = -48, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
132218123218123151\bar{3}^2 \quad \Delta = -48, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
13221812321812315131\bar{3} \text{ }_{[\# = 3]} \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
13151321812321812231\bar{3} \text{ }_{[\# = 3]} \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
13221812321812315131\bar{3} \text{ }_{[\# = 2]} \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
13151321812321812231\bar{3} \text{ }_{[\# = 2]} \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\text{Base: } 22218123218123151321 & \\
22218123218123151321\bar{3}^2 \quad \Delta = -40, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
22218123218123151321\bar{3}^2 \quad \Delta = -40, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
22218123218123151321\bar{3} \text{ }_{[\# = 5]} \quad \Delta = -48, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
22218123218123151321\bar{3} \text{ }_{[\# = 2]} \quad \Delta = -48, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
22218123218123151321\bar{3}^2 \quad \Delta = -40, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
22218123218123151321\bar{3}^2 \quad \Delta = -40, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
22218123218123151321\bar{3}^2 \quad \Delta = -40, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
22218123218123151321\bar{3}^2 \quad \Delta = -40, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
22218123218123151321\bar{3} \text{ }_{[\# = 3]} \quad \Delta = -48, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
22218123218123151321\bar{3} \text{ }_{[\# = 4]} \quad \Delta = -48, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
22218123218123151321\overset{\textcolor{red}{3}}{3} \text{ [#=9]} \quad \Delta = -48, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
22218123218123151321\overset{\textcolor{red}{3}}{3} \text{ [#=8]} \quad \Delta = -48, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
22218123218123151321\overset{\textcolor{red}{3}}{3} \text{ [#=7]} \quad \Delta = -48, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
22218123218123151321\overset{\textcolor{red}{3}}{3} \text{ [#=6]} \quad \Delta = -48, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix}
\end{aligned}$$

Base: 61315132181232181223

$$\begin{aligned}
61315132181232181223 \quad \Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
61315132181232181223 \quad \Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
61315132181232181223 \quad \Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
61315132181232181223 \quad \Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
61315132181232181223 \quad \Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
61315132181232181223 \quad \Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
61315132181232181223 \quad \Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix}
\end{aligned}$$

Base: 13161232181232181222

$$\begin{aligned}
13161232181232181222 \quad \Delta = -67, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
13161232181232181222 \quad \Delta = -67, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
13161232181232181222 \quad \Delta = -67, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
13161232181232181222 \quad \Delta = -67, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
13161232181232181222 \quad \Delta = -67, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
13161232181232181222 \quad \Delta = -67, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
22218123218123216131\overset{\textcolor{red}{3}}{3} \text{ [#=9]} \quad \Delta = -75, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
22218123218123216131\overset{2}{3} \text{ }_{[\# = 8]} \quad \Delta = -75, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
22218123218123216131\overset{2}{3} \text{ }_{[\# = 2]} \quad \Delta = -75, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
22218123218123216131\overset{2}{3} \text{ }_{[\# = 3]} \quad \Delta = -75, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
22218123218123216131\overset{2}{3} \text{ }_{[\# = 4]} \quad \Delta = -75, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
22218123218123216131\overset{2}{3} \text{ }_{[\# = 5]} \quad \Delta = -75, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
22218123218123216131\overset{2}{3} \text{ }_{[\# = 6]} \quad \Delta = -75, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
22218123218123216131\overset{2}{3} \text{ }_{[\# = 7]} \quad \Delta = -75, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix}
\end{aligned}$$

Base: 22171232181232181222

$$\begin{aligned}
22171232181232181222 \quad \Delta = -86, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
22171232181232181222 \quad \Delta = -86, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
22171232181232181222 \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
22171232181232181222 \quad \Delta = -86, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
22171232181232181222 \quad \Delta = -86, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
22171232181232181222 \quad \Delta = -86, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
22171232181232181222 \quad \Delta = -86, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{21}{1}71232181232181222$

$$\begin{aligned}
\overset{21}{1}71232181232181222 \quad \Delta = -86, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\overset{21}{1}71232181232181222 \quad \Delta = -86, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\overset{21}{1}71232181232181222 \quad \Delta = -86, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\overset{21}{1}71232181232181222 \quad \Delta = -86, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{21}{1} \overset{2}{7} 1232181232181222 \quad \Delta = -86, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{21}{1} 7 1232181232181222 \quad \Delta = -86, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 16123218123218122315

$$\begin{aligned}
& 16123218123218122315 \quad \Delta = -120, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 16123218123218122315 \quad \Delta = -120, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 16123218123218122315 \quad \Delta = -120, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 16123218123218122315 \quad \Delta = -120, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 16123218123218122315 \quad \Delta = -120, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 50 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& 16123218123218122315 \quad \Delta = -120, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

Base: 21612321812321812231

$$\begin{aligned}
& 21612321812321812231 \quad \Delta = -68, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& 21612321812321812231 \quad \Delta = -68, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 21612321812321812231 \quad \Delta = -68, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 21612321812321812231 \quad \Delta = -68, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 21612321812321812231 \quad \Delta = -76, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 21612321812321812231 \overset{3}{3} \text{ [#=2]} \quad \Delta = -76, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
& 21612321812321812231 \overset{3}{3} \text{ [#=5]} \quad \Delta = -76, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 21612321812321812231 \overset{3}{3} \text{ [#=4]} \quad \Delta = -76, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 21612321812321812231 \overset{3}{3} \text{ [#=3]} \quad \Delta = -76, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& 21612321812321812231 \quad \Delta = -68, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

$$21612321812321812231^{\mathbf{2}} \quad \Delta = -68, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: $1612321812321812231^{\mathbf{1}}$

$$1612321812321812231^{\mathbf{1}} \mathbf{3}^{\mathbf{1}}_{[\#=3]} \quad \Delta = -76, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1612321812321812231^{\mathbf{1} \mathbf{2}} \quad \Delta = -68, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1612321812321812231^{\mathbf{1}} \mathbf{2}^{\mathbf{2}} \quad \Delta = -68, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1612321812321812231^{\mathbf{1}} \mathbf{2}^{\mathbf{2}} \quad \Delta = -68, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1612321812321812231^{\mathbf{1}} \mathbf{2}^{\mathbf{2}} \quad \Delta = -68, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1612321812321812231^{\mathbf{1}} \mathbf{2}^{\mathbf{2}} \quad \Delta = -68, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1612321812321812231^{\mathbf{1}} \mathbf{3}^{\mathbf{1}}_{[\#=2]} \quad \Delta = -76, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$1612321812321812231^{\mathbf{1}} \mathbf{3}^{\mathbf{1}}_{[\#=5]} \quad \Delta = -76, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1612321812321812231^{\mathbf{1}} \mathbf{3}^{\mathbf{1}}_{[\#=4]} \quad \Delta = -76, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: 31321812321812315131

$$31321812321812315131^{\mathbf{1}} \mathbf{3}^{\mathbf{1}}_{[\#=3]} \quad \Delta = -64, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$31321812321812315131^{\mathbf{1}} \mathbf{3}^{\mathbf{1}}_{[\#=4]} \quad \Delta = -64, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$31321812321812315131^{\mathbf{1}} \mathbf{3}^{\mathbf{1}}_{[\#=5]} \quad \Delta = -64, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$31321812321812315131^{\mathbf{2}} \quad \Delta = -58, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$31321812321812315131^{\mathbf{2}} \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$31321812321812315131^{\mathbf{3}} \quad \Delta = -64, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 21 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$31321812321812315131^{\mathbf{1}} \mathbf{3}^{\mathbf{1}}_{[\#=2]} \quad \Delta = -64, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$31321812321812315131^{\mathbf{2}} \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\begin{aligned}
31321812321812315131^2 \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
31321812321812315131^2 \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
31321812321812315131^3_{[\# = 6]} \quad \Delta = -64, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
31321812321812315131^3_{[\# = 2]} \quad \Delta = -64, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 93 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
31321812321812315131^3_{[\# = 8]} \quad \Delta = -64, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
31321812321812315131^3_{[\# = 9]} \quad \Delta = -64, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
31321812321812315131^3_{[\# = 7]} \quad \Delta = -64, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
31321812321812315131^2 \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
31321812321812315131^2 \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
31321812321812315131^2 \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\text{Base: } 13218123218123151315 & \\
13218123218123151315^2 \quad \Delta = -100, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
13218123218123151315^2 \quad \Delta = -100, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
13218123218123151315^2 \quad \Delta = -100, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
13218123218123151315^2 \quad \Delta = -100, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
13218123218123151315^2 \quad \Delta = -100, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
13218123218123151315^2 \quad \Delta = -100, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
13218123218123151315^2 \quad \Delta = -100, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
13218123218123151315^2 \quad \Delta = -101, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
51315132181232181231^3_{[\# = 2]} \quad \Delta = -108, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
51315132181232181231\overset{3}{3} \text{ }_{[\# = 3]} \quad \Delta = -108, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
51315132181232181231\overset{3}{3} \text{ }_{[\# = 4]} \quad \Delta = -108, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
51315132181232181231\overset{3}{3} \text{ }_{[\# = 5]} \quad \Delta = -108, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\text{Base: } 14132181232181231512 & \\
21513218123218123141\overset{3}{3} \text{ }_{[\# = 3]} \quad \Delta = -76, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
14132181232181231\overset{2}{5}12 \quad \Delta = -68, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21513218123218123141\overset{3}{3} \text{ }_{[\# = 2]} \quad \Delta = -76, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
14132181232181231\overset{3}{5}12 \quad \Delta = -76, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
141\overset{2}{3}2181232181231512 \quad \Delta = -68, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
14132181232181231\overset{2}{5}12 \quad \Delta = -68, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
141\overset{2}{3}2181232181231512 \quad \Delta = -68, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
141\overset{2}{3}2181232181231512 \quad \Delta = -68, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
14132181232181231\overset{2}{5}12 \quad \Delta = -68, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
14132181232181231\overset{2}{5}12 \quad \Delta = -68, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
141\overset{2}{3}2181232181231512 \quad \Delta = -70, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\text{Base: } 14132181232181231\overset{1}{5}1 & \\
14132181232181231\overset{1}{5}1 \quad \Delta = -68, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1\overset{3}{1}4132181232181231\overset{1}{5}1 \text{ }_{[\# = 3]} \quad \Delta = -76, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
141\overset{2}{3}2181232181231\overset{1}{5}1 \quad \Delta = -70, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1\overset{3}{1}4132181232181231\overset{1}{5}1 \text{ }_{[\# = 2]} \quad \Delta = -76, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
14132181232181231\overset{1}{5}1 \quad \Delta = -68, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1413218123218123151^1 \quad \Delta = -68, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1413218123218123151^2 \quad \Delta = -68, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1413218123218123151^{21} \quad \Delta = -68, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1413218123218123151^2 \quad \Delta = -68, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 41232181232181232151

$$\begin{aligned}
41232181232181232151^3 \quad \Delta = -80, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 41 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
41232181232181232151^2 \quad \Delta = -72, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
41232181232181232151^2 \quad \Delta = -72, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
41232181232181232151^2 \quad \Delta = -72, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
41232181232181232151^2 \quad \Delta = -72, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
41232181232181232151^2 \quad \Delta = -72, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
41232181232181232151^2 \quad \Delta = -72, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 14132181232181232151

$$\begin{aligned}
151232181232181231413^{3} \text{ [#=3]} \quad \Delta = -78, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
14132181232181232151^2 \quad \Delta = -70, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
14132181232181232151^2 \quad \Delta = -70, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
14132181232181232151^2 \quad \Delta = -70, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
14132181232181232151^2 \quad \Delta = -70, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
14132181232181232151^2 \quad \Delta = -70, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
14132181232181232151^2 \quad \Delta = -72, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
15123218123218123141\overset{\textcolor{red}{3}}{3}_{[\# = 2]} \quad \Delta = -78, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
14132181232181232151\overset{\textcolor{red}{2}}{2} \quad \Delta = -70, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\text{Base: } 41232181232181231512 & \\
41232181232181231512\overset{\textcolor{red}{3}}{3} \quad \Delta = -78, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
\overset{\textcolor{red}{3}}{3}41232181232181231512 \quad \Delta = -78, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 41 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
41232181232181231512\overset{\textcolor{red}{2}}{2} \quad \Delta = -70, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
41232181232181231512\overset{\textcolor{red}{2}}{2} \quad \Delta = -70, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
41232181232181231512\overset{\textcolor{red}{2}}{2} \quad \Delta = -70, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\overset{\textcolor{red}{2}}{2}41232181232181231512 \quad \Delta = -70, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
41232181232181231512\overset{\textcolor{red}{2}}{2} \quad \Delta = -70, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
41232181232181231512\overset{\textcolor{red}{2}}{2} \quad \Delta = -70, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
41232181232181231512\overset{\textcolor{red}{2}}{2} \quad \Delta = -70, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\text{Base: } 4123218123218123151\overset{1}{1} & \\
\overset{\textcolor{red}{3}}{3}4123218123218123151\overset{1}{1} \quad \Delta = -78, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 41 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
\overset{\textcolor{red}{2}}{2}4123218123218123151\overset{1}{1} \quad \Delta = -70, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
4123218123218123151\overset{\textcolor{red}{2}}{2}\overset{1}{1} \quad \Delta = -70, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
4123218123218123151\overset{\textcolor{red}{2}}{2}\overset{1}{1} \quad \Delta = -70, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
4123218123218123151\overset{\textcolor{red}{2}}{2}\overset{1}{1} \quad \Delta = -70, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
4123218123218123151\overset{\textcolor{red}{2}}{2}\overset{1}{1} \quad \Delta = -70, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
4123218123218123151\overset{\textcolor{red}{2}}{2}\overset{1}{1} \quad \Delta = -70, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\text{Base: } 21612321812321812313 &
\end{aligned}$$

$$\begin{aligned}
21612321812321812313^2 \quad \Delta = -76, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21612321812321812313^2 \quad \Delta = -78, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
21612321812321812313^2 \quad \Delta = -76, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21612321812321812313^2 \quad \Delta = -76, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
21612321812321812313^2 \quad \Delta = -76, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21612321812321812313^2 \quad \Delta = -76, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21612321812321812313^2 \quad \Delta = -76, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
21612321812321812313^2 \quad \Delta = -76, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21612321812321812313^3 \quad \Delta = -84, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21612321812321812313^3 \quad \Delta = -84, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
21612321812321812313^3_{[\# = 2]} \quad \Delta = -84, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1612321812321812313^1

$$\begin{aligned}
1612321812321812313^1 \quad \Delta = -84, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1612321812321812313^1_{[\# = 2]} \quad \Delta = -84, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1612321812321812313^1 \quad \Delta = -76, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1612321812321812313^1 \quad \Delta = -76, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1612321812321812313^1 \quad \Delta = -76, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1612321812321812313^1 \quad \Delta = -76, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1612321812321812313^1 \quad \Delta = -76, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1612321812321812313^1 \quad \Delta = -78, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{l}
\begin{array}{c} 1 \\ 1612321812321812313 \end{array} \quad \Delta = -76, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\\
\text{Base: } 2^{\begin{array}{c} 13151321812321812 \\ 3 \end{array}} 1 \\
\\
\begin{array}{c} \textcolor{red}{3}13151321812321812 \\ 2 \quad 3 \end{array} 1 \text{ [#=5]} \quad \Delta = -58, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\\
\begin{array}{c} \textcolor{red}{3}13151321812321812 \\ 2 \quad 3 \end{array} 1 \text{ [#=3]} \quad \Delta = -58, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\\
\begin{array}{c} \textcolor{red}{3}13151321812321812 \\ 2 \quad 3 \end{array} 1 \text{ [#=9]} \quad \Delta = -58, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
\\
\begin{array}{c} 13151321812321812 \\ 2 \quad 3 \end{array} \textcolor{red}{1}3 \text{ [#=2]} \quad \Delta = -58, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 41 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
\\
\begin{array}{c} \textcolor{red}{3}13151321812321812 \\ 2 \quad 3 \end{array} 1 \text{ [#=4]} \quad \Delta = -58, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\\
\begin{array}{c} 13151321812321812 \\ 2 \quad 3 \end{array} \textcolor{red}{1}3 \text{ [#=3]} \quad \Delta = -58, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 96 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
\\
\begin{array}{c} 13151321812321812 \\ 2 \quad 3 \end{array} \textcolor{red}{1}2 \quad \Delta = -50, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\\
\begin{array}{c} \textcolor{red}{3}13151321812321812 \\ 2 \quad 3 \end{array} 1 \text{ [#=2]} \quad \Delta = -58, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\\
\begin{array}{c} \textcolor{red}{3}13151321812321812 \\ 2 \quad 3 \end{array} 1 \text{ [#=7]} \quad \Delta = -58, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
\\
\begin{array}{c} \textcolor{red}{3}13151321812321812 \\ 2 \quad 3 \end{array} 1 \text{ [#=6]} \quad \Delta = -58, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\\
\begin{array}{c} \textcolor{red}{3}13151321812321812 \\ 2 \quad 3 \end{array} 1 \text{ [#=8]} \quad \Delta = -58, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
\\
\begin{array}{c} 13151321812321812 \\ 2 \quad 3 \end{array} \textcolor{red}{1}4 \quad \Delta = -78, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\\
\begin{array}{c} 13151321812321812 \\ 2 \quad 3 \end{array} 1 \quad \Delta = -50, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\\
\begin{array}{c} 13151321812321812 \\ 2 \quad 3 \end{array} 1 \quad \Delta = -50, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\\
\begin{array}{c} 13151321812321812 \\ 2 \quad 3 \end{array} 1 \quad \Delta = -50, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\\
\begin{array}{c} 13151321812321812 \\ 2 \quad 3 \end{array} 1 \quad \Delta = -50, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\\
\begin{array}{c} 13151321812321812 \\ 2 \quad 3 \end{array} 1 \quad \Delta = -50, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\\
\begin{array}{c} 13151321812321812 \\ 2 \quad 3 \end{array} 1 \quad \Delta = -50, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{array}$$

Base: 2161232181232181232¹

$$2161232181232181232 \quad \Delta = -70, \quad n_+ = 1, \quad n_- = 20, \quad n_0 = 0, \quad |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$2161232181232181232 \quad \Delta = -70, \quad n_+ = 1, \quad n_- = 20, \quad n_0 = 0, \quad |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$2161232181^{2}232181232^1 \quad \Delta = -70, \quad n_+ = 1, \quad n_- = 20, \quad n_0 = 0, \quad |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$2161232181232181232 \quad \Delta = -70, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$21612321812321812\overset{41}{3}2 \quad \Delta = -98, \, n_+ = 1, \, n_- = 20, \, n_0 = 0, \, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$2161^2232181232181232^1 \quad \Delta = -70, \quad n_+ = 1, \quad n_- = 20, \quad n_0 = 0, \quad |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$2161232181232181232 \quad \Delta = -70, \quad n_+ = 1, \quad n_- = 20, \quad n_0 = 0, \quad |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$21612321812321812\overset{21}{3}2 \quad \Delta = -70, \quad n_+ = 1, \quad n_- = 20, \quad n_0 = 0, \quad |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$2161232181232181232 \quad \Delta = -78, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$21612321812321812\overset{31}{3}2_{[\# = 2]} \quad \Delta = -78, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$21612321812321812\overset{31}{3}2_{[\# = 3]} \quad \Delta = -78, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: $\overset{1}{1}6\overset{1}{1}232181232181232$

$$1612321812321812\overset{41}{3}2 \quad \Delta = -98, \quad n_+ = 1, \quad n_- = 20, \quad n_0 = 0, \quad |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$1612321812321812\overset{31}{3}2_{[\#=3]} \quad \Delta = -78, \quad n_+ = 1, \quad n_- = 20, \quad n_0 = 0, \quad |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1612321812321812\overset{31}{3}2_{[\# = 2]} \quad \Delta = -78, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1612321812321812\overset{21}{3}2 \quad \Delta = -70, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$161232181232181232 \quad \Delta = -70, \quad n_+ = 1, \quad n_- = 20, \quad n_0 = 0, \quad |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$161232181232181232 \quad \Delta = -70, \quad n_+ = 1, \quad n_- = 20, \quad n_0 = 0, \quad |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$${}^1161232181232181232 \quad \Delta = -70, \quad n_+ = 1, \quad n_- = 20, \quad n_0 = 0, \quad |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$16123218123218\overset{2}{1}232 \quad \Delta = -70, \, n_+ = 1, \, n_- = 20, \, n_0 = 0, \, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$${}^{1,2}_{161232181232181232} \quad \Delta = -70, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

Base: ${}^1_{1812321613161316131}$

$${}^1_{1812321613161316131} \textcolor{red}{3} \text{ [#=3]} \quad \Delta = -103, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$${}^1_{1812321613161316131} \textcolor{red}{3} \text{ [#=4]} \quad \Delta = -103, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$${}^1_{1812321613161316131} \textcolor{red}{2} \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$${}^1_{1812321613161316131} \textcolor{red}{2} \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$${}^1_{1812321613161316131} \textcolor{red}{2} \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$${}^1_{1812321613161316131} \textcolor{red}{2} \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$${}^1_{1812321613161316131} \textcolor{red}{2} \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$${}^{1,2}_{1812321613161316131} \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$${}^1_{1812321613161316131} \textcolor{red}{2} \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 168 & 26 \\ 2 & 26 & 8 \end{pmatrix}$$

$${}^1_{1812321613161316131} \textcolor{red}{3} \text{ [#=5]} \quad \Delta = -103, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$${}^1_{1812321613161316131} \textcolor{red}{3} \text{ [#=6]} \quad \Delta = -103, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$${}^1_{1812321613161316131} \textcolor{red}{3} \text{ [#=7]} \quad \Delta = -103, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$${}^1_{1812321613161316131} \textcolor{red}{3} \text{ [#=8]} \quad \Delta = -103, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$${}^1_{1812321613161316131} \textcolor{red}{3} \text{ [#=9]} \quad \Delta = -103, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$${}^1_{1812321613161316131} \textcolor{red}{3} \text{ [#=2]} \quad \Delta = -103, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

Base: ${}^1_{1812315132181232161}$

$${}^1_{1812315132181232161} \textcolor{red}{2} \quad \Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$${}^1_{1812315132181232161} \textcolor{red}{2} \quad \Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\begin{array}{ll}
{}^1_{1812315132181232161} \quad \Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
{}^1_{1812315132181232161} \quad \Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
{}^{12}_{1812315132181232161} \quad \Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
{}^1_{1812315132181232161} \quad \Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix}
\end{array}$$

Base: ${}^1_{1812321812315131613}$

$$\begin{array}{ll}
{}^1_{1812321812315131613} \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
{}^{12}_{1812321812315131613} \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^1_{1812321812315131613} \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
{}^1_{1812321812315131613} \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 144 & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
{}^1_{1812321812315131613} \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 196 & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
{}^1_{1812321812315131613} \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
{}^1_{1812321812315131613} \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{array}$$

Base: ${}^1_{4132181232171232181}$

$$\begin{array}{ll}
{}^{21}_{4132181232171232181} \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
{}^{21}_{4132181232171232181} \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
{}^{21}_{4132181232171232181} \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 163 & 25 \\ 2 & 25 & 8 \end{pmatrix} \\
{}^{21}_{4132181232171232181} \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
{}^{21}_{4132181232171232181} \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^{21}_{4132181232171232181} \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
{}^{21}_{4132181232171232181} \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{array}$$

Base: ${}^{21}_{612315132181232181}$

$$\begin{aligned}
6123151321812321\overset{221}{8}1 \quad \Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
6123151321812321\overset{2}{8}1^{21} \quad \Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
6123151321812321\overset{2}{8}1^{21} \quad \Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
6123151321812321\overset{2}{8}1^{21} \quad \Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
6123151321812321\overset{2}{8}1^{21} \quad \Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
6123151321812321\overset{21}{8}1 \quad \Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
6123151321812321\overset{2}{8}1^{21} \quad \Delta = -118, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{21}{181232181231513161}$

$$\begin{aligned}
\overset{21}{181232181231513161} \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
\overset{21}{181232181231513161} \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
\overset{21}{181232181231513161} \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
\overset{21}{181232181231513161} \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
\overset{21}{181232181231513161} \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\overset{21}{181232181231513161} \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\overset{212}{181232181231513161} \quad \Delta = -117, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{21}{181232171232181231}$

$$\begin{aligned}
\overset{21}{181232171232181231}\overset{3}{\text{[#}=4]} \quad \Delta = -103, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\overset{21}{181232171232181231}\overset{3}{\text{[#}=2]} \quad \Delta = -103, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
\overset{21}{181232171232181231}\overset{3}{\text{[#}=5]} \quad \Delta = -103, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\overset{212}{181232171232181231} \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{21}{1\ 8\ 1232171232181231} \quad \Delta = -95, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{21}{1\ 8\ 1232171232181231} \quad \Delta = -95, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{21}{1\ 8\ 1232171232181231} \quad \Delta = -95, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{21}{1\ 8\ 1232171232181231} \quad \Delta = -95, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& \overset{21}{1\ 8\ 1232171232181231} \quad \Delta = -95, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{21}{1\ 8\ 1232171232181231} \overset{2}{3} \text{ [#=3]} \quad \Delta = -103, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{21}{1\ 8\ 1232181232161315}$

$$\begin{aligned}
& \overset{21}{1\ 8\ 1232181232161315} \quad \Delta = -119, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{21}{1\ 8\ 1232181232161315} \quad \Delta = -119, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{21}{1\ 8\ 1232181232161315} \quad \Delta = -119, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{21}{1\ 8\ 1232181232161315} \quad \Delta = -119, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{21}{1\ 8\ 1232181232161315} \quad \Delta = -119, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{21}{1\ 8\ 1232181232161315} \quad \Delta = -119, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{21}{1\ 8\ 1232181232161315} \quad \Delta = -119, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{221}{1\ 8\ 123218123151321}$

$$\begin{aligned}
& \overset{221}{1\ 8\ 123218123151321} \quad \Delta = -40, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{221}{1\ 8\ 123218123151321} \quad \Delta = -40, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{221}{1\ 8\ 123218123151321} \quad \Delta = -40, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{221}{1\ 8\ 123218123151321} \overset{2}{3} \text{ [#=7]} \quad \Delta = -48, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
& \overset{221}{1\ 8\ 123218123151321} \overset{2}{3} \text{ [#=2]} \quad \Delta = -48, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{221}{1\ 8\ 123218123216131} \quad \Delta = -76, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{221}{1\ 8\ 123218123216131\textcolor{red}{3}}_{[\# = 3]} \quad \Delta = -75, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{221}{1\ 8\ 123218123216131\textcolor{red}{3}}_{[\# = 8]} \quad \Delta = -75, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{221}{1\ 8\ 123218123216131\textcolor{red}{3}}_{[\# = 9]} \quad \Delta = -75, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{221}{1\ 8\ 123218123216131\textcolor{red}{3}}_{[\# = 5]} \quad \Delta = -75, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{221}{1\ 8\ 123218123216131\textcolor{red}{3}}_{[\# = 2]} \quad \Delta = -75, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{221}{1\ 8\ 123218123217122}$

$$\begin{aligned}
& \overset{221}{1\ 8\ 123218123217122} \quad \Delta = -86, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{221\textcolor{red}{2}}{1\ 8\ 123218123217122} \quad \Delta = -86, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{221}{1\ 8\ 1232181\textcolor{red}{2}23217122} \quad \Delta = -86, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{221}{1\ 8\ 1232181\textcolor{red}{2}23217122} \quad \Delta = -86, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{221}{1\ 8\ 123218123217122} \quad \Delta = -95, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{221}{1\ 8\ 123218123217122\textcolor{red}{3}} \quad \Delta = -95, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
& \overset{221}{1\ 8\ 123218123217122} \quad \Delta = -86, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{221}{1\ 8\ 123218123217122\textcolor{red}{2}} \quad \Delta = -86, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{21}{1\ 7\ 12321812321}\overset{221}{8\ 1}$

$$\begin{aligned}
& \overset{21}{1\ 7\ 12321812321}\overset{221}{8\ 1} \quad \Delta = -86, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{21\textcolor{red}{2}}{1\ 7\ 12321812321}\overset{221}{8\ 1} \quad \Delta = -86, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{21}{1\ 7\ 12321812321}\overset{221}{8\ 1} \quad \Delta = -86, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{21}{1\ 7\ 12321812321}\overset{221}{8\ 1} \quad \Delta = -95, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
& \overset{21}{1\ 7\ 12321812321}\overset{221}{8\ 1} \quad \Delta = -86, \ n_+ = 1, \ n_- = 20, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{l} \overset{21}{1}712321812321\overset{221}{8}1 \quad \Delta = -86, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \end{array}$$

$$\begin{array}{l} \overset{21}{1}712321812321\overset{221}{8}1 \quad \Delta = -86, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \end{array}$$

Base: 613151321812321³²¹81

$$\begin{array}{l} \overset{2}{6}13151321812321\overset{321}{8}1 \quad \Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \end{array}$$

$$\begin{array}{l} 613151321812321\overset{321}{8}1 \quad \Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \end{array}$$

$$\begin{array}{l} \overset{2}{6}13151321812321\overset{321}{8}1 \quad \Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \end{array}$$

$$\begin{array}{l} 613151321812321\overset{321}{8}1\overset{2}{2} \quad \Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \end{array}$$

$$\begin{array}{l} \overset{2}{6}13151321812321\overset{321}{8}1 \quad \Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \end{array}$$

$$\begin{array}{l} 613151321812321\overset{2321}{8}1 \quad \Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \end{array}$$

$$\begin{array}{l} \overset{2}{6}13151321812321\overset{321}{8}1 \quad \Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \end{array}$$

$$\begin{array}{l} 613151321812321\overset{2}{8}1\overset{321}{1} \quad \Delta = -126, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \end{array}$$

Base: 2218123218123218122¹

$$\begin{array}{l} 22181\overset{2}{2}321\overset{1}{8}123218122 \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \end{array}$$

$$\begin{array}{l} \overset{3}{2}218123218123218122 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \end{array}$$

$$\begin{array}{l} 221812321\overset{21}{8}123218122 \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \end{array}$$

$$\begin{array}{l} 22181\overset{2}{2}321\overset{1}{8}123218122 \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \end{array}$$

$$\begin{array}{l} 221812321\overset{21}{8}123218122 \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \end{array}$$

Base: 2181232181232181223¹

$$\begin{array}{l} 21812321\overset{21}{8}1232181223 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \end{array}$$

$$\begin{array}{l} 21812321\overset{1}{8}12321\overset{2}{8}1223 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \end{array}$$

$$\begin{array}{l} 21812321\overset{12}{8}1232181223 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \end{array}$$

$$21812321\overset{2}{8}1232181223 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$21812321\overset{2}{8}1232181223 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$21812321\overset{1}{8}12321\overset{2}{8}1223 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$21812321\overset{21}{8}1232181223 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\text{Base: } 1 \overset{12321}{8} 1232181231512$$

$$1 \overset{312321}{8} 1232181231512 \text{ [#=9]} \quad \Delta = -50, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix}$$

$$1 \overset{312321}{8} 1232181231512 \text{ [#=2]} \quad \Delta = -50, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1 \overset{312321}{8} 1232181231512 \text{ [#=8]} \quad \Delta = -50, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix}$$

$$1 \overset{12321}{8} 1232181231\overset{3}{5}12 \quad \Delta = -50, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$1 \overset{312321}{8} 1232181231512 \text{ [#=7]} \quad \Delta = -50, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix}$$

$$1 \overset{312321}{8} 1232181231512 \text{ [#=5]} \quad \Delta = -50, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix}$$

$$1 \overset{312321}{8} 1232181231512 \text{ [#=6]} \quad \Delta = -50, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

$$1 \overset{312321}{8} 1232181231512 \text{ [#=4]} \quad \Delta = -50, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$1 \overset{312321}{8} 1232181231512 \text{ [#=3]} \quad \Delta = -50, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1 \overset{12321}{8} 1232181231\overset{2}{5}12 \quad \Delta = -42, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{12321}{8} 1232181231\overset{2}{5}12 \quad \Delta = -42, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{12321}{8} 1232181231\overset{2}{5}12 \quad \Delta = -42, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{12321}{8} 1232181231\overset{2}{5}12 \quad \Delta = -42, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1 \overset{12321}{8} 1232181231512 \quad \Delta = -42, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1 \overset{12321}{8} 1232181231\overset{2}{5}12 \quad \Delta = -42, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: $1 \overset{1321}{8} 12321812315131$

$$1 \overset{1321}{8} 12321812315131 \overset{2}{3} [\#=3] \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1 \overset{1321}{8} 12321812315131 \quad \Delta = -48, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{1321}{8} 12321812315131 \overset{2}{8} \quad \Delta = -48, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{1321}{8} 12321812315131 \overset{2}{13} \quad \Delta = -48, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{1321}{8} 12321812315131 \quad \Delta = -48, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1 \overset{1321}{8} 12321812315131 \overset{2}{13} \quad \Delta = -48, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1 \overset{1321}{8} 12321812315131 \overset{2}{3} [\#=9] \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$1 \overset{1321}{8} 12321812315131 \overset{2}{6} \quad \Delta = -48, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{31321}{8} 12321812315131 [\#=3] \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1 \overset{1321}{8} 12321812315131 \overset{2}{3} [\#=2] \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{31321}{8} 12321812315131 [\#=4] \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1 \overset{1321}{8} 12321812315131 \overset{2}{3} [\#=8] \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1 \overset{1321}{8} 12321812315131 \overset{2}{3} [\#=7] \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$1 \overset{31321}{8} 12321812315131 [\#=2] \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{1321}{8} 12321812315131 \overset{2}{3} [\#=5] \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1 \overset{31321}{8} 12321812315131 [\#=5] \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$1 \overset{1321}{8} 12321812315131 \overset{2}{3} [\#=4] \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{1321}{8} 12321812315131 \overset{2}{3} [\#=6] \quad \Delta = -56, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: $1 \overset{12321}{8} \overset{1}{123218123151}$

$$\begin{aligned}
& \overset{12321}{1} \overset{21}{8} 123218123151 \quad \Delta = -42, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{12321}{1} \overset{2}{8} 123218123151 \quad \Delta = -42, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{12321}{1} \overset{2}{8} 123218123151 \quad \Delta = -42, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{312321}{1} \overset{1}{8} 123218123151 \text{ [#=2]} \quad \Delta = -50, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{312321}{1} \overset{1}{8} 123218123151 \text{ [#=3]} \quad \Delta = -50, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{312321}{1} \overset{1}{8} 123218123151 \text{ [#=4]} \quad \Delta = -50, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& \overset{312321}{1} \overset{1}{8} 123218123151 \text{ [#=5]} \quad \Delta = -50, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
& \overset{312321}{1} \overset{1}{8} 123218123151 \text{ [#=9]} \quad \Delta = -50, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
& \overset{312321}{1} \overset{1}{8} 123218123151 \text{ [#=8]} \quad \Delta = -50, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
& \overset{312321}{1} \overset{1}{8} 123218123151 \text{ [#=7]} \quad \Delta = -50, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
& \overset{312321}{1} \overset{1}{8} 123218123151 \text{ [#=6]} \quad \Delta = -50, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
& \overset{12321}{1} \overset{1}{8} 123218123151 \quad \Delta = -42, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{12321}{1} \overset{2}{8} 123218123151 \quad \Delta = -42, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{6}123218123218123151$

$$\begin{aligned}
& \overset{2}{6}12321 \overset{1}{8}123218123151 \quad \Delta = -120, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 612321 \overset{21}{8}123218123151 \quad \Delta = -120, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 612321 \overset{1}{8} \overset{2}{123218123151} \quad \Delta = -120, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& 612321 \overset{12}{8}123218123151 \quad \Delta = -120, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& 612321 \overset{21}{8}123218123151 \quad \Delta = -120, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& 612321 \overset{1}{8}1232181 \overset{2}{23151} \quad \Delta = -120, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

$$6123218123218123151 \quad \Delta = -120, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

Base: 4132181232181232171

$$4132181232181232171 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$4132181232181232171 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$4132181232181232171 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$4132181232181232171 \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$4132181232181232171 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$4132181232181232171 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$4132181232181232171 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$4132181232181232171 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 2171232181232181231

$$2171232181232181231 \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$2171232181232181231 \text{ } \# = 4 \quad \Delta = -103, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$2171232181232181231 \text{ } \# = 2 \quad \Delta = -103, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$2171232181232181231 \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$2171232181232181231 \text{ } \# = 3 \quad \Delta = -103, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$2171232181232181231 \text{ } \# = 5 \quad \Delta = -103, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$2171232181232181231 \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$2171232181232181231 \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$2171232181232181231 \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\begin{aligned}
2171232181232181231 \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
2171232181232181231 \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

Base: 171232181232181231

$$\begin{aligned}
171232181232181231 \text{ \textcolor{red}{3} } [\# = 5] \quad \Delta = -103, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
171232181232181231 \text{ \textcolor{red}{3} } [\# = 2] \quad \Delta = -103, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
171232181232181231 \text{ \textcolor{red}{3} } [\# = 3] \quad \Delta = -103, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
171232181232181231 \text{ \textcolor{red}{3} } [\# = 4] \quad \Delta = -103, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
171232181232181231 \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
171232181232181231 \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
171232181232181231 \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
171232181232181231 \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
171232181232181231 \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
171232181232181231 \quad \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1^{12321} 8^{12321} 1232181232151$

$$\begin{aligned}
1^{12321} 8^{12321} 1232181232151 \quad \Delta = -44, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1^{312321} 8^{12321} 1232181232151 [\# = 4] \quad \Delta = -52, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1^{312321} 8^{12321} 1232181232151 [\# = 7] \quad \Delta = -52, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
1^{312321} 8^{12321} 1232181232151 [\# = 6] \quad \Delta = -52, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
1^{312321} 8^{12321} 1232181232151 [\# = 5] \quad \Delta = -52, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
1^{312321} 8^{12321} 1232181232151 [\# = 8] \quad \Delta = -52, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1 \overset{12321}{8} \overset{2}{1232181232151} \quad \Delta = -44, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{12321}{8} \overset{2}{1232181232151} \quad \Delta = -44, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1 \overset{12321}{8} \overset{2}{1232181232151} \quad \Delta = -44, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \overset{312321}{8} 1232181232151 \text{ [#=2]} \quad \Delta = -52, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1 \overset{312321}{8} 1232181232151 \text{ [#=3]} \quad \Delta = -52, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1 \overset{12321}{8} 1232181232151 \quad \Delta = -44, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1 \overset{312321}{8} 1232181232151 \text{ [#=9]} \quad \Delta = -52, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix}
\end{aligned}$$

Base: 6123218123218123215

$$\begin{aligned}
6123218123218123215 \overset{2}{1} \quad \Delta = -122, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
6123218123218123215 \overset{1}{2} \quad \Delta = -122, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 \quad & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
6123218123218123215 \overset{1}{2} \quad \Delta = -122, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
6123218123218123215 \overset{21}{8} \quad \Delta = -122, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
6123218123218123215 \overset{1}{2} \quad \Delta = -122, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
6123218123218123215 \overset{2}{1} \quad \Delta = -122, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
6123218123218123215 \overset{21}{8} \quad \Delta = -122, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 100 \quad & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1612321812321812321

$$\begin{aligned}
1612321812321812321 \overset{1}{3} \text{ [#=9]} \quad \Delta = -78, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
1612321812321812321 \overset{1}{3} \text{ [#=2]} \quad \Delta = -78, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1612321812321812321 \overset{1}{2} \quad \Delta = -70, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1612321812321812321 \overset{12}{8} \quad \Delta = -70, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1612321\overset{21}{8}12321812321 \quad \Delta = -70, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1612321\overset{2}{8}12321812321 \quad \Delta = -70, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1612321\overset{1}{8}12321812321\overset{3}{\text{[#}=3]} \quad \Delta = -78, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1612321\overset{1}{8}12321812321\overset{3}{\text{[#}=4]} \quad \Delta = -78, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1612321\overset{1}{8}12321812321\overset{3}{\text{[#}=5]} \quad \Delta = -78, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1612321\overset{1}{8}12321812321\overset{3}{\text{[#}=6]} \quad \Delta = -78, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1612321\overset{1}{8}12321812321\overset{3}{\text{[#}=7]} \quad \Delta = -78, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
1612321\overset{1}{8}12321812321\overset{3}{\text{[#}=8]} \quad \Delta = -78, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1612321\overset{21}{8}12321812321 \quad \Delta = -70, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1612321\overset{1}{8}12321812321\overset{2}{\text{[#}=9]} \quad \Delta = -70, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{51321}{8} 1232181232161 & \\
1 \overset{51321}{8} 1232181232161 \quad \Delta = -120, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
51321\overset{21}{8}1232181232161 \quad \Delta = -120, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{51321}{8} 1232181232161 \quad \Delta = -120, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 100 \quad & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 168 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
1 \overset{51321}{8} 12321812321\overset{2}{61} \quad \Delta = -120, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1 \overset{51321}{8} 12321812321\overset{2}{61} \quad \Delta = -120, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
1 \overset{51321}{8} 12321812321\overset{2}{61} \quad \Delta = -120, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 \quad & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 196 & 28 \\ 2 & 28 & 8 \end{pmatrix} \\
1 \overset{51321}{8} 12321812321\overset{2}{61} \quad \Delta = -120, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{1321}{8} 12321812321612 & \\
1 \overset{31321}{8} 12321812321612 \text{ [#}=5] \quad \Delta = -76, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{1321}{1} \overset{3}{8} 12321812321612 \quad \Delta = -76, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
& \overset{31321}{1} \overset{3}{8} 12321812321612 \text{ [#=4]} \quad \Delta = -76, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& \overset{31321}{1} \overset{3}{8} 12321812321612 \text{ [#=2]} \quad \Delta = -76, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{1321}{1} \overset{2}{8} 12321812321612 \quad \Delta = -68, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{1321}{1} \overset{2}{8} 12321812321612 \quad \Delta = -68, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{1321}{1} \overset{2}{8} 12321812321612 \quad \Delta = -68, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{1321}{1} \overset{2}{8} 12321812321612 \quad \Delta = -68, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{1321}{1} \overset{2}{8} 12321812321612 \quad \Delta = -68, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{1321}{1} \overset{2}{8} 12321812321612 \quad \Delta = -68, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{31321}{1} \overset{3}{8} 12321812321612 \text{ [#=3]} \quad \Delta = -76, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \text{Base: } \overset{1}{16} \overset{1}{1232181232181231} \\
& \overset{1}{16} \overset{2}{1232181232181231} \quad \Delta = -68, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{1}{16} \overset{2}{1232181232181231} \quad \Delta = -68, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{1}{16} \overset{2}{1232181232181231} \quad \Delta = -68, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{1}{16} \overset{2}{1232181232181231} \quad \Delta = -68, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{1}{16} \overset{2}{1232181232181231} \quad \Delta = -68, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{1}{16} \overset{1}{1232181232181231} \overset{3}{3} \text{ [#=5]} \quad \Delta = -76, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{1}{16} \overset{1}{1232181232181231} \overset{3}{3} \text{ [#=4]} \quad \Delta = -76, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{1}{16} \overset{1}{1232181232181231} \overset{3}{3} \text{ [#=3]} \quad \Delta = -76, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{1}{16} \overset{1}{1232181232181231} \overset{3}{3} \text{ [#=2]} \quad \Delta = -76, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{1}812321\overset{1}{8}1232181223$

$$\begin{aligned}
\overset{1}{1}812321\overset{1}{8}12321\overset{2}{8}1223 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
\overset{1}{1}812321\overset{21}{8}1232181223 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\overset{1}{1}812321\overset{1}{8}12321\overset{2}{8}1223 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\overset{1}{1}812321\overset{21}{8}1232181223 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
\overset{1}{1}812321\overset{12}{8}1232181223 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
\overset{12}{1}812321\overset{1}{8}1232181223 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{21}{1}812321\overset{1}{8}123218122$

$$\begin{aligned}
\overset{21}{1}812321\overset{12}{8}123218122 \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\overset{21}{1}812321\overset{1}{8}12321\overset{2}{8}122 \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\overset{212}{1}812321\overset{1}{8}123218122 \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\overset{21}{1}812321\overset{21}{8}123218122 \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\overset{21}{1}812321\overset{21}{8}123218122 \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\overset{21}{1}812321\overset{1}{8}12321\overset{2}{8}122 \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\overset{21}{1}812321\overset{1}{8}123218122 \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\overset{21}{1}812321\overset{1}{8}12321\overset{3}{8}122 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}
\end{aligned}$$

Base: $\overset{321}{1}812321\overset{1}{8}12321812$

$$\begin{aligned}
\overset{3212}{1}812321\overset{1}{8}12321812 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
\overset{321}{1}812321\overset{21}{8}12321812 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
\overset{321}{1}812321\overset{12}{8}12321812 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
\overset{321}{1}812321\overset{1}{8}12321\overset{2}{8}12 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

$$321 \overset{21}{8} 12321 \overset{1}{8} 12321812 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\overset{321}{1} \overset{1}{8} 12321812321812 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\overset{321}{1} \overset{1}{8} 12321812321 \overset{2}{8} 12 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\overset{321}{1} \overset{21}{8} 12321812321812 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: $\overset{2321}{1} \overset{1}{8} 1232181232181$

$$\overset{2321}{1} \overset{2}{8} 1232181232181 \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix}$$

$$\overset{2321}{1} \overset{21}{8} 1232181232181 \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 225 \quad \begin{pmatrix} 0 & 15 & 2 \\ 15 & 163 & 25 \\ 2 & 25 & 8 \end{pmatrix}$$

$$\overset{2321}{1} \overset{1}{8} 12321812321 \overset{2}{8} 12 \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$\overset{2321}{1} \overset{21}{8} 1232181232181 \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\overset{2321}{1} \overset{1}{8} 1232181232181 \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$\overset{2321}{1} \overset{21}{8} 1232181232181 \quad \Delta = -125, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

Base: $\overset{1}{18} 12321 \overset{1}{8} 12321 \overset{1}{8} 123$

$$\overset{1}{18} 12321 \overset{21}{8} 12321 \overset{1}{8} 123 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$\overset{1}{18} 12321 \overset{1}{8} 12321 \overset{21}{8} 123 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\overset{1}{18} 12321 \overset{21}{8} 12321 \overset{1}{8} 123 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\overset{1}{18} 12321 \overset{12}{8} 12321 \overset{1}{8} 123 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$\overset{1}{18} 12321 \overset{1}{8} 12321 \overset{21}{8} 123 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$\overset{12}{18} 12321 \overset{1}{8} 12321 \overset{1}{8} 123 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\overset{1}{18} 12321 \overset{1}{8} 12321 \overset{12}{8} 123 \quad \Delta = -123, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: $\overset{21}{18} 12321 \overset{1}{8} 12321 \overset{21}{8} 1$

$$\overset{21}{18} 12321 \overset{21}{8} 12321 \overset{21}{8} 1 \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$\begin{aligned}
& \overset{21}{1} \overset{21}{8} 12321 \overset{21}{8} 12321 \overset{21}{8} 1 \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{21}{1} \overset{1}{8} 12321 \overset{21}{8} 12321 \overset{21}{8} 1 \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{21}{1} \overset{21}{8} 12321 \overset{1}{8} 12321 \overset{21}{8} 1 \quad \Delta = -114, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{21}{21} \overset{21}{8} 12321 \overset{21}{8} 123151321$

$$\begin{aligned}
& \overset{21}{21} \overset{21}{8} 12321 \overset{21}{8} 123151321 \overset{21}{3} \text{ [#=8]} \quad \Delta = -48, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& \overset{21}{21} \overset{21}{8} 12321 \overset{21}{8} 123151321 \overset{21}{3} \text{ [#=9]} \quad \Delta = -48, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
& \overset{21}{21} \overset{21}{8} 12321 \overset{21}{8} 123151321 \overset{21}{3} \text{ [#=7]} \quad \Delta = -48, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
& \overset{21}{21} \overset{21}{8} 12321 \overset{21}{8} 123151321 \overset{21}{3} \text{ [#=6]} \quad \Delta = -48, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{21}{21} \overset{21}{8} 12321 \overset{21}{8} 123151321 \overset{21}{3} \text{ [#=5]} \quad \Delta = -48, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{21}{21} \overset{21}{8} 12321 \overset{21}{8} 123151321 \overset{21}{3} \text{ [#=4]} \quad \Delta = -48, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{21}{21} \overset{21}{8} 12321 \overset{21}{8} 123151321 \overset{21}{3} \text{ [#=3]} \quad \Delta = -48, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{21}{21} \overset{21}{8} 12321 \overset{21}{8} 123151321 \overset{21}{3} \text{ [#=2]} \quad \Delta = -48, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{21}{21} \overset{21}{8} 12321 \overset{21}{8} 123151321 \overset{21}{3} \text{ [#=1]} \quad \Delta = -48, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 1 \quad \begin{pmatrix} 0 & 1 & 1 \\ 1 & 18 & 5 \\ 1 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{21}{21} \overset{21}{8} 12321 \overset{21}{8} 123216131$

$$\overset{21}{21} \overset{21}{8} 12321 \overset{21}{8} 123216131 \overset{21}{3} \text{ [#=3]} \quad \Delta = -75, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$\begin{array}{ll}
21\bar{8}1232181\overset{2}{2}3216131 & \Delta = -67, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21\bar{8}123218123216131\overset{21}{3}_{[\# = 7]} & \Delta = -75, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21\bar{8}123218123216131\overset{21}{3}_{[\# = 9]} & \Delta = -75, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21\bar{8}123218123216131\overset{21}{3}_{[\# = 8]} & \Delta = -75, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21\bar{8}123218123216131\overset{21}{3}_{[\# = 6]} & \Delta = -75, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
21\bar{8}123218123216131\overset{21}{3}_{[\# = 5]} & \Delta = -75, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21\bar{8}123218123216131\overset{21}{3}_{[\# = 4]} & \Delta = -75, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21\bar{8}123218123216131\overset{21}{3}_{[\# = 2]} & \Delta = -75, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
21\bar{8}123218123216131\overset{21}{3} & \Delta = -67, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\overset{221}{21}\bar{8}123218123216131 & \Delta = -67, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21\bar{8}123218123216131\overset{21}{3} & \Delta = -67, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
21\bar{8}123218123216131\overset{21}{3} & \Delta = -67, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
21\bar{8}123218123216131\overset{21}{3} & \Delta = -67, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\text{Base: } 21\bar{8}123218123217122 & \\
21\bar{8}123218123217122\overset{21}{3} & \Delta = -95, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
21\bar{8}123218123217122\overset{21}{3} & \Delta = -86, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
21\bar{8}123218123217122\overset{21}{3} & \Delta = -86, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
21\bar{8}123218123217122\overset{21}{3} & \Delta = -86, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21\bar{8}123218123217122\overset{21}{3} & \Delta = -86, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\overset{221}{21}\bar{8}123218123217122 & \Delta = -86, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{array}$$

$$^{21}218123218123217\overset{2}{1}22 \quad \Delta = -86, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: $^{21}1712321812321\overset{21}{8}12$

$$^{21}1712321812321\overset{221}{8}12 \quad \Delta = -86, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$^{21}\overset{2}{1}712321812321\overset{21}{8}12 \quad \Delta = -86, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$^{21}1712321812321\overset{2}{8}12 \quad \Delta = -86, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$^{21}1712321812321\overset{212}{8}12 \quad \Delta = -86, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$^{21}1712321812321\overset{21}{8}12 \quad \Delta = -86, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$^{21}1712321812321\overset{2}{8}12 \quad \Delta = -86, n_+ = 1, n_- = 20, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$T = 21$ (2520 bases)

Base: 1212231414141414141413

$$1212231414141414\overset{2}{1}4141413 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$121223141414\overset{2}{1}41414141413 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$1212231414141414141414\overset{2}{1}3 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$^{21}121223141414141414141413 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 130 & 22 \\ 2 & 22 & 8 \end{pmatrix}$$

$$121223\overset{2}{1}41414141414141413 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 289 \quad \begin{pmatrix} 0 & 17 & 2 \\ 17 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

$$12122314\overset{2}{1}414141414141413 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 225 \quad \begin{pmatrix} 0 & 15 & 2 \\ 15 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$121223141414\overset{2}{1}41414141413 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$1212231414141414\overset{2}{1}4141413 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$121223141414141414\overset{2}{1}413 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1212231414141414141414\overset{3}{1}3 \quad \Delta = -133, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: 1414141414141413221121

$$11212231414141414141\textcolor{red}{3}_{[\# = 3]} \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$11212231414141414141\textcolor{red}{3}_{[\# = 2]} \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1414141414141413221121 \quad \Delta = -117, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$141414141414141413221121 \quad \Delta = -117, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$141414141414141\overset{2}{3}221121 \quad \Delta = -117, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 225 \quad \begin{pmatrix} 0 & 15 & 2 \\ 15 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix}$$

[illegible]

$$14141\overset{2}{4}1414141413221121 \quad \Delta = -117, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1414\overset{2}{1}41414141413221121 \quad \Delta = -117, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$141\overset{2}{4}141414141413221121 \quad \Delta = -117, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1414141\overset{2}{4}14141413221121 \quad \Delta = -117, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

Base: 2231513161315132211212

$$2231513161315132211212 \quad \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 196 \quad \begin{pmatrix} 0 & 14 & 2 \\ 14 & 236 & 30 \\ 2 & 30 & 8 \end{pmatrix}$$

$$2231513161315132211212^2 \quad \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$2231513161315132211212 \quad \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 256 \quad \begin{pmatrix} 0 & 16 & 2 \\ 16 & 302 & 34 \\ 2 & 34 & 8 \end{pmatrix}$$

$$2231513161315132211212 \quad \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 289 \quad \begin{pmatrix} 0 & 17 & 2 \\ 17 & 338 & 36 \\ 2 & 36 & 8 \end{pmatrix}$$

$$2231513161315132211212 \quad \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix}$$

$$2231513161315132211212 \quad \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 108 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$2231513161315132211212 \quad \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$2231513161315132211212 \quad \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 206 & 28 \\ 2 & 28 & 8 \end{pmatrix}$$

Base: 1414141322112122315131

$$\begin{aligned}
1414141322112122315131^2 \quad \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1414141322112122315131^3_{[\# = 7]} \quad \Delta = -82, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1414141322112122315131^3_{[\# = 4]} \quad \Delta = -82, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1414141322112122315131^2 \quad \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1414141322112122315131^2 \quad \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1414141322112122315131^2 \quad \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1414141322112122315131^2 \quad \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1414141322112122315131^2 \quad \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1414141322112122315131^2 \quad \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1315132211212231414141^3_{[\# = 3]} \quad \Delta = -82, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1414141322112122315131^3_{[\# = 5]} \quad \Delta = -82, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1414141322112122315131^3_{[\# = 6]} \quad \Delta = -82, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1414141322112122315131^3_{[\# = 2]} \quad \Delta = -82, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1414141322112122315131^3_{[\# = 3]} \quad \Delta = -82, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1315132211212231414141^3_{[\# = 2]} \quad \Delta = -82, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1414141322112122315131^3_{[\# = 9]} \quad \Delta = -82, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1414141322112122315131^3_{[\# = 8]} \quad \Delta = -82, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

Base: 4131513221121223151314

$$\begin{aligned}
4131513221121223151314^2 \quad \Delta = -107, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
4131513221121223151314^2 \quad \Delta = -106, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
4131513221121223151314 \quad \Delta = -106, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
4131513221121223151314 \quad \Delta = -106, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
4131513221121223151314 \quad \Delta = -114, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
4131513221121223151314 \quad \Delta = -106, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 3141413221121223151321

$$\begin{aligned}
3141413221121223151321 \textcolor{red}{3} [\# = 3] \quad \Delta = -63, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
3141413221121223151321 \textcolor{red}{3} [\# = 2] \quad \Delta = -63, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
3141413221121223151321 \textcolor{red}{3} [\# = 4] \quad \Delta = -63, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
3141413221121223151321 \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
3141413221121223151321 \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
3141413221121223151321 \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
3141413221121223151321 \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
3141413221121223151321 \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
3141413221121223151321 \textcolor{red}{3} [\# = 9] \quad \Delta = -63, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
3141413221121223151321 \textcolor{red}{3} [\# = 8] \quad \Delta = -63, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
3141413221121223151321 \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
3141413221121223151321 \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
3141413221121223151321 \textcolor{red}{3} [\# = 6] \quad \Delta = -63, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
3141413221121223151321 \textcolor{red}{3} [\# = 7] \quad \Delta = -63, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
3141413221121223151321 \quad \Delta = -63, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
3141413221121223151321\overset{2}{3} \text{ [#}=5] \quad \Delta = -63, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\text{Base: } 1231513221121223151232 & \\
2321513221121223151321\overset{2}{3} \text{ [#}=3] \quad \Delta = -62, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
2321513221121223151321\overset{2}{3} \text{ [#}=7] \quad \Delta = -62, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
2321513221121223151321\overset{2}{3} \text{ [#}=4] \quad \Delta = -62, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
2321513221121223151321\overset{2}{3} \text{ [#}=8] \quad \Delta = -62, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
2321513221121223151321\overset{2}{3} \text{ [#}=6] \quad \Delta = -62, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
1231513221\overset{2}{1}121223151232 \quad \Delta = -54, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
2321513221121223151321\overset{2}{3} \text{ [#}=5] \quad \Delta = -62, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
2321513221121223151321\overset{2}{3} \text{ [#}=9] \quad \Delta = -62, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
2321513221121223151321\overset{2}{3} \text{ [#}=2] \quad \Delta = -62, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
123151322112122315\overset{2}{1}232 \quad \Delta = -54, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1231513221121\overset{2}{2}223151232 \quad \Delta = -54, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
123151322112122315\overset{2}{1}232 \quad \Delta = -54, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
12315\overset{2}{1}3221121223151232 \quad \Delta = -54, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
12315\overset{2}{1}3221121223151232 \quad \Delta = -54, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\text{Base: } 2314132211212231513221 & \\
2314132211212231513221\overset{2}{3} \text{ [#}=6] \quad \Delta = -36, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
2314132211212231513221\overset{2}{3} \text{ [#}=8] \quad \Delta = -36, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
2314132211212231513221\overset{2}{3} \text{ [#}=5] \quad \Delta = -36, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
23141\overset{2}{3}2211212231513221 \quad \Delta = -28, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
23141\overset{2}{3}2211212231513221 \quad \Delta = -30, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2314132211212231513221\overset{2}{3} \text{ [#=4]} \quad \Delta = -36, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
2314132211212231513221\overset{2}{3} \text{ [#=7]} \quad \Delta = -36, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
2314132211212231513221\overset{2}{3} \text{ [#=9]} \quad \Delta = -36, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
2314132211212231513221\overset{2}{3} \text{ [#=2]} \quad \Delta = -36, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
2314132211212231513221 \quad \Delta = -28, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
231413221121\overset{2}{2}2231513221 \quad \Delta = -28, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
2314132211212231513221\overset{2}{3} \text{ [#=3]} \quad \Delta = -36, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
2314132211212231513221 \quad \Delta = -28, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2314132211212231513221 \quad \Delta = -28, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
23141\overset{2}{3}2211212231513221 \quad \Delta = -28, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\text{Base: } 1414132211212231513215 & \\
5123151322112122314141\overset{2}{3} \text{ [#=2]} \quad \Delta = -107, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
5123151322112122314141\overset{2}{3} \text{ [#=3]} \quad \Delta = -107, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
141413221121\overset{2}{2}2231513215 \quad \Delta = -99, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1414132211212231513215 \quad \Delta = -99, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
14141\overset{2}{3}2211212231513215 \quad \Delta = -99, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
141413221121\overset{2}{2}2231513215 \quad \Delta = -99, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1414132211212231513215 \quad \Delta = -99, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1414132211212231513215^2 \quad \Delta = -99, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1414132211212231513215^2 \quad \Delta = -99, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 3215132211212231513215

$$\begin{aligned}
3215132211212231513215^2 \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
3215132211212231513215^2 \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
3215132211212231513215^2 \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
3215132211212231513215^2 \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
3215132211212231513215^2 \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
3215132211212231513215^2 \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
3215132211212231513215^2 \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 4141413221121223151315

$$\begin{aligned}
4141413221121223151315^2 \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
4141413221121223151315^2 \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
4141413221121223151315^2 \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
4141413221121223151315^2 \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
4141413221121223151315^2 \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
4141413221121223151315^2 \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
4141413221121223151315^2 \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
4141413221121223151315^2 \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 144 & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
4141413221121223151315^2 \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 89 & 19 \\ 2 & 19 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1414132211212231513151

$$\begin{array}{ll}
1513151322112122314141\textcolor{red}{3} \text{ [#=2]} & \Delta = -106, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1513151322112122314141\textcolor{red}{3} \text{ [#=3]} & \Delta = -106, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
14141\textcolor{red}{2}32211212231513151 & \Delta = -98, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
141413221121\textcolor{red}{2}2231513151 & \Delta = -98, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1414132211212231\textcolor{red}{5}13151 & \Delta = -98, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
141413221121223151\textcolor{red}{3}151 & \Delta = -98, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
141413221121223151\textcolor{red}{3}151 & \Delta = -98, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
14141\textcolor{red}{2}32211212231513151 & \Delta = -98, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
141413221121\textcolor{red}{2}2231513151 & \Delta = -98, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
141413221121223151\textcolor{red}{3}151 & \Delta = -99, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\textbf{Base: } 3215132211212231513151 & \\
321513221121223151\textcolor{red}{3}151 & \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
321\textcolor{red}{5}132211212231513151 & \Delta = -103, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
3215132211212231\textcolor{red}{5}13151 & \Delta = -103, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
321\textcolor{red}{5}132211212231513151 & \Delta = -103, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
321513221121\textcolor{red}{2}2231513151 & \Delta = -103, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
321513221121223151\textcolor{red}{3}151 & \Delta = -103, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
321513221121\textcolor{red}{2}2231513151 & \Delta = -103, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
3215132211212231\textcolor{red}{5}13151 & \Delta = -103, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\textbf{Base: } 2151322112122315131513 &
\end{array}$$

$$\begin{array}{ll}
2151322112122315131513 & \Delta = -102, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
215132211212231513^21513 & \Delta = -103, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
215132211212231513^21513 & \Delta = -102, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
215132211212231513^21513 & \Delta = -102, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
215132211212231513^21513 & \Delta = -102, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
215132211212231513^21513 & \Delta = -102, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
2151322112122315131513^3 & \Delta = -110, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2151322112122315131513^2 & \Delta = -102, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
2151322112122315131513^2 & \Delta = -102, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
2151322112122315131513^2 & \Delta = -102, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{array}$$

Base: $1 \begin{smallmatrix} 3151315132211212231 \\ 5 \end{smallmatrix} 1$

$$\begin{array}{ll}
1 \begin{smallmatrix} 3151315132211212231 \\ 5 \end{smallmatrix} 1 & \Delta = -103, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \begin{smallmatrix} 3151315132211212231 \\ 5 \end{smallmatrix} 1 & \Delta = -102, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1 \begin{smallmatrix} 3151315132211212231 \\ 5 \end{smallmatrix} 1 & \Delta = -102, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \begin{smallmatrix} 3151315132211212231 \\ 5 \end{smallmatrix} 1 & \Delta = -102, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1 \begin{smallmatrix} 3151315132211212231 \\ 5 \end{smallmatrix} 1 & \Delta = -102, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1 \begin{smallmatrix} 3151315132211212231 \\ 5 \end{smallmatrix} 1 & \Delta = -102, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1 \begin{smallmatrix} 3151315132211212231 \\ 5 \end{smallmatrix} 1 & \Delta = -102, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1 \begin{smallmatrix} 3151315132211212231 \\ 5 \end{smallmatrix} 1 & \Delta = -102, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{array}$$

Base: $31 \begin{smallmatrix} 512315132211212231 \\ 5 \end{smallmatrix} 1$

Base: 12315132211212231¹5132

$$231 \begin{smallmatrix} \textcolor{red}{3}12315132211212231 \\ 5 \end{smallmatrix} 1 \text{ [#}=7] \quad \Delta = -60, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix}$$

$$231 \begin{smallmatrix} \textcolor{red}{3}12315132211212231 \\ 5 \end{smallmatrix} 1 \text{ [#}=9] \quad \Delta = -60, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix}$$

$$231 \begin{smallmatrix} \textcolor{red}{3}12315132211212231 \\ 5 \end{smallmatrix} 1 \text{ [#}=8] \quad \Delta = -60, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix}$$

$$12315132211212231\textcolor{red}{5}^{1,2}132 \quad \Delta = -52, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$231 \begin{smallmatrix} \textcolor{red}{3}12315132211212231 \\ 5 \end{smallmatrix} 1 \text{ [#}=5] \quad \Delta = -60, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix}$$

$$231 \begin{smallmatrix} \textcolor{red}{3}12315132211212231 \\ 5 \end{smallmatrix} 1 \text{ [#}=2] \quad \Delta = -60, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$12315132211212231\textcolor{red}{5}^{2,1}132 \quad \Delta = -52, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$12315132211212231\textcolor{red}{5}^{2,1}132 \quad \Delta = -52, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$12315132211212231\textcolor{red}{5}^{2,1}132 \quad \Delta = -52, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$12315132211212231\textcolor{red}{5}^{2,1}132 \quad \Delta = -52, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$12315132211212231\textcolor{red}{5}^{2,1}132 \quad \Delta = -52, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$231 \begin{smallmatrix} \textcolor{red}{3}12315132211212231 \\ 5 \end{smallmatrix} 1 \text{ [#}=6] \quad \Delta = -60, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

$$12315132211212231\textcolor{red}{5}^{3,1}132 \quad \Delta = -60, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$12315132211212231\textcolor{red}{5}^{2,1}132 \quad \Delta = -52, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$231 \begin{smallmatrix} \textcolor{red}{3}12315132211212231 \\ 5 \end{smallmatrix} 1 \text{ [#}=3] \quad \Delta = -60, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$231 \begin{smallmatrix} \textcolor{red}{3}12315132211212231 \\ 5 \end{smallmatrix} 1 \text{ [#}=4] \quad \Delta = -60, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

Base: 6122315132211212231413

$$6122315132211212231\textcolor{red}{4}^{3,1}13 \quad \Delta = -113, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 63 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$6122315132211212231\textcolor{red}{4}^{2,1}13 \quad \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

6122315132211212231413 ³	$\Delta = -113, n_+ = 1, n_- = 21, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$
6122315132211212231413 ²	$\Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
6122315132211212231413 ²	$\Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$
6122315132211212231413 ²	$\Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$
6122315132211212231413 ²	$\Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
6122315132211212231413 ²	$\Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$
6122315132211212231413 ²	$\Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$
Base: 3141322112122315132161		
3141322112122315132161 ²	$\Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$
3141322112122315132161 ³	$\Delta = -113, n_+ = 1, n_- = 21, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 21 & 9 \\ 2 & 9 & 8 \end{pmatrix}$
3141322112122315132161 ²	$\Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$
3141322112122315132161 ²	$\Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$
3141322112122315132161 ²	$\Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$
3141322112122315132161 ²	$\Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
3141322112122315132161 ²	$\Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 76 & 18 \\ 2 & 18 & 8 \end{pmatrix}$
3141322112122315132161 ²	$\Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 58 & 16 \\ 2 & 16 & 8 \end{pmatrix}$
Base: 1413221121223151321613		
1413221121223151321613 ²	$\Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$
1413221121223151321613 ²	$\Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$
1413221121223151321613 ²	$\Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, \det = 64$	$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$

$$\begin{aligned}
3161231513221121223141\overset{3}{3} \quad \Delta = -113, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
3161231513221121223141\overset{3}{3} \quad \Delta = -113, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
14132211212231\overset{2}{5}1321613 \quad \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
14132211212231\overset{2}{5}1321613 \quad \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
14132211212231\overset{2}{5}1321613 \quad \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
14132211212231\overset{2}{5}1321613 \quad \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\text{Base: } 3141322112122315131612 & \\
3141322112122315131\overset{3}{6}12 \quad \Delta = -112, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
3141322112122315131612 \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 58 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
3141322112122315131612 \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
3141322112122315131612 \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 76 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
3141322112122315131612 \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
314132211212231\overset{2}{5}131612 \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
3141322112122315131\overset{2}{6}12 \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
3141322112122315131\overset{2}{6}12 \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
3141322112122315131\overset{2}{6}12 \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
3141322112122315131612 \quad \Delta = -112, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 21 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\text{Base: } 1413221121223151316123 & \\
14132211212231\overset{2}{5}1316123 \quad \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
14132211212231\overset{2}{5}1316123 \quad \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
14132211212231\overset{2}{5}1316123 \quad \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1413221121223151316123^2 \quad \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1413221121223151316123^2 \quad \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1413221121223151316123^2 \quad \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1413221121223151316123^2 \quad \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
32161315132211212231413^{\#3} \quad \Delta = -113, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
32161315132211212231413^{\#2} \quad \Delta = -113, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1316131513221121223132

$$\begin{aligned}
1316131513221121223132^2 \quad \Delta = -85, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1316131513221121223132^3 \quad \Delta = -93, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1316131513221121223132^2 \quad \Delta = -85, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1316131513221121223132^2 \quad \Delta = -85, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1316131513221121223132^2 \quad \Delta = -85, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1316131513221121223132^2 \quad \Delta = -85, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1316131513221121223132^2 \quad \Delta = -85, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1316131513221121223132^2 \quad \Delta = -85, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1316131513221121223132^2 \quad \Delta = -87, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
23132211212231513161313^{\#3} \quad \Delta = -93, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
23132211212231513161313^{\#7} \quad \Delta = -93, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
23132211212231513161313^{\#6} \quad \Delta = -93, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
23132211212231513161313^{\#5} \quad \Delta = -93, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
2313221121223151316131\overset{3}{3} \text{ }_{[\# = 4]} \quad \Delta = -93, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
2313221121223151316131\overset{3}{3} \text{ }_{[\# = 8]} \quad \Delta = -93, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
2313221121223151316131\overset{3}{3} \text{ }_{[\# = 9]} \quad \Delta = -93, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
2313221121223151316131\overset{3}{3} \text{ }_{[\# = 2]} \quad \Delta = -93, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: 4131613151322112122313

$$\begin{aligned}
413161315132211212231\overset{2}{3} \quad \Delta = -112, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
413161315132211212231\overset{2}{3} \quad \Delta = -112, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
413161\overset{2}{3}151322112122313 \quad \Delta = -112, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
413161315132211212231\overset{2}{3} \quad \Delta = -112, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
413161315132211212231\overset{2}{3} \quad \Delta = -112, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
413161315132211212231\overset{2}{3} \quad \Delta = -112, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
413161315132211212231\overset{2}{3} \quad \Delta = -112, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
413161315132211212231\overset{2}{3} \quad \Delta = -112, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
413161315132211212231\overset{3}{3} \text{ }_{[\# = 2]} \quad \Delta = -120, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
413161315132211212231\overset{3}{3} \quad \Delta = -120, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
413161315132211212231\overset{3}{3} \quad \Delta = -120, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1513161315132211212222

$$\begin{aligned}
1513161\overset{2}{3}15132211212222 \quad \Delta = -119, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
151316131513221121222\overset{2}{2} \quad \Delta = -119, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
151316131513221121222\overset{2}{2} \quad \Delta = -119, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
151316131513221121222^2 \quad \Delta = -119, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
151316131513221121222^2 \quad \Delta = -119, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
151316131513221121222^2 \quad \Delta = -119, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
151316131513221121222^2 \quad \Delta = -119, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: 3151316131513221121222

$$\begin{aligned}
3151316131513221121222^2 \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
3151316131513221121222^2 \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
3151316131513221121222^2 \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
3151316131513221121222^2 \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
3151316131513221121222^2 \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
3151316131513221121222^2 \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
3151316131513221121222^2 \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 70 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
3151316131513221121222^2 \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix}
\end{aligned}$$

Base: 2315131613151322112122

$$\begin{aligned}
2315131613151322112122^2 \quad \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 196 & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 236 & 30 \\ 2 & 30 & 8 \end{pmatrix} \\
2315131613151322112122^2 \quad \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 88 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
2315131613151322112122^2 \quad \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
2315131613151322112122^2 \quad \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 206 & 28 \\ 2 & 28 & 8 \end{pmatrix} \\
2315131613151322112122^2 \quad \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2315131613151322112122^2 \quad \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix}
\end{aligned}$$

$$2315131613151322112122 \quad \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix}$$

$$2315131613151322112122 \quad \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: 161315132211212231413

$$161315132211212231413 \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$161315132211212231413 \quad \Delta = -112, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$161315132211212231413 \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$161315132211212231413 \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$161315132211212231413 \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$161315132211212231413 \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$161315132211212231413 \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$161315132211212231413 \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 16131513221121223141

$$316131513221121223141 \quad \Delta = -112, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$161315132211212231413_{[\# = 2]} \quad \Delta = -112, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$161315132211212231413_{[\# = 3]} \quad \Delta = -112, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}$$

$$16131513221121223141 \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$16131513221121223141 \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 117 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$16131513221121223141 \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 150 & 26 \\ 2 & 26 & 8 \end{pmatrix}$$

$$316131513221121223141 \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$16131513221121223141 \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 110 & 22 \\ 2 & 22 & 8 \end{pmatrix}$$

$$\begin{array}{lcl}
{}^{31}_1 6 \overset{2}{1} 3 1 5 1 3 2 2 1 1 2 1 2 2 3 1 4 1 & \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
{}^{31}_1 6 \overset{2}{1} 3 1 5 1 3 2 2 1 1 2 1 2 2 3 1 4 1 & \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 103 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
{}^{31}_1 6 \overset{2}{1} 3 1 5 1 3 2 2 1 1 2 1 2 2 3 1 4 1 & \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{array}$$

Base: 1712231513221121223132

$$\begin{array}{lcl}
1712231513221121223132 & \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
1712231513221121223132 & \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1712231513221121223132 & \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1712231513221121223132 & \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1712231513221121223132 & \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1712231513221121223132 & \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1712231513221121223132 & \Delta = -113, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1712231513221121223132 & \Delta = -107, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{array}$$

Base: 2171231513221121223132

$$\begin{array}{lcl}
2171231513221121223132 & \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
2171231513221121223132 & \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
2171231513221121223132 & \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
2171231513221121223132 & \Delta = -113, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2171231513221121223132 & \Delta = -107, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
2171231513221121223132 & \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2171231513221121223132 & \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
2171231513221121223132^2 \quad \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
2171231513221121223132^2 \quad \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 3217123151322112122313

$$\begin{aligned}
3217123151322112122313^2 \quad \Delta = -113, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
3217123151322112122313^2 \quad \Delta = -113, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
3217123151322112122313^2 \quad \Delta = -113, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
3217123151322112122313^2 \quad \Delta = -113, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
3217123151322112122313^3 \quad \Delta = -121, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
3217123151322112122313^3_{[\# = 2]} \quad \Delta = -121, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
3217123151322112122313^2 \quad \Delta = -113, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
3217123151322112122313^2 \quad \Delta = -113, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
3217123151322112122313^2 \quad \Delta = -113, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}
\end{aligned}$$

Base: 2321712315132211212231

$$\begin{aligned}
23217123151322112122313_{[\# = 2]} \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
23217123151322112122313_{[\# = 3]} \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
23217123151322112122313_{[\# = 4]} \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
23217123151322112122313_{[\# = 5]} \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
2321712315132211212231^2 \quad \Delta = -107, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 219 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
2321712315132211212231^2 \quad \Delta = -107, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
2321712315132211212231^2 \quad \Delta = -107, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
2321712315132211212231 \quad \Delta = -107, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 70 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
2321712315132211212231 \quad \Delta = -107, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
2321712315132211212231 \quad \Delta = -107, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 171231513221121223132

$$\begin{aligned}
171231513221121223132 \quad \Delta = -113, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
171231513221121223132 \quad \Delta = -107, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
171231513221121223132 \quad \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
171231513221121223132 \quad \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
171231513221121223132 \quad \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
171231513221121223132 \quad \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
171231513221121223132 \quad \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
171231513221121223132 \quad \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{aligned}$$

Base: 2181223151322112122222

$$\begin{aligned}
2181223151322112122222 \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
2181223151322112122222 \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
2181223151322112122222 \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
2181223151322112122222 \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2181223151322112122222 \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
2181223151322112122222 \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: 2218123151322112122222

$$2218123151322112122222 \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$\begin{array}{ll}
2218123151322112122222^2 & \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
2218123151322112122222^2 & \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
2218123151322112122222^3 & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2218123151322112122222^2 & \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
2218123151322112122222^2 & \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
2218123151322112122222^2 & \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{array}$$

Base: 3218122315132211212222

$$\begin{array}{ll}
3218122315132211212222^2 & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 174 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
3218122315132211212222^2 & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
3218122315132211212222^2 & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
3218122315132211212222^2 & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
3218122315132211212222^2 & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
3218122315132211212222^2 & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}
\end{array}$$

Base: 3221812315132211212222

$$\begin{array}{ll}
3221812315132211212222^2 & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
3221812315132211212222^2 & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
3221812315132211212222^2 & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
3221812315132211212222^2 & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
3221812315132211212222^2 & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
3221812315132211212222^2 & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}
\end{array}$$

Base: 2321812231513221121222

$$\begin{aligned}
2321812231513221121222^2 \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2321812231513221121222^2 \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 256 & \begin{pmatrix} 0 & 16 & 2 \\ 16 & 334 & 36 \\ 2 & 36 & 8 \end{pmatrix} \\
2321812231513221121222^2 \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
2321812231513221121222^2 \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
2321812231513221121222^2 \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 70 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
2321812231513221121222^2 \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}
\end{aligned}$$

Base: 181223151322112122222^1

$$\begin{aligned}
181223151322112122222^1 \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
181223151322112122222^1 \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
181223151322112122222^1 \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
181223151322112122222^1 \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
181223151322112122222^{12} \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{aligned}$$

Base: $18123151322112122222^{21}$

$$\begin{aligned}
18123151322112122222^{21} \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
18123151322112122222^{21} \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
18123151322112122222^{21} \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
18123151322112122222^{212} \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
18123151322112122222^{21} \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
18123151322112122222^{21} \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1812315132211212222^{321}$

$$1812315132211212222^{321} \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$\begin{array}{ll}
\overset{321}{1} \overset{2}{8} 1231513221121222 & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
\overset{321}{1} \overset{2}{8} 1231513221121222 & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\overset{321}{1} \overset{2}{8} 1231513221121222 & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
\overset{21}{321} 8 1231513221121222 & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\overset{321}{1} \overset{2}{8} 123151322112 \overset{2}{1} 2222 & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\overset{321}{1} \overset{2}{8} 1231513221121222 & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{array}$$

Base: $\overset{2321}{1} \overset{2}{8} 1231513221121222$

$$\begin{array}{ll}
\overset{2321}{1} \overset{2}{8} 1231513221121222 & \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
\overset{21}{2321} 8 1231513221121222 & \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\overset{2321}{1} \overset{2}{8} 1231513221121222 & \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
\overset{2321}{1} \overset{2}{8} 1231513221121222 & \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 70 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
\overset{2321}{1} \overset{2}{8} 1231513221121222 & \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 219 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
\overset{2321}{1} \overset{2}{8} 1231513221121222 & \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
\overset{2321}{1} \overset{2}{8} 1231513221121222 & \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{array}$$

Base: $\overset{1}{11} 2122315131613151322$

$$\begin{array}{ll}
\overset{1}{11} 2122315131613151322 & \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
\overset{1}{11} 2122315131613151322 & \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 289 \quad \begin{pmatrix} 0 & 17 & 2 \\ 17 & 236 & 30 \\ 2 & 30 & 8 \end{pmatrix} \\
\overset{1}{11} 2122315131613151322 & \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 196 \quad \begin{pmatrix} 0 & 14 & 2 \\ 14 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
\overset{1}{11} \overset{2}{2} 2122315131613151322 & \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 88 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
\overset{1}{11} 2122315131613151322 & \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
& \overset{1}{112122315131613151322} \quad \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{1}{112122315131613151322} \quad \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 256 \quad \begin{pmatrix} 0 & 16 & 2 \\ 16 & 206 & 28 \\ 2 & 28 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{21}{11212231513161315132}$

$$\begin{aligned}
& \overset{21}{11212231513161315132} \quad \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 196 \quad \begin{pmatrix} 0 & 14 & 2 \\ 14 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
& \overset{21}{11212231513161315132} \quad \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{21}{11212231513161315132} \quad \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{21}{11212231513161315132} \quad \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
& \overset{21}{11212231513161315132} \quad \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& \overset{21}{11212231513161315132} \quad \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
& \overset{21}{11212231513161315132} \quad \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 70 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& \overset{21}{11212231513161315132} \quad \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{131513221}{4131513221} \overset{12}{12} 1$

$$\begin{aligned}
& \overset{3131513221}{4131513221} \overset{12}{12} 1 \text{ [#=8]} \quad \Delta = -86, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{2}{4131513221} \overset{131513221}{12} 1 \quad \Delta = -79, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{2131513221}{4131513221} \overset{12}{12} 1 \quad \Delta = -78, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{2}{4131513221} \overset{131513221}{12} 1 \quad \Delta = -78, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{2}{4131513221} \overset{131513221}{12} 1 \quad \Delta = -78, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{2}{4131513221} \overset{131513221}{12} 1 \quad \Delta = -78, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{131513221}{4131513221} \overset{12}{12} 1 \quad \Delta = -78, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{131513221}{4131513221} \overset{12}{12} 1 \quad \Delta = -78, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
4131513221 \overset{131513221}{12} 1 \quad \Delta = -78, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
4131513221 \overset{3131513221}{12} 1 \text{ [#=9]} \quad \Delta = -86, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
4131513221 \overset{3}{12} \overset{131513221}{12} 1 \quad \Delta = -86, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
4131513221 \overset{3131513221}{12} 1 \text{ [#=7]} \quad \Delta = -86, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
4131513221 \overset{3131513221}{12} 1 \text{ [#=6]} \quad \Delta = -86, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
4131513221 \overset{3131513221}{12} 1 \text{ [#=3]} \quad \Delta = -86, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
4131513221 \overset{3131513221}{12} 1 \text{ [#=5]} \quad \Delta = -86, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
4131513221 \overset{3131513221}{12} 1 \text{ [#=4]} \quad \Delta = -86, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
4131513221 \overset{3131513221}{12} 1 \text{ [#=2]} \quad \Delta = -86, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{1231513221}{12} 122314141 & \\
1 \overset{1231513221}{12} 122314141 \text{ } 2 \quad \Delta = -47, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
1 \overset{31231513221}{12} 122314141 \text{ [#=2]} \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \overset{31231513221}{12} 122314141 \text{ [#=8]} \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
1 \overset{31231513221}{12} 122314141 \text{ [#=4]} \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1 \overset{31231513221}{12} 122314141 \text{ [#=5]} \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
1 \overset{1231513221}{12} 122314141 \text{ } 3 \text{ [#=2]} \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
1 \overset{31231513221}{12} 122314141 \text{ [#=6]} \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1 \overset{31231513221}{12} 122314141 \text{ [#=3]} \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1 \overset{31231513221}{12} 122314141 \text{ [#=7]} \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
1 \overset{1231513221}{12} 122314141 \text{ } 3 \text{ [#=3]} \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& 1^{\overline{31231513221}} 12^{122314141} [\#=9] \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix} \\
& 1^{\overline{1231513221}} 12^{122314141} \quad \Delta = -47, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& 1^{\overline{1231513221}} 12^{122314141} \quad \Delta = -47, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 1^{\overline{1231513221}} 12^{122314141} \quad \Delta = -47, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 1^{\overline{1231513221}} 12^{\overline{2}} 122314141 \quad \Delta = -47, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 1^{\overline{1231513221}} 12^{\overline{2}} 122314141 \quad \Delta = -47, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \text{Base: } 1^{\overline{321513221}} 12^{1223151321} \\
& 1^{\overline{321513221}} 12^{1223151321\overline{3}} [\#=3] \quad \Delta = -60, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& 1^{\overline{321513221}} 12^{1223151321\overline{3}} [\#=4] \quad \Delta = -60, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& 1^{\overline{321513221}} 12^{1223151321\overline{3}} [\#=6] \quad \Delta = -60, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
& 1^{\overline{321513221}} 12^{1223151321\overline{3}} [\#=9] \quad \Delta = -60, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
& 1^{\overline{321513221}} 12^{1223151321\overline{3}} [\#=7] \quad \Delta = -60, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
& 1^{\overline{321513221}} 12^{1223151321} \quad \Delta = -52, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 1^{\overline{321513221}} 12^{1223151321\overline{3}} [\#=2] \quad \Delta = -60, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& 1^{\overline{321513221}} 12^{1223151321\overline{3}} [\#=8] \quad \Delta = -60, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
& 1^{\overline{321513221}} 12^{1223151321\overline{3}} [\#=5] \quad \Delta = -60, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
& 1^{\overline{321513221}} 12^{1223151321} \quad \Delta = -52, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 1^{\overline{321513221}} 12^{\overline{2}} 1223151321 \quad \Delta = -52, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 1^{\overline{321513221}} 12^{\overline{2}} 1223151321 \quad \Delta = -52, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 1^{\overline{321513221}} 12^{1223151321} \quad \Delta = -52, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{l}
\begin{array}{l} \text{321513221}^{\text{2}} \\ 1 \quad 12 \quad 1223151321 \end{array} \quad \Delta = -52, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\\
\text{Base: } \begin{array}{l} \text{12231513221} \\ 1 \quad 12 \quad 12231413 \end{array} \\
\begin{array}{l} \text{312231513221} \\ 1 \quad 12 \quad 12231413 \end{array} [\#=5] \quad \Delta = -35, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
\\
\begin{array}{l} \text{12231513221} \\ 1 \quad 12 \quad 12231413 \end{array} \quad \Delta = -27, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\\
\begin{array}{l} \text{12231513221} \\ 1 \quad 12 \quad 12231413 \end{array} \quad \Delta = -27, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\\
\begin{array}{l} \text{12231513221} \\ 1 \quad 12 \quad 12231413 \end{array} \quad \Delta = -27, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\\
\begin{array}{l} \text{12231513221} \\ 1 \quad 12 \quad 12231413 \end{array} \quad \Delta = -27, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\\
\begin{array}{l} \text{12231513221} \\ 1 \quad 12 \quad 12231413 \end{array} \quad \Delta = -27, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\\
\begin{array}{l} \text{12231513221}^{\text{2}} \\ 1 \quad 12 \quad 12231413 \end{array} \quad \Delta = -27, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\\
\begin{array}{l} \text{312231513221} \\ 1 \quad 12 \quad 12231413 \end{array} [\#=3] \quad \Delta = -35, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
\\
\begin{array}{l} \text{312231513221} \\ 1 \quad 12 \quad 12231413 \end{array} [\#=9] \quad \Delta = -35, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix} \\
\\
\begin{array}{l} \text{312231513221} \\ 1 \quad 12 \quad 12231413 \end{array} [\#=8] \quad \Delta = -35, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
\\
\begin{array}{l} \text{312231513221} \\ 1 \quad 12 \quad 12231413 \end{array} [\#=7] \quad \Delta = -35, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
\\
\begin{array}{l} \text{312231513221} \\ 1 \quad 12 \quad 12231413 \end{array} [\#=6] \quad \Delta = -35, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
\\
\begin{array}{l} \text{12231513221} \\ 1 \quad 12 \quad 12231413 \end{array} \quad \Delta = -35, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\\
\begin{array}{l} \text{312231513221} \\ 1 \quad 12 \quad 12231413 \end{array} [\#=2] \quad \Delta = -35, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\\
\begin{array}{l} \text{312231513221} \\ 1 \quad 12 \quad 12231413 \end{array} [\#=4] \quad \Delta = -35, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
\\
\text{Base: } \begin{array}{l} \text{51231513221} \\ 1 \quad 12 \quad 12231512 \end{array} \\
\begin{array}{l} \text{51231513221} \\ 1 \quad 12 \quad 12231512 \end{array} \quad \Delta = -95, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\\
\begin{array}{l} \text{51231513221} \\ 1 \quad 12 \quad 12231512 \end{array} \quad \Delta = -95, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{ll}
1^{51231513221\textcolor{red}{2}}_{12} 12231512 & \Delta = -95, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1^{51231513221}_{12} 12231512 & \Delta = -95, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{51231513221}_{12} 12231512^{\textcolor{red}{2}} & \Delta = -95, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1^{51231513221}_{12} 12231512^{\textcolor{red}{3}} & \Delta = -103, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1^{51231513221}_{12} 12231512 & \Delta = -95, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1^{51231513221}_{12} 12231512 & \Delta = -95, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\textbf{Base: } 1^{15131513221}_{12} 12231512 & \\
1^{15131513221}_{12} 12231512 & \Delta = -94, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1^{15131513221}_{12} 12231512 & \Delta = -95, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1^{15131513221}_{12} 12231512^{\textcolor{red}{2}} & \Delta = -94, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1^{15131513221}_{12} 12231512 & \Delta = -94, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1^{15131513221\textcolor{red}{2}}_{12} 12231512 & \Delta = -94, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1^{15131513221}_{12} 12231512 & \Delta = -94, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{15131513221}_{12} 12231512^{\textcolor{red}{2}} & \Delta = -94, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{15131513221}_{12} 12231512 & \Delta = -94, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1^{15131513221}_{12} 12231512^{\textcolor{red}{3}} & \Delta = -102, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\textbf{Base: } 1^{51231513221}_{12} 1223151^1 & \\
1^{51231513221}_{12} 1223151^1 & \Delta = -95, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1^{51231513221}_{12} 1223151^1 & \Delta = -95, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{51231513221}_{12} 1223151^1 & \Delta = -95, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{lll}
1^{51231513221} 12^{21} 1223151 & \Delta = -95, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{51231513221} 12^{11} 1223151 & \Delta = -95, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1^{51231513221} 12^{21} 1223151^1 & \Delta = -95, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{array}$$

Base: $\begin{matrix} & 15131513221 & & 1 \\ 1 & & 12 & & 1223151 \end{matrix}$

$$1^{15131513221} 12^{1223151} \Delta = -94, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1^{15131513221} 12^{1223151} \Delta = -94, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1^{15131513221} 12^{1513221} 1223151 \quad \Delta = -94, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1^{15131513221} 12^{1223151} \Delta = -94, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1^{15131513221} 12^{1} 1223151 \quad \Delta = -95, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1^{1513}12^{1513221}1223151^{\textcolor{red}{2}1} \quad \Delta = -94, \, n_+ = 1, \, n_- = 21, \, n_0 = 0, \, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1^{15131513221} 12^{1223151} \Delta = -94, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: $1^{31513221} 12^{1223151321}$

$$1^{31513221} 12^{1223151321} \mathbf{3}_{[\# = 2]} \quad \Delta = -59, \, n_+ = 1, \, n_- = 21, \, n_0 = 0, \, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1^{31513221}{}_{12}{}_{1223151321} \quad \Delta = -51, \quad n_+ = 1, \quad n_- = 21, \quad n_0 = 0, \quad |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1^{31513221} 12^{1223151321} \textcolor{red}{3} \text{ [#}=4] \quad \Delta = -59, \, n_+ = 1, \, n_- = 21, \, n_0 = 0, \, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$1^{31513221}{}_{12}{}_{1223151321\textcolor{red}{3}}_{[\# = 5]} \quad \Delta = -59, \, n_+ = 1, \, n_- = 21, \, n_0 = 0, \, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix}$$

$$1^{31513221} 12^{12} 1223151321 \mathbf{3} \text{ }_{[\# = 6]} \quad \Delta = -59, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

$$1^{31513221}{}_{12}{}_{1223151321\textcolor{red}{3}}_{[\# = 3]} \quad \Delta = -59, \, n_+ = 1, \, n_- = 21, \, n_0 = 0, \, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1^{31513221} 12^{22} 1223151321^2 \quad \Delta = -51, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1^{31513221} 12^{1223151321} \quad \Delta = -59, \quad n_+ = 1, \quad n_- = 21, \quad n_0 = 0, \quad |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$\begin{aligned}
& \overset{31513221}{1 \quad 12 \quad 1223151321} \overset{\textcolor{red}{3}}{[\# = 9]} \quad \Delta = -59, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
& \overset{31513221}{1 \quad 12 \quad 1223151321} \overset{\textcolor{red}{3}}{[\# = 8]} \quad \Delta = -59, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
& \overset{31513221}{1 \quad 12 \quad 1223151321} \overset{\textcolor{red}{3}}{[\# = 7]} \quad \Delta = -59, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
& \overset{31513221}{1 \quad 12 \quad 1223151321} \quad \Delta = -51, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{31513221}{1 \quad 12 \quad 1223151321} \overset{\textcolor{red}{2}}{\quad} \quad \Delta = -51, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{31513221}{1 \quad 12 \quad 1223151321} \quad \Delta = -51, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{31513221}{1 \quad 12 \quad 1223151321} \quad \Delta = -51, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{31513221}{1 \quad 12 \quad 1223151321} \overset{\textcolor{red}{2}}{\quad} \quad \Delta = -51, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: $6122315132211\overset{1}{2}1223141$

$$\begin{aligned}
& 61223151\overset{\textcolor{red}{2}}{3}2211\overset{1}{2}1223141 \quad \Delta = -97, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 6122315132211\overset{\textcolor{red}{2}}{2}1223141 \quad \Delta = -97, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 6122315132211\overset{1}{2}\overset{\textcolor{red}{2}}{2}1223141 \quad \Delta = -97, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 6122315132211\overset{1}{2}12231\overset{\textcolor{red}{2}}{4}1 \quad \Delta = -97, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 6122315132211\overset{\textcolor{red}{2}}{2}1223141 \quad \Delta = -97, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& 6122315132211\overset{\textcolor{red}{2}}{2}\overset{1}{2}1223141 \quad \Delta = -97, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 6122315132211\overset{\textcolor{red}{3}}{2}1223141 \quad \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 63 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& 6122315132211\overset{1}{2}1223141\overset{\textcolor{red}{3}}{[\# = 3]} \quad \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 6122315132211\overset{1}{2}1223141\overset{\textcolor{red}{3}}{[\# = 2]} \quad \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: $612315132211\overset{1}{2}12231414$

$$6123151\overset{\textcolor{red}{2}}{3}2211\overset{1}{2}12231414 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$\begin{aligned}
61231513221\overset{21}{1}212231414 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
61231513221\overset{1}{1}\overset{2}{2}212231414 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
61231513221\overset{1}{1}21223\overset{2}{1}414 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
61231513221\overset{1}{1}2122314\overset{2}{1}4 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
61231513221\overset{2}{1}212231414 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
61231\overset{2}{5}13221\overset{1}{1}212231414 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
61231513221\overset{2}{1}\overset{1}{2}212231414 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{aligned}$$

Base: 1413221¹12122315132161

$$\begin{aligned}
161231513221\overset{1}{1}21223141\overset{3}{3} \text{ [#=3]} \quad \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1413221\overset{1}{1}\overset{2}{2}22315132161 \quad \Delta = -97, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1413221\overset{1}{1}21223\overset{2}{1}5132161 \quad \Delta = -97, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1413221\overset{1}{1}2122315\overset{2}{1}32161 \quad \Delta = -97, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1413221\overset{1}{1}212231513\overset{2}{2}161 \quad \Delta = -97, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1413221\overset{2}{1}\overset{1}{2}22315132161 \quad \Delta = -97, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
141\overset{2}{3}221\overset{1}{1}2122315132161 \quad \Delta = -97, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
161231513221\overset{1}{1}21223141\overset{3}{3} \text{ [#=2]} \quad \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: 41413221¹1212231513161

$$\begin{aligned}
41413221\overset{21}{1}212231513161 \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
41413221\overset{2}{1}212231513161 \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
4141\overset{2}{3}221\overset{1}{1}212231513161 \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{ll}
414132211212231513161 & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
414132211212231513161 & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
414132211212231513161 & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
414132211212231513161 & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
414132211212231513161 & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\text{Base: } 141322112122315131612 & \\
141322112122315131612 & \Delta = -96, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
2161315132211212231413_{[\# = 2]} & \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
2161315132211212231413_{[\# = 3]} & \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
141322112122315131612 & \Delta = -96, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
141322112122315131612 & \Delta = -96, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
141322112122315131612 & \Delta = -96, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
141322112122315131612 & \Delta = -96, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
141322112122315131612 & \Delta = -96, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
141322112122315131612 & \Delta = -96, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
141322112122315131612 & \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\text{Base: } 1^{313221} 12 \quad 1223151316131 & \\
1^{313221} 12 \quad 12231513161313_{[\# = 4]} & \Delta = -92, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1^{313221} 12 \quad 12231513161313_{[\# = 5]} & \Delta = -92, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1^{313221} 12 \quad 1223151316131 & \Delta = -84, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
& \overset{313221}{1 \ 12 \ 1223151316131} \overset{\textcolor{red}{3}}{3} \text{ [#=8]} \quad \Delta = -92, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{313221}{1 \ 12 \ 1223151316131} \overset{\textcolor{red}{3}}{3} \text{ [#=9]} \quad \Delta = -92, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{313221}{1 \ 12 \ 1223151316131} \quad \Delta = -84, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{313221}{1 \ 12 \ 1223151316131} \overset{\textcolor{red}{3}}{3} \text{ [#=6]} \quad \Delta = -92, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{313221}{1 \ 12 \ 1223151316131} \quad \Delta = -84, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{313221}{1 \ 12 \ 1223151316131} \overset{\textcolor{red}{2}}{2} \quad \Delta = -84, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{313221}{1 \ 12 \ 1223151316131} \overset{\textcolor{red}{2}}{2} \quad \Delta = -84, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{313221}{1 \ 12 \ 1223151316131} \overset{\textcolor{red}{2}}{2} \quad \Delta = -84, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{313221}{1 \ 12 \ 1223151316131} \overset{\textcolor{red}{2}}{2} \quad \Delta = -84, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& \overset{313221}{1 \ 12 \ 1223151316131} \overset{\textcolor{red}{3}}{3} \text{ [#=3]} \quad \Delta = -92, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{313221}{1 \ 12 \ 1223151316131} \overset{\textcolor{red}{3}}{3} \text{ [#=2]} \quad \Delta = -92, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
& \overset{313221}{1 \ 12 \ 1223151316131} \text{ [#=2]} \quad \Delta = -92, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 93 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& \overset{313221}{1 \ 12 \ 1223151316131} \quad \Delta = -92, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 21 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{313221}{1 \ 12 \ 1223151316131} \overset{\textcolor{red}{3}}{3} \text{ [#=7]} \quad \Delta = -92, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \textbf{Base: } 41316131513221 \overset{13221}{12 \ 1} \\
& \overset{\textcolor{red}{3}}{41316131513221} \overset{13221}{12 \ 1} \quad \Delta = -112, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& 41316131513221 \overset{\textcolor{red}{3}}{12 \ 1} \overset{313221}{1} \text{ [#=2]} \quad \Delta = -112, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
& \overset{\textcolor{red}{2}}{41316131513221} \overset{13221}{12 \ 1} \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 41316131513221 \overset{\textcolor{red}{3}}{12 \ 1} \overset{313221}{1} \text{ [#=3]} \quad \Delta = -112, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{\textcolor{red}{2}}{41316131513221} \overset{13221}{12 \ 1} \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{ll}
41316131513221 \overset{2}{12} 1 \overset{13221}{} & \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
41316131513221 \overset{313221}{12} 1 \text{ [#=4]} & \Delta = -112, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
41316131513221 \overset{313221}{12} 1 \text{ [#=5]} & \Delta = -112, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
41316131513221 \overset{13221}{12} 1 & \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
41316131513221 \overset{213221}{12} 1 & \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
41316131513221 \overset{2}{12} 1 \overset{13221}{} & \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
41316131513221 \overset{2}{12} 1 \overset{13221}{} & \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
\text{Base: } \overset{2221}{1} \overset{2}{12} 122315131613151 & \\
\overset{2221}{1} \overset{2}{12} 122315131613151 & \Delta = -119, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\overset{2221}{1} \overset{2}{12} 122315131613151 & \Delta = -119, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\overset{2221}{1} \overset{2}{12} 122315131613151 & \Delta = -119, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\overset{2221}{1} \overset{2}{12} 122315131613151 & \Delta = -119, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\overset{2221}{1} \overset{2}{12} 122315131613151 & \Delta = -119, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\overset{2221}{1} \overset{2}{12} 122315131613151 & \Delta = -119, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\overset{2221}{1} \overset{2}{12} 122315131613151 & \Delta = -119, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\text{Base: } \overset{221}{112} 1223151316131513 & \\
\overset{221}{112} 1223151316131513 & \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\overset{221}{112} 1223151316131513 & \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\overset{221}{112} 1223151316131513 & \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\overset{221}{112} \overset{2}{12} 1223151316131513 & \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{lcl}
\begin{array}{c} 221 \\ 1121223151316131513 \end{array} \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\begin{array}{c} 221 \\ 1121223151316131513 \end{array} \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
\begin{array}{c} 221 \\ 1121223151316131513 \end{array} \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
\begin{array}{c} 221 \\ 1121223151316131513 \end{array} \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{array}$$

Base: $\begin{array}{c} 1 \\ 16131513221121223141 \end{array}$

$$\begin{array}{lcl}
\begin{array}{c} 1 \\ 16131513221121223141 \end{array} \text{[#=3]} \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\begin{array}{c} 1 \\ 16131513221121223141 \end{array} \text{[#=2]} \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\begin{array}{c} 1 \\ 16131513221121223141 \end{array} \quad \Delta = -96, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\begin{array}{c} 1 \\ 16131513221121223141 \end{array} \quad \Delta = -96, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\begin{array}{c} 12 \\ 16131513221121223141 \end{array} \quad \Delta = -96, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\begin{array}{c} 1 \\ 16131513221121223141 \end{array} \quad \Delta = -96, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\begin{array}{c} 1 \\ 16131513221121223141 \end{array} \quad \Delta = -96, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\begin{array}{c} 1 \\ 16131513221121223141 \end{array} \quad \Delta = -96, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{array}$$

Base: $\begin{array}{c} 313221 \\ 1 \text{ } 12 \text{ } 1223151322171 \end{array}$

$$\begin{array}{lcl}
\begin{array}{c} 313221 \\ 1 \text{ } 12 \text{ } 1223151322171 \end{array} \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
\begin{array}{c} 313221 \\ 1 \text{ } 12 \text{ } 1223151322171 \end{array} \text{[#=2]} \quad \Delta = -112, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 93 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
\begin{array}{c} 313221 \\ 1 \text{ } 12 \text{ } 1223151322171 \end{array} \quad \Delta = -112, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 21 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\begin{array}{c} 313221 \\ 1 \text{ } 12 \text{ } 1223151322171 \end{array} \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
\begin{array}{c} 313221 \\ 1 \text{ } 12 \text{ } 1223151322171 \end{array} \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\begin{array}{c} 313221 \\ 1 \text{ } 12 \text{ } 1223151322171 \end{array} \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{array}$$

$$1 \overset{313221}{12} \overset{2}{1223151322171} \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1 \overset{313221}{12} \overset{2}{1223151322171} \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: $1 \overset{313221}{12} 1223151321712$

$$1 \overset{313221}{12} \overset{2}{1223151321712} \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{313221}{12} 1223151321712 \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1 \overset{313221}{12} \overset{2}{1223151321712} \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1 \overset{313221}{12} 1223151321712 \quad \Delta = -112, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 21 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1 \overset{313221}{12} 1223151321712 \text{ [#=2]} \quad \Delta = -112, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 93 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$1 \overset{313221}{12} \overset{2}{1223151321712} \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1 \overset{313221}{12} \overset{2}{1223151321712} \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$1 \overset{313221}{12} 1223151321712 \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{313221}{12} \overset{2}{1223151321712} \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: $1 \overset{13221}{12} 12231513217123$

$$1 \overset{313221}{12} 12231513217123 \text{ [#=3]} \quad \Delta = -113, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{313221}{12} 12231513217123 \text{ [#=4]} \quad \Delta = -113, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1 \overset{13221}{12} \overset{2}{12231513217123} \quad \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$1 \overset{13221}{12} 12231513217123 \quad \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1 \overset{13221}{12} \overset{2}{12231513217123} \quad \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{13221}{12} \overset{2}{12231513217123} \quad \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1 \overset{313221}{12} 12231513217123 \text{ [#=2]} \quad \Delta = -113, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$1^{13221} \quad 1 \quad 12 \quad 12231513217123 \quad [\# = 5] \quad \Delta = -113, \quad n_+ = 1, \quad n_- = 21, \quad n_0 = 0, \quad |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1^{13221} \overset{2}{12} 12231513217123 \quad \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1^{13221} 12^{12} 122315^{2} 13217123 \quad \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: $\overset{1}{1}7\overset{1}{1}23151322112122313$

$$1^1 17 12315132211^1 2122313^3 \text{ }_{[\# = 2]} \quad \Delta = -112, \ n_+ = 1, \ n_- = 21, \ n_0 = 0, \ |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 57 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$17123151322112122313 \quad \Delta = -104, \quad n_+ = 1, \quad n_- = 21, \quad n_0 = 0, \quad |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$17123151322112122313 \quad \Delta = -104, \quad n_+ = 1, \quad n_- = 21, \quad n_0 = 0, \quad |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$1^1 1712315^2 13221^1 12122313 \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1712315\overset{2}{1}3221\overset{1}{1}2122313 \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1^1 17 123 15 1322 1^2 112 1223 13 \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1\overset{1}{7}123151322\overset{1}{1}2\overset{2}{1}22313 \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$17123151322112122313 \quad \Delta = -112, \quad n_+ = 1, \quad n_- = 21, \quad n_0 = 0, \quad |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: $1^{22221} 12^{12231513221812}$

$$1 \overset{22221}{12} \overset{2}{12231513} 221812 \quad \Delta = -115, \, n_+ = 1, \, n_- = 21, \, n_0 = 0, \, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1^{22221} 12^{12231513221} 1812^{\textcolor{red}{2}} \quad \Delta = -115, \, n_+ = 1, \, n_- = 21, \, n_0 = 0, \, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1 \overset{22221}{12} 12231513221812 \quad \Delta = -115, \, n_+ = 1, \, n_- = 21, \, n_0 = 0, \, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1 \overset{22221}{12} 12231513221812^{\textcolor{red}{2}} \quad \Delta = -115, \, n_+ = 1, \, n_- = 21, \, n_0 = 0, \, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1^{22221} 12^{12} 12231513221812 \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$1 \overset{22221}{12} \overset{2}{12231513} 221812 \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: $1^{22221} 12 12231513218122$

$$1 \overset{22221}{12} 12231513218122 \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$1^{22221} 12^{2} 12231513218122$	$\Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$
$1^{22221} 12^{2} 12231513218122$	$\Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$
$1^{22221} 12^{2} 12231513218122$	$\Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
$1^{22221} 12^{2} 12231513218122$	$\Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$
$1^{22221} 12^{2} 12231513218122$	$\Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$
$1^{22221} 12^{3} 12231513218122$	$\Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, \det = 4$	$\begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$

Base: $1\ 12\ 122315132218123$

$1 \overset{2221}{12} 122315132218123$	$\Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$
$1 \overset{2221}{12} 122315132218123$	$\Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$
$1 \overset{2221}{12} 122315132218123$	$\Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, \det = 121$	$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$
$1 \overset{2221}{12} 122315132218123$	$\Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, \det = 64$	$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
$1 \overset{2221}{12} 122315132218123$	$\Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$
$1 \overset{2221}{12} 122315132218123$	$\Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$

Base: $1\ 12\ 122315132181223$

$1 \overset{2221}{12} 122315132181 \overset{2}{223}$	$\Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$
$1 \overset{2221}{12} \overset{2}{223} 15132181223$	$\Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
$1 \overset{2221}{12} 122315132181 \overset{2}{223}$	$\Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$
$1 \overset{2221}{12} 122315132181223 \overset{2}{223}$	$\Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, \det = 64$	$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
$1 \overset{2221}{12} 122315132181 \overset{2}{223}$	$\Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, \det = 81$	$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$
$1 \overset{2221}{12} 122315132181223$	$\Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$

Base: 1121223151322181232

$^{221}_{1121223151322181232}$	$\Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, \det = 81$	$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$
$^{221}_{1121223151322181232}$	$\Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, \det = 121$	$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$
$^{221}_{1121223151322181232}$	$\Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, \det = 256$	$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 174 & 26 \\ 2 & 26 & 8 \end{pmatrix}$
$^{221}_{1121223151322181232}$	$\Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$
$^{221}_{1121223151322181232}$	$\Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, \det = 64$	$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
$^{221}_{1121223151322181232}$	$\Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, \det = 64$	$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$
Base: $^{1}_{18122315132211212222}$		
$^{12}_{18122315132211212222}$	$\Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$
$^{1}_{18122315132211212222}$	$\Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
$^{1}_{18122315132211212222}$	$\Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$
$^{1}_{18122315132211212222}$	$\Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$
$^{1}_{18122315132211212222}$	$\Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
Base: $^{21}_{1812315132211212222}$		
$^{21}_{1812315132211212222}$	$\Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$
$^{21}_{1812315132211212222}$	$\Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
$^{21}_{1812315132211212222}$	$\Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
$^{21}_{1812315132211212222}$	$\Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$
$^{21}_{1812315132211212222}$	$\Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, \det = 4$	$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$
$^{21}_{1812315132211212222}$	$\Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$
Base: $^{321}_{1812315132211212222}$		
$^{321}_{1812315132211212222}$	$\Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$

$$\begin{aligned}
& \overset{21}{321} 8 1231513221 \overset{1}{121222} \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{321}{1} 8 1231513221 \overset{2}{121222} \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{321}{1} 8 1231513221 \overset{2}{121222} \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{321}{1} 8 1231513221 \overset{2}{121222} \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
& \overset{321}{1} 8 1231513221 \overset{2}{121222} \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
& \overset{321}{1} 8 1231513221 \overset{2}{121222} \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{2321}{1} 8 \overset{221}{1231513221} 121$

$$\begin{aligned}
& \overset{2321}{1} 8 \overset{2}{1231513221} \overset{221}{121} \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
& \overset{2321}{1} 8 \overset{2}{1231513221} \overset{221}{121} \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& \overset{2321}{1} 8 \overset{2}{1231513221} \overset{221}{121} \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 70 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& \overset{2321}{1} 8 \overset{2}{1231513221} \overset{221}{121} \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 219 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
& \overset{2321}{1} 8 \overset{2}{1231513221} \overset{221}{121} \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{2321}{1} 8 \overset{2}{1231513221} \overset{221}{121} \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
& \overset{2321}{1} 8 \overset{2}{1231513221} \overset{221}{121} \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{21}{211} 21223151316131513$

$$\begin{aligned}
& \overset{21}{211} 2122315131613151 \overset{2}{3} \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{2}{211} 21223151316131513 \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{21}{211} 21223151316131513 \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{21}{211} 21223151316131513 \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{21}{211} 21223151316131513 \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{21}{21121223151316131513} \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& \overset{21}{21121223151316131513} \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
& \overset{21}{21121223151316131513} \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{21}{21121223151322181232}$

$$\begin{aligned}
& \overset{21}{21121223151322181232} \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{21}{21121223151322181232} \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{21}{21121223151322181232} \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{21}{21121223151322181232} \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
& \overset{21}{21121223151322181232} \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 256 \quad \begin{pmatrix} 0 & 16 & 2 \\ 16 & 174 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
& \overset{21}{21121223151322181232} \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{2321}{8} 1231513221 \overset{21}{1212}$

$$\begin{aligned}
& 1 \overset{2321}{8} 1231513221 \overset{21}{1212} \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& 1 \overset{2321}{8} 1231513221 \overset{21}{1212} \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 70 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& 1 \overset{2321}{8} 1231513221 \overset{21}{1212} \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
& 1 \overset{2321}{8} 1231513221 \overset{21}{1212} \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 219 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
& 1 \overset{2321}{8} 1231513221 \overset{21}{1212} \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 1 \overset{2321}{8} 1231513221 \overset{21}{1212} \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
& 2321 \overset{21}{8} 1231513221 \overset{21}{1212} \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: $21 \overset{131513221}{12} 122315131$

$$\begin{aligned}
& 21 \overset{131513221}{12} 122315131 \quad \Delta = -50, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 21 \overset{131513221}{12} 122315131 \overset{131513221}{3} [\# = 2] \quad \Delta = -58, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
21^{131513221} 12^{122315131} \textcolor{red}{3} [\#=3] \quad \Delta = -58, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21^{131513221} 12^{122315131} \textcolor{red}{3} [\#=9] \quad \Delta = -58, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
21^{131513221} 12^{122315131} \textcolor{red}{3} [\#=4] \quad \Delta = -58, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
21^{131513221} 12^{122315131} \textcolor{red}{2} \quad \Delta = -50, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21^{131513221} 12^{122315131} \textcolor{red}{3} [\#=8] \quad \Delta = -58, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
21^{131513221} 12^{122315131} \textcolor{red}{3} [\#=7] \quad \Delta = -58, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
21^{131513221} 12^{122315131} \textcolor{red}{3} [\#=6] \quad \Delta = -58, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
21^{131513221} 12^{122315131} \textcolor{red}{3} [\#=5] \quad \Delta = -58, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
21^{131513221} 12^{122315131} \textcolor{red}{2} \quad \Delta = -50, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21^{131513221} 12^{122315131} \textcolor{red}{2} \quad \Delta = -50, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\textbf{Base: } 21^{21513221} 12^{1223151321} & \\
21^{21513221} 12^{1223151321} \textcolor{red}{3} [\#=5] \quad \Delta = -51, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
21^{21513221} 12^{1223151321} \textcolor{red}{3} [\#=6] \quad \Delta = -51, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
21^{21513221} 12^{1223151321} \quad \Delta = -43, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21^{\textcolor{red}{2}21513221} 12^{1223151321} \quad \Delta = -43, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21^{21513221} 12^{1223151321} \textcolor{red}{2} \quad \Delta = -43, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21^{21513221} 12^{1223151321} \textcolor{red}{2} \quad \Delta = -43, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21^{21513221} 12^{1223151321} \textcolor{red}{2} \quad \Delta = -43, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21^{21513221} 12^{1223151321} \quad \Delta = -43, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21^{21513221} 12^{1223151321} \quad \Delta = -43, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
21^{21513221}_{12} 1223151321\mathbf{3}_{[\#=9]} \quad \Delta = -51, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
21^{21513221}_{12} 1223151321\mathbf{3}_{[\#=8]} \quad \Delta = -51, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
21^{21513221}_{12} 1223151321\mathbf{3}_{[\#=7]} \quad \Delta = -51, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
21^{21513221}_{12} 1223151321\mathbf{3}_{[\#=3]} \quad \Delta = -51, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
21^{21513221}_{12} 1223151321 \quad \Delta = -51, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
21^{21513221}_{12} 1223151321\mathbf{3}_{[\#=2]} \quad \Delta = -51, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
21^{21513221}_{12} 1223151321\mathbf{3}_{[\#=4]} \quad \Delta = -51, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
\text{Base: } 21^{12231513221}_{12} 1223141 & \\
21^{12231513221}_{12} 1223141 \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
21^{12231513221}_{12} 1223141 \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21^{\mathbf{3}12231513221}_{12} 1223141_{[\#=2]} \quad \Delta = -27, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 25 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
21^{\mathbf{3}12231513221}_{12} 1223141_{[\#=6]} \quad \Delta = -27, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 249 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
21^{12231513221}_{12} 1223141\mathbf{3}_{[\#=3]} \quad \Delta = -27, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
21^{\mathbf{3}12231513221}_{12} 1223141_{[\#=3]} \quad \Delta = -27, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 60 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
21^{12231513221}_{12} 1223141\mathbf{3}_{[\#=2]} \quad \Delta = -27, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
21^{\mathbf{3}12231513221}_{12} 1223141_{[\#=9]} \quad \Delta = -27, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 564 & 62 \\ 2 & 62 & 8 \end{pmatrix} \\
21^{\mathbf{3}12231513221}_{12} 1223141_{[\#=4]} \quad \Delta = -27, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 109 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
21^{\mathbf{3}12231513221}_{12} 1223141_{[\#=8]} \quad \Delta = -27, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 445 & 55 \\ 2 & 55 & 8 \end{pmatrix} \\
21^{12231513221}_{12} 1223141\mathbf{2} \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
21^{12231513221}_{12} 1223141\mathbf{2} \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{2}{21} \overset{12231513221}{12} 1223141 \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 21 \overset{12231513221}{12} 1223141 \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 21 \overset{3}{12231513221}{12} 1223141 \text{ [#=5]} \quad \Delta = -27, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 172 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
& 21 \overset{3}{12231513221}{12} 1223141 \text{ [#=7]} \quad \Delta = -27, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 340 & 48 \\ 2 & 48 & 8 \end{pmatrix} \\
& \text{Base: } 21 \overset{1513221}{12} 1223151321 \\
& 21 \overset{1513221}{12} 1223151321 \overset{3}{3} \text{ [#=3]} \quad \Delta = -51, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& 21 \overset{1513221}{12} 1223151321 \overset{3}{3} \text{ [#=2]} \quad \Delta = -51, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& 21 \overset{1513221}{12} 1223151321 \quad \Delta = -43, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 21 \overset{1513221}{12} 1223151321 \quad \Delta = -43, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 21 \overset{1513221}{12} 1223151321 \overset{3}{3} \text{ [#=7]} \quad \Delta = -51, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
& 21 \overset{1513221}{12} 1223151321 \overset{3}{3} \text{ [#=6]} \quad \Delta = -51, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
& 21 \overset{1513221}{12} 1223151321 \overset{3}{3} \text{ [#=8]} \quad \Delta = -51, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
& 21 \overset{1513221}{12} 1223151321 \overset{2}{2} \quad \Delta = -43, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 21 \overset{1513221}{12} 1223151321 \overset{3}{3} \text{ [#=5]} \quad \Delta = -51, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
& 21 \overset{1513221}{12} 1223151321 \overset{3}{3} \text{ [#=9]} \quad \Delta = -51, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
& \overset{2}{21} \overset{1513221}{12} 1223151321 \quad \Delta = -43, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 21 \overset{1513221}{12} 1223151321 \overset{2}{2} \quad \Delta = -43, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 21 \overset{1513221}{12} 1223151321 \overset{2}{2} \quad \Delta = -43, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 21 \overset{1513221}{12} 1223151321 \overset{3}{3} \text{ [#=4]} \quad \Delta = -51, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& \text{Base: } 61231513221 \overset{1513221}{12} 12
\end{aligned}$$

$$\begin{array}{ll}
61231513221 \overset{2}{1513221} 12 \quad \Delta = -121, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
61231513221 \overset{1513221}{12} 12 \quad \Delta = -121, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
61231513221 \overset{1513221}{12} \overset{2}{12} \quad \Delta = -121, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
61231513221 \overset{2}{1513221} 12 \quad \Delta = -121, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
61231513221 \overset{2}{1513221} 12 \quad \Delta = -121, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
61231513221 \overset{2}{1513221} 12 \quad \Delta = -121, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
61231513221 \overset{1513221}{12} 12 \quad \Delta = -121, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\text{Base: } 6131513221 \overset{31513221}{12} 12 & \\
6131513221 \overset{2}{31513221} 12 \quad \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
6131513221 \overset{2}{31513221} 12 \quad \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 256 & \begin{pmatrix} 0 & 16 & 2 \\ 16 & 174 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
6131513221 \overset{2}{31513221} 12 \quad \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
6131513221 \overset{31513221}{12} \overset{2}{12} \quad \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
6131513221 \overset{31513221}{12} 12 \quad \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
6131513221 \overset{2}{31513221} 12 \quad \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
6131513221 \overset{31513221}{12} 12 \quad \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 196 & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
6131513221 \overset{31513221}{12} 12 \quad \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\text{Base: } 21 \overset{1513221}{12} 12231513161 & \\
21 \overset{2}{1513221} 12 12231513161 \quad \Delta = -120, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 24 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21 \overset{1513221}{12} 12231513161 \quad \Delta = -120, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
21 \overset{1513221}{12} 12231513161 \quad \Delta = -120, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
21^{1513221} 12^{2231513161} \quad \Delta = -120, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
21^{1513221} 12^{2231513161} \quad \Delta = -120, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21^{1513221} 12^{2231513161} \quad \Delta = -120, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
21^{1513221} 12^{2231513161} \quad \Delta = -120, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 174 & 26 \\ 2 & 26 & 8 \end{pmatrix}
\end{aligned}$$

Base: $21^{513221} 12^{22315131613}$

$$\begin{aligned}
21^{513221} 12^{22315131613} \quad \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 289 & \begin{pmatrix} 0 & 17 & 2 \\ 17 & 185 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
21^{513221} 12^{22315131613} \quad \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
21^{513221} 12^{22315131613} \quad \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
21^{513221} 12^{22315131613} \quad \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 144 & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
21^{513221} 12^{22315131613} \quad \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
21^{2513221} 12^{22315131613} \quad \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
21^{513221} 12^{22315131613} \quad \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21^{513221} 12^{22315131613} \quad \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: $21^{13221} 12^{223151316131}$

$$\begin{aligned}
21^{13221} 12^{223151316131} \quad \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
21^{13221} 12^{223151316131} \quad \Delta = -84, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
21^{313221} 12^{223151316131} \quad \Delta = -84, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21^{13221} 12^{223151316131} \quad \Delta = -84, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
21^{313221} 12^{223151316131} \quad \Delta = -84, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21^{13221} 12^{223151316131} \quad \Delta = -84, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{ll}
21^{13221} 12 1223151316131\bar{3} \text{ [#=4]} & \Delta = -84, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
21^{\bar{2}13221} 12 1223151316131 & \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21^{13221} 12 1223151316131^{\bar{2}} & \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21^{13221} 12 1223151316131^{\bar{2}} & \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
21^{\bar{3}13221} 12 1223151316131 \text{ [#=4]} & \Delta = -84, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
21^{13221} 12 1223151316131^{\bar{2}} & \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
21^{13221} 12 1223151316131^{\bar{2}} & \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
21^{13221} 12 1223151316131 & \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
21^{13221} 12 1223151316131\bar{3} \text{ [#=9]} & \Delta = -84, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21^{13221} 12 1223151316131\bar{3} \text{ [#=8]} & \Delta = -84, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
21^{13221} 12 1223151316131\bar{3} \text{ [#=7]} & \Delta = -84, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
21^{13221} 12 1223151316131\bar{3} \text{ [#=6]} & \Delta = -84, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
21^{\bar{3}13221} 12 1223151316131 \text{ [#=5]} & \Delta = -84, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
\text{Base: } 21^{3221} 12 12231513161315 & \\
21^{3221} 12 12231513161315^{\bar{2}} & \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
21^{3221} 12 12231513161315^{\bar{2}} & \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
21^{3221\bar{2}} 12 12231513161315 & \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 76 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
21^{3221} 12 12231513161315^{\bar{2}} & \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 63 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
21^{3221} 12 12231513161315^{\bar{2}} & \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 58 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
21^{\bar{2}3221} 12 12231513161315 & \Delta = -128, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{array}$$

$${}^{3221}_{21\ 12\ 12231513161315} \quad \Delta = -128, \ n_+ = 1, \ n_- = 21, \ n_0 = 0, \ |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$${}^{3221}_{21\ 12\ 12231513161315} \quad \Delta = -128, \ n_+ = 1, \ n_- = 21, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

Base: ${}^{221}_{21\ 12\ 122315131613151}$

$${}^{221}_{21\ 12\ 122315131613151} \quad \Delta = -119, \ n_+ = 1, \ n_- = 21, \ n_0 = 0, \ |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$${}^{221}_{21\ 12\ 122315131613151} \quad \Delta = -119, \ n_+ = 1, \ n_- = 21, \ n_0 = 0, \ |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$${}^{221}_{21\ 12\ 122315131613151} \quad \Delta = -119, \ n_+ = 1, \ n_- = 21, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$${}^{221}_{21\ 12\ 122315131613151} \quad \Delta = -119, \ n_+ = 1, \ n_- = 21, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$${}^{221}_{21\ 12\ 122315131613151} \quad \Delta = -119, \ n_+ = 1, \ n_- = 21, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$${}^{221}_{21\ 12\ 122315131613151} \quad \Delta = -119, \ n_+ = 1, \ n_- = 21, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$${}^{221}_{21\ 12\ 122315131613151} \quad \Delta = -119, \ n_+ = 1, \ n_- = 21, \ n_0 = 0, \ |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$${}^{221}_{21\ 12\ 122315131613151} \quad \Delta = -119, \ n_+ = 1, \ n_- = 21, \ n_0 = 0, \ |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: ${}^{21}_{41322112122315132217}$

$${}^{21}_{41322112122315132217} \quad \Delta = -124, \ n_+ = 1, \ n_- = 21, \ n_0 = 0, \ |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$${}^{21}_{41322112122315132217} \quad \Delta = -124, \ n_+ = 1, \ n_- = 21, \ n_0 = 0, \ |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 5 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$${}^{21}_{41322112122315132217} \quad \Delta = -124, \ n_+ = 1, \ n_- = 21, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$${}^{21}_{41322112122315132217} \quad \Delta = -124, \ n_+ = 1, \ n_- = 21, \ n_0 = 0, \ |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$${}^{21}_{41322112122315132217} \quad \Delta = -124, \ n_+ = 1, \ n_- = 21, \ n_0 = 0, \ |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$${}^{21}_{41322112122315132217} \quad \Delta = -124, \ n_+ = 1, \ n_- = 21, \ n_0 = 0, \ |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$${}^{21}_{41322112122315132217} \quad \Delta = -124, \ n_+ = 1, \ n_- = 21, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

Base: ${}^{13221}_{21\ 12\ 1223151322171}$

$${}^{13221}_{21\ 12\ 1223151322171} \quad \Delta = -96, \ n_+ = 1, \ n_- = 21, \ n_0 = 0, \ |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$\begin{aligned}
& \overset{3}{21} \overset{13221}{12} 1223151322171 \text{ }_{[\# = 4]} \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{3}{21} \overset{13221}{12} 1223151322171 \text{ }_{[\# = 2]} \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{3}{21} \overset{13221}{12} 1223151322171 \text{ }_{[\# = 3]} \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{3}{21} \overset{13221}{12} 1223151322171 \text{ }_{[\# = 5]} \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& \overset{2}{21} \overset{13221}{12} 1223151322171 \quad \Delta = -96, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{13221}{21} \overset{12}{12} 1223151322171 \quad \Delta = -96, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{13221}{21} \overset{12}{12} \overset{2}{1223151322171} \quad \Delta = -96, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& \overset{13221}{21} \overset{12}{12} \overset{2}{1223151322171} \quad \Delta = -96, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{13221}{21} \overset{12}{12} \overset{2}{1223151322171} \quad \Delta = -96, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \text{Base: } \overset{21}{41322112122315132171} \\
& \overset{21}{41322112122315132171} \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{21}{41322112122315132171} \overset{2}{122315132171} \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{21}{41322112122315132171} \overset{2}{122315132171} \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{21}{41322112122315132171} \overset{2}{122315132171} \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& \overset{2}{41322112122315132171} \overset{21}{122315132171} \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{21}{41322112122315132171} \overset{2}{122315132171} \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{2}{41322112122315132171} \overset{21}{122315132171} \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \text{Base: } \overset{13221}{21} \overset{12}{12} 1223151321712 \\
& \overset{13221}{21} \overset{12}{12} \overset{2}{1223151321712} \quad \Delta = -96, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{3}{21} \overset{13221}{12} 1223151321712 \text{ }_{[\# = 5]} \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& 21 \overset{3}{1} 12 \overset{3}{1} 223151321712 \text{ [#=4]} \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& 21 \overset{3}{1} 12 \overset{3}{1} 223151321712 \text{ [#=2]} \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 21 \overset{3}{1} 12 \overset{3}{1} 223151321712 \text{ [#=3]} \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 21 \overset{1}{1} 12 \overset{2}{2} 223151321712 \quad \Delta = -96, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& 21 \overset{1}{1} 12 \overset{2}{2} 223151321712 \quad \Delta = -96, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 21 \overset{2}{1} 12 \overset{2}{2} 223151321712 \quad \Delta = -96, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 21 \overset{1}{1} 12 \overset{2}{2} 223151321712 \quad \Delta = -96, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 21 \overset{1}{1} 12 \overset{2}{2} 223151321712 \quad \Delta = -96, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& 21 \overset{1}{1} 12 \overset{2}{2} 223151321712 \quad \Delta = -96, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \text{Base: } \overset{1}{1} 171231513221 \overset{21}{1} 1212231 \\
& 171231513221 \overset{21}{1} 1212231 \overset{2}{2} \quad \Delta = -96, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 171231513221 \overset{21}{1} 1212231 \overset{1}{2} \quad \Delta = -96, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& 171231513221 \overset{21}{1} 1212231 \overset{3}{3} \text{ [#=5]} \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& 171231513221 \overset{21}{1} 1212231 \overset{3}{3} \text{ [#=4]} \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 171231513221 \overset{21}{1} 1212231 \overset{3}{3} \text{ [#=2]} \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 171231513221 \overset{21}{1} 1212231 \overset{3}{3} \text{ [#=3]} \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 171231513221 \overset{21}{1} 1212231 \quad \Delta = -96, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& 171231513221 \overset{21}{1} 1212231 \quad \Delta = -96, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 171231513221 \overset{21}{1} 1212231 \quad \Delta = -96, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 171231513221 \overset{21}{1} 1212231 \quad \Delta = -96, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: 2221 21 12 12231513221812

$$\begin{aligned}
^{2221} 21 \overset{2}{1} 2 12231513221812 \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
^{2221} 21 \overset{2}{1} 2 12231513221812 \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
^{2221} 21 \overset{2}{1} 2 12231513221812 \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
^{2221} 21 \overset{2}{1} 2 12231513221812 \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
^{2221} 21 \overset{2}{1} 2 12231513221812 \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
^{2221} 21 \overset{2}{1} 2 12231513221812 \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
^{2221} 21 \overset{2}{1} 2 12231513221812 \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: 2221 21 12 12231513218122

$$\begin{aligned}
^{2221} 21 \overset{2}{1} 2 12231513218122 \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
^{2221} 21 \overset{2}{1} 2 12231513218122 \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
^{2221} 21 \overset{2}{1} 2 12231513218122 \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
^{2221} 21 \overset{2}{1} 2 12231513218122 \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
^{2221} 21 \overset{2}{1} 2 12231513218122 \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
^{2221} 21 \overset{2}{1} 2 12231513218122 \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
^{2221} 21 \overset{2}{1} 2 12231513218122 \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
^{2221} 21 \overset{3}{1} 2 12231513218122 \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}
\end{aligned}$$

Base: 221 21 12 122315132218123

$$\begin{aligned}
^{221} 21 \overset{2}{1} 2 122315132218123 \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
^{221} 21 \overset{2}{1} 2 122315132218123 \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
^{221} 21 \overset{2}{1} 2 122315132218123 \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{ll}
{}^{221}{}^2 2112122315132218123 & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
{}^{221}{}^2 2112122315132218123 & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
{}^{221}{}^2 2112122315132218123 & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
{}^{221}{}^2 2112122315132218123 & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{array}$$

Base: ${}^{221} 2112122315132181223$

$$\begin{array}{ll}
{}^{221}{}^2 2112122315132181223 & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
{}^{221}{}^2 2112122315132181223 & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
{}^{221} 2112122315132181223 & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
{}^{221}{}^2 2112122315132181223 & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
{}^{221}{}^2 2112122315132181223 & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
{}^{221}{}^2 2112122315132181223 & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
{}^{221}{}^2 2112122315132181223 & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{array}$$

Base: ${}^1 1812231513221121222$

$$\begin{array}{ll}
{}^1 1812231513221121222 & \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
{}^{12}{}^{21} 1812231513221121222 & \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
{}^1{}^2{}^{21} 1812231513221121222 & \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
{}^1{}^2{}^{21} 1812231513221121222 & \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
{}^1{}^{21}{}^2 1812231513221121222 & \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
{}^1{}^{21}{}^2 1812231513221121222 & \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{array}$$

Base: ${}^{21} 181231513221121222$

$${}^{21}{}^2 181231513221121222 \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$\begin{array}{lcl}
\begin{smallmatrix} 21 \\ 1\ 8\ 1231513221121222 \end{smallmatrix} & \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\begin{smallmatrix} 21 \\ 1\ 8\ 1231513221121222 \end{smallmatrix} & \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\begin{smallmatrix} 21\ 2 \\ 1\ 8\ 1231513221121222 \end{smallmatrix} & \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
\begin{smallmatrix} 21 \\ 1\ 8\ 12315132211\ 2\ 21222 \end{smallmatrix} & \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\begin{smallmatrix} 21 \\ 1\ 8\ 1231513221121\ 2\ 222 \end{smallmatrix} & \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
\begin{smallmatrix} 21\ 2 \\ 1\ 8\ 1231513221121222 \end{smallmatrix} & \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{array}$$

Base: $\begin{smallmatrix} 321 \\ 1\ 8\ 12315132211212 \end{smallmatrix}$

$$\begin{array}{lcl}
\begin{smallmatrix} 21 \\ 321\ 8\ 12315132211212 \end{smallmatrix} & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\begin{smallmatrix} 321\ 2 \\ 1\ 8\ 12315132211212 \end{smallmatrix} & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
\begin{smallmatrix} 321 \\ 1\ 8\ 12315132211212 \end{smallmatrix} & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\begin{smallmatrix} 321\ 2 \\ 1\ 8\ 12315132211212 \end{smallmatrix} & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
\begin{smallmatrix} 321 \\ 1\ 8\ 123151\ 2\ 32211212 \end{smallmatrix} & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
\begin{smallmatrix} 321 \\ 1\ 8\ 1231513221121\ 2 \end{smallmatrix} & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\begin{smallmatrix} 321 \\ 1\ 8\ 12315132211212 \end{smallmatrix} & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\begin{smallmatrix} 321 \\ 1\ 8\ 12315132211\ 2\ 221 \end{smallmatrix} & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{array}$$

Base: $\begin{smallmatrix} 3221 \\ 221\ 12\ 1223151322181 \end{smallmatrix}$

$$\begin{array}{lcl}
\begin{smallmatrix} 3221 \\ 3221\ 12\ 1223151322181 \end{smallmatrix} & \Delta = -133, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\begin{smallmatrix} 3221 \\ 221\ 12\ 1223151322181 \end{smallmatrix} & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
\begin{smallmatrix} 2\ 3221 \\ 221\ 12\ 1223151322181 \end{smallmatrix} & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 55 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
\begin{smallmatrix} 3221 \\ 221\ 12\ 12231513221\ 2 \end{smallmatrix} & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{lll}
221 \overset{3221}{12} 1223151322181 & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
221 \overset{3221}{12} 1223151322181 & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
221 \overset{3221}{12} 1223151322181 & \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 174 & 26 \\ 2 & 26 & 8 \end{pmatrix}
\end{array}$$

Base: $221^{221}1212231513221812$

$$221\overset{221}{1}2122315\overset{2}{1}3221812 \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$221\overset{221}{12}12231513221\overset{2}{18}12 \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$221\overset{221}{1}212231513221812 \quad \Delta = -115, \, n_+ = 1, \, n_- = 21, \, n_0 = 0, \, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$221\overset{2}{1}1212231513221812 \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$221\overset{221}{1}21223151322\overset{2}{1}812 \quad \Delta = -115, \, n_+ = 1, \, n_- = 21, \, n_0 = 0, \, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$221\overset{221}{12}1223\overset{2}{15}13221812 \quad \Delta = -115, \, n_+ = 1, \, n_- = 21, \, n_0 = 0, \, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: $221^{3221} 12 1223151321812$

$$3221 \overset{3221}{12} 1223151321812 \quad \Delta = -133, \ n_+ = 1, \ n_- = 21, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$221 \overset{3221}{12} 1223151321812 \overset{2}{\Delta} = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$221 \overset{3221}{12} 1223151321812 \quad \Delta = -124, \ n_+ = 1, \ n_- = 21, \ n_0 = 0, \ |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$221 \overset{3221}{12} 1223151321812 \quad \Delta = -124, \quad n_+ = 1, \quad n_- = 21, \quad n_0 = 0, \quad |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 174 & 26 \\ 2 & 26 & 8 \end{pmatrix}$$

$$221 \overset{3221}{12} 1223 \overset{2}{15} 1321812 \quad \Delta = -124, \ n_+ = 1, \ n_- = 21, \ n_0 = 0, \ |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

$$221 \overset{3221}{12} 122315\overset{2}{13}21812 \quad \Delta = -124, \, n_+ = 1, \, n_- = 21, \, n_0 = 0, \, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$221 \overset{3221}{12} 1223151321\overset{2}{18}12 \quad \Delta = -124, \quad n_+ = 1, \quad n_- = 21, \quad n_0 = 0, \quad |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$221 \overset{2}{1} \overset{3}{2} 221 \quad \Delta = -124, \quad n_+ = 1, \quad n_- = 21, \quad n_0 = 0, \quad |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 55 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

Base: $221^{221}1212231513218122$

$$221^{221}1212231513218122^{\textcolor{red}{3}} \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$\begin{aligned}
^{221}2211212231513218122 \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
^{221}2211212231513218122 \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
^{2221}2211212231513218122 \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
^{221}2211212231513218122 \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
^{221}2211212231513218122 \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
^{2212}2211212231513218122 \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{aligned}$$

Base: $^{112}181223151322112122$

$$\begin{aligned}
^{12}181223151322112122 \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
^{12}181223151322112122 \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
^{12}181223151322112122 \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
^{12}181223151322112122 \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
^{12}181223151322112122 \quad \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: $^{112}181231513221121223$

$$\begin{aligned}
^{112}181231513221121223 \quad \Delta = -133, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
^{112}181231513221121223 \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
^{112}181231513221121223 \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
^{112}181231513221121223 \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
^{112}181231513221121223 \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
^{112}181231513221121223 \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
^{112}181231513221121223 \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: $^{21}18123151322112122$

$$\begin{array}{ll}
1 \overset{21}{8} 1231513 \overset{2}{221} 12122 & \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \overset{21}{8} 1231513 \overset{2}{221} 12122 & \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1 \overset{21}{8} 1231513 \overset{221}{221} 12122 & \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \overset{21}{8} 1231513 \overset{221}{221} 12122 & \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \overset{21}{8} 1231513 \overset{2}{221} 12122 & \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1 \overset{21}{8} 1231513 \overset{2}{221} 12122 & \Delta = -115, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{array}$$

Base: $1 \overset{1,131513221}{12} 122315131$

$$\begin{array}{ll}
1 \overset{1,131513221}{12} \overset{2}{122315131} & \Delta = -50, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1 \overset{1,131513221}{12} 122315131 \overset{3}{\# = 9} & \Delta = -58, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
1 \overset{1,131513221}{12} 122315131 \overset{3}{\# = 2} & \Delta = -58, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \overset{1,131513221}{12} 122315131 \overset{3}{\# = 5} & \Delta = -58, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1 \overset{1,131513221}{12} \overset{2}{122315131} & \Delta = -50, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1 \overset{1,131513221}{12} \overset{2}{122315131} & \Delta = -50, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \overset{1,131513221}{12} 122315131 \overset{3}{\# = 8} & \Delta = -58, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
1 \overset{1,131513221}{12} 122315131 \overset{3}{\# = 7} & \Delta = -58, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
1 \overset{1,131513221}{12} 122315131 \overset{3}{\# = 6} & \Delta = -58, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
1 \overset{1,131513221}{12} 122315131 \overset{3}{\# = 4} & \Delta = -58, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1 \overset{1,131513221}{12} 122315131 \overset{3}{\# = 3} & \Delta = -58, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{array}$$

Base: $1 \overset{1,21513221}{12} 1223151321$

$$1 \overset{1,21513221}{12} 1223151321 \overset{3}{\# = 7} \quad \Delta = -51, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix}$$

$$\begin{array}{l}
1^{1,21513221} \quad 1 \quad 12 \quad 1223151321\mathbf{3}_{[\# = 6]} \quad \Delta = -51, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
1^{1,21513221} \quad 1 \quad 12 \quad 1223151321\mathbf{3}_{[\# = 5]} \quad \Delta = -51, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
1^{1,21513221} \quad 1 \quad 12 \quad 1223151321\mathbf{3}_{[\# = 9]} \quad \Delta = -51, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
1^{1,21513221} \quad 1 \quad 12 \quad 1223151321\mathbf{3}_{[\# = 8]} \quad \Delta = -51, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
1^{1,21513221} \quad 1 \quad 12 \quad 1223151321 \quad \Delta = -51, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1^{1,21513221} \quad 1 \quad 12 \quad 1223151321 \quad \Delta = -43, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1^{1,21513221} \quad 1 \quad 12 \quad 1223151321\mathbf{3}_{[\# = 4]} \quad \Delta = -51, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1^{1,21513221} \quad 1 \quad 12 \quad 1223151321\mathbf{3}_{[\# = 3]} \quad \Delta = -51, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1^{1,21513221} \quad 1 \quad 12 \quad 1223151321 \quad \Delta = -43, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1^{1,21513221} \quad 1 \quad 12 \quad 1223151321 \quad \Delta = -43, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{1,21513221} \quad 1 \quad 12 \quad 1223151321\mathbf{2} \quad \Delta = -43, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1^{1,21513221} \quad 1 \quad 12 \quad 1223151321\mathbf{3}_{[\# = 2]} \quad \Delta = -51, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1^{1,21513221} \quad 1 \quad 12 \quad 1223151321\mathbf{2} \quad \Delta = -43, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1^{1,21513221} \quad 1 \quad 12 \quad 1223151321\mathbf{2} \quad \Delta = -43, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\mathbf{Base:} \quad 1^{1,1513221} \quad 1 \quad 12 \quad 1223151321 \\
1^{1,1513221} \quad 1 \quad 12 \quad 1223151321\mathbf{3}_{[\# = 7]} \quad \Delta = -51, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
1^{1,1513221} \quad 1 \quad 12 \quad 1223151321\mathbf{3}_{[\# = 4]} \quad \Delta = -51, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1^{1,1513221} \quad 1 \quad 12 \quad 1223151321\mathbf{3}_{[\# = 3]} \quad \Delta = -51, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1^{1,1513221} \quad 1 \quad 12 \quad 1223151321\mathbf{3}_{[\# = 5]} \quad \Delta = -51, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
1^{1,1513221} \quad 1 \quad 12 \quad 1223151321\mathbf{3}_{[\# = 6]} \quad \Delta = -51, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
& 1^{1,1513221} \quad 1 \quad 12 \quad 1223151321\textcolor{red}{3} \quad [\# = 9] \quad \Delta = -51, \quad n_+ = 1, \quad n_- = 21, \quad n_0 = 0, \quad |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
& 1^{1,1513221} \quad 1 \quad 12 \quad 1223151321\textcolor{red}{3} \quad [\# = 8] \quad \Delta = -51, \quad n_+ = 1, \quad n_- = 21, \quad n_0 = 0, \quad |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
& 1^{1,1513221} \quad 1 \quad 12 \quad 1223151321\textcolor{red}{2} \quad \Delta = -43, \quad n_+ = 1, \quad n_- = 21, \quad n_0 = 0, \quad |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 1^{1,1513221} \quad 1 \quad 12 \quad 1223151321 \quad \Delta = -43, \quad n_+ = 1, \quad n_- = 21, \quad n_0 = 0, \quad |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 1^{1,1513221} \quad 1 \quad 12 \quad 1223151321 \quad \Delta = -43, \quad n_+ = 1, \quad n_- = 21, \quad n_0 = 0, \quad |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 1^{1,1513221} \quad 1 \quad 12 \quad 1223151321\textcolor{red}{3} \quad [\# = 2] \quad \Delta = -51, \quad n_+ = 1, \quad n_- = 21, \quad n_0 = 0, \quad |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& 1^{1,1513221} \quad 1 \quad 12 \quad 1223151321\textcolor{red}{2} \quad \Delta = -43, \quad n_+ = 1, \quad n_- = 21, \quad n_0 = 0, \quad |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 1^{1,1513221} \quad 1 \quad 12 \quad 1223151321\textcolor{red}{2} \quad \Delta = -43, \quad n_+ = 1, \quad n_- = 21, \quad n_0 = 0, \quad |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \textbf{Base: } 12314141414141414132^2 \\
& 2^{\textcolor{red}{8}123141414141414141} \quad 3^2 \quad \Delta = -13, \quad n_+ = 1, \quad n_- = 21, \quad n_0 = 0, \quad |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \textbf{Base: } 132161316131613151321 \\
& 132161316131613151321\textcolor{red}{3} \quad [\# = 4] \quad \Delta = 0, \quad n_+ = 1, \quad n_- = 21, \quad n_0 = 0, \quad |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 123151316131613161231\textcolor{red}{3} \quad [\# = 4] \quad \Delta = 0, \quad n_+ = 1, \quad n_- = 21, \quad n_0 = 0, \quad |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& 132161316131613151321\textcolor{red}{3} \quad [\# = 5] \quad \Delta = 0, \quad n_+ = 1, \quad n_- = 21, \quad n_0 = 0, \quad |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& 132161316131613151321\textcolor{red}{3} \quad [\# = 9] \quad \Delta = 0, \quad n_+ = 1, \quad n_- = 21, \quad n_0 = 0, \quad |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
& 132161316131613151321\textcolor{red}{3} \quad [\# = 3] \quad \Delta = 0, \quad n_+ = 1, \quad n_- = 21, \quad n_0 = 0, \quad |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 123151316131613161231\textcolor{red}{3} \quad [\# = 3] \quad \Delta = 0, \quad n_+ = 1, \quad n_- = 21, \quad n_0 = 0, \quad |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& 132161316131613151321\textcolor{red}{3} \quad [\# = 6] \quad \Delta = 0, \quad n_+ = 1, \quad n_- = 21, \quad n_0 = 0, \quad |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& 132161316131613151321\textcolor{red}{3} \quad [\# = 8] \quad \Delta = 0, \quad n_+ = 1, \quad n_- = 21, \quad n_0 = 0, \quad |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& 132161316131613151321\textcolor{red}{3} \quad [\# = 2] \quad \Delta = 0, \quad n_+ = 1, \quad n_- = 21, \quad n_0 = 0, \quad |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

123151316131613161231 $\bar{3}$ [$\# = 2$]	$\Delta = 0, n_+ = 1, n_- = 21, n_0 = 0, \det = 4$	$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix}$
123151316131613161231 $\bar{3}$ [$\# = 5$]	$\Delta = 0, n_+ = 1, n_- = 21, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix}$
132161316131613151321 $\bar{3}$ [$\# = 7$]	$\Delta = 0, n_+ = 1, n_- = 21, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}$
Base: 123216131613161315131		
131513161316131612321 $\bar{3}$ [$\# = 4$]	$\Delta = -1, n_+ = 1, n_- = 21, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix}$
123216131613161315131 $\bar{3}$ [$\# = 7$]	$\Delta = -1, n_+ = 1, n_- = 21, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$
131513161316131612321 $\bar{3}$ [$\# = 7$]	$\Delta = -1, n_+ = 1, n_- = 21, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix}$
123216131613161315131 $\bar{3}$ [$\# = 6$]	$\Delta = -1, n_+ = 1, n_- = 21, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$
131513161316131612321 $\bar{3}$ [$\# = 2$]	$\Delta = -1, n_+ = 1, n_- = 21, n_0 = 0, \det = 4$	$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix}$
123216131613161315131 $\bar{3}$ [$\# = 4$]	$\Delta = -1, n_+ = 1, n_- = 21, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$
131513161316131612321 $\bar{3}$ [$\# = 8$]	$\Delta = -1, n_+ = 1, n_- = 21, n_0 = 0, \det = 64$	$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix}$
131513161316131612321 $\bar{3}$ [$\# = 3$]	$\Delta = -1, n_+ = 1, n_- = 21, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix}$
123216131613161315131 $\bar{3}$ [$\# = 2$]	$\Delta = -1, n_+ = 1, n_- = 21, n_0 = 0, \det = 4$	$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$
131513161316131612321 $\bar{3}$ [$\# = 6$]	$\Delta = -1, n_+ = 1, n_- = 21, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix}$
123216131613161315131 $\bar{3}$ [$\# = 3$]	$\Delta = -1, n_+ = 1, n_- = 21, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$
131513161316131612321 $\bar{3}$ [$\# = 5$]	$\Delta = -1, n_+ = 1, n_- = 21, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix}$
123216131613161315131 $\bar{3}$ [$\# = 8$]	$\Delta = -1, n_+ = 1, n_- = 21, n_0 = 0, \det = 64$	$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}$
131513161316131612321 $\bar{3}$ [$\# = 9$]	$\Delta = -1, n_+ = 1, n_- = 21, n_0 = 0, \det = 81$	$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix}$
123216131613161315131 $\bar{3}$ [$\# = 9$]	$\Delta = -1, n_+ = 1, n_- = 21, n_0 = 0, \det = 81$	$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$
123216131613161315131 $\bar{3}$ [$\# = 5$]	$\Delta = -1, n_+ = 1, n_- = 21, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$

Base: 313161316131613151321

$$313161316131613151321 \quad \Delta = -7, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 30 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$313161316131613151321 \textcolor{red}{3} \text{ [#=5]} \quad \Delta = -7, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$313161316131613151321 \textcolor{red}{3} \text{ [#=3]} \quad \Delta = -7, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 294 & 38 \\ 2 & 38 & 8 \end{pmatrix}$$

$$313161316131613151321 \textcolor{red}{3} \text{ [#=9]} \quad \Delta = -7, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix}$$

$$313161316131613151321 \textcolor{red}{3} \text{ [#=4]} \quad \Delta = -7, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$313161316131613151321 \textcolor{red}{3} \text{ [#=3]} \quad \Delta = -7, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$313161316131613151321 \textcolor{red}{4} \quad \Delta = -27, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 29 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$313161316131613151321 \textcolor{red}{3} \text{ [#=6]} \quad \Delta = -7, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$313161316131613151321 \textcolor{red}{3} \text{ [#=2]} \quad \Delta = -7, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 129 & 25 \\ 2 & 25 & 8 \end{pmatrix}$$

$$313161316131613151321 \textcolor{red}{3} \text{ [#=7]} \quad \Delta = -7, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$313161316131613151321 \textcolor{red}{3} \text{ [#=2]} \quad \Delta = -7, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$313161316131613151321 \textcolor{red}{3} \text{ [#=8]} \quad \Delta = -7, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

Base: 141316131613161315131

$$141316131613161315131 \textcolor{red}{3} \text{ [#=4]} \quad \Delta = -26, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$141316131613161315131 \textcolor{red}{3} \text{ [#=7]} \quad \Delta = -26, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$141316131613161315131 \textcolor{red}{3} \text{ [#=5]} \quad \Delta = -26, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$141316131613161315131 \textcolor{red}{3} \text{ [#=6]} \quad \Delta = -26, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$131513161316131613141 \textcolor{red}{3} \text{ [#=2]} \quad \Delta = -26, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$141316131613161315131 \textcolor{red}{3} \quad \Delta = -26, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 30 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$\begin{aligned}
141316131613161315131\overset{\textcolor{red}{3}}{3}_{[\# = 2]} \quad \Delta = -26, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
141316131613161315131\overset{\textcolor{red}{3}}{3}_{[\# = 8]} \quad \Delta = -26, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
141316131613161315131\overset{\textcolor{red}{3}}{3}_{[\# = 3]} \quad \Delta = -26, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
141316131613161315131\overset{\textcolor{red}{3}}{3}_{[\# = 9]} \quad \Delta = -26, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1413161\overset{\textcolor{red}{2}}{3}1613161315131 \quad \Delta = -18, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1413161\overset{\textcolor{red}{2}}{3}1613161315131 \quad \Delta = -18, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
141316131613161315131\overset{\textcolor{red}{2}}{3} \quad \Delta = -18, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
141316131613161315131\overset{\textcolor{red}{2}}{3} \quad \Delta = -18, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
131513161316131613141\overset{\textcolor{red}{3}}{3}_{[\# = 3]} \quad \Delta = -26, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
141316131613161315131\overset{\textcolor{red}{2}}{3} \quad \Delta = -18, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
141316131613161315131\overset{\textcolor{red}{2}}{3} \quad \Delta = -18, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1413161316131\overset{\textcolor{red}{2}}{6}161315131 \quad \Delta = -18, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
141316131613161315131\overset{\textcolor{red}{2}}{3} \quad \Delta = -18, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
141316131613161315131\overset{\textcolor{red}{2}}{3} \quad \Delta = -18, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\textbf{Base: } 215131613161316131512 \\
215131613161316131512\overset{\textcolor{red}{3}}{3} \quad \Delta = -42, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 15 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
215131613161316131512\overset{\textcolor{red}{2}}{3} \quad \Delta = -34, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
215131613161316131512\overset{\textcolor{red}{2}}{3} \quad \Delta = -34, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
2151316131\overset{\textcolor{red}{2}}{6}161316131512 \quad \Delta = -34, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
215131613161316131512\overset{\textcolor{red}{2}}{3} \quad \Delta = -34, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

$$215131613161316131512 \quad \Delta = -34, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: $1 \overset{215131613161316131}{5} 1$

$$1 \overset{215131613161316131}{5} 1 \quad \Delta = -34, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$1 \overset{215131613161316131}{5} 1 \quad \Delta = -34, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1 \overset{215131613161316131}{5} 1 \quad \Delta = -42, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

$$1 \overset{215131613161316131}{5} 1 \quad \Delta = -34, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{215131613161316131}{5} 1 \quad \Delta = -34, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$1 \overset{215131613161316131}{5} 1 \quad \Delta = -34, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1 \overset{215131613161316131}{5} 1 \quad \Delta = -34, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1 \overset{215131613161316131}{5} 1 \quad \Delta = -34, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{215131613161316131}{5} 1 \quad \Delta = -34, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$1 \overset{215131613161316131}{5} 1 \quad \Delta = -34, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

Base: $1 \overset{15131613161316131}{5} 1$

$$1 \overset{15131613161316131}{5} 1 \quad \Delta = -34, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1 \overset{15131613161316131}{5} 1 \quad \Delta = -34, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{15131613161316131}{5} 1 \quad \Delta = -34, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$1 \overset{15131613161316131}{5} 1 \quad \Delta = -34, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: 221613161316131612321

$$221613161316131612321 \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 36 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$221613161316131612321 \quad \Delta = -11, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$221613161316131612321 \overset{3}{3} \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\begin{aligned}
221613161316131612321\overset{3}{\color{red}}\text{ }_{[\# = 8]} \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
221613161316131612321\overset{3}{\color{red}}\text{ }_{[\# = 6]} \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
221613161\overset{2}{\color{red}}316131612321 \quad \Delta = -11, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
221613161316131612321\overset{3}{\color{red}} \quad \Delta = -20, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 15 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
221613161316131612321\overset{3}{\color{red}}\text{ }_{[\# = 7]} \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
22161316131613161\overset{2}{\color{red}}2321 \quad \Delta = -11, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
221613161316131612321\overset{2}{\color{red}} \quad \Delta = -11, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
221613161316131612321\overset{2}{\color{red}} \quad \Delta = -11, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 37 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
22161316131613161\overset{2}{\color{red}}2321 \quad \Delta = -11, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
221613161316131612321\overset{2}{\color{red}} \quad \Delta = -11, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
221613161316131612321\overset{2}{\color{red}} \quad \Delta = -11, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
221613161316131612321\overset{3}{\color{red}}\text{ }_{[\# = 4]} \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
221613161316131612321\overset{3}{\color{red}}\text{ }_{[\# = 2]} \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
221613161316131612321\overset{3}{\color{red}}\text{ }_{[\# = 9]} \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
221613161316131612321\overset{3}{\color{red}}\text{ }_{[\# = 5]} \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\text{Base: } 132161316131613161231 & \\
132161316131613161231\overset{3}{\color{red}}\text{ }_{[\# = 3]} \quad \Delta = -26, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
132161316131613161231\overset{3}{\color{red}}\text{ }_{[\# = 5]} \quad \Delta = -26, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
132161316131613161231\overset{3}{\color{red}}\text{ }_{[\# = 4]} \quad \Delta = -26, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
132161316131613161231\overset{3}{\color{red}}\text{ }_{[\# = 2]} \quad \Delta = -26, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
132161316131613161231 \quad \Delta = -18, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
132161316131613161231 \quad \Delta = -18, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
132161316131613161231 \quad \Delta = -18, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
132161316131613161231 \quad \Delta = -18, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 37 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

Base: 141316131613161316122

$$\begin{aligned}
141316131613161316122 \quad \Delta = -36, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
141316131613161316122 \quad \Delta = -36, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 37 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
141316131613161316122 \quad \Delta = -36, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
141316131613161316122 \quad \Delta = -36, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
141316131613161316122 \quad \Delta = -36, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
141316131613161316122 \quad \Delta = -36, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
141316131613161316122 \quad \Delta = -36, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
141316131613161316122 \quad \Delta = -36, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
141316131613161316122 \quad \Delta = -36, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 40 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
141316131613161316122 \quad \Delta = -45, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
141316131613161316122 \quad \Delta = -44, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 39 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
141316131613161316122 \quad \Delta = -44, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
2216131613161316131413_{[\# = 2]} \quad \Delta = -44, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
2216131613161316131413_{[\# = 3]} \quad \Delta = -44, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix}
\end{aligned}$$

Base: 132161316131613161313

$$132161316131613161313 \quad \Delta = -53, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 9 & 5 \\ 5 & 9 \end{pmatrix}$$

$$\begin{aligned}
313161316131613161231\overset{3}{3} \text{ }_{[\# = 3]} \quad \Delta = -33, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
313161316131613161231\overset{3}{3} \text{ }_{[\# = 2]} \quad \Delta = -33, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
132161316131613161313 \overset{2}{\quad} \quad \Delta = -25, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 37 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
132161316131613161313 \overset{2}{\quad} \quad \Delta = -25, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
132161316131613161313 \overset{2}{\quad} \quad \Delta = -25, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
132161316131613161313 \overset{2}{\quad} \quad \Delta = -25, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
132161316131613161313 \overset{2}{\quad} \quad \Delta = -25, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
132161316131613161313 \overset{2}{\quad} \quad \Delta = -25, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
132161316131613161313 \overset{2}{\quad} \quad \Delta = -25, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
132161316131613161313 \overset{2}{\quad} \quad \Delta = -25, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
132161316131613161313 \overset{2}{\quad} \quad \Delta = -25, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
132161316131613161313 \overset{3}{\quad} \quad \Delta = -33, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
313161316131613161231\overset{3}{3} \text{ }_{[\# = 5]} \quad \Delta = -33, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
313161316131613161231\overset{3}{3} \text{ }_{[\# = 4]} \quad \Delta = -33, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
132161316131613161313 \overset{3}{\quad} \text{ }_{[\# = 3]} \quad \Delta = -33, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
132161316131613161313 \overset{3}{\quad} \text{ }_{[\# = 2]} \quad \Delta = -33, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\text{Base: } 313161316131613161313 \\
313161316131613161313 \overset{3}{\quad} \text{ }_{[\# = 3]} \quad \Delta = -40, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 375 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
313161316131613161313 \overset{2}{\quad} \quad \Delta = -32, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
313161316131613161313 \overset{2}{\quad} \quad \Delta = -32, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{2}{3}13161316131613161313 \quad \Delta = -32, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 40 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& 313161316131613161313 \overset{2}{3} \quad \Delta = -32, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 37 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& 313161316131613161313 \overset{2}{3} \quad \Delta = -32, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& \overset{3}{3}13161316131613161313 \quad \Delta = -40, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 39 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{3}{3}13161316131613161313 \overset{21}{\text{Base: } 1613161316131612321} \quad \Delta = -40, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 165 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
& \overset{4}{3}13161316131613161313 \quad \Delta = -60, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 38 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{21}{16}13161316131612321 \overset{21}{2} \quad \Delta = -11, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& \overset{21}{16}13161316131612321 \overset{3}{3} \overset{21}{\text{Base: } 1613161316131612321} \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& \overset{21}{16}13161316131612321 \overset{3}{3} \overset{21}{\text{Base: } 1613161316131612321} \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{21}{16}13161316131612321 \overset{3}{3} \overset{21}{\text{Base: } 1613161316131612321} \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{21}{16}13161316131612321 \overset{3}{3} \overset{21}{\text{Base: } 1613161316131612321} \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{21}{16}13161316131612321 \overset{3}{3} \overset{21}{\text{Base: } 1613161316131612321} \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{21}{16}13161316131612321 \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 15 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{21}{16}13161316131612321 \overset{3}{3} \overset{21}{\text{Base: } 1613161316131612321} \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{21}{16}13161316131612321 \overset{3}{3} \overset{21}{\text{Base: } 1613161316131612321} \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{21}{16}13161316131612321 \overset{3}{3} \overset{21}{\text{Base: } 1613161316131612321} \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{21}{16}13161316131612321 \overset{2}{3} \quad \Delta = -11, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{21}{16}13161316131612321 \quad \Delta = -11, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{21}{16}13161316131612321 \overset{2}{3} \quad \Delta = -11, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{21}{1} \overset{2}{6} 13161316131612321 \quad \Delta = -11, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{21}{1} \overset{2}{6} 13161316131612321 \quad \Delta = -11, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{21}{1} \overset{2}{6} 13161316131612321 \quad \Delta = -11, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{21}{1} \overset{2}{6} 13161316131612321 \quad \Delta = -11, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{21}{1} 6 13161316131613141$

$$\begin{aligned}
& \overset{21}{1} \overset{2}{6} 13161316131613141 \quad \Delta = -36, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 37 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{21}{1} 6 13161316131613141 \overset{3}{3} \text{ [#=3]} \quad \Delta = -44, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& \overset{21}{1} 6 13161316131613141 \overset{3}{3} \text{ [#=2]} \quad \Delta = -44, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& \overset{21}{1} 6 13161316131613141 \overset{3}{3} \quad \Delta = -44, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{21}{1} 6 13161316131613141 \quad \Delta = -44, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 19 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{21}{1} \overset{2}{6} 13161316131613141 \quad \Delta = -36, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& \overset{21}{1} \overset{2}{6} 13161316131613141 \quad \Delta = -36, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 40 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& \overset{21}{1} 6 13161316131613141 \overset{2}{2} \quad \Delta = -36, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{21}{1} 6 13161316131613141 \overset{2}{2} \quad \Delta = -36, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{21}{1} 6 13161316131613141 \overset{2}{2} \quad \Delta = -36, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{21}{1} 6 13161316131613141 \overset{2}{2} \quad \Delta = -36, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{21}{1} 6 13161316131613141 \overset{2}{2} \quad \Delta = -36, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{21}{1} 6 13161316131613141 \quad \Delta = -36, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 20 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{131}{1} 6 1316131613161231$

$$\overset{3131}{1} \overset{6}{6} 1316131613161231 \text{ [#=7]} \quad \Delta = -25, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix}$$

$$\begin{aligned}
& \overset{3131}{1 \ 6 \ 1316131613161231} \text{ }_{[\#=3]} \quad \Delta = -25, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 1316131613161231} \overset{3}{\text{ }} \text{ }_{[\#=3]} \quad \Delta = -25, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 1316131613161231} \overset{2}{\text{ }} \quad \Delta = -17, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 1316131613161231} \overset{2}{\text{ }} \quad \Delta = -17, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 6 \ 1316131613161231} \text{ }_{[\#=9]} \quad \Delta = -25, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 6 \ 1316131613161231} \text{ }_{[\#=6]} \quad \Delta = -25, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 1316131613161231} \overset{3}{\text{ }} \text{ }_{[\#=5]} \quad \Delta = -25, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 6 \ 1316131613161231} \text{ }_{[\#=5]} \quad \Delta = -25, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 1316131613161231} \overset{3}{\text{ }} \text{ }_{[\#=4]} \quad \Delta = -25, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 1316131613161231} \overset{2}{\text{ }} \quad \Delta = -17, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 1316131613161231} \overset{2}{\text{ }} \quad \Delta = -17, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 1316131613161231} \quad \Delta = -17, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 1316131613161231} \overset{2}{\text{ }} \quad \Delta = -17, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 1316131613161231} \overset{2}{\text{ }} \quad \Delta = -17, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 1316131613161231} \overset{2}{\text{ }} \quad \Delta = -17, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 6 \ 1316131613161231} \text{ }_{[\#=4]} \quad \Delta = -25, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 6 \ 1316131613161231} \text{ }_{[\#=2]} \quad \Delta = -25, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 1316131613161231} \overset{3}{\text{ }} \text{ }_{[\#=2]} \quad \Delta = -25, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 6 \ 1316131613161231} \text{ }_{[\#=8]} \quad \Delta = -25, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{131}{1 \ 6 \ 1316131613161313}$

$$\begin{aligned}
& \overset{3131}{1 \ 6 \ 1316131613161313} \text{ [#=7]} \quad \Delta = -32, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 1316131613161313} \overset{3}{} \quad \Delta = -32, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 6 \ 1316131613161313} \text{ [#=2]} \quad \Delta = -32, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 1316131613161313} \overset{3}{} \text{ [#=2]} \quad \Delta = -32, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 6 \ 1316131613161313} \text{ [#=8]} \quad \Delta = -32, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 6 \ 1316131613161313} \text{ [#=6]} \quad \Delta = -32, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 6 \ 1316131613161313} \text{ [#=3]} \quad \Delta = -32, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 1316131613161313} \overset{3}{} \text{ [#=3]} \quad \Delta = -32, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 6 \ 1316131613161313} \text{ [#=9]} \quad \Delta = -32, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 6 \ 1316131613161313} \text{ [#=5]} \quad \Delta = -32, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
& \overset{3131}{1 \ 6 \ 1316131613161313} \text{ [#=4]} \quad \Delta = -32, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 1316131613161313} \overset{4}{} \quad \Delta = -52, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 9 & 5 \\ 5 & 9 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 1316131613161313} \overset{2}{} \quad \Delta = -24, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 1316131613161313} \overset{2}{} \quad \Delta = -24, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 1316131613161313} \quad \Delta = -24, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 1316131613161313} \overset{2}{} \quad \Delta = -24, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 1316131613161313} \overset{2}{} \quad \Delta = -24, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 1316131613161313} \overset{2}{} \quad \Delta = -24, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{131}{1 \ 6 \ 1316131613161313} \overset{2}{} \quad \Delta = -24, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{l}
\begin{array}{c} 131 \\ 1 \ 6 \end{array} \begin{array}{c} \textcolor{red}{2} \\ 1316131613161313 \end{array} \quad \Delta = -24, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\begin{array}{c} 131 \\ 1 \ 6 \end{array} \begin{array}{c} \textcolor{red}{2} \\ 1316131613161313 \end{array} \quad \Delta = -24, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{array}$$

Base: $\begin{array}{c} 131 \\ 1 \ 6 \end{array} 131613161316131$

$$\begin{array}{c} \textcolor{red}{3}131 \\ 1 \ 6 \end{array} \begin{array}{c} 1 \\ 131613161316131 \end{array} \text{ [#=9]} \quad \Delta = -24, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix}$$

$$\begin{array}{c} 131 \\ 1 \ 6 \end{array} \begin{array}{c} \textcolor{red}{2} \\ 131613161316131 \end{array} \quad \Delta = -16, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\begin{array}{c} 131 \\ 1 \ 6 \end{array} \begin{array}{c} 1 \\ 131613161316131 \end{array} \quad \Delta = -16, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\begin{array}{c} 131 \\ 1 \ 6 \end{array} \begin{array}{c} \textcolor{red}{2} \\ 131613161316131 \end{array} \quad \Delta = -16, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$\begin{array}{c} 131\textcolor{red}{2} \\ 1 \ 6 \end{array} \begin{array}{c} 1 \\ 131613161316131 \end{array} \quad \Delta = -16, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$\begin{array}{c} \textcolor{red}{3}131 \\ 1 \ 6 \end{array} \begin{array}{c} 1 \\ 131613161316131 \end{array} \text{ [#=4]} \quad \Delta = -24, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix}$$

$$\begin{array}{c} \textcolor{red}{3}131 \\ 1 \ 6 \end{array} \begin{array}{c} 1 \\ 131613161316131 \end{array} \text{ [#=8]} \quad \Delta = -24, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix}$$

$$\begin{array}{c} \textcolor{red}{3}131 \\ 1 \ 6 \end{array} \begin{array}{c} 1 \\ 131613161316131 \end{array} \text{ [#=7]} \quad \Delta = -24, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix}$$

$$\begin{array}{c} \textcolor{red}{3}131 \\ 1 \ 6 \end{array} \begin{array}{c} 1 \\ 131613161316131 \end{array} \text{ [#=6]} \quad \Delta = -24, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix}$$

$$\begin{array}{c} \textcolor{red}{3}131 \\ 1 \ 6 \end{array} \begin{array}{c} 1 \\ 131613161316131 \end{array} \text{ [#=5]} \quad \Delta = -24, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

$$\begin{array}{c} \textcolor{red}{3}131 \\ 1 \ 6 \end{array} \begin{array}{c} 1 \\ 131613161316131 \end{array} \text{ [#=3]} \quad \Delta = -24, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$\begin{array}{c} \textcolor{red}{3}131 \\ 1 \ 6 \end{array} \begin{array}{c} 1 \\ 131613161316131 \end{array} \text{ [#=2]} \quad \Delta = -24, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

Base: $\begin{array}{c} 1 \\ 6131613161316131 \end{array}$

$$6131\begin{array}{c} \textcolor{red}{2}1 \\ 6 \end{array} 131613161316131 \quad \Delta = -94, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$613161\begin{array}{c} 1 \\ 131613161316131 \end{array} \begin{array}{c} \textcolor{red}{2} \\ 1316131 \end{array} \quad \Delta = -94, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$613161\begin{array}{c} 1 \\ 131613161316131 \end{array} \begin{array}{c} \textcolor{red}{2} \\ 1316131 \end{array} \quad \Delta = -94, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$\begin{array}{c} \textcolor{red}{2} \\ 613161 \end{array} \begin{array}{c} 1 \\ 131613161316131 \end{array} \quad \Delta = -94, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\begin{array}{ll}
6131\overset{21}{6}131613161316131 & \Delta = -94, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
6131\overset{12}{6}131613161316131 & \Delta = -94, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
6131\overset{1}{6}13161\overset{2}{3}161316131 & \Delta = -94, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
6131\overset{1}{6}13161\overset{2}{3}161316131 & \Delta = -94, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
6131\overset{1}{6}13161\overset{2}{3}161316131 & \Delta = -94, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
6131\overset{1}{6}13161\overset{2}{3}161316131 & \Delta = -94, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
6131\overset{1}{6}131613161316131\overset{3}{3} \text{ [#=9]} & \Delta = -102, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
6131\overset{1}{6}131613161316131\overset{3}{3} \text{ [#=8]} & \Delta = -102, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
6131\overset{1}{6}131613161316131\overset{3}{3} \text{ [#=7]} & \Delta = -102, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
6131\overset{1}{6}131613161316131\overset{3}{3} \text{ [#=6]} & \Delta = -102, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
6131\overset{1}{6}131613161316131\overset{3}{3} \text{ [#=5]} & \Delta = -102, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
6131\overset{1}{6}131613161316131\overset{3}{3} \text{ [#=4]} & \Delta = -102, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
6131\overset{31}{6}131613161316131 & \Delta = -102, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
6131\overset{1}{6}131613161316131\overset{3}{3} \text{ [#=2]} & \Delta = -102, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
6131\overset{1}{6}131613161316131\overset{3}{3} \text{ [#=3]} & \Delta = -102, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\text{Base: } 31\overset{31}{6}1316131613151321 & \\
31\overset{31}{6}1316131613151321\overset{3}{3} \text{ [#=5]} & \Delta = -7, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 197 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
31\overset{31}{6}1316131613151321\overset{3}{3} \text{ [#=4]} & \Delta = -7, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
31\overset{31}{6}1316131613151321\overset{3}{3} \text{ [#=6]} & \Delta = -7, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 285 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
31\overset{31}{6}1316131613151321\overset{5}{5} & \Delta = -51, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
& 31 \text{ } 3161316131613151321\textcolor{red}{3} \text{ }_{[\# = 7]} \quad \Delta = -7, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 196 \quad \begin{pmatrix} 0 & 14 & 2 \\ 14 & 389 & 55 \\ 2 & 55 & 8 \end{pmatrix} \\
& 31 \text{ } 3161316131613151321\textcolor{red}{4} \quad \Delta = -27, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 31 \text{ } 3161316131613151321\textcolor{red}{3} \text{ }_{[\# = 3]} \quad \Delta = -7, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& 31 \text{ } 3161316131613151321\textcolor{red}{3} \text{ }_{[\# = 2]} \quad \Delta = -7, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& 31 \text{ } 3161316131613151321\textcolor{red}{3} \quad \Delta = -7, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 31 \text{ } 3161316131613151321\textcolor{red}{3} \text{ }_{[\# = 8]} \quad \Delta = -7, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 256 \quad \begin{pmatrix} 0 & 16 & 2 \\ 16 & 509 & 63 \\ 2 & 63 & 8 \end{pmatrix} \\
& 31 \text{ } 3161316131613151321\textcolor{red}{7} \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 31 \text{ } 3161316131613151321\textcolor{red}{6} \quad \Delta = -77, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 31 \text{ } 3161316131613151321\textcolor{red}{3} \text{ }_{[\# = 9]} \quad \Delta = -7, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 324 \quad \begin{pmatrix} 0 & 18 & 2 \\ 18 & 645 & 71 \\ 2 & 71 & 8 \end{pmatrix} \\
& 31 \text{ } 3161316131613151321\textcolor{red}{8} \quad \Delta = -132, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \textbf{Base: } 31 \text{ } 3161316131613161231 \\
& 31 \text{ } 3161316131613161231\textcolor{red}{4} \quad \Delta = -53, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 31 \text{ } 3161316131613161231\textcolor{red}{3} \quad \Delta = -33, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 31 \text{ } 3161316131613161231\textcolor{red}{2} \quad \Delta = -27, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 31 \text{ } 3161316131613161231\textcolor{red}{2} \quad \Delta = -25, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& 31 \text{ } 3161316131613161231\textcolor{red}{2} \quad \Delta = -25, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 46 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
& 31 \text{ } 3161316131613161231\textcolor{red}{2} \quad \Delta = -25, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \textcolor{red}{2}31 \text{ } 3161316131613161231 \quad \Delta = -25, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 31 \text{ } 3161316131613161231\textcolor{red}{5} \quad \Delta = -77, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 31 \text{ } 3161316131613161231\textcolor{red}{2} \quad \Delta = -25, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 37 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
^{31}3161316131613161231 \quad \Delta = -25, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
^{31}3161316131613161231 \quad \Delta = -25, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
^{31}3161316131613161231 \quad \Delta = -25, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
^{31}3161316131613161231 \quad \Delta = -25, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
^{31}31613161316131612313_{[\# = 4]} \quad \Delta = -33, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 109 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
^{31}31613161316131612313_{[\# = 5]} \quad \Delta = -33, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 172 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
^{31}31613161316131612313_{[\# = 3]} \quad \Delta = -33, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 60 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
^{31}31613161316131612313_{[\# = 2]} \quad \Delta = -33, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{aligned}$$

Base: $^{31}3161316131613161313$

$$\begin{aligned}
^{31}3161316131613161313 \quad \Delta = -40, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 48 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
^{31}3161316131613161313 \quad \Delta = -32, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 37 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
^{31}3161316131613161313 \quad \Delta = -32, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
^{31}3161316131613161313 \quad \Delta = -32, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
^{31}3161316131613161313 \quad \Delta = -32, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
^{231}3161316131613161313 \quad \Delta = -32, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
^{31}3161316131613161313 \quad \Delta = -32, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
^{31}3161316131613161313 \quad \Delta = -32, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 49 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
^{31}3161316131613161313 \quad \Delta = -60, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 47 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
^{31}3161316131613161313_{[\# = 2]} \quad \Delta = -40, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 144 & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 201 & 37 \\ 2 & 37 & 8 \end{pmatrix} \\
^{31}3161316131613161313_{[\# = 3]} \quad \Delta = -40, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 324 & \begin{pmatrix} 0 & 18 & 2 \\ 18 & 456 & 56 \\ 2 & 56 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{ll}
{}^{31}3161316131613161313 \quad \Delta = -32, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 46 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
{}^{31}3161316131613161313 \quad \Delta = -32, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^{31}31613161316131613132 \quad \Delta = -33, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\text{Base: } {}^{31}316131613161316131 & \\
{}^{31}316131613161316131 \quad \Delta = -24, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
{}^{31}3161316131613161313 \quad \Delta = -32, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
{}^{31}316131613161316131 \quad \Delta = -32, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
{}^{31}316131613161316131 \quad \Delta = -24, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
{}^{31}3161316131613161312 \quad \Delta = -24, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
{}^{31}3161316131613161313_{[\# = 3]} \quad \Delta = -32, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 60 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
{}^{31}3161316131613161313_{[\# = 4]} \quad \Delta = -32, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 109 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
{}^{31}3161316131613161313_{[\# = 2]} \quad \Delta = -32, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
{}^{31}3161316131613161313_{[\# = 9]} \quad \Delta = -32, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 324 & \begin{pmatrix} 0 & 18 & 2 \\ 18 & 564 & 62 \\ 2 & 62 & 8 \end{pmatrix} \\
{}^{31}3161316131613161316 \quad \Delta = -102, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
{}^{31}316131613161316131 \quad \Delta = -24, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
{}^{31}316131613161316131 \quad \Delta = -24, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^{31}3161316131613161313_{[\# = 5]} \quad \Delta = -32, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 172 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
{}^{31}316131613161316131 \quad \Delta = -24, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 20 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
{}^{31}316131613161316131 \quad \Delta = -24, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
{}^{31}316131613161316131 \quad \Delta = -24, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{array}$$

$$31 \overset{1}{6} 131613161316131 \overset{1}{3} \text{ [#}=8] \quad \Delta = -32, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 256 \quad \begin{pmatrix} 0 & 16 & 2 \\ 16 & 445 & 55 \\ 2 & 55 & 8 \end{pmatrix}$$

$$31 \overset{1}{6} 131613161316131 \overset{1}{4} \quad \Delta = -52, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\overset{231}{31} \overset{1}{6} 131613161316131 \quad \Delta = -24, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$31 \overset{1}{6} 131613161316131 \overset{1}{5} \quad \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$31 \overset{1}{6} 131613161316131 \overset{1}{3} \text{ [#}=7] \quad \Delta = -32, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 196 \quad \begin{pmatrix} 0 & 14 & 2 \\ 14 & 340 & 48 \\ 2 & 48 & 8 \end{pmatrix}$$

$$31 \overset{1}{6} 131613161316131 \overset{1}{3} \text{ [#}=6] \quad \Delta = -32, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 249 & 41 \\ 2 & 41 & 8 \end{pmatrix}$$

$$31 \overset{1}{6} 131613161316131 \overset{2}{1} \quad \Delta = -24, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 37 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

Base: 712315131613161316131

$$712315 \overset{2}{1} 31613161316131 \quad \Delta = -96, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 142 & 24 \\ 2 & 24 & 8 \end{pmatrix}$$

$$71231513 \overset{2}{1} 613161316131 \quad \Delta = -96, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 168 & 26 \\ 2 & 26 & 8 \end{pmatrix}$$

$$7123151316 \overset{2}{1} 3161316131 \quad \Delta = -96, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$\overset{2}{71} 2315131613161316131 \quad \Delta = -96, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 89 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$712315131613161316 \overset{2}{1} 31 \quad \Delta = -96, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$712315 \overset{2}{1} 31613161316131 \quad \Delta = -96, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 207 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

$$712315131613161316 \overset{2}{1} 31 \quad \Delta = -96, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$712315131613161316131 \overset{1}{3} \text{ [#}=2] \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$712315131613161316131 \overset{1}{3} \text{ [#}=3] \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$712315131613161316131 \overset{1}{3} \text{ [#}=4] \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$712315131613 \overset{2}{1} 61316131 \quad \Delta = -96, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$712315131613161316 \overset{2}{1} 31 \quad \Delta = -96, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$\begin{aligned}
712315131613161316131\overset{3}{3} \text{ [#=5]} \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
712315131613161316131\overset{3}{3} \text{ [#=6]} \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
712315131613161316131\overset{3}{3} \text{ [#=7]} \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
712315131613161316131\overset{3}{3} \text{ [#=8]} \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
712315131613161316131\overset{3}{3} \text{ [#=9]} \quad \Delta = -104, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 712321613161316131612

$$\begin{aligned}
71232161316131613161\overset{2}{2} \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
71232161316131613161\overset{2}{2} \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
71232161316131613161\overset{2}{2} \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 89 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
71232161316131613161\overset{2}{2} \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 168 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
71232161316131613161\overset{2}{2} \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
71232161316131613161\overset{2}{2} \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
71232161316131613161\overset{2}{2} \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 297 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
71232161316131613161\overset{2}{2} \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
71232161316131613161\overset{2}{2} \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 207 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
71232161316131613161\overset{3}{3} \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}
\end{aligned}$$

Base: 71232161316131613161¹

$$\begin{aligned}
71232161316131613161\overset{2}{2}\overset{1}{1} \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
71232161316131613161\overset{2}{2}\overset{1}{1} \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 207 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
71232161316131613161\overset{2}{2}\overset{1}{1} \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 297 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
71232161316131613161\overset{2}{2}\overset{1}{1} \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 168 & 26 \\ 2 & 26 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
71232161316131613161^1 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
71232161316131613161^1 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
71232161316131613161^{21} \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
71232161316131613161^1 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 89 & 19 \\ 2 & 19 & 8 \end{pmatrix}
\end{aligned}$$

Base: 812231513161316131612

$$\begin{aligned}
812231513161316131612^3 \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
812231513161316131612^2 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
812231513161316131612^2 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
812231513161316131612^2 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
812231513161316131612^2 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
812231513161316131612^2 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
812231513161316131612^2 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 118 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
812231513161316131612^2 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 142 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
812231513161316131612^2 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 207 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
812231513161316131612^2 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 168 & 26 \\ 2 & 26 & 8 \end{pmatrix}
\end{aligned}$$

Base: 81223151316131613161^1

$$\begin{aligned}
81223151316131613161^1 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 118 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
81223151316131613161^1 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
81223151316131613161^1 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
81223151316131613161^{21} \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
81223151316131613161^1 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{ll}
81223151316131613161^1 & \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 168 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
81223151316131613161^1 & \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 142 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
81223151316131613161^1 & \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 207 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
\text{Base: } 812315131613161316123 & \\
812315131613161316123 & \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
812315131613161316123 & \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
812315131613161316123 & \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
812315131613161316123 & \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
812315131613161316123 & \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
812315131613161316123 & \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
812315131613161316123 & \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 196 \quad \begin{pmatrix} 0 & 14 & 2 \\ 14 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
812315131613161316123 & \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
812315131613161316123 & \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\text{Base: } 812321613161316131513 & \\
812321613161316131513 & \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
812321613161316131513 & \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
812321613161316131513 & \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 196 \quad \begin{pmatrix} 0 & 14 & 2 \\ 14 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
812321613161316131513 & \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 256 \quad \begin{pmatrix} 0 & 16 & 2 \\ 16 & 206 & 28 \\ 2 & 28 & 8 \end{pmatrix} \\
812321613161316131513 & \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
812321613161316131513 & \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
812321613161316131513 \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
812321613161316131513 \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
812321613161316131513 \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

Base: 8123151316131613161³¹

$$\begin{aligned}
8123151316131613161 \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
8123151316131613161 \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 196 & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
8123151316131613161 \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
8123151316131613161 \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
8123151316131613161 \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
8123151316131613161 \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
8123151316131613161 \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
8123151316131613161 \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
8123151316131613161 \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
8123151316131613161 \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: 216131613161315132181

$$\begin{aligned}
216131613161315132181 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
216131613161315132181 \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
216131613161315132181 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
216131613161315132181 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
216131613161315132181 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

216131613161315132181	$\Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, \det = 121$	$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix}$
216131613161315132181	$\Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$
216131613161315132181	$\Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$
216131613161315132181	$\Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
216131613161315132181	$\Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, \det = 81$	$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$
Base: 151316131613161232181		
151316131613161232181	$\Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$
151316131613161232181	$\Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$
151316131613161232181	$\Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, \det = 100$	$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix}$
151316131613161232181	$\Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
151316131613161232181	$\Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$
151316131613161232181	$\Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, \det = 169$	$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 206 & 28 \\ 2 & 28 & 8 \end{pmatrix}$
151316131613161232181	$\Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, \det = 121$	$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix}$
151316131613161232181	$\Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, \det = 64$	$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$
Base: 16131613161315132181		
16131613161315132181	$\Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$
16131613161315132181	$\Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$
16131613161315132181	$\Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$
16131613161315132181	$\Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, \det = 81$	$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$
16131613161315132181	$\Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, \det = 121$	$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix}$

$$\begin{array}{lcl}
{}^1_{16131613161315132181} & \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
{}^1_{16131613161315132181} & \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
{}^{12}_{16131613161315132181} & \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{array}$$

Base: 132171232181231513221

$$\begin{array}{lcl}
132171232181231513221\mathbf{3}_{[\# = 7]} & \Delta = 0, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
132171232181231513221\mathbf{3}_{[\# = 6]} & \Delta = 0, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
122315132181232171231\mathbf{3}_{[\# = 2]} & \Delta = 0, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
132171232181231513221\mathbf{3}_{[\# = 4]} & \Delta = 0, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
122315132181232171231\mathbf{3}_{[\# = 4]} & \Delta = 0, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
132171232181231513221\mathbf{3}_{[\# = 8]} & \Delta = 0, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
132171232181231513221\mathbf{3}_{[\# = 9]} & \Delta = 0, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
122315132181232171231\mathbf{3}_{[\# = 5]} & \Delta = 0, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
132171232181231513221\mathbf{3}_{[\# = 5]} & \Delta = 0, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
132171232181231513221\mathbf{3}_{[\# = 3]} & \Delta = 0, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
122315132181232171231\mathbf{3}_{[\# = 3]} & \Delta = 0, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
132171232181231513221\mathbf{3}_{[\# = 2]} & \Delta = 0, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{array}$$

Base: 123217123218123151321

$$\begin{array}{lcl}
123217123218123151321\mathbf{3}_{[\# = 8]} & \Delta = -2, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
123151321812321712321\mathbf{3}_{[\# = 3]} & \Delta = -2, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
123217123218123151321\mathbf{3}_{[\# = 4]} & \Delta = -2, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
123151321812321712321\overset{2}{3} \text{ [#=5]} \quad \Delta = -2, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
123217123218123151321\overset{2}{3} \text{ [#=9]} \quad \Delta = -2, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
123151321812321712321\overset{2}{3} \text{ [#=9]} \quad \Delta = -2, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
123151321812321712321\overset{2}{3} \text{ [#=6]} \quad \Delta = -2, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
123151321812321712321\overset{2}{3} \text{ [#=8]} \quad \Delta = -2, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
123217123218123151321\overset{2}{3} \text{ [#=7]} \quad \Delta = -2, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
123151321812321712321\overset{2}{3} \text{ [#=2]} \quad \Delta = -2, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
123217123218123151321\overset{2}{3} \text{ [#=2]} \quad \Delta = -2, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
123151321812321712321\overset{2}{3} \text{ [#=4]} \quad \Delta = -2, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
123217123218123151321\overset{2}{3} \text{ [#=3]} \quad \Delta = -2, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
123151321812321712321\overset{2}{3} \text{ [#=7]} \quad \Delta = -2, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
123217123218123151321\overset{2}{3} \text{ [#=5]} \quad \Delta = -2, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
123217123218123151321\overset{2}{3} \text{ [#=6]} \quad \Delta = -2, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix}
\end{aligned}$$

Base: 812231513218123217123

$$\begin{aligned}
812231513218123217123\overset{2}{2} \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
812231513218123217123\overset{2}{2} \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
812231513218123217123\overset{2}{2} \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
812231513218123217123\overset{2}{2} \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
812231513218123217123\overset{2}{2} \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
812231513218123217123\overset{2}{2} \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$812231513218123217123 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

Base: 812321712321812315132

$$812321712321812315132 \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$812321712321812315132 \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$812321712321812315132 \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$812321712321812315132 \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 225 \quad \begin{pmatrix} 0 & 15 & 2 \\ 15 & 163 & 25 \\ 2 & 25 & 8 \end{pmatrix}$$

$$812321712321812315132 \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 256 \quad \begin{pmatrix} 0 & 16 & 2 \\ 16 & 206 & 28 \\ 2 & 28 & 8 \end{pmatrix}$$

$$812321712321812315132 \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 361 \quad \begin{pmatrix} 0 & 19 & 2 \\ 19 & 283 & 33 \\ 2 & 33 & 8 \end{pmatrix}$$

$$812321712321812315132 \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 289 \quad \begin{pmatrix} 0 & 17 & 2 \\ 17 & 219 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

Base: 812315132181232171232

$$812315132181232171232 \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix}$$

$$812315132181232171232 \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$812315132181232171232 \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$812315132181232171232 \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 196 \quad \begin{pmatrix} 0 & 14 & 2 \\ 14 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix}$$

$$812315132181232171232 \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$812315132181232171232 \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 289 \quad \begin{pmatrix} 0 & 17 & 2 \\ 17 & 219 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

$$812315132181232171232 \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 225 \quad \begin{pmatrix} 0 & 15 & 2 \\ 15 & 163 & 25 \\ 2 & 25 & 8 \end{pmatrix}$$

Base: 217123218123151322181

$$217123218123151322181 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$217123218123151322181 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$217123218123151322181 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

$$\begin{aligned}
217123218123151322181 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
217123218123151322181 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
217123218123151322181 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
217123218123151322181 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 17123218123151322181

$$\begin{aligned}
17123218123151322181 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
17123218123151322181 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
17123218123151322181 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
17123218123151322181 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
17123218123151322181 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
17123218123151322181 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}
\end{aligned}$$

Base: 321712321812315132181

$$\begin{aligned}
321712321812315132181 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
321712321812315132181 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 152 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
321712321812315132181 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 163 & 25 \\ 2 & 25 & 8 \end{pmatrix} \\
321712321812315132181 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 54 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
321712321812315132181 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
321712321812315132181 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 219 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
321712321812315132181 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix}
\end{aligned}$$

Base: 181232171232181231513

$$181232171232181231513 \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$\begin{aligned}
181232171232181231513 \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
181232171232181231513 \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 225 & \begin{pmatrix} 0 & 15 & 2 \\ 15 & 163 & 25 \\ 2 & 25 & 8 \end{pmatrix} \\
181232171232181231513 \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
181232171232181231513 \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
181232171232181231513 \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
181232171232181231513 \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}
\end{aligned}$$

Base: 217123218123151321812

$$\begin{aligned}
217123218123151321812 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
217123218123151321812 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
217123218123151321812 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
217123218123151321812 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
217123218123151321812 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
217123218123151321812 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
217123218123151321812 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
217123218123151321812 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix}
\end{aligned}$$

Base: 218123217123218123151

$$\begin{aligned}
218123217123218123151 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
218123217123218123151 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
218123217123218123151 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
218123217123218123151 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
218123217123218123151 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
218123217123218123151 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
218123217123218123151 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 17123218123151321812

$$\begin{aligned}
17123218123151321812 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
17123218123151321812 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
17123218123151321812 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
17123218123151321812 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
17123218123151321812 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
17123218123151321812 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
17123218123151321812 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}
\end{aligned}$$

Base: 171232181231513218123

$$\begin{aligned}
171232181231513218123 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
171232181231513218123 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 196 & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
171232181231513218123 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
171232181231513218123 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 144 & \begin{pmatrix} 0 & 12 & 2 \\ 12 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
171232181231513218123 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
171232181231513218123 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
171232181231513218123 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: 321812321712321812315

$$321812321712321812315 \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\begin{aligned}
321812321712321812315^2 \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
321812321712321812315^2 \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 196 & \begin{pmatrix} 0 & 14 & 2 \\ 14 & 68 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
321812321712321812315^2 \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 53 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
321812321712321812315^2 \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 289 & \begin{pmatrix} 0 & 17 & 2 \\ 17 & 117 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
321812321712321812315^2 \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 324 & \begin{pmatrix} 0 & 18 & 2 \\ 18 & 124 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
321812321712321812315^2 \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: 132181232171232181231

$$\begin{aligned}
132181232171232181231^3_{[\# = 4]} \quad \Delta = -83, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
132181232171232181231^3_{[\# = 3]} \quad \Delta = -83, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
132181232171232181231^3_{[\# = 5]} \quad \Delta = -83, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
132181232171232181231^3_{[\# = 2]} \quad \Delta = -83, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
132181232171232181231^2 \quad \Delta = -75, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
132181232171232181231^2 \quad \Delta = -75, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
132181232171232181231^2 \quad \Delta = -75, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}
\end{aligned}$$

Base: 132171232181232171231

$$\begin{aligned}
132171232181232171231^3_{[\# = 2]} \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
132171232181232171231^3_{[\# = 3]} \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
132171232181232171231^3_{[\# = 4]} \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
132171232181232171231^3_{[\# = 5]} \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
132171232181232171231^2 \quad \Delta = -47, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

$$132171232181232171231 \quad \Delta = -47, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$132171232181232171231 \quad \Delta = -47, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

Base: 131612321812321712321

$$131612321812321712321 \quad \Delta = -21, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1232171232181232161313_{[\# = 4]} \quad \Delta = -29, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$1316123218123217123213_{[\# = 6]} \quad \Delta = -29, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$1316123218123217123213_{[\# = 4]} \quad \Delta = -29, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$131612321812321712321 \quad \Delta = -21, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1232171232181232161313_{[\# = 5]} \quad \Delta = -29, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix}$$

$$131612321812321712321 \quad \Delta = -21, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$131612321812321712321 \quad \Delta = -21, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1232171232181232161313_{[\# = 6]} \quad \Delta = -29, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix}$$

$$1316123218123217123213_{[\# = 5]} \quad \Delta = -29, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$1232171232181232161313_{[\# = 9]} \quad \Delta = -29, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix}$$

$$1232171232181232161313_{[\# = 3]} \quad \Delta = -29, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$1316123218123217123213_{[\# = 2]} \quad \Delta = -29, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1232171232181232161313_{[\# = 8]} \quad \Delta = -29, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix}$$

$$1316123218123217123213_{[\# = 8]} \quad \Delta = -29, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$$1232171232181232161313_{[\# = 2]} \quad \Delta = -29, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$131612321812321712321 \quad \Delta = -21, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\begin{aligned}
131612321812321712321\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -29, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
131612321812321712321\textcolor{red}{3} \text{ [#=7]} \quad \Delta = -29, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
123217123218123216131\textcolor{red}{3} \text{ [#=7]} \quad \Delta = -29, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
131612321812321\textcolor{red}{2}712321 \quad \Delta = -21, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
131612321812321712321\textcolor{red}{3} \text{ [#=3]} \quad \Delta = -29, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\textbf{Base: } 612321712321812321712 \\
612321\textcolor{red}{2}712321812321712 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 100 \quad & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 168 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
612321712321812321\textcolor{red}{2}712 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
612321\textcolor{red}{2}712321812321712 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
612321712321812321\textcolor{red}{2}712 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
612321712321812321\textcolor{red}{2}712 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
61232171232181\textcolor{red}{2}2321712 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
612321\textcolor{red}{2}712321812321712 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 \quad & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 196 & 28 \\ 2 & 28 & 8 \end{pmatrix} \\
\textbf{Base: } 221712321812321712321 \\
221712321812321712321\textcolor{red}{3} \text{ [#=4]} \quad \Delta = -48, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
221712321812321712321 \quad \Delta = -40, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
221\textcolor{red}{2}712321812321712321 \quad \Delta = -40, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
221712321812321\textcolor{red}{2}712321 \quad \Delta = -40, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
221712321812321712321\textcolor{red}{3} \text{ [#=9]} \quad \Delta = -48, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
221712321812321712321\textcolor{red}{3} \text{ [#=8]} \quad \Delta = -48, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
221712321812321712321\overset{3}{3} \text{ }_{[\# = 3]} \quad \Delta = -48, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
221712321812321712321\overset{3}{3} \text{ }_{[\# = 7]} \quad \Delta = -48, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
221712321812321712321\overset{3}{3} \text{ }_{[\# = 5]} \quad \Delta = -48, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
221712321812321712321\overset{3}{3} \text{ }_{[\# = 6]} \quad \Delta = -48, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
221712321812321712321\overset{3}{3} \text{ }_{[\# = 2]} \quad \Delta = -48, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
221712321812321712321\overset{2}{2} \quad \Delta = -40, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
221712321812321712321\overset{2}{2} \quad \Delta = -40, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
221712321812321712321\overset{3}{3} \quad \Delta = -49, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
221712321812321712321\overset{2}{2} \quad \Delta = -40, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 712321712321812321613

$$\begin{aligned}
712321712321812321613\overset{2}{2} \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
712321712321812321613\overset{2}{2} \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 225 & \begin{pmatrix} 0 & 15 & 2 \\ 15 & 163 & 25 \\ 2 & 25 & 8 \end{pmatrix} \\
712321712321812321613\overset{2}{2} \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
712321712321812321613\overset{2}{2} \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
712321712321812321613\overset{2}{2} \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
712321712321812321613\overset{2}{2} \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
712321712321812321613\overset{2}{2} \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: 161232181232171232171

$$\begin{aligned}
161232181232171232171\overset{2}{2} \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
161232181232171232171\overset{2}{2} \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
161232181232171232171 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
161232181232171232171 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
161232181232171232171 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 163 & 25 \\ 2 & 25 & 8 \end{pmatrix} \\
161232181232171232171 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix}
\end{aligned}$$

Base: 61232171232181232171

$$\begin{aligned}
61232171232181232171 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 168 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
61232171232181232171 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
61232171232181232171 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
61232171232181232171 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 196 & 28 \\ 2 & 28 & 8 \end{pmatrix} \\
61232171232181232171 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
61232171232181232171 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: $17^{21}12321812321712321$

$$\begin{aligned}
17^{21}12321812321712321 \quad \Delta = -40, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
17^{21}12321812321712321 \quad \Delta = -40, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
17^{21}12321812321712321 \quad \Delta = -40, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
17^{21}12321812321712321 \quad \Delta = -48, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
17^{21}12321812321712321 \quad \Delta = -40, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
17^{21}12321812321712321 \quad \Delta = -48, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
17^{21}12321812321712321 \quad \Delta = -40, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
17^{21}12321812321712321 \quad \Delta = -40, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{21}{17}12321812321712321\overset{\color{red}3}{\text{}}_{[\# = 4]} \quad \Delta = -48, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{21}{17}12321812321712321\overset{\color{red}3}{\text{}}_{[\# = 9]} \quad \Delta = -48, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{21}{17}12321812321712321\overset{\color{red}3}{\text{}}_{[\# = 8]} \quad \Delta = -48, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{21}{17}12321812321712321\overset{\color{red}3}{\text{}}_{[\# = 5]} \quad \Delta = -48, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{21}{17}12321812321712321\overset{\color{red}3}{\text{}}_{[\# = 6]} \quad \Delta = -48, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{21}{17}12321812321712321\overset{\color{red}3}{\text{}}_{[\# = 7]} \quad \Delta = -48, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix}
\end{aligned}$$

Base: 218123217123218123215

$$\begin{aligned}
& 21812321712321812321\overset{\color{red}2}{5} \quad \Delta = -120, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 21812321712321812321\overset{\color{red}2}{5} \quad \Delta = -120, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& 21812321712321812321\overset{\color{red}2}{5} \quad \Delta = -120, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& 21812321712321812321\overset{\color{red}2}{5} \quad \Delta = -120, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& 21812321712321812321\overset{\color{red}2}{5} \quad \Delta = -120, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 21812321712321812321\overset{\color{red}2}{5} \quad \Delta = -120, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 21812321712321812321\overset{\color{red}2}{5} \quad \Delta = -120, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

Base: 221812321712321812321

$$\begin{aligned}
& 221812321712321812321\overset{\color{red}3}{\text{}}_{[\# = 5]} \quad \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& 221812321712321812321\overset{\color{red}3}{\text{}}_{[\# = 7]} \quad \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& 221812321712321812321\overset{\color{red}3}{\text{}}_{[\# = 6]} \quad \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 221812321712321812321\overset{\color{red}3}{\text{}}_{[\# = 2]} \quad \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 221812321712321812321\overset{\color{red}2}{\text{}} \quad \Delta = -68, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
221812321712321812321^2 \quad \Delta = -68, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
221812321712321812321^3_{[\# = 8]} \quad \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
221812321712321812321^2 \quad \Delta = -68, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
221812321712321812321^2 \quad \Delta = -68, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
221812321712321812321^2 \quad \Delta = -68, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
221812321712321812321^3 \quad \Delta = -77, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
221812321712321812321^2 \quad \Delta = -68, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
221812321712321812321^3_{[\# = 9]} \quad \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
221812321712321812321^3_{[\# = 4]} \quad \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
221812321712321812321^3_{[\# = 3]} \quad \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\text{Base: } 123218123151321812321 \\
123218123151321812321^3_{[\# = 6]} \quad \Delta = -30, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
123218123151321812321^2 \quad \Delta = -22, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
123218123151321812321^2 \quad \Delta = -22, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
123218123151321812321^2 \quad \Delta = -22, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
123218123151321812321^3_{[\# = 2]} \quad \Delta = -30, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
123218123151321812321^3_{[\# = 3]} \quad \Delta = -30, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
123218123151321812321^3_{[\# = 9]} \quad \Delta = -30, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
123218123151321812321^3_{[\# = 4]} \quad \Delta = -30, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
123218123151321812321^3_{[\# = 8]} \quad \Delta = -30, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix}
\end{aligned}$$

$$123218123151321812321\overset{2}{3} \text{ [#}=7] \quad \Delta = -30, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix}$$

$$123218123151321812321\overset{2}{3} \text{ [#}=5] \quad \Delta = -30, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix}$$

Base: 712321812315132181232

$$712321\overset{2}{8}12315132181232 \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix}$$

$$7123218123151321\overset{2}{8}1232 \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 256 \quad \begin{pmatrix} 0 & 16 & 2 \\ 16 & 174 & 26 \\ 2 & 26 & 8 \end{pmatrix}$$

$$7123218123151321\overset{2}{8}1232 \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 289 \quad \begin{pmatrix} 0 & 17 & 2 \\ 17 & 202 & 28 \\ 2 & 28 & 8 \end{pmatrix}$$

$$71232181231\overset{2}{5}132181232 \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 324 \quad \begin{pmatrix} 0 & 18 & 2 \\ 18 & 232 & 30 \\ 2 & 30 & 8 \end{pmatrix}$$

$$71232181\overset{2}{2}315132181232 \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 361 \quad \begin{pmatrix} 0 & 19 & 2 \\ 19 & 264 & 32 \\ 2 & 32 & 8 \end{pmatrix}$$

$$7123218123151321\overset{2}{8}1232 \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$7123218123151321\overset{2}{8}1232 \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 38 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

Base: 316131612321812321812

$$316131612321\overset{2}{8}12321812 \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix}$$

$$316131612321812321\overset{2}{8}12 \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 196 \quad \begin{pmatrix} 0 & 14 & 2 \\ 14 & 264 & 32 \\ 2 & 32 & 8 \end{pmatrix}$$

$$316131612321812321\overset{2}{8}12 \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 202 & 28 \\ 2 & 28 & 8 \end{pmatrix}$$

$$316131612321812321\overset{2}{8}12 \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix}$$

$$316131612321812321\overset{2}{8}12 \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 324 \quad \begin{pmatrix} 0 & 18 & 2 \\ 18 & 448 & 42 \\ 2 & 42 & 8 \end{pmatrix}$$

$$316131612321812321\overset{2}{8}12 \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$316131612321812321\overset{2}{8}12 \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$316131612321812321\overset{2}{8}12 \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: 161316123218123218122

$$16131612321812321\overset{2}{8}122 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

161316123218123218122 ²	$\Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$
161316123218123218122 ³	$\Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, \det = 4$	$\begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$
161316123218123218122 ²	$\Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$
161316123218123218122 ²	$\Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, \det = 81$	$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix}$
161316123218123218122 ²	$\Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, \det = 121$	$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 196 & 28 \\ 2 & 28 & 8 \end{pmatrix}$
161316123218123218122 ²	$\Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix}$
161316123218123218122 ²	$\Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$
Base: 712231513218123218122		
712231513218123218122 ²	$\Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 89 & 19 \\ 2 & 19 & 8 \end{pmatrix}$
712231513218123218122 ²	$\Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix}$
712231513218123218122 ²	$\Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$
712231513218123218122 ²	$\Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$
712231513218123218122 ²	$\Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, \det = 64$	$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix}$
712231513218123218122 ²	$\Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$
712231513218123218122 ²	$\Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, \det = 81$	$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix}$
712231513218123218122 ³	$\Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, \det = 4$	$\begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$
Base: 171231513218123218122		
171231513218123218122 ³	$\Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, \det = 4$	$\begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$
171231513218123218122 ²	$\Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$
171231513218123218122 ²	$\Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$
171231513218123218122 ²	$\Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, \det = 81$	$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix}$

$$\begin{aligned}
171231513218123218122^2 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
171231513218123218122^2 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
171231513218123218122^2 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
171231513218123218122^2 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{aligned}$$

Base: 513217123218123218122

$$\begin{aligned}
513217123218123218122^2 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
513217123218123218122^2 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
513217123218123218122^2 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 168 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
513217123218123218122^2 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
513217123218123218122^2 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
513217123218123218122^2 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
513217123218123218122^2 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 196 & 28 \\ 2 & 28 & 8 \end{pmatrix} \\
513217123218123218122^3 \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}
\end{aligned}$$

Base: 132218123218123151321

$$\begin{aligned}
132218123218123151321^3_{[\# = 4]} \quad \Delta = -28, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
123151321812321812231^3_{[\# = 5]} \quad \Delta = -28, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
132218123218123151321^3_{[\# = 6]} \quad \Delta = -28, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
123151321812321812231^3_{[\# = 4]} \quad \Delta = -28, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
132218123218123151321^3_{[\# = 3]} \quad \Delta = -28, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
123151321812321812231^3_{[\# = 2]} \quad \Delta = -28, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
132218123218123151321^3_{[\# = 5]} \quad \Delta = -28, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
132218123218123151321\bar{3} \text{ }_{[\# = 8]} \quad \Delta = -28, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
132218123218123151321\bar{3} \text{ }_{[\# = 7]} \quad \Delta = -28, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
132218123218123151321\bar{3} \text{ }_{[\# = 2]} \quad \Delta = -28, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
132218123218123151321\bar{3} \text{ }_{[\# = 9]} \quad \Delta = -28, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
123151321812321812231\bar{3} \text{ }_{[\# = 3]} \quad \Delta = -28, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
13221812321\bar{2}18123151321 \quad \Delta = -20, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
13221812321\bar{2}18123151321 \quad \Delta = -20, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1322181232181231\bar{2}51321 \quad \Delta = -20, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1322181232181231\bar{2}51321 \quad \Delta = -20, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
132218123218123151\bar{2}321 \quad \Delta = -20, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1322181\bar{2}23218123151321 \quad \Delta = -20, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 222181232181231513221

$$\begin{aligned}
222181232181231513221\bar{3} \text{ }_{[\# = 7]} \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
222181232181231513221\bar{3} \text{ }_{[\# = 6]} \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
222181232181231513221\bar{3} \text{ }_{[\# = 3]} \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
222181232181231513221\bar{3} \text{ }_{[\# = 5]} \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
222181232181231513221\bar{3} \text{ }_{[\# = 9]} \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
222181232181231513221\bar{3} \text{ }_{[\# = 2]} \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
222181232181231513221\bar{3} \text{ }_{[\# = 8]} \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
222181232181\bar{2}231513221 \quad \Delta = -11, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
222181232181231513221 \quad \Delta = -11, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
222181232181231513221 \quad \Delta = -11, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
222181232181231513221 \quad \Delta = -11, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
222181232181231513221 \quad \Delta = -11, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
222181232181231513221 \quad \Delta = -11, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
222181232181231513221 \textcolor{red}{3}_{[\# = 4]} \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

Base: 712315132181232181223

$$\begin{aligned}
712315132181232181223 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
712315132181232181223 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
712315132181232181223 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
712315132181232181223 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
712315132181232181223 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
712315132181232181223 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
712315132181232181223 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: 132171232181232181222

$$\begin{aligned}
222181232181232171231 \textcolor{red}{3}_{[\# = 5]} \quad \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
132171232181232181222 \quad \Delta = -66, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
132171232181232181222 \quad \Delta = -66, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
132171232181232181222 \quad \Delta = -66, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
132171232181232181222 \quad \Delta = -66, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
132171232181232181222^2 \quad \Delta = -66, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
222181232181232171231^3_{[\# = 4]} \quad \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
132171232181232181222^2 \quad \Delta = -66, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
222181232181232171231^3_{[\# = 3]} \quad \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
222181232181232171231^3_{[\# = 2]} \quad \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
\text{Base: } 131612321812321812231 & \\
131612321812321812231^2 \quad \Delta = -47, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
131612321812321812231^2 \quad \Delta = -47, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
131612321812321812231^2 \quad \Delta = -47, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
131612321812321812231^2 \quad \Delta = -47, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
131612321812321812231^2 \quad \Delta = -47, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
131612321812321812231^2 \quad \Delta = -47, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
132218123218123216131^3_{[\# = 7]} \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
131612321812321812231^3_{[\# = 5]} \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
132218123218123216131^3_{[\# = 2]} \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
131612321812321812231^3_{[\# = 2]} \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
132218123218123216131^3_{[\# = 5]} \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
132218123218123216131^3_{[\# = 9]} \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
132218123218123216131^3_{[\# = 8]} \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
132218123218123216131^3_{[\# = 6]} \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
132218123218123216131\overset{2}{3} \text{ [#=3]} \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
131612321812321812231\overset{2}{3} \text{ [#=3]} \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
132218123218123216131\overset{2}{3} \text{ [#=4]} \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
131612321812321812231\overset{2}{3} \text{ [#=4]} \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: 613161232181232181223

$$\begin{aligned}
613161232181232181223 \overset{2}{} \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
613161232181232181223 \overset{2}{} \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
613161232181232181223 \overset{2}{} \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
613161232181232181223 \overset{2}{} \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
613161232181232181223 \overset{2}{} \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 225 & \begin{pmatrix} 0 & 15 & 2 \\ 15 & 163 & 25 \\ 2 & 25 & 8 \end{pmatrix} \\
613161232181232181223 \overset{2}{} \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
613161232181232181223 \overset{2}{} \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix}
\end{aligned}$$

Base: 217123218123218122315

$$\begin{aligned}
217123218123218122315 \overset{2}{} \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
217123218123218122315 \overset{2}{} \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
217123218123218122315 \overset{2}{} \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
217123218123218122315 \overset{2}{} \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
217123218123218122315 \overset{2}{} \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
217123218123218122315 \overset{2}{} \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
217123218123218122315 \overset{2}{} \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 50 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{aligned}$$

Base: 221712321812321812231

$$\begin{array}{ll}
221712321812321812231^2 & \Delta = -66, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
221712321812321812231^2 & \Delta = -66, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
221712321812321812231^2 & \Delta = -66, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
221712321812321812231^2 & \Delta = -66, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
221712321812321812231^3 & \Delta = -75, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
221712321812321812231^2 & \Delta = -66, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
221712321812321812231^3_{[\# = 2]} & \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
221712321812321812231^3_{[\# = 3]} & \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
221712321812321812231^3_{[\# = 4]} & \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
221712321812321812231^3_{[\# = 5]} & \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
221712321812321812231^2 & \Delta = -66, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\text{Base: } 17123218123218122315^1 & \\
17123218123218122315^1 & \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
17123218123218122315^{12} & \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
17123218123218122315^1 & \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
17123218123218122315^1 & \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 50 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
17123218123218122315^1 & \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
17123218123218122315^1 & \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\text{Base: } 1712321812321812231^{21} & \\
1712321812321812231^{21} & \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{array}$$

$$1\overset{21}{7}12321812321812231 \quad \Delta = -66, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1\overset{21}{7}12321812321812231 \quad \Delta = -66, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1\overset{21}{7}12321812321812231 \quad \Delta = -66, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1\overset{21}{7}12321812321812231\overset{3}{\text{[}\#\!=\!4\text{]}} \quad \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1\overset{21}{7}12321812321812231 \quad \Delta = -66, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1\overset{21}{7}12321812321812231 \quad \Delta = -66, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$1\overset{21}{7}12321812321812231\overset{3}{\text{[}\#\!=\!3\text{]}} \quad \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1\overset{21}{7}12321812321812231\overset{3}{\text{[}\#\!=\!2\text{]}} \quad \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$1\overset{21}{7}\overset{2}{1}2321812321812231 \quad \Delta = -66, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

Base: 313218123218123151321

$$\overset{3}{3}13218123218123151321 \quad \Delta = -36, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 21 & 9 \\ 2 & 9 & 8 \end{pmatrix}$$

$$\overset{3}{3}13218123218123151321 \text{ [}\#\!=\!2\text{]} \quad \Delta = -36, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 93 & 19 \\ 2 & 19 & 8 \end{pmatrix}$$

$$313218123218123151321\overset{3}{\text{[}\#\!=\!2\text{]}} \quad \Delta = -36, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$313218123218123151321\overset{3}{\text{[}\#\!=\!6\text{]}} \quad \Delta = -36, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

$$313218123218123151321\overset{3}{\text{[}\#\!=\!3\text{]}} \quad \Delta = -36, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$313218123218123151321\overset{3}{\text{[}\#\!=\!4\text{]}} \quad \Delta = -36, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$$

$$313218123218123151321\overset{3}{\text{[}\#\!=\!5\text{]}} \quad \Delta = -36, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$313218123218123151321 \quad \Delta = -28, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$313218123218123151321\overset{3}{\text{[}\#\!=\!7\text{]}} \quad \Delta = -36, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$313218123218123151321 \quad \Delta = -28, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$\begin{aligned}
313218123218123151321^2 \quad \Delta = -28, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
313218123218123151321^2 \quad \Delta = -30, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
313218123218123151321^2 \quad \Delta = -28, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
313218123218123151321^2 \quad \Delta = -28, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
313218123218123151321^3_{[\# = 8]} \quad \Delta = -36, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
313218123218123151321^3_{[\# = 9]} \quad \Delta = -36, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
313218123218123151321^2 \quad \Delta = -28, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
313218123218123151321^2 \quad \Delta = -28, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\text{Base: } 612315132181232181231 \\
612315132181232181231^2 \quad \Delta = -98, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
612315132181232181231^3_{[\# = 4]} \quad \Delta = -106, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
612315132181232181231^3_{[\# = 5]} \quad \Delta = -106, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
612315132181232181231^3_{[\# = 2]} \quad \Delta = -106, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
612315132181232181231^3_{[\# = 3]} \quad \Delta = -106, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
612315132181232181231^2 \quad \Delta = -98, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
612315132181232181231^2 \quad \Delta = -98, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
612315132181232181231^2 \quad \Delta = -98, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
612315132181232181231^2 \quad \Delta = -98, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
612315132181232181231^2 \quad \Delta = -98, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
612315132181232181231^2 \quad \Delta = -98, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{aligned}$$

Base: 161315132181232181231

$$161315132181232181231\overset{\textcolor{red}{3}}{[\# = 3]} \quad \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$161315132181232181231\overset{\textcolor{red}{3}}{[\# = 2]} \quad \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$161315132181232181231\overset{\textcolor{red}{3}}{[\# = 4]} \quad \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$161315132181232181231\overset{\textcolor{red}{3}}{[\# = 5]} \quad \Delta = -105, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$161315132181232181231\overset{\textcolor{red}{2}}{\quad} \quad \Delta = -97, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$161315132181232181231\overset{\textcolor{red}{2}}{\quad} \quad \Delta = -97, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$161315132181232181231\overset{\textcolor{red}{2}}{\quad} \quad \Delta = -97, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

$$161315132181232181231\overset{\textcolor{red}{2}}{\quad} \quad \Delta = -97, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix}$$

$$161315132181232181231\overset{\textcolor{red}{2}}{\quad} \quad \Delta = -97, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix}$$

$$161315132181232181231\overset{\textcolor{red}{2}}{\quad} \quad \Delta = -97, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$161315132181232181231\overset{\textcolor{red}{2}}{\quad} \quad \Delta = -97, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 215132181232181231512

$$215132181232181231512\overset{\textcolor{red}{2}}{\quad} \quad \Delta = -63, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$215132181232181231512\overset{\textcolor{red}{2}}{\quad} \quad \Delta = -63, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$215132181232181231512\overset{\textcolor{red}{3}}{\quad} \quad \Delta = -71, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$215132181232181231512\overset{\textcolor{red}{2}}{\quad} \quad \Delta = -63, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$215132181232181231512\overset{\textcolor{red}{2}}{\quad} \quad \Delta = -63, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 141321812321812315131

$$141321812321812315131\overset{\textcolor{red}{3}}{[\# = 3]} \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$141321812321812315131\overset{\textcolor{red}{2}}{\quad} \quad \Delta = -47, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\begin{aligned}
141321812321812315131 \quad \Delta = -47, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
141321812321812315131 \quad \Delta = -47, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
141321812321812315131 \quad \Delta = -47, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
141321812321812315131 \quad \Delta = -47, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1315132181232181231413_{[\# = 3]} \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1413218123218123151313_{[\# = 2]} \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
141321812321812315131 \quad \Delta = -49, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1413218123218123151313_{[\# = 4]} \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1413218123218123151313_{[\# = 5]} \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1413218123218123151313_{[\# = 7]} \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1315132181232181231413_{[\# = 2]} \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1413218123218123151313_{[\# = 9]} \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
1413218123218123151313_{[\# = 8]} \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
141321812321812315131 \quad \Delta = -47, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1413218123218123151313_{[\# = 6]} \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
141321812321812315131 \quad \Delta = -47, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\text{Base: } 1 \begin{matrix} 215132181232181231 \\ 5 \end{matrix} 1 & \\
1 \begin{matrix} 215132181232181231 \\ 5 \end{matrix} 1 \quad \Delta = -63, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \begin{matrix} 215132181232181231 \\ 5 \end{matrix} 1 \quad \Delta = -63, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1 \begin{matrix} 215132181232181231 \\ 5 \end{matrix} 1 \quad \Delta = -63, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

$$1 \overset{215132181232181231}{5} 1 \quad \Delta = -63, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$1 \overset{215132181232181231}{5} 1 \quad \Delta = -63, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$1 \overset{215132181232181231}{5} 1 \quad \Delta = -63, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$1 \overset{215132181232181231}{5} 1 \quad \Delta = -63, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1 \overset{215132181232181231}{5} 1 \quad \Delta = -71, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$$

Base: $1 \overset{15132181232181231}{5} 1$

$$1 \overset{15132181232181231}{5} 1 \quad \Delta = -63, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$1 \overset{15132181232181231}{5} 1 \quad \Delta = -63, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$1 \overset{15132181232181231}{5} 1 \quad \Delta = -63, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: 413218123218123151316

$$413218123218123151316 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$413218123218123151316 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$413218123218123151316 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 16 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$413218123218123151316 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$413218123218123151316 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 50 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$$413218123218123151316 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$413218123218123151316 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$413218123218123151316 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$413218123218123151316 \quad \Delta = -127, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: 513161232181232181231

$$513161232181232181231 \quad \Delta = -99, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

$$\begin{aligned}
513161232181232181231 \quad \Delta = -99, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 196 & 28 \\ 2 & 28 & 8 \end{pmatrix} \\
513161232181232181231 \quad \Delta = -99, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
513161232181232181231 \quad \Delta = -99, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
513161232181232181231 \quad \Delta = -99, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
513161232181232181231 \quad \Delta = -99, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
513161232181232181231 \quad \Delta = -99, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
5131612321812321812313_{[\# = 2]} \quad \Delta = -107, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
5131612321812321812313_{[\# = 5]} \quad \Delta = -107, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
5131612321812321812313_{[\# = 4]} \quad \Delta = -107, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
5131612321812321812313_{[\# = 3]} \quad \Delta = -107, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: 151232181232181232151

$$\begin{aligned}
151232181232181232151 \quad \Delta = -67, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
151232181232181232151 \quad \Delta = -67, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
151232181232181232151 \quad \Delta = -67, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: 151232181232181231512

$$\begin{aligned}
151232181232181231512 \quad \Delta = -65, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
151232181232181231512 \quad \Delta = -65, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
151232181232181231512 \quad \Delta = -65, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
151232181232181231512 \quad \Delta = -65, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
151232181232181231512 \quad \Delta = -65, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
151232181232181231512^2 \quad \Delta = -65, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
151232181232181231512^2 \quad \Delta = -65, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
151232181232181231512^3 \quad \Delta = -73, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}
\end{aligned}$$

Base: 412321812321812315131

$$\begin{aligned}
412321812321812315131^3_{[\# = 5]} \quad \Delta = -57, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
412321812321812315131^2 \quad \Delta = -49, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
412321812321812315131^2 \quad \Delta = -49, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
412321812321812315131^2 \quad \Delta = -49, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
412321812321812315131^2 \quad \Delta = -49, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
412321812321812315131^3_{[\# = 7]} \quad \Delta = -57, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
412321812321812315131^3_{[\# = 6]} \quad \Delta = -57, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
412321812321812315131^2 \quad \Delta = -49, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
412321812321812315131^3_{[\# = 2]} \quad \Delta = -57, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
412321812321812315131^3_{[\# = 3]} \quad \Delta = -57, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
412321812321812315131^3_{[\# = 4]} \quad \Delta = -57, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
412321812321812315131^3_{[\# = 8]} \quad \Delta = -57, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
412321812321812315131^3_{[\# = 9]} \quad \Delta = -57, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
412321812321812315131^3 \quad \Delta = -57, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 41 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
412321812321812315131^2 \quad \Delta = -49, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
412321812321812315131^2 \quad \Delta = -49, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: 123218123218123151315

$$\begin{aligned}
513151321812321812321\overset{2}{3} \text{ [#=3]} \quad \Delta = -81, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
513151321812321812321\overset{2}{3} \text{ [#=2]} \quad \Delta = -81, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
1232181232181231\overset{2}{5}1315 \quad \Delta = -73, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
513151321812321812321\overset{2}{3} \text{ [#=5]} \quad \Delta = -81, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1232181\overset{2}{2}3218123151315 \quad \Delta = -73, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1232181232181231\overset{2}{5}1315 \quad \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1232181232181231\overset{2}{5}1315 \quad \Delta = -73, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1232181232181\overset{2}{2}23151315 \quad \Delta = -73, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1232181232181231\overset{2}{5}1315 \quad \Delta = -73, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1232181232181231\overset{2}{5}1315 \quad \Delta = -73, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1232181\overset{2}{2}3218123151315 \quad \Delta = -73, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
513151321812321812321\overset{2}{3} \text{ [#=9]} \quad \Delta = -81, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
513151321812321812321\overset{2}{3} \text{ [#=4]} \quad \Delta = -81, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
513151321812321812321\overset{2}{3} \text{ [#=6]} \quad \Delta = -81, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
513151321812321812321\overset{2}{3} \text{ [#=8]} \quad \Delta = -81, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
513151321812321812321\overset{2}{3} \text{ [#=7]} \quad \Delta = -81, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{151232181232181231}{5} 1 & \\
1 \overset{151232181232181231}{5} 1 \quad \Delta = -65, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1 \overset{151232181232181231}{5} 1 \quad \Delta = -65, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \overset{151232181232181231}{5} 1 \quad \Delta = -65, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{lll}
1^{151232181232181231}_5 1 & \Delta = -65, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 22 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1^{151232181232181231}_5 1 & \Delta = -65, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1^{151232181232181231}_5 1 & \Delta = -65, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{array}$$

Base: 412321812321812321612

412321812321812321612	$\Delta = -77, n_+ = 1, n_- = 21, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 57 & 17 \\ 2 & 17 & 8 \end{pmatrix}$
412321812321812321612 ³	$\Delta = -77, n_+ = 1, n_- = 21, n_0 = 0, \det = 4$	$\begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix}$
412321812321812321612 ²	$\Delta = -69, n_+ = 1, n_- = 21, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$
412321812321812321612 ²	$\Delta = -69, n_+ = 1, n_- = 21, n_0 = 0, \det = 4$	$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$
412321812321812321612 ²	$\Delta = -69, n_+ = 1, n_- = 21, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 58 & 18 \\ 2 & 18 & 8 \end{pmatrix}$
412321812321812321612 ²	$\Delta = -69, n_+ = 1, n_- = 21, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$
412321812321812321612 ²	$\Delta = -69, n_+ = 1, n_- = 21, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 50 & 16 \\ 2 & 16 & 8 \end{pmatrix}$
412321812321812321612 ²	$\Delta = -69, n_+ = 1, n_- = 21, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$
412321812321812321612 ²	$\Delta = -69, n_+ = 1, n_- = 21, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$

Base: 141321812321812321612

141321812321812321612 ²	$\Delta = -67, n_+ = 1, n_- = 21, n_0 = 0, \det = 4$	$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$
141321812321812321612 ²	$\Delta = -67, n_+ = 1, n_- = 21, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$
141321812321812321612 ²	$\Delta = -67, n_+ = 1, n_- = 21, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$
141321812321812321612 ²	$\Delta = -69, n_+ = 1, n_- = 21, n_0 = 0, \det = 4$	$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$
141321812321812321612 ²	$\Delta = -67, n_+ = 1, n_- = 21, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 31 & 13 \\ 2 & 13 & 8 \end{pmatrix}$
141321812321812321612 ²	$\Delta = -67, n_+ = 1, n_- = 21, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 50 & 16 \\ 2 & 16 & 8 \end{pmatrix}$
141321812321812321612 ²	$\Delta = -67, n_+ = 1, n_- = 21, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$

$$\begin{aligned}
141321812321812321612 \quad \Delta = -67, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
141321812321812321612 \quad \Delta = -75, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
2161232181232181231413_{[\# = 2]} \quad \Delta = -75, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
2161232181232181231413_{[\# = 3]} \quad \Delta = -75, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\text{Base: } 131612321812321812313 & \\
3132181232181232161313_{[\# = 2]} \quad \Delta = -63, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
131612321812321812313 \quad \Delta = -57, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
131612321812321812313 \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
131612321812321812313 \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
131612321812321812313 \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
131612321812321812313 \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
131612321812321812313 \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
131612321812321812313 \quad \Delta = -63, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
3132181232181232161313_{[\# = 4]} \quad \Delta = -63, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
3132181232181232161313_{[\# = 3]} \quad \Delta = -63, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
3132181232181232161313_{[\# = 5]} \quad \Delta = -63, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
3132181232181232161313_{[\# = 9]} \quad \Delta = -63, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
131612321812321812313_{[\# = 2]} \quad \Delta = -63, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
3132181232181232161313_{[\# = 7]} \quad \Delta = -63, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
3132181232181232161313_{[\# = 8]} \quad \Delta = -63, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& 313218123218123216131\overset{3}{\color{red}}_{[\#=6]} \quad \Delta = -63, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
& 131612321812321812313 \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& 131612321812321812313 \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \text{Base: } 41232181232181232161^1 \\
& 41232181232181232161^{\overset{2}{\color{red}}} \quad \Delta = -69, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 41232181232181232161^{\overset{2}{\color{red}}} \quad \Delta = -69, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 58 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
& 41232181232181232161^{\overset{2}{\color{red}}} \quad \Delta = -69, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 50 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& 41232181232181232161^{\overset{2}{\color{red}}} \quad \Delta = -69, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& 41232181232181232161^{\overset{3}{\color{red}}} \quad \Delta = -77, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 57 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& 41232181232181232161^{\overset{2}{\color{red}}} \quad \Delta = -69, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 41232181232181232161^{\overset{2}{\color{red}}1} \quad \Delta = -69, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \text{Base: } 16123218123218123141^1 \\
& 16123218123218123141^{\overset{2}{\color{red}}} \quad \Delta = -69, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 16123218123218123141^{\overset{2}{\color{red}}} \quad \Delta = -67, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& 16123218123218123141^{\overset{1}{\color{red}}2} \quad \Delta = -67, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 58 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
& 16123218123218123141^{\overset{2}{\color{red}}} \quad \Delta = -67, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 16123218123218123141^{\overset{2}{\color{red}}} \quad \Delta = -67, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 16123218123218123141^{\overset{2}{\color{red}}} \quad \Delta = -67, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 16123218123218123141\overset{3}{\color{red}}_{[\#=3]} \quad \Delta = -75, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix} \\
& 16123218123218123141\overset{3}{\color{red}}_{[\#=2]} \quad \Delta = -75, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 8 \end{pmatrix}
\end{aligned}$$

$$16123218123218123141 \quad \Delta = -67, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 50 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

Base: 221712321812321812313

$$221712321812321812313 \quad \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$221712321812321812313 \quad \Delta = -82, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$221712321812321812313 \quad \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$221712321812321812313 \quad \Delta = -83, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$221712321812321812313 \quad \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$221712321812321812313 \quad [\# = 2] \quad \Delta = -82, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$221712321812321812313 \quad \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$221712321812321812313 \quad \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$221712321812321812313 \quad \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$221712321812321812313 \quad \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$221712321812321812313 \quad \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

Base: $17^{21}12321812321812313$

$$17^{21}12321812321812313 \quad \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$17^{21}12321812321812313 \quad [\# = 2] \quad \Delta = -82, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$17^{21}12321812321812313 \quad \Delta = -82, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$17^{21}12321812321812313 \quad \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

$$17^{21}12321812321812313 \quad \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$17^{21}12321812321812313 \quad \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$\begin{aligned}
& \begin{matrix} 21 \\ 17 \end{matrix} 12321812321812313 \quad \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \begin{matrix} 21 \\ 17 \end{matrix} 12321812321812313 \quad \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \begin{matrix} 21 \\ 17 \end{matrix} 12321812321812313 \quad \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \begin{matrix} 21 \\ 17 \end{matrix} 12321812321812313 \quad \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: $2 \begin{matrix} 123151321812321812 \\ 3 \end{matrix} 1$

$$\begin{aligned}
& 2 \begin{matrix} 123151321812321812 \\ 3 \end{matrix} 1 \quad \Delta = -22, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 2 \begin{matrix} 123151321812321812 \\ 3 \end{matrix} 1 \quad \Delta = -22, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 2 \begin{matrix} 123151321812321812 \\ 3 \end{matrix} 1 \quad \Delta = -22, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 2 \begin{matrix} 123151321812321812 \\ 3 \end{matrix} 1 \quad \Delta = -22, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& 2 \begin{matrix} 123151321812321812 \\ 3 \end{matrix} 1 \quad \Delta = -22, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 2 \begin{matrix} 123151321812321812 \\ 3 \end{matrix} 12 \quad \Delta = -22, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& 2 \begin{matrix} 3123151321812321812 \\ 3 \end{matrix} 1 \quad [\# = 9] \quad \Delta = -30, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
& 2 \begin{matrix} 123151321812321812 \\ 3 \end{matrix} 1 \quad \Delta = -22, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 2 \begin{matrix} 3123151321812321812 \\ 3 \end{matrix} 1 \quad [\# = 8] \quad \Delta = -30, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& 2 \begin{matrix} 123151321812321812 \\ 3 \end{matrix} 14 \quad \Delta = -50, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 2 \begin{matrix} 3123151321812321812 \\ 3 \end{matrix} 1 \quad [\# = 7] \quad \Delta = -30, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
& 2 \begin{matrix} 123151321812321812 \\ 3 \end{matrix} 13 \quad [\# = 2] \quad \Delta = -30, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 41 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& 2 \begin{matrix} 3123151321812321812 \\ 3 \end{matrix} 1 \quad [\# = 6] \quad \Delta = -30, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& 2 \begin{matrix} 3123151321812321812 \\ 3 \end{matrix} 1 \quad [\# = 2] \quad \Delta = -30, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 2 \begin{matrix} 3123151321812321812 \\ 3 \end{matrix} 1 \quad [\# = 5] \quad \Delta = -30, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& 2^{\overline{3}123151321812321812} \quad 3 \quad 1 \quad [\# = 4] \quad \Delta = -30, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 2^{\overline{1}23151321812321812} \quad 3 \quad \overline{1}3 \quad [\# = 3] \quad \Delta = -30, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 96 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
& 2^{\overline{3}123151321812321812} \quad 3 \quad 1 \quad [\# = 3] \quad \Delta = -30, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \text{Base: } 13161232181232181232^{\overline{1}} \\
& 13161232181232181232^{\overline{2}} \quad 1 \quad \Delta = -49, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& 13161232181232181232^{\overline{4}1} \quad \Delta = -77, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
& 2^{\overline{3}131612321812321812} \quad 3 \quad 1 \quad [\# = 7] \quad \Delta = -57, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
& 2^{\overline{3}131612321812321812} \quad 3 \quad 1 \quad [\# = 8] \quad \Delta = -57, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix} \\
& 2^{\overline{3}131612321812321812} \quad 3 \quad 1 \quad [\# = 9] \quad \Delta = -57, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix} \\
& 13161232181232181232^{\overline{2}} \quad 1 \quad \Delta = -49, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& 13161232181232181232^{\overline{2}} \quad 1 \quad \Delta = -49, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 13161232181232181232^{\overline{2}} \quad 1 \quad \Delta = -49, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 13161232181232181232^{\overline{2}1} \quad \Delta = -49, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 2^{\overline{3}131612321812321812} \quad 3 \quad 1 \quad [\# = 2] \quad \Delta = -57, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& 13161232181232181232^{\overline{3}1} \quad [\# = 2] \quad \Delta = -57, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 2^{\overline{3}131612321812321812} \quad 3 \quad 1 \quad [\# = 3] \quad \Delta = -57, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& 13161232181232181232^{\overline{3}1} \quad [\# = 3] \quad \Delta = -57, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 2^{\overline{3}131612321812321812} \quad 3 \quad 1 \quad [\# = 6] \quad \Delta = -57, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
& 13161232181232181232^{\overline{2}} \quad 1 \quad \Delta = -49, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 13161232181232181232^{\overline{2}} \quad 1 \quad \Delta = -49, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{l}
\begin{array}{c} \textcolor{red}{3}131612321812321812 \\ 2 \quad \quad \quad 3 \quad \quad \quad 1 \end{array} \begin{array}{c} \text{[#}=5] \end{array} \quad \Delta = -57, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
\begin{array}{c} \textcolor{red}{3}131612321812321812 \\ 2 \quad \quad \quad 3 \quad \quad \quad 1 \end{array} \begin{array}{c} \text{[#}=4] \end{array} \quad \Delta = -57, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix}
\end{array}$$

Base: 22171232181232181232

$$\begin{array}{l}
22171232181232181232 \quad \Delta = -68, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
22171232181232181232 \quad \Delta = -68, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
22171232181232181232 \quad \Delta = -68, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
22171232181232181232 \quad \Delta = -68, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
22171232181232181232 \quad \Delta = -96, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
22171232181232181232 \begin{array}{c} \textcolor{red}{3}1 \\ \text{[#}=2] \end{array} \quad \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
22171232181232181232 \quad \Delta = -68, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
22171232181232181232 \quad \Delta = -68, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
22171232181232181232 \quad \Delta = -68, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
22171232181232181232 \begin{array}{c} \textcolor{red}{3}1 \\ \text{[#}=3] \end{array} \quad \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
22171232181232181232 \quad \Delta = -77, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{array}$$

Base: 171232181232181232

$$\begin{array}{l}
171232181232181232 \quad \Delta = -68, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
171232181232181232 \quad \Delta = -96, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
171232181232181232 \quad \Delta = -68, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
171232181232181232 \quad \Delta = -68, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
171232181232181232 \quad \Delta = -68, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
171232181232181232 \quad \Delta = -68, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
& \overset{21}{1} \overset{21}{7} 12321812321812\overset{21}{3}2 \quad \Delta = -68, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{21}{1} \overset{21}{7} 12321812321812\overset{2}{3}\overset{1}{2} \quad \Delta = -68, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{21}{1} \overset{21}{7} 12321812321812\overset{31}{3}2_{[\# = 2]} \quad \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{21}{1} \overset{21}{7} 12321812321812\overset{31}{3}2_{[\# = 3]} \quad \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: 222181232181232181222

$$\begin{aligned}
& 222181\overset{2}{2}32181232181222 \quad \Delta = -85, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& 222181\overset{2}{2}32181232181222 \quad \Delta = -85, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 222181\overset{2}{2}32181232181222 \quad \Delta = -85, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 50 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{18}123151321812321712$

$$\begin{aligned}
& \overset{1}{18}12315132181\overset{2}{2}321712 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& \overset{1}{18}12315132181\overset{2}{2}321712 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
& \overset{1}{18}12315132181\overset{2}{2}321712 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{1}{18}12315132181\overset{2}{2}321712 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
& \overset{1}{18}123151\overset{2}{3}21812321712 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
& \overset{1}{18}12315132181\overset{2}{2}321712 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{12}{18}123151321812321712 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 64 & 16 \\ 2 & 16 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{18}123217123218123151$

$$\begin{aligned}
& \overset{12}{18}123217123218123151 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{1}{18}1232171232181\overset{2}{2}3151 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{1}{18}1232171232181\overset{2}{2}3151 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& \overset{1}{18}1232171\overset{2}{2}3218123151 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}
\end{aligned}$$

$${}^1_{18123217123218123151} \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$${}^1_{18123217123218123151} \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

Base: ${}^1_{1712321812315132181}$

$${}^1_{1712321812315132181} \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}$$

$${}^1_{1712321812315132181} \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$${}^1_{1712321812315132181} \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

$${}^1_{1712321812315132181} \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix}$$

$${}^{12}_{1712321812315132181} \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$${}^1_{1712321812315132181} \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix}$$

Base: ${}^1_{18123218123216131613}$

$${}^{12}_{18123218123216131613} \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$${}^1_{18123218123216131613} \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 144 \quad \begin{pmatrix} 0 & 12 & 2 \\ 12 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix}$$

$${}^1_{18123218123216131613} \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$${}^1_{18123218123216131613} \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$${}^1_{18123218123216131613} \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix}$$

$${}^1_{18123218123216131613} \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 196 \quad \begin{pmatrix} 0 & 14 & 2 \\ 14 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix}$$

$${}^1_{18123218123216131613} \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 324 \quad \begin{pmatrix} 0 & 18 & 2 \\ 18 & 196 & 28 \\ 2 & 28 & 8 \end{pmatrix}$$

Base: ${}^1_{18123217123218123215}$

$${}^{12}_{18123217123218123215} \quad \Delta = -120, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$${}^1_{18123217123218123215} \quad \Delta = -120, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$${}^1_{18123217123218123215} \quad \Delta = -120, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\begin{array}{lll}
{}^1 18123217123218123215 & \Delta = -120, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
{}^1 18123217123218123215 & \Delta = -120, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
{}^1 18123217123218123215 & \Delta = -120, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
\text{Base: } 7122315132181232181 & & \\
7122315132181232181 & \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
7122315132181232181 & \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
7122315132181232181 & \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 102 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
7122315132181232181 & \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
7122315132181232181 & \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
{}^2 7122315132181232181 & \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 89 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
{}^2 7122315132181232181 & \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 82 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
\text{Base: } 1812321812315132171 & & \\
{}^{21} 1812321812315132171 & \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
{}^{21} 1812321812315132171 & \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
{}^{21} 1812321812315132171 & \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
{}^{21} 1812321812315132171 & \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
{}^{21} 1812321812315132171 & \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
{}^{21} 1812321812315132171 & \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
{}^{21} 1812321812315132171 & \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
\text{Base: } 1812321812321613161 & & \\
{}^{21} 1812321812321613161 & \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
& \overset{21}{18}1232181232161\overset{2}{3}161 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{21}{18}12321812321613161 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{21}{18}1232181232161\overset{2}{3}161 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{21}{18}12321812321613161 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& \overset{21}{18}1232181232161\overset{2}{3}161 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 163 & 25 \\ 2 & 25 & 8 \end{pmatrix} \\
& \overset{21}{18}1232181232161\overset{2}{3}161 \quad \Delta = -116, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{21}{18}12321812321712315$

$$\begin{aligned}
& \overset{21}{18}12321812321712315 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{21}{18}12321812321712315 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{21}{18}12321812321712315 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{21}{18}12321812321712315 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{21}{18}12321812321712315 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{21}{18}12321812321712315 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{21}{18}12321812321712315 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{21}{18}12321712321812321$

$$\begin{aligned}
& \overset{21}{18}12321712321812321 \quad \Delta = -68, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{21}{18}12321712321812321 \quad \Delta = -68, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& \overset{21}{18}12321712321812321 \quad \Delta = -68, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{21}{18}12321712321812321 \quad \Delta = -68, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& \overset{21}{18}12321712321812321 \quad \Delta = -68, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{21}{1} \overset{2}{8} 12321712321812321 \quad \Delta = -68, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{21}{1} \overset{2}{8} 12321712321812321 \overset{3}{\# = 3} \quad \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{21}{1} \overset{2}{8} 12321712321812321 \overset{3}{\# = 9} \quad \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{21}{1} \overset{2}{8} 12321712321812321 \overset{3}{\# = 7} \quad \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& \overset{21}{1} \overset{2}{8} 12321712321812321 \overset{3}{\# = 6} \quad \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{21}{1} \overset{2}{8} 12321712321812321 \overset{3}{\# = 5} \quad \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{21}{1} \overset{2}{8} 12321712321812321 \overset{3}{\# = 4} \quad \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{21}{1} \overset{2}{8} 12321712321812321 \overset{3}{\# = 2} \quad \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{21}{1} \overset{2}{8} 12321712321812321 \overset{3}{\# = 8} \quad \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{221}{1} \overset{2}{8} 1232181231513221$

$$\begin{aligned}
& \overset{221}{1} \overset{2}{8} 1232181231513221 \quad \Delta = -11, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{221}{1} \overset{2}{8} 1232181231513221 \overset{3}{\# = 2} \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{221}{1} \overset{2}{8} 1232181231513221 \quad \Delta = -11, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{221}{1} \overset{2}{8} 1232181231513221 \quad \Delta = -11, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{221}{1} \overset{2}{8} 1232181231513221 \quad \Delta = -11, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{221}{1} \overset{2}{8} 1232181231513221 \quad \Delta = -11, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{221}{1} \overset{2}{8} 1232181231513221 \overset{3}{\# = 3} \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{221}{1} \overset{2}{8} 1232181231513221 \overset{3}{\# = 4} \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{221}{1} \overset{2}{8} 1232181231513221 \overset{3}{\# = 5} \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& \overset{221}{1} \overset{2}{8} 1232181231513221 \overset{3}{\# = 6} \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{l}
\begin{array}{l} 221 \\ 1 \ 8 \ 1232181231513221 \end{array} \begin{array}{l} \textcolor{red}{3} \ [\# = 8] \\ \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \end{array} \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
\begin{array}{l} 221 \\ 1 \ 8 \ 1232181231513221 \end{array} \begin{array}{l} \textcolor{red}{3} \ [\# = 9] \\ \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \end{array} \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
\begin{array}{l} 221 \\ 1 \ 8 \ 1232181 \end{array} \begin{array}{l} \textcolor{red}{2} \\ 1231513221 \end{array} \Delta = -11, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
\begin{array}{l} 221 \\ 1 \ 8 \ 1232181231513221 \end{array} \begin{array}{l} \textcolor{red}{3} \ [\# = 7] \\ \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \end{array} \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
\begin{array}{l} 221 \\ 1 \ 8 \ 1232181231513221 \end{array} \Delta = -20, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
\text{Base: } \begin{array}{l} 221 \\ 1 \ 8 \ 1232181232171231 \end{array} \\
\begin{array}{l} 221 \\ 1 \ 8 \ 1232181232171231 \end{array} \begin{array}{l} \textcolor{red}{3} \ [\# = 2] \\ \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \end{array} \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
\begin{array}{l} 221 \textcolor{red}{2} \\ 1 \ 8 \ 1232181232171231 \end{array} \Delta = -66, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\begin{array}{l} 221 \\ 1 \ 8 \ 1232181232171231 \end{array} \begin{array}{l} \textcolor{red}{3} \ [\# = 4] \\ \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \end{array} \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\begin{array}{l} 221 \\ 1 \ 8 \ 1232181232171231 \end{array} \begin{array}{l} \textcolor{red}{3} \ [\# = 3] \\ \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \end{array} \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\begin{array}{l} 221 \\ 1 \ 8 \ 1232181232171231 \end{array} \begin{array}{l} \textcolor{red}{3} \ [\# = 5] \\ \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \end{array} \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\begin{array}{l} 221 \\ 1 \ 8 \ 1232181232171231 \end{array} \Delta = -75, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\begin{array}{l} 221 \\ 1 \ 8 \ 1232181 \end{array} \begin{array}{l} \textcolor{red}{2} \\ 1232171231 \end{array} \Delta = -66, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\begin{array}{l} 221 \\ 1 \ 8 \ 1232181232171231 \end{array} \Delta = -66, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\begin{array}{l} 221 \\ 1 \ 8 \ 1232181 \end{array} \begin{array}{l} \textcolor{red}{2} \\ 1232171231 \end{array} \Delta = -66, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\begin{array}{l} 221 \\ 1 \ 8 \ 1232181232171231 \end{array} \Delta = -66, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\begin{array}{l} 221 \\ 1 \ 8 \ 1232181232171231 \end{array} \begin{array}{l} \textcolor{red}{2} \\ \Delta = -66, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \end{array} \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\text{Base: } \begin{array}{l} 221 \\ 1 \ 8 \ 1232181232181222 \end{array} \\
\begin{array}{l} 221 \\ 1 \ 8 \ 1232181232181222 \end{array} \begin{array}{l} \textcolor{red}{2} \\ \Delta = -85, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \end{array} \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\begin{array}{l} 221 \\ 1 \ 8 \ 1232181232181 \end{array} \begin{array}{l} \textcolor{red}{2} \\ 222 \end{array} \Delta = -85, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
& \overset{221}{1\ 8\ 1232181232181222} \quad \Delta = -85, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{221}{1\ 8\ 1232181232181222} \quad \Delta = -94, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{221}{1\ 8\ 1232181\overset{2}{232}181222} \quad \Delta = -85, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{221\overset{2}{2}}{1\ 8\ 1232181232181222} \quad \Delta = -85, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{221}{1\ 8\ 1232181232181\overset{2}{222}} \quad \Delta = -85, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{221}{1\ 8\ 12321812321}\overset{221}{8\ 1}$

$$\begin{aligned}
& \overset{221}{1\ 8\ 12321812321}\overset{221}{8\ 1} \quad \Delta = -85, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{221\overset{2}{2}}{1\ 8\ 12321812321}\overset{221}{8\ 1} \quad \Delta = -85, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{221}{1\ 8\ 12321812321}\overset{221}{8\ 1} \quad \Delta = -94, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{221}{1\ 8\ 1232181\overset{2}{232}1}\overset{221}{8\ 1} \quad \Delta = -85, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{aligned}$$

Base: $7123151321812321\overset{321}{8\ 1}$

$$\begin{aligned}
& 7123151321812321\overset{321}{8\ 1}\overset{2}{12} \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 7123151321812321\overset{2\overset{321}{2}}{8\ 1} \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& 7123151321812321\overset{2}{8\ 1}\overset{321}{1} \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& 7123151321812321\overset{2}{8\ 1}\overset{321}{1} \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
& 7123151\overset{2}{321}812321\overset{321}{8\ 1} \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 86 & 18 \\ 2 & 18 & 8 \end{pmatrix} \\
& 7123151\overset{2}{321}812321\overset{321}{8\ 1} \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& 7123151321812321\overset{2}{8\ 1}\overset{321}{1} \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 7123151321812321\overset{321}{8\ 1} \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

Base: $6131612321812321\overset{321}{8\ 1}$

$$6131612321812321\overset{321}{8\ 1}\overset{2}{12} \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$\begin{aligned}
6131612321812321 \overset{321}{8} 1 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
6131612321812321 \overset{2}{8} 1 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
6131612321812321 \overset{2}{8} 1 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
6131612321812321 \overset{2}{8} 1 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
6131612321812321 \overset{2}{8} 1 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 169 & \begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
6131612321812321 \overset{2}{8} 1 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
6131612321812321 \overset{2}{8} 1 \quad \Delta = -125, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 225 & \begin{pmatrix} 0 & 15 & 2 \\ 15 & 163 & 25 \\ 2 & 25 & 8 \end{pmatrix}
\end{aligned}$$

Base: $1 \overset{12321}{8} 12321812315131$

$$\begin{aligned}
1 \overset{12321}{8} 12321812315131 \overset{3}{\text{red}} [\# = 3] \quad \Delta = -29, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1 \overset{12321}{8} 12321812315131 \quad \Delta = -21, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1 \overset{12321}{8} 12321812315131 \overset{2}{\text{red}} \quad \Delta = -21, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1 \overset{12321}{8} 12321812315131 \overset{2}{\text{red}} \quad \Delta = -21, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1 \overset{12321}{8} 12321812315131 \overset{2}{\text{red}} \quad \Delta = -21, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{12321}{8} 12321812315131 \overset{2}{\text{red}} \quad \Delta = -21, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
1 \overset{12321}{8} 12321812315131 \overset{2}{\text{red}} \quad \Delta = -21, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1 \overset{312321}{8} 12321812315131 [\# = 5] \quad \Delta = -29, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
1 \overset{312321}{8} 12321812315131 [\# = 7] \quad \Delta = -29, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 242 & 34 \\ 2 & 34 & 8 \end{pmatrix} \\
1 \overset{12321}{8} 12321812315131 \overset{3}{\text{red}} [\# = 6] \quad \Delta = -29, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 69 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \overset{12321}{8} 12321812315131 \overset{3}{\text{red}} [\# = 7] \quad \Delta = -29, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 95 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
1 \overset{12321}{8} 12321812315131 \overset{3}{\text{red}} [\# = 8] \quad \Delta = -29, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 125 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

$1 \overset{312321}{8} 12321812315131$ $[\# = 4]$	$\Delta = -29, n_+ = 1, n_- = 21, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix}$
$1 \overset{312321}{8} 12321812315131$ $[\# = 3]$	$\Delta = -29, n_+ = 1, n_- = 21, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$
$1 \overset{312321}{8} 12321812315131$ $[\# = 2]$	$\Delta = -29, n_+ = 1, n_- = 21, n_0 = 0, \det = 4$	$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix}$
$1 \overset{12321}{8} 12321812315131 \color{red}{3}$ $[\# = 5]$	$\Delta = -29, n_+ = 1, n_- = 21, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix}$
$1 \overset{312321}{8} 12321812315131$ $[\# = 6]$	$\Delta = -29, n_+ = 1, n_- = 21, n_0 = 0, \det = 36$	$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 177 & 29 \\ 2 & 29 & 8 \end{pmatrix}$
$1 \overset{12321}{8} 12321812315131 \color{red}{3}$ $[\# = 4]$	$\Delta = -29, n_+ = 1, n_- = 21, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}$
$1 \overset{12321}{8} 12321812315131 \color{red}{3}$ $[\# = 2]$	$\Delta = -29, n_+ = 1, n_- = 21, n_0 = 0, \det = 4$	$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix}$
$1 \overset{12321}{8} 12321812315131 \color{red}{3}$ $[\# = 9]$	$\Delta = -29, n_+ = 1, n_- = 21, n_0 = 0, \det = 81$	$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 159 & 17 \\ 2 & 17 & 8 \end{pmatrix}$
$1 \overset{312321}{8} 12321812315131$ $[\# = 9]$	$\Delta = -29, n_+ = 1, n_- = 21, n_0 = 0, \det = 81$	$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 402 & 44 \\ 2 & 44 & 8 \end{pmatrix}$
$1 \overset{312321}{8} 12321812315131$ $[\# = 8]$	$\Delta = -29, n_+ = 1, n_- = 21, n_0 = 0, \det = 64$	$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 317 & 39 \\ 2 & 39 & 8 \end{pmatrix}$
Base: $1 \overset{1321}{8} 123218123151321$		
$1 \overset{1321}{8} 123218123151321 \color{red}{3}$ $[\# = 9]$	$\Delta = -28, n_+ = 1, n_- = 21, n_0 = 0, \det = 81$	$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix}$
$1 \overset{1321}{8} 123218123151321 \color{red}{3}$ $[\# = 8]$	$\Delta = -28, n_+ = 1, n_- = 21, n_0 = 0, \det = 64$	$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix}$
$1 \overset{1321}{8} 123218123151321 \color{red}{3}$ $[\# = 7]$	$\Delta = -28, n_+ = 1, n_- = 21, n_0 = 0, \det = 49$	$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}$
$1 \overset{1321}{8} 123218123151321 \color{red}{3}$ $[\# = 4]$	$\Delta = -28, n_+ = 1, n_- = 21, n_0 = 0, \det = 16$	$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix}$
$1 \overset{31321}{8} 123218123151321$ $[\# = 5]$	$\Delta = -28, n_+ = 1, n_- = 21, n_0 = 0, \det = 25$	$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}$
$1 \overset{1321}{8} 123218123151321 \color{red}{3}$ $[\# = 3]$	$\Delta = -28, n_+ = 1, n_- = 21, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix}$
$1 \overset{31321}{8} 123218123151321$ $[\# = 2]$	$\Delta = -28, n_+ = 1, n_- = 21, n_0 = 0, \det = 4$	$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix}$
$1 \overset{1321}{8} 123218123151321 \color{red}{3}$ $[\# = 2]$	$\Delta = -28, n_+ = 1, n_- = 21, n_0 = 0, \det = 4$	$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}$
$1 \overset{31321}{8} 123218123151321$ $[\# = 3]$	$\Delta = -28, n_+ = 1, n_- = 21, n_0 = 0, \det = 9$	$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix}$

$$\begin{aligned}
& \overset{1321}{1} \overset{2}{8} 123218123151321 \quad \Delta = -20, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{1321}{1} \overset{2}{8} 123218123151321 \overset{3}{\# = 6} \quad \Delta = -28, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{1321}{1} \overset{2}{8} 123218123151321 \quad \Delta = -20, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{1321}{1} \overset{2}{8} 1232181 \overset{2}{23} 151321 \quad \Delta = -20, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{1321}{1} \overset{2}{8} 123218123151321 \quad \Delta = -20, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{1321}{1} \overset{2}{8} 123218123151 \overset{2}{32} 1 \quad \Delta = -20, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{1321}{1} \overset{2}{8} 123218123151 \overset{2}{32} 1 \quad \Delta = -20, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{1321}{1} \overset{2}{8} 123218123151321 \overset{3}{\# = 5} \quad \Delta = -28, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{3}{1} \overset{1321}{8} 123218123151321 \overset{3}{\# = 4} \quad \Delta = -28, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix}
\end{aligned}$$

Base: $71232181 \overset{1}{1232181231513}$

$$\begin{aligned}
& 712321 \overset{2}{8} \overset{1}{1232181231513} \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 128 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
& 712321 \overset{1}{8} \overset{2}{1232181231513} \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 169 \quad \begin{pmatrix} 0 & 13 & 2 \\ 13 & 115 & 21 \\ 2 & 21 & 8 \end{pmatrix} \\
& 712321 \overset{1}{8} \overset{2}{1232181231513} \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 225 \quad \begin{pmatrix} 0 & 15 & 2 \\ 15 & 163 & 25 \\ 2 & 25 & 8 \end{pmatrix} \\
& 712321 \overset{2}{8} \overset{1}{1232181231513} \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 40 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& 712321 \overset{1}{8} 1232181231 \overset{2}{513} \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 712321 \overset{1}{8} 1232181 \overset{2}{23} 1513 \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 712321 \overset{1}{8} 1232181 \overset{2}{23} 1513 \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& 712321 \overset{2}{8} \overset{1}{1232181231513} \quad \Delta = -126, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

Base: $171232181 \overset{1}{123218123151}$

$$1712321 \overset{1}{8} \overset{2}{123218123151} \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix}$$

$$\begin{aligned}
17123218123218123151 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
17123218123218123151 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
17123218123218123151 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
17123218123218123151 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
17123218123218123151 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
17123218123218123151 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix}
\end{aligned}$$

Base: 41321812321812321812

$$\begin{aligned}
41321812321812321812 \quad \Delta = -122, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
41321812321812321812 \quad \Delta = -122, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
41321812321812321812 \quad \Delta = -122, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
41321812321812321812 \quad \Delta = -122, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
41321812321812321812 \quad \Delta = -122, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
41321812321812321812 \quad \Delta = -122, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
41321812321812321812 \quad \Delta = -122, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
41321812321812321812 \quad \Delta = -122, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
41321812321812321812 \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix}
\end{aligned}$$

Base: 22181232181232181231

$$\begin{aligned}
22181232181232181231 \quad \Delta = -102, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
22181232181232181231 \quad \Delta = -94, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
22181232181232181231 \quad \Delta = -94, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{array}{ll}
221812321\overset{21}{8}1232181231 & \Delta = -94, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
221812321\overset{2}{18}1232181231 & \Delta = -94, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
221812321\overset{1}{18}1232181231 & \Delta = -94, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
221812321\overset{21}{18}1232181231 & \Delta = -94, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
221812321\overset{2}{18}1232181231 & \Delta = -94, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
221812321\overset{1}{18}1232181231\overset{3}{\text{}}_{[\# = 5]} & \Delta = -102, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
221812321\overset{1}{18}1232181231\overset{3}{\text{}}_{[\# = 4]} & \Delta = -102, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
221812321\overset{1}{18}1232181231\overset{3}{\text{}}_{[\# = 2]} & \Delta = -102, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
221812321\overset{3}{18}1232181231 & \Delta = -103, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\text{Base: } 612321\overset{1}{18}1232181232161 & \\
612321\overset{1}{18}12321812321\overset{2}{6}1 & \Delta = -119, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
612321\overset{21}{8}1232181232161 & \Delta = -119, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
612321\overset{2}{18}1232181232161 & \Delta = -119, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
612321\overset{21}{18}1232181232161 & \Delta = -119, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
612321\overset{1}{18}12321812321\overset{2}{6}1 & \Delta = -119, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
612321\overset{1}{18}12321812321\overset{2}{6}1 & \Delta = -119, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
612321\overset{12}{18}1232181232161 & \Delta = -119, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\text{Base: } 1\overset{12321}{8}12321812321612 & \\
1\overset{312321}{8}12321812321612_{[\# = 2]} & \Delta = -49, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 21 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1\overset{12321}{8}12321812321\overset{2}{6}1 & \Delta = -41, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{array}$$

$$\begin{array}{ll}
\begin{array}{l} 12321 \\ 1 \ 8 \ 12321812321612 \end{array} \quad \Delta = -41, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\begin{array}{l} 12321 \\ 1 \ 8 \ 12321812321612 \end{array} \quad \Delta = -41, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\begin{array}{l} 12321 \\ 1 \ 8 \ 12321812321612 \end{array} \quad \Delta = -41, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
\begin{array}{l} 12321 \\ 1 \ 8 \ 12321812321612 \end{array} \quad \Delta = -41, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\begin{array}{l} 12321 \\ 1 \ 8 \ 12321812321612 \end{array} \quad \Delta = -41, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\begin{array}{l} 312321 \\ 1 \ 8 \ 12321812321612 \end{array} \quad [\#=9] \quad \Delta = -49, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 483 & 53 \\ 2 & 53 & 8 \end{pmatrix} \\
\begin{array}{l} 312321 \\ 1 \ 8 \ 12321812321612 \end{array} \quad [\#=8] \quad \Delta = -49, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 381 & 47 \\ 2 & 47 & 8 \end{pmatrix} \\
\begin{array}{l} 12321 \\ 1 \ 8 \ 12321812321612 \end{array} \quad \Delta = -49, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
\begin{array}{l} 312321 \\ 1 \ 8 \ 12321812321612 \end{array} \quad [\#=7] \quad \Delta = -49, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 291 & 41 \\ 2 & 41 & 8 \end{pmatrix} \\
\begin{array}{l} 312321 \\ 1 \ 8 \ 12321812321612 \end{array} \quad [\#=3] \quad \Delta = -49, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 51 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
\begin{array}{l} 312321 \\ 1 \ 8 \ 12321812321612 \end{array} \quad [\#=4] \quad \Delta = -49, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 93 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
\begin{array}{l} 312321 \\ 1 \ 8 \ 12321812321612 \end{array} \quad [\#=6] \quad \Delta = -49, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 213 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
\begin{array}{l} 312321 \\ 1 \ 8 \ 12321812321612 \end{array} \quad [\#=5] \quad \Delta = -49, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 147 & 29 \\ 2 & 29 & 8 \end{pmatrix} \\
\text{Base: } \begin{array}{l} 1321 \\ 1 \ 8 \ 123218123216131 \end{array} & \\
\begin{array}{l} 1321 \\ 1 \ 8 \ 123218123216131 \end{array} \quad [\#=8] \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 61 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
\begin{array}{l} 1321 \\ 1 \ 8 \ 123218123216131 \end{array} \quad [\#=2] \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix} \\
\begin{array}{l} 1321 \\ 1 \ 8 \ 123218123216131 \end{array} \quad [\#=9] \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 78 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\begin{array}{l} 31321 \\ 1 \ 8 \ 123218123216131 \end{array} \quad [\#=5] \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
\begin{array}{l} 1321 \\ 1 \ 8 \ 123218123216131 \end{array} \quad [\#=4] \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
\begin{array}{l} 31321 \\ 1 \ 8 \ 123218123216131 \end{array} \quad [\#=4] \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
& \overset{1321}{1} \overset{2}{8} 123218123216131 \quad \Delta = -47, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{31321}{1} \overset{2}{8} 123218123216131 \text{ [#=3]} \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{1321}{1} \overset{2}{8} 123218123216131 \quad \Delta = -47, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{1321}{1} \overset{2}{8} 123218123216131 \overset{3}{3} \text{ [#=7]} \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 46 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{1321}{1} \overset{2}{8} 123218123216131 \overset{3}{3} \text{ [#=6]} \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 33 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{1321}{1} \overset{2}{8} 123218123216131 \overset{3}{3} \text{ [#=3]} \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \overset{1321}{1} \overset{2}{8} 123218123216131 \overset{3}{3} \text{ [#=5]} \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{31321}{1} \overset{2}{8} 123218123216131 \text{ [#=2]} \quad \Delta = -55, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{1321}{1} \overset{2}{8} 123218123216131 \quad \Delta = -47, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{1321}{1} \overset{2}{8} 123218123216131 \quad \Delta = -47, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{1321}{1} \overset{2}{8} 123218123216131 \quad \Delta = -47, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{1321}{1} \overset{2}{8} 123218123216131 \quad \Delta = -47, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{16}12321812321\overset{1}{8}12321$

$$\begin{aligned}
& \overset{1}{16}12321812321\overset{1}{8}12321 \overset{3}{3} \text{ [#=7]} \quad \Delta = -49, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
& \overset{1}{16}12321812321\overset{1}{8}12321 \overset{3}{3} \text{ [#=2]} \quad \Delta = -49, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{1}{16}12321812321\overset{1}{8}12321 \overset{3}{3} \text{ [#=8]} \quad \Delta = -49, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
& \overset{1}{16}12321812321\overset{1}{8}12321 \overset{3}{3} \text{ [#=9]} \quad \Delta = -49, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
& \overset{1}{16}12321812321\overset{1}{8}12321 \overset{3}{3} \text{ [#=6]} \quad \Delta = -49, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& \overset{1}{16}12321812321\overset{1}{8}12321 \overset{3}{3} \text{ [#=3]} \quad \Delta = -49, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{1}{16}12321812321\overset{2}{8}12321 \quad \Delta = -41, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
1612321812321812321 \quad \Delta = -41, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
1612321812321812321 \quad \Delta = -41, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1612321812321812321 \quad \Delta = -41, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
16123218123218123213 \text{ [#=4]} \quad \Delta = -49, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
1612321812321812321 \quad \Delta = -41, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 & \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
16123218123218123213 \text{ [#=5]} \quad \Delta = -49, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix}
\end{aligned}$$

Base: 17123218123218123215

$$\begin{aligned}
17123218123218123215 \quad \Delta = -120, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
17123218123218123215 \quad \Delta = -120, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
17123218123218123215 \quad \Delta = -120, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
17123218123218123215 \quad \Delta = -120, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 100 & \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
17123218123218123215 \quad \Delta = -120, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
17123218123218123215 \quad \Delta = -120, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 & \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
17123218123218123215 \quad \Delta = -120, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 21712321812321812321

$$\begin{aligned}
217123218123218123213 \text{ [#=6]} \quad \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 & \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
21712321812321812321 \quad \Delta = -68, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 & \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
217123218123218123213 \text{ [#=9]} \quad \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 & \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
217123218123218123213 \text{ [#=8]} \quad \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 & \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
217123218123218123213 \text{ [#=7]} \quad \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 & \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& 21712321812321812321\overset{1}{3} \text{ [#=5]} \quad \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& 21712321812321812321\overset{1}{3} \text{ [#=4]} \quad \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& 21712321812321812321\overset{1}{3} \text{ [#=3]} \quad \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 21712321812321812321\overset{2}{21} \quad \Delta = -68, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 21712321812321812321\overset{21}{8} \quad \Delta = -68, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 21712321812321812321\overset{2}{8} \quad \Delta = -68, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 21712321812321812321\overset{1}{8}\overset{2}{21} \quad \Delta = -68, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 21712321812321812321\overset{12}{8} \quad \Delta = -68, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& 21712321812321812321\overset{1}{3} \text{ [#=2]} \quad \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 21712321812321812321\overset{21}{8} \quad \Delta = -68, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \text{Base: } 1 \overset{51321}{8} 12321812321712 \\
& 51321\overset{21}{8} 12321812321712 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& 1 \overset{51321}{8} \overset{2}{12321812321712} \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 124 & 22 \\ 2 & 22 & 8 \end{pmatrix} \\
& 1 \overset{51321}{8} \overset{2}{12321812321712} \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& 1 \overset{51321}{8} \overset{2}{12321812321712} \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 196 & 28 \\ 2 & 28 & 8 \end{pmatrix} \\
& 1 \overset{51321}{8} \overset{2}{12321812321712} \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& 1 \overset{51321}{8} 12321812321712 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& 1 \overset{51321}{8} 12321812321712 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 168 & 26 \\ 2 & 26 & 8 \end{pmatrix} \\
& 1 \overset{51321}{8} \overset{2}{12321812321712} \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix} \\
& \text{Base: } 1 \overset{1321}{8} 123218123217122
\end{aligned}$$

$$\begin{array}{l}
1 \overset{1321}{8} 123218123217122 \overset{3}{\Delta} = -75, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 4 & 4 \\ 4 & 9 \end{pmatrix} \\
1 \overset{31321}{8} 123218123217122 \text{ [#=5]} \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 122 & 24 \\ 2 & 24 & 8 \end{pmatrix} \\
1 \overset{31321}{8} 123218123217122 \text{ [#=4]} \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 77 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
1 \overset{31321}{8} 123218123217122 \text{ [#=3]} \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1 \overset{31321}{8} 123218123217122 \text{ [#=2]} \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 17 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1 \overset{1321}{8} 123218123217122 \overset{2}{\Delta} = -66, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1 \overset{1321}{8} 123218123217122 \overset{2}{\Delta} = -66, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1 \overset{1321}{8} 123218123217122 \overset{2}{\Delta} = -66, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
1 \overset{1321}{8} 123218123217122 \overset{2}{\Delta} = -66, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
1 \overset{1321}{8} 123218123217122 \overset{2}{\Delta} = -66, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
1 \overset{1321}{8} 123218123217122 \overset{2}{\Delta} = -66, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\text{Base: } 1 \overset{1}{7} 12321 \overset{1}{8} 12321812321 \\
1 \overset{1}{7} 12321 \overset{1}{8} 12321812321 \overset{3}{\Delta} \text{ [#=4]} \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
1 \overset{1}{7} 12321 \overset{1}{8} 12321812321 \overset{2}{\Delta} = -68, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
1 \overset{1}{7} 12321 \overset{1}{8} 12321812321 \overset{2}{\Delta} = -68, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
1 \overset{1}{7} 12321 \overset{21}{8} 12321812321 \overset{1}{\Delta} = -68, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
1 \overset{12}{7} 12321 \overset{1}{8} 12321812321 \overset{1}{\Delta} = -68, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
1 \overset{1}{7} 12321 \overset{21}{8} 12321812321 \overset{1}{\Delta} = -68, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1 \overset{1}{7} 12321 \overset{1}{8} 12321812321 \overset{2}{\Delta} = -68, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
1 \overset{1}{7} 12321 \overset{1}{8} 12321812321 \overset{3}{\Delta} \text{ [#=2]} \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{array}$$

$$\begin{aligned}
& \overset{1}{17}12321\overset{1}{8}12321812321\overset{1}{3} \text{ [#=3]} \quad \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{1}{17}12321\overset{1}{8}12321812321\overset{1}{3} \text{ [#=5]} \quad \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{1}{17}12321\overset{1}{8}12321812321\overset{1}{3} \text{ [#=6]} \quad \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 105 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& \overset{1}{17}12321\overset{1}{8}12321812321\overset{1}{3} \text{ [#=7]} \quad \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 144 & 20 \\ 2 & 20 & 8 \end{pmatrix} \\
& \overset{1}{17}12321\overset{1}{8}12321812321\overset{1}{3} \text{ [#=8]} \quad \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 189 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& \overset{1}{17}12321\overset{1}{8}12321812321\overset{1}{3} \text{ [#=9]} \quad \Delta = -76, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 240 & 26 \\ 2 & 26 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{1}{17}12321812321\overset{1}{8}12315$

$$\begin{aligned}
& \overset{1}{17}1232181\overset{2}{2}321\overset{1}{8}12315 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{1}{17}1232181\overset{2}{2}321\overset{1}{8}12315 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{1}{17}12321812321\overset{12}{2}315 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{12}{17}12321812321\overset{1}{8}12315 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{1}{17}12321812321\overset{21}{8}12315 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{1}{17}12321812321\overset{21}{8}12315 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{1}{17}12321812321\overset{1}{8}1\overset{2}{2}315 \quad \Delta = -118, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{21}{17}12321812321\overset{1}{8}1231$

$$\begin{aligned}
& \overset{21}{17}1232181\overset{2}{2}321\overset{1}{8}1231 \quad \Delta = -66, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
& \overset{21}{17}12321812321\overset{1}{8}1231\overset{1}{3} \text{ [#=5]} \quad \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 72 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{21}{17}12321812321\overset{1}{8}1231\overset{1}{3} \text{ [#=4]} \quad \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 45 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{21}{17}12321812321\overset{1}{8}1231 \quad \Delta = -66, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{21}{17}12321812321\overset{12}{2}31 \quad \Delta = -66, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{21}{1} \overset{2}{7} 1232181232181231 \quad \Delta = -66, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
& \overset{21}{1} 712321812321 \overset{21}{1} 81231 \quad \Delta = -66, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{21}{1} 71232181232181231 \overset{1}{3} \text{ [#=2]} \quad \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{21}{1} 71232181232181231 \overset{1}{3} \text{ [#=3]} \quad \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 24 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
& \overset{21}{1} 712321812321 \overset{2}{8} 1231 \quad \Delta = -66, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: 4132181232181232181

$$\begin{aligned}
& 41321812321 \overset{21}{8} 1232181 \quad \Delta = -122, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& 41321812321 \overset{1}{8} 12321 \overset{1}{8} 1 \quad \Delta = -124, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& 41321812321 \overset{1}{8} 12321 \overset{1}{8} 1 \quad \Delta = -122, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 2 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& 4132181 \overset{2}{2} 321 \overset{1}{8} 12321 \overset{1}{8} 1 \quad \Delta = -122, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 43 & 13 \\ 2 & 13 & 8 \end{pmatrix} \\
& 41321812321 \overset{21}{8} 12321 \overset{1}{8} 1 \quad \Delta = -122, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 121 \quad \begin{pmatrix} 0 & 11 & 2 \\ 11 & 75 & 17 \\ 2 & 17 & 8 \end{pmatrix} \\
& 41321812321 \overset{12}{8} 12321 \overset{1}{8} 1 \quad \Delta = -122, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 100 \quad \begin{pmatrix} 0 & 10 & 2 \\ 10 & 68 & 16 \\ 2 & 16 & 8 \end{pmatrix} \\
& 41321812321 \overset{1}{8} 12321 \overset{1}{8} 1 \quad \Delta = -122, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& 41321812321 \overset{1}{8} 12321 \overset{21}{8} 1 \quad \Delta = -122, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 28 & 10 \\ 2 & 10 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{21}{1} 81232181232181231$

$$\begin{aligned}
& \overset{21}{1} 812321 \overset{21}{8} 1232181231 \quad \Delta = -94, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{21}{1} 812321 \overset{1}{8} 1232181231 \overset{1}{3} \text{ [#=2]} \quad \Delta = -102, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{21}{1} 81232181232181231 \overset{1}{3} \text{ [#=3]} \quad \Delta = -102, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{21}{1} 812321 \overset{1}{8} 1232181231 \overset{1}{3} \text{ [#=5]} \quad \Delta = -102, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 47 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{21}{1} 81232181232181231 \overset{1}{3} \text{ [#=4]} \quad \Delta = -102, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 29 & 7 \\ 2 & 7 & 8 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& \overset{21}{1} \overset{1}{8} 1232181232181231 \quad \Delta = -94, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
& \overset{21}{1} \overset{21}{8} 1232181232181231 \quad \Delta = -94, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{21}{1} \overset{1}{8} \overset{2}{1232181232181231} \quad \Delta = -94, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 19 & 9 \\ 2 & 9 & 8 \end{pmatrix} \\
& \overset{21}{1} \overset{1}{8} 1232181 \overset{2}{232181231} \quad \Delta = -94, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix} \\
& \overset{21}{1} \overset{1}{8} 12321812321 \overset{2}{181231} \quad \Delta = -94, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{21}{1} \overset{2}{8} \overset{1}{1232181232181231} \quad \Delta = -94, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}
\end{aligned}$$

Base: $\overset{21}{21} \overset{1}{8} 1232181231513221$

$$\begin{aligned}
& \overset{21}{21} \overset{2}{8} 1232181231513221 \quad \Delta = -11, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{21}{21} \overset{2}{8} 1232181231513221 \quad \Delta = -11, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 1 & 3 \\ 2 & 3 & 8 \end{pmatrix} \\
& \overset{21}{21} \overset{2}{8} 1232181231513221 \quad \Delta = -11, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
& \overset{221}{21} \overset{2}{8} 1232181231513221 \quad \Delta = -11, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 6 \\ 2 & 6 & 8 \end{pmatrix} \\
& \overset{21}{21} \overset{1}{8} 1232181231513221 \overset{3}{\text{[#}=7]} \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 49 \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 193 & 27 \\ 2 & 27 & 8 \end{pmatrix} \\
& \overset{21}{21} \overset{1}{8} 1232181231513221 \overset{3}{\text{[#}=6]} \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 36 \quad \begin{pmatrix} 0 & 6 & 2 \\ 6 & 141 & 23 \\ 2 & 23 & 8 \end{pmatrix} \\
& \overset{21}{21} \overset{1}{8} 1232181231513221 \overset{3}{\text{[#}=5]} \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 97 & 19 \\ 2 & 19 & 8 \end{pmatrix} \\
& \overset{21}{21} \overset{1}{8} 1232181231513221 \overset{3}{\text{[#}=4]} \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 61 & 15 \\ 2 & 15 & 8 \end{pmatrix} \\
& \overset{21}{21} \overset{1}{8} 1232181231513221 \overset{3}{\text{[#}=8]} \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 64 \quad \begin{pmatrix} 0 & 8 & 2 \\ 8 & 253 & 31 \\ 2 & 31 & 8 \end{pmatrix} \\
& \overset{21}{21} \overset{1}{8} 1232181231513221 \overset{3}{\text{[#}=3]} \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 33 & 11 \\ 2 & 11 & 8 \end{pmatrix} \\
& \overset{21}{21} \overset{1}{8} 1232181231513221 \overset{3}{\text{[#}=2]} \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 13 & 7 \\ 2 & 7 & 8 \end{pmatrix} \\
& \overset{21}{21} \overset{1}{8} 1232181231513221 \overset{3}{\text{[#}=9]} \quad \Delta = -19, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 81 \quad \begin{pmatrix} 0 & 9 & 2 \\ 9 & 321 & 35 \\ 2 & 35 & 8 \end{pmatrix} \\
& \overset{21}{21} \overset{2}{8} 1232181231513221 \quad \Delta = -11, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 8 \end{pmatrix}
\end{aligned}$$

$$^{21}_{21} 8^{21} 1232181231513221 \quad \Delta = -11, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

Base: $^{21}_{21} 81232181232171231$

$$^{21}_{21} 81232181232171231^3_{[\# = 2]} \quad \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 8 \end{pmatrix}$$

$$^{21}_{21} 81232181232171231^2 \quad \Delta = -66, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

$$^{21}_{21} 81232181232171231^2 \quad \Delta = -66, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$^{221}_{21} 81232181232171231 \quad \Delta = -66, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$^{21}_{21} 81232181232171231^2 \quad \Delta = -66, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$^{21}_{21} 81232181232171231^3_{[\# = 5]} \quad \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 25 \quad \begin{pmatrix} 0 & 5 & 2 \\ 5 & 22 & 4 \\ 2 & 4 & 8 \end{pmatrix}$$

$$^{21}_{21} 81232181232171231^3_{[\# = 4]} \quad \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 3 \\ 2 & 3 & 8 \end{pmatrix}$$

$$^{21}_{21} 81232181232171231^2 \quad \Delta = -66, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$^{212}_{21} 81232181232171231 \quad \Delta = -66, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$^{21}_{21} 81232181232171231^3_{[\# = 3]} \quad \Delta = -74, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 2 \\ 2 & 2 & 8 \end{pmatrix}$$

Base: $^{21}_{21} 81232181232181222$

$$^{21}_{21} 81232181232181222^2 \quad \Delta = -85, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}$$

$$^{21}_{21} 81232181232181222^2 \quad \Delta = -85, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 10 & 6 \\ 2 & 6 & 8 \end{pmatrix}$$

$$^{212}_{21} 81232181232181222 \quad \Delta = -85, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix}$$

$$^{221}_{21} 81232181232181222 \quad \Delta = -85, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$^{21}_{21} 81232181232181222^2 \quad \Delta = -85, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix}$$

$$^{21}_{21} 81232181232181222^2 \quad \Delta = -85, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix}$$

Base: $^{221}_{1} 812321812321812$

$$^{221}_{1} 812321812321812^{21} \quad \Delta = -94, n_+ = 1, n_- = 21, n_0 = 0, |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix}$$

$$\begin{array}{ll}
\begin{array}{c} 221 \\ 1 \ 8 \ 12321812321 \end{array} \begin{array}{c} 21 \ 2 \\ 8 \ 12 \end{array} & \Delta = -85, \ n_+ = 1, \ n_- = 21, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 8 \end{pmatrix} \\
\begin{array}{c} 221 \\ 1 \ 8 \ 12321812321 \end{array} \begin{array}{c} 2 \\ 8 \ 12 \end{array} \begin{array}{c} 21 \\ 8 \ 12 \end{array} & \Delta = -85, \ n_+ = 1, \ n_- = 21, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 26 & 10 \\ 2 & 10 & 8 \end{pmatrix} \\
\begin{array}{c} 221 \\ 1 \ 8 \ 12321812321 \end{array} \begin{array}{c} 21 \\ 8 \ 12 \end{array} \begin{array}{c} 21 \\ 8 \ 12 \end{array} & \Delta = -85, \ n_+ = 1, \ n_- = 21, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 18 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\begin{array}{c} 221 \\ 1 \ 8 \ 12321812321 \end{array} \begin{array}{c} 2 \\ 8 \ 12 \end{array} \begin{array}{c} 21 \\ 8 \ 12 \end{array} & \Delta = -85, \ n_+ = 1, \ n_- = 21, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\begin{array}{c} 221 \ 2 \\ 1 \ 8 \ 12321812321 \end{array} \begin{array}{c} 21 \\ 8 \ 12 \end{array} & \Delta = -85, \ n_+ = 1, \ n_- = 21, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\begin{array}{c} 221 \\ 1 \ 8 \ 12321812321 \end{array} \begin{array}{c} 21 \\ 8 \ 12 \end{array} & \Delta = -85, \ n_+ = 1, \ n_- = 21, \ n_0 = 0, \ |\det| = 9 \quad \begin{pmatrix} 0 & 3 & 2 \\ 3 & 28 & 12 \\ 2 & 12 & 8 \end{pmatrix} \\
\textbf{Base:} \begin{array}{c} 21 \\ 21 \ 8 \ 12321812321 \end{array} \begin{array}{c} 21 \\ 8 \ 12 \end{array} & \\
\begin{array}{c} 21 \ 2 \\ 21 \ 8 \ 12321812321 \end{array} \begin{array}{c} 21 \\ 8 \ 12 \end{array} & \Delta = -85, \ n_+ = 1, \ n_- = 21, \ n_0 = 0, \ |\det| = 4 \quad \begin{pmatrix} 0 & 2 & 2 \\ 2 & 12 & 8 \\ 2 & 8 & 8 \end{pmatrix} \\
\begin{array}{c} 21 \ 2 \\ 21 \ 8 \ 12321812321 \end{array} \begin{array}{c} 21 \\ 8 \ 12 \end{array} & \Delta = -85, \ n_+ = 1, \ n_- = 21, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 42 & 14 \\ 2 & 14 & 8 \end{pmatrix} \\
\begin{array}{c} 21 \\ 21 \ 8 \ 12321812321 \end{array} \begin{array}{c} 2 \\ 8 \ 12 \end{array} \begin{array}{c} 21 \\ 8 \ 12 \end{array} & \Delta = -85, \ n_+ = 1, \ n_- = 21, \ n_0 = 0, \ |\det| = 16 \quad \begin{pmatrix} 0 & 4 & 2 \\ 4 & 34 & 12 \\ 2 & 12 & 8 \end{pmatrix}
\end{array}$$